GRAND CANYON NATIONAL PARK
GRAND CANYON, ARIZONA

Rehabilitate North Rim Infrastructure

GRCA 318719

SPECIFICATIONS

Volume C (Divisions 26-46)

CONSTRUCTION DOCUMENTS

May 2026

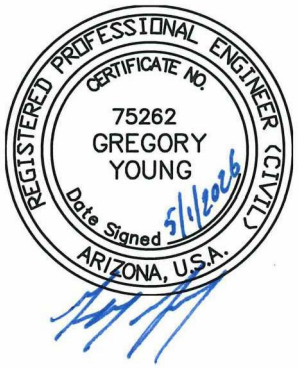
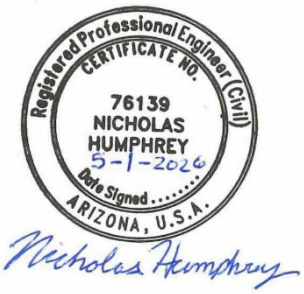
NATIONAL PARK SERVICE
DENVER SERVICE CENTER

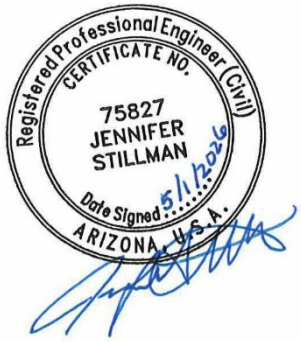
This page intentionally left blank.

SEALS AND SIGNATURES - Volume C

It is a violation of applicable laws and regulations governing professional licensing and registration for any person, unless acting under the direction of the licensed and registered design professional(s) indicated above, to alter in any way the Specifications in this project manual.

	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Juan Coronado, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 32 32 16, 41 22 23, 46 13 15
	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Adam Putman, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 33 52 10, 33 56 13, 33 56 16, 40 42 00
	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Travis Moore, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 26 05 00, 26 05 19, 26 05 26, 26 05 33, 26 05 36, 26 05 48, 26 08 13, 26 09 13, 26 09 16, 26 09 43, 26 22 13, 26 23 00, 26 24 13, 26 24 16, 26 24 19, 26 25 00, 26 26 13, 26 27 26, 26 28 00, 26 28 02, 26 28 16, 26 29 23, 26 32 14, 26 35 26, 26 41 13, 26 43 13, 26 50 00, 40 41 13

	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Tou Vang, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 26 05 13, 26 05 43, 26 05 46, 26 05 73, 26 12 19, 26 13 16, 26 13 23, 26 31 00, 26 33 00, 33 71 16.23, 33 71 23.17
	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Gregory Young, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 31 10 00, 31 22 00, 31 23 00, 31 25 00, 32 12 00, 32 13 13, 32 15 40, 32 16 23, 32 31 13, 32 91 13, 32 92 21, 33 12 19
	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Nicholas Humphrey, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 31 23 33, 33 01 30.11, 33 05 05.33, 33 05 15, 33 05 31.11, 33 05 61, 33 31 11, 33 05 15, 33 05 24, 33 05 28, 33 05 33.23, 33 11 13, 33 16 33, 40 05 24, 40 05 25, 40 05 63, 40 05 65, 40 05 66, 43 21 00, 43 23 22, 43 41 45, 46 61 73

	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Brian Stephens-Hotopp, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 33 01 30.11, 33 05 05.33, 33 05 31.11, 33 05 61, 33 31 11
	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Jennifer Stillman, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 40 05 00, 40 05 07, 40 05 19, 40 05 23, 40 05 31, 40 05 51, 40 05 52, 40 05 61, 40 05 62, 40 05 64, 43 23 31.29, 43 24 17, 43 41 15, 43 41 16, 46 33 11, 46 41 11, 46 41 17, 46 61 33, 46 61 33.A
	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Josie Brainard, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 27 10 00, 33 81 19, 40 05 57, 40 61 13, 40 61 21, 40 61 43, 40 61 93, 40 61 93.A, 40 61 93.B, 40 61 96, 40 61 96.A, 40 61 96.B, 40 62 05, 40 63 43, 40 66 05, 40 66 40, 40 67 00, 40 67 63, 40 71 00, 40 72 00, 40 73 00, 40 74 00, 40 75 00, 40 76 00, 40 78 00

	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Nate Soule, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Section 31 71 00
	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Nick LaFronz, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 31 32 19, 31 68 13

END OF SEALS AND SIGNATURES

TABLE OF CONTENTS

DIVISION 26 — ELECTRICAL
26 00 00 - DIVISION 26 DIVIDER
26 05 00 - COMMON WORK RESULTS FOR ELECTRICAL
26 05 13 - MEDIUM VOLTAGE CABLE
26 05 19 - LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES
26 05 26 - GROUNDING AND BONDING
26 05 33 - RACEWAYS AND BOXES
26 05 36 - CABLE TRAY
26 05 43 - ELECTRICAL - EXTERIOR UNDERGROUND
26 05 46 - OVERHEAD POWER LINES
26 05 48 - ELECTRICAL SEISMIC RESTRAINT SYSTEMS
~~26 05 73~~ - NOT USED
26 08 13 - ACCEPTANCE TESTING
26 09 13 - ELECTRICAL METERING DEVICES
26 09 16 - CONTROL EQUIPMENT ACCESSORIES
26 09 43 - NETWORKED LIGHTING CONTROL SYSTEM
26 12 19 - DISTRIBUTION TRANSFORMERS
26 13 16 - MEDIUM VOLTAGE METAL ENCLOSED LOAD INTERRUPTER SWITCHGEAR
26 13 23 - SECTIONALIZING EQUIPMENT
26 22 13 - LOW-VOLTAGE DISTRIBUTION TRANSFORMERS
26 23 00 - GENERATOR PARALLELING SWITCHGEAR
26 24 16 - PANELBOARDS
26 24 19 - MOTOR CONTROL EQUIPMENT
26 25 00 - LOW VOLTAGE BUSWAY
26 26 13 - PACKAGE POWER SUPPLY
26 27 26 - WIRING DEVICES
26 28 00 - OVERCURRENT AND SHORT CIRCUIT PROTECTIVE DEVICES
26 28 02 - COORDINATED POWER SYSTEM PROTECTION
26 28 16 - SAFETY SWITCHES
26 29 23 - VARIABLE FREQUENCY DRIVES - LOW VOLTAGE
26 31 00 - PHOTOVOLTAIC POWER SYSTEM
26 32 14 - ENGINE GENERATOR - DIESEL
26 33 00 - BATTERY AND BATTERY CHARGING SYSTEMS
26 35 26 - HARMONIC CORRECTION EQUIPMENT
26 41 13 - LIGHTNING PROTECTION SYSTEM
26 43 13 - SURGE PROTECTION DEVICES FOR LOW-VOLTAGE ELECTRICAL POWER CIRCUITS
26 50 00 - INTERIOR AND EXTERIOR LIGHTING

DIVISION 27 — COMMUNICATIONS
27 00 00 - DIVISION 27 DIVIDER
27 10 00 - STRUCTURED CABLING

DIVISION 31 — EARTHWORK
31 00 00 - DIVISION 31 DIVIDER
31 10 00 - SITE CLEARING
31 22 00 - SITE GRADING
31 23 00 - EARTHWORK
31 23 33 - UTILITIES - TRENCHING AND BACKFILLING

31 25 00 - SOIL EROSION AND SEDIMENT CONTROL
31 32 19 - SLOPE ARMORING SYSTEM
31 68 13 - ROCK FOUNDATION ANCHORS
31 71 00 - DIRECTIONAL DRILLING AND CARRIER PIPE INSTALLATION

DIVISION 32 — EXTERIOR IMPROVEMENTS
32 00 00 - DIVISION 32 DIVIDER
32 12 00 - FLEXIBLE PAVING
32 13 13 - CONCRETE PAVING
32 15 40 - CRUSHED STONE SURFACING
32 16 23 - CONCRETE SIDEWALK AND STEPS
32 32 16 - PRECAST MODULAR BLOCK RETAINING WALL
32 91 13 - FINISH GRADING
32 92 21 - NATIVE GRASS SEEDING

DIVISION 33 — UTILITIES
33 00 00 - DIVISION 33 DIVIDER
33 01 30.11 - TELEVISION INSPECTION OF SEWERS
33 05 05.33 - INFILTRATION AND EXFILTRATION TESTING
33 05 15 - PRECAST CONCRETE UTILITY STRUCTURES
33 05 24 - STEEL UTILITY PIPE
33 05 28 - PIPE - SPOOLABLE COMPOSITE
33 05 31.11 - POLYVINYL CHLORIDE GRAVITY SEWER PIPE
33 05 33.23 - HIGH DENSITY POLYETHYLENE (HDPE) PIPE
33 05 61 - CONCRETE MANHOLES
33 11 13 - WATER MAIN CONSTRUCTION
33 12 19 - FIRE HYDRANT
33 16 33 - HYDROPNEUMATIC TANKS
33 31 11 - GRAVITY SEWER PIPELINE AND MANHOLE CONSTRUCTION
33 52 10 - SERVICE PIPING - FUEL SYSTEMS
33 56 13 - ABOVEGROUND FUEL STORAGE TANKS
33 56 16 - FUEL DAY TANK SYSTEMS
33 71 16.23 - TUBULAR STEEL POLES
33 71 23.17 - MICROPILE FOUNDATIONS
33 81 19 - FOLD OVER ANTENNA TOWER ASSEMBLY

DIVISION 40 — PROCESS INTERCONNECTIONS
40 00 00 - DIVISION DIVIDER
40 05 00 - PIPE AND PIPE FITTINGS - BASIC REQUIREMENTS
40 05 07 - PIPE SUPPORT SYSTEMS
40 05 19 - PIPE - DUCTILE IRON PROCESS PIPE
40 05 23 - PIPE - STAINLESS STEEL PROCESS PIPE AND TUBING
40 05 24 - PIPE - STEEL
40 05 25 - PIPE - PREFABRICATED, INSULATED AND JACKETED
40 05 31 - PIPE - PLASTIC
40 05 51 - COMMON REQUIREMENTS FOR PROCESS AND UTILITY VALVES
40 05 52 - MISCELLANEOUS VALVES
40 05 57 - ACTUATORS FOR PROCESS VALVES AND GATES
40 05 61 - GATE VALVES
40 05 62 - PLUG VALVES
40 05 63 - BALL VALVES
40 05 64 - BUTTERFLY VALVES
40 05 65 - GLOBE VALVES
40 05 66 - CHECK VALVES
40 41 13 - HEAT TRACING CABLE
40 42 00 - PIPE, DUCT AND EQUIPMENT INSULATION
40 61 13 - PROCESS CONTROL SYSTEM GENERAL REQUIREMENTS

40 61 21 - PROCESS CONTROL SYSTEM TESTING
40 61 43 - SURGE PROTECTION DEVICES FOR INSTRUMENTATION AND CONTROL EQUIPMENT
40 61 93 - PROCESS CONTROL SYSTEM INPUT-OUTPUT LIST
40 61 93.A - RSPH IO List
40 61 93.B - NR WTP IO List
40 61 96 - PROCESS CONTROL DESCRIPTIONS
40 61 96.A - VOLUME 5 ROARING SPRINGS PUMPHOUSE PROCESS CONTROL DESCRIPTIONS
40 61 96.B - VOLUME 6 NORTH RIM WTP PROCESS CONTROL DESCRIPTIONS
40 62 05 - COMPUTER SYSTEM HARDWARE AND ANCILLARIES
40 63 43 - PROGRAMMABLE LOGIC CONTROLLERS
40 66 05 - NETWORK EQUIPMENT
40 66 40 - TELEMETRY SYSTEMS
40 67 00 - CONTROL SYSTEM EQUIPMENT PANELS AND RACKS
40 67 63 - UNINTERRUPTIBLE POWER SUPPLY
40 71 00 - FLOW INSTRUMENTATION
40 72 00 - LEVEL INSTRUMENTATION
40 73 00 - PRESSURE INSTRUMENTATION
40 74 00 - TEMPERATURE INSTRUMENTATION
40 75 00 - PROCESS LIQUID ANALYTICAL MEASUREMENT
40 76 00 - PROCESS GAS ANALYTICAL MEASUREMENT
40 78 00 - PANEL MOUNTED INSTRUMENTS

DIVISION 41 — MATERIAL PROCESSING AND HANDLING EQUIPMENT
41 00 00 - DIVISION DIVIDER
41 22 23 - HOISTS

DIVISION 43 — PROCESS GAS AND LIQUID HANDLING, PURIFICATION AND STORAGE EQUIPMENT
43 00 00 - DIVISION DIVIDER
43 21 00 - PUMPING EQUIPMENT - BASIC REQUIREMENTS
43 23 22 - PUMPING EQUIPMENT - HORIZONTAL MULTI-STAGE CENTRIFUGAL PUMPS
43 23 31.29 - PUMPING EQUIPMENT - VERTICAL INLINE CENTRIFUGAL PUMPS
43 24 17 - PUMPING EQUIPMENT - SAMPLE PUMPS
43 41 15 - FIELD ERECTED WELDED STEEL TANKS
43 41 16 - CARBON STEEL TANKS
43 41 45 - FIBERGLASS REINFORCED PLASTIC TANKS

DIVISION 44 — POLLUTION AND WASTE CONTROL EQUIPMENT
44 00 00 - DIVISION 44 DIVIDER

DIVISION 46 — WATER AND WASTEWATER EQUIPMENT
46 00 00 - DIVISION 46 DIVIDER
46 13 15 - ALUMINUM COVERS - FORMED PANEL TYPE
46 33 11 - CHEMICAL FEED - LIQUID SYSTEMS
46 41 11 - VERTICAL MIXERS
46 41 17 - STATIC MIXER
46 61 33 - UF MEMBRANE EQUIPMENT
46 61 33.A - UF MEMBRANE EQUIPMENT SPARE PARTS LIST
46 61 73 - AUTOMATIC SELF-CLEANING FILTER

This page intentionally left blank.

DIVISION 26

ELECTRICAL

This page intentionally left blank.

SECTION 26 05 00
COMMON WORK RESULTS FOR ELECTRICAL

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Basic requirements for electrical systems.

B. Related Specification Sections include but are not necessarily limited to:
   1. Division 03 - Concrete.
   2. Section 01 61 03 - Equipment - Basic Requirements.
   3. Section 03 15 19 - Anchorage to Concrete.
   4. Section 10 14 00 - Identification Devices.
   5. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables.
   6. Section 26 05 33 - Raceways and Boxes.
   7. Section 26 05 48 - Seismic Bracing Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Aluminum Association (AA):
      a. ADM, Aluminum Design Manual.
   2. American Institute of Steel Construction (AISC):
      a. Steel Construction Manual.
   3. American National Standards Institute (ANSI).
   4. ASTM International (ASTM):
      a. A36/A36M, Standard Specification for Carbon Structural Steel.
      b. A123/A123M, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
      c. A153/A153M, Standard Specification for Zinc Coating (Hot-Dip Galvanized) on Iron and Steel Hardware.
   5. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. C2, National Electrical Safety Code (NESC).
   6. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   7. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. ICS 6, Industrial Control and Systems Enclosures.
   8. Underwriters Laboratories, Inc. (UL).

B. Products to be listed by a Nationally Recognized Testing Laboratory (NRTL) in accordance with applicable product standards.
   1. Applicable product standards including, but not limited to, ANSI, FM, IEEE, NEMA and UL.
   2. NRTL includes, but is not limited to, CSA Group Testing and Certification (CS), FM Approvals LLC (FM), Intertek Testing Services NA, Inc. (ETL), and Underwriters Laboratories, Inc. (UL).

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

B. For the purposes of providing materials and installing electrical work the following definitions shall be used.

1. Outdoor area: Exterior locations where the equipment is normally exposed to the weather and including below grade structures, such as vaults, manholes, handholes and in-ground pump stations.
2. Architecturally finished interior area: Offices, laboratories, conference rooms, restrooms, corridors and other similar occupied spaces.
3. Non-architecturally finished interior area: Pump, chemical, mechanical, electrical rooms and other similar process type rooms.
4. Highly corrosive and corrosive area: Areas identified on the Drawings where there is a varying degree of spillage or splashing of corrosive materials such as water, wastewater or chemical solutions; or chronic exposure to corrosive, caustic or acidic agents, chemicals, chemical fumes or chemical mixtures.
5. Hazardous areas: Class I, II or III areas as defined in NFPA 70.
6. Shop fabricated: Manufactured or assembled equipment for which a UL test procedure has not been established.

C. The North Rim electrical distribution system is served by Garkane Energy and owned by the National Park Service. The point of demarcation between Garkane Energy and NPS is at the northern park boundary. The electrical distribution system inside the park beyond the demarcation point is owned and operated by the National Park Service.

New electrical distribution equipment and systems are being designed and shall be constructed to match the standards of Garkane Energy.

Coordinate installation of distribution circuits and equipment, and transformers and electrical metering at the North Rim WTP, and Roaring Springs Pumphouse the NPS Park Staff.

1. The serving utility for this Project is Garkane Energy Cooperative, Inc.
   a. Contact:
      1) Kanab Office: Email: neal.brown@garkane.com
         Phone: 800-747-5403
2. NPS North Rim Park Staff:
   a. Contact
      1) Park Electrician Name: Josiah Brown
         Email: Josiah_Brown@nps.gov
3. Refer to the Drawings for Contractor work scope listing. Generally:
   a. North Rim WTP:
      1) New distribution circuit to the WTP site, new padmount distribution transformers with foundation and accessories, new padmounted switchgear with foundation and accessories.
      2) Installation of required electrical equipment in new WTP Building
      3) Power and control wiring to raw and finished water tanks.
   b. North Rim WWTP:
      1) New distribution circuit to the WWTP area.
   c. Roaring Springs PS:
      1) New padmount distribution transformer, new transformer primary circuit from the new overhead distribution line.
      2) Installation of required equipment in Roaring Springs Pumphouse
   d. Power Distribution Center:
      1) Replace selected electrical equipment in the yard as indicated on the Drawings.
      2) Install new generator and parallelling switchgear in Power Distribution Center as indicated on the Drawings.
      3) Install accessory power and control circuits to bulk fuel system.
   e. North Rim:
      1) Replace selected electrical equipment on the North Rim 25kV distribution power system.
      2) Replace 12.47kV overhead power line from the North Rim to Roaring Springs PS
   f. Inner Canyon:

1) Replace 12.47kV 3-Phase Power line from Roaring Springs PS to Manzanita Ranger Station.
2) Install new 3-Phase sectionalizing enclosure.

g. Cottonwood Campground
1) Replace non-functional solar system with new solar shade structure and battery storage system.

h. Other site locations: Replace selected electrical equipment as indicated on the Drawings.

D. Refer to **Garkane Electrical Service Requirement Manual** for general service requirements:
1. Coordinate with Garkane to obtain Garkane Construction Standards for reference.

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
1. See Specification Section 01 61 03 and individual specification sections for submittal requirements for products defined as equipment.
2. General requirements:
a. Provide manufacturer's technical information on products to be used, including product descriptive bulletin.
b. Include data sheets that include manufacturer's name and product model number.
1) Clearly identify all optional accessories.
c. Acknowledgement that products are NRTL listed or are constructed utilizing NRTL recognized components.
d. Manufacturer's delivery, storage, handling and installation instructions.
e. Product installation details.
f. Short Circuit Current Rating (SCCR) nameplate marking per NFPA 70, include any required calculations.
g. See individual specification sections for any additional requirements.
3. Product technical data:
a. Provide submittal data for all products specified in PART 2 of this Specification Section.
4. Fabrication and/or layout drawings:
a. Concrete and reinforcing steel, per Division 03 requirements.

C. Operation and Maintenance Manuals:
1. See Specification Section 01 78 23 for requirements for:
a. The mechanics and administration of the submittal process.
b. The content process of Operation and Maintenance Manuals.

D. When a Specification Section includes products specified in another Specification Section, each Specification Section shall have the required Shop Drawing transmittal form per Specification Section 01 33 23 and all Specification Sections shall be submitted simultaneously.

1.5 DELIVERY, STORAGE, AND HANDLING

A. See Specification Section 01 67 00 for storage requirements.

B. Protect nameplates on electrical equipment to prevent defacing.

1.6 AREA DESIGNATIONS

A. Designation of an area will determine the NEMA rating of the electrical equipment enclosures, types of conduits and installation methods to be used in that area.
1. Outdoor areas:
a. Wet.
b. Also, corrosive and/or hazardous when specifically designated on the Drawings or in the Specifications.

2. Indoor areas:
    a. Dry.
    b. Also, wet, corrosive and/or hazardous when specifically designated on the Drawings or in the Specifications.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Subject to compliance with the Contract Documents, refer to specific Electrical Specification Sections and specific material paragraphs below for acceptable manufacturers.

B. Materials and equipment shall be the standard products of a manufacturer regularly engaged in the manufacture of the products and shall be the manufacturer's latest standard design that has been in satisfactory use for at least one year prior to installation.
    1. Materials and equipment shall conform to the respective publications and other requirements specified below and to the application requirements of NFPA 70
    2. Components shall be UL or third-party certified.

C. Submit request for substitution in accordance with Section 01 67 00.

D. Provide all components of a similar type by one (1) manufacturer.

2.2 MATERIALS

A. Electrical Equipment Support Pedestals and/or Racks:
    1. Material requirements:
        a. Modular strut:
            1) Galvanized steel (hot-dipped): ASTM A123/123M or ASTM A153/A153M.
            2) Stainless steel: AISI Type 316.
            3) PVC coated galvanized steel: ASTM A123/A123M or ASTM A153/A153M and 20 mil PVC coating.
            4) Aluminum: AA Type 6063-T6.
            5) Fiberglass: Fire-retardant polyester or vinylester resin, ASTM E84, UL 94.
        b. Structural members (e.g., I beams, L and C channels):
            1) Galvanized steel (hot-dipped): ASTM A36/A36M steel with galvanizing per ASTM A123/A123M.
            2) Aluminum: AA Type 6061-T6 or 6063-T6.
        c. Mounting plates:
            1) Galvanized steel (hot-dipped): ASTM A36/A36M steel with galvanizing per ASTM A123/A123M.
            2) Aluminum: AA Type 6063-T6.
        d. Mounting hardware:
            1) Galvanized steel (hot-dipped).
            2) Stainless steel.
        e. Anchorage per Specification Section 03 15 19.
        f. Concrete and reinforcing steel: See Division 03 specifications.

B. Equipment pads (interior and exterior):
    1. Concrete and reinforcing steel: See Division 03 specifications.

C. Field touch-up of galvanized surfaces.
    1. Zinc-rich primer.
        a. One coat, 3.0 mils, ZRC by ZRC Products.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install and wire all equipment, including prepurchased equipment, and perform all tests necessary to assure conformance to the Drawings and Specification Sections and ensure that equipment is ready and safe for energization.

B. Install equipment in accordance with the requirements of:
   1. NFPA 70.
   2. IEEE C2.
   3. The manufacturer's instructions.

C. In general, conduit routing is not shown on the Drawings.
   1. The Contractor is responsible for routing all conduits including those shown on one-line and control block diagrams and home runs shown on floor plans.
   2. Conduit routings and stub-up locations that are shown are approximate; exact routing to be as required for equipment furnished and field conditions.

D. When complete branch circuiting is not shown on the Drawings:
   1. A homerun indicating panelboard name and circuit number will be shown and the circuit number will be shown adjacent to the additional devices (e.g., light fixture and receptacles) on the same circuit.
   2. The Contractor is to furnish and install all conduit and conductors required for proper operation of the circuit.
   3. The indicated home run conduit and conductor size shall be used for the entire branch circuit.
   4. See Section 26 05 19 for combining multiple branch circuits in a common conduit.

E. Do not use equipment that exceed dimensions or reduce clearances indicated on the Drawings or as required by the NFPA 70.

F. Install equipment plumb, square and true with construction features and securely fastened.

G. Install electrical equipment, including pull and junction boxes, minimum 6 inches from process, gas, air and water piping and equipment.

H. Install equipment so it is readily accessible for operation and maintenance, is not blocked or concealed and does not interfere with normal operation and maintenance requirements of other equipment.

I. Device Mounting Schedule:
   1. Unless indicated otherwise on the Drawings, mounting heights are as indicated below:
      a. Light switch (to center): 46 inches.
      b. Receptacle in architecturally finished areas (to center): 18 inches.
      c. Receptacle on exterior wall of building (to center): 18 inches.
      d. Receptacle in non-architecturally finished areas (to center): 46 inches.
      e. Telephone outlet in architecturally finished areas (to center): 18 inches.
      f. Telephone outlet for wall-mounted phone (to center): 46 inches.
      g. Safety switch (to center of operating handle): 54 inches.
      h. Separately mounted motor starter (to center of operating handle): 54 inches.
      i. Pushbutton or selector switch control station (to center): 46 inches.
      j. Panelboard (to top): 72 inches.

J. Avoid interference of electrical equipment operation and maintenance with structural members, building features and equipment of other trades.
   1. When it is necessary to adjust the intended location of electrical equipment, unless specifically dimensioned or detailed, the Contractor may make adjustments of up to 6 inches in equipment location with the Contracting Officer's approval.

K. Provide electrical equipment support system per the following area designations:

1. Dry areas:
    a. Galvanized system consisting of galvanized steel channels and fittings, nuts and hardware.
    b. Aluminum system consisting of aluminum channels and fittings with stainless steel nuts and hardware.
    c. Field touch-up cut ends and scratches of galvanized components with the specified primer during the installation, before rust appears.
2. Wet areas:
    a. Galvanized system consisting of galvanized steel channels and fittings, nuts and hardware.
    b. Field touch-up cut ends and scratches of galvanized components with the specified primer during the installation, before rust appears.
3. Corrosive areas:
    a. Aluminum system consisting of aluminum channels and fittings with stainless steel nuts and hardware.
    b. Stainless steel system consisting of stainless steel channels and fittings, nuts and hardware.
    c. PVC coated steel system consisting of PVC coated steel channels and fittings with stainless steel nuts and hardware.
    d. Fiberglass system consisting of fiberglass channels and fittings, nuts and hardware and.
4. Highly corrosive areas:
    a. PVC coated steel system consisting of PVC coated steel channels and fittings with stainless steel nuts and hardware.
    b. Fiberglass system consisting of fiberglass channels and fittings, nuts and hardware.

L. Provide all necessary anchoring devices and supports rated for the equipment load based on dimensions and weights verified from approved submittals, or as recommended by the manufacturer.
    1. See Specification Section 03 15 19.
    2. Do not cut, or weld to, building structural members.
    3. Do not mount safety switches or other equipment to equipment enclosures, unless enclosure mounting surface is properly braced to accept mounting of external equipment.

M. Provide non-metallic corrosion resistant spacers to maintain 1/4 inches separation between metallic equipment and/or metallic equipment supports and mounting surface in wet areas, on below grade walls and on walls of liquid containment or processing areas such as Basins, Clarifiers, Digesters, Reservoirs, etc.

N. Do not place equipment fabricated from aluminum in direct contact with earth or concrete.

O. Screen or seal all openings into equipment mounted outdoors to prevent the entrance of rodents and insects.

P. Do not use materials that may cause the walls or roof of a building to discolor or rust.

Q. Identify electrical equipment and components in accordance with Specification Section 10 14 00.

R. Provide field markings and/or documentation of available short-circuit current (available fault current) and related information for equipment as required by the NFPA 70 and other applicable codes.

S. Provide equipment or control panels with Short Circuit Current Rating (SCCR) labeling as required by NFPA 70 and other applicable codes.
    1. Determine the SCCR rating by one of the following methods:
        a. Method 1: SCCR rating meets or exceeds the available fault current of the source equipment when indicated on the Drawings.
        b. Method 2: SCCR rating meets or exceeds the source equipment's Amp Interrupting Current (AIC) rating as indicated on the Drawings.

c. Method 3: SCCR rating meets or exceeds the calculated available short circuit current at the control panel.

2. The source equipment is the switchboard, panelboard, motor control center or similar equipment where the equipment or control panel circuit originates.
3. For Method 3, provide calculations justifying the SCCR rating. Utilize source equipment available fault current or AIC rating as indicated on the Drawings.

T. NPS will provide labels and tags for medium voltage equipment (transformers, switch, splice vaults, pull vaults).

3.2 MEDIUM VOLTAGE POWER EQUIPMENT INSTALLATION SEQUENCING

A. Medium Voltage (MV) power plan below only includes MV equipment replacement.

B. Contractor to provide separate construction sequencing plan for all related work required for other areas.

C. Contractor to adjust installation sequencing plan as needed for other areas of work related to the project.

D. Prepare all new equipment, power cable conductor, and appurtenances prior to installation.

E. Set all items to be installed at all locations prior to installation and demolition.

F. Trench and install all new power cable conductor prior to demolition and cutover.

G. Coordinate all required power outages with NPS staff.

H. Utilize temporary generators as needed and as indicated below. Additional temporary generation usage not indicated below shall receive approval from the Contracting Officer Representative prior to installation.

I. Sequence:
1. Park Entrance:
   a. Transformer T-38
      1) Pull MV elbow termination at ESB-1 to isolate T-38 from utility power.
      2) Pull MV elbow termination at T-38.
      3) Disconnect secondary service power cables.
      4) Provide temporary service generation and connect to service.
      5) Replace T-38 and reconnect MV power cable termination at T-38.
      6) Disconnect generation and reconnect existing secondary service power cables to transformer.
      7) Connection MV elbow terminations at ESB-1 to energize T-38 from utility.
2. Power Distribution Center (PDC):
   a. Junction Enclosure ESB-30
      1) Move Park and Concessions circuit to generation service at PTS-1 and PTS-2.
      2) Ensure Normal Closed Switch positions at PS-1, PTS-1, and PTS-2 are in the Open position isolating ESB-30 and power from Utility.
      3) Disconnect all MV power elbow terminations at ESB-30 and replace ESB-30.
      4) Reconnect all MV power elbow terminations at ESB-30.
      5) Move Park and Concessions circuit back to Utility Power at PTS-1 and PTS-2.
      6) Ensure Normal Closed Switch positions at PS-1, PTS-1, and PTS-2 are in the Closed position reenergizing power from ESB-30 and from Utility
   b. Transformer T-36
      1) Trench conduit from PDC to T-36.
      2) Open MCC switch in Generation Building to T-36 and leave switch position at PTS-2 in the Normal Open position to isolate T-36.
      3) Disconnect existing MV power cable terminations and secondary terminations at T-36.
      4) Remove transformer and core drill existing concrete pad for new conduit stubs from PDC building.

5) Pull power cable from MCC to T-36.
6) Set new transformer and terminate existing MV power cable terminations and secondary terminations.
7) Close output switch from MCC in generation building to T-36.
8) T-45 to be used for both Park and Concessions emergency generation during T-36 replacement.

c. Switchgear ESB-12A
1) Coordinate the installation of ESB-12A with the replacement of ESB-12.
2) Trench cable route to ESB-12 and Point of Interception with ESB-14.
3) Excavate area for Switch ESB-12A.
4) Start ESB-12 replacement.
5) Coordinate temporary generation installation and transition to localize generation at ESB-13 and at ESB-14 with ESB-12 Replacement.
6) Confirm MV power feeds to ESB-14 and ESB-13 have been disconnected, and line is deenergized.
7) Install ESB-12A Switch.
8) Install MV power cable from Switch ESB-12 and terminate at ESB-12A.
9) Splice MV power cable at Point of interception with ESB-14 and install cable from splice point to ESB-12A. Terminate cable at ESB-12A.
10) Locate, expose, and reroute MV power cable to ESB-13. Terminate cable in ESB-12A.
11) Leave all switch positions in the normal open position.
12) After ESB-12 has been replaced:
    a) Disconnect temporary generation at ESB-14 and reconnect MV Line from PDC.
    b) At ESB-12A, close switch position connecting ESB-12 and switch position connecting ESB-14 to reenergize ESB-14.
13) At ESB-12A, leave switch connecting ESB-13 in the normal open position until MV power work has been completed at Temporary Wastewater Treatment Plant.

d. Switchgear ESB-12
1) Trench cable to ESB-12A, Interception point to ESB-16, T-35, T-49, and WWTP Area Intercept Point “D”.
2) Open NPS system at PTS-2.
3) Disconnect MV power cable at ESB-16 to open power line from T-35.
4) Close Normal Open at PS-2 to feed NPS circuit from Concessions.
5) Splice MV power cable at interception point to ESB-16 and pull cable to ESB-12 location.
6) Disconnect MV power cable at ESB-14 to open power line from existing ESB-12.
7) Connect temporary generation to junction bar to energize ESB-14.
8) Disconnect MV power cable at ESB-13.
9) Connect temporary generation to ESB-13 to energize WWTP area.
10) Start ESB-12A installation and cut over.
11) Remove existing junction enclosure ESB-12 and install ESB-12 (6-way switchgear).
12) Terminate power cable from PTS-2, T-49, T-35, ESB-16, ESB-12A, and WWTP area.
13) Leave switch connections to WWTP area and T-49 in the Normal Open position until MV power installation at WWTP area and WTP has been completed.
14) Leave switch connection to T-35 in the Normal Open position.
15) Open Normal Open at PS-2 and reconnect MV power line from PDC to ESB-16.
16) Close NPS system at PTS-2 to restore normal operation service to ESB-16, T-35, and ESB-12A.

e. Transformer T-35
1) Coordinate replacement with ESB-12 for temporary power shut off to ESB-16.

2) After MV power cable to ESB-16 has been deenergized, remove power cable connections and secondary power cable connections.
3) Replace T-35.
4) Install and terminate MV power cable from ESB-12 and service power cables into generation building.
5) Install temporary generation to new T-35 until ESB-12 has been replaced.
6) After ESB-12 switchgear has been installed, remove temporary generation and reconnect MV cable from ESB-12.
7) Close position at ESB-12 to restore utility service to T-35.

3. WWTP Area:
   a. Trench and install new MV power cables from PDC to point of intercept "D" near the WWTP area.
   b. Install cold shrink end caps at point of intercept "D" and protect cable in place for future extension into WWTP area.
   c. Leave switch position at ESB-12 in the Normal Open position. Tag and lock with warning label to not close position.
4. WTP Area:
   a. Set all proposed equipment prior to demolition and removal of existing MV equipment
   b. Excavate equipment vaults for T-27, T-49, and ESB-18A.
   c. Trench and pull new MV power cables from ESB-18A to T-49.
   d. Trench and pull new MV power cable from ESB-18A to new T-27 location.
   e. Trench and pull new MV power cable from ESB-18A to existing T-48.
   f. Coordinate equipment installation and replacement with ESB-12 and ESB-12A installation.
   g. Switchgear ESB-18A (New)
      1) Install switchgear ESB-18A.
      2) Terminate MV power cable from T-49, existing T-48, and new T-27.
      3) Leave connection to ESB-18, T-48, and T-27 in the normal open position.
      4) Set switch connection to T-49 in the normal closed position.
   h. Transformer T-49
      1) Install T-49.
      2) Terminate MV power cable from PDC and MV power cable from new ESB-18A.
      3) Close switch position at ESB-12 to energize T-49 and new ESB-18A.
   i. Transformer T-27 (New)
      1) Install T-27 and terminate MV power cable from ESB-18A.
      2) Begin demo of existing T-27 after new T-27 has been installed and new ESB-18A has been installed and energized from PDC.
      3) Open NPS system at ESB-12.
      4) Disconnect power cable elbows at existing ESB-18A to isolate existing T-27.
      5) Close NPS system at ESB-12 to reenergize NPS feed to existing ESB-18A.
      6) Disconnect existing secondary power cable at existing T-27 and expose secondary power cable back to service entrance.
      7) Reroute existing secondary power cables to new T-27 and terminate secondary power cables.
      8) Close switch at new ESB-18A to energize T-27 restoring power.
      9) Remove existing T-27 and abandon existing MV power cable from existing ESB-18A.
   j. Existing Transformer T-48
      1) Begin cable replacement after new MV power cable from new ESB-18A switchgear has been trenched and terminated at new switch ESB-18A.
      2) Open NPS system at ESB-12 deenergizing power cable to ESB-18.
      3) At ESB-18, disconnect power cable to existing junction enclosure ESB-18A to isolate existing ESB-18A.
      4) Close system at ESB-12 to reenergize NPS feed to junction enclosure ESB-18.

5) Remove existing cable terminations at T-48 and terminate new MV power cable from new switchgear ESB-18A.
6) Close switch position at ESB-18A to energize T-48.
7) Cut and abandon existing MV cable from existing ESB-18A.

k. Existing ESB-18A
1) After MV power cable feed from T-48 has been transferred to new switchgear ESB-18A, begin demo of existing ESB-18A.
2) Disconnect all power cable elbows as existing ESB-18A.
3) Locate, expose, reroute, and terminate MV cable from ESB-18 to new switch ESB-18A.
4) Leave switch position at new ESB-18A in the normal open position.

5. Concessionaire Area:
a. Junction Enclosure ESB-18
1) Begin replacement after MV equipment at WTP has been installed and energized.
2) At ESB-12, open NPS system to denergize ESB-18.
3) At ESB-17, disconnect MV Power cable elbows to ESB-18.
4) At ESB-21, disconnect MV Power cable elbows to ESB-18.
5) At ESB-12, close NPS system to reenergize ESB-17.
6) At PS-2, close Normal Open to feed NPS circuit form Concessions to reenergize system up to ESB-21.
7) At ESB-18, disconnect all MV power cable terminations and protect in place.
8) Replace ESB-18.
9) Reconnect MV power cable terminations to T-42, switchgear ESB-18A, ESB-21, and ESB-17.
10) At PS-2, open normal open position to deenergize NPS from Concessions.
11) At ESB-12, open NPS system to deenergize ESB-17.
12) At ESB-17 and ESB-21, reconnect MV power cables from ESB-18.
13) At ESB-12, close NPS system to reenergize ESB-17 and ESB-18.
14) After ESB-18 has been replaced:
a) At switch ESB-18A, open switch connection to T-49 and close switch connection from ESB-18.
b) T-49 to be connected directly to PDC.
c) T-48 and T-27 to be fed from NPS circuit.
15) Note: Transformer T-42 will not have power during replacement.
a) Confirm with NPS Park staff and Contracting Officer Representative if temporary generation is required for Transformer T-42 and provided as needed.

b. Shunt Reactor SR7
1) Open concessions circuit at PS-2.
2) At ESB-28, disconnect MV power cable terminations to isolate SR7 from power system.
3) Close concessions circuit at PS-2 to reenergize system.
4) Replace SR7 and reconnect MV power cables at SR7.
5) Open Concessions circuit at PS-2.
6) At ESB-28, reconnect MV power cable terminations to SR7.
7) Close Concessions circuit at PS-2 to reenergize power SR7.

6. Inner Canyon Power Outage:
a. The following MV equipment replacements are to take place during Inner Canyon power outage for the Overhead Power replacement. Additional coordination to maintain inner canyon power will be needed if the following MV equipment are replaced outside of the inner canyon power outage
b. Junction Enclosure ESB-23
1) At ESB-12, open NPS system energizing ESB-21 to deenergize ESB-23.
2) At ESB-21, disconnect MV power cable terminations to ESB-23.

3) At ESB-12, close NPS system to reenergize NPS feed to ESB-21. Leaving ESB-23 deenergized.
4) Replace ESB-23 and reconnect MV power cables to T-41, T-14 and ESB-23A.
5) At PS-2, disconnect MV Power cable from T-14 to isolate inner canyon power. Do not reconnect power cables until all inner canyon work is completed and ready to be energized.
6) At ESB-12, open NPS system energizing ESB-21 to deenergize ESB-21.
7) At ESB-21, reconnect MV power cable terminations to ESB-23.
8) At ESB-12, close NPS system reenergizing ESB-23.

c. Switchgear ESB-15
1) Begin replacement after MV power cable at PS-2 has been disconnected from T-14 and no power is flowing into the Inner Canyon.
2) Remove existing junction enclosure and disconnect all terminations. Protect cable terminations.
3) Install new switchgear ESB-15 and reconnect MV power cable terminations from T-16 and MV power cable terminations to North Rim power poles.
4) Leave all switch positions in the closed position.

d. Transformer T-16
1) Begin replacement after MV power cable at PS-2 has been disconnected from T-14 and no power is flowing into the Inner Canyon.
2) Disconnect MV power cable terminations to ESB-15 and PS-2 and protect cable terminations.
3) Replace transformer T-16.
4) Reconnect MV power cable terminations to ESB-15 and PS-2.

e. Transformer T-67
1) Begin replacement after MV power cable at PS-2 has been disconnected from T-14 and no power is flowing into the Inner Canyon.
2) Disconnect existing MV power cables and protect cable termination.
3) Disconnect existing secondary power cables.
4) Replace transformer T-67
5) Reconnect existing MV power cable terminations.
6) Reconnect existing secondary power cables to service Roaring Springs Pumphouse.
7) Coordinate secondary power cable replacement with Roaring Springs Pumphouse rehabilitation.

f. Overhead Power Replacement (Volume 10)
1) Begin replacement after MV power cable at PS-2 has been disconnected from T-14 and no power is flowing into the Inner Canyon.
2) Leave switch position at PS-2 in the Open position to isolate MV power to the inner canyon.
3) Begin replacement after power to the inner canyon has been isolated.
4) Take NPS outage and reconnect MV power cables at PS-2 after all Inner Canyon work has been completed.

g. Inner Canyon Power Replacement (Volume 11)
1) Coordinate replacement with power outage for overhead power replacement.
2) At existing junction enclosures at Roaring Springs Pumphouse, disconnect existing MV power cables to isolate power cable to Manzanita.
3) Replace power cable.
4) Reconnect MV power cable to existing junction enclosures at Roaring Springs Pumphouse.

7. Concessions:
a. The following MV equipment replacements should be completed during the off season to limit the effect the intermittent required power outages will have on guest and park staff.
b. Transformer T-63

1) Coordinate replacement with Transformer T-65 and T-62
2) Open Concessions system at PTS-1
3) At ESB-25, disconnect MV power cable to T-63 to isolate T-63
4) Close Concessions system at PTS-1 to reenergize Concessions feed to ESB-25.
5) Replace T-63, re-terminate MV power cable terminations, and reconnect secondary power connections.
6) Open Concessions system at PTS-1
7) At ESB-25, reconnect MV power cable to T-63
8) Close Concessions system at PTS-1 to reenergize Concessions feed to T-63

c. Transformer T-65
1) Coordinate replacement with Transformer T-63 and T-62
2) Open Concessions system at PTS-1
3) At ESB-24, disconnect MV power cable to T-65 to isolate T-65
4) Close Concessions system at PTS-1 to reenergize Concessions feed to ESB-24.
5) Replace T-65, re-terminate MV power cable terminations, and reconnect secondary power connections.
6) Open Concessions system at PTS-1
7) At ESB-24, reconnect MV power cable to T-65
8) Close Concessions system at PTS-1 to reenergize Concessions feed to T-65

d. Transformer T-62
1) Coordinate replacement with Transformer T-63 and T-65
2) Open Concessions system at PTS-1.
3) At ESB-24, disconnect MV power cable to T-62 to isolate T-62.
4) Close Concessions system at PTS-1 to reenergize Concessions feed to ESB-24.
5) Replace T-62, re-terminate MV power cable terminations, and reconnect secondary power connections.
6) Open Concessions system at PTS-1.
7) At ESB-24, reconnect MV power cable to T-62.
8) Close Concessions system at PTS-1 to reenergize Concessions feed to T-62.

e. Junction Enclosure ESB-22
1) Begin replacement before overhead power replacement starts or after it has been completed.
2) Open Concessions system at PTS-1.
3) At ESB-20, disconnect MV power cables to ESB-22.
4) At ESB-26, disconnect MV power cables to ESB-22.
5) Close Concessions system at PTS-1 to reenergize Concessions feed to ESB-20.
6) Close Normal Open at PS-2 to feed Concessions from NPS circuit to reenergize ESB-26.
7) Replace ESB-22 and reconnect power cable to T-47, ESB-22A, ESB-20, and ESB-26.
8) Open Normal Open at PS-2 to isolate NPS system from Concessions System.
9) Open Concessions system at PTS-1.
10) At ESB-20, reconnect MV power cable to ESB-22.
11) At ESB-26, reconnect MV power cables to ESB-22.
12) Close Concessions system to PTS-1 to restore normal operations.

f. Junction Enclosure ESB-19
1) Begin replacement before overhead power replacement starts or after it has been completed.
2) Open Concessions system at PTS-1.
3) At ESB-20, disconnect MV power cables to ESB-19.
4) Close Normal Open at PS-2 to feed Concessions from NPS circuit to reenergize ESB-20.
5) Replace ESB-19 and reconnect power cable to T-26, T-24, T-25, ESB-20, and PTS-1.
6) Open Normal Open at PS-2 to isolate NPS system from Concessions System.

7) At ESB-20, reconnect MV power cable to ESB-19.
8) Close Concessions system to PTS-1 to restore normal operations.

3.3 FIELD QUALITY CONTROL

A. Verify exact rough-in location and dimensions for connection to electrified equipment, provided by others.
   1. See Specification Section 01 73 20 for openings and penetrations in structures.

B. Replace equipment and systems found inoperative or defective and re-test.

C. Cleaning:
   1. See Specification Section 01 73 40.

D. The protective coating integrity of support structures and equipment enclosures shall be maintained.
   1. Repair galvanized components utilizing a zinc rich paint.
   2. Repair painted components utilizing touch up paint provided by or approved by the manufacturer.
   3. Repair PVC coated components utilizing a patching compound, of the same material as the coating, provided by the manufacturer of the component.
   4. Repair surfaces which will be inaccessible after installation prior to installation.
   5. See Specification Section 26 05 33 for requirements for conduits and associated accessories.

E. Replace nameplates damaged during installation.

3.4 DEMONSTRATION

A. Demonstrate equipment in accordance with Specification Section 01 75 00.

END OF SECTION

This page intentionally left blank.

SECTION 26 05 13
MEDIUM VOLTAGE CABLE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for:
      a. Medium voltage cable (1000 V and above).
      b. Cable terminations and splices.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables.
   3. Section 26 08 13 - Acceptance Testing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Association of Edison Illuminating Companies (AEIC).
      a. CS8, Specification for Extruded Dielectric Shielded Power Cables Rated 5 Through 46 kV.
   2. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. 48, Standard for Test Procedures and Requirements for Alternating-Current Cable Terminations Used on Shielded Cables Having Laminated Insulation Rated 2.5 kV through 765 kV or Extruded Insulation Rated 2.5 kV through 500 kV.
      b. 386, Standard for Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV.
      c. 400, Guide for Field Testing and Evaluation of the Insulation of Shielded Power Cable Systems Rated 5 kV and Above.
      d. 400.3, Guide for Partial Discharge Testing of Shielded Power Cable Systems in a Field Environment.
      e. 404, Standard for Extruded and Laminated Dielectric Shielded Cable Joints Rated 2.5 kV to 500 kV.
   3. National Electrical Manufacturers Association/Insulated Cable Engineers Association (NEMA/ICEA):
      a. WC 74/S-93-639, 5 - 46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy.
   4. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   5. Underwriters Laboratories, Inc. (UL):
      a. 1072, Standard for Medium-Voltage Power Cables.
      b. 651A, High Density Polyethylene (HDPE) Condui

B. Qualifications:
   1. Cable technician:
      a. Three years of experience in handling, terminating and splicing medium voltage cables.
      b. Successfully completed minimum 18 medium voltage terminations and underground power cable splices.
      c. Specifically trained by a factory representative on the terminations and splices to be used on the project.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product data:

a. Provide submittal data for all products specified in PART 2 of this Specification Section.
b. See Section 26 05 00 for additional requirements.

2. Fabrication and/or layout drawings:
   a. Cable pulling plan.

B. Informational Submittals:
1. Cable pulling tension measurements or photograph of breakaway swivel attached to cable.
2. Submit the following before terminating cables:
   a. Cable Technician qualifications.
3. Submit the following before testing:
   a. Independent testing organization qualifications.
   b. Engineering Technician qualifications.
   c. Sample of testing log sheets to be utilized.
4. Cable test report(s) shall include, at a minimum, the following:
   a. Summary of Project.
   b. Date tests were performed.
   c. Description of equipment/components tested.
   d. Visual inspection report.
   e. Description of tests.
   f. Test results including appropriate test forms.
   g. Conclusions and recommendations for corrective action as appropriate.
   h. Identification of test equipment used.
   i. Identification of company and person that performed the tests.
   j. Reports signed by the contracting officer representative interpreting the data.

1.4 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.5 DELIVERY, STORAGE, AND HANDLING

A. Ship cable with removable watertight end seals, and store in dry place.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Contractor to confirm all equipment is rated for installed elevation and provide final equipment ratings for review. Proposed equipment shall be compatible with existing equipment interface.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Armored Wire and cable:
   a. The Okonite Company.
   b. Or approved equal.
2. Non-Armored Wire and Cable.
   a. Southwire Company.
   b. Or approved equal.
3. Lugs, connectors and terminations:

a. 3M.
b. Or approved equal.

4. Fault indicators:
   a. Schweitzer Engineering Laboratories.
   b. Or approved equal.
5. Rock Wall Cable Support Clamps
   a. hvGrid tech.
   b. Or approved equal.
6. Cable Support Grips:
   a. Eaton Cooper Power Systems.
   b. Or approved equal.

2.2 MEDIUM VOLTAGE CABLES

A. Ratings:
   1. Voltage class as indicated on the Drawings.

B. Standards:
   1. NEMA/ICEA WC 74/S-93-639.
   2. AEIC CS8.
   3. UL 1072.

C. Conductor Material:
   1. Compact or compressed concentric stranded aluminum.

D. Insulation:
   1. Temperature rating: Type MV-105 per NFPA 70.
   2. Ethylene-propylene-rubber (EPR).
   3. 133% insulation level.

E. Shielding:
   1. Shielding on cables rated above 2 kV consists of:
      a. Semiconductor conductor screen.
      b. Semiconductor insulation screen.
      c. Copper concentric neutral

F. Neutrals:
   1. All cables to have internal concentric neutral.
      a. Cables shall have 1/3 concentric neutral size for 3-phase installations.
      b. Cables shall have full-size concentric neutral for 1-phase installations.

G. Grounds:
   1. Insulated copper per Section 26 05 19.
   2. Direct buried cables shall be rated for direct bury.
   3. Jacket: Polyvinyl chloride (PVC)

H. Jackets:
   1. Direct buried cables shall be rated for direct bury.
   2. Jacket: Polyethylene (PE).

I. Conduit Fill
   1. MV cable to be installed to meet NEC conduit fill requirements
      a. Single cable – 53 percent filled
      b. Three or more conductors – 40 percent filled
   2. Conduit size as indicated in the drawings.

J. Armored Conductor - Three Phase:
   1. Material: Compressed Aluminum
   2. Ground: Internal Copper Grounding

a. 1/3 neutral size
3. Conductor size: As shown on drawings
4. Sheath: Close fitting, impervious, continuous, corrugated aluminum, armored cable, with outer polyethylene (PE) jacket.
5. Multiconductor cable, Type MV-105
6. Cable Exterior Color:.
   a. Tan
7. Rated for surface-mounted installation.

2.3 CABLE ACCESSORIES

A. Lugs and Connectors:
   1. Lugs:
      a. Compression type.
      b. Standard: UL 486B for aluminum and copper cables.
      c. Voltage rating: Up to 35 kV.
      d. Current rating: Continuous operation at the rating of the cable.
      e. Material: Tin-plated copper or Aluminum with bronze underplating, tin outerplating and aluminum anti-oxide paste.
      f. Number of holes: Two, except one on motor leads.
   2. Splice connectors:
      a. Standard: UL 486B for aluminum and copper cables.
      b. Voltage rating: Up to 35 kV.
      c. Current rating: Continuous operation at the rating of the cable.
      d. Material: Tin-plated copper or Aluminum with bronze underplating, tin outerplating and aluminum anti-oxide paste.

B. Terminations:
   1. End caps:
      a. Cold or hot shrink.
      b. Used to environmentally seal and mechanically protect exposed cable ends.
      c. Sized to fit cable
   2. Cold shrink kits:
      a. Standard: IEEE 48, Class 1 termination.
      b. Voltage rating: Same as the cable rating.
      c. Current rating: Continuous operation at the rating of the cable.
      d. One-piece design, where high-dielectric constant stress control is integrated within a skirted insulator made of silicone rubber.
      e. Suitable for contaminated indoor and outdoor locations.
   3. Molded rubber kit:
      a. Standard: IEEE 48.
      b. Voltage rating: Same as the cable rating.
      c. Current rating: Continuous operation at the rating of the cable.
      d. One-piece design or modular with stress cone and skirts, where high-dielectric constant stress control is integrated within a skirted insulator made of EPDM rubber.
      e. Suitable for contaminated indoor and outdoor locations.
   4. Elbow connectors:
      a. Standard: IEEE 386.
      b. Voltage rating: Same as the cable rating.
      c. Current rating:
         1) 200A Loadbreak
      d. One-piece design, comprised of an insulation shield, insulation layer and an outer shield constructed of EPDM rubber.
      e. Deadfront, loadbreak type with:
         1) Hot stick pulling eye.
         2) Grounding tab.

3) Test point.

f. Accessories to be constructed in a similar manner as the elbow connector:
   1) Bushing inserts.
   2) Bushing well plugs.
   3) Feed thru inserts.
   4) Protective caps.

C. Splices:
   1. Cold shrink kits:
      a. Standard: IEEE 404.
      b. Voltage rating: Same as the cable rating.
      c. Current rating: Continuous operation at the rating of the cable.
      d. One-piece design, comprised of an insulation shield, insulation layer and a silicone rubber body.
      e. Suitable for indoor, direct burial, or submersible applications.
   2. Molded rubber kit:
      a. Standard: IEEE 386 or IEEE 404.
      b. Voltage rating: Same as the cable rating.
      c. Current rating: Continuous operation at the rating of the cable.
      d. One- or multi-piece design, comprised of an insulation shield, insulation layer and an outer shield constructed of EPDM rubber.
      e. Suitable for indoor, direct burial or submersible applications.
   3. Splice Protection
      a. All medium voltage armored cable splices shall be protected with HDPE pipe used as a sleeve around the entire splice and extending 2 feet beyond both ends of splice.
         1) HDPE shall be minimum standard dimension ratio (SDR) 19 and maximum SDR 13.5.
         2) See HDPE specifications 33 05 33.23 for additional details.
      b. HDPE pipe shall be clamped at both ends with full circle repair clamps (See detail drawings in Volume 11 for visual representation).
         1) The full circle repair clamp shall have an 18-8 Type 304 stainless steel band.
         2) Minimum of 7.5" in width.
         3) The lugs shall be made of high strength ductile iron per ASTM A536 GR80-55-06 and have a fusion bonded epoxy coating. The lugs shall be attached to the band by hemming the band and sliding it into the lug slot. The slot shall be cast into the lug base. The lug shall have a friction fit with the band, preventing it from sliding off, yet allowing for easy removal and reattachment in case of installing the clamp in restricted spaces.
         4) The gasket shall be made of Nitrile (Buna N) specially compounded to resist water, oil, acids, alkalies, most (aliphatic) hydrocarbon fluids, and many other chemicals. The temperature range of the gasket shall be -20 Deg F thru +180 Deg F. The gasket must have a bridge plate that helps the end of the clamp band to seal across the gap.
         5) The bridge plate shall be made of quarter-hard 304 stainless steel. It shall be recessed and bonded into the gasket. The gasket shall have a gridded pattern.
         6) The gasket shall be tapered on the ends and overlap for range capability. All bolts shall be at least 5/8" x 6" high strength low alloy per ANSI A21.11 and B18.10. At least one 5/8" x 6 7/8" bolt with a taller nut shall be furnished to help facilitate installation of clamps up thru 4.5" O.D.
         7) Heavy semi-finished nuts meeting ASTM A563 shall be provided with the bolts. Multi-band clamps shall have all closed lugs except for one set of open and closed lugs to allow for easy opening and installation of the clamp. OPTIONAL - Stainless steel type 304 bolts and nuts. The nuts shall be fluoropolymer coated to prevent galling.
         8) Modular pipe sealer (used for bushing between HDPE pipe and medium voltage cable at pipe clamp):

a) Hardware: S316 Stainless Steel
b) Pressure plate: reinforced nylon polymer
c) Rubber seal element: EPDM
d) Temperature range: -40°F to 250°F
e) Pressure resistant to 20psig

4. Multi-point Cable Junctions:
   a. Standard: IEEE 386.
   b. Voltage Class: 15 kV.
   c. Configuration: As indicated on the Drawings.
   d. Current rating: 200A loadbreak
   e. Bushing: One piece design, comprised of an insulation shield, insulation layer and an outer shield constructed of EPDM rubber.
   f. Bus: Structural grade plated aluminum.
   g. Wall mount bracket: Adjustable stainless steel with ground lug and parking stands.

D. Cable Shield Grounding Adapters:
   1. Type: Molded rubber with constant force spring and solder-blocked tinned copper braid pigtail.
   2. Waterproof, providing a positive seal for the cable jacket.
   3. May be integral with termination of splice device with contracting officer representative's approval.

E. Rock Wall Cable Support Clamp:
   1. Type: Galvanized Steel
   2. Direct sunlight application
   3. Match clamp range to fit armored cable diameter
   4. Minimum 50lb holding capacity
   5. Install cable clamp to manufacturer recommendations

F. Fault Indicators:
   1. Standards: IEEE 495.
   2. Type: Manual reset.
      a. Nominal trip rating: 400 A.
      b. Reset delay: 4 hours.
   3. Integral indicator type: LED.
   4. When required, battery shall be a long-life lithium ion.

G. Cable Support Grips
   1. Type: Stainless Steel
   2. Double eye, Double Weave
   3. Match cable support diameter range to conductor cable diameter.
   4. One cable grip per conductor
   5. Ensure working load of grip is suitable for application

H. Conduit
   1. Aluminum conduit shall be used for all exposed installations. (For example, all riser pole conduit)
   2. HDPE conduit shall only be used for cable pull protection. The intent is to use HDPE conduit as a protection layer during installation of armored cable.
      a. HDPE pipe for electrical power to meet UL 651A
      b. See Section 33 05 33.23 for more information.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Do not install cable during wet conditions.

1. Prior to pulling cables, drain or pump out equipment vaults and other low points if standing water is present.
2. Blow out conduits with dried compressed air if moisture is present in conduits.
3. Install end caps immediately on all cut ends of cable prior to pulling, and maintain end caps while pulling in cable.
    a. If end caps are damaged, remove and install new end caps.
    b. Do not remove end caps until ready to terminate or splice cable.

B. Cable Installation in Equipment Vaults:
1. Arc-proof all cables in Equipment Vaults.
    a. Apply in spiral, half-overlap fashion to full exposed length of each cable in manhole.
    b. Secure in place with glass cloth electrical tape.
        1) Apply in reverse spiral to arc-proofing tape, at maximum interval of 9 inches and double wrapped at each end.

C. Do not install conductors when ambient temperature is near minimum as recommended by manufacturer for installation of the type of conductor insulation.

D. Provide components in kit form, complete with instructions, supplied by a single approved manufacturer and suitable for each shielded cable termination.
1. Select correct termination to match cable diameter and construction.
2. Form and install terminations in strict accordance with instructions of cable manufacturer and termination manufacturer.

E. Splices:
1. Provide components in kit form, complete with instructions, supplied by a single approved manufacturer and suitable for the type of cable being used.
2. Prepare cable ends, provide materials and follow all application steps in accordance with manufacturer's instructions.
    a. As a minimum requirement:
        1) The cable ends shall be cut squarely.
        2) The insulation shall be free from nicks or burrs after removal of jacket.
        3) The conductors shall be cleaned and an oxide inhibitor applied.
        4) For splices, connector indents shall be filled with insulating putty to eliminate voids or prepared per manufacturer's instructions.
        5) Attach grounding lead to system ground.
3. Splices shall be avoided whenever possible.
    a. No more than one splice is permitted between termination points without contracting officer representative's approval.
    b. Do not pull splices into duct banks or conduits or leave them under tension.

F. The ground shield grounding adaptors shall be grounded:
1. Shirted and elbow terminators: Grounded to ground bar or cable loop in equipment.
2. Connect with insulated, stranded #6 AWG wire.

3.2 FIELD QUALITY CONTROL

A. Cable pulling plan.
1. Include proposed splice points, cable pulling direction for each pull, and estimated pulling tensions (tension calculation).
2. Provide measurement of tensions during the pull.
    a. Do no exceed manufacturer's suggested maximum pulling tension.

B. When estimated pulling tension is greater than 75% of cables maximum allowed tension, provide measurement of tensions during the pull or use breakaway swivel of a strength less than the cable's maximum allowed tension.

C. See Specification Section 26 08 13 for acceptance testing requirements.

END OF SECTION

This page intentionally left blank.

SECTION 26 05 19
LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for:
      a. Building wire.
      b. Power cable.
      c. Control cable.
      d. Shielded VFD cable.
      e. Instrumentation cable.
      f. Wire connectors.
      g. Insulating tape.
      h. Pulling lubricant.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 08 13 - Acceptance Testing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Insulated Cable Engineers Association (ICEA):
      a. S-58-679, Standard for Control Cable Conductor Identification.
   2. National Electrical Manufacturers Association (NEMA):
      a. ICS 4, Industrial Control and Systems: Terminal Blocks.
   3. National Electrical Manufacturers Association/Insulated Cable Engineers Association (NEMA/ICEA):
      a. WC 57/S-73-532, Standard for Control Cables.
      b. WC 70/S-95-658, Non-Shielded Power Cables Rated 2000 Volts or Less for the Distribution of Electrical Energy.
   4. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
      b. 262, Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air-Handling Spaces.
   5. Telecommunications Industry Association/Electronic Industries Alliance (TIA/EIA):
      a. 598-C, Optical Fiber Cable Color Coding.
   6. Telecommunications Industry Association/Electronic Industries Alliance/American National Standards Institute (TIA/EIA/ANSI):
      a. 568, Commercial Building Telecommunications Cabling Standard.
   7. Underwriters Laboratories, Inc. (UL):
      a. 44, Standard for Safety Thermoset-Insulated Wires and Cables.
      b. 83, Standard for Safety Thermoplastic-Insulated Wires and Cables.
      c. 467, Standard for Safety Grounding and Bonding Equipment.
      d. 486A, Standard for Safety Wire Connectors and Soldering Lugs for use with Copper Conductors.
      e. 486C, Standard for Safety Splicing Wire Connections.
      f. 510, Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape.
      g. 1277, Standard for Safety Electrical Power and Control Tray Cables with Optional Optical-Fiber Members.
      h. 1581, Standard for Safety Reference Standard for Electrical Wires, Cables, and Flexible Cords.
      i. 2250, Standard for Safety Instrumentation Tray Cable.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

B. Cable: Multi-conductor, insulated, with outer sheath containing either building wire or instrumentation wire.

C. Instrumentation Cable:
   1. Multiple conductor, insulated, twisted or untwisted, with outer sheath.
   2. The following are specific types of instrumentation cables:
      a. Analog signal cable:
         1) Used for the transmission of low current (e.g., 4-20mA DC) or low voltage (e.g., 0-10 VDC) signals, using No. 16 AWG and smaller conductors.
         2) Commonly used types are defined in the following:
            a) TSP: Twisted shielded pair.
            b) TST: Twisted shielded triad.
      b. Digital signal cable: Used for the transmission of digital signals between computers, PLC's, RTU's, etc.

D. Power Cable: Multi-conductor, insulated, with outer sheath containing building wire, No. 8 AWG and larger.

E. Shielded VFD Cable: Multi-conductor, insulated, with shield, drain wire and building wires, No. 12 and larger.

F. Control Cable: Multi-conductor, insulated, with outer sheath containing building wires, No. 14, No. 12 or No. 10 AWG.

G. Building Wire: Single conductor, insulated, with or without outer jacket depending upon type.

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data:
      a. Provide submittal data for all products specified in PART 2 of this Specification Section except:
         1) Wire connectors.
         2) Insulating tape.
         3) Cable lubricant.
      b. See Section 26 05 00 for additional requirements.
   2. Miscellaneous:
      a. Cable pulling plan and data for circuits in underground conduits or duct banks longer than 100FT.

1.5 DELIVERY, STORAGE, AND HANDLING

A. See Section 26 05 00.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Building wire, power and control cable:
      a. Southwire Company.
      b. Or Equal.
   2. Shielded VFD cable:
      a. Okonite Company.
      b. Or Equal.
   3. Metal Clad - Hazardous Location (MC-HL) Cable:
      a. Okonite Company.
      b. Or Equal.
   4. Instrumentation cable:
      a. Analog cable:
         1) General Cable.
         2) Or Equal.
   5. Wire connectors:
      a. 3M Co.
      b. Or Equal.
   6. Insulating and color coding tape:
      a. 3M Co.
      b. Or Equal.

2.2 MANUFACTURED UNITS

A. Building Wire:
   1. Conductor shall be copper with 600 V rated insulation.
   2. Conductors shall be stranded, except for conductors used in lighting and receptacle circuits which may be stranded or solid.
   3. Surface mark with manufacturer's name or trademark, conductor size, insulation type and UL label.
   4. When direct buried, UL listed and marked as suitable for direct bury.
   5. Conform to NEMA/ICEA WC 70/S-95-658 and UL 83 for type THHN/THWN and THHN/THWN-2 insulation for lighting and receptacle circuits.
   6. Conform to NEMA/ICEA WC 70/S-95-658 and UL 44 for type XHHW-2 insulation for power circuits and all other uses.

B. Power Cable:
   1. Conductor shall be copper with 600 V rated insulation.
   2. Surface mark with manufacturer's name or trademark, conductor size, insulation type and UL label.
   3. Conform to NEMA/ICEA WC 70/S-95-658 and UL 44 and UL 1277 for type XHHW-2 insulation with an overall PVC or cross-linked polyethylene (XLPE) jacket.
   4. Number of conductors as required, including a bare ground conductor.
   5. Individual conductor color coding:
      a. ICEA S-58-679, Method 4.
      b. See PART 3 of this Specification Section for additional requirements.
   6. When direct buried, UL listed and marked as suitable for direct bury.
   7. Conform to NFPA 70 Type TC and IEEE 1202 or CSA FT-4.

C. Control Cable:
   1. Conductor shall be **tinned copper** with 600 V rated insulation.
   2. Surface mark with manufacturer's name or trademark, conductor size, insulation type and UL label.

3. Conform to NEMA/ICEA WC 57/S-73-532 and UL 44 and UL 1277 for type XHHW-2 insulation with an overall PVC jacket.
4. Number of conductors as required, provided with or without bare ground conductor of the same AWG size.
    a. When a bare ground conductor is not provided, an additional insulated conductor shall be provided and used as the ground conductor (e.g., 6/c No. 14 w/g and 7/c No. 14 are equal).
5. Individual conductor color coding:
    a. ICEA S-58-679, Method 1, Table E-2.
    b. See PART 3 of this Specification Section for additional requirements.
6. Conform to NFPA 70 Type TC and IEEE 1202, CSA FT-4 or NFPA 262.

D. Electrical Equipment Control Wire:
1. Conductor shall be **tinned copper** with 600 V rated insulation.
2. Conductors shall be stranded.
3. Surface mark with manufacturer's name or trademark, conductor size, insulation type and UL label.
4. Conform to UL 44 for Type SIS insulation.
5. Conform to UL 83 for Type MTW insulation.

E. Metal Clad - Hazardous Location (MC-HL) Cable:
1. Sheath: Continuous welded corrugated copper-free aluminum.
2. Jacket: Low temperature, sun light resistant PVC.
3. Conductors:
    a. Building wire as specified herein.
    b. Material: Copper or aluminum as indicated on the Drawings.
    c. Size and quantity: As indicated on the Drawings, including ground wire.
    d. Individual conductor color coding or marking per ICEA S-58-679, Method 4.
        1) See PART 3of this Specification Section for additional requirements.
4. Temperature Ratings:
    a. Normal operation: 90 degrees C.
    b. Emergency overload: 130 degrees C.
    c. Short circuit: 250 degrees C.
5. Surface mark with manufacturer's name or trademark, conductor size, insulation type and UL label.
6. Use manufacturer approved fittings.
7. Conform to NFPA 70, Type MC-HL, UL 1569, UL 2225, IEEE 1202, IEEE 1580.

F. Shielded VFD Cable:
1. Conductor shall be copper, stranded with 600 V rated insulation.
2. Surface mark with manufacturer's name or trademark, conductor size, insulation type and UL label.
3. Cables No. 1 AWG and less:
    a. Conform to NEMA/ICEA WC 70/S-95-658 and UL 44 and UL 1277 for type RHW-2 or XHHW-2 insulation with an overall PVC jacket.
    b. Shielding: 85% tinned copper braid, full size tinned copper drain wire and 100% foil shield.
    c. Number of conductors: 3 PH and 1 full size ground.
4. Cables No. 12 through 750 kcmil:
    a. Conform to NEMA/ICEA WC 70/S-95-658 and UL 44 type XHHW-2 insulation.
    b. Shielding: Continuous corrugated copper-free aluminum sheath covered with a PVC jacket.
    c. Number of conductors: 3 PH and 3 equally spaced ground conductors.
5. Individual conductor color coding:
    a. ICEA S-58-679, Method 4.
    b. See PART 3 of this Specification Section for additional requirements.
6. When installed exposed outdoors, UL listed and marked as sunlight resistant.

7. For continuously corrugated cable, use manufacturer approved fittings.
8. Conform to NFPA 70, Type TC and IEEE 1202 or CSA FT-4.

G. Instrumentation Cable:
1. Surface mark with manufacturer's name or trademark, conductor size, insulation type and UL label.
2. Analog cable:
   a. **Tinned copper** conductors.
   b. 600 V PVC insulation with PVC jacket.
   c. Twisted with 100% foil shield coverage with drain wire.
   d. Six (6) twists per foot minimum.
   e. Individual conductor color coding: ICEA S-58-679, Method 1, Table E-2.
   f. Conform to UL 2250, UL 1581 and NFPA 70 Type ITC.
3. Digital cable:
   a. As recommended by equipment (e.g., PLC, RTU) manufacturer.
   b. Horizontal voice and data cable:
      1) Category 6 per TIA/EIA/ANSI 568.
      2) Cable shall be label-verified.
      3) Cable jacket shall be factory marked at regular intervals indicating verifying organization and performance level.
      4) Conductors: No. 24 AWG solid untinned copper.
      5) Rated CMP per NFPA 70.
   c. Conform to IEEE 1202 NFPA 262 and NFPA 70 Type ITC.

H. Wire Connectors:
1. Twist/screw on type:
   a. Insulated pressure or spring type solderless connector.
   b. 600 V rated.
   c. Ground conductors: Conform to UL 486C and/or UL 467 when required by local codes.
   d. Phase and neutral conductors: Conform to UL 486C.
2. Compression and mechanical screw type:
   a. 600 V rated.
   b. Ground conductors: Conform to UL 467.
   c. Phase and neutral conductors: Conform to UL 486A.
3. Terminal block type:
   a. High density, screw-post barrier-type with white center marker strip.
   b. 600 V and ampere rating as required, for power circuits.
   c. 600 V, 20 ampere rated for control circuits.
   d. 600 V, 15 ampere rated for instrumentation circuits.
   e. Conform to NEMA ICS 4 and UL 486A.
4. Disallowed types of wire connectors:
   a. Butt connectors.
   b. Automotive-type wire connectors.
   c. Push-on, lever, or "wago" type connectors.

I. Insulating and Color Coding Tape:
1. Pressure sensitive vinyl.
2. Premium grade.
3. Heat, cold, moisture, and sunlight resistant.
4. Thickness, depending on use conditions: 7, 8.5, or 10 mil.
5. For cold weather or outdoor location, tape must also be all-weather.
6. Color:
   a. Insulating tape: Black.
   b. Color coding tape: Fade-resistant color as specified herein.
7. Comply with UL 510.

J. Cold Shrink Insulation:

1. Factory expanded sleeve with removable core.
2. With core removed, the insulation will shrink over splice connector providing a water-resistant seal.
3. Material: EPDM Rubber.
4. Voltage: 1000V.
5. Fungus-resistant per ASTM G21.

K. Electrical Equipment Enclosure and Cable Sealing Bushing:
1. Galvanized malleable or ductile iron.
2. Zinc electroplated locknut.
3. Neoprene sealing ring.

L. Pulling Lubricant: Cable manufacturer's standard containing no petroleum or other products which will deteriorate insulation.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Permitted Usage of Insulation Types:
1. Type XHHW-2:
a. Building wire and power and control cable in all areas.
b. Building wire and power and control cable in conduit in outdoor areas and below grade.
2. Type THHN/THWN and THHN/THWN-2:
a. Lighting and receptacle circuits in indoor locations only.
3. Type SIS and MTW:
a. **Tinned copper**, for the wiring of control equipment within control panels and field wiring of control equipment within switchgear, switchboards, motor control centers.

B. Shielded VFD Cable:
1. For wiring between a VFD and motor when routing in cable trays or conduit other than RGS.

C. Conductor Size Limitations:
1. Feeder and branch power conductors shall not be smaller than No. 12 AWG unless otherwise indicated on the Drawings.
2. Control conductors shall not be smaller than No. 14 AWG unless otherwise indicated on the Drawings.
3. Instrumentation conductors shall not be smaller than No. 16 AWG unless otherwise indicated on the Drawings.

D. Color Code All Wiring as Follows:
1. Building wire:

	240 V, 208 V, 240/120 V, 208/120 V	480 V, 480/277 V
Phase 1	Black	Brown
Phase 2	Red *	Orange
Phase 3	Blue	Yellow
Neutral	White	White or Gray
Ground	Green	Green

* Orange when it is a high leg of a 120/240 V Delta system.

a. Conductors No. 6 AWG and smaller: Insulated phase, neutral and ground conductors shall be identified by a continuous colored outer finish along its entire length.
b. Conductors larger than No. 6 AWG:

1) Insulated phase and neutral conductors shall be identified by one of the following methods:
   a) Continuous colored outer finish along its entire length.
   b) 3 inches of colored tape applied at the termination.
2) Insulated grounding conductor shall be identified by one of the following methods:
   a) Continuous green outer finish along its entire length.
   b) Stripping the insulation from the entire exposed length.
   c) Using green tape to cover the entire exposed length.
3) The color coding shall be applied at all accessible locations, including but not limited to: Junction and pull boxes, wireways, manholes and handholes.

2. Power cables ICEA S-58-679, Method 4 with:
   a. Phase and neutral conductors identified with 3 inches of colored tape, per the Table herein, applied at the terminations.
   b. Ground conductor: Bare.
3. Shielded VFD cable ICEA S-58-679, Method 4 with:
   a. Phase conductors identified with 3 inches of colored tape, per the Table herein, applied at the terminations.
   b. Ground conductor: Green color insulation or bare.
4. Control cables ICEA S-58-679, Method 1, Table E-2:
   a. When a bare ground is not provided, one of the colored insulated conductors shall be re-identified by stripping the insulation from the entire exposed length or using green tape to cover the entire exposed length.
   b. When used in power applications the colored insulated conductors used as phase and neutral conductors may have to be re-identified with 3 inches of colored tape, per the Table herein, applied at the terminations.

E. Install all wiring in raceway unless otherwise indicated on the Drawings, including all circuits, grounding conductors, and grounding electrode conductors.

F. Feeder, branch, control and instrumentation circuits shall not be combined in a raceway, cable tray, junction or pull box, except as permitted in the following:
   1. Where specifically indicated on the Drawings.
   2. Where field conditions dictate and written permission is obtained from the Contracting Officer.
   3. Control circuits shall be isolated from feeder and branch power and instrumentation circuits but combining of control circuits is permitted.
      a. The combinations shall comply with the following:
         1) 12 VDC, 24 VDC and 48 VDC may be combined.
         2) 125 VDC shall be isolated from all other AC and DC circuits.
         3) AC control circuits shall be isolated from all DC circuits.
   4. Instrumentation circuits shall be isolated from feeder and branch power and control circuits but combining of instrumentation circuits is permitted.
      a. The combinations shall comply with the following:
         1) Analog signal circuits may be combined.
         2) Digital signal circuits may be combined but isolated from analog signal circuits, unless explicitly shown on the Drawings, or with written permission from the Contracting Officer.
   5. Multiple branch circuits for similar loads may be combined in a common raceway, such as multiple lighting circuits or multiple receptacle circuits or other 120Vac circuits. Do not combine lighting and receptacle circuits.
      a. Do not combine control device circuits with lighting or receptacle circuits.
      b. Contractor is responsible for making the required adjustments in conductor and raceway size, in accordance with all requirements of the NFPA 70, including but not limited to:
         1) Up sizing conductor size for required ampacity de-ratings for the number of current carrying conductors in the raceway.
         2) The neutral conductors may not be shared.

3) Up sizing raceway size for the size and quantity of conductors.

G. Ground the drain wire of shielded instrumentation cables at one end only.
   1. The preferred grounding location is at the load (e.g., control panel), not at the source (e.g., field mounted instrument).

H. Cable Prepration:
   1. Heat-shrink ends of multiconductor control and instrument cables after the outer sheath is removed. Heat shrink shall be centered over the cut end of the outer cable sheath.

I. Splices and terminations for the following circuit types shall be made in the indicated enclosure type using the indicated method.
   1. Feeder and branch power circuits:
      a. Device outlet boxes:
         1) Twist/screw on type connectors.
      b. Junction and pull boxes and wireways:
         1) Twist/screw on type connectors for use on No. 8 and smaller wire.
         2) Compression, mechanical screw or terminal block or terminal strip type connectors for use on No. 6 AWG and larger wire.
      c. Motor terminal boxes:
         1) Twist/screw on type connectors for use on No. 10 AWG and smaller wire.
         2) Insulated mechanical screw type connectors for use on No. 8 AWG and larger wire.
         3) Bolted lugs with explicit permission from Contracting Officer.
      d. Manholes or handholes:
         1) No splices permitted in manholes or handholes.
   2. Control circuits:
      a. Junction and pull boxes: Terminal block type connector.
      b. Manholes or handholes: No splices are permitted in manholes or handholes.
      c. Control panels and motor control centers: Terminal block or strips provided within the equipment or field installed within the equipment by the Contractor.
   3. Instrumentation circuits:
      a. No splicing of instrumentation circuits is permitted, except in control panels and motor control centers.
      b. Control panels and motor control centers: Terminal block or strip provided within the equipment or field installed within the equipment by the Contractor.
   4. Non-insulated compression and mechanical screw type connectors shall be insulated with tape or hot or cold shrink type insulation to the insulation level of the conductors.
   5. Splices or terminations shall not be made with any type of disallowed wire connector.

J. Insulating Tape Usage:
   1. For insulating connections of No. 8 AWG wire and smaller: 7 mil vinyl tape.
   2. For insulating splices and taps of No. 6 AWG wire or larger: 10 mil vinyl tape.
   3. For insulating connections made in cold weather or in outdoor locations: 8.5 mil, all weather vinyl tape.

K. Color Coding Tape Usage: For color coding of conductors.

3.2 FIELD QUALITY CONTROL

A. Acceptance Testing:
   1. See Specification Section 26 08 13.

END OF SECTION

SECTION 26 05 26
GROUNDING AND BONDING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for grounding and bonding system(s).

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 10 14 00 - Identification Devices.
   2. Section 26 05 00 - Common Work Results for Electrical.
   3. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables.
   4. Section 26 05 33 - Raceways and Boxes.
   5. Section 26 08 13 - Acceptance Testing.
   6. Section 26 41 13 - Lightning Protection System.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. B8, Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft.
   2. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. 837, Standard for Qualifying Permanent Connections Used in Substation Grounding.
   3. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   4. Underwriters Laboratories, Inc. (UL):
      a. 467, Grounding and Bonding Equipment.

B. Assure ground continuity is continuous throughout the entire Project.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data.
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. See Section 26 05 00 for additional requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Ground rods and bars and grounding clamps, connectors and terminals:
      a. ERICO by Pentair.
      b. Or Equal.
   2. Exothermic weld connections:
      a. ERICO by Pentair - Cadweld.
      b. Or Equal.
   3. Prefabricated composite test stations:
      a. Armorcast Products Company.
      b. Or Equal.

2.2 COMPONENTS

A. Wire and Cable:
   1. Bare conductors: Soft drawn stranded copper meeting ASTM B8.
   2. Insulated conductors: Color coded green, per Section 26 05 19.

B. Conduit: As specified in Section 26 05 33.

C. Ground Bars:
   1. Solid copper:
      a. 1/4 inches thick.
      b. 2 or 4 inches wide.
      c. 24 inches long minimum in main service entrance electrical rooms, 12 inches long elsewhere.
   2. Predrilled grounding lug mounting holes.
   3. Stainless steel or galvanized steel mounting brackets.
   4. Insulated standoffs.

D. Ground Rods:
   1. Unless otherwise specified in the drawings, ground rods to be 3/4 inches x 10 feet, copper-clad:
      a. 10 mil minimum uniform coating of electrolytic copper molecularly bonded to a rigid steel core.
      b. Corrosion resistant bond between the copper and steel.
      c. Hard drawn for a scar-resistant surface.

E. Grounding Clamps, Connectors and Terminals:
   1. Mechanical type:
      a. Standards: UL 467.
      b. High copper alloy content.
   2. Compression type for interior locations:
      a. Standards: UL 467.
      b. High copper alloy content.
      c. Non-reversible.
      d. Terminals for connection to bus bars shall have two bolt holes.
   3. Compression type suitable for direct burial in earth or concrete:
      a. Standards: UL 467, IEEE 837.
      b. High copper alloy content.
      c. Non-reversible.
      d. Factory filled with oxide inhibiting compound.

F. Exothermic Weld Connections:
   1. Copper oxide reduction by aluminum process.
   2. Molds properly sized for each application.

G. Prefabricated Composite Material Test Stations:

1. Body and cover: Fiberglass reinforced polymer concrete conforming to all test provisions of SCTE 77.
2. Minimum load ratings: SCTE 77 Tier 15.
3. Open bottom.
4. Stackable design as required for 3 feet depth.
5. Cover:
    a. Engraved legend of "GROUND".
    b. Lay-in non-bolt down.
6. Size: 12 inches round or 12 inches square.

PART 3 - EXECUTION

3.1 INSTALLATION

A. General:
1. Install products in accordance with manufacturer's instructions.
2. Size grounding conductors and bonding jumpers in accordance with NFPA 70, Article 250, except where larger sizes are indicated on the Drawings.
3. Remove paint, rust, or other non-conducting material from contact surfaces before making ground connections. After connection, apply manufacturers approved touch-up paint to protect metallic surface from corrosion.
4. Where ground conductors pass through floor slabs or building walls provide nonmetallic sleeves and install sleeve per Specification Section 01 73 20.
    a. Seal the sleeve interior to stop water penetration.
5. Grounding conductors, grounding electrode conductors, and similar conductors shall be protected from damage, and shall be contained in raceway where not buried in soil or encased in concrete.
6. Do not splice grounding electrode conductors except at ground rods.
7. Install ground rods and grounding electrode conductors in undisturbed, firm soil.
    a. Provide excavation required for installation of ground rods and conductors.
    b. Use driving studs or other suitable means to prevent damage to threaded ends of sectional rods.
    c. Unless otherwise specified, connect conductors to ground rods with compression type connectors or exothermic weld.
    d. Provide sufficient slack in conductor to prevent conductor breakage during backfill or due to ground movement.
    e. Backfill excavation completely, thoroughly tamping to provide good contact between backfill materials and ground rods and conductors.
8. Do not use exothermic welding if it will damage the structure the grounding conductor is being welded to.

B. Grounding Electrode System:
1. Provide a grounding electrode system in accordance with NFPA 70, Article 250 and as indicated on the Drawings.
    a. All grounding electrode conductors terminate on a main ground bar located adjacent to the service entrance equipment.
2. Grounding electrode conductor terminations:
    a. Ground bars mounted on wall: Use a two-hole compression type conductor terminal and bolt it to the ground bar with two bolts.
    b. Ground bars in electrical equipment: Use compression type conductor terminal and bolt it to the ground bar or manufacture's provided mechanical type termination device.
    c. Piping systems: Use mechanical type connections.
    d. Building steel, below grade and encased in concrete: Use compression type connector or exothermic weld.
    e. Building steel, above grade: Use a two-hole compression type conductor terminal and bolt to the steel with two bolts or exothermic weld.

f. Ground rod: Compression type or exothermic weld, unless otherwise specified.
g. At all above grade terminations, the conductors shall be labeled per Specification Section 10 14 00.

3. Ground ring grounding system:
   a. Ground ring consists of ground rods and a conductor looped around the structure.
   b. Placed at a minimum of 10 feet from the structure foundation and 2 feet-6 inches below grade.
   c. Provide a minimum of four ground rods placed at the corners of the structure and additional rods so that the maximum distance between ground rods does not exceed 50 FT.
   d. Building/Structure grounding:
      1) Bond building/structure metal support columns to the ground ring at all corners of the structure.
   e. Grounding conductor: Bare conductor, size as indicated on the Drawings.
   f. Ground rod test stations:
      1) Provided where indicated on the Drawings.
      2) Grounding conductors connected to ground rod with removable ground clamps.
   g. Grounding conductor shall connect the ground rods, primary cable neutral wires, secondary neutral, and all groundable parts of the equipment.

C. Supplemental Grounding Electrode:
1. Provide the following grounding in addition to the equipment ground conductor supplied with the feeder conductors whether or not shown on the Drawings.
   a. See Grounding Electrode System paragraph for conductor termination requirements.
2. Metal light poles:
   a. Connect metal pole and pole base reinforcing steel to a ground rod.
   b. Grounding conductor: Bare #6 AWG minimum.
3. Equipment support rack and pedestals mounted outdoors:
   a. Connect metallic structure to a ground rod.
   b. Grounding conductor: #6 AWG minimum.
4. Engine generator:
   a. Connect generator frame to the ground ring at two locations (opposite corners of frame).
   b. Grounding conductor: Bare conductor, size as indicated on the Drawings.

D. Other Bonding Requirements:
1. Other metal piping:
   a. Connect in a daisy chain or radial fashion: Interior hot and cold water piping system , process piping , fire sprinkler system service water pipe to the main ground bus as indicated on the Drawings.
   b. See Grounding Electrode System paragraph for conductor termination requirements.
2. Lightning protection system:
   a. Connect to ground ring.
   b. See Specification Section 26 41 13.

E. Transformer Separately Derived Grounding System:
1. Install the System Bonding Jumper at the transformer. At the first disconnect, ensure the neutral is isolated from ground.
2. Structures with a single electrical room/area:
   a. Connect grounding electrode conductor to the Grounding Electrode System main ground bar.
3. Structures with multiple electrical rooms/areas:
   a. Provide a ground bar mounted in each electrical room/area.
   b. Interconnect all ground bars in a daisy chain or radial fashion to the main ground bar as indicated on the Drawings.
   c. Connect grounding electrode conductor to the electrical room/area ground bar.
4. See Grounding Electrode System paragraph for conductor termination requirements.

F. Raceway Bonding/Grounding:
   1. Install all metallic raceway so that it is electrically continuous.
   2. Provide an equipment grounding conductor in all raceways with insulation identical to the phase conductors, unless otherwise indicated on the Drawings.
   3. NFPA 70 required grounding bushings shall be of the insulating type.
   4. Provide Myers' hubs or double locknuts at all panels, per Section 26 05 33.
   5. Bond all conduits, at entrance and exit of equipment, to the equipment ground bus or lug.
   6. Provide bonding jumpers if conduits are installed in concentric knockouts.
   7. Make all metallic raceway fittings and grounding clamps tight to ensure equipment grounding system will operate continuously at ground potential to provide low impedance current path for proper operation of overcurrent devices during possible ground fault conditions.

G. Equipment Grounding:
   1. Ground all utilization equipment with an equipment grounding conductor.

H. Cable Tray Grounding:
   1. Make every metal cable tray electrically continuous by one of the following methods:
      a. Tray sections and fittings suitable for grounding purposes.
      b. Provide bonding jumpers at discontinuous joints.
      c. Lay a grounding conductor within the tray for bonding of each individual tray section.
         1) Provide a minimum of one ground lug per tray section.
         2) Grounding conductor: Insulated #4 AWG minimum with green insulation or green marking tape every 10 feet.
         3) Securely tie the grounding conductor to the cable tray every 5 feet. Bond the grounding conductor to the cable tray run a minimum of every 10 ft with a UL listed connector.
   2. Bond the tray or tray grounding conductor to every electrical equipment ground bus or telecom backboard ground bus where conductors terminate.
   3. Bond all conduits that extend conductors to field equipment to the tray.

I. Manhole and Handhole Grounding:
   1. Provide a ground rod and ground bar, when indicated or as needed, in each manhole and handhole with exposed metal parts.
      a. Expose a minimum of 4 inches of the rod above the floor for field connections to the rod.
   2. Connect all exposed metal parts (e.g., conduits and cable racks) to the ground rod.

J. Pad Mounted Switchgear and Sectionalizing Equipment Grounding:
   1. Per details on Drawings.
   2. Provide a ground rod(s) in each enclosure with a minimum of 4IN of rode exposed above the floor for connection to the rod.
   3. Connect metal frame to the ground rods.
   4. Connect medium-voltage cable shields to the rods.
   5. Connect surge arrestor to the ground rods.

K. Pad Mounted Transformer Grounding:
   1. Per detail on the Drawings.
   2. Provide a ground ring around transformer with:
      a. A minimum of two ground rods located at the corners.
      b. A ground rod connected to the ring and located in the conduit stub up area.
      c. Or as indicated on the Drawings.
   3. Connect medium voltage cable shields to the ground rod.
   4. Connect surge arrestor to the ground rod.
   5. Connect elbow terminations to ground ring.
   6. Connect the XO and ground pad to the ground rod.
      a. Isolate the XO bushing from ground where indicated in the Drawings.
   7. Connect exposed metallic conduits to the ground rod.

L. Prefabricated Composite Material Test Station:
   1. Place test station on a foundation of compacted 1/4 to 1/2 inches crushed rock or gravel a minimum of 8 inches thick and 6 inches larger than handholes footprint on all sides.
   2. Provide concrete encasement ring around test station per manufacturers installation instructions (minimum of 10 inches wide x 12 inches deep).

3.2 FIELD QUALITY CONTROL

A. Leave grounding system uncovered until observed by Contracting Officer Representative.

B. Acceptance testing:
   1. See Specification Section 26 08 13.

C. Provide a continuity test on the components of the grounding electrode system.

D. Complete grounding system: Resistance of 5 ohms or less.

E. Test resistance of installed ground system after backfilling and before connection to any other grounded system including underground piping, utility services or other building ground systems.
   1. Test ground grid resistance by fall-of-potential method.
   2. Perform test at the main ground bar.

END OF SECTION

SECTION 26 05 33
RACEWAYS AND BOXES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for:
      a. Conduits.
      b. Conduit fittings.
      c. Conduit supports.
      d. Wireways.
      e. Outlet boxes.
      f. Pull and junction boxes.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 05 19 - Low Voltage Electrical Power Conductors and Cables.
   3. Section 26 05 43 - Electrical - Exterior Underground.
   4. Section 26 27 26 - Wiring Devices.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Iron and Steel Institute (AISI).
   2. ASTM International (ASTM):
      a. A123/A123M, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
      b. A153/A153M, Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
      c. D2564, Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems.
   3. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. RN 1, Polyvinyl Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit.
      c. TC 2, Electrical Polyvinyl Chloride (PVC) Tubing and Conduit.
      d. TC 3, Polyvinyl Chloride (PVC) Fittings for Use with Rigid PVC Conduit and Tubing.
      e. TC 14.AG, Aboveground Reinforced Thermosetting Resin Conduit and Fittings.
      f. TC 14.BG, Belowground Reinforced Thermosetting Resin Conduit and Fittings.
   4. National Electrical Manufacturers Association/American National Standards Institute (NEMA/ANSI):
      a. C80.1, Electric Rigid Steel Conduit (ERSC).
      b. C80.3, Steel Electrical Metallic Tubing (EMT).
      c. OS 1, Sheet-Steel Outlet Boxes, Device Boxes, Covers, and Box Supports.
   5. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   6. Underwriters Laboratories, Inc. (UL):
      a. 1, Standard for Flexible Metal Conduit.
      b. 6, Electrical Rigid Metal Conduit - Steel.
      c. 50, Enclosures for Electrical Equipment, Non-Environmental Considerations.
      d. 360, Standard for Liquid-Tight Flexible Metal Conduit.
      e. 467, Grounding and Bonding Equipment.
      f. 514A, Metallic Outlet Boxes.
      g. 514B, Conduit, Tubing, and Cable Fittings.

h. 651, Standard for Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings.
i. 651A, High Density Polyethylene (HDPE) Conduit
j. 797, Electrical Metallic Tubing - Steel.
k. 870, Standard for Wireways, Auxiliary Gutters, and Associated Fittings.
l. 1203, Standard for Explosion-Proof and Dust-Ignition-Proof Electrical Equipment for Use in Hazardous (Classified) Locations.
m. 2420, Belowground Reinforced Thermosetting Resin Conduit (RTRC) and Fittings.
n. 2515, Aboveground Reinforced Thermosetting Resin Conduit (RTRC) and Fittings.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 00 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
1. Product technical data:
a. Provide submittal data for all products specified in PART 2 of this Specification Section.
b. See Section 26 05 00 for additional requirements.
2. Fabrication and/or layout drawings:
a. Proposed routing of raceways buried under concrete floors and embedded in concrete walls.
1) Identify conduit by tag number of equipment served or by circuit schedule number.
b. Proposed routing and details of construction, including raceway and rebar, for raceways embedded in floor slabs, walls and columns.
c. Proposed location and details of construction for openings in slabs and walls for raceway runs.
d. Identify dimensional size of pull and junction boxes to be used.
e. Seismic location installation details.

1.5 DELIVERY, STORAGE, AND HANDLING

A. See Section 26 05 00.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Rigid metal conduits and electrical metallic tubing:
a. Allied Tube and Conduit.
b. Or Equal.
2. PVC coated rigid metal conduits:
a. Robroy Industries.
b. Or Equal.
3. Rigid nonmetallic conduit:

a. Prime Conduit.
b. Or Equal.

4. Flexible conduit:
   a. Southwire Company, LLC.
   b. Or Equal.
5. Wireway:
   a. Square D by Schneider Electric.
   b. Or Equal.
6. Underfloor raceways.
   a. Walkerduct by Legrand.
   b. Or Equal.
7. Conduit fittings and accessories:
   a. Appleton by Emerson Electric Co.
   b. Or Equal.
8. Support systems:
   a. Unistrut by Atkore International, Inc.
   b. Or Equal.
9. Outlet, pull and junction boxes:
   a. Appleton by Emerson Electric Co.
   b. Or Equal.

2.2 RIGID METAL CONDUITS

A. Rigid Galvanized Steel Conduit (RGS):
   1. Mild steel with continuous welded seam.
   2. Metallic zinc applied by hot-dip galvanizing or electro-galvanizing.
   3. Threads galvanized after cutting.
   4. Internal coating: Baked lacquer, varnish or enamel for a smooth surface.
   5. Standards: NFPA 70 Type RMC, NEMA/ANSI C80.1, UL 6.

B. PVC-Coated Rigid Steel Conduit (PVC-RGS):
   1. Nominal 40 mil Polyvinyl Chloride Exterior Coating:
      a. Coating: Bonded to hot-dipped galvanized rigid steel conduit conforming to NEMA/ANSI C80.1.
      b. The bond between the PVC coating and the conduit surface: Greater than the tensile strength of the coating.
   2. Nominal 2 mil, minimum, urethane interior coating.
   3. Urethane coating on threads.
   4. Conduit: Epoxy prime coated prior to application of PVC and urethane coatings.
   5. Female Ends:
      a. Have a plastic sleeve extending a minimum of one pipe diameter or 2 inches, whichever is less beyond the opening.
      b. The inside diameter of the sleeve shall be the same as the outside diameter of the conduit to be used with it.
   6. Pass high temperature water PVC coating adhesion test and be ETL Verified PVC-001.
   7. Standards: NFPA 70 Type RMC, NEMA/ANSI C80.1, UL 6, NEMA RN 1.

2.3 ELECTRICAL METALLIC TUBING (EMT)

A. Mild steel with continuous welded seam.

B. Metallic zinc applied by hot-dip galvanizing or electro-galvanizing.

C. Internal coating: Baked lacquer, varnish, or enamel for a smooth surface.

D. Standards: NFPA 70 Type EMT, NEMA/ANSI C80.3, UL 797.

2.4 RIGID NONMETALLIC CONDUIT

A. Schedules 40 (PVC-40) and 80 (PVC-80):

1. Polyvinyl-chloride (PVC) plastic compound which includes inert modifiers to improve weatherability and heat distribution.
2. Rated for direct sunlight exposure.
3. Fire retardant and low smoke emission.
4. Shall be suitable for use with 90 degrees C wire and shall be marked "maximum 90 degrees C".
5. Standards: NFPA 70 Type PVC, NEMA TC 2, UL 651.

B. Type EB (PVC-EB):
1. Polyvinyl-chloride (PVC) plastic compound.
2. Rated for concrete encasement.
3. Suitable for use with 90 degrees C wire.
4. Standards: ASTM F512, NFPA 70 Type PVC, UL 651A.

C. High-Density Polyethylene (HDPE) Pipe:
1. Meets UL 651A
2. Pressure Rating: 160 psi or higher.
3. Natural Density: 0.949 gm/cc
4. Melt Index (190°C/2.16 kg): 0.08 gm/10 min
5. Tensile Strength @ Yield: 3,500 psi
6. Ultimate Elongation: >800%
7. Flexural Modulus: 150,000 psi
8. Vicat Softening Temperature: 255°F
9. PPI HYDROSTATIC DESIGN BASIS (HDB): 1,600 psi @ 73.4°F

2.5 FLEXIBLE CONDUIT

A. Flexible Galvanized Steel Conduit (FLEX):
1. Formed of continuous, spiral wound, hot-dip galvanized steel strip with successive convolutions securely interlocked.
2. Standard: NFPA 70 Type FMC, UL 1.

B. PVC-Coated Flexible Galvanized Steel (liquid-tight) Conduit (FLEX-LT):
1. Core formed of continuous, spiral wound, hot-dip galvanized steel strip with successive convolutions securely interlocked.
2. Extruded PVC outer jacket positively locked to the steel core.
3. Liquid and vaportight.
4. Standard: NFPA 70 Type LFMC, UL 360.

2.6 FIBER OPTTIC MICRODUCT

A. See Specification Section 27 10 00.

2.7 WIREWAY

A. General:
1. Suitable for lay-in conductors.
2. Designed for continuous grounding.
3. Covers:
   a. Hinged or removable in accessible areas.
   b. Non-removable when passing through partitions.
4. Finish: Rust inhibiting primer and manufacturer's standard paint inside and out except for stainless steel type.
5. Standards: UL 870, NEMA 250.

B. General Purpose (NEMA 1 rated) Wireway:
1. 14 or 16 gage steel without knockouts.
2. Cover: Solid, non-gasketed and held in place by captive screws.

C. Watertight (NEMA 4X rated) Wireway:

1. 14 GA Type 304 or 316 stainless steel bodies and covers without knockouts and 10 GA stainless steel flanges.
2. Cover: Fully gasketed and held in place with captive clamp type latches.
3. Flanges: Fully gasketed and bolted.

D. Dusttight (NEMA 12 rated) Wireway:
1. 14 GA steel bodies and covers without knockouts and 10 GA steel flanges.
2. Cover: Fully gasketed and held in place with captive clamp type latches.
3. Flanges: Fully gasketed and bolted.

2.8 CONDUIT FITTINGS AND ACCESSORIES

A. Fittings for Use with RGS:
1. General:
   a. Finish: Electrostatically applied epoxy powder coat.
   b. In hazardous locations listed for use in Class I, Groups C and D locations.
2. Locknuts:
   a. Threaded steel or malleable iron.
   b. Gasketed or non-gasketed.
   c. Grounding or non-grounding type.
3. Bushings:
   a. Threaded, insulated metallic.
   b. Grounding or non-grounding type.
4. Hubs: Threaded, insulated and gasketed metallic for raintight connection.
5. Couplings:
   a. Threaded straight type: Same material and finish as the conduit with which they are used on.
   b. Threadless type: Gland compression or self-threading type, concrete tight.
6. Unions: Threaded galvanized steel or zinc plated malleable iron.
7. Conduit bodies (ells and tees):
   a. Body: Zinc plated cast iron with threaded hubs.
   b. Standard and mogul size.
   c. Cover:
      1) Clip-on type with stainless steel screws.
      2) Gasketed or non-gasketed galvanized steel or zinc plated cast iron.
8. Conduit bodies (round):
   a. Body: Zinc plated cast iron with threaded hubs.
   b. Cover: Threaded screw on type, gasketed, galvanized steel or zinc plated cast iron.
9. Sealing fittings:
   a. Body: Zinc plated cast iron or cast copper free aluminum with threaded hubs.
   b. Standard and mogul size.
   c. With or without drain and breather.
   d. Fiber and sealing compound: UL listed for use with the sealing fitting.
10. Hazardous location flexible coupling (HAZ-FLEX):
   a. Liquid tight and arc resistant.
   b. Electrically conductive so no bonding jumper is required.
   c. Dry and wet areas:
      1) Bronze braided covering over flexible brass core.
      2) Bronze end fittings.
      3) Zinc-plated steel or malleable iron unions and nipples.
   d. Corrosive areas:
      1) Stainless steel braided covering over flexible stainless steel core.
      2) Stainless steel end fittings.
      3) Aluminum unions and nipples.
11. Expansion couplings:
   a. 2 inches nominal straight-line conduit movement in either direction.
   b. Galvanized steel with insulated bushing.

c. Gasketed for wet locations.
d. Internally or externally grounded.

12. Expansion/deflection couplings:
    a. 3/4 inches nominal straight-line conduit movement in either direction.
    b. 30 degrees nominal deflection from the normal in all directions.
    c. Metallic hubs, neoprene outer jacket and stainless steel jacket clamps.
    d. Internally or externally grounded.
    e. Watertight, raintight and concrete tight.
13. Standards: UL 467, UL 514B, UL 1203.

B. Fittings for Use with PVC-RGS:
1. The same material and construction as those fittings listed under paragraph "Fittings for Use with RGS " and coated as defined under paragraph "PVC Coated Rigid Steel Conduit (PVC-RGS)."

C. Fittings for Use with EMT:
1. Connectors:
    a. Straight, angle and offset types furnished with locknuts.
    b. Zinc plated steel.
    c. Insulated gland compression type.
    d. Concrete and raintight.
2. Couplings:
    a. Zinc plated steel.
    b. Gland compression type.
    c. Concrete and raintight.
3. Conduit bodies (ells and tees):
    a. Body: Copper free aluminum with threaded hubs.
    b. Standard and mogul size.
    c. Cover:
        1) Screw down type with steel screws.
        2) Gasketed or non-gasketed galvanized steel or copper free aluminum.
4. Standard: UL 514B.

D. Fittings for Use with FLEX:
1. Connector:
    a. Zinc plated malleable iron.
    b. Squeeze or clamp-type.
2. Standard: UL 514B.

E. Fittings for Use with FLEX-LT:
1. Connector:
    a. Straight or angle type.
    b. Metal construction, insulated and gasketed.
    c. Composed of locknut, grounding ferrule and gland compression nut.
    d. Liquid tight.
2. Standards: UL 467, UL 514B.

F. Fittings for Use with Rigid Nonmetallic PVC Conduit:
1. Coupling, adapters and conduit bodies:
    a. Same material, thickness, and construction as the conduits with which they are used.
    b. Homogeneous plastic free from visible cracks, holes or foreign inclusions.
    c. Bore smooth and free of blisters, nicks or other imperfections which could damage the conductor.
2. Solvent cement for welding fittings shall be supplied by the same manufacturer as the conduit and fittings.
3. Standards: ASTM D2564, NEMA TC 3, UL 651, UL 514B.

G. Weather and Corrosion Protection Tape:

1. PVC based tape, 10 mils thick.
2. Protection against moisture, acids, alkalis, salts and sewage and suitable for direct bury.
3. Used with appropriate pipe primer.

2.9 ALL RACEWAY AND FITTINGS

A. Mark Products:
   1. Identify the nominal trade size on the product.
   2. Stamp with the name or trademark of the manufacturer.

2.10 OUTLET BOXES

A. Metallic Outlet Boxes:
   1. Hot-dip galvanized steel.
   2. Conduit knockouts and grounding pigtail.
   3. Styles:
      a. 2 inches x 3 inches rectangle.
      b. 4 inches square.
      c. 4 inches octagon.
      d. Masonry/tile.
   4. Accessories:
      a. Flat blank cover plates.
      b. Barriers.
      c. Extension, plaster or tile rings.
      d. Box supporting brackets in stud walls.
      e. Adjustable bar hangers.
   5. Standards: NEMA/ANSI OS 1, UL 514A.

B. Cast Outlet Boxes:
   1. Zinc plated cast iron with manufacturer's standard finish.
   2. Threaded hubs and grounding screw.
   3. Styles:
      a. "FS" or "FD".
      b. "Bell".
      c. Single or multiple gang and tandem.
      d. "EDS" or "EFS" for hazardous locations.
   4. Accessories: 40 mil PVC exterior coating and 2 mil urethane interior coating.
   5. Standards: UL 514A, UL 1203.

C. Nonmetallic Outlet Boxes:
   1. Polyvinyl-chloride (PVC) plastic compound.
   2. Rated for direct sunlight exposure.
   3. Fire retardant and low smoke emission.
   4. Suitable for use with 90 degrees C wire.
   5. Styles:
      a. “FS” or “FD”.
      b. Single or multiple gang.
   6. Standards: UL 514A, NEMA TC 3.

D. See Specification Section 26 27 26 for wiring devices, wallplates and coverplates.

2.11 PULL AND JUNCTION BOXES

A. NEMA 1 Rated:
   1. Body and cover: 14 GA minimum, galvanized steel or 14 GA minimum, steel finished with rust inhibiting primer and manufacturers standard paint inside and out.
   2. With or without concentric knockouts on four sides.
   3. Flat cover fastened with screws.

B. NEMA 4X Rated (metallic):

1. Body and cover: 14 GA Type 304 or 316 stainless steel.
2. Seams continuously welded and ground smooth.
3. No knockouts.
4. External mounting flanges.
5. Hinged door and stainless steel screws and clamps.
6. Door with oil-resistant gasket.

C. NEMA 4X Rated (Nonmetallic):
1. Body and cover: Ultraviolet light protected fiberglass-reinforced polyester boxes.
2. No knockouts.
3. External mounting flanges.
4. Hinged door with quick release latches and padlocking hasp.
5. Door with oil resistant gasket.

D. NEMA 12 Rated:
1. Body and cover:
a. 14 GA steel finished with rust inhibiting primer and manufacturers standard paint inside and out.
2. Seams continuously welded and ground smooth.
3. No knockouts.
4. External mounting flanges.
5. Non-hinged cover held closed with captivated cover screws threaded into sealed wells or hinged cover held closed with stainless steel screws and clamps.
6. Flat door with oil resistant gasket.

E. Miscellaneous Accessories:
1. Rigid handles for covers larger than 9 square feet or heavier than 25 pounds.
2. Split covers when heavier than 25 pounds.
3. Weldnuts for mounting optional panels and terminal kits.
4. Tamper proof screws.
5. Terminal blocks: Screw-post barrier-type, rated 600 volt and 20 ampere minimum.

F. Standards: NEMA 250, UL 50.

2.12 SPECIAL PURPOSE BOXES

A. Pedestal-Type Floor-Mounted or Counter-Mounted Duplex Receptacles:
1. Horizontal design housing with threaded conduit fittings in base with satin chromium finish.

B. Flush in Floor Duplex Receptacles:
1. Dual level, fully adjustable box with power fitting and brass carpet flange.

2.13 SUPPORT SYSTEMS

A. Multi-conduit Surface or Trapeze Type Support and Pull or Junction Box Supports:
1. Material requirements.
a. Galvanized steel: ASTM A123/A123M or ASTM A153/A153M.
b. Stainless steel: AISI Type 316.
c. PVC coat galvanized steel: ASTM A123/A123M or ASTM A153/A153M and 20 mil PVC coating.
d. Aluminum: AA Type 6063-T6.
e. Fiberglass: Fire-retardant polyester or vinylester resin, ASTM E84, UL 94.

B. Single Conduit and Outlet Box Support Fasteners:
1. Material requirements:
a. Zinc plated steel.
b. Stainless steel.
c. Malleable iron.
d. PVC coat malleable iron or steel: 20 mil PVC coating.
e. Steel protected with zinc phosphate and oil finish.

2.14 OPENINGS AND PENETRATIONS IN WALLS AND FLOORS

A. Sleeves, smoke and fire stop fitting through walls and floors:
   1. See Specification Section 01 73 20.

B. Smoke and Fire Stop Fittings:
   1. 3 hour UL rating.
   2. Flanged and segmented.
   3. Cast malleable iron mounting frame and pressure plate.
      a. Elastomeric sealing material.
      b. Steel clamping hardware.

C. Sleeves through walls and floors:
   1. Uncoated or galvanized iron or steel:
   2. Wall thickness: Not less than standard Schedule 40 pipe.

PART 3 - EXECUTION

3.1 RACEWAY INSTALLATION - GENERAL

A. Shall be in accordance with the requirements of:
   1. NFPA 70.
   2. Manufacturer instructions.

B. Size of Raceways:
   1. Raceway sizes are shown on the Drawings, if not shown on the Drawings, then size in accordance with NFPA 70.
   2. Unless specifically indicated otherwise, the minimum raceway size shall be:
      a. Conduit: 3/4 inches.
      b. Wireway: 2-1/2 inches x 2-1/2 inches.

C. Field Bending and Cutting of Conduits:
   1. Utilize tools and equipment recommended by the manufacturer of the conduit, designed for the purpose and the conduit material to make all field bends and cuts.
   2. Do not reduce the internal diameter of the conduit when making conduit bends.
   3. Prepare tools and equipment to prevent damage to the PVC coating.
   4. Degrease threads after threading and apply a zinc rich paint.
   5. Debur interior and exterior after cutting.

D. Male threads of conduit systems shall be coated with an electrically conductive anti-seize compound.

E. The protective coating integrity of conduits, fittings, outlet, pull and junction boxes and accessories shall be maintained.
   1. Repair galvanized components utilizing a zinc rich paint.
   2. Repair painted components utilizing touch up paint provided by or approved by the manufacturer.
   3. Repair PVC coated components utilizing a patching compound, of the same material as the coating, provided by the manufacturer of the conduit; or a self-adhesive, highly conformable, cross-linked silicone composition strip, followed by a protective coating of vinyl tape.
      a. Total nominal thickness: 40 mil.
   4. Repair surfaces which will be inaccessible after installation prior to installation.
   5. Wrap metallic conduits with protective tape where conduits pass through above-grade penetrations where PVC-RGS is not required.

F. Remove moisture and debris from conduit before wire is pulled into place.
   1. Pull mandrel with diameter nominally 1/4 inches smaller than the interior of the conduit, to remove obstructions.
   2. Swab conduit by pulling a clean, tight-fitting rag through the conduit.

3. Tightly plug ends of conduit with tapered wood plugs or plastic inserts until wire is pulled.

G. Only nylon or polyethylene rope shall be used to pull wire and cable in conduit systems.

H. Seal all conduits that pass from the interior of the building or structure to the exterior of the building or structure at both ends using duct seal putty.
   1. Conduits that terminate in manholes or handholes shall be left unsealed at the end in the manhole or handhole to allow for drainage.

I. Fill openings in walls, floors, and ceilings and finish flush with surface.
   1. See Specification Section 01 73 20.

3.2 RACEWAY ROUTING

A. Raceways shall be routed in the field unless otherwise indicated.
   1. Conduit and fittings shall be installed, as required, for a complete system that has a neat appearance and is in compliance with all applicable codes.
   2. Run in straight lines parallel to or at right angles to building lines.
   3. Do not route conduits:
      a. Through areas of high ambient temperature or radiant heat.
      b. In suspended concrete slabs.
      c. In concrete members including slabs, slabs on grade, beams, walls, and columns unless specifically located and detailed on structural Drawings.
   4. Locate sleeves or conduits penetrating floors, walls, and beams so as not to significantly impair the strength of the construction. Do not place conduit penetrations in columns.
   5. Conduit shall not interfere with, or prevent access to, piping, valves, ductwork, or other equipment for operation, maintenance and repair.
   6. Provide pull boxes or conduit bodies as needed so that there is a maximum of 360 degrees of bends in the conduit run or in long straight runs to limit pulling tensions.

B. All conduits within a structure shall be installed exposed except as follows:
   1. As indicated on the Drawings.
   2. Concealed above gypsum wall board or acoustical tile suspended ceilings.
   3. Conduits in architecturally finished areas shall be concealed.

C. Maintain minimum spacing between parallel conduit and piping runs in accordance with the following when the runs are greater than 30 feet:
   1. Between instrumentation and telecommunication: 1 inch.
   2. Between instrumentation and 125 V, 48 V and 24 VDC: 2 inches.
   3. Between instrumentation and 600 V and less AC power or control: 6 inches.
   4. Between instrumentation and greater than 600 VAC power: 12 inches.
   5. Between telecommunication and 125 V, 48 V and 24 VDC, 2 inches.
   6. Between telecommunication and 600 V and less AC power or control: 6 inches.
   7. Between telecommunication and greater than 600 VAC power: 12 inches.
   8. Between 125 V, 48 V and 24 VDC and 600 V and less AC power or control: 2 inches.
   9. Between 125 V, 48 V and 24 VDC and greater than 600 VAC power: 2 inches.
   10. Between 600 V and less AC and greater than 600 VAC: 2 inches.
   11. Between process, gas, air and water pipes: 6 inches.

D. Conduits shall be installed to eliminate moisture pockets.
   1. Where water cannot drain to openings, provide drain fittings in the low spots of the conduit run.

E. Conduit shall not be routed on the exterior of structures except as specifically indicated on the Drawings.

F. Where sufficient room exists within the housing of roof-mounted equipment, the conduit shall be stubbed up inside the housing.

G. Provide all required openings in walls, floors, and ceilings for conduit penetration.
   1. See Specification Section 01 73 20.

3.3 RACEWAY APPLICATIONS

A. Permitted Raceway Types Per Wire or Cable Types:
   1. Power wire or cables: All raceway types.
   2. Control wire or cables: All raceway types.
   3. Instrumentation cables: Metallic raceway except nonmetallic may be used underground.
   4. Motor leads from a VFD: RGS, or shielded VFD cables in all other raceways.
   5. Telecommunication cables: All raceway types.

B. Permitted Raceway Types Per Area Designations:
   1. Dry areas:
      a. RGS.
   2. Wet areas:
      a. Indoors: RGS.
      b. Outdoors, vaults, manholes, handholes: PVC-RGS.
   3. Corrosive areas:
      a. PVC-RGS.
   4. Highly corrosive areas:
      a. PVC-RGS.
   5. NFPA 70 hazardous areas:
      a. RGS.

C. Permitted Raceway Types Per Routing Locations:
   1. In stud framed walls:
      a. EMT.
   2. In concrete block or brick walls:
      a. PVC-40.
      b. PVC-80
   3. Above acoustical tile ceilings:
      a. EMT.
      b. NEMA 1 rated wireway.
   4. Embedded in poured concrete walls and floors:
      a. PVC-40.
      b. PVC-RGS when emerging from concrete in any location – extend PVC coating minimum 6 inches above finished floor surface.
   5. Beneath floor slab-on-grade:
      a. PVC-40.
      b. RGS for VFD motor lead circuits, and where indicated on the Drawings.
   6. Through floor penetrations, see Specification Section 01 73 20:
      a. PVC-RGS - extend PVC coating minimum 6 inches above and below finished surfaces.
   7. Direct buried conduits and ductbanks:
      a. PVC-80.
      b. 90 degrees elbows for transitions to above grade:
         1) PVC-RGS – extend PVC coating minimum 6 inches above finished surface.
      c. Long sweeping bends greater than 15 degrees:
         1) PVC-RGS.
   8. Concrete encased ductbanks:
      a. PVC-40.
      b. 90 degree elbows for transitions to above grade:
         1) PVC-RGS – extend PVC coating minimum 6 inches above finished surface.
      c. Long sweeping bends greater than 15 degrees:
         1) RGS for sizes 2 inches and larger.

D. FLEX conduits shall be installed for connections to light fixtures, HVAC equipment and other similar devices above the ceilings.
   1. The maximum length shall not exceed:

a. 6 feet to light fixtures.
b. 3 feet to all other equipment.

E. FLEX-LT conduits shall be installed as the final conduit connection to light fixtures, dry type transformers, motors, electrically operated valves, instrumentation primary elements, and other electrical equipment that is liable to vibrate.
1. The maximum length shall not exceed:
a. 6 feet to light fixtures.
b. 3 feet to motors.
c. 2 feet to all other equipment.

F. HAZ-FLEX coupling shall be installed as the final conduit to motors, electrically operated valves, instrumentation primary elements and electrical equipment that is liable to vibrate.
1. The maximum length shall not exceed:
a. 3 feet to motors.
b. 2 feet to all other equipment.

G. NEMA 1 Rated Wireway:
1. Surface mounted in electrical rooms.
2. Surface mounted above removable ceilings tiles of an architecturally finished area.

H. NEMA 4X Rated Wireway:
1. Surface mounted outdoors and in areas designated as wet and or corrosive.

I. NEMA 12 Rated Wireway:
1. Surface mounted in areas designated as dry in architecturally and non-architecturally finished areas.

J. Underground Conduit: See Specification Section 26 05 43.

3.4 CONDUIT FITTINGS AND ACCESSORIES

A. Conduit Seals:
1. Installed in conduit systems located in hazardous areas as required by the NFPA 70.
2. Fill plug and drain shall be accessible.
3. Pour the conduit seals in a two-step process.
a. Pour the seal and leave cover off.
b. After seal is dry, inspect for proper sealing, install cover and mark (for example, paint or permanent marker) as complete.

B. Rigid nonmetallic conduit and fittings shall be joined utilizing solvent cement.
1. Immediately after installation of conduit and fitting, the fitting or conduit shall be rotated 1/4 turn to provide uniform contact.

C. Install Expansion Fittings:
1. Where conduits are exposed to the sun and conduit run is greater than 200 feet.
2. Elsewhere as identified on the Drawings.

D. Install Expansion/Deflection Fittings:
1. Where conduits enter a structure.
a. Except electrical manholes and handholes.
b. Except where the ductbank is tied to the structure with rebar.
2. Where conduits span structural expansions joints.
3. Elsewhere as identified on the Drawings.

E. Threaded connections shall be made wrench-tight.

F. Conduit joints shall be watertight:
1. Where subjected to possible submersion.
2. In areas classified as wet.
3. Underground.

G. Terminate Conduits:
   1. In metallic outlet boxes:
      a. RGS:
         1) Conduit hub and locknut.
         2) Insulated bushing and two locknuts.
         3) Use grounding type locknut or bushing when required by NFPA 70.
      b. EMT: Compression type connector and locknut.
   2. In NEMA 1 rated enclosures:
      a. RGS:
         1) Conduit hub and locknut.
         2) Insulated bushing and two locknuts.
         3) Use grounding type locknut or bushing when required by NFPA 70.
      b. EMT: Compression type connector and locknut.
   3. In NEMA 12 rated enclosures:
      a. Watertight, insulated and gasketed hub and locknut.
      b. Use grounding type locknut or bushing when required by NFPA 70.
   4. In NEMA 4X rated enclosures:
      a. Watertight, insulated and gasketed hub and locknut.
      b. Use grounding type bushing when required by NFPA 70.
   5. When stubbed up through the floor into floor mount equipment:
      a. With an insulated grounding bushing on metallic conduits.
      b. With end bells on nonmetallic conduits.

H. Threadless couplings shall only be used to join new conduit to existing conduit when the existing conduit end is not threaded and it is not practical or possible to cut threads on the existing conduit with a pipe threader.

3.5 CONDUIT SUPPORT

A. Permitted multi-conduit surface or trapeze type support system per area designations and conduit types:
   1. Dry or wet and/or hazardous areas:
      a. Galvanized system consisting of: Hot-dipped galvanized steel channels and fittings, nuts and hardware and conduit clamps.
   2. Corrosive areas:
      a. Stainless steel system consisting of: Stainless steel channels and fittings, nuts and hardware and conduit clamps.
      b. PVC coated steel system consisting of: PVC coated galvanized steel channels and fittings and conduit clamps with stainless steel nuts and hardware.
   3. Highly corrosive areas:
      a. PVC coated steel system consisting of: PVC coated galvanized steel channels and fittings and conduit clamps with stainless steel nuts and hardware.
      b. Fiberglass system consisting of: Fiberglass channels and fittings, nuts and hardware and conduit clamps.
   4. Conduit type shall be compatible with the support system material.
      a. Galvanized steel system may be used with RGS and EMT.
      b. Stainless steel system may be used with RGS and PVC-RGS.
      c. PVC coated galvanized steel system may be used with PVC-RGS and PVC-40 and PVC-80.
      d. Fiberglass system may be used with PVC-40 and PVC-80 and PVC-RGS and Fiberglass.

B. Permitted single conduit support fasteners per area designations and conduit types:
   1. Architecturally finished areas:
      a. Material: Zinc plated steel, or steel protected with zinc phosphate and oil finish.
      b. Types of fasteners: Spring type hangers and clips, straps, hangers with bolts, clamps with bolts and bolt-on beam clamps.

c. Provide anti-rattle conduit supports when conduits are routed through metal studs.

2. Dry or wet and/or hazardous areas:
   a. Material: Zinc plated steel, stainless steel and malleable iron.
   b. Types of fasteners: Straps, hangers with bolts, clamps with bolts and bolt on beam clamps.
3. Corrosive areas:
   a. Material: Stainless steel and PVC coat malleable iron or steel.
   b. Types of fasteners: Straps, hangers with bolts, clamps with bolts and bolt on beam clamps.
4. Highly corrosive areas:
   a. Material: PVC coat malleable iron or steel.
   b. Types of fasteners: Straps, hangers with bolts, clamps with bolts and bolt on beam clamps.
5. Conduit type shall be compatible with the support fastener material.
   a. Zinc plated steel, steel protected with zinc phosphate and oil finish and malleable iron fasteners may be used with RGS and EMT.
   b. Stainless steel system may be used with RGS and PVC-RGS.
   c. PVC coated fasteners may be used with PVC-RGS and PVC-40 and PVC-80.
   d. Nonmetallic fasteners may be used with PVC-40, PVC-80 and fiberglass.

C. Conduit Support General Requirements:
1. Maximum spacing between conduit supports per NFPA 70.
2. Support conduit from the building structure.
3. Do not support conduit from process, gas, air or water piping; or from other conduits.
4. Provide hangers and brackets to limit the maximum uniform load on a single support to 25 LBS or to the maximum uniform load recommended by the manufacturer if the support is rated less than 25 pounds.
   a. Do not exceed maximum concentrated load recommended by the manufacturer on any support.
   b. Conduit hangers:
      1) Continuous threaded rods combined with struts or conduit clamps: Do not use perforated strap hangers and iron bailing wire.
   c. Do not use suspended ceiling support systems to support raceways.
   d. Hangers in metal roof decks:
      1) Utilize fender washers.
      2) Not extend above top of ribs.
      3) Not interfere with vapor barrier, insulation, or roofing.
5. Conduit support system fasteners:
   a. Use sleeve-type expansion anchors as fasteners in masonry wall construction.
   b. Do not use concrete nails and powder-driven fasteners.

3.6 OUTLET, PULL AND JUNCTION BOX INSTALLATION

A. General:
1. Install products in accordance with manufacturer's instructions.
2. See Section 26 05 00 and the Drawings for area classifications.
3. Fill unused punched-out, tapped, or threaded hub openings with insert plugs.
4. Size boxes to accommodate quantity of conductors enclosed and quantity of conduits connected to the box.

B. Outlet Boxes:
1. Permitted uses of metallic outlet boxes:
   a. Housing of wiring devices:
      1) Recessed in all stud framed walls and ceilings.
      2) Recessed in poured concrete, concrete block and brick walls of architecturally finished areas and exterior building walls.
   b. Pull or junction box:

1) Above gypsum wall board or acoustical tile ceilings.
2) Above 10 feet in an architecturally finished area where there is no ceiling.

2. Permitted uses of cast outlet boxes:
   a. Housing of wiring devices surface mounted in non-architecturally finished dry, wet, corrosive, highly corrosive and hazardous areas.
   b. Pull and junction box surface mounted in non-architecturally finished dry, wet, corrosive and highly corrosive areas.
3. Mount device outlet boxes where indicated on the Drawings and at heights as scheduled in Section 26 05 00.
4. Set device outlet boxes plumb and vertical to the floor.
5. Outlet boxes recessed in walls:
   a. Install with appropriate stud wall support brackets or adjustable bar hangers so that they are flush with the face of the wall.
   b. Locate in ungrouted cell of concrete block with bottom edge of box flush with bottom edge of block and flush with the face of the block.
6. Place barriers between switches in boxes with 277 V switches on opposite phases.
7. Back-to-back are not permitted.
8. When an outlet box is connected to a PVC coated conduit, the box shall also be PVC coated.

C. Pull and Junction Boxes:
1. Install pull or junction boxes in conduit runs where indicated or required to facilitate pulling of wires or making connections.
   a. Make covers of boxes accessible.
2. Permitted uses of NEMA 1 enclosure:
   a. Pull or junction box surface mounted above removable ceiling tiles of an architecturally finished area.
3. Permitted uses of NEMA 4X metallic enclosure:
   a. Pull or junction box surface mounted in areas designated as wet and/or corrosive.
4. Permitted uses of NEMA 4X non-metallic enclosure:
   a. Pull or junction box surfaced mounted in areas designated as wet and/or highly corrosive.
5. Permitted uses of NEMA 12 enclosure:
   a. Pull or junction box surface mounted in areas designated as dry.

END OF SECTION

This page intentionally left blank.

SECTION 26 05 36
CABLE TRAY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Cable tray and associated fittings and supports.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 07 84 00 - Firestopping.
   2. Section 10 14 00 - Identification Devices.
   3. Section 26 05 00 - Common Work Results for Electrical.
   4. Section 26 05 26 - Grounding.
   5. Section 26 05 33 - Raceways and Boxes for Electrical Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. A123/A123M, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
      b. A510/A510M, Standard Specification for General Requirements for Wire Rods and Coarse Round Wire, Carbon Steel, and Alloy Steel.
      c. A1011/A1011M, Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength.
      d. B633, Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.
   2. National Electrical Manufacturers Association (NEMA):
      a. VE-1, Metal Cable Tray Systems.
      b. VE-2, Cable Tray Installation Guidelines.
   3. Underwriters Laboratories, Inc. (UL).
   4. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

B. Types of Cable Tray:
   1. Ladder: A prefabricated metal structure consisting of two longitudinal side rails connected by individual transverse members of rungs.
   2. Ventilated trough: A prefabricated metal structure consisting of two longitudinal side rails and a ventilated bottom with clear openings no greater than 4 inches.
   3. Channel: A prefabricated metal structure consisting of a one-piece ventilated or solid bottom channel section not exceeding 6 inches in width.

1.4 SYSTEM DESCRIPTION

A. The following is a brief description of the types of the trays to be used.
   1. Power Distribution Center: Aluminum cable tray in indoor locations as indicated on the drawings. Trays shall be covered to 8FT above floors and access areas.

B. Miscellaneous:
   1. Cable tray systems are sized on the Drawings.

2. When cable tray system size is not shown on the Drawings or scheduled, the cable tray shall be sized in accordance with the NFPA 70 and the requirements of this Specification Section.
3. Cable tray runs, where shown, are diagrammatic and intended to be used as a guide, unless otherwise indicated on the Drawings.
    a. Site conditions may affect actual routing.
    b. Coordinate routing and measurement with other trades and with equipment suppliers to avoid interference with equipment, piping, ductwork, etc.
4. When not specifically shown on the drawings, the Contractor can provide cable tray per the following, with approval by the Engineer:
    a. Ladder tray.
    b. Tray size and fill per NFPA 70.
    c. Cable separation per NFPA 70 and as shown.
    d. Submit layout to Contracting Officer.

1.5 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
1. Product technical data.
    a. Provide submittal data for all products specified in PART 2 of this Specification Section.
    b. See Section 26 05 00 for additional requirements.
2. Fabrication and/or layout drawings:
    a. Routing, size and fittings.
    b. Calculation of weight of cables and trays, and necessary supports.
    c. Calculation of tray fill, and arrangement of cables in each tray.
    d. Seismic location installation details.
    e. Submit cable tray drawings showing cable trays, and joint and support locations separately for each installation site.

C. Informational Submittals:
1. Cable tray fill calculations.
2. Cable schedule of cables in cable trays.

1.6 DELIVERY, STORAGE, AND HANDLING

A. See Section 26 05 00.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Metallic and nonmetallic cable tray systems:
    a. B-Line by Eaton.
    b. Or Equal.
2. Cable tray conduit and ground clamps and brackets:
    a. B-Line by Eaton.

b. Or Equal.

2.2 COMPONENTS

A. Ladder Tray:
1. Materials:
a. Aluminum:
1) Side channels: Heat treatable alloy Type 6063-T6.
2) Transverse elements: Heat-treated alloy Type 6063-T6 (solid channel) or Type 6063-NHT (tubular).
3) Hardware: Stainless steel.
2. Fabrication:
a. Standard:
1) Metallic: NEMA VE-1.
b. The working (allowable) load capacity: NEMA VE-1 designation: 20C, minimum 103LB/FT load capacity at 20FT span. .
1) Provide heavier load capacity if required to support the cables placed in the tray.
c. Side rails:
1) I-beam or channel.
2) Flange in or out (full width top opening).
3) Useable clear nominal loading depth: 5IN (6IN side rail height). .
d. Transverse elements: Solid bar, tube, or channel with stiffened flanges.
e. Useable clear nominal loading width: Manufacturer standard for the trade width of the tray.
f. Maximum centerline rung spacing on straight sections: 9 inches.
g. Metallic trays shall be UL classified per NFPA 70 as an equipment grounding conductor.

B. Ventilated Trough Tray:
1. Material:
a. Aluminum:
1) Side channels: Heat treatable alloy Type 6063-T6.
2) Transverse elements: Heat-treated alloy Type 6063-T6 (solid channel) or Type 6063-NHT (tubular).
2. Fabrication:
a. Standard: NEMA VE-1.
b. The working (allowable) load capacity: NEMA VE-1 designation 20C, minimum 103LB/FT load capacity at 20FT span..
1) Provide heavier load capacity if required to support the cables placed in the tray.
c. Side rails:
1) I-beam or channel.
2) Flange in or out (full width top opening).
3) Useable clear nominal loading depth: 5IN (6IN siderail height) .
d. Transverse elements: One-piece, corrugated, ventilated bottom.
e. Useable clear nominal loading width: Manufacturer-standard for trade width of the tray product.
f. Maximum open spacing between elements is 4 inches as measured in a direction parallel to the side rails.
g. Metallic trays shall be UL classified per NFPA 70 as an equipment grounding conductor.

C. Channel Tray:
1. Material:
a. Aluminum: Heat treatable alloy Type 6063-T6.
2. Fabrication:
a. Standard: NEMA VE-1.
b. Minimum loading depth: 1-1/4 inches.

c. Loading nominal width: Manufacturer standard for the trade width of the tray product..
d. UL classified per NFPA 70 as an equipment grounding conductor.

D. Fittings:
1. Radius of bends: As required for cable layout in tray. Maintain manufacturer minimum bend radii for all cables installed in cable tray.
2. Degrees of arc for elbows: As required for cable tray layout.

2.3 ACCESSORIES

A. Accessories including but not limited to wall penetration kits, seal kits splice plates, barrier strips, drop outs, box connector, end plate and conduit clamps to be the same material as the tray or other compatible material.

B. Covers and Associated Accessories:
1. Material: Same base material as tray.
2. Types: Solid flanged (flat).
3. Clamps to secure covers to cable trays.

C. Cable Tray Ground Clamps:
1. Extruded aluminum with stainless steel screws.
2. Serrated edges to bite into and bond to the cable tray system.

D. Support system:
1. Material: See Section 26 05 00 for material specifications.
2. See PART 3 of this Specification Section for material type.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with NEMA VE-2 and as recommended by the manufacturer's instructions unless otherwise indicated on the Drawings.

B. Install cable tray, fittings and accessories, as required, for a complete system that has a neat appearance and is in compliance with all applicable codes.

C. Install cable tray systems as close as practical to the locations and elevations shown on the Drawings.
1. Minor changes (12 inches or less) in location or elevation may be made to avoid interference with piping, ductwork and equipment.
2. Obtain Contracting Officer's approval prior to making major changes (greater than 12 inches) in location or elevation.
3. When cable tray is located adjacent to, beneath or near large piping or major equipment, or terminates at equipment; do not install cable tray until the installation of such piping and equipment is complete.
4. Ensure openings are provided in walls that cable tray will penetrate.
5. Fire stop penetrations in fire rated walls per Specification Section 07 84 00.
a. Use a removable fire stopping system such as pillows.

D. Cable Tray Supports:
1. Provide supports at required locations to provide the loading capacity as indicated herein.
a. All support locations shall be determined by Contractor to coordinate with field locations, locations of joints between cable tray sections, and to provide support to all trays per NEMA VE-2 requirements.
b. Support cable trays minimum every 10FT.
c. Install supports and arrange joints in straight cable tray sections such that supports between cable sections fall at approximately the quarter-point of the tray segment, per NEMA VE-2 requirements.

d. Support cable tray bends, angles, and changes of elevation per NEMA VE-2 requirements.
e. Coordinate all supports with other trades, equipment, and building features.

2. Cantilever bracket type when cable tray is installed adjacent to a wall.
3. Trapeze type hangers for all other applications.
4. For vertical runs in excess of 20 feet support the tray by angles bolted through the side rails and suspended from hanger rod, which is connected to a cantilevered support.
5. In seismic locations provide required supports and/or sway bracing per local building codes.

E. Permitted prefabricated bracket or trapeze type support system per area designations and tray material:
1. Dry or wet areas:
a. Aluminum system consisting of: Aluminum channels and fittings with stainless steel nuts and hardware and conduit clamps.
2. Corrosive areas:
a. Stainless steel system consisting of: Stainless steel channels and fittings, nuts and hardware and conduit clamps.
b. PVC coated steel system consisting of: PVC coated galvanized steel channels and fittings and conduit clamps with stainless steel nuts and hardware.
c. Fiberglass system consisting of: Fiberglass channels and fittings, nuts and hardware and conduit clamps.
3. Highly corrosive areas:
a. PVC coated steel system consisting of: PVC coated galvanized steel channels and fittings and conduit clamps with stainless steel nuts and hardware.
b. Fiberglass system consisting of: Fiberglass channels and fittings, nuts and hardware and conduit clamps.
4. Tray material shall be compatible with the support system material.
a. Stainless steel system may be used with aluminum trays.
b. Aluminum system may be used with aluminum trays.

F. Whenever cable tray system spans a structural expansion joint provide one of the following:
1. Expansion connector allowing a minimum of 1 inch straight-line movement of sections.
2. A 2 inch discontinuity (gap) in the cable tray to allow horizontal and vertical movement.

G. Maintain electrical continuity of the cable tray system.
1. Bolt connectors to each section or fitting.
2. Span expansion connectors by a bonding jumper.
3. Use one of the following to bond conduits to the tray:
a. Conduits connected to the cable tray system by a one-piece conduit clamp shall be considered bonded to the cable tray system.
b. Terminate conduits connected to the cable tray system by a bracket and clamp assembly in an insulated grounding bushing and bond to the cable tray system.
4. Tighten all bolted connections to manufacturer's recommendations to ensure electrical continuity.

H. Cable Tray System Grounding:
1. See Specification Section 26 05 26.
a. Install an insulated grounding conductor in the cable tray as specified in Section 26 05 26.
b. Grounding cable shall be continuous across all tray discontinuities.

I. Secure cables, in vertical runs of cable tray, with cable clamps or ties as near as practical to the top and bottom of the vertical run and at a maximum of 5 feet spacing over the length of the run.
1. Plastic wire ties to be UV resistant type.

J. All cables shall be neatly arranged and secured in cable trays.

K. Carefully form cables into a “sweep” transition from one cable tray to another with long-radius bends.

L. Conduit terminating at a cable tray system:
   1. See Specification Section 26 05 33.

M. Use flanged fittings to terminate cable tray systems at switchgear, motor control centers, and other equipment, unless indicated otherwise on the Drawings.

N. Install flat, solid, flanged covers on vertical cable tray to a minimum 8 feet above finished floor and walkways.

O. Cover cable tray systems passing under open grating type walkways and platforms.
   1. Peak flanged.
      a. Install clamps to secure covers to cable tray.

P. Install barrier strips in cable tray systems containing both power and control wiring to physically separate the control cables from the power cables.

Q. Cable Installation:
   1. Conductors #4/0 AWG and less, shall be in a multi-conductor cable.
   2. Conductors 250 KCMIL and larger, shall be single conductor cables bundled together, Phase A, B, C and neutral and/or ground when required.

3.2 FIELD QUALITY CONTROL

A. Tray Fill Calculations:
   1. Cable tray fill shall not exceed NFPA 70 requirements.
      a. Coordinate the installation of all cables and maintain cable fill calculations and schedule of cables in the trays.

B. Remove trash and accumulated dirt from the entire cable tray system at the completion of the project and install covers where applicable.

C. Tagging and warning signs:
   1. See Specification Section 10 14 00.
      a. Install “DO NOT STEP” and other signage on cable trays as specified.

END OF SECTION

SECTION 26 05 43
ELECTRICAL - EXTERIOR UNDERGROUND

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for:
      a. Manholes.
      b. Handhole.
      c. Underground conduits and ductbanks.

B. Related Specification Sections include but are not necessarily limited to:
   1. Division 03 - Concrete.
   2. Section 03 31 30 - Concrete, Materials and Proportioning.
   3. Section 10 14 00 - Identification Devices.
   4. Section 26 05 26 - Grounding.
   5. Section 26 05 33 - Raceways and Boxes.
   6. Section 31 23 33 - Trenching and Backfilling.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Association of State Highway and Transportation Officials (AASHTO):
      a. HB-17, Standard Specifications for Highway Bridges.
   2. ASTM International (ASTM):
      a. A536, Standard Specification for Ductile Iron Castings.
   3. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   4. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   5. Society of Cable Telecommunications Engineers (SCTE):
      a. 77, Specifications for Underground Enclosure Integrity.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

B. Direct-Buried Conduit(s):
   1. Individual (single) underground conduit.
   2. Multiple underground conduits, arranged in one or more planes, in a common trench.

C. Concrete Encased Ductbank: An individual (single) or multiple conduit(s), arranged in one or more planes, encased in a common concrete envelope.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data:
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. Provide signed and sealed design computations for manholes and handholes indicating compliance with design load rating.
   2. Fabrication and/or layout drawings:
      a. Provide dimensional drawings of each manhole and handhole indicating all specified accessories and conduit entry locations.

b. Submittals:
    1) Organize and submit required product information and fabrication/layout drawings into discrete packages as follows:
        a) North Rim Water Treatment Plant
        b) Roaring Springs Pump Station
        c) Power Distribution Facility
        d) Other areas not listed above.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
    1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
    1. Sidewalk type handhole:
        a. Crouse-Hinds by Eaton.
        b. Or Equal.
    2. Prefabricated composite handholes:
        a. Armorcast Products Company.
        b. Or Equal.
    3. Precast manholes and handholes:
        a. Lister Industries Ltd.
        b. Or Equal.
    4. Manhole and handhole and ductbank accessories:
        a. Cantex, Inc.
        b. Or Equal.

2.2 MANHOLES AND HANDHOLES

A. Sidewalk Type Handholes:
    1. Cast-iron box and cover, hot-dip galvanized.
    2. Flange for flush mounting.
    3. Checkered cover with neoprene gasket, pry bar slots and stainless steel screws.
    4. Drilled and tapped holes.
    5. Watertight NEMA 4 classification.

B. Prefabricated Composite Material Handholes:
    1. Handhole body and cover: Fiberglass reinforced polymer concrete conforming to all test provisions of SCTE 77.
    2. Minimum load ratings: SCTE 77 Tier 8 (8000lb vertical design load).
        a. SCTE Tier 15 (15,000lb vertical design load) for areas with vehicle access.
    3. Open bottom.
    4. Stackable design as required for specified depth.
    5. Cover:
        a. Engraved legend of "ELECTRIC" or "COMMUNICATIONS".
        b. Gasketed bolt down with stainless steel penta head bolts.
        c. Lay-in non-bolt down, when cover is over 100 pounds.
        d. One or multiple sections so the maximum weight of a section is 125 pounds.
    6. Cover lifting hook: 24 inches minimum in length.

C. Precast Manholes and Handholes:
   1. Fiberglass reinforced polymer concrete or steel reinforced cement concrete structures:
   2. AASHTO live load rating: H-20 for full deliberate vehicle traffic.
   3. Mating edges: Tongue and groove type.
   4. Solid bottom with a 12 inches x 12 inches or 12 inches diameter french drain in the bottom of each manhole.
   5. Gasketed removable top slab with lifting eyes and cast in frame for cover.
   6. Cover extension rings as required.
   7. Cable pulling eyes opposite all conduit entrances.
      a. Coordinate exact location with installation contractor.

D. Cast-in-Place Manholes:
   1. Comply with Specification Section 03 31 30.

2.3 CONCRETE MANHOLE AND HANDHOLE ACCESSORIES

A. Cover and Frame:
   1. Cast ductile iron: ASTM A536.
   2. AASHTO live load rating: H-20.
   3. Diameter: 36 inches.
   4. Cast the legend "ELECTRICAL" or "COMMUNICATIONS" into manhole and handhole covers.
      a. Stainless steel penta-head bolts.

B. Cable Racks and Hooks:
   1. Material: Heavy-duty nonmetallic (glass reinforced nylon).
   2. Hook loading capacity: 400 pounds minimum.
   3. Rack loading capacity: Four hooks maximum.
   4. Hook deflection: 0.25 inches maximum.
   5. Hooks: Length, as required, with positive locking device to prevent upward movement.
   6. Mounding hardware: Stainless steel.

C. Cable Pulling Irons:
   1. 7/8 inches diameter hot-dipped galvanized steel.
   2. 6000 pound minimum pulling load.

D. Ground Rods and Grounding Equipment: See Specification Section 26 05 26.

2.4 UNDERGROUND CONDUIT AND ACCESSORIES

A. Concrete and reinforcing steel: See Division 03 Specifications.

B. Concrete and reinforcing steel: See Specification Section 03 11 13, Specification Section 03 21 00, Specification Section 03 31 30 and Specification Section 03 35 00.

C. Duct Terminators:
   1. Window type.
   2. ABS plastic.
   3. Provide for conduit entrance.
   4. Designed for installation into manhole or handhole walls for a watertight seal.
   5. Sufficient space between terminator walls to allow for placement of rebar and concrete.

D. Conduit: See Specification Section 26 05 33.

E. Duct Spacers/Supports:
   1. High density polyethylene or high impact polystyrene.
   2. Interlocking web or mesh design.
   3. Non-interlocking type provided with adequate holes for concrete to flow.
   4. Provide 3 inches minimum spacing between conduits.
   5. Accessories, as required:
      a. Hold down bars.

b. Ductbank strapping.

PART 3 - EXECUTION

3.1 GENERAL

A. Drawings indicate the intended location of manholes and handholes and routing of ductbanks and direct buried conduit.
   1. Field conditions may affect actual routing.

B. Manhole and Handhole Locations:
   1. Approximately where shown on the Drawings.
   2. As required for pulling distances.
   3. As required to keep pulling tensions under allowable cable tensions.
   4. As required for number of bends in ductbank routing.
   5. Shall not be installed in a swale or ditch.
   6. Determine the exact locations after careful consideration has been given to vehicle and traffic patterns, the location of other utilities, grading, and paving.
   7. Locations are to be approved by the Contracting Officer prior to excavation and placement or construction of manholes and handholes.

C. Install products in accordance with manufacturer's instructions.

D. Install manholes and handholes in conduit runs where indicated or as required to facilitate pulling of wires or making connections.

E. Comply with Specification Section 31 23 33 for trenching, backfilling and compacting.

3.2 MANHOLES AND HANDHOLES

A. Prefabricated Composite Material Handholes:
   1. For use in areas subjected to occasional non-deliberate vehicular traffic.
   2. Place handhole on a foundation of compacted 1/4 to 1/2 inches crushed rock or gravel a minimum of 8 inches thick and 6 inches larger than handholes footprint on all sides.
   3. Provide concrete encasement ring around handhole per manufacturers installation instructions (minimum of 10 inches wide x 12 inches deep).
   4. Install so that the surrounding grade is 1 inch lower than the top of the handhole.
   5. Size: As indicated on the Drawings or as required for the number and size of conduits.
   6. Provide cable rails and pulling eyes as needed.

B. Precast Manholes and Handholes:
   1. For use in driveways, parking lots, and vehicular and non-vehicular traffic areas.
   2. Construction:
      a. Grout or seal all joints, per manufacturer's instructions.
      b. Support cables on walls by cable racks:
         1) Provide a minimum of two racks, install symmetrically on each wall of manholes and handholes.
            a) Provide additional cable racks, as required, so that both ends of cable splices will be supported horizontally.
         2) Equip cable racks with adjustable hooks: Quantity of cable hooks as required by the number of conductors to be supported.
      c. In each manhole and handhole, drive 3/4 inches x 10 feet long copper clad ground rod into the earth with approximately 6 inches exposed above finished floor.
         1) Drill opening in floor for ground rod.
         2) Connect all metallic components to ground rod by means of #8 AWG minimum copper wire and approved grounding clamps.
         3) Utilize a ground bar in the manhole or handhole if the quantity of ground wires exceeds three.
            a) Connect ground bar to ground rod with a #2/0 AWG minimum copper wire.

3. Place manhole or handhole on a foundation of compacted 1/4 to 1/2 inches crushed rock or gravel a minimum of 8 inches thick and 6 inches larger than manholes or handholes footprint on all sides.
4. Install so that the top of cover is 1 inch above finished grade.
    a. Where existing grades are higher than finished grades, install sufficient number of courses of curved segmented concrete block between top of handhole and manhole frame to temporarily elevate manhole cover to existing grade level.
5. After installation is complete, backfill and compact soil around manholes and handholes.
6. Handhole size:
    a. As indicated on the Drawings or as required for the number and size of conduits entering or as indicated on the Drawings.
    b. Minimum floor dimension of 4 feet x 4 feet and minimum depth of 4 feet.
7. Manhole size:
    a. As indicated on the Drawings or as required for the number and size of conduits entering or as indicated on the Drawings.
    b. Minimum floor dimension of 6 feet x 6 feet and a minimum depth of 6 feet.

3.3 UNDERGROUND CONDUITS

A. General Installation Requirements:
1. Ductbank types per location:
    a. Duct banks on all sites shall be reinforced concrete. All duct banks shall be formed.
    b. Direct-buried conduit(s):
        1) As indicated in the Ductbank Schedule.
2. Do not place concrete or soil until conduits have been observed by the Contracting Officer.
3. Ductbanks shall be sloped a minimum of 4 inches per 100 feet or as detailed on the Drawings.
    a. Low points shall be at manholes or handholes.
4. During construction and after conduit installation is complete, plug the ends of all conduits.
5. Provide conduit supports and spacers.
    a. Place supports and spacers for rigid nonmetallic conduit on maximum centers as indicated for the following trade sizes:
        1) 1 inches and less: 3 feet.
        2) 1-1/4 to 3 inches: 5 feet.
        3) 3-1/2 to 6 inches: 7 feet.
    b. Place supports and spacers for rigid steel conduit on maximum centers as indicated for the following trade sizes:
        1) 1 inches and less: 10 feet.
        2) 1-1/4 to 2-1/2 inches: 14 feet.
        3) 3 inches and larger: 20 feet.
    c. Securely anchor conduits to supports and spacers to prevent movement during placement of concrete or soil.
6. Stagger conduit joints at intervals of 6 inches vertically.
7. Make conduit joints watertight and in accordance with manufacturer's recommendations.
8. Accomplish underground changes in direction of runs exceeding a total of 15 degrees by long sweep bends having a minimum radius of 10 feet.
    a. Sweep bends may be made up of one or more curved or straight sections or combinations thereof.
9. Furnish manufactured elbows at end of runs as the conduit transitions to above grade.
    a. Minimum radius of 18 inches for conduits less than 3 inches trade size and 36 inches for conduits 3 inches trade size and larger.
10. Field cuts requiring tapers shall be made with the proper tools and shall match factory tapers.
11. After the conduit run has been completed:
    a. Prove joint integrity and test for out-of-round duct by pulling a test mandrel through each conduit.

1) Test mandrel:
a) Length: Not less than 12 inches.
b) Diameter: Approximately 1/4 inches less than the inside diameter of the conduit.

b. Clean the conduit by pulling a heavy duty wire brush mandrel followed by a rubber duct swab through each conduit.

12. Pneumatic rodding may be used to draw in lead wire.
a. Install a heavy nylon cord free of kinks and splices in all unused new ducts.
b. Extend cord 3 feet beyond ends of conduit.
13. Transition from rigid nonmetallic conduit to rigid metallic conduit, per Specification Section 26 05 33, prior to entering a structure or going above ground.
a. Except rigid nonmetallic conduit may be extended directly to manholes, handholes, pad mounted transformer boxes and other exterior pad mounted electrical equipment where the conduit is concealed within the enclosure.
b. Terminate rigid PVC conduits with end bells.
c. Terminate steel conduits with insulated bushings.
14. Place warning tape in trench directly over ductbanks, direct-buried conduit, and direct-buried wire and cable in accordance with Specification Section 10 14 00.
15. Placement of conduits stubbing into handholes and manholes shall be located to allow for proper bending radiuses of the cables.

B. Concrete Encased Ductbank:
1. Ductbank system consists of conduits completely encased in minimum 3 inches of concrete and with separations between different cabling types as required in Specification Section 26 05 33 or as detailed on the Drawings.
2. Install so that top of concrete encased duct, at any point:
a. Is not less than 30inches below grade.
b. Is below pavement sub-grading.
3. The concrete shall be reinforced.
a. The reinforcement shall consist of #4 bars and #4 ties placed 12 inches on center, in accordance with Division 03 Specification Sections or as detailed on the Drawings.
b. Conduit supports to be staggered to minimize weak vertical shear point.
4. Conduit supports shall provide a uniform minimum clearance of 3 inches between the bottom of the trench and the bottom row of conduit.
5. Conduit separators shall provide a uniform minimum clearance of 3 inches between conduits or as required in Specification Section 26 05 33 for different cabling types.

C. Direct-Buried Conduit(s):
1. See Drawings for burial depths details of direct buried conduits.
2. Install so that the top of the uppermost conduit, at any point:
a. Is below pavement sub-grading.
3. Provide a uniform minimum clearance of 3 inches between conduits or as required in Specification Section 26 05 33 for different cabling types.
a. Maintain the separation of multiple planes of conduits by one of the following methods:
1) Install multilevel conduits with the use of conduit supports and separators to maintain the required separations and backfill with flowable fill (100 psi) per Specification Section 31 23 33 or concrete per Division 03 specifications.
2) Install the multilevel conduits one level at a time.
a) Each level is backfilled with the appropriate amount of soil and compaction, per Specification Section 31 23 33, to maintain the required separations.

END OF SECTION

SECTION 26 05 46
OVERHEAD POWER LINES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Construction of overhead power lines.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. A475, Standard Specification for Zinc-Coated Steel Wire Strand.
      b. B232/B232M, Standard Specification for Concentric-Lay-Stranded Aluminum Conductors, Coated-Steel Reinforced (ACSR).
   2. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. 524, Guide to the Installation of Overhead Transmission Line Conductors.
      b. C2, National Electrical Safety Code (NESC).
      c. C62.11, Standard for Metal-Oxide Surge Arresters for AC Power Circuits (>1 kV).
   3. National Electrical Manufacturers Association/American National Standards Institute (NEMA/ANSI):
      a. C29.2, Insulators - Wet Process Porcelain and Toughened Glass - Transmission Suspension Type.
      b. C29.3, Wet-Process Porcelain Insulators - Spool Type-.
      c. C29.4, Wet-Process Insulators - Strain Type-.
      d. C29.5, Wet-Process Porcelain Insulators - Low and Medium Voltage Types.
      e. C119.4, Electric Connectors - Connectors for Use between Aluminum-to-Aluminum and Aluminum-to Copper Conductors Designed for Normal Operation at or Below 93 degrees C and Copper-to-Copper Conductors Designed for Normal Operation at or Below 100 degrees C.
   4. National Electrical Manufacturers Association/Insulated Cable Engineers Association (NEMA/ICEA):
      a. WC 70/S-95-658, Power Cables Rated 2000 V or Less for the Distribution of Electrical Energy.
   5. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).

B. Qualifications:
   1. Erection and installation by personnel skilled and with minimum 5 years of experience in power line construction for the electric utility industry, experienced in construction projects of similar size and character as required for this Work.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data:
      a. A complete list of all materials to be supplied by the Contractor by description and manufacturer's catalog number.
      b. Contractor to provide a list of miscellaneous materials needed to complete the job.
         1) Miscellaneous items are defined as nuts, bolts, washers, etc., for complete installation.
   2. Test reports:
      a. Pole Inspection Reports by independent Agency.

B. Operation and Maintenance Manuals.

1. See Specification Section 01 78 23 for requirements for:
    a. The mechanics and administration of the submittal process.
    b. The content of Operation and Maintenance Manuals.

C. Informational Submittals:
1. Record of measurements made during sagging operations.
2. Record of clearance measurements.
3. Record of final grounding measurements at each structure (if required).

1.4 PROJECT/SITE CONDITIONS

A. Field Measurements:
1. Obtain all necessary field dimensions.
2. Prospect for and locate all subgrade facilities which are either part of the Work, or are nearby and must be protected.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Contractor to confirm all equipment is rated for installed elevation and provide final equipment ratings for review. Proposed equipment shall be compatible with existing equipment interface.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Spacer Cable and Components
    a. Hendrix (Basis of Design).
    b. Or Equal.
2. Bare Conductor ACCR and Components
    a. 3M (Basis of Design).
    b. Or Equal.
3. Optical Ground Wire (OPGW) and Components
    a. AFL (Basis of Design)
    b. Or Equal

D. Submit request for product substitution in accordance with Specification Section 01 33 23

2.2 COMPONENTS

A. All material used shall be rated for 9,000ft elevation minimum.

B. Poles:
1. See 33.71.16.23 TUBULAR STEEL POLES

C. Micropile Specification
1. See MICROPILE specification 33 71 23.17 for performance criteria

D. Bare Conductors:
1. Aluminum conductor composite supported core
    a. Size: 183 kcmill

b. Rated Tensile Strength: 32,600 lb strength.
c. Weight: 0.415 lbs/ft.
d. Diameter: 0.719 in.
e. AC resistance shall be 0.3510 ohms/mile at 25°C.
f. Thermal elongation coefficient of 9.3 (10^{-6}/°C) of complete cable.

E. Insulated Conductors:
1. Low voltage insulated conductors (single conductor):
a. Rated for 600 V.
b. Conductors shall be copper with regular stranding.
c. Conductor insulation shall be polyethylene per NEMA/ICEA WC 70/S-95-658 and shall be black in color.
d. See Drawings for size.
2. Spacer Cable:
a. Phase Cable.
1) Rated for 15kV
2) Maximum Cable Weight: 0.215 lbs/ft
3) Cable diameter: 0.666 in
4) Spacer cable phase conductors shall be:
a) 7 stranded hard drawn 1/0 aluminum and covered with three extruded layers.
5) UV Resistant Outer insulation layer
6) Outdoor installation
7) Conductors shall be suspended from the messenger cable using polyethylene spacers.
a) Spacer to have internal support clamps to prevent slipping on steep elevation drops.
b) Spacer to have ability to clamp to two messenger cables.
8) Maximum cable reel and cable weights for helicopter lifting to be determined by Contractor prior to ordering cable.
9) Basic Lightning Impulse level (BIL): 110 kV minimum.

b. Messenger Wire:
1) 37 NO. 8 aluminum clad steel. Cladding thickness shall be minimum 10% of the radius of the wire.
2) Weight: 1.398 lbs/ft
c. Cable, spacers and components all by same manufacturer.

F. Optical Ground Wire (OPGW)
1. Aluminum Clad Steel Wire.
2. Internal Tube with optical fibers.
3. 144 Fiber Count
4. Approximate cable weight: 0.705 lbs/ft
5. Approximate Overall Diameter: 0.724 inches
6. Approximate Calculated Breaker Strength: 34,454 lbs
7. Short circuit rating: 239 (KA)^2SEC

G. Marker Balls:
1. Marker balls shall be constructed from aluminum for direct attachment to high corona, high voltage aluminum conductor lines and are rated up to 482° F and 500 kV.
2. 36” diameter
3. Armor rods shall be matched at time of order to specific size, type, and diameter of wire.
4. Furnish marker balls in yellow, white, and orange. See drawing package for color coordination installation.
5. Marker ball shall have clamshell assembly to enable uncomplicated helicopter installation.
6. Marker ball shall have internal clamping component to prevent slipping.
7. Weight: 20lbs maximum.

H. Insulators (Porcelain):
   1. Suspension insulators per NEMA/ANSI C29.2, Class 56-2.
   2. Medium and low voltage post insulators per NEMA/ANSI C29.5, Class 55-5.
   3. Insulators rated 6000 V and above shall be interference proof.
   4. Sky gray color.

I. Surge Arresters:
   1. Metal oxide type.
   2. Manufactured per IEEE C62.11.
   3. See Drawings for Voltage and MCOV ratings.
   4. Equipped with mounting brackets suitable for the indicated installations.
   5. Type: Outdoor, Gapless Metal Oxide High Voltage Surge Arrestor.
   6. Mounting: self-supporting and shall have a mounting base of cast (ductile) iron, structural steel, or aluminum.
   7. Insulator: Porcelain or Polymer.
   8. System Configuration: Solidly Grounded
   9. Basic Lightning Impulse level (BIL): 110 kV minimum.
   10. Ambient Temperature: -40 to + 112 Degrees F.
   11. Altitude: 10,000 feet.
   12. Metal Components: All metal components exposed to weather and corrosion shall be zinc coated by the hot dip galvanizing process.
   13. Metal Oxide Disks: Gapless construction of metal oxide varistor elements.
   14. Finish:
      a. Manufacturer's standard corrosion protection system in accordance with IEEE C57.13.
   15. Acceptance Testing
      a. Perform inspections and tests per NETA ATS 7.19.2 for medium and high voltage surge arrestors.
      b. Perform all optional NETA ATS test per 7.19.2

J. Anchors:
   1. Furnish expanding type rock anchor for anchoring in solid rock.
   2. Rod shall be hot-dip galvanized with eye contoured to prevent damage to guy(s).
   3. Expanding rock anchors which shall be painted.
   4. See Drawings for details..

K. Connectors:
   1. Copper alloys for copper conductors, aluminum alloys for aluminum composition conductors, type designed to minimize galvanic corrosion for copper to aluminum composition.
   2. For bare connections, conform to NEMA/ANSI C119.4 for aluminum composition and aluminum composition to copper.
   3. All connections shall be compression type, except where specifically required otherwise.
   4. See Drawings for sizes.

L. Optical Ground Wire (OPGW) Components
   1. OPGW bonding ground wire
      a. Material: aluminum
      b. Connectors to be pre-compressed onto cable at factory
   2. Bolted dead end for OPGW.
      a. Sustained load equivalent to 95% of cable rated breaking strength
      b. Ultimate mechanical strength of dead end components: 40,000 lbs
      c. Cable guide to train Optical Ground Wire down or around structure
   3. Coil bracket for OPGW and ADSS
      a. Aluminum Material
      b. Bolted to structure
      c. Minimum 400 LBS loading capacity

4. Bonding Clamps for OPGW – Parallel Groove Clamp
    a. High strength aluminum
    b. Extruded parallel groove clamp
    c. Clamp grooves are coated with NO-OX-ID and prefilled with electrical joint compound
    d. Groove Ranges:
        1) 0.322 in – 0.336 in
        2) 0.721 in – 0.735 in
        3) See Drawings for additional details.
5. Optical Splice Enclosure with Connector Kit:
    a. Pressurizing Valve
    b. Casing and Drawer with protective drawer cover
    c. Approximate Height: 40 in
    d. Pole Mounted

M. Miscellaneous Components
1. Clevis:
    a. Y Clevis Clevis:
        1) Ultimate Strength: 30,000 lbs
        2) Approximate Weight: 2.50 lbs
    b. Hot Line Clevis Eye:
        1) Ultimate Strength: 30,000 lbs
        2) Approximate Weight: 2.5 lbs
        3) Length: 7 in
    c. Clevis Clevis:
        1) Ultimate Strength: 30,000lbs
        2) Approximate Weight: 5.586
        3) Length: 5.5 in
    d. Heavy Duty Thimble Clevis (Spacer Cable Component):
        1) See Drawing for additional details.
        2) Ultimate Load: 40,000lbs
            a) Approximate Weight: 2.6 lbs
            b) Width: 3.25 in
        3) Ultimate Load: 80,000lbs
            a) Approximate Weight: 7.17 lbs
            b) Width: 3.88 in
2. Plates:
    a. Dead End Link Plate (OPGW Component):
        1) Ultimate Strength: 40,000 lbs
        2) Approximate Weight: 4.40 lbs
        3) Length: 12 in
    b. Yoke Plate:
        1) Delta,
        2) Type YPD
        3) Ultimate Strength: 60,000lbs
        4) Approximate Weight: 18.13lbs
        5) Ductile Iron

3. Anchor shackle:
    a. Ultimate Rated Strength: 30,000 lbs
        1) Weight: 1.08 lbs
        2) Length: 3.01 in
        3) Width: 1.76
    b. Ultimate Rated Strength: 60,000 lbs
        1) Weight: 3.0 lbs
        2) Length: 5 in

3) Radius: 1 in
c. Ultimate Rated Strength: 80,000lbs
1) Weight: 4.10 lbs
2) Length: 3.75 in
3) Radius: 1.125 in
d. See Drawings for Additional details and usage location.

4. Chain Link:
a. Ultimate Rated Strength: 80,000lbs
b. Length: 4”
c. Weight: 2.15 lbs
d. Thickness: 0.875 lbs

5. Standoff Bracket:
a. 2-Position Cutout/Arrester Configuration
b. 24 in Arm length
c. 15 Degree
d. Minimum Ultimate Strength: 1,900 lbs

6. Compression Deadend ACCR
a. Sized for 183 kCMIL
b. Single Tongue Clevis Attachment

7. Deadend Insulator:
a. 37 No 8 Messenger dead end grip
1) Diameter Range: 0.869-1.024 in
2) Nominal Strand Size: 1
b. Spacer Cable
1) 15kV
2) Length: 12.5
3) Rated Tensile Load: 7,500 lbs
4) Fiberglass Core, Polymetric Sheath

8. Cable grips – Closed Mesh – Neutral
a. See Drawings for cable size. Match grip cable holding range with cable size.
b. Heavy duty, single eye, closed mesh, tin coated bronze.
c. Breaking strength: 500lbs
d. Working load: 50lbs

9. Cable grips – Split mesh - Fiber
a. See Drawings for cable size. Match grip cable holding range with cable size.
b. Heavy duty, offset eye, lace closing,
c. Breaking strength: 2,820lbs
d. Working load: 282lbs
e. Strand: stainless steel
f. Eye: stainless steel
g. Lace: stainless steel

10. Cable grips – Split mesh - Phase
a. See Drawings for cable size. Match grip cable holding range with cable size.
b. Heavy duty, offset eye, lace closing,
c. Breaking strength: 8,950lbs
d. Working load: 895lbs
e. Strand: stainless steel
f. Eye: stainless steel

g. Lace: stainless steel

11. Galvanized Rigid Conduit
    a. Coatings: Hot-dip galvanized inside and out providing galvanic corrosion protection and top coated with a compatible organic layer to protect against white rust
    b. Couplings: Electro-galvanized inside and out to provide galvanic corrosion protection
    c. See Drawings for conduit size.

12. PVC Conduit and Bends
    a. Schedule 40
    b. Rated for direct buried
    c. See Drawings for conduit size.

13. Cable Terminations
    a. Meet the requirements of IEEE standard 48–1990, for class 1 terminations
    b. Skirted Termination
    c. Size cable termination to fit underground power cable show in Drawings.
    d. UV resistant
    e. Ultimate tensile strength: 1200 psi
    f. Electrical Properties
        1) Dielectric Constant (K) (ASTM D150)
            a) 23°C (73°F): 3.4
            b) 90°C (194°F): 3.0
            c) 130°C (266°F): 2.7
        2) Dissipation Factor (ASTM D150)
            a) 23°C (73°F): 0.4%
            b) 90°C (194°F): 1.3%
            c) 130°C (266°F): 1.2%
        3) Dielectric Strength (ASTM D149)
            a) 0.075inch thick slab: 507 volts/mil
        4) Track Resistance (ASTM 2303)
            a) 2.5 kV: 10 hrs.
            b) 3.5 kV: 1 hr.

14. Glass Suspension Insulators (ANSI CLASS 52-6-H)
    a. Combined mechanical and electrical strength: 30,000lbs
    b. Impact strength: 400in-lbs
    c. Tension proof: 15,000lbs
    d. Diameter: 10in
    e. Spacing: 5 3/4in
    f. Leakage Distance: 12 5/8in
    g. Low frequency dry flashover: 80kV
    h. Low frequency wet flashover: 50kV
    i. Critical Impulse flashover positive: 125kV
    j. Critical Impulse flashover negative: 130kV
    k. Weight: 9lbs

15. Parallel Grove Connectors
    a. Galvanized steel carriage bolt, nut, and lock wather
    b. Dual rated for aluminum and copper
    c. Prefilled with oxide inhibitor
    d. See Drawings for cable sizes

PART 3 - EXECUTION

3.1 PREPARATION

A. Pole Locating:
   1. Notify contracting officer representative for inspection and approval of staking locations immediately after staking line and prior to beginning construction.
   2. Locate structures by stake at center of pole hole.
   3. Locate anchors by stake at location where anchor rod, in installed position, shall protrude from the ground.
   4. Relocate and replace any stakes which are missing or damaged prior to installation of the facilities.

B. Determine phasing of the existing system.

3.2 DEMOLITION

A. General:
   1. Remove the facilities designated in the Contract Documents.
   2. Cleanup practices shall be in accordance with the General Conditions.
   3. No facilities designated for removal shall be reused in this Work.

B. Removal:
   1. Use procedures which are safe and guarantee no hazard to personnel or other facilities.
   2. For poles:
      a. Completely remove, including pole butts.
      b. Fill holes with dry, clean soil, pneumatically or hydraulically tamped in a manner which will prevent future settlement, 6 inches lifts maximum. (top 18 inches to be top soil).
   3. Anchors shall be completely removed if possible; otherwise, cut off at least 3 feet below the ground line.
   4. For removal of concrete, neatly cut boundary of area designated for removal with suitable power saw.

C. Disposition:
   1. The following items shall become property of Contractor for disposal in accordance with local ordinances:
      a. All material and equipment designated for removal.
      b. Concrete rubble.
      c. Excess backfill from holes.
   2. Not be required to dispose of any equipment containing oil with PCB content of 50 ppm or greater.

3.3 INSTALLATION

A. Contractor will provide coordination with helicopter contractor for integration between helicopter operation and other contractors.

B. Structure Framing and Assembly:
   1. Frame in accordance with the Drawings.
   2. Completely assemble structure prior to setting except the grounding conductor, if installed on standoff brackets, shall be installed after setting pole.
   3. The loose end of the grounding down leads shall be looped over the pole top and temporarily secured prior to wire stringing (for protecting pole from lightning until shield wire is installed).
   4. Bolts, nuts and washers:
      a. Nuts to be drawn tight so as to obtain full strength of all hardware fittings.
      b. Spring washers and lockwashers, where required, to be fully compressed.
      c. Install locknut with each nut, eyenut or other fastener on all bolts and threaded hardware (such as insulator pins, upset bolts, double arming bolts, etc.).

d. Bolts shall not extend less than 2 threads or more than 2 inches beyond nuts.

C. Structure Erection:
1. See Mircropile Performance Specification and 33.71.16.23 – Tubular Steel Pole
2. Remove any surplus excavation material from site and dispose of in manner suitable to contracting officer representative.

D. Anchor Installation:
1. Install at locations shown in the Contract Documents in accordance with manufacturer's recommendations.
2. Anchor rods to be aligned with direction of tension..

E. Grounding:
1. See specification 26 05 26 Grounding and Bonding
2. Install according to the Drawings.
3. Install ground lead in manner which prevents whipping in wind, and also undue stress during cold temperatures.
4. Drive ground rods full length in undisturbed earth, with top of rod not less than 1 foot below grade.
5. For sectional ground rods, use steel driving studs which do not damage the threads.

F. Conductor Stringing:
1. Install all conductors (phase, shield, neutral, messenger, etc.) indicated in the Drawings.
2. Follow recommendations of conductor manufacturer.
3. Furnish all ropes and other necessary equipment.
4. Provide reliable communication facilities for crews, suitable to contracting officer representative.
5. Install any necessary temporary guying.
6. Install temporary stringing and landing platforms as need for construction stringing. Contractor to coordinate and receive approval from contracting officer representative before installation and vegetation management
7. Contractor to remove temporary platforms after installation has been completed and return disturbed areas to previous conditions.
8. Do not string conductors if adverse weather conditions are predicted before sagging and clipping-in can be completed.
9. In the event of storm with high winds while conductor is in stringing sheaves:
   a. Conductor subject to contracting officer representative's inspection.
   b. Assist by providing personnel and equipment as requested.
   c. Repair or replace any damaged conductor to contracting officer’s satisfaction.
10. Conductors, after installation, shall have no cuts, kinks, abrasions or other injuries.
   a. Continuously inspect conductor as it leaves reel.
   b. Report any damaged conductor to contracting officer representative and stop installation until a repair has been identified and approved by the contracting officer representative..
   c. Advise contracting officer's representative of any repairs made after receiving approval for repairs from contracting officer representative.
11. Protect existing facilities.
   a. Install sufficient guard structures of adequate strength as required, in opinion of Contractor, to maintain adequate clearance over buildings, roads, railroads, communication and power lines, etc.
   b. Provide warning signs and lighting as necessary to prevent a hazard to public.
12. Use stringing equipment acceptable to contracting officer representative:
   a. Stringing sheaves:
      1) Sheave diameter and groove construction in accordance with recommendations of IEEE 524.
      2) Roller or ball bearings with provisions for lubrication.
   b. Cable grips:

1) Swivel connection to pulling line.
2) Stocking type grip.

c. Tensioning equipment (for controlled tension stringing):
1) Double bull-wheel type.
2) Bullwheel diameter and groove construction in accordance with recommendations of IEEE 524.
3) Neoprene-lined grooves.
4) Capable of maintaining preset tensions not less than 10,000 pounds.
5) Failsafe type braking system.
6) Equipped with tension indicating and limiting devices.

d. Provide contracting officer representative complete manufacturer's data on equipment prior to beginning of Work.

13. Provide adequate grounding for stringing equipment and conductors and use procedures which guarantee safety.
14. Pulling tension to be minimum required to provide clearance to ground and obstructions (not to exceed 90% of sagging tension).
15. No overload strains, drilling of temporary holes, or abrasion of any permanent structure.
16. Mounting of stringing sheaves on structure:
a. Sheaves shall hang freely without binding.
b. All sheaves in stringing section shall hold conductor at uniform distance, within 2 feet above or below, of location it will be when clipped-in.
c. Use two sheaves in tandem if conductor angle exceeds 30 degrees.
17. Splices not permitted to be pulled through stringing sheaves.
18. Controlled tension stringing:
a. Use where stringing by slack method is not practical.
b. Use for any other conductor at the Contractor’s option and approval from the Contracting Officer Representative.
c. Use no equipment or procedure which would cause outer layer of conductor strands to loosen as it passes over bullwheel.
d. Provide light and steady back tension on conductor reel to prevent overrun.
e. Observe all clearances during pulling.
f. Do not allow conductors to contact ground, guard structures or any other object which, in opinion of contracting officer representative, could cause abrasion or damage.
g. When left unattended, provide safe clearance between conductors in string sheaves and ground or other objects.
h. The slope of the conductor between the stringing equipment and the stringing sheave at the first structure not to be steeper than three horizontal to one vertical.
19. Slack stringing:
a. Use for any conductors where controlled tension stringing is not required.
b. Pay out conductor from rotating reel.
c. Control speed of reel during pay out in a manner which prevents backlash and overrunning.
d. Application of breaking to conductor reels to provide tension for pulling specifically prohibited.
e. Do not pull conductor over rocks, pavement, fences, crossarms, or any other abrasive material or sharp object.
f. Provide suitable guards, supports or other precautions necessary to prevent damage.
g. Do not allow trucks, trailers or other vehicles to be driven over conductor.
h. Do not allow conductor to be pulled through mud, wet ground, water, etc.
i. Raise conductors to stringing sheaves as soon as possible.
j. Do not leave conductors on the ground overnight.
20. For spacer cable, use stringing method recommended by cable manufacturer.
21. Terminations: Contractor shall provide permanent wildlife guards at all cable terminations and approved by the Contracting Officer’s Representative.

G. Conductor Sagging:

1. Follow recommendations of conductor manufacturer.
2. Sag conductors in accordance with the sag and tension charts shown in drawings.
    a. Contractor to provide Contracting Officer Representative sagging means and methods prior to sagging conductor for approval.
3. Sag only during good weather conditions.
    a. Relatively constant temperature, no excessive wind.
    b. Contracting officer representative reserves the right to stop sagging operations if he considers weather conditions to be excessive.
4. Measure air temperature with certified etched-glass thermometer.
    a. Place in basket made from conductor (remove internal strands as necessary to permit insertion of thermometer).
    b. Suspend from structure in position near and parallel to the conductors being sagged for at least 1/2 hour before sagging.
5. Conductor to hang freely in stringing sheaves, clear of the ground, a minimum of 2 hours before being sagged.
6. Properly tension conductors by measuring sags in the following spans:
    a. For sagging five spans or less:
        1) Select one level span near the end remote from pulling point, so that length is equal to or greater than the ruling span.
7. Record all data regarding sagging operations including:
    a. Specific spans measured.
    b. Tension and sag measurements.
    c. Temperature corresponding to measurements.
8. Sags will be verified by contracting officer's representative.
    a. Not necessarily the same spans used for sagging.
    b. Target or stopwatch method, at contracting officer representative option.
    c. Assist by providing personnel and equipment as requested.
    d. Re-sagging of incorrectly sagged spans to be done at expense of Contractor.
9. Sags in all spans shall be as required by the sagging charts, except the following tolerances are permitted:
    a. No minus tolerance.
    b. Plus tolerance not to exceed 1 inch per 100 feet of span length, with maximum of 6 inches.
    c. All conductors in span having same required sag must not differ by more than 1-1/2 inches.
10. Spacer Cable (aerial cable systems) shall be installed per manufacturer's recommendations.
    a. Taps and deadends shall be installed per instructions.

H. Jumpers:
1. Install in neat and workmanlike manner.
2. Install with adequate clearance from structure.
3. Provide sufficient slack to allow free movement of conductors to which the jumpers are attached.
4. Form smooth loop with no sharp bends, kinks or "cork screws".
5. All jumpers should have current carrying capability equal or larger than main conductors.

I. Installation of Insulators, Conductor Hardware and Accessories:
1. Insulators:
    a. Inspect before and after installation.
    b. Remove damaged units from work site immediately.
    c. Clean all porcelain surfaces and wipe to a bright finish.
    d. When raising suspension strings, use procedure which guarantees no damage to sockets or pins.
    e. No climbing on insulators.
    f. Pin and post insulators to be tight on pins or brackets.
    g. Insulator groove (clamp) to be in line with conductor.

h. For orientation of retaining pins and cotter pins, see below.

2. Conductor Hardware:
   a. Handle in manner to prevent contact with ground.
   b. Horizontal bolts or pins to have the nut or cotter on the pole side of the connection.
   c. Vertical bolts or pins to have the nut or cotter on underside of the connection.
   d. Horizontal ball and socket connections to have the cotter pins with eyes on top.
   e. Vertical ball and socket connections to have the cotter pins with eyes toward the pole.
   f. Hump back cotters to have ends spread to a minimum opening of 45 degrees.
   g. Straight cotters to have ends spread and turned back to an opening of 180 degrees.
   h. Ball hooks for horizontal strings to be installed so that open end of hook faces up.
   i. Tighten "U" bolts of suspension and dead end clamps to torques recommended by clamp manufacturer, using suitable torquing equipment.
   j. Tighten all other bolted fittings so that nuts are drawn tight so as to obtain full strength of fitting.
   k. All fasteners to be installed with appropriate locking devices.
3. Connections:
   a. Follow exactly the recommendations of connector manufacturer for the type of materials being connected (including installation, bolting hardware, joint compound, etc.).
   b. When connecting different material, use connectors which have no galvanic action.
   c. Tighten all nuts to required torques using suitable torquing equipment.
   d. Compression dies to be as recommended by manufacturer.
   e. Tape all connections to medium voltage spacer cable, except where specified otherwise (use materials and procedures recommended by cable manufacturer).
4. Formed ties:
   a. Follow manufacturer's recommendations.
   b. Do not use any tie a second time.
5. Tie wires:
   a. #6 strong aluminum alloy tie on 1/0 ACSR and smaller.
   b. #4 strong aluminum alloy tie on 2/0 ACSR and larger.
   c. Primary conductor shall be bare with weatherproofing removed when bare ties are used.
6. Armor rods:
   a. Follow manufacturer's recommendations.
7. Armor grip supports:
   a. Follow manufacturer's recommendations.
8. Dampers:
   a. Follow manufacturer's recommendations.
9. Spacer Cable :
   a. Follow manufacturer's recommendations for installation.
   b. Spacers shall hang plumb.
10. Spacers (for bundled conductors):
    a. Follow manufacturer's recommendations.
    b. Spacer shall not exert torque on any subconductor or create a difference in sag between subconductors.

J. Tree Trimming (if required):
   1. Contractor shall receive approval from Contracting Officer Representative prior to any tree trimming or vegetation removal.
   2. Take all precautions necessary for protection of existing facilities and structures.
   3. Use the best modern practice available for the type of trimming, topping and shaping required.
   4. Use personnel who are experienced and adequately trained in tree trimming.
   5. Disposal of vegetation:
      a. Remove large limbs from site.
      b. Dispose of all materials in manner acceptable to Contracting Officer Representative.
      c. Burning at site will not be permitted.

K. Phasing:
   1. Contractor to confirm phasing with Contracting Officer Representative prior to energizing.

L. Clearances:
   1. All clearances must meet or exceed NFPA 70 and IEEE C2 requirements.
      a. Verify that all required clearances are obtained.
      b. Notify contracting officer or its representative, in writing, of any locations where clearances cannot be obtained.

M. Conduit Fill: Any conductor pulled through conduit shall not exceed conduit fill beyond 53% for single cable in one conduit, 40% for three cables in one conduit.

END OF SECTION

This page intentionally left blank.

SECTION 26 05 48
ELECTRICAL SEISMIC RESTRAINT SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. The design and installation of seismic bracing and anchorage required for electrical equipment, conduit, cable tray, and bus ducts.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 81 10 – Wind and Seismic Design Criteria
   2. Division 26 - Electrical

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Civil Engineers (ASCE):
      a. 7, Minimum Design Loads and Associated Criteria for Buildings and Other Structures.
   2. ASTM International (ASTM):
      a. A36/A36M, Standard Specification for Carbon Structural Steel.
      b. A307, Standard Specification Carbon Steel Bolts, Studs, and Threaded Rod, 60,000 psi Tensile Strength.
      c. A500/A500M, Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes.
      d. A588/A588M, Standard Specification for High-Strength Low-Alloy Structural Steel, up to 50 ksi (345 MPa) Minimum Yield Point, with Atmospheric Corrosion Resistance.
      e. A992/A992M, Standard Specification for Structural Steel Shapes.
      f. F1554, Standard Specification for Anchor Bolts, Steel, 36, 55, and 105 ksi Yield Strength.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SYSTEM DESCRIPTION

A. Contractor is responsible for design and installation of seismic bracing and anchorage systems.

B. Description of Systems:
   1. Transverse and longitudinal bracing for seismic forces on suspended electrical systems including conduit, cable tray, bus duct, and equipment.
   2. Anchorage of floor and roof mounted electrical equipment.

C. Seismic Design Requirements:
   1. Seismic design criteria: Provide bracing and anchoring for equipment, conduit, cable tray, bust duct, designed, constructed, and installed to resist stresses produced by lateral forces.

D. Design and install seismic anchorage and bracing for all floor or roof mounted equipment weighing 400 pounds or more and all suspended or wall mounted equipment weighing 20 pounds or more.

E. The following components are exempt from the requirements of this Specification Section:
   1. Electrical components in structures assigned to Seismic Design Category C provided that the importance factor (I_p) is equal to 1.0.

2. Electrical components in Seismic Design Categories D, E, and F where I_p = 1.0 and flexible connections between the components and associated conduit are provided and that are mounted at 4 feet (1.22 M) or less above a floor level and weigh 400 LBS (1780 N)or less.
3. Electrical components in Seismic Design Categories D, E, and F weighing 20 pounds (95 N) or less where I_p = 1.0 and flexible connections between the components and conduit are provided, or for distribution systems, weighing 5 pounds/FT (7 N/M) or less.

F. Seismic forces shall be presumed to act through the center of mass of the equipment in a direction that will produce the largest single anchor force.

1.5 SUBMITTALS

A. See Specification Section 01 33 00 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
1. Product technical data:
   a. Seismic control devices.
2. Fabrication and/or layout drawings:
   a. Layout and mounting detail drawings showing system and proposed brace locations for all systems including pre-engineered systems.
   b. The specific detail for each type of brace or anchor must be referenced on a plan that identifies the required location.
      1) Supplying a book of details without referencing the proper detail to a specific location on a plan is not acceptable.
   c. Structural calculations for required lateral force level for each component.
   d. All submittals, including pre-approved systems, shall be signed and sealed by a licensed engineer, licensed in the state in which the project is located.

1.6 PROJECT CONDITIONS

A. Wind and Seismic Loads:
1. Comply with Section 01 81 10 – Wind and Seismic Design Criteria

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Pre-engineered suspended bracing systems:
   a. Unistrut by Atkore International, Inc.
   b. Or Equal.
2. Custom engineered systems designed using specified criteria and common building materials.

2.2 EQUIPMENT ANCHORS AND SUPPORTS

A. Drilled-in-place concrete anchors shall have an approved ICBO Evaluation Services Report.

B. Cast-in-place anchors shall comply with ASTM A36, ASTM A307, or ASTM F1554, 36 ksi.

C. Anchors permanently exposed to weather or corrosive environments shall be stainless steel or hot-dipped galvanized.

D. Structural steel for supports: ASTM A36, A588, A992 or A500.

E. Cold formed metal and connection material: Unistrut, or equal.

F. Any details provided are based on assumed equipment and arrangement.
   1. The Contractor shall be responsible for design and acquiring approval for support and anchorage of equipment and arrangement which varies from equipment and arrangement assumed in detail provided.

PART 3 - EXECUTION

3.1 GENERAL REQUIREMENTS

A. Every run which requires bracing shall have a minimum of two transverse braces and one longitudinal brace.
   1. A "run" is defined as suspended pipe, conduit, cable tray, bus ductor trapeze rack having a minimum 10 feet straight run length.

B. Brace spacing shall not exceed the maximum allowable brace spacing as engineered by the manufacturer or custom bracing designer.

C. Bracing may be omitted from conduit, cable tray and bus duct runs less than 10 feet in length.

D. Bracing may be omitted from conduit, cable tray and bus duct runs where rod hung supports of less than 12 inches (305 mm) in length are required.
   1. All unbraced suspended utility systems having 2 inches conduit and larger or systems weighing more then 5 pounds/FT shall be installed with a minimum 6 inches clearance to suspended ceiling vertical hanger wires.
   2. The conduit, cable tray, or bus duct shall be installed such that the lateral motion of the members will not cause damaging impact with other systems or structural members or loss of vertical support.

E. A longitudinal brace at a 90 degrees change in direction may act as a transverse brace if it is located within 2 feet of the change in direction.

F. A transverse brace may act as a longitudinal brace if it is located within 2 feet of a change in direction and if the brace arm and anchorage have been sized to meet or exceed the requirements of the longitudinal brace.

G. When bracing equipment or a utility system that is suspended from an overhead deck, brace back to the overhead deck or to the supporting structure supporting the deck.
   1. Do not brace to another element of the structure which may respond differently during a seismic event.

H. Obtain approval from the Contracting Officer prior to attaching any brace elements to structural steel or wood framing.

I. When utilizing cable bracing, tension the cable to remove slack without inducing uplift of the suspended element.
   1. Tension seismic bracing system prior to system start-up and adjust if necessary after equipment start-up.

J. As a general rule, do not mix rigid bracing with cable bracing in the same run.
   1. However, once bracing has transitioned a 90 degrees change in run direction, the bracing may switch from rigid to cable or vice versa if required due to a significant change in overhead deck elevation or to provide an implementable bracing scheme in a congested area.

K. Install brace members at an angle of 45 degrees from horizontal within a tolerance of +2-1/2 DEG or -45 degrees provided the brace length is accounted for in design.
   1. Brace angle may be increased to 60 degrees provided the brace spacing is reduced to 1/2 that required for a 45 degrees brace.

L. Seismic bracing may not pass through a building separation joint.
   1. Utility systems that pass through a separation joint must be seismically restrained no greater than 5 feet from the point of connection.
   2. Any hardware designed to accommodate seismic movement across the span of the separation joint shall be installed per manufacturer's installation and listing instructions.

M. With approval of the Contracting Officer, utility systems that are suspended from the overhead deck may be braced to load bearing concrete or CMU (concrete masonry) walls provided that the walls and the overhead decks will respond similarly during a seismic event.

N. Each layer of a multiple layer trapeze rack shall be braced individually based on the weight of the individual layer.

O. Conduit, cable tray, or bus duct constructed of non ductile material (plastic or fiberglass), shall have brace spacing reduced to 1/2 of the spacing allowed for ductile materials.

P. Where brace elements are through-bolted, the mounting hole in the element is to be no more than 1/16 inches in diameter larger then the bolt or threaded rod.

Q. Seismic braces shall directly brace the support and not the hanger.

3.2 SUSPENDED ELECTRICAL SYSTEMS

A. Install seismic bracing for all conduit 2-1/2 inches trade size or greater.

B. All trapeze assemblies supporting conduits, cable trays or bus ducts shall be braced considering the total weight of the elements on the trapeze.
   1. For the purposes of calculating weight, all conduits are to be treated as full.

C. Brace all trapeze racks which support conduit 2-1/2 inches trade size or larger.
   1. Brace all other conduit rack, cable tray or bus duct trapezes having a minimum weight in excess of 10 pounds/LF.
   2. Include a minimum 10% additional capacity for future additions.

D. Seismic bracing may be omitted from cable trays, conduit and bus ducts suspended by rod hung supports 12 inches or less in length from the top of the element to the bottom of the structural attachment of the hanger provided lateral motion will not cause damaging impacts to other systems or loss of system vertical support.

E. For steel and aluminum bus ducts, conduit and cable trays:
   1. Provide transverse bracing at 40 feet maximum spacing unless otherwise noted.
   2. Provide longitudinal bracing at 80 feet maximum spacing unless otherwise noted.

F. All vertical risers involving conduit 2-1/2 inches in diameter or larger shall include lateral restraint at maximum 30 feet intervals and at the top and bottom of the riser.

G. Make provisions to eliminate seismic impact between components.

3.3 FLOOR OR ROOF MOUNTED EQUIPMENT

A. Provide one anchor on each leg or corner.
   1. Support with a minimum of three 3/8 inches diameter anchors.

B. Friction shall be neglected when designing anchors for shear.

C. Vertical seismic forces, when required, shall be presumed to act concurrently with horizontal seismic forces.

D. Batteries on racks or for generators shall have wraparound restraints to ensure that the batteries will not slide laterally. Spacers shall be used between the restraints and cases to prevent damage to the cases.

E. Electrical cabinet design shall comply with the applicable NEMA standards. Cutouts in the lower shear panel that have not been made by the manufacturer and reduce significantly the strength of the cabinet shall be specifically evaluated.

END OF SECTION

This page intentionally left blank.

SECTION 26 08 13
ACCEPTANCE TESTING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Basic requirements for acceptance testing.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 - Equipment - Basic Requirements.
   2. Section 26 05 00 – Common Work Results for Electrical.
   3. Section 26 23 00 – Generator Paralleling Switchgear.
   4. Section 26 32 14 - Engine Generator - Diesel.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. 400, Guide for Field Testing and Evaluation of the Insulation of Shielded Power Cable Systems.
      b. 400.3, Guide for Partial Discharge Testing of Power Cable Systems in a Field Environment.
   2. InterNational Electrical Testing Association (NETA):
      a. ATS, Standard for Acceptance Testing Specifications for Electric Power Equipment and Systems.
   3. Nationally Recognized Testing Laboratory (NRTL).
   4. Telecommunications Industry Association/Electronic Industries Alliance/American National Standards Institute (TIA/EIA/ANSI):
      a. 455-78-B, Optical Fibres - PART 1-40: Measurement Methods and Test Procedures - Attenuation.

B. Qualifications:
   1. Testing firm:
      a. An independent firm performing, as the sole or principal part of its business for a minimum of 10 years, the inspection, testing, calibration, and adjusting of systems.
      b. Must have an established monitoring and testing equipment calibration program with accuracy traceable in an unbroken chain, according to NIST.
   2. Field personnel:
      a. Minimum of one year field experience covering all phases of electrical equipment inspection, testing, and calibration.
      b. Relay test technician having previous experience with testing and calibration of relays of the same manufacturer and type used on project and proficient in setting and testing the types of protection elements used.
      c. Supervisor certified by NETA or NICET.
         1) As an alternative, supervising technician may be certified by the equipment manufacturer
   3. Analysis personnel:
      a. Minimum three years combined field testing and data analysis experience.
      b. Supervisor certified by NETA or NICET.
         1) As an alternative, supervising technician may be certified by the equipment manufacturer.

C. Phasing Diagram:
   1. Coordinate with Utility Company and NPS staff for phase rotations and Phase A, B and C markings.

a. Create a phasing diagram showing the coordinated phase rotations of power distribution systems, distribution transformers and switches/switchgear, with generators and motors through the transformers.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Equipment Monitoring and Testing Plan.
   2. See Specification 01 61 03 for electrical equipment and connection testing plan submittal requirements.

C. Informational Submittals:
   1. Prior to energizing equipment:
      a. Coordinated phasing diagram.
      b. Photocopies of continuity tests.
   2. Within two weeks after successful completion of Demonstration Period (Commissioning Period) at each project site:
      a. Single report containing information including:
         1) Summary of Project.
         2) Information from pre-energization testing.
         3) Testing and monitoring reports.
      b. Project sites:
         1) Electrical work contained in each Volume of project Drawings shall be considered a “site”.

PART 2 - PRODUCTS

2.1 FACTORY QUALITY CONTROL

A. Provide Electrical equipment with all factory tests required by the applicable industry standards or NRTL.

B. Factory testing will not be accepted in lieu of specified field acceptance testing requirements.

PART 3 - EXECUTION

3.1 FIELD QUALITY CONTROL

A. General:
   1. Complete electrical testing in three phases:
      a. Pre-energization testing phase.
      b. Equipment energized with no load.
      c. Equipment energized under load.
   2. Perform testing in accordance with this Specification Section and NETA ATS.
   3. Provide field setting and programming of all adjustable protective devices and meters to settings as determined by the approved coordination study.
   4. Equip testing and analysis personnel with all appropriate project related reference material required to perform tests, analyze results, and provide documentation including, but not limited to:
      a. Contract Drawings and Specifications.
      b. Related construction change documentation.

c. Approved Shop Drawings.
d. Approved Operation and Maintenance Manuals.
e. Other pertinent information as required.

B. Equipment Monitoring and Testing Plan:
1. Approved in accordance with Shop Drawing submittal schedule.
2. Included as a minimum:
a. Qualifications of firm, field personnel, and analysis personnel doing the Work.
b. List and description of testing and analysis equipment to be utilized.
c. List of all equipment to be tested, including:
1) Name and tag numbers identified in the Contract Documents.
2) Manufacturer's serial numbers.
3) Other pertinent manufacturer identification,

C. Instruments Used in Equipment and Connections Quality Control Testing:
1. Minimum calibration frequency:
a. Field analog instruments: Not more than 6 months.
b. Field digital instruments: Not more than 12 months.
c. Laboratory instruments: Not more than 12 months.
d. If instrument manufacturer's calibration requirements are more stringent, those requirements shall govern.
2. Carry current calibration status and labels on all testing instruments.
3. See individual testing programs for additional instrumentation compliance requirements.

D. Testing and Monitoring Program Documentation:
1. Provide reports with tabbed sections for each piece of equipment tested.
2. Include all testing results associated with each piece of equipment under that equipment's tabbed section.
a. Include legible copies of all forms used to record field test information.
3. Prior to start of testing, submit one copy of preliminary report format for Contracting Officer review and comment
a. Include data gathering and sample test report forms that will be utilized.
4. In the final report, include as a minimum, the following information for all equipment tested:
a. Equipment identification, including:
1) Name and tag numbers identified in the Contract Documents.
2) Manufacturer's serial numbers.
3) Other pertinent manufacturer identification,
b. Date and time of each test.
c. Ambient conditions including temperature, humidity, and precipitation.
d. Visual inspection report.
e. Description of test and referenced standards, if any, followed while conducting tests.
f. Results of initial and all retesting.
g. Acceptance criteria.
h. "As found" and "as left" conditions.
i. Corrective action, if required, taken to meet acceptance.
j. Verification of corrective action signed by the Contractor, equipment supplier, and Contracting Officer Representative.
k. Instrument calibration dates of all instruments used in testing.
5. Provide three (3) bound final reports prior to Project final completion.

E. Electrical Equipment and Connections Testing Program:
1. See individual Division 26 Specification Sections for equipment specific testing requirements.
2. Test all electrical equipment.
a. Perform all required NETA testing.

b. Perform all required NETA testing plus the optional testing identified with each specific type of equipment in Article 3.2 of this Specification Section.

3.2 SPECIFIC EQUIPMENT TESTING REQUIREMENTS

A. Switchgear and Switchboards:
   1. Perform inspections and tests per NETA ATS 7.1.
   2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.

B. Transformers - Small Dry Type:
   1. Perform inspections and tests per NETA ATS 7.2.1.1.
   2. Perform the following additional tests:
      a. Record phase-to-phase, phase-to-neutral, and neutral-to-ground voltages at no load after energizing, and at operating load after startup.
   3. Adjust tap connections as required to provide secondary voltage within 2-1/2% of nominal under normal load after approval of Contracting Officer.
   4. Record as-left tap connections.

C. Transformers - Liquid Filled:
   1. Perform inspections and tests per NETA ATS 7.2.2., including:
      a. Bushing power factor.
      b. Core insulation resistance.
      c. Dissolved gas analysis.
   2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.
   3. Perform the following additional tests:
      a. Record phase-to-phase, phase-to-neutral, and neutral-to-ground voltages at no load after energizing, and at operating load after start-up.
   4. Adjust tap changer setting as required to provide secondary voltage within 2-1/2% of nominal under normal load after approval of Contracting Officer.
   5. Record as-left tap changer setting.

D. Cable - Low Voltage:
   1. Perform inspections and tests per NETA ATS 7.3.2.
   2. DC High Potential test for all 480V power circuits.
   3. Submit results as an informational submittal.

E. Cable - Medium Voltage:
   1. Perform inspections and tests per NETA ATS 7.3.3.
   2. Perform Dielectric Withstand Test.
   3. Perform tan delta or partial discharge Baseline Diagnostic Tests.
   4. Prior to energization, perform a very low frequency (VLF) test following IEEE 400.2 Guide for Field Testing of Shielded Power Cable systems for baseline data for future VLF maintenance testing:
      a. VLF testing shall include:
         1) Withstand test to evaluate whether the cable can handle the test voltage (See IEEE 400.2 Table 3).
         2) Tangent delta test, including differential tangent delta and tangent delta stability, for baseline comparison:
         3) VLF testing shall be completed by technicians trained to use testing equipment with at least three years of experience in performing VLF testing on medium voltage cables.
      b. Results for new cable shall be compared to Table G.2 in IEEE 400.2.
         1) Discrepancies between field test results and Table G.2 shall be notated and discussed with testing equipment manufacturer to determine validity of results.
   5. Shield continuity test.
   6. Cable test report(s) shall include the following at minimum

a. Summary of Project.
b. Date tests were performed.
c. Description of equipment/components tested.
d. Visual inspection report.
e. Description of tests.
f. Test results, including appropriate test forms.
g. Conclusions and recommendations for corrective action as appropriate.
h. Identification of test equipment used.
i. Identification of company and person that performed the tests.
j. Reports signed by the engineering technician or engineer interpreting the data.

7. Non-destructive partial discharge (PD) test:
   a. After energization, perform a partial discharge test for baseline data for future partial discharge maintenance testing.
   b. Perform the work while the medium voltage circuits and equipment are energized.
      1) The cables shall not be disconnected or de-energized and the testing shall not expose the cables to voltages that exceed normal operating voltage.
   c. Use a frequency domain detection process incorporating a spectrum analyzer with radio frequency current transformer (RF CT) sensors.
      1) The detection system, including spectrum analyzer, RF CT's and interconnecting cable, shall have a partial discharge detection range that at least covers the frequency range of 10 kHz to 300 MHz.
      2) Testing shall be performed in a manner that complies with the requirements of IEEE 400 and IEEE 400.3.

F. Busway and Busduct:
   1. Perform inspections and tests per NETA ATS 7.4.
   2. Components: Test all components per applicable paragraphs of this Section and NETA ATS.
   3. Confirm phasing and phase connections before energizing.

G. Cable - Optical Fiber:
   1. Perform inspections on tests per TIA/EIA/ANSI 455-78-B, including:
      a. Optical time domain reflectometer test.
      b. Power attenuation test.
      c. Gain margin test.

H. Air Interrupter Switches:
   1. Perform inspections and tests per NETA ATS 7.5 and NETA ATS 7.6.
   2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.
   3. Perform the following optional tests per NETA ATS on all medium voltage switches:
      a. Insulation resistance phase-to-phase, phase-to-ground in open and closed positions and across each open pole.

I. Medium Voltage Source Transfer System:
   1. Perform applicable inspections and test per:
      a. NETA ATS for Interrupter Switches.
      b. Manufacturer's instructions.

J. Medium Voltage Circuit Breakers:
   1. Perform inspections and tests per NETA ATS 7.6.2.
   2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.
   3. Perform the following optional tests per NETA ATS:
      a. Control wiring insulation resistance.
      b. Minimum trip and close voltage.
      c. Overpotential.
      d. Circuit breaker timing test.

e. Power factor.
4. Perform the following additional tests:
a. High-potential vacuum integrity test per manufacturer's recommendations.

K. Low Voltage Power Circuit Breakers:
1. Perform inspections and tests per NETA ATS 7.6.1.2.
a. Tests shall include primary current injection testing of all breakers at final settings.
b. Where short-time or instantaneous settings on large frame breakers are beyond the current capability of field testing, primary injection tests at reduced currents shall be permitted if combined with secondary injection calibration test of trip unit at final settings.
2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.
3. Perform the following additional tests:
a. Shunt trip devices minimum tripping voltage.
4. Record as-left settings.

L. Low Voltage Molded Case Circuit Breakers:
1. Perform inspections and tests per NETA ATS 7.6.1.1.
2. Components:
a. Test all components per applicable paragraphs of this Specification Section and NETA ATS.
b. Thermal magnetic breakers: Visual and mechanical inspection per NETA ATS only.
c. Solid state trip type: Visual and mechanical inspection and electrical tests per NETA ATS.
3. Record as-left settings.

M. Protective Relays:
1. Perform inspections and tests per NETA ATS 7.9.
a. Tests to be performed using secondary injection of 3 PH current and potential at final settings.
b. Test at manufacturer's recommended test points and critical timing points identified on relay setting sheet.
2. Perform all tests identified as optional per NETA ATS.
3. Perform the following additional tests:
a. Verification of direct trip of associated lockout relay or circuit breaker(s) by using relay test function or shorting trip contact at relay case.
b. Microprocessor-based relays:
1) Complete commissioning procedure per manufacturer's instructions, followed by tests of each relay element at final settings.
2) Verification of all internally-programmed logic.
c. Verification of all auxiliary input and output signals.
d. Verification of power supply/self-diagnostic alarm contact and remote annunciation.
4. Record as-left settings.

N. Instrument Transformers:
1. Perform inspections and tests per NETA ATS 7.10.
2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.
3. Perform the following optional tests per NETA ATS:
a. Dielectric withstand test on potential transformers.

O. Metering:
1. Perform inspections and tests per NETA ATS 7.11.
2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.

P. Grounding:

1. Perform inspections and tests per NETA ATS 7.13.
2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.

Q. Ground Fault Protection:
1. Perform inspections and tests per NETA ATS 7.14.
2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.
3. Perform the following optional tests per NETA ATS:
   a. Control wiring insulation resistance.
4. Perform the following additional tests for four-wire systems:
   a. Primary current injection into switchgear bus with test set configured to simulate transformer source and high current jumper used to simulate unbalanced load and ground fault conditions.
   b. Verify no tripping for unbalanced load on each feeder and each main breaker.
   c. Verify no tripping for unbalanced load across tie breaker for dual-source schemes.
   d. Verify tripping for ground fault on load side of feeder each feeder and on each main bus.
   e. Verify tripping for ground fault on a single feeder and on each main bus through tie breaker(s) for multiple-source schemes.

R. Motors:
1. Perform inspections and tests per NETA ATS 7.15.

S. Motor Controllers:
1. Perform inspections and tests per NETA ATS 7.16.
2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.

T. Generators:
1. Perform inspections and tests per NETA ATS 7.15.2.
2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.
3. Perform the following additional tests:
   a. Load and cycle crank test per Specification Section 26 32 14.

U. DC Power Systems:
1. Perform inspections and tests per NETA ATS 7.18.
2. Components: Test all components per applicable paragraphs of this Specification Section and NETA ATS.
3. Perform the following optional tests per NETA ATS:
   a. Cell impedance test.

V. Control System Functional Test:
1. Perform test upon completion of equipment acceptance tests.
2. The test is to prove the correct interaction of all sensing, processing and action devices.
3. Develop a test plan and parameters for the purpose of evaluating the performance of the system.
4. Perform the following tests:
   a. Verify the correct operation of all interlock safety devices for fail-safe functions in addition to design function.
   b. Verify the correct operation of all sensing devices, alarms and indicating devices.
5. Systems to be tested:
   a. Protective Relaying Systems.
   b. Digital Metering System.
   c. Paralleling Switchgear Schemes.
   d. Standby Generator Systems.
   e. PLC Automatic Throwover System.

f. Low Voltage Lighting Control System.

END OF SECTION

SECTION 26 09 13
ELECTRICAL METERING DEVICES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Digital metering equipment installed integral to electrical distribution equipment.
   2. The metering equipment and devices described in this Section are for Government metering.

B. Related Sections include but are not necessarily limited to:
   1. Section 26 05 00 – Common Work Results for Electrical.
   2. Section 26 08 13 - Acceptance Testing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. C57.13, Standard Requirements for Instrument Transformers.
   2. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   3. National Electrical Manufacturers Association/American National Standards Institute (NEMA/ANSI):
      a. C12.20, For Electricity Meter - 0.2 and 0.5 Accuracy Classes.
   4. Underwriters Laboratories, Inc. (UL):
      a. 508, Standard for Safety Industrial Control Equipment.

1.3 DEFINIITONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data including:
      a. Provide submittal data for all products specified in PART 2 of this Specification:
      b. See Section 26 05 00 for additional requirements.
   2. Fabrication and/or layout drawings.
      a. Electrical wiring/connection diagrams.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
      b. Content of Operation and Maintenance Manual:
         1) Data sheet of the meter’s electrical parameters, configuration and characteristics including a complete model number and associated equipment connected too.
         2) Operating instructions for the meter(s) supplied.
         3) Operating instructions for the Power Management software.
         4) Maintenance instructions.
         5) As-constructed electrical wiring/connection diagrams.
         6) As-commissioned programming, settings, and parameters.
         7) Acceptance testing data.

D. Refer to Section 01 81 33 – Cyber Security Requirements for required cyber security related submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Eaton.
   2. Or Equal

2.2 DIGITAL METERING DEVICES

A. Refer to Section 01 81 33 - Cyber Security Requirements for cyber security related product requirements.

B. General:
   1. Direct reading metered or calculated values.
   2. Microprocessor based.
   3. Integral LED or LCD display.
   4. Current and potential transformers as required.
   5. Integral fusing.
   6. Operating temperature: 0 degrees F to 150 degrees F.
   7. Standards:
      a. NEMA/ANSI C12.20.
      b. UL 508.

C. Type 'C' High Range Meter:
   1. Eaton "PXM2270" or better, or approved equal.
   2. Display the following minimum electrical parameters (accuracy):
      a. RMS current per phase (+0.2% full scale).
      b. RMS voltage line-to-line and line-to-neutral (+0.2% full scale).
      c. Real power (W): 3 PH total (+0.4% full scale).
      d. Apparent power (VA): 3 PH total (+0.4% full scale).
      e. Reactive power (VAR): 3 PH total (+0.4% full scale).
      f. Power factor (+1.0%).
      g. Frequency (+0.04%).
      h. Percent current individual harmonic and total harmonic distortion (50th).
      i. Percent voltage individual harmonic and total harmonic distortion (50th).
      j. Watt-hours (0.5%).
      k. VAR-hours (1.0%).
      l. VA-hours (0.5%).
      m. Ampere demand (+0.2% full scale).
      n. Watt demand (+0.4% full scale).
      o. VAR demand (+0.4% full scale).
      p. VA demand (+0.4% full scale).
   3. Data logging:
      a. 128 KB.
      b. Selectable for parameters listed above for display.

c. Software for configuration, retrieval, and trending.
4. NEMA/ANSI C12.20, Class 0.2 revenue accuracy.
5. Communication ports and protocols: Ethernet TCP/IP.
6. Supply voltage: 120 VAC.

2.3 TRANSFORMERS

A. Current Transformer (CT):
1. Standard: IEEE C57.13
2. Current ratios: As indicated on the Drawings, or as required for system operation.
3. Window type current transformers shall be accessible from the front of the cubicle to permit changing or adding, without disconnecting bud joints.
4. Minimum ANSI metering accuracy class of 0.3 through B-0.5
5. Mount and brace to withstand mechanical stresses resulting from short circuit currents.
6. Equip with a shorting terminal block.

B. Potential Transformers (PT):
1. Standard: C57.13
2. Voltage ratio: As required.
3. Primary current-limiting fuses.
4. Secondary fuses or circuit breaker.
5. Accuracy class: 0.3 at burden W through Z, and 0.6 at burden ZZ.
6. Thermal burden rating: Exceed maximum connected burden.

2.4 ACCESSORIES

A. Power Management and Control Software:
1. Manufacturer’s standard, see Part 3 for minimum functionality.

2.5 MAINTENANCE MATERIALS

A. Three sets of spare fuses for each metering package.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install as indicated and in accordance with manufacturer's recommendations and instructions.
1. Provide all equipment as necessary to provide a complete and functioning system.
2. Coordinate with Contracting Officer on final computer screen layouts, trending requirements and printouts.

B. Meter Type Application:
1. Type A meters: Not used.
2. Type B meters: Not used.
3. Type C meters: Integral to equipment as indicated on the Drawings.

C. Communication Configuration:
1. As indicated on the Drawings.
a. Ethernet communication to the Plant Control System.
1) Provide all bridges, gateways, and other accessories as required for a complete and fully-functional Ethernet communications interface.

D. Current transformers shall be connected to shorting type terminal blocks.

E. Potential transformers shall have primary and secondary fuses with a disconnecting means.

3.2 FIELD QUALITY CONTROL

A. Acceptance Testing: See Section 26 08 13.

B. Manufacturer shall configure and program all metering equipment and accessories, and confirm that all meters are providing correct and accurate readings. Programming and configuration shall include the interface to the Plant Control System.

3.3 TRAINING

A. A qualified factory-trained manufacturer's representative shall provide 8 hours of on-site training in the operation and maintenance of the metering system and its components.

B. See Section 01 79 00 for Operator training requirements.

3.4 CLOSEOUT ACTIVITIES

A. Refer to Section 01 81 33 – Cyber-Security Requirements for cyber security related closeout requirements.

B. Provide field training at each project site
   1. North Rim Water Treatment Plant
   2. Roaring Springs Pump Station

END OF SECTION

SECTION 26 09 16
CONTROL EQUIPMENT ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Operator control devices (selector switches, pushbuttons, indicator lights, etc.).
   2. Control devices (timers, relays, contactors, etc.).
   3. Operator Control Stations.

B. Related Sections include but are not necessarily limited to:
   1. Section 01 81 33 – Cyber Security Requirements.
   2. Section 26 05 00 - Common Work Results for Electrical.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. ICS 2, Industrial Control and System Controllers, Contactors and Overload Relays Rated 600 Volts.
      c. ICS 5, Control Circuit and Pilot Devices.
   2. Underwriters Laboratories, Inc. (UL):
      a. 508, Standard for Safety Industrial Control Equipment.
      b. 508A, Standard for Industrial Control Panels.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SYSTEM DESCRIPTION

A. This Specification specifies components used within other equipment as referenced in other technical specifications.

B. This Specification is used to specify the components and construction of Operator Control Stations as shown in the Drawings.

1.5 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data:
      a. Provide submittal data for all products specified in PART 2 of this Specification:
         1) When components are used within equipment specified in another Section, submittal data for components specified herein shall be included with the submittal for the equipment the components are used in.
      b. Industrial Control Panel bill of material.
      c. Control Station bill of material.
      d. See Section 26 05 00 for additional requirements.
   2. Fabrication and/or layout drawings.
      a. Operator Control Station:
         1) Interior (if applicable) and exterior layout.
         2) Wiring/connection diagrams.

b. Associate Operator Control Stations with associated equipment name and tagging.

C. Informational Submittals:
1. Functional Test Plan.

D. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
b. Content of Operation and Maintenance Manual:
1) Product technical data of components used within Operator Control Stations.
2) As-constructed wiring/connection diagrams for Operator Control Stations.
3) Operating instructions for adjustable and programmable devices.
4) Functional Test Report.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Pilot devices, relays, contactors, and termination equipment:
a. Eaton.
b. Or Equal.
2. Photocells and time clocks:
a. See Section 26 09 43 for requirements for photocells and time clocks.
3. Alarm devices:
a. Edwards Signaling by United Technologies Corp.
b. Or Equal.
4. Enclosures:
a. B-Line by Eaton.
b. Or Equal.

C. Submit request for substitution in accordance with Section 01 67 00.

2.2 PILOT DEVICES

A. General Requirements:
1. Standards: NEMA ICS 5, UL 508.
2. Heavy-duty NEMA 4/13 watertight/oiltight.
3. Heavy-duty NEMA 4/4X corrosion resistant.
4. Heavy-duty factory sealed, explosion-proof and dust ignition-proof (Class I and II).
5. Mounting hole: 30.5 mm.
6. Contact blocks: 10 amp, NEMA A600 rated, number as required to fulfill functions shown or specified.
7. Legend plate marked as indicated on Drawings or specified.

B. Selector Switches:
1. Two, three- or four-position rotary switch as required to fulfill functions shown or specified.
2. Maintained contact type.
3. Knob or lever type operators.

C. Pushbuttons:
   1. Non-illuminated type:
      a. Protective boot.
      b. Momentary contact.
      c. Standard flush and mushroom operators.
      d. Green colored buttons for START or ON and red color for STOP or OFF.
      e. Emergency stop pushbuttons: Red mushroom head operator and maintained contact.
   2. Illuminating type:
      a. Protective boot.
      b. Momentary contact.
      c. Standard flush operator.
      d. Serves as both pushbutton control and indicating light.
      e. Green colored lenses: START or ON.
      f. Red colored lenses: STOP or OFF.
      g. Resistor-type full voltage light unit with lens and panel gasket.

D. Indicating Lights:
   1. Allowing replacement of bulb without removal from control panel.
   2. Lamp: LED, 120 V or 24 V as required.
   3. Full voltage type.
   4. Push-to-test indicating lights.
   5. Plastic lens.
   6. Color code lights as follows:
      a. Green: ON or running; valve open.
      b. Amber: Standby; auto mode; ready.
      c. Red: OFF or stopped; valve closed.

2.3 RELAYS

A. General Requirements:
   1. Standards: NEMA ICS 5, UL 508.

B. Control Relays:
   1. General purpose (ice cube) type:
      a. Plug-in housing.
      b. Clear polycarbonate dust cover with clip fastener.
      c. Coil voltage: 120 VAC or as required.
      d. Contacts:
         1) 10 amp continuous.
         2) Silver cadmium oxide.
         3) Minimum of 3 SPDT contacts.
      e. Sockets: DIN rail mounted.
      f. Internal neon or LED indicator is lit when coil is energized.
      g. Manual operator switch.
   2. Industrial type:
      a. Coil voltage: 120 VAC or as required.
      b. Contacts:
         1) 10 amp, NEMA A600 rated.
         2) Double break, silver alloy.
         3) Convertible from normally open to normally closed or vice versa, without removing any wiring.
         4) Expandable from 2 poles to 12 poles.
      c. Provide contacts for all required control plus two spares.

C. Time Delay Relays:
   1. General purpose type:
      a. Timing modes: On and Off delay, interval, one shot and repeat cycle.
      b. Plug-in housing.

c. Polycarbonate dust cover with clip fastener.
d. Coil voltage: 120 VAC or as required.
e. Contacts:
1) 10 amp continuous.
2) Silver cadmium oxide.
3) Two normally open and two normally closed DPDT contacts.
f. Sockets: DIN rail mounted.
g. External timing adjustment knob.
h. Timing ranges: 0.05 seconds to 16.65 hours.
i. Repeat accuracy: +1%.

2. Solid State industrial type:
a. Timing modes: On and Off delay and repeat cycle.
b. Industrial housing.
c. Coil voltage: 120 VAC or as required.
d. Contacts:
1) 5 amp, NEMA B150 rated.
2) Silver alloy.
3) Convertible On Delay and Off Delay contacts.
4) One normally open and one normally closed timed contacts.
5) One normally open and one normally closed instantaneous contacts.
e. Furnish with "on" and "timing out" indicators.
f. External timing adjustment knob.
g. Timing ranges: 0.05 seconds to 10 hours.
h. Repeat accuracy: +1%.

3. Mechanical industrial type:
a. Timing modes: On and Off delay.
b. Coil voltage: 120 VAC or as required.
c. Contacts:
1) 10 amp, NEMA A600 rated.
2) Double break, silver alloy.
3) Convertible On Delay and Off Delay contacts.
4) Convertible normally open and normally closed timed contacts.
5) Convertible normally open instantaneous contacts.
d. External timing adjustment knob.
e. Timing ranges: 0.2 - 60 seconds or 5 - 180 seconds.
f. Repeat accuracy: Greater than +10%.

2.4 CONTACTORS

A. General Requirements:
1. Standards: NEMA ICS 2, UL 508.

B. Lighting and Remote Control Switches:
1. See Section 26 09 43 for requirements for lighting and remote control contactor switches.

C. Definite Purpose:
1. Coil voltage: 120 VAC or as required.
2. Contacts: Totally enclosed, double-break silver-cadmium-oxide.
3. Resistive load and horsepower rated.
4. Number of poles, continuous ampere rating and voltage, as indicated on Drawings or as specified.
5. Auxiliary contacts, as indicated on Drawings or as specified.

2.5 PHOTOCELLS AND TIME CLOCKS

A. Photocells:
1. See Section 26 09 43 for requirements for photocells.

B. General Requirements for Time Clocks:

1. See Section 26 09 43 for requirements for time clocks.

2.6 ALARM DEVICES

A. Alarm Horns:
   1. Vibrating horn type.
   2. PLC compatible as required.
   3. Heavy-duty die cast housing with corrosion resistant finish.
   4. Adjustable volume: 78 to 103 dB at 10 feet.
   5. Voltage: 120 VAC or as required.
   6. Enclosures/mountings:
      a. Flush wall or panel mounting in dry areas.
      b. NEMA 4X panel mounting in wet areas.
      c. Surface mounting in dry areas.
      d. NEMA 4X surface mounting in wet areas.
      e. NEMA 4X, hazardous location surface mounting in wet and hazardous areas.
         1) Fixed volume: 97 dB at 10 feet.

B. Alarm Lights:
   1. Panel mounted:
      a. Strobe type.
      b. Shatter resistant polycarbonate lens and base.
      c. Lens color as indicated on Drawings.
      d. NEMA 4X enclosure.
      e. PLC compatible.
      f. Voltage: 120 VAC.
   2. Wall mounted:
      a. Heavy-duty strobe type.
      b. Weatherproof shatter resistant polycarbonate lens and cast base.
      c. Optically designed fresnel lens with color as indicated on Drawings.
      d. Immune to shock and vibration, no moving parts.
      e. Xenon flash tube providing a minimum of 65 single flashes per minute.
      f. Mounting: Wall or corner wall brackets.
   3. Hazardous and corrosive locations:
      a. Heavy-duty strobe type.
      b. Weatherproof and rated for the indicated hazardous location.
      c. Body: Zinc plated cast iron or cast copper free aluminum and/or coated with 20 mils of PVC.
      d. High impact glass dome with guard.
      e. Shatter resistant polycarbonate lens with color as indicated on Drawings.
      f. Immune to shock and vibration, no moving parts.
      g. Xenon flash tube providing a minimum of 65 single flashes per minute.
      h. Mounting: Wall bracket or pendant.

2.7 MISCELLANEOUS DEVICES

A. Run Time Meters:
   1. Six-digit wheels including a 1/10 digit.
   2. Non-reset type.
   3. No batteries required for operation.
   4. Time range in hours.
   5. Automatic recycle at zero.
   6. Accuracy: 1%.
   7. Sealed against dirt and moisture.
   8. Tamperproof.

B. Control Power Transformer:
   1. Primary voltage: 480 V.

2. Secondary voltage: 120 V.
3. Sized for 125% of required load.
4. Fused on primary and secondary.
5. Standard: NEMA ST 1.

2.8 TERMINATION EQUIPMENT

A. General Requirements:
1. Modular type with screw compression clamp.
2. Screws: Stainless steel.
3. Current bar: Nickel-plated copper alloy.
4. Thermoplastic insulation rated for -40 to +90 degrees C.
5. Wire insertion area: Funnel-shaped to guide all conductor strands into terminal.
6. End sections and end stops at each end of terminal strip.
7. Machine-printed terminal markers on both sides of block.
8. Spacing: 6 mm.
9. Wire size: 22-12 AWG.
10. Rated voltage: 600 V.
11. DIN rail mounting.

B. Standard-Type Block:
1. Rated current: 30 A.
2. Color: Gray body.

C. Bladed-Type Disconnect Block:
1. Terminal block with knife blade disconnect which connects or isolated the two sides of the block.
2. Rated current: 10 A.
3. Color:
a. Panel control voltage leaves enclosure - normal: Gray body, orange switch.
b. Foreign voltage entering enclosure: Orange body, orange switch.

D. Grounded-Type Block:
1. Electrically grounded to mounting rail.
2. Terminal ground wires and analog cable shields.
3. Color: Green and yellow body.

E. Fuse Holders:
1. Blocks can be ganged for multi-pole operation.
2. Spacing: 9.1 mm.
3. Wire size: 30-12 AWG.
4. Rated voltage: 300 V.
5. Rated current: 12 A.
6. Fuse size: 1/4 x 1-1/4.
7. Blown fuse indication.
8. DIN rail mounting.

2.9 ENCLOSURES

A. Operator Control Stations:
1. NEMA 4X rated:
a. Type 304 or 316 stainless steel body.
b. Gasketed Type 304 or 316 stainless steel cover.
c. Number of device mounting holes as required.

2.10 FABRICATION

A. Supplier of Industrial Control Panels shall build control panel under the provisions of UL 508A or UL 698A.

1. Entire assembly shall be affixed with a UL 508A or UL 698A label "Listed Enclosed Industrial Control Panel" prior to shipment to the jobsite.
2. Provide equipment or control panels with Short Circuit Current Rating (SCCR) labeling as required by NFPA 70 and other applicable codes.
    a. Determine the SCCR rating by one of the following methods:
        1) Method 1: SCCR rating meets or exceeds the available fault current of the source equipment when indicated on the Drawings.
        2) Method 2: SCCR rating meets or exceeds the source equipment's Amp Interrupting Current (AIC) rating as indicated on the Drawings.
        3) Method 3: SCCR rating meets or exceeds the calculated available short circuit current at the control panel.
    b. The source equipment is the switchboard, panelboard, motor control center or similar equipment where the control panel circuit originates.
3. For Method 3, provide calculations justifying the SCCR rating. Utilize source equipment available fault current or AIC rating as indicated on the Drawings.

2.11 MAINTENANCE MATERIALS

A. Provide 100% replacement lamps for indicating lights.

B. Provide 10% replacement caps for indicating lights.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install as indicated and in accordance with manufacturer's recommendations and instructions.

B. Operator Control Stations:
    1. Permitted uses of NEMA 4X enclosure:
        a. Surface mounted in all areas.

3.2 FIELD QUALITY CONTROL

A. See Section 26 05 00.

B. Operator Control Station Functional Test:
    1. The test is to prove the correct interaction of all sensing, processing and action devices.
    2. Develop a test plan and parameters for the purpose of evaluating the performance of the system.
        a. Plan shall have witness signature lines for the contractor and Contracting Officer Representative and submitted when system passes the test.
    3. Perform the following tests:
        a. Verify functionality of all control states.
        b. Verify the correct operation of all interlock safety devices for fail-safe functions
        c. Verify the correct operation of all sensing devices, alarms and indicating devices.

3.3 CLOSEOUT ACTIVITIES

A. Refer to Section 01 81 33 – Cyber-Security Requirements for cyber security related closeout requirements.

END OF SECTION

This page intentionally left blank.

SECTION 26 09 43
NETWORKED LIGHTING CONTROL SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Networked Lighting Control System, as indicated, in accordance with provisions of Contract Documents.

B. Completely coordinate with work of other trades.

1.2 SYSTEM DESCRIPTION

A. This specification describes the design, engineering, programming, hardware, software, ancillary devices and associated technical services required to provide a building-wide networked lighting control system.

B. Networked Lighting Control System is specified to perform scheduled and automated lighting control sequences.

C. Networked Lighting Control System includes but is not limited to the following components:
1. Wall stations.
2. Touch screen panels.
3. Dimmers.
4. Switches.
5. Occupant sensors.
6. Photo sensors.
7. Daylight harvesting lighting controls.
8. Time clock.
9. System contact closures (input and output devices).
10. Scene setting and selection controllers.
11. Distributed relay lighting controllers.
12. Relay-based lighting and receptacle control panels.
13. Energy control units for load shedding.
14. System server.
15. Lighting control system software – graphical user interface-based.
16. Communication wiring.
17. Interface to fire alarm system.
18. Interface to security system.
19. Interface to building management system (BMS).

D. The system shall perform, at a minimum, the following functions:
1. Networking of Luminaires and Devices.
    a. The capability of luminaires and control devices to exchange digital data with other luminaires and control devices on the system.
2. Occupancy Sensing.
    a. The capability to affect the operation of lighting or other equipment based upon detecting the presence or absence of people in a space.
3. Daylight Harvesting.
    a. The capability to automatically affect the operation of lighting or other equipment based on the amount of daylight and/or ambient light that is present in a space.
4. High-End Trim (also known as Task-Tuning).
    a. The capability to set the maximum light output of an individual or group of luminaires at the time of installation and/or commissioning. The high-end trim feature must be field configurable.
5. Low-End Trim.

a. The capability to set the minimum light output of an individual or group of dimmable luminaires at the time of installation and/or commissioning. The low-end trim feature must be field configurable and is designed to mitigate flicker when luminaires are dimmed.

6. Zoning.
   a. The capability to group luminaires and form unique lighting control zones for a control strategy. Zoning is required for occupancy sensing, high-end trim, and daylight harvesting control strategies.
7. Device Addressability.
   a. The ability to uniquely identify and/or address each individual sensor, controller, and user interface device in the lighting system, allowing for configuration and reconfiguration of devices and control zones independent of electrical circuiting.
8. Continuous Dimming.
   a. The capability of the control system to provide control with sufficient resolution (100+ steps) to support light level changes perceived as smooth (as opposed to step dimming with a limited number of discrete levels).
9. User interface.
   a. The control software shall permit users with proper password access to read and adjust settings during system start-up, commissioning, and/or ongoing operation.
10. Scheduling.
    a. The ability to affect the operation of lighting and/or other equipment based on time of day or astronomical event.
11. Plug Load Control.
    a. The ability to control power delivered to receptacles through scheduling or occupancy sensing.
12. Energy Monitoring.
    a. The ability of a system, luminaire, or device to report its own energy consumption, or the energy consumption of any controlled device via direct measurement or other methodology.
13. Device Monitoring / Remote Diagnostics.
    a. The ability to monitor, diagnose, and report operational performance including system and/or component failures.

E. Related Information:
   1. Division 26 Section "Electrical General Requirements".
   2. Division 26 Section "Lighting Control Devices" for line voltage lighting controls.
   3. Division 26 Section "Building Lighting" for luminaires controlled by lighting control systems.
   4. Division 26 Section “Site Lighting” for luminaires controlled by lighting control system.

1.3 THIRD PARTY SYSTEM INTEGRATION

A. Pre-Bid Coordination: Third party interface with the lighting control system shall be coordinated with all parties to assign responsibility for equipment and installation of all equipment necessary for a fully functional system as described. Provide statement with bid indicating each manufacturer this specification has been reviewed and coordinated with.

B. Building Management System (BMS):
   1. The lighting control system shall integrate with Building Management System as follows:
      a. Allow one-way communication for the BMS system to monitor lighting control system status.
   2. The following lighting control system information should be mapped to the Building Management System:
      a. Monitor (One-Way Communication):
         1) Occupant sensor detection status.
         2) Aggregate occupant sensor status for each room.
         3) Wall station button state.

4) Load status.
5) Schedule State – normal or after-hours.
6) Electrical current (when available).
7) Log the hours of operation (cumulative run time) for each relay or load controller. Provide the ability to re-set.
8) Total watts per controller.
9) Demand response – enabled or disabled.
10) Lighting watts/sq ft by room.
11) Total lighting and plug loads by room.

C. Fire Alarm System: The lighting control system shall integrate with the facility fire alarm system to receive dry contact output from tire alarm system upon general alarm to activate predetermined scene(s).

1.4 REFERENCES

A. National Fire Protection Association (NFPA):
1. NFPA 70 National Electrical Code

B. Underwriters Laboratories (UL):
1. UL 20 Plug Load Controls
2. UL 508 Industrial Control Equipment
3. UL 916 Energy Management Equipment
4. UL 924 Emergency Lighting
5. UL 1008 Emergency Transfer Switching Devices

1.5 DEFINITIONS

A. Contracting Officer: Include the Contracting Officer (CO) and Contracting Officer Representative (COR).

1.6 SUBMITTALS

A. Timing:
1. Provide submittals after successful review of luminaire shop drawings. Coordinate driver types with control devices.
2. Provide shop drawings and product data concurrently. Submittals provided individually will be rejected without review.

B. Specification Compliance Review: Provide an annotated copy of this specification section to demonstrate compliance. This may be submitted prior to shop drawing review or concurrent.
1. The following letters shall be marked in the margins to identify compliance, deviations, or exceptions:
a. “C” – Comply with no exceptions.
b. “D” – Comply with deviations. Provide footnote that indicates reason for deviation and how the intent of this performance specification is satisfied.
c. “E” – Exception, does not comply. Provide footnote that indicates reason for non-compliance, impact on system performance, and possible alternatives means of compliance.
2. Unless deviations or exceptions are specifically noted, it will be assumed the Contractor and equipment manufacturer is in complete compliance with the Drawings and Specifications.
3. Deviations and exceptions are subject to review and approval.
4. Deviations or exceptions taken in other documents or by other Contractors do not release the Division 26 contractor from compliance unless specifically noted in this Specification Section Compliance Review document.

C. Shop Drawings:
1. Connection Schematics: Composite wiring, schematic diagram, or both, of each control circuit as proposed for installation.
2. Plans: Show device locations for each area on reflected ceiling plans.

a. Identify every system component with exact nomenclature indicated for that device on Drawings, and every component shall be labelled with corresponding room number.
b. Provide a schedule of every control station indicating all standard and custom wall station engraving.

3. Sequence of Operation: Provide a sequence of operation for each lighting zone, including those connected to emergency power. Refer to Part 3 – Execution.
4. System Riser Diagram: Include building levels indicated. Include network cable specification and end-of-line termination details, if required. Illustrate points of connection to integrated systems.
5. Compatibility Statement with Third Party Systems: Provide a compatibility statement from third party system manufactures certifying lighting control system compatibility with required integrations.
6. Identify all power supply connections on riser diagrams. Indicate load type, normal or emergency, voltage, and amperage. Division 26 contractor is responsible for supplying power to all devices in accordance with manufacturer's instructions; power supply connections are not shown on drawings.

D. Product Data:
1. Manufacturer's literature demonstrating compliance with requirements for each type of product required for complete distributed lighting control system.

E. Refer to Section 01 81 33 - Cyber Security Requirements for required cyber security related submittals.

F. Contract Closeout Information:
1. Operating and Maintenance Data.
a. See Section 01 78 23.
2. Project Record Documents.
a. See Section 01 78 39.
3. Warranty.
a. See Section 01 67 00.

1.7 QUALITY ASSURANCE

A. Manufacturer Qualification:
1. Manufacturer of relay lighting controls with minimum five (5) years satisfactory manufacturing and support of components comparable to basis of design system.

B. Project Conditions:
1. Environmental Conditions Range.
a. Temperature: 32 to 104 deg F.
b. Relative Humidity: 10 – 90 percent, noncondensing.

1.8 WARRANTY

A. Manufacturer's warranty agreeing to repair or replace components of lighting controls system that fail in materials or workmanship within specified warranty period following substantial completion.
1. Warranty Period: One year.
2. Software and firmware updates and onsite software and firmware support: One (1) year.
3. Technical hotline support: Unlimited.
4. Response time for on-site system maintenance: 72 hours.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient

characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Networked lighting control system:
   1. Acuity Brands: nLight.
   2. Or Equal.

2.2 DIGITAL LOCAL NETWORK

A. Refer to Section 01 81 33 - Cyber Security Requirements for cyber security related product requirements.

B. The digital local network uses free topology lighting control physical connections and communication protocol.

C. Local Network:
   1. Automatic recognition of devices connected.
   2. Replacement of any device in local network shall not require significant re-configuration or setup.
   3. System with automatic configuration must permit the ability to change and override the automatic settings, including binding and load parameters without tools, using only the buttons on the digital devices in the local network.
   4. If used, two-way infrared communication with handheld remote controllers, and configuration by handheld tool including adjusting load parameters, sensor configuration and binding, must be limited to within a line of sight of up to 30 feet from a sensor, wall switch or IR receiver.

D. Digital Cabling:
   1. Provide factory pre-terminated connectors and devices.
   2. Field terminated cables are not permitted without prior approval.

E. If manufacturer’s pre-terminated cables are not used for the installation, the contractor is responsible for testing each cable following installation and supplying manufacturer with test results.

2.3 DIGITAL ROOM CONTROLLERS

A. Digital controllers for lighting loads bind room loads to connected devices in the space and have the ability to assign and bind any device in the network. Provide room controllers to match luminaire dimming driver specifications and lighting control sequence of operations.

B. Control units will include following features:
   1. Ability for automatic room configuration to most energy-efficient, code-compliant, sequence of operation based upon the devices in the room.
   2. Ability for custom programming and assignment of sequence of operation for each control zone.
   3. Simple replacement and re-assignment of device settings to replacement devices. A room controller may be replaced without the need for factory re-programming.
   4. Multiple room controllers connected together in a local network must automatically arbitrate with each other, without requiring any configuration or setup, so that individual load numbers are sequentially assigned using each controller’s device IDs from highest to lowest.
   5. Device Status LEDs to indicate:
      a. Data transmission.
      b. Device has power.
      c. Status for each load.
      d. Configuration status.
   6. Quick installation features:

a. Standard junction box mounting.
b. Quick low voltage connections using standard RJ-45 patch cable.

7. Based on individual configuration, each critical and normal load shall be capable of being assigned one of the following behaviors upon power up following the loss of normal power:
   a. Turn on to 100 percent.
   b. Turn off.
   c. Return to prior state.
8. Each emergency load shall be capable of bypassing control status and turning on to 100 percent upon loss of normal power.
9. Each load shall at a minimum be configurable to operate in following sequences based on occupancy:
   a. Auto-ON/Auto-OFF.
   b. Manual-ON/Auto-OFF.
   c. Auto-Partial ON/Manual adjust/Auto-OFF.
   d. Auto-ON/Auto-DIM (Never off).
10. UL 2043 plenum rated.
11. Manual override ON and OFF with LED indication for each load.
12. Dual voltage 120/277 VAC, 60 Hz, or 347 VAC, 60 Hz. 120/277 volt models rated for 20A total load, de-rating to 16A required for dimmed loads, forward phase dimming; 347 VAC models rated for 15A total load; plug load controllers carry application-specific UL 20 rating for receptacle control.
13. Zero cross circuitry for each load.
14. All digital parameter data programmed into an individual room controller or plug load controller shall be retained in non-volatile FLASH memory within the controller itself. Memory shall have an expected life of no less than 10 years.

C. ON/OFF Room Controllers:
   1. Single or multiple relay configuration.
   2. Integral switching power supply.
   3. Integral digital network terminals or ports.
   4. Real time current monitoring.
   5. Logging of cumulative operating hours with the ability to manually re-set via software.

D. ON/OFF/Dimming Room Controllers:
   1. Real time current monitoring.
   2. Single or multiple relay configurations.
   3. Integral switching power supply.
   4. Logging of cumulative operating hours with the ability to manually re-set via software.
   5. Integral digital network terminals or ports.
   6. One dimming output per relay.
      a. 0-10V Dimming:
         1) One 0-10 volt analog output per relay for control of compatible LED drivers.
         2) Emergency controllers shall shunt control signal upon loss of normal power to assure full light output during emergency conditions.
      b. Line Voltage, Phase-Cut Dimming:
         1) Where indicated, one magnetic low voltage (MLV) or electronic low voltage (ELV) phase-cut dimming output per relay compatible with connected LED drivers.
      c. Dimming output channel:
         1) Independently configurable minimum and maximum trim level to set dynamic range of output generated by switch within 0-100 percent dimming range defined by minimum and maximum calibration trim.
      d. Set calibration and trim levels digitally per output channel.
         1) Devices that set calibration or trim levels per controller or multiple zone groupings are not acceptable.

2.4 DIGITAL WALL OR CEILING MOUNTED OCCUPANCY SENSOR

A. Wall or ceiling mounted (as shown on drawings) passive infrared (PIR), ultrasonic or dual technology (passive infrared and ultrasonic OR passive infrared and microphonic) occupancy sensor.

B. Occupancy Sensors shall include the following:
   1. Calibration and configuration for the following variables:
      a. Sensitivity – 0-100 percent in 10 percent increments.
      b. Time delay – 1-30 minutes in 1 minute increments.
      c. Test mode – Five second time delay.
      d. Detection technology assignment for ON and OFF signals when using dual-technology sensors. ON/OFF trigger modes can be configured to an individual technology, either (either technology detects/fails to detect presence), or both (both technologies must detect/fail to detect presence).
      e. Walk-through mode.
   2. Load parameters including Auto/Manual-ON, blink warning, and daylight enable/disable when photosensors are integral to occupancy sensor.
   3. Programmable control functionality including:
      a. Each sensor may be programmed to control specific loads within a local network.
      b. Sensor shall be capable of activating programmable lighting scenes.
      c. Adjustable retrigger time period for returning to automatic operation after manual override.
         1) On dual technology sensors: If either technology detects presence the lights will turn on. Both technologies must fail to detect presence for lights to turn off.
      d. Independently configurable sensitivity settings for passive infrared and ultrasonic technologies.
   4. Manual override of controlled loads.
   5. All digital parameter data programmed into an individual occupancy sensor shall be retained in non-volatile FLASH memory within the sensor itself. Memory shall have an expected life of no less than 10 years.

C. Multiple occupancy sensors may be installed in a room by simply connecting them to the free topology local network. No additional configuration will be required.

2.5 DIGITAL WALL SWITCHES WITH OCCUPANCY SENSOR

A. Wall box mounted passive infrared PIR or dual technology (passive infrared and ultrasonic OR passive infrared and microphonic) occupancy sensor with 1 or 2 load switch buttons.

B. Occupancy Sensors shall include the following:
   1. Calibration and configuration for the following variables:
      a. Sensitivity – 0-100 percent in 10 percent increments.
      b. Time delay – 1-30 minutes in 1 minute increments.
      c. Test mode – Five second time delay.
      d. Detection technology – assignment for ON and OFF signals when using dual-technology sensors. ON/OFF trigger modes can be configured to an individual technology, either (either technology detects/fails to detect presence), or both (both technologies must detect/fail to detect presence).
      e. Walk-through mode.
      f. Load parameters including Auto/Manual-ON, blink warning, and daylight enable/disable when photosensors are integral.
   2. Programmable control functionality including:
      a. Each manual load switch or sensor may be programmed to control specific loads within a local network.
      b. Sensor or manual load switch shall be capable of activating programmable lighting scenes.

c. Adjustable retrigger time period for retuning to automatic operation after manual override.
d. On dual technology sensors If either technology detects presence the lights will turn on. Both technologies must fail to detect presence for lights to turn off.

3. Independently configurable sensitivity settings.
4. Assignment of any occupancy sensor to a specific load within the room or to a specific dimming level.
5. Assignment of local buttons to specific loads within the room.
6. Manual override of controlled loads.
7. All digital parameter data programmed into an individual wall switch sensor shall be retained in non-volatile FLASH memory within the wall switch sensor itself. Memory shall have an expected life of no less than 10 years.

C. Each Load switch shall include and ON and OFF button, clearly labeled.

2.6 DIGITAL WALL SWITCHES

A. Low voltage momentary pushbutton switches. Wall switches shall include the following features:
1. Removable buttons for field replacement with engraved buttons and/or alternate color buttons. Button replacement may be completed without removing the switching device from the wall.
2. Two button switches shall have top button engraved “ON” and bottom button engraved “OFF” unless otherwise noted.
3. Programmable control functionality including:
a. Programmable time-out operation.
1) Two button timed switches shall have top button engraved “TIMED ON” and bottom button engraved “OFF.” Pressing top button shall trigger the programmable count-down time period assigned to the device.
2) Multi-button timed switches shall have various time settings assigned to each switch and shall have buttons engraved to reflect assigned time.
3) Timed switches shall have optional blink warning.
4) Two button override switches shall have top button labeled ON and bottom button labeled OFF.
5) Multi-button scene switches shall be engraved as specified on drawings. Refer to drawings for scene selection settings.
b. Scene settings, once set, may be locked to prevent overwriting of the preset levels.
4. All digital parameter data programmed into an individual wall switch shall be retained in non-volatile FLASH memory within the wall switch itself. Memory shall have an expected life of no less than 10 years.

B. Multiple wall switches may be installed in a room by simply connecting them to the local network to achieve multi-way switching and dimming.

2.7 DIGITAL PHOTOCELL SENSORS

A. Digital photocell sensors shall work with load controllers and relays to provide automatic switching and dimming capabilities for assigned networked relays. Photocell sensors shall be interchangeable without the need for rewiring.
1. Open loop sensors measure incoming daylight in the space, and generally located in close proximity to glazing.
2. Closed loop sensors measure the ambient light in the space, both daylight and electric, and generally are located away from glazing and outside of the field of indirect distribution from pendant luminaires.

B. Digital photocell sensors shall include the following features:
1. The sensor’s internal photodiode shall only measure light waves within the visible spectrum. The photodiode’s spectral response curve shall closely match the entire photopic curve. The photodiode shall not measure energy in either the ultraviolet or infrared spectrums. The

photocell shall have a sensitivity of less than 5 percent for any wavelengths less than 400 nanometers or greater than 700 nanometers.
2. Sensor light level range shall be from 1-6,500 footcandles (fc).
3. Sensor shall have programmable dead band setting to prevent lights from cycling.

C. Daylight harvesting systems shall be programmed to include the following features:
1. The capability to turn ON/OFF individual or multiple zones.
2. The ability to dim and adjust intensity relative to the amount the daylight.
3. A networked dimming switch shall allow occupants to override the photosensor adjusted dimming set point lower but may not increase the dimming level established by the photocell.
4. Any load or group of loads in the room can be assigned to a daylighting zone.
5. Each load or relay within a daylighting zone can be individually enabled or disabled, and individually dimmed for discrete control.
6. All digital parameter data programmed into a photosensor shall be retained in non-volatile FLASH memory within the photosensor itself. Memory shall have an expected life of no less than 10 years.

D. Photocell sensors shall be capable of reverse daylight harvesting, or increasing electric illumination when high levels of daylight are present to accommodate adaptation, as noted in lighting control zone sequence of operation.

2.8 LIGHTING MANAGEMENT SEGMENT NETWORK

A. Refer to Section 01 81 33 - Cyber Security Requirements for cyber security related product requirements.

B. The segment network shall be a linear topology, BACnet-based MS/TP subnet to connect local devices and relays for centralized control.
1. Each segment network shall include a network bridge. The network bridge interconnects room-based devices with the segment network.
2. Multiple segment networks may be connected to form a networked building control system that may consist of a single building or multiple buildings.
3. Network bridges, relay panels and segment managers shall include terminal blocks, with provisions for separate “in” and “out” terminations, for segment network connections.
4. The segment network shall utilize cable supplied by the lighting control manufacturer. The maximum cable run for each segment is 4,000 feet.
5. Network signal integrity requires that each conductor and ground wire be correctly terminated at every connected device. Manufacturer shall provide termination instructions.
6. Substitution of manufacturer-supplied cable must be pre-approved by lighting controls manufacturer.
7. Segment networks shall be capable of connecting to BACnet-compliant BAS (provided by others) either directly, via MS/TP, or through NB-ROUTERs, via BACnet/IP or BACnet/Ethernet. Gateways may be utilized to limit BAS system transmissions not related to shared lighting points.

2.9 NETWORK BRIDGE

A. Refer to Section 01 81 33 - Cyber Security Requirements for cyber security related product requirements.

B. Each local network shall include a network bridge component to provide a connection to the local network room devices. The network bridge shall use industry standard BACnet MS/TP network communication and an optically isolated EIA/TIA RS-485 transceiver.
1. The network bridge shall be provided as a separate module connected on the local network.
2. Provide automatic discovery of room devices connected to the local network and make all device parameters visible to the segment manager via the segment network. .

2.10 SEGMENT MANAGER

A. Refer to Section 01 81 33 - Cyber Security Requirements for cyber security related product requirements.

B. It shall be capable of serving up a graphical user interface via a standard web browser utilizing either unencrypted TCP/IP traffic via a configurable port (default is 80) or 256 bit AES encrypted SSL TCP/IP traffic via a configurable port (default is 443).

C. Each segment manager shall have integral support for at least three segment networks. Segment networks may alternately be connected to the segment manager via external BACnet-to-IP interface routers and switches, using standard Ethernet structured wiring. Each router shall accommodate one segment network. Provide the quantity of routers and switches as needed for a complete and operable system as described.

D. Operational features of the Segment Manager shall include the following:
   1. Connection to PC or LAN via standard Ethernet TCP/IP via standard Ethernet TCP/IP with the option to use SSL encrypted connections for all traffic.
   2. Easy to learn and use graphical user interface. Provide PC access to lighting control software with password protected access through local server. .
   3. Log in security capable of restricting some users to view-only or other limited operations.
   4. Segment Manager shall provide two main sets of interface screens – those used to initially configure the unit (referred to as the config screens) and those used to allow users to dynamically monitor the performance of their system and provide a centralized scheduling interface.
   5. Capabilities using the Config Screens shall include:
      a. Automatic discovery of devices and relay panels on the segment networks. Commissioning beyond activation of the discovery function shall not be required to provide communication, monitoring or control of all local networks and lighting control panels.
      b. Information entered into the lighting management software should not have to be re-entered manually via keystrokes into the Segment Manager.
      c. After discovery, all rooms and panels shall be presented in a standard navigation tree format. Selecting a device from the tree will allow the device settings and operational parameters to be viewed and changed by the user.
      d. Ability to view and modify device operational parameters. It shall be possible to set device parameters independently for normal hours and after hours operation including sensor time delays and sensitivities, and load response to sensor including Manual-ON or Auto-ON.
      e. One-way communication with BMS using BACnet protocol may use an “Export Table” that will provide any integrator with only the data they need, and by using the Export Table effectively create a firewall between the integrator’s request for info and the overall system performance.
   6. Segment Managers shall be integrated into a larger control network by the addition of a Campus View or Network Supervisor package. The Supervisor is a server level computer interfacing the local system software with dedicated communication and networking capability, able to pull information so individual Segment Managers can be accessed and stored on the Supervisor’s hard drive.
   7. The Segment Manager shall allow access and control of the overall system database via Native Niagara AX FOX connectivity. Systems that must utilize a Tridium Niagara controller in addition to the programming, scheduling and configuration server are not acceptable.

E. Segment Manager shall support multiple segment networks, network bridges and rooms.

2.11 PROGRAMMING, CONFIGURATION AND DOCUMENTATION SOFTWARE

A. PC-native application for programming of relay operation and device settings, including all parameters accessible or inaccessible via BACnet. Software must be capable of accessing room-level parameter information.

B. System shall include a central atomic time clock, with automatic daylight savings adjustment and astronomic 24-hr programming.

C. Additional parameters exposed through networked software include but are not limited to:
   1. Ability to assign both normal-hour and after-hour settings.
   2. Ability to provide Blink Warn prior to turning off, including the ability to adjust grace time.
   3. Separate fade time adjustments per load for both normal and after hours, adjustable from 0 - 4 hours.
   4. Configurable occupancy sensor re-trigger grace period from 0 - 4 minutes separate for both normal hours and after hours.
   5. Per-load DR (demand response) shed level in units of dimming percentage.
   6. Ability to establish the dimmer command level at which point a switched load closes or opens its relay to allow for ON/OFF operation of switched loads in response to a dimmer.

D. Generation of reports at the whole file, partial file, or room level. Reports include but are not limited to:
   1. Device list report: All devices in a project listed by type.
   2. Load binding report: All load controller bindings showing interaction with sensors, switches, and daylighting.
   3. BACnet points report: Per room identify Device ID and associated BACnet point mapping.
   4. Device parameter report: Per-room lists of all configured parameters for all control devices for use with O&M documentation.
      a. Include time delay and sensitivity settings for all occupancy sensors.
      b. Include measured room illumination target and associated sensor illumination readings along with dead-band settings for all daylight harvesting photocells.
   5. Scene setting reports: All project scene setting values associated with each named scene.

E. Network-wide programming of parameter data on an individual or group basis, including but not limited to the following operations:
   1. Set, copy/paste sensor time delays.
   2. Set, copy/paste sensor sensitivity settings.
   3. Search based on room name and text labels.
   4. Filter by product type to allow parameter set by product.
   5. Filter by parameter value to search for product with specific configurations.

F. Network-wide firmware upgrading ability.

PART 3 - EXECUTION

3.1 PRE-INSTALLATION MEETING

A. The successful bidder shall review required installation and pre-start-up procedures with the factory-authorized manufacturer's representative during pre-installation meetings.
   1. The pre-installation site visit shall be scheduled after receipt of approved submittals:
      a. Confirm location and mounting of digital devices, with special attention to placement of occupancy and daylighting sensors.
      b. Review specifications for low voltage control wiring and termination.
      c. Discuss functionality and configuration of products, including sequences of operation, per design requirements.
      d. Discuss requirements for interface with other trades.

3.2 IT NETWORK INFRASTRUCTURE COORDINATION

A. The successful bidder is required to coordinate with the Contracting Officer to secure all required network connections to the NPS IT network infrastructure.
   1. The bidder shall provide to the Contracting Officer all network infrastructure requirements of the networked lighting control system.
   2. The bidder shall provide to the factory-authorized manufacturer's representative all necessary contacts pertaining to the NPS IT infrastructure to ensure that the system is properly connected and started up.

3.3 INSTALLATION

A. General:

B. Coordinate, receive, mount, connect, and place into operation all equipment.
   1. Provide conduit, wire, connectors, hardware, and other incidental items necessary for properly functioning lighting control as described herein and shown in Drawings including but not limited to system field devices, power supplies and communication wire.
   2. Coordinate all dimming drivers provided with approved lighting fixtures.
   3. Maintain performance criteria stated by manufacturer without defects, damage, or failure.

C. Wiring:
   1. Comply with Section 26 05 19 - Low Voltage Electrical Power Conductors and Cables.
   2. Test branch load circuits for operation before connecting loads to sensor system load terminals.
      a. De-energize circuits before installation.
   3.
   4. Install devices and wiring and tagline voltage connections to indicate circuit and switched legs.
   5. Low voltage wiring topology must comply with manufacturer's specifications.
   6. Route network wiring as shown in submittal drawings.
   7. Document final wiring location, routing, and topology on record drawings.

D. Sensors:
   1. Calibrate sensor time delays and sensitivity to guarantee proper detection of occupants and energy savings.
      a. Adjust time delay so controlled area remains lighted while occupied.
      b. Adjust sensitivity or install masking on sensor devices to avoid false "ON's" from occurring. Install masking only after all furnishings are installed.
   2. Vacancy sensor configurations shall be made via the system's control software.
   3. Install occupancy sensors on vibration-free stable surfaces.
   4. Coordinate the location and quantity of all occupancy sensors based on the manufacturer's coverage patterns and coordination will all trades.
   5. Do not locate wall switches and occupancy sensors behind door swings, shelving, or tall equipment.

E. Software:
   1. Install and program software with initial settings of adjustable values.
   2. Generate backup copies of software and user-supplied values.
   3. Provide current site licenses for software.

F. Provide written or computer generated documentation of room-by-room configuration of system including:
   1. Sensor parameters, time delays, sensitivities, and daylighting set points.
   2. Sequence of operation, e.g., manual ON, Auto OFF.
   3. Load Parameters, e.g., blink warning.

3.4 DOCUMENTATION

A. The installing contractor shall be responsible for documenting installed location of all networked devices, including networked luminaires. This includes responsibility to provide record drawings showing device addresses corresponding to locations of installed equipment.

B. The installing contractor is also responsible for the following additional documentation to the manufacturer's representative if visualization / graphical floor plan software is provided as part of bid package:
   1. Record plan drawings showing daisy-chain wired network control zones outlined, in addition to device address locations required above. Documentation shall remain legible when reproducing\scanning drawing files for electronic submission.
   2. Record electrical lighting drawings (reflected ceiling plan) in PDF and CAD format. Architectural floor plans shall be based on as-built conditions.
      a. CAD files shall have layers already turned on/off as desired to be shown in the floor plan background images. The following CAD elements are recommended to be hidden to produce an ideal background graphic image:
         1) Title block.
         2) Text- Inclusive of room names and numbers, fixture tags and drawings notes.
         3) Fixture wiring and homeruns.
         4) Control devices.
         5) Hatching or poche of light fixtures or architectural elements.

3.5 SYSTEM START-UP

A. Upon completion of installation by the installer, including completion of all required verification and documentation required by the manufacturer, the system shall be started up and programmed by a factory-authorized manufacturer's representative.
   1. Low voltage network cable testing shall be performed prior to system start-up.

B. System start-up and programming shall include:
   1. Verifying operational communication to all system devices (wall stations, sensors, etc.).
   2. Programming the network devices into functional control zones to meet the required sequence of operation.
   3. Programming and verifying sequence of operations.
   4. Customization of NPS software interfaces and applications.

C. Initial start-up and programming are to occur on-site.
   1. The manufacturer's representative must walk the site with the Contracting Officer to confirm the programming of typical spaces prior to the final programming of all similar spaces.
   2. Daylight harvesting settings shall maintain specified illumination levels as indicated on drawings or as described below if values are not provided on drawings:
      a. Corridors and Lobbies – maintain an average of 5 fc on floor.
      b. Offices and Conference rooms – maintain an average of 50 fc at 30"above finished floor.
      c. Laboratories – maintain an average of 90fc at 36" above finished floor.
      d. Collaboration and Break areas – maintain an average of 30 fc at 30" above finished floor.

3.6 TRAINING

A. The provider of the system shall train the facility staff or end users responsible for managing the building lighting. Training shall include how to install and access lighting control software, including how to adjust, operate, utilize, troubleshoot, and monitor lighting status. Review field selectable device settings as well as programmable settings.
   1. The initial training session shall be for a maximum of 8 hours and shall follow project close-out.

2. After the installation has been occupied by NPS Staff for a period of 6 months, a follow-up on-site training session shall occur. The factory-authorized manufacturer's representative shall provide training to NPS Staff for a maximum of 2 hours, and a programmer, either on-site or remotely, shall spend a maximum of 6 hours confirming all daylight harvesting system settings and making programming adjustments to the system as requested/identified by NPS Staff.

B. See Section 01 79 00 for Operator training requirements.

3.7 CLOSEOUT ACTIVITIES

A. Refer to Section 01 81 33 - Cyber-Security Requirements for cyber security related closeout requirements.

END OF SECTION

SECTION 26 12 19
DISTRIBUTION TRANSFORMERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Distribution pad-mounted transformers.
      a. Transformers for installation at Roaring Springs and other Inner-canyon locations will require transportation to the installation site by helicopter.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 08 13 - Acceptance Testing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI).
   2. FM Global (FM).
   3. Institute of Electronic and Electronics Engineers, Inc. (IEEE):
      a. 386, Standard for Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV.
      b. C57.12.00, Standard General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers.
      c. C57.12.28, Standard for Pad-Mounted Equipment - Enclosure Integrity.
      d. C57.12.34, Standard Requirements for Pad-Mounted, Compartmental-Type, Self-Cooled, Three-Phase Distribution Transformers (2500 kVa and Smaller) - High-Voltage: 34 500 GrdY/19 920 Volts and Below; Low-Voltage: 480 Volts and Below.
      e. C57.12.70, Standard for Standard Terminal Markings and Connections for Distribution and Power Transformers.
      f. C57.12.80, Standard Terminology for Power and Distribution Transformers.
      g. C57.12.90, Standard Test Code for Liquid-Immersed Distribution, Power, and Regulating Transformers.
      h. C62.11, Standard for Metal-Oxide Surge Arresters for Alternating Current Power Circuits (>1 kV).
   4. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).

1.3 DEFINITIONS

1. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR)

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data:
      a. Provide submittal data for all products including subcomponents specified in PART 2 of this Specification Section.
      b. See Specification Section 26 05 00 for additional requirements.
   2. Fabrication and/or layout Drawings.
      a. Outline Drawing including dimensions, weight and identification of all components and features.

b. Nameplate Drawing.
c. Weights, center of gravity, lifting points.
d. Enclosure paint color chart for Contracting Officer selection of final exterior color.

3. Certifications:
   a. Letter stating compliance with current Department of Energy efficiency requirements.

C. Contract Closeout Information:
1. Operation and Maintenance Data:
   a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
   b. Content of Operation and Maintenance Manual:
      1) Instruction and maintenance manual.
      2) Product technical data provided in the submittal.
      3) Outline drawing updated for as-built conditions.
      4) Nameplate Drawing updated for as-built conditions.
      5) Actual impedance data.
      6) Factory test report.
      7) Acceptance testing report.

D. Informational Submittals:
1. Factory test report.

1.5 DELIVERY, STORAGE, AND HANDLING

A. See Specification Section 26 05 00.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Contractor to confirm all equipment is rated for installed elevation and provide final equipment ratings for review. Proposed equipment shall be compatible with existing equipment interface.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Eaton.
2. Or Equal.

2.2 THREE-PHASE TRANSFORMER

A. General:
1. Transformer to be designed and constructed in accordance with:
   a. IEEE C57.12.00, IEEE C57.12.28, IEEE C57.12.34, IEEE C57.12.70, IEEE C57.12.80.
   b. FM Global approved.
2. Transformers for installation at Roaring Springs and other Inner-canyon locations must be transported to the installation site by helicopter.
   a. The maximum weight of individual components and assemblies is 14,000LB to facilitate transport by helicopter long-line.

B. Ratings and Configurations:
1. Type: Outdoor, pad-mounted, liquid-immersed, self-cooled, compartmental type.
2. Operation and application: Step-down operation and Step-up operation.

3. Configuration:
    a. Dead-front, loop-feed primary.
4. Voltage and kVA Ratings: As specified on the Drawings.
5. Number of phases: Three (3).
6. Frequency: 60 Hz.
7. Polarity: ANSI standard.
8. Percent impedance: ANSI standard.
9. Basic impulse level (BIL):
    a. BIL voltage ratings shown are for equipment installations at 9000 Feet. Contractor to confirm final elevation height and provide final BIL equipment ratings for review.
    b. Equipment cable termination bushings to be compatible with existing cable termination BIL ratings.
    c. 25kV Medium Voltage
        1) Primary: 150 kV.
        2) Secondary: 30kV.
    d. 15kV Medium Voltage
        1) Primary: 110 kV.
        2) Secondary: 30Kv.
10. Temperature rise: 65 degrees C.
11. Efficiency: Meet all current Department of Energy rules.
12. Sound Rating: Per NEMA TP 1.
13. Transformers size and BIL ratings meet IEEE recommendations for operational altitudes at and above 8,000 feet.
14. Connections:
    a. Delta or Wye: As indicated on the Drawings.
    b. Primary: 200 ampere bushing wells and insert for loadbreak elbows.
    c. Secondary (208/120V):
        1) Less than 750 kVA: Six-hole, tin-plated, spade-type minimum or as required.
        2) 750 kVA and greater: Twelve-hole, tin-plated, spade-type minimum or as required.
        3) Molded case circuit breaker, ratings as indicated on the Drawings.
        4) Removable ground strap on neutral terminal.
    d. Secondary (480/277V):
        1) Less than 500 kVA: Four-hole, tin plated, spade-type minimum or as required.
        2) 750 - 1500 kVA: Six-hole, tin plated, spade-type minimum or as required.
        3) Molded case circuit breaker, ratings as indicated on the Drawings.
        4) Removable ground strap on neutral terminal.
    e. Step-Down Secondary (12.47/7.2kV):
        1) 200 ampere bushing wells and insert for loadbreak elbows.
15. Tap-changer: De-energized type on H-winding, five total with:
    a. Two 2.5% above and two 2.5% below nominal tap.
16. Sectionalizing oil immersed load break primary switch:
    a. Rating: 200A.
    b. Four position, 'T' blade, with the switch positions labeled:
        1) "SOURCE A" (Loop is open and the coil is energized by Source A).
        2) "SOURCE B" (Loop is open and the coil is energized by Source B).
        3) "SOURCE A & B" (Loop is closed and the coil is energized by both sources).
        4) "LOOP CLOSED & XFMR OPEN" (Loop is closed and the coil is de-energized).

C. Components:
1. Core and Coil:
    a. Coil material: Aluminum or copper.
    b. Windings designed to reduce losses.
    c. Insulating materials rated for 120 degrees C class.
    d. For grounded wye to grounded wye application, the core assembly shall be a 5-legged distributed-gap, wound core.

1) For ratings above 2500 kVA, a stacked core design may be utilized.

e. Core material: High-grade, grain-oriented, non-aging silicon core steel with high magnetic permeability, low hysteresis and eddy current losses.
   1) Magnetic flux densities are to be kept well below saturation to allow for a minimum 10% overvoltage excitation.
   2) Properly annealed to reduce stresses induced during the manufacturing processes and reduce core losses.

f. Core frame:
   1) Designed to provide maximum support for the core and coil assembly.
   2) Welded or bolted to ensure maximum short-circuit strength.

g. Designed and manufactured to meet the short circuit requirements of ANSI C57.12.90.

h. Vacuum processed and energized or baked in an oven prior to tanking to set the epoxy coating on the insulating paper and remove moisture from the insulation prior to vacuum filing.

2. Tank and Terminal Cabinet Enclosure:
   a. Carbon steel reinforced with external, internal or sidewall braces, with all seams and joints continuously welded.
   b. Design the tank and attached components to withstand pressures greater than the required operating design value without permanent deformation.
   c. Sealed-tank construction:
      1) 1000kVA and below: Bolted main tank cover.
      2) 1500kVA and above: Welded main cover and bolted tamper-resistant handhole.
   d. Lifting lugs welded to tank.
      1) Lugs shall have manufacturer-drilled holes or other means for secure connection of bolted shackles for use in transport by helicopter.
   e. Steel divider between high-voltage (left side) and low-voltage (right side) compartments.
   f. No exposed screws, bolts, or other fastening devices that are externally removable.
   g. Tamper resistant per IEEE C57.12.28.
   h. Cabinet depth: IEEE standard or 20 inches minimum.
   i. 1 inches upper fill plug.
   j. 1 inches drain valve with sampling device mounted in an enclosure outside of the main terminal cabinet.
   k. Automatic pressure relief device.
   l. Ground pad(s):
      1) 500kVA and below: Two steel pads with one hole.
      2) Above 500kVA: Two stainless steel pads with two holes.
   m. Door:
      1) Each compartment will have removable, three-point latching hinged doors equipped for latching in the open position.
      2) The high-voltage compartment door will have a fastening device that is accessible only through the low-voltage compartment.
      3) The hinge assemblies made of corrosion-resistant material.
         a) Provide stainless-steel hinge pins of 3/8 inches minimum diameter.
      4) Both compartment doors capable of securing with a single padlock having a maximum 1/2 inches diameter shackle.
      5) Door secured with a hex-head bolt.
      6) Provisions for mounting fault indicator.

3. Overcurrent protection:
   a. Less than 750kVA
      1) Internal partial-range current limiting backup fuse.
      2) Provide full range protection two fuse system.
      3) Partial-range current limiting backup fuse.
         a) Interrupting rating: 50,000A.
         b) Under oil, non-field replaceable.

4) Current sensing, expulsion-type, Bay-O-Net fuse.
a) Under oil, field-replaceable

b. 750kVA or larger
1) 38kV Bay-O-Net
a) Under oil, field-replaceable
2) Provide full range protection two fuse system.
3) Partial-range current limiting backup fuse.
a) Interrupting rating: 50,000A.
b) Under oil, non-field replaceable.
4) Current sensing, expulsion-type, 38kV Bay-O-Net fuse.
a) Under oil, field-replaceable

4. Finish:
a. Manufacturer's standard corrosion protection system in accordance with IEEE C57.12.28.
b. Dark "padmount" green, or other color as selected by the Contracting Officer.
5. Dielectric Fluid:
a. Tested for compatibility with the transformer.
b. Less-flammable, natural or synthetic ester, biodegradable per US EPA OPPTS 835.3100 and certified to comply with US EPA Environmental Technology Verification requirements.
c. High fire point silicone approved by FM and UL.
d. Permanently affix nameplate stamped "Non PCB".
6. Accessories:
a. Liquid level gage.
b. Dial-type thermometer gage.
c. Pressure vacuum gage.
d. Stainless steel or laser-scribed anodized aluminum nameplate, with date of manufacturer and mounted on exterior of transformer.
1) Provide an identical duplicate nameplate and permanently affix to the exterior of the transformer terminal compartment door to allow viewing without opening the door.
e. Cable parking stands, one per bushing.

2.3 SINGLE PHASE TRANSFORMERS

A. General:
1. Transformer to be designed and constructed in accordance with IEEE C57.12.00, IEEE C57.12.25, IEEE C57.12.28, IEEE C57.12.70, IEEE C57.12.80.

B. Ratings and configurations:
1. Type: Outdoor, pad-mounted, liquid-immersed, self-cooled, compartmental type.
2. Operation and application: Step-down operation.
3. Configuration:
a. Dead-front, loop-feed primary.
b. ANSI Type II.
4. Voltage and kVA Ratings: As specified on the Drawings.
5. Number of phases: One (1).
6. Frequency: 60 Hz.
7. Polarity: ANSI standard.
8. Percent impedance: ANSI standard.
9. Basic Impulse Level (BIL):
a. BIL voltage ratings shown are for equipment installations at 9000 Feet. Contractor to confirm final elevation height and provide final BIL equipment ratings for review.
b. Equipment cable termination bushings to be compatible with existing cable termination BIL ratings.
1) Primary: 150 kV.
2) Secondary: 30 kV.

10. Temperature rise: 65 degrees C.
11. Efficiency: Meet all current Department of Energy rules.
12. Sound Rating: Per NEMA TP 1.
13. The installed altitude shall be operable above 8,000 feet.
14. Connections:
    a. Primary: 200 ampere bushing wells and insert for loadbreak elbows.
    b. Secondary: Four-hole, tin-plated, spade type.
        1) Molded case circuit breaker, ratings as indicated on the Drawings.
        2) Fully insulated neutral bushing with removable ground strap on neutral terminal.
15. Tap-changer: De-energized type on H-winding, five total with:
    a. Two 2.5% above and two 2.5% below nominal tap.
16. Sectionalizing oil immersed load break primary switch:
    a. Rating: 200A.
    b. Four position, 'T' blade, with the switch positions labeled:
        1) "SOURCE A" (Loop is open and the coil is energized by Source A).
        2) "SOURCE B" (Loop is open and the coil is energized by Source B).
        3) "SOURCE A & B" (Loop is closed and the coil is energized by both sources).
        4) "LOOP CLOSED & XFMR OPEN" (Loop is closed and the coil is de-energized).

C. Components:
1. Core and Coil:
    a. Coil material: Aluminum or Copper.
    b. Windings designed to reduce losses.
    c. Insulating materials rated for 120 degrees C class.
    d. Core material: High-grade, grain-oriented, non-aging silicon core steel with high magnetic permeability, low hysteresis and eddy current losses.
        1) Magnetic flux densities are to be kept well below saturation to allow for a minimum 10% overvoltage excitation.
        2) Properly annealed to reduce stresses induced during the manufacturing processes and reduce core losses.
    e. Core frame:
        1) Designed to provide maximum support for the core and coil assembly.
        2) Welded or bolted to ensure maximum short-circuit strength.
    f. Designed and manufactured to meet the short circuit requirements of ANSI C57.12.90.
    g. Vacuum processed and energized or baked in an oven prior to tanking to set the epoxy coating on the insulating paper and remove moisture from the insulation prior to vacuum filing.
2. Tank and Terminal Cabinet Enclosure:
    a. Carbon steel reinforced with external, internal or sidewall braces, with all seams and joints continuously welded.
    b. Stainless steel reinforced with external, internal or sidewall braces, with all seams and joints continuously welded.
    c. Design the tank and attached components to withstand pressures greater than the required operating design value without permanent deformation.
    d. Sealed-tank construction.
    e. No exposed screws, bolts, or other fastening devices that are externally removable.
    f. No openings through which foreign objects such as sticks, rods, or wires may contact live parts.
    g. Recessed, stainless steel lifting lug receptacles.
    h. Tamper resistant per IEEE C57.12.28.
    i. 16 inches deep cabinet (minimum)
    j. Oil level / fill plug.
    k. Oil drain plug with sampling device.
    l. Automatic pressure relief device.
    m. Minimum of two steel tapped ground pads.
3. Door:

a. Hinged flip-top.
b. Removable stainless steel hinge pins.
c. Pad lock provisions.
d. Door secured with a hex-head bolt.
e. Provisions for mounting fault indicator.

4. Overcurrent protection:
   a. Internal partial-range current limiting backup fuse.
   b. Provide full range protection two fuse system.
   c. Partial-range current limiting backup fuse.
      1) Interrupting rating: 50,000A.
      2) Under oil, non-field replaceable.
   d. Current sensing, expulsion-type, bay-O-Net fuse.
      1) Under oil, field-replaceable
5. Dielectric Fluid:
   a. Tested for compatibility with the transformer.
   b. Less-flammable, natural or synthetic ester, biodegradable per US EPA OPPTS 835.3100 and certified to comply with US EPA Environmental Technology Verification requirements.
   c. High fire point silicone approved by FM and UL.
   d. Permanently affix nameplate stamped "Non PCB".
6. Finish:
   a. Manufacturer's standard corrosion protection system in accordance with IEEE C57.12.28.
   b. Dark "padmount" green, or other color as selected by the Contracting Officer.
7. Accessories:
   a. Cable parking stands.
   b. Stainless steel or laser-scribed anodized aluminum nameplate, with date of manufacturer and mounted on exterior of transformer.

2.4 THREE-PHASE SHUNT REACTOR

A. General:
   1. Shunt Reactor to be designed and constructed in accordance with:
      a. IEEE C57.12.00, IEEE C57.12.26, IEEE C57.12.28, IEEE C57.12.90,
      b. FM Global approved.

B. Ratings and Configurations:
   1. Type: Outdoor, pad-mounted, liquid-immersed, self-cooled, compartmental type.
   2. Configuration:
      a. Dead-front, loop-feed primary.
   3. Voltage Rating: 24.9GY/14.4 kV
   4. kVA Rating: As specified on the Drawings.
   5. Number of phases: Three (3).
   6. Frequency: 60 Hz.
   7. Polarity: ANSI standard.
   8. Percent impedance: ANSI standard.
   9. Basic impulse level (BIL):
      a. BIL voltage ratings shown are for equipment installations at 9000 Feet. Contractor to confirm final elevation height and provide final BIL equipment ratings for review.
      b. Equipment cable termination bushings to be compatible with existing cable termination BIL ratings.
         1) Primary: 150 kV.
   10. Temperature rise: 65 degrees C.
      a. The average temperature rise of the windings, measured by the resistance method, shall be 65° C operated at rated kVAR. The shunt reactor shall be capable of being operated in a 30° C average, 40° C maximum ambient, as defined by IEEE C57.12.00.
   11. The installed altitude shall be operable above 8,000 feet.

12. Efficiency: Meet all current Department of Energy rules.
13. Sound Rating: Per NEMA TP 1.
14. Connections:
    a. As indicated on the Drawings.
    b. Primary: 200 ampere bushing wells and insert for loadbreak elbows.
15. Tap-changer: De-energized type on H-winding, five total with:
    a. Two 2.5% above and two 2.5% below nominal tap.
16. Sectionalizing oil immersed load break primary switch:
    a. Comply with IEEE C37.71 or C37.74.
    b. Rating: 300A.
    c. Short-time rating: 12 kA minimum.
    d. Basic Impulse Level (BIL):
        1) 24.9 kV system: 125 kV minimum.
    e. Switch type: Break before make.
    f. Two position, with the switch positions labeled:
        1) "XFMR OPEN" when the coil is de-energized.
        2) "XFMR CLOSED" when the coil is energized.

C. Components:
1. Core and Coil:
    a. Coil material: Aluminum or copper.
    b. Windings designed to reduce losses.
    c. Insulating materials rated for 120 degrees C class.
    d. For grounded wye to grounded wye application, the core assembly shall be a 5-legged distributed-gap, wound core.
    e. Core material: High-grade, grain-oriented, non-aging silicon core steel with high magnetic permeability, low hysteresis and eddy current losses.
        1) Magnetic flux densities are to be kept well below saturation to allow for a minimum 10% overvoltage excitation.
        2) Properly annealed to reduce stresses induced during the manufacturing processes and reduce core losses.
    f. Core frame:
        1) Designed to provide maximum support for the core and coil assembly.
        2) Welded or bolted to ensure maximum short-circuit strength.
    g. Designed and manufactured to meet the short circuit requirements of ANSI C57.12.90.
    h. Vacuum processed and energized or baked in an oven prior to tanking to set the epoxy coating on the insulating paper and remove moisture from the insulation prior to vacuum filing.
2. Tank and Terminal Cabinet Enclosure:
    a. Carbon steel reinforced with external, internal or sidewall braces, with all seams and joints continuously welded.
    b. Stainless steel reinforced with external, internal or sidewall braces, with all seams and joints continuously welded.
    c. Design the tank and attached components to withstand pressures greater than the required operating design value without permanent deformation.
    d. Sealed-tank construction:
        1) Bolted main tank cover.
    e. Lifting lugs welded to tank.
    f. Steel divider between high-voltage (left side) and low-voltage (right side) compartments.
    g. No exposed screws, bolts, or other fastening devices that are externally removable.
    h. Tamper resistant per IEEE C57.12.28.
    i. Cabinet depth: IEEE standard or 20 inches minimum.
    j. 1 inches upper fill plug.
    k. 1 inches drain valve with sampling device mounted in an enclosure outside of the main terminal cabinet.

l. Automatic pressure relief device.
m. Ground pad(s):
    1) 500kVA and below: Two steel pads with one hole.
n. Door:
    1) Each compartment will have removable, three-point latching hinged doors equipped for latching in the open position.
    2) The high-voltage compartment door will have a fastening device that is accessible only through the low-voltage compartment.
    3) The hinge assemblies made of corrosion-resistant material.
        a) Provide stainless-steel hinge pins of 3/8 inches minimum diameter.
    4) Both compartment doors capable of securing with a single padlock having a maximum 1/2 inches diameter shackle.
    5) Door secured with a hex-head bolt.
    6) Provisions for mounting fault indicator.

3. Overcurrent protection:
    a. Internal partial-range current limiting backup fuse.
    b. Provide full range protection two fuse system.
    c. Partial-range current limiting backup fuse.
        1) Interrupting rating: 50,000A.
        2) Under oil, non-field replaceable.
    d. Current sensing, expulsion-type, Bay-O-Net fuse.
        1) Under oil, field-replaceable
4. Finish:
    a. Manufacturer's standard corrosion protection system in accordance with IEEE C57.12.28.
    b. Dark "padmount" green, or other color as selected by the Contracting Officer.
5. Dielectric Fluid:
    a. Tested for compatibility with the transformer.
    b. Less-flammable, natural or synthetic ester, biodegradable per US EPA OPPTS 835.3100 and certified to comply with US EPA Environmental Technology Verification requirements.
    c. Permanently affix nameplate stamped "Non PCB".
6. Accessories:
    a. Liquid level gage.
    b. Dial-type thermometer gage.
    c. Pressure vacuum gage.
    d. Stainless steel or laser-scribed anodized aluminum nameplate, with date of manufacturer and mounted on exterior of transformer.
    e. Cable parking stands, one per bushing.

2.5 INSULATING SECONDARY COVERS

A. Insulated secondary covers shall be included on all secondary terminals.

B. Covers shall be installed over terminals and cable, without the need to remove cables.

C. Rated for 600V minimum.

2.6 SURGE ARRESTORS

A. Surge arrestors shall be included on all radial fed or normal open looped transformers, or where indicated on the Drawings.

B. Standards: IEEE 386 and IEEE C62.11.

C. MOV gapless elbow type.

D. Elbow Connector:

1. One-piece design, comprised of an insulation layer and outer shield constructed of EPDM rubber.
2. 200A, dead front, load break type with hot stick pulling eye and grounding tab.
3. #4 Copper ground or larger
4. One arrester per phase

E. Rating:
1. Nominal 24.9kV line to line: 18kV (17.0 MCOV)
2. Nominal 12.47kV line to line: 10kV (8.4 MCOV)

2.7 SOURCE QUALITY CONTROL

A. Factory Tests: At a minimum, provide all routine tests as specified in IEEE C57.12.00 inches accordance with IEEE C57.12.90.

2.8 MAINTENANCE MATERIALS

A. Touch-up paint, two separate one quart containers per color.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install on pad as detailed on the Drawings and in accordance with NFPA 70 and manufacturer's instructions.

B. Transformer locations as shown on the Drawings are intended to be used as a guide.
1. Field conditions may affect actual transformer location.
2. Coordinate final location with contracting officer representative.
3. Coordinate final metering equipment and metering location with Contracting Officer.

C. Install three-phase transformers on concrete pad per detail on the Drawings.

D. Install single phase transformers on concrete pad per detail on the Drawings.

3.2 FIELD QUALITY CONTROL

A. Acceptance Testing: See Specification Section 26 08 13.

END OF SECTION

SECTION 26 13 16
MEDIUM VOLTAGE METAL ENCLOSED LOAD INTERRUPTER SWITCHGEAR

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Padmounted Load interrupter switchgear.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 08 13 - Acceptance Testing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI).
   2. Institute of Electronic and Electronics Engineers, Inc. (IEEE):
      a. C37.20.3, Standard for Metal-Enclosed Interrupter Switchgear (1 kV–38 kV).
      b. C62.11, Standard for Metal-Oxide Surge Arresters for Alternating Current Power Circuits (>1 kV).
      c. C37.74, Standard for Load Interrupter Switchgear and Fused Load interrupter Switchgear for Alternating Current systems up to 38kV.
   3. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   4. Underwriters Laboratories, Inc. (UL):
      a. 44, Thermoset-Insulated Wires and Cables.

1.3 SUBMITTALS

A. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

B. Informational Submittals:
   1. Service equipment marking and documentation.
   2. Record of test results, inspections and procedures witnessed or performed by factory service representative.

C. Shop Drawings:
   1. Product technical data including:
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. Nameplate data for all equipment.
      c. Mounting details and loading information for concrete foundation design.
      d. Installation instructions and procedures.
      e. See Specification Section 26 05 00 for additional requirements.
   2. Fabrication and/or layout drawings:
      a. General arrangement plan view showing door swings, cable entrance locations, shipping splits, etc.
      b. Cross sections, elevations and details.
      c. Complete single-line diagram.
      d. Auxiliary and control system wiring diagrams (e.g., heaters).
   3. Test reports:
      a. Certified reports of all factory production tests.

D. Refer to Section 01 81 33 – Cyber Security Requirements for required cyber security related submittals.

1.4 DELIVERY, STORAGE AND HANDLING

A. See Specification Section 26 05 00.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Contractor to confirm all equipment is rated for installed elevation and provide final equipment ratings for review. Proposed equipment shall be compatible with existing equipment interface.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Eaton (Basis of Design).
   2. Or Equal.

2.2 LOAD INTERRUPTER SWITCHGEAR

A. Ratings:
   1. Voltage: As indicated on the Drawings.
   2. Amperage: 600 Amp Source, 200 Amp Tap
   3. Number of phases: Three (3).
   4. Number of wires: Three (3).
   5. Frequency: 60 Hz.
   6. Grounding system: Solid.
   7. Short circuit:
      a. Fault closing and momentary: 12.5 kA.
      b. Short time (2 second): 12.5 kA.
      c. RMS Symmetrical: 12.5 kA.
   8. Basic impulse level: 150 kV.

B. Construction:
   1. Standards: IEEE C37.74., IEEE C37.20.2
   2. Liquid filled, vacuum, solid state, or gas unit with tamperproof bolted tank cover designed housing load interrupter switches.
   3. Front and rear access or Front access.
   4. Hinged, full-length front doors latching padlockable handles.
   5. Visible Break Switches:
      a. All ways
      b. High impact viewing window to view switch blades through the closed door.
   6. Switches:
      a. Provisions to padlock switch in open or closed position.
      b. OPEN-CLOSED switch position indicators.
      c. Grounding point to allow grounding cables to be installed with the switch in the open position and the switch door closed and locked.
      d. Fuse storage provisions when section contains fuses.

7. Enclosure:
    a. NEMA 3R non-walk-in, for exterior locations.
        1) Sloped roof.
        2) Screened opening to prevent entrance of rodents.
8. Bushings
    a. Source:
        1) Deadfront Type
        2) 600 Amps Bushings.
            a) With 600 Amp to 200 Amp bushing adapters
            b) Cabinet to allow space for 200 Amp elbow termination either via back of 600 Amp termination or 600 Amp to 200 Amp reducer.
    b. Tap:
        1) Load Break Type
        2) 200 Amps Bushings
9. External nameplate with switchgear ratings, manufacturer name and date of manufacture.
10. Interior and exterior steel surfaces cleaned and painted with rust inhibiting primer and manufacturer's standard paint.
    a. Finish color:
        1) Dark “padmount” green.
11. Resettable fault interrupter

C. Buses:
1. Material: Tin-plated aluminum or silver-plated copper.
2. Bus supports, stand-off insulators and sleeves: Porcelain, glass polyester or epoxy.
3. Ground bus: Sized to carry the rated 2-second short circuit current.
4. Provisions for future extension.

D. Load Interrupter Switches - Source:
1. Vacuum interruption
2. Three-pole gang operated.
3. Two-position (open/closed).
4. Manual quick-make, quick-break utilizing a heavy-duty coil spring to provide openings and closing energy.
5. Switch mechanism shaft is driven by a metal-to-metal linkage from the operating handle.
6. The opening and close of the switch shall be independent of the speed at which the handle is moved.
7. Circuit interruption shall take place within an arc compressor or chute.
8. Insulating barriers between each phase and/or enclosure per manufacturer standards.

E. Fault Interrupter Switches - Taps:
1. Fault Interrupter with programmable protective device
2. Three-pole gang operated.
3. Three-position (open/closed/tripped)
4. Manual quick-make, quick-break utilizing a heavy-duty coil spring to provide openings and closing energy.
5. Switch mechanism shaft is driven by a metal-to-metal linkage from the operating handle.
6. The opening and close of the switch shall be independent of the speed at which the handle is moved.
7. One main contact per phase or a separate main and auxiliary break or arcing contacts per phase.
8. Circuit interruption shall take place within an arc compressor or chute.
9. Insulating barriers between each phase and/or enclosure per manufacturer standards.

F. Insulating Medium: Selected switchgear to contain one of the following insulating mediums.
1. Dielectric Fluid:
    a. Tested for compatibility with the switchgear.

b. Less-flammable, non-toxic, non-bioaccumulating, biodegradable per US EPA OPPTS 835.3100 and certified to comply with US EPA Environmental Technology Verification requirements.
c. Permanently affix nameplate stamped “Non-PCB”
d. 1” drain valve with sampler
e. Operatable at cold temperatures (-30°C)

2. Insulated Gas:
a. Tested for compatibility with the switchgear
b. Non-flammable, odorless, colorless, non-SF6
c. Gas Refill Kit
d. Operatable at cold temperatures (-30°C)

3. Solid Dielectric:
a. Modules coated with semi-conductor epoxy layer
b. Operatable at cold temperatures (-30°C)

G. Programmable Protective Device:
1. Mechanically linked trip mechanisms with electronic control set so that overcurrent condition on any phase shall simultaneously trip all three phases.
2. Minimum trip selection for each phase element and for ground element shall be independently settable.
3. Control to have field replaceable TCC modules
4. Ground Trip Control
5. Control settings shall be field selectable.
6. Minimum temperature operating temperature of -30°C
7. Control and its enclosure to be mounted on the inside of the cabinet door.
8. Power and sensing shall be supplied by internally mounted current transformers.

H. Control Wire:
1. Conductor shall be copper with 600 V rated insulation.
2. Conductors shall be stranded.
3. Minimum #12 AWG.
4. Surface mark with manufacturer's name or trademark, conductor size, insulation type and UL label.
5. Conform to UL 44 for type SIS or MTW insulation.

I. Wire Terminators:
1. Ratings: 600 V.
2. Tin plated high strength copper alloy.
3. Solderless, non-insulated, ring type.

2.3 SWITCH BOX PAD

A. Box pad:
1. Fiberglass or pre-cast concrete construction
2. 18 inch minimum depth
3. Sized to fit switchgear enclosure
4. Adequately reinforced to support switchgear enclosure and ancillary equipment.

2.4 SOURCE QUALITY CONTROL

A. Switchgear factory tests in accordance with IEEE and ANSI standards.

2.5 MAINTENANCE MATERIALS

A. One set of three grounding jumpers with storage bag.

B. Touch-up paint.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install in accordance with manufacturer's instructions.

B. Installation to be outdoor:
   1. Install on box pad, align all sides of the switchgear 3 inches from top edge of pad chamfer and secure to pad.
   2. Location as shown on the Drawings are intended to be used as a guide.

C. Miscellaneous:
   1. Paint any scratched surfaces with touch-up paint.
   2. Tag switchgear and all switches and components per Specification Section 26 05 00.

D. Coordination:
   1. Contractor to coordinate with equipment supplier to provide complete installation and startup.
      a. Coordinate final testing and startup with equipment supplier.

3.2 FIELD QUALITY CONTROL

A. A qualified factory-trained manufacturer’s representative shall certify in writing that the equipment has been installed, adjusted and tested in accordance with the manufacturer’s recommendations.

B. Acceptance Tests: See Specification Section 26 08 13.

C. See Section 01 79 00 for Operator training requirements.

END OF SECTION

This page intentionally left blank.

SECTION 26 13 23
SECTIONALIZING EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Pad mounted sectionalizing junction enclosure.

B. Related Specification Sections include but are not necessarily limited to:
   1. Division 01 - General Conditions.
   2. Section 26 05 00 - Electrical - Basic Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Institute of Electronic and Electrical Engineers (IEEE):
      a. 386, Standard for Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV.
      b. C57.12.28, Standard for Pad-Mounted Equipment - Enclosure Integrity.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. Nameplate data.
      c. See Specification Section 26 05 00 for additional requirements.
   2. Fabrication and/or layout drawings:
      a. General arrangement plan view showing door swings, cable entrance locations, etc.
   3. Test reports:
      a. Certified reports of all factory production tests.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

C. Informational Submittals:
   1. Record of test results, inspections and procedures witnessed or performed by factory service representative.

1.4 DELIVERY, STORAGE AND HANDLING

A. See Specification Section 26 05 00.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Contractor to confirm all equipment is rated for installed elevation and provide final equipment ratings for review. Proposed equipment shall be compatible with existing equipment interface.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Cooper Power Systems by Eaton.
   2. Or Equal.

2.2 PADMOUNT TYPE SECTIONALIZING JUNCTION ENCLOSURE

A. Configuration:
   1. Three phase.
   2. Three, Four-way junctions

B. Enclosure:
   1. Material: Heavy-duty stainless steel or fiberglass.
   2. Tamperproof per IEEE and ANSI standards.
   3. Top hinged cover.
   4. Universal cable junction mounting plates.
   5. Parking stands.
   6. Ground nuts.
   7. Interior and exterior steel surfaces cleaned and painted with rust inhibiting primer and manufacturer's standard paint.
      a. Finish color: Munsell green.
   8. Ground sleeve to increase cable termination height.
      a. Material: Fiberglass Reinforced Composites (FRC) incorporating fire retardant polyester resins and a UV stabilizer coating.
      b. Height: 18 inches minimum.

C. Multi-point Variable Cable Junctions:
   1. Standard: IEEE 386.
   2. Voltage Class: 25 kV.
   3. Current rating: 200A dead-front, loadbreak.
   4. Bushing: One piece design, comprised of an insulation shield, insulation layer and an outer shield constructed of EPDM rubber.
   5. Bus: Structural grade plated aluminum.
   6. Bracket: Stainless steel U-strap.

2.3 SOURCE QUALITY CONTROL

A. Switchgear factory tests in accordance with IEEE and ANSI standards.

2.4 MAINTENANCE MATERIALS

A. Touch-up paint.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install in accordance with manufacturer's instructions.

B. Arrange as shown on the Drawings.

C. Padmount Type Sectionalizing Junction Enclosure: Install on ground sleeve.

D. Miscellaneous: Paint any scratched surfaces with touch-up paint.

3.2 FIELD QUALITY CONTROL

A. A qualified factory-trained manufacturer’s representative shall certify in writing that the equipment has been installed, adjusted and tested in accordance with the manufacturer’s recommendations.

3.3 TRAINING

A. A qualified factory-trained manufacturer's representative shall provide the contracting officer with 2 hours of on-site training in the operation and maintenance of the unit substation and its components.

B. See Section 01 79 00 for Operator training requirements.

END OF SECTION

This page intentionally left blank.

SECTION 26 22 13
LOW-VOLTAGE DISTRIBUTION TRANSFORMERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Dry-type distribution transformers, 1000 kVA and less.

B. Related Requirements: Include, but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 05 26 - Grounding.
   3. Section 26 08 13 - Acceptance Testing.

1.2 REFERENCES

A. Reference Standards: Standards referenced in this section include, but are not necessarily limited to, the following:
   1. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. C57.96 - Guide for Loading Dry-Type Distribution and Power Transformers.
   2. National Electrical Manufacturers Association (NEMA):
      a. 250 - Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. ST 20 - Dry Type Transformers for General Applications.
   3. Underwriters Laboratories, Inc. (UL):
      a. 506 - Specialty Transformers.
      b. 1561 - Dry-Type General Purpose and Power Transformers.

1.3 QUALITY ASSURANCE

A. Regulatory Requirements:
   1. Applicable electrical code is indicated 01 42 00 - References.
   2. The Work of this Section shall comply with 10 CFR 431, Energy Efficiency Program For Certain Commercial and Industrial Equipment, administered by the United States Department of Energy.

1.4 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.5 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Action Submittals. Submit the following:
   1. Shop Drawings:
      a. Schedule (in tabular format), indicating low-voltage transformers proposed. Indicate for each proposed location, whether the item will be installed within a panel such as a motor control center or other panel, voltage, manufacturer and model, and other pertinent data.
      b. Fabrication drawings for each size and type of equipment to be furnished. Show and indicate dimensions, arrangement, points of connection, parts and elements, and materials.
      c. Nameplate drawing.
   2. Product data:
      a. Acknowledgement or indication that items submitted comply with requirements of standards referenced in this Section.

b. Low-voltage transformer manufacturer's published product data for each type of transformer and accessory required, including catalogs and similar published literature, marked to indicate items to be furnished for the Work and employing cross outs to indicate items not proposed and that will not be furnished.
c. Sound level certifications.
d. Comply with Section 26 05 00 - Common Work Results for Electrical for additional requirements concerning Submittals.

C. Informational Submittals. Submit the following:
1. Supplier Instructions:
a. Serial numbers of items furnished, equipment nameplate information, and similar information for all items furnished.
b. Instructions for handling, storing, installing, and startup.
2. Source Quality Control Submittals:
a. Upon Contracting Officer's request, submit results of tests, inspections, and other quality control activities required by the Contract Documents and performed at the place of production or fabrication.

D. Closeout Submittals. Submit the following:
1. Operation and Maintenance Data:
a. Submit in accordance with Section 01 78 23 - Operation and Maintenance Data.

PART 2 - PRODUCTS

2.1 GENERAL PURPOSE DRY-TYPE TRANSFORMERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Manufacturers: Subject to compliance with the Contract Documents, furnished required quantity of items from one of the following:
1. Eaton.
2. Or equal.

C. Description:
1. Furnish low-voltage distribution transformers in accordance with this Section, other provisions of the Contract Documents, and Laws and Regulations.
2. Transformers shall be ventilated or non-ventilated, air-cooled, two-winding type.
3. Transformers shall comply with the following: IEEE C57.96, NEMA ST 20, UL 506, UL 1561.

D. Performance Criteria:
1. Ratings: 60 Hertz, voltage, kVA and phase shall be as indicated on the Drawings.
2. During operation, sound levels shall not exceed the following, measured three feet from the associated transformer:
a. Average sound levels for ventilated units:
1) Up to 9 kVA: 40 dB.
2) 10 to 50 kVA: 45 dB.
3) 51 to 150 kVA: 50 dB.
4) 151 to 300 kVA: 55 dB.
5) 301 to 500 kVA: 60 dB.
6) 500 to 700 kVA: 62 dB.
7) 701 to 1000 kVA: 64 dB.

b. Average sound levels for non-ventilated units:
1) Up to 9 kVA: 45 dB.
2) 10 to 50 kVA: 50 dB.
3) 51 to 150 kVA: 55 dB.

3. Efficiency shall be not less than the following:
a. Ventilated:
1) Single-phase, 15 to 333 kVA: In accordance with 10 CFR 431.
2) Three-phase, 15 to 1000 kVA: In accordance with 10 CFR 431.

4. K-Factor rated:
a. Suitable for circuits with harmonic profile equal to or less than K-factor of 4 or 13 without exceeding allowable temperature rise.
b. Neutral conductor and lugs rated for 200 percent of normal phase current.

E. Materials and Fabrication:
1. Cores:
a. High-grade, non-aging silicon steel with high magnetic permeability, and low hysteresis and eddy current losses.
b. During operation, magnetic flux densities shall be well below saturation point.

2. Coils:
a. Continuous wound with **copper windings**.

3. Ventilated Transformer Units:
a. Core and coils assembly shall be impregnated with non-hygroscopic, thermosetting varnish and cured to reduce hot spots and seal out moisture, completely isolated from enclosure using vibration dampening pads.
b. Enclosure: Drip-proof, NEMA 1 in accordance with NEMA 250, finished steel enclosure for indoor locations.
c. Weather-resistant, finished steel enclosure that restricts entry of rain, sleet, or snow using baffles, grills, or barriers on ventilation openings.

4. Non-ventilated Transformer Units:
a. Core and coil assembly shall be encapsulated in proportioned mixture of resin and aggregate providing moisture-proof, shock resistant seal.
b. Enclosure:
1) Totally enclosed, NEMA 3R in accordance with NEMA 250, weather-resistant finished steel enclosure.

5. Taps: Provide Taps for Transformers as follows:
a. Single-phase, 2 kVA and below: None.
b. Single-phase, 3 to 25 kVA: Two, 5 percent FCBN.
c. Single-phase, 25 kVA and above: Two, 2.5 percent FCAN and four, 2.5 percent FCBN.
d. Three-phase, 3 to 15 kVA: Two, 5 percent FCBN.
e. Three-phase, 15 kVA and above: Two, 2.5 percent FCAN and four, 2.5 percent FCBN.

6. Insulating Material (600 V and below):
a. 3 to 10 kVA units: 185 degrees C insulation system with 115 degrees C rise.
b. 15 kVA and above units: 220 degrees C insulation system with 150 degrees C rise.

7. Nameplates: Provide for each transformer furnished corrosion-resistant nameplate and connection diagram in accordance with NEMA ST 20 including date of manufacture.

8. Provide the following when so indicated on the Drawings:
a. Electrostatic shielded transformer: Provide shield consisting of a single turn of thin sheet of aluminum or copper, extending over full width of windings, installed between primary and secondary winding, and grounded.
b. Isolation transformer: Primary and secondary windings shall be electrically isolated from each other.
c. Shielded isolation transformer: Provide system Incorporating both electrostatic shield and isolation features indicated above.
d. SCR drive shielded isolation transformer:
1) Specially intended and fabricated for use as drive isolation transformer.

2) Core bracing shall withstand current reversals and short circuits associated with SCR drive circuitry.
3) Incorporate both electrostatic shield and isolation features as indicated above.

F. Finishes:
1. Provide for each transformer furnished finish protecting against corrosion.
2. Appropriately clean and prepare surfaces to be finished in accordance with written instructions of finish system manufacturer.
3. Provide protective coating of paint or enameling, inside and out.

2.2 SOURCE QUALITY CONTROL

A. Tests and Inspections:
1. Perform manufacturer's routine tests and inspections, in accordance with NEMA ST 20.
2. Comply with Section 26 08 13 – Acceptance Testing.
3. Remedy defects prior to shipping items from assembly and testing location.
4. Upon Contracting Officer's request, submit to Contracting Officer written results of successful completion of required tests and inspections.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Installing the Work constitutes Contractor's approval of underlying work, related, connected systems, and field conditions prevailing at the time of the Work.

3.2 INSTALLATION

A. Installation Requirements:
1. Install items in accordance with the Contract Documents, manufacturer's written instructions, and Laws and Regulations. Where such requirements conflict, obtain interpretation or clarification from Contracting Officer prior to commencing the associated Work.
2. Indoor Locations:
a. Provide ventilated transformers for 15 kVA units and above.
b. Provide non-ventilated transformers for 9 kVA units and below and at other locations shown or indicated on the Drawings.
c. Mount 9 kVA units and below on wall.
d. Mount 15 kVA units and above on either (1) chamfered, four-inch-high concrete housekeeping pad, or (2) supported from wall or ceiling, seven feet above finished floor, by providing equipment support brackets in accordance with Section 26 05 00 - Common Work Results for Electrical.
3. Provide rubber vibration isolation pads.
4. Ground in accordance with Section 26 05 26.

3.3 FIELD QUALITY CONTROL

A. Field Tests and Inspections:
1. Comply with Section 26 08 13 - Acceptance Testing.
2. Field-verify proper operation and functionality of each transformer installed.

END OF SECTION

SECTION 26 23 00
GENERATOR PARALLELING SWITCHGEAR

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Low voltage metal-enclosed switchgear, SWGR-1.
   2. PLC based automatic throwover (ATO) system for switching and synchronizing multiple Generator breakers.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 08 13 - Acceptance Testing.
   3. Section 26 09 13 - Electrical Power Monitoring.
   4. Section 26 09 16 – Control Equipment Accessories.
   5. Section 26 28 00 - Overcurrent and Short Circuit Protective Devices.
   6. Section 26 43 13 - Surge Protection Devices for Low-Voltage Electrical Power Circuits.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI).
   2. Institute of Electrical and Electronic Engineers, Inc. (IEEE):
      a. C37.20.1, Standard for Metal-Enclosed Low-Voltage (100 VAC and below, 3200 VDC and below) Power Circuit Breaker Switchgear.
   3. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   4. Underwriters Laboratories, Inc. (UL):
      a. 1558, Standard for Metal-Enclosed Low-Voltage Power Circuit Breaker Switchgear.

B. Verify the space required for the switchgear is equal to or less than the space allocated.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SYSTEM DESCRIPTION

A. A new switchgear lineup will be used at the North Rim Power Distribution Center (PDC) to automatically control operation of four diesel engine-generator sets, and transfer switching to connect and synchronize the generator system to the North Rim power distribution system to supply standby power during periods of utility power outage.

B. Incorporate an automatic throwover system within the indicated switchgear lineup at the PDC.

C. Provide all control equipment and integration necessary to control a throwover scheme for controlling multiple Generator breakers.

D. Switchgear and transfer control system shall include:
   a. Monitoring loss of utility power via existing outdoor medium-voltage primary-source transfer switches (PTS-1, PTS-2).
   b. Control starting/stopping of engine-generator sets in response to utility power conditions.
   c. Synchronous control of motor-operated circuit breakers to connect generator(s) to the power distribution network.
   d. Monitoring of status of engine-generator sets.

e. Integration of digital power metering packages in the switchgear.
f. Retransmission of signals and data to the plant control system for monitoring purposes via network and discrete signal connections.

1.5 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
1. Product technical data:
a. Provide submittal data for all products specified in PART 2 of this Specification Section.
b. Nameplate data for all equipment.
c. Mounting details and equipment weights.
d. Installation instructions and procedures.
e. Written instructions on operation of the throwover schemes, including detailed sequences of operation.
f. See Specification Section 26 05 00 for additional requirements
2. Fabrication and/or layout drawings:
a. General arrangement plan view showing dimensions, door swings, cable entrance locations, shipping splits, lifting points, weights, etc.
b. Cross sections, elevations and details.
c. Mimic bus layout.
d. Complete single-line and three-line diagrams.
e. AC and/or DC schematics of breaker control, metering, etc.
f. Point-to-point/terminal block wiring diagrams.
g. HMI graphics for all HMI screens.

C. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
2. Fabrication and/or layout drawings updated with as-built conditions.
3. Final “as-commissioned” sequence of operations and operational narrative.

D. Informational Submittals:
1. Equipment marking and documentation.
2. Installation certification report signed by the manufacturer's representative.
3. Ground fault protection system test reports signed by the projects supervising electrical foreman.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Eaton.
a. Basis of design is the Eaton “Magnum PXR” line of switchgear.
2. Or Equal.

2.2 SWITCHGEAR

A. Ratings:
   1. Voltage, number of phases, number of wires, and main bus current rating as indicated on the Drawings.
   2. Assembly short circuit current and circuit breaker fault interrupting rating as indicated on the Drawings.
   3. Bus system with a minimum ANSI 4-cycle short circuit withstand rating of 100 KAIC.
   4. Service Entrance Equipment rated when indicated on the Drawings.
   5. 100% rated when 100% rated breakers are installed.

B. Construction:
   1. Standards: IEEE C37.20.1, UL 1558.
   2. Completely enclosed, dead-front, self-supporting metal structure.
   3. Vertical panel sections bolted together.
   4. Each vertical panel section is a self-contained housing with individual breaker or instrument compartments, a centralized bus compartment and a rear cabling compartment.
   5. Individual circuit breaker compartments are segregated from adjacent compartments and sections, including the bus compartment, by means of steel barriers.
   6. Bus protected by appropriate metal or nonmetallic barriers, shields and shutters.
   7. NEMA 1 rated enclosure.
   8. Interior and exterior steel surfaces cleaned and painted with rust inhibiting primer and manufacturer's standard paint.

C. Buses:
   1. Material: Tin-plated copper.
   2. Main horizontal bus:
      a. Fully rated and continuous over length of switchboard with all three phases arranged in the same vertical plane.
      b. Sufficient size to limit temperature rise to 65 degrees C over average air temperature outside the enclosure of 40 degrees C.
      c. Insulated with a minimum of 5 mil thickness of epoxy resin coating.
         1) Removable non-PVC boots used to give access to the cross bus joints.
   3. Neutral bus: None.
   4. Ground bus: 1/4 x 2 inches copper, continuous over length of switchboard, solidly grounded to each vertical section structure and meet the short time withstand rating of the largest breaker.
   5. Bus joints connected using through bolts and conical spring-type washers for maximum conductivity.
   6. Low-voltage busway will be used to carry power from the four (4) generators to the switchgear. Coordinate connections between the busway assemblies and the switchgear.

D. Overcurrent and Short Circuit Protective Devices:
   1. Generator (incoming) overcurrent protective device:
      a. Drawout low voltage electrically-operated power circuit breaker.
   2. Utility feeder (outgoing) overcurrent protective devices:
      a. Drawout low voltage electrically-operated power circuit breaker.
   3. See Specification Section 26 28 00 for overcurrent and short circuit protective device requirements.
   4. Factory installed.
   5. Means to padlock all main and feeder devices in the open position.

E. Switchgear Dimensions:
   1. Switchgear will be installed in an existing electrical room. Space available is fixed and the electrical room cannot be enlarged.
   2. The design dimensions for the switchgear are:
      a. Depth: 60 inches.
      b. Width: 155 inches (seven 22 inch structures)

c. Height: Manufacturer-standard.

F. Surge Protective Device: Integrally mounted, see Specification Section 26 43 13.

G. Metering:
1. Utility:
a. No utility metering is required.
2. Power monitor:
a. One power monitor package for each outgoing power feeder, as shown on the Drawings.
1) Power monitor packages shall be monitored by the ATO system for display on the HMI and retransmitting to the plant control system.
b. Separate compartment with hinged door.
c. See Specification Section 26 09 13 for meter requirements.

H. Accessories:
1. Coordinate installation methods and provide lay down brackets or other hardware to facilitate the installation of the gear without damage.
2. Thermograph viewing windows for infrared surveys without opening doors:
a. Provide windows at manufacturer's recommended locations to view cable connections and bus connections.
3. A mimic bus shall be installed on the front face of the switchgear lineup to illustrate paths of power flow and internal connections of buses, circuit breakers and other power system components.
a. Fabricate the mimic bus from phenolic material and install on the face of the switchgear using screws.
4. 24VDC “Best Source” Control Power System
a. 24VDC UPS internal to the switchgear, fed from a dedicated external 120V circuit.
b. Connections to the battery systems of all four (4) engine-generators.
c. The “Best Source” system shall automatically select among the five (5) 24VDC power sources to supply 24VDC control power to switchgear controls and components.
5. Interposing Relays:
a. Provide 4PDT interposing relays for signals from the existing primary transfer switches (PTS-1, PTS-2), and other signals to produce a complete and fully-functional system.
6. Generator Emergency-Stop Pushbuttons:
a. Emergency-stop pushbuttons shall be provided to comply with 2023 NEC 445.19(B).
b. One button shall be provided for each generator, and shall be located prominently on the face of the switchgear.
c. The buttons shall be red mushroom-head type and shall latch in the “stopped” position until released by operations personnel.

I. PLC Automatic Throwover System:
1. Manufacturer’s standard programmable logic controller (PLC) assembly, containing:
a. Power supply.
b. CPU.
c. Input and output modules.
d. Human-Machine-Interface (HMI).
2. Requirements for pilot devices, relays and terminal equipment: See Specification Section 26 09 16.
3. Control power: Internal 120 VAC UPS dedicated to the PLC control system.
4. Power quality meter(s) that evaluates the line voltage for undervoltage, phase imbalance, phase loss and phase reversal.
5. Generator relay(s) that contains an over/under frequency relay and overvoltage relay and along with the power quality relay indicate to the PLC that the generator source is available.
6. Sync-check relay(s) that close a contact when the two sources are sufficiently similar in phase angle, magnitude and frequency for a closed transition system.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install switchgear in accordance with manufacturer's instructions.

B. Arrange switchgear as shown on the Drawings.

C. Indoor Locations:
   1. NEMA 1 enclosure.
   2. Install on concrete housekeeping pad.
      a. Align front of switchgear with top edge of pad chamfer and securely bolt to floor sills (C channel) set level (within 1/8 inches) and embedded in the concrete.

D. Equipment Marking and Documentation:
   1. Provide labeling per NFPA 70 and other applicable codes.
   2. Service equipment:
      a. Arc-flash hazard warning label. (Ref. NFPA 70 Article 110.16(A) and (B))
      b. Available fault current label and documentation of the calculations made for compliance with marking requirements. (Ref. NFPA 70 Article 110.24)
   3. Other than service equipment:
      a. Arc-flash hazard warning label. (Ref. NFPA 70 Article 110.16(A))
      b. Available fault current label. (Ref. NFPA 70 Article 408.6)
   4. Identify (tag) all equipment and equipment components.
   5. Provide labels and tags in accordance with Section 10 14 00.
   6. Available fault current and other required label data from Coordinated Power System Study as required by the contract documents.

E. Miscellaneous:
   1. Provide circuit protective devices and other associated equipment as indicated on the Drawings.
   2. Neatly lace all control wires and have flexibility at hinge locations.

3.2 SEQUENCE OF OPERATION

A. The following is a generalized description of the control requirements of the Automatic Throwover System.
   1. It is the design intent to use a manufacturers "standard" automatic throwover system to the largest extent practical, with modifications to meet the intent of the following descriptions.
   2. Four diesel generators are available to supply standby power to the North Rim facilities during a utility power outage. Based on historical power demand information, one or two generators are expected to be adequate to supply backup power to the North Rim distribution system. Thus, the ATO control system must be able to select and control the appropriate number of generators to supply system power during a utility power outage.
   3. Transfer switching between utility and standby power sources is accomplished through operation of pad-mounted primary transfer switches (PTS-1 and PTS-2) located in the yard outside the generator building. When utility power is lost, the primary transfer switches will signal the ATO to initiate the standby power sequence and will switch from utility source to generator power produced by the generator system.
   4. Connection of generator(s) to the distribution system will be accomplished by ATO-controlled motor-operated generator breakers in the switchgear lineup. If multiple generators are required, then the ATO must perform synchronous switching to connect the second (or subsequent) generator to the distribution network.
      a. The generators do not have a frequency control system, therefore synchronized switching will be accomplished via passive synchronization of the generators.
   5. Upon a call to the ATO system for standby power via a dry contact signal from PTS-1 and/or PTS-2:
      a. The ATO shall call all available generators to start, and parallels/synchronizes the available generators.

b. Upon sensing that standby power is available, PTS-1 and/or PTS-2 will switch from the "Utility" source to the "Standby" source.
c. The ATO control system shall monitor system power demand, and modulate the quantity of operating generators to follow system power demand.
    1) If aggregate system load drops below a preset minimum threshold for a defined amount of time, then a generator will be disconnected and begin a cooldown sequence.
    2) If aggregate system load rises above a preset maximum threshold for a defined amount of time, then an additional generator will be started and synchronized.
    3) The selection of which generator to start or stop is based on cumulative engine runtimes as monitored by the ATO system, with an objective to maintain balanced runtimes among available generators.
d. The HMI shall include a screen to allow selection of the following by authorized users:
    1) Enable/disable generator(s) for maintenance or other purposes. If a generator is disabled from the ATO, then it is automatically removed from the starting order. The generator is automatically returned to the starting order when the generator is enabled.
    2) Setpoints:
        a) Power demand (kW, or manufacturer-recommended parameter) above operating generator(s) aggregate capacities at which an additional generator is called to operate/synchronize.
        b) Time delay before calling an additional generator to allow for transient system events.
        c) Power demand (kW, or manufacturer-recommended parameter) below operating generator(s) aggregate capacities at which a generator is disconnected and enters cool-down mode.
        d) Time delay before disconnecting a generator to allow for transient system events.
        e) Additional time delays, including:
            (1) Generator warm-up time before synchronizing to the power system.
            (2) Generator cool-down period following disconnection from the power system.
            (3) Other system operational time delays as recommended by the manufacturer.
e. The ATO shall include an exercise function to exercise the generators on a predetermined schedule.
    1) Exercise schedule shall allow generators be scheduled for exercise:
        a) Specific day of the month.
        b) Specific day of the week.
        c) Specific time of day.
        d) Adjustable exercise and cool-down durations.
        e) Independent exercise schedules for each generator, such that the generators do not exercise at the same time.

B. General features of the control system.
1. All programming and configuration of the ATO, PLC control system, HMI, power metering packages, input/output, relays, and other devices related to the switchgear shall be performed by the system manufacturer.
2. Human-Machine Interface (HMI):
    a. The ATO HMI shall include a graphical representation of the system, including generators, distribution system, switchgear, circuit breakers and other devices and equipment being controlled or monitored.
        1) HMI shall monitor and display all information on the switchgear/ATO, signals from PTS-1, PTS-2, and generators.
3. Modes of operation:
    a. Automatic, i.e., no operator input for transfers or retransfers.

b. Semi-automatic, i.e., automatic transfers and operator input to indicate when the automatic controls are to perform a retransfer.
c. Manual, i.e., operator to select to close or open a selected circuit breaker for operational or maintenance purposes.
d. Test, i.e., operator to select to test the diesel generators using Plant load by paralleling with the utility.

4. Control scenarios:
   a. Scenario 1: Connect generator(s) using the generator feeder breakers in the switchgear lineup to energize all North Rim facilities in the event of utility power loss.
5. Protection features common to all modes of operation and control scenarios:
   a. Protection from excess cycling of the transfer schemes.
   b. Interlocks for fault conditions (bell alarm).
      1) Create an alarm if the automatic control system is requesting a tripped breaker to be closed.
   c. Interlocks to prevent the operation of the breaker mounted close button, which would result in uncontrolled and non-synchronous paralleling of two sources.
   d. Monitor each breaker's 52a and 52b contact.
      1) If they disagree, the system shall go into alarm and suspend the automatic operation.
   e. Monitor each breaker's 52TOC.
      1) Indicate that the breaker is racked out and unavailable for automatic control.
      2) Create an alarm if the automatic control system is requesting a racked-out breaker to be closed.
   f. A scheme to protect the system when the diesel generator(s) is/are lost and provisions to bring the system back to normal conditions.
   g. Provide an indicator light to indicate the status of each breakers control power.
      1) When control power is lost, an alarm shall be created.
   h. Provide a common system alarm dry contact to be monitored by the Plant's Control System
6. Breaker positions during normal conditions.
   a. Generators 1, 2, 3, 4: Open.
   b. Concessions Feeder (via transformer T-45): Closed.
   c. NPS Facilities Feeder (via transformer T-36): Closed.

3.3 FIELD QUALITY CONTROL

A. A qualified factory-trained manufacturer's representative shall certify in writing that the equipment has been installed, adjusted and tested in accordance with the manufacturer's recommendations.

B. Test the ground fault protection system as indicated in Specification Section 26 28 00.

3.4 STARTUP AND COMMISSIONING

A. Manufacturer shall actively participate in startup, testing and commissioning of the switchgear, generators, and associated systems.
   1. Field assistance shall be provided as required by the project commissioning team, but is expected to be the following minimum:
      a. Five (5) separate 8-hour workdays on the project site.

3.5 TRAINING

A. A qualified factory-trained manufacturer's representative shall provide the NPS Staff with 4 hours of on-site training in the operation and maintenance of the switchgear and its' components.

B. See Section 01 79 00 for Operator training requirements

END OF SECTION

This page intentionally left blank.

SECTION 26 24 16
PANELBOARDS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Branch circuit panelboards.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 10 14 00 – Identification Devices.
   2. Section 26 05 00 - Common Work Results for Electrical.
   3. Section 26 28 00 - Overcurrent and Short Circuit Protective Devices.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. PB 1, Panelboards.
   2. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   3. Underwriters Laboratories, Inc. (UL):
      a. 50, Enclosures for Electrical Equipment, Non-Environmental Considerations.
      b. 67, Standard for Panelboards.

1.3 DEFINITIONS

A. Branch Circuit Panelboard: Bus rating of 400A and less or where labeled as Branch Circuit Panelboard on the Drawings.

B. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data.
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. See Section 26 05 00 for additional requirements.
   2. Fabrication and/or layout drawings:
      a. Panelboard layout with alphanumeric designation, branch circuit breakers size and type, as indicated in the panelboard schedules.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
   2. Panelboard schedules updated with as-built conditions.

D. Informational Submittals:
   1. Service equipment marking and documentation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Eaton.
   2. Or Equal.

2.2 MANUFACTURED UNITS

A. Standards: NEMA PB 1, NFPA 70, UL 50, UL 67.

B. Ratings:
   1. Current, voltage, number of phases, number of wires as indicated on the Drawings.
   2. Short Circuit Current Rating (SCCR) and/or Ampere Interrupting Current (AIC) ratings equal to or greater than the interrupting rating indicated on the Drawings or in the schedule.
      a. Series rating is not acceptable.
      b. When fault current or minimum interrupting rating is not indicated, use rating of upstream equipment or infinite bus calculation of transformer secondary.
   3. Service Entrance Equipment rated when indicated on the Drawings or when shown to be fed from a utility source.

C. Construction:
   1. Interiors factory assembled and designed such that switching and protective devices can be replaced without disturbing adjacent units and without removing the main bus connectors.
   2. Multi-section panelboards: Feed-through or sub-feed lugs.
   3. Main lugs: Solderless type approved for copper and aluminum wire.

D. Bus Bars:
   1. Main bus bars:
      a. Tin-plated copper sized to limit temperature rise to a maximum of 65 degrees C above an ambient of 40 degrees C.
      b. Drilled and tapped and arranged for sequence phasing of the branch circuit devices.
   2. Ground bus and isolated ground bus, when indicated on the Drawings: Solderless mechanical type connectors.
   3. Neutral bus bars: Insulated 100% rated or 200% rated, when indicated on the Drawings and with solderless mechanical type connectors.

E. Overcurrent and Short Circuit Protective Devices:
   1. Main overcurrent protective device:
      a. Molded case circuit breaker.
   2. Branch overcurrent protective devices:
      a. Molded case circuit breaker.
   3. See Specification Section 26 28 00 for overcurrent and short circuit protective device requirements.
   4. Factory installed.

F. Integral surge protective device:
   1. Provide for panelboards where indicated on the Drawings or where required by National Electrical Code or local code official.
   2. See Specification Section 26 43 13.

G. Enclosure:
   1. Boxes: Code gage galvanized steel, furnish without knockouts.
   2. Trim assembly: Code gage steel finished with rust inhibited primer and manufacturers standard paint inside and out.
   3. Branch circuit panelboard:
      a. Trims supplied with hinged door-in-door construction.
         1) Outer door:
            a) Allows access to the interior of the enclosure.
            b) Hinged to the enclosure.
            c) Opened by removal of screws or by operating a mechanical latch located behind the inner door.
         2) Inner door:
            a) Allows access to breakers (non-live parts).
            b) Hinged to outer door.
            c) Opened by operation of a keyed corrosion resistant chrome-plated combination lock and catch. Locks for all branch circuit panelboards keyed alike.
      b. Trims for surface mounted panelboards, same size as box.
      c. Trims for flush mounted panelboards, overlap the box by 3/4 inches on all sides.
      d. Nominal 20 inches wide and 5-3/4 inches deep with gutter space in accordance with NFPA 70.
      e. Clear plastic cover for directory card mounted on the inside of each door.
      f. Where NEMA 3R or NEMA 12 rating is indicated: Door gasketed.
      g. Where NEMA 4X is indicated: Stainless Steel.

H. Lockout Provisions
   1. All panelboard operating at 480V shall include permanently-installed devices to allow breakers feeding motor-operated equipment, heating/cooling equipment, or other equipment requiring lock-out/tag-out to be locked in the open position.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install as indicated on the Drawings, in accordance with the NFPA 70, and in accordance with manufacturer's instructions.

B. Support panelboard enclosures from wall studs or modular channels support structure, per Section 26 05 00.

C. Provide NEMA rated enclosure as indicated on the Drawings. Where enclosure type is not indicated, provide enclosure rating suitable for the atmosphere where equipment is installed.

D. Equipment Marking and Documentation:
   1. Provide labeling per NFPA 70 and other applicable codes.
   2. Service equipment:
      a. Arc-flash hazard warning label. (Ref. NFPA 70 Article 110.16(A) and (B))
      b. Available fault current label and documentation of the calculations made for compliance with marking requirements. (Ref. NFPA 70 Article 110.24)
   3. Other than service equipment:
      a. Arc-flash hazard warning label. (Ref. NFPA 70 Article 110.16(A))
      b. Available fault current label. (Ref. NFPA 70 Article 408.6)
   4. Identify (tag) all equipment and equipment components.
   5. Provide labels and tags in accordance with Section 10 14 00.
   6. Available fault current and other required label data from Coordinated Power System Study as required by the contract documents.

E. Provide each panelboard with a typed directory:

1. Identify all circuit locations in each panelboard with the load type and location served.
2. Use Contracting Officer-furnished mechanical equipment designation if different than designation indicated on the Drawings.
3. Use final building room names and numbers as identified by the Contracting Officer if different than designation indicated on the Drawings.
4. Identify spare overcurrent devices.

END OF SECTION

SECTION 26 24 19
MOTOR CONTROL EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Motor control centers.
   2. Separately mounted motor starters (including those supplied with equipment).
   3. Manual motor starters.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 08 13 - Acceptance Testing.
   3. Section 26 29 23 - Variable Frequency Drives - Low Voltage.
   4. Section 26 28 00 - Overcurrent and Short Circuit Protective Devices.
   5. Section 26 43 13 - Surge Protection Devices for Low-Voltage Electrical Power Circuits.
   6. Section 26 09 13 - Electrical Metering Devices.
   7. Section 26 09 16 - Control Equipment Accessories.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. International Electrotechnical Commission (IEC).
   2. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volt Maximum).
      b. ICS 2, Controllers, Contactors and Overload Relays Rated 600 V.
      c. ICS 18, Motor Control Centers.
   3. Underwriters Laboratories, Inc. (UL):
      a. 508, Standard for Industrial Control Equipment.
      b. 845, Motor Control Centers.

B. Miscellaneous:
   1. Verify motor horsepower loads, other equipment loads, and controls from approved shop drawings and notify Contracting Officer of any discrepancies.
   2. Verify the required instrumentation and control wiring for a complete system and notify Contracting Officer of any discrepancies.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data:
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. See Section 26 05 00 for additional requirements.
   2. Fabrication and/or layout drawings:
      a. Motor control center:
         1) Elevation drawing with overall dimensions.

2) Starter and component schedule.
3) Identification of units and their location in the MCC.
4) Location of incoming line terminals.
5) Mounting dimensions.
6) Available conduit entrance areas.
7) Nameplate schedule.
8) Assembly ratings (amps, volts, short circuit, etc.).
9) Unit ladder logic wiring for each unit depicting electrical interlocking and wiring between units (NEMA ICS 18 Class II) and identification of terminals where field devices or remote control signals are to be terminated (NEMA ICS 18 Class II-S) including industry standard symbology of the field devices as indicated on the Drawings, product specification and/or loop descriptions. Drawings indicate basic control functionality, provide diagrams for the manufacturer's product(s) meeting the required functionality.

b. Separately mounted combination starters:
1) Unit ladder logic wiring for each unit depicting electrical wiring and identification of terminals where field devices or remote control signals are to be terminated including industry standard symbology of the field devices as indicated on the Drawings, specification and/or loop descriptions. Drawings indicate basic control functionality, provide diagrams for the manufacturer's product(s) meeting the required functionality.
2) Short Circuit Current Rating (SCCR) nameplate marking per NFPA 70, include any required calculations.

C. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
b. Fabrication and/or layout drawings updated with as-built conditions.
2. Electronic files in CAD format of all drawings and diagrams.

D. Informational Submittals:
1. Equipment marking and documentation.
2. Ground fault protection system test report signed by the projects supervising electrical foreman.

E. Refer to Section 01 81 33 – Cyber Security Requirements for required cyber security related submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Eaton.
2. Or Equal.

2.2 MOTOR CONTROL CENTERS

A. Refer to Section 01 81 33 - Cyber Security Requirements for cyber security related product requirements.

B. Ratings:
   1. 600 V class, 3 PH, 60 Hz with operating voltage and number of wires as indicated on the Drawings.
   2. Assembly short circuit current and interrupting device rating as indicated on the Drawings.
      a. Equipment must be fully rated for indicated short circuit current. Series ratings are not acceptable.
   3. Service Entrance Equipment rated when indicated on the Drawings.

C. Construction:
   1. Standards: UL 845.
   2. Totally enclosed, dead front, free standing assemblies, bolted together to form a single assembly.
   3. Fabricate of not less than 14 GA steel with 16 GA steel doors in standardized units.
   4. Nominal size per section: 20 inches wide, 20 or 21 inches deep, and 90 inches high.
   5. Enclosure:
      a. NEMA 1 gasketed.
      b. NEMA 12:
         1) Dust-tight and drip-proof.
         2) Gasketed material round all doors, door cutouts, cover plates, side, top and back sheets.
         3) Gasketed bottom plate.
      c. NEMA 3R non-walk-in:
         1) Not Used.
   6. Horizontal wireways:
      a. At the top, isolated from the main bus.
      b. At the bottom.
      c. Easily accessible.
      d. Full length of the MCC.
   7. Vertical wireway:
      a. Located in each MCC section that accepts plug-in units.
      b. Connect to top and bottom wireways.
      c. Isolated from the unit interiors.
      d. Accessible through a separate hinged door.
      e. Cable tie supports to hold wiring in place.
   8. Unit doors:
      a. Formed round corners and rolled edges.
      b. Minimum of two heavy-duty hinges or continuous piano hinge.
      c. Held closed by means of captive fasteners.
      d. Fabricate to be a part of the structure and not part of the starter.
   9. Unit cubicles:
      a. Draw-out type for motor starters through NEMA Size 5, breakers through 400A frame and fusible switches through 400A.
      b. Guide rails for supporting and aligning starters.
         1) Larger motor starters, breakers and fusible switches to be bused or cabled to the main horizontal bus.
      c. Operating handle:
         1) With the unit stabs engaged and door closed the handle mechanism allows complete ON/OFF control of the unit disconnect and clear indication of the disconnect status.
         2) Circuit breaker and MCP operators include a separate TRIPPED position.
         3) Mechanical interlock to prevent the opening of the door when the disconnect is in the ON position with a defeater mechanism.

4) Mechanical interlock to prevent the placement of the disconnect in the ON position with the door open with a defeater mechanism.
5) Non-defeatable interlock to prevent the installation or removal of a unit unless the disconnect is in the OFF position.
6) Padlockable in the OFF position.

d. Control panel:
1) Provide control devices (selector switch, indicating devices, etc.) as indicated on the Drawings per Specification Section 26 09 16.

e. Control power:
1) Control power transformer:
a) 120 V secondary.
b) Fused on primary and secondary side.
c) Sized for 140% of required load.

f. Minimum of one full size space unit (12 inches) for any combination magnetic motor starter or starter without overload relay.

g. One-half full size space unit (6 inches) for circuit breakers 100 A and less, or one half of a 12 inch space containing two circuit breakers.

h. Effectively baffled to isolate any ionized gases which may occur within unit starter.

10. Externally mounted overload relay pushbutton.
11. Assemblies effectively ventilated to allow relocation of starters and other components:
a. Within the assembly and with the same load.
b. Without having to compensate for changes in location.
12. Finish: Rust inhibited primer and manufacturer's standard paint inside and out.
13. Provide ample unrestricted space for conduit entry from the bottom.
14. Wiring: NEMA ICS 18 Class II, Type B-D.

D. Buses:
1. Material: Tin-plated copper.
2. Main horizontal bus:
a. 600 A unless otherwise indicated on the Drawings.
b. Extend the full-length of the MCC with provisions for splicing additional sections to either end.
3. Vertical buses:
a. 300 A minimum. Larger vertical bus size when required by the loads in the same structure.
b. Securely bolted to the horizontal main bus with joint easily accessible for maintenance.
c. Completely isolated and insulated by means of a barrier.
d. Extended full length of vertical section to distribute incoming power to each circuit breaker and starter in structure.
1) Starters NEMA Size 5 and larger and certain other components may be cable connected to the main bus with the approval of the Contracting Officer.
e. Extend Vertical bus to spaces provided for future equipment.
4. Ground bus:
a. Extend the full-length of the MCC with provisions for splicing additional sections to either end.
b. 300 A tin-plated copper.
c. Solidly grounded to each structure.
d. Locate near bottom of structure.
e. Provide for lug connection of equipment ground wires.

E. Overcurrent and Short Circuit Protective Devices:
1. Main device:
a. Molded case circuit breaker.
2. Feeder devices:
a. Molded case circuit breaker.
3. Motor protection with full voltage starters:

a. Motor circuit protector.
4. See Specification Section 26 28 00 for overcurrent and short circuit protective device requirements.
5. Factory installed.

F. Motor Starters: See requirements within this Specification Section.

G. Surge Protective Device: Integrally mounted, see Specification Section 26 43 13.

H. Power Monitor Metering:
1. Separate compartment.
2. See Specification Section 26 09 13 for meter requirements.

I. Miscellaneous:
1. See Drawings for items provided by other but factory installed (e.g., submersible motor temperature/leak controller, control system gateways or switches).

2.3 SEPARATELY MOUNTED COMBINATION STARTERS

A. Standards:
1. NEMA 250, NEMA ICS 2.
2. UL 508.

B. Enclosure:
1. NEMA 4X rated:
a. Body and cover: Type 304 or 316 stainless steel.
b. No knockouts, external mounting flanges, hinged and gasketed door.
2. NEMA 12 rated:
a. Body and cover: Sheet steel finished with rust inhibiting primer and manufacturer's standard paint inside and out.
b. No knockouts, external mounting flanges, hinged and gasketed door.

C. Operating Handle:
1. With the door closed the handle mechanism allows complete ON/OFF control of the unit disconnect and clear indication of the disconnect status.
2. Circuit breaker and MCP operators includes a separate TRIPPED position.
3. Mechanical interlock to prevent to prevent the opening of the door when the disconnect is in the ON position with a defeater mechanism for use by authorized personnel.
4. Mechanical interlock to prevent the placement of the disconnect in the ON position with the door open with a defeater mechanism for use by authorized personnel.
5. Padlockable in the OFF position.

D. External mounted overload relay pushbutton.

E. Control Devices:
1. Provide control devices as indicated on the Drawings per Specification Section 26 09 16.
2. Devices will be accessible with the door closed.

F. Control Power Transformer:
1. 120V secondary.
2. Fused on primary and secondary side.
3. Sized for 140% of required load.

G. Fault Current Withstand Rating: Equal to the rating of the electrical gear from which it is fed.

H. Motor Starters: See requirements within this Specification Section.

I. Disconnect Switch, Overcurrent and Short Circuit Protective Devices:
1. Motor circuit protector.
2. See Specification Section 26 28 00 for overcurrent and short circuit protective device requirements.
3. Factory installed.

2.4 MOTOR STARTERS

A. Standards:
   1. NEMA ICS 2.
   2. UL 508.

B. Full Voltage Non-Reversing (FVNR) Magnetic Starters:
   1. NEMA full size rated contactor.
      a. NEMA half sizes and IEC contactors are not permitted.
   2. Double-break silver alloy contacts.
   3. Overload relays:
      a. Ambient insensitive, adjustable solid-state type with phase-loss protection phase imbalance protection and manual reset.
   4. Interlock and auxiliary contacts, wired to terminal blocks:
      a. Holding circuit contact, normally open.
      b. Overload alarm contact, normally open.
      c. Normally open auxiliary contact, for remote run status.
      d. Additional field replaceable auxiliary contacts as required per the Sequence of Operation.
      e. Two additional normally open spare field replaceable auxiliary contacts.

C. Variable Frequency Drives: See Specification Section 26 29 23.

2.5 MANUAL MOTOR STARTERS

A. Standards:
   1. NEMA 250, NEMA ICS 2.
   2. UL 508.

B. Quick-make, quick-break toggle mechanism that is lockable in the OFF position.

C. Types:
   1. Horsepower rated, for ON/OFF control.
   2. Horsepower rated, for ON/OFF control and thermal overload protection.
      a. Switch to clearly indicate ON, OFF, and TRIPPED position.

D. Voltage and current ratings and number of poles as required for the connected motor.

E. Enclosures:
   1. NEMA 4X rated:
      a. Type 304 or 316 stainless steel.
      b. No knockouts, external mounting flanges.
   2. NEMA 12 rated:
      a. Body and cover: Sheet steel finished with rust inhibiting primer and manufacturer's standard paint inside and out.
      b. No knockouts, external mounting flanges.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install as indicated on the Drawings and in accordance with manufacturer's recommendations and instructions.

B. Mounting height for surface mounted equipment: See Section 26 05 00.

C. Mount MCC on 4 inch high concrete pad:
   1. Install two, 4 inch wide channel sills flush in pads to support and maintain alignment of the MCC.
   2. Align front of MCC with top edge of pad chamfer.

D. Equipment Marking and Documentation:
   1. Provide labeling per NFPA 70 and other applicable codes.
   2. Service equipment:
      a. Arc-flash hazard warning label. (Ref. NFPA 70 Article 110.16(A) and (B))
      b. Available fault current label and documentation of the calculations made for compliance with marking requirements. (Ref. NFPA 70 Article 110.24)
   3. Other than service equipment:
      a. Arc-flash hazard warning label. (Ref. NFPA 70 Article 110.16(A))
      b. Available fault current label. (Ref. NFPA 70 Article 408.6)
   4. Identify (tag) all equipment and equipment components.
   5. Provide labels and tags in accordance with Section 10 14 00.
   6. Available fault current and other required label data from Coordinated Power System Study as required by the contract documents.

E. Provide separately mounted combination starters with Short Circuit Current Rating (SCCR) labeling as required by NFPA 70 and other applicable codes.
   1. Determine the SCCR rating by one of the following methods:
      a. Method 1: SCCR rating meets or exceeds the available fault current of the source equipment when indicated on the Drawings.
      b. Method 2: SCCR rating meets or exceeds the source equipment's Amp Interrupting Current (AIC) rating as indicated on the Drawings.
      c. Method 3: SCCR rating meets or exceeds the calculated available short circuit current at the control panel.
   2. The source equipment is the switchboard, panelboard, motor control center or similar equipment where the equipment or control panel circuit originates.
   3. For Method 3, provide calculations justifying the SCCR rating. Utilize source equipment available fault current or AIC rating as indicated on the Drawings.

F. Overload Heaters:
   1. Size for actual motor full load current of the connected motor.
   2. For motors with power factor correction capacitors, size to compensate for the capacitors effect on load current.

G. Combination and Manual Starter Enclosures:
   1. Permitted uses of NEMA 4X enclosure:
      a. Surface mounted in areas designated as wet and/or corrosive.
   2. Permitted uses of NEMA 12 enclosure:
      a. Surface mounted in areas designated as dry.

3.2 FIELD QUALITY CONTROL

A. Acceptance Testing: See Specification Section 26 08 13.

B. Ground Fault System Testing: See Specification Section 26 28 00.

3.3 TRAINING

A. Refer to Specification Section 01 79 00 for requirements for training of Contracting Officer's personnel.

B. Provide classroom and field training at each project site:
   1. North Rim Water Treatment Plant.
   2. Roaring Springs Pump Station.

3.4 CLOSEOUT ACTIVITIES

A. Refer to Section 01 81 33 – Cyber-Security Requirements for cyber security related closeout requirements.

END OF SECTION

This page intentionally left blank.

SECTION 26 25 00
LOW VOLTAGE BUSWAY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for:
      a. Low voltage (600 V and less) busway for connection of generators to switchgear SWGR1- in the Power Distribution Center.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 08 13 - Acceptance Testing.
   3. Section 26 23 00 – Generator Paralleling Switchgear
   4. Section 26 28 00 - Overcurrent and Short Circuit Protective Devices.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. National Electrical Manufacturers Association (NEMA).
   2. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   3. Underwriters Laboratories, Inc. (UL):
      a. 857, Busways.

B. Qualifications:
   1. Installer shall have installed five busway systems of similar size.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data:
      a. Electrical ratings.
      b. Dimensions and weights of all components.
      c. See Section 26 05 00 for additional requirements.
   2. Fabrication and/or layout drawings:
      a. Complete coordinated three-line phasing diagrams.
      b. Physical layout and installation drawings including dimensioned plans and elevations.
         1) Include isometric diagram, mounting heights, through-wall transition details, and details on termination of busway at both ends.
      c. Mounting details.
      d. Bus termination details at switchgear and end cable tap boxes.
   3. Certifications:
      a. Installer's qualifications.
   4. Test reports:
      a. Certified reports of all factory production tests.
      b. Certified insulation performance tests.

C. Contract Closeout Information:

1. Operation and Maintenance Data:
    a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

D. Informational Submittals:
    a. Certified installation report from Factory Representative.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
    1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
    1. Eaton.
    2. Or Equal.

2.2 MANUFACTURED UNITS

A. Standards: UL 857.

B. Types: Feeder, as indicated on the Drawings.

C. Ratings:
    1. Voltage: 480V.
    2. Amperage: 1200A, or as indicated on the Drawings.
    3. Number of phases: Three.
    4. Number of wires: 3-wire with:
        a. 50% housing and 50% internal ground bus.
    5. Short circuit rating: 100kA, to match switchgear bus rating.

D. Construction:
    1. Capable of being mounted flatwise, edgewise or vertical without derating.
    2. Feeder busways to be interchangeable at the same rating without the use of adaptors or special splices.
    3. Feeder busways fittings shall be identical and interchangeable.
    4. Self-supporting between hangers.

E. Bus:
    1. Material: Insulated copper.
    2. Insulation:
        a. NEMA Class B epoxy.
        b. UL rated as self-extinguishing and impervious to acids, alkalis, acetones, machine oils and lubricants.
        c. Tested for impact resistance, chemical resistance and expected life of 50 years.
    3. Rectangular with full round edges.
    4. Temperature rise shall not exceed 55 degrees C above ambient temperature when operating at full load.
    5. Joints to be fastened with a single-bolt design.

F. Housing:
    1. Total enclosed.
    2. Material: Aluminum.

3. Weatherproof (IP 65/66) in indoor and outdoor applications.
4. Interior and exterior surfaces cleaned and painted with rust inhibiting primer and manufacturer's standard paint.

G. Accessories:
1. End cable tap boxes:
a. Provide end cable tap boxes for connection to generators. Coordinate size and quantity of lugs with generators and cable connections.
2. Switchgear interface:
a. Provide end cable tap boxes or flanges for top-entry connection into switchgear SWGR-1 in the Power Distribution Center.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Coordinate the routing of the busway with other trades, and with new and existing building elements and openings, mechanical and generator equipment, and electrical equipment and systems.

B. Final field measurements to be made before fabrication.

C. Provide all required fittings, hanger supports, wall flanges, transition termination cubicles and mounting hardware for proper installation.

D. Support busway as required by the NEC and in accordance with manufacturer's instructions.

E. Install expansion joints whenever busway span across a structural expansion joint or as required by the manufacturer.

3.2 FIELD QUALITY CONTROL

A. A factory representative shall:
1. Train the installers on proper installation.
2. Supervise the installation until the representative is assured the installation will be done correctly.
3. Certify the installation when complete.

B. Acceptance Testing: See Specification Section 26 08 13.

END OF SECTION

This page intentionally left blank.

SECTION 26 26 13
PACKAGE POWER SUPPLY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Package power supply consisting of a transformer and panelboard.
      a. Indicated as “Mini-Power Zone” (MPZ) on the Drawings.

B. Related Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 05 26 - Grounding.
   3. Section 26 28 00 - Overcurrent and Short Circuit Protective Devices.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. C57.96, Guide for Loading Dry-Type Distribution and Power Transformers.
   2. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. PB 1, Panelboards.
      c. ST 20, Dry Type Transformers for General Applications.
   3. Underwriters Laboratories, Inc. (UL):
      a. 67, Standard for Panelboards.
      b. 1561, Standard for Safety Dry-Type General Purpose and Power Transformers.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data:
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. See Section 26 05 00 for additional requirements.
   2. Fabrication and/or layout drawings:
      a. Nameplate drawing.
      b. Panelboard layout with alphanumeric designation, branch circuit breakers size and type, as indicated in the panelboard schedules.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

A. Product requirements are to conform with the following salient characteristics.

2.2 MANUFACTURERS

A. Materials and equipment shall be the standard products of a manufacturer regularly engaged in the manufacture of the products and shall be the manufacturer's latest standard design that has been in satisfactory use for at least one year prior to installation.
   1. Materials and equipment shall conform to the respective publications and other requirements specified below and to the applicable requirements of NFPA 70.
   2. Components shall be UL or third-party certified.

2.3 PACKAGED POWER SUPPLY

A. General:
   1. Standards: IEEE C57.96, NEMA PB 1, NEMA ST 20, UL 67 and UL 1561.
   2. Package power supply includes a main primary circuit breaker, an encapsulated dry-type transformer and a secondary panelboard with main circuit breaker.

B. Ratings:
   1. Single or three-phase as indicated on the Drawings.
   2. KVA and voltage ratings as indicated on the Drawings.
   3. Suitable for use as service entrance equipment.

C. Transformer:
   1. Non-ventilated, air cooled, two winding type.
   2. Core and coil assembly encapsulated in a proportioned mixture of resin and aggregate to provide a moistureproof, shock resistant seal.
   3. Cores:
      a. High grade, non-aging silicon steel with high magnetic permeability, and low hysteresis and eddy current losses.
      b. Magnetic flux densities are to be kept well below the saturation point.
   4. Coils: Continuous wound with electrical grade aluminum and grounded to the enclosure.
   5. Insulation system: 185 degrees C with a 115 degrees C rise.
   6. Taps: Two, 5% FCBN.
   7. Sound levels:
      a. Manufacturer shall guarantee not to exceed the following:
         1) 9 kVA and less: 40 dB.
         2) 10 to 30 kVA: 45 dB.

D. Panelboard and Protective Devices:
   1. Bus: **Copper**.
   2. Factory installed wiring between primary breaker and transformer, secondary breaker and transformer and distribution section.
   3. 480 VAC primary circuit breaker: 65,000 amp minimum interrupting rating.
      a. Equipment must be fully-rated for indicated short circuit. Series ratings are not acceptable.
   4. 240 VAC or less secondary circuit breaker: 10,000 amp minimum interrupting rating.
   5. Feeder breakers:
      a. Plug-in type with 10,000 amp minimum interrupting rating.
      b. See Section 26 28 00 for additional requirements.

E. Enclosure:
   1. Main, secondary and feeder circuit breakers enclosed with a padlockable hinged door.
   2. Wiring compartment suitable for conduit entry and large enough to allow convenient wiring.
   3. Totally enclosed, non-ventilated, NEMA 3R, steel finished with a rust inhibitor primer and manufacturer's standard paint for indoor locations.
   4. Totally enclosed, non-ventilated, NEMA 4X, stainless steel for outdoor locations and where indicated on the Drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Ground in accordance with Section 26 05 26 or as indicated on the Drawings.

END OF SECTION

This page intentionally left blank.

SECTION 26 27 26
WIRING DEVICES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for:
      a. Wall switches.
      b. Receptacles.
      c. Device wallplates and coverplates.
      d. Plug-in raceway.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 05 33 - Raceways and Boxes.
   3. Section 26 24 19 - Motor Control Equipment.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. WD 1, General Color Requirements for Wiring Devices.
      c. WD 6, Wiring Devices - Dimensional Requirements.
   2. Underwriters Laboratories, Inc. (UL):
      a. 20, General-Use Snap Switches.
      b. 498, Standard for Attachment Plugs and Receptacles.
      c. 514A, Metallic Outlet Boxes.
      d. 894, Standard for Switches for Use in Hazardous (Classified) Locations.
      e. 943, Ground-Fault Circuit-Interrupters.
      f. 1010, Standard for Receptacle-Plug Combinations for Use in Hazardous (Classified) Locations.
      g. 1310, Standard for Class 2 Power Units.

1.3 DEFINITIONS

1. Contracting Officer: Includes Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data:
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. See Section 26 05 00 for additional requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient

characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Wall switches and receptacles:
      a. Cooper Wiring Devices by Eaton.
      b. Or Equal.

2.2 WALL SWITCHES

A. Basic requirements unless modified in specific requirements paragraph of switches per designated areas or types:
   1. Industrial Specification Grade.
   2. Quiet action, snap switch.
   3. Self-grounding with grounding terminal.
   4. Back and side wired.
   5. Solid silver cadmium oxide contacts.
   6. Rugged thermoplastic and/or nylon housing and one-piece switch arm.
   7. Ratings: 20 A, 120/277 VAC.
   8. Switch handle type: Toggle.
   9. Switch handle color:
      a. Architecturally-finished areas: White.
      b. Non-architecturally-finished areas: Gray.
   10. Types as indicated on the Drawings:
      a. Single-pole.
      b. Double-pole.
      c. 3-way.
      d. 4-way.
      e. Momentary contact.
   11. Standards: UL 20, UL 514A, NEMA WD 1, NEMA WD 6.

B. Specialty Switch Specific Requirements:
   1. Momentary Contact Switch for Lighting Control:
      a. SPDT, 3-position, center “off”.
      b. Rating: 15A, 120/277VAC.

C. Architecturally Finished Area Specific Requirements:
   1. Commercial Specification Grade.
   2. Ratings: 15A, 120/277V.
   3. Switch handle type: Decorator rocker or paddle.
   4. Switch handle color: White.
   5. Wallplate:
      a. 302 or 304 brushed finish stainless steel.
      b. Single or multiple gang as required.

D. Dry Non-architecturally Finished Area Specific Requirements:
   1. Coverplate for use on surface mounted outlet boxes:
      a. 302 or 304 brushed finish stainless steel.
      b. Single or multiple gang as required.
   2. Wallplate for use on recessed outlet boxes:
      a. 302 or 304 brushed finish stainless steel.
      b. Single or multiple gang as required.

E. Wet or Damp Non-Architecturally Finished or Exterior Area Specific Requirements:
   1. Coverplate:

a. Cast iron alloy, gasketed, stainless steel hardware, galvanized and factory painted finish.
b. Operator type:
1) Side mounted rocker type handle to operate snap switch.
2) Front mounted lever type handle to operate snap switch.
3) Push/pull operator to operate snap switch.
4) Spring type door to cover snap switch.
c. Wet location rated.
d. Single or multiple gang as required.

F. Corrosive and Dry Area Specific Requirements:
1. Corrosion resistant nickel plated metal parts.
2. Coverplate for use on metallic outlet boxes:
a. Cast aluminum, stainless steel hardware, natural, lacquer, or factory painted finish.
b. Single or multiple gang as required.
3. Coverplate for use on non-metallic outlet boxes:
a. High impact thermoplastic or nylon, color to match handle, stainless steel screws.
b. Single or multiple gang as required.

G. Corrosive and Wet or Damp Area Specific Requirements:
1. Corrosion resistant nickel plated metal parts.
2. Coverplate for use on metallic outlet boxes:
a. Cast iron alloy, gasketed, stainless steel hardware, galvanized and factory painted finish.
b. Operator type:
1) Front mounted lever to operate snap switch.
2) Push/pull operator to operate snap switch.
3) Spring type door to cover snap switch.
c. Wet location rated.
d. Single or multiple gang as required.
3. Coverplate for use on non-metallic outlet boxes:
a. High impact thermoplastic, gasketed, stainless steel screws.
1) Front mounted lever to operate snap switch.
2) Spring type door to cover snap switch.
b. Wet location rated.
c. Single or multiple gang as required.

H. Highly Corrosive and Dry Area Specific Requirements:
1. Corrosion resistant nickel plated metal parts.
2. Coverplate for use on PVC coated metallic outlet boxes:
a. PVC coated galvanized cast iron alloy, stainless steel hardware.
b. PVC coated cast aluminum, stainless steel hardware.
c. Single or multiple gang as required.
3. Coverplate for use on non-metallic outlet boxes:
a. High impact thermoplastic, stainless steel hardware.
1) Front mounted lever to operate snap switch.
2) Spring type door to cover snap switch.
b. Single or multiple gang as required.

I. Highly Corrosive and Wet or Damp Area Specific Requirements:
1. Corrosion resistant nickel plated metal parts.
2. Coverplate for use on PVC coated metallic outlet boxes:
a. PVC coated galvanized iron alloy, gasketed, stainless steel hardware.
b. PVC coated aluminum, gasketed, stainless steel hardware.
c. Operator Type:
1) Front mounted lever to operate snap switch.
2) Push/pull operator to operate snap switch.

3) Spring type door to cover snap switch.
d. Wet location rated.
e. Single or multiple gang as required.
3. Coverplate for use on non-metallic outlet boxes:
a. High impact thermoplastic, gasketed, stainless steel hardware.
1) Front mounted lever to operate snap switch.
2) Spring type door to cover snap switch.
b. Wet location rated.
c. Single or multiple gang as required.

2.3 RECEPTACLES

A. Basic requirements unless modified in specific requirements paragraph of receptacles and per designated areas:
1. Industrial Specification Grade.
2. Straight blade.
3. Brass triple wipe line contacts.
4. One-piece grounding system with double wipe brass grounding contacts and self-grounding strap with grounding terminal.
5. Back and side wired.
6. Rating: 20 A, 125 VAC.
7. High impact nylon body.
8. Receptacle body color:
a. Normal power:
1) Architecturally-finished areas: White
2) Non-architecturally-finished areas: Gray.
b. Generator or UPS power: Red.
9. Duplex or simplex as indicated on the Drawings.
10. Configuration: NEMA 5-20R.
11. Standards: UL 498, UL 514A, NEMA WD 1, NEMA WD 6.

B. Receptacle Type Specific Requirements:
1. Basic receptacles:
a. Weather-resistant when located in exterior locations or interior damp or wet areas as indicated on the Drawings.
1) Identification: Letters “WR” on face of receptacle.
2. Ground Fault Circuit Interrupter (GFCI):
a. Specification Grade.
b. Class A protection.
c. Feed through type.
d. Test and reset buttons.
e. Self-testing.
f. Visual indicator light.
g. Weather-resistant when located in exterior locations or interior damp or wet areas as indicated on the Drawings.
1) Identification: Letters “WR” on face of receptacle.
h. Additional standards: UL 943.
3. Plug load (PL) control receptacle.
a. Commercial Specification Grade.
b. Dual controlled (PLD) or half controlled (PLH) as indicated on the Drawings.
c. Identification: NEMA approved controlled receptacle marking on face of receptacle.
4. Combination receptacle and USB charging station:
a. Commercial Specification Grade.
b. Compatible with USB 2.0 and 3.0 devices.
c. USB ports ratings: 5 VDC, 2.1A minimum total charging capacity.
d. Configurations as indicated on the drawings:
1) Simplex receptacle with two (2) USB ports.

2) Duplex receptacle with two (2) USB ports.
e. Additional standards: UL 1310.

5. USB charging station:
   a. Compatible with USB 2.0 and 3.0 devices.
   b. USB ports ratings: 5 VDC, 4.2A minimum total charging capacity.
   c. Configuration: Four (4) USB ports.
   d. Additional standards: UL 1310.

C. Architecturally Finished Areas Specific Requirements:
   1. Commercial Specification Grade.
   2. Ratings: 15A, 125V.
   3. Configuration: NEMA 5-15R.
   4. Decorator style.
   5. Body color: White.
   6. Wallplate:
      a. 302 or 304 brushed finish stainless steel.
      b. Single or multiple gang as required.

D. Dry Non-Architecturally Finished Areas Specific Requirements:
   1. Coverplate for use on surface mounted outlet boxes:
      a. 302 or 304 brushed finish stainless steel.
      b. Single or multiple gang as required.
   2. Wallplate for use on recessed outlet boxes:
      a. 302 or 304 brushed finish stainless steel.
      b. Single or multiple gang as required.

E. Damp Non-Architecturally Finished Areas Specific Requirements:
   1. Coverplate:
      a. Cast iron alloy, gasketed, self-closing cover, stainless steel hardware, galvanized and factory painted finish.
      b. Weatherproof when receptacle is covered.
      c. Single or multiple gang as required.

F. Wet Non-architecturally Finished Areas Specific Requirements:
   1. Coverplate:
      a. Extra-duty rated, weatherproof (NEMA 3R) while in use, gasketed, stainless steel hardware, copper-free aluminum, 3.2 inches minimum cover depth for #12 AWG cords.

G. Exterior Locations Specific Requirements:
   1. Coverplate:
      a. Extra-duty rated, weatherproof (NEMA 3R) while in use, gasketed, stainless steel hardware, copper-free aluminum, 3.2 inches minimum cover depth for #12 AWG cord.

H. Corrosive and Dry Area Specific Requirements:
   1. Corrosion resistant nickel plated metal parts.
   2. Receptacle body color: Gray.
   3. Coverplate for use on metallic outlet boxes:
      a. Cast aluminum, stainless steel hardware, natural, lacquer or factory painted finish.
      b. Single or multiple gang as required.
   4. Coverplate for use on non-metallic outlet boxes:
      a. High impact thermoplastic or nylon, gray in color, stainless steel screws.
      b. Single or multiple gang as required.

I. Corrosive and Wet or Damp Area Specific Requirements:
   1. Corrosion resistant nickel plated metal parts.
   2. Receptacle body color: Gray.
   3. Weather-resistant.
      a. Identification: Letters “WR” on face of receptacle.
   4. Ground Fault Circuit Interrupter (GFCI):

a. Standards: UL 943.

5. Coverplate for use on metallic outlet boxes:
   a. Extra-duty rated, weatherproof (NEMA 3R) while in use, gasketed, stainless steel hardware, copper-free aluminum, 3.2 inches minimum cover depth for #12 AWG cord.

J. Highly Corrosive and Dry Area Specific Requirements:
1. Corrosion resistant nickel plated metal parts.
2. Receptacle body color: Gray.
3. Coverplate for use on PVC coated metallic outlet boxes:
   a. PVC coated galvanized cast iron alloy, stainless steel hardware.
   b. PVC coated cast aluminum, stainless steel hardware.
4. Coverplate for use on non-metallic outlet boxes:
   a. High impact thermoplastic, self-closing cover, stainless steel hardware.

K. Highly Corrosive and Wet or Damp Area Specific Requirements:
1. Corrosion resistant nickel plated metal parts.
2. Receptacle body color: Gray.
3. Weather-resistant.
   a. Identification: Letters “WR” on face of receptacle.
4. Ground Fault Circuit Interrupter (GFCI):
   a. Standards: UL 943.
5. Coverplate for use on PVC coated metallic outlet boxes:
   a. Extra-duty rated, weatherproof (NEMA 3R) while in use, gasketed, stainless steel hardware, flame retardant, UV stabilized polycarbonate, 3.2 inches minimum cover depth for #12 AWG cords.

L. Special Purpose Receptacles:
1. NEMA configuration as indicated on the Drawings.
2. Coverplate: See requirements per area designations herein.

2.4 MISCELLANEOUS WIRING DEVICES

A. Manual Motor Starters: Horsepower rated with or without thermal overloads, see Specification Section 26 24 19.

B. Plug-In Raceway:
1. Surface dual-channel, aluminum raceway plug-in strip with single 15 A, 125 V, 3-wire grounding-type receptacles spaced 24 inches on-center and single data port outlet spaced 72 inches on-center.
2. Prewired with two #12 TW and one #12 TW green insulated ground per branch circuit shown on plans and cat6 cable per data port outlet.
3. Minimum 4-1/2 inches wide x 3/4 inches deep.
4. Suitable fittings and snap-in cover.
5. Finish: Brushed Aluminum.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Mount devices where indicated on the Drawings and as scheduled in Section 26 05 00.

C. See Specification Section 26 05 33 for device outlet box requirements.

D. Where more than one receptacle is installed in a room, they shall be symmetrically arranged.

E. Provide blank plates for empty outlets.

END OF SECTION

SECTION 26 28 00
OVERCURRENT AND SHORT CIRCUIT PROTECTIVE DEVICES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Low voltage circuit breakers.
   2. Low voltage fuses.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 08 13 - Acceptance Testing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. 242, Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems (Buff Book).
      b. 399, Recommended Practice for Industrial and Commercial Power Systems Analysis (Brown Book).
      c. C37.13, Standard for Low-Voltage AC Power Circuit Breakers Used in Enclosures.
      d. C37.16, Low-Voltage Power Circuit Breakers and AC Power Circuit Protectors - Preferred Ratings, Related Requirements, and Application Recommendations.
      e. C37.17, Trip Devices for AC and General Purpose DC Low Voltage Power Circuit Breakers.
   2. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   3. Underwriters Laboratories, Inc. (UL):
      a. 248-8, Low-Voltage Fuses - Part 8: Class J Fuses.
      b. 489, Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit-Breaker Enclosures.
      c. 943, Standard for Safety for Ground-Fault Circuit-Interrupters.
      d. 1053, Standard for Ground-Fault Sensing and Relaying Equipment.
      e. 1066, Standard for Low-Voltage AC and DC Power Circuit Breakers Used in Enclosures.
      f. 1699, Standard for Arc-Fault Circuit-Interrupters.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data including:
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. See Section 26 05 00 for additional requirements.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:

a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

D. Informational Submittals:
1. Ground fault protection system test reports signed by the projects supervising electrical foreman.
2. Reports:
a. As-left condition of all circuit breakers that have adjustable settings.

E. Refer to Section 01 81 33 – Cyber Security Requirements for required cyber security related submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Circuit breakers:
a. Eaton.
b. Or Equal.
2. Fuses:
a. Eaton Bussmann, Inc.
b. Or Equal.

2.2 CIRCUIT BREAKERS

A. Molded Case Type:
1. General:
a. Standards: UL 489.
b. Unit construction.
c. Over-center, toggle handle operated.
d. Quick-make, quick-break, independent of toggle handle operation.
e. Manual and automatic operation.
f. All poles open and close simultaneously.
g. Three position handle: On, off and tripped.
h. Molded-in ON and OFF markings on breaker cover.
i. One-, two- or three-pole as indicated on the Drawings.
j. Current and interrupting ratings as indicated on the Drawings.
2. Thermal magnetic type:
a. Inverse time overload and instantaneous short circuit protection by means of a thermal magnetic element.
b. Frame size 150 amp and below:
1) Non-interchangeable, non-adjustable thermal magnetic trip units.
c. Frame sizes 225 to 400 amp (trip settings less than 400A):
1) Interchangeable and adjustable instantaneous thermal magnetic trip units.
d. Ground Fault Circuit Interrupter (GFCI) Listed:
1) Standard: UL 943.
2) One- or two-pole as indicated on the Drawings.
3) Class A ground fault circuit.

4) Trip on 5 mA ground fault (4-6 mA range).
e. Ground Fault Equipment Protective Circuit Interrupter (GFEPCI) Listed:
1) Standard: UL 1053.
2) Trip on 30 mA ground fault (6-50 mA range).
f. Arc-Fault Circuit Interrupter (AFCI):
1) Standard: UL 1699.
g. Switching Duty (SWD) Listed:
1) 15 and 20 amp rated at 347 V and less.
2) Suitable for switching fluorescent lighting loads on a regular basis
h. Switching Duty (HID) Listed:
1) 50 amp maximum rated at 480 V and less.
2) Suitable for switching high intensity discharge lighting loads on a regular basis
3. Solid state trip type:
a. Inverse time overload, instantaneous short circuit and ground fault protection by means of a solid state trip element, associated current monitors and flux shunt trip mechanism.
b. Frame size 400 amp to 1200 amp (trip settings between 400 and 1200A):
1) Standard rating.
2) Interchangeable current sensor or rating plug.
3) Adjustable long time pick-up setting.
a) Adjustable from 50 to 100% of the current sensor or rating plug.
4) Adjustable short time pick-up setting.
5) Adjustable instantaneous pick-up.
6) Fixed ground fault pick-up, when indicated on the Drawings.
c. Adjustable arc energy-reducing maintenance system.
1) Frame size 1200A or when indicated on the Drawings for less than 1200A frame size.
2) Activate and deactivate without opening door and exposing operators to energized parts.
3) System status indicator adjacent to activation switch.
d. Frame size 1600 amp and above:
1) 100% rated.
2) Interchangeable current sensor or rating plug.
3) Adjustable long time pick-up setting.
a) Adjustable from 50 to 100% of the current sensor or rating plug.
4) Adjustable long time delay setting.
5) Adjustable short time pick-up setting.
6) Adjustable instantaneous pick-up setting.
7) Adjustable ground fault pick-up setting, when indicated on the Drawings.
8) Adjustable ground fault delay setting, when indicated on the Drawings.
e. Adjustable arc energy-reducing maintenance system.
1) Activate and deactivate without opening door and exposing operators to energized parts.
2) System status indicator adjacent to activation switch.
4. Motor circuit protector:
a. Adjustable instantaneous short circuit protection by means of a magnetic or solid state trip element.
b. Sized for the connected motor.

B. Low Voltage Power Type:
1. Inverse time overload, instantaneous short circuit and ground fault protection by means of a solid state trip element, associated current monitors and two-step stored energy trip mechanism.
2. Standards: IEEE C37.13, IEEE C37.16, IEEE C37.17, UL 1066.
3. 100% rated.
4. Electrically operated (EO) or manually operated (MO) as indicated on the Drawings.
5. Manually operated breakers:

a. Close/Open pushbuttons.
b. Red and green indicators to indicate breaker position.
c. Trip unit power: Internal CPT of the electrical gear.

6. Electrically operated breakers:
   a. Close/open pushbuttons.
   b. Red and green indicators to indicated breaker position.
   c. Trip unit power: Internal CPT of the electrical gear.
   d. Trip/close signal: 24VDC, from source internal to switchgear.
   e. Charging motor power: Internal CPT of the electrical gear.
   f. Charge springs after a close.
7. Motor driven operator for charging mechanism with open, close and charge push button.
8. Stored energy mechanism position indicator.
9. Contact position indicator: Reversible 52a/52b contacts, quantity as indicated on the Drawings or required for system operation.
10. Truck operated cell switch (52TOC) when indicated on the Drawings or required for system operation.
11. Bell alarm with manual reset when indicated on the Drawings.
12. Draw out construction:
    a. Roll out type operated by removable crank handle and interlocked with the door.
    b. Four positions: Connected, test, disconnected and removed.
    c. Cell switches to short out ground fault relay when main or tie breaker is drawn out.
13. Current and interrupting ratings as indicated on the Drawings and a 30-cycle short-time withstand ratings equal to their symmetrical interrupting ratings, regardless of whether equipped with instantaneous trip protection or not.
14. Current limiters:
    a. Integrally mounted on 2000 amp frames and less and separate draw-out limiter on 3200 amp and larger.
    b. Coordinated with the trip unit to avoid unnecessary blowing of the current limiters.
    c. Anti-single-phasing device to trip breaker in the event of a blown limiter, indicator to indicate which limiter is blown and prevent breaker from being re-closed on a single-phase condition.
    d. Current limiting fuses: 200,000 amp RMS symmetrical interrupting capacity.
15. Frame size 800 amp and above:
    a. Interchangeable current sensor or rating plug:
    b. Adjustable long time pick-up setting.
       1) Adjustable from 50 to 100% of the current sensor or rating plug.
    c. Adjustable long time delay setting.
    d. Adjustable short time pick-up setting.
    e. Adjustable instantaneous pick-up setting.
    f. Adjustable ground fault pick-up setting, when indicated on the Drawings.
    g. Adjustable ground fault delay setting, when indicated on the Drawings.
    h. Adjustable arc energy-reducing maintenance system.
       1) Frame size 1200A or larger or when indicated on the Drawings for 800A frame size.
       2) Activate and deactivate without opening door and exposing operators to energized parts.
       3) System status indicator adjacent to activation switch.
    i. Unit status indicator.
    j. Cause of trip indicator.
    k. Current display.

2.3 FUSES

A. UL Class L fuses:
   1. Standard: UL 248-1 and UL 248-10.
   2. Dual-element time-delay and current limiting type.
   3. Ratings: 600 V, 601-6000 amps and 200,000 RMS AIC symmetrical.

B. UL Class J fuses:
   1. Standard: UL 248-1 and UL 248-8.
   2. Dual-element time-delay and current limiting rejection type.
   3. Ratings: 600 V, 0-600 amps and 200,000 RMS AIC symmetrical.

C. UL Class RK-1 fuses:
   1. Standard: UL 248-1 and UL 248-12.
   2. Single-element fast-acting and current limiting rejection type.
   3. Dual-element time-delay and current limiting rejection type.
   4. Ratings: 250 and 600 V, 1/10-600 amps and 200,000 RMS AIC symmetrical.

D. UL Class RK-5 fuses:
   1. Standard: UL 248-1 and UL 248-12.
   2. Dual-element time-delay and current limiting rejection type.
   3. Ratings: 250 and 600 V, 1/10-600 amps and 200,000 RMS AIC symmetrical.

E. UL Class CC fuses:
   1. Standard: UL 248-1 and UL 248-4.
   2. Single-element fast-acting and current limiting rejection type.
   3. Ratings: 250 and 600 V, 1/10-30 amps and 200,000 RMS AIC symmetrical.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Current and interrupting ratings as indicated on the Drawings.

B. Series rated systems not acceptable.

C. Devices shall be ambient temperature compensated.

D. Circuit Breakers:
   1. Molded case circuit breakers shall incorporate the following, unless indicated otherwise on the Drawings:
      a. Frame sizes 400 amp and less with trip setting less than 400A shall be thermal magnetic type.
      b. Frame sizes 400 amp and larger shall be solid state trip type.
      c. Motor circuit protectors sized for the connected motor.
   2. Low Voltage Power circuit breakers shall incorporate the following, unless indicated otherwise on the Drawings:
      a. Set current sensor or rating plugs long time pick-up setting so that the indicated trip level is near the 75% trip point.

E. Fuses:
   1. UL Class J: Use for feeder devices 600 amps and smaller.

3.2 FIELD QUALITY CONTROL

A. Adjustable Circuit Breakers:
   1. Set all circuit breaker adjustable taps and other adjustable protective devices as recommended in the Coordinated Power System Protection Studies and approved by the Contracting Officer, except adjust motor circuit protectors per the motor nameplate and NFPA 70 requirements.

B. Fuses:
   1. Install fuses (type rating, speed) as recommended in the Coordinated Power System Protection Studies and approved by the Contracting Officer. Provide recommended sizes for all fuses in each study.

C. Programmable Protective Devices:

1. Contractor shall program and configure all programmable protective devices with settings and parameters determined in the Coordinated Power System Protection Studies as approved by the Contracting Officer.

D. Ground Fault Protection System:
   1. Single source system:
      a. Main breaker using the residual sensing method system.
      b. Main and feeder breakers: Utilize four individual current sensors; the phase sensors are integral to the circuit breaker and the neutral sensor is external to the circuit breaker.

E. Testing:
   1. Acceptance testing: See Specification Section 26 08 13.
   2. Adjustable circuit breakers:
      a. Test and verify all circuit breaker trip functions using a test set provided by the manufacturer for that purpose for circuit breakers 1200 A and above.
   3. Ground fault protection system testing:
      a. Performance test installed ground fault protection system as required by the NFPA 70.
      b. Use high current injection method to test system.
      c. Test report shall indicate:
         1) Device settings.
         2) Tripping time in cycles for each device.
         3) Test current.
         4) Date of test.
         5) Name of certified testing firm that performed the test.
   4. Contracting Office shall have the option to witness the on-site commissioning and field testing at their own expense. Notice of testing shall be provided minimum 15 working days before the test is to be conducted.

3.3 CLOSEOUT ACTIVITIES

A. Refer to Section 01 81 33 – Cyber-Security Requirements for cyber security related closeout requirements.

END OF SECTION

SECTION 26 28 02
COORDINATED POWER SYSTEM PROTECTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Coordinated system protection studies for electrical systems in the Project:
      a. Short-circuit studies.
      b. Protective coordination studies.
      c. Arc-flash hazard studies.
      d. Circuit breaker and protective device settings.
      e. Associated reports.
      f. Separate studies and reports for:
         1) North Rim Water Treatment Plant.
         2)
         3) Roaring Springs Pump House.
         4) Power Distribution Center.
         5) Electrical medium voltage power distribution system.

B. Related Sections include but are not necessarily limited to:
   1. Section 26 05 00 – Common Work Results for Electrical.
   2. Section 26 08 13 - Acceptance Testing.
   3. Section 26 28 00 – Overcurrent and Short Circuit Protective Devices.

C. Completely coordinate with work of other trades.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. 242, Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems (Buff Book).
      b. 399, Recommended Practice for Industrial and Commercial Power Systems Analysis (Brown Book).
   2. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC)
      b. 70E, Standard for Electrical Safety in the Workplace

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Reports:
   a. Short circuit study report.
   b. Protective coordination study report.
   c. Arc-flash hazard study.
   d. As-left condition of all circuit breakers that have adjustable settings.
   e. Provide separate studies for each project site:
      1) North Rim Water Treatment Plant.

2) Roaring Springs Pump House.
3) Power Distribution Center (PDC).
4) Electrical medium voltage power distribution system.

PART 2 - PRODUCTS – NOT USED.

PART 3 - EXECUTION

3.1 COORDINATED POWER SYSTEM PROTECTION

A. Coordinated Power System Protection:
1. Prepare studies to demonstrate that the equipment and system constructed meet the specified requirements for equipment ratings, coordination, and protection. Each study shall include a short circuit study report, system protective coordination study report, arc-flash study, and associated report(s) and protective device settings.
2. Prepare a separate study for each project site, including:
   a. North Rim Water Treatment Plant.
   b. Roaring Springs Pump House
   c. Electrical medium voltage power distribution system.
3. The studies shall begin at the locations indicated and extend down the system through all buses.
   a. North Rim Water Treatment Plant: Begin the study at the first overcurrent device upstream from the park transformer that feeds the plant. The study shall include all downstream buses, equipment, and devices.
   b. Roaring Springs Pump House: Begin the study at the first overcurrent device upstream from the park transformer that feeds the plant. The study shall include all downstream buses, equipment, and devices.
   c. Power Distribution Center: Begin at the generator step-up transformers (T-36 and T-45) and include all 480V and lower voltage equipment and generators at the Power Distribution Center.
   d. Electrical power distribution system: Begin the study at the Garkane Electric substation circuit breaker upstream from the connection point at the North Rim Entrance existing 3-phase power line that extends from the North Rim Park entrance to Roaring Springs Pump House. Analysis shall include generator step-up transformers at the Power Distribution Center (T-36 and T-45) as input to the study for the PDC, and as interface between the PDC and the medium voltage distribution system for analysis scenarios under generator power.
4. Perform the studies in accordance with IEEE 242 and IEEE 399.
5. Include the name of the software developer, software package and software version number in the computer generated studies.
6. System short circuit study report:
   a. Begin each study at the location indicated above and extend down the system through all buses. Perform a balanced three-phase fault, bolted line-to-line fault and line-to-ground fault study.
   b. Prepare a one-line diagram to show the electrical system buses, transformers and all sources of fault current including generators and motors.
   c. Utilize manufacturer's data for the actual proposed equipment (e.g., transformer impedance).
   d. Coordinate the available utility fault current and existing protective equipment settings with the power utility company and by inspection of existing equipment. The power utility is Garkane Energy Cooperative (Garkane). Contact information for Garkane and NPS staff includes:
      1) Garkane Energy - Kanab:
         Neal.brown@garkane.com
         800-747-5403

2) NPS North Rim Park Staff:
Josiah Brown
Josiah_Brown@nps.gov

e. Preliminary short circuit information (provided by Garkane on 23 May 2024) includes:
1) North Rim Entrance Station:
a) Line-Line service voltage: 24.9kV
b) Maximum 3-phase fault current: 236.8A
c) Maximum 1-phase fault current: 182.6A
d) Positive sequence X/R ratio: Contractor obtain from Garkane
e) Zero-sequence X/R ratio: Contractor obtain from Garkane
f) First overcurrent device upstream: Contractor obtain from Garkane.
g) Confirm all electric information used in the studies with Garkane at the time the studies are performed.

f. Show input data in tabular form in the report and/or on the one-line diagram.
1) Input data shall include but is not limited to:
a) Utility fault current or MVA and X/R ratio.
b) Bus voltages.
c) Conductor sizes and type of conduit.
d) Generator and motor sizes and contributions.
e) Transformer sizes and impedances.

g. Show available fault current at each bus in tabular form in the report and/or on the one-line diagram.
h. Perform studies for both normal power and emergency/standby power scenarios.

7. System protective coordination study report:
a. Begin each study at the location indicated above and extend down the system through all buses as required to ensure a coordinated power system.
b. Include utility power and generator power scenarios.
c. Demonstrate that the maximum possible degree of selectivity has been obtained between devices specified for the protection of equipment and conductors from damage from overloads and fault conditions.
1) Where necessary, an appropriate compromise shall be made between system protection and service continuity.
2) Consider system protection and service continuity to be of equal importance.
d. Prepare a one-line diagram to show the electrical system buses, transformers and protective devices.
e. Utilize manufacturer's data for the actual proposed protective devices.
f. Summarize the coordination study, conclusions and recommendations.
1) As a minimum, include the following:
a) The manufacturer's information used to prepare the study.
b) Assumptions made during the study.
c) Recommended taps and settings of all adjustable devices in tabulated form.
d) Composite coordination time-current curves on log-log paper showing:
(1) That the settings for each protective device will provide protection and selectivity.
(2) Identify each curve.
(3) Cable and equipment damage points.
(4) Circuit interrupting device operating and interrupting times.
(5) One-line sketch of the part of the system being investigated.
(6) Include as many curves as possible on a graph while maintaining readability.
e) Ground fault curves shall be separated from the standard time-current curves but shall be on log-log paper showing:
(1) That the settings for each protective device will provide protection and selectivity.
(2) Identify each curve.

(3) Cable and equipment damage points.
(4) Circuit interrupting device operating and interrupting times.
(5) One-line sketch of the part of the system being investigated.
(6) Include as many curves as possible on a graph while maintaining readability.

f) Position time-current curves for each device to provide for maximum selectivity to minimize system disturbances during fault clearing.

g) Advise the Contracting Officer of potential coordination problems discovered during the study and include recommendations to resolve the problem.

h) Submit the report for approval 90 days prior to equipment energization.

8. Arc Flash Study and Labeling
   a. Using the results of the short circuit study, prepare a detailed Arc Flash Study for each project location.
   b. The study shall include normal source and generator-powered scenarios.
   c. Analysis shall identify any areas with a hazard level above 2 (>8 Cal/cm^2). The study shall be re-run to ensure the software is looking at clearing times of upstream devices and mis-coordinated devices, use of ARM switches, etc.
   d. If there are remaining areas with a hazard level above 2 (>8 Cal/cm^2) after re-running the results, prepare a list of recommendations to address the problem. Some typical suggestions include:
      1) Arc flash reduction maintenance (ARM) switches on breakers to reduce time delays or settings during maintenance.
      2) Arc flash detection relays.
      3) Multiple labels describing different arc flash hazards specific to the electrical configuration (i.e. on generator power).
   e. Arc Flash shall be performed in accordance with latest AHJ approved versions of the NFPA 70 and 70E requirements.
   f. Labeling Requirements:
      1) Provide updated arc flash labels for all equipment at each project location. All labels shall include required information per NFPA 70 & 70E.
      2) Labels shall be permanent machine printed labels that are rated for the environment the electrical is mounted in. Outdoor labels shall be sunlight and weatherproof.
      3) Arc flash labels shall be affixed to all electrical switchgear, MCCs and other electrical equipment.
9. The final report and associated software files for all the Coordinated Power System shall be available to the Government for their use, without further royalty, usage fees or other compensation. The reports shall be transmitted via electronic media, CDROM or other typically acceptable electronic file transfer methods at the conclusion of the project.
10. All submittals, resubmittals and reports shall be signed and stamped by a Registered Professional Electrical Engineer in the state of Arizona.

B. Adjustable Circuit Breakers:
   1. Provide recommended settings for all adjustable circuit breakers and other adjustable protective devices.

C. Fuses
   1. Provide recommended sizes for all fuses in each study. Include recommendation for fuse type, rating, and speed.

D. Programmable Protective Devices.
   1. Provide recommended settings for all programmable protective devices.

END OF SECTION

SECTION 26 28 16
SAFETY SWITCHES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Safety switches.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 28 00 - Overcurrent and Short Circuit Protective Devices.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. KS 1, Heavy Duty Enclosed and Dead-Front Switches (600 Volts Maximum).
   2. Underwriters Laboratories, Inc. (UL):
      a. 98, Enclosed and Dead-Front Switches.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Product technical data:
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. Provide a Summary Table or use Exhibit A that associates the safety switch features with connected equipment tag number. Exhibit A indicates minimum data required.
      c. See Section 26 05 00 for additional requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following safety switch manufacturers are acceptable:
   1. Eaton.
   2. Or Equal.

2.2 SAFETY SWITCHES

A. General:
   1. Non-fusible or fusible as indicated on the Drawings.
   2. Suitable for service entrance when required.
   3. NEMA Type HD heavy-duty construction.
   4. Switch blades will be fully visible in the OFF position with the enclosure door open.
   5. Quick-make/quick-break operating mechanism.
   6. Deionizating arc chutes.
   7. Manufacture double-break rotary action shaft and switchblade as one common component.
   8. Clear line shields to prevent accidental contact with line terminals.
   9. Operating handle:
      a. Red and easily recognizable.
      b. Padlockable in the OFF position.
      c. Interlocked to prevent door from opening when the switch is in the ON position with a defeater mechanism.

B. Ratings:
   1. Horsepower rated for connected motor.
   2. Voltage and amperage: As indicated on the Drawings.
   3. Short circuit withstand:
      a. Non-fused: 10,000A.
      b. Fused: 200,000A.

C. Accessories, when indicated in PART 3 of this Specification Section or on the Drawings:
   1. Neutral kits.
   2. Ground lug kits.
   3. Auxiliary contact kits:
      a. Opens before main switch.
      b. Rated 10A at 125/250 VAC.
      c. One N.O. and one N.C. contact.

D. Enclosures:
   1. NEMA 4X rated (metallic):
      a. Body and cover: Type 304 or 316 stainless steel.
      b. No knockouts, external mounting flanges, hinged and gasketed door.
   2. NEMA 4X rated (nonmetallic):
      a. Body and cover: Ultraviolet light protected fiberglass-reinforced polyester boxes.
      b. No knockouts, external mounting flanges, hinged, gasketed and lockable door.
   3. NEMA 12 rated:
      a. Body and cover: Sheet steel finished with rust inhibiting primer and manufacturers standard paint inside and out.
      b. No knockouts, external mounting flanges, hinged and gasketed door.

E. Overcurrent and short circuit protective devices:
   1. Fuses.
   2. See Specification Section 26 28 00 for overcurrent and short circuit protective device requirements.

F. Standards: NEMA KS 1, UL 98.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install as indicated and in accordance with manufacturer's instructions and recommendations.

B. Install switches adjacent to the equipment they are intended to serve unless otherwise indicated on the Drawings.

C. Provide auxiliary contact kit on local safety switches for motors being controlled by a variable frequency drive.
   1. The VFD is to be disabled when the switch is in the open position.

D. Permitted uses of NEMA 4X metallic enclosure:
   1. Surface mounted outdoors and in areas designated as wet and/or corrosive.

E. Permitted uses of NEMA 4X nonmetallic enclosure:
   1. Surface mounted in areas designated as corrosive.
      a. Areas containing sodium hypochlorite.

F. Permitted uses of NEMA 12 enclosure:
   1. Surface mounted in areas designated as dry in non-architecturally finished areas.

END OF SECTION

EXHIBIT A

Safety Switch Summary Table					
Equipment Tag	Switch Model Number	Rated Amps	Fused / Non-fused	Enclosure Type	Accessories
Example	Per MFR	60A	NF	NEMA 4X non-metallic	Ground lug, Aux Contact

SECTION 26 29 23
VARIABLE FREQUENCY DRIVES - LOW VOLTAGE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Variable frequency drives (VFDs) for operation of inverter duty motors.
      a. Roaring Springs Pumphouse: Active front-end VFDs.
      b. North Rim Water Treatment Plant: 6-pulse VFDs.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 - Equipment - Basic Requirements.
   2. Section 01 79 00 – Demonstration and Training.
   3. Section 10 14 00 - Identification Devices.
   4. Section 26 05 00 - Common Work Results for Electrical.
   5. 26 24 19 – Motor Control Equipment.
   6. Section 40 61 96 - Process Control Descriptions.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI).
   2. ETL Testing Laboratories (ETL).
   3. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. 399, Recommended Practice for Industrial and Commercial Power Systems Analysis.
      b. 519, Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems.
      c. C62.41, Recommended Practice for Surge Voltages in Low-Voltage AC Power Circuits.
   4. National Electrical Manufacturer's Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. MG 1, Motors and Generators.
   5. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC):
         1) Article 430, Motors Motor Circuits, and Controllers.
   6. Occupational Safety and Health Administration (OSHA).
   7. Underwriters Laboratory, Inc. (UL):
      a. 508, Standard for Industrial Control Equipment.
      b. 508A, Standard for Industrial Control Panels.

B. Qualifications:
   1. Provide drives that are listed and labeled by UL, ETL, or other Nationally Recognized Testing Laboratory (NRTL) as defined by OSHA regulations, or that have been inspected and subsequent field-labeled by such NRTL.
   2. Where listed drives and other components are installed in a common enclosure, the assembly shall be listed and labeled per UL 508 and UL 508A or equivalent NRTL standard.
      a. Entire assembly shall be affixed with a UL 508A label "Listed Enclosed Industrial Control Panel" or equivalent NRTL label prior to shipment to the jobsite.
   3. VFD Supplier shall maintain an authorized service organization within 300 miles of the Project Site.

C. Coordination:
   1. The intent of this Specification Section is to allow the VFD manufacturer to provide the best solution for the harmonic and motor protection outlined herein.

a. This solution shall include, but not be limited to, all aspects of the distribution system including standby generation, motor feeder cable type and available floor space.

2. Motor and VFD coordination: See Specification Section 01 61 03.
3. VFD shall be supplied complete with all required control components.
   a. Provide control as indicated:
      1) On the electrical drawings.
      2) As specified in this Specification Section.
      3) As specified in the process control system loop descriptions.
         a) See Specification Section 40 61 96.
   b. VFD manufacturer shall review the application and provide, at no additional cost to the Contracting Officer, the hardware and software necessary to allow the VFD to control the driven equipment motor over its required operating range.
      1) These may include, but are not limited to, analog and digital interface modules, communication interface modules, switches, lights and other devices.
   c. Coordinate control devices with devices furnished with driven equipment such as vibration switches, thermal sensors, leak detectors, etc.
4. Verify plan dimensions with equipment space requirements as indicated on the Drawings.
   a. Equipment which exceeds the allotted maximum dimensions may not be acceptable.
   b. Equipment which reduces clear working space below the minimums established by the NFPA 70 will not be acceptable.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

B. Variable Torque (VT):
   1. Defines a load characteristic in which the torque delivered from the motor to the load is reduced as speed is reduced below full rated.
   2. This type of load permits the VFD and the motor to operate at reduced output current at reduced speed.

C. Constant Torque (CT):
   1. Defines a load characteristic in which the torque delivered from the motor to the load remains constant as speed is varied.
   2. This type of load requires the VFD to be able to continuously deliver rated output current over the entire speed range.

D. Constant Horsepower:
   1. Defines a load characteristic in which the torque delivered from the motor to the load is reduced as the speed is increased.
   2. This characteristic is required for operation of the VFD and motor above rated frequency to maintain output current within the rated value.

E. Inverter Duty Motor: An AC induction motor complying with all requirements of NEMA MG 1 Part 31 for definite-purpose inverter-fed motors.

F. Standard Motor: An AC induction motor that fails to comply with one or more requirements of NEMA MG 1 Part 31.

G. Low Voltage: 600 VAC or less.

1.4 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Provide a schedule for each VFD including the following information:
      a. Equipment Tag Number.
      b. VFD Complete Catalog Number.

c. VFD Amp Frame Size.
d. Variable or Constant Torque Rating Basis.
e. Rated Input Current.
f. Rated Continuous Output Current.
g. Rated Short Circuit Current.
h. VFD cable type specified (shielded or non-shielded).
i. VFD Maximum Motor Lead Length for the type of cable used.
j. Motor Manufacturer.
k. Motor Frame Size.
l. Motor Full Load Amps.
m. Motor Service Factor.
n. As installed motor Lead Length.
o. VFD options provided to meet harmonic or motor protection specifications.

2. Submit VFD Shop Drawings concurrently with driven equipment and motor Shop Drawings.
3. Product technical data:
   a. Complete electrical ratings and performance specifications confirming compliance with specified ratings and performance.
   b. Maximum rate of heat rejection from VFD and all related components and associated cooling requirements.
   c. Manufacturer's installation instructions.
   d. Manufacturer's programming and operating instructions.
   e. See Section 26 05 00 for additional requirements.
4. Fabrication and/or layout drawings:
   a. Top, front and side exterior views, with details showing maximum overall dimensions of enclosure, mounting provisions and conduit/cable entry provisions including top, bottom, and side-entry locations.
   b. Equipment weight, center of gravity, and lifting requirements.
   c. Identify minimum clearances from other VFDs or electrical equipment required for proper cooling at top, bottom, side and back of enclosure.
   d. Three-line diagrams showing AC schematic of VFD, input, output and bypass devices including device ratings.
   e. Interior layout drawings showing location of all components within enclosure, field wiring terminal boards, and power and grounding connections.
   f. Field wiring diagrams showing locations and sizes of all electrical connections, ground terminations, and requirements for shielded wire usage or any other special installation considerations.
   g. Short Circuit Current Rating (SCCR) nameplate marking per NFPA 70, include any required calculations.
5. Certifications:
   a. Submit with Shop Drawings:
      1) Identification and location of closest authorized service organization.
      2) Harmonic analysis at each PCC per Harmonic Protection Requirements Article.
   b. Submit prior to shipment:
      1) Certified factory test reports confirming compliance with specified requirements.
   c. Submit after installation:
      1) Certified field service reports showing:
         a) Each VFD is operational.
         b) Each VFD and its driven equipment motor are compatible.
         c) Each VFD responds correctly to the input control signals.
         d) Critical frequencies of the drive system and that the VFD has been set to lockout these frequencies.
         e) Measured harmonic levels per Harmonic Protection Requirements Article.
         f) Measured motor terminal peak voltages per Motor Protection Requirements Article.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
   2. Approved copy of VFD schedule per Submittals Article.
   3. Manufacturer's instruction manuals.
   4. Troubleshooting procedures with a cross-reference between symptoms and corrective recommendations.
   5. Connection data to permit removal and installation of recommended smallest field-replaceable parts.
   6. Recommended spare parts list.
   7. Commissioning sheets showing "as-left" values of all user-programmable or adjustable drive parameters.

D. Refer to Section 01 81 33 – Cyber Security Requirements for required cyber security related submittals.

1.5 WARRANTY

A. Full VFD warranty shall be provided by the manufacturer for a period of three (3) years following VFD commissioning.

B. See Specification Section 01 67 00 – Product Requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS – VOLUME 5 (ROARING SPRINGS PUMPHOUSE)

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable to satisfy the limited space constraints of the existing Roaring Springs Pumphouse building:
   1. Schneider Electric (Basis of Design for Roaring Springs Pumphouse).
   2. Or Equal.

2.2 MANUFACTURERS – ALL OTHER VOLUMES

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Eaton (Basis of Design for all other Volumes).
   2. Or Equal.

2.3 GENERAL

A. VFDs shall consist of a rectifier-DC bus-inverter combination producing a sine-coded pulse-width-modulated (PWM) output voltage waveform.

B. VFDs, whether installed in motor control center (MCC) construction or separately mounted, shall constitute a complete combination motor controller per NFPA 70, Article 430 and shall provide the following per the requirements of that article without the addition of any external components or devices.
   1. Motor control.
   2. Motor overload protection.
   3. Motor and motor branch circuit short circuit and ground fault protection.
   4. Motor and controller disconnecting means.

C. It is the intent of this Specification that VFDs shall be an "engineered" or "configured" drive package in which the VFD chassis, all input, output and bypass power devices, VFD accessories, ancillary switches, contactors, relays, and related control devices are selected, furnished, factory-assembled and factory-tested by the VFD manufacturer in a single enclosure requiring only connection of the power supply circuit, motor branch circuit, and external control wiring in the field.

D. VFDs in Volume 5 (Roaring Springs Pumphouse) shall maintain a maximum footprint of 60 inches wide by 26 inches deep to satisfy space constraints of existing building.

2.4 PERFORMANCE AND DESIGN REQUIREMENTS

A. Application:
   1. VFD(s) shall be of sufficient capacity and shall provide a quality of output waveform for stepless motor control from 10 to 100% of base speed of the driven equipment.
   2. VFDs shall be compatible with:
      a. Inverter duty induction motors.
      b. Standard induction motors.
      c. Special purpose submersible induction motor/pump.
   3. VFDs shall be suitable for Constant Torque (CT) or Variable Torque (VT) applications.
      a. VFD manufacturer shall coordinate with the manufacturer of the driven equipment to identify CT and VT applications.
   4. VFDs shall be designed to operate successfully under the following site conditions:
      a. Ambient:
         1) Temperature: 0-40 degrees C.
         2) 95% non-condensing relative humidity.
      b. Elevation:
         1) Roaring Springs Pumphouse – 4900 feet above MSL
         2) Water Treatment Plant – 8930 feet above MSL
      c. Power supply characteristics:
         1) 480Vac, 3 PH, 60 Hz, 3 wire, (±10%).
         2) Effectively grounded.

B. Ratings and Performance Specifications:
   1. Voltage rating:
      a. Nominal: 460 or 480 VAC, 3 PH, 60 Hz.
      b. Range for continuous full load operation: ±10% of nominal.
      c. Voltage imbalance tolerance for full load operation: 3% minimum.
   2. Current ratings:
      a. Continuous (including altitude derating for site elevations):
         1) Equal to or greater than the motor nameplate full load.
      b. Short-term overload:
         1) VT: 110% for 1 minute.
         2) CT: 150% for 1 minute.
         3) Permissible for 1 minute every 10 minutes continuously.
      c. Short circuit:
         1) 65,000 A RMS SYM, minimum.

2) Where a short circuit rating is not indicated or specified for individual VFDs, each VFD shall have a rating not less than indicated on the Drawings for the MCC, switchboard or panelboard the VFD is supplied from.
3) Where specified short circuit rating indicates additional input impedance is required to protect semiconductors, provide input AC line reactors, whether required to meet harmonic performance specifications or not.

3. Efficiency:
   a. 97%, minimum, at full speed and full load.
   b. 93%, minimum at 1/2 speed and full load.
4. Displacement power factor:
   a. 95%, minimum from 50% to 100% speed and load.
5. Efficiency and power factor criteria apply from the input terminals to the output terminals of the VFD alone, excluding losses of input and output power circuit accessories.
6. Frequency drift:
   a. +0.5% of set frequency.
7. Speed regulation (motor dependent): 3%.
8. Speed range: 10:1.
9. Control type:
   a. Volts/Hertz ratio; constant over the entire operating range of the VFD except:
      1) When operating under voltage boost.
      2) At frequencies over 60 Hz.
   b. Open Loop Sensorless vector.

C. Operational Features:
1. Insensitive to input phase sequence.
2. Continued operation with momentary voltage dips of 25% of rated voltage, or single phase condition: 4 seconds, minimum.
3. Controls power loss ride-through: 500 MSEC, minimum.
4. Electronic reversing.
5. DC injection braking.
6. Anti-windmilling: Synchronization of VFD starting frequency with spinning or coasting load, forward or reverse.
7. Critical frequency band lockout:
   a. Minimum of three settings.
   b. Adjustable bandwidth, 1 - 5 Hz.
8. Capable of operating without the motor connected for start-up and troubleshooting.
9. Provide drive protection as required to protect the drive components when the connected motor has become a generator by surges in the water lines forcing the pump to turn in the reverse direction.

D. The VFD shall be provided with the following minimum user-programmable parameters:
1. Carrier frequency.
2. Independent maximum and minimum speeds for forward and reverse operation.
3. Start frequency and hold time.
4. Independent linear acceleration and deceleration time.
5. Preset “jog” speed.
6. Three critical frequency bands.
7. One preset speed selectable by logic input.
8. Volts/Hertz ratio.
9. Voltage boost, magnitude and frequency range.
10. Process controller gain, offset and bias.
11. Current limit.
12. Overcurrent pickup.
13. Overcurrent delay.
14. Ground fault pickup.
15. DC injection level and time.

E. The VFD shall be designed such that the power circuit components are fully protected from line side disturbances and load side faults:
   1. General:
      a. Shutdown conditions associated with supply circuit conditions which can be corrected external to the VFD-motor system shall be provided with automatic reset, with shutdown cause logged in memory:
         1) Input under voltage.
         2) Input over voltage.
         3) Input under frequency.
         4) Input over frequency.
         5) Input Phase loss.
         6) DC Bus under voltage.
      b. Shutdown conditions which indicate overload or fault within the VFD, the output circuit, or the motor shall require local manual reset at the VFD, requiring operator intervention.
         1) Over temperature.
         2) Blown fuse.
         3) Component failure.
         4) Overload.
         5) Short circuit.
         6) Ground fault.
         7) DC Bus over voltage.
         8) External safety input (e.g., motor thermal protection).
         9) Logic fault.
      c. When automatic shutdown occurs, VFD shall restart only when remote run signal is removed and reapplied.
      d. VFD shall hold cause of trip data for a minimum of four shutdowns in memory.
         1) Data to be accessible through the keypad, local communication link and remotely.
   2. Input protection:
      a. Input circuit breaker or current-limiting fuses with externally operable disconnect.
         1) Fault current interrupting rating equal to or greater than the specified withstand rating of the VFD.
         2) Handle padlockable in the OFF position.
      b. Provide full protection for semiconductors integral to the VFD; units requiring current-limiting fuses or circuit breakers in the supply circuit are not acceptable.
      c. Incoming line transient suppression.
         1) 6000V peak per IEEE C62.41.
         2) Phase-to-phase and phase-to-ground protection.
      d. Sustained over voltage trip.
   3. Internal protection:
      a. Surge suppression and power device snubbers.
      b. Power devices rated at 2.5 times line voltage.
      c. Instantaneous over current trip.
      d. DC bus over voltage trip.
      e. Power device over temperature trip.
      f. Control logic circuit malfunction trip.
   4. Output protection:
      a. Inverse-time overload trip:
         1) UL Class 10 characteristic.
      b. Over voltage trip.
      c. Over frequency trip.
      d. Short circuit trip.
         1) Line to line and line to ground.
      e. Ground fault trip.

2.5 OPERATOR AND REMOTE CONTROL INTERFACE

A. Drive controls shall be microprocessor-based with on-board human machine interface and both local and remote digital communications capability.
   1. All monitoring and control functions, other than those shutdowns specified to be manual reset only, shall be available both locally and remotely.

B. Control circuits shall be 120 VAC or 24 VAC or 24 VDC.
   1. 120 VAC supplied by CPT in the VFD.
      a. CPT shall have minimum additional capacity of 150 VA greater than that required by control devices.
      b. CPT shall have two fuses on the primary side and one fuse on the secondary side.
      c. CPT shall have surge protection on the primary side independent of any other surge protection in the VFD.
   2. 24 VAC or 24 VDC supplied by Class 2 power supply in the VFD.
      a. Power supply shall have minimum additional capacity of 33% greater than that required by control devices.
      b. Provide two current-limiting fuses on the AC supply to the power supply.
      c. Power supply shall have surge protection on the primary side independent of any other surge protection in the VFD.

C. Operator Interface:
   1. Door mounted sealed keypad, membrane type with LED or LCD display.
      a. Messages shall be in English and engineering units.
      b. Drive operating parameters shall be programmable.
      c. Menu driven.
      d. Password security.
      e. Display fault and diagnostic data.
      f. Operating parameters, fault and diagnostic data maintained in non-volatile memory with historic log of fault and diagnostic data.
      g. Gold plated plug-in contacts.
   2. Provide indication and control interface, integral in the keypad, as required in the sequence of operation and Drawings.
      a. Minimum indications:
         1) Run.
         2) Stop.
         3) Ready.
         4) Alarm.
         5) Fault.
         6) Local control.
         7) Remote control.
         8) Control source local.
         9) Control source remote.
         10) Speed indication.
      b. Minimum control functions:
         1) Local/Remote switch.
         2) Stop button.
         3) Start button.
         4) Reset button.
         5) Speed control buttons.
   3. Diagnostic indicators located externally on the face of the drive shall show the type of fault responsible for drive warning, shutdown or failure.
      a. On occurrence of more than one condition, each shall be recorded or indicated by the diagnostics.

D. Remote Control Interface:
   1. Local portable computer interface via RS232/RS242 serial communications port:

a. Capability to:
    1) Start-Stop VFD.
    2) Control VFD Speed.
    3) Access fault and diagnostic data.
2. Analog and discrete inputs:
    a. Speed reference (setpoint) signal 4-20 mA DC.
    b. Drive RUN command.
3. Analog and discrete outputs:
    a. 4-20 mA DC output for remote speed indication, as a function of frequency, calibrated 0 to 100%.
    b. Drive FAULT contacts.
    c. Drive RUNNING contacts.
    d. Drive selector switch in AUTO status contacts.
    e. Additional dry contacts to indicate status signals as indicated on the Drawings.
    f. All contacts for signaling external to the VFD shall be through auxiliary relays with contacts rated minimum 10A and shall be wired to field terminal blocks.
4. Contacts:
    a. Contacts shall be rated 2 A inductive at 120 VAC.
    b. All contacts shall be wired to field wiring terminal boards.
5. Drive shutdown on external fault input:
    a. Provide isolated input for dry contact from external motor or system safety devices to cause immediate shutdown of VFD.
    b. Safety shutdown to be operable in all operating modes of drive, including local operation from keypad.
    c. For submersible pump/motors incorporate the temperature/leak detection monitor and shut down.
    d. Local safety switch, to driven equipment, auxiliary contact to lock-out VFD from running when safety switch is open.
6. Network communications capability:
    a. Provide VFD with communication card, protocol and required programming for digital communication of all VFD program and operational parameters to plant control system via:
        1) Modbus TCP/IP.
        2) Ethernet IP (if interfacing with an Allen Bradley PLC).
    b. Refer to Drawings for additional requirements.

2.6 HARMONIC PROTECTION REQUIREMENTS

A. All VFDs shall be capable of satisfactory operation from a source having voltage distortion and notch characteristics identified as acceptable for a “dedicated system” in IEEE 519 Table 10.2.

B. With all VFDs operating under worst-case harmonic current conditions, and the facility supplied from either or both the utility and generator sources, the VFDs shall not produce harmonic effects in excess of the following limits at any point of common coupling (PCC).
    1. Voltage distortion and notch characteristics: IEEE 519 Table 10.2 for General System.
    2. Current distortion: IEEE 519 Table 10.3 based on calculated I_{SC}/I_L at each PCC.

C. PCC shall be considered:
    1. Building service entrance switchgear, switchboard or MCC.

D. The minimum mitigation measures required to meet the specified harmonic criteria are one of the following topologies:
    1. Volume 5 (Roaring Springs Pumphouse):
        a. Active front end in each VFD.
    2. All other volumes:
        a. 6-pulse rectifier topology with input line reactors or DC link reactors, minimum impedance 5% on drive kVA base.

b. 18-pulse rectifier topology for each VFD with input current ratings in excess of 100 HP.

E. VFD manufacturer shall determine, for their proposed equipment, uncorrected harmonic distortion levels and mitigation techniques required to meet the specified limits and shall furnish the VFD types and all accessory items and equipment necessary to do so, whether specified herein or not.

F. VFD manufacturer shall provide a harmonic analysis of the distribution system based on their proposed specific equipment characteristics and mitigation techniques confirming that the specified levels are not exceeded.
1. Analysis shall be based on the methodology of IEEE 519 and IEEE 399.
2. Power system data for analysis shall be taken from the electrical drawings and approved equipment submittals.
a. VFDs provided in a package with equipment specified elsewhere, shall be included in the analysis.

G. Following start-up, with facility at full load operation, provide measurement of harmonic voltage, current and notch characteristics at each PCC according to the requirements of IEEE 519 Section 9.
1. Values in excess of specified limits require correction by contractor and re-measurement.
2. Provide certification of compliant measurements as part of Field Service Engineer's final report.

2.7 MOTOR PROTECTION REQUIREMENTS

A. The VFD shall produce a quality of output waveform adequate to allow the motor to produce rated torque at rated RPM continuously without exceeding the temperature rise given in NEMA MG 1 Table 31-2.

B. Provide motor overload, short circuit and ground fault protection integral to drive electronics.

C. The VFD shall not produce voltage spikes in excess of the following values at the motor terminals when operated with the feeder types shown on the Drawings and the actual installed feeder lengths.
1. If unmitigated voltage peaks exceed the specified limits, provide output line reactors, filters, or other devices as required to meet the specified limits:
a. Inverter duty motors: 1280 V.
b. Standard motors: 800 V.
c. Rise time shall be greater than or equal to 0.1 microsecond.
d. Motor lead length and data shall be determined by the Contractor based on the actual routing of the conductors.

D. Following start-up, provide measurement of peak voltage at the terminals of each motor, unless the lead lengths are 10% shorter than the manufacturers published literature for maximum lead length for the type of cable installed.
1. Values in excess of specified limits require correction by contractor and re-measurement.
2. Provide certification of compliant measurements as part of Field Service Engineer's final report.

2.8 EQUIPMENT CONSTRUCTION

A. Fabrication and Assembly:
1. Each VFD system shall be factory-assembled for installation in an MCC or in an enclosure for stand-alone mounting, and shall utilize interchangeable plug-in printed circuit boards and power conversion components wherever possible.
a. Factory assembly shall be performed by the VFD manufacturer or authorized agent.
b. Systems fabricated or assembled in whole or in part by parties other than the VFD manufacturer or authorized agent will not be acceptable.

c. Indicated VFDs shall be mounted within standard motor control center sections, and included as part of the overall motor control center.
2. Reactors and/or filters, where required, shall be mounted within or in an ancillary enclosure adjacent to the drive enclosure, or with the Contracting Officer's permission may be mounted in a separate enclosure.
3. Cooling fans, as required, shall be provided to run when drive is running.
4. Enclosures for separately mounted VFD's:
   a. NEMA Type 1 for installation in Electrical Rooms.
   b. NEMA Type 12 for installation in other unclassified areas.
   c. Enlosures must fit in the allocated space as indicated on the Drawings.

B. Wiring:
1. The wiring in the VFD shall be neatly installed in wire ways or with wire ties where wire ways are not practical.
   a. Where wire ties are used, the wire bundles are to be held at the back panel with a screw-mounted wire tie mounting base.
   b. Bases with a self-sticking back will not be allowed.
2. All plug-in contacts shall be gold-plated.
3. Provide terminal boards for all field wiring and inter-unit connections, including analog signals.
   a. Provide terminals for shield continuity where required.
4. Terminal blocks shall be complete with marking strip, covers and pressure connectors.
   a. Non-brittle, interlocking, track-mounted type.
   b. Screw terminals will not be allowed.
   c. A terminal for each conductor of external circuits plus one ground for each shielded cable.
   d. For free-standing panels, 8 inches of clearance shall be provided between terminals and the panel base for conduit and wiring space.
   e. Not less than 25% spare terminals shall be provided.
   f. Terminals shall be labeled to agree with identification indicated on the suppliers submittal drawings.
   g. Individually fuse each control loop or system and all fuses or circuit breakers shall be clearly labeled and located for easy maintenance.
5. All grounding wires shall be attached to the enclosure sheet metal with a ring tongue terminal.
   a. The surface of the sheet metal shall be prepared to assure good conductivity and corrosion protection.
6. Wiring shall not be kinked or spliced and shall have markings on both ends or be color coded.
   a. Markings or color code shall match the manufacturer's drawings.
7. With the exception of electronic circuits, all interconnecting wiring and wiring to terminals for external connection shall be stranded **tinned copper**, type MTW or SIS, insulated for not less than 600 V, with a moisture-resistant and flame-retardant covering rated for not less than 90 DegC.

C. Nameplates:
1. All devices mounted on the face of the drive shall be provided with a suitable nameplate as specified in Specification Section 10 14 00.
2. Push buttons, selector switches, and pilot lights shall have the device manufacturer's standard legend plate.
3. Relays, terminals and special devices inside the control enclosure shall have permanent markings to match identification used on manufacturer's wiring diagrams.

D. Painting: Enclosure, after being phosphate washed, shall be thoroughly cleaned and given at least one (1) coat of rust-inhibiting primer on all inner surfaces prior to fabrication.

2.9 COMPONENTS AND ACCESSORIES

A. Input Reactors:
   1. For use on 6-pulse VFDs.
   2. Impedance: 5%.
   3. Continuous current: Not less than drive rating.
   4. Current overload: 150% for 1 minute.
   5. Insulation temperature rating: 180 degrees C.
   6. Copper windings.
   7. Saturation current rating: 3.5 to 5 times rated current.
   8. Hi-potential rating: 2500 VAC line to ground and line to line, for 1 minute.
   9. Noise reduction features:
      a. Epoxy over cast coil.
      b. Extra dips and bakes of varnish over continuous wound coil.

2.10 SOURCE QUALITY CONTROL

A. Factory Tests:
   1. Conduct all standard tests in accordance with NEMA and ANSI standards to ensure conformance to Specification requirements.

2.11 MAINTENANCE MATERIALS

A. Provide manufacturer's recommended renewable spare parts (e.g., power and control fuses).

B. Spare parts utilized during pre-start-up or start-up and demonstration testing shall be immediately restocked, at no cost to the Contracting Officer.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions and as indicated on the Drawings.

B. Provide separately mounted VFD enclosure with Short Circuit Current Rating (SCCR) labeling as required by NFPA 70 and other applicable codes.
   1. Determine the SCCR rating by one of the following methods:
      a. Method 1: SCCR rating meets or exceeds the available fault current of the source equipment when indicated on the Drawings.
      b. Method 2: SCCR rating meets or exceeds the source equipment’s Amp Interrupting Current (AIC) rating as indicated on the Drawings.
      c. Method 3: SCCR rating meets or exceeds the calculated available short circuit current at the control panel.
   2. The source equipment is the switchboard, panelboard, motor control center or similar equipment where the equipment or control panel circuit originates.
   3. For Method 3, provide calculations justifying the SCCR rating. Utilize source equipment available fault current or AIC rating as indicated on the Drawings.

C. Verify the installed motor nameplate electrical requirements do not exceed the VFD capacity.

D. Provide services of manufacturer's representative to perform start-up services.

E. The selection of input and output harmonic and voltage spike protection shall also be made on the available physical space.
   1. The space available on the Drawings shall not be exceeded.

3.2 START UP

A. Pre-start-up Services:
   1. Shall be completed a minimum of 30 days prior to the start-up and demonstration period described in Specification Section 01 75 00.

2. Shall consist of:
    a. Physical and electrical installation check.
    b. Final adjustments and calibration of drive parameters.
    c. VFD operation from simulated input signals.
3. Shall be complete when VFD(s) are fully operational.

B. Field Quality Control:
1. Perform field measurement of harmonics at each PCC per Harmonic Protection Requirements Article.
    a. For each individual VFD.
    b. For the maximum number of VFDs that will be operational at the same time.
    c. When all loads are at 75% load minimum.
    d. Duration: 1 hour minimum.
2. Perform field measurement of the maximum voltage peak at the terminals of each motor fed from a VFD per Motor Protection Requirements Article.
    a. Use a high speed oscilloscope to produce a plot of Voltage (Y axis) versus Time (X axis).
        1) Time shall be measured in microseconds.
    b. Tests shall be performed at full:
        1) Full voltage and speed.
        2) Loaded to 75% minimum.
        3) Duration: 1 hour minimum.
3. Record all data necessary for the preparation of required test reports.

C. Start-up and Demonstration Services:
1. Supervise start-up of all units including recheck of settings made during the pre-start-up tests.
    a. Perform all work in the presence of the Contracting Officer's designated representatives.
2. Setup all VFDs with carrier frequency at minimum value consistent with proper operation; inform Contracting Officer of carrier frequencies set in excess of 5 kHz and reason for setting.
3. Simulate operation of the VFD and its associated control and instrumentation system in both the manual and automatic modes.
    a. Ensure compatibility of VFD with associated control and instrumentation signals.
4. Simulate VFD failures and demonstrate troubleshooting aids.

D. Specific Commissioning Requirements
1. All startup and commissioning shall be performed by Manufacturer-certified technicians such that all warranty provisions remain in full effect.
2. Manufacturer-certified technicians shall perform all programming, configuration, and adjustments necessary to fully commission the VFDs and support successful integration with the plant control system.
3. Set the pump-starting ramp-rate for the pump VFDs at Roaring Springs Pumphouse to a "slow" ramp rate in order to slowly ramp up discharge pressure and mitigate water-hammer and effects on the electrical system.
    a. Coordinate ramp settings with Pump Manufacturer and Contracting Officer before startup and commissioning commences.

E. Instruct Contracting Officer's designated personnel:
1. Minimum of 8 hours at the jobsite.
    a. Roaring Springs Pumphouse
    b. North Rim Water Treatment Plant
2. Include both field and classroom instruction.
3. Instructions shall include proper operation and maintenance procedures including, but not limited to:
    a. Lubrication.

b. Troubleshooting.
c. Repair and replacement.
d. Parts inventory.
e. Maintenance records.

4. See Section 01 79 00 for Operator training requirements.

3.3 CLOSEOUT ACTIVITIES

A. Refer to Section 01 81 33 – Cyber-Security Requirements for cyber security related closeout requirements.

END OF SECTION

SECTION 26 31 00
PHOTOVOLTAIC POWER SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for:
      a. Photovoltaic Modules
      b. Photovoltaic Inverter System

1.2 SYSTEM DESCRIPTION

A. Contractor shall provide PV modules to attain the nominal maximum power rating (Pmax) DC capacity specified as determined by Contractor's module selection and the requirements in the Project Drawings.

B. The PV system design is based on the QCells, Q.TRON BLK M-G2+, 430W module. Alternate, equivalent panels are acceptable per Contracting Officer's approval. Alternate modules may require system re-design and require Contracting Officer Representative's approval.

C. Contractor shall provide PV module racking and associated design. Racking shall be compatible with module selection, shade structure design, and the requirements in the Project Drawings.

D. Inverter system shall be an "all-in-one" system containing DC breaker panel, charge controller(s), inverter(s), AC breaker panel, and communication hub.

E. The inverter system design is based on the Outback Power FLEXPower Two FXR inverter system. Alternate equipment is acceptable per Contracting Officer's approval. Alternate equipment may require system re-design and require Contracting Officer Representative's approval.

F. All proposed equipment shall work together to form a fully functioning system without adjustments, derates, or field modification.

G. Contractor shall propose a recommended quantity of spare modules in excess of proposed project installed DC capacity.

H. The PV system is intended to operate in parallel with the battery energy storage system. Contractor shall coordinate with the battery and inverter manufacturer to ensure system compliance.

I. The PV system shall be designed to provide for automatic, unattended operation of the power system. The system shall provide for local manual operation.

J. This system is intended to connect to the existing Building electrical system at one point, the point of common coupling (POCC)

K. Related Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 33 00 – Battery and Battery Charging Systems

1.3 QUALITY ASSURANCE

A. Referenced Standards:
   1. ACC Global Compliance
      a. CE Mark of Conformity
   2. ETL Testing Laboratories (ETL)
   3. International Electrotechnical Commission (IEC):
      a. 61215, Crystalline Silicon Terrestrial Photovoltaic (PV) Modules.

b. 61646, Thin-Film Terrestrial Photovoltaic (PV) Modules.
c. 61730, Photovoltaic (PV) Module Safety Qualification.

4. Institute of Electrical and Electronics Engineers (IEEE):
   a. 1547, Standard for Interconnecting Distributed Resources with Electric Power Systems.
   b. 1526, Recommended Practice for Testing the Performance of Stand-Alone Photovoltaic systems.
   c. National Electric Safety Code (NESC).
5. National Electrical Manufacturers Association (NEMA):
   a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
6. National Fire Protection Association (NFPA)
   a. 70, National Electrical Code (NEC)
7. Underwriters Laboratories, Inc. (UL):
   a. 790, Standard Test Methods for Fire Tests of Roof Coverings.
   b. 1703, Standard for Safety of Flat-Plate Photovoltaic Modules and Panels.
   c. 1741, Inverters, Converts, Controllers and Interconnection System Requirements for Use with Distributed Energy Resources.

B. Miscellaneous:
1. Install per manufacturer guidelines and applicable NEC requirements for compliant installation, for the listed purpose.
2. Components are installed as part of the project completion and workmanship review.
3. Product Shall be listed for its intended use. Listing must be to a UL standard, tested and certified by and OSHA recognized NRTL.
4. Design manuals for system engineering and planning purposes.
5. Appropriate packing for delivery and procedures for equipment un-packaging and event reporting.

1.4 SUBMITTALS

A. Shop Drawings:
1. See other Specification Sections for requirements for the mechanics and administration of the submittal process.
2. Product technical data including:
   a. Acknowledgement that products meet requirements of standards references.
3. Supplier to provide the following information:
   a. Manufacture specification and certification sheets.
   b. Design or planning documents.
   c. Installation and un-packing/handling instructions.
   d. Module flash data and other pertinent electrical data for PV design confirmation.
   e. Typical shop drawings for inverter units.
   f. Typical shop drawings for pre-packaged solutions.
   g. Manufacturing quality control documentation.
   h. All PV modules shall come with appropriate label listing information and attach to each module.
   i. Applicable value engineering documentation.

B. Contract Closeout Documentation
1. O&M Documentation
2. Warranty Information

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government

B. Contractor to confirm all equipment is rated for installed elevation and provide final equipment ratings for review. Proposed equipment shall be compatible with existing equipment.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. PV Modules
      a. QCells (Basis of Design)
      b. Or Equal
   2. Inverter/Charge Controller Unit
      a. Outback Power (Basis of Design)
      b. Or Equal
   3. Submit request for product substitution in accordance with Specification Section 01 33 23.

2.2 PV MODULES:

A. PV Module Ratings:
   1. All PV Products for delivery to the United States shall meet the following specifications:
      a. Certified to UL 1703 and listed by an OSHA recognized NRTL, including the Class C Fire Rating pursuant to UL 790 and UL 1703.
      b. IEC 61215 (for crystalline modules) or IEC 61646 (for thin film modules).
      c. IEC 61730.
      d. UL 790, Class C.
      e. All applicable NEC requirements, including, without limitation, series fuse rating requirements.
      f. CEC approved and listed.
   2. CE Mark of Conformity.
   3. PV modules shall be supplied with a copy of UL and IEC certification reports indicating compliance for the model numbers used and indicating the certification is in force as of the dates of module manufacture.

B. PV modules shall be new and certified to IEC 61730 - Photovoltaic (PV) module safety qualification and IEC 61215, shall be UL listed and shall meet any certifications / listings as required by the authority/(ies) having jurisdiction.

C. PV modules shall be supplied with documentation confirming that the PV modules have been manufactured in accordance with the manufacturer's specifications. PV modules must be of the same manufacturer and model number and consistent sub-components.

D. PV module supplier shall be certified to ISO9001:2000 and ISO140001:2004.

E. PV modules and ancillary materials shall be rated to withstand and be capable of continuous operation for the full range of "Ambient Temperature Rating Range" design conditions as listed on sheet 121S001.

F. The module/module framing shall have load ratings that are compatible with the site design conditions listed on sheet 120S001 including wind and snow loads

G. Wire leads and connectors shall be factory-installed with lengths appropriate for the module connection scheme. Connectors shall be locking, Amphenol or equivalent, and consistent throughout the design. Provide USE-2 or RHH or RHW-2 wire, and sunlight- resistant wire when exposed to sunlight.

H. PV modules shall be crystalline silicon. Modules shall incorporate anti-reflective coated (ARC) glass with an aluminum frame, bypass diodes and connector

I. Loading:
   1. Frames shall be capable of withstanding loads of not less than the design conditions listed in sheet 121S001.

J. PV modules shall have a power tolerance of +5W/-0W.

K. PV module binning and ratings shall be clearly marked on packaging to facilitate efforts of matching PV module bin class rating within strings to minimize mismatch losses for the project.

L. PV manufacturer shall provide factory flash test results in accordance with IEC 61853- 1, and temperature coefficients at: STC, nominal operating cell temperature (NOTC), low irradiance conditions (LIC), high temperature conditions (HTC), and low temperature conditions (LTC).

M. PV modules shall have a minimum rating of 1,000V DC.

N. PV modules shall be rated as a Top Performer according to the latest PVEL Reliability Scorecard, or shall be tested in accordance with the full PVEL PQP testing program and receive Top Performer ratings, or other Contracting Officer approved equivalent testing and certification program. Supplied PV modules shall be manufactured to the PVEL tested Bill of Materials (BOM) specification to ensure performance, durability, and reliability match that of the tested modules.

O. Contractor shall be responsible for the performance of all inspection and testing activities as set forth in, and as required to demonstrate compliance with, the drawings, specifications, referenced standards, applicable codes and industry practices.

P. PV modules shall not utilize top or back sheet coatings made from fluoropolymers such as per- and polyfluoroalkyl substances (PFAS) or GenX. PV module manufacturer to provide evidence that PFAS, GenX, or similar chemicals are not used in the module manufacturing process. Letter confirmation from the module manufacturer is sufficient.

Q. PV modules shall have a minimum 25-year manufacturer's warranty covering defect and workmanship.

R. PV modules shall have a minimum 25-year manufacturer's warranty covering product performance that incorporates a degradation guarantee with peak power rating at year 25 not less than 90% of the original peak power rating identified on the product nameplate.

S. The contractor shall provide information from the PV module manufacturer identifying the loading basis in accordance with IEC 61215.

T. PV modules will be installed on a fixed-tilt, roof mount system.

U. Grounding of PV modules will be performed by use of WEEB system or a self- grounding racking system that meets the applicable requirements of UL 3703 and UL 2703 and their referenced requirements for grounding. PV module design offered shall be compatible with this grounding method.

2.3 INVERTER/CHARGER CONTROLLER

A. Inverters shall be compliant with IEEE 929-2000 and IEEE-1547-2003 and IEEE 519.

B. All same-sized MPPT charge controllers supplied must be of the same manufacturer and model number.

C. Provide MPPT charge controller utilizing a wall-mount or support structure mounting system.

D. Overcurrent protection and ground-fault protection must comply with the requirements of NFPA 70.

E. Provide charge controller with self-diagnostics routines, and remote and local display of operating status and remote monitoring capabilities.

F. Charge controllers shall have network capability.

G. Match charge controller DC input to the design of the PV module array outputs

H. Array-to-controller kW ratio must not exceed manufacturer limitations.

I. Efficiency: CEC efficiency as verified by CEC SAND2007-5036 to have a weighted average power conversion efficiency of 98% or greater.

J. Full power operating temperature range: -40 DegC to 60 DegC.
   1. If other, specify operating range.

K. Integrated DC voltage surge protection

L. DC interface: specify maximum number of conductors and size per phase.
   1. All terminations to be crimp on conductors and bolt on to inverter bus.
   2. All termination rated 90DegC for copper or aluminum.
   3. Specify neutral/grounding requirements if applicable.

M. Enclosure:
   1. Casing must be rated for outdoor installation.
   2. Casing must be lockable and mountable on modular equipment rack.
   3. Inverter/Charge Controller unit to be located within enclosure.

N. Specify maximum operating elevation at full power.
   1. Specify derate curve for higher elevations.

O. Assembly and all components listed for use with max 600 Vdc and continuous duty.

P. Specify MPPT operating range.

Q. Array ground configuration: negative ground.

R. DC input monitoring for voltage and current at each DC input.

S. DC conduit entry: Bottom entry only.

T. Communications protocol: Ethernet/IP to coordinate with integrated communication hub.

U. Where a circuit grounding connection is not designed to be automatically interrupted as part of the ground-fault protection system required by Article 690.5, a switch or circuit breaker used as a disconnecting means shall not have a pole in the grounded conductor.

V. Means shall be provided to disconnect equipment such as inverters, batteries, charge controllers, and the like from all ungrounded conductors of all sources. If the equipment is energized from more than one source, the disconnecting means shall be grouped and identified.

W. The Controller shall have a minimum 5-year manufacturer's warranty covering defects and workmanship that provides for all parts and labor required to replace or repair warranty-eligible failures.

X. Materials, equipment and parts comprising the units specified herein shall be new and unused, of current manufacture, of highest grade, and assembled in a workmanlike manner.

Y. The Controller for the proposed solar electric system shall be identified for use in solar photovoltaic systems. All equipment shall be UL approved or equivalent.

PART 3 - EXECUTION

3.1 INSTALLATION

A. PV Modules
   1. Module Installation shall be in accordance with manufacturer's recommendations.
   2. Contractor shall be responsible for the performance of all inspection and testing activities to demonstrate compliance with the drawings, specifications, referenced standards, applicable codes, and industry practices.

3. Ensure wire bending radius for all conductors meets NEC requirements from the modules.
4. Ensure electrical installation is secured and properly grounded.
5. Bolted connections are torqued per manufacturer and NEC requirements.
6. Source circuit wires should be organized and bent in a uniform manner

B. Inverter/Charge Controller System
1. Installation shall meet or exceed all applicable federal, state and local requirements, referenced standards and conform to codes and ordinances of authorities having jurisdiction.
2. Contractor shall inspect all packages for damage upon delivery. Any damaged packages shall be opened so the inverter system and equipment can be inspected. Any damage to the packaging or equipment shall be documented and reported to the Contracting Officer immediately. Inverter system to be stored shall be properly replaced in the packaging for storage.
3. Inverter system shall be stored in a secure and clean location per the manufacturer's recommendations and documentation. Inverter system shall be protected from the harsh environment such as excessive heat, cold, moisture, dust, snow, etc.
4. Inverter system shall be handled and installed in accordance with the manufacturer's recommendations and documentation. All instructions and reference documents shall be reviewed and understood by the contractor prior to handling and unpacking the equipment.
   a. Reference the manufacturer's instructions for unpacking the equipment. Inverter system shall be transported by means outlined in the manufacturer's documentation only.
5. PV array DC grounding configurations may vary by manufacturer and technology.
6. The grounding configuration shall be noted by the Contractor for safety and proper installation.
7. Warning and safety labels properly placed.
8. Inverter system components are labeled and identified to match the naming convention shown on the plans.
9. Provide sufficient service access and equipment setbacks per NEC 110.26. Ensure enclosure doors are able to swing a full 180 degrees at installed location.
10. The contractor is to ensure that working clearances meet the requirements of all applicable codes and the manufacturer's requirements. Any discrepancies shall be reported to the Contracting Officer immediately.
11. Inverter system shall be secured to the enclosure utilizing all of the provided mounting points. Reference the manufacturer's documentation for location and size of mounting points.
12. All disconnect switches shall be in the open position during installation and shall remain in the open position until proper testing, inspection, and commissioning has been completed.
13. All fasteners shall be torqued per the manufacturer's documentation.
14. It is prohibited to modify the inverter system or install equipment not explicitly recommended by the manufacturer. Do not store documents, instructions, plans, or any other foreign material not intended to be part of the system inside the controller cabinets (when applicable).
15. Components of the charge controllers may be damaged by electrostatic discharge (ESD). When handling the electrical components observe all ESD safety regulations.
16. All conductors shall be connected to the inverter system per the manufacturer's recommendations and documentation making note of recommended terminations, torque values, and bolt stack up details if provided. All buss bars, conductors, and terminations shall be clean prior to making, the connection.

3.2 COMISSIONING

A. PV manufacturer shall provide factory flash test results in accordance with IEC 61853- 1, and temperature coefficients at: STC, nominal operating cell temperature (NOTC), low irradiance conditions (LIC), high temperature conditions (HTC), and low temperature conditions (LTC).

B. Provide documentation for all test results and certifications in accordance with this specification and industry standards.

C. After installation, perform IV curve tracing for each string circuit. Conduct a full energy system functional test.

D. Provide documentation for all test results and certifications in accordance with this specification and industry standards and in accordance with manufacturer recommendations.

END OF SECTION

This page intentionally left blank.

SECTION 26 32 14
ENGINE GENERATOR - DIESEL

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Engine generator equipment.
      a. One new engine-generator system for indoor installation at the Power Distribution Center (PDC), to operate in parallel with three existing engine-generators in the PDC to provide standby power for Park facilities.
   2. Fuel systems to interface with systems provided in other Specification Sections..
   3. Engine generator accessories.
   4. Engine exhaust system.

B. Related Requirements: Include, but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 33 52 10 – Service Piping – Fuel Systems
   3. Section 33 56 13 – Aboveground Fuel Storage Tanks
   4. Section 33 56 16 – Fuel Day Tank Systems

1.2 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.3 REFERENCES

A. Reference Standards:
   1. Environmental Protection Agency (EPA):
      a. 40 CFR Part 60, Subpart IIII, Protection of Environment, Standards of Performance for New Stationary Sources, Standards for Performance for Stationary Compression Ignition Internal Combustion Engines.
   2. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. MG 1, Motors and Generators.
   3. National Fire Protection association (NFPA):
      a. 37, Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines.
      b. 70, National Electrical Code (NEC):
         1) Article 701, Legally Required Standby Systems.
      c. 211, Standard for Chimneys, Fireplaces, Vents, and Solid Fuel–Burning Appliances.
   4. Underwriters Laboratories, Inc. (UL):
      a. 2200, Standard for Stationary Engine Generator Assemblies.

1.4 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Regarding delegated design Work required by this Section, comply with “Coordination” requirements of Contract Documents.

B. Pre-Design Conference:
   1. Provisions of Contract Documents, notwithstanding, a pre-design conference is not required for the Work of this Section.

1.5 QUALITY ASSURANCE

A. Qualifications:

1. Supplier:
    a. The engine generator equipment manufacturer or other authorized Supplier has unit responsibility for furnishing all components of required engine generator systems and proper, initial operation of systems required by this Section.
2. Delegated Design Professional:
    a. Comply with qualifications requirements indicated in Contract Documents.

1.6 SUBMITTALS

A. Action Submittals: Submit the following:
1. Shop Drawings:
    a. Schedule: Where the Work includes furnishing more than one engine generator system, submit schedule of engine generator systems, indicating for each engine generator system: identification or tag number, engine generator size, location, enclosure type, fuel storage tank capacity, and other pertinent data.
    b. Dimensioned plan, elevation, sections, and detail drawings for engine generator equipment, enclosures, and other components furnished by engine generator manufacturer.
    c. Weights, centers of gravity, lifting points.
    d. Fuel system schematic. Coordinate with bulk fuel systems provided under other Sections.
    e. Fuel storage tank.
    f. Wire interconnection drawings.
    g. Control panel layout drawings and internal wiring diagrams.
2. Product Data:
    a. Manufacturer's literature and published data for all components and accessories of engine generator systems.
    b. Comply with Section 26 05 00 - Common Work Results for Electrical, regarding required Submittals.
    c. Engine/generator performance curves.
    d. Provide manufacturer cut sheets for engine exhaust piping and components including exhaust pipe, silencer, roof thimble and expansion joints.
        1) Provide detailed design drawings from the exhaust pipe manufacturer. The system shall be designed to compensate for all exhaust gas induced thermal expansion.
        2) Provide manufacturer's computations verifying stack size.
3. Testing Plans, Procedures, and Testing Limitations:
    a. Testing procedure, apparatus, and limitations of apparatus and procedure, for:
        1) Source quality control activities indicated in this Section.
        2) Field quality control activities indicated in this Section.

B. Informational Submittals: Submit the following:
1. Certificates:
    a. Generator equipment manufacturer's documentation of engine USEPA certification including USEPA family name and generator equipment model designation.
2. Supplier Instructions:
    a. Serial numbers of items furnished, equipment nameplate information, and similar information for all items furnished.
    b. Instructions for handling, installing, and startup.
3. Source Quality Control Submittals:
    a. Results of tests, inspections, and other quality control activities required by the Contract Documents and performed at the place of production or fabrication.
4. Field Quality Control Submittals:
    a. Results of tests, inspections, and other quality control activities required by the Contract Documents and performed at the Site.
5. Supplier Site Visit Reports:

a. Report of each visit to the Site by Supplier, summarizing purpose of visit, activities while onsite, problems encountered, advice given to Contractor or Subcontractor, and actions taken.

6. Qualifications:
   a. Contractor's, Subcontractor's, or Supplier's delegated design professional.

C. Closeout Submittals: Submit the following:
   1. Operation and Maintenance Data:
      a. Submit in accordance with Section 01 78 23 - Operation and Maintenance Data.
   2. Post-Startup Statement:
      a. Equipment installation, startup, and operational statement for each engine generator system provided.

D. Maintenance Material Submittals:
   1. Furnish the following items and submit documentation of delivery to and acceptance of such items by Contracting Officer or facility manager (if any), in accordance with Section 01 78 23 – Operations and Maintenance Data:
      a. Spare Parts and Extra Materials:
         1) Furnish spare parts and extra materials sufficient for one year of operation as recommended by generator system manufacturer, for each generator system furnished.

E. Refer to Section 01 81 33 - Cyber Security Requirements for required cyber security related submittals.

1.7 FIELD CONDITIONS

A. Ambient Conditions:
   1. Ambient Air Temperature:
   2. Minimum: -20 degrees F.
   3. Maximum: 100 degrees F.

B. Existing Conditions:
   1. Site Elevation: 8000 feet above sea level.
   2. Seismic Forces:
      a. Seismic use group: II.
      b. S_s: 0.3.
      c. S_1: 0.110.
      d. Site Class: C.
      e. Seismic Design Category: C.
      f. Comply with Section 26 05 48 - Electrical Seismic Restraint Systems.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Generator Assembly
      a. Engine generator unit:
         a) Caterpillar "C18" (basis of design, to match existing equipment).
         b) No substitutions allowed.
            (1) New generator must be compatible with existing generators for parallel operation.
      b. Silencers:
         1) Generator equipment manufacturer's standard.
      c. Battery charger:
         1) Generator equipment manufacturer's standard.
      d. Governor:

1) Generator equipment manufacturer's standard.
e. Radiator:
1) Generator equipment manufacturer's standard.
f. Vibration isolators:
1) Generator equipment manufacturer's standard.
2. Day Tank and Engine Exhaust Pipe:
a. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1) Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
2) Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.
a) Engine exhaust pipe:
(1) Metal-Fab, model IPIC.
(2) Or Equal.

2.2 DESCRIPTION

A. Engine generators will be used and rated for:
1. Legally required standby power during a utility power outage, NFPA 70, Article 701.
2. The generator will be installed in the PDC, to supplement three existing generators.
a. The new generator may be required to operate in parallel in synchronized operation with one or more additional generators.
b. Generator paralleling operations will be controlled by the switchgear to which all generators are connected.
c. The switchgear will control generator start/stop, exercise periods, and synchronizing operations.

2.3 EQUIPMENT

A. Emissions Requirements:
1. Each generator system's exhaust emissions shall comply with Laws and Regulations. Federal, State and Local government requirements, including but not limited to:
a. Environmental Protection Agencies (USEPA) New Source Performance Standards (NSPS), 40 CFR Part 60, Subpart IIII.

B. Performance and Configuration Requirements:
1. The Unit shall be capable of providing standby power.
2. The Unit shall have the following nominal manufacturer's size rating, as indicated on the Drawings:
a. 750 kVA / 600 kW
3. Output voltage: 480Y/277V.
a. Alternator neutral bonded to ground with a removable strap on the generator skid.

2.4 COMPONENTS

A. Engine Generator Unit General:
1. Diesel engine direct-connected to alternating current generator mounted on suitable rigid steel skid supports.
2. Mount unit on skid suitable for installation on concrete foundation.
3. Base rating on operation at rated RPM when equipped with all operating accessories.
4. Standards: UL 2200.

B. Engine:
1. Four-cycle, full compression ignition, single acting, solid-injection unit, either vertical or V-type pistons turbo charged with inner and after cooling.

2. Fuel supply: No. 2 Diesel.
3. Removable full wet-type cylinder liners of close grained alloy iron, heat treated for proper hardness to obtain maximum life.
4. Capable of operating at idle or light loads for extended periods of time.

C. Injection Pumps and Valves:
1. Type not requiring adjustment in service, which may be individually removed and replaced.
2. Individual injection pump and valve for each cylinder.
3. Fuel injection pumps: Positive action, constant-stroke, actuated by cam driven by gears from engine crankshaft.
4. Fuel lines between injection pumps and valves: Heavy seamless steel tubing.
5. Flexible fuel line connectors for supply and return connections at pump.

D. Oil Pump:
1. Gear-type lubricating oil pump to supply oil under pressure to main bearings, crank pin bearings, pistons, timing gears, camshaft bearings and valve rocker mechanism.
2. Spray cool and lubricate pistons.
3. Oil filters so located that lubricating oil is continuously filtered, except during periods when oil is automatically by-passed to protect vital parts when filters are clogged.
4. Filter elements accessible and easily removable.
5. Filter elements: Effective full flow, replaceable resin-impregnated cellulose type.
6. Equip filter system with spring-loaded by-pass valve.
7. Oil cooler: Water-cooled, engine-mounted.

E. Fuel System:
1. Coordinate with day tank provided under another Specification Section.
2. Fuel pump: Built-in gear-type, engine-driven fuel transfer pump.
3. Equip fuel system with replaceable fuel filter elements arranged for easy removal without breaking any fuel line connections or disturbing fuel pumps or any other part of engine.
4. Locate all fuel filters in an accessible housing, ahead of injection pumps to thoroughly filter fuel before it reaches the pump.
5. Use no screens or filters requiring cleaning or replacement of injection pumps or valve assemblies.

F. Governor: Fully enclosed electronic type governor with actuator capable of providing accurate speed control within 1% of rated speed, complete with panel-mounted electronic assembly with ramp generator and speed-sensing modules.

G. Air Cleaners: Engine-mounted, dry type air cleaners of sufficient capacity.

H. Electric Starting System:
1. Sufficient capacity to crank at speed which will start engine under normal operating conditions.
2. Controls to provide automatic cranking of engine when generator is called to start.
3. Prevent excessive cranking which could damage cranking motor.
4. Automatic stop controls.
5. Starter motors with positive-engagement feature.

I. Cooling System:
1. Capacity for cooling engine at the specified operating conditions.
2. Engine driven, centrifugal type water circulating pump and thermostatic valve to maintain the engine at recommended temperature level.
3. Unit mounted radiator.
    a. Core guard flexible duct adapter.
    b. Site glass at top of unit.
    c. Engine driven blower fan.
    d. Low water level cutoff switch.
4. Provide fan guards.

J. Heater:
   1. Thermostatically controlled jacket water heater(s) to maintain cooling jacket at the manufacturer's recommended temperature at the specified low ambient temperature.
   2. 208 V, single phase.

K. Silencer:
   1. Suitable type for industrial silencing.
   2. Seamless, stainless steel, flexible, exhaust adapter for exhaust outlet to silencer.

L. Engine Instruments and Controls:
   1. Engine-mounted instruments:
      a. Oil pressure gage.
      b. Water temperature gage.
      c. Run time meter.
      d. Battery voltage meter.
   2. Automatic cycle cranking and over-crank protection.
   3. Safety controls: Equip engine with automatic safety controls to shut down engine in event of low lubricating oil pressure, high jacket water temperature, overspeed or overcrank.
   4. Auxiliary control devices: Either integral with specified engine instruments, control, and safety devices or as separate devices as required to operate various signal circuits specified for remote annunciator panel.
   5. Three NO auxiliary contacts for interface with louvers, fans or other miscellaneous equipment.
      a. Contacts shall close when generator is started.

M. Main Fuel Tank: See Specification section 33 56 13.

N. Fuel Day Tank: See Specification section 33 56 16.

O. Batteries:
   1. Lead acid type.
   2. Furnish electrolyte separately for use when installation is complete and unit is ready for testing.

P. Battery Charger:
   1. Output current rating of at least 1/20th of ampere hour capacity of battery and capable of automatically switching between low rate (float) mode and high rate (equalize) mode.
   2. Solid state rectifiers, DC voltmeter and ammeter, fuse input and output, and 115 VAC input.
   3. Malfunction alarm contacts (minimum): low and high battery voltage, weak battery and charger failure.

Q. Generator:
   1. Brushless, 4-pole drip-proof revolving field type with permanent magnet, 2/3 pitch stator, direct-coupled rotor, Class H insulation.
      a. Match generator/alternator to existing engine-generators to allow parallel operation of generators.
   2. Minimum continuous standby ratings:
      a. Specified herein, substantiated by manufacturer's standard published curves and conform to NEMA MG 1 specification.
      b. Special ratings or maximum ratings are not acceptable.
   3. Rated to serve up to 50% non-linear load without exceeding rated temperature rise.
   4. Minimum efficiency: 92% at 50 to 110% of nominal standby rating, less than 30% instantaneous voltage dip at full load and rated power factor and suitable for simultaneous operation with other future units connected in parallel.
   5. Stator and rotor: 800 degrees C temperature rise with minimum Class H insulated with 100 percent epoxy impregnation and overcoat of resilient insulating material to reduce possible fungus and/or abrasive deterioration.
   6. Directly connect stator to engine flywheel housing.
   7. Drive rotor through semiflexible driving flange to ensure permanent alignment.

8. Self-ventilating with suitable blower, air inlet and outlet openings.
9. Provide terminal box of adequate size for entrance of conduit and termination of conductors.
10. Generator drive free from critical torsional vibration within operating range.
11. Provide generator mounted main circuit breaker:
    a. Solid state molded case type.
    b. Electronic trip unit, with minimum of Long, Short, and Instantaneous trip adjustments.
    c. Ratings as indicated.

R. Voltage Regulator:
1. SCR type, to maintain 2% voltage regulation from 0 to full load with steady state modulation not exceeding plus 1/2% including cross-current compensation to provide maximum of 5% unbalance in kVA load sharing between this unit and possible future generators.
2. Automatic protection against short circuits on system.
3. Permit unit to operate at no load below rated frequency for engine start up and shut down procedures.
4. Provide voltage level and gain controls for normal operating adjustments.
5. Provide voltage level control with minimum range of plus or minus 5% from rated voltage.
6. Mount regulator, volts per hertz type, in generator housing on suitable vibration isolators.

S. Generator Instruments and Controls:
1. Generator mounted NEMA 1 type, illuminated vibration isolated instrument and control panel(s).
2. AC voltmeter and phase selector switch.
3. AC ammeter and phase selector switch.
4. Frequency meter.
5. Run-off-auto engine, start-stop control switch.
6. Emergency stop buttons:
    a. One mounted on control panel, and one mounted on the face of the synchronizing switchgear.
7. Run time meter – 6 digit type, requiring no batteries for operation.
8. Governor control rheostat.
9. Voltage level adjustment rheostat.
10. Cool down time delay 0-15 minute adjustable.
11. Cycle cranking control.
12. Minimum red shut down indicating lights as follows:
    a. Overcrank.
    b. Overspeed.
    c. Low lubricating oil pressure.
    d. High engine water temperature.
13. Minimum amber alarm indicator lights as follows:
    a. Control switch not in auto position.
    b. Low engine water temperature (less than 70 degrees F).
    c. Low fuel in day tank.
    d. Day tank leak.
    e. Battery charger malfunctioning.
    f. Low battery voltage.
14. Minimum amber prealarm indicator lights as follows:
    a. High engine water temperature.
    b. Low lubricating oil pressure.
15. Common dry contact and audible alarm to indicate when one or more alarm or prealarm conditions exist.

T. The generator control system shall include a network-based connection (Modbus) to allow full integration and monitoring of the generator by plant operators through the plant control system. The generator supplier shall provide a Modbus gateway with copper-cable and fiber optic

connections and all interface devices to allow monitoring of all generator feature, status, alarms, fuel level, and other operating conditions from the plant control system.

U. The generator control panel shall include an emergency-stop pushbutton. The emergency-stop circuit shall be wired to accept additional emergency-stop pushbuttons located in the Power Distribution Center.

V. The generator control system shall include the following minimum signals for monitoring by the plant control system:
   1. Machine Status via Form C contacts:
      a. Generator Running
      b. Generator Alarm
      c. High Temperature
      d. Low Fuel Level
      e. Low Oil Level
      f. Emergency-stop system activated (in the event that any of the emergency-stop pushbuttons are activated).
   2. Machine status via Modbus (through manufacture-supplied Modbus gateway to fiber optic connection):
      a. Machine operating parameters as determined by the Contracting Officer.

W. Vibration Isolators: Vibration system shall consist of engine and generator mount isolators with or without additional mechanical spring isolators rubber pads to control both high and low frequency vibrations between major components, sub-base and structural foundation and to provide required vibration isolation for the seismic zone of the Project.

2.5 ACCESSORIES

A. Provide interposing relays (24 VDC to 120 VAC) as required for interfacing with customer's 120 Vac monitoring system.

B. Provide relays (10A at 120VAC) to interface with existing PDC air inlet/outlet motorized damper systems.

C. Generator remote annunciator panel:
   1. Surface mounted NEMA 1 enclosure. Flush mounted enclosure with flush plate.
   2. Circuits:
      a. 24 VDC powered from starting batteries.
      b. Verify circuit voltage to match battery voltage.
   3. Provide red and green signal lamps, buzzer, silencing switch, lamp test switch, relays, solid-state components, and engraved function identifications.
   4. Annunciator functions:
      a. Green light "ON" to indicate generator is operating to supply power to load.
      b. Separate red light for each shutdown or alarm condition and amber light for each prealarm condition and common buzzer with silence/ acknowledge switch.
      c. Shut down indicating lights as follows:
         1) Overcrank.
         2) Overspeed.
         3) Low lubricating oil pressure.
         4) High engine water temperature.
      d. Alarm indicator lights as follows:
         1) Control switch not in auto position.
         2) Low engine water temperature (less than 70 degrees F).
         3) Low fuel in day tank.
         4) Fuel in day tank rupture basin.
         5) Battery charger malfunctioning.
         6) Low battery voltage.
      e. Prealarm indicator lights as follows:
         1) High engine water temperature.

2) Low lubricating oil pressure.

2.6 ENGINE GENERATOR EXHAUST

A. Double wall piping systems:
   1. Positive pressure, insulated, pre-fabricated system:
   2. UL listed for flue gases at temperatures up to1000°F deg F under continuous operation, and 1400°F intermittent operation.
   3. Codes and standards:
      a. UL 103 listed.
      b. Installation and fabrication shall comply with NFPA 37 and NFPA 211.
   4. Ceramic fiber insulation, 2-inch thickness, installed between inner pipe and outer casing.
   5. Materials:
      a. Inner carrier pipe: Type 304 stainless steel.
      b. External casing pipe: Type 304 stainless steel.
   6. Designed and installed to be gas tight, zero leakage, using high temperature sealants and gasketing.
   7. Designed to compensate for all exhaust gas induced thermal expansion.
   8. Provide piping supports, roof and wall penetrations, terminations, adapters, drain fittings and expansion joints as required for a complete exhaust system as indicated on the drawings.
   9. Refer to the mechanical drawings for additional details.

2.7 SOURCE QUALITY CONTROL

A. Individually test each prime mover.
   1. Apply derating factors for the proposed site to test data.
   2. Continuously test for a period not less than two hours.
   3. Test procedure shall be as follows:
      a. Start prime mover and upon reaching rated RPM, pick up 100 percent of nameplate KW rating at rated power factor in one step.
      b. Observe and record the cranking time(s) required to start and run for each prime mover.
      c. Observe and record the time required to come up to operating speed for each prime mover.
      d. Record voltage and frequency overshoot for each prime mover.
      e. Record voltage, frequency and amperes.
      f. Record oil pressure, water temperature where applicable and battery charge rate at first load acceptance and at 15 minute intervals thereafter for each prime mover.

2.8 MAINTENANCE MATERIALS

A. Spare Parts:
   1. Provide manufacturer’s recommended spare parts.
   2. Provide three sets of all fuses for the generator.
   3. Provide two sets of all filters for the generator.
   4. Provide two sets of all belts for the generator.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install all components as indicated and in accordance with manufacturer's recommendations and instructions and Specification Section 26 05 48.

B. Fill cooling system with solution of 50-50 water and ethylene glycol anti-freeze to prevent freezing at temperatures as low as minus 30 degrees F.

C. Provide fuel to fill all tanks (outdoor bulk storage tank, indoor day tank). Refill tanks following all testing before handover to the Contracting Officer.

D. Install all wiring to engine in conduit.
   1. Control wiring on engine may be factory installed in high temperature loom.

E. Provide control wiring in conduit between generator control panel, remote annunciator panel(s)and remote devices as described under generator instrument and controls paragraph and remote annunciator paragraph of this Specification.

F. Mount on existing concrete pad utilizing vibration/seismic isolators.

G. Engine generator exhaust:
   1. All products shall be installed with the required minimum clearances to combustibles as identified within the terms of their listings.
   2. Install engine exhaust system in accordance with manufacturer's installation instructions, per NFPA 211, NFPA 37 and approved shop drawings.
   3. Store materials inside and protected from the weather.
   4. Protect incomplete stack installations by attaching temporary closures over open ends of sections.
   5. Clean all stacks and breechings of dust and debris prior to final connection to appliances or equipment.

3.2 FIELD QUALITY CONTROL

A. Field Tests and Inspections:
   1. Provide two load tests and one cycle crank test.
   2. Tests one and two shall be for continuous period of no less than two hours each.
   3. Contracting Officer shall be notified seven days prior to testing.
   4. Test number one:
      a. With prime mover(s) in a "cold start" condition and emergency load at normal operating level, initiate a normal power failure by opening all switches or breakers supplying normal power to facility.
      b. Observe and record the time delay on engine start.
      c. Observe and record the cranking time(s) required to start and run for each prime mover.
      d. Observe and record the time required to come up to operating speed for each prime mover.
      e. Record voltage and frequency overshoot for each prime mover.
      f. Observe and record time required to achieve steady-state condition with all switches transferred to emergency position.
      g. Record voltage, frequency and amperes.
      h. Record oil pressure, water temperature where applicable and battery charge rate at 5-minute intervals for the first 15 minutes and at 15 minute intervals thereafter for each prime mover.
      i. Return normal power to facility, record time delay on retransfer to normal for each switch and cooldown time delay for each prime mover.
   5. Test number two:
      a. Immediately after completion of test number one, start prime mover and upon reaching rated RPM, pick up 100% of nameplate KW rating in one step.
         1) Unity power factor is acceptable for on-site testing
      b. Observe and record the cranking time(s) required to start and run for each prime mover.
      c. Observe and record the time required to come up to operating speed for each prime mover.
      d. Record voltage and frequency overshoot for each prime mover.
      e. Observe and record time required to achieve steady-state condition.
      f. Record voltage, frequency and amperes.
      g. Record oil pressure, water temperature where applicable and battery charge rate at first load acceptance and at 15 minute intervals thereafter for each prime mover.
   6. Cycle crank test:
      a. Perform test for each prime mover.

1) Utilize any method recommended by manufacturer to prevent prime mover(s) from running.
2) Put control switch into "run" position to cause prime mover to crank.

b. A complete cranking cycle shall consist of an automatic crank period of approximately 15 seconds duration followed by a rest period of approximately 15 seconds duration.
1) Upon starting and running of the prime mover, further cranking shall cease.
2) Two means of cranking termination shall be utilized so that one will act as a backup to the other to prevent inadvertent starter engagement.
3) Cranking limiter time shall be 75 seconds for cycle crank.

7. Furnish load banks of required ratings necessary for tests.
8. Record engine fuel consumption by means of test equipment.
9. Test all safeties specified for generator instruments, controls, and generator remote annunciator panel as recommended by manufacturer and as required to verify proper operation.
10. The Contractor shall be responsible for fuel and all consumables use during the test.

B. Supplier’s onsite services: Employ and pay for services of equipment manufacturer's field service representative(s) to:
1. Inspect equipment covered by this Specification Section.
2. Supervise pre-startup adjustments and installation checks.
3. Conduct initial startup of equipment and perform operational checks.
4. Provide Contracting Officer written statement that manufacturer's equipment has been installed properly, started up, tested, and is ready for operation by facilities operations and maintenance personnel.
5. Provide training of facility operation and maintenance personnel in accordance with Section 01 79 00 – Demonstration and Training.

C. Programming assistance: Manufacturer shall provide Contracting Officer with on-site assistance with programming and configuration of the generator and all control systems and integration with plant control system.
1. Provide 16 hours of assistance in two separate sessions (two 8-hour sessions) for each engine-generator set.

3.3 CLOSEOUT ACTIVITIES

A. Refer to Section 01 81 33 - Cyber-Security Requirements for cyber security related closeout requirements.

END OF SECTION

This page intentionally left blank.

SECTION 26 33 00
BATTERY AND BATTERY CHARGING SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for:
      a. Batteries.
      b. Battery charging system.

B. Related Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. D635, Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position.
   2. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   3. Underwriters Laboratories, Inc. (UL):
      a. 94, Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances.
      b. 1012, Standard for Power Units Other Than Class 2.
   4. NFPA
      a. 855, Standard for the Installation of Stationary Energy Storage Systems

B. Qualifications:
   1. Battery system design to be by personnel skilled in the design and construction of the system.
      a. Experienced in design projects of similar size and character as required for this work.
      b. Battery system furnished shall use standard components assembled in a general configuration having acceptable history of service in similar applications for more than three uninterrupted years.

1.3 SYSTEM DESCRIPTION

A. The Battery and Battery Charging System shall provide 48 VDC power for:
   1. Distributed AC loads through associated charge controller and inverter.

B. Battery charging capability to be included through external inverter/charger system. Charger and battery capacity to be sized to meet the capacity requirements of the design.

C. Inverter/charger system manufacturer and battery provider shall coordinate to calculate the appropriate size/ratings of the system in accordance with the design documents.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Provide submittal data for all products specified in PART 2 of this Specification:
      b. Calculations used to size the batteries and chargers in accordance with the specified duty and criteria.
      c. See Section 26 05 00 for additional requirements.

2. Fabrication and/or layout drawings:
    a. General arrangement drawing.
    b. Cross sections, elevations and details.
    c. Battery cell layout drawing to facilitate the installation process.
3. Certifications:
    a. Guarantee of performance.

B. Contract Closeout Information:
1. Operation and Maintenance Data:
    a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

C. Informational Submittals:
1. Qualifications of system designer.
2. Record of test results, inspections and procedures witnessed or performed by factory service representative.

1.5 PROJECT CONDITIONS

A. Environmental Requirements:
1. Refer to sheet 121S001 for site conditions

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government

B. Contractor to confirm all equipment is rated for installed elevation and provide final equipment ratings for review. Proposed equipment shall be compatible with existing equipment interface.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Briggs & Stratton (Basis of Design)
2. Or Equal

D. Submit request for product substitution in accordance with Specification Section 01 33 23.

2.2 MANUFACTURED UNITS

A. Lithium Iron Phosphate (LFP) Battery
1. General:
    a. LFP Chemistry
    b. Rack Mountable
2. Battery ratings:
    a. Nominal voltage: 48VDC.
    b. Power Output: Refer to project Drawing Set for power minimum.
    c. Battery life: 10 years minimum.
3. Each cell shipped completely assembled:
    a. Fully charged within one week of shipment to at least 90% capacity and ready for service.
4. Battery housing:

a. Individual modules to come in 3.8kWh capacity minimum
b. Modules to be stackable or rack mounted.
c. Heat resistant and flame retardant.
d. Mark each container with month/year of manufacture.

B. Battery Charger:
1. Controller:
a. Controller to be provided by same manufacturer as battery units.
b. Microprocessor controlled:
1) Programmable customized alarms.
2) Display both volts and current simultaneously.
2. Rating:
a. Input voltage: 48VDC.
1) Voltage deviation: 48-56 VDC.
b. Output voltage: 48 VDC.
c. Used in a DC system consisting of LFP batteries and used to maintain the battery in fully charged condition by float charge and to supply DC power to continuous load.
d. Service:
1) Used to supply Constant Current Constant Voltage (CCCV) to the battery while simultaneously supplying DC power to the continuous loads.
a) The equalizing charge rate is to be sufficient to fully charge the battery.
b) Sized to meet the requirements of the specification, from its fully discharged condition in 4 hours while also energizing the continuous load.
c) For any setting of the output voltage within the "equalizing charge voltage setting range", the output voltage shall not vary beyond +0.5% under full load condition and input supply voltage and frequency variations.
2) The transfer from charging to discharging service is to be automatic.
3) The return to charging from discharging is to be automatic.
e. Current limiting circuitry:
1) Limit the output current to a value which will not cause any damage or blow the fuses.
2) The charger voltage is to automatically return to normal value when the overload or short circuit is removed.
f. The charger shall be self-protecting against all AC and DC transients.
1) The charger shall be designed with or without the option of a blocking diode to prevent power feedback from the batteries during input AC supply failure or a charger internal failure.
2) Upon restoration of AC supply, the pre-set charging rate shall be resumed automatically.
3. Charger enclosure:
a. Charger/Inverter to be installed within NEMA 3R enclosure
b. Hinged door on which controls and instruments are mounted.
c. Provisions for mounting on modular strut.
4. Protective devices:
a. Input circuit breaker with thermal magnetic trip element, operable for the front.
b. Two-pole output DC circuit breaker with thermal magnetic trip element, operable from the front.
5. Provide with the following:
a. Alarm light for each alarm relay.
b. Input AC circuit breaker.
c. Output DC circuit breaker.
d. Output DC voltmeter and ammeter.
e. Input AC failure alarm.
f. Ground fault detector.
g. Output DC failure alarm.
h. High charge volt alarm.

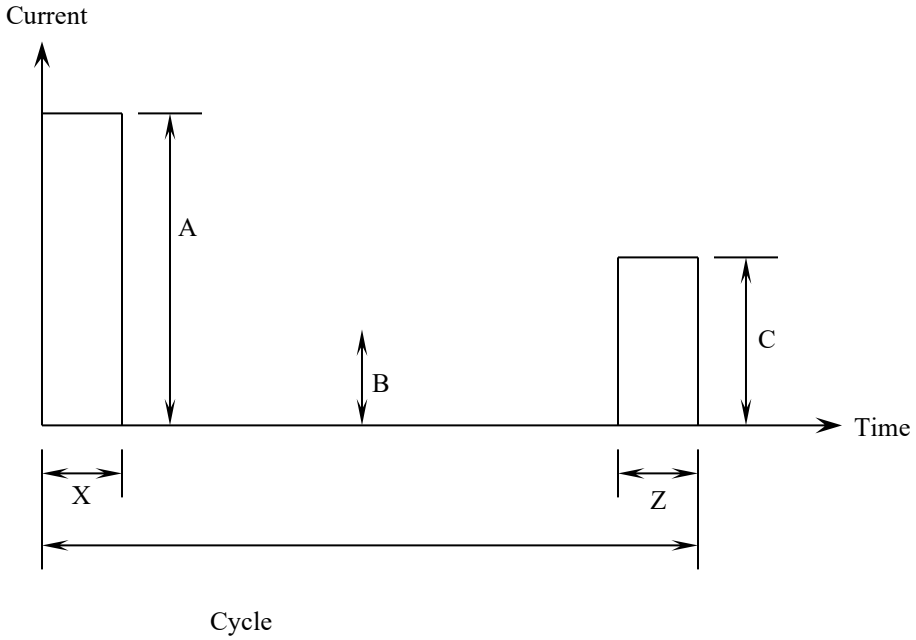
i. High DC voltage shutdown.
j. Low charge volt alarm.
k. Load current display.
l. Adjustable timer.
m. Input voltage lightning arrestors.

6. Standards: UL 1012NEMA PE 5.

C. Battery Enclosure and Accessories:
1. Enclosure:
a. Dimensions: 36” W x 16” D x 40” H
1) Size should be ample to fit battery modules, conductors, battery specific communication equipment, and accessories.
2) Where necessary, battery modules shall be divided between several identical enclosures.
b. Construction:
1) Welded carbon steel construction
2) NEMA Rating: 3R
c. Doors:
1) Minimum of 1 access door.
2) Stainless steel handles with 3-point locking system, lockable
3) Aluminum hinges
2. Accessories
1) No-Oxide grease in sufficient quantity to coat all posts and connectors
2) Anti-corrosion compound for battery connections.
3) Inter cell connectors (including interstep connectors).
4) Individual cell number labels.
5) Forces air equipment capable of maintaining internal temperature below 120 F.
a) Fans
6) Internal copper busbar

2.3 PERFORMANCE AND DESIGN REQUIREMENTS

A. Battery capacity shall be sized to support the entire load at the end of battery life with 20% degradation.
1. Design factors:
a. Aging factor: 1.25.
b. Design factor: 1.10.
2. Battery sized with the capability of supplying the specified emergency duty cycle throughout its life with the initial discharge capacity of the battery at 80% of rated discharge capacity.
3. Battery capacity includes the voltage drop across the connectors.
4. Battery sized to support the above duty cycle with a minimum electrolyte temperature of 0 degrees C.
5. Standards: IEEE 1679.1.



2.4 SOURCE QUALITY CONTROL

A. The battery and charger are to be inspected and tested for conformance with the specifications prior to shipment.
   1. A Contracting Officer may be present for all witness testing and inspections.
   2. The representative shall notify Contracting Officer of all tests and inspections at least three weeks prior to conducting said test or inspection.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install in accordance with manufacturer's instructions.

B. Furnish and install all supplementary or miscellaneous items, appurtenances and devices incidental to or necessary for a complete installation.

3.2 FIELD QUALITY CONTROL

A. Field Tests:
   1. Inspect the general assembly of the equipment to insure that the battery and charger conform to the Contract Drawings, procedures, and specifications.
   2. Provide inspection and test procedures for the battery and charger in accordance with the manufacturer's instructions.
   3. The battery and charger performance test are to be conducted in accordance with the manufacturer's instructions.
   4. Standards: IEEE P2962

B. Provide field services to handle correction of errors, discrepancies or omissions in the equipment and materials furnished.

C. See Section 01 79 00 for Operator training requirements.

END OF SECTION

SECTION 26 35 26
HARMONIC CORRECTION EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Harmonic correction equipment including Active Harmonic Filters.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 08 13 - Acceptance Testing.
   3. Section 26 09 16 - Control Equipment Accessories.
   4. Section 26 24 19 - Motor Control Equipment.
   5. Section 26 29 23 - Variable Frequency Drives - Low Voltage.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Institute of Electrical and Electronics Engineers (IEEE):
      a. C62.41, Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits.
      b. 519, Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems.
   2. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).

1.3 DEFINITIONS:

1. Contracting Officer: Includes Contracting Officer (CO) and Contracting Officer Representative (COR).

1.4 SYSTEM DESCRIPTION

A. General:
   1. Voltage: 480 V, 3 PH, 3-wire, plus ground.
   2. Output Load Capacity: Rated capacity shall be the specified current capacity at the voltage required as indicated within this Specification and shown on the respective Electrical Drawings included within this Document.
   3. Current Transducers:
      a. Split core type current transducers shall be installed as defined herein and shown in the Electrical Drawings contained in these Documents.
      b. Two current transducers per power correction system location are required and mounted on Phases A & B.
      c. Current ratings of the current transducers shall be according to full load current of the circuit on which installed.
      d. Current transducers rated for 400 Hz shall be used.

B. Modes of Operation:
   1. The power correction system shall be designed to electronically inject harmonic current to cancel load produced harmonic current such that the upstream power harmonic current and voltage are reduced. Refer to Drawings herein for specific locations.
   2. Reactive current compensation (aka displacement power factor correction) shall be activated via a digital keypad/display mounted on the door of the enclosure. When reactive current compensation is activated, the power correction system shall first perform harmonic current correction and then use the remaining capacity to inject reactive current compensation to attain the specified level.

C. Performance Requirements:

1. Input Power:
    a. Voltage: 480V, 3 PH plus ground.
    b. Voltage Tolerance: ±10% of nominal.
    c. Frequency: 60 Hz, ±3 Hz.
    d. Surge Withstand Capability: IEEE C62.41-1991 without damage.
    e. Input Fuses: Rated at 200,000 AIC (amperes interrupting capacity), Class T.
2. Output Performance:
    a. Performance of the power correction system shall be independent of the impedance of the power source. All performance levels shall be attained whether on the AC lines, backup generator, or output of UPS.
    b. Harmonic Correction:
        1) Limit the 2nd through 50th order harmonic current to < 5% TDD each installed location indicated herein. Harmonic levels for individual harmonic orders shall comply with respective levels established in IEEE 519-1992, Table 10.3.
        2) Limit the THD(V) added to the electrical system immediately upstream of the power correction system location(s) to less than or equal to 5%. The power correction system shall not correct for utility supplied voltage distortion levels.
        3) Reactive Current Compensation: To 0.95 lagging displacement power factor. Leading power factor is not permitted.
3. Current Transducers:
    a. Current transducers shall be rated for the total rated rms current of the total load at each installed location or higher.
    b. Two current transducers, mounted on Phases A and B, are to be installed per location.
    c. Each current transducer shall have a current output of 5 amperes. Current capacity of each current transducer shall be as required for the electrical system where installed.
    d. Each current transducer shall be rated for 400 Hz.
4. Environmental Conditions:
    a. The power correction system shall be able to withstand the following environmental conditions without damage or degradation of operating characteristics or life:
        1) Operating Ambient Temperature: 0 degrees C to 40 degrees C.
        2) Storage Temperature: -40 degrees C to 65 degrees C.
        3) Relative Humidity: 0 to 95%, non-condensing.
        4) Audible Noise: Generated by power correction system not to exceed 65 dbA measured 1 meter from surface of unit.

1.5 SUBMITTALS

A. Shop Drawings:
1. Product technical data including:
    a. Acknowledgement that products submitted meet requirements of standards referenced.
    b. Manufacturer's installation instructions.
2. Fabrication and/or layout drawings.
3. Certifications.
4. Test reports.

B. Contract Closeout Information:
1. Operation and Maintenance Data:
    a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

C. Approval Submittals:
1. Submittals for approvals shall include:
    a. System configuration in a single line diagram.
    b. Size and weights of shipping units to be handled by installer.
    c. Dimensional drawings with installed weight and heat dissipation for each size provided.
    d. Detailed layouts of customer power and control connections.
    e. Detailed installation drawings including all terminal locations.

f. Technical brochure detailing the standard features of the power correction system.

D. Certifications:
   a. Submit with Shop Drawings:
      1) Compliance with IEEE 519.
      2) Compliance with "Factory Tests".
   2. Prior to shipment, the manufacturer shall fully test the performance at full current and voltage while functioning as a harmonic correction device to assure compliance with equipment specifications defined herein. A certified report shall be provided to the Contracting Officer of successful completion of performance tests.

E. Programmed Settings:
   1. For each device with programmable requirements, provide the final "as-built" settings for all devices.
   2. Submit on 8.5 inches x 11 inches sheets in 3-ring binders.
   3. Provide in accordance with the submittals section.

F. Software:
   1. Provide two copies of unrestricted software for each equipment type.
   2. Provide software license for NPS Staff usage.
   3. Provide two cables of each type for connection from each equipment type to a laptop computer.

G. Refer to Section 01 81 33 – Cyber Security Requirements for required cyber security related submittals.

1.6 DELIVERY, STORAGE AND HANDLING

A. See Section 26 05 00.

1.7 WARRANTY

A. The power correction system manufacturer shall warrant products against defects in material and workmanship for a period of two years from date of shipment.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Eaton, Cutler-Hammer (Basis of Design).
   2. Or Equal.

C. Submit request for substitution in accordance with Specification Section 01 67 00.

2.2 EQUIPMENT, PERFORMANCE AND DESIGN REQUIREMENTS

A. Enclosure:
   1. Each power correction unit shall be provided in a NEMA Type 1 enclosure.

2. Units rated 100 amperes and under for total output current shall be designed for wall mounting with cable entry through the bottom, or placed within a Motor Control Center line-up. Units rated greater than 100 amperes for total output current shall be freestanding for mounting to the floor with cable entry though the top.
3. Freestanding units shall include a door-interlocked disconnect that provides power interruption when the door is opened. Disconnect shall be lockable in the power-off position. Wall mount units shall be disconnected from the power source by a disconnect device or circuit breaker contained in the power distribution center.
4. Freestanding units shall include lifting provisions by forklift truck and lifting lugs. Wall mount units weighing more than 80 pounds shall have lifting lugs.
5. All units shall include 200,000 AIC rated fuses with Class T actuation.
6. All units shall be provided with a grounding lug. Grounding by the Contractor is to be performed according to local and national standards.
7. The paint shall be the manufacturer's standard type and color.

B. Operator and Controls and Interface:
1. All units shall include a digital interface model (DIM) that includes an alphanumeric display consisting of 2 lines with 20 characters per line. All information shall be in English. Display shall be easily viewed under all lighting conditions, including sunlight, as found inside buildings.
2. Operators include run, stop, setup, enter, and up/down scroll.
3. The display shall provide operating data while functioning. Standard operating parameters available for display are AC line voltage, total rms load current, harmonic current of load, reactive current of load, output harmonic and reactive current of power correction system.
4. When the output of the power correction unit is at full rated capacity, the display shall indicate at-maximum capacity and actuate an at-maximum capacity relay.
5. All fault conditions shall be displayed as they occur. Diagnostic information shall be provided in English and clearly indicate the nature of the fault.
6. The run pushbutton shall include a green LED. LED shall be lighted when unit is operating.
7. Contacts shall be provided for operator information for power-on, run, fault and at-maximum capacity. One form C contact shall be provided for each relay.
8. A RS485 serial communication port shall be provided for remote control and diagnostic information.

C. Design:
1. All power correction units shall be defined as a power electronic device consisting of power semiconductors that switch into the AC lines to modulate its output to cancel detrimental harmonic and/or reactive currents. A DC bus shall store power for power semiconductor switching. A microprocessor shall control the operation of the power converter.
2. Each unit shall be designed with a current limiting function to protect the semiconductors. When this level is attained, a message shall be displayed indicating the output capacity is at-maximum capacity and actuate the at-maximum capacity relay. Operation shall continue indefinitely at this level without trip off or destruction of the power correction unit.
3. Each unit shall incorporate an over-temperature output roll back that reduces the total output current in order to maintain maximum current correction within the electrical system.
4. Two distinct levels of faults shall be employed. Non-critical level faults will provide automatic restart and a return to normal operation upon automatic fault clearance. Critical level faults stop the function of the unit and await operator action.
    a. Faults such as AC line over voltage, AC line under-voltage, AC line power loss, and AC line phase imbalance shall be automatically restarted. Upon removal of these fault conditions, the power correction system shall restart without user action. Automatic restart will not occur if 5 faults have occurred in less than 5 minutes. During the fault condition, except line loss, the display shall state the type of fault and indicate that automatic restart will occur. The run relay and run LED shall be disabled. The fault relay shall not be enabled unless time out occurs. Upon AC line loss, the power-on relay shall be disabled and no display shall be provided.

b. All other types of faults shall be considered critical and stop the power correction system. The display shall indicate the fault condition and "stop." The run LED and relay shall be disabled and the fault relay enabled. User shall be required to initiate a power reset (turn power off and on) to restart the power correction system.

5. The logic of the power correction system shall monitor the load current by utilizing two current transducers (CTs) mounted on Phases A and B to direct the function of the power electronic converter. The ratio of the CTs must be entered into the logic via the digital keypad/display to calibrate the operation of the power correction system. The output of the current transducers shall be 5 amperes.
6. Up to five power correction units may be installed in parallel to inject current according to the information received from one set of CTs. The units will function independently. If one unit is stopped or faulted, the remaining units will adjust accordingly to maintain optimum harmonic cancellation levels up to the capacity of the remaining units.

PART 3 - EXECUTION

3.1 FIELD QUALITY CONTROL

A. The following inspections and test procedures shall be performed by factory trained field service personnel during startup:
   1. Visual Inspection:
      a. Inspect equipment for signs of damage.
      b. Verify installation per Drawings.
      c. Verify ground conductors are properly sized and configured.
      d. Verify current transducer orientation and wiring to power correction system.
      e. Verify logic setup corresponds to performance specifications defined herein.
   2. Mechanical Inspection:
      a. Check all control wiring connections for tightness.
      b. Check all power wiring connections for tightness.
      c. Check all terminal screws, nuts, and/or spade lugs for tightness.
   3. Electrical Inspection:
      a. Check all fuses for continuity.
      b. Confirm input voltage and verify unit match.
      c. Check current transducers for proper phase location and orientation.

B. Provide training of NPS personnel as specified in Specification Section 01 75 00.

C. Startup and Site Testing: The manufacturer shall provide field test personnel to initialize all equipment. A 3-phase harmonic analyzer shall be used to measure TDD and THD(V) levels to verify performance to the specifications defined herein. Test reports shall be prepared for each point of test. Test reports shall be documented, signed, and dated. All tests shall be submitted to the site manager.

D. Testing shall be included in the overall testing requirements as defined in Section 26 29 23.

3.2 CLOSEOUT ACTIVITIES

A. Refer to Section 01 81 33 – Cyber-Security Requirements for cyber security related closeout requirements.

END OF SECTION

This page intentionally left blank.

SECTION 26 41 13
LIGHTNING PROTECTION SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Lightning protection systems using lightning rods, including lightning protection systems for:
      a. Buildings and Structures.

B. Scope:
   1. Contractor shall provide all labor, materials, equipment, tools, services, and incidentals necessary and required to provide lightning protection systems of the type indicated.
   2. Contractor shall furnish the services of individual possessing required qualifications, as indicated in this Section, to prepare the final design of the lightning protection systems required, in accordance with this Section and other Contract Documents. Such services are not delegated design.
   3. Provide lightning protection systems for the following buildings and structures:
      a. North Rim Water Treatment Plant.
         1) Water Treatment Plant Building.
         2) Raw Water Tanks No.1 and No. 2
      b. Roaring Springs Pump Station

C. Related Requirements include but are not necessarily limited to:
   1. Section 26 05 26 - Grounding and Bonding.
   2. Section 26 05 33 – Raceways and Boxes

1.2 DEFINITIONS

A. Contracting Officer: Includes Contracting Officer (CO) and Contracting Officer Representative (COR).

1.3 REFERENCES

A. Terminology: In this Section and relative to lightning protection system Work, the terminology indicated below has the following meaning, regardless of whether indicated with initial capital letters:
   1. Classification of Buildings in accordance with NFPA 780:
      a. Class I: Any commercial, industrial, or residential building less than 75 feet in height.
      b. Class II: Any commercial, industrial, or residential building 75 feet or taller.
      c. Heavy-duty stacks: Any smokestack or venting stack with a flue cross-section area greater than 500 square inches and stack height greater than 75 feet.

B. Reference Standards:
   1. ASTM International (ASTM):
      a. B8 - Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft.
   2. Lightning Protection Institute (LPI):
      a. 175 - Standard for the Design - Installation - Inspection of Lightning Protection Systems.
   3. National Fire Protection Association (NFPA):
      a. 780 - Standard for the Installation of Lightning Protection Systems.
   4. National Electrical Manufacturers Association (NEMA):
      a. TC 2 - Electrical Polyvinyl Chloride (PVC) Tubing and Conduit.
   5. Underwriters Laboratories, Inc. (UL):
      a. 96 - Standard for Safety Lightning Protection Components.

b. 96A - Standard for Installation Requirements for Lightning Protection Systems.
c. 651 - Standard for Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings.

1.4 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
1. Coordination - General:
a. Contractor shall coordinate the services of lightning protection system designer with all other elements of the Work.
b. Contractor has full responsibility for scheduling lightning protection system designs and all related Work.
c. Allow sufficient time in Progress Schedule for performance of lightning protection system services, including requests for interpretation or clarification between lightning protection system designer and Contractor and between Contractor and Contracting Officer.
2. Coordination of Lightning Protection System Work's Connections to Other Work:
a. Where lightning protection system Work connects to other Work, existing construction, or both, the lightning protection system Work shall be consistent with the other Work and existing construction to which lightning protection system Work connects, and adjacent construction.
b. Submit details, loading, anchorage, and other coordinating information necessary for the lightning protection system Work to properly interface with other Work.
c. Changes in the Work or existing construction, necessary as a result of the lightning protection system, are ineligible for increase in Contract Price or Contract Times, unless: (1) otherwise agreed by both Contracting Officer and Government, or (2) expressly indicated otherwise elsewhere in the Contract Documents for the lightning protection system Work.
d. Changes requiring extra compensation, time, or both arising from lightning protection system design aspects needed for convenience of Contractor, Subcontractor, or Supplier, are not grounds for increase in Contract Price or Contract Times.
3. Coordination of Submittals, Fabrication, Production, and Shipment:
a. Do not release for raw materials procurement, fabrication, production, and shipment to the Site materials, equipment, or systems designed by lightning protection system designer until the associated lightning protection system designer has completed the design and Contracting Officer has reviewed and approved all associated Shop Drawings, product data Submittals, Samples, and such Submittals have been delivered to and accepted by Contracting Officer.
b. Allow sufficient time in the Progress Schedule for required Submittals and required actions by lightning protection system designer and Contracting Officer.

1.5 QUALITY ASSURANCE

A. Qualifications:
1. Lightning Protection System Designer:
a. One individual, acceptable to Contracting Officer, shall design or directly supervise preparation of the final design of all lightning protection systems required for the Work.
b. Individual preparing, or directly supervising preparation of, final design of lightning protection systems shall possess current, valid "Designer Inspector" (DI) certification from Lightning Protection Institute (LPI) or "Lightning Protection Master Certification" (LPMC) from Underwriters Laboratories (UL).
c. In addition, individual performing, or directly supervising preparation of, final design of lightning protection system shall be able to furnish documentation indicating possession of not less than ten years' experience personally designing or supervising the preparation of lightning protection systems similar to that required for the Work.
d. Designer may be employee of lightning protection system Supplier or individual or entity retained by lightning protection system Supplier.

e. Submit to Contracting Officer copy of current certification as LPI DI or UL LPMC and record documenting experience of lightning protection system designer. Indicate designer's current employer and employment history.
f. Upon Contracting Officer's request, submit information for not less than five prior lightning protection systems, similar in type, scope, and extent to that required for the Work. Construction of each lightning protection system on each such project shall have been substantially complete for not less than one year at the time such documentation is submitted to Contracting Officer. Such information shall include, for each project, the following: project name and location (city, state or province, country); project owner; designer's employer and their role in the project (prime contractor, subcontractor, or supplier); name of project's design professional (company name) and location; year lightning protection system was completed; and general description of scope and extent of lightning protection system work. .

2. Installer:
a. Installer of lightning protection system may be Contractor or Subcontractor.
b. Throughout the Work, installer shall be a current, "UL Listed Lightning Protection Contractor".
c. Installer's project manager or site superintendent shall possess a current, valid certification as "Master Installer" from LPI. Such individual shall be actively involved in managing and supervising installation of the lightning protection system Work.
d. Installer shall furnish valid "UL Master Label" for the substantially completed lightning protection system Work.
e. In addition, installer shall possess not less than five years' relevant experience performing lightning protection system construction and shall be able to document not less than five completed, prior projects or similar scope and complexity to the lightning protection system Work within the most-recent five-year period.
f. Submit to Contracting Officer documentation of installer's current status as "UL Listed Lightning Protection Contractor" and copy of required LPI "Master Installer" certification.
g. Upon Contracting Officer's request, submit documentation of required experience.

1.6 SUBMITTALS

A. Action Submittals: Submit the following:
1. Shop Drawings:
a. Drawings of entire lightning protection system for each building and structure for which lightning protection system is required. Include plans, sections, schematics, and details as appropriate.
b. Plan drawings showing type, size, and locations of all lightning protection materials and equipment.Coordinate locations of air terminals and other devices with steel tank supplier and radio tower supplier.
c. Roof penetration detail drawings.
d. Submit schedules (tables) of materials and equipment as appropriate, indicating type, size, materials, and location of each, together with other pertinent information.
e. "Certificate of Compliance" by lightning protection system designer, in accordance with this Section's Article, "Responsibilities of Lightning Protection System Designer".

2. Product Data:
a. Data for all materials and equipment required by "Part 2 – Products" of this Section. Submit material and equipment manufacturers' published product data such as catalog pages, standard drawings, expected performance data, specifications, and the like.
b. Submit data sheets that include manufacturer's name and product model number. Clearly identify all optional accessories.
c. Certification that materials and equipment proposed submitted are in accordance with applicable standards of LPI or UL.

B. Informational Submittals: Submit the following:
1. Certifications and Notices:

a. Roofing system manufacturer written consent to installer's proposed penetrations (if any) through roofing system.

2. Roofing System Manufacturer's Instructions:
   a. When Contractor proposes mechanically fastening lightning protection system components to, or providing penetrations through, roofing, obtain and submit to Contracting Officer the roofing system manufacturer's:
   b. Written concurrence with proposed installation methods.
   c. Written instructions for sealing penetrations into or through roofing system.
3. Supplier's Instructions:
   a. Manufacturer's written instructions for handling, storage, and installation for all materials and equipment furnished.
4. Field Quality Control Results:
   a. Submit results of field quality control activities required in this Section.
5. Supplier's Reports:
   a. Submit written report of each visit to the Site by manufacturer's factory-trained representative.
   b. Submit written report of each visit to the Site by lightning protection system designer.
6. Qualifications Statements:
   a. Lightning protection system designer.
   b. Installer.

C. Closeout Submittals: Submit the following:
1. Post-Installation Certifications:
   a. Installer's written certification that the substantially completed lightning protection system Work complies with the Contract Documents.
   b. "UL Master Label" certificate.
2. Operation and Maintenance Data:
   a. Manufacturer's operation and maintenance manual for each building's or structure's lightning protection system provided or modified. Comply with Section 01 78 23 - Operation and Maintenance Data.
   b. Indicate requirements for, and frequency of, periodic inspections.
3. Record Documents:
   a. Prepare and submit drawings, including plans of roof or top of each building and structure for which lightning protection system was provided or modified, indicating as-constructed conditions. Include appropriate sections, details, and schematics.
   b. Record drawings shall expressly indicate their status as "as-constructed" drawings with an appropriate revision date.

1.7 RESPONSIBILITIES OF LIGHTNING PROTECTION SYSTEM DESIGNER

A. Responsibilities of lightning protection system designer employed on the Work include, but are not necessarily limited to, the following, unless specifically indicated otherwise in the associated elements of the Contract Documents where the lightning protection system is required:
1. Ethical Conduct and Professionalism: Comply with Laws and Regulations, and standards and guidelines regarding codes of ethics and codes of conduct published by relevant industry organizations, including LPI and UL.
2. Comply with Laws and Regulations and relevant design standards applicable to the subject lightning protection system Work.
3. Performance and Design Criteria Indicated in the Contract Documents and Other Information:
   a. Review performance and design criteria, indicated in the Contract Documents, that the lightning protection system Work must satisfy.
   b. Prepare written requests for interpretations or clarifications of performance or design criteria.
   c. Review existing information about the Site that constitutes Technical Data (if any, applicable to the subject lightning protection system Work), as indicated in the Supplementary Conditions.

4. Site Information and Investigations: With Contractor, obtaining all other necessary dimensions, field information, and other information necessary for preparing lightning protection system Shop Drawings.
5. Design Services: Personally, perform and prepare, or actively exercise direct, personal, supervisory control over others performing or preparing:
    a. Necessary design evaluations of conditions, materials, and equipment.
    b. Prepare the Shop Drawings and product data Submittals, and related design documents such as calculations, for the subject lightning protection system Work.
    c. Assist Contractor with applying for and obtaining permits and approvals (not previously obtained by Contracting Officer or those for whom Contracting Officer is responsible) necessary for the lightning protection system Work.
    d. Preparing modifications of the lightning protection system design documents as necessary.
6. Certification of Compliance by Lightning protection system designer:
    a. Through Contractor, submit to Contracting Officer, lightning protection system designer's written certification indicating:
    b. General Information: (1) Project name and designation, (2) Contractor name and Contract designation, (3) Subcontractor or Supplier name (when applicable), (4) full name of lightning protection system designer's business entity under which the lightning protection system services were performed, (5) full name and certification number of the individual responsible for the final design of the subject lightning protection system Work, (6) specific elements of lightning protection system Work to which the certification applies, and (7) lightning protection system designer's signature, and date of signature; apply lighting protection designer's seal when applicable.
    c. Explicit certification that the subject lightning protection system complies with:
        1) All applicable performance and design criteria indicated in the Contract Documents. Expressly indicate on certification of compliance the specific performance and design criteria used in the lightning protection system design.
        2) All Laws and Regulations.
        3) Applicable design standards commonly applicable to such types of construction. Expressly indicate such design standards on the certification of compliance.
7. Progress and Quality of Construction of Lightning protection system Work:
    a. Where appropriate for the subject lightning protection system Work, periodically visit the Site at appropriate intervals to observe the progress and quality of the subject lightning protection system Work.
    b. Where lightning protection system designer does not visit the Site during construction, keep informed of the progress and quality of the subject lightning protection system Work via discussions with Contractor, Subcontractor, and Suppliers, via photographic documentation, and other means acceptable to lightning protection system designer.
    c. Advise Contractor in writing when the subject lightning protection system Work is not in accordance with the lightning protection system designer's design documents (approved by Contracting Officer) and related Submittals approved by lightning protection system designer.
    d. Furnish to entity that retained lightning protection system designer copy of lightning protection system designer's written report of each visit to the Site.
8. Modifications to Design:
    a. Design appropriate modifications to the lightning protection system Work, including preparing new or revised certifications, reports, design drawings, sketches, design specifications, and calculations, as appropriate.
    b. Such design documents and calculations shall be submitted to Contracting Officer through Contractor to same extent original design documents Submittals and calculations, if any, where required by the Contract Documents for the subject lightning protection system Work.
9. Other services, as mutually agreed upon by lightning protection system designer and its client, or as required elsewhere in the Contract Documents.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. A/C Lightning Protection, Inc.
   2. Or equal.

2.2 PERFORMANCE AND DESIGN CRITERIA.

A. Provide lightning protection systems Work in accordance with NFPA 780, UL 96A, and LPI 175.

B. Size of air terminals, main conductors, and bonding conductors shall be in accordance with NFPA 780.

2.3 MATERIALS

A. Unless otherwise expressly indicated in the Contract Documents, lightning protection system materials shall be compatible with materials of construction for the building or structure being protected and meet the requirements of UL 96.

B. Provide lightning protection system with Class I or Class II materials, in accordance with NFPA 780.

C. Ground Rods:
   1. 5/8-inch diameter by 10 feet long.
   2. Copper-clad:
      a. Uniform coating, not less than 10 mils thick, of electrolytic copper molecularly bonded to rigid steel core.
      b. Corrosion resistant bond between copper and steel.
      c. Hard drawn, scar-resistant surface.

D. Material for conductor fasteners, connector fittings, bonding fittings, conductor splicers and through-wall or through-roof assemblies shall be cast bronze, brass, or copper, with bolt pressure connectors.

E. Material for bolts, nuts, and screws shall be stainless steel.

F. Underground conductors shall be bare, soft-drawn, stranded copper in accordance with ASTM B8.
   1. Downleads shall be copper from underground to minimum 3FT above grade in all locations, or higher depending on installation conditions, at which point the downlead material can transition to aluminum.
      a. Aluminum downleads shall not contact earth or concrete at any location, in order to avoid corrosion of the aluminum cables.

G. Material for air terminals, air terminal bases, main conductors and downlead conductors to the point of connection to copper underground conductors: Aluminum.

H. Raceways:
   1. Indicated as Schedules 40 ("PVC-40") or Schedule 80 ("PVC-80"):

a. PVC thermoplastic, with inert modifiers to improve weatherability and heat distribution.
b. Rated for direct sunlight exposure.
c. Fire retardant and low smoke emission.
d. In accordance with NFPA 70 Type PVC, NEMA TC 2, UL 651.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verification of Conditions:
1. When lightning protection systems will be installed on existing buildings or structures, carefully evaluate and become familiar with existing conditions, and tailor lightning protection system to be appropriate for the Work and the existing facility, in accordance with the Contract Documents.
2. When lightning protection systems will be installed on new buildings or structures, coordinate such Work with other elements of the associated building or structure and provide complete system in accordance with the Contract Documents.

B. Installing the Work constitutes Contractor's approval of underlying work, substrates, and field conditions prevailing at the time of the Work.

3.2 INSTALLATION

A. Installation – General:
1. Design and provide lightning protection system Work in accordance with LPI 175, NFPA 780, and UL 96A.
2. Install lightning protection system in accordance with the Contract Documents and Shop Drawings approved by Contracting Officer.
3. Roofing System and Installation of Lightning Protection System:
a. Adhesively fasten lightning protection system components to roof system unless expressly shown or indicated otherwise in the Contract Documents.
b. Do not mechanically fasten lightning protection system components to metal roof panels, metal coping or penetrate roof membrane without written consent of both roofing system manufacturer and Contracting Officer. Submit to Contracting Officer written consent of roofing system manufacturer.
c. Seal all penetrations in accordance with roofing system manufacturer's written instructions and details. Obtain such instructions and details from roofing system manufacturer.
4. Surge protective devices to be provided, by others, as shown on the Drawings.

B. Structures and Buildings:
1. Lightning protection system shall include:
a. Roof-mounted air terminals.
b. Interconnecting conductors.
1) Steel roof beams and trusses may be utilized as the main and secondary conductors.
c. Downleads: Provide the following type(s) of conductor downleads:
1) Conductors surface-mounted on the building's or structure's exterior wall. Route conductor in Schedule 80 PVC conduit for not less than one foot below finished grade and four feet above finished grade.
d. Ground terminations.
e. Bonding of other grounded systems of building or structure.
f. Bonding to grounding electrode system.
2. Connect downleads to the grounding electrode system ground ring.
3. Underground connections shall be via exothermic weld.

3.3 FIELD QUALITY CONTROL

A. Field Tests and Inspections:
   1. Perform fall-of-potential ground resistance test on each ground rod or ground ring.
      a. When resistance exceeds 25 ohms for individual ground rod:
         1) Provide additional ground rod so they are 20 feet apart, interconnect with no. 4/0 AWG and retest.
         2) If retest exceeds 25 ohms, contact Contracting Officer.
      b. When resistance exceeds 25 ohms for a ground ring, contact Contracting Officer.
   2. Perform continuity test for system elements concealed with in structure.
   3. Perform continuity test for system elements utilized structural steel as downconductors.
   4. Submit to Contracting Officer written results of required field quality control activities expressly indicating building or structure name, and Project, type of test or inspection performed, results obtained, problems observed or noted during the test or inspection, and whether results obtained indicate successful completion (i.e., acceptance). The Work will not be eligible for inspection for Substantial Completion until such results are submitted to and accepted by Contracting Officer.

3.4 CLOSEOUT

A. Documentation and Nameplates:
   1. Prior to requesting inspection for Substantial Completion of lightning protection system Work, provide the following for each lightning protection system provided or modified:
      a. Nameplate, securely fastened to building or structure, indicating: “Lightning protection system provided by”, followed by company name and address of lightning protection system installer. Name plate shall be resistant to corrosion and deterioration (to which it may be subject at its installed location), including deterioration and fading due to exposure to sunlight.
      b. Submit “UL Master Label” certification.
      c. Deliver final operation and maintenance manuals for the lightning protection system.
   2. Prior to requesting final payment, deliver to Contracting Officer the following for each lightning protection system provided or modified:
      As-constructed record drawings, in accordance with this Section’s “Submittals” Article. Such as-constructed drawings shall be developed from Shop Drawings approved by Contracting Officer and those of the Drawings showing the lightning protection system.

END OF SECTION

SECTION 26 43 13
SURGE PROTECTION DEVICES FOR LOW-VOLTAGE ELECTRICAL POWER CIRCUITS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Type PC1 SPD - High exposure locations (switchgear, switchboard, panelboard or motor control center), integrally mounted.
   2. Type PC3 SPD - Medium exposure locations (switchboard, panelboard and motor control center), integrally mounted.
   3. Type PC5 SPD - Medium or low exposure locations at individual equipment locations, external, parallel connection.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. C62.41, Recommended Practice for Surge Voltages in Low-Voltage AC Power Circuits.
      b. C62.41.1, Guide on the Surge Environment in Low-Voltage (1000V and Less) AC Power Circuits.
      c. C62.41.2, Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits.
      d. C62.45, Recommended Practice on Surge Testing For Equipment Connected to Low-Voltage (1000V and Less) AC Power Circuits.
   2. Military Standard:
      a. MIL-STD-220B, Method of Insertion Loss Measurement.
   3. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   4. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   5. Underwriters Laboratories, Inc. (UL):
      a. 1283, Standard for Electromagnetic Interference Filters.
      b. 1449, Standard for Surge Protective Devices.

B. Qualifications:
   1. Provide devices from a manufacturer who has been regularly engaged in the development, design, testing, listing and manufacturing of SPDs of the types and ratings required for a period of 10 years or more and whose products have been in satisfactory use in similar service.
      a. Upon request, suppliers or manufacturers shall provide a list of not less than three customer references showing satisfactory operation.

1.3 DEFINITIONS

A. Contracting Officer:
   1. Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

B. Clamping Voltage:
   1. The applied surge shall be induced at the 90 degrees phase angle of the applied system frequency voltage.
   2. The voltage measured at the end of the 6 inches output leads of the SPD and from the zero voltage reference to the peak of the surge.

C. Let-Through Voltage:

1. The applied surge shall be induced at the 90 degrees phase angle of the applied system frequency voltage.
2. The voltage measured at the end of the 6 inches output leads of the SPD and from the system peak voltage to the peak of the surge.

D. Maximum Continuous Operating Voltage (MCOV): The maximum steady state voltage at which the SPD device can operate and meet its specification within its rated temperature.

E. Maximum Surge Current:
1. The maximum 8 x 20 microsecond surge current pulse the SPD device is capable of surviving on a single-impulse basis without suffering either performance degradation or more than 10% deviation of clamping voltage at a specified surge current.
2. Listed by mode, since number and type of components in any SPD may very by mode.

F. MCC: Motor Control Center.

G. Protection Modes: This parameter identifies the modes for which the SPD has directly connected protection elements, i.e., line-to-neutral (L-N), line-to-line (L-L), line-to-ground (L-G), neutral-to-ground (N-G).

H. Surge Current per Phase:
1. The per phase rating is the total surge current capacity connected to a given phase conductor.
   a. For example, a wye system surge current per phase would equal L-N plus L-G; a delta system surge current per phase would equal L-L plus L-G.
   b. The N-G mode is not included in the per phase calculation.

I. System Peak Voltage: The electrical equipment supply voltage sine wave peak (i.e., for a 480/277 V system the L-L peak voltage is 679V and the L-N peak voltage is 392 V).

1.4 SUBMITTALS

A. Shop Drawings:
1. Product technical data including:
   a. Manufacturer's qualifications.
   b. Standard catalog cut sheet.
   c. Electrical and mechanical drawing showing unit dimensions, weights, mounting provisions, connection details and layout diagram of the unit.
   d. Testing procedures and testing equipment data.
   e. Create a Product Data Sheet for each different model number of SPD provided (i.e., Model XYZ with disconnect and Model XYZ without disconnect, each require a Product Data Sheet).
      1) Data in the Product Data Sheet heading:
         a) SPD Type Number per PART 2 of the Specification.
         b) Manufacturer’s Name.
         c) Product model number.
      2) Data in the Product Data Sheet body:
         a) Column one: Specified value/feature of every paragraph of PART 2 of the Specification.
         b) Column two: Manufacturer’s certified value confirming the product meets the specified value/feature.
         c) Name of the nationally recognized testing laboratory that preformed the tests.
         d) Warranty information.
      3) Data in the Product Data Sheet closing:
         a) Signature of the manufacturer’s official (printed and signed).
         b) Title of the official.
      4) Date of signature.

B. Operation and Maintenance Manuals:
1. See Specification Section 01 78 23 for requirements for:

a. The mechanics and administration of submittal process.
b. The content of the Operation and Maintenance Manuals.
2. Warranty.

1.5 WARRANTY

A. Minimum of a five year Warranty from date of shipment against failure when installed in compliance with applicable national/local electrical codes and the manufacturer's installation, operation and maintenance instructions.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

A. Materials and equipment shall be the standard products of a manufacturer regularly engaged in the manufature of the products and shall be the manufacturer’s latest standard design that has been in satisfactory use for at least one year prior to installation.
1. Materials and equipment shall conform to the respective publications and other requirements specified below and to the applicable requirements of NFPA 70.
2. Components shall be UL or third-party certified.

2.2 GENERAL

A. Standards: IEEE C62.41.1, IEEE C62.41.2, IEEE C62.45, MIL-STD 220B, UL 1283, UL 1449.

2.3 TYPE PC1 SPD

A. Product:
1. SPD tag number or electrical equipment tag number SPD is connected to.
2. Integrally mounted in switchgear, switchboards or MCCs.
3. Hybrid solid-state high performance suppression system.
a. Do not use a suppression system with gas tubes, spark gaps or other components which might short or crowbar the line resulting in interruption of normal power flow to connected loads.
4. Do not connect multiple SPD modules in series to achieve the specified performance.
5. Designed for parallel connection.
6. Field connection: Use mechanical or compression lugs for each phase, neutral and ground that will accept bus bar or #10 through #1/0 conductors.
7. Device monitor:
a. Long-life, solid state, externally visible indicators and Form C dry contact(s) that monitors the on-line status of each mode of the units suppression filter system and power loss in any of the phases.
b. A fuse status only monitor system is not acceptable.

B. Operating Voltage: The nominal unit operating voltage and configuration as indicated on Drawings.

C. Modes of Protection: All modes.
1. Three phase (delta): L-L, L-G.
2. Three phase (wye): L-N, L-L, L-G and N-G.
3. Single phase (2-pole): L-L, L-N, L-G and N-G.
4. Single phase: L-N, L-G and N-G.

D. Maximum Continuous Operating Voltage: Less than 130% of system peak voltage.

E. Operating Frequency: 45 to 65 Hz.

F. Short Circuit Rating: Equal to or greater than rating of equipment SPD is connected to.

G. Maximum Surge Current: 240,000 A per phase, 120,000 A per mode minimum.

H. Minimum Repetitive Surge Current Capacity: 4000 IEEE C High waveform impulses with no degradation greater than 10% deviation of the clamping voltage.

I. SPD Protection:
   1. Integral unit level and/or component level overcurrent fuses and sustained overvoltage thermal cutout device.
   2. An IEEE C High waveforms shall not cause the fuse to open and render the SPD inoperable.

J. Maximum Clamping Voltages: Dynamic test at the 90 degree phase angle including 6 inches lead length and measured from the zero voltage reference:

System Voltage	Test Mode	IEEE C62.41		UL 1449
		C High V & I Wave	B Combination Wave	
L-L < 250 V **L-N < 150 V**	L-L	1470 V	1000 V	800 V
	L-N	850 V	600 V	500 V
	L-G	1150 V	800 V	600 V
	N-G	1150 V	800 V	600 V
L-L > 250 V **L-N > 150 V**	L-L	2700 V	2000 V	1800 V
	L-N	1500 V	1150 V	1000 V
	L-G	2000 V	1550 V	1200 V
	N-G	2000 V	1550 V	1200 V

K. EMI-RFI Noise Rejection: Attenuation greater than 30 dB for frequencies between 100 kHz and 100 MHz.

2.4 TYPE PC3 SPD

A. Product:
   1. SPD-1 is connected to MCC-1 at the North Rim Water Treatment Plant.
   2. Integrally mounted in a switchboard, panelboards or motor control centers.
   3. Hybrid solid state high performance suppression system.
      a. Do not use gas tubes, spark gaps or other components in suppression system which might short or crowbar the line resulting in interruption of normal power flow to connected loads.
   4. Do not connect multiple SPD modules in series to achieve the specified performance.
   5. Designed for parallel connection.
   6. Field connection: Use mechanical or compression lugs for each phase, neutral and ground that will accept bus bar or #10 through #1/0 conductors.
   7. Device monitor:
      a. Long-life, solid state, externally visible indicators and Form C contact(s) that monitor the on-line status of each mode of the units suppression filter system or power loss in any of the phases.
      b. A fuse status only monitor system is not acceptable.

B. Operating Voltage: The nominal unit operating voltage and configuration as indicated on the Drawings.

C. Modes of Protection: All modes.
   1. Three phase (delta): L-L, L-G.
   2. Three phase (wye): L-N, L-L, L-G and N-G.
   3. Single phase (2 pole): L-L, L-N, L-G and N-G.
   4. Single phase: L-N, L-G and N-G.

D. Maximum Continuous Operating Voltage: Less than 130% of system peak voltage.

E. Operating Frequency: 45 to 65 Hz.

F. Short Circuit Rating: Equal to or greater than rating of equipment SPD is connected to.

G. Maximum Surge Current: 160,000 A per phase, 80,000 A per mode minimum.

H. Minimum Repetitive Surge Current Capacity: 4000 IEEE C High or B combination waveform impulses with no degradation of more than 10% deviation of the clamping voltage.

I. SPD Protection:
   1. Integral unit level and/or component level overcurrent fuses and sustained overvoltage thermal cutout device.
   2. An IEEE B combination wave shall not cause the fuse to open and render the SPD inoperable.

J. Maximum Clamping Voltages: Dynamic test at the 90 degrees phase angle including 6 inches lead length and measured from the zero voltage reference:

System Voltage	Test Mode	IEEE C62.41		UL 1449
		B Comb. Wave	B3 Ring Wave	
L-L < 250 V **L-N < 150 V**	L-L	1000 V	700 V	800 V
	L-N	600 V	400 V	500 V
	L-G	800 V	550 V	600 V
	N-G	800 V	550 V	600 V
L-L > 250 V **L-N > 150 V**	L-L	2000 V	1400 V	1800 V
	L-N	1150 V	800 V	1000 V
	L-G	1550 V	1000 V	1200 V
	N-G	1550 V	1000 V	1200 V

K. EMI-RFI Noise Rejection: Attenuation greater than 30 dB for frequencies between 100 kHz and 100 MHz.

2.5 TYPE PC5 SPD

A. Product:
   1. SPD tag number or electrical equipment tag number SPD is connected to.
   2. Externally mounted next to equipment or internally to control panel for point-of-use loads.
   3. Hybrid solid state high performance suppression system.
      a. Do not use gas tubes, spark gaps or other suppression system components which might short or crowbar the line resulting in interruption of normal power flow to connected loads.
   4. Designed for parallel connection.
   5. Enclosure:
      a. Metallic NEMA 4 or 12 for interior locations.
      b. Metallic NEMA 4 or 4X for exterior locations.
   6. Field connection:
      a. Mechanical or compression lugs for each phase, neutral and ground that will accept #10 through #1/0 conductors. OR
      b. Preinstalled lead conductors: Size per manufacturer, length as required with a maximum of 2 feet.
   7. Device monitor:

a. Long-life, solid state, externally visible indicators and Form C dry contact(s) that monitor the on-line status of each mode of the units suppression filter system or power loss in any of the phase.
b. A fuse status only monitor system is not acceptable.

8. Accessories (when specifically specified): Unit mounted disconnect switch.

B. Operating Voltage: Nominal unit operating voltage and configuration as indicated on the Drawings.

C. Modes of Protection: All modes.
1. Three phase (delta): L-L, L-G.
2. Three phase (wye): L-N, L-L, L-G and N-G.
3. Single phase (2 pole): L-L, L-N, L-G and N-G.
4. Single phase: L-N, L-G and N-G.

D. Maximum Continuous Operating Voltage: Less than 130% of system peak voltage.

E. Operating Frequency: 45 to 65 Hz.

F. Short Circuit Rating: Equal to or greater than rating of equipment SPD is connected to.

G. Maximum Surge Current: 120,000 A per phase, 60,000 A per mode minimum.

H. Minimum Repetitive Surge Current Capacity: 4000 IEEE C High or B combination waveform impulses with no degradation of more than 10% deviation of the clamping voltage.

I. SPD Protection:
1. Integral unit level and/or component level overcurrent fuses and sustained overvoltage thermal cutout device.
2. An IEEE B combination wave shall not cause the fuse to open and render the SPD inoperable.

J. Maximum Clamping Voltages: Dynamic test at the 90 degrees phase angle including 6 inches lead length and measured from the zero voltage reference:

System Voltage	Test Mode	IEEE C62.41		UL 1449
		B Comb. Wave	B Ring Wave	
L-L < 250 V **L-N < 150 V**	L-L	1000 V	700 V	800 V
	L-N	600 V	400 V	500 V
	L-G	800 V	550 V	600 V
	N-G	800 V	550 V	600 V
L-L > 250 V **L-N > 150 V**	L-L	2000 V	1400 V	1800 V
	L-N	1150 V	8000 V	1000 V
	L-G	1550 V	1000 V	1200 V
	N-G	1550 V	1000 V	1200 V

K. EMI-RFI Noise Rejection: Attenuation greater than 30 dB for frequencies between 100 kHz and 100 MHz.

2.6 SOURCE QUALITY CONTROL

A. SPD approvals and ratings shall be obtained by manufacturers from nationally recognized testing laboratories.

B. The SPD are to be tested as a complete SPD system including:
1. Integral unit level and/or component level fusing.

2. Neutral and ground shall not be bonded during testing.
3. 6 inches lead lengths.
4. Integral disconnect switch when provided.

C. The “as installed” SPD system including the manufacturers recommended circuit breaker, the SPD is connected to, will not open when tested with a IEEE C3 combination waveform.

D. Tests to be performed in accordance with IEEE C62.45:
1. Clamping voltage performance testing using IEEE C62.41 Category waveforms.
2. Single pulse surge current capacity test.
3. Repetitive surge current capacity testing.
4. Spectrum analysis for EMI-RFI noise rejection.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Type PC1 and PC3 SPD:
1. Connected in parallel to the equipment.
2. Install in dedicated electrical equipment compartment, bucket or panelboard box at the factory before shipment.
3. Provide leads that are as short and straight as possible.
4. Maximum lead length: 12 inches.
5. Minimum lead size: #2 stranded AWG or bus bar.
6. Connect leads to the equipment to be protected by one of the following means:
   a. Through a circuit breaker or molded case switch mounted in the equipment.
   b. Use manufacturer recommended circuit breaker size.
   c. Circuit breaker or switch to be operable from the equipment exterior or from behind a hinged door.

C. Type PC5 SPD:
1. Mounting options:
   a. On wall or support structure adjacent to the equipment to be protected with leads routed through conduit.
   b. Nipple connection directly to the equipment to be protected.
2. Install leads as short and straight as possible.
3. Maximum lead length: 5 feet.
4. Minimum lead size:
   a. Type PC5: #10 stranded AWG.
5. When conduit connection is used, provide a minimum of four twists per foot in the lead conductors and install in NFPA 70 sized conduit.
6. Connect leads to the equipment to be protected by one of the following means:
   a. Through a circuit breaker or molded case switch mounted in the equipment.
      1) Use manufacturer recommended circuit breaker size.
   b. Directly to the protected equipment bus, when SPD has integral disconnect switch.
   c. To the load side of field mounted equipment’s local disconnect switch.
      1) Provide taps or lugs as required to provide a UL and NFPA 70 compliant connection.

END OF SECTION

This page intentionally left blank.

SECTION 26 50 00
INTERIOR AND EXTERIOR LIGHTING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for:
      a. Interior building and exterior building mounted luminaires.
      b. Exterior luminaires.
      c. LEDs.
      d. Drivers.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI):
      a. C78.377, Specification for the Chromaticity of Solid State Lighting Products.
   2. Federal Communications Commission (FCC):
      a. Code of Federal Regulations (CFR), 47 CFR 18, Industrial, Scientific and Medical Equipment.
   3. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. C62.41, Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits.
   4. Illuminating Engineering Society of North America (IESNA):
      a. LM-79, Electrical and Photometric Measurements of Solid-State Lighting Products.
      b. LM-80, Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules.
   5. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. 410, Performance Testing for Lighting Controls and Switching Devices with Electronic Drivers and Discharge Ballasts.
      c. LE 4, Recessed Luminaires, Ceiling Compatibility.
   6. National Electrical Manufacturers Association/American National Standards Institute (NEMA/ANSI):
      a. SSL 1, Electronic Drivers for LED Devices, Arrays or Systems.
   7. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
      b. 101, Life Safety Code.
   8. Underwriters Laboratories, Inc. (UL):
      a. 248-4, Low-Voltage Fuses - Part 4: Class CC Fuses.
      b. 844, Standard for Luminaires for Use in Hazardous (Classified) Locations.
      c. 924, Standard for Emergency Lighting and Power Equipment.
      d. 1012, Power Units Other Than Class 2.
      e. 1310, Standard for Class 2 Power Units.
      f. 1598, Luminaires.
      g. 8750, Standard for Light Emitting Diode (LED) Equipment for Use in Lighting Products.
   9. United States Department of Energy (USDOE):
      a. EPAct, the National Energy Policy Act.

1.3 DEFINITIONS

A. Contracting Officer: Includes Contracting Officer (CO) and Contracting Officer Representative (COR).

B. Useful Life for LED luminaire light sources:
   1. The operating hours before reaching 70% of the initial rated lumen output (L70) with no catastrophic failures under normal operating conditions.
   2. This is also known as 70% "Rated Lumen Maintenance Life" as defined in IESNA LM-80.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data:
      a. Provide submittal data for all products specified in PART 2 of this Specification Section.
      b. Identify luminaire by Luminaire Schedule designation.
      c. Luminaire data sheet:
         1) Name of manufacturer.
         2) Complete order information (catalog number).
         3) Description of construction and optics.
         4) Total input wattage.
         5) Luminous efficacy (lumens/Watt).
         6) Photometric performance data including candlepower distribution and coefficient of utilization (CU) table.
         7) Dimensional size.
         8) Weight.
         9) UL nameplate data for luminaires used in Class 1, Division 1 and 2 areas.
         10) Effective Projected Areas (EPA) for pole mounted luminaires.
      d. Solid state Luminaire additional data:
         1) Voltage.
         2) Initial and IES L70 lumens.
         3) Luminous efficacy (lumens/Watt).
         4) Correlated Color Temperature (CCT).
         5) Color Rendering Index (CRI).
         6) Total Harmonic Distortion (THD).
         7) Lamp life.
         8) Driver manufacturer and model number.
         9) Driver life.
         10) Driver type (0-10V, constant voltage, constant current).
         11) Dimming range and control device compatibility.
         12) Remote driver: Maximum wire length to luminaire.
         13) Emergency battery driver:
            a) Compatibility with lighting module.
            b) Lumen output of lighting module in emergency operation.
            c) Battery life.
            d) Description of testing.
            e) Ambient operating temperature.
         14) Toxicity Characteristic Leaching Procedure (TCLP) compliance.
         15) DesignLights Consortium (DLC) Listing.
         16) Warranty information.
      e. Photometric calculation for manufacturers not listed in the Luminaire Schedule:
         1) Location: Per Photometric Performance Statistics Schedule on the Drawings.
            a) Calculation method: Point-to-Point.
            b) Required footcandle level: Per Photometric Performance Statistics Schedule on the Drawings.

c) Light loss factor: Per Photometric Performance Statistics Schedule Table on the Drawings.

f. See Section 26 05 00 for additional requirements.

2. Test Reports:
   a. IESNA LM-79 Test Report for Solid-State Luminaire.
   b. IESNA LM-80 Test Report Solid-State Light Source.
3. Certifications: Solid-state Luminaire Useful Life Certificate.

B. Contract Closeout Information:
1. Operation and Maintenance Data:
   a. See Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
   b. Submittal data for each component covered by warranty.
   c. Warranty.

1.5 WARRANTY

A. Minimum of a five year Warranty from date of manufacture against failure for solid-state luminaire including LED arrays, LED drivers and integral control devices. The solid-state product is considered defective if more than 15% of the individual light emitting diodes fail to illuminate.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Luminaires:
   a. Per Luminaire Schedule.
2. Solid State Light Sources:
   a. Cree.
   b. Luminaire manufacturer's proprietary system.
   c. Or Equal.
3. LED Driver:
   a. Luminaire manufacturer's standard.
   b. Or Equal.
4. Emergency ballasts:
   a. Iota Engineering.
   b. Or Equal.
5. Emergency transfer devices:
   a. Philips Bodine.
   b. Or Equal.

2.2 GENERAL REQUIREMENTS

A. Luminaires complete with LED modules and drivers.

B. Rated for area classification as indicated on the Drawings.

1. In Class I, Division 1 and 2 areas, the temperature rating of the luminaires and LED combination shall not exceed the auto-ignition temperature of the atmosphere in which the Luminaire is used.

C. Provide all recessed luminaires with gaskets of rubber, fiberglass, or equivalent material to prevent light leaks around flush trim.
   1. Provide recessed luminaires with trim gaskets cemented in proper position.

D. Provide standard plaster frame for all recessed luminaires installed in plaster walls or ceilings.
   1. Design, finish and fabricate material to preclude possibility of rust stain in plaster.

E. Coordinate luminaire mounting where recessed into building canopies prior to Submitting Shop Drawings. Confirm clearances and luminaire flange compatibility with construction.

F. Electrical components of recessed luminaires shall be accessible and removable through luminaire without having to remove luminaire from ceiling.

G. No live parts normally exposed to contact.

H. When intended for use in wet areas: Mark luminaire "Suitable for wet locations."

I. When intended for use in damp areas: Mark luminaire "Suitable for damp locations" or "Suitable for wet locations."

2.3 LUMINAIRES

A. Standards and Listings:
   1. DesignLights Consortium (DLC).
   2. UL 1598.
   3. UL 844 for hazardous locations.
   4. NEMA LE 4 for recessed locations.

B. Housings:
   1. As indicated in the Luminaire Schedule and the following:
      a. Troffer luminaires:
         1) Minimum 22 gage sheet steel.
         2) Integral end plates and trim flanges to suit ceiling construction.
         3) Wire way covers with captive retainers to allow access to electrical components without use of tools.
      b. Down Light luminaires:
         1) Minimum 22 gage sheet steel, or minimum 16 gage sheet aluminum, unless noted otherwise.
         2) Auxiliary junction box secured to mounting frame.
      c. Extruded aluminum housings, where scheduled, shall be at least 1/8 inches thick.
      d. Punch and form housings prior to finishing (post-paint).

C. Trim (Recessed Mounted):
   1. As indicated in the Luminaire Schedule and the following:
      a. For square and rectangular luminaires, miter and continuously weld corners.
      b. Miter perimeter inverted T-Bar angles at corners.
      c. Do not butt or overlap squared ends.
      d. Finish joints smooth.

D. Castings:
   1. As indicated in the Luminaire Schedule and the following:
      a. Uniform quality, free from imperfections affecting strength and appearance.
      b. Exterior surfaces, if not receiving a finish coat, shall be smooth and match adjacent surfaces. At least one coat of clear methacrylate lacquer shall be applied unless a painted finish is specified.

E. Fasteners:
   1. As indicated in the Luminaire Schedule and the following:

a. Aluminum or steel luminaires: Zinc-Nickel plated, stainless steel, or equivalent.
b. Stainless steel luminaires: Stainless steel.
c. Bronze luminaires: Bronze or stainless steel.
d. Non-metallic luminaires: Stainless steel.

F. Finishes:
1. As indicated in the Luminaire Schedule and the following:
a. Painted surfaces:
1) Manufacturer's standard metal pretreatment and baked or air-dried, light-stabilized enamel finish, acrylic, alkyd, epoxy, polyester, or polyurethane.
2) White finishes shall have minimum 85% reflectance.
b. Unpainted surfaces:
1) Interior: Clear anodic coating, satin finish.
2) Exterior: Clear anodic coating.

G. Lens/Louver Frames:
1. As indicated in the Luminaire Schedule and the following:
a. Extruded aluminum with mitered corners.
b. Hinging or other normal motion shall not cause lens or louver to drop out.
c. No light leak between frame and housing.

H. Lenses:
1. As Indicated in the Luminaire Schedule and the Following:
a. 100% virgin, UV stabilized acrylic.
b. Held securely in place but must also be removable for cleaning and servicing.
c. Luminaires with directional lenses shall include a lens orientation device to ensure that lens installation provides light distribution as designed.
d. No light leaks between the lens and the luminaire.

I. Reflectors:
1. As Indicated in the Luminaire Schedule and the Following:
a. Down Light Reflector and Baffle Finishes: First-quality "Alzak" anodized specular finish.
b. Troffer reflector finish: Integral reflectors shall be painted white after fabrication with a minimum reflectance value of 90%.

J. Gaskets:
1. As Indicated in the Luminaire Schedule and the Following:
a. Gaskets at face plates or frames of recessed luminaires which serve as ceiling trim and allow interior access.
b. Moisture seal gaskets at exterior locations and in other designated wet areas.
c. Secure frames to luminaire bodies with screws or other means, to result in tight installation, without light leaks.

K. Ventilation:
1. Ventilation openings of adequate size and quantity to permit operation of driver without affecting rated output or life expectancy. Include wire mesh screens.

L. Wiring:
1. Factory-wired to be compatible with the project electrical and controls systems.

M. Mounting Accessories:
1. Provide appropriate mounting accessories for each luminaire, compatible with various structural conditions encountered.
2. All luminaires with adjustable beam angles shall have a locking device to ensure that the beam distribution is not effected during servicing or cleaning.
3. Recessed Luminaires:

a. Plaster Frames: Provide frames for luminaires installed in gypsum board and concealed suspension system ceiling tile. Make frames of non-ferrous metal or suitably rustproof after fabrication.
b. Baffles and Gaskets: As required to prevent light leakage.
c. Flanged luminaires are required in all ceiling systems except exposed grid lay-in panel type.

4. Luminaire Suspension Material:
   a. Unfinished Spaces:
      1) 1/2 inches minimum diameter swivel stem, unless otherwise noted.
      2) Safety chain on high bay type.
   b. Finished Spaces: Unless otherwise noted.
      1) Manufactured cable or stem and outlet box canopy.
         a) Contemporary design with swivel self-aligning features.
         b) Size canopy to cover outlet box, minimize size of canopy not associated with outlet box.
         c) Finish to match luminaire.
      2) Coordinate pendant location with ceiling tiles/ceiling grid.
         a) Submit coordinated mounting accessories as part of Shop Drawing submission.
      3) Luminaires mounted on suspended ceiling grids should be provided with outlet box designed for grid mounting with direct cord entry and supported by outlet box.

2.4 SOLID-STATE LUMINAIRES - ADDITIONAL REQUIREMENTS

A. Standards:
   1. IESNA LM-79, IESNA LM-80.
   2. NEMA SSL 1.
   3. UL 1012, 1310, and 8750.
   4. UL 844 for hazardous locations.

B. Solid state modules and driver to be provided and warrantied by luminaire manufacturer.

C. Solid-State Modules:
   1. Uniform color temperature of 4000K unless otherwise noted on the Luminaire schedule.
      a. Color temperature measurement shall have a maximum 3 SDCM on the MacAdam Ellipse for frosted lensed luminaires, and 2 SDCM for other luminaire types (ANSI C78.377).
   2. Minimum color rendering index (CRI) of 80 for indoor and 70 for outdoor applications.
   3. LED module light output and efficacy: Measured in accordance with IESNA LM-79 standards.
   4. LED useful life and lumen maintenance: Measured in accordance with IESNA LM-80 standards.
   5. Driver and LED module: Minimum useful life of 50,000 hours (L70).
   6. Individual LEDs connected such that a failure of one LED will not result in a light output loss of the entire luminaire.

D. Driver:
   1. Compatible with solid-state modules and control devices specified.
   2. Operate from 60 Hz input source of 120V through 277V with sustained variations of ±10% (voltage and frequency).
   3. Input current Total Harmonic Distortion (THD): Less than 20% when operated at nominal line voltage.
   4. Power Factor: Greater than 0.90.
   5. Avoid interference with infrared devices and eliminate visible flicker.
   6. Comply with ANSI C62.41 Category A for Transient protection.
   7. Comply with the requirements of the Federal Communications Commission (FCC) rules and regulations, Title 47 CFR part 18, Non-Consumer (Class A) for EMI/RFI (conducted and radiated).

8. Dimmable drivers capable of continuous dimming over a range of 100% to 10% of rated lumen output, unless otherwise specified in Luminaire Schedule. Dimming controlled by a 0 - 10 VDC signal, unless otherwise specified in Luminaire Schedule.
9. Control device must be compatible with type of driver and coordinated prior to submission of Shop Drawings. List of compatible dimming controllers must include the range of perceived brightness. No visible flicker throughout the dimming range.
10. Remote-mounting:
    a. Provide maximum allowable distances for secondary wire runs to luminaires.
    b. Provide remote mounting hardware and enclosures as required.
11. Operating temperature range must be suitable for site temperature conditions within exterior and gasketed luminaires.

E. Emergency Battery Driver:
1. UL 924.
2. Confirm compatibility with LED modules utilized.
3. Consist of a high temperature, maintenance-free nickel cadmium battery, charger, and electronic circuitry.
4. A solid state charging indicator light to monitor the charger and battery.
5. Single-pole test switch.
6. The following product family shall be selected based on coordination with LED lamp type:
    a. Philips Bodine “BSL23C”: can operate up to 4.5W at 410mA.
    b. Philips Bodine “BSL26C”: can operate up to 5.1W at 265mA.
    c. Philips Bodine “BSL722 inches: can operate up to 23W at 770mA.
    d. Philips Bodine “BSL23C”: can operate up to 23W at 770mA in operating conditions ranging from -20 degrees C (-4 degrees F) to 60 degrees C (140 degrees F).
    e. Alternate manufacturer: Iota.

F. Luminaire properly heat sinked to assure LED junction temperature ratings are not exceeded.
1. Provide ambient operating temperature range for which product is warrantied.

2.5 EXIT SIGNS AND EMERGENCY LIGHTING UNITS

A. Standards:
1. UL 924.
2. NFPA 101.
3. Local State or City requirements.

B. Exit Signs:
1. Housing and finish: As indicated in the Luminaire Schedule.
2. LED illuminated with integral driver.
3. AC powered or AC and battery powered: As indicated in the Luminaire Schedule.
4. Battery powered units:
    a. Battery type: As indicated in the Luminaire Schedule.
    b. Self-testing/self-diagnostic.
        1) Electronic circuitry automatically test emergency lighting for a minimum of 30 seconds every 30 days and 90 minutes once a year.
    c. Consist of battery, charger, and electronic circuitry.
    d. Solid state charging indicator light to monitor the charger and battery.
    e. Single-pole test switch.
    f. A user selectable audible alarm. The alarm shall be engaged unless noted otherwise on the Drawings.

C. Emergency Lighting Units:
1. Housing: As indicated in the Luminaire Schedule.
2. Lamps: As indicated in the Luminaire Schedule.
3. Battery type: As indicated in the Luminaire Schedule.
4. Self-testing/self-diagnostic.

a. Electronic circuitry automatically test emergency lighting for a minimum of 30 seconds every 30 days and 90 minutes once a year.
5. Consist of battery, charger, and electronic circuitry.
6. Solid state charging indicator light to monitor the charger and battery.
7. Single-pole test switch.
8. A user selectable audible alarm. The alarm shall be engaged unless noted otherwise on the Drawings.

2.6 MAINTENANCE MATERIALS

A. Furnish a minimum of 10% of total of each type and amperage of fuses for fixtures indicated to be fused.

B. Spare parts are to be stored in a box clearly labeled as to its contents.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Coordinate Luminaire Types with Ceiling Construction:
1. Provide mounting hardware for the ceiling system in which the luminaire is to be installed.

B. Fasten luminaires supported by suspended ceiling systems to ceiling framing system with hold down clips.

C. Provide mounting brackets and/or structural mounting support for wall-mounted luminaires.
1. Do not support luminaire from conduit system.
2. When luminaire is supported from outlet boxes, install per NFPA 70.
3. Supports for luminaire mounted on exterior walls shall not be attached to exterior face of the wall.

D. Support surface mounted luminaires from the building structure and not from the ceiling suspension system.
1. Luminaires up to 4 feet wide and 4 feet long: A minimum of four supporting points, one at each corner.
2. Luminaires 8 feet long: A minimum of five support points, one at center of luminaire and one at each corner.
3. Luminaires smaller than 2 feet in length: A minimum of two supporting points.

E. Provide pendant luminaires with swivel hangers which will allow luminaire to swing in any direction but will not permit stem to rotate.
1. Provide hangers with enclosure rating (NEMA 1, 4, or 7) equal to enclosure requirements of area in which they are installed.
2. Swivel hangers for luminaires in mechanical equipment areas: Shock absorbing type.
3. Secure low and high bay luminaires with safety chain or safety aircraft cable to the building structure.
a. Chain or cable to prevent luminaire from falling more than 3 inches before the luminaire is caught by the chain or cable.

F. Pendant Mounted, Open, Industrial Luminaire:
1. Not in continuous rows:
a. Supported by conduit or by approved chains or cable:
b. Hardwired to ceiling mounted junction box.
2. In continuous rows:
a. Supported rigidly with conduit and fasten luminaire to each other or mount on continuous metal channel per Section 26 05 00.
b. Hardwired to ceiling mounted junction box.
c. Provide reflector alignment clips.

G. Provide access panels for recessed luminaires that require access for maintenance when such access is not provided for in design of luminaire.
   1. Locate luminaires in accordance with reflected ceiling plans.

H. Locate luminaire in exact center of ceiling tile unless otherwise indicated.
   1. Relocate incorrectly installed luminaire and replace damaged ceiling materials.

I. Mount luminaire at heights indicated in Section 26 05 00 or per Luminaire Schedule or as indicted on the Drawings.

J. Install exterior luminaires so that water cannot enter or accumulate in the wiring compartment.

K. Luminaires with Emergency Battery:
   1. Where luminaires with emergency battery are shown controlled via switching device, connect the emergency battery to corresponding unswitched circuit so emergency battery will not operate when normal power is available and switching device turns lights off. Upon failure of normal power, luminaire will operate in emergency mode regardless of switch position.
   2. Luminaire manufacturer to supply the emergency battery with luminaire.

L. Ground luminaire.

3.2 LIGHTING CONTROL

A. See Specification Section 26 09 43 for lighting control equipment.

B. Exterior wall mounted fixtures controlled as detailed on the Drawings.

3.3 ADJUST AND CLEAN

A. See Section 01 74 00.

B. Aim all emergency lighting units, so that, the path of egress is illuminated.

END OF SECTION

This page intentionally left blank.

DIVISION 27

COMMUNICATIONS

This page intentionally left blank.

SECTION 27 10 00
STRUCTURED CABLING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Material and installation requirements for the structural cabling system such as:
      a. Cabinets, racks, frames, and enclosures.
      b. Patch panels.
      c. Cable management.
      d. Rack mounted power protection and power strips.
      e. Backbone cabling.
      f. Horizontal cabling.
      g. Patch cords.
      h. Fiber Optic MicroDuct.

B. Related Sections include but are not necessarily limited to:
   1. Section 07 84 00 - Firestopping.
   2. Section 10 14 00 - Identification Devices.

1.2 REFERENCES

A. Reference Standards: Standards referenced in this section include, but are not necessarily limited to the following:
   1. Building Industry Consulting Service International (BICSI)
   2. Electronics Components Industry Association (ECA):
      a. 310, Cabinets, Racks, Panels and Associated Equipment.
   3. National Electrical Manufacturers Association (NEMA):
      a. WC 66, Standard for Category 6 and 6A, 100 Ohm, Individually Unshielded Twisted Pairs, Indoor Cables (With Or Without An Overall Shield) For Use In LAN Communication Wiring Systems.
   4. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   5. Telecommunications Industry Association/Electronic Industries Alliance/American National Standards Institute (TIA/EIA/ANSI):
      a. TIA 526-7 – Measurement of Optical Power Loss of Installed Single-Mode Fiber Cable Plant.
      b. TIA 568-0-E – Generic Telecommunications Cabling for Customer Premises.
      c. TIA 568-1-E – Commercial Building Telecommunications Cabling Standard Part 1: General Requirements.
      d. TIA 568-2-D – Balanced Twisted-Pair Telecommunications Cabling and Components Standards.
      e. TIA 568-3-D – Optical Fiber Cabling Components Standard.
      f. TIA 569-E – Telecommunications Pathways and Spaces.
      g. TIA-606-D Administration Standard for the Telecommunications Infrastructure.
      h. TIA/EIA 604-2 – FOCIS 2 Fiber Optic Connector Intermateability Standard, Type ST.
      i. TIA/EIA 604-3 – FOCIS 3 Fiber Optic Connector Intermateability Standard, Type SC and SC-APC.
      j. TIA/EIA 604-10 – FOCIS 10 Fiber Optic Connector Intermateability Standard, Type LC.

1.3 QUALITY ASSURANCE

A. Qualifications:
   1. Installers:

a. Comply with TIA-569E Standard and the BICSI Telecommunications Distribution Methods Manual (TDMM) for the installation of Communication cabling.

B. Qualifications:
1. Telecommunications Contractor:
a. Shall be regularly and professionally engaged in the business of the applications, installation, and testing of telecommunications systems and equipment.
b. Include three references of similar scope jobs completed in the last two years.
c. Supervisors and Installers shall be Building Industry Consulting Services International (BICSI) Registered Cabling Installers, Technician Level.
d. In lieu of BICSI Certification, Supervisors and Installers assigned to the installation of this system or any of its components shall have:
1) A minimum of five years of experience in the installation of the specified copper and fiber optic cable and components.
2) Factory or factory approved certification from each equipment manufacturer indicating that they are qualified to install and test the provided products.
2. Manufacturer:
a. Company specializing in manufacturing products specified in this Section with minimum 10 years documented experience in the manufacturing, assembly, and factory testing of components which comply with TIA-568-1-E, TIA-568-2-D, and TIA-568-3-D.

1.4 SUBMITTALS

A. Action Submittals:
1. Shop Drawings:
a. Fabrication and/or layout drawings, sealed and approved by a Registered Communications Distribution Designer (RCDD):
1) Layout of complete building per floor:
a) Building area boundaries, backbone systems and horizontal pathways.
2) Building area drawings:
a) Drop locations and cable identifications in accordance with TIA-606-D.
3) Telecommunications space drawings:
a) Telecommunication rooms plan views, pathway layout, mechanical/electrical layout, and cabinet, rack, backboard, and wall elevations.
4) Typical detail drawings:
a) Faceplate labeling, faceplate types, and firestopping.
5) Layout between sites across all volumes:
a) Building area boundaries, backbone systems and horizontal pathways.
2. Product Data:
a. Submittal data for all products specified in PART 2 of this Specification.
b. See Specification Section 26 05 00 for additional requirements.

B. Informational Submittals: Submit the following:
1. Certificates:
a. Telecommunications Contractor and Installers qualifications.
b. Key Personnel qualifications.
c. Manufacturer qualifications.
2. Test reports:
a. Testing plan and procedures.
b. Telecommunications cabling test results for all horizontal and backbone links.
3. Coordination between Fiber Optic cable and MicroDuct vendors. Show calculations for fill ratio that complies with MicroDuct sizing software.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

A. All Materials and equipment shall be:
   1. Appropriate for the intended use.
   2. Recognized as such by a Nationally Recognized Testing Laboratory (NRTL) such as Underwriters Laboratories (UL), ETL SEMCO (ETL), the Canadian Standards Association (CSA) or the American National Standards Institute (ANSI).
   3. NFPA 70 Compliant.
   4. Permitted by the Authority Having Jurisdiction (AHJ).

B. All products shall be new, of the latest version at time of bid, and brought to the job site in original manufacturer's packaging. Used equipment and damaged materials will be rejected.

C. Cable lubricants specifically designed for installing communications cable may be used as needed to reduce pulling tension when pulling cable into conduit.

D. All copper products shall be from a single manufacturer so that a single performance warranty covers all applications on horizontal links/channel.

2.2 EQUIPMENT RACKS

A. Floor mounted 2-post equipment racks shall be welded steel or aluminum relay racks with uprights to mount industry standard 19 inches components and equipment.
   1. Uprights shall be 3 inches deep channel, 1-1/4 inches wide, drilled and tapped 12 - 24 inches standard EIA pattern.
   2. Racks shall be provided with a standard top cross member, and predrilled base plate to allow floor fastening.
   3. Open frame equipment racks shall be 7 feet tall, 45U capacity.
   4. UL Listed.
   5. Load Capacity: 1500 pounds.
   6. Equipment mounting positions marked and numbered.
   7. Equipment racks shall be black powder coated.
   8. Chatsworth Products Inc., 48353-703 or approved equivalent.

B. Cable Management:
   1. Cable management shall be specifically manufactured for the purpose of routing cables, wires, and patch cords horizontally across 19 inches equipment racks.
      a. Horizontal Cable Manager, Front Only, 1u x 3.5 inches deep: Panduit WMPFSE or equal.
      b. Horizontal Cable Manager, Front Only, 2u x 3.5 inches deep: Panduit WMPFS1E or equal.
   2. Cable management shall be specifically manufactured for the purpose of routing cables, wires, and patch cords vertically adjacent to equipment racks.
      a. Vertical Cable Manager, NetRunner, Front/Rear, 45U, 4 x 5 inch, Panduit WMPV45E or equal.
      b. Vertical Cable Manager, NetRunner, Front/Rear, 45U, 6 x 6 inch, Panduit WMPVHC45E or equal.
   3. Cable management shall consist of ring or bracket-like devices with removable covers.
   4. Cable management shall mount to racks by screws and/or nuts and lock washers.

C. Relay Rack Grounding Strip.
   1. Provide Vertical Grounding Strip on the rear of the Relay Rack (right vertical rail).
   2. Grounding Strip shall be same Rack Unit height as Relay Rack.
   3. Panduit RGS134-1Y or approved equivalent.

2.3 TELEPHONE TELECOMMUNICATIONS OUTLET BOXESTERMINAL CABINETS

A. Factory painted or galvanized steel NEMA 3R enclosure with hinged door.
   1. Used to house building entrance protector and terminal blocks.

2.4 TELECOMMUNICATIONS CABLE RUNWAY

A. Provide cable runways with widths as shown on Drawings.
   1. Materials:
      a. Steel:
         1) Side stringers: 3/8 IN wide by 1-1/2 IN high tubular steel with 0.065 IN wall thickness.
         2) Cross members: 1 IN wide by ½ IN high tubular steel with 0.065 IN wall thickness, spaced 9 IN apart on center. There will be a maximum of 8 IN open space between each cross member.
         3) Black powder coat finish.

2.5 PATCH PANELS

A. Patch panels mounted in equipment racks with sufficient ports to accommodate all installed cable plus 25% spare:
   1. Copper Patch Panel:
      a. 48-port modular jack (2U maximum) with rear mounted type 110 insulation displacement connectors.
      b. Panel shall have incoming cable strain relief and cable management guides.
      c. Jack pin/pair configuration shall be T568B.
      d. Jacks shall be unkeyed.
   2. Fiber Optic Patch Panel:
      a. Panel shall accommodate 24 duplex connectors (2U minimum).
      b. Panel shall have cable management tray, incoming cable strain relief and cable management guides.
      c. Connectors shall be duplex LC.
      d. Connectors shall be unkeyed.
      e. Provide dust covers for unused connectors.
   3. Fiber Optic Distribution Panel:
      a. Panel shall accommodate 144 duplex connectors (6U maximum).
      b. Panel shall have cable management tray(s), incoming cable strain relief and cable management guides.
      c. Connectors shall be duplex LC.
      d. Connectors shall be unkeyed.
      e. Provide dust covers for unused connectors.

2.6 FIBER OPTIC ENCLOSURES

A. Manufacturers:
   1. Corning is Basis of Design.

B. Rack Mount Fiber Optic Enclosure:
   1. Enclosure sized to accommodate 2, 4, 6 or 12 Fiber Adapter Panels.
   2. Each Fiber Adapter Panel shall accommodate up to 6 duplex connectors (12 fibers).
   3. Adapters shall be duplex LC style.
   4. Adapters shall be unkeyed.
   5. Provide dust covers for unused adapters.
   6. Housing shall have cable management tray, incoming cable strain relief and cable management guides.
   7. Size Enclosure as shown on drawings.
   8. Fiber Optic Adapter Panels, shall hold 12 fibers or less, duplex style, LC style., Part Numbers:
      a. CCH-CP-12-XX Closet Connector Housing (CCH) 12 Fiber Panel. (XX = adapter code).
         1) A9 = LC Duplex Adapters, Single Mode (OS2).

C. Wall Mount Fiber Optic Enclosure:
   1. Enclosure shall accommodate 2, 4 or 6 Fiber Adapter Panels.

2. Each Fiber Adapter Panel shall accommodate up to 6 duplex connectors (12 fibers).
3. Adapters shall be duplex LC style.
4. Adapters shall be unkeyed.
5. Provide dust covers for unused adapters.
6. Housing shall have cable management tray, incoming cable strain relief and cable management guides.
7. Size Enclosure as shown on drawings, Part Numbers:
    a. WCH-02P Wall Mountable Closet Housing, holds two CCH connector panels/cassettes.
    b. WCH-04P Wall Mountable Closet Housing, holds four CCH connector panels/cassettes.
    c. WCH-06P Wall Mountable Closet Housing, holds six CCH connector panels/cassettes.
8. Fiber Optic Adapter Panels, shall hold 12 fibers or less, duplex style, LC style, Part Numbers:
    a. CCH-CP-12-XX Closet Connector Housing (CCH) 12 Fiber Panel. (XX = adapter code).
    b. A9 = LC Duplex Adapters, Single Mode (OS2).

2.7 BACKBONE CABLING SYSTEM

A. Fiber Optic Outside Plant Cable (OSP):
1. Optical fiber backbone cabling system shall provide connections between Entrance Facilities / Equipment Room of one building to another.
2. Size and configuration of fiber optic cables shall be as shown on Drawings.
3. Fiber shall be armored where shown on Drawings.
4. The buffered fibers shall be grouped in subunits of 6 or 12 fibers.
5. Install all cables in accordance with project Drawings.
6. Provide pull-tape in unused conduits.
7. Leave pull-tape in used conduits for future cables as needed.
8. Leave a 10-meter-long maintenance loop at each end of the link, neatly managed in the room containing Fiber Distribution Enclosure.
9. Install the fiber backbone in continuous length between Fiber Distribution Enclosures (no splices permitted).
10. Corning Indoor / Outdoor Rated Cable (Plenum):
    a. 012E8F-31131-29 FREEDM One Cable, OFNP, TB 12F, 9µm single mode (OS2)
    b. 024E8F-31131-29 FREEDM One Cable, OFNP, TB 24F, 9µm single mode (OS2)
11. Corning Outdoor Rated Cable (must terminate within 50 feet of entering building)
    a. 024ZU4-T4F22D20 ALTOS Cable, Gel-Free, 24F, 10µm single mode (OS2)
    b. 144ZU4-T4F22D20 ALTOS Cable, Gel-Free, 144F, 10µm single mode (OS2)

B. Fiber Optic Inside Plant Cable (ISP):
1. Optical fiber backbone cabling system shall provide connections between Entrance Facilities / Equipment Room and Telecommunications Rooms.
2. Size and configuration of fiber optic cables shall be as shown on Drawings.
3. The buffered fibers shall be grouped in subunits of 6 or 12 fibers.
4. Install all cables in accordance with project Drawings.
5. Provide pull-tape in unused conduits.
6. Leave pull-tape in used conduits for future cables is needed.
7. Leave a 10 meter long maintenance loop at each end of the link, neatly managed in the room containing Fiber Distribution Enclosure.
8. Install the fiber backbone in continuous length between Fiber Distribution Enclosures (no splices permitted)

C. Fiber Optic Connectors:
1. Optical Fiber Connector Intermateability Standards (FOCIS) compliant with TIA/EIA 604-10 (LC), TIA/EIA604-3(SC and SC-APC), TIA/EIA 604-2(ST), Comply with TIA/EIA 568-3-D.
2. No Epoxy/No Polish fiber optic connector.

3. Field installed, Mechanical splice technology.
4. Ceramic ferrule.
5. Fiber Optic Connectors to match Fiber grade installed.
6. Corning 95-050-99-X Unicam LC Connector MM, OM3/OM4.

2.8 HORIZONTAL CABLING SYSTEM

A. Category 6 Unshielded High Performance Twisted Pair Cable – All Work Areas. Comply with TIA 568-0-E.
   1. Panduit TX6000 High Performance Category 6 UTP Copper Cable.
   2. In addition, Category 6 UTP Copper Cable must meet the following mechanical and performance criteria:
      a. Exceeds requirements for TIA-562.2-D Category 6 and ISO 11801 2nd Edition Class E channel standards.
      b. Exceeds requirements of TIA-568.2-D and IEC 61156-5 Category 6 component standards.
      c. Meets requirements of IEEE 802.3af, IEEE 802.at and IEEE 802.bt for PoE applications.
      d. Third party tested to comply with TIA-568.2-D.
      e. Installation temperature range: 32 to 122 degrees F (0 to 50 degrees C).
      f. Operating temperature range: -4 to 194 degrees F (-20 to 90 degrees C).
      g. Cable shall be UL listed "LP" Limited Power for 0.05A. – LP (0.5A) shall be identified on the cable.
   3. Environmental Space:
      a. Plenum (CMP), Diameter .225 inches (6.2mm) nominal.
   4. Cable Color:
      a. Work Area Outlet Data/Voice – Blue.
      b. Work Area Outlet Data/Voice – White.
      c. Wireless Access Point – Green.
      d. Security Cameras – Yellow.
      e. SCADA – Orange, Red.
      f. Access Control – Orange.
      g. Lighting / Building Management Systems (BMS) – Gray.
      h. Power Management Systems (PMS) – Violet.

B. Category 6 Unshielded Twisted Pair Cable – All Work Areas.
   1. Panduit TX6 Category 6 UTP Copper Cable.
   2. In addition, Category 6 UTP Copper Cable must meet the following mechanical and performance criteria:
      a. Exceeds requirements for TIA-562.2-D Category 6 and ISO 11801 2nd Edition Class E channel standards.
      b. Exceeds requirements of TIA-568.2-D and IEC 61156-5 Category 6 component standards.
      c. Meets requirements of IEEE 802.3af, IEEE 802.at and IEEE 802.bt for PoE applications.
      d. Third party tested to comply with TIA-568.2-D.
      e. Installation temperature range: 32 to 122 degrees F (0 to 50 degrees C).
      f. Operating temperature range: -4 to 194 degrees F (-20 to 90 degrees C).
      g. Cable shall be UL listed "LP" Limited Power for 0.05A. – LP (0.5A) shall be identified on the cable.
   3. Environmental Space:
      a. Plenum (CMP), Diameter .224 inches (6.2mm) nominal.
   4. Cable Color:
      a. Work Area Outlet Data/Voice – Blue.
      b. Work Area Outlet Data/Voice – White.
      c. Wireless Access Point – Green.
      d. Security Cameras – Yellow.

e. SCADA – Orange, Red.
f. Access Control – Orange.
g. Lighting / Building Management Systems (BMS) – Gray.
h. Power Management Systems (PMS) – Violet.

C. Category 6A Unshielded Twisted Pair High Density (HD) Cable – All Work Areas.
1. Panduit TX6A UTP Copper with Vari-MaTriX HD Technology.
2. In addition, Category 6A UTP Copper Cable must meet the following mechanical and performance criteria:
a. The UTP cable must include an electrically discontinuous metallic barrier (Vari-MaTriX) to suppress the effect of alien cross talk allowing 10 Gb/s transmission. The metallic barrier is not bonded/grounded. The continuity of the metallic barrier is to be partitioned along the cable's length into segments of varying length such that EMI immunity is uncompromised.
b. Exceeds requirements for TIA-562.2-D Category 6A and ISO 11801 Class EA channel standards in a 4-connector configuration up to 100 meters.
c. Exceeds requirements of TIA-568.2-D and IEC 61156-5 Category 6A component standards.
d. Meets requirements of IEEE 802.3af, IEEE 802.at and IEEE 802.bt for PoE applications.
e. Third party tested to comply with TIA-568.2-D.
f. Installation temperature range: 32 to 140 degrees F (0 to 60 degrees C).
g. Operating temperature range: -4 to 221 degrees F (-20 to 105 degrees C).
h. Cable shall be UL listed and identified on the cable:
1) "CMP-LP (0.7A)" Limited Power rating of (0.7A).
3. Environmental Space:
a. Plenum (CMP), Diameter .230 in. (5.842mm) nominal.
4. Cable Color:
a. Work Area Outlet Data/Voice – Blue.
b. Work Area Outlet Data/Voice – White.
c. Wireless Access Point – Green.
d. Security Cameras – Yellow.
e. SCADA – Orange, Red.
f. Access Control – Orange.
g. Lighting / Building Management Systems (BMS) – Gray.
h. Power Management Systems (PMS) – Violet.

D. Category 6A High Performance Unshielded Twisted Pair Cable – All Work Areas.
1. Panduit TX6A UTP Copper with Vari-MaTriX Technology.
2. In addition, Category 6A UTP Copper Cable must meet the following mechanical and performance criteria:
a. The UTP cable must include an electrically discontinuous metallic barrier (Vari-MaTriX) to suppress the effect of alien cross talk allowing 10 Gb/s transmission. The metallic barrier is not bonded/grounded. The continuity of the metallic barrier is to be partitioned along the cables length into segments of varying length such that EMI immunity is uncompromised.
b. Exceeds requirements for TIA-562.2-D Category 6A and ISO 11801 Class EA channel standards in a 4-connector configuration up to 100 meters.
c. Exceeds requirements of TIA-568.2-D and IEC 61156-5 Category 6A component standards.
d. Meets requirements of IEEE 802.3af, IEEE 802.at and IEEE 802.bt for PoE applications.
e. Third party tested to comply with TIA-568.2-D.
f. Installation temperature range: 32 to 140 degrees F (0 to 60 degrees C).
g. Operating temperature range CMP: -4 to 194 degrees F (-20 to 90 degrees C).
h. Operating temperature range CMR: -4 to 167 degrees F (-20 to 75 degrees C).

i. Cable shall be UL listed and identified on the cable:
1) "CMP-LP (0.7A)" Limited Power rating of (0.7A).

3. Environmental Space:
a. Plenum (CMP), Diameter .250 inches (6.4mm) nominal.

4. Cable Color:
a. Work Area Outlet Data/Voice – Blue.
b. Work Area Outlet Data/Voice – White.
c. Wireless Access Point – Green.
d. Security Cameras – Yellow.
e. SCADA – Orange, Red.
f. Access Control – Orange.
g. Lighting / Building Management Systems (BMS) – Gray.
h. Power Management Systems (PMS) – Violet.

2.9 TELECOMMUNICATIONS OUTLETS

A. Faceplates.
1. Faceplates shall have the following attributes:
a. Be single-gang or double-gang.
b. Made by the same manufacturer as the jacks.
c. Supplied in colors and finishes coordinated with the Division 26 27 26 Wiring Devices.
d. Have the capability for integral labeling and identification.
e. Provide capacity for a maximum of:
1) Six individual jacks for single-gang applications.
2) Up to 12 individual jacks for double-gang applications.

2. Manufacturer:
a. Panduit CFPL2**Y Classic Faceplate, Single Gang with Label, 2-port.
b. Panduit CFPL4**Y Classic Faceplate, Single Gang with Label, 4-port.
c. Panduit KWPY Wall Phone Plate, Single Gang.

B. Copper Jacks.
1. Category 6 UTP Jacks.
a. Install category 6 jacks at each workstation location and match in the TR at the patch panel.
b. Category 6 jacks at the work area shall be color:
1) PANDUIT - CJ688TGBL (Non-Color Coordinated – Black).
2) PANDUIT – CJ688TGBU (Work Area Outlet Data/Voice – Blue).
3) PANDUIT – CJ688TGWH (Work Area Outlet Data/Voice – White).
4) PANDUIT – CJ688TGYL (Security Cameras – Yellow).
5) PANDUIT – CJ688TGGR (Wireless Access Points – Green).
6) PANDUIT – CJ688TGOR (Access Control, SCADA – Orange).
7) PANDUIT – CJ688TGRD (SCADA – Red).
8) PANDUIT – CJ688TGVL (Power Management Systems PMS – Violet).
9) PANDUIT – CJ688TGIG (Lighting/Building Management Systems BMS – Gray).
c. In addition, Category 6 Copper Jacks must meet the following mechanical and performance criteria:
1) Exceed TIA-568.2-D Category 6 and ISO 11801 2nd Edition Class EA standards.
2) Meets requirements of IEEE 802.3af, IEEE 802.3at and IEEE 802.3bt for PoE applications.
3) Be 100% tested to ensure NEXT and RL performance and be individually serialized for traceability.
4) Color-coded, keyed jack modules mechanically and visually distinguish connections to prevent unintentional mating with unlike keyed or non-keyed modular plugs accommodating more discrete networks.
5) Utilize patented enhanced Giga-TX ™Technology for jack terminations which optimizes performance by maintaining cable pair geometry and eliminating conductor untwist.

6) Have contacts plated with 50 micro inches of gold for superior performance.
7) 3rd party certified to meet the mechanical endurance to the standard requirement of IEC 60512-99-001 for support of remote power applications with test current of 2 Amperes per conductor (for future PoE++ applications).
8) Have guaranteed ability to be re-terminated a minimum of twenty times without measurable degradation of performance.
9) Have range to terminate 4-pair, 22 – 26 AWG, 100 Ohm, solid or stranded twisted pair cable.
10) Accept 6 and 8-position modular plugs without damage to conductor pins.
11) Identified options that include optional labels and icons.
12) Be compatible with Mini-Com ® Modular Patch Panels, Faceplates, and Surface Mount Boxes.
13) Have available optional RJ45 block-out device that blocks out unauthorized access to jack modules and potentially harmful foreign objects, saving time and money associated with data security breaches, network downtime, repair, and hardware replacement.

2. Category 6A UTP Jacks.
   a. Install category 6A jacks at each workstation location and match in the TR at the patch panel.
   b. Category 6A jacks at the work area shall be color:
      1) PANDUIT - CJ6X88TGBL (Non-Color Coordinated - Black).
      2) PANDUIT – CJ6X88TGBU (Work Area Outlet Data/Voice – Blue).
      3) PANDUIT – CJ6X88TGWH (Work Area Outlet Data/Voice – White).
      4) PANDUIT – CJ6X88TGYL (Security Cameras).
      5) PANDUIT – CJ6X88TGGR (Wireless Access Points – Green).
      6) PANDUIT – CJ6X88TGOR (Access Control, SCADA – Orange).
      7) PANDUIT – CJ688TGRD (SCADA – Red).
      8) PANDUIT – CJ6X88TGVL (Power Management Systems PMS – Violet).
      9) PANDUIT – CJ6X88TGIG (Lighting/Building Management Systems BMS – Gray).
   c. In addition, Category 6A Copper Jacks must meet the following mechanical and performance criteria:
      1) Exceed TIA-568.2-D Category 6A and ISO 11801 Class EA standards.
      2) Meets requirements of IEEE 802.3af, IEEE 802.3at and IEEE 802.3bt for PoE applications.
      3) Be 100% tested to ensure NEXT and RL performance and be individually serialized for traceability.
      4) Include MaTriX split foil tape to suppress the effects of alien crosstalk, allowing 10 Gb/s transmission even in high density 48-port, 1RU patch panels.
      5) Utilize patented enhanced Giga-TX ™Technology for jack terminations which optimizes performance by maintaining cable pair geometry and eliminating conductor untwist.
      6) Meets TIA-1096-A contacts plated with 50 microinches of gold for superior performance.
      7) 3rd party certified to meet the mechanical endurance to the standard requirement of IEC 60512-99-001 for support of remote power applications with test current of 2 Amperes per conductor (for future PoE++ applications).
      8) Have available a high-volume “gun-style” optional termination tool (TGJT) that reduces termination time by 25% and is ideal for high volume installations.
      9) Have guaranteed ability to be re-terminated a minimum of twenty times without measurable degradation of performance.
      10) Have range to terminate 4-pair, 22 – 26 AWG, 100 ohm, solid or stranded twisted pair cable.
      11) Utilize a universal termination cap is color-coded for T568A and T568B wiring schemes for flexibility across installations.

12) Accept 6 and 8-position modular plugs without damage to conductor pins.
13) Identified options that include optional labels and icons.
14) Be compatible with Mini-Com ® Modular Patch Panels, Faceplates, and Surface Mount Boxes.
15) Have available optional RJ45 block-out device that blocks out unauthorized access to jack modules and potentially harmful foreign objects, saving time and money associated with data security breaches, network downtime, repair, and hardware replacement.

2.10 PATCH CORDS

A. Supply patch cords equal to 1.1 times the number of cables terminated in the communication room(s).
   1. Provide for installed copper and fiber-optic systems.
   2. Length: As required for connection without strain and ability to service equipment.

B. Patch Cords, Copper:
   1. Assemblies consisting of flexible, twisted pair stranded wire with eight-position plugs at each end.
   2. Cable shall be label-verified.
   3. Cable jacket shall be factory marked at regular intervals indicating verifying organization and performance level.
   4. Patch cords shall be wired straight through; pin numbers shall be identical at each end and shall be paired to match T568B patch panel jack wiring.
   5. Patch cords shall be unkeyed.
   6. Patch cords shall be factory assembled.
   7. Patch cords shall match the color of the installed system.

C. Patch Cords, Singlemode Fiber:
   1. Assemblies consisting of flexible, 9um/125µm OS2 rated cable with duplex LC connectors at each end.
   2. Cable shall be label-verified.
   3. Cable jacket shall be factory marked at regular intervals indicating verifying organization and performance level.
   4. Patch cords shall have crossover orientation.
   5. Patch cords shall be unkeyed.
   6. Patch cords shall be factory assembled.
   7. Patch cords shall be yellow.

2.11 DEVICE CABLES

A. Supply Device cables equal to 1.1 times the number of jacks installed.
   1. Provide for installed copper and fiber-optic systems.
   2. Length: 15 foot, or as required for connection without strain and ability to service equipment for Wireless Access Points / Cameras.

B. Device Category 6 Cables, Copper:
   1. Assemblies consisting of flexible, twisted pair stranded wire with eight-position plugs at each end.
   2. Cable shall be label-verified.
   3. Cable jacket shall be factory marked at regular intervals indicating verifying organization and performance level.
   4. Device cables shall be wired straight through; pin numbers shall be identical at each end and shall be paired to match T568B patch panel jack wiring.
   5. Device cables shall be unkeyed.
   6. Device cables shall be factory assembled.
   7. Device cables shall match the color of the installed system.
   8. Panduit: UTP28SP15xx Cat 6 28 AWG UTP Patch Cord (xx = BL, BU, GR, IG, OR, RD, VL, YL).

C. Device Category 6A Cables, Copper:
   1. Assemblies consisting of flexible, twisted pair stranded wire with eight-position plugs at each end.
   2. Category 6A/Class EA channel and component performance.
   3. Exceeds all TIE-568-2-D Category 6A and ISO 11801 Class EA electrical performance requirements for all frequencies from 1 to 500 MHz.
   4. FCC and ANSI compliance: Meets TIA/EIA-1096-A; contacts plated with 50 micro inches of gold for superior performance.
   5. IEC compliance: Meets IEC 60603-7.
   6. PoE compliance: Meets IEEE 802.3af and IEEE 802.3 at for PoE application in bundle sizes up to 48 cables.
   7. Cable jacket shall be factory marked at regular intervals indicating verifying organization and performance level.
   8. Device cables shall be wired straight through; pin numbers shall be identical at each end and shall be paired to match T568B patch panel jack wiring.
   9. Device cables shall be unkeyed.
   10. Device cables shall be manufactured by the same connectivity manufacturer.
   11. Device cables shall match the color of the installed system.
   12. Panduit: UTP28X15xx Cat 6 28 AWG UTP Patch Cord (xx = BL, BU, GR, IG, OR, RD, VL, YL).

2.12 LABELING AND COLOR CODING

A. Labels shall be developed by the contractor and approved by the Contracting Officer.
   1. Labels shall be machine printed on opaque or clear tape, stenciled onto adhesive labels.
   2. Handwritten labeling is unacceptable.

B. Cable and Jacks:
   1. Voice: White.
   2. Data: Blue.

2.13 FIBER OPTIC MICRODUCT

A. Dura-Line FuturePath is Basis of Design.

B. Specifications:
   1. General:
      a. Armored.
      b. 2 IN nominal OD.
      c. Minimum 0.39 IN nominal ID per microduct.
      d. Minimum 7 ducts.
      e. Unique color for each duct.
      f. Rated for direct burial.
         1) Exposed above-grade microduct shall be installed inside 3” conduit.
      g. Manufacturer markings every foot.
      h. 20 gauge locate wire.
      i. Low-friction interior to aid in fiber pulling.
         1) Ribbed Ultra Low Friction Silicone.
      j. Rip cord to air in opening of inner sheath.
      k. Bend radius:
         1) 19 IN supported.
         2) 38 IN unsupported.

C. Materials:
   1. Outer and inner sheaths: Polyethylene.
   2. Armor: Steel.

PART 3 - EXECUTION

3.1 INSTALLATION

A. System components and appurtenances shall be installed in accordance with NFPA 70, manufacturer's instructions and as shown.

B. Necessary interconnections, services, and adjustments required for a complete and operable signal distribution system shall be coordinated with the local communications provider(s).

C. Components shall be labeled in accordance with TIA 606-D.

D. Penetrations in fire-rated construction shall be firestopped.

E. Wiring shall be installed in accordance with TIA/EIA/ANSI Standards.
   1. Wiring, and terminal blocks and outlets shall be marked in accordance with TIA 606-D.

F. Cables shall not be installed in the same cable tray, utility pole compartment, or floor trench compartment with AC power cables.

G. Equipment Racks:
   1. Open frame equipment racks shall be bolted to the floor.
   2. Cable guides shall be bolted or screwed to racks.
   3. Racks shall be installed level.
   4. Wall mounted racks shall be secured to the mounting surface to prevent fully loaded racks from separating from the mounting surface.

H. Rack Mounted Equipment: Equipment to be rack mounted shall be securely fastened to racks by means of the manufacturer's recommended fasteners.

I. Terminal Blocks:
   1. Terminal blocks shall be mounted in orderly rows and columns.
   2. Adequate vertical and horizontal wire routing areas shall be provided between groups of blocks.
   3. Industry standard wire routing guides shall be utilized.

J. Pathway System:
   1. Provide in accordance with TIA 569-E and NFPA 70.
   2. Provide conduits in accordance with 26 05 33 – Raceways and Boxes.
   3. Provide bonding of raceways and cable tray in accordance with TIA 607-D and NFPA 70.

K. Unshielded Twisted Pair Patch Panels:
   1. Patch panels shall be mounted in equipment racks with sufficient modular jacks to accommodate the installed cable plant plus 25% spares.
   2. Cable guides shall be provided above, below and between each panel.

L. Fiber Optic Patch Panels:
   1. Patch panels shall be mounted in equipment racks with sufficient connectors to accommodate the installed cable plant plus 25% spares.
   2. Cable guides shall be provided above, below and between each panel.

M. Backbone and Horizontal Distribution Cable:
   1. Cable pulling tension shall not be exceeded.
      a. 110N (25 pounds-force) for copper cabling.
   2. Cable shall not be stressed such that twisting, stretching, or kinking occurs.
   3. Cable shall not be spliced.
   4. All backbone cable shall be installed in conduit or cable tray.
   5. All horizontal cable shall be installed in an appropriate telecommunications pathway.
   6. Cable shall not be run through structural members or in contact with conduits, pipes, ducts, or other potentially damaging items.
   7. Placement of cable parallel to power conductors shall be avoided, if possible; a minimum separation of 12 inches shall be maintained when such placement cannot be avoided.

8. Cables shall be terminated; no cable shall contain unterminated elements.
9. Minimum bending radius shall not be exceeded during installation or once installed.
10. Only fabric hook and loop fasteners shall be used to wrap cables, 1/2 inches width minimum. Plastic or nylon cable ties shall not be used.
11. Coordination with MicroDuct vendor for optimal duct/MicroDuct fill ratio.

N. Telecommunications Outlets:
   1. Faceplates: As a minimum each jack shall be labeled as to its function and a unique number to identify cable link.
   2. Cables:
      a. Unshielded twisted pair cables shall have a minimum of 12 inches of slack cable loosely coiled into the telecommunications outlet boxes.
      b. Minimum manufacturers bend radius for each type of cable shall not be exceeded.

O. Fiber Optic Microduct
   1. Microduct shall be installed per manufacturer standards.
   2. Microducts must be connected to the same colors within a pull box. Push-Lock couplers made for direct burial must be utilized for all microducts within a pull box. For splices of the microduct not in a pull box a manufacturers splice kit must be utilized.
   3. Coordination with fiber optic cable for an optimal duct/MicroDuct fill ratio.

3.2 TERMINATION

A. Cables and conductors shall sweep into termination areas; cables and conductors shall not bend at right angles.
   1. Manufacturer's minimum bending radius shall not be exceeded.
   2. When there are multiple system type drops to individual workstations, relative position for each system shall be maintained on each system termination block or patch panel.
   3. Unshielded Twisted Pair Cable:
      a. Each pair shall be terminated on appropriate outlets, terminal blocks, or patch panels.
      b. No cable shall be unterminated or contain unterminated elements.
      c. Pairs shall remain twisted together to within the proper distance from the termination as specified in the TIA/EIA/ANSI 568B Series.
      d. Conductors shall not be damaged when removing insulation.
      e. Wire insulation shall not be damaged when removing outer jacket.
   4. Fiber Optic Cable:
      a. Each pair shall be terminated with appropriate connectors.
      b. No cable shall be unterminated or contain unterminated elements.

3.3 GROUNDING

A. Signal distribution system ground shall be installed in the telecommunications entrance facility and in each telecommunication closet in accordance with TIA607-D.
   1. Equipment racks shall be connected to the electrical safety ground.

3.4 LABELING

A. All cables will be labeled using color labels on both ends per TIA 606-D.

B. All workstation and patch panel connections will be labeled using color coded labels per TIA 606-D.

3.5 TESTING

A. Testing shall conform to the TIA/ANSI Standards for all test parameters.
   1. All test data sheets shall be downloaded from the tester, printed out and provided to the Contracting Officer.
   2. Provide Contracting Officer with all test results as Electronic Documents (in portable document format, PDF files) by Electronic Means.
   3. Tester shall be capable of testing parameters for the warranted system.

B. Materials and documentation to be furnished under this Specification are subject to inspections and tests.
   1. All components shall be terminated prior to testing.
   2. Equipment and systems will not be accepted until the required inspections and tests have been made, demonstrating that the signal distribution system conforms to the specified requirements, and that the required equipment, systems, and documentation have been provided.

C. Unshielded Twisted Pair Tests:
   1. All metallic cable pairs shall be tested for proper identification and continuity.
   2. All opens, shorts, crosses, grounds, and reversals shall be corrected.
   3. Correct color coding and termination of each pair shall be verified in the communications closet and at the outlet.
   4. Horizontal wiring shall be tested from and including the termination device in the communications closet to and including the modular jack in each room.
   5. These tests shall be completed, and all errors corrected before any other tests are started.

D. Category 6 Circuits:
   1. Perform Category 6 link tests in accordance with TIA-568-1-E and TIA-568-2-D. Tests shall include wire map, length, insertion loss, NEXT, PSNEXT, ELFEXT, PSELFEXT, return loss, propagation delay, and delay skew.
   2. Cables which contain failed circuits shall be replaced and retested to verify the standard is met.

E. Fiber Optic Cable:
   1. Unless stated otherwise, tests shall be performed from both ends of each circuit. per TIA-526-7 and TIA-526-14-C.
   2. All terminations shall be visually inspected for scratches, pits or chips and shall be reterminated if any of these conditions exist.
   3. Each link shall be tested for insertion loss using a light source similar to that used for the intended communications equipment.
   4. High-resolution Optical Time Domain Reflectometer (OTDR) tests shall be performed for each fiber.
   5. Scale of the OTDR trace shall be such that the entire circuit appears over a minimum of 80% of the X-axis.

F. Fiber Optic Microduct:
   1. Pressure test each microduct to 100 PSI for a period of at least 15 minutes per microduct. Maximum pressure loss is limited to 10 PSI.
   2. Mandrel test each microduct path with a mandrel to prove each path is clear, not kinked and able to blow fiber in the line. It is allowable to perform testing between pull boxes or terminations. If this option is chosen, the line must be reconnected and pressure tested after mandrel testing is successful to prove all connectors are reattached.

END OF SECTION

DIVISION 31

EARTHWORK

This page intentionally left blank.

SECTION 31 10 00
SITE CLEARING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Site clearing Work, including:
      a. General provisions for site clearing Work.
      b. Preparation for site clearing Work.
      c. Clearing and grubbing.
      d. Tree protection, selective removals, and selective trimming.
      e. Disposal and cleaning.
      f. Stripping and stockpiling of topsoil.

B. Scope:
   1. The work under this Specification includes removal of existing vegetation within the limits of disturbance as shown on Project Plans.
      a. See
   2. The contractor is advised that the recommended season for tree removal is August 15 to April 1.
      a. If tree removal is performed between April 1 and August 15, the contractor shall survey for active bird nests.
      b. If an active nest is discovered, work within one half mile of that nest will be prohibited until the birds have fledged.
      c. Survey work shall be performed by a qualified biologist.

C. Related Requirements: Include but are not necessarily limited to:
   1. Division 00 – Procurement and Contracting Requirements
   2. Division 01 – General Requirements
   3. Section 31 22 00 - Site Grading
   4. Section 31 23 00 - Earthwork
   5. Section 31 23 33 – Utilities - Trenching and Backfilling.

1.2 REFERENCES

A. Terminology:
   1. Terms indicated below are not defined terms indicated with initial capital letters but, when used in this Section, have the meanings indicated below:
      a. “Arborist” means Subcontractor retained to perform Work requiring a specialist in trees, shrubs, brush, other vegetation of the types existing at the Site, and possessing not less than the required qualifications indicated in this Section.
      b. “Clearing and grubbing” means removing and disposing of all: (1) trees, brush, and other vegetation, logs, and similar items (“clearing”); and (2) stumps, roots, logs, rubbish, and debris on or in the soil (“grubbing”) after Clearing. Clearing and grubbing includes grinding and removing of stumps. When clearing and grubbing, topsoil stripping and stockpiling, and required demolition Work is complete, the Site will be ready for grading and other new construction.
      c. “Demolition” means removal, whether in whole or in part, of existing human-made construction, such as removal of buildings, structures, and building systems; site work (such as pavement, curbs, sidewalks, gutters) and the like; Underground Facilities; and other existing construction.
      d. “Selective removal” means removal of specific trees, shrubs, brush, and other vegetation, whether as shown or indicated in the Contract Documents or as directed at the Site by Contracting Officer’s Representative.

e. “Selective trimming” means removal of selected parts of trees, shrubs, brush, and other vegetation, performed by arborist, for the purpose of either: (1) allowing installation of new construction adjacent to or through the tree, shrub, brush, or vegetation, or 2) removing damaged or unhealthy growth, to allow balance of the subject tree, shrub, brush, or vegetation to continue normal, healthy growth.
f. “Site clearing” means all the Work required by this Section and related Drawings.
g. “Topsoil” means existing material at the Site, visible after clearing and grubbing, to be stripped, when such material is friable, clay loam, surface soil present in depth of not less than four inches. Topsoil shall be free of subsoil, clay lumps, stones, and other objects over two-inch diameter and other objectionable material. Topsoil required for planting and landscaping Work in the Specifications of Division 32 may differ from the meaning indicated in this Section.

B. Reference Standards:
1. ASTM International (ASTM):
a. C700, Standard Specification for Vitrified Clay Pipe, Extra Strength, Standard Strength, and Perforated.
b. D448, Standard Classification for Sizes of Aggregate for Road and Bridge Construction.
2. American National Standards Institute (ANSI):.
a. A300, Standard for Tree Care Operations – Tree, Shrub, and Other Woody Plant Maintenance – Standard Practices.
3. International Society of Arboriculture (ISA):
a. Container Rootball Shaving.
b. Crown Correction.
c. Balled and Burlapped Root Correction.
d. Container Root Correction.
e. High Branched Crown Observation.
f. Low Branched Crown Observation.
g. Multiple Low Branches Crown Observation.
h. Balled and Burlapped Root Observation.
i. Container Root Observation - Tree.
j. Protection Maintenance.

1.3 ADMINISTRATIVE REQUIREMENTS

A. Pre-Clearing Conference:
1. Prior to commencing site clearing Work, actively participate in conference regarding site clearing Work. Date and time of conference will be mutually established by Contractor and Contracting Officer.
2. Location: Conference will be held at the Site in a location and time determined by the Contractor and the Contracting Officer or be by online video conference established by Contracting Officer.
3. Participants:
a. Required Participants: Contractor’s project manager, Contractor’s site superintendent, arborist, and the Contracting Officer Representative.
b. Potential Participants: Contracting Officer
4. Contracting Officer or Contracting Officer’s Representative will chair the conference, prepare and distribute an agenda, and prepare and distribute a record of the conference.
5. Participants shall be prepared to discuss:
a. Project requirements and the Contract requirements for site clearing Work.
b. Submittals required for site clearing Work.
c. Schedule for site clearing Work and availability of resources.
d. Locations and types of site clearing Work, with specific attention to areas of selective removals, selective trimming, and notable species. Limits of clearing and grubbing. Proposed locations for stockpiles and temporary storage of cuttings and debris.
e. Environmental and other conditions at the Site.

f. Growth, vegetation, and existing construction requiring protection.
g. Required observations and field quality control for site clearing Work.

6. If additional information affecting site clearing Work subsequently comes to light, reconvene the conference at earliest opportunity.

1.4 QUALITY ASSURANCE

A. Qualifications:
1. Arborist:
a. Where selective removal, selective trimming, or both are required or necessary, retain the services of, and employ in the Work, accredited arborist, possessing qualifications acceptable to Contract Officer.
b. Arborist shall be skilled, trained, and possess documented experience in successfully protecting, trimming, and restorative care of trees and shrubs of the types necessary for the Work.
c. Certification: Arborist shall be certified by either International Society of Arboriculture (ISA) or American Society of Consulting Arborists (ASCA).
d. Arborist shall use in selective removal and selective trimming Work only workers with specific skill and successful experience in this type of Work required. Such workers shall work under the direct, personal supervision of arborist.
e. Submit to Contract Officer names, employer(s), certifications, other relevant qualifications and record of relevant experience, as indicated below, for not less than three successful projects involving same species of trees, shrubs, and vegetation as involved in the site clearing Work:
1) Names and telephone numbers of site owners, architects or engineers responsible for projects.
2) Approximate contract price of the selective removal and selective trimming of trees, shrubs, and vegetation.
3) Approximate time of year work was performed on each referenced project.
4) Approximate quantity and types of selective removal and selective trimming of performed.
5) General indication of species of trees, shrubs, and vegetation involved.

2. Biologists performing site survey as described above shall be qualified by U.S Fish and Wildlife Service for this purpose.

1.5 SUBMITTALS

A. Action Submittals: Submit the following:
1. Shop Drawings:
a. Limits of Site Clearing Work: When limits of site clearing Work are not expressly shown on the Drawings, or where Contractor proposes alternative limits of site clearing Work, submit drawings, developed from the Drawings, clearly indicating proposed limits of site clearing Work, with dimensions indicated where appropriate.
b. Where proposed limits of clearing and grubbing, selective removals, and topsoil stripping are not identical with each other, clearly and expressly indicate limits of each on the Shop Drawings.

B. Informational Submittals: Submit the following:
1. Permits and Approvals:
a. Submit copy of each permit required and obtained for site clearing Work, issued by authority having jurisdiction.
b. Where Government permission or approval is required for selected site clearing activities, submit copy of Contract Officer’s written permission or approval for such activity.
c. When approval of Contracting Officer of property (other than Contracting Officer) is required for selected site clearing activity, submit to Contracting Officer written copy of such approval.
2. Certificates:

a. Arborist's certification that trees and shrubs shown or indicated to remain were protected during the site clearing Work in accordance with the Contract Documents.
b. Where trees or shrubs (shown or indicated to remain) were damaged during site clearing Work, submit arborist's certification that such trees and shrubs were promptly and properly treated or, where successful treatment was not feasible, were replaced.

3. Qualifications Statements:
   a. Arborist qualifications.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 MATERIALS

A. Tree Wound Dressing:
   1. Provide tree wound dressing, waterproof, adhesive and elastic with an antiseptic, free from kerosene, coal tar, creosote, and other material injurious to life of tree or other plant.

B. Filter Fabric:
   1. In accordance with requirements for temporary erosion and sediment controls indicated in Section 01 50 00 – Temporary Facilities and Controls.

C. Drainage Fill:
   1. Selected crushed stone, or crushed or uncrushed gravel, washed.
   2. Size 24 (in accordance with ASTM D448), with 90 to 100 percent passing 2.5-inch sieve and not more than 10 percent passing 3/4-inch sieve.

D. Burlap:
   1. Jute, not less than 7.2 ounces per square yard.

E. For other materials needed for site clearing Work, such as stone, topsoil, or other, comply with requirements of the Contract Documents.

PART 3 - EXECUTION

3.1 SITE CLEARING – GENERAL

A. Limits of Site Clearing Work:
   1. Limits of site clearing Work are shown and indicated on the Drawings.

B. General Provisions for Site Clearing:
   1. Contractor shall provide all labor, materials, equipment, tools, services, and incidentals necessary and required to perform site clearing Work in accordance with the Contract Documents.
   2. Perform site clearing Work to avoid creating nuisances, pollution, and preventable adverse effects on the environment.
   3. Comply with Section 01 73 40 - Execution, and this Section's Article, "Tree Protection, Selective Removals, and Selective Trimming".
   4. Excavation required for site clearing shall comply with Section 31 23 00 - Earthwork.
   5. Requirements of Authorities Having Jurisdiction:
      a. Perform site clearing Work in accordance with Laws and Regulations.
      b. Obtain, pay for, and comply with permits required for site clearing Work. Obtain from authorities having jurisdiction and furnish copy of each permit as a Submittal.
   6. Traffic Control:
      a. Where site clearing Work is along or adjacent to travelled ways subject to public traffic, comply with Section 01 50 00 – Temporary Facilities and Controls.
      b. Establish such controls prior to and maintain in place throughout site clearing Work that has potential to affect traffic.

7. Site clearing Work shall comply with ANSI A300 and applicable ISA standards indicated in this Section's "References" Article, unless the Contract Documents indicate more-stringent requirements.
8. Demolition Work is indicated in Section 02 41 00 - Demolition, including removal of site improvements and Underground Facilities.

3.2 PREPARATION

A. Permits and Approvals:
   1. Do not commence site clearing Work until necessary permits and approvals are obtained and copies furnished to Contracting Officer as Submittals.

B. Delineation of Limits of Site Clearing Work Areas:
   1. Locate and clearly flag at the Site:
      a. Limits of site clearing Work.
      b. Trees, shrubs, vegetation, and other materials to remain within limits of site clearing Work.
      c. Trees, shrubs, and other vegetation to be selectively trimmed.
      d. Trees, shrubs, and vegetation to be selectively removed.
      e. Salvageable trees, shrubs, and vegetation (to be relocated) within limits of site clearing Work.
   2. Flagging:
      a. Flagging shall be high-visibility type. Where necessary, provide lath or stakes driven into the ground, with flagging, to clearly delineate limits.
      b. Provide different-colored flagging for each type of delineation required by this Article.
      c. Promptly replace lost, moved, or destroyed flagging until Contracting Officer's Representative concurs that flagging is no longer needed.
   3. Review with Contracting Officer:
      a. Before starting site clearing Work, other than flagging, review at the Site with Contracting Officer or Contracting Officer's Representative.
      b. Make corrections as necessary.
      c. Review all trees, 5inch DBH (diameter at breast height) and larger, to be removed or selectively trimmed.
      d. Review with Contracting Officer's Representative at the Site trees and shrubs to be selectively trimmed, to reach mutual agreement on extent of selective trimming required.

C. Protection: Establish protection of trees, shrubs, and vegetation to remain, in accordance with this Section's Article, "Tree Protection, Selective Removals, and Selective Trimming", and other applicable provisions of the Contract Documents.

D. Temporary Erosion and Sediment Controls:
   1. Provide applicable temporary erosion and sediment controls before commencing clearing and grubbing and topsoil stripping Work.
   2. Comply with temporary erosion and sediment control requirements of Section 01 57 23 - Temporary Stormwater Pollution Prevention and Section 31 25 00 Erosion and Sedimentation Controls.
   3. Continue providing temporary erosion and sediment controls as clearing and grubbing and topsoil stripping and stockpiling Work progresses into previously uncleared, ungrubbed areas of the Site...

3.3 CLEARING AND GRUBBING

A. Clearing and Grubbing – General:
   1. Prior to the contractor performing clearing operations, NPS staff will perform vegetation salvage activities within the limits of work including seed collection and ponderosa sapling harvesting. The NPS will transport and store salvaged vegetation.

a. Contractor to coordinate with the Contracting Officer Representative and confirm vegetation salvage work has been completed by the NPS prior to site clearing operations.

2. Following review and approval of limits of disturbance and installation of temporary fencing, contractor shall clear from within the approved limits of disturbance all vegetation not marked to remain.
   a. Include shrubs, brush, downed timber, rotten wood, heavy growth of grass and weeds, vines, rubbish, structures and debris.
3. Grub from within limits of disturbance all stumps, roots, root mats, logs, and debris encountered. Cavities resulting from the removal of stumps or other materials shall be backfilled with material approved by the Contracting Officer.

B. Trees and Shrubs Improperly Destroyed or Damaged:

C. Where, in the opinion of the Contracting Officer, vegetation to be preserved is significantly damaged by clearing and grubbing operations, the contractor shall be required to cute and remove from site damaged vegetation at no additional cost. Trees and Shrubs to Remain:
   1. Trees and shrubs to remain shall be protected, and trimmed where necessary or required, in accordance with this Section's Article, "Tree Protection, Selective Removals, and Selective Trimming".

3.4 TREE PROTECTION, SELECTIVE REMOVALS, AND SELECTIVE TRIMMING

A. General Provisions for Tree Protection, Selective Removals, and Selective Trimming:
   1. Provide temporary fencing, barricades, or guarding measures, as recommended by arborist, outside drip line of trees and shrubs to remain.
   2. Protect root systems from damage caused by noxious materials, storm water runoff, site clearing, planting and landscaping, other Work, and storage of materials and equipment. Protect root systems from flooding, erosion, and excessive wetting resulting from dewatering of excavations, drainage of tanks, and other construction activities.
   3. Do not store matter resulting from site clearing or demolition, topsoil, or other excavated material within drip line of trees and shrubs to remain. Vehicles are not allowed within drip line without authorization from the Contracting Officer. Restrict foot, vehicle and equipment traffic to prevent compaction of soil over root systems. Where such activities are unavoidable, and only as acceptable to Contracting Officer, provide temporary, continuous, heavy-duty wood planking effectively fastened together and capable of distributing loads from such activities. Temporary planking shall be underlaid by layer of filter fabric covered with two-inch layer of gravel.
   4. Cut branches and roots, when necessary, with sharp pruning instruments; do not break or chop. Fully paint cuts 1/2-inch and larger in size with tree wound dressing.

B. Excavation and Protection of Trees and Shrubs:
   1. Excavate within drip line of trees only where shown.
   2. Where trenching for utilities is required within drip line, tunnel under or around main lateral feeder roots by drilling, auger boring, pipe jacking, or digging by hand under supervision of arborist. Do not cut main lateral roots or tap roots; cut smaller roots, which interfere with installation of the Work.
   3. Where excavation for the Work is required within drip line of trees or shrubs, hand excavate to minimize damage to root systems. Perform excavation under supervision of arborist. Provide temporary shoring or other protective support systems at excavations, to minimize sloping and benching of excavations. Use narrow tine spading forks and comb excavated material to expose roots.
   4. Relocate roots in backfill areas wherever possible. If large, main lateral roots are encountered, expose beyond excavation limits as required to bend and relocate without breaking. If encountered immediately adjacent to location of construction and relocation is not practical, cut roots approximately three inches back from construction.

5. Do not allow exposed roots to dry out before permanent backfill is provided; provide temporary earth cover. Water and maintain in moist condition and temporarily support and protect from damage until permanently relocated and covered with earth.

C. Grade Adjustments and Tree Protection:
   1. Maintain existing grade within drip line of trees, unless otherwise shown or indicated.
   2. Lowering of Preconstruction Grade:
      a. Where required finish grade is below preconstruction grade around trees and shrubs, grade beyond drip line. Maintain preconstruction grade within drip line of trees and shrubs.
      b. Prune tree and shrub roots exposed during grade lowering, or provide permanent protections as recommended by arborist. Do not cut main or lateral roots or tap-roots; cut only smaller roots.
      c. Compensate for loss of roots and prune branches to stimulate root growth.
      d. Provide extended service through completion of the Contract correction period as recommended by arborist.
   3. Minor Fills:
      a. Where preconstruction grade is six inches or less below elevation of finish grade shown, fill with topsoil complying with quality requirements of Specifications for finish grading and landscaping.
      b. Place in single layer and do not compact.
      c. Hand-grade to required finish elevations.
   4. Moderate Fills:
      a. Where preconstruction grade is more than six inches, but less than 16 inches, below finish grade elevation, provide layer of drainage fill and filter fabric on preconstruction grade prior to placing topsoil that complies with quality requirements of Specifications on finish grading and landscaping.
      b. Carefully place drainage fill against trunk of tree or shrub approximately two inches above elevation of finish grade and extend not less than 1.5 feet from tree or shrub trunk on all sides.
      c. Provide filter fabric with edges overlapping by 6 inches, minimum.
      d. Place fill layer of topsoil to finish grade elevation.
      e. Do not compact stone, gravel or topsoil layers. Hand-grade to required finish elevations.
   5. Deep Fills:
      a. Provide an open dry circular well of durable, unmortared stone, located not less than two feet s from trunk of associated tree or shrub.
      b. To facilitate proper drainage, provide eight to 10 continuous runs of four-inch drain tiles horizontally on the preconstruction ground surface under the complete spread of branches in radial pattern around the tree or shrub.
      c. Slope drain tiles away from tree or shrub.
      d. Provide drainage fill on preconstruction ground surface in depth of two inches under and six inches over drain tiles.
      e. Provide filter fabric over entire top surface of drainage fill.
      f. Provide eight to 10 drainage tiles, each four-inch diameter, vertically in radial pattern around tree or shrub, located five feet from trunk of tree or shrub. Extend vertical drainage tiles from the filter fabric layer above the horizontal drainage tiles to postconstruction finished grade
      g. Hold drainage tiles in place with drainage fill.
      h. Hand-grade to required finish grade elevation.

D. Tree and Shrub Pruning (Selective Trimming):
   1. Pruning necessary for the project and future periodic maintenance in the area, will adhere to the Park's pruning guidelines with the goal of retaining health and integrity of trees and shrubs treated. Damage to trees or roots in or adjacent to project areas during construction will be avoided as much as possible

2. Tree pruning shall be conducted in accordance with *Grand Canyon National Park Pruning Protocols* available from the Contracting Officer's Representative.
    a. Pruning shall occur at the branch collar and the branch collar shall remain intact.
    b. No more than 20% of the total tree canopy will be pruned (the green leaf or needle-bearing area of branches).
    c. Tree roots shall be pruned if they are exposed during the project. No more than 33% of the tree's roots shall be removed and not more than 20% from one side shall be removed.

3.5 DISPOSAL AND CLEANING

A. Disposal – General:
    1. Do not use cleared, grubbed, or trimmed material as fill, backfill, or in embankments.
    2. Dispose of cleared, grubbed, and trimmed material, and other materials, rubbish, and debris, in accordance with Laws and Regulations.
    3. Pay all costs associated with transporting and disposing of materials and debris resulting from site clearing Work.
    4. Do not sell removed materials at the site. If materials, equipment or debris will be sold by contractor, remove the items from the park boundary and perform the sale or transaction elsewhere, in accordance with Laws and Regulations.

B. Tree Removal
    1. Contractor shall salvage all 6 IN diameter trees or larger. These trees shall be limbed and cut into 4 FT logs.
        a. North Rim: Contactor shall transport all logs to a final off-site location as indicated in Section 01 11 00. Slash and limbs shall be removed from the Park by the Contractor.
        b. Innercanyon: Contractor may dispose of logs at innercanyon locations approved by the Contracting Officer's Representative.
    2. Stumps not removed due to construction shall be flush cut, not to exceed 6 IN in height above the ground.

C. Cleaning:
    1. Perform progress cleaning and other cleaning Work, and disposal of resulting materials and debris, in accordance with Section –01 73 40 - Execution.

D. Burning:
    1. Burning of trees, shrubs, brus, other vegetation, and other materials at or adjacent to the Park is not allowed.
    2. Do not burn combustible materials on site.

3.6 TOPSOIL STRIPPING AND STOCKPILING

A. Topsoil Removal:
    1. Remove required topsoil prior to all other soil disturbing operations
    2. Remove the topsoil in one pass, do not mix topsoil with subsoil, and do not compact or drive upon topsoil during removal.
    3. Extents of topsoil removal shall be approved by Contracting Officer's Representative prior to all other soil disturbing operations.

B. Stripping:
    1. Before commencing topsoil stripping:
        a. Perform clearing and grubbing and selective removals.
        b. Remove grass and other vegetation that may remain following clearing and grubbing.
        c. Provide necessary and required temporary erosion and sediment controls.
    2. Strip topsoil to depths encountered or as specified within the geotechnical report, 4 IN minimum.
    3. Before stockpiling, separate objectionable material from topsoil.

4. Live vegetation less than 3 FT in height and limbs less than 1 IN in diameter shall be incorporated as topsoil in the stockpiles.
5. Do not strip topsoil from within drip line of trees and shrubs to remain as part of the completed Project.

C. Stockpiling:
1. Construct and maintain topsoil stockpiles in accordance with Section 01 67 00 - Product Requirements, at locations in accordance with Section 01 11 00 - Summary of Work.

D. Reuse of Stripped Topsoil:
1. Reuse in the finish grading and landscaping Work topsoil that complies with the Contract Documents for such Work.
2. Where topsoil stripped from the Site does not comply with the Contract Documents relative to quality required for use in finish grading and landscaping Work, provide appropriate soil amendment material, properly and fully mixed into topsoil stripped from the Site, so that amended material complies with quality requirements for topsoil required for finish grading and landscaping Work.

E. Disposal of Excess Topsoil:
1. Topsoil in excess of quantity required for the finished Project is Government's property and shall be stockpiled onsite at location designated by Government, in accordance with stockpiling requirements of Section 01 67 00 - Product Requirements.

END OF SECTION

This page intentionally left blank.

SECTION 31 22 00
SITE GRADING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Earthwork - site excavation, grading, compaction, and other Earthwork related work.

B. Related Specification Sections include but are not necessarily limited to:
   1. Division 01 - General Requirements.
   2. Section 31 23 00 - Earthwork
   3. Section 32 15 40 - Crushed Stone Surfacing.
   4. Section 31 25 00 – Soil Erosion and Sediment Controls

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. C33, Standard Specification for Concrete Aggregates.
      b. D698, Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 FT-LBF/CUFT).
      c. D1241, Standard Specification for Material for Soil-Aggregate Subbase, Base, and Surface Courses.
      d. D1557, Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 FT-LBF/CUFT (2,700 kN-M/M)).
      e. D2487, Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System).
      f. D3786, Standard Test Method for Bursting Strength of Textile Fabrics--Diaphragm Bursting Strength Tester Method.
      g. D4253, Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table.
      h. D4254, Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density.
      i. D4632, Standard Test Method for Grab Breaking Load and Elongation of Geotextiles.
   2. American Association of State Highway and Transportation Officials (AASHTO)
      a. M 43, Standard Specification for Sizes of Aggregate for Road and Bridge Construction.
      b. M 57, Standard Specification for Materials for Embankment and Subgrades.
      c. M 147, Standard Specification for Materials for Aggregate and Soil-Aggregate Subbase, Base, and Surface Courses.
   3. Federal Regulations:
      a. Occupational Safety and Health Administration (OSHA):
         1) 29 CFR Part 1926.650, Occupational Safety and Health Standards, referred to herein as OSHA Standards.

1.3 DEFINITIONS

A. Geotechnical Engineer: Independent geotechnical specialist, hired by the Contractor, providing field quality control for the project.

B. Finish Grade: The earth or soil layer immediately below the surfacing material of the site.

C. Rough Grade: The earth or soil layer immediately below the finish grade surface.

D. Subgrade: The earth or soil layer immediately below foundation bearing elevation, subbase material, fill material, backfill material, or topsoil materials.

E. Unauthorized Excavation:
   1. Consists of removal of materials beyond indicated subgrade elevations or dimensions without specific direction of Contracting Officer.
      a. Unauthorized excavation, as well as associated remedial work as directed by Contracting Officer or Geotechnical Engineer, shall be at Contractor's expense.
   2. Unsuitable Soil Materials: Soil materials encountered at or below subgrade elevation of insufficient strength and stiffness to support construction as determined by the Geotechnical Engineer.

1.4 SUBMITTALS

A. Product Data:
   1. Acknowledgement that products submitted meet requirements of standards referenced.
   2. Manufacturer's installation instructions.
   3. Certifications.

B. Samples:
   1. Coordinate samples and testing for approval of off-site materials with the Geotechnical Engineer.

1.5 PROJECT CONDITIONS

A. Salvageable Items: Carefully remove items to be salvaged, and store on Contracting Officer's premises unless otherwise directed.

B. Dispose of waste materials, legally, off site.
   1. Burning, as a means of waste disposal, is not permitted.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Fill:
   1. Selected on-site material from site excavation or from off-site borrow approved by the Contracting officer or Geotechnical Engineer.

B. Aggregate Base:
   1. Granular Fill Material:
      a. Clean, Granular material.
      b. Minimum of 100% passing the ¾ inch (19 mm) sieve.
      c. Less than 25% passing the No. 50 (0.297 mm) sieve.
      d. Less than 5% fines passing the No. 200 sieve.
      e. ASTM D1241/AASHTO M147 Grading "B" or other material acceptable to the Contracting Officer or Geotechnical Engineer, hired by the Contractor.

PART 3 - EXECUTION

3.1 PROTECTION

A. Erosion Control:
   1. See Specification Section 31 25 00.
   2. See Specification Section 01 35 05.
   3. See Specification Section 31 23 00.

3.2 SITE GRADING

A. Excavation and Grading:
   1. See Specification 31 23 00

B. Grading Tolerances: As shown on Drawings.

C. Disposition:
   1. North Rim: Remove excess material and dispose of at the ballfield per Volume 1 drawing 013C102.
   2. Innercanyon:
      a. Pipeline construction (Volume 05): Dispose of excess material with trail reconstruction process. Refer to trail reconstruction requirements.
      b. RSPH Facility (Volume 05): Dispose of excess material within limits of work in locations approved by the Contracting Officer's Representative.

3.3 COMPACTION DENSITY REQUIREMENTS

A. Obtain approval from Contracting Officer with regard to suitability of soils and acceptable subgrade prior to subsequent operations.

B. Remove frozen, loose, wet, or soft material and replace with approved material as directed by the Contracting Officer.

C. Assure by results of testing that compaction densities comply with the following requirements:
   1. Sitework:

LOCATION	COMPACTION DENSITY	MOISTURE CONTENT
Rough Grade:		
Cohesive soils	95% of ASTM D698	-2 to +2% of optimum
Cohesionless soils	75% relative density per ASTM D4253 and ASTM D4254	
Finish Grade:		
Cohesive soils	95% of ASTM D698	-2 to +2% of optimum
Cohesionless soils	75% relative density per ASTM D4253 and ASTM D4254	

3.4 FIELD QUALITY CONTROL

A. All excavation, trenching, and related sheeting, bracing, etc. shall comply with the requirements of OSHA standards 29 CFR Part 1926.650 Subpart P, and state requirements. Where conflict between OSHA and state regulations exists, the more stringent requirements shall apply.

B. Special Inspection and testing:
   1. See Section 01 40 00.

C. Responsibilities of Special Inspector:
   1. Review proposed materials for rough grade and finish grade around the site and structures.
   2. All testing, observation and work indicated as being performed by the Geotechnical Engineer, hired by the Contractor, in this Specification Section.
   3. Services will include verification and documentation of satisfactory soil materials, subgrade quality, sampling, placement, moisture conditioning, compaction and testing of proposed soil materials, and field testing for quality control.
   4. Moisture density relations, to be established by the Geotechnical Engineer required for all materials to be compacted.
   5. Extent of compaction testing will be as necessary to assure compliance with specifications.
   6. Prepare and submit inspection and test reports to Contracting Officer.
      a. Coordinate such work with other Special Inspectors.
   7. Test reports to include the following:
      a. Report and certification of aggregate fill and drainage fill.
      b. Test reports on borrow material.

c. Verification of suitability of each footing subgrade material, in accordance with specified requirements.
d. Field reports; in-place soil density and moisture tests.
e. One optimum moisture-maximum density curve for each type of soil encountered.
f. Other documentation necessary for the Contracting Officer to approve earthwork.
g. Assist Contracting Officer to determine corrective measures necessary for defective work.

D. Responsibilities of Testing Agency for Site Rough Grading and Finish Grading:
1. All testing, observation and work indicated as being performed by the Contractor's Geotechnical Engineer in other than Article 3.4 of this Specification Section.
2. Services will include verification and documentation of satisfactory soil materials, subgrade quality, sampling, placement, moisture conditioning, compaction and testing of proposed soil materials, and field testing for quality control.
3. Moisture density relations, to be established by the Contractor's Geotechnical Engineer required for all materials to be compacted.
4. Extent of compaction testing will be as necessary to assure compliance with specifications.

END OF SECTION

SECTION 31 23 00
EARTHWORK

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Earthwork - excavation, backfilling, grading, placement of fill, compaction, preparation of subgrade below pavements, disposal of waste and surplus materials, placing crushed stone, construction of berms, sheeting, bracing, dewatering, placing structural fill and other Earthwork related work.
   2. Rock Removal – removal of rock boulders using mechanical methods, including drilling, or by expansive demolition grout application, disposal of waste and surplus materials, and other Rock Removal related work.

B. See Geotechnical Investigation Reports referenced in Section 01 11 00.
   1. A copy of these reports is included with Contract Documents.
   2. Where discrepancies between recommendations contained in these reports and these specifications arise, these specifications shall take precedent. Contractor shall immediately bring this matter to the attention of the Contracting Officer.

C. Related Specification Sections include but are not necessarily limited to:
   1. Division 00 - Procurement and Contracting Requirements.
   2. Division 01 - General Requirements.
   3. Section 31 23 33 – Utilities – Trenching and Backfilling.
   4. Section 31 25 00 - Soil Erosion and Sediment Controls.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. C33/C33M, Standard Specification for Concrete Aggregates.
      b. D698, Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 FT-LBF/FT^3).
      c. D1557, Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 FT-LBF/FT^3(2,700 kN-M/M^3)).
      d. D2487, Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System).
      e. D3786, Standard Test Method for Bursting Strength of Textile Fabrics--Diaphragm Bursting Strength Tester Method.
      f. D4253, Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table.
      g. D4254, Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density.
      h. D4632, Standard Test Method for Grab Breaking Load and Elongation of Geotextiles.
   2. Occupational Safety and Health Administration (OSHA):
      a. 29 CFR Part 1926.650, Safety and Health Regulations for Construction - Excavations, referred to herein as OSHA Standards.

1.3 DEFINITIONS

A. Excavation:
   1. Consists of removal of material encountered to subgrade elevations required or indicated.
   2. Includes excavation of soils; pavements and other obstructions visible on surface; underground structures, utilities, and other items indicated to be demolished and removed; boulders; and rock.

B. Rock Removal:
   1. Classification of rock removal shall include solid ledge rock in its natural location that requires systematic quarrying, drilling, and/or expansive grout application for its removal and also large boulders that cannot be easily dislodged as described in the conditions below.
   2. Rock removal:
      a. North Rim: Rock removal includes any material which cannot be dislodged by a D-8 Caterpillar tractor, or equivalent, equipped with a hydraulically operated power ripper, or by a Caterpillar 330 hydraulic excavator, or equivalent, without the use of drilling.
      b. Innercanyon: Rock removal includes any material which cannot be dislodged by the largest piece of excavation equipment that can be transported to the worksite via helicopter without the use of drilling.

C. Foundations: Footings, base slabs, foundation walls, mat foundations, grade beams, piers and any other support placed directly on soil or rock.

D. Geotechnical Engineer: Independent geotechnical specialist, hired by the Contractor, providing field quality control for the project.

E. Non-Structural Fill/Backfill: Soil materials placed and compacted to achieve finish grade elevations that do NOT support foundations, slabs, paving, or other flatwork.

F. Structure: Buildings, foundations, slabs, tanks, curbs, or other man-made stationary features occurring above or below ground surface.

G. Subgrade: The earth or soil layer immediately below foundation bearing elevation, subbase material, fill material, backfill material, or topsoil materials.

H. Unauthorized Excavation:
   1. Consists of removal of materials beyond indicated subgrade elevations or dimensions without specific direction of Contracting Officer.
      a. Unauthorized excavation, as well as associated remedial work as directed by the Contracting Officer or Contractor’s Geotechnical Engineer, shall be at Contractor’s expense.
   2. Unsuitable Soil Materials: Soil materials encountered at or below subgrade elevation of insufficient strength and stiffness to support construction as determined by the Contractor’s Geotechnical Engineer.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
   3. Certifications.
      a. Contractor shall submit certification, to the Contracting Officer, that sources of rock, sand, earth and other natural materials imported into the project construction limits are noxious weed free.

B. Samples:
   1. Coordinate samples and testing for approval of off-site materials with the Contractor’s Geotechnical Engineer and with the Contracting Officer.

1.5 PROJECT CONDITIONS

A. See Geotechnical Investigation Reports referenced in Section 01 11 00.

B. Salvageable Items: Carefully remove items to be salvaged, and store on Contracting Officer’s premises unless otherwise directed.

C. Dispose of waste materials, legally, off site.

1. Burning, as a means of waste disposal, is not permitted.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 MATERIALS

A. General
   1. Do not import into the project limits rock, sand, gravel, earth, subsoil, or other natural materials from a Contractor-selected non-commercial materials source that have not been certified free of noxious weeds.
   2. Materials imported into the project limits, which do not include a noxious weed free certification may be rejected and ordered by the Contracting Officer to be removed from the project limits. Any fill material coming from off-site shall be inspected to reduce the chance for introduction of exotic plant species. Offsite source for any fill material shall be also inspected prior to material coming into the Park.
   3. The Contracting Officer has the discretion of requesting inspection of certified materials by a third party and rejecting the use of the source if noxious weeds or seeds thereof are found to be present.
   4. Only Clean fill, preferably from a site within the Park, shall be used. Any fill coming from off-site shall be certified weed seed-free soil and The source of the material shall be inspected and approved by GRCA's Vegetation Program Manager (NPS) prior to use.
   5. The source of fill for this project is from project-related excavations. There are no other borrow pits available on-site for Contractor sourcing of fill material.

B. Select Fill and Backfill, to be used under Building foundations, slabs, equipment pads, and for wall backfill and trench backfill:
   1. Native soil and/ or excavated material to be used as Select fill:
      a. Native soil and/or excavated fill material encountered during excavation that is desired to be reused as backfill shall not contain unsuitable material as determined by a qualified Geotechnical Engineer, hired by the Contractor.
      b. Processing onsite material shall include removing frozen soil, wood, trash, organic material, cobbles and boulders larger than 3 inches in longest dimension, and other deleterious materials.
      c. If other fill materials are preferred for any of the presented applications, Contractor shall present to Contracting Officer for review.
      d. Excavated bedrock should not be used as backfill.
   2. Selected material approved by Contractor's Geotechnical Engineer and submitted to the Contracting Officer from off-site borrow.
   3. Select Fill:
      a. May be low volume change cohesive or granular soil at Contractor's option.
      b. Free of organic matter, frozen material and debris, or other deleterious materials.
      c. Free of cobbles and boulders greater than 3-inches diameter
      d. Low volume change cohesive soil:
         1) ASTM D2487 classification: CL-ML or CL.
         2) Liquid limit: Less than 40.
         3) Maximum plasticity index: 20.
      e. Granular soil:
         1) ASTM D2487 classification: GW, GP, GM, GC, SW, SP, SM or SC.
   4. Alternate Select fill, to be used for wall backfill only:
      a. Arizona Department of Transportation (ADOT) Class 4 aggregate
      b. Maximum plasticity index: 5
   5. Non-Structural Fill:
      a. ASTM D2487 classification: GW, GP, GM, GC, SC, SW, SP, SM, CL-ML or CL.
      b. Liquid limit: Less than 40.
      c. Maximum plasticity index: 15.

C. Granular Fill Under Manholes and Handholes:
   1. Clean, granular material.
   2. Less than 5 PCT fines passing the No. 200 sieve.
   3. ASTM C33/C33M gradation size No. 67, 3/4 IN to No. 4 or other material acceptable to Contractor's Geotechnical Engineer.

D. Geotextile Filter Fabric:
   1. Nonwoven type.
   2. Equivalent opening size: 50 - 100 (U.S. Standard Sieve).
   3. Permeability coefficient (cm/second): 0.07 minimum, 0.30 maximum.
   4. Grab strength: 157 LBS minimum in either direction in accordance with ADOT Standards, 1014-9.
   5. Mullen burst strength: 125 PSI minimum in accordance with ASTM D3786 requirements.

E. Flowable Fill: See Section 03 31 30.

F. Expansive Demolition Grout
   1. Powder-based type.
   2. Non-toxic material.
   3. Expansive strength: 18,000 PSI minimum.
   4. Selected model shall be rated for concrete and rock temperature range expected during rock removal operation.

PART 3 - EXECUTION

3.1 PROTECTION

A. Protect existing vegetation that is to be preserved as noted in Section 02 41 00. Protect surface and subsurface features on-site and adjacent to site as follows:
   1. Provide barricades, coverings, or other types of protection necessary to prevent damage to existing items indicated to remain in place.
   2. Protect and maintain bench marks, monuments or other established reference points and property corners.
      a. If disturbed or destroyed, replace at own expense to full satisfaction of Contracting Officer and controlling agency.
   3. Verify location of utilities.
      a. Omission or inclusion of utility items does not constitute nonexistence or definite location.
      b. Take necessary precautions to protect existing utilities from damage due to any construction activity.
         1) If utilities are indicated to remain in place, provide adequate means of support and protection during earthwork operations.
         2) Do not interrupt existing utilities serving facilities occupied by Contracting Officer or others, or otherwise in use by the Park during occupied hours, except when permitted in writing by Contracting Officer and then only after acceptable temporary utility services have been provided.
         3) Obtain Contracting Officer's approval prior to disconnecting any utility service.
      c. Repair damages to utility items at own expense.
      d. In case of damage, notify Contracting Officer at once so required protective measures may be taken.
   4. Maintain free of damage, existing fences, sidewalks, structures, and pavement, not indicated to be removed.
      a. Protect new and existing structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.
      b. Any item known or unknown or not properly located that is inadvertently damaged shall be repaired to original condition.

c. All repairs to be made and paid for by Contractor.

5. Provide full access to public and private premises, fire hydrants, street crossings, sidewalks and other points as designated by Contracting Officer.
   a. See also Section 01 73 50 – Trail Access Management
6. Maintain stockpiles and excavations in such a manner to prevent inconvenience or damage to structures on-site or on adjoining property.
7. Avoid surcharge or excavation procedures which can result in heaving, caving, or slides.
   a. Contractor shall note recommendations provided in Section 6.1, Temporary Excavations, of Geotechnical Report.

B. Protection of trees to remain:
   1. Where trees are near the outside of the limit of disturbance and have been determined to be preserved in place during clearing and grubbing operations, place temporary fencing as noted on Project Plans.

3.2 SITE EXCAVATION AND GRADING

A. Excavation and Grading:
   1. Prepare subgrades under asphaltic concrete pavement, concrete pavement and those areas where aggregate base course and crushed aggregate are used as finish grade materials.
   2. Perform as required for construction of storm water drainage improvements. Project Plans indicate both existing grade and finished grade required for construction of Project.
      a. Stake all units, structures, piping, roads, parking areas and walks and establish their elevations.
      b. Perform other layout work required.
      c. Replace property corner markers to original location if disturbed or destroyed.
   3. Preparation of ground surface for embankments or fills:
      a. Before fill is started, scarify to a minimum depth of 8 IN in all proposed embankment and fill areas.
      b. Where ground surface is steeper than one vertical to four horizontal, plow surface in a manner to bench and break up surface so that fill material will bind with existing surface. Benching shall be in accordance with OSHA recommendations.
   4. Where trees are near the outside of the limit of disturbance and have been determined to be preserved in place during clearing and grubbing operations, perform excavation by hand within dripline of these trees.
      a. Remove organic matter and unconsolidated soils with rakes and shovels within dripline where pavement is to be placed.
      b. Compact subgrade with a hand tamper.
      c. Do not use machinery.
   5. Protection of finish grade:
      a. During construction, shape and drain embankment and excavations.
      b. Maintain ditches and drains to provide drainage at all times.
      c. Protect graded areas against action of elements prior to acceptance of work.
      d. Reestablish grade where settlement or erosion occurs.

B. Borrow:
   1. Provide necessary amount of approved fill compacted to density equal to that indicated in this Specification.
   2. Include cost of all borrow material in original proposal.

C. Construct embankments and fills as required by the Project Plans:
   1. Construct embankments and fills at locations and to lines of grade indicated.
      a. Completed fill shall correspond to shape of typical cross section or contour indicated regardless of method used to show shape, size, and extent of line and grade of completed work.
   2. Provide approved fill material which is free from unsuitable soil materials including roots, organic matter, trash, frozen material, and stones having maximum dimension greater than 6 IN.

a. Ensure that stones larger than 4 IN are not placed in upper 6 IN of fill or embankment.
b. Do not place material in layers greater than 8 IN loose thickness.
c. Place layers horizontally and compact each layer prior to placing additional fill.

3. Compact soils as required to obtain specified density. Selection of appropriate equipment is the Contractor's responsibility.
   a. In general, compact cohesive soils by sheepsfoot, and granular soils by pneumatic rollers, vibrators, or by other equipment as required to obtain specified density.
   b. Control moisture for each layer necessary to meet requirements of compaction as defined herein.

D. Grading Tolerances: ±2 IN

E. Topsoil Placement
1. Prior to placing the topsoil, the Contractor will disk or scarify all disturbed areas to a depth of 8 IN in a direction perpendicular to the natural flow of water.
   a. Prior to topsoil placement, if the Contracting Officerdetermines that subsoil is excessively compacted by the Contractor's operations, the Contractor will scarify the subsoils to a depth of 24 IN to de-compact and break the interface between particles.
2. When applying topsoil, the Contractor will avoid mixing the topsoil with subsoil material.
3. The Contractor will spread conserved topsoil to a depth of 4 IN over all disturbed soil areas.
4. Topsoil will be spread using the lightest weight equipment applicable.
5. If the Contractor's Geotechnical Engineer determines that topsoil is excessively compacted by the Contractor's operations, the Contractor will scarify the topsoil to a depth of 8 IN to de-compact and break the interface between particles.
6. Under no circumstances should placed topsoil be allowed to remain without erosion protection (i.e., hydromulch, erosion wattles, log erosion barriers) for more than seven days or when heavy rain is predicted prior to erosion protection placement.
7. In areas where proposed grades are not shown within the contract documents, the Contractor shall scarify compacted soils and reestablish original grades if disturbed by construction activities.

F. Disposition:
1. North Rim: Remove excess material and dispose at the ballfield per Volume 1 drawing 011 L101.
2. stockpile at Marble Flat.
3. Innercanyon:
   a. Pipeline construction (Volume 05): Dispose of excess material with trail reconstruction process. Refer to trail reconstruction requirements.
   b. RSPH Facility (Volume 05): Dispose of excess material within limits of work in locations approved by the Contracting Officer's Representative.

3.3 BEDROCK EXCAVATION

A. Bedrock excavation will be necessary.
1. Bedrock is anticipated to be difficult to excavate with increasing resistance with depth.
2. Based on this range, rippability per the Caterpillar Handbook of Ripping is between Rippable to Marginally Rippable using a D11 dozer.
3. Additional methods for excavation of rock include an impact hammer attachment to an excavator and specialized rock trenching equipment.
4. Refer to Geotechnical Report for depth to bedrock encountered during soils testing.

3.4 ROCK REMOVAL

A. General:
1. Contractor is solely responsible for method of rock removal employed, safety of all persons, and protection of property in accordance with Contract Documents.
2. Avoid damaging or weakening foundations, buildings, structures, Underground Facilities, wells, sheeting, bracing, and other facilities.

3. Coordinate rock removal with other Work to avoid damage to new construction and possible decrease in concrete strength placed less than 28-days prior to rock removal operations.
4. Perform rock removal adjacent to Underground Facilities and above-ground utilities, life-safety facilities, and facilities that may pose hazards to health and safety, with utmost care, after properly notifying and coordinating with Contracting Officer.
5. Contractor shall promptly remedy property damaged during or as a result of rock removal operations, in manner acceptable to Contracting Officer at no additional cost to Government.
6. Where conditions of hazard exist, or clearances with existing facilities, piping, or structures are very small, or where the potential for damage to persons or property is strong.

B. When rock is encountered:
   1. Strip free of earth.
   2. Employ an independent surveyor to determine rock quantities before operations begin.
   3. In computing the volumetric content of rock excavation for payment, the pay lines shall be taken as follows:
      a. For Structures: 3 FT outside the exterior limits of foundation and from rock surface to 6 IN below bottom of foundations.
      b. For piping and utilities: A width 18 IN wider that the outside diameter of the pipe or conduit and from rock surface to 6 IN below bottom exterior surface of the pipe or conduit.
      c. For rough or finish grading: 2 FT outside the exterior limits of excavation and rock surface to 6 IN below bottom of finish grade surface.

C. Expansive Demolition Grout Application
   1. Where mechanical methods are not capable of quarrying rock boulders, employ expansive demolition grout, a non-explosive alternative to rock demolition.
   2. Procedure:
      a. Pre-drill holes in rock to be removed.
         1) Hole depth: 80 to 90-percent of total material depth.
         2) Material shall not be applied to rock with a depth less than 6 in.
      b. Mix power-based grout with water and apply slurry to holes.
      c. Remove rock debris once grout has expanded and cracked material.
         1) Process may take several hours

D. Disposition:
   1. North Rim: Remove excess material and dispose outside of Park limits.
   2. Innercanyon:
      a. Pipeline construction (Volume 05): Dispose of excess material with trail reconstruction process. Refer to trail reconstruction requirements.
      b. RSPH Facility (Volume 05): Dispose of excess material within limits of work in locations approved by the Contracting Officer's Representative.
   3. Do not use rock removed by Contractor as fill or backfill.

3.5 USE OF EXPLOSIVES

A. Reference 01 35 25 Explosives.

3.6 COMPACTION DENSITY REQUIREMENTS

A. See also Geotechnical Report.

B. Backfill placement shall occur in maximum 8-IN. loose lifts. Each lift shall be tested for proper compaction.

C. CO shall be consulted as to the adequacy of whether onsite soils are suitable as fill material.

D. Provide dewatering system necessary to successfully complete compaction and construction requirements.

E. Remove frozen, loose, wet, or soft material and replace with approved material as directed by Contractor's Geotechnical Engineer.

F. Stabilize subgrade with well graded granular materials as directed by Contractor's Geotechnical Engineer.

G. Assure by results of testing that compaction densities comply with the following requirements:
   1. Sitework:

LOCATION	COMPACTION DENSITY	MOISTURE CONTENT
Under Paved Areas, Sidewalks and Piping:		
Cohesive soils	95 PCT per ASTM D698	-1 to +3 PCT of optimum
Cohesionless soils	95 PCT relative density per ASTM D698	-2 to +2 PCT of optimum
Pipe Bedding	65 PCT relative density per ASTM D4253 and ASTM D4254	
Unpaved Areas:		
Cohesive soils	95 PCT of ASTM D698	-1 to +3 PCT of optimum
Cohesionless soils	95 PCT per ASTM D698	-2 to +2 PCT of optimum

   2. Structures:

LOCATION	COMPACTION DENSITY	MOISTURE CONTENT
Inside of structures under foundations, under equipment support pads, under slabs-on-grade and scarified existing subgrade under fill material	98 PCT per ASTM D698	-1 to +3 PCT of optimum
Outside structures next to walls, piers, columns and any other structure exterior member	95 PCT per ASTM D698	-2 to +2 PCT of optimum

   3. Specific areas:

LOCATION	COMPACTION DENSITY	MOISTURE CONTENT
Outside structures under equipment support foundations	98 PCT per ASTM D698	-1 to +3 PCT of optimum
Under void	85 PCT per ASTM D698	-1 to +3 PCT of optimum
Granular fill under manholes	65 PCT relative density per ASTM D4253 and ASTM D4254 or 98 PCT of ASTM D698	

3.7 EXCAVATION, FILLING, AND BACKFILLING FOR STRUCTURES

A. General:
   1. In general, work includes, but is not necessarily limited to, excavation structures and retaining walls, removal of underground obstructions and undesirable material, backfilling, filling, and fill, backfill, and subgrade compaction.
   2. Obtain fill and backfill material necessary to produce grades required.
      a. Materials and source to be approved by Contractor's Geotechnical Engineer.

b. Excavated material approved by Contractor's Geotechnical Engineer may also be used for fill and backfill.

3. In the paragraphs of this Specification Section, the word "soil" also includes any type of rock subgrade that may be present at or below existing subgrade levels.

B. Excavation Requirements for Structures:

1. General:
   a. Do not commence excavation for foundations for structures until:
      1) The removal of topsoil and other unsuitable and undesirable material from existing subgrade.
      2) Density and moisture content of site area compacted fill material meets requirements of specifications.
      3) Site surcharge or mass fill material can be removed from entire construction site or portion thereof.
      4) Contractor's Geotechnical Engineer grants approval to begin excavations.
2. Dimensions:
   a. Excavate to elevations and dimensions indicated or specified.
   b. Allow additional space as required for construction operations and inspection of foundations.
   c. Slope sides of excavations to comply with local codes, ordinances, and requirements of agencies having jurisdiction.
   d. Maintain sides and slopes of excavations in safe condition until completion of backfilling.
3. Removal of obstructions and undesirable materials in excavation includes, but is not necessarily limited to, removal of old foundations, existing construction, unsuitable subgrade soils, expansive type soils, and any other materials which may be concealed beneath present grade, as required to execute work indicated on Project Plans.
   a. If undesirable material and obstructions are encountered during excavation, remove material and replace as directed by Contractor's Geotechnical Engineer.
   b. Remove unsuitable subgrade soils located below foundations. The bottom of the overexcavation shall be located outside the exterior limits of foundations around the perimeter of structure the following horizontal distance, whichever is greater:
      1) Distance equal to depth of overexcavation below bottom of foundations. See Geotechnical Report.
   c. When excavation has reached required subgrade elevations, notify Contractor's Geotechnical Engineer, who shall make an inspection of conditions.
      1) If Contractor's Geotechnical Engineer determines that bearing materials at required subgrade elevations are unsuitable, provide Subgrade Stabilization as specified herein.
4. Level off bottoms of excavations to receive foundations, floor slabs, equipment support pads, or compacted fill.
   a. Remove loose materials and bring excavations into approved condition to receive concrete or fill material.
   b. Where compacted fill material must be placed to bring subgrade elevation up to underside of construction, scarify existing subgrade upon which fill material is to be placed to a depth of 8 IN and then compact to density stated in this Specification Section before fill material can be placed thereon.
   c. Do not carry excavations lower than shown for foundations except as directed by Contractors' Geotechnical Engineer.
   d. If any part of excavations is carried below required depth without authorization, notify Contractor's Geotechnical Engineer and correct unauthorized excavation as directed. Corrections may include:
      1) Under soil supported footings, foundation bases, or retaining walls, fill unauthorized excavation by extending indicated bottom elevation of footing or base to excavation bottom, without altering required top elevation.
         a) Concrete fill may be used to bring elevations to proper position.

2) In locations other than those above, including slabs on grade and pile supported foundations, backfill and compact unauthorized excavations as specified for authorized excavations of same classification, unless otherwise directed by Contractor's Geotechnical Engineer.
3) No extra compensation will be made to Contractor for correcting unauthorized excavations.

5. Make excavations large enough for working space, forms, dampproofing, waterproofing, and inspection.
6. Notify Contractor's Geotechnical Engineer and Contracting Officer as soon as excavation is completed in order that subgrades may be inspected.
   a. Do not commence further construction until subgrade under compacted fill material, under foundations, under floor slabs-on-grade, under equipment support pads, and under retaining wall footings has been inspected and approved by the Contractors' Geotechnical Engineer as being free of undesirable material, being of compaction density required by this specification, and being capable of supporting the allowable foundation design bearing pressures and superimposed foundation, fill, and building loads to be placed thereon.
   b. Contractor's Geotechnical Engineer shall be given the opportunity to inspect subgrade below fill material both prior to and after subgrade compaction.
   c. Place fill material, foundations, retaining wall footings, floor slabs-on-grade, and equipment support pads as soon as weather conditions permit after excavation is completed, inspected, and approved and after forms and reinforcing are inspected and approved.
   d. Before concrete or fill material is placed, protect approved subgrade from becoming loose, wet, frozen, or soft due to weather, construction operations, or other reasons.
7. Subgrade stabilization:
   a. If subgrade under foundations, fill material, floor slabs-on-grade, or equipment support pads is in a frozen, loose, wet, or soft condition before construction is placed thereon, remove frozen, loose, wet, or soft material and replace with approved compacted material as directed by Contractor's Geotechnical Engineer.
   b. Provide compaction density of replacement material as stated in this Specification Section.
   c. Loose, wet, or soft materials, when approved by Contractor's Geotechnical Engineer, may be stabilized by a compacted working mat of well graded crushed stone meeting the requirements defined in this specification section. If the base of the excavation is still loose, wet, or soft, place geotextile fabric meeting the requirements defined in this specification section at the base of the excavation prior to placement of crushed stone.Compact stone mat thoroughly into subgrade to avoid future migration of fines into the stone voids.
   d. Remove and replace frozen materials as directed by Contractor's Geotechnical Engineer.
   e. Method of stabilization shall be performed as directed by Contractor's Geotechnical Engineer.
   f. Do not place further construction on the repaired subgrades, until the subgrades have been approved by the Contractor's Geotechnical Engineer.
8. Do not place floor slabs-on-grade including equipment support pads until subgrade below has been approved, piping has been tested and approved, reinforcement placement has been approved, and Contractor receives approval to commence slab construction.
   a. Do not place building floor slabs-on-grade including equipment support pads when temperature of air surrounding the slab and pads is or is expected to be below 40 DEGF before structure is completed and heated to a temperature of at least 50 DEGF.
9. Protection of structures:
   a. Prevent new and existing structures from becoming damaged due to construction operations or other reasons.

b. Prevent subgrade under new and existing foundations from becoming wet and undermined during construction due to presence of surface or subsurface water or due to construction operations.

10. Shoring:
    a. Shore, slope, or brace excavations as required to prevent them from collapsing.
    b. Remove shoring as backfilling progresses but only when banks are stable and safe from caving or collapse.
    c. Construct shoring that is required to retain water as part of the dewatering system, using non-permeable details such as interlock sealant for sheet piles.
11. Drainage:
    a. Control grading around structures so that ground is pitched to prevent water from running into excavated areas or damaging structures.
    b. Maintain excavations where foundations, floor slabs, equipment support pads or fill material are to be placed free of water.
    c. Provide pumping required to keep excavated spaces clear of water during construction.
    d. Should any water be encountered in the excavation, notify Contracting Officer and Contractor's Geotechnical Engineer.
    e. Provide free discharge of water by trenches, pumps, wells, well points, or other means as necessary and drain to point of disposal that will not damage existing or new construction or interfere with construction operations.
12. Frost protection:
    a. Do not place foundations, slabs-on-grade, equipment support pads, or fill material on frozen ground.
    b. When freezing temperatures may be expected, do not excavate to full depth indicated, unless foundations, floor slabs, equipment support pads, or fill material can be placed immediately after excavation has been completed and approved.
    c. Protect excavation from frost if placing of concrete or fill is delayed.
    d. Where a concrete slab is a base slab-on-grade located under and within a structure that will not be heated, protect subgrade under the slab from becoming frozen until final acceptance of the Project by the Contracting Officer.
    e. Protect subgrade under foundations of a structure from becoming frozen until structure is completed and heated to a temperature of at least 50 DEGF.

C. Fill and Backfill Inside of Structure and Below Foundations, Base Slabs, Floor Slabs, Equipment Support Pads and Piping:
1. General:
    a. Subgrade to receive fill or backfill shall be free of undesirable material as determined by Contractor's Geotechnical Engineer and scarified to a depth of 8 IN and compacted to density specified herein.
    b. Surface may be stepped by at not more than 12 IN per step or may be sloped at not more than 2 PCT.
    c. Do not place any fill or backfill material until subgrade under fill or backfill has been inspected and approved by Contractor's Geotechnical Engineer as being free of undesirable material and compacted to specified density.
2. Obtain approval of fill and backfill material and source from Contractor's Geotechnical Engineer prior to placing the material.
3. Vapor barrier: Install a continuous vapor barrier under floor slabs-on-grade as shown on Contract Drawings.
4. Fill and backfill placement:
    a. Prior to placing fill and backfill material, optimum moisture and maximum density properties for proposed material shall be obtained from Contractor's Geotechnical Engineer.
    b. Place fill and backfill material in maximum 8IN lifts.
    c. Compact material by means of equipment of sufficient size and proper type to obtain specified density.

d. Use hand operated equipment for compacting fill within 5 FT of walls and less than 3 FT above pipes.
    1) Compaction equipment exceeding 3000 LBS dead weight shall not be used within 5 FT of the wall as a minimum
    2) Contractor is responsible for method of compaction so as not to damage wall.
e. Do not place fill and backfill when the temperature is less than 40 DEGF and when subgrade to receive fill and backfill material is frozen, wet, loose, or soft.
f. Use vibratory equipment to compact granular material; do not use water to compact soil by flooding or jetting. Moisture conditioning for compaction by vibratory equipment is allowed.

5. Where fill material is required below foundations, place fill material, conforming to the required density and moisture content as required to fill the specified overexcavation to bottom of foundation.

D. Filling and Backfilling Outside of Structures:
1. This paragraph of this Specification applies to fill and backfill placed outside of structures above bottom level of both foundations and piping but not under paving.
2. Provide material as approved by Contractor's Geotechnical for filling and backfilling outside of structures.
3. Fill and backfill placement:
    a. Prior to placing fill and backfill material, obtain optimum moisture and maximum density properties for proposed material from Contractor's Geotechnical Engineer.
    b. Place fill and backfill material to maximum allowable lift thickness indicated in this Section.
    c. Compact material with equipment of proper type and size to obtain density specified.
    d. Use hand operated equipment for compacting fill within 5 FT of walls and less than 3 FT above pipes.
        1) Compaction equipment exceeding 3000 LBS dead weight shall not be used within 5 FT of the wall as a minimum
        2) Contractor is responsible for method of compaction so as not to damage wall.
    e. Do not place fill or backfill material when temperature is less than 40 DEGF and when subgrade to receive material is frozen, wet, loose, or soft.
4. Backfilling against walls:
    a. Do not backfill around any part of structures until each part has reached specified 28-day compressive strength and backfill material has been approved.
    b. Do not start backfilling until concrete forms have been removed, trash removed from excavations, pointing of masonry work, concrete finishing, dampproofing and waterproofing have been completed.
    c. Do not place fills against walls until floor slabs at top, bottom, and at intermediate levels of walls are in place and have reached 28-day required compressive strength to prevent wall movement.
        1) See Contract Drawings for specific exceptions.
    d. Bring backfill and fill up uniformly around the structures and individual walls, piers, or columns.

E. Backfilling Outside of Structures Under Piping or Paving:
1. Preparation of subgrade under pavements:
    a. Concrete pavement shall be constructed atop a minimum 8 IN. thick zone of moisture-conditioned and compacted subgrade.
    b. Aggregate base course as used as subgrade for asphaltic concrete pavement shall be installed atop a minimum 8 IN. thick zone of moisture-conditioned and compacted subgrade.
    c. Aggregate base course as used as finish surface for the contractor access road shall be constructed atop a minimum 8 IN. thick zone of moisture-conditioned and compacted subgrade.

d. Aggregate base course as used as finish surface for the construction trailers and staging area at the South Rim Water Treatment Facilities shall be constructed atop a minimum 8 IN. thick zone of moisture-conditioned and compacted subgrade.

2. When backfilling outside of structures requires placing backfill material under piping or paving, the material shall be placed from bottom of excavation to underside of piping or paving at the density required for fill under piping or paving as indicated in this Specification Section.
3. This compacted material shall extend transversely to the centerline of piping or paving a horizontal distance each side of the exterior edges of piping or paving equal to the depth of backfill measured from bottom of excavation to underside of piping or paving.
4. Provide special compacted bedding or compacted subgrade material under piping or paving as required by other Specification Sections for the Project.

F. Mechanical compaction is required for all materials placed as fill during construction.
1. Compaction of cohesive materials is best accomplished with equipment such as a jumping jack or padfoot roller.
2. Noncohesive soils are best compacted with a vibratory plate or vibratory smooth-drum roller.
3. Compaction utilizing any flooding type technique is not recommended. The resulting subgrades shall be proof-rolled to verify stability.
4. If the proof-roll reveals soft or loose soils, the subgrade shall be stabilized.
a. Stabilization shall be accomplished by over excavating existing soil to bedrock and replacing with compacted, moisture conditioned Engineered Fill or by using geotextile stabilization under 12 IN. of Engineered Fill.

3.8 FIELD QUALITY CONTROL

A. All excavation, trenching, and related sheeting, bracing, etc. shall comply with the requirements of OSHA Standards, and state requirements. Where conflict between OSHA and state regulations exists, the more stringent requirements shall apply.

B. Special Inspection and Testing:
1. See Section 01 40 00.

C. Responsibilities of Special Inspector:
1. Review proposed materials for fill and backfill around structures.
2. All testing, observation and work indicated as being performed by the Contractor's Geotechnical Engineer inArticle 3.5 of this Specification Section .
3. Services will include verification and documentation of satisfactory soil materials, subgrade quality, sampling, placement, moisture conditioning, compaction and testing of proposed soil materials, and field testing for quality control.
4. Moisture density relations, to be established by the Contractor's Geotechnical Engineer required for all materials to be compacted.
5. Extent of compaction testing will be as necessary to assure compliance with specifications. Minimum test requirement is equal to 1 test per 2000 sf per lift.
6. Prepare and submit inspection and test reports to Contracting Officer.
a. Coordinate such work with other Special Inspectors.
7. Test reports to include the following:
a. Report and certification of aggregate fill and drainage fill.
b. Test reports on borrow material.
c. Verification of suitability of each footing subgrade material, in accordance with specified requirements.
d. Field reports; in-place soil density and moisture tests.
e. One optimum moisture-maximum density curve for each type of soil encountered.
f. Report of actual unconfined compressive strength and/or results of bearing tests of each strata tested.
g. Other documentation necessary for Contractor's Geotechnical Engineer to approve earthwork.

h. Assist Contracting Officer to determine corrective measures necessary for defective work.

D. Responsibilities of Testing Agency for Site Excavation and Grading:
   1. All testing, observation and work indicated as being performed by the Contractor's Geotechnical Engineer.
   2. Services will include verification and documentation of satisfactory soil materials, subgrade quality, sampling, placement, moisture conditioning, compaction and testing of proposed soil materials, and field testing for quality control.
   3. Moisture density relations, to be established by the Contractor's Geotechnical Engineer required for all materials to be compacted.
   4. Extent of compaction testing will be as necessary to assure compliance with specifications.

END OF SECTION

SECTION 31 23 33
UTILITIES - TRENCHING AND BACKFILLING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Requirements for excavating, filling, and other earthwork for linear, buried utility projects, including:
      a. Field conditions.
      b. Soil and aggregate materials.
      c. Test pits.
      d. Excavation.
      e. Preparation of subgrades.
      f. Filling and compacting.
      g. Rough grading.
      h. Disposal of excavated spoil materials.

B. Related Requirements Include, but are not necessarily limited to, the following:
   1. Section 31 10 00 - Site Clearing.
   2. Section 32 91 13 - Finish Grading.
   3. Section 31 23 00 – Earthwork.

1.2 PRICE AND PAYMENT PROCEDURES

A. Measurement:
   1. Contractor shall provide all labor, materials, construction equipment and machinery, tools, and incidentals required or necessary for performing all trenching, backfilling, and related earthwork for the Contract, including all work related to excavating, preparing subgrades, providing and stockpiling materials, filling with required materials, compacting, grading, proper disposal of excess spoil materials, and all other work required by this Section. This Section includes trenching, backfilling, and related earthwork required for constructing: buried utilities, including piping and, where applicable, other Underground Facilities; site improvements (including pavement, sidewalks, curbs, and the like); subgrade preparation for other work; and all other trenching, backfilling, and related earthwork required by the Contract Documents and not addressed under other Specifications sections.
   2. Unless expressly indicated otherwise in the Contract Documents, such as Section 01 27 00 - Definition of Contract Line Items, the Work of this Section is included in the various lump sum and unit price bid/pay items for the Contract.
   3. All Work under this Section is unclassified and covered by this Section and the associated bid/payment item(s). No separate payment will be made for removing, handling, or disposing of rock (unless the Contract includes a separate bid/pay item specifically for rock excavation or rock removal), stones, concrete (whether or not reinforced), debris, stumps, and other material, regardless of moisture content, unless such material constitutes an unforeseen Hazardous Environmental Condition and Work associated with such Hazardous Environmental Condition is not included in the Contract.

1.3 REFERENCES

A. Terminology:
   1. The following terms are not defined and are not indicated with initial capital letters but, when used in this Section, have the following meaning:
      a. “Borrow” means material required for trenching, backfilling, and related earthwork in excess of quantity of suitable material available from required grading, cuts, and excavations at the Site. Borrow may be necessary even though not otherwise expressly required by the Contract Documents.

b. “Cohesive soil” means clay (fine grained soil), or soil with a high clay content, which has cohesive strength. Cohesive soil does not crumble, can be excavated with vertical or near-vertical sideslopes, and is plastic when moist. Cohesive soil is difficult to break up when dry, and exhibits significant cohesion when submerged. Cohesive soils include clayey silt, sandy clay, silty clay, clay and organic clay.
c. “Cohesionless soil” means fill material that does not clump together; its grains remain separate and apart from each other. Cohesionless soils typically have a low proportion of fine (no. 200 mesh size and less) particles and may be soil material or granular material. The terms “cohesionless soils”, “cohesionless materials”, “non-cohesive soils”, and the like have the same meaning.
d. “Fill” means soil material, granular material, or other material indicated in this Section required for backfilling excavations or as material for achieving required rough grading. Unless indicated otherwise in the context of a specific provision, the terms “fill” and “backfill” have the same meaning, as do derivative terms such as, “filling” and “backfilling”. When used by itself, the term “backfill” means material required for refilling an excavation.
e. “Foundations” means bottoms of manholes, chambers, vaults, and similar structures; base slabs (slabs-on-grade); bearing directly on soil, granular material, or rock.
f. “Granular material” means gravel, sand, or silt, (coarse grained soil) with little or no clay content. Granular soil has no cohesive strength. Some moist granular soils exhibit apparent cohesion. Granular soil cannot be molded when moist and crumbles easily when dry. The terms “granular material”, “aggregate material”, and “granular soil” have the same meaning. Detailed requirements for specific types of granular material are indicated in Article 2.1 of this Section.
g. “Subgrade” means the uppermost surface of native soil material unmoved from cuts; the bottom of excavation.
h. “Trenching” means excavating for buried utilities, other Underground Facilities, and related excavation work. In this Section, the words, “trenching” and “excavating” have the same meaning. In this Section, the word “excavation”, whether singular or plural, is general and includes trenching.
i. “Unauthorized excavation” means removing materials beyond limits shown or indicated in the Contract Documents or otherwise directed by Contracting Officer’s Representative.
j. “Unclassified”, when used in referring to trenching, excavating, or excavations, means excavation and disposal of all materials, of any description whatsoever, encountered in performing the Work, unless otherwise shown or indicated in the Contract Documents.
k. “Unsuitable soil materials” means soil or other materials encountered at or below subgrade elevation of insufficient strength, stiffness, or characteristics to support the Work in accordance with the Contract Documents or as determined by Contracting Officer’s Representative. The terms “unsuitable soil material”, “unsuitable soil”, and “unsuitable material” have the same meaning.

B. Reference Standards: Standards referenced in this section include, but are not necessarily limited to, the following:
1. American Association of State Highway and Transportation Officials (AASHTO):
a. AMRL R18 - AASHTO Accreditation Procedure Manual.
2. ASTM International (ASTM):
a. C33/C33M - Standard Specification for Concrete Aggregates.
b. D698 - Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft^3 (600 kN-m/m^3)).
c. D1556 - Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method.
d. D1557 - Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft^3 (2,700 kN-m/m^3)).
e. D2166 - Standard Test Method for Unconfined Compressive Strength of Cohesive Soil.

f. D2216 - Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass.
g. D2434 - Standard Test Methods for Measurement of Hydraulic Conductivity of Coarse-Grained Soils.
h. D2487 - Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System).
i. D2850 - Standard Test Method for Unconsolidated-Undrained Triaxial Compression Test on Cohesive Soils.
j. D3740 - Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction.
k. D4253 - Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table.
l. D4254 - Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density.
m. D4318 - Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils.
n. D6913 - Standard Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis.
o. D6938 - Standard Test Methods for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth).
p. D7928 - Standard Test Method for Particle-Size Distribution (Gradation) of Fine-Grained Soils Using the Sedimentation (Hydrometer) Analysis.
q. E329 - Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection.

1.4 QUALITY ASSURANCE

A. Regulatory Requirements:
1. Areas requiring concrete or asphalt paving work are owned and under the jurisdiction of the Government, unless indicated otherwise elsewhere in the Contract Documents.

B. Qualifications:
1. Contractor's Testing Laboratory:
a. Retain the services of independent testing laboratory to perform testing and determine compliance with the Contract Documents of materials specified in this Section.
b. When Contractor is expressly required, by this Section's "Field Quality Control" Article or elsewhere in the Contract Documents, to retain testing laboratory for tests required under this Section's "Field Quality Control" Article, retain one testing laboratory for quality assurance testing and field quality control testing.
c. Testing laboratory shall possess current, valid accreditation from the AASHTO Materials Reference Laboratory (AMRL), in accordance with AASHTO AMRL R18, for ASTM D3740, for tests required for quality assurance testing indicated in this Article and field quality control testing indicated to be Contractor's responsibility.
d. Testing laboratory shall comply with ASTM E329.
e. Testing laboratory shall be experienced in the types of testing required.
f. Selection of testing laboratory is subject to Contracting Officer's acceptance. Upon Contracting Officer's request, submit qualifications statement for testing laboratory, including name of entity, location, copies of applicable certifications, summary of entity's experience, and names and qualifications of personnel who will perform the subject sampling and testing.

C. Quality Assurance Testing:
1. Quality assurance testing, which may also be regarded as source quality control testing, is in addition to field quality control testing required by this Section's "Field Quality Control" Article.

2. Materials used in the Work may require testing and retesting, as directed by Contracting Officer's Representative, during the Project. Allow free access to onsite material stockpiles, borrow sources, and other sources of aggregate and soil materials at all times. Tests not specifically indicated to be performed at Government's expense (if any), including retesting of defective materials and defective work, shall be at Contractor's cost.
3. Contractor's Testing Laboratory Scope for Quality Assurance Sampling and Testing:
    a. Obtain samples and perform testing of proposed fill materials in the laboratory, at the Site, and at borrow locations (if any) to provide assurance that the Work will comply with the Contract Documents.
    b. Testing laboratory shall perform testing required to obtain data for selecting moisture content for placing and compacting fill materials.
    c. Submit to Contracting Officer's Representative, via Contractor, written results of each test.
4. Required Quality Assurance Material Testing by Contractor's Testing Laboratory:
    a. Perform the following quality assurance tests on materials to be incorporated into the Work. Submit results of such tests to Contracting Officer's Representative and obtain Contracting Officer's Representative's acceptance thereof prior to incorporating the subject materials into the Work.
    b. Gradation in accordance with ASTM D6913 and ASTM D7928. Perform one test for every 1,000 cubic yards of each material to be incorporated into the Work.
    c. Atterberg limits in accordance with ASTM D4318. Perform one test for every 1,000 cubic yards of the following types of materials to be incorporated into the Work: suitable soil material, and soil structural fill.
    d. Moisture/density relations for fill in accordance with ASTM D698, ASTM D1557, ASTM D4253, or ASTM D4254, as applicable. Perform one test for every 1,000 cubic yards of each material to be incorporated into the Work.
    e. Moisture content of stockpiled or borrow material in accordance with ASTM D2216. Perform one test for every 1,000 cubic yards of each material to be incorporated into the Work.

1.5 SUBMITTALS

A. Action Submittals: Submit the following:
    1. Shop Drawings:
        a. Schedule (table or written narrative) indicating, for each specific area of the Work, type of fill materials proposed, together with indication of whether source of materials is onsite or borrow source. Identification of each type of material shall employ material names/designations identical to those in the Contract Documents.
    2. Product Data:
        a. Borrow Soil Materials: Name and address of Supplier of borrow soil materials, gradation, and Supplier's certification that materials are sufficiently free of Constituents of Concern so that a Hazardous Environmental Condition will not be created or exacerbated. If no soil materials will be obtained from any borrow location, expressly so indicate to Contracting Officer's Representative in writing.
        b. Aggregate: Supplier's published description of aggregate materials, including name and address of Supplier, certification of material's compliance with AASHTO and state department of transportation standards, and gradation.

B. Informational Submittals: Submit the following:
    1. Procedure Submittals:
        a. Excavation plan indicating:
            1) Excavation Method(s): Indicate high-level methods to be used for excavating, such as use of backhoe or other earth moving construction equipment, trenchless excavating methods, micro-trenching, hydro-excavating (when allowed), air vacuum excavating (when allowed), or other (use of explosives is not allowed). Where different methods will be used, indicate locations of each in the Work. Purpose of indicating high-level excavating methods is for Contracting Officer's

information regarding effect of excavating method on Government property and operations.

b. Compaction procedure and compaction equipment proposed for use. Where different procedures or equipment will be used for compacting different types of material or at different locations at the Site, indicate where each procedure and equipment item will be used.

c. Contracting Officer's acceptance of excavation plan does not, in any way, modify Contractor's sole responsibility for complying with the Contract Documents, construction means, methods, procedures, techniques, or sequences, and safety and protection measures incident thereto.

2. Quality Assurance Test Results Submittals:
   a. Submit results of quality assurance testing performed in accordance with this Section's "Quality Assurance" Article, unless included as part of another submittal under this Section. Submit results for the following quality assurance testing:
      1) Tests on materials from onsite and borrow sources.
      2) Optimum moisture – maximum dry density curve for each type of fill material.
3. Field Quality Control Submittals:
   a. Submit results of testing and inspection performed in accordance with the field quality control Article in Part 3 of this Section, including:
      1) Indication of soil classification in accordance with ASTM D2487.
      2) Field density testing.
      3) Proctor Compaction Test to determine optimum moisture for soil and maximum dry density.
4. Qualifications Statements:
   a. Quality assurance testing laboratory, when requested by Contracting Officer.

1.6 FIELD CONDITIONS

A. Existing Conditions:
   1. See geotechnical reports
   2. Subsurface Conditions:
      a. Subsurface Information: The Supplementary Conditions indicate information (if any) that is Technical Data on subsurface conditions at the Site. Such information and data are not intended as a representation or warranty of continuity of conditions between soil borings or test pits, nor of groundwater levels at dates and times other than date and time when measured, nor that purpose of obtaining the information and data were appropriate for use by Contractor. Contracting Officer and Government are not responsible for interpretations or conclusions drawn therefrom by Contractor.
      b. Contractor's Explorations: Soil borings and other exploratory operations may be made by Contractor, at no additional cost to Government. Coordinate Contractor-performed test borings and other exploratory operations with Contracting Officer. Perform such explorations without disrupting or otherwise adversely affecting operations of Government and others.
   3. Where subsurface conditions or locations of Underground Facilities and structures are unforeseen or differ substantively from that shown or indicated in the Contract Documents, comply with Contract requirements for giving notice and other actions.
   4. Existing Structures and Underground Facilities:
      a. The Contract Documents show or indicate certain structures and Underground Facilities adjacent to the Work. Such information was obtained from existing records and is not guaranteed to be correct or complete. Contractor shall explore ahead of the excavation to determine the exact location of all existing structures and Underground Facilities. Existing structures and Underground Facilities shall be supported and protected from damage by Contractor. Immediately repair and restore existing structures and Underground Facilities damaged by Contractor without additional cost to Government.
      b. Movement or operation of construction equipment and machinery over Underground Facilities shall be at Contractor's sole risk. When required by owner of the

Underground Facility, prepare and submit to owner of such Underground Facility, and obtain their acceptance, of Contractor's plan presenting Contractor's analysis of the loads to be imparted during construction (including transient loading by construction equipment, machinery, and vehicles) and Contractor's proposed measures to protect structures and Underground Facilities during the Work.

c. Acceptance of or approval by Contracting Officer of Contractor's proposed evaluation and method(s) for protecting Underground Facilities in no way modifies or reduces Contractor's sole responsibility for complying with the Contract Documents, means, methods, procedures, techniques, and sequences of construction, and the safety and protection measures incident thereto.

d. Coordinate with utility owners for shut off of services in active piping and conduits. When required by utility owner, Government will assist Contractor with utility owner notifications. Completely remove buried piping and conduits indicated for removal and not otherwise indicated as being abandoned or to remain in place.

e. In general, service lines and laterals to individual residences, businesses, buildings, and structures, are not shown; however, Contractor shall assume that a service exists for each utility owner to each residence, business, building, structure, and property.

f. Do not interrupt existing utilities serving facilities occupied and used by Government or others, except when such interruption is indicated in the Contract Documents or when allowed in writing by Contracting Officer after acceptable temporary utility services are provided by Contractor for the affected structure or property.

PART 2 - PRODUCTS

2.1 MATERIALS

1. Provide granular under-slab fill under foundations, manholes, chambers, vaults, handholes, and similar work unless other under-slab material is required by the Contract Documents.
2. ASTM C33/C33M gradation size no. 67, or other material acceptable to Contracting Officer's Representative.
3. Material shall be uniformly-graded, washed, evenly graded mixture of crushed stone, or crushed or uncrushed gravel, free of organic and deleterious material, complying with the following:
   a. 100 percent passing a 1.5-inch sieve.
   b. Not more than five percent passing a no. 4 sieve.

B. Granular Structural Fill:
1. Granular structural fill, sometimes called "select granular fill", is required for improving subgrade (after removal of unsuitable material), as backfill for excavations in areas that, after restoration, will be subject to vehicular traffic, and at other locations shown or indicated for such material in the Contract Documents.
2. Material shall be well-graded, crushed aggregate, free of organic and deleterious material, complying with ADOT Class 4 Aggregate, with a maximum PI of 5.
3. For unstable subgrade conditions backfill shall consist of coarse granular material, free of organic and deleterious material, complying with ADOT Class 1 or 2 base course.

C. Soil Structural Fill:
1. At Contractor's option, soil structural fill may be low-volume change cohesive soil material or granular soil material.
2. Free of organic and deleterious matter, frozen material and debris.
3. Low-volume change cohesive soil material:
   a. ASTM D2487 classification: CL-ML or CL.
   b. Liquid limit: Less than 40.
   c. Maximum plasticity index: 20.
4. Granular soil material:
   a. ASTM D2487 classification:.

5. Previously excavated materials complying with the Contract Documents requirements for soil structural fill may be used for soil structural fill.
6. When onsite materials are unsuitable for use as soil structural fill, provide granular structural fill or approved borrow material as soil structural fill. Prior to using borrow material as soil structural fill, furnish product data Submittal and quality assurance Submittal for proposed material and obtain Contract Officer's approval and acceptance (as applicable) of the proposed material.

D. Suitable Soil Material as Non-Structural Fill:
1. Provide suitable soil material as fill for the Work, unless other material is required.
2. Material shall be:
   a. Free of rock larger than three inches in any dimension, debris, waste, frozen materials, organic material, and other deleterious material.
   b. ASTM D2487 classification: GW, GP, GM, GC, SC, SW, SP, SM, CL-ML or CL.
   c. Liquid limit: Not greater than 40.
   d. Plasticity index: not greater than 20.
3. Previously-excavated materials complying with the Contract Documents requirements for suitable soil material may be used for suitable soil material.
4. When onsite materials are unsuitable for use as suitable soil material, provide or approved borrow material as suitable soil material. Prior to using borrow material as suitable soil material, furnish product data Submittal and quality assurance Submittal for proposed material and obtain Contracting Officer's approval and acceptance (as applicable) of the proposed material.

E. Subbase Material:
1. Subbase material is granular material underlying pavement, sidewalks, and other surfaces expressly indicated in the Contract Documents.
2. Material shall be naturally- or artificially-graded mixture of natural or crushed gravel, crushed stone, or natural or crushed sand. Slag is unacceptable.
3. Material shall be ADOT Class 4 Aggregate, with a maximum PI of 5.

F. Pipe Bedding Material:
1. Materials: Unless otherwise expressly required in the Contract Documents, pipe bedding material shall be crushed stone and gravel, free of: rock or gravel larger than one-inch in any dimension, debris, waste, frozen materials, organic material and other deleterious matter.
   a. Materials shall be ASTM C33 No. 67 coarse aggregate with less than 5 percent fines.

G. Flowable Fill, also known as Controlled Low Strength Material (CLSM):
1. Unless expressly required otherwise by the Contract Documents, flowable fill may be used, at Contractor's option (following Contracting Officer's approval), for backfilling in lieu of the following materials:, pipe bedding material, soil structural fill, and suitable soil material used as non-structural fill.
2. Comply with Section 03 31 30- Concrete Materials and Proportioning.
3. Material provided shall be excavatable by ordinary mechanical excavation equipment, such as backhoes.

H. Trench Plugs:
1. Materials: Trench plugs shall be constructed from compacted clay soils with an USCS classification of CL or with at least 60 percent fines and a plasticity index of 15 or greater. Trench plugs may also be constructed from reinforced concrete, lean concrete, controlled low-strength material (CLSM), or site soils processed with bentonite.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verification of Conditions:

1. Provide Contracting Officer's Representative with sufficient notice and with means to examine areas and conditions under which trenching, backfilling, other related earthwork, and grading will be performed. Contracting Officer will advise Contractor in writing when Contracting Officer's Representative is aware of conditions that may be detrimental to proper and timely completion of the Work. Do not proceed with the Work until unsatisfactory conditions are corrected.
2. This Article, however, in no way modifies or reduces Contractor's sole responsibility for complying with the Contract Documents, for construction means, methods, procedures, techniques, and sequences, and for the safety and protection measures incident thereto.
3. Nothing in this Article, including action by Contracting Officer (if any), is for benefit of Contractor.

B. Installing the Work constitutes Contractor's approval of underlying work, subgrades, and field conditions prevailing at the time of the Work.

3.2 TEST PITS

A. Test Pits - General:
   1. Test Pit Locations Not Shown or Indicated in the Contract Documents and Test Pits Not Part of a Dedicated Bid/Pay Item:
      a. In advance of other construction, excavate test pits, make observations and measurements, and promptly fill test pits.
      b. Purpose of test pits is to determine existing subsurface conditions and locations of Underground Facilities and structures.
      c. Perform all work required in connection with excavating, stockpiling, maintaining, temporarily supporting excavations, filling, and replacing pavement for test pits. Contractor is responsible for the definite location of each existing Underground Facility involved within the area of excavation for the Work. Exercise care during such location work to avoid damaging and disrupting affected Underground Facility or structure. Contractor is responsible for repairing, at Contractor's cost, damage to Underground Facilities or structure caused during the Work.

B. Payment for Test Pits:
   1. All payment for test pits shown or indicated in the Contract Documents will be part of the bid/pay item Contract Price.

C. Results of Test Pits:
   1. Document results of all test pits excavated by Contractor.
   2. Submit to Contracting Officer's Representative written results of test pits expressly required by the Contract Documents or directed by Contracting Officer's Representative. Furnish such Submittal within three days of excavation of associated test pit, unless Contracting Officer's Representative allows more time. When Contracting Officer's Representative requests written results of test pits performed for Contractor's own use, promptly furnish such information as Informational Submittal.
   3. Required documentation of test pits shall include:
      a. Identification of Project, Site, and Contractor.
      b. Date and time of test pit excavation, indication of weather conditions, and names of personnel present and employer of each.
      c. Detailed description of location of test pit, including, where necessary, measurements to landmarks shown on the Drawings. Where appropriate, include sketch of location from the Drawings.
      d. Indication of approximate dimensions of test pit, including length, width, and depth.
      e. Verbal (written) description of observed subsurface conditions, including apparent strata, rock (if any), groundwater conditions, and approximate depth at which each was observed.
      f. Verbal (written) description of unusual conditions observed, including potential existence of Hazardous Environmental Condition, if any. Include pertinent data, such as readings from instrumentation used.

g. Description and sketch, with appropriate dimensions, of Underground Facilities and structures in the test pit. Sketches shall be clearly legible with clear, well-written annotations and dimensions.
h. Other pertinent information and data from observations made while test pit was open.
i. Brief written description of backfilling and restoration of test pit, including time operations were completed.

3.3 PREPARATION

A. Site Preparation:
1. Comply with Section 31 10 00 - Site Clearing.
2. Comply with this Section's "Erosion and Sediment Control During Construction" Article.

B. Dust Control:
1. Relative to controlling dust during construction, comply with Section 01 50 00 – Temporary Facilities and Controls and Section –01 73 40 - Execution.

C. Maintenance and Protection of Traffic:
1. Comply with Section 01 50 00 – Temporary Facilities and Controls.

3.4 EROSION AND SEDIMENT CONTROLS DURING CONSTRUCTION

A. Temporary Erosion and Sediment Controls – General:
1. Provide temporary erosion and sediment controls in accordance with Section 01 57 23 – Temporary Stormwater Pollution Prevention, Section 31 25 00 – Soil Erosion and Sediment Control, and other requirements of the Contract Documents.

3.5 EXCAVATION

A. Excavation – General:
1. Perform all excavation work, including trenching and backfilling, required to complete the Work as shown, specified, and required. Excavating work includes removing, handling, and proper disposal of earth, sand, clay, expansive-type soils, other unsuitable soil materials, gravel, hardpan, soft, weathered or decomposed rock, concrete and masonry (reinforced or unreinforced), pavements, rubbish, debris, stumps, and other materials, regardless of moisture content, within the excavation limits.
2. Trenching and backfilling work shall comply with the Contract Documents, including required grades, elevations, general dimensions, and alignment shown or indicated on the Drawings including cuts and fills, as applicable.
3. Protect existing buildings and structures against undermining and destabilization. If adjacent trenching affects foundations of one or more buildings or structures, promptly advise Contracting Officer in writing.
4. Where excavation includes rock or other material that requires drilling or specialized equipment for removal, remove rock in accordance with Section 31 23 00 - Earthwork.
5. If unsuitable material and obstructions are encountered during excavation, remove material and replace as directed by Contracting Officer.

B. Excavation Methods:
1. Traditional Excavation Methods: Excavation via traditional methods, such as backhoe, bulldozer, other earth moving construction equipment or machinery, or excavation using hand tools, is acceptable for all excavating work on the North Rim. Excavation within the Innercanyon is limited to earth moving machinery that can be safely transported via helicopter to designated landing zones or excavation using hand tools..
2. Use of Explosives:
a. Use of explosives is not allowed.
3. Micro-trenching: Installation of buried utilities using micro-trenching equipment and techniques is acceptable when the associated Underground Facility is of a size and depth appropriate for such construction method. Submit to Contracting Officer request for interpretation or clarification for requirements for resin backfilling of micro-trenches. Contracting Officer will issue such requirements via Field Order.

4. Trenchless: Where Contractor proposes trenchless construction techniques, such as directional drilling, micro-tunneling, or others, comply with the Contract Documents regarding such construction methods. When the Contract Documents do not include requirements for the proposed trenchless construction method, submit to Contracting Officer request for interpretation or clarification to obtain appropriate requirements, which will be issued via appropriate Contract modification.
5. Hydro-excavating (also known as Hydro-Vacuum Excavating):
    a. Hydro-excavating is acceptable only for excavating test pits.
    b. Avoid creating nuisances, unsafe conditions, and inconvenience to Government, owners and occupants of adjacent property, and the public when hydro-excavating is allowed and used.
6. Air Vacuum Excavating:
    a. Air vacuum excavating is acceptable only for excavating test pits.
    b. Avoid creating sandblasting outside of excavations, dust emissions, other nuisances, unsafe conditions, and inconvenience to Government, owners and occupants of adjacent property, and the public when air vacuum excavating is allowed and used.

C. Excavation Protection:
1. Provide excavation protection system(s) in accordance with Laws and Regulations to prevent injury to persons and property, including Underground Facilities.
2. Provide safe access to excavations, including emergency exit, and safe access for construction equipment and machinery, where applicable, in accordance with laws and Regulations.

D. Excavations for Manholes, Chambers, and Vaults:
1. Refer to this Section, Section 01 27 00 - Definition of Contract Line Items, and applicable details (if any) on the Drawings for requirements on excavation pay limits for manholes, chambers, vaults, and similar items.
2. Elevation of bottom of manholes, chambers, and vaults may be changed only via duly issued Contract modification. Such modifications may be directed by Contracting Officer to ensure a stable bearing surface. Elevations of piping, conduit, and similar other Underground Facilities shall be as shown or indicated on the Contract Documents.
3. Extend excavations sufficiently on each side of manholes, chambers, vaults, or similar construction only as necessary for installing precast concrete structures and appurtenances, including temporary supports of excavations, observation and inspection of the Work, and safe sloping of sides of excavations, as necessary.

E. Subgrades – General:
1. Provide firm, dense, thoroughly compacted, consolidated subgrades, free of mud, muck, and other soft and unsuitable materials. Subgrades shall remain firm and intact under all construction operations. Reinforce subgrade by providing granular structural fill on subgrades that are otherwise solid but become soft or muddy on top due to construction operations. Finished elevation of stabilized subgrades shall not be above subgrade elevations shown on the Drawings.
2. If, in Contracting Officer's Representative's opinion, subgrade becomes soft, muddy, or both, because of construction delays, failure to dewater properly, or other cause within Contractor's control, excavate subgrade to firm material, trim the subgrade, and backfill with granular structural fill material at Contractor's cost.
3. Remove unsuitable subgrade materials below foundations. Limits of required removal of unsuitable material, when not otherwise expressly shown or indicated in the Contract Documents, shall be outside exterior limits of foundations around perimeter of manhole, chamber, vault, or similar item, by not less than the greater of the following horizontal distances:
    a. Distance equal to depth of required removal of unsuitable material, below bottom of foundations.
    b. Three feet.

c. As directed by Contracting Officer's Representative, in consultation with Contractor's geotechnical engineering consultant (if any).

4. Preparation of Subgrades for Subsequent Work:
   a. Level off subgrades to receive foundations or compacted fill.
   b. Where subgrades are to receive subsequent fill or manholes, chambers, vaults, or similar items, remove loose materials and bring subgrades into compliance with the Contract Documents.
   c. Where compacted fill material is required to bring subgrade up to underside of construction, scarify existing subgrade upon which fill material will be placed to depth of six inches and then compact to density required in this Section before fill material is placed thereon.
   d. Do not excavate lower than required for buried piping or foundations, unless directed otherwise by Contracting Officer's Representative.
   e. Where excavation extends below required elevation without authorization, promptly advise Contracting Officer's Representative and remedy unauthorized excavation in accordance with the Contract Documents. Corrections may include:
      1) Under soil-supported foundations or soil-supported retaining walls:
         a) Filling entire vertical distance between excavated subgrade and material required under foundation or other structure with compacted granular structural fill.
         b) Concrete fill is acceptable for bringing elevations to proper position.
      2) For locations other than those indicated above including slabs-on-grade:
         a) Backfill and compact unauthorized excavations as indicated for authorized excavations of same classification, unless otherwise directed by Contracting Officer's Representative.
      3) Contractor is not eligible for change in Contract Price or Contract Times for remedying unauthorized excavations.
5. Evaluation and Protection of Subgrade:
   a. Advise Contracting Officer's Representative as soon as excavation is completed for Contracting Officer's Representative's observation of subgrades and, where necessary or required, inspection and testing of subgrades.
   b. If Contracting Officer's Representative determines that bearing materials at required subgrade elevations are unsuitable, provide subgrade stabilization as indicated in this Article.
   c. Do not commence further construction until subgrade under compacted fill material, under foundations and retaining walls is acceptable (in accordance with the Contract Documents) to Contracting Officer's Representative, who may consider results of required testing, and who may elect to consult with geotechnical consultant (if any).
   d. Contracting Officer's Representative shall have sufficient opportunity to observe subgrade, after successful completion of tests and inspections required by the Contract Documents.
   e. Place fill material and perform subsequent construction after subgrade is acceptable in accordance with the Contract Documents.
   f. Protect subgrade from becoming loose, wet, frozen, or soft due to weather, construction operations, or other reasons.
6. Subgrade stabilization:
   a. If subgrade under foundations or fill material, is frozen, loose, wet, or soft before construction is placed thereon, remove frozen, loose, wet, or soft material and replace with approved compacted material as directed by Contracting Officer's Representative, who may consult with the Contractor's geotechnical consultant.
   b. Loose, wet, or soft materials, when approved by Contracting Officer's Representative, may be stabilized by a compacted working mat of well graded crushed stone.
   c. Stabilization shall be in accordance with the Contract Documents. Where the Contract Documents do not address subgrade stabilization, promptly submit to Contracting Officer written request for clarification or interpretation, or other written instrument in

accordance with the Contract Documents. In such event, do not proceed until response is obtained from Contracting Officer.

F. Pipe Trench Preparation:
   1. Not more than 150 feet of trench may be opened in advance of installing pipe in trench.
   2. Trench width shall be minimized to greatest extent practical, and shall comply with the following:
      a. Trench width shall be sufficient to provide space for installing, jointing and inspecting piping. The Drawings show additional requirements for trenches. In no case shall trench be wider at top of pipe than pipe barrel's outside diameter (OD) plus one foot, unless otherwise shown or indicated.
      b. Enlargement of trench width at pipe joints may be made when required and approved by Contracting Officer's Representative.
      c. For excavating and backfilling for manholes, chambers, and vaults associated with buried piping, excavation at widest point of manhole, chamber, or vault shall be not wider than two feet greater than widest outside dimension (including bottom footings or flanges, if any, of precast concrete structures) of the associated manhole, chamber, or vault.
      d. Trench width shall be sufficient for Contractor's means of temporarily supporting the excavation and dewatering.
      e. Trench width shall be sufficient to allow thorough compaction of fill adjacent to bottom half of pipe.
      f. Do not use construction equipment or machinery that necessitates trench to be excavated to excessive width.
   3. Depth of trench shall be as shown or indicated in the Contract Documents. If required and approved by Contracting Officer in writing (such as a duly issued Contract modification), depths may be revised.
   4. Where Contracting Officer's Representative considers existing subgrade (beneath intended bedding material) unsuitable, remove and replace such unsuitable material with granular structural fill material for full width of trench, down to suitable material.

G. Freeze Protection:
   1. Do not place foundations or fill material on frozen subgrade.
   2. When freezing temperatures are expected, based on widely recognized, reputable source for weather forecasts at or near the Site, do not excavate to full depth indicated, unless foundations or fill material can be placed immediately after excavation has been completed and is acceptable in accordance with the Contract Documents.
   3. Protect from freezing excavation, subgrade, and fill materials when subsequent Work thereon is delayed.

H. Construction Traffic on Soils:
   1. Furnish and use appropriate backhoe or other low-contact pressure equipment to remove existing materials.
   2. Minimize construction traffic on existing materials and saturated soils and avoid soil disturbance.
   3. Repair disturbed subgrade materials prior to performing subsequent Work. Remedial work shall be in accordance with the Contract Documents. When not addressed in the Contract Documents, remedial Work shall be as directed by Contracting Officer via appropriate Contract modification.
   4. Allow only minimal foot traffic on bearing soils prepared for subsequent installation of fill or foundations.

3.6 FILLING AND COMPACTING

A. Filling and Compacting – General:
   1. Provide and compact all fill required for the finished grades as shown and as indicated in this Section.

B. Place fill as promptly as progress of the Work allows, but not until completing the following:
   1. Obtaining Contracting Officer's Representative's concurrence, after observation by Contracting Officer's Representative of construction below finish grade, including removal of trash and debris from excavations, waterproofing (where required), and related work.
   2. Successful completion of required field quality control activities for work that will be covered or concealed by filling, including field quality control testing of buried piping, conduits that will convey fluid or gas, and exfiltration testing (where performed or required) of manholes, chambers, vaults, and similar items. For manholes, chambers, and vaults, where infiltration testing or vacuum testing is performed or required, such testing may be performed following backfilling.
   3. Completion of proper documentation of as-constructed conditions and existing conditions (such as locations of Underground Facilities) in accordance with Section 01 77 00 – Closeout Procedures.
   4. Removal of temporary supports of excavations.
   5. Remove fill that includes organic materials or other unacceptable material and replace with fill material complying with the Contract Documents.

C. Placement – General:
   1. Place fill to the grades shown or indicated. Bring up evenly on all sides fill around manholes, chambers, vaults, and similar items, and Underground Facilities.
   2. Management of Moisture Content:
      a. Place fill materials at moisture content and density as indicated in this Article's requirements on compaction density. Prior to placing fill material, optimum moisture and maximum density properties for proposed material shall be obtained from Contracting Officer's Representative, who may consult with Contractor's geotechnical consultant, if any. Furnish and use construction equipment and machinery capable of adding measured amounts of water to fill materials to bring fill materials to a condition within required moisture content range. Control moisture for each lift necessary for required compaction.
      b. Furnish and use construction equipment and machinery capable of discing, aerating, and mixing fill materials to ensure reasonable uniformity of moisture content throughout fill materials, and to reduce moisture content of borrow materials by air drying, when necessary.
      c. When subgrade or fill materials requires moisture-conditioning before compaction, fill material shall be sufficiently mixed or worked on the subgrade to ensure uniform moisture content throughout the lift of material to be compacted.
      d. Materials at moisture content in excess of specified limit shall be dried by appropriate means, such as aeration or stockpiling for gravity drainage and air drying.
   3. Compaction:
      a. Compaction Equipment:
         1) Perform compaction with construction equipment and machinery suitable for the type of fill material placed. Unless otherwise required, compact cohesive soils using sheepsfoot compactor, and compact granular materials using pneumatic rollers, vibrators, or other construction equipment or machinery to obtain required density.
         2) Select and use equipment capable of providing the minimum density required in the Contract Documents.
         3) Use light compaction equipment and machinery, with equipment or machinery gross weight not exceeding 7,000 pounds within horizontal distance of 10 feet from the wall of completed, below-grade manholes, chambers, vaults, and similar items, and walls of adjacent buildings or structures.
         4) Furnish and use construction equipment and machinery capable of compacting in restricted areas next to structures and around piping and Underground Facilities.
      b. Compaction Test Area:

1) Effectiveness of construction equipment and machinery selected by Contractor for compacting, shall be tested at start of compacted fill work by constructing a small section of fill within the area where fill will be placed.
2) If tests on the test section of fill indicate that required compaction is not obtained, do one or more of the following: increase the quantity of coverages, decrease the lift thicknesses, or use different compactor equipment or machinery.

4. Placement of Lifts and Compaction:
   a. Place fill materials in horizontal, loose lifts, not exceeding specified thickness prior to compacting.
   b. Place fill in manner ensuring uniform lift thickness after placing.
   c. Preparation of surfaces for placement of fill:
      1) Before commencing placing of fill, scarify underlying material to depth of six inches.
      2) Where ground surface is steeper than one vertical to four horizontal ("1V:4H"), plow surface to bench and break up surface so that fill material will bind with existing surface. For slopes steeper than 1V:4H and less than 1V:1.5H, bench shall be six feet wide. For slopes steeper than or equal to 1V:1.5H and less than 1V:0.5H, bench shall be four feet wide.
   d. Mechanically compact each lift, by not less than two complete coverages of compactor. One coverage is defined as the conditions reached when all portions of the fill lift have been subjected to the direct contact of compactor's compacting surface.
   e. Compaction of fill materials by inundation with water is unacceptable.
5. Restrictions:
   a. Do not place fill materials when standing water is present on surface of area where fill will be placed.
   b. Do not compact fill when standing water is present on the fill to be compacted.
   c. Do not place or compact fill in frozen condition or on top of frozen material.
   d. Fill containing organic or deleterious materials or other unacceptable material previously described shall be removed and replaced prior to compaction.
6. If required densities are not obtained because of improper control of placement or compaction procedures, or because of inadequate or improperly functioning compaction equipment or machinery, perform all work necessary to provide required densities. Such work shall include, at no additional cost to Government, complete removal of unacceptable fill areas and replacement and re-compaction until acceptable fill is provided.
7. Repair, at Contractor's cost, observed or measured settlement. Make repairs and replacements as necessary and required within 30 days after being so advised by Contracting Officer's Representative, unless more-timely action is necessary to avoid or reduce the potential for safety hazards or damage to property.

D. Fill Placed in Pipe Trenches:
1. Piping trenches may be backfilled prior to testing of piping, unless nature of test requires observation of pipe or manholes, chambers, vaults, or similar items during testing. Do not construct building or structure over piping until piping has been successfully tested and passed.
2. Pipe Bedding: Pipe bedding material shall be as follows:
   a. Unless otherwise shown or indicated, install buried piping on not less than six-inch layer of aggregate pipe bedding material underneath piping.
   b. Pipe bedding material shall extend upward from elevation required immediately above to 6 inches above crown of associated buried piping.
   c. Pipe bedding material shall be spread and the surface graded to provide a uniform and continuous support beneath piping at all points between bell holes or pipe joints.
   d. Install initial layer of pipe bedding material on prepared subgrade and compact as required. Provide appropriate divot in the bedding material to accommodate each pipe bell or joint. Install piping so that pipe barrel carries the weight of each pipe segment, rather than pipe bells or joints bearing the pipe's weight.

e. Slight disturbance of installed pipe bedding material surface during withdrawal of pipe slings or other lifting tackle is acceptable.
f. After each pipe's bedding material has been graded, and the piping has been aligned, joined in accordance with the Contract Documents, and placed in final position on bedding material, provide and compact sufficient pipe trench fill material under and around each side of pipe and back of bell or end thereof to hold piping in proper position and maintain alignment during subsequent pipe jointing and embedment operations.
g. Subsequent to forming pipe joint, place additional pipe bedding material, in required lifts, to required top elevation of pipe bedding material. Ensure bedding material is properly and fully placed under pipe "haunches" for proper support of piping. Compact lifts of pipe bedding material in accordance with this Article prior to placing subsequent backfill material.

3. Placing and Compacting Pipe Trench Fill: Unless otherwise shown, placing and compacting of pipe trench fill materials shall comply with the following:
a. Deposit and compact pipe trench fill material uniformly and simultaneously on each side of piping to prevent lateral displacement of piping.
b. Each layer of pipe trench fill material shall be compacted by not less than two complete coverages of all portions of surface of each lift using appropriate compaction equipment or machinery.
c. Method of compaction and compaction equipment and machinery used shall be appropriate for material to be compacted and shall not transmit damaging shocks to piping.

E. Installation of Flowable Fill, also known as Controlled Low Strength Material (CLSM):
1. Discharge flowable fill from ready-mix delivery vehicle into excavation, or other space to be backfilled with such material, via chutes, pumping, or hand-carried in buckets, in accordance with the Contract Documents (including the Specifications of Division 03 - Concrete) and requirements of flowable fill ready-mix Supplier.
2. Subgrade on which flowable fill is placed shall be free of disturbed or softened material and water.
3. Place flowable fill in lifts not greater than four feet depth.
4. Placing of each lift of flowable fill shall be a continuous operation, to the extent practicable.
5. Place flowable fill in manner that prevents flotation of piping or other damage to the Work, other Underground Facilities, and adjacent property.
6. Cold Weather Placing:
a. Do not place flowable fill on frozen subgrade.
b. Flowable fill batching, mixing, and placing may be performed when weather conditions are favorable, and ambient air temperature is 34 degrees F and rising.
c. At the time of placing, flowable fill must have a temperature of not less than 40 degrees F.
d. Discontinue placing flowable fill when ambient air temperature is 38 degrees F or less and falling.
7. After placing flowable fill, allow not less than 12 hours before placing subsequent lift or approved otherwise by Contracting Officer's Representative.
8. Prevent contact with flowable fill by construction vehicles and equipment, or other traffic, for not less than 24 hours after placing, or until flowable fill is sufficiently hard to prevent rutting or other damage by vehicles and construction equipment.

F. Fill Required Under Subbase for Pavement and Other Traveled Ways:
1. Where bottom of pavement or surfacing of other traveled way will be at elevation equal to or less than preconstruction ground surface, after required excavation and preparation of subgrade, provide required subbase material to required depth. At underside of pavement or other surfacing of traveled way, subbase material shall extend horizontally six inches beyond edges of pavement or other surfacing. Exterior edges of subbase material shall be

placed and compacted at uniform slope of one foot vertical-to-one foot horizontal away from top edge of subbase down to subgrade.

2. Where bottom of pavement or surfacing of other traveled way will be at elevation greater than preconstruction ground surface, after required preparation of subgrade, provide required fill material and subbase material to required depth. At underside of pavement or other surfacing of traveled way, subbase material shall extend horizontally six inches beyond edges of pavement or other surfacing. Exterior edges of subbase material and fill material shall be placed and compacted at uniform slope of one foot vertical-to-one foot horizontal away from top edge of subbase down to subgrade.

G. Temporary Pavement:
   1. Provide not less than 1.5 inches of temporary bituminous pavement immediately after filling excavations in paved roads and other pavement subject to vehicular traffic, where such pavement will remain permanently.
   2. Maintain surface of paved area over the fill in good and safe condition during progress of the Work, and promptly fill depressions over and adjacent to the fill area caused by settlement of fill.
   3. Permanent replacement pavement shall be equal to that of the existing roadways plus one inch, unless otherwise shown or specified.

H. Compaction Density Requirements:
   1. Compaction Requirements for Sitework (including civil, grading, and Underground Facilities):

LOCATION	COMPACTION DENSITY	MOISTURE CONTENT
Under Pavement, Other Traveled Ways, Curbs, Gutters, Sidewalks, and Buried Utilities (including manholes, chambers, vaults, and similar items):		
Cohesive soils	95% per ASTM D698	-2 to +2% of optimum
Cohesionless soils	75% relative density per ASTM D4253 and ASTM D4254	
Sitework Not Covered by the Above Category or Classification:		
Cohesive soils	90% of ASTM D698	-2 to +2% of optimum
Cohesionless soils	65% relative density per ASTM D4253 and ASTM D4254	

   2. Compaction Requirements for Materials Supporting Foundations and Adjacent to Buildings and Structures:

LOCATION	COMPACTION DENSITY	MOISTURE CONTENT
Fill supporting foundations and fill adjacent to exterior of buildings and structures, adjacent to walls, piers, and columns	95% per ASTM D698	-2 to +2% of optimum

   3. Compaction Density and Moisture Requirements for Specific Areas:

LOCATION	COMPACTION DENSITY	MOISTURE CONTENT
Outside buildings and structures, under slabs-on-grade	98% per ASTM D698	-2 to +2% of optimum

   4. Fill Lift Thickness:

a. Unless expressly shown or indicated otherwise, place fill in trenches, below piping, below foundations, or under paved areas in horizontal uncompacted layers not exceeding eight inches deep, and thoroughly compact each before next layer is placed.
b. In pipe trenches, above elevation of bottom of pipe, horizontal uncompacted layers shall be six inches deep.

5. Fill Moisture Content: Fill shall be wetted and thoroughly mixed to achieve moisture content to within the required range indicated in the applicable table, above, with the following exception(s):
   a. On-site clayey soils: Optimum to plus two percent (moisture less than optimum is unacceptable).
6. Replace natural, undisturbed soils or compacted soil subsequently disturbed or removed by construction operations with materials compacted as indicated.
7. Field quality control testing for density; to verify that specified density was obtained, will be performed during each day of compaction work. Responsibility for field quality control testing is indicated in this Section's "Field Quality Control" Article.
8. When field quality control testing indicates unsatisfactory compaction, provide additional compaction necessary to obtain the required compaction. Perform additional compaction work at no additional cost to Government until required compaction is achieved. Such work includes complete removal of unacceptable (as determined by Contracting Officer's Representative) fill areas and replacement and re-compaction until acceptable fill is provided in accordance with the Contract Documents.

I. Replacement of Unacceptable Excavated Materials: In cases where over-excavation to replace unacceptable soil materials is required, backfill the excavation to required subgrade with granular structural fill material and thoroughly compact in accordance with applicable table in this provision. Slope the sides of excavation in accordance with the maximum allowable inclinations specified for each element of the Work.

3.7 UNAUTHORIZED EXCAVATION

A. Unauthorized Excavation – General:
   1. Excavations outside lines and grades shown or indicated and that are not approved by Contracting Officer's Representative (including excavations below required subgrade elevation, unless Contracting Officer's Representative directs removal of unsuitable subgrade material), together with removing and disposing of the associated material, shall be at Contractor's cost and risk.
   2. Fill unauthorized excavations with properly compacted granular structural fill material at Contractor's cost.

3.8 PAVEMENT SUBBASE COURSE

A. Pavement Subbase – General:
   1. Place subbase material, in layers of specified thickness, over subgrade to support pavement base course.
   2. Prepare subgrade underlying subbase course in accordance with this Section's general requirements for subgrades.

B. Grade Control:
   1. During construction, maintain lines and grades including crown and cross-slope of subbase course.
   2. Comply with Section 01 73 40 - Execution, regarding staking and elevation controls.

C. Placing of Pavement Subbase Course:
   1. Provide subbase material where required to the limits shown or indicated.
   2. Place subbase course material on prepared subgrade in layers of uniform thickness, in accordance with indicated cross-section and thickness. If the Contract Documents do not show or indicate required section of replacement paving, replacement subbase course shall be of thickness equal to existing subbase course.

3. Maintain optimum moisture content for compacting subbase material during placing operations.
4. After completing compaction, other than that necessary for bringing material for the next course, do not haul or drive over compacted subbase.
5. To prevent softening of subgrade, do not install pavement subbase in excess of 500 feet in length without compacting.
6. If subgrade material becomes churned up into or mixed with subbase material, remove the mixed material and replace with clean, compacted subbase material.

3.9 GRADING

A. Grading – General:
   1. Uniformly grade areas within limits of grading under this Section, including adjacent transition areas.
   2. Smooth subgrade surfaces within specified tolerances, compact with uniform levels or slopes between points where elevations are shown, or between such points and existing grades.

B. Grading Outside Building Lines:
   1. Grade areas adjacent to building lines to drain away from buildings and structures.
   2. Grade to prevent ponding of water. Provide grades and slopes as shown on the Drawings and to allow effective drainage to proper drainage routes, such as catch basins, ditches or swales intended to convey storm water, and other storm water drainage routes. Where intended grade for effective drainage is unclear, promptly submit to Contracting Officer request for interpretation and obtain Contracting Officer's Representative's interpretation or clarification before proceeding,
   3. Provide final surfaces free of irregular surface changes, in accordance with the following:
      a. Grassed Areas and Areas Covered with Gravel, Stone, Wood Chips, or Other Special Cover: Finish areas to receive topsoil or special cover to within not more than one inch above or below required subgrade elevations. Requirements for topsoil and finish grading are in Section 32 91 13 - Finish Grading.
      b. Sidewalks: Shape surface of areas under sidewalks to line, grade, and cross section, with finish surface not more than one inch above or below required subgrade elevation.
      c. Pavement (including hard-surfaced pavement and areas with stone surfacing): Shape surface of areas under pavement to line, grade, and cross section, with finish surface not more than 0.5-inch above or below required subgrade elevation.

C. Grading Surface of Fill Under Concrete Slabs-on-Grade:
   1. Grade smooth and even, free of voids, compacted as specified, and to required elevation.
   2. Provide final grades within tolerance of 0.5-inch when tested with a ten-foot straight edge.

3.10 DISPOSAL OF EXCAVATED SPOIL MATERIALS

A. Spoil Disposal - General:
   1. Contractor shall haul away excavated material that does not comply with requirements for fill, or is in excess of quantity required for fill.
   2. Disposal of materials shall comply with Laws and Regulations, at no additional cost to Government.

3.11 FIELD QUALITY CONTROL

A. Site Tests and Inspections:
   1. Testing Laboratory:
      a. Employ a testing laboratory to perform field quality control testing.
      b. Testing Laboratory Scope:
         1) Perform field moisture content tests (in accordance with ASTM D2216 or ASTM D6938) and density tests (in accordance with ASTM D1556 or ASTM D6938) to verify that required compaction of fill materials was provided.

2) Tests of actual unconfined compressive strength or bearing tests (in accordance with ASTM D2850 or ASTM D2166) on each stratum.
3) Furnish to Contracting Officer's Representative and Contractor written results of each test.

c. Authority and Duties of Testing Laboratory:
1) Testing laboratory representatives ("technician") shall inspect the materials in the field, perform testing, and report findings to Contracting Officer's Representative and Contractor. When the Work is defective, technician will direct attention of Contracting Officer's Representative and Contractor to such defective work.
2) Technician shall not supervise performance of any of Contractor's Work. Technician shall not perform other duties for Contractor.
3) Work will be tested and inspected by testing laboratory as the Work progresses, but failure to detect defective work (including defective materials) shall not, in any way, prevent later rejection when defect is discovered, nor shall it obligate Contracting Officer for Substantial Completion or final acceptance. Technicians are not authorized to revoke, alter, relax, enlarge, or release requirements of the Contract Documents, or to approve or accept any portion of the Work.

2. Responsibilities and Duties of Contractor:
a. Use of testing laboratory or geotechnical consultant in no way relieves Contractor of responsibility to provide work in compliance with the Contract Documents.
b. When Government furnishes services of testing laboratory, geotechnical consultant, or any combination thereof, for trenching, backfilling, and related earthwork, Government and Contracting Officer's Representative do not represent that the sampling and testing performed by Government-hired testing laboratory or observations by Government-hired geotechnical consultants, is sufficient for Contractor's purposes, or to monitor all aspects of the trenching, backfilling, and related earthwork. If additional quality control is needed to verify that the work performed and materials delivered to the Site are in accordance with the Contract Documents, Contractor may retain its own, independent testing laboratory to perform additional sampling and testing for Contractor's purposes.
c. To facilitate testing laboratory and geotechnical consultant, Contractor shall advise testing laboratory and geotechnical consultant not less than two days in advance of filling operations to allow for completion of field quality control testing and for assignment of personnel.
d. Contractor shall be responsible for providing required compaction for fill and other earthwork. Contractor shall control construction operations by performing confirming tests to verify that Contractor has complied with the Contract Documents relative to compaction and moisture content.
e. Contractor shall demonstrate adequacy of compaction equipment and procedures before exceeding one or more of the following quantities of earthwork. Each test location shall include tests for each layer, type, or class of fill to finish grade.
1) 200 linear feet of trench fill.
2) 10 cubic yards of granular structural fill.
3) 100 cubic yards of suitable soil material.
4) 50 cubic yards of subbase material.

3. Testing laboratory will inspect and indicate acceptable subgrades and fill layers before construction work is performed thereon. Testing of subgrades and fill layers shall be as follows:
a. Trenches for Underground Facilities (including buried ductbanks):
1) In Open Fields: Two locations every 1,000 linear feet.
2) Along Dirt or Gravel Roads or Off Traveled Right-of-Way: Two locations every 500 linear feet.
3) Crossing Paved Roads: Two locations along each crossing.
4) Under Pavement Cuts or Within Two Feet of Pavement Edges: One location every 400 linear feet.

b. For granular structural fill: Once per lift for each 1,000 square feet placed and compacted.
c. For soil structural fill:
    1) For backfilling trenches directly adjacent to buildings or structures, on 30-foot intervals on all sides of the building or structure for every compacted lift, but not less than one per lift on each side of the building or structure for buildings or structures less than 60 feet long on a side.
    2) For material not used as fill against buildings or structures, once per lift for each 1,000 square feet placed and compacted.
d. For Suitable Soil Material: One per 1,000 square feet on every compacted lift.
e. Subbase Material: One per 1,000 square feet on every compacted lift.

4. Periodic compliance tests will be made by Contracting Officer's Representative to verify that compaction is in accordance with the Contract Documents, at no cost to Contractor. Contractor shall remove overburden above the level at which Contracting Officer's Representative wishes to test and shall fill and re-compact the material after testing is complete.

B. Defective Work:
1. If testing laboratory reports or inspections indicate subgrade, fill, or bedding compaction less than specified density, Contractor shall remove unacceptable materials as necessary and replace with specified materials and provide additional compaction at Contractor's cost until subgrades, bedding, and fill are acceptable.
2. Costs for retesting of subgrade, fill, or bedding materials that did not originally comply with the Contract Documents, including required density, shall be paid by Contractor.

3.12 RESTORATION

A. Restoration – General:
1. Perform finish grading, including providing topsoil, in accordance with Section 32 91 13 - Finish Grading.
2. Provide replacement paving, sidewalks, curbs, gutters, and similar items in accordance with the Contract Documents including the Specifications of Division 32 - Exterior Improvements.
3. Restore landscaping including lawn and meadow areas, in accordance with the Contract Documents including the Specifications of Division 32 - Exterior Improvements.

END OF SECTION

SECTION 31 25 00
SOIL EROSION AND SEDIMENT CONTROL

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Soil erosion and sediment control.

B. Related Specification Sections include but are not necessarily limited to:
   1. Division 00 - Procurement and Contracting Requirements.
   2. Division 01 - General Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Erosion control standards: Standards and Specifications for Soil Erosion and Sediment Control in Developing Areas by the United Sates Department of Agriculture (USDA), Soil Conservation Service, College Park, Maryland.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 MATERIALS

A. Erosion Control Logs, Wattles, and Matts:
   1. Aspen fiber, not straw, shall be used for all erosion-control products such as erosion control logs and wattles.
   2. Coconut fiber materials will be used, rather than jute or other fabrics, for erosion-control blankets on slopes greater than 3:1.

B. Reinforced Rock Berm:
   1. Crushed Rock:
      a. Crushed rock shall be fractured face (all sides) and shall have a gradation of 1-1/2 IN -.
      b. Recycled concrete meeting gradation requirement may be used
   2. Wire Mesh:
      a. 10-gauge wire twisted into a mesh with a maximum opening of 1.0 IN. Roll width shall be 48-IN.
      b. Wire mesh shall be secured using “Hog Rings” or wire ties at 6 IN centers along all joints and at 2 IN centers on ends of berm.

C. Silt Fence:
   1. Synthetic filter fabric shall conform to the following specifications:

Fabric Properties	Minimum Requirements	Test Method
Grab tensile strength (lbs.)	100	ASTM D4632
Elongation @ Failure (%)	15	ASTM D4632

Mullen Burst Strength (psi)	250	ASTM D3786
Puncture Strength (lbs.)	50	ASTM D4833
Apparent Opening Size (mm)	0.60	ASTM D6461
Ultraviolet Radiation Stability (%)	90	ASTM D4355

2. Synthetic filter fabric shall contain ultraviolet ray inhibitors and stabilizers to provide a minimum of six months of expected usable construction life at a temperature range of 0 to 120 degrees Fahrenheit.
3. Wood or steel fence posts shall be a minimum length of 4 feet. Steel posts will be standard "T" or "U" posts weighing not less than 1.33 pounds per lineal foot.
4. Wire fencing support, if needed, shall be at least 2.5 feet high and minimum 14-1/2 gauge with a maximum 6 inch mesh opening.

PART 3 - EXECUTION

3.1 PREPARATION

A. Prior to General Stripping Topsoil and Excavating:
   1. Install perimeter dikes and swales.
   2. Excavate and shape sediment basins and traps.
   3. Construct pipe spillways and install stone filter where required.
   4. Machine compact all berms, dikes and embankments for basins and traps.
   5. Install silt fence and erosion control logs where indicated, per manufacturer's specifications.

B. Construct sediment traps where indicated on Drawings during rough grading as grading progresses.

C. Temporarily seed basin slopes and topsoil stockpiles:
   1. Rate: 1/2 LB/1000 SQFT.
   2. Reseed as required until good stand of grass is achieved.

3.2 DURING CONSTRUCTION PERIOD

A. Maintain Basins, Dikes, Traps, Stone Filters, Silt Fence, Erosion Control Logs, etc.:
   1. Inspect regularly especially after rainstorms.
   2. Repair or replace damaged or missing items.

B. After rough grading, sow temporary grass cover over all exposed earth areas not draining into sediment basin or trap.

C. Construct inlets as soon as possible.
   1. Excavate and tightly secure straw bales completely around inlets as detailed on Drawings.

D. Provide necessary swales and dikes to direct all water towards and into sediment basins and traps.

E. Do not disturb existing vegetation (grass and trees).

F. Excavate sediment out of basins and traps when capacity has been reduced by 50 PCT.
   1. Remove sediment from behind bales to prevent overtopping.

G. Topsoil and Fine Grade Slopes and Swales, etc.: Seed and mulch as soon as areas become ready.

3.3 NEAR COMPLETION OF CONSTRUCTION

A. Eliminate basins, dikes, traps, etc.

B. Grade to finished or existing grades.

C. Fine grade all remaining earth areas, then seed and mulch.

END OF SECTION

SECTION 31 32 19
SLOPE ARMORING SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Installation all components of a slope armoring system, including the rockfall protection mesh, rock foundation anchors, anchorage assemblies, boundary ropes, and miscellaneous materials.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 31 68 13 – Rock Foundation Anchors
   2. Section 31 23 00 – Earthwork
   3. Section 09 06 00 – Color Schedule

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Association of State Highway and Transportation Officials (AASHTO):
      a. M111, Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
      b. M232, Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
   2. American Galvanizers Association.
   3. ASTM International (ASTM):
      a. A123, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
      b. A153/A153M, Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
      c. A641, Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire.
      d. B117, Standard Practice for Operating Salt Spray (Fog) Apparatus.
      e. B571, Standard Practice for Qualitative Adhesion Testing of Metallic Coatings.
      f. E3, Standard Guide for Preparation of Metallographic Specimens.
   4. Federal Specification RR-W-410F, Wire Rope and Strand

B. Qualifications:
   1. Provide a Professional Engineer, registered in the State of Arizona, with at least three years of experience in the design and construction of slope armoring systems. The Professional Engineer shall have experience gained on no less than two projects in successful design and supervision of similar slope armoring applications. The Professional Engineer shall design and supervise the installation of the slope armoring system. The Professional Engineer shall certify that the installation of the slope armoring system is in accordance with his design and this section.
   2. Provide a specialty subcontractor with experience in the design and construction of slope armoring systems. Submit to the Contracting Officer's Representative a project reference list of at least three separate permanent slope armoring projects successfully completed within the last five years with conditions similar to those indicated and/or proposed. Include a brief description of each project and the client's contact person's name and current phone number for each project listed. Provide written proof of experience with all specified procedures and proposed equipment to be used for this application. The specialty subcontractor shall provide an on-site representative who shall supervise the installation of the slope armoring system.

1.3 SUBMITTALS

A. Shop Drawings:

1. Product technical data, including:
    a. General Notes. Necessary information on the design and construction of the slope armoring system.
    b. Materials and Quantity Summary List. All required items for the slope armoring system, including any incidental items.
    c. Plan and Elevation Views. The slope armoring system alignment and offset with respect to Plan layout.
    d. Structural and Geometric Details. Complete structural and geometric details including, but not limited to wire mesh; wire mesh lap; end, lateral, and intermediate support boundary wire ropes; rock foundation anchors and anchor connections; and boundary wire rope connections.
    e. Galvanized Surface Colorant:
        1) A copy of the manufacturer's product Safety Data Sheet.
        2) Independent lab tests demonstrating that the material is environmentally safe.
        3) Bend Test results demonstrating that the galvanized steel, treated by the reactive color treatment, exhibits no lifting or separation from mechanical stress as per ASTM B571.
        4) Results of the cross-sectional metallographic examination demonstrating that the application surface has an average conversion depth of 0.1 MIL and no decrease in total thickness, per ASTM E3.
        5) Corrosion resistance demonstrating the performance of the galvanized steel, treated with the reactive color treatment, meets or exceeds the corrosion resistance of untreated galvanized steel after 50 hours of salt spray when tested in accordance with ASTM B117.
        6) Reactive color treatment must not include volatile organic compounds (VOCs) or chromate compounds.

B. Samples:
    1. Galvanized Surface Colorant Sample Section:
        a. Apply colorant to a minimum 12 IN by 12 IN sample section of galvanized metal. In the event that more than one sample section is required by the Contracting Officer, each additional sample section will be paid for as change order work.
        b. Use the Contracting Officer approved sample as the standard of comparison in determining acceptability of final colorant.

C. Field Installation Manual:
    1. Installation manual explaining the slope armoring system details, installation steps and details of special cases (e.g. cutting openings for trees and landscaping).
    2. System drawings, showing all the elements required for the correctly installed slope armoring system (e.g., rock foundation anchor pattern, wire mesh boundaries, and anchoring plates).
    3. Detailed connections scheme, allowing transmission of the full applied forces for each design condition.

B. Informational Submittals:
    1. Construction sequence and schedule.
    2. Descriptions of procedures, means, methods, equipment, and materials.
    3. Stamped calculations for the slope armoring system design.
    4. Supporting technical data for all manufactured items.
    5. Names of the proposed on-site supervisor and on-site representative.

PART 2 - PRODUCTS

Product materials are to conform with the following.

2.1 COMPONENTS

A. Wire Mesh:

1. High-tensile steel wire
    a. Tensile strength greater than or equal to 256,719 PSI (1,770 N/MM^2).
    b. Wire diameter: 0.0157 IN. (4.0 MM)
    c. Mesh opening width: 2.48 IN. (63 MM) (+/- 3%)
    d. Wire shall be galvanized in accordance with ASTM A 641 Class C or equivalent galvanizing

B. Anchor plates:
1. Steel spike plate
    a. Longitudinal bending resistance greater than or equal to 1,844 LB-FT (2.5 kN-M).
    b. Diamond-shaped geometry with built-in spikes that fix the mesh at least in 3 points.
    c. Length of spikes: minimum of 0.787 IN. (20 MM)
    d. Hole diameter compatible with rock anchor tendon diameter.
    e. Plates shall be Grade 36 (minimum) and shall be hot-dip galvanized in accordance with AASHTO M 111 (ASTM A123) or AASHTO M 232 (ASTM A153). The minimum galvanized coating thickness shall be 2.0 MILS (50 MICRONS).
    f. Hardware (including nuts, washers, and fasteners) shall be Grade 60 (minimum). Hardware to be compatible with rock foundation anchor tendon type.

C. Wire Mesh and Anchor Plate Finish:
    a. Galvanized Surface Colorant:
        1) The colorant must be a soluble solution (powdered form is not allowed) and must contain organic acids to slightly etch into galvanized surfaces without affecting the protection of the galvanized layer.
        2) Organic acids used shall have no negative effects on the environment.
        3) The product must react with the target surface to produce a rustic brown/weathered finish. See section 09 06 00 – Color Schedule.

D. Rock foundation anchors:
1. Refer to Section 31 68 13 – Rock Foundation Anchors.
2. Rock foundation anchors shall be sized for a minimum insertion of 10-FT into competent undisturbed rock, based on an allowable grout-to-rock bond strength of 15 PSI recommended for the Bright Angel Shale rock unit within the project area.

E. Boundary Rope
1. Rope diameter: 12 MM (0.5 IN)
2. Minimum break strength of 20,000 LBS (9,072 KG).
3. Independent Wire Rope Core construction.
4. Galvanized in accordance with Federal Specification RR-W-410F or equivalent.

F. Miscellaneous Materials and Accessories
1. All miscellaneous materials and accessories associated with the slope armoring system, such as anchors, wire rope connection clips, press claws, thimbles, etc. shall be compatible with the slope armoring system and supplied by the same manufacturer as provided the wire mesh. Steel miscellaneous materials and accessories shall be hot-dip galvanized in accordance with AASHTO M 111 (ASTM A123) or AASHTO M 232 (ASTM A153).

2.2 SOURCE QUALITY CONTROL

1. Testing
    a. Provide proof of the load bearing mode of the full system based on a 1:1 scale test in a size that the wire mesh with four surrounding rock foundation anchors demonstrates the expected behavior without boundary effects. The test shall be instrumented with force and deformation measurement devices (e.g. laser scan). The test shall be supervised and reported by an independent laboratory or testing facility. Test conditions shall be repeatable and comparable.
    b. For single elements quality and performance control, the following test reports are requested to be executed and provided by an independent laboratory or testing facility:

1) Wire mesh tensile strength / strain tests on min. 3 FT by 3 FT (1.0 M x 1.0 M) sample.
2) Slope parallel resistance of wire mesh with plate connected to rock foundation anchor.
3) Shearing-off resistance of wire mesh with plate on soil (puncture test).
4) Bending test on plate.
5) Wire mesh to mesh connection tests.

PART 3 - EXECUTION

3.1 INSTALLATION

A. General installation guidelines:
1. Galvanized Surface Colorant application:
   a. Preparation:
      1) Target galvanized surfaces to be treated must be free of excessive oils, dirt, and other contaminants. All surfaces must be dry before application of colorant.
   b. Colorant application:
      1) After areas to be treated have been prepared and the surface colorant sample approved, apply colorant to target galvanized surfaces. Apply product according to the colorant Manufacturer′s instructions to achieve a color consistent with the approved sample.
      2) Colorant must be applied uniformly. Irregularities (above and beyond those that the product naturally produces) must be corrected according to the product manufacturer′s recommendations.
      3) Treated surfaces must be kept dry for a period of 5 days following the application of colorant or additional coats may be required.
      4) Material should be applied outdoors with daytime temperatures above 60 degrees F and nighttime temperatures above 45 degrees F (follow colorant Manufacturer's recommendations regarding allowable range in ambient temperature during application).
      5) For most applications, Manufacturer shall apply colorant at their facility.
2. Wire mesh placement:
   a. Lay wire mesh on the slope by unrolling down the slope.
   b. Overlap adjacent wire mesh panels by a minimum of 1 FT.
   c. Rolls can be shortened or lengthened as necessary by removing or adding sections.
   d. Fasten the overlapped wire mesh panels with wire rope connection clips in accordance with the Plans, the approved Shop Drawings, and the approved Field Installation Manual.
3. Boundary rope placement:
   a. Install required boundary ropes along edge of wire mesh, in line with outermost anchor plate assemblies.
   b. Fasten the wire mesh to boundary ropes in accordance with the approved Shop Drawings and approved Field Installation Manual.
   c. Tighten boundary ropes and pull wire mesh tight against the ground.
4. Anchorage assembly placement:
   a. Install rock foundation anchors per Section 31 68 13 – Rock Foundation Anchors
      1) Anchoring pattern to be in accordance with the Plans, the approved Shop Drawings, and the approved Field Installation Manual.
   b. Place anchor plates (spike plates) onto the rock foundation anchors.
      1) Using a hydraulic wrench, tighten the nuts and push the spike plate and wire mesh into the hollow in order to tension the mesh to 10,000 LBS (10 KIPS).
      2) Torque the nuts in accordance with the approved Shop Drawings and Manufacturer's recommendations.

3.2 STORAGE

A. Surfaces treated with galvanized surface colorant must be properly stored at the construction yard after delivery and before, and up to, time of installation in accordance with American Galvanizers Association (AGA) Guidelines.
   1. If components are stacked, spacers should be placed between rows to allow for necessary airflow.
   2. Components should be stacked perpendicular to the ground slope to allow for proper drainage.
   3. Components should be elevated on timber cribbing or equivalent to prevent direct contact with the ground surface or plant matter.

END OF SECTION

This page intentionally left blank.

SECTION 31 68 13
ROCK FOUNDATION ANCHORS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Rock anchors.

B. Related Specification Sections include but are not necessarily limited to:
   1. Division 00 – Procurement and Contracting Requirements
   2. Division 01 – General Requirements
   3. Section 05 50 00 - Metal Fabrications

C. All anchors indicated on the Drawings or specified to be rock anchors obtaining their required tension load carrying capacity from embedment into rock

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Institute of Steel Construction (AISC):
      a. Steel Construction Manual.
   2. ASTM International (ASTM):
      a. A29, Standard Specification for General Requirements for Steel Bars, Carbon and Alloy, Hot-Wrought.
      b. A36, Standard Specification for Carbon Structural Steel.
      c. A153, Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
      d. A311, Standard Specification for Cold-Drawn, Stress-Relieved Carbon Steel Bars Subject to Mechanical Property Requirements.
      e. A572, Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel.
      f. A615, Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement.
      g. A722, Standard Specification for Uncoated High-Strength Steel Bar for Prestressing Concrete.
      h. C31, Standard Practice for Making and Curing Concrete Test Specimens in the Field.
      i. C33, Standard Specification for Concrete Aggregates.
      j. C39, Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens.
      k. C150, Standard Specification for Portland Cement.
      l. F436, Standard Specification for Hardened Steel Washers.
      m. F3125, Standard Specification for High Strength Structural Bolts, Steel and Alloy Steel, Heat Treated, 120 KSI (830 MPa) and 150 KSI (1040 MPa) Minimum Tensile Strength, Inch and Metric Dimensions.
   3. Post-Tensioning Institute (PTI):
      a. Post-Tensioning Manual:
         1) Recommendations for Prestressed Rock and Soil Anchors, 5th edition, PTI DC35.1-14.

B. Qualifications:
   1. Anchor installer to have a minimum of five years of experience in the installation and construction of the type of anchors shown on the Drawings and required by the Specification Sections.

2. Contractor's Engineer and Geotechnical Engineer to have a minimum of five years of experience in the installation and construction of the type of anchors shown on the Drawings and required by the Specification Sections.
    a. Contractor's Engineer and Geotechnical Engineer shall be registered professional engineers licensed in the State of Arizona.

1.3 DEFINITIONS

A. Anchor Bond Length: That length of anchor which is bonded to a passive subgrade zone, and which transmits the required anchor tension load to the structure or subgrade.

B. Anchor Embedment Length: That portion of the total length of anchor extending into and surrounded by subgrade rock material.

C. Production Anchors: Anchors indicated on Drawings or otherwise required for support of structural members.

D. Installer or Applicator:
1. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
2. Installer and applicator are synonymous.

1.4 SUBMITTALS

A. Shop Drawings:
1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
2. Product technical data including:
    a. Manufacturer and type of proposed anchor along with proposed anchor corrosion protection.
    b. Proposed sizes and types of steel anchor bar and base plates, washers and nuts, and other hardware.
    c. Material certifications for all materials used and shown on Shop Drawings.
    d. Shop Drawings showing all bracing members and their connections.
    e. Manufacturer and type of proposed epoxy or polyester resin grout.
    f. Manufacturer and type of proposed fast setting leveling grout.
    g. Cement anchor grout mix design (for soil bond anchors only).

B. Qualifications:
1. Submit record of three of Installer's past successful installations of required type of anchor under similar soil and rock conditions.

C. Informational Submittals:
1. Submit results of all anchor load tests for Engineer information.
2. Submit strength test results of grout placed in the field.
3. Submit recommended anchor embedment and bond length into rock as established by anchor load testing by Contractor's Geotechnical Engineer for Contracting Officer Information.
4. Submit calculations for anchor embedment and bond length into rock for each type of rock strata sealed by Contractor's Geotechnical Engineer to Contracting Officer for information.
5. Submit anchor installation report for all in place production anchors.
    a. Submittal is for information only.

1.5 DELIVERY, STORAGE AND HANDLING

A. Deliver and handle fabrications to avoid damage.

B. Store above ground on skids or other supports to keep items free of dirt and other foreign debris and to protect against corrosion.

1.6 SITE CONDITIONS

A. Do not begin rock anchor installation until the earthwork in the area where anchors are to be installed has been completed as shown on the Drawings and as indicated in this Specification Section.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements.
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Rock Anchors:
      a. Dywidag threadbar reinforcing steel as manufactured by Dywidag Systems International, USA, Inc. (Basis of Design)
      b. Or Equal
   2. Fast setting leveling grout under steel anchor or base plates:
      a. Super Por-Rok as manufactured by CGM, Inc. (Basis of Design)
      b. Or Equal
   3. Polyester or epoxy resin grout for bonding anchors to subgrade:
      a. FASLOC resin cartridges as manufactured by Dywidag Systems International, USA, Inc. (Basis of Design)
      b. Or Equal

2.2 ANCHOR CAPACITY

A. Allowable working load carrying capacity for each required production rock anchor to be as indicated in the Drawings.

2.3 MATERIALS

A. Rock Anchors:
   1. Minimum anchor bar diameter to be as indicated in the Drawings.
      a. Minimum yield stress: 60 KSI.

B. Steel Anchor or Base Plates: ASTM A36 or ASTM A572; Hot dip galvanized: ASTM A153 Class B-1.

C. Washers Under Anchor Locking Nuts:
   1. Beveled washers: Steel or malleable iron of sufficient strength to prevent gouging of washer while tightening locking nut.
   2. Flat washers: ASTM F436.

D. Fast setting leveling grout under steel anchor or base plates: Minimum 28-day compressive strength to be 5,000 PSI.

E. Resin anchor grout for corrosion-protecting and bonding anchors to subgrade:
   1. Polyester or epoxy resin grout as specified in this Specification Section.

F. Prior to beginning installation of rock anchors, submit the proposed resin grout material to Contracting Officer for review.

G. Cement anchor grout for corrosion-protecting and bonding anchors to soil subgrade (not allowed for rock anchors):
   1. Cement: ASTM C150, Type I, II, or III.
   2. Aggregates:
      a. ASTM C33 well graded from fine to coarse.
      b. Aggregates to be natural not manufactured.
   3. Water: Potable, clean, free of oils, acids, and organic matter.
   4. Water content to be minimum necessary for proper pumping of grout with a maximum water:cement ratio by weight of 0.45.
   5. Minimum 28-DAY compressive strength to be 5,000 PSI with a higher 28-DAY compressive strength provided if required to allow anchors to be post tensioned to specified tension loads.
   6. Do not use expansive agents or admixtures in the grout mix.

H. Prior to beginning installation of anchors, submit the proposed cement grout mix design to Contracting Officer for review.
   1. Mix design to contain the following information:
      a. Sieve analysis and source of aggregates.
      b. Test for aggregate organic impurities.
      c. Proportioning of all materials.
      d. Type of cement along with mill certificate for the cement.
      e. One, seven DAY and two, 28-DAY compression test results using 6 IN diameter by 12 IN high cylinders for testing.

PART 3 - EXECUTION

3.1 INSPECTION

A. Furnish and pay for inspection services indicated herein to be performed by Contractor's Geotechnical Engineer.

3.2 LINES AND LEVELS

A. Complete necessary excavation and furnish all lines and levels necessary for completion of rock anchor installation.

B. See Section 01 40 00 Quality Requirements and Structural Tests and Special Inspections (attached to 01 40 00) for more information.

C. Test results and communications shall be provided to the Contracting Officer.

3.3 DETERMINATION OF ANCHOR LENGTH(S) TO BE INSTALLED

A. Perform no less than five pullout load tests on rock anchors to determine required anchor bond length and embedment length into subgrade necessary to provide required allowable working tension load carrying capacity.
   1. Perform load tests prior to installing production anchors for each type of rock strata.
      a. Install test anchors at locations with anchor bond length and embedment length into subgrade all as determined and directed by Contractor's Geotechnical Engineer.
   2. All components and details of the test anchors to be the same as proposed and required for use in production anchors.
   3. When approved, test anchors may be placed at locations required for production anchors.
      a. If test anchors are placed at these locations and for any reason are deemed unacceptable, install additional anchor or anchors adjacent to the unacceptable anchor as directed by Engineer.
      b. Provide additional structural members as required to transfer load due to the location of an additional anchor or anchors and as directed by Engineer.
      c. Additional anchors and structural members to be provided and installed at no additional expense to Government.

B. Entire anchor load testing program to be under the direction and supervision of the Contractor's Geotechnical Engineer.
   1. When establishing anchor bond length and anchor embedment length into subgrade for test anchors, account for the reduced tension capacity of an anchor due to "group action" caused by the proximity to the locations of adjacent production anchors.
   2. Apparatus for load testing anchors to be designed such that the load-carrying capacity of the anchor and the subgrade material surrounding the anchor are tested.
   3. Contractor to design test anchor setup.
      a. Entire apparatus for load testing anchors to be supplied by Contractor.

C. Load test anchors in accordance with the PTI "Recommendations for Prestressed Rock and Soil Anchors."
   1. Provide a calibration chart not more than 30 DAYS old for the hydraulic jack, pump and pressure gauge(s) used for the load tests. Calibrate the hydraulic jack, pump and pressure gauge as a system.
   2. Provide current, valid calibrations for the dial gauges and electronic load cell used for the load tests.
   3. Entire load test setup to have adequate capacity for testing anchors to 200 PCT of their specified allowable working tension load carrying capacity with sufficient factor of safety against overload.
   4. Provide a dial gauge(s) to measure anchor movement during load testing.
      a. All readings, taken by Contractor's Geotechnical Engineer, to be verified and signed by Contractor, and a copy sent to Contracting Officer for information.
   5. Provide an electronic load cell to measure the applied load (independent of the pump pressure gauge) during load testing.
   6. Do not begin load tests until material used to bond rock anchors to subgrade has cured sufficiently to be capable of transferring the required test loads from the anchor to the subgrade.

D. Load test anchors to a minimum of 200 PCT of the specified allowable working tension load carrying capacity as specified in this Specification Section.

E. Criteria for establishing failure of test anchors will be as determined by Contractor's Geotechnical Engineer.

F. In the event of failure of test anchor apparatus setup, repeat load test at no additional expense to Government.

G. Based on the results of the anchor load tests, the anchor bond length and anchor embedment length into subgrade for production anchors to be installed at required locations shall be as directed by the Contractor's Geotechnical Engineer.
   1. Send anchor load test results to Engineer for information.

H. If testing cannot be performed at the site of each type of rock strata as specified above, Contractor shall propose alternative testing and acceptance criteria, prepared and sealed by Contractor's Geotechnical Engineer, and submitted for approval by Contracting Officer.

I. If, during the installation of the production anchors, subgrade conditions are found to be different from those encountered during installation of the test anchors, or if production anchors are not capable of being post tensioned to the required loads as specified herein, revise the anchor bond length and/or the anchor embedment length into subgrade for the anchor being installed as directed by the Contractor's Geotechnical Engineer.
   1. If anchor is found to be not capable of being post tensioned to the required loads, install additional anchor or anchors as directed by Contracting Officer with anchor bond length and embedment length as directed by Contractor's Geotechnical Engineer.
   2. Install additional structural members required by placement of additional anchor or anchors as directed by the Contractor's Geotechnical Engineer.
      a. Allow Contractor's Geotechnical Engineer to witness installation of all test anchors and production anchors.

3.4 ADDITIONAL ANCHOR LOAD TESTS

A. In addition to anchor load test requirements for Base Bid, Contractor's Geotechnical Engineer may require installation and testing of additional test anchors loaded and tested in accordance with requirements of this Specification Section.

3.5 ANCHOR INSTALLATION AND DETAIL REQUIREMENTS

A. Equipment:
   1. Drilling equipment shall be adequate for providing the required drill holes in the subgrade encountered.
   2. All drilling is to be done by rotary or rotary-percussion drilling methods such that the drill hole sidewalls are sufficiently rough and not smooth.
   3. Grouting equipment to be capable of continuous mechanical mixing that will produce a uniform and thoroughly mixed grout at the required pressures.
   4. Post tensioning of anchors to be accomplished by use of a hydraulic jack and pump properly calibrated, capable of applying the required tension loads and including appropriate pressure and dial gauges and electronic load cell to indicate the load applied and to measure anchor elongation and movements.

B. Anchor Corrosion Protection:
   1. All installed anchors for their entire length shall be hot-dip galvanized in accordance with ASTM A153 Class B-1.
   2. Any anchors which have been field cut and/or galvanized coating has been damaged, the repairs shall be performed following Specification Section 05 50 00.

C. Anchor Placement:
   1. Before anchor placement, each anchor to be thoroughly cleaned to permit adequate bonding of the anchor to the bonding material.
   2. Do not install any anchors which are bent, cracked, of insufficient length, of reduced cross section due to any reason, or damaged in any way which would decrease the tension load carrying capacity of the anchor.
   3. Install anchors at the indicated locations, to the embedment length as directed by the Contractor's Geotechnical Engineer and at the indicated angle of inclination.
      a. Where the angle of inclination is not shown on the Drawings, an anchor shown horizontally shall be placed at 90 DEG with true vertical and an anchor shown vertically shall be placed at 90 DEG with true horizontal.
      b. A tolerance of ±3 DEG will be allowed on the indicated or specified angle of inclination.
   4. All anchors to be continuous full length without splices.
   5. Provide a washer under the anchor locking nut where indicated on the Drawings.
      a. Washer to be beveled where anchor is not 90 DEG to anchor base plate.
   6. When indicated on the Drawings or otherwise required, locking nuts shall be hexagonal head of the heavy-duty type, conforming to the requirements of or in accordance with the Manufacturer's specification for special nuts to be used with anchors supplied.
      a. Locking nuts to be capable of developing an ultimate strength of not less than 125 PCT of the minimum yield strength of the anchor threadbar to which it is attached.

D. Epoxy or Polyester Resin Grout for Bonding Anchors to Subgrade:
   1. The resin grout to be packaged in cartridge form with cartridges placed in the drill hole prior to placement of the anchor.
   2. Install sufficient number of cartridges into the drill hole so that the entire embedded length and circumference of anchor is entirely coated with the resin.
   3. Use fast setting resin for required anchor bond length and a slow setting resin for remainder of embedment length to accommodate stressing and to provide corrosion protection.
   4. Install anchors through the resin cartridges to full length of required embedment in accordance with the anchor Manufacturer's instructions.

5. Do not post tension anchors until resin grout has cured sufficiently to be capable of transferring the required post tensioning loads from the anchor to the adjacent subgrade material.

E. Cement Grout for Coating and Bonding Anchors to Subgrade (for anchors bonded in soil):
   1. Place grout continuously from bottom of drill hole to top of drill hole using tremie method until grout overflows top of hole.
      a. Place grout in such a manner to ensure that grout completely fills the drill hole and completely coats the anchor.
   2. The entire required anchor bond length to be pressure grouted using sufficient pressure to allow the required post tensioned loads to be placed on the anchors.
      a. Do not place required post tensioning loads on anchors until cement grout has obtained sufficient strength to be capable of transferring the post tensioning loads from the anchor to the adjacent subgrade material.
   3. The drill holes for soil anchors to be steel encased as required to prevent the intrusion of soil, water or any other material into the holes that would prevent the required bonding of the anchors to the subgrade.
      a. Withdraw steel casing slowly during cement grout placement in such a manner that cement grout completely fills the bore hole and completely coats the anchor.
   4. Make three, 6 IN DIA x 12 IN high cement grout test cylinders for each consecutive group of 10 production anchors installed.
      a. Make cylinders in accordance with ASTM C31.
      b. Test one cylinder at seven days and two cylinders at 28 DAYS.
         1) Perform tests in accordance with ASTM C39.
      c. Send test results indicating production anchors they represent to Contracting Officer's Representative.

F. Steel Anchor or Baseplates with Leveling Grout:
   1. Provide anchor or baseplates of sufficient size and thickness to be capable of transferring the required post tensioning loads to the anchors without exceeding the specified allowable bearing stress on the leveling grout when such grout is required beneath the anchor or baseplate, and without exceeding the allowable bending stress for the steel plate as stated in AISC 325.
   2. When indicated on the Drawings or otherwise required, fully bed all anchor or baseplates on a minimum 1 IN thick layer of leveling grout.
      a. Leveling grout to be a fast-setting cement grout.
   3. Required post tensioning anchor loads to not impose a bearing stress on the leveling grout which exceeds 0.3 times the compressive strength of the grout at the time of post tensioning.
      a. Minimum compressive strength of leveling grout at time of post tensioning to be 3,000 PSI.

G. Application, Measurement, and Magnitude of Post Tensioning Forces Required for Production Anchors:
   1. All production anchors indicated on Drawings or specified (except load test anchors) to be post tensioned to 150 PCT of the required allowable working tension load carrying capacity indicated for each production anchor as stated in this Section.
   2. Maintain this 150 PCT load on each production anchor for a minimum period of five MIN.
      a. Measure anchor movement by dial gauge during the application of the 150 PCT load and while maintaining this load for the minimum 5 MIN period.
      b. If during the 5 MIN period, the anchor movement is less than 0.03 IN, the 150 PCT load can be reduced and the production anchor locked-off at the required post tensioning force stated below.
      c. If the anchor movement is greater than 0.03 IN during the 5 MIN period, then the 150 PCT load to be maintained until the anchor movement decreases to an acceptable value as determined by the Contractor's Geotechnical Engineer.
   3. When approval to do so is obtained from the Contractor's Geotechnical Engineer, reduce the 150 PCT load at each production anchor to a post tensioning force which will remain

permanently in each production anchor equal to 110 PCT of the required allowable working tension load carrying capacity giving full consideration to all possible losses of the applied post tensioning force.

4. Allow the installation and post tensioning of all production anchors to be witnessed and approved by the Contractor's Geotechnical Engineer.
5. All anchors to be post tensioned by means of a hydraulic jack.
   a. Post tensioning forces to be determined by both of the following methods:
   b. Measurement of anchor elongation.
      1) Required anchor elongation to be determined from average load elongation curves for type and size of anchors being installed.
   c. Observation of post tensioning force indicated on a calibrated gauge attached to the hydraulic jack applying the post tensioning force and by the independent electronic load cell.
      1) Discrepancies of more than 5 PCT between elongation and gauge reading to be investigated and remedied.
      2) Movement of the structural system during the application of the post tensioning forces to be taken into account.

3.6 ROCK ANCHORS

A. Place each rock anchor in drilled hole which is 3/4 IN larger than the diameter of the anchor threadbar and in accordance with the Manufacturer's requirements.
   1. Allow Contractor's Geotechnical Engineer to witness the drilling of all drill holes.

B. Completely coat and bond all rock anchors to rock subgrade with resin grout for entire anchor embedment length.
   1. See this Specification Section for resin grout requirements.
   2. Prior to placing resin grout cartridges in drill holes, ensure that drill holes are free of water, loose rock particles or any other item that might weaken the bonding of the anchor to the rock subgrade.
   3. Clean drill holes as necessary using brushes and compressed air.

3.7 ANCHOR REPORT

A. During the installation of the production anchors, provide an anchor report for each production anchor installed recording for each anchor the following information:
   1. Anchor type, manufacturer, diameter and length.
   2. Total anchor embedment length into rock.
   3. Anchor drill hole diameter.
   4. Whether or not temporary steel casing was required for drill hole.
   5. Anchor bond length.
   6. Quantity of resin grout placed in drill hole.
   7. Quantity of cement grout placed in drill hole.
   8. Hydraulic jacking force readings for pressure gauge and electronic load cell for load test at each increment of applied load up to 200 PCT of required allowable working tension load carrying capacity with all accompanying anchor movement measurements.

B. The anchor report to be signed by the Contractor and Contractor's Geotechnical Engineer with copies sent to Contracting Officers Representative.

3.8 FIELD QUALITY CONTROL

A. Contractor shall include the cost of testing and inspection services in their bid cost.

B. Contractor is responsible for all testing, inspections and reports as specified herein. See Section 01 40 00, Article 1.6.

C. Scope of Contractor's Testing and Inspection:
   1. Observe and direct anchor load testing program.
   2. Establish test criteria for failure of test anchors.

3. Revise anchor bond length and/or the anchor embedment length into rock as required by changed field conditions.
4. Witness drilling of all drill holes.
5. Witness installation of all test anchors and production anchors.

END OF SECTION

This page intentionally left blank.

SECTION 31 71 00
DIRECTIONAL DRILLING AND CARRIER PIPE INSTALLATION

PART 1 - GENERAL

1.1 WORK INCLUDED

A. This is a performance specification. Minimum requirements are included herein. The Contractor is responsible for successful completion of the work subject to the requirements of this specification.

B. Procure and transport all suitable equipment, labor, and materials for the Work to and from the job site.

C. Prepare the site as specified herein.

D. Construct two cased borings of dimensions and locations as shown on the Drawings.

E. Install the Carrier Pipe into the Casing and secure in place. The selected materials must meet all project requirements and be approved by the Contracting Officer's Representative.

F. Cleanup and perform final restoration of all work areas.

1.2 RELATED WORK

A. Related documents and Specification Sections include but are not limited to:
1. 01 27 00 – Definition of Contract Line Items
2. 01 50 00 – Temporary Facilities and Controls
3. 03 00 00 – Concrete
4. 33 11 13 – Water Main Construction
5. 31 23 00 – Earthwork
6. 33 05 24 – Steel Utility Pipe
7. Inner Canyon Geotechnical Investigation Report (GIR), GEI Consultants, Inc.
8. North Rim Geotechnical Investigation Report (GIR), GEI Consultants, Inc.
9. Geologic Interpretive Summary Report (GISR), GEI Consultants, Inc.

1.3 DEFINITIONS

A. ANNULAR SPACE: The void space between the Borehole walls and the outside diameter of the Casing Pipe. The void space between the inner diameter of the final Casing Pipe and the outer diameter of the Carrier Pipe.

B. AREA OF UNCERTAINTY: An area on the ground surface around the Borehole exit location within which the Borehole is anticipated to exit. Defined by the Radii of Uncertainty.

C. BOREHOLE: The elongated cavity stabilized with Drilling Mediums mixed with native materials created through the Directional Drilling process.

D. BOREHOLE A: The Borehole located within 40 feet to the north of Borehole B. See Contract Drawings for indicative location. Contractor to survey entry location prior to the start of drilling. The exact location of Borehole A and order in which the Boreholes are drilled are to be at the Contractor's discretion.

E. BOREHOLE B: The southernmost of the two Boreholes whose entry location and planned exit location have already been surveyed. See Contract Drawings for exact location. The exact order in which the Boreholes are drilled is to be at the Contractor's discretion.

F. BOTTOM HOLE ASSEMBLY (BHA): Lowest part of a drill string. Provides weight and stability to the drill bit. Includes tools for drilling, steering, and locating.

G. CARRIER PIPE: The proposed production tubing to be used for conveyance of pressurized water. Also see "TUBING".

H. CASING PIPE: Steel Casing Pipe used for lining the completed Borehole. Installed in telescopic segments of decreasing diameter: conductor casing, surface casing, intermediate casing, and final/production casing. Casing Pipe provides Borehole stability, improves circulation, seals off areas of instability or lost circulation, and protects the Carrier Pipe.

I. CLOSED-LOOP DRILLING: A Drilling Medium management method where fluid circulation pits (mud pits) are replaced with a series of closed storage tanks, Fluid Recycling System equipment including shakers, mud cleaners, and centrifuges, and spoil collection containers.

J. CONTRACTING OFFICER (CO): The National Park Service (NPS)

K. CONTRACTOR: Drilling Contractor responsible for all aspects of the Directionally Drilled Borehole, Casing Pipe installation, and Carrier Pipe installation within the Directionally Drilled segment of the alignment.

L. DIRECTIONAL DRILLING: The process of drilling a Borehole along a planned trajectory. The trajectory often includes a curve and need not follow a straight or vertical trajectory.

M. DRILLING MEDIUM: Conveyance fluids, including air, mist, foam, water, and mud, which are pumped downhole to cool downhole equipment, stabilize the Borehole, and facilitate removal of cuttings. Consists of combinations of air, water, polymers, surfactants, bentonite, and approved non-toxic additives depending on the specific Drilling Medium used.

N. ELECTRONIC DRILLING RECORDER (EDR): A network of sensors and instruments which collect, record, and display data related to the drilling operation. Data can be from the BHA (e.g. hole depth, down hole pressures, etc.), from the drill rig (e.g. torque, thrust, etc.) and from general drilling progress (e.g. rate of penetration, drill time, etc.).

O. FILTER CAKE: The wall cake that forms from the platelets in bentonite-based Drilling Mediums. The Filter Cake provides a barrier between the Borehole and the formation, helping to reduce loss of Drilling Medium to the formation and infiltration of groundwater to the Borehole. May also be referred to as a “mud cake”.

P. FLUID RECYCLING SYSTEM: A specialized system designed to separate suspended solids or excavated soil or rock materials from Drilling Mediums during the Directional Drilling installation.

Q. GYROSCOPE: A tool mounted within the BHA as part of MWD or run separately in a multi-shot survey that measures Pitch/Inclination and azimuth in real time. The Gyroscope combines this data with measured distance to calculate the location of the BHA and determine vertical and horizontal offsets from the design alignment.

R. INADVERTENT RETURN (IR): When the Drilling Medium pressure exceeds the strength and confining pressure of the surrounding formation and reaches the surface or waterway. Also known as hydrofracture or “frac-out.”

S. INCLINATION: The deviation angle of the Borehole from vertical is measured as Inclination. Industry standard defines 0° as vertical and 90° as horizontal.

T. IN-FIELD REFERENCING (IFR): A technique which measures the local magnetic field strength declination, and dip angle in the vicinity of drilling activity to create a more accurate model of the local magnetic field. This model improves the performance of Magnetic MWD by reducing positional errors associated with local magnetic anomalies and reduces the Radii of Uncertainty for Magnetic MWD.

U. KICK-OFF POINT: The location in the Borehole where the Inclination first deviates from vertical (i.e. 0°) and the curve begins building.

V. MAGNETIC RANGING: A method by which the distance between the BHA and an existing adjacent Borehole is determined. The magnetic field may be emitted from equipment lowered into the adjacent Borehole (active ranging) or from the magnetic field generated by the steel Casing Pipe within the adjacent Borehole (passive ranging).

W. MAXIMUM ALLOWABLE DOWNHOLE PRESSURE: The Drilling Medium pressure at which higher pressures will theoretically cause an IR to occur. Estimates of this pressure for a given location along the Borehole profile use subsurface engineering properties, Drilling Medium properties, drilling means and methods, and Directional Drilling alignment geometry.

X. MEASUREMENT WHILE DRILLING (MWD): The process of continuously measuring pitch/Inclination and azimuth of the BHA. Measurements are sent to the surface via telemetry and used to locate and steer the Directional Drill. Tooling used for MWD may include magnetometers, accelerometers, and/or Gyroscopes.

Y. PRODUCT PIPE: See "CARRIER PIPE" and "TUBING".

Z. RADII OF UNCERTAINTY: The radius within which the BHA may be at any given distance along the Borehole path during drilling. It is the result of measurement errors in the vertical (i.e. elevation) and horizontal (i.e. left and right of the planned Borehole azimuth) directions during drilling and is defined by a number of standard deviations away from the expected location of the Borehole.

AA. TUBING: Steel tubular goods (pipes) commonly used in the oil and gas industry and conform to specifications set forth by the American Petroleum Institute.

BB. VOID: Gaps or cavities within the drilled formation. Commonly associated with karstification but may be formed through other geologic processes (faulting, folding). Intersections with Voids often result in loss of Drilling Medium circulation and/or loss of weight on bit depending on the size, persistence, and orientation with respect to the Borehole.

1.4 REFERENCE STANDARDS

A. American Petroleum Institute (API)
   1. API STD 1104 – Welding of Pipelines and Related Facilities
   2. API RP 582 – Welding Guidelines for the Chemical, Oil, and Gas Industries.
   3. API Spec 5CT – Casing and Tubing.
   4. API Spec 5B – Threading, Gauging, and Thread Inspection of Casing, Tubing, and Line Pipe Threads.
   5. API Spec 6A – Wellhead and Tree Equipment
   6. API TR 5C3 – Calculating Performance Properties of Pipe Used as Casing or Tubing.
   7. API RP 5C1 – Recommended Practice for Care and use of Casing and Tubing.
   8. API RP 13B-1 – Field Testing Water-Based Drilling Fluids.
   9. API RP 5L8 – Recommended Practice for Field Inspection of New Line Pipe.
   10. API RP 10B-2 – Recommended Practice for Testing Well Cements
   11. API RP 51R – Environmental Protection for Onshore Oil and Gas Production Operations and Leases.

B. ASTM International:
   1. ASTM C403 – Standard Test Method for Time Resistance of Setting of Concrete Mixtures by Penetration Resistance.
   2. ASTM C495 – Standard test Method for Compressive Strength of Lightweight Insulating Concrete
   3. ASTM C109 – Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or 50-mm Cube Specimens)

C. Bureau Veritas
   1. Standard DS-1

D. NSF International Standard / American National Standard (ANSI)
   1. NSF/ANSI 61 – Drinking Water System Components – Health Effects
   2. ANSI/AWS D1.1 – Structural Welding Code – Steel

1.5 GENERAL REQUIREMENTS

A. Be responsible for drafting an independent Directional Drilling Program, construction, and performance of the Directional Drilling work as specified herein.

B. Minimum design and construction requirements are presented herein and in the Drawings. It is the Contractor's responsibility to review and confirm design to fulfill the specified design requirements necessary for completion of the work.

C. Perform the Directional Drill installation in accordance with all Governing Agency(s) applicable permit requirements and industry standards. In the case of differing Governing Agency(s) specifications or requirements, the appropriate requirements shall apply as determined by the Contracting Officer.

D. Provide means of immediate verbal communication between entry and exit locations when work is being performed at both ends of the alignment.

E. Be responsible for containing, collecting, cleaning up, and disposal for all Drilling Medium and drilling spoils produced for the Project.

1.6 QUALIFICATIONS

A. Contractor personnel shall meet the following qualifications:

1. List of previously completed projects in which the Contractor has completed at least two (2) installations of parallel Boreholes or Boreholes in close proximity to neighboring Boreholes (spudding within 40 feet horizontally) for installation with a minimum length of 9,500 feet each within the past 5 years. At least one of the projects shall have used Magnetic Ranging. Qualifying projects must have been completed by the Contractor. For each of the two projects include the following:
   a. Project name,
   b. Measured distance,
   c. Diameter,
   d. Casing and Tubing material,
   e. Geological conditions for the drill
   f. Project references, with project descriptions, and current name and phone number for the references,
   g. Project budget,
   h. Contracting Officer or Government
2. Names and experience resumes for key field personnel. Key staff shall demonstrate on their resumes that they meet the following qualifications:
   a. Drilling Manager: Minimum three years' experience as a drilling manager and have completed at least three projects of similar length and size within the past three years.
   b. Drilling Supervisor: Minimum three years' experience as a drilling supervisor and have completed at least three projects of similar length and size within the past three years.
   c. Drilling Engineer: Minimum three years' experience as a drilling engineer and have completed at least two projects of similar length and size within the past three years.
   d. Geologist: Minimum three years' experience as a geologist and have completed at least three projects of similar length and size within the past three years.
   e. Directional Driller: Minimum three years' experience as a Directional Driller and have completed at least three projects of similar length and size within the past three years and at least one project where magnetic ranging was used.
   f. Cement Supervisor: Minimum two years' experience as a cement supervisor and have completed at least two projects of similar length and size within the past three years.
   g. Casing Crew Supervisor: Minimum two years' experience as a casing crew supervisor and have completed at least two projects of similar length and size, within the past three years.
   h. Certifications for welders per ANSI/AWS D1.1.
      1) Welders constructing wellhead to be certified per API 6A.

i. Drilling Fluids Engineer: Minimum two years' experience as a drilling fluids engineer and have completed at least two projects of similar length and size within the past three years.

B. All personnel employed by the Contractor in the work shall be experienced and competent in their respective tasks and shall work only under the direct control of a suitably experienced Superintendent.

1.7 SUBMITTALS

A. Preconstruction: Submit to the Contracting Officer's Representative the following a minimum of twelve weeks before the scheduled start of the applicable activity. Each numbered submittal below is to be submitted as a single package, i.e. submit all components of the Directional Drilling Program as a single submittal package.
   1. Qualifications
      a. Submit notification that qualifications provided in the Bid are still applicable. Include any changes in staff, equipment, or experience that differ from the Bid. Substantial changes may delay approval to start work, subject to review by the Contracting Officer's Representative.
   2. Grading and drainage plans (with corresponding SWPPP) at the Ballfield and the bore exit.
   3. Directional Drilling Program: The Contractor is to submit an independent and complete Drilling Program detailing, at a minimum, the following:
      a. Borehole General Information: Provide a summary of the drilling operation including:
         1) Borehole A surface hole (entry) and planned bottom hole (exit) locations to be confirmed and surveyed prior to the start of drilling.
         2) Borehole B surface hole (entry) and planned bottom hole (exit) locations to be confirmed and surveyed prior to the start of drilling.
         3) A description of which Borehole will be drilled first.
         4) A description of how the drill rig will move from the first Borehole location to the second.
         5) Planned Measured Depth, True Vertical Depth, North Coordinate, East Coordinate, Vertical Section and Dogleg Severity.
         6) Kick off Point(s) and curve build rates (degrees per 100 feet)
         7) Drilling Medium(s) to be used. If multiple Drilling Mediums are to be used describe at what locations along the Borehole path each will be used.
         8) Drilled Hole Sizes.
         9) Casing Pipe diameters (inner, outer, and drift), grade, pipe body yield strength, collapse resistance, and internal yield pressure.
         10) Borehole Schematic Drawing
         11) Description of drill rig to be used including at a minimum footprint size, rated horsepower, hook load, mast height, and method for moving from one Borehole location to another.
      b. Directional Drilling schedule. Include a critical path schedule for Directional Drilling-related activities. The schedule shall be updated weekly. At minimum the schedule shall have the following tasks:
         1) Pre-drilling tasks including work site and labor camp setup and mobilization.
         2) Tasks related to drilling the first Borehole including the start of drilling (spud date), estimated start and completion of each hole section, start and completion of each casing installation, start and completion of cementing for each casing, Borehole exit date, and Carrier Pipe installation.
         3) Tasks related to moving the drill rig and associated equipment from the first Borehole location to the second Borehole location and any additional design work that is required to be done prior to the start of drilling for the second Borehole.
         4) Tasks related to drilling the second Borehole including surveying in the second Borehole entry location, start of drilling (spud date), estimated start and completion of each hole section, start and completion of each casing installation, start and

completion of cementing for each casing, Borehole exit date, Carrier Pipe installation site cleanup, and demobilization.

c. Working drawings for project layout including:
   1) For the Casing Pipe and Carrier Pipe, this includes work, storage, staging, and assembly areas.
   2) Drilling equipment, ancillary equipment, and materials staging and layout for the Directional Drill operation.
   3) Equipment, materials, storage, and staging areas which will be implemented at the Borehole exit locations.
   4) Temporary spoil storage location.
   5) Pollution prevention equipment, materials, and layout.

d. Planned Formation Evaluation – provide equipment and procedure to perform rock formation evaluation during drilling.

e. Detailed Borehole Design
   1) Borehole section summary including Borehole path, Borehole inclinations, geologic profile, setting depths for all Casing Pipes, lengths for each Casing Pipe, and points of curvature and points of tangency for the respective beginnings and ends of all curves.
   2) Casing Design. At a minimum detail:
      a) Perform design of each casing, including torque drag analysis, and provide results of the design including safety factors and assumptions used.
      b) Proposed set depth for each casing.
      c) Proposed API casing specifications and manufacturers data. Include at a minimum: thread type, weight, grade, outer diameter, inner diameter, internal drift, allowable axial force, allowable yield stress, allowable burst stress, allowable collapse stress, and ultimate tensile strength.

f. Detailed Cement Program
   1) Include description of cementing plan for each Casing Pipe, noting cement mix design, volume of cement per Casing, tops of cement per stage, anticipated Maximum Down-Hole Pressures during cementing, and, if used, the rate of N^2 added to the cement.
   2) Include description of accessories used for each Casing including packers, downhole valves (DV), and stage tools. Describe where these accessories will be located for each Casing and their purpose.
   3) Include description of methods, materials, and equipment used to cement Annular Space between the final Casing Pipe and Borehole walls at bore exit locations. In accordance with Contract Drawings.
   4) Include method for verifying the top of cement for Casings where the full length of Annular Space is not cemented.

g. Detailed Drilling Medium Procedures
   1) Include all proposed or potential Drilling Mediums to be used and the locations along the Borehole they are planned for use, their mix designs based on anticipated subsurface conditions, and percentages of additives. Include manufacturer's information and purpose of any additives used in the mix designs. The Drilling Medium shall be non-toxic to the surface and subsurface environment.
   2) Procedures for monitoring Drilling Medium properties.
      a) For drilling fluid/mud
         (1) Density
         (2) Water content
         (3) Viscosity
      b) For air/mist/foam
         (1) Standpipe pressure
         (2) Air rate
         (3) Volume of water added
         (4) Amount of any additives to the water (i.e. surfactants, corrosion inhibitors, etc.)

3) Detailed Solids Control Procedures
    a) Describe methods and equipment used for solids control as applicable, which may include but are not limited to dilution plans, shale shakers screens, de-sanders, de-silters, mud cleaners, centrifuges, etc.
4) Detailed Drill Medium Disposal Procedures
    a) Describe how residual Drilling Medium will be temporarily stored on-site prior to being hauled off for disposal.
    b) Describe where residual Drilling Medium will be disposed of.
    c) Describe any defoaming agents if used when foam is the Drilling Medium.
5) Detailed Drill Cuttings Disposal Procedures
    a) Describe how drill cuttings will be temporarily stored on-site prior to being hauled off for disposal.
    b) Describe where drill cuttings will be disposed of.
    c) Estimate the quantity of drill cuttings to be produced in drilling the Boreholes and how many truckloads will be required to haul cuttings off site for disposal.

h. Detailed description of Closed-Loop system to be used when drilling with approved Drilling Mediums
i. Borehole Exit Plan
    1) Describe what methods will be used for the following:
        a) Estimating Borehole exit location in advance of exiting through the canyon wall.
        b) Demonstrating that the final Borehole section is clean, and free of cuttings and residual Drilling Medium prior to exiting.
    2) List what equipment and materials will be required for work at the Borehole exit locations.
        a) Include estimated weights for helicopter transport.
        b) Include estimated area required for temporary storage at bore exit location.
    3) Indicate where materials and equipment will be temporarily stored when not in use at Borehole exit locations.
    4) Describe the plan for constructing a working area at the Borehole exit location including any earthwork that will be performed.
    5) Identify the size and configuration for areas used in containment of spills and fluids exiting the Borehole. Describe the materials and equipment which will be used to construct containment measures.
j. Tracking and Survey Plan
    1) Describe what methods will be included for MWD and telemetry, including any methods used to improve MWD such as IFR. Verify that these methods work when drilling with air, mist, and/or foam
    2) For the first Borehole, Contractor to conduct validation survey and provide an anticipated ellipse of uncertainty for at a minimum the kick-off point, Borehole exit, and 4 evenly spaced locations between the kick-off point and Borehole exit.
        a) Contractor to describe how many standard deviations they are using when calculating the Radii of Uncertainty and associated ellipses of uncertainty.
    3) For the second Borehole, describe the method of Magnetic Ranging to be used and associated tooling. Verify that the method works when drilling with air, mist, and/or foam Drilling Mediums.
    4) Include a description of any other tracking methods or tools which the Contractor plans to use on this project, such as Gyroscopic surveys.
k. Bits, Hydraulics, and Drilling Parameters
l. BHA Details
    1) List anticipated BHA components, including lengths, weights, and type, for each casing and hole size.
m. Float shoes, float collars, and valves to be used in Borehole construction.
n. Pressure Testing Procedures for the Casing Pipes and the Carrier Pipe
o. Detailed Drilling Procedures
    1) Team members roles and responsibilities

2) Detailed rig mobilization plan
3) At minimum, for each hole and casing size describe:
    a) Drilling plan summary
    b) Cement plan summary
    c) Equipment to be used and the anticipated equipment supplier. At a minimum, the plan must address:
        (1) Casing running equipment (crane, tongs, elevators, etc.)
        (2) Welding detail
        (3) Emergency slips and seals
        (4) Float collars
        (5) Centralized joints
        (6) Centralized float shoes
        (7) Downhole valve (DV), stage, and packer tools
    d) Torque drag analysis for drill pipe and BHA.
4) Detailed contingency plan for each of the following scenarios for each Borehole and casing size. At a minimum, the contingency plan must describe the observations that may indicate the occurrence of the scenario and procedures that may be taken to mitigate or remedy the scenario:
    a) Tight hole or drill pipe
    b) Hole cleaning procedures for collapsed Borehole
    c) Lost circulation/ lost returns
    d) Lost cement during cementing
    e) Hole washout
    f) Drilling deviation
    g) Well controls/water influx
    h) Lost tooling down hole
    i) Void encountered during drilling
    j) Void encountered during casing advancement
    k) Stuck drill pipe following exit from canyon wall.

4. Carrier Pipe Completion Plan. The Carrier Pipe Completion Plan may be combined with the Directional Drilling Work Plan or provided as separate submittals.
    a. Carrier Pipe.
        1) Torque drag analysis for running Carrier Pipe into the final Casing Pipe.
        2) Proposed API Tubing specifications and manufacturers data. Include at a minimum: outer diameter, inner diameter, weight, grade, drift, allowable axial force, yield stress, allowable burst stress, allowable collapse stress, ultimate tensile strength, and connection type.
    b. Centralizers
        1) Provide proposed centralizer design including manufacturers technical data showing complete information on material composition, physical properties, dimensions and standoff percentages.
    c. Wellhead Design
        1) Manufacturer’s technical data showing complete information on material composition, physical properties, and dimensions of the wellhead and fittings. Manufacturer’s recommendations for transport, handling, and storage of the wellhead shall be included.
        2) Contractor’s planned cellar dimensions to accommodate the wellhead.
        3) Written description of wellhead construction sequence including concrete footing.
        4) Mix design for concrete used in footing.
    d. Carrier Pipe Installation Procedure
        1) Carrier Pipe connection and testing procedure detailing how the Carrier Pipes will be joined and how the joints will be tested.
        2) Carrier Pipe centralizer installation procedure detailing how the centralizers will be installed, at what spacing interval they will be installed, and how the Contractor will prevent the loss of centralizers during installation.

3) Carrier Pipe handling and lowering procedure detailing how the Carrier Pipe will be handled, centralized within the Casing Pipe, and lowered into the Casing Pipe during installation.

e. Carrier Pipe Stabilization Procedure
   1) Details of how the Carrier Pipe will be affixed to and hung from the wellhead.
   2) Details of how the Carrier Pipe will be affixed at the Borehole terminus.

5. Submit an example Daily Drilling Report which would be in accordance with section 3.25.B.
6. Submit an example Drill and Drilling Medium Logs Report which would be in accordance with section 3.25.C.

B. Construction Submittals shall be digital and submitted to the Contracting Officer's Representative within 24 hours of the day the event occurs and shall include at minimum:
1. Should a scenario submitted in the preconstruction contingency plan become true during construction the Contractor shall review the submitted contingency plan, tailor their mitigation approach to the specific scenario encountered, and submit the tailored plan to the Contracting Officer.
2. At minimum, submit construction tracking and survey data detailing the location of the Borehole face and showing the Borehole path twice per day while drilling.
3. Prior to the exit of the first Borehole, provide the anticipated exit location and associated Area of Uncertainty.
   a. If anticipated exit location is outside of stated tolerances or within a ditch or drainage, submit a plan to make corrections based on response from the Contracting Officer.
      1) If the Contracting Officer accepts the exit location outside of tolerances, then submit plan describing modifications to the bore exit area including access, equipment and material staging.
      2) If the Contracting Officer rejects the exit location outside of tolerances, then submit a plan for steering the Borehole back to the desired location and for cementing the abandoned Borehole.
4. The as-built profile for the first Borehole shall be submitted prior to the start of the second Borehole.
5. Provide a surveyed second Borehole entry location and survey of the planned second Borehole exit location upon completion of the first Borehole.
   a. These locations may differ from those submitted in the Directional Drilling Program based on the constructed locations of the first Borehole.
6. Drill and Drill Medium logs.
   a. Air and Foam Logs
   b. Mud Reports (when drilling with mud)
   c. Mud Logs (when drilling with mud)
7. Casing Pipe segment connection torque records.
8. Carrier pipe testing results.
9. Casing cementing records.
10. Pressure test results for Carrier Pipe per Specification Section 33 11 13 Water Main Construction.
11. Construction records as noted in Part 3.25.
12. Prior to the exit of the second Borehole, provide the anticipated exit location and associated Area of Uncertainty.
    a. If anticipated exit location is outside of stated tolerances, submit a plan to make corrections based on response from the Contracting Officer.
       1) If the Contracting Officer accepts the exit location outside of tolerances, then submit plan describing modifications to the bore exit area including access, equipment and material staging.
       2) If the Contracting Officer rejects the exit location outside of tolerances, then submit a plan for steering the Borehole back to the desired location and for cementing the abandoned Borehole.

13. Submit any modifications to the drilling program made during construction for approval by the Contracting Officer Representative.
14. EDR data shall be collected and submitted to Contracting Officer Representative in accordance with Sections 3.08 and 3.25.

C. Postconstruction:
1. Submit an end of well report for each Borehole in accordance with Section 3.25.

1.8 QUALITY ASSURANCE

A. Perform work in accordance with the following:
1. Contractor is to follow the submitted and approved pre-construction submittals. Any desire to deviate from approved submittals shall be documented, requested, and approved in writing by the Contracting Officer prior to implementation.
2. Industry best practices as determined by the American Petroleum Institute.
3. Contractor to provide real time rig operational and performance data to the Contracting Officer as per Sections 3.08 and 3.25.

1.9 DELIVERY, STORAGE, AND HANDLING

A. Transport, handle, and store Casing and Carrier Pipe and fittings as recommended by the manufacturer.

B. Casing Pipe, Carrier Pipe, and fittings that are damaged before or during installation shall be repaired or replaced, as recommended by the manufacturer or required by the Contracting Officer Representative. Be responsible for costs of such repair or replacement. Repair or replace damaged Casing Pipe, Carrier Pipe, and fittings prior to proceeding with the project.

C. Deliver, store, and handle other materials as required to prevent damage. Repair or replace materials that are damaged or lost at no additional expense to the Government.

D. Protect piping system pieces from entry of foreign materials and water by temporary covers, completing sections of work, and isolating parts of completed system.

E. Accept products on site in manufacturer’s original containers or configuration. Inspect for damage.

1.10 ENVIRONMENTAL REQUIREMENTS

A. Conduct operations so as not to interfere with, interrupt, damage, destroy, or endanger integrity of surface or subsurface structures or utilities, and landscape in immediate or adjacent areas.

B. Comply with all other Federal, State, and local environmental requirements including, but not limited to, storm water runoff, dust mitigation, construction dewatering, disposal of Drilling Medium, and hazardous waste management and disposal.

C. All materials used in the construction of the Boreholes which are used down-hole including Drilling Medium, additives, and lubricants shall be non-toxic. Contractor to provide SDS sheets for all materials.

1.11 COORDINATION

A. Coordinate work with the Contracting Officer, and utilities within construction area.

PART 2 - EQUIPMENT, PRODUCTS, AND MATERIALS

2.1 EQUIPMENT:

A. Provide the equipment necessary for installation of all segments and sizes of Casing Pipe, excavation of the Borehole, installation of the Carrier Pipe and to complete all work associated with this section.

B. TRACKING SYSTEM
1. Shall include MWD to provide real-time measurements for the drill bit location.

2. Magnetic Ranging shall be used on the second Borehole to verify its position with respect to the first Borehole and avoid collisions.
3. Shall provide immediate information on the tool face orientation, Inclination, and azimuth. The Tracking System shall provide continuous electronic monitoring and recording capabilities.
4. Tracking System data shall be recorded digitally. Handwritten data will not be accepted.

C. ELECTRONIC DRILLING RECORDER
1. The Electronic Drilling Recorder used for this construction shall be capable of recording the following drilling parameters at one second intervals:
   a. Hole depth (ft)
   b. Bit depth(ft)
   c. Block or carriage height (ft)
   d. Rotary torque (psi)
   e. Thrust push/pull (psi)
   f. Weight on bit (lbf)
   g. Rotations per minute (RPM)
   h. Rate of penetration (ft/hour)
   i. Total pumping output (TPO) (gpm)
   j. Standpipe pressure (psi)
   k. Differential pressure (psi)
   l. Annular pressure (psi)
   m. Pipe pressure (psi)
2. Provide necessary unit conversion between hydraulic pressure, rotary torque, and pull/push force.

D. DRILLING MEDIUM SPILL KIT
1. Shall be adequate to contain, collect, and dispose of Drilling Medium that is either spilled on site or surfaced to the ground or waterbody through an IR. A Drilling Medium Spill Kit shall be located at each end of the Borehole path during all Directional Drilling operations.

E. DIRECTIONAL DRILLING RIG
1. Supply a drill rig, top drive, and additional equipment suitable for drilling the boring, installing the Casing Pipes, and installing the final Carrier Pipes as required for the means and methods included in the Contractor's drilling program.
2. Drill Rig shall have a minimum of 1,500 HP
3. Top drive shall be capable of the following torques, at minimum:
   a. Continuous torque – 37,000 ft-lbs
   b. Breakout torque – 60,000 ft-lbs
   c. Makeup torque – 50,000 ft-lbs

F. BOTTOM HOLE ASSEMBLY
1. Have minimum of 3 mud motors and 1 rotary steerable system available on site to account for different steering needs and potential damages to motors.
   a. Contractor shall have the ability to procure additional mud motors and rotary steerable systems as needed such that construction delays are not realized.
2. Tooling shall be suitable for the site conditions as presented in the Contract Drawings.

G. DRILL BITS
1. Maintain an inventory of drill bits on-site suitable for replacing damaged bits and/or switching bits out depending on the formation encountered.
2. The Contractor may use a different drill bit sequence from that shown in the Contract Drawings provided they describe it in their drilling program (see 1.07.A) and demonstrate that it remains appropriate for installing the minimum 4.5-inch outer diameter Carrier Pipe.

H. DRILL MEDIUM MANAGEMENT AND SEPARATION
1. Obtain and utilize equipment suitable for generating sufficient downhole pressure to remove cuttings. Surface equipment shall be sufficient to separate cuttings from Drill Medium and shall include at a minimum:

a. Air Compressors
b. Fluid Pumps
c. Blowout Preventer
d. Fluid Recycling System
e. Shakers
f. Mud Tanks

I. CEMENTING EQUIPMENT
1. Conductor Casing
a. Have sufficient 1-inch diameter minimum cementing lines to cement the Conductor Casing via a top job.
2. Surface Casing
a. Have sufficient 1-inch diameter minimum cementing lines to cement the Surface Casing should the Contractor desire to do so or should other cementing methods be ineffective.
3. Borehole exit
a. Obtain all necessary materials and equipment for cementing operations at the Borehole exit location as shown on the Contract Drawings and described in Section 3.10.

2.2 PRODUCTS AND MATERIALS

A. Provide the products and materials necessary for the excavation of the Borehole path and installation of the Casing Pipe, installation of the Carrier Pipe and to complete all work associated with this section.

B. DRILLING MEDIUM
1. Shall be designed to circulate cuttings from the Borehole and increase Borehole stability.
2. Have the materials on site to allow for multiple Drilling Mediums to be implemented. At a minimum, materials to allow for air, mist, foam, and mud drilling should be available.
3. All Drilling Mediums, additives, and lubricants are to be non-toxic and NSF certified.

C. WELLHEAD
1. Wellhead components and appurtenances shall be in accordance with API standards.
2. Wellhead components and appurtenances shall be acquired from Cameron Wellhead or approved equivalent.

D. CASING PIPE
1. The Contractor may use a different Casing Pipe sequence from that shown in the Contract Drawings contingent upon approval of the Contractor's submitted drilling program. All Casing Pipe must be installed without damage and be capable of housing the specified minimum 4.5-inch outer diameter Carrier Pipe.

E. WATER SOURCE
1. Be responsible for obtaining water required for conducting the Work of this section. All water sources shall be approved in advance by the Contracting Officer. Water is anticipated to be sourced from outside the National Park, as the North Rim may not have water supply available during construction. See Specification 01 50 00 Temporary Facilities and Controls for additional information.

F. CASING CEMENT
1. Portland Cement and/or additives.
2. Typical minimum compressive strength of 500 psi in 24 hours at 60 degrees Fahrenheit when tested in accordance with API Specification 10B-2.
3. Cement shall have a unit weight greater than Drilling Medium.
4. Use only potable water. Alternative water sources may be utilized with the approval of the Contracting Officer.

G. CARRIER PIPE
1. Carrier Pipe used in both Boreholes shall have the following properties:
a. Outer diameter of 4.5 inches

b. Nominal inner diameter of 3.826 inches
c. API grade 5CT 15.1 L-80 Type 1

2. Carrier Pipe used for finished water shall be in accordance with Specification 33 05 24 Steel Utility Pipe

H. BORE EXIT
1. Impermeable Membrane to contain fluids exiting the Borehole at the exit locations shall possess a maximum hydraulic conductivity of $1x10^{-9}$ inch/sec and a minimum puncture resistance of 300 lbs.
2. Protective Matting shall consist of a blast mat sized accordingly to cover the Area of Uncertainty for the largest Area of Uncertainty between the two Borehole exits.
a. Blast matting shall be able to conform to uneven ground and be able to contain fly rock while allowing pressurized gases to escape.
b. Blast matting shall be capable of withstanding multiple blasts without a reduction in efficacy.
c. Matting shall contain and mitigate dust as the bore exits the rock.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify connection to existing piping system size, location, and invert elevations are in accordance with the Drawings.

B. Do not begin work until required preconstruction submittals listed in Part 1.7A have been accepted by the Contracting Officer and notice to proceed has been provided by the Contracting Officer.

3.2 PREPARATION

A. Call the Local Utility Line Information (811) service not less than three working days, and as required by conflicting utility owners, before performing Work.
1. Request underground utilities to be located and marked within and surrounding construction areas.
2. Maintain active utility clearances for the duration of the Work.

B. Locate, identify, and protect utilities indicated to remain from damage.

C. Identify required lines, levels, contours, and datum locations.

D. Protect benchmarks, survey control points, existing structures, fences, sidewalks, trails, paving, curbs, and trees not identified for removal from excavating equipment and vehicular traffic.

E. Perform grading and site modifications necessary to facilitate the Work and control and contain Drilling Mediums and all other fluids.

F. PRE-PLANNING
1. Contractor shall perform a pre-planning “dry run” bore prior to commencing drilling. This process should include the following:
a. Development of bore sketch.
b. Summary of the bore overview.
c. Summary of lessons learned. Discuss what happens when subsurface conditions trigger one of the contingencies. Discuss communication protocol.
d. Schematic of depth vs target days and type of drilling fluid to be used.
e. Detailed breakdown of each casing, required hole diameter, depth, drilling fluid, cement, bottom hole assembly, drill head bits, carrier pipe, volumes, weights, etc.
f. Summary of how drilling head will be steered and located.
g. Discussion of the geologic formations and challenges associated with each.
h. Summary of how the second bore will be drilled. Plan for ranging and exit location.

2. All day meeting to review and go over the drilling plan. Contracting Officer's Agent and A/E will be invited to attend. Meeting will be onsite at the North Rim.
    a. Discussion of safety plan for each of the drilling phases.

G. Following the first Borehole exiting from the canyon wall and prior to the start of the second Borehole, the second Borehole entry location is to be identified in the field and surveyed. Final entry location of the second Borehole shall be used to refine the Borehole path as needed. Final entry location, Borehole path, and exit area are to be approved by the Contracting Officer in writing prior to beginning the second Borehole.

3.3 DEWATERING

A. Intercept and divert surface drainage, precipitation, and groundwater away from excavation through use of dikes, curb walls, ditches, pipes, sumps, or other means.

B. Comply with Arizona State and NPS Federal standards and requirements for dewatering to any watercourse, prevention of stream degradation, and erosion and sediment control.

3.4 GENERAL:

A. Provide the Contracting Officer unrestricted access and provide necessary assistance to review rig telemetry, data, measurements, observations, and other records including but not limited to visual access to real-time operator controls, locator's screens, gauges, and indicators during drilling operations.

B. Control Drilling Medium pressures, Drilling Medium weights, drilling speed, and other operational factors to reduce Drilling Medium loss to the surrounding formation and prevent IR.

C. Calibrate all survey and tracking systems prior to beginning each Borehole.

D. Check that all parts of the BHA are in good working condition prior to beginning each Borehole.
    1. Inspect BHA and drill string components in accordance with DS-1 Category 4.

E. Perform Borehole excavation operations in a manner that will minimize loss of ground and minimize settlement of the ground surface, structures, and utilities above and adjacent to the alignment.

F. Install and operate Directional Drilling equipment and support utilities as necessary for the proposed construction and the safety of workers in accordance with project requirements and all Federal, State, and local laws, regulations, and codes.

G. Use an electronic monitoring system to record Drill Log and Tracking System data. Maintain a log of the electronic data and make it available for review by the Contracting Officer Representative.

H. Guide and monitor the Borehole using the accepted tracking and survey methods described in Section 3.09.

I. Should the Contractor realize a scenario submitted in the Contractor's contingency plan submittal, the Contractor shall review their contingency plan and tailor their mitigation approach as necessary for the specific scenario encountered. The tailored plan shall be provided to the Contracting Officer.

3.5 SOUND MITIGATION:

A. The following sound mitigation measures shall be implemented for Directional Drilling work.
    1. High-specification enclosures and mufflers shall be used on generators, pumps, compressors, etc.
    2. A shroud or enclosure shall be assembled around the top drive of the drilling rig.
    3. A screening or barrier shall be assembled around the drill floor.

B. Sound mitigation methods shall be implemented in accordance with API RP 51R and follow construction recommendations included in the National Park Service's Acoustical Toolbox: Recommendations for Reducing Noise Impacts in National Parks.

3.6 EXCAVATION:

A. Construct drilling pad, cellar, and other earthwork required for the drill rig in accordance with shop drawings, as site conditions require, and in accordance with Specification 31 23 00 Earthwork.

B. Final grading and leveling for drilling equipment to be planned and executed by Contractor within the limits of construction shown on the Contract Drawings and per Specification 31 23 00 Earthwork.

C. Always maintain clean working conditions at the job site. All excavated soil, and any other material not required for construction, shall be removed from the excavation area in a timely manner.

D. Safety fencing shall be installed around the perimeter of any excavations and access limited to what is necessary for continuous operation.

E. Restore areas after completion of Directional Drilling and final Carrier Pipe installation.

3.7 DRILLING MEDIUM

A. Mix Drilling Mediums including air, mist, foam, water, and/or mud to suit the proposed drilling application, anticipated ground conditions, calculated pressures, and the subsurface materials to be encountered. Drilling Mediums shall be used and mixed in full accordance with the Drilling Medium manufacturer's instructions.

B. Drilling shall be conducted with controlled Drilling Medium and advancement rates to minimize Borehole annular pressures and prevent excavating at a rate unable to be maintained by the Drilling Medium circulation system.

C. Continuously monitor Drilling Medium pumping rate and pressure while drilling each Borehole and installing Carrier Pipe to ensure adequate removal of rock cuttings and stabilization of the Borehole.
   1. At minimum for drilling mud, monitor sand content, unit weight or density, plastic viscosity, pH, salinity, filtrate loss, and yield point of the Drilling Medium no less than once every four hours during Directional Drilling operations.
   2. At minimum for air/mist, monitor standpipe pressure, air rate, volume of water added, and the amount of any additives added to the water no less than once every four hours during Directional Drilling operations.

D. Air, mist, foam, water, and mud are approved Drilling Mediums for both the Conductor and Surface casings.

E. The Intermediate Casing shall be drilled with air, mist, or foam Drilling Mediums.

F. The Final Casing shall be drilled with air or mist Drilling Medium.
   1. See Section 3.16.C for the final 200 feet of drilling.

G. A Closed-Loop Drilling methodology for managing and recycling Drilling Medium shall be used. Open fluid circulation pits, mud pits, earthen pits or other open Drilling Medium collection pits dug into the surface shall not be permitted.

3.8 ELECTRONIC DRILLING RECORDING

A. Use Electronic Drilling Recording as described in Section 2.01.C. At a minimum the Contractor is to monitor hole depth, bit depth, block or carriage height, rotary torque, thrust push/pull, rotations per minute, advance rate, pumping rates, and standpipe, differential, annular, and pipe pressure.
   1. Store data collected with EDR on a cloud-based server by the Contractor which can be accessed by the Contracting Officer Representative, Contracting Officer, and third parties for additional data processing.
   2. Store EDR data in the cloud-based server as .csv format.
      a. Upload EDR data to the cloud-based server at a minimum of twice per day.

B. The Contractor or a 3rd party employed by the Contractor shall provide statistical analysis and correlations along the Borehole path throughout each stage of drilling for key drilling parameters including: drill time, push/pull, rig rotary torque, rig rpm, motor torque (if applicable), advance rate, annular pressure, and mechanical specific energy, based on the EDR system on a per joint basis.
   1. Results and visualizations from the statistical analysis of the EDR data are to be shared with the Contracting Officer Representative and Contracting Officer as they become available and are updated.

C. Modify their drilling program accordingly should EDR data indicate a change in approach is necessary. Modifications to the Contractor's drilling program shall be submitted and approved by the Contracting Officer.

D. All data shall be correlated with the date, time of day, measured distance, and the activity of which data is being recorded.

3.9 TRACKING AND SURVEY

A. Utilize tracking and surveying methods to achieve an exit within tolerances dictated in section 3.22 of this specification.
   1. Tracking and surveying methods which may be employed to achieve these tolerances include but are not limited to:
      a. Magnetic MWD throughout the entirety of the first Borehole.
      b. Gyroscopic MWD throughout the entirety of the first Borehole.
      c. Use of In-Field Referencing to refine the model used in magnetic MWD.
      d. Use of multi-shot Gyroscopic surveys to refine location periodically.

B. Contractor shall utilize Magnetic Ranging when drilling the second Borehole and shall keep the second Borehole within ranging distance of the first Borehole at all points during drilling.

C. For both Boreholes:
   1. Record and plot the x, y, and z coordinates of the alignment at least once per each drill pipe, including position of the BHA relative to the ground surface. Update daily at a minimum during the Borehole advancement.
   2. Adjust BHA orientation to maintain design alignment as indicated in the approved drilling program.
   3. Calibrate and verify electronic monitor accuracy in presence of Contracting Officer Representative. When specified tolerances are not met, calibrate equipment or provide new equipment capable of meeting specified tolerances.

3.10 CEMENTING

A. Fully cement the Conductor Casing back to the surface.

B. Cement the Surface Casing according to the following minimum lengths:
   1. 500 feet back from the Casing shoe.
   2. From a depth of 200 feet back to the surface.

C. Cement the Intermediate Casing for a minimum distance from the casing shoe of 500 feet.

D. The Contractor is to cement the Final Casing with enough cement to reach a minimum of 100 feet past the shoe of the Intermediate Casing.
   1. Cementing shall be done by, first, pumping into the annular space between the final casing and Borehole wall from the Borehole exit per Contract Drawings. Second, either:
      a. Pumping cement through the annular space between the intermediate and final Casings from the Borehole entry location.
      b. Setting an ACP, mechanically open stage collar, and pump cement to fill the annular space between the intermediate and final Casings required to be cemented.
      c. The Contractor may submit alternative methods for cementing the Final Casing. These shall be subject to approval by the Contracting Officer's Representative.

E. For cementing operations where the full length of Annular Space is not cemented, verify the top of cement through a bond log, temperature survey, or other approved method.

3.11 DRILLING OBSTRUCTIONS

A. When obstructions are encountered during drilling, notify the Contracting Officer's Representative immediately. Do not proceed around obstruction without Contracting Officer's Representative's approval.

B. Maintain adjusted Borehole alignment within limits of construction.

3.12 DRILLING VOIDS

A. When encountering a suspected Void during drilling the Contractor will document the following:
   1. Hook load.
   2. Weight on bit.
   3. Drilling Medium circulation observations (loss of Drilling Medium).
   4. Bottom hole pressure.
   5. The location along the Borehole path at which the suspected Void was encountered.
   6. The inclination or azimuth of the Borehole path and deviation from the design Borepath, if any.

3.13 WATER INFLUX

A. Monitor and record Drilling Medium returns such that additional groundwater not inherent to the Drilling Medium mixing with the Drilling Medium can be detected. Influxes of groundwater int the Drilling Medium shall be documented through measurements included but not limited to the following:
   1. Increase in bottom hole pressure
   2. Additional water returning to the surface with cuttings
   3. Change in Drilling Medium density

B. When the Contractor identifies additional groundwater in the Drilling Medium and drilling progress is impeded, Contractor may switch to a denser Drilling Medium to manage the influx of water.
   1. Permitted changes in Drilling Medium include switching from air to mist, mist to foam, and air directly to foam.
      a. Switching from air, mist, or foam Drilling Medium to water or mud will not be permitted past the surface casing as the high pressures generated by the Drilling Medium result in unacceptable risk of IR

C. If foam is insufficient to mitigate groundwater influx below the surface casing, the Contractor is to make an attempt to squeeze off the water with cement in the zone of infiltration.

3.14 DRILLING MEDIUM & CUTTINGS REMOVAL AND DISPOSAL

A. Contain excess Drilling Mediums and cuttings at Borehole entry and exit sites.

B. Remove, transport, and dispose of cuttings in accordance with NPS regulations.
   1. Do not discharge drilling spoils into sanitary sewers, storm sewers, or other drainage systems.

C. Should a Drilling Medium spill or IR occur, immediately contain the Drilling Medium and barricade area from vehicular and pedestrian travel before resuming Directional Drilling operations.
   1. Cleanup spilled Drilling Medium at end of each shift, excepting the Fluid Recycling System.

3.15 BOREHOLE ENTRY

A. The anticipated Borehole A and B entry locations are shown on the Contract Drawings, with Borehole B located south of Borehole A. The Contractor may drill either Borehole first so long as the Contractor's means and methods are suitable for magnetically ranging from the first

Borehole while drilling the second Borehole and there is no more than 40 feet between the two entry locations.

3.16 BOREHOLE EXIT

A. Exit locations for Borehole A and B shall adhere to the tolerances described in Section 3.21.

B. Prior to either Borehole exiting from the canyon wall the Contractor shall perform the following:
   1. Clean the Borehole with air to remove residual Drilling Medium and cuttings from the Borehole.
   2. Position materials and equipment at Borehole exit around Area of Uncertainty after determining anticipated Borehole exit location. Materials and equipment are as shown in the Contract Drawings and consist of the following:
      a. Clean the borehole with air to remove residual Drilling Medium and cuttings from the Borehole.
      b. Position materials and equipment at the Borehole exit around the Area of Uncertainty after determining the anticipated Borehole exist location. Materials and equipment at the Borehole exit are to be positioned as shown in the Contract Drawings.
         1) Where otherwise not indicated, impermeable membrane embedded in sandbag embankments shall reach a minimum height of 3 feet.

C. The final 200 feet of the Borehole path for each Borehole is to be drilled with only air (i.e. dust drilling)
   1. Within final 200 feet of the Borehole the BHA is to be advanced 10 feet at a time then partially retracted to swab the drill hole so that cuttings remain moving and can be circulated and residual Drilling Medium can be removed.
   2. The drill bit is not to exit through the canyon wall until the Contractor demonstrates the hole is cleared of Drilling Medium and cuttings.

D. The Contractor is to finalize the location for the second Borehole exit after the first Borehole exits from the canyon wall and submit this planned location per Section 1.07.B.

E. Contractor to temporarily store equipment and materials required at Borehole exit locations within the limits of construction and in a position that does not impede helicopter access or work being performed in the area by others.

3.17 PRESSURE AND LEAKAGE TESTING

A. Pressure test.
   1. The Carrier Pipe shall be pressure tested per Specification Section 33 11 13 Water Main Construction.

B. The Carrier Pipe will not be accepted without passing a pressure test.

3.18 CARRIER PIPE INSTALLATION

A. The Carrier Pipe shall be installed into the Final Casing Pipe from the Borehole entry location to the respective Borehole exit location as shown on the Contract Drawings. Carrier Pipe minimum inner diameter shall be as described in Section 2 of this Specification.

3.19 WELLHEAD

A. Wellhead and associated vault shall be constructed such that the entire wellhead assembly is below the final grade.

3.20 ANNULAR SPACE NITROGEN PURGING

A. The Annular Space between the final Casing Pipe and Carrier Pipe is to be purged with nitrogen gas after installation of the Carrier Pipe to displace residual air and moisture.

B. Nitrogen gas is to be injected into the Annular Space from the Borehole entry location through an airtight seal to prevent gas from escaping.

C. An amount of nitrogen gas equivalent to five Annular Space volumes is to be injected from the top of each Borehole to fully displace residual air and moisture out of the bottom of each Borehole.

3.21 BORE TERMINUS

A. Final Casing Pipe shall extend far enough past the ground surface such that a minimum of 2 feet of clearance can be achieved between the end of the Casing Pipe and the ground surface.

B. Carrier pipe shall be welded to the Final Casing Pipe using a 0.5-inch thick steel face plate and steel flanges as shown in the Contract Drawings.

3.22 INSTALLATION TOLERANCES:

A. For the Borehole A exit location:
   1. Maximum variation from Contractor submitted and approved Borehole A exit location:
      a. Horizontal position: 20 feet.
      b. Vertical elevation: 20 feet.

B. For the Borehole B exit location:
   1. Maximum variation from Borehole B exit location shown on Contract Drawings:
      a. Horizontal position: 20 feet.
      b. Vertical elevation: 20 feet.

C. Both Borehole exit locations shall be separated by a maximum of 20 feet and a minimum of 10 feet.

D. Note that distances stated in 3.22.A – 3.22.C shall be measured perpendicular to the azimuth direction of the respective Boreholes. Actual measured distances on the ground surface may differ due to local topography.

E. Under no circumstances shall the installed Casing Pipes be outside of established Limits of Construction as shown in the Contract Drawings.

F. If Borehole deviations exceed or are anticipated to exceed specified tolerances the Contractor is to immediately inform the Contracting Officer's Representative. Deviations beyond the specified tolerances will require the Contractor to complete the following at their sole expense as deemed sufficient by the Contracting Officer:
   1. If anticipated Borehole exit is approved by the Contracting Officer
      a. Amend access routes to the Borehole exit as approved by the Contracting Officer.
      b. Relocate all Borehole exit supplies, equipment, and materials to the projected exit location.
   2. If anticipated Borehole exit is rejected by the Contracting Officer
      a. Retract the BHA and redrill such that the Borehole can exit within the specified tolerances. Any abandoned sections of Borehole shall be backfilled with cement or flowable fill material.

3.23 SAFETY REQUIEMENTS:

A. Methods of construction shall ensure the safety of the work, project participants, the public, third parties, and adjacent properties, whether public or private. All work shall conform to the requirements of all Federal, State, and local laws and regulations. The Contractor is solely responsible for maintaining safe working conditions at the site at all times.

B. Designate a Safety Officer. The Safety Officer shall administer an accident prevention program and shall prepare a code of safe practices and an emergency plan. Hold safety meetings at the beginning of each shift or once daily, whichever is most frequent, and provide safety instruction for new employees.

C. Fire Prevention and Control. All construction shall be performed in accordance with the applicable fire prevention and control requirements of all Federal, State, and local agencies.

3.24 CLEANUP AND RESTORATION:

A. Upon completion of Directional Drilling and Carrier Pipe installation, remove drilling spoils, Drilling Mediums, debris, and unacceptable material from the drill pad and surrounding areas disturbed by Directional Drilling operations. Clean up slurry and Drilling Medium from the ground to preconstruction conditions or better. Dispose of Drilling Mediums and cuttings at an approved site or facility. All fees and costs related to the collection, transport, and disposal of fluids, cuttings, and debris and any related permit costs shall be the Contractor's responsibility. During transport, all fluids shall be contained in steel tanks or other suitable vessels.

B. Restore drill pad and surrounding areas disturbed by Directional Drilling operations to original condition.

C. Remove temporary facilities used during drilling operations.

D. Remove all equipment and debris from the site at the end of the job. Restoration shall follow construction as the work progresses and shall be completed as soon as possible. Any property or improvements damaged or destroyed shall be restored to a condition equal to or better than existing prior to construction at no additional cost to the Government as approved by the Contracting Officer. Restoration shall be completed immediately if a third party or the Contracting Officer is inconvenienced by the damage, and in no case later than 30 days after the damage is discovered. The provision for restoration shall include all property which was affected by the construction operations.

3.25 CONSTRUCTION RECORDS:

A. Real Time Data Collection.
   1. Identify and measure drilling conditions and performance metrics with EDR.
   2. The real time data recorded by EDR must be available to Contracting Officer.
   3. The EDR shall conform to the Wellsite Information Transfer Standard Markup Language (WITSML) and/or Energistics Transfer Protocol (ETP) standards.
   4. Minimum data to be recorded include:
      a. Hole depth (ft)
      b. Bit depth(ft)
      c. Block or carriage height (ft)
      d. Rotary torque (psi)
      e. Thrust push/pull (psi)
      f. Weight on bit (lbf)
      g. Rotations per minute (RPM)
      h. Rate of penetration (ft/hour)
      i. Total pumping output (TPO) (gpm)
      j. Standpipe pressure (psi)
      k. Differential pressure (psi)
      l. Annular pressure (psi)
      m. Pipe pressure (psi)
   5. Data collected with EDR shall be collected, stored, and provided to other parties according to 3.08.A.

B. Maintain Daily Drilling Reports for all components and phases of the Work. The reports shall be signed by the site superintendent or shift foreman, submitted to the Contracting Officer Representative within 24 hours of completed daily work, and shall contain the following:
   1. Contractors onsite with their personnel and equipment, and the work performed. The reports shall be broken down into work time and down/standby time for each Contractor.
   2. Work crews and equipment onsite, and the work performed. The reports shall be broken down into work time and down/standby time for each crew and piece of equipment.
   3. Progress made for each construction stage.
   4. Problems or unusual conditions encountered, and actions taken to address these situations.
   5. Record of safety meetings conducted.
   6. List of visitors to construction site.

C. Maintain Drill and Drilling Medium Logs submitted to the Contracting Officer Representative within 24 hours of the day the event occurs and shall include at a minimum:
   1. The station of the excavation and advance distance with respect to drill pipe count and date and time.
      a. At minimum, submit construction tracking and survey data detailing the location of the Borehole face, deviations from the design trajectory, and showing the completed Borehole path twice per day while drilling.
   2. Drilling Medium properties noted a minimum of once every four hours during drilling.
      a. For drilling fluid/mud
         1) Density
         2) Sand Content
         3) Plastic Viscosity
         4) Yield Point
         5) pH
         6) Salinity
         7) Filtrate Loss
      b. For air/mist/foam
         1) Standpipe pressure
         2) Air rate
         3) Volume of water added
         4) Amount of any additives to the water (i.e. surfactants, corrosion inhibitors, etc.)

D. Keep and maintain at the construction site a complete set of field drawings for the recording of as-built conditions. It shall be marked or noted thereon all field information, properly dated, recording up-to-date as-built conditions. As-built drawings shall be submitted and accepted by the Contracting Officer prior to recommending final payment. As-built drawings shall be submitted in AutoCAD and PDF format.

E. Pressure test logs for Casing and Carrier Pipe with testing data and results.

F. Provide an end of well report for each Boring once complete which documents the following:
   1. An executive summary of the project noting, at a minimum, schedule, operation comments, challenges, and sololutions.
   2. An as-built plan and profile for the Borehole path. At minimum, include the following:
      a. A survey data point shall be obtained for the as-built Borehole path at a maximum spacing equivalent to one stand of drill pipe.
      b. Depths at which each Casing Pipe is set.
      c. Days vs Depth plot for each bore.
   3. A record of products and materials used in the creation of the Borehole including the following:
      a. Talley of Casing Pipes used.
      b. Record of drill bits used.
      c. Record of mud motors and rotary steerable systems used.
      d. Record of amount of cement used.
   4. Identify and describe unexpected variations to subsurface conditions.
   5. Show depth and location of any abandoned Borehole paths and lost downhole tooling.
   6. A copy of all daily drilling reports.

3.26 PROGRESS MEETINGS:

A. Contractor’s onsite foreman to attend weekly hour-long construction meetings in person during a time frame of two weeks before assembly of the drill rig operations to two weeks after completion of Wellhead construction and completion of the Borehole exit terminus assemblies.

END OF SECTION

This page intentionally left blank.

DIVISION 32

EXTERIOR IMPROVEMENTS

This page intentionally left blank.

SECTION 32 12 00
FLEXIBLE PAVING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Hot-mix, hot-laid, flexible paving, also known as asphalt concrete paving, used for roadways, driveways, parking areas, asphalt concrete sidewalks, and other areas where flexible paving is required by the Contract Documents, including:
      a. Preparation, such as sawcutting, milling where shown or indicated, cleaning, and other preparation for installing flexible paving.
      b. Asphalt concrete paving materials.
      c. Prime coating.
      d. Tack coats.
      e. Joint adhesives.
   2. Pavement markings where shown or indicated.
      a. Quality control and testing.

B. Related Requirements: Include, but are not necessarily limited to:
   1. Section 31 23 33 – Utilities - Trenching and Backfilling.

1.2 PRICE AND PAYMENT PROCEDURES

A. Measurement and Payment:
   1. Flexible paving Work is part of the lump sum bid/pay item indicated on the Bid Form, and part of the Contract, and in Section 01 27 00 - Definition of Contract Line Items.

1.3 REFERENCES

A. Terminology:
   1. Terminology indicated below are not defined terms and are not indicated with initial capital letters, but when used in this section have the meaning indicated below:
      a. “Flexible paving” means a composite material commonly used in construction projects such as road surfaces, driveways, parking areas, and similar paved surfaces. Flexible paving consists of asphalt (used as a binder) and a mineral aggregate (which may be augmented by recycled materials, such as recycled asphalt pavement or “RAP”) mixed together, installed in layers (courses), and compacted. Flexible paving may be divided into different types of materials, such as base course, binder course, and top course. Appurtenant materials, such as prime coats, tack coats, joint adhesives, and similar Work, are part of flexible paving Work. The terms “flexible paving”, “asphalt paving”, “asphalt concrete paving”, “bituminous paving”, and derivative terms, such as when the word “pavement” is used in lieu of the word, “paving”, have the same meaning. In this Section, the term “top course” is used, which has the same meaning as the terms ”wearing course”, and the final “surface course”.
      b. “Milling” means shaping and removing portions of existing surfaces by the cold milling process and subsequent cleaning.
      c. “Subbase” means the layer of compacted aggregate material installed on either the subgrade or, when required, geotextile separation fabric. Courses of flexible paving material are subsequently installed on top of the subbase.
      d. “Subgrade” means the uppermost surface of native soil material unmoved from cuts; the bottom of excavation on which either geotextile or subbase is installed.

B. Reference Standards: Standards referenced in this section include, but are not necessarily limited to, the following:
   1. American Association of State Highway and Transportation Officials (AASHTO):

a. AMRL - AASHTO Accreditation Procedures Manual for the Accreditation of Construction Materials Testing Laboratories.
b. M320 - Specification for Performance-Graded Asphalt Binder.
c. MP1 - Standard Specification for Performance Graded Asphalt Binder.

2. Asphalt Institute (AI):
   a. MS-2 - Mix Design Methods.
3. ASTM International (ASTM):
   a. D242/D242M - Standard Specification for Mineral Filler for Asphalt Mixtures.
   b. D692/D692M - Standard Specification for Coarse Aggregate for Asphalt Paving Mixtures.
   c. D946/D946M - Standard Specification for Penetration-Graded Asphalt Binder for Use in Pavement Construction.
   d. D977 - Standard Specification for Emulsified Asphalt.
   e. D1073 - Standard Specification for Fine Aggregate for Asphalt Paving Mixtures.
   f. D1188/D1188M - Standard Test Method for Bulk Specific Gravity and Density of Compacted Asphalt Mixtures Using Coated Samples.
   g. D2726/D2726M - Standard Test Method for Bulk Specific Gravity and Density of Non-Absorptive Compacted Asphalt Mixtures.
   h. D2950/D2950M - Standard Test Method for Density of Asphalt Mixtures in Place by Nuclear Methods.
   i. D3549/D3549M - Standard Test Method for Thickness or Height of Compacted Asphalt Mixture Specimens.
   j. D3666 - Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
   k. D6690 - Standard Specification for Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements.
   l. E329 - Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection.
4. Federal Specifications (FS):
   a. TT-B-1325D - Beads (Glass Spheres) Retro-Reflective
   b. TT-P-1952F - Paint, Traffic and Airfield Marking, Waterborne.

1.4 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Review installation procedures under this and other Specifications and coordinate installation of items that must be installed with, or before, flexible paving Work.

B. Notify other contractors, utility owners, and owner's of transportation facilities (as applicable) in advance of installing flexible paving to provide such other entities with sufficient time for installing items included in their contracts or projects, as applicable, that must be installed with, or before, flexible paving Work. Sequencing and Scheduling:
   1. Schedule and perform flexible paving Work when local hot-mix asphalt concrete plants are open. Contractor's Progress Schedule shall include appropriate float, relative to flexible paving Work, to accommodate potential early and late seasonal closures and openings of hot-mix asphalt concrete plants and potential effect of weather on the Work.
   2. Perform flexible paving Work after other necessary and required Work has been performed, such as providing required traffic controls, adjusting rim elevations of buried items such as manhole covers, valve box covers, and stormwater inlets, and coordination with authorities having jurisdiction.

1.5 QUALITY ASSURANCE

A. Regulatory Requirements:
   1. Areas requiring flexible paving Work are owned and under the jurisdiction of Government, unless indicated otherwise elsewhere in the Contract Documents.

B. Qualifications:

1. Asphalt Concrete Production Facility:
    a. Production facility for hot-mix asphalt concrete, prime coat material, tack coat material, and other bitumastic materials shall be certified by the Contractor's Geotechnical Engineer for furnishing such materials for highway construction within jurisdiction of such entity.
    b. Submit to Contracting Officer's Representative documentation of compliance with requirements of this provision, either prior to, or together with, product data Submittals for materials required under this Section.
2. Contractor's Testing Laboratory:
    a. Retain the services of independent testing laboratory to perform testing and determine compliance with the Contract Documents of materials specified in this Section.
    b. When Contractor is expressly required, by this Section's "Field Quality Control" Article or elsewhere in the Contract Documents, to retain testing laboratory for tests required under this Section's "Field Quality Control" Article, retain one testing laboratory for quality assurance testing and field quality control testing.
    c. Testing laboratory shall possess current, valid accreditation from the AASHTO Materials Reference Laboratory (AMRL), in accordance with AASHTO AMRL Procedures Manual for the Accreditation of Construction Materials Testing Laboratories, or ASTM D3666, for tests required for quality assurance testing indicated in this Article and field quality control testing indicated to be Contractor's responsibility.
    d. Testing laboratory shall comply with ASTM E329.
    e. Testing laboratory shall be experienced in the types of testing required.
    f. Selection of testing laboratory is subject to Contracting Officer's Representative's acceptance. Upon Contracting Officer's Representative's request, submit qualifications statement for testing laboratory, including name of entity, location, copies of applicable certifications, summary of entity's experience, and names and qualifications of personnel who will perform the subject sampling and testing.
3. Installer of Flexible Paving:
    a. Installer shall be experienced in providing flexible paving of the types required
    b. to, or greater than, those required for the Project.
    c. Installer shall be experienced in performing flexible paving installation for the state department of transportation having jurisdiction at the Site.
    d. Upon Contracting Officer's Representative's request, submit documentation of the proposed installer's compliance with qualifications requirements of this Article.

C. Quality Assurance Testing:
1. Quality assurance testing is in addition to source quality control testing, when required, and field quality control testing required in the "Field Quality Control" Article of this Section.
2. Materials used in the Work may require testing and retesting, as directed by Contracting Officer, during the Project. Allow free access to stockpiles of materials used in producing flexible paving and material production facilities during the Work. Tests not specifically indicated to be performed at Government's expense, including retesting of defective materials and installed Work, shall be performed at Contractor's expense.
3. Contractor's Quality Assurance Testing Laboratory Scope:
    a. Use of testing laboratory shall not relieve Contractor of responsibility for providing materials and the Work in compliance with the Contract Documents.
    b. Quality assurance testing laboratory shall perform the following, unless evidence of material compliance with reference specifications indicated in the various provisions of material requirements in "Part 2 – Products" of this Section, is submitted to Contracting Officer's Representative by Contractor and asphalt concrete production facility:
        1) Test in accordance with reference specifications indicated in requirements for materials in "Part 2 – Products" of this Section. In lieu of quality assurance testing, submit evidence and certification of material compliance with reference specifications. When evidence of such compliance submitted is not acceptable to Contracting Officer, perform quality assurance testing.

c. To facilitate testing services, Contractor shall:
   1) Secure and deliver to testing laboratory (when requested by Contracting Officer) representative Samples of materials that Contractor proposes to furnish and that are required to be tested.
   2) Furnish such labor as is necessary to obtain and handle Samples at the Site or at asphalt concrete production facility and other material sources.
   3) Advise testing laboratory and Contracting Officer sufficiently in advance of operations to allow for completion of quality assurance tests and for assignment of personnel.

1.6 SUBMITTALS

A. Action Submittals: Submit the following:
   1. Shop Drawings:
      a. Submit proposed asphalt concrete mix design for each asphalt concrete material, and other bituminous materials, required under this Section, providing complete data on materials, including location in the Work, source, material content and percentages, temperatures, and other pertinent data. Indicate proportion of bituminous material from recycled asphalt pavement.
      b. Proposed gradation for each aggregate required in flexible paving. Submit gradation test results for identical material furnished on a previous project. Indicate proportion of recycled asphalt pavement.
      c. In lieu of information required in the two paragraphs immediately above this paragraph, submit certificates of compliance with reference specifications indicated in the requirements for asphalt concrete materials in "Part 2 – Products" of this Section, for each of the following:
         1) Each mix design required.
         2) Bituminous materials required.
         3) Aggregates required in flexible paving, from each material source and each required gradation.
         4) Density of uncompacted asphalt concrete material.
         5) Density of previously-compacted, previously-tested asphalt concrete material.
         6) Density and voids analysis for each asphalt concrete material test specimen.
         7) Evidence of asphalt concrete plant inspection and compliance with reference specifications indicated in "Part 2 – Products" of this Section.
         8) Proportion of recycled asphalt pavement in bituminous materials and aggregate.
   2. Product Data:
      a. Product manufacturer's published data sheet, written instructions for handling and installation, and similar published information for joint adhesive material.
      b. Manufacturer's complete product data on pavement marking materials proposed for use, including product literature, specifications, recommended application techniques, and other installation data.

B. Informational Submittals: Submit the following:
   1. Quality Assurance Test Data Submittals and Source Quality Control Submittals:
      a. Submit for quality assurance tests and source quality control tests required.
   2. Delivery Tickets:
      a. Submit copy of delivery ticket for each load of asphalt concrete, prime coat materials, tack coat materials, crack sealant material, and other materials obtained from asphalt concrete production facility, signed by Contractor upon delivery to the Site.
   3. Field Quality Control Submittals:
      a. Submit results of required field quality control tests and inspections that are Contractor's responsibility.
   4. Qualifications:
      a. Asphalt concrete production facility.
      b. Contractor's testing laboratory, when required by Contracting Officer's Representative.
      c. Flexible paving installer, when required by Contracting Officer.

1.7 FIELD CONDITIONS

A. Restrictions on temperature and weather conditions for installing flexible paving are indicated in the "Installation" Article in "Part 3 – Execution" of this Section.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 ASPHALT CONCRETE MATERIALS:

A. Description:
   1. Provide all labor, materials, construction equipment and machinery, tools, services, and incidentals necessary and required to furnish and install flexible paving.
   2. Provide flexible paving at locations shown on the Drawings. When locations of flexible paving are not shown, provide flexible paving to restore areas where flexible paving existed on the Effective Date of the Contract where such areas were disturbed, damaged, or removed by Contractor.

B. Bituminous Materials:
   1. Bituminous Materials for Asphalt Concrete:
      a. Bituminous materials for asphalt concrete shall comply with the reference specifications indicated in the "Mixes" provision of this Article, for the associated flexible paving mix.
      b. Bituminous Materials from Recycled Asphalt Pavement (RAP): When use of RAP in asphalt concrete materials is acceptable, comply with requirements for RAP in the "Mixes" provision of this Article.
   2. Prime Coat:
      a. Prime coat shall be low-viscosity, emulsified asphalt.
      b. Provide prime coat in accordance with Arizona Department of Transportation.
   3. Tack Coat:
      a. Tack coat shall be emulsified asphalt.
      b. Provide tack coat in accordance with Arizona Department of Transportation.
   4. Joint Adhesive:
      a. Joint adhesive shall be a hot-applied asphalt product furnished as a solid and subsequently heated and applied to joints in asphalt concrete.

C. Aggregate Materials:
   1. Aggregates for Asphalt Concrete – General:
      a. Aggregate materials for asphalt concrete used in flexible paving shall be in accordance with the reference specifications indicated in the "Mixes" provision of this Article, for the associated asphalt concrete mix.
   2. Recycled Asphalt Pavement (RAP):
      a. Processed material obtained by milling or full depth removal of existing flexible pavement may be used as aggregate in asphalt concrete base course and binder course .
      b. When RAP is used, comply with the Contract Documents relative to applicable asphalt concrete mix, bituminous materials, and aggregates.

D. Mixes:
   1. Asphalt Concrete Mixtures: Provide the following materials designed and manufactured in accordance with Contract Documents:
      a. SuperPave Gyratory Design Revolution of 75.
      b. Base Course: 3/4-inch (per MAG, 2021 Gradation) with an asphalt cement binder grade PG 58-22, thickness of 2.5 inches to 4 inches.
      c. Top Course: ½-inch (per MAG, 2021 Gradation) with an asphalt binder grade PG 58-22, with a thickness of 1.5 inchs to 3 inches.
      d. Intermediate Course: ½-inch asphalt concrete intermediate course may be utilized if desired.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verification of Conditions:
   1. Examine subbase and base on which flexible paving will be installed.
   2. Advise Contracting Officer in writing of conditions detrimental to proper and timely completion of the Work.
   3. Do not proceed with flexible paving Work until unsatisfactory conditions are corrected.

B. Installing the Work constitutes Contractor's approval of underlying work, surfaces upon which flexible paving will be installed and field conditions prevailing at the time of the Work.

3.2 PREPARATION

A. General Preparation: Before starting installation of flexible paving, perform the following:
   1. Grade Control: Establish and maintain throughout flexible paving Work required lines and grades, including crown and cross-slope for each asphalt concrete course during construction.
   2. Coordinate placement of flexible paving with new and existing curbs, gutters, sidewalks, drainage structures (such as catch basins, drop inlets, and similar items), manholes and associated frames and covers, valve boxes and associated covers, sensors and similar devices associated with traffic controls, signage, pavement markings and markers, and similar items. Where necessary, adjust rim elevations of manhole frames and covers, valve boxes and covers, and stormwater drainage inlets within areas of flexible paving Work to required final elevations.
   3. Provide appropriate maintenance and protection of traffic measures during pavement installation and related Work.

B. Subgrade and Subbase:
   1. Prepare subgrade and provide subbase for flexible paving in accordance with 31 23 00 Earthwork. Before installing flexible paving Work, obtain Contracting Officer concurrence that subgrade and subbase are suitable for installing flexible paving.
   2. Prime Coat:
      a. Provide and use application equipment in accordance with requirements, in the "Installation" Article of this Section, for tack coat.
      b. Application Temperature: Material shall be 140 to 180 degrees F.
      c. Application Rate: 0.3 to 0.7 gallons per square yard, in accordance with applicable Arizona Department of Transportation standards.
      d. Uniformly distribute prime coat at junction of adjacent applications.
      e. Following application, do not disturb prime coat. Allow prime coat material to penetrate treated surface of subbase and treated surface to absorb prime coat material until the prime coat material is completely cured.
      f. Apply additional prime coat materials to uncoated or lightly coated areas.
      g. Twenty-four hours after applying prime coat, cover areas evidencing excessive prime coat material with sufficient dry fine aggregate to blot up or remove excess prime coat material.
      h. Protection of Treated Surface: Maintain and protect treated surface against damage. Remedy damaged areas to Contacting Officer's satisfaction before installing subsequent Work.

C. Milling:
   1. Mill existing pavement where shown, indicated, or otherwise required.
   2. Milling Equipment:
      a. Milling equipment shall be power operated, self-propelled machines capable of removing the desired thickness of existing surfaces. Equipment shall have sufficient power, traction, and stability to accurately maintain depth of cut and slope. Equipment shall produce finished profile and cross-slope to within 1/4-inch of that required and

shall produce uniform surface texture free of gouges and ridges greater than 3/8-inch deep.

b. Equipment shall include means of controlling dust and other particulate matter created by cutting and milling.
c. Furnish and use equipment that removes millings as quickly as rate of milling.
d. Furnish and use vacuum trucks, street sweepers, or power brooms to clean milled surfaces.

3. Milling Operations:
   a. Perform milling so that when final course of flexible paving is placed, required elevations and grades are provided. Where necessary, establish a taut reference string line to control line and grade of milling.
   b. Control and Tolerances:
      1) Milling equipment shall control profile and cross-slope with a moving reference not less than 30 feet in length. Moving reference may be a floating beam, ski, or other suitable type, such that resulting milled surface is sufficiently even.
      2) In areas accessible to milling equipment, provide milled surface to tolerance not greater than 1/2-inch.
      3) When subsequently installed flexible paving is determined to not comply with tolerances indicated in this provision, based on visual observation or apparent riding quality, Contracting Officer's Representative will direct Contractor to provide and test surface with 15 foot straight edge or string line parallel to pavement centerline and with 10 foot straight edge or string line placed transversely to pavement centerline on any portion of pavement.
      4) Variations exceeding 1/2-inch shall be appropriately remedied at no additional cost to Government.
   c. To the extent practicable, minimize time between milling and placement of flexible paving on milled surface.
   d. Areas not accessible to milling equipment, such as around or adjacent to drainage structures, manholes, curbs, and transverse joints on structures, may be removed using smaller milling equipment, by hand, other appropriate method, or combination thereof.
   e. Remove milled material as soon as it is milled. Remove fines and other material prior to opening milled area to traffic. Control dust emissions resulting from milling to extent practicable. Comply with requirements of Section 01 50 00 – Temporary Facilities and Controls. When traffic has been allowed onto milled surfaces, or when more than 48 hours have elapsed since milling, clean milled surfaces again prior to applying tack coat and performing other flexible paving Work.
   f. Appropriately maintain drainage to drainage inlets and other drainage structures.
   g. Properly dispose of milled material at location away from the Site, in accordance with Laws and Regulations, the Contract Documents, and requirements of owner of the disposal location.
   h. Milled surfaces that will receive flexible paving, shall be covered with not less than one course of asphalt concrete before end of paving season and closure of hot-mix asphalt concrete plants in the vicinity of the Site.
   i. Damage to milled surfaces resulting from traffic or other causes such as, but not limited to, raveling, petroleum spillage, or other contaminants that may potentially inhibit bonding of asphalt concrete to the milled surface, shall be appropriately repaired or re-milled by Contractor.

D. Potholes in Existing Paved Areas:
   1. Before placing top course, correct irregularities in binder course. Repair potholes and gouges greater than one inch deep.
   2. For potholes and abrupt depressions three inches and less in depth:
      a. Repair such defects manually with required asphalt concrete top course.
      b. Preparation:

1) Contracting Officer's Representative will mark locations of such potholes and other surface defects requiring repair, at the Site, using appropriate lumber crayon. Contractor shall furnish necessary lumber crayons.
2) Cut out and patch marked areas. Ensure each patch has an area of not less than one square foot.
3) Excavate marked area to sound, stable material. Using milling or planing equipment is acceptable. Cut all edges vertically.
4) After removing debris and unsound material, thoroughly clean area to be patched and adjacent pavement.
5) Thoroughly dry surfaces of area to be patched.
6) Apply tack coat to bottom and vertical surfaces of area to be patched as specified in the "Installation" Article of this Section.

c. Patching:
1) After tack coat is cured, place asphalt concrete top course material. Compact with steel wheeled roller or other appropriate compaction equipment. After compaction, surface of patch shall have surface elevation equal to final, required surface elevation of completed top course minus required depth of new top course overlay. Fill excavated areas before close of each work day.
2) Potholes extending into shoulder do not require excavation beyond edge of roadway.

3. For potholes and abrupt depressions greater than three inches deep:
a. Repair such defects manually with required asphalt top course material and asphalt binder course material.
b. Perform preparation for patching as indicated above in this provision for potholes and abrupt depressions three inches and less in depth.
c. Patching:
1) After tack coat is cured, place asphalt binder course in lifts not exceeding three inches in uncompacted depth. After leveling each lift, compact using appropriate mechanical tamper.
2) After compaction, surface of patch shall have surface elevation equal to final, required surface elevation of completed top course minus required depth of new top course overlay.
3) Compact last lift with steel wheeled roller or other appropriate compaction equipment. Fill excavated areas before close of each work day, except that required top course may be installed on following day.

E. Surface Preparation:
1. Repair surface defects in existing pavement to provide uniform surface to receive new pavement.
2. Provide crack sealant to completely fill cracks more than 1/16-inch wide in areas shown or indicated on the Drawings.
3. Clean existing surfaces over which asphalt concrete pavement will be installed, by removing from surface foreign material, excess asphalt concrete, excess joint sealant, crack filler, and other undesirable matter.
4. Where new flexible paving will abut existing paving, sawcut edges of existing paving in accordance with the "Installation" Article of this Section.
5. Provide tack coat as indicated in the "Installation" provision of this Section.

3.3 INSTALLATION

A. Installation Requirements for Asphalt Concrete:
1. Flexible Paving Installation - General:
a. Provide final surfaces of flexible paving with uniform texture, at required locations and grades, with required cross-sections, as shown and indicated.
b. Where new flexible paving top course (wearing course or surface course) is provided and connects to existing pavement, provide new top course that smoothly and evenly transitions to existing paved surfaces and grades.

2. Joining of Pavements:
    a. When new flexible paving will connect to or abut existing or previously-installed pavement, existing or previously-installed pavement shall be neatly and carefully edged to allow overlapping and feathering of subsequent course of asphalt concrete material.
    b. Sawcutting of Edges:
        1) Where new flexible paving will meet existing pavement, existing pavement shall be sawcut and notched.
        2) Sawcut existing pavement with equipment and machinery specifically intended and assembled for sawcutting pavements of the type required.
        3) Edges of existing pavement shall be cleanly cut. Edges shall be linear and have vertical cut face.
        4) Where cut edges are not smooth, even, and straight, repair cut edge to remain to provide a smooth, even appearance.
        5) Sawcutting of concrete pavement, concrete curbs, concrete gutters, and sidewalks shall comply with Section 01 73 29 - Cutting and Patching, and Section 02 41 00 - Demolition.
    c. Where flexible paving will meet existing asphalt pavement, remove existing pavement 12 inches onto undisturbed existing pavement course at edges where new flexible paving will meet existing pavement.
    d. Provide tack coat in accordance with requirements of this Article.
3. Tack Coat:
    a. Provide tack coat material at surface of existing or previously-installed pavement course over which new flexible paving will be installed, prior to installing subsequent pavement course. With Contracting Officer's Representative's concurrence, tack coat may be omitted when subsequent course of asphalt concrete will be installed over freshly-installed asphalt pavement course subjected to very little or no traffic.
    b. Tack Coat Installation: Install tack coat immediately prior to installing asphalt concrete. Install asphalt concrete while tack coat is wet. Apply tack coat in accordance with applicable Arizona Department of Transportation standards.
4. Weather Limitations for Installing Asphalt Concrete:
    a. Temperature:
        1) For base course and binder course paving lifts equal to or greater than two inches thick, ambient temperature shall be 40 degrees F and rising.
        2) For top course paving or other pavement courses in lifts less than two inches thick, temperature of surface on which pavement will be placed shall be 50 degrees F or greater.
    b. Prohibitions:
        1) Do not install flexible paving materials during foggy conditions or during precipitation.
        2) Do not install flexible paving materials when base on which material will be installed contains moisture in excess of optimum.
        3) Install flexible paving materials only when Contracting Officer concurs weather conditions are suitable.
5. Installation of Asphalt Concrete:
    a. Asphalt concrete mixture shall be transported to the Site and installed as soon as possible after mixing.
    b. Do not pave over existing manhole frames and covers, valve box covers, stormwater inlets, or similar items in areas of flexible paving Work. Where new asphalt concrete will be installed in areas of existing detectors for traffic controls, remove and properly reinstall such detectors in accordance with operation and maintenance instructions of detector's manufacturer. Subsequent to reinstallation, verify proper operation of detectors and associated traffic controls.
    c. Placement of each asphalt concrete course shall be completed over full width of section under construction during each day's flexible paving Work.

d. Spread and finish asphalt concrete courses using self-propelled, mechanical spreading and finishing equipment. Initially placed thickness of courses shall not exceed 150 percent of required, compacted thickness unless approved in writing by Contracting Officer.

e. Compaction:
   1) Provide and use compacting rollers to properly compact and finish asphalt concrete. Provide not less than two rollers at the Site at all times during installation of asphalt concrete. One roller may be a pneumatic-tire roller.
   2) During compaction, roller speed shall not exceed three miles per hour. When sufficient quantity of rollers is unavailable, reduce quantity of asphalt concrete installed to accommodate quantity of available rollers.
   3) Required rollers shall be at the Site, in proper, operating condition, prior to placing asphalt concrete.
   4) Using vibratory rollers in lieu of steel-wheeled rollers is acceptable, however, when thickness of asphalt concrete is one-inch or less, rolling shall be via static mode.
   5) Rolling of initially placed asphalt concrete material, or breakdown rolling, shall begin as soon as asphalt concrete will bear roller without undue displacement.
   6) Rolling shall be longitudinal, overlapping on successive trips by not less than one-half roller rear wheel width, and not more than three-quarters of roller rear wheel width. Alternate trips of roller shall be of slightly different lengths.
   7) Roller motion shall be slow enough to avoid displacing asphalt concrete.
   8) Operate rollers continuously from breakdown of laid asphalt concrete through finish rolling.
   9) Perform finish rolling using a steel-wheeled roller or a vibratory steel-wheel roller operating in static mode.
   10) Perform rolling with consecutive passes to achieve even, smooth finish without roller marks.
   11) At each location not accessible to roller, thoroughly compact asphalt concrete using tampers and finish, where necessary, with hot smoothing iron to provide uniform, smooth layer over entire area so compacted.
   12) Criteria for acceptance of installed density of asphalt concrete is indicated in the "Field Quality Control" Article of this Section.

f. Placement of Adjacent Strips of New Asphalt Concrete:
   1) When more than one width of asphalt concrete material will be placed, six-inch wide strip of asphalt concrete adjacent to area on which future material will subsequently be placed, shall not be rolled until such subsequently-placed material is installed.
   2) Do not leave unrolled strip unrolled for more than two hours after placing, unless six-inch unrolled strip is first heated with a joint heater.
   3) After first strip or width of asphalt concrete is compacted, place, finish, and compact second width or strip as required for first width, except that rolling shall be extended to include six-inch strip of first width not previously compacted.

6. Joint Adhesive:
   a. Provide joint adhesive at the following locations:
      1) At edges where new flexible paving will connect to existing or previously-installed pavement, including transverse and longitudinal joints.
      2) Where new pavement will abut curbing, concrete gutters, drainage structures and frames, manhole cover frames, valve boxes, and similar items.
   b. Preparation: Prior to applying joint adhesive, properly clean surfaces to receive joint adhesive. Clean using pneumatic air discharge. Provide suitable traps or devices on compressed air equipment to prevent moisture and oil from contaminating surfaces to receive joint adhesive.
   c. Equipment: Heat and melt joint adhesive in melter constructed either as double boiler filled with heat-transfer medium between inner and outer shells, or with internal tubes

or coils carrying joint adhesive through heated oil bath and into heated double wall hopper. Insulate or provide heat tracing on hose used for conveying joint adhesive between melter or hopper and surfaces to receive joint adhesive. Equipment shall include appropriate temperature sensor and indicator to facilitate monitoring of material temperature during application.

d. Application:
   1) Apply joint adhesive at temperatures and using procedures in accordance with material manufacturer's written instructions.
   2) Apply joint adhesive to entire face of transverse and longitudinal joints in flexible paving.
   3) Apply joint adhesive to full depth of top course where top course abuts curbing, concrete gutters, drainage structures and frames, manhole cover frames, valve boxes, and similar items.

7. Construction Joints:
   a. Provide construction joints in flexible paving to provide neat junction, thorough compaction, and bond throughout.
   b. Longitudinal joints in asphalt concrete top course shall correspond with edges of proposed traffic lanes. Other joint arrangements require Contracting Officer's Representative's approval.
   c. Provide transverse joint, perpendicular to centerline of asphalt concrete installed, extending across full width of strip being laid, at end of each workday, and at other times when placement of hot-mix asphalt concrete will be suspended for a period of time allowing asphalt concrete mixture to chill. Transverse joints in adjacent lanes shall be staggered at not less than 10 feet.
   d. Thoroughly compact, by rolling forward, end of freshly laid asphalt concrete before asphalt concrete becomes chilled. When work is resumed, vertically cut for full depth of course.
8. Curing:
   a. Do not allow traffic onto newly installed pavement until obtaining Contracting Officer's concurrence. Do not allow traffic on flexible paving until surface temperature is less than 140 degrees F.
   b. Hold construction traffic on new pavement to a minimum.
9. Paver and Equipment Cleaning:
   a. Tools and equipment used for asphalt installation shall not be cleaned on pavement surface, or near streams, ponds, drainage structures, or other areas that are tributary to drainage routes or surface waters. Designated area, at location acceptable to Contracting Officer, shall be used for cleaning paving equipment and tools.
   b. When petroleum products are used for cleaning, use tarpaulins, sand pails, other types of pails, or other methods to collect liquids produced by cleaning, to prevent contamination spills and release to the environment.
   c. Hand sprayers or other similar devices may be used to minimize the quantity of petroleum product applied. Sand, soil, and collected petroleum products resulting from cleaning shall be properly disposed of as petroleum contaminated soil in accordance with Laws and Regulations.

B. Installation Tolerances:
   1. Milling: Tolerances for milling, are indicated in the "Milling" provision of the "Preparation" Article of this Section.
   2. Thickness:
      a. Asphalt concrete provided as a truing and leveling course, if any, and conditioning of existing surface will not be considered in pavement thickness determinations.
      b. Allowable tolerance for asphalt concrete required for each separate type of asphalt concrete mixture required is as follows:
         1) Nominal pavement lift thicknesses of four inches or less, 1/4-inch.
         2) Nominal pavement lift thicknesses greater than four inches, 1/2-inch.

c. Tolerance for total thickness of all asphalt concrete pavement courses shall be 1/4-inch. When asphalt concrete is installed on newly constructed subbase, additional tolerance of 1/4-inch is acceptable, both in nominal thickness of course installed directly on subbase, and total pavement thickness. No payment will be made for material provided in excess of indicated tolerance.

3. Surface Trueness:
   a. "Surface trueness", as used in this Section refers to new pavement constructed to required surface cross-section, with even surface, without abrupt changes or transitions in elevation or grade, and without ruts or raveling. Appropriately installed pavement results in no ponding of water and properly conveys stormwater runoff to appropriate drainage routes and stormwater inlets.
   b. Contractor shall construct each pavement course to 1/4-inch surface tolerance.
   c. Contracting Officer's Representative will evaluate surface trueness and compliance with surface tolerance by directing Contractor to provide 16-foot straight edge or string line placed parallel to centerline of pavement and with 10-foot straight edge or string line placed transversely to centerline of pavement. Variations exceeding indicated tolerance shall be corrected or pavement remedied and replaced at no additional cost to Government.
   d. Crowned Surfaces:
      1) Test crowned surfaces with a crown template, centered and at right angles to the crown.
      2) Surfaces will be acceptable when variance is equal to or less than 1/4-inch from the template.
4. Elevation: Finished surface of pavement shall be within plus or minus 1/2-inch of elevations shown or indicated. New pavement shall be equal in elevation to frames and covers of manholes, valve boxes and similar items, and allow appropriate stormwater drainage into stormwater inlets.

3.4 APPLICATION OF PAVEMENT MARKINGS:

A. Pavement Markings: Provide pavement markings where shown or indicated.
   1. Pavement Marking Installation – General
      a. Provide pavement markings, of the types required, at locations shown and indicated in the Contract Documents.
      b. Comply with paint manufacturer's written installation instructions.
      c. Where public traffic will be allowed to transit newly installed pavement prior to application of permanent pavement markings, provide appropriate temporary markings sufficient for safe use of the paved area during wet and dry conditions and during all hours of the day.
      d. When the Work includes providing temporary pavement or alterations of existing pavement markings during construction, provide appropriate temporary pavement markings and signage, in accordance with requirements of the applicable state department of transportation standard specifications.
   2. Preparation:
      a. Sweep surface with power broom supplemented by hand brooms to remove loose material and dirt. Ensure surfaces to receive pavement markings are of appropriate cleanliness to receive pavement markings, in accordance with paint manufacturer's written instructions.
      b. Do not begin marking bituminous concrete pavement until obtaining Contracting Officer's Representative's concurrence.
      c. Surfaces to receive pavement markings shall be completely dry at the time markings are applied.
      d. When reflective glass beads are required, mix with paint prior to paint application.
   3. Application:
      a. Do not apply pavement markings not less than five days from time surface is completed. Longer period when reasonably directed by Contracting Officer.

b. Do not apply pavement markings on wet surfaces, during wet or damp weather, or when ambient temperature is below 40 degrees F.
c. Using mechanical equipment, provide uniform, straight edges in two separate coats. Apply in accordance with paint manufacturer's written instructions.
d. Lay out markings and striping in accordance with the Contract Documents. Width of pavement striping for lane markings and parking spaces shall be four inches, unless otherwise shown or indicated on the Drawings.

4. Protection:
a. Do not allow vehicles or construction equipment or machinery to traverse pavement markings until applied markings are completely cured in accordance with manufacturer's written instructions and is dry.
b. Properly remove defective pavement markings, including those damaged by untimely passage of vehicles, equipment, or machinery, and replace with required pavement marking in accordance with the Contract Documents.

3.5 FIELD QUALITY CONTROL

A. Field Tests and Inspections:
1. Contracting Officer's Representative's Field Quality Control Activities:
a. Contracting Officer's Representative may determine compliance with installation tolerances for surface trueness, and depth and trueness of milling (where required), as indicated elsewhere in this Section. Contractor shall furnish implements required for evaluating and determining trueness of asphalt concrete courses and milling.
b. Contracting Officer's Representative may identify, by visual observation, areas of apparent defective flexible paving Work and appropriately advise or notify Contractor of such defects.
2. Responsibilities of Testing Laboratory:
a. Government's Field quality control testing laboratory will:
1) Test proposed materials for compliance with the Contract Documents, as directed by Contracting Officer.
2) Perform field density tests, as indicated below in this provision, to verify required compaction of asphalt materials was provided obtained.
3) When directed by Contracting Officer, perform coring of installed and compacted flexible paving to determine compliance with required course thickness and tolerances. Submit written results of tests to Contracting Officer and Contractor.
b. Density:
1) Test Method: ASTM D2950; test one sample every 250 square yards of pavement. Test for each asphalt concrete course installed.
2) In addition, when directed by Contracting Officer, compare density of in place flexible paving materials against laboratory specimen (as determined during required quality assurance testing) or certificates on same asphalt pavement mixture, using nuclear density device.
3) Criteria for Acceptance: Density of in-place, compacted asphalt pavement shall be not less than 90 percent of recorded laboratory specimen (as determined during required quality assurance testing) or certificate density. Compacted density shall be not be greater than 98 percent.
c. Thickness of Asphalt Concrete Pavement:
1) Contractor shall provide work zone traffic control and facilitate obtaining pavement cores by testing laboratory.
2) Measure in accordance with ASTM D3549; obtain and test one core sample from every 1,000 square yards of compacted pavement. Comply with thickness tolerance specified in Article 3.3 of this Section.
3) Contracting Officer's Representative may use another acceptance method such as yield calculations to determine final thickness for acceptability and payment.
4) Contractor shall repair holes from test specimens in accordance with this Section's requirements for repairing defective Work.

d. Authority of Field Quality Control Testing Laboratory:
    1) Technicians representing testing laboratory will inspect materials at the Site and perform required testing. When materials furnished or Work performed do not comply with the Contract Documents, field quality control testing laboratory technician shall promptly direct attention of Contracting Officer's Representative and Contractor to such non-compliance.
    2) Testing laboratory personnel shall not act as foreman or perform other duties for, or on behalf of, Contractor. The Work will be checked as it progresses, but failure to detect defective Work shall not in any way prevent later rejection of such defective Work when defect is discovered. Failure to detect defective Work as it occurs does not obligate Contracting Officer to final acceptance. Testing laboratory personnel are not authorized to revoke, alter, relax, enlarge, or release requirements of the Contract Documents, nor to approve or accept any portion of the Work.

3. Asphalt Concrete Mix Temperature: Measure temperature at time of placement, record, and submit to Contracting Officer's Representative.
4. Drainage:
    a. Upon completion and curing of top course provided under the Contract. Contracting Officer's Representative will direct Contractor to spray clean water (whether potable or non-potable, and without observable solid material, biological matter, and Constituents of Concern) thoroughly over top course of flexible paving Work, in presence of Contracting Officer's Representative.
    b. Spray sufficient quantity of water over all top course areas to result in visually obvious runoff to drainage routes.
    c. In lieu of water spray, Naturally occurring precipitation sufficient for visually observable runoff is acceptable.
    d. Immediately following conclusion of spraying or precipitation event, as applicable, Contracting Officer's Representative will mark areas of ponding, if any, with appropriate lumber crayon of high visibility color contrasting with pavement. Contractor shall furnish necessary quantity of lumber crayons. Immediately following marking, obtain digital photographic documentation of each marked area, in accordance with Section 01 32 33 - Project Photographic Documentation, and promptly submit to Contracting Officer's Representative. Such photographic documentation does not count toward quantity of photographs required by Section 01 32 33 - Project Photographic Documentation.
    e. Contracting Officer's Representative may redesignate, re-mark, or both, areas of ponding as such areas become apparent to Contracting Officer's Representative or Government.
5. Surface Trueness:
    a. Contractor's obligations regarding assisting Contracting Officer's Representative in determining compliance with milling tolerances are indicated in the "Milling" provision of Article 3.2 of this Section.
    b. Contractor's obligations regarding assisting Contracting Officer's Representative in determining compliance with tolerance for trueness of installed asphalt concrete courses are indicated in the "Installation Tolerances" provision of Article 3.3 of this Section.

B. Defective Paving Work:
1. The following constitutes defective flexible paving Work: ruts, ridges, roller marks, cracking, tearing, segregation, bleeding, ponding of stormwater, or other irregularities. Flexible paving Work that does not comply with required tolerances or other requirements of the Contract Documents, or that is damaged while protection is Contractor's responsibility will be considered as defective Work.
2. When directed by Contracting Officer, remove and replace defective flexible paving Work. Cut out such areas of defective pavement and fill with fresh asphalt concrete materials, compacted to required density. When performing such repairs, provide tack coats and joint adhesive as required elsewhere in this Section, and compact as required elsewhere in this

Section. When pavement is cut out or removed for such remedies, the area removed shall be of regular shape with right angles. To extent practicable, cut-out areas shall be square or rectangular. Surface of repaired areas shall be flush with adjacent pavement.

3. Completely fill holes in flexible paving, resulting from test specimens, using required asphalt concrete top course material, together with required tack coats and joint adhesive, compacted to required density.

3.6 CLEANING

A. Cleaning: After completing flexible paving installation, completely clean surfaces of excess and spilled bituminous materials, excess asphalt concrete, and foreign matter. Appropriately dispose of such material.

3.7 PROTECTION

A. Protect finished pavement until pavement is properly cured, hard, and cool.

B. Cover openings of drainage structures, manholes, valve boxes, and similar items in paved areas until permanent coverings are provided.

END OF SECTION

This page intentionally left blank.

SECTION 32 13 13
CONCRETE PAVING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Reinforced, “traditional” Portland cement concrete paving.
2. Surface treatment applied to hardened concrete pavement.
3. Construction joints in concrete pavement.
4. Expansion joints (also known as “pressure relief joints”) and isolation joints in concrete pavement, together with associated joint sealants.

B. Related Requirements: Include, but are not necessarily limited to:
1. Section 02 41 00 - Demolition.
2. Section 03 01 30 - Repair and Rehabilitation of Existing Construction.
3. Section 03 05 05 - Concrete Testing and Inspection.
4. Section 03 11 13 - Formwork.
5. Section 03 15 19 - Anchorage to Concrete.
6. Section 03 21 00 - Reinforcement.
7. Section 03 31 30 - Concrete, Materials and Proportioning.
8. Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.
9. Section 03 35 00 - Concrete Finishing and Repair of Surface Defects.
10. Section 31 23 33 - Trenching and Backfilling.
11. 31 23 00 - Earthwork

1.2 PRICE AND PAYMENT PROCEDURES

A. Measurement and Payment:
1. Concrete paving Work is part of the lump sum bid/pay item indicated on the Bid Form, and part of the Contract, and in Section 01 27 00 - Definition of Contract Line Items.

B. Payment Adjustment for Deficient Thickness of Concrete Pavement:
1. When thickness of new concrete paving provided is determined, based on requirements of this Section’s “Field Quality Control” Article to be more than 0.25-inch less than thickness required by the Contract Documents, and less than 0.5-inch less than the thickness required by the Contract Documents, unless there is other basis for Contracting Officer to reject the new concrete paving as defective Work, Contracting Officer will accept the subject concrete paving upon an associated reduction in the Contract Price. Such reduction in the Contract Price shall be determined as indicated in this provision.
2. Relative to this provision, potential adjustment in Contractor’s compensation for new concrete pavement will be determined for each separate traffic lane (or, for parking areas and other, similar open areas of concrete paving, by apportioning the subject concrete paving into strips, each 12 feet wide), divided into segments each 400 feet in length and each remaining segment less than 400 feet. Contracting Officer will determine the apportionment of segments per lane or lane-equivalent. Such reductions in the Contract Price shall, in all cases, be for the full width of the associated lane or lane-equivalent for which field quality control activities determined concrete pavement to be of thickness less than required by the Contract.
3. Government or Contracting Officer may set-off such reductions from amounts due Contractor under the Contract.
4. Reduction of Contractor’s compensation for new concrete paving shall be as indicated in the following table:

Deficiency in Depth Determined by Cores inches	Proportional Part of Contract Price Allowed
0.0 to 0.25	100%
0.26 to 0.30	95%
0.31 to 0.35	90%
0.36 to .040	85%
0.41 to 0.45	80%
0.46 to .050	70%
Greater than 0.50	Defective Work

1.3 REFERENCES

A. Terminology:
   1. Terminology indicated below are not defined terms and are not indicated with initial capital letters, but when used in this section have the meaning indicated below:
      a. "Average pavement thickness" means the weighted average of the acceptable pavement, taking into consideration the length of the pavement after eliminating defective pavement Work.
      b. "Concrete" means Portland cement concrete, in accordance with the Specifications of Division 03 - Concrete. In this Section, the term, "concrete" does not mean asphalt concrete used in flexible paving, unless the term "asphalt concrete", is expressly indicated.
      c. "Milling" means shaping and removing portions of existing surfaces by the cold milling process and subsequent cleaning.
      d. "Subbase" means the layer of compacted aggregate material installed on either the subgrade or, when required, geotextile separation fabric. Concrete paving is subsequently installed on top of subbase, unless new concrete paving is expressly required to be installed over other, existing pavement to remain.
      e. "Subgrade" means the uppermost surface of native soil material unmoved from cuts; the bottom of excavation on which either geotextile or subbase is installed.

B. Reference Standards: Standards referenced in this section include, but are not necessarily limited to, the following:
   1. American Association of State Highway and Transportation Officials (AASHTO):
      a. M31 - Standard Specification for Deformed and Plain Carbon and Low-Alloy Steel Bars for Concrete Reinforcement (ASTM A615/A615M).
      b. M153 - Standard Specification for Preformed Sponge Rubber and Cork Expansion Joint Fillers for Concrete Paving and Structural Construction (ASTM D1752).
      c. M213 - Standard Specification for Preformed Expansion Joint Fillers for Concrete Paving and Structural Construction (Non-extruding and Resilient Bituminous Types) (ASTM D1751).
      d. M224 - Standard Specification for Use of Protective Sealers for Portland Cement Concrete.
      e. M233 - Standard Specification for Boiled Linseed Oil Mixture for Treatment of Portland Cement Concrete.
   2. ASTM International (ASTM):
      a. A615/A615M - Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement (AASHTO M31).
      b. A706 - Standard Specification for Deformed and Plain Low-Alloy Steel Bars for Concrete Reinforcement.
      c. A996 - Standard Specification for Rail-Steel and Axle-Steel Deformed Bars for Concrete Reinforcement.
      d. C31/C31M - Standard Practice for Making and Curing Concrete Test Specimens in the Field.
      e. C39/C39M - Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens.

f. C42/C42M - Standard Test Method of Obtaining and Testing Drilled Cores and Sawed Beams of Concrete.
g. C78/C78M - Standard Test Method for Flexural Strength of Concrete (Using Simple Beam with Third-Point Loading).
h. C138/C138M - Standard Test Method for Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete.
i. C174/C174M - Standard Test Method for Measuring Thickness of Concrete Elements Using Drilled Concrete Cores.
j. D1751 - Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Non-extruding and Resilient Asphalt Types) (AASHTO M213).
k. D1752 - Standard Specification for Preformed Sponge Rubber Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction (AASHTO M153).
l. D3666 - Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials.
m. D6690 - Standard Specification for Joint and Crack Sealants.
n. E329 - Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection.

3. Federal Specification (FS):
   a. TT-B-1325D - Beads (Glass Spheres) Retro-Reflective.
   b. TT-P-1952F - Paint, Traffic and Airfield Marking, Waterborne.
4. American Concrete Institute (ACI):
   a. 305R, Guide to Hot Weather Concreting
   b. 306R, Guide to Cold Weather Concreting

1.4 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Review installation procedures under this and other Specifications and coordinate installation of items that must be installed with, or before, concrete paving Work.

1.5 QUALITY ASSURANCE

A. Regulatory Requirements:
   1. Areas requiring concrete paving Work are owned and under the jurisdiction of Government, unless indicated otherwise elsewhere in the Contract Documents.

B. Qualifications:
   1. Ready-mix Concrete Supplier:
      a. Comply with requirements of Section 03 31 31 - Concrete, Mixing, Placing, Jointing, and Curing.
   2. Installer of Concrete Paving:
      a. Installer shall be experienced in providing concrete paving of the types required for the Project, and experienced in providing concrete paving in quantities similar to, or greater than, those required for the Project.
      b. Upon Contracting Officer's request, submit documentation of the proposed installer's compliance with qualifications requirements of this Article.

1.6 SUBMITTALS

A. Submittals Required Under Other Sections:
   1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. In addition to Submittals required in this Section, furnish Submittals required by the Specifications of Division 03 - Concrete, including, but not necessarily limited to, the following:
      a. Concrete materials and supplemental materials, such as admixtures and curing compounds. Mixes for concrete paving are specified under this Section.

b. Reinforcing steel required for concrete paving may be submitted under this Section, but shall comply with requirements for reinforcing steel Submittals as indicated in the Specifications of Division 03 - Concrete.
c. Joint locations. Shop Drawings and product data Submittals for joint materials and sealants for concrete paving shall be submitted under this Section.
d. Delivery tickets.
e. Field quality control Submittals. Additional requirements for field quality control Submittals specific to concrete paving are indicated in this Section.

B. Action Submittals: Submit the following:
1. Shop Drawings:
a. Mix designs for concrete paving, in accordance with requirements for mix design Submittals indicated in the Specifications of Division 03 - Concrete, but demonstrating compliance with mix design required in this Section.
b. Drawings showing and indicating locations of concrete pavement joints, including longitudinal joints and transverse joints, including construction joints and expansion joints (pressure relief joints), and isolation joints. Submit details of each type of joint required, including details for unique conditions.
2. Product Data: For each item or material specified in this Section, submit manufacturer's published product information, including indication of materials, performance requirements, compliance with reference standards indicated in this Section, manufacturer's published specifications, application rates, manufacturer's written instructions for storage, preparation, and installation, and similar information.
a. Surface treatment material applied to new concrete pavement.
b. Expansion joint and isolation joint materials, including sealants.
3. Samples
a. Samples of fabricated jointing materials, when requested by Contracting Officer's Representative. When furnished, such Samples shall be of sufficient size to demonstrate materials, color, texture, arrangement, features, thickness, and height of joint materials proposed for use.

C. Informational Submittals: Submit the following:
1. Special Procedure Submittals:
a. Submit manufacturer's description and characteristics for traveling form paver, if proposed for use.
b. Contracting Officer's review, comments, and acceptance of special procedure Submittals required is not for Contractor's benefit and, in no way reduces or mitigates Contractor's sole responsibility for construction means, methods, techniques, procedures, and sequences, and associated safety and protection measures.
2. Field Quality Control Submittals:
a. Submit written results of required field quality control tests and inspections, in accordance with the Specifications of Division 03 - Concrete.
b. Submit written results of required field quality control tests and inspections required in the "Field Quality Control" Article of this Section.
3. Qualifications Statements:
a. Concrete ready-mix Supplier: In addition to documentation of qualifications requirements in the Specifications of Division 03 - Concrete, submit documentation of compliance with qualifications requirements of this Section:
b. Installer, when requested by Contracting Officer.

1.7 FIELD CONDITIONS

A. Restrictions on temperature and weather conditions for installing concrete paving are indicated in the Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 MATERIALS

A. Portland Cement: ASTM C150/C150M, Type I or II.

B. Aggregates:
   1. ASTM C33/C33M, gradation size #67, ¾ IN to #4.
   2. Clean, crushed gravel

C. Water: Potable quality.

D. Admixtures: Comply with Specification Section 03 31 30.

E. Reinforcing Bars: ASTM A615/A615M, Grade 60.

F. Welded Wire Reinforcement:
   1. ASTM A1064/A1064M.
   2. Flat.
   3. Clean, free from dirt, scale, rust.

G. Preformed Joint Filler:
   1. Non-extruding cork, self-expanding cork, sponge rubber or cork rubber.
   2. AASHTO M153 or AASHTO M213.

H. Hot-Poured Joint Sealing Material:
   1. FS SS-S-1614.

I. Membrane Curing Compound: ASTM C309.

J. Cover Materials for Curing:
   1. Burlap:
      a. AASHTO M182.
      b. Minimum Class 2, 8 OZ material (1 YD x 42 IN).
   2. Polyethylene film, AASHTO M171.

K. Paper Subgrade Cover: Polyethylene film, AASHTO M171.

L. Concrete Treatment:
   1. Boiled linseed oil mixture.
   2. AASHTO M233.

M. Forms:
   1. Steel or wood.
   2. Size and strength to resist movement during concrete placement and able to retain horizontal and vertical alignment.
   3. Free of distortion and defects.
   4. Full depth.
   5. Metal forms:
      a. Minimum 7/32 IN thick.
      b. Depth equal to edge thickness of concrete.
      c. Flat or rounded top minimum 1-3/4 IN wide.
      d. Based 8 IN wide or equal to height, whichever is less.
      e. Maximum deflection 1/8 IN under center load of 1,700 LBS.
      f. Use flexible spring steel forms or laminated boards to form radius bends.

2.2 MIXES

A. Mix design to provide 4,000 PSI 28-day compressive strength, 1 IN slump, 6 PCT air.

2.3 CONCRETE PAVING – GENERAL

A. Description:
   1. Provide all labor, materials, construction equipment and machinery, tools, services, and incidentals necessary and required to furnish and install Portland cement concrete paving.
   2. Provide concrete paving at locations shown on the Drawings. When locations of concrete paving are not shown, provide concrete paving to restore areas where concrete paving existed on the Effective Date of the Contract where such areas were disturbed, damaged, or removed by Contractor.
   3. This Section does not address concrete curbing, gutters, sidewalks, or concrete driveway aprons, which are, when part of the Work, addressed under other sections of the Specifications.

2.4 CONCRETE PAVING MATERIALS SPECIFIED ELSEWHERE

A. Requirements for materials for concrete paving are generally indicated in the Specifications of Division 03 - Concrete, unless expressly indicated in Article 2.3 or Article 2.4 of this Section. Items and materials required for concrete paving indicated in the Specifications of Division 03 include, but are not necessarily limited to, the following:
   1. Portland cement and supplementary cementitious materials (SCM).
   2. Aggregates.
   3. Water used in concrete mixes and installation.
   4. Admixtures.
   5. Reinforcing steel and other reinforcing materials required.
   6. Waterstops.
   7. Membrane-Forming Curing Compound.
   8. Temporary cover materials for curing concrete.
   9. Preformed Joint Filler.
   10. Hot-Poured Joint Sealing Material.
   11. Paper Subgrade Cover
   12. Concrete Treatment
   13. Forms

2.5 CONCRETE PAVING MATERIALS AND MIXES

A. Expansion Joint (Pressure-Relief Joint) and Isolation Joint Filler Material.
   1. Filler material is required in transverse joints in concrete pavement, as a means of controlling cracking caused by temperature and other movement (i.e., expansion joints). In addition, filler material is typically required between concrete pavement and adjacent items or features, where it may be called “isolation joints”. Isolation joints are typically required between concrete paving and manhole cover frames, valve boxes, stormwater inlets, curbing, gutters, and the like.
   2. Providenon-extruding cork, self-expanding cork, sponge rubber, or cork rubber. Joint filler modified to include polyurethane bonded recycled rubber, consisting of preformed strips of a durable composite of ground recycled rubber from automobile tires bonded together with polyurethane adhesive, is acceptable.
   3. Comply with AASHTO M153 (ASTM D1752) or AASHTO M213 (ASTM D1751).
   4. Size Requirements:
      a. When provided for transverse joints, as well as isolation joints, provide lengths equal to width of traffic lane or ramp with thickness of not less than one inch. Provide smooth top surface and holes punched for reinforcing dowels, providing a snug fit without loss in thickness of joint filler material.
      b. When provided for longitudinal joints, provide continuous filler joint material, in lengths of not less than eight feet, and join securely.

B. Joint Sealants for Expansion Joint (Pressure-Relief Joint) and Isolation Joint Filler Material.

1. Provide one of the following types of joint sealant material for expansion joints (pressure relief joints) and isolation joints. Joint sealant provided shall be compatible with joint filler material to which it is applied.
2. Silicone Joint Sealing Material:
    a. Sikasil 728 NS, by Sika.
    b. Or equal.
3. Rubberized Joint Sealing Material (ASTM D6690-Type II and IV)
    a. RoadSaver 231, or RoadSaver 221, or Flex-A-Fill 9005, or Deery 102, or Deery 101, by Crafco
    b. Or equal.
4. Rubberized Joint Sealing Material (ASTM D6690-Type I)
    a. RoadSaver 111 or Deery 103, by Crafco
    b. Or equal.
5. Preformed Neoprene Compression Seals: Provide both sealant material and lubricant adhesive as follows:
    a. Sealant Material:
        1) E-1006, or E-816, or V-1625 Delastic Preformed Pavement Seal, by D. S. Brown.
        2) Or equal.
    b. Lubricant Adhesive:
        1) Adhesive Lubricant 105 or Lube Plus 4070, by ChemMasters.
        2) Or equal.
6. Preformed Closed Cell Polyethylene Joint Filler
    a. Preformed Closed Cell Polyethylene Joint Filler, by W. R. Meadows.
    b. Or equal.
7. Asphalt Rubber Sealing Compound
    a. Asphalt Rubber Type 2, Part No. 34232, or PolyFlex 601, or PolyFlex Type 2, Part No. 34518, or Deery 5078, by Crafco.
    b. Or equal.
8. Accessories: In addition to joint sealant, provide the following:
    a. Joint cleaner, primer, bond breaker: As recommended by sealant manufacturer.
    b. Sealant backer rod or compressible filler: Closed cell polyethylene, polyethylene jacketed polyurethane foam, or other flexible, nonabsorbent, non-bituminous material recommended by joint sealant manufacturer to perform the following: control joint depth, break bond of sealant at bottom of joint, provide proper shape of sealant bead, and serve as expansion joint filler.
        1) Epoxy Coated Reinforcing Steel, in accordance with Section 03 21 00 - Reinforcement.

C. Concrete Surface Treatment Material:
    1. Provide boiled linseed oil mixture complying with AASHTO M233.

D. Mixes
    1. Provide concrete for concrete paving with the following characteristics:
        a. Compressive strength: not less than 4,000 psi at 28 days. Test specimens shall be prepared in accordance with ASTM C31 and testing for compressive strength shall be in accordance with ASTM C39.
        b. Flexural strength: not less than 500 psi at seven days, and 600 psi at 28 days. When high-early-strength cement is provided, flexural strength shall be not less than 550 psi at seven days, and 600 psi at 28 days. Test specimens shall be prepared in accordance with ASTM C31 and testing for flexural strength shall be in accordance with ASTM C78.
        c. Coarse, dry aggregate shall not exceed 85 percent of loose volume of concrete.
        d. Slump: Not less than one inch and not greater than four inches, unless otherwise indicated based on type of concrete placing equipment used.
        e. Air Content: five to seven percent air by volume.
        f. Water-to-Cement Ratio: 0.37 to 0.45.

g. Cement Content per Cubic Yard: 517 to 611 pounds.
h. Supplementary Cementitious Materials (SCM): If fly ash or ground granulated blast furnace slag (GGBFS) is proposed for use, weight of fly ash or GGBFS plus weight of Portland cement shall be considered in total for compliance with target cement content. Maximum acceptable proportion of SCM shall not exceed the following: (1) fly ash: 35 to 40 percent by weight; (2) GGBSF: 65 percent by weight; (3) silica fume: 2.4 multiplied by LBA, where LBA (in pounds per cubic yard of concrete) equals cement content of the concrete in pounds per cubic yard, multiplied by alkali content of cement, divided by 100; for example, for concrete containing 500 pounds of cement per cubic yard of concrete, with an alkali content of 0.81 percent (sodium oxide equivalent), then LBA equals 500 x 0.81/100 = 4.05 pounds per cubic yard. Thus, in this example, the limit of silica fume would be 2.4 multiplied by 4.05 equals 9.72 pounds per cubic yards of concrete.

2. Unless otherwise required in this Section, Concrete mixes for concrete paving shall comply with the following:
   a. Section 03 31 30 - Concrete Materials and Proportioning, and Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.

2.6 PAVEMENT MARKING MATERIALS

A. Description:
   1. Provide pavement markings as shown on the Drawings. If not shown on the Drawings and concrete paving is either an extension of existing paving with markings, or restoration of paved areas with preconstruction markings, provide pavement markings equal to preconstruction markings in areas of pavement restoration and, in new pavement (in areas not paved prior to the Effective Date of the Contract) similar to pavement markings in adjacent, paved areas existing prior to the Effective Date of the Contract.

B. Materials:
   1. Pavement Marking Paint:
      a. Provide low VOC, ready-mixed, one-component, 100 percent acrylic waterborne paint, suitable for marking pavement on highways and airfields.
      b. Suitable for application on Portland cement concrete paving, asphalt concrete paving, asphalt tar, and previously painted areas of such surfaces.
      c. Materials shall be non-bleeding and shall comply with FS TT-P-1952F, Type III.
      d. Apply paint at thicknesses required in FS TT-P-1952F.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verification of Conditions:
   1. Examine subbase and base on which concrete paving will be installed.
   2. Advise Contracting Officer in writing of conditions detrimental to proper and timely completion of the Work.
   3. Do not proceed with concrete paving Work until unsatisfactory conditions are corrected.

B. Installing the Work constitutes Contractor's approval of underlying work, surfaces upon which concrete paving will be installed, and field conditions prevailing at the time of the Work.

3.2 PREPARATION

A. General Preparation: Before starting installation of concrete paving, perform the following:
   1. Grade Control: Establish and maintain throughout concrete paving Work required lines and grades, including crown and cross-slope during construction. Comply with Section 01 73 40 – Execution.
   2. Coordinate placement of concrete paving with new and existing curbs, gutters, sidewalks, drainage structures (such as catch basins, drop inlets, and similar items), manholes and associated frames and covers, valve boxes and associated covers, sensors and similar

devices associated with traffic controls, signage, pavement markings and markers, and similar items. Where necessary, adjust rim elevations of manhole frames and covers, valve boxes and covers, and stormwater drainage inlets within areas of concrete paving Work to required final elevations.

3. Provide appropriate maintenance and protection of traffic measures during pavement installation and related Work.

B. Subgrade and Subbase:
   1. Prepare subgrade and provide subbase for concrete paving in accordance with 31 23 00 - Earthwork. Before installing concrete paving Work, obtain Contracting Officer's concurrence that subgrade and subbase are suitable for installing concrete paving.

C. Surface Preparation:
   1. Where new concrete paving will abut existing paving, sawcut edges of existing paving in accordance with the "Installation" Article of this Section.
   2.

3.3 INSTALLATION

A. Installation Requirements:
   1. Concrete Paving Installation – General:
      a. Unless indicated otherwise elsewhere in this Section, this Article applies to installation of new concrete pavement.
      b. Unless indicated otherwise in this Section, comply with concrete installation requirements in the Specifications of Division 03 - Concrete.
      c. Provide final surfaces of concrete pavement with uniform texture, at required locations and grades, with required cross-sections, as shown and indicated.
      d. Where new concrete pavement is provided and connects to existing pavement, provide new concrete surface that smoothly and evenly transitions to existing paved surfaces and grades.
   2. Sawcutting of Edges:
      a. Where new concrete pavement will meet existing pavement, existing pavement shall be sawcut.
      b. Furnish and use saw with diamond-tipped blade, blade guard, alignment guides, water cooling system, and cut-depth controls for saw cutting.
      c. Do not allow cooling water, slurry, and dust from sawing to enter any lane opened to traffic.
      d. Edges of existing pavement shall be cleanly cut. Edges shall be linear and have vertical cut face.
      e. Do not break back underside of existing pavement. If break back or spalling occurs, make new full depth transverse sawcut beyond area of break back or spalling. Provide full-depth replacement concrete pavement at Contractor's cost.
      f. Sawcutting of concrete pavement, concrete curbs, concrete gutters, and sidewalks shall comply with Section 02 41 00 - Demolition.
   3. Construction Equipment and Machinery:
      a. Subbase Planer:
         1) Furnish and use subbase planer with adjustable cutting blades to trim subbase to exact section shown on the Drawings.
         2) Mount planer on visible rollers that ride on forms.
         3) Planer frame shall have sufficient weight to remain on forms, and shall possess strength and rigidity that, under tests made by changing support from wheels to center, planer shall not deflect more than 1/8-inch.
         4) Equipment used for pulling planers shall not produce ruts or indentations in subbase.
         5) When slip-form paving is performed, operate subbase planer on prepared, graded track, or control planer using electronic sensor system operated from string line, to establish horizontal alignment and elevation of subbase.

b. Subbase Template:
    1) Provide template for checking contour of subbase.
    2) Template shall be of sufficient length to rest on side forms and shall have sufficient strength and rigidity, when supported at center, so that deflection shall not exceed 1/8-inch.
    3) Provide template with accurately adjustable rods projecting downward at one-foot intervals. Adjust rods to gauge cross-sections of bottom of concrete when template is resting on side forms.
c. Machine Finisher:
    1) Furnish and use power-driven, transverse finishing machine designed and operated to strike off and consolidate concrete.
    2) Machine shall have two screeds accurately adjusted to required crown of pavement and with frame riding on forms.
    3) Furnish and use finishing machine with rubber tires when operating on concrete pavement.
d. Hand Finishing:
    1) Provide mechanical strike and tamping template two feet longer than width of pavement to be finished. Shape template to required pavement section.
    2) Provide two bridges, riding on forms, spanning pavement for finishing expansion joints and construction joints. Provide floats and necessary edging and finishing tools.
e. Burlap Drag for Finishing Concrete:
    1) Furnish and use four plies of 10-ounce burlap material fastened to bridge, forming continuous burlap strip across full width of pavement. Three foot width of burlap material shall contact concrete surface across full width of lane or pavement.
    2) Keep burlap drags clean and free of encrusted concrete material.
f. Vibrators:
    1) Furnish and use mechanically operated, synchronized vibrators mounted on tamping bar, riding on forms, and hand-manipulated mechanical vibrators.
    2) Furnish and use vibrators with frequency of vibration to provide required consolidation of concrete without segregation of concrete's constituents.
g. Traveling Form Paver:
    1) Traveling form paver, suitable for use on projects of the state department of transportation having jurisdiction at or adjacent to the site, may be used in lieu of construction methods requiring forms, and consolidating, finishing, and floating equipment as otherwise required in this Section.
    2) When traveling form paver does not provide concrete paving in accordance with requirements of the Contract Documents, immediately discontinue using such equipment and perform the Work using conventional construction equipment, machinery, and methods.
    3) When used, traveling form paver shall have longitudinal transangular finishing float adjustable to required crown and grade. Float mechanism shall be of sufficient length to extend across pavement to side forms or edge of concrete. Traveling form paver shall continuously deposit concrete to minimize starting and stopping.
    4) Use conventional construction equipment, machinery, and methods to provide concrete paving at locations inaccessible to traveling form pavers, and for concrete paving with required horizontal or vertical curvature that precludes using traveling form pavers.
    5) Where tie bars are required for adjacent paving, securely tie and support bars to prevent displacement. Installing tie bars using mechanical bar inserter installed on traveling form paver is acceptable. Replace concrete paving provided as part of the Work where tie bars' final position is other than that required by the Contract Documents.

4. Forming:

a. Unless otherwise indicated in this Section, formwork shall comply with Section 03 11 13 - Formwork.
b. Side Forms:
    1) Furnish and use metal forms of acceptable shape and section. Aluminum forms are unacceptable. Forms eight inches or more high shall have width, at base, not less than eight inches.
    2) Depth of form shall be equal to required edge thickness of pavement.
    3) Forms with depths greater or less than required edge thickness are acceptable, provided difference between form depth and required edge thickness when such vertical difference is not greater than one-inch. Forms with depth less than required thickness of pavement edge are acceptable when (a) vertical difference between top of form and required concrete surface at pavement edge is not greater than one-inch, and (b) required thickness of edge of pavement can be provided by providing securely attaching temporary wood or metal strips to bottom of form, or by grouting under form.
    4) Maximum deflection: 1/8-inch under center load of 1,700 pounds.
    5) Thickness: Not less than 7/32-inch thick.
    6) Top: Flat or rounded, not less than 1.75-inches wide.
    7) Bottom flange of form shall be same size as required thickness of edge of pavement.
    8) Length of form sections shall be not less than 10 feet. Each section of forms shall provide for staking at required position using not less than three temporary pins.
    9) Furnish and use flexible spring steel forms or laminated boards for forming radius bends. Furnish and use curved forms of wood or metal, of proper radius, for curves of 200-foot radius and less. Provide sufficient forms for satisfactory placement of concrete. For short radius curves, forms less than 10 feet in length or curved forms may be used.
    10) Forms shall have sufficient strength and possess adequate devices for secure, temporary installation, so that, when installed forms shall withstand, without visible springing or settling, impact and vibration of finishing machine and other construction-related stresses to which they will be subjected.
    11) Forms shall be free from warp, bends, and kinks. Forms shall be sufficiently true to provide straight edge on concrete. Top of each form section, when tested with straight edge, shall comply with requirements specified for surface of completed pavement.
c. Installation of Forms:
    1) Install forms directly on subbase. Do not adjust forms to required elevation using matter such as stones, earthen material, or similar items. Accurately set forms to required elevation, grade, and alignment. During concrete installation and finishing, forms' installed location shall not deviate more than 1/8-inch in 10 feet from required location.
    2) Do not remove forms for at least eight hours after completing finishing of installed concrete. Provide sufficient quantity of forms for orderly and continuous placing of concrete. Install forms and verify proper location, elevation, and alignment for 300 feet ahead of concrete installation, unless otherwise acceptable to Contracting Officer.
    3) Adjacent, existing concrete that will abut the required concrete upon completion may be used in lieu of temporary forms, provided that existing concrete is properly protected from potential damage by finishing equipment. When new concrete will be placed adjacent to other concrete Work provided as part of the Project, adjacent concrete shall not be used in lieu of temporary forms until such adjacent concrete is not less than seven days old.

5. Reinforcing:
    a. Unless otherwise indicated in this Section, reinforcing shall comply with Section 03 21 00 - Reinforcement.

b. Provide reinforcing steel, securely installed as shown and indicated on the Drawings. Locate longitudinal edge bars between three and six inches from edge of slab.
c. At intersections and splices of reinforcing steel, wire reinforcing bars securely together. Lap nonwelded bars not less than one foot.
d. Securely tie ends of transverse reinforcing of upper layer of reinforcing to prevent curving.
e. Lap pavement slabs one full space.
f. Bars and coatings shall be free of rust, dirt and other foreign matter when concrete is placed.
g. Properly vertically support reinforcing steel using "chairs" in accordance with the Specifications of Division 03 - Concrete.
h. Provide sufficient quantity of "chairs" supporting reinforcing to maintain position of bars within allowable tolerances. In plane of steel parallel to nearest surface of concrete, reinforcing shall not vary from required locations by more than 1/12 of spacing between bars. In plane of steel perpendicular to nearest surface of concrete, reinforcing shall not vary from required locations by more than 1/4-inch.

6. Joints:
   a. Joints – General:
      1) Unless otherwise indicated in this Section, joints in concrete paving shall comply with Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.
      2) Provide joints at locations shown or indicated in the Contract Documents. When approved Shop Drawings indicate deviations from locations of joints shown or indicated in the Contract Documents, obtain from Contracting Officer's Representative appropriate Contract Modification, such as a Field Order, before proceeding with installing joints. Provide joints at locations and alignment to within 1/4-inch of that required.
      3) Position pavement joint assemblies at required locations and elevations, and rigidly secure in position. Secure in required position to prevent displacement during placing and finishing of concrete.
      4) Provide dowel bars in joint assemblies, each parallel to pavement surface and to center line of pavement, as shown, except in isolation joints.
      5) Cut header boards, joint filler, and other material used for forming joints as necessary for each dowel bar.
      6) Drill dowels into existing pavement, secure with epoxy, and provide paving headers as required to provide rigid pavement sections.
      7) At joints, finish concrete surfaces adjacent to previously installed concrete to within plus 1/8-inch, with tooled radius of 1/4-inch.
   b. Expansion Joints:
      1) Properly position and stake in place load-transferring elements for expansion joints, consisting of dowels.
      2) Provide in each expansion joint expansion joint filler material and sealant, as shown and indicated in details on the Drawings.
      3) Install expansion joint filler material in accordance with manufacturer's written instructions.
      4) Following installation of filler material, thoroughly clean joint and install sealant in accordance with sealing manufacturer's written instructions. Upper surface of joint sealant shall be within 1/8-inch of finished surface of adjacent concrete pavement.
      5) Remove excess sealant from joint and surfaces of adjacent pavement.
   c. Construction Joints:
      1) Provide construction joints at end of each day's work and where concrete placing is interrupted for 30 minutes or more.
      2) Provide construction joints at open ends of concrete pavement.
      3) Extend construction joint full width of pavement. Provide construction joint with same crown as crown required for concrete pavement.
      4) Tooled joints:

a) Form groove in freshly placed concrete with tooling device.
b) Groove dimensions shall be 3/8-inch wide at surface, and 1/4-inch wide at root.

5) Sawcut Joints:
a) Sawcut groove, 1/4-inch wide, in recently placed concrete.
b) Commence sawcutting as soon as concrete is sufficiently hard to withstand sawcutting without chipping, spalling, tearing, or disturbance of aggregate, regardless of time of day and weather conditions.
c) Immediately prior to sawcutting, thoroughly wet surface of concrete to protect cured membrane. Immediately following sawcutting, apply membrane curing compound to sawcut area.
d) Complete sawcutting before shrinkage stresses result in cracking of concrete.

d. Isolation Joints:
1) Provide isolation joints where concrete paving abuts existing construction or facilities not part of the concrete paving, such as manhole cover frames, valve boxes, catch basins and other stormwater inlets, curbs, gutters, sidewalks, and other structures.
2) Provide in each isolation joint expansion joint filler material and sealant, in accordance with the Contract Documents, including details on the Drawings.
3) Install expansion joint filler material in accordance with manufacturer's written instruction.
4) Following installation of filler material, thoroughly clean joint and install sealant in accordance with sealing manufacturer's written instructions. Upper surface of joint sealant shall be within 1/8-inch of finished surface of adjacent concrete pavement.
5) Remove excess sealant from joint and surfaces of adjacent pavement.

e. Metal Keyway Joints:
1) Form by providing metal joint strip, to remain in place.
2) Stake and support in manner identical to side forms.
3) Provide dowels or tie bars, as required.

7. Placing Concrete:
a. Unless otherwise indicated in this Section, concrete placing shall comply with Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing, including (but not limited to) requirements for hot weather and cold weather placing.
b. Advise Contracting Officer in writing of scheduled date and time for commencement of concrete placing not less than 24 hours prior to commencing concrete placing.
c. Place concrete within 90 minutes after initial water was added. Remove and dispose of concrete not placed within such period.
d. Concrete slump during placing shall be in accordance with Article 2.3 of this Section unless traveling-form paver is used, in which case slump shall be not more than two inches.
e. Place concrete continuously in successive batches. Place concrete in manner necessitating the least extent of handling possible.
f. Where spreading by hand is necessary, install concrete using shovels, buckets, or other appropriate methods. Use concrete rakes for distributing concrete.
g. Provide transverse construction joints when concrete placing is interrupted for more than 30 minutes. When distance between construction joints would be 10 feet or less, remove concrete placed in such section so that spacing between construction joints is greater than 10 feet.
h. Carefully place concrete against forms and at longitudinal and transverse joints and prevent honeycombing. Voids at edges of finished concrete pavement will constitute defective Work.
i. Washout of Concrete Delivery Vehicles and Equipment:
1) Provide appropriate temporary facilities for washout (cleaning) of concrete delivery vehicles and equipment following discharge of concrete. Comply with Laws and Regulations.

2) Provide washout area not less than 50 feet from open stormwater conveyance routes including ditches and swales, catch basins and other stormwater inlets, wetlands and other sensitive wildlife habitats, surface waters, and water supply wells.
3) Provide washout areas conveniently accessible to concrete delivery vehicles and equipment in all weather conditions.
4) Provide multiple washout areas, or washout areas larger than required in this provision, when necessary for the Project, the Work being performed, and site and weather conditions.
5) Prefabricated units shall be not less than four feet by four feet by one foot deep with not less than four-mil polyethylene plastic liner. Constructed concrete washout areas shall be not less than six feet by six feet by three feet deep, with not less than 10-mil polyethylene liner, with side slopes of one foot vertical to two feet horizontal, and surrounded by one foot high by one foot wide berm of compacted fill material.
6) Liners shall be free of tears or holes and placed over smooth surfaces to prevent puncturing. For constructed washouts, anchor liners underneath berm, or over top of berm and held in place by sandbags or concrete blocks.
7) Provide temporary signage designating washout area and, for large construction sites, provide signage throughout the Site directing traffic to washout areas.
8) Allow washed out concrete mixture to harden through evaporation of contaminated wash water. When facility is 75 percent of capacity, remove hardened concrete by reusing broken aggregate onsite, recycling, or properly disposing offsite.
9) Provide new liners before reusing stations for additional washouts.

8. Cold and Hot Weather Concreting:
   a. Cold Weather:
      1) Cease concrete placing when descending air temperature in shade falls below 40 DEGF.
      2) Do not resume until ambient temperature rises to minimum 40 DEGF.
      3) If placing below 40 DEGF is authorized by Contracting Officer, maintain temperature of mix between 60 and 80 DEGF.
      4) Heat aggregates or water or both.
      5) Water temperature may not exceed 175 DEGF.
      6) Aggregate temperature may not exceed 150 DEGF.
      7) Remove and replace frost damaged concrete.
      8) Salt or other antifreeze is not permitted.
      9) Comply with ACI 306R.
   b. Hot Weather:
      1) Case concrete placing when plastic mix temperature cannot be maintained under 90 DEGF.
      2) Aggregates or water or both may be cooled.
      3) Cool water with crushed ice.
      4) Cool aggregates by evaporation of water spray.
      5) Never batch cement hotter than 160 DEGF.
      6) Comply with ACI 305R.
9. Consolidation of Concrete:
   a. Unless otherwise indicated in this Section, consolidate concrete in accordance with Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.
   b. Consolidate concrete using appropriate mechanical vibrators. Extend vibratory unit across pavement perpendicular to intended traffic flow, and along entire longitudinal direction of concrete pavement.
   c. Consolidate concrete pavement uniformly, with appropriate spacing of vibratory units. During consolidation, vibrators shall not contact forms, reinforcing, or pavement joints.

d. Furnish and use sufficient quantity of hand-manipulated mechanical vibrators for proper consolidation of concrete along forms, at joints and in areas not consolidated using mechanically controlled vibrators.

10. Finishing:
    a. Unless otherwise indicated in this Section, concrete finishing shall comply with Section 03 35 00 - Concrete Finishing and Repair of Surface Defects.
    b. Finish concrete pavement with power-driven transverse finishing machines or by hand finishing. Regardless of finishing method used, maintain uniform ridge full width with first pass of first screed.
    c. Transverse Finishing Machine:
        1) Furnish and use transverse finishing machine to perform not less than two passes over installed concrete pavement. Perform final pass in continuous run of not less than 40 feet.
        2) Strike off concrete with strike-off screed. Move strike-off screed forward with combined transverse and longitudinal motion in direction work is progressing, maintaining screed in contact with forms, and maintaining slight excess of materials in front of cutting edge.
        3) Perform floating in accordance with requirements, below.
    d. Hand Finishing:
        1) Hand finish using mechanical strike-and-tamping template. Template shall extend across entire width of concrete pavement being finished. Shape template to pavement section required by the Contract Documents.
        2) Move strike template forward in direction of concrete placing, maintaining slight excess of material in front of template's cutting edge. Perform not less than two passes over each area.
        3) Screed pavement surface to required elevation and grade. Work screed with combined transverse and longitudinal motion in direction finishing Work is progressing. Maintain screed in contact with forms.
        4) Perform floating in accordance with requirements, below.
    e. On narrow strips and transitions, finish concrete pavement by hand. Thoroughly work concrete around reinforcing and embedded items, such as manhole cover frames, valve boxes, catch basins, and similar items.
    f. Floating - General:
        1) Tamp concrete with tamping template. Use longitudinal float to level concrete surface.
        2) Float by longitudinally reciprocating float, passing gradually from edge to edge.
        3) Ensure successive advances do not exceed half the length of the float.
        4) Test level of concrete with straightedge not less than 10 feet.
        5) Fill depressions with fresh concrete, consolidate, and refinish.
        6) Cut down high areas and retest.
        7) Tool pavement edges with suitable edger.
    g. After completing screeding and floating, perform first pass of burlap drag before water sheen has disappeared from surface. Subsequently perform as many passes as necessary to provide required texture and amplitude. Time between passes shall be as short as practicable. Burlap drag material shall be maintained in wet, clean condition during dragging.
    h. Retest concrete surface with straightedge and, if pavement is determined to deviate from Contract requirements by more than 1/8-inch in 10 feet, remedy the defective Work.

11. Curing:
    a. Unless otherwise indicated in this Section, concrete curing shall comply with Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.
    b. Provide membrane-forming curing compound in accordance with manufacturer's written instructions , at rate not less than 200 square feet per gallon.

c. Apply curing compound within four hours after finishing and as soon as moisture has dissipated from concrete surface.
d. Cure for not less than seven days.
e. When average daily ambient temperature is less than 50 degrees F, provide temporary insulation of not less than one foot thick loose dry straw, or equivalent, for 10 days. When straw is used for insulating, remove and properly dispose of straw at location other than the Site.

12. Concrete Surface Treatment:
    a. After concrete paving has cured for 30 days, provide concrete surface treatment by providing concrete surface treatment material. Such material is specified in Article 2.3 of this Section.
    b. Prior to applying surface treatment material, clean surface of concrete paving in accordance with surface treatment material manufacturer's written instructions.
    c. Apply concrete surface treatment material to all new, repaired, and rehabilitated concrete pavement in accordance with AASHTO M224.
    d. Quantity of Applications and Rates: Provide two separate applications of concrete surface treatment, at the following rates:
        1) First application at rate of 67 square yards per gallon.
        2) Second application to dry surface at rate of 40 square yards per gallon.
13. Protection of Concrete:
    a. Refer to the "Protection" Article in this Section.
14. Opening to Traffic:
    a. Refer to the "Protection" Article in this Section.

B. Installation Tolerances:
1. Milling: Tolerances for milling are indicated in the "Milling" provision of the "Preparation" Article of this Section.
2. Surface Trueness:
    a. "Surface trueness", as used in this Section refers to new pavement constructed to required surface cross-section, with even surface, without abrupt changes or transitions in elevation or grade, and without ruts or raveling. Appropriately installed pavement results in no ponding of water and properly conveys stormwater runoff to appropriate drainage routes and stormwater inlets.
    b. Contract Officer Representative will evaluate surface trueness and compliance with surface tolerance by directing Contractor to provide 16-foot straightedge or string line placed parallel to centerline of pavement and with 10-foot straight edge or string line placed transversely to centerline of pavement. Variations exceeding tolerance indicated in the "Finishing" provision of this Article shall be corrected or pavement remedied and replaced at no additional cost to the Government.
    c. Crowned Surfaces:
        1) Test crowned surfaces with a crown template, centered and at right angles to the crown.
        2) Surfaces will be acceptable when variance is equal to or less than 1/4-inch from the template.
3. Elevation: Finished surface of pavement shall be within plus or minus 1/2-inch of elevations shown or indicated. New pavement shall be equal in elevation to frames and covers of manholes, valve boxes and similar items, and allow appropriate stormwater drainage into stormwater inlets.

3.4 PAVEMENT GROOVING

A. TransverseGrooving of Existing Concrete Pavement:
1. Description:
    a. Transverse grooving is cutting of transverse grooves in existing Portland cement concrete pavement, using groove-cutting equipment appropriate for the intended purpose. Transverse grooves are perpendicular to the intended direction of traffic flow.
    b. Required limits of transverse grooving are shown or indicated on the Drawings.

c. Do not cut longitudinal grooves on bridge decks.

2. Equipment:
   a. Furnish and use grooving equipment with diamond-edged cutters, appropriate for the intended purpose and suitable for use on projects of the state department of transportation having jurisdiction at or adjacent to the Site.
   b. Do not use flailing type equipment.
3. Cutting Grooves:
   a. Cut grooves 3/32-inch to 5/32-inch wide, not less than 3/16-inch deep, spaced at 3/4-inch center-to-center.
   b. Do not cut grooves within six inches of outside edge or centerline of pavement joints.
   c. Do not overlap grooves.
   d. Continuously remove slurry or residue resulting from grooving. Slurry discharging across lanes or into gutters, catch basins or other stormwater inlets, or other stormwater drainage routes is unacceptable.
   e. Properly clean pavement surface after grooving and before opening to traffic.

3.5 APPLICATION OF PAVEMENT MARKINGS

A. Pavement Markings: Provide pavement markings where shown or indicated.
   1. Pavement Marking Installation – General:
      a. Provide pavement markings, of the types required, at locations shown and indicated in the Contract Documents.
      b. Comply with paint manufacturer's written installation instructions, and application rates as indicated in Article 2.4 of this Section.
      c. Where public traffic will be allowed to transit newly installed pavement prior to application of permanent pavement markings, provide appropriate temporary markings sufficient for safe use of the paved area during wet and dry conditions and during all hours of the day and night.
      d. When the Work includes providing temporary pavement or alterations of existing pavement markings during construction, provide appropriate temporary pavement markings and signage, in accordance with requirements of the applicable state department of transportation standard specifications.
   2. Preparation:
      a. Allow concrete paving to cure sufficiently before commencing application of pavement markings.
      b. Sweep surface with power broom supplemented by hand brooms to remove loose material and dirt. Ensure surfaces to receive pavement markings are of appropriate cleanliness to receive pavement markings, in accordance with paint manufacturer's written instructions.
      c. Do not begin marking concrete pavement until obtaining Contracting Officer's concurrence.
      d. Surfaces to receive pavement markings shall be completely dry at the time markings are applied.
      e. When reflective glass beads are required, mix with paint prior to paint application.
   3. Application:
      a. Apply pavement markings not less than five days following completion of concrete surface treatment.
      b. Do not apply pavement markings on wet surfaces, during wet or damp weather, or when ambient temperature is below 40 degrees F.
      c. Using mechanical equipment, provide uniform, straight edges in two separate coats. Apply in accordance with paint manufacturer's written instructions.
      d. Lay out markings and striping in accordance with the Contract Documents. Width of pavement striping for lane markings and parking spaces shall be four inches, unless otherwise shown or indicated on the Drawings.
   4. Protection:

a. Do not allow vehicles or construction equipment or machinery to traverse pavement markings until applied markings are completely cured in accordance with manufacturer's written instructions and is dry.
b. Properly remove defective pavement markings, including those damaged by untimely passage of vehicles, equipment, or machinery, and replace with required pavement marking in accordance with the Contract Documents.

3.6 FIELD QUALITY CONTROL

A. Field Tests and Inspections:
1. Contracting Officer's Field Quality Control Activities:
a. Contracting Officer may determine compliance with installation tolerances for locations of reinforcing steel and pavement joints, surface trueness, and depth and trueness of milling (where required), as indicated elsewhere in this Section. Contractor shall furnish implements required for evaluating and determining trueness of concrete pavement and milling.
b. Contracting Officer may identify, by visual observation, areas of apparent defective concrete paving Work and appropriately advise or notify Contractor of such defects.
2. Responsibilities of Testing Laboratory:
a. Government's field quality control testing laboratory will:
1) Test proposed materials for compliance with the Contract Documents, in accordance with this Article and the Specifications of Division 03 - Concrete.
2) When directed by Contracting Officer's Representative, perform coring of installed concrete paving to determine compliance with required thickness and tolerances. Submit written results of tests to Contracting Officer's Representative and Contractor.
b. Authority of Field Quality Control Testing Laboratory:
1) Technicians representing testing laboratory will inspect materials at the Site and perform required testing. When materials furnished or Work performed do not comply with the Contract Documents, field quality control testing laboratory technician shall promptly direct attention of Contracting Officer's Representative and Contractor to such non-compliance.
2) Testing laboratory personnel shall not act as foreman or perform other duties for, or on behalf of, Contractor. The Work will be checked as it progresses, but failure to detect defective Work shall not in any way prevent later rejection of such defective Work when defect is discovered. Failure to detect defective Work as it occurs does not obligate Contracting Officer's Representative to final acceptance. Testing laboratory personnel are not authorized to revoke, alter, relax, enlarge, or release requirements of the Contract Documents, nor to approve or accept any portion of the Work.
3. Concrete Testing – General:
a. Testing laboratory will test concrete for compressive strength, slump, water content, air content, and temperature in accordance with Section 03 05 05 - Concrete Testing and Inspection, unless otherwise indicated in this Section.
4. Compressive Strength Testing of Concrete:
a. Four specimens for compressive strength testing will be obtained in accordance with ASTM C31 for each 150 cubic yards and fraction thereof of concrete pavement placed in one day.
b. Two specimens will be tested at seven days. Remaining two specimens will be tested at 28 days.
c. Specimens will be tested in accordance with ASTM C39. Compression strength test results will be acceptable when:
1) Seven-day compressive strength: not less than 3,000 psi.
2) Twenty-eight-day compressive strength not less than compressive strength required in the "Mixes" provision of Article 2.3 of this Section.

d. When compression test indicates failure, yield test will be performed in accordance with ASTM C138 to determine cement content per cubic yard of concrete. If cement content is less than required, increase quantity of cement and SCM in subsequent batches until quantity of cement in concrete complies with requirements.

5. Flexural Strength Testing of Concrete:
   a. When directed by Contracting Officer's Representative testing laboratory technician will obtain additional samples in accordance with ASTM C31 for flexural strength testing, in accordance with ASTM C78.
6. Concrete Pavement Thickness:
   a. Thickness of installed concrete will be determined by obtaining not less than one 4-inch diameter core at representative locations, determined by Contracting Officer, for each 1,000 linear feet of traffic lane or 500 square yards of concrete pavement provided.
   b. Depth shall be measured in accordance with ASTM C174/C174M. Each core may also be tested for 28-day compressive strength in accordance with ASTM C42.
   c. Contractor may, at Contractor's cost, request three additional cores in vicinity of cores that resulted in deficient depth. For such additional locations, in-place depth shall be average depth of four cores.
   d. Completely fill locations where cores were obtained with non-shrink grout, with surface elevation equal to surface of installed concrete.
   e. Categories of Coring: In determining the average thickness of acceptable pavement for which payment will be made, the following core categories are established:

CATEGORY NUMBER	CORE THICKNESS IN RELATION TO REQUIRED THICKNESS	CORE DEPTH USED IN CALCULATING CHANGE IN COMPENSATION
1	1 inches or more deficiency	NOT USED
2	Less than 1 inch deficiency through 1/2 inches excess	Actual Core Thickness
3	More than 1/2 inches excess	Design Thickness plus 1/2 inches

   f. Should measured thickness at any location be 0.5-inch or greater less than required thickness, two additional cores shall be obtained. First additional core shall be five feet from original core, parallel to direction of intended traffic flow. Second additional core shall be obtained five feet from original core and 180 degrees opposite first additional core.
   g. If both additional cores indicate pavement thickness in accordance with the Contract Documents, within acceptable tolerances, further additional coring will not be performed for the subject zone of the Work.
   h. If either, or both, additional cores are not within 0.5-inch of required pavement thickness, further cores shall be obtained in the following order on each side of the original (deficient) core parallel to intended direction of traffic flow:
      1) At intervals of 25 feet and 50 feet from the original (deficient) core, and thereafter at subsequent intervals of 50 feet from each successive core, until thickness deficiency of less than 0.5-inch is determined, in each direction from original (deficient) core.
      2) Contractor and Contracting Officer's Representative may agree to perform additional coring to determine, more precisely, locations where pavement thickness becomes deficient by less than 0.5-inch. If Contractor and Contracting Officer's Representative cannot so agree, then outermost of the further cores indicating thickness deficient by less than 0.5-inch will be considered as location where thickness was deficient by less than 0.5-inch.
7. Drainage:
   a. Upon completion and curing of concrete pavement provided under the Contract, Contracting Officer's Representative will direct Contractor to spray clean water

(whether potable or non-potable, and without observable solid material, biological matter, and Constituents of Concern) thoroughly over surface of concrete pavement, in presence of Contracting Officer's Representative.

b. Spray sufficient quantity of water over concrete pavement to result in visually obvious runoff to drainage routes.

c. In lieu of water spray, Naturally occurring precipitation sufficient for visually observable runoff is acceptable.

d. Immediately following conclusion of spraying or precipitation event, as applicable, Contracting Officer's Representative will mark areas of ponding, if any, with appropriate lumber crayon of high visibility color contrasting with pavement. Contractor shall furnish necessary quantity of lumber crayons. Immediately following marking, obtain digital photographic documentation of each marked area, in accordance with Section 01 32 33 - Photo Documentation, and promptly submit to Contracting Officer's Representative. Such photographic documentation does not count toward quantity of photographs required by Section 01 32 33 - Photo Documentation.

e. Contracting Officer's Representative. may redesignate, re-mark, or both, areas of ponding as such areas become apparent to Contracting Officer's Representative. .

8. Surface Trueness:

a. Contractor's obligations regarding assisting Contracting Officer's Representative. in determining compliance with milling tolerances are indicated in the "Milling" provision of Article 3.2 of this Section.

b. Contractor's obligations regarding assisting Contracting Officer's Representative. in determining compliance with tolerance for trueness of installed concrete pavement are indicated in the "Installation Tolerances" provision of Article 3.3 of this Section.

B. Defective Paving Work:

1. The following constitutes defective concrete paving Work: ruts, ridges, roller marks, cracking, tearing, segregation, ponding of stormwater, or other irregularities. Concrete paving Work that does not comply with required tolerances or other requirements of the Contract Documents, or that is damaged while protection is Contractor's responsibility, will be considered as defective Work.

2. When directed by Contracting Officer, remove and replace defective concrete paving Work. Cut out full depth of such areas of defective pavement and fill with new reinforced concrete. When pavement is cut out or removed for such remedies, the area removed shall be of regular shape with right angles. To extent practicable, cut-out areas shall be square or rectangular. Surface of repaired areas shall be flush with adjacent pavement.

3. Remove and replace areas of pavement thinner than required thickness by more than 0.5-inch, or that fail either compressive strength tests or flexural strength tests, with concrete of required thickness and quality.

4. Nonconforming pavement sections shall be replaced at no additional cost to Government.

3.7 CLEANING

A. Cleaning: After completing concrete paving installation, completely clean surfaces of excess and spilled materials, excess concrete, and foreign matter. Appropriately dispose of such material.

B. Assure clean up work is completed within two weeks after pavement has been opened to traffic.

C. No new work will begin until clean up work has been completed, or is maintained within two weeks after pavement has been opened to traffic.

3.8 PROTECTION

A. Protection of Installed Concrete Paving:

1. Do not drive equipment, machinery, or vehicles on concrete unless such equipment, machinery, or vehicle (as applicable), is rubber-tired and only when concrete is designed for, and capable of, sustaining associated loading. Do not drive over new or existing concrete with tracked vehicles or equipment.

2. Protect concrete surfaces and appurtenances from traffic for not less than 14 days after concrete placing and finishing.
3. Protect finished pavement until pavement is opened for non-construction traffic.
4. Cover openings of drainage structures, manholes, valve boxes, and similar items in paved areas until permanent coverings are provided.
5. Remedy concrete surfaces damaged by traffic, or other causes, occurring prior to Substantial Completion.
6. Maintain concrete paving in good condition until Substantial Completion.
7. When pavement markings are a part of the Work, comply with requirements for protecting pavement markings in the "Application of Pavement Markings" Article of this Section.

B. Opening Pavement to Traffic:
1. On those sections of pavement to be opened to traffic, seal joints, clean pavement, and place earth against pavement edges (when pavement edges are not otherwise protected or abutted by gutters, curbing, or shoulders) before opening to traffic. Such opening of pavement to traffic shall not relieve Contractor from responsibility for the Work.
2. After 14 days following concrete placing and finishing, pavement may, at Contracting Officer's Representative's discretion (who may elect to consult with Contracting Officer), be opened to traffic when the Project's cured concrete test cylinders have attained compressive strength of not less than 3,000 psi.

END OF SECTION

This page intentionally left blank.

SECTION 32 15 40
CRUSHED STONE SURFACING

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Crushed Stone Surfacing in accordance with provisions of Contract Documents.

B. Completely coordinate with work of other trades.

C. Related Requirements:
   1. Division 00 – Procurement and Contracting Requirements.
   2. Division 01 – General Requirements.
   3. Division 31 – Earthwork.

1.2 QUALITY ASSURANCE

A. Reference Standards:
   1. ASTM C117 Standard Test Method for Materials Finer than 75-μm (No. 200) Sieve in Mineral Aggregates by Washing.
   2. ASTM C136 Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates.
   3. ASTM D4491 Standard Test Methods for Water Permeability of Geotextiles by Permittivity.
   4. ASTM D4751 Standard Test Methods for Determining Apparent Opening Size of a Geotextile.
   5. ASTM D6241 Standard Test Method for Static Puncture Strength of Geotextiles and Geotextile-Related Products Using a 50-mm Probe.

1.3 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Product Information:
   1. Acknowledgement that products submitted meet requirements of standards referenced.
   2. Crushed stone source information and sieve analysis.
   3. Source tests: submit certified test reports and service records to determine acceptability and application of stone material.

C. Samples:
   1. 1 QT sample of crushed stone.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 MATERIALS

A. Crushed Stone:
   1. Material shall be crushed stone or crushed gravel meeting ADOT Section 303 and AASHTO M 147
   2. Los Angeles wear test shall meet AASHTO T96.
   3. Liquid limit shall meet AASHTO T89.
   4. Plastic index shall meet AASHTO T90.
   5. Sieve analysis:

Crushed Stone	
Sieve Size	Percent Passing
3/4 inches	100
No. 4	50-78
No. 8	37-67
#100	13-35
#200	4-15
Plasticity Index	4-12

B. Geotextile Filter Fabric:
   1. Base: Mirafi 140 N.
   2. Nonwoven polypropylene.
   3. CBR Puncture: 310 pound.
   4. Apparent opening size: US Sieve 70.
   5. Water flow rate: 110 gal/min/sf.

C. Metal Edging
   1. See Drawings.

D. Stone Edging
   1. See Drawings for information regarding size, type, color, style of cut, and finish.

2.2 SOURCE QUALITY CONTROL

A. Furnish material from a single source.

B. Obtain samples per ASTM D75 from a local quarry that typically provides material to meet the specification requirements.

C. On-Site Aggregate Processing.

PART 3 - EXECUTION

3.1 PREPARATION

A. Verify that subgrades have been removed, recompacted, and rough graded to lines and grades to within 0.10 feet plus/minus to accept crushed stone surfacing. Compact sub-base to 95% Standard Proctor Density.

B. Install flush edging or curbs to the lines and grades and in the locations shown on the Drawings.

C. Thoroughly clean sub-base of debris, loose dirt and other extraneous materials before installing filter fabric or crushed stone. Do not install crushed stone when sub-base is wet or muddy.

D. Trim and dress all areas to required cross sections.

E. Bring areas that are below allowable minus tolerance limit to grade by filling with material similar to adjacent material.

F. Compact to density specified for backfill in accordance with Section 31 23 00 – Earthwork.

G. Do not place any stone on soft, muddy, or frozen material.

H. Prevent contamination of existing surfacing stone during excavation activities.

I. Do not place any stone material on prepared base prior to approval by Contracting Officer.

3.2 INSTALLATION

A. Place filter fabric under crushed stone areas while overlapping fabric a minimum of 6 inches and extending edges up edging or curbs a minimum of 2 inches.

B. Install crushed stone in two lifts:
   1. Compacting:
      a. Compact with a water ballast hand roller.
      b. Thoroughly compact to a minimum 90% Standard Proctor Density.
      c. Spray water onto the crusher fines while compacting.
      d. Do not overwork crusher fines. Any areas determined to be overworked shall be replaced at no cost to Government.
      e. After compacting, screed smooth.
   2. Grading:
      a. When surface areas have been rolled, add a thin layer of material to bring the surface to finished grade or to meet adjacent walking surface.
      b. Rake the previously rolled or compacted area to provide a bond with the added material.
      c. Provide a uniform texture and smooth, even surface to the finished paved area.
      d. Place to required thickness and grades:
         1) Depth: -0 IN, +2 IN.

3.3 CLEAN-UP

A. Keep areas of work clean, neat and orderly.

B. Clean up and remove debris from the work area prior to Final Acceptance.

END OF SECTION

This page intentionally left blank.

SECTION 32 16 23
CONCRETE SIDEWALK AND STEPS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Concrete sidewalk and steps.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 31 30 - Concrete, Materials and Proportioning.
   2. Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.
   3. Section 03 05 05 – Concrete Testing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Association of State Highway and Transportation Officials (AASHTO):
      a. M153, Preformed Sponge Rubber and Cork Expansion Joint Fillers for Concrete Paving and Structural Construction.
      b. M171, Sheet Materials for Curing Concrete.
      c. M182, Burlap Cloth Made from Jute or Kenaf.
      d. M213, Preformed Expansion Joint Fillers for Concrete Paving and Structure Construction (Nonextruding and Resilient Bituminous Types).
      e. M224, Use of Protective Sealers for Portland Cement Concrete.
      f. M233, Boiled Linseed Oil Mixture for Treatment of Portland Cement Concrete.
   2. American Concrete Institute (ACI):
      a. 305R, Hot Weather Concreting.
      b. 306R, Cold Weather Concreting.
   3. ASTM International (ASTM):
      a. A185, Standard Specification for Steel Welded Wire Reinforcement, Plain, for Concrete.
      b. A615, Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement.
      c. A1064, Standard Specification for Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete.
      d. C33, Standard Specification for Concrete Aggregates.
      e. C150, Standard Specification for Portland Cement.
      f. C309, Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete.
      g. D698, Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 feet-LBF/FT^3).
      h. D4253, Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table.
      i. D4254, Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density.
   4. Federal Specification (FS):
      a. SS-S-1614, Sealants, Joint, Jet-Fuel-Resistant, Hot-Applied for Portland Cement and Tar Concrete Pavements.
      b. TT-S 00227 E(3), Sealing Compound: Elastomeric Type, Multi-Component (for Calking, Sealing, and Glazing in Buildings and Other Structures).

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:

a. Acknowledgement that products submitted meet requirements of standards referenced.
2. Mix design(s) in accordance with Specification Section 03 31 30 and Specification Section 03 05 05.
3. Qualifications of concrete installer.
4. Drawings detailing all reinforcing.
5. Concrete cylinder test results from field quality control.

B. Samples:
1. Samples of fabricated jointing materials and devices.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 MATERIALS

A. Portland Cement:
1. ASTM C150, Type I or II.

B. Aggregates:
1. ASTM C33, gradation size #67, 3/4 inches to #4.

C. Water:
1. Potable quality.

D. Admixtures:
1. Comply with Specification Section 03 31 30.

E. Reinforcing Bars:
1. ASTM A615, Grade 60.

F. Welded Wire Reinforcement:
1. ASTM A185 or ASTM A1064.
2. Flat.
3. Clean, free from dirt, scale, rust.

G. Preformed Joint Filler:
1. Nonextruding cork, self-expanding cork, sponge rubber or cork rubber.
2. Meet AASHTO M153 or AASHTO M213.

H. Hot-Poured Joint Sealing Material:
1. FS SS-S-1614.

I. Sidewalk Joint Sealant:
1. Two compound, polyurethane sealant.
2. Class A, Type 1.
3. Self-leveling.
4. Nontracking.
5. FS TT-S 00227 E(3).

J. Membrane Curing Compound:
1. ASTM C309.

K. Cover Materials for Curing:
1. Burlap:
a. AASHTO M182.
b. Minimum Class 2, 8 oz material (1 yard x 42 inches).
2. Polyethylene film:
a. AASHTO M171.

L. Paper Subgrade Cover:
1. Polyethylene film, AASHTO M171.

M. Concrete Treatment:
   1. Boiled linseed oil mixture.
   2. Meets AASHTO M233.

N. Forms:
   1. Steel or wood.
   2. Size and strength to resist movement during concrete placement and to retain horizontal and vertical alignment.
   3. Free of distortion and defects.
   4. Full depth.
   5. Metal Side Forms:
      a. Minimum 7/32 inches thick.
      b. Depth equal to edge thickness of concrete.
      c. Flat or rounded top minimum 1-3/4 inches wide.
      d. Base 8 inches wide or equal to height, whichever is less.
      e. Maximum deflection 1/8 inches under center load of 1700 pounds.
      f. Use flexible spring steel forms or laminated boards to form radius bends.

2.2 MIXES

A. Mix design to provide 4,000 psi 28-day compressive strength, 1-1/2 inches +1 inches slump, 6% air.

B. Comply with Specification Section 03 31 30 and Specification Section 03 31 31.

PART 3 - EXECUTION

3.1 PREPARATION

A. Subgrade Preparation:
   1. Prepare using methods, procedures, and equipment necessary to attain required compaction densities, elevation and section.
   2. Scarify and recompact top 6 inches of fills and embankments which will be sidewalk and step areas.
   3. Remove soft or spongy areas.
      a. Replace with aggregate material.
   4. Compact to the following densities:
      a. Cohesive soils: 95% per ASTM D698.
      b. Noncohesive soils: 75% relative per ASTM D4253 and ASTM D4254.
   5. Assure moisture content is within limits prescribed to achieve required compaction density.
   6. Following compaction, trim and roll to exact cross section.
      a. Check with approved grading template.
   7. Perform density tests on subgrade to determine that subgrade complies with the specification.

B. Aggregate Course:
   1. Place material in not more than 6 inches thick layers.
   2. Spread, shape, and compact all material deposited on the subgrade during the same day.
   3. Compact to 75% relative per ASTM D4253 and ASTM D4254.

C. Loose and Foreign Material:
   1. Remove loose and foreign material immediately before application of paving.

D. Appurtenance Preparation:
   1. Block out or box out curb inlets and curb returns.
   2. Provide for joint construction as detailed and dimensioned on Drawings.
   3. Adjust manholes, inlets, valve boxes and any other utility appurtenances to design grade.
      a. Secure to elevation with concrete.
      b. Place concrete up to 5 inches below design grade.

4. Clean and oil forms.

3.2 ERECTION, INSTALLATION AND APPLICATION

A. Concrete Production:
   1. Comply with Specification Section 03 31 31.

B. Forms:
   1. Form support:
      a. Compact soil foundation and cut to grade to support forms.
      b. Use bearing stakes driven flush with bottom of form to supplement support as necessary.
      c. Do not use earth pedestals.
   2. Staking forms:
      a. Joint forms neatly and tightly.
      b. Stake and pin securely with at least three pins for each 10 feet section.
   3. Clean and oil forms prior to placement of concrete.
   4. Set forms sufficiently in advance of work (minimum 2 hours) to permit proper inspection.
   5. Previously finished pavement or curb and gutter contiguous with new work may serve as side form when specifically approved.

C. Reinforcing:
   1. Lap mats one full space.
   2. Tie end transverse member of upper mat securely to prevent curling.
   3. Lap nonwelded bars 12 inches minimum.
   4. Support:
      a. Place bars securely on chairs at called-for height.
      b. Place other fabric on the first of a two-course pour and cover promptly with final pour, or place fabric by a fabric-placer if procedure is reviewed and approved by Contracting Officer’s Representative.

D. Joints:
   1. Hold locations and alignment to within + 1/4 inches.
   2. Finish concrete surface adjacent to previous section to within + 1/8 inches, with tooled radius of 1/4 inches.
   3. Metal keyway joints:
      a. Form by installing metal joint strip, left in place.
      b. Stake and support like side form.
      c. Provide dowels or tie bars.
   4. Weakened plane joints:
      a. Locate at 6 feet intervals.
      b. Tool groove in freshly placed concrete with tooling device.
      c. Groove dimensions shall be 3/8 inches at surface and 1/4 inches at root.
   5. Install construction joints at end of day's work or wherever concreting must be interrupted for 30 minutes or more.
   6. Expansion joints:
      a. Locate at 48 feet intervals and at all intersection curb returns.
      b. Stake in place load transfer device consisting of dowels.
      c. Supporting and spacing means and premolded joint filler as per Drawing details.
      d. Provide preformed joint filler at all junctions with existing sidewalks, steps, or other structures.
   7. Thoroughly clean and fill joints with joint sealing material as specified.
   8. Upper surface of filled joint to be flush to 1/8 inches below finish surface.

E. Place Concrete:
   1. Comply with Specification Section 03 31 31.
   2. Construct driveway openings and other features as per Drawing details.

F. Cold and Hot Weather Concreting:

1. Cold weather:
    a. Cease concrete placing when descending air temperature in shade falls below 40 degrees F.
    b. Do not resume until ambient temperature has risen to 40 degrees F.
    c. If placing is authorized below 40 degrees F by Contracting Officer's Representative, maintain temperature of mix between 60 and 80 degrees F.
    d. Heat aggregates or water or both.
    e. Water temperature may not exceed 175 degrees F.
    f. Aggregate temperature may not exceed 150 degrees F.
    g. Remove and replace frost damaged concrete.
    h. Salt or other antifreeze is not permitted.
    i. Comply with ACI 306R.
2. Hot weather:
    a. Cease concrete placing when plastic mix temperature cannot be maintained under 90 degrees F.
    b. Aggregates or water or both may be cooled.
    c. Cool water with crushed ice.
    d. Cool aggregates by evaporation or water spray.
    e. Never batch cement hotter than 160 degrees F.
    f. Comply with ACI 305R.

G. Finishing:
1. As soon as placed, strike off and screed to crown and cross section, slightly above grade, so that consolidation and finishing will bring to final Drawing elevations.
2. Maintain uniform ridge full width with first pass of first screed.
3. Test with 6 feet straightedges equipped with long handles and operated from sidewalk.
4. Draw excess water and laitance off from surface.
5. Float finish so as to leave no disfiguring marks but to produce a uniform granular or sandy texture.
6. Broom finish after floating.
7. Tool edges with suitable edger.
8. Provide exposed aggregate surfaces in areas indicated on the Drawings.
9. Provide method such as abrasive blasting, bush hammering, or surface retarder acceptable to the Contracting Officer.

H. Curing:
1. Apply membrane curing compound complying with ASTM C309, and in accordance with manufacturer's directions but at a rate of minimum 200 square feet per gallon.
2. Apply curing compound within 4 hours after finishing or as soon as surface moisture has dissipated.
3. Cure for minimum of seven days.
4. When average daily temperature is below 50 degrees F, provide insulative protection of 12 inches minimum thickness loose dry straw, or equivalent, for 10 days.
5. Linseed oil sealant:
    a. For concrete sidewalk and step, seal surface with linseed oil.
    b. Apply linseed oil to clean surface as per AASHTO M224 after concrete has cured for 1 month.
    c. Apply first application at rate of 67 square yards per gallon.
    d. Apply second application to a dry surface at rate of 40 square yards per gallon.

I. Protection of Concrete:
1. Protect new sidewalk, steps, and their appurtenances from traffic for a minimum of 14 days.
2. Repair or replace parts of sidewalk and steps damaged by traffic, or other causes, prior to final acceptance.

J. Opening to Traffic:

1. After 14 days, area may, at Government's discretion, be opened to traffic if job cured cylinders have attained a compressive strength of 3000 pounds per square inch when tested in accordance with ASTM standard methods.
2. Prior to opening to traffic, clean and refill joints as required with specified filler material.

K. Clean Up:
1. Assure clean-up work is completed within two weeks after sidewalk has been opened to traffic.
2. No new work will begin until clean-up work has been completed, or is maintained within 2 weeks after sidewalk has been opened to traffic.

L. Handrails:
1. Provide handrails where required and as per Drawing details.

3.3 FIELD QUALITY CONTROL

A. Provide test cylinders in accordance with Specification Section 03 05 05 for each 50 CUYD of placed concrete.

END OF SECTION

SECTION 32 32 16
PRECAST MODULAR BLOCK RETAINING WALL

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes furnishing all materials and labor required for the design and construction of a precast concrete modular block (PMB) retaining wall with or without geosynthetic reinforcement. Precast modular block retaining wall blocks under this section shall be cast utilizing a wet-cast concrete mix and exhibit a final handling weight in excess of 1,000 pounds (450 kg) per unit and may utilize concrete-reinforcing steel.

B. Scope of Work: The work shall consist of furnishing materials, labor, equipment and supervision for the construction of a precast modular block (PMB) retaining wall structure in accordance with the requirements of this section and in acceptable conformity with the lines, grades, design and dimensions shown in the project site plans.

C. Drawings and General Provisions of the Contract, including General and Supplementary Conditions and Division 31, Division 32 and Division 33 also apply to this Section.

1.2 PRICE AND PAYMENT PROCEDURES

A. Allowances. No allowance shall be made in the price of the retaining wall for excavation beyond the limits required for retaining wall construction as shown on the project plans. The cost of excavation for the purposes of site access shall be the responsibility of the General Contractor. Removal of unsuitable soils and replacement with select fill shall be as directed and approved in writing by the Contracting Officer and shall be paid under separate pay items.

B. Unit Prices. In addition to a lump sum price pursuant to completion of the scope of work described in Part 1.1 of this Section, the General Contractor shall provide a unit price per square foot of vertical wall face that shall be the basis of compensation for up to a ten (10) percent increase or reduction in the overall scope of the retaining wall work.

C. Measurement and Payment.
   1. The unit of measurement for furnishing the precast modular block retaining wall system shall be the vertical area of the wall face surface as measured from the top of the leveling pad to the top of the wall including coping. The final measured quantity shall include supply of all material components and the installation of the precast modular block system.
   2. The final accepted quantities of the precast modular block retaining wall system will be compensated per the vertical face area as described above. The quantities of the precast modular block retaining wall as shown on the plans and as approved by the Contracting Officer shall be the basis for determination of the final payment quantity. Payment shall be made per square foot of vertical wall face.

1.3 REFERENCES

A. Where the specification and reference documents conflict, the Contracting Officer's Representative will make the final determination of the applicable document.

B. Definitions:
   1. Precast Modular Block (PMB) Unit – machine-placed, "wet cast" concrete modular block retaining wall facing unit.
   2. Geotextile – a geosynthetic fabric manufactured for use as a separation and filtration medium between dissimilar soil materials.
   3. Geogrid – a geosynthetic material comprised of a regular network of tensile elements manufactured in a mesh-like configuration of consistent aperture openings. When connected to the PMB facing units and placed in horizontal layers in compacted fill, the geogrid prevents lateral deformation of the retaining wall face and provides effective tensile reinforcement to the contiguous reinforced fill material.

4. Drainage Aggregate – clean, crushed stone placed within and immediately behind the precast modular block units to facilitate drainage and reduce compaction requirements immediately adjacent to and behind the precast modular block units.
5. Unit Core Fill – clean, crushed stone placed within the hollow vertical core of a precast modular block unit. Typically, the same material used for drainage aggregate as defined above.
6. Foundation Zone – soil zone immediately beneath the leveling pad and the reinforced zone.
7. Retained Zone – soil zone immediately behind the drainage aggregate and wall infill for wall sections designed as modular gravity structures. Alternatively, in the case of wall sections designed with geosynthetic soil reinforcement, the retained zone is the soil zone immediately behind the reinforced zone.
8. Reinforced Zone – structural fill zone within which successive horizontal layers of geogrid soil reinforcement have been placed to provide stability for the retaining wall face. The reinforced zone exists only for retaining wall sections that utilize geosynthetic soil reinforcement for stability.
9. Reinforced Fill – structural fill placed within the reinforced zone.
10. Leveling Pad – hard, flat surface upon which the bottom course of precast modular blocks is placed. The leveling pad may be constructed with crushed stone or cast-in-place concrete. A leveling pad is not a structural footing.
11. Wall Infill – the fill material placed and compacted between the drainage aggregate and the excavated soil face in retaining wall sections designed as modular gravity structures.

C. Reference Standards
   1. Design
      a. AASHTO LRFD Bridge Design Specifications, 7th Edition, 2014.
      b. Minimum Design Loads for Buildings and Other Structures – ASCE/SEI 7-10.
      c. International Building Code, 2018 Edition.
      d. FHWA-NHI-10-024 Volume I and GEC 11 Design of Mechanically Stabilized Earth Walls and Reinforced Soil Slopes.
      e. FHWA-NHI-10-025 Volume II and GEC 11 Design of Mechanically Stabilized Earth Walls and Reinforced Soil Slopes.
      f. National Concrete Masonry Association (NCMA) Design Manual for Segmental Retaining Walls (ASD), 3rd Edition
   2. Precast Modular Block Units
      a. ACI 201 – Guide to Durable Concrete
      b. ACI 318 – Building Code Requirements for Structural Concrete
      c. ASTM A615 – Steel Bars for Concrete Reinforcement
      d. ASTM A767 – Galvanized Steel Bars for Concrete Reinforcement
      e. ASTM A775 – Epoxy-Coated Steel Reinforcing Bars
      f. ASTM C33 – Standard Specification for Concrete Aggregates
      g. ASTM C39 – Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
      h. ASTM C94 – Standard Specification for Ready-Mixed Concrete.
      i. ASTM C136 – Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates.
      j. ASTM C143 – Standard Test Method for Slump of Hydraulic-Cement Concrete.
      k. ASTM C150 – Standard Specification for Portland Cement
      l. ASTM C231 - Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method.
      m. ASTM C260 – Standard Specification for Air-Entraining Admixtures for Concrete.
      n. ASTM C494 – Standard Specification for Chemical Admixtures for Concrete.
      o. ASTM C595 - Standard Specification for Blended Hydraulic Cements.
      p. ASTM C618 - Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete.
      q. ASTM C666 – Standard Test Method for Concrete Resistance to Rapid Freezing and Thawing.

r. ASTM C845 - Standard Specification for Expansive Hydraulic Cement.
s. ASTM C920 – Standard Specification for Elastomeric Joint Sealants.
t. ASTM C989 - Standard Specification for Slag Cement for Use in Concrete and Mortars.
u. ASTM C1116 – Standard Specification for Fiber-Reinforced Concrete.
v. ASTM C1157 - Standard Performance Specification for Hydraulic Cement.
w. ASTM C1218 - Standard Test Method for Water-Soluble Chloride in Mortar and Concrete.
x. ASTM C1240 - Standard Specification for Silica Fume Used in Cementitious Mixtures.
y. ASTM C1611 – Standard Test Method for Slump Flow of Self-Consolidating Concrete.
z. ASTM C1776 – Standard Specification for Wet-Cast Precast Modular Retaining Wall Units.
aa. ASTM D6638 – Standard Test Method for Determining Connection Strength Between Geosynthetic Reinforcement and Segmental Concrete Units (Modular Concrete Blocks).
bb. ASTM D6916 – Standard Test Method for Determining Shear Strength Between Segmental Concrete Units (Modular Concrete Blocks).

3. Geosynthetics
   a. AASHTO M 288 – Geotextile Specification for Highway Applications.
   b. ASTM D3786 – Standard Test Method for Bursting Strength of Textile Fabrics Diaphragm Bursting Strength Tester Method.
   c. ASTM D4354 – Standard Practice for Sampling of Geosynthetics for Testing.
   d. ASTM D4355 – Standard Test Method for Deterioration of Geotextiles
   e. ASTM D4491 – Standard Test Methods for Water Permeability of Geotextiles by Permittivity.
   f. ASTM D4533 – Standard Test Method for Trapezoid Tearing Strength of Geotextiles.
   g. ASTM D4595 – Standard Test Method for Tensile Properties of Geotextiles by the Wide-Width Strip Method.
   h. ASTM D4632 – Standard Test Method for Grab Breaking Load and Elongation of Geotextiles.
   i. ASTM D4751 – Standard Test Method for Determining Apparent Opening Size of a Geotextile.
   j. ASTM D4759 – Standard Practice for Determining Specification Conformance of Geosynthetics.
   k. ASTM D4833 – Standard Test Method for Index Puncture Resistance of Geomembranes and Related Products.
   l. ASTM D4873 – Standard Guide for Identification, Storage, and Handling of Geosynthetic Rolls and Samples.
   m. ASTM D5262 – Standard Test Method for Evaluating the Unconfined Tension Creep and Creep Rupture Behavior of Geosynthetics.
   n. ASTM D5321 – Standard Test Method for Determining the Coefficient of Soil and Geosynthetic or Geosynthetic and Geosynthetic Friction by the Direct Shear Method.
   o. ASTM D5818 – Standard Practice for Exposure and Retrieval of Samples to Evaluate Installation Damage of Geosynthetics.
   p. ASTM D6241 – Standard Test Method for the Static Puncture Strength of Geotextiles and Geotextile-Related Products Using a 50-mm Probe.
   q. ASTM D6637 – Standard Test Method for Determining Tensile Properties of Geogrids by the Single or Multi-Rib Tensile Method.
   r. ASTM D6706 – Standard Test Method for Measuring Geosynthetic Pullout Resistance in Soil.
   s. ASTM D6992 – Standard Test Method for Accelerated Tensile Creep and Creep-Rupture of Geosynthetic Materials Based on Time-Temperature Superposition Using the Stepped Isothermal Method.
4. Soils
   a. AASHTO M 145 – AASHTO Soil Classification System.

b. AASHTO T 104 – Standard Method of Test for Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate.
c. AASHTO T 267 – Standard Method of Test for Determination of Organic Content in Soils by Loss of Ignition.
d. ASTM C33 – Standard Specification for Concrete Aggregates.
e. ASTM D448 – Standard Classification for Sizes of Aggregates for Road and Bridge Construction.
f. ASTM D698 – Standard Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort. (12,400 ft-lbf/ft (2,700 kN-m/m)).
g. ASTM D1241 – Standard Specification for Materials for Soil-Aggregate Subbase, Base and Surface Courses.
h. ASTM D1556 – Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method.
i. ASTM D1557 – Standard Test Method for Laboratory Compaction Characteristics of Soil Using Modified Effort. (56,000 ft-lbf/ft (2,700 kN-m/m)).
j. ASTM D2487 – Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System).
k. ASTM D2488 – Standard Practice for Description and Identification of Soils (Visual-Manual Procedure).
l. ASTM D3080 – Standard Test Method for Direct Shear Test of Soils Under Consolidated Drained Conditions.
m. ASTM D4254 – Standard Test Method for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density.
n. ASTM D4318 – Standard Test Method for Liquid Limit, Plastic Limit, and Plasticity Index of Soils.
o. ASTM D4767- Test Method for Consolidated-Undrained Triaxial Compression Test for Cohesive Soils.
p. ASTM D4972 – Standard Test Method for pH of Soils.
q. ASTM D6913 – Standard Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis.
r. ASTM D6938 – Standard Test Method for In-Place Density and Water Content of Soil and Aggregate by Nuclear Methods (Shallow Depth).
s. ASTM G51 – Standard Test Method for Measuring pH of Soil for Use in Corrosion Testing.
t. ASTM G57 – Standard Test Method for Field Measurement of Soil Resistivity Using the Wenner Four-Electrode Method.

5. Drainage Pipe
   a. ASTM D3034 – Standard Specification for Type PSM Poly (Vinyl Chloride) (PVC) Sewer Pipe and Fittings.
   b. ASTM F2648 – Standard Specification for 2 to 60 inch (50 to 1500 mm) Annular Corrugated Profile Wall Polyethylene (PE) Pipe and Fittings for Land Drainage Applications.

1.4 ADMINISTRATIVE REQUIREMENTS

A. Preconstruction Meeting. As directed by the Government, the General Contractor shall schedule a preconstruction meeting at the project site prior to commencement of retaining wall construction. Participation in the preconstruction meeting shall be required of the Contracting Officer, General Contractor, Retaining Wall Design Engineer (RWDE), Retaining Wall Installation Contractor (RWIC), Grading Contractor, and Special Inspector. The General Contractor shall provide notification to all parties at least 10 calendar days prior to the meeting.
   1. Preconstruction Meeting Agenda:
      a. The RWDE shall explain all aspects of the retaining wall construction drawings.
      b. The RWDE shall explain the required bearing capacity of soil below the retaining wall structure and the shear strength of in-situ soils assumed in the retaining wall design to the Special Inspector.

c. The RWDE shall explain the required shear strength of fill soil in the reinforced, retained and foundation zones of the retaining wall to the Special Inspector.
d. The RWDE shall explain any measures required for coordination of the installation of utilities or other obstructions in the reinforced or retained fill zones of the retaining wall.
e. The RWIC shall explain all excavation needs, site access and material staging area requirements to the General Contractor and Grading Contractor.

1.5 SUBMITTALS

A. Product Data. At least 14 days prior to construction, the General Contractor shall submit the retaining wall product submittal package to the Contracting Officer's Representative for review and approval. The submittal package shall include technical specifications and product data from the manufacturer for the following:
   1. Precast Modular Block System brochure
   2. Precast Modular Block concrete test results specified in paragraph 2.01, subparagraph B of this section as follows:
      a. 28-day compressive strength
      b. Air content
      c. Slump or Slump Flow (as applicable)
   3. Drainage Pipe
   4. Geotextile
   5. Geosynthetic Soil Reinforcement (if required by the retaining wall design). The contractor shall provide certified manufacturer test reports for the geosynthetic soil reinforcement material in the manufactured roll width specified. The test report shall list the individual roll numbers for which the certified material properties are valid.

B. Installer Qualification Data. At least 14 days prior to construction, the General Contractor shall submit the qualifications of the business entity responsible for installation of the retaining wall, the RWIC, per paragraph 1.07, subparagraph A of this section.

C. Retaining Wall Design Calculations and Construction Shop Drawings. At least 14 days prior to construction, the General Contractor shall furnish electronic versions construction shop drawings and the supporting structural calculations report to the Contracting Officer for review and approval. This submittal shall include the following:
   1. Signed, sealed and dated drawings and engineering calculations prepared in accordance with specification 01 35 73 – Delegated Design Procedures.
   2. Qualifications Statement by the RWDE summarizing their Experience.
   3. Certificate of Insurance of the RWDE as specified in section1.6, subparagraph B of this section.

1.6 CONSTRUCTION SHOP DRAWING PREPARATION

A. The RWDE shall coordinate the retaining wall construction shop drawing preparation with the project Civil Engineer, project Geotechnical Engineer and Contracting Officer's Representative. The General Contractor shall furnish the RWDE the following project information required to prepare the construction shop drawings. This information shall include, but is not limited to, the following:
   1. Current versions of the site, grading, drainage, utility, erosion control, landscape, and irrigation plans; electronic CAD file of the civil site plans listed in (1); report of geotechnical investigation and all addenda and any supplemental reports;
   2. Recommendations of the project Geotechnical Engineer regarding effective stress shear strength and total stress shear strength (when applicable) parameters for in-situ soils in the vicinity of the proposed retaining wall(s) and for any fill soil that may potentially be used as backfill in retained and/or foundation zones of the retaining wall.

B. The RWDE shall provide the Contracting Officer with a certificate of professional liability insurance verifying the minimum coverage limits of $1 million per claim and $1 million aggregate.

C. Design of the precast modular block retaining wall shall satisfy the requirements of this section. Where the 2024 International Building Code exceed these specifications, these requirements shall also be satisfied.

D. The RWDE shall note any exceptions to the requirements of this section by listing them at the bottom right corner of the first page of the construction shop drawings.

E. Approval or rejection of the exceptions taken by the Retaining Wall Design Engineer (RWDE) will be made in writing as directed by the Contracting Officer.

F. The RWDE shall determine the appropriate standard(s) to be utilized, and to which the precast modular block design shall be based upon, except as noted herein. Refer to Part 1.3, Paragraph C, Part 1.

G. In the event that a conflict is discovered between these specifications and a reasonable interpretation of the design specifications and methods referenced in paragraph F above, these specifications shall prevail. If a reasonable interpretation is not possible, the conflict shall be resolved per the requirements in paragraph 1.03, subparagraph A of this section.

H. Soil Shear Parameters. The RWDE shall prepare the construction shop drawings based upon soil shear strength parameters from the available project data and the recommendations of the project Geotechnical Engineer. If insufficient data exists to develop the retaining wall design, the RWDE shall communicate the specific deficiency of the project information or data to the Contracting Officer in writing.

I. Allowable bearing pressure requirements for each retaining wall shall be clearly shown on the construction drawings.

J. Global Stability. Overall (global) stability shall be evaluated in accordance with the principals of limit equilibrium analysis as set forth in FHWA-NHI-10-024 Volume I and FHWA-NHI-10-025 Volume II GEC 11 Design of Mechanically Stabilized Earth Walls and Reinforced Soil Slopes, or other methods, as determined by the RWDE, as referenced in Section 1.6, Part F. The minimum factors of safety shall be as follows:

Normal Service (Static)	1.3
Seismic	1.1
Rapid Drawdown (if applicable)	1.2

Note: RWDE to select appropriate FOS

K. Seismic Stability. Seismic loading shall be evaluated in accordance with AASHTO Load and Resistance Factor Design (LRFD) methodology, or NCMA (ASD) methodology as determined by the RWDE as referenced in Section 1.6, Part F.

1.7 QUALITY ASSURANCE

A. RWIC Qualifications. In order to demonstrate basic competence in the construction of precast modular block walls, the RWIC shall possess the following:
   1. Experience.
      a. Construction experience with a minimum of 3,000 square feet (280 square meters) of the proposed precast modular block retaining wall system.
      b. Construction of at least three (3) precast modular block (large block) retaining wall structures within the past three (3) years.
      c. Construction of at least 5,000 square feet (465 square meters) of precast modular block (large block) retaining walls within the past five (5) years.
   2. RWIC experience documentation for each qualifying project shall include:
      a. Project name and location
      b. Date (month and year) of construction completion
      c. Contact information of General Contractor
      d. Type (trade name) of precast modular block system used

e. Maximum height of the wall constructed
f. Face area of the wall constructed

3. In lieu of these specific requirements, the contractor may submit alternate documentation demonstrating competency in Precast Modular Block retaining wall construction.

B. RWDE (RWDE) Qualifications and Statement of Experience. The RWDE shall submit a written statement affirming that he or she has the following minimum qualifications and experience.
1. The RWDE shall be licensed to practice as a Professional Civil or Structural Engineer in the State of Arizona.
2. The RWDE shall be independently capable of performing all internal and external stability analyses, including those for seismic loading, compound stability, rapid draw-down and deep-seated, global modes of failure.
3. The RWDE shall affirm in writing that he or she has personally supervised the design of the retaining walls for the project, that the design considers all the requirements listed in paragraph 1.06 and that he or she accepts responsibility as the design engineer of record for the retaining walls constructed on the project.
4. The RWDE shall affirm in writing that he or she has designed a minimum of approximately 3,000 face square feet (280 face square meters) of modular block earth retaining walls within the previous five (5) years.
5. In lieu of these specific requirements, the engineer may submit alternate documentation demonstrating competency in Precast Modular Block retaining wall design.

C. The Contracting Officer reserves the right to reject the services of any engineer, engineering firm, or contractor who, in the sole opinion of the Owner, does not possess the requisite experience or qualifications.

1.8 QUALITY CONTROL

A. The Contracting Officer's Representative shall review all submittals for materials, design, RWDE qualifications and the RWIC qualifications.

B. The General Contractor shall retain the services of a Special Inspector who is experienced with the construction of precast modular block retaining wall structures to perform inspection and testing. The cost of inspection shall be the responsibility of the General Contractor. Inspection shall be continuous throughout the construction of the retaining walls.

C. The Special Inspector shall perform the following duties:
1. Inspect the construction of the precast modular block structure for conformance with construction shop drawings and the requirements of this specification.
2. Verify that soil or aggregate fill placed and compacted in the reinforced, retained and foundation zones of the retaining wall conforms with paragraphs 2.04 and 2.05 of this section and exhibits the shear strength parameters specified by the RWDE.
3. Verify that the shear strength of the in-situ soil assumed by the RWDE is appropriate.
4. Inspect and document soil compaction in accordance with these specifications:
a. Required dry unit weight
b. Actual dry unit weight
c. Allowable moisture content
d. Actual moisture content
e. Pass/fail assessment
f. Test location – wall station number
g. Test elevation
h. Distance of test location behind the wall face
5. Verify that all excavated slopes in the vicinity of the retaining wall are bench-cut as required.
6. Notify the RWIC of any deficiencies in the retaining wall construction and provide the RWIC a reasonable opportunity to correct the deficiency.

7. Notify the General Contractor, Contracting Officer, and RWDE of any construction deficiencies that have not been corrected in a timely manner.
8. Document all inspection results and provide reports to Contracting Officer, RWDE, and RWIC.
9. Test compacted density and moisture content of the retained backfill with the following frequency:
    a. At least once every 500 square feet (45 square meters) (in plan) per vertical lift, and
    b. At least once per every 18 inches (460 mm) of vertical wall construction.

D. The General Contractor's engagement of the Special Inspector does not relieve the RWIC of responsibility to construct the proposed retaining wall in accordance with the approved construction shop drawings and these specifications.

E. The RWIC shall inspect the on-site grades and excavations prior to construction and notify the RWDE and General Contractor if on-site conditions differ from the elevations, assumptions, and grading conditions depicted in the retaining wall construction shop drawings.

1.9 DELIVERY, STORAGE AND HANDLING

A. The RWIC shall inspect the materials upon delivery to ensure that the proper type, grade and color of materials have been delivered.

B. The RWIC shall store and handle all materials in accordance with the manufacturer's recommendations as specified herein and in a manner that prevents deterioration or damage due to moisture, temperature changes, contaminants, corrosion, breaking, chipping, UV exposure or other causes. Damaged materials shall not be incorporated into the work.

C. Geosynthetics
1. All geosynthetic materials shall be handled in accordance with ASTM D4873. The materials should be stored off the ground and protected from precipitation, sunlight, dirt and physical damage.

D. Precast Modular Blocks
1. Precast modular blocks shall be stored in an area with positive drainage away from the blocks. Be careful to protect the block from mud and excessive chipping and breakage. Precast modular blocks shall not be stacked more than three (3) units high in the storage area.

E. Drainage Aggregate and Backfill Stockpiles
1. Drainage aggregate or backfill material shall not be piled over unstable slopes or areas of the project site with buried utilities.
2. Drainage aggregate and/or reinforced fill material shall not be staged where it may become mixed with or contaminated by poor draining fine-grained soils such as clay or silt.

PART 2 - MATERIALS

2.1 PRECAST MODULAR BLOCK RETAINING WALL UNITS

A. Acceptable Manufactuerers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturer listed, and other manufactuers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Construction Documents, the following manufactuerers are acceptable:
1. Precast Modular Block Retaining Wall Units:

a. Redi-Rock
b. Or Equal.

C. All units shall be wet-cast precast modular retaining wall units conforming to ASTM C1776.

D. All units for the project shall be obtained from the same manufacturer. The manufacturer shall be licensed and authorized to produce the retaining wall units by the precast modular block system patent holder/licensor and shall document compliance with the published quality control standards of the proprietary precast modular block system licensor for the previous three (3) years or the total time the manufacturer has been licensed, whichever is less.

E. Concrete used in the production of the precast modular block units shall be first-purpose, fresh concrete. It shall not consist of returned, reconstituted, surplus or waste concrete. It shall be an original production mix meeting the requirements of ASTM C94 and exhibit the properties as shown in the following table:

Concrete Mix Properties

Freeze Thaw Exposure Class[1]	Minimum 28-Day Compressive Strength[2]	Maximum Water Cement Ratio	Nominal Maximum Aggregate Size	Aggregate Class Designation[3]	Air Content[4]
Very Severe	4,500 psi (30.0 MPa)	0.40	1 inch (25 mm)	4S	6.0% +/- 1.5%
Maximum Water-Soluble Chloride Ion (Cl^-) Content in Concrete, Percent by Weight of Cement[5,6]					0.15
Maximum Chloride as Cl^- Concentration in Mixing Water, Parts Per Million					1000
Maximum Percentage of Total Cementitious Materials By Weight [7,9] (Very Severe Exposure Class Only):					
Fly Ash or Other Pozzolans Conforming to ASTM C618					25
Slag Conforming to ASTM C989					50
Silica Fume Conforming to ASTM C1240					10
Total of Fly Ash or Other Pozzolans, Slag, and Silica Fume[8]					50
Total of Fly Ash or Other Pozzolans and Silica Fume[8]					35
Alkali-Aggregate Reactivity Mitigation per ACI 201					
Slump (Conventional Concrete) per ASTM C143[10]			5 inches +/- 1½ inches (125 mm +/- 40 mm)		
Slump Flow (Self-Consolidating Concrete) per ASTM C1611			18 inches – 32 inches (450 mm – 800 mm)		

[1]Exposure class is as described in ACI 318. "Moderate" describes concrete that is exposed to freezing and thawing cycles and occasional exposure to moisture. "Severe" describes concrete that is exposed to freezing and thawing cycles and in continuous contact with moisture. "Very Severe" describes concrete that is exposed to freezing and thawing cycles and in continuous contact with moisture and exposed to deicing chemicals. Exposure class should be specified by owner/purchaser prior to order placement.
[2]Test method ASTM C39.
[3]Defined in ASTM C33 Table 3 *Limits for Deleterious Substances and Physical Property Requirements of Coarse Aggregates for Concrete*.
[4]Test method ASTM C231.
[5]Test method ASTM C1218 at age between 28 and 42 days.
[6]Where used in high sulfate environments or where alkali-silica reactivity is an issue, water soluble chloride shall be limited to no more than trace amounts (from impurities in concrete-making components, not intended constituents.)
[7]The total cementitious material also includes ASTM C150, C595, C845, C1157 cement. The maximum percentages shall include:
(a) Fly ash or other pozzolans in type IP, blended cement, ASTM C595, or ASTM C1157.
(b) Slag used in the manufacture of an IS blended cement, ASTM C595, or ASTM C1157.
(c) Silica fume, ASTM C1240, present in a blended cement.
[8]Fly ash or other pozzolans and silica fume shall constitute no more than 25 and 10 percent, respectively, of the total weight of the cementitious materials.
[9]Prescriptive limits shown may be waived for concrete mixes that demonstrate excellent freeze/thaw durability in a detailed and current testing program.
[10]Slump may be increased by a high-range water-reducing admixture.

F. Concrete reinforcing steel, when required for the specified block, shall conform to ASTM A615 and have a minimum yield strength of 60,000 psi. When required by the Contracting Officer to be galvanized or epoxy-coated, reinforcing steel shall conform to ASTM A767 or ASTM A775, respectively, and have a minimum yield strength of 60,000 psi.

G. At least 1 inch of concrete cover shall be maintained over all reinforcing steel bars.

H. Each concrete block shall be cast in a single continuous pour without cold joints. With the exception of half-block units, corner units and other special application units, the precast modular block units shall conform to the nominal dimensions listed in the table below and be produced to the dimensional tolerances shown.

Block Type	Dimension	Nominal Value	Tolerance
28" (710 mm) Block	Height	18" (457 mm)	+/- 3/16" (5 mm)
	Length	46-1/8" (1172 mm)	+/- 1/2" (13 mm)

	Width*	28” (710 mm)	+/- 1/2” (13 mm)
41” (1030 mm) Block	Height	18” (457 mm)	+/- 3/16” (5 mm)
	Length	46-1/8” (1172 mm)	+/- 1/2” (13 mm)
	Width*	40-1/2” (1030 mm)	+/- 1/2” (13 mm)
60” (1520 mm) Block	Height	18” (457 mm)	+/- 3/16” (5 mm)
	Length	46-1/8” (1172 mm)	+/- 1/2” (13 mm)
	Width*	60” (1520 mm)	+/- 1/2” (13 mm)
52” (1320 mm) XL Block	Height	36” (914 mm)	+/- 3/16” (5 mm)
	Length	46-1/8” (1172 mm)	+/- 1/2” (13 mm)
	Width*	60” (1520 mm)	+/- 1/2” (13 mm)
72” (1830 mm) XL Block	Height	36” (914 mm)	+/- 3/16” (5 mm)
	Length	46-1/8” (1172 mm)	+/- 1/2” (13 mm)
	Width*	60” (1520 mm)	+/- 1/2” (13 mm)
96” (2440 mm) XL Block	Height	36” (914 mm)	+/- 3/16” (5 mm)
	Length	46-1/8” (1172 mm)	+/- 1/2” (13 mm)
	Width*	60” (1520 mm)	+/- 1/2” (13 mm)

* Block tolerance measurements shall exclude variable face texture

I. With the exception of half-block units, corner units and other special application units, the precast modular block units shall have two (2), circular dome shear knobs that are 10 inches (254 mm), 7.5 inches (190 mm), or 6.75 inches (171 mm) in diameter and 4 inches (102 mm) or 2 inches (51 mm) in height. The shear knobs shall fully index into a continuous semi-cylindrical shear channel in the bottom of the block course above. ThePeak interlock shear between any two (2) vertically stacked precast modular block units, with 10-inch (254 mm) diameter shear knobs, measured in accordance with ASTM D6916 shall exceed 6,500 lb/ft (95 kN/m) at a minimum normal load of 500 lb/ft (7kN/m). as well as an ultimate peak interface shear capacity in excess of 11,000 lb/ft (160 kN/m). The peak interlock shear between any two (2) vertically stacked precast modular block units, with 7.5-inch (190 mm) or 6.75-inch (171 mm) diameter shear knobs, measured in accordance with ASTM D6916 shall exceed 1,850 lb/ft (27 kN/m) at a minimum normal load of 500 lb/ft (7kN/m) as well as an ultimate peak interface shear capacity in excess of 10,000 lb/ft (146 kN/m).Test specimen blocks tested under ASTM D6916 shall be actual, full-scale production blocks of known compressive strength. The interface shear capacity reported shall be corrected for a 4,000 psi (27.6 MPa) concrete compressive strength. Regardless of precast modular block configuration, interface shear testing shall be completed without the inclusion of unit core infill aggregate.

J. The 28-inch (710 mm) and 41-inch (1030 mm) precast modular block units may be cast with a continuous vertical core slot that will permit the insertion of a 12-inch (305 mm) inch wide strip of geogrid reinforcement to pass completely through the block. When installed in this manner, the geogrid reinforcement shall form a non-normal load dependent, positive connection between the block unit and the reinforcement strip. The use of steel for the purposes of creating the geogrid-to-block connection is not acceptable.

K. Without field cutting or special modification, the precast modular block units shall be capable of achieving a minimum radius of 14 ft - 6 in (4.42 m).

L. The precast modular block units shall be manufactured with an integrally cast shear knobs that establishes a standard horizontal set-back for subsequent block courses. The precast modular block system shall be available in the standard horizontal set-back facing batter options listed below:

18-inch High Blocks		36-inch High Blocks	
Horizontal	Max.	Horizontal	Max.
Set-Back/Blk. Course	Facing Batter	Set-Back/Blk. Course	Facing Batter
3/8” (10 mm)	1.2˚	3-1/4” (83 mm)	5.2˚
1-5/8” (41 mm)	5.2˚		
9-3/8” (238 mm)	27.5˚		
16-5/8” (422 mm)	42.7˚		

1. The precast modular block units shall be furnished with the required shear knobs that provide the facing batter required in the construction shop drawings.

M. The precast modular block unit face texture shall be selected by the Contracting Officer's Representative from the available range of textures available from the precast modular block manufacturer. Each textured block facing unit shall be a minimum of 5.76 square feet (0.54 square meters) with a unique texture pattern that repeats with a maximum frequency of once in any 15 square feet (1.4 square meters) of wall face.

N. The block color shall be selected by the Contracting Officer's Representative from the available range of colors available from the precast modular block manufacturer.

O. All precast modular block units shall be sound and free of cracks or other defects that would interfere with the proper installation of the unit, impair the strength or performance of the constructed wall. PMB units to be used in exposed wall construction shall not exhibit chips or cracks in the exposed face or faces of the unit that are not otherwise permitted. Chips smaller than 1.5" (38 mm) in its largest dimension and cracks not wider than 0.012" (0.3 mm) and not longer than 25% of the nominal height of the PMB unit shall be permitted. PMB units with bug holes in the exposed architectural face smaller than 0.75" (19 mm) in its largest dimension shall be permitted. Bug holes, water marks, and color variation on non-architectural faces are acceptable. PMB units that exhibit cracks that are continuous through any solid element of the PMB unit shall not be incorporated in the work regardless of the width or length of the crack.

P. Preapproved Manufacturers.
1. Manufacturers of Redi-Rock Retaining Wall Systems as licensed by Redi-Rock International, LLC, 05481 US 31 South, Charlevoix, MI 49720 USA; telephone (866) 222-8400; website www.redi-rock.com.

Q. Substitutions. Technical information demonstrating conformance with the requirements of this specification for an alternative precast modular block retaining wall system must be submitted for preapproval at least 14 calendar days prior to the bid date. Acceptable alternative PMB retaining wall systems, otherwise found to be in conformance with this specification, shall be approved in writing by the owner 7 days prior to the bid date. The Contracting Officer's Representative reserves the right to provide no response to submissions made out of the time requirements of this section or to submissions of block retaining wall systems that are determined to be unacceptable to the owner.

R. Value Engineering Alternatives. The owner may evaluate and accept systems that meet the requirements of this specification after the bid date that provide a minimum cost savings of 20% to the Government. Construction expediency will not be considered as a contributing portion of the cost savings total.

2.2 GEOGRID REINFORCEMENT

A. Geogrid reinforcement shall be a woven or knitted PVC coated geogrid manufactured from high-tenacity PET polyester fiber with an average molecular weight greater than 25,000 ($M_n >$ 25,0000) and a carboxyl end group less than 30 (CEG < 30). The geogrid shall be furnished in prefabricated roll widths of certified tensile strength by the manufacturer. The prefabricated roll width of the geogrid shall be 12" (300 mm) +/- 1/2" (13 mm). No cutting of geogrid reinforcement down to the 12" (300 mm) roll width from a larger commercial roll width will be allowed under any circumstances.

B. The ultimate tensile strength (T_{ult}) of the geogrid reinforcement shall be measured in accordance with ASTM D6637.

C. Geogrid – Soil Friction Properties
1. Friction factor, F*, shall be equal to 2/3 Tan ϕ, where ϕ is the effective angle of internal friction of the reinforced fill soil.
2. Linear Scale Correction Factor, α, shall equal 0.8.

D. Long-Term Tensile Strength (T_{al}) of the geogrid reinforcement shall be calculated in accordance with Section 3.5.2 of FHWA-NHI-10-024 and as provided in this specification.
   1. The creep reduction factor (RF_{CR}) shall be determined in accordance with Appendix D of FHWA-NHI-10-025 for a minimum 75 year design life.
   2. Minimum installation damage reduction factor (RF_{ID}) shall be 1.25. The value of RF_{ID} shall be based upon documented full-scale tests in a soil that is comparable to the material proposed for use as reinforced backfill in accordance with ASTM D5818.
   3. Minimum durability reduction factor (RF_D) shall be 1.3 for a soil pH range of 3 to 9.

E. Connection between the PMB retaining wall unit and the geogrid reinforcement shall be determined from short-term testing per the requirements of FHWA NHI-10-025, Appendix B.4 for a minimum 75-year design life.

F. The minimum value of T_{al} for geogrid used in design of a reinforced precast modular block retaining wall shall be 2,000 lb/ft (29 kN/m) or greater.

G. The minimum length of geogrid reinforcement shall be the greater of the following:
   1. 0.7 times the wall design height, H.
   2. 6 feet (1.83 m).
   3. The length required by design to meet internal stability requirements, soil bearing pressure requirements and constructability requirements.

H. Constructability Requirements. Geogrid design embedment length shall be measured from the back of the precast modular block facing unit and shall be consistent for the entire height of a given retaining wall section.

I. Geogrid shall be positively connected to every precast modular block unit. Design coverage ratio, Rc, as calculated in accordance with AASHTO LRFD Bridge Design Specifications Figure 11.10.6.4.1-2 shall not exceed 0.50.

J. Preapproved Geogrid Reinforcement Products.
   1. Miragrid XT Geogrids as manufactured by TenCate Geosynthetics of Pendergrass, Georgia USA and distributed by Manufacturers of the Redi-Rock Retaining Wall System.

K. Substitutions. No substitutions of geogrid reinforcement products shall be allowed.

2.3 GEOTEXTILE

A. Nonwoven geotextile fabric shall be placed as indicated on the retaining wall construction shop drawings. Additionally, the nonwoven geotextile fabric shall be placed in the v-shaped joint between adjacent block units on the same course. The nonwoven geotextile fabric shall meet the requirements Class 3 construction survivability in accordance with AASHTO M 288.

B. Preapproved Nonwoven Geotextile Products
   1. Mirafi 140N
   2. Propex Geotex 451
   3. Skaps GT-142
   4. Thrace-Linq 140EX
   5. Carthage Mills FX-40HS
   6. Stratatex ST 142
   7. Or approved equal.

2.4 DRAINAGE AGGREGATE AND WALL INFILL

A. Drainage aggregate (and wall infill for retaining walls designed as modular gravity structures) shall be a durable crushed stone conforming to No. 57 size per ASTM C33 with the following particle-size distribution requirements per ASTM D6913:

U.S. Standard Sieve Size	% Passing
1-½" (38 mm)	100

1” (25 mm)	95-100
½” (13 mm)	25-60
No. 4 (4.76 mm)	0-10
No. 8 (2.38 mm)	0-5

2.5 REINFORCED FILL

A. Material used as reinforced backfill material in the reinforced zone (if applicable) shall be a granular fill material meeting the requirements of USCS soil type GW, GP, SW or SP per ASTM D2487 or alternatively by AASHTO Group Classification A-1-a or A-3 per AASHTO M 145. The backfill shall exhibit a minimum effective internal angle of friction, $\phi = 34$ degrees at a maximum 2% shear strain and meet the following particle-size distribution requirements per ASTM D6913.

U.S. Standard Sieve Size	% Passing
3/4” (19 mm)	100
No. 4 (4.76 mm)	0-100
No. 40 (0.42 mm)	0-60
No. 200 (0.07 mm)	0-15

B. The reinforced backfill material shall be free of sod, peat, roots or other organic or deleterious matter including, but not limited to, ice, snow or frozen soils. Materials passing the No. 40 (0.42 mm) sieve shall have a liquid limit less than 25 and plasticity index less than 6 per ASTM D4318. Organic content in the backfill material shall be less than 1% per AASHTO T-267 and the pH of the backfill material shall be between 5 and 8.

C. Soundness. The reinforced backfill material shall exhibit a magnesium sulfate soundness loss of less than 30% after four (4) cycles, or sodium sulfate soundness loss of less than 15% after five (5) cycles as measured in accordance with AASHTO T-104.

D. Reinforced backfill shall not be comprised of crushed or recycled concrete, recycled asphalt, bottom ash, shale or any other material that may degrade, creep or experience a loss in shear strength or a change in pH over time.

2.6 LEVELING PAD

A. The precast modular block units shall be placed on a leveling pad constructed from crushed stone or unreinforced concrete. The leveling pad shall be constructed to the dimensions and limits shown on the retaining wall design drawings prepared by the RWDE.

B. Crushed stone used for construction of a granular leveling pad shall meet the requirements of the drainage aggregate and wall infill in section 2.04 or a preapproved alternate material.

C. Concrete used for construction of an unreinforced concrete leveling pad shall satisfy the criteria for AASHTO Class B. The concrete should be cured a minimum of 12 hours prior to placement of the precast modular block wall retaining units and exhibit a minimum 28-day compressive strength of 2,500 psi (17.2 MPa).

2.7 DRAINAGE

A. Drainage Pipe
   1. Drainage collection pipe shall be a 4” (100 mm) diameter, 3-hole perforated, HDPE pipe with a minimum pipe stiffness of 22 psi (152 kPa) per ASTM D2412.
   2. The drainage pipe shall be manufactured in accordance with ASTM D1248 for HDPE pipe and fittings.

B. Preapproved Drainage Pipe Products
   1. ADS 3000 Triple Wall pipe as manufactured by Advanced Drainage Systems, or Equal.

PART 3 - EXECUTION

3.1 GENERAL

A. All work shall be performed in accordance with OSHA safety standards, state and local building codes and manufacturer's requirements.

B. The General Contractor is responsible for the location and protection of all existing underground utilities. Any new utilities proposed for installation in the vicinity of the retaining wall, shall be installed concurrent with retaining wall construction. The General Contractor shall coordinate the work of subcontractors affected by this requirement.

C. New utilities installed below the retaining wall shall be backfilled and compacted to a minimum of 98% maximum dry density per ASTM D698 standard proctor.

D. The General Contractor is responsible to ensure that safe excavations and embankments are maintained throughout the course of the project.

E. All work shall be inspected by the Special Inspector as directed by the Contracting Officer.

3.2 EXAMINATION

A. Prior to construction, the General Contractor, Grading Contractor, RWIC and Special Inspector shall examine the areas in which the retaining wall will be constructed to evaluate compliance with the requirements for installation tolerances, worker safety and any site conditions affecting performance of the completed structure. Installation shall proceed only after unsatisfactory conditions have been corrected.

3.3 PREPARATION

A. Fill Soil.
   1. The Special Inspector shall verify that reinforced backfill placed in the reinforced soil zone satisfies the criteria of this section.
   2. The Special Inspector shall verify that any fill soil installed in the foundation and retained soil zones of the retaining wall satisfies the specification of the RWDE as shown on the construction drawings.

B. Excavation.
   1. The Grading Contractor shall excavate to the lines and grades required for construction of the precast modular block retaining wall as shown on the construction drawings. The Grading Contractor shall minimize over-excavation. Excavation support, if required, shall be the responsibility of the Grading Contractor.
   2. Over-excavated soil shall be replaced with compacted fill in conformance with the specifications of the RWDE and "Division 31, Section 31 23 00 - Earthwork" of these project specifications.
   3. Embankment excavations shall be bench cut as directed by the project Geotechnical Engineer and inspected by the Special Inspector for compliance.

C. Foundation Preparation.
   1. Prior to construction of the precast modular block retaining wall, the leveling pad area and undercut zone (if applicable) shall be cleared and grubbed. All topsoil, brush, frozen soil and organic material shall be removed. Additional foundation soils found to be unsatisfactory beyond the specified undercut limits shall be undercut and replaced with approved fill as directed by the project Geotechnical Engineer. The Inspection Engineer shall ensure that the undercut limits are consistent with the requirements of the project Geotechnical Engineer and that all soil fill material is properly compacted in accordance with project specifications. The Special Inspector shall document the volume of undercut and replacement.
   2. Following excavation for the leveling pad and undercut zone (if applicable), the Special Inspector shall evaluate the in-situ soil in the foundation and retained soil zones.

a. The Special Inspector shall verify that the shear strength of the in-situ soil assumed by the RWDE is appropriate. The Special Inspector shall immediately stop work and notify the Contracting Officer if the in-situ shear strength is found to be inconsistent with the retaining wall design assumptions.
b. The Special Inspector shall verify that the foundation soil exhibits sufficient ultimate bearing capacity to satisfy the requirements indicated on the retaining wall construction shop drawings per paragraph 1.06 I of this section.

D. Leveling Pad.
1. The leveling pad shall be constructed to provide a level, hard surface on which to place the first course of precast modular block units. The leveling pad shall be placed in the dimensions shown on the retaining wall construction drawings and extend to the limits indicated.
2. Crushed Stone Leveling Pad. Crushed stone shall be placed in uniform maximum lifts of 6" (150 mm). The crushed stone shall be compacted by a minimum of 3 passes of a vibratory compactor capable of exerting 2,000 lb (8.9 kN) of centrifugal force and to the satisfaction of the Special Inspector.
3. Unreinforced Concrete Leveling Pad. The concrete shall be placed in the same dimensions as those required for the crushed stone leveling pad. The RWIC shall erect proper forms as required to ensure the accurate placement of the concrete leveling pad according to the retaining wall construction drawings.

3.4 PRECAST MODULAR BLOCK WALL SYSTEM INSTALLATION

A. The precast modular block structure shall be constructed in accordance with the construction drawings, these specifications and the recommendations of the retaining wall system component manufacturers. Where conflicts exist between the manufacturer's recommendations and these specifications, these specifications shall prevail.

B. Drainage components. Pipe, geotextile and drainage aggregate shall be installed as shown on the construction shop drawings.

C. Precast Modular Block Installation
1. The first course of block units shall be placed with the front face edges tightly abutted together on the prepared leveling pad at the locations and elevations shown on the construction drawings. The RWIC shall take special care to ensure that the bottom course of block units are in full contact with the leveling pad, are set level and true and are properly aligned according to the locations shown on the construction drawings.
2. Backfill shall be placed in front of the bottom course of blocks prior to placement of subsequent block courses. Nonwoven geotextile fabric shall be placed in the V-shaped joints between adjacent blocks. Drainage aggregate shall be placed in the V-shaped joints between adjacent blocks to a minimum distance of 12" (300 mm) behind the block unit. If stone infill of hollow core blocks exceeds 45% of the block design volume, drainage aggregate will not be required to extend beyond the back of the blocks, with the approval of the RWDE.
3. Drainage aggregate shall be placed in 9-inch maximum lifts and compacted by a minimum of three (3) passes of a vibratory plate compactor capable exerting a minimum of 2,000 lb (8.9 kN) of centrifugal force.
4. Unit core fill shall be placed in the precast modular block unit vertical core slot. The core fill shall completely fill the slot to the level of the top of the block unit. The top of the block unit shall be broom-cleaned prior to placement of subsequent block courses. No additional courses of precast modular blocks may be stacked before the unit core fill is installed in the blocks on the course below.
5. Base course blocks for gravity wall designs (without geosynthetic soil reinforcement) may be furnished without vertical core slots. If so, disregard item 4 above, for the base course blocks in this application.

6. Nonwoven geotextile fabric shall be placed between the drainage aggregate and the retained soil (gravity wall design) or between the drainage aggregate and the reinforced fill (reinforced wall design) as required on the retaining wall construction drawings.
7. Subsequent courses of block units shall be installed with a running bond (half block horizontal course-to-course offset). With the exception of 90-degree corner units, the shear channel of the upper block shall be fully engaged with the shear knobs of the block course below. The upper block course shall be pushed forward to fully engage the interface shear key between the blocks and to ensure consistent face batter and wall alignment. Geogrid, drainage aggregate, unit core fill, geotextile and properly compacted backfill shall be complete and in-place for each course of block units before the next course of blocks is stacked.
8. The elevation of retained soil fill shall not be less than 1 block course (18 inches (457 mm)) below the elevation of the reinforced backfill throughout the construction of the retaining wall.
9. If included as part of the precast modular block wall design, cap units shall be secured with an adhesive in accordance with the precast modular block manufacturer's recommendation.

D. Geogrid Reinforcement Installation (if required)
1. Geogrid reinforcement shall be installed at the locations and elevations shown on the construction drawings on level fill compacted to the requirements of this specification.
2. Continuous 12" (300 mm) wide strips of geogrid reinforcement shall be passed completely through the vertical core slot of the precast modular block unit and extended to the embedment length shown on the construction plans. The strips shall be staked or anchored as necessary to maintain a taut condition.
3. Reinforcement length (L) of the geogrid reinforcement is measured from the back of the precast modular block unit. The cut length (L_c) is two times the reinforcement length plus additional length through the block facing unit. The cut length is calculated as follows:

L_c = 2*L + 3 ft (2*L + 0.9 m) (28" (710 mm) block unit)
L_c = 2*L + 5 ft (2*L + 1.5 m) (41" (1030 mm) block unit)

4. The geogrid strip shall be continuous throughout its entire length and may not be spliced. The geogrid shall be furnished in nominal, prefabricated roll widths of 12" (300 mm)+/- ½" (13 mm). No field modification of the geogrid roll width shall be permitted.
5. Neither rubber tire nor track vehicles may operate directly on the geogrid. Construction vehicle traffic in the reinforced zone shall be limited to speeds of less than 5 mph (8 km/hr) once a minimum of 9 inches (230 mm) of compacted fill has been placed over the geogrid reinforcement. Sudden braking and turning of construction vehicles in the reinforced zone shall be avoided.

E. Construction Tolerance. Allowable construction tolerance of the retaining wall shall be as follows:
1. Deviation from the design batter and horizontal alignment, when measured along a 10' (3 m) straight wall section, shall not exceed 3/4" (19 mm).
2. Deviation from the overall design batter shall not exceed 1/2" (13 mm) per 10' (3 m) of wall height.
3. The maximum allowable offset (horizontal bulge) of the face in any precast modular block joint shall be 1/2" (13 mm).
4. The base of the precast modular block wall excavation shall be within 2" (50 mm) of the staked elevations, unless otherwise approved by the Special Inspector.
5. Differential vertical settlement of the face shall not exceed 1' (300 mm) along any 200' (61 m) of wall length.
6. The maximum allowable vertical displacement of the face in any precast modular block joint shall be 1/2" (13 mm).
7. The wall face shall be placed within 2" (50 mm) of the horizontal location staked.

3.5 WALL INFILL AND REINFORCED BACKFILL PLACEMENT

A. Backfill material placed immediately behind the drainage aggregate shall be compacted as follows:
   1. 98% of maximum dry density at ± 2% optimum moisture content per ASTM D698 standard proctor or 85% relative density per ASTM D4254.

B. Compactive effort within 3’ (0.9 m) of the back of the precast modular blocks should be accomplished with walk-behind compactors. Compaction in this zone shall be within 95% of maximum dry density as measured in accordance with ASTM D698 standard proctor or 80% relative density per ASTM D 4254. Heavy equipment should not be operated within 3’ (0.9 m) of the back of the precast modular blocks.

C. Backfill material shall be installed in lifts that do not exceed a compacted thickness of 9” (230 mm).

D. At the end of each work day, the RWIC shall grade the surface of the last lift of the granular wall infill to a 3% ± 1% slope away from the precast modular block wall face and compact it.

E. The General Contractor shall direct the Grading Contractor to protect the precast modular block wall structure against surface water runoff at all times through the use of berms, diversion ditches, silt fence, temporary drains and/or any other necessary measures to prevent soil staining of the wall face, scour of the retaining wall foundation or erosion of the reinforced backfill or wall infill.

3.6 OBSTRUCTIONS IN THE INFILL AND REINFORCED FILL ZONE

A. The RWIC shall make all required allowances for obstructions behind and through the wall face in accordance with the approved construction shop drawings.

B. Should unplanned obstructions become apparent for which the approved construction shop drawings do not account, the affected portion of the wall shall not be constructed until the RWDE can appropriately address the required procedures for construction of the wall section in question.

3.7 COMPLETION

A. For walls supporting unpaved areas, a minimum of 12” (300 mm) of compacted, low-permeability fill shall be placed over the granular wall infill zone of the precast modular block retaining wall structure. The adjacent retained soil shall be graded to prevent ponding of water behind the completed retaining wall.

B. For retaining walls with crest slopes of 5H:1V or steeper, silt fence shall be installed along the wall crest immediately following construction. The silt fence shall be located 3’ to 4’ (0.9 m to 1.2 m) behind the uppermost precast modular block unit. The crest slope above the wall shall be immediately seeded to establish vegetation. The General Contractor shall ensure that the seeded slope receives adequate irrigation and erosion protection to support germination and growth.

C. The General Contractor shall confirm that the as-built precast modular block wall geometries conform to the requirements of this section. The General Contractor shall notify the Government of any deviations.

END OF SECTION

SECTION 32 91 13
FINISH GRADING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Topsoil quality and placing.
   2. Finish grading.
   3. Topsoil salvage, storage, and replacement

B. Related Requirements: Include, but are not necessarily limited to, the following:
   1. Section 32 92 21 – Native Grass Seeding

1.2 PRICE AND PAYMENT PROCEDURES

A. Measurement and Payment:
   1. Perform the Work of this Section for all areas within limits of grading and all areas outside limits of grading disturbed during construction.
   2. Perform the Work of this Section as part of the Lump Sum bid/pay item(s) indicated in the Contract.

1.3 REFERENCES

A. Reference Standards:
   1. American Association of State Highway and Transportation Officials (AASHTO):
      a. T194, Standard Method of Test for Determination of Organic Content of Soils by Loss on Ignition.
   2. ASTM International (ASTM):
      a. D2974, Standard Test Methods for Moisture, Ash, and Organic Matter of Peat and Other Organic Soils.
      b. D4221, Standard Test Method for Dispersive Characteristics of Clay Soil by Double Hydrometer.
      c. D4972, Standard Test Methods for pH of Soils.
      d. D5268, Standard Specification for Topsoil Used for Landscaping and Construction Purposes.
      e. D6913, Standard Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis.
      f. D7928, Standard Test Method for Particle-Size Distribution (Gradation) of Fine-Grained Soils Using the Sedimentation (Hydrometer) Analysis.
      g. E329, Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection.

1.4 QUALITY ASSURANCE

A. Qualifications:
   1. Contractor's Testing Laboratory:
      a. Retain the services of independent testing laboratory to perform testing and determine compliance with the Contract Documents of materials specified in this Section.
      b. When Contractor is expressly required, by this Section's "Quality Assurance" Article, "Source Quality Control" Article or elsewhere in the Contract Documents, to retain testing laboratory for tests required under this Section's "Quality Assurance" Article or "Source Quality Control" Article, retain one testing laboratory for quality assurance testing and field quality control testing.
      c. Testing laboratory shall comply with ASTM E329.
      d. Testing laboratory shall be experienced in the types of testing required.

e. Selection of testing laboratory is subject to Contracting Officer's Representative's acceptance. Upon Contracting Officer's Representative's request, submit qualifications statement for testing laboratory, including name of entity, location, copies of applicable certifications, summary of entity's experience, and names and qualifications of personnel who will perform the subject sampling and testing.

B. Quality Assurance Testing:
   1. Quality assurance testing, which may also be regarded as source quality control testing when topsoil to be installed will be topsoil originally stripped from the Site, is in addition to field quality control activities required by this Section's "Field Quality Control" Article.
   2. Allow free access to material stockpiles and borrow areas at all times. Tests shall be at Contractor's cost.
   3. Contractor's Testing Laboratory Scope:
      a. Obtain samples and perform testing of proposed topsoil materials, whether obtained from offsite borrow sources or stockpiled topsoil originally stripped from the Site, to provide assurance that the Work will comply with the Contract Documents.
   4. Required Quality Assurance Material Testing by Contractor's Testing Laboratory:
      a. Perform the following quality assurance tests for every 1,000 cubic yards of topsoil, or part thereof, to be incorporated into the Work. Submit results of such tests to Contracting Officer's Representative and obtain Contracting Officer's Representative's acceptance thereof prior to incorporating the subject topsoil into the Work.
         1) Gradation in accordance with ASTM D6913 and D7928.
         2) Clay content in accordance with ASTM D4221.
         3) pH in accordance with ASTM D4972.
         4) Organic content in accordance with ASTM D2974 or AASHTO T194.

1.5 SUBMITTALS

A. Informational Submittals: Submit the following:
   1. Quality Assurance Test Results Submittals:
      a. Submit results of quality assurance testing performed in accordance with this Section's "Quality Assurance" Article, unless included as part of another submittal under this Section.
         1) Tests on materials from onsite and borrow sources.
   2. Qualifications Statements:
      a. Quality assurance testing laboratory, when requested by Contracting Officer's Representative.
   3. Soil Survey Report
      a. Contractor will access the USDA NRCS Web Soil Survey, delineate project areas, and produce customized soil reports specific to each project area where soil disturbance will occur.
         1) Soils report to include soil profile descriptions, soil mapping units, characteristics, topsoil depth, soil textures, rock content, soil depth, available water holding capacity, permeability rates, and drainage classes.
      b. The contractor will meet with the Contracting Officer and GRCA Vegetation Program Manager 120 days before the start of topsoil salvage to discuss soils report and finalize topsoil management actions per project area.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 MATERIALS

A. Topsoil:
   1. All soil accepted as topsoil, whether obtained from onsite or offsite sources, shall comply with requirements of this Section.

2. Topsoil Source: Reuse surface soil stockpiled on Site, where possible. Verify suitability of stockpiled surface soil to produce topsoil, as specified. Clean surface soil of roots, plants, sod, stones, clay lumps, and other extraneous materials harmful to plant growth.
    a. Supplement acceptable onsite soil with manufactured topsoil from offsite sources, when quantities available on-Site are insufficient to complete the Work.
    b. When existing soil material stripped from the Site and stockpiled does not comply with requirements of this Section, such soil may be provided as topsoil when addition of amendments and fertilizers will bring such soil material into compliance with this Section's topsoil requirements. Furnish to Contracting Officer written advisory of Contractor's intent to amend existing soil materials to be compliant with topsoil requirements.
3. Provide fertile, friable, natural loam, surface soil, capable of sustaining vigorous plant growth; free of any admixture of subsoil, clods of hard earth, plants or roots, sticks, stones larger than one inch diameter, or other extraneous material harmful to plant growth, in compliance with ASTM D 5268.
4. Provide topsoil of the following characteristics:
    a. Required gradation:

Sieve Size	Percent Passing
3/4 inch mesh	100
No. 4-sieve	90 to 100
No. 200-sieve	0 to 10

    b. Clay content of material passing No. 200-sieve shall be not greater than 60 percent, as determined by hydrometer tests.
    c. pH-adjusted with ferrous sulphate or ground limestone to provide pH 5.5 to 7.0 at time of installation of lawns and meadow areas, unless particular species of planting requires a different pH for optimal growth.
    d. Electrical conductivity of a 1:2 soil-water suspension shall not exceed 1.0 milliohm per centimeter and with less than 200 parts per million of extractable aluminum.
    e. Cation Exchange Capacity: 5, minimum.
    f. Organic content not less than five percent, as determined by ignition loss of oven-dried samples passing No. 10-sieve (Muffle Furnace Temperature: 110 plus or minus five degrees C for eight hours).
    g. Free of pests and pest larvae.
    h. All topsoil must be confirmed as "weed free" by the supplier if sourced from offsite or by the NPS Vegetation Program if sourced onsite.
    i. Relative to Constituents of Concern, topsoil shall be in accordance with Laws and Regulations. Constituents of Concern, if any, in topsoil furnished shall not constitute or result in a Hazardous Environmental Condition.

B. Topsoil Amendments and Fertilizer
    1. Fertilizer prohibited by NPS.

2.2 SOURCE QUALITY CONTROL

A. Tests and Inspections at Source of Supply:
    1. Perform quality assurance testing, and submit results to Contracting Officer's Representative, in accordance with the "Quality Assurance" Article in Part 1 of this Section.

PART 3 - EXECUTION

3.1 PREPARATION

A. Protection of In-place Conditions:

1. Protect structures, utilities, sidewalks, pavements, and other facilities, and lawns and existing plants from damage caused by finish grading.

B. Topsoil Salvage
   1. Contractor to remove all topsoil prior to all other soil disturbances.
   2. Contractor to remove the topsoil in one pass, will not mix topsoil with subsoil, and will not compact or drive upon topsoil during removal
   3. Topsoil salvage depth will be a minimum of 6 inches of material in all cases, with additional topsoil salvageable in some locations.
   4. Live vegetation less than 3 feet in height and limbs less than 1 inch in diameter shall be incorporated as topsoil in the stockpiles.
   5. The contractor will properly dispose or bury all topsoil contaminated with invasive non-native plant species as directed by the Contract Officer and GRCA Vegetation Program Manager.

C. Topsoil Stockpiling
   1. Salvaged topsoil will be separated from the sub-soil and stockpiled in windrowed piles no higher than three feet, and no wider than three feet.
   2. The contractor will identify removal areas and stockpiles so that topsoil may be replaced in the same locations.
   3. If salvaged topsoil must be stored longer than 3 weeks, topsoil stockpiles will be hydroseeded and hyrdomulched by the contractor with locally collected seed provided by NPS.

D. Preparation – General:
   1. In accordance with Section 01 50 00 – Temporary Facilities and Controls, provide temporary measures for controlling erosion and sedimentation, airborne dust, and fugitive dust emissions from the Site.
   2. Adjust rough grading to within 0.1 foot of required elevations. Properly compact sub-grades. Rough grading shall provide for appropriate drainage in accordance with the Contract Documents.
   3. Correct, adjust and/or repair rough graded areas.
      a. Cut off mounds and ridges.
      b. Fill gullies and depressions.
      c. Perform other necessary repairs.
      d. Bring all sub-grades to specified contours, even and properly compacted.
   4. Before applying topsoil:
   5. Scarify all disturbed areas to a depth of 6” in a direction perpendicular to the natural flow of water.
   6. If the Contracting Officer and GRCA Vegetation Program Manager determines that subsoil is excessively compacted by the NPS or contractors’ operations, the contractor will scarify the subsoil to a depth of 24” to de-compact and break the interface between particles.

3.2 INSTALLATION

A. Installation Requirements:
   1. Required Depth of Topsoil:
      a. Spread conserved topsoil to a depth of 4 inches or greater (if additional topsoil is available) over all disturbed areas.
      b. Contractor to spread using lightest weight equipment possible.
      c. If the Contracting Officer and GRCA Vegetation Program Manager determine that topsoil is excessively compacted by the NPS or contractors’ operations, the contractor will scarify the topsoil to a depth of 8” to de-compact and break the interface between particles.
   2. Contractor to avoid mixing topsoil with subsoil materials.
   3. Do not place topsoil when subgrade is muddy, wet or frozen enough to cause clodding.

4. Provide finished surface smooth and true to required grades, in accordance with this Article's "Installation Tolerances" provision. Provide finished surface of topsoil equal to required finished elevations after light rolling and natural settlement.
5. Grade topsoil to a smooth, uniform surface plane with loose, uniformly fine texture. Roll and rake, remove ridges, and fill depressions to provide required finish grades.
6. Restore topsoil if eroded or otherwise disturbed after finish grading and before planting.
7. Topsoil Quantity Differentials:
    a. If quantity of required topsoil exceeds the quantity, if any, of topsoil stockpiled at the Site and available for use by Contractor, provide the additional required quantity from appropriate offsite borrow sources at no additional cost to Government.
    b. When quantity of topsoil obtained from offsite borrow sources exceeds the required quantity of topsoil, remove excess from the Site, unless Government allows such excess to be stockpiled at an appropriate location at the Site.
    c. When quantity of required topsoil is less than quantity of onsite stockpiles, following installation of topsoil, properly grade remaining stockpile for appropriate drainage without undue erosion and seed stockpile as required for other areas of lawn restoration in accordance with the Contract Documents.

B. Installation Tolerances:
    1. Finish Grading Tolerance: ±0.1 foot from finish elevation required by the Contract Documents. Ensure adequate drainage.

3.3 FIELD QUALITY CONTROL

A. Tests and Inspections:
    1. Upon completion of topsoil installation, obtain Contracting Officer's concurrence regarding grades and drainage.
    2. Where directed by Contracting Officer's Representative, provide test holes to demonstrate that required depth of topsoil was provided. After observation and measurement by Contracting Officer's Representative, properly refill test holes.
    3. Where insufficient depth of topsoil is revealed, remedy by providing required depth and quantity in accordance with the Contract Documents. When additional topsoil is added after initial establishment of plant growth, re-establish required plantings to provide compliance with the Contract Documents.

3.4 PROTECTION

A. Commencing immediately after topsoil installation, protect installed topsoil from erosion and damage from other causes, including personnel, vehicles, and equipment. Promptly perform hydro-seeding and hydro-mulching to permanently stabilize soils and prevent erosion, in accordance with Section 32 92 21 – Native Grass Seeding.

END OF SECTION

This page intentionally left blank.

SECTION 32 92 21
NATIVE GRASS SEEDING

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Native Grass Seeding by way of hydro-seeding and hydro-mulching, as indicated, in accordance with provisions of Contract Documents.
   1. NPS is responsible for providing all seed for the project, to be propagated by GRCA vegetation program.

B. Completely coordinate with work of other trades.

1.2 SUBMITTALS

A. Contract Closeout Information:
   1. Maintenance data.

1.3 PRODUCT DELIVERY, STORAGE, AND HANDLING

A. Deliver seed in standard sealed containers labeled with producer's name and seed analysis, and in accordance with US Department of Agriculture Rules and Regulations under Federal Seed Act.

B. Deliver fertilizer in original containers, labeled with content analysis.

1.4 JOB CONDITIONS

A. Perform hydro-seeding and hydro-mulching between October 1 and November 15, depending on weather and soil conditions.

B. Fall seeding, conducted when the possibility of frost is near, is acceptable; but do not water until the following growing season.

C. No work shall be done when the ground is frozen, snow covered, too wet or in an otherwise unsuitable condition for planting.
   1. Special conditions may exist that warrant a variance in the specified planting dates or conditions.
   2. Submit a written request to Contracting Officer's Representative stating the special conditions and proposed variance.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 MATERIALS

A. Seed:
   1. NPS is responsible for providing certified seed of locally adapted strains.
      a. GRCA Vegetation Program has initiated seed collection efforts to provide locally-collected seed for propagation of plant materials.

B. Mulch:
   1. All slopes steeper than 3:1 shall receive mulch in addition to hydro-seeding.
   2. Bonded fiber mulch composed of strong tackifiers with crimped wood fibers
   3. Mulch shall we designed to assist with site stabilization for 1 to 1.5 years.

C. Fertilizer:
   1. Prohibited by NPS.

D. Cover crop: Sterile; Composed of Sorghum, soybeans, milo, millet or sudangrass, etc.

E. Water: Provide clean, fresh water free from impurities injurious to vegetation.

PART 3 - EXECUTION

3.1 SEEDING

A. Hydro-seeding:
   1. Contracted hydro-seeder/hydro-mulcher to apply a slurry of seed and water to application areas using a centrifugal pump.
      a. GRCA Vegetation Program staff to determine seed mixture and rate of application.
   2. Apply slurry evenly in a minimum of two (2) passes over prepared soil area.
      a. First pass: apply 50% of material from the bottom of the slope, directed upward on the slope, applying hydro-seed materials to the entire area.
      b. Second pass: apply 50% of material from the top of the slope, directed downwards on the slope, applying hydro-seed materials to the entire area.
      c. Using the color of the seed as a metering agent, spray the slurry mixture uniformly over the designated seeded area, taking care to avoid bare spots in the application of the material.
   3. Do not hydro-seed in the presence of surface water.
   4. Thoroughly clean site amenities after seeding.
   5. Hydraulic seeding equipment and accessories (i.e., centrifugal pump): Obtain Contracting Officer's approval for use.

B. Hydro-mulching:
   1. Contracted hydro-seeder/hydro-mulcher to apply a slurry of bonded fiber mulch and water to all hydro-seeded areas with slopes greater than 3:1.
      a. GRCA Vegetation Program staff to determine seed mixture and rate of application.
   2. Apply slurry evenly in a minimum of two (2) passes over hydro-seeded area.
      a. First pass: apply 50% of material from the bottom of the slope, directed upward on the slope, applying hydro-mulch materials to the entire area.
      b. Second pass: apply 50% of material from the top of the slope, directed downwards on the slope, applying hydro-mulch materials to the entire area.
      c. Using the color of the mulch as a metering agent, spray the slurry mixture uniformly over the designated seeded area, taking care to avoid bare spots in the application of the material.
   3. Do not hydro-mulch in the presence of surface water.
   4. Thoroughly clean site amenities after mulching.
   5. Hydraulic mulching equipment and accessories (i.e., centrifugal pump): Obtain Contracting Officer's approval for use.

C. Quality Assurance:
   1. Contractor shall coordinate with the Contracting Officer's Representative for inspection of the first test area of hydro-seeding and hydro-mulching to verify application technique, coverage, and adherence to slope requirements.

3.2 MAINTENANCE

A. Maintain seeded areas for one full growing season.

B. Do not mow native grass.

C. When weeds are shading out native grass seedlings, shred weeds back to height of seedlings with rotary shredder.
   1. Before August of growing season, shred weeds at least 2 times.

END OF SECTION

DIVISION 33

UTILITIES

This page intentionally left blank.

SECTION 33 01 30.11

TELEVISION INSPECTION OF SEWERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Pipeline flushing and cleaning.
   2. Television inspection of sewer pipelines.
   3. Audio-video recording of pipeline interior.

1.2 DEFINITIONS

A. Limit list of definitions to terms unique to this Section and not provided elsewhere.

B. DVD: An optical disc storage format, offering higher storage capacity than compact discs (CDs) while having the same dimensions.

1.3 SUBMITTALS

A. Only request submittals needed to verify compliance with Project requirements.

B. DVDs:
   1. Three copies of completed narrated color DVDs identified by Project name, street name, right-of-way property name, and manhole numbers.
   2. DVDs become property of the National Park Service.

C. Inspection Logs:
   1. Submit cleaning and television inspection logs for each section of sewer line to be rehabilitated.
   2. Include following minimum information:
      a. Stationing and location of lateral services, wyes, or tees.
      b. Date and clock time references.
      c. Pipe joints.
      d. Infiltration/inflow defects.
      e. Cracks.
      f. Leaks.
      g. Offset joints.

D. Proposed bypass pumping system, including written description of plan addressing schedule, quantity, capacity, and location of pumping equipment.

E. Spill plan to address any spills that might occur.

F. Field Quality-Control Submittals: Indicate results of Contractor-furnished tests and inspections.

G. Qualifications Statement:
   1. Coordinate following Subparagraph with requirements specified in QUALITY ASSURANCE Article.
   2. Qualifications for applicator.

1.4 QUALITY ASSURANCE

A. Include this Article to specify compliance with overall reference standards affecting products and installation.

B. In following Paragraph insert "State of _____ Department of Transportation," "Municipality of _____ Department of Public Works," or other agency as appropriate.

C. Perform Work according to State of Arizona standards.

D. Applicator: Company specializing in performing Work of this Section with minimum three years' documented experience.

PART 2 - PRODUCTS

2.1 DVDS

A. Description: Digital video formatted discs.

B. Audio track containing simultaneously recorded narrative commentary and evaluations of videographer, describing in detail condition of pipeline interior.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify location of sewer pipelines to be inspected.

3.2 PREPARATION

A. Flush and clean pipeline to remove sludge, dirt, sand, stone, grease, and other materials to ensure clear view of interior conditions.

B. Debris:
1. Intercept flushed debris at next downstream manhole using weir or screening device.
2. Remove and dispose of debris off Site.

C. Bypassing:
1. Furnish temporary bypass pumping system around Work area for time required to complete television inspection.
2. Provide standby pump of equal or greater capacity at bypass location.
3. Provide safety precautions, including barricades, lights, and flaggers.

3.3 APPLICATION

A. Closed-Circuit Television (CCTV) Camera System:
1. Use cameras specifically designed and constructed for closed-circuit sewer line inspection.
2. Use camera equipment with pan-and-tilt capability to view each lateral connection at multiple angles.
3. Use camera capable of moving both upstream and downstream with minimum 250 feet horizontal distance within one setup and direct-reading cable position meter.

3.4 FIELD QUALITY CONTROL

A. Pipeline Inspection:
1. Audio-video record sections of sewer pipeline between designated manholes.
2. Identify and record locations of flat grades, dips, deflected joints, open joints, broken pipe, protrusions into pipeline, and points of infiltration.
3. Locate and record service connections.
4. Record locations of pipeline defects, connection horizontal distance in feet, and direction from manholes.
5. Video-record with pipe section plugged to view 100 percent of pipe ID.
6. Use flow-control methods as specified for bypass pumping system to eliminate surcharging and to reduce flow.

END OF SECTION

SECTION 33 05 05.33

INFILTRATION AND EXFILTRATION TESTING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Testing of Gravity Sewer Piping:
      a. Exfiltration testing.
      b. Infiltration testing.
   2. Testing of Manholes: Exfiltration testing.

1.2 SUBMITTALS

A. Submit following items prior to start of testing:
   1. Testing procedures.
   2. List of test equipment.
   3. Testing sequence schedule.
   4. Provisions for disposal of flushing and test water.
   5. Certification of test gage calibration.

B. Test and Evaluation Reports: Indicate results of manhole and piping tests.

C. Qualifications Statement:
   1. Qualifications for applicator.

1.3 QUALITY ASSURANCE

A. Perform Work according to State of Arizona standards.

B. Applicator: Company specializing in performing Work of this Section with minimum three years' documented experience.

PART 2 - PRODUCTS

2.1 EXFILTRATION TESTING

A. Equipment:
   1. Plugs.
   2. Pump.
   3. Measuring device.

2.2 INFILTRATION TESTING

A. Equipment: Weirs.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify that manholes and piping are ready for testing.

B. Verify that trenches are backfilled.

3.2 PREPARATION

A. Lamping:
   1. Lamp gravity piping after flushing and cleaning.
   2. Perform lamping operation by shining light at one end of each pipe section between manholes.
   3. Observe light at other end.
   4. Pipe not installed with uniform line and grade will be rejected.
   5. Remove and reinstall rejected pipe sections.
   6. Clean and lamp until pipe section is installed to uniform line and grade.

B. Plugs:
   1. Plug outlets, wye branches, and laterals.
   2. Brace plugs to resist test pressures.

3.3 FIELD QUALITY CONTROL

A. Exfiltration Testing of Pipes Larger Than 36 Inches in Diameter:
   1. Perform exfiltration testing not exceeding 100 gal. for each inch of pipe diameter for each mileper day for each reach of piping undergoing testing.
   2. Perform testing with minimum positive head of 2 feet.

B. Infiltration Testing:
   1. Maximum Allowable Infiltration: 100 gal./in.of pipe diameter for each mile per day for reach of piping undergoing testing.
   2. Include allowances for leakage from manholes.
   3. Perform testing with minimum positive head of 2 feet.

C. Manhole Testing:
   1. Repair both outside and inside of joint to ensure permanent seal.
   2. Test manholes with manhole frame set in place.
   3. Plug pipes in manhole.
   4. Remove water from manhole.
   5. Observe plugs over period of not less than two hours to ensure that there is no leakage into manhole.
   6. Determine ground water level outside manhole.
   7. Fill manhole with water within 4 inchesof top of cover frame.
   8. Prior to testing, allow manhole to soak from minimum of four hours to maximum of 72 hours.
   9. After soak period, adjust water level inside manhole to within 4 inchesof top of cover frame.
   10. Measure water level from top of manhole frame.
   11. At end of four-hour testing period, again measure water level from top of manhole frame; compute drop in water level during testing period.
   12. Manhole exfiltration test is considered satisfactory when drop in water level is less than following:
      a. Manhole Depth of 4 Feet:
         1) Diameter of 4 feet: 0.11 inch.
         2) Diameter of 5 feet: 0.14 inch.
         3) Diameter of 6 feet: 0.17 inch.
      b. Manhole Depth of 6 Feet:
         1) Diameter of 4 feet: 0.17 inch.
         2) Diameter of 5 feet: 0.21 inch.
         3) Diameter of 6 feet: 0.26 inch.
      c. Manhole Depth of 8 Feet:
         1) Diameter of 4 feet: 0.23 inch.
         2) Diameter of 5 feet: 0.29 inch.
         3) Diameter of 6 feet: 0.35 inch.
      d. Manhole Depth of 10 Feet:

1) Diameter of 4 feet: 0.28 inch.
2) Diameter of 5 feet: 0.35 inch.
3) Diameter of 6 feet: 0.42 inch.

e. Manhole Depth of 12 Feet:
1) Diameter of 4 feet: 0.34 inch.
2) Diameter of 5 feet: 0.43 inch.
3) Diameter of 6 feet: 0.51 inch.

f. Manhole Depth of 14 Feet:
1) Diameter of 4 feet: 0.40 inch.
2) Diameter of 5 feet: 0.50 inch.
3) Diameter of 6 feet: 0.60 inch.

g. Manhole Depth of 16 Feet:
1) Diameter of 4 feet: 0.45 inch.
2) Diameter of 5 feet: 0.56 inch.
3) Diameter of 6 feet: 0.68 inch.

h. Manhole Depth of 18 Feet:
1) Diameter of 4 feet: 0.51 inch.
2) Diameter of 5 feet: 0.64 inch.
3) Diameter of 6 feet: 0.77 inch.

i. Manhole Depth of 20 Feet:
1) Diameter of 4 feet: 0.57 inch.
2) Diameter of 5 feet: 0.71 inch.
3) Diameter of 6 feet: 0.86 inch.

j. Manhole Depth of 22 Feet:
1) Diameter of 4 feet: 0.62 inch.
2) Diameter of 5 feet: 0.78 inch.
3) Diameter of 6 feet: 0.93 inch.

k. Manhole Depth of 24 Feet:
1) Diameter of 4 feet: 0.68 inch.
2) Diameter of 5 feet: 0.85 inch.
3) Diameter of 6 feet: 1.02 inch.

l. Manhole Depth of 26 Feet:
1) Diameter of 4 feet: 0.74 inch.
2) Diameter of 5 feet: 0.93 inch.
3) Diameter of 6 feet: 1.11 inch.

m. Manhole Depth of 28 Feet:
1) Diameter of 4 feet: 0.79 inch.
2) Diameter of 5 feet: 0.99 inch.
3) Diameter of 6 feet: 1.19 inch.

n. Manhole Depth of 30 Feet
1) Diameter of 4 feet: 0.85 inch.
2) Diameter of 5 feet: 1.06 inch.
3) Diameter of 6 feet: 1.28 inch.

13. If unsatisfactory testing results are achieved, repair manhole and retest until result meets criteria.
14. Repair visible leaks regardless of quantity of leakage.

END OF SECTION

This page intentionally left blank.

SECTION 33 05 15
PRECAST CONCRETE UTILITY STRUCTURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Precast concrete utility structures, non-circular in plan, and appurtenant items.
      a. Valve and meter vaults.
   2. Design and fabrication of precast concrete utility structures.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 08 31 00 - Access Doors.
   2. Section 09 96 00 - High Performance Industrial Coatings.
   3. Section 31 23 00 - Earthwork
   4. Section 31 23 33 – Utilities - Trenching and Backfilling.
   5. Section 33 05 61 – Concrete Manholes

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. C857, Standard Practice for Minimum Design Loading for Underground Precast Concrete Utility Structures.
      b. C858, Standard Specification for Underground Precast Concrete Utility Structures.
      c. C890, Standard Practice for Minimum Structural Design Loading for Monolithic or Sectional Precast Concrete Water and Wastewater Structures.
      d. C990, Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants.
      e. D1227, Standard Specification for Emulsified Asphalt Used as a Protective Coating for Roofing.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
   2. Concrete mix design(s):
      a. Include submittal information defined in Specification Section 03 31 30.
      b. Certification in accordance with ASTM C858, Section 12.
   3. Fabrication and/or layout drawings:
      a. Include detailed diagrams of utility structures showing typical components and dimensions, reinforcement, and other details.
      b. Itemize, on separate schedule, elevations or sectional breakdown of each utility structure with all components and refer to drawing identification number or notation.
      c. Indicate required penetration details for all piping entering each structure.
   4. Drawings and calculations: All Drawings, including layout drawings, certifications and calculations shall be sealed by a Professional Engineer registered in the State of Arizona.
      a. Provide certification stating that calculations provided have been prepared specifically for this Project and that they match and pertain to the Shop Drawings provided.
      b. Provide a summary document as part of the above certification listing the design criteria used for precast design including:
         1) Codes and standards.
         2) Soil load.
         3) Exterior groundwater load.

4) Live loads.
5) Other loads.

5. Test Reports:
   a. Copies of source quality control tests, including compressive strength and air content, for units provided.

1.4 SITE CONDITIONS

A. Design groundwater elevation for precast structure design shall be the 100-year flood elevation shown on the Contract Documents.
   1. If the 100-year flood elevation is not shown on the Contract Documents, the design groundwater elevation shall be equal to the ground surface elevation at the structure.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Access doors:
      a. Conform to requirements of Specification Section 08 31 00.
   2. Manhole rings, covers and frames:
      a. Conform to requirements of Specification Section 33 05 61 for nonpressure type frames.
   3. Premolded joint sealant:
      a. Ram-Nek, Henry Co.
      b. Or Equal.
   4. Elastomeric joint seals:
      a. Kent Seal.
      b. Or Equal.
   5. External joint wrap.
      a. RUBR-Nek, Henry Co.
      b. Or Equal.
   6. Emulsified fibrated asphalt compound:
      a. BASF MasterSeal 615.
      b. Or equal.
   7. Polymer Concrete Vault
      a. Armorcast
      b. Or equal.

2.2 PRECAST UTILITY STRUCTURE COMPONENTS

A. For Volumes 1, 3, and Volume 5 (Polymer Concrete Vaults):
   1. Provide utility structures with interior dimensions as shown on the Drawings.
   2. Load rating: 20K
   3. Provide the following components for each utility structure:
      a. Open bottom.
      b. Precast vault.
      c. Polymer concrete cover with non-skid surface.
   4. Provide openings and appurtenances as shown on Drawings.

a. Access doors:
    1) Spring assisted.
    2) Tamper proof.

B. For Volume 1, 2, and 6 (Precast Concrete Vaults):
1. Provide utility structures with interior dimensions as shown on the Drawings.
2. Provide the following components for each utility structure:
    a. Precast base section with integral or cast in place base slab.
    b. Precast wall section(s).
    c. Precast flat top.
        1) Where reinforcement is shown for top slab, furnish slab with reinforcing as designed, but not less than reinforcing shown on Drawings.
3. Provide openings and appurtenances as shown on Drawings.
    a. Access doors:
        1) Cast access doors into top slab.
        2) Where access door frames have drainage channels, cast PVC drain lines in top slab to drain location shown on Drawings. If no drain location is shown, drain frames to outer edge of top slab.
        3) Protect doors and frames from damage during concrete placement and shipping.
        4) See Specification Section 08 31 00.
    b. Manhole frames and covers:
        1) Furnish and install in accordance with Specification Section 33 05 61.
        2) Cast frames into top slab.
4. Concrete:
    a. Conform to requirements of Specification Section 03 31 30.
    b. Conform to requirements of ASTM C858, where stricter than Specification Section 03 31 30.
        1) Minimum 28-day compressive strength 4500 psi.
        2) All portions of precast utility structure are considered to be exposed to freeze-thaw cycles.
5. Joints:
    a. Joints of precast riser and top sections:
        1) Preformed flexible joint sealants: ASTM C990.
        2) Exterior joint wrap.
    b. Pipe and conduit entry for utility structures.
        1) In accordance with Specification Section 01 73 20.
6. Coatings:
    a. Vertical wall surfaces:
        1) Emulsified fibrated asphalt compound meeting ASTM D1227 Type I for all vertical wall exterior surfaces.

2.3 DESIGN

A. General Design Requirements:
1. Design precast units and appurtenances in accordance with ASTM C858.
    a. Notify Contracting Officer and furnish cast-in-place structures if sizes of precast utility structures shown on Drawings can not be designed or fabricated.

B. Design loads:
1. Design precast units for all loads and load cases described in ASTM C857, with the following values and selections:
    a. Minimum uniform live load for exposed roof slabs shall be 150 PSF.
    b. Wheel loads shall be considered.
        1) Use wheel load designation A-16 (HS20-44) as shown in ASTM C857, Table 1.
            a) For Volumes1, 3, and 5 add a 30% load increase.
        2) Wheel loads and uniform live load do not act concurrently.

c. Unit weight of soil W shall be taken as no less than 130 LB/CUFT.
d. Minimum lateral soil pressure coefficient (K_0):0.48.

C. Specific Design Requirements:
1. Out-of-plane shear:
a. Out-of-plane shear shall be shown in the calculations.
b. Wall thickness shall be determined based on meeting design requirements for out-of-plane shear resulting from soil and groundwater loads.
c. Wall sections shall be designed as one-way spans between corners for calculation of out-of-plane shear. Transfer of shear or bending load shall not be considered to be transferred across joints between precast units or between walls and slabs, unless unit as integrally cast together.
d. Use of shear steel reinforcement to increase out-of-plane shear capacity shall be prohibited.
2. The distribution of moments in adjacent walls of different lengths in rectangular structures shall be considered.
3. Design precast units taking into account reduced cross section at openings and penetrations.
4. Structure shall be checked for buoyancy.
a. The minimum factor of safety for uplift with the design groundwater elevation shall be 1.25, unless a larger factor of safety is required by the local governing body or building code.
b. The minimum factor of safety for uplift at the fully submerged condition shall be no less than 1.0.
c. If the buoyant weight of soil above base slab extensions beyond the external dimensions of the structure is used to resist uplift, the volume of soil considered to resist uplift shall be limited to soil within the vertical projection of the edge of the base slab extensions.

PART 3 - EXECUTION

3.1 PRECAST UTILITY STRUCTURE CONSTRUCTION

A. General:
1. Prepare subgrade for base as required by Specification Section 31 23 33.
a. For precast base slabs, place and compact 6 inches of Granular Fill or Bedding Material as shown on Drawings.
b. For cast-in-place concrete base slabs, support base section and prepare bottom joint with preformed strip-type hydrophilic waterstop in accordance with Specification Section 03 31 31.
c. Confirm that base is level and fully supported by stable material.
2. For structures with open pipe flow, make inverts in accordance with Section 33 05 61.
3. Ensure accurate vertical placement and leveling prior to placement of interior grout.
a. Provide vertical alignment tolerance of maximum 1 inch horizontal to 10 feet vertical.

B. Build each structure to dimensions shown on plans and at such elevation that pipe sections built into wall of structure will be true line of pipe extensions.

C. For all horizontal mating surfaces between precast concrete units, apply premolded flexible joint sealant to clean mating surfaces in accordance with sealant manufacturer's written instructions. Apply sufficient pressure to each concrete unit to seat unit in sealant.

D. Seal all pipe penetrations in manhole.
1. Where post-installed seals are permitted, form pipe openings smooth and well shaped.
2. After installation, seal exterior of penetration with non-shrink grout.
3. After grout cures, wire brush smooth and apply two coats emulsified fibrated asphalt compound to minimum wet thickness of 1/8 inches to ensure complete seal.

E. Set top slab level to elevation shown on Drawings.

3.2 FIELD QUALITY CONTROL

A. Any proposed repairs of precast components or structures shall be submitted to Contracting Officer for approval.

B. Structures shall be observed for signs of leakage during periods of high groundwater.

C. No leakage that includes visible flow through joints between precast concrete sections or through pipe penetrations shall be permitted.

D. Damp spots on interior wall surfaces shall be considered leakage and shall not be permitted.
   1. Damp spots shall be defined as spots where moisture from a source outside the structure can be picked up on a dry hand.
   2. Locate the source of water movement through the wall and permanently seal.

E. Dampness on the top of the base slab will not be construed as leakage.

END OF SECTION

This page intentionally left blank.

SECTION 33 05 24
STEEL UTILITY PIPE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Coated and welded steel pipe, specials and fittings.
      a. Fittings and Specials: Including, but not limited to fittings, closure pieces, bends, elbows, reducers, tees, wyes, bifurcations, crosses, outlets, manifolds, nozzles, wall sleeves, bulkheads, vent pipes, and other piping and appurtenances fabricated from steel plate, sheet, or coil as required to provide the Work, complete.
      b. Specials also include piping above ground or inside structures.
      c. F_Y: Material Yield Stress, PSI

A. Furnish all materials, tools, labor, supervision, and appurtenances for and properly install, connect, adjust, test, disinfect potable systems, and place in continuous satisfactory service all welded steel pipe and fittings at the locations and to the elevations indicated on the Drawings, specified, or required for the proper completion of the Work.

B. Welded steel pipe and fittings are not necessarily completely indicated or detailed on the Drawings.
   1. Drawings are schematic only and indicate pipe and fittings in a general way.
   2. Furnish all materials, pipe and fittings required.

C. Provide an installation, complete in every detail, whether indicated on the Drawings, or specified herein.

D. Related Sections include but are not necessarily limited to:
   1. Section 09 96 00 - High Performance Industrial Coatings.
   2. Section 10 14 00 - Identification Devices.
   3. Section 31 23 33 – Utilities - Trenching and Backfilling.
   4. Section 33 11 13 – Water Main Construction.

1.2 QUALITY ASSURANCE

A. Reference Standards: Standards referenced in this section include, but are not necessarily limited to, the following:
   1. American Society of Mechanical Engineers (ASME):
      a. B1.1, Unified Inch Screw Threads (UN and UNR Thread Form).
      b. B1.20.1, Pipe Threads, General Purpose (Inch).
      c. B16.5, Pipe Flanges and Flanged Fittings and Other Special Alloys.
      d. B16.9, Factory Made Wrought Steel Buttwelding Fittings.
      e. B16.11, Forged Steel Fittings, Socket-Welding and Threaded.
      f. B18.2.1 Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series).
      g. B18.2.2, Nuts for General Applications: Machine Screw Nuts; and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series).
      h. B36.10, Welded and Seamless Wrought Steel Pipe.
   2. ASTM International (ASTM):
      a. A36, Standard Specification for Carbon Structural Steel.
      b. A53, Standard Specification for Pipe, Steel, Black and Hot Dipped, Zinc Coated, Welded and Seamless.
      c. A139, Standard Specification for Electric Fusion Arc – Welded Steel Pipe (Sizes 4-IN and Over).

d. A181, Standard Specification for Forged or Rolled Steel Pipe Flanges, Forged Fittings, and Valves and Parts for General Service.
e. A283, Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates, Shapes, and Bars.
f. A307, Standard Specification for Carbon Steel Externally and Internally Threaded Standard Fasteners.
g. A370, Standard Test Methods and Definitions for Mechanical Testing of Steel Products.
h. A568, Standard Specification for General Requirements for Steel, Sheet, Carbon, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled.
i. A370, Standard Test Methods and Definitions for Mechanical Testing of Steel Products.
j. A435/A435M, Standard Specification for Straight Beam Ultrasonic Examination of Steel Plates.
k. A1011, Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability.
l. B6, Standard Specification for Zinc.

3. American Welding Society (AWS):
   a. A2.4, Standard Symbols for Welding, Brazing, and Nondestructive Examination.
   b. A3.0, Standard Welding Terms and Definitions.
   c. B2.1, Specification for Welding Procedure and Performance Qualification.
   d. D1.1, Structural Welding Code-Steel.
   e. QC1, Standard for AWS Certification of Welding Inspectors.
4. American Water Works Association (AWWA):
   a. C200, Steel Water Pipe 6 inch and Larger.
   b. C206, Field Welding of Steel Water Pipe.
   c. C207, Steel Pipe Flanges for Waterworks Service Sizes 4 inch through 144 inch.
   d. C208, Dimensions for Fabricated Steel Water Pipe Fittings.
   e. C209, Tape Coatings for Steel Water Pipe and Fittings.
   f. C214, Tape Coating Systems for the Exterior of Steel Water Pipelines.
   g. C216, Heat-Shrinkable Cross-Linked Polyolefin Coatings for Steel Water Pipe and Fittings.
   h. C217, Microcrystalline Wax and Petrolatum Tape Coating Systems for Steel Water Pipe and Fittings.
   i. C219, Bolted Sleeve-Type Couplings for Plain-End Pipe.
   j. C222, Polyurethane Coatings for the Interior and Exterior of Steel Water Pipe and Fittings.
   k. C604, Installation of Steel Water Pipe – 4 In. (100 mm) and Larger.
   l. C606, Grooved and Shouldered Joints.
   m. M11 Steel Water Pipe – A Guide for Design and Installation.
5. National Association of Corrosion Engineers (NACE):
   a. RP 02-74, High-Voltage Electrical Inspection of Pipeline Coatings.
   b. TM 01-70, Visual Standard for Surfaces of New Steel Air Blast Cleaned with Sand Abrasive.
   c. TM 01-75, Visual Comparator for Surfaces of New Steel Centrifugally Blast Cleaned with Steel Grot and Shot.
6. National Sanitation Foundation (NSF):
   a. 60, Drinking Water Treatment Chemicals - Health Effects.
   b. 61, Drinking Water System Components - Health Effects.
7. Society for Protective Coatings (SPCC):
   a. SP1, Solvent Cleaning.
   b. SP2, Hand Tool Cleaning.
   c. SP3, Power Tool Cleaning.
   d. SP6/NACE No.3, Commercial Blast Cleaning.

e. SP10/NACE No.2, Near-White Metal Blast Cleaning.
f. SP11, Power Tool Cleaning to Bare Metal.

B. Qualifications
1. Supplier qualifications:
a. Minimum five years experience in supplying spiral weld pipe between 2 IN and 12 IN DIA.
b. ISO-9001 or SPFA certified.
2. Application of coating materials including preparation of surfaces, priming, and lining and coating of pipe, fittings, and specials, in shop, repairs of any damage to lining or coating occurring during shipment or any other time, and field lining and coating of ends where linings or coatings have been held back for welded field joints, shall be done by established and recognized pipe company acceptable to Engineer.
3. Use only certified welders meeting procedures and performance outlined in ASME Section IX, AWWA C200 Section 3.3.3 and other codes and requirements per local building and utility requirements.

1.3 SUBMITTALS

A. Action Submittals.
1. Shop Drawings
2. Product data for the following:
a. Pipe.
b. Fittings.
c. Coatings and Linings.
d. Flanged Joints.
e. Mechanical Joints.

B. Informational Submittals.
1. Certificates.
a. Manufacturer's Certificate of Compliance that products furnished meet requirements of this Specification.
b. Steel Pipe Installation: Certificate that training has been provided to Contractor's installation crews regarding proper pipe handling and storage procedures.
2. Statements of Qualification.
a. Pipe manufacturer.
b. Fittings and specials fabricator.
c. Welders or Welding Operators.
d. Pipe Coating manufacturer.
3. Reports.
1) Factory Test Reports
b. Field Quality Control Test Reports.
4. Field Testing Plan: Submit at least 15 days prior to testing and include at least the following information:
a. Testing dates.
b. Piping system and sections to be tested.
c. Method of isolation.
d. Method of conveying water from source to system being tested.
e. Method of disposal of testing water if required by Government.
f. Calculation of maximum allowable leakage for piping sections to be tested.

1.4 DELIVERY, HANDLING, AND STORAGE

A. Delivery.
1. Securely bulkhead or otherwise seal ends of pipe, specials, and fittings prior to loading at manufacturing site.
2. Keep pipe ends sealed until installation.
3. Inspect each pipe and fitting for damage.

4. Repair damage to pipe, fittings, or specials, including linings and coatings, found upon delivery to jobsite or remove from site and replace at no additional cost to the Government.
    a. Remove or smooth out any burrs, gouges, weld splatter or other small defects prior to laying the pipe.

B. Storage.
1. Support pipe securely to prevent accidental rolling and to avoid contact with mud, water, or other deleterious materials.
    a. Support width and placement per pipe manufacturer's recommendations.
2. Carefully handle and protect pipe, fittings, and specials against damage to lining and coating/interior and exterior surfaces, impact shocks, and free fall.
    a. Move pipe with support using sling widths and pick point placements as recommended by pipe manufacturer to limit deflection during handling.
    b. Do not place pipe directly on rough ground but support on sand, bedding or earth berms free of frozen material and rock exceeding 3 inches in diameter per pipe manufacturer's recommendations.
    c. Support in a manner which will protect the pipe against injury whenever stored at the trench site or elsewhere.
3. Repair damage to pipe, fittings, or specials, including linings and coatings, found in stored pipe in accordance with manufacturer's instructions or remove from site and replace at no additional cost to the Government.
4. Gasket Storage:
    a. Store rubber gaskets in cool, well-ventilated place, and do not expose to direct rays of sun.
    b. Do not allow contact with oils, fuels, petroleum, or solvents.
5. Pipe and Specials Protection:
    a. Protect the openings of pipe and specials with suitable bulkheads where the pipe and specials have been cement-mortar lined in the shop to maintain a moist atmosphere and to prevent unauthorized access by persons, animals, water or any undesirable substance.
    b. Maintain bulkheads, fix tears or replace bulkheads damaged.
    c. Introduce water into the pipe to keep the mortar moist where moisture has been lost due to damaged bulkheads.

PART 2 - PRODUCTS

2.1 GENERAL

A. Manufacture, test, inspect and mark steel pipe, fittings, and specials to comply with AWWA C200, C208 and additional requirements of these Contract Documents.

B. Do not fabricate fittings or specials with material that includes the skelp end weld.

2.2 MANUFACTURERS

A. General:
1. Product requirements are to conform with the specified salient characteristics listed in this Specification Section.

B. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Mechanical couplings:

a. Victaulic, Style 809N
b. Or Equal

2. Factory-applied epoxy coatings:
   a. 3M Company, Scotchkote 6233P
   b. Or Equal
3. Liquid Epoxy for welded joints and holiday repairs (buried joints):
   a. Denso Protal 7250
   b. Or Equal
4. Rock shield:
   a. Tuff-N-Nuff
   b. Or Equal

D. Submit request for substitution in accordance with Specification Section 01 33 23.

2.3 MATERIALS

A. All materials used in steel piping systems defined in Section 33 05 24 shall meet or exceed pressure test requirements specified

B. Steel Pipe (Fabricated Type):
1. AWWA C200 and as follows:
   a. Minimum Yield Strength: 42,000 PSI.
   b. Minimum Tensile Strength: 60,000 PSI.
   c. Maximum Measured Yield Strength: Less <=77 percent of measured tensile strength.
   d. Minimum tensile to yield ration of 1.3
   e. Minimum Elongation in 2-inch Gauge Length: 21 percent.
   f. Weldability: Maximum carbon equivalent of 0.45, as measured using AWS D1.1, Annex XI , Guideline on Alternative Methods for Determining Preheat formula: CE=C+(Mn+Si)/6+(Cr+Mo+V)/5+(Ni+Cu)/15.
   g. Maximum carbon content not to exceed 0.25 percent.
   h. Maximum sulfur content not to exceed 0.015 percent.

C. Steel Pipe (Mill Type):
1. ASTM A53, Type E or S

D. Fittings and Specials (For Fabricated Pipe):
1. AWWA C208

E. Fittings and Specials (For Mill Type Pipe):
1. ASTM A234.
2. ASME B16.3, ASME B16.5, ASME B16.9, ASME B16.11.

F. Steel Butt-Weld Fittings:
1. 24 inch and Smaller: In accordance with ASME B16.9 with material conforming to ASTM A234.

G. Steel API Tubing:
   a. Grade P110
   b. Minimum Yield Strength: 110,000 PSI.
   c. Minimum Tensile Strength: 125,000 PSI.

2.4 FABRICATION

A. Provide piping (mill or fabricated) for use in this Project with minimum wall thicknesses as indicated on Drawings
1. Design pipe in accordance with operating pressures shown in drawings for a design stress limited to 50% of yield.

B. Steel Pipe (Mill Type): ASTM A53, Type E or S, with minimum thickness as indicated on drawings.

C. Pipe Linings: None

D. Fabricated Fittings:
   1. AWWA C208.
   2. Assure ratio of radius of bend to diameter of pipe equal to or greater than 1.0.

E. Protective Coatings:
   1. Factory Applied Coating: See Specification Section 09 96 00 – High Performance Industrial Coatings
   1. Field paint pipe in accordance with Section 09 96 00 – High Performance Industrial Coatings.

2.5 JOINTS

A. Field Joints.
   1. Single welded-butt joint where specified or shown on the drawings.
   2. Mechanically coupled or flanged where shown on drawings.
   3. Butt-straps where specified or required for closures.
   4. Minimum pressure rating of joints: equal to pressure rating of adjoining pipe.

B. Shop Joints.
   1. Fabricate in accordance with AWWA C200 as modified herein.
   2. Utilize complete joint penetration (CJP) butt joints for longitudinal, girth, and spiral welds, unless otherwise indicated.
   3. Do not join lengths of pipe in the shop joined using lap joints.

C. Preparation of Joints.
   1. Butt Joint Welded: Plain ends beveled as required by AWWA C200 and Contractor's field Welding Procedure Specification.

D. Beveled Ends (Mitered-End Cuts)
   1. Moderate deflections and long radius curves may be made using beveled ends with the maximum total allowable angle as indicated.
   2. Use only with singed welded-butt joints, unless specifically approved in writing by Contracting Officer's Representative.
   3. Maximum Total Allowable Angle:
      a. 5 degrees per pipe joint where for welded joints are utilized on adjacent joints.
   4. Combining miters and field joint pulls with shop fabrication will not be accepted without prior Contracting Officer's Representative approval.
   5. Provide miter-cut that is cold expanded square with face of miter-cut on bell ends only.

E. Flanged Joints:
   1. Flange classifications per criteria below.
   2. Flange Coating and Lining.
      a. See Specification Section 09 96 00 – High Performance Industrial Coatings
   3. Provide flanges of ASME B16.5 Pressure Class as indicated in Drawings:
      a. Conform to ASME B16.5, flanges, bolting materials, and flange gaskets.
      b. Do not expose ASME B16.5 flanges to test pressures greater than 150 percent of pressure rating.
      c. If mating to a flat face flange, remove raised face and provide flat face finish in accordance with ASME B16.5.
      d. Gaskets:
         1) Inner Ring: 304 Stainless Steel.
         2) Spiral Winding: 304 Stainless Steel
         3) Filler: PTFE
         4) Outer Ring: Carbon Steel.
      e. Bolts: ASTM A193, B7 stud bolts with two ASTM A194 Grade 2H heavy hex nuts field coated with AWWA C217 wax tape.
   4. Insulating Components:
      a. Dielectric Flange Kits:
         1) Type E - Flat faced.

2) 1/8 IN thick dielectric gasket, phenolic, non-asbestos.
3) Match or exceed pressure rating of flange
4) Maximum temperature: 210 DEG F
5) 1/32 IN wall thickness bolt sleeves
6) 1/8 IN thick phenolic insulating washers.

5. Blind Flanges.
   a. Blind flanges: in accordance with the appropriate standard determined by the maximum operating pressure as described above.
   b. AWWA C606.
   c. Flanged joints: AWWA C207.

F. Dielectric unions:
   a. Screwed end connections
   b. Match or exceed pressure rating of piping system
   c. Maximum temperature: 210 DEG F
   d. Provide dielectric gaskets suitable for continuous operation at union rated temperature and pressure.

2.6 COATINGS

A. See Specification Section 09 96 00 – High Performance Industrial Coatings

B. Buried Pipe:
   1. Factory applied coated with fusion bonded epoxy to a thickness of between 14 and 20 MILS.
      a. Color:
         1) Sherwin Williams SW4003 Pallet Tan
         2) Or Equal
   2. Wrap pipe with rock shield material anywhere bedding materials may be sharp edged or exceed 3/8 IN in size.
   3. Coat field weld locations with epoxy after welding, 20-50 MILS
      a. Color:
         1) Sherwin Williams SW4003 Pallet Tan
         2) Or Equal
   4. Inspect buried pipe coatings with a holiday detector and repair any defects prior to placement of pipe in trench.
      a. Recoat coating defects with liquid coating system to a thickness of 20-50 MILS.
         1) Color:
            a) Sherwin Williams SW4003 Pallet Tan
            b) Or Equal

C. Exposed Pipe:
   1. Factory coated with primer and top coated with liquid epoxy in the field per Specification Section 09 96 00 – High Performance Industrial Coatings
   2. Color:
      a. Sherwin Williams SW4003 Pallet Tan
      b. Or Equal

D. Field Coating of Joints.
   1. Coat buried welded joints with a 2-layer heat-shrink coating system in accordance with AWWA C216
      a. Type I or Type II:
         1) Layer 1: Heat-activated sealant adhesive.
         2) Layer 2: Cross-linked polyolefin backing.
         3) Type I: Tubular type installed prior to joining the pipe ends.
         4) Type II: Wraparound type wrapped circumferentially around the area to be coated; include a separate or pre-attached closure to secure overlap during heating.
   2. Thickness: 60-mils, minimum.

3. Width: 4-inch overlap on existing coating, minimum.
4. Provide a heat-shrink sleeve compatible with the existing coating in accordance with the Manufacturer's recommendations.
5. Consider sleeve shrinkage during heating in determining sizing.

2.7 ACCESSORIES

A. Mechanical couplings and fittings.
   1. Shouldered:
      a. Victaulic Style 809N
      b. Or Equal
   2. Provide bolts, nuts, and washers (when required) of suitable quality, workmanship, and yield strength to ensure compatibility with the coupling design and rated pressure.
      a. Select materials to minimize the possibility of galvanic corrosion.
   3. Coating: See Section 09 96 00 – High Performance Industrial Coatings.

B. Rock Shield:
   1. Install rock shield wrap where steel pipe is buried
   2. Impact Strength: No failure at 6 IN rock using ASTM D792 test method
   3. Resistance to Static Puncture: No failure at 6 IN rock using ASTM G13 test method
   4. Puncture Strength (ASTM D6241): 150 lbs
   5. Compressive Strength (ASTM D-1621): 2.2 lbs/in^2 at 10% deflection
   6. Tear Strength (ASTM D4533): 150 lbs/in
   7. Grab Strength (ASTM D-4632): 99 lbs

C. Test Station, Pipeline Markers and Pipe Locating Tape: See Section 10 14 00 – Identification Devices.

2.8 SOURCE QUALITY CONTROL (PIPE MANUFACTURING)

A. Shop Hydrostatic Pressure Test: In accordance with AWWA C200 Section 5.2, except as follows:
   1. General: Unless specified otherwise, perform testing of pipe, fittings, and specials before lining and coating is applied.
   2. Pipe: Maintain test pressure for minimum of 2 minutes.
   3. Fittings and Specials:
      a. If fabricated from untested straight pipe, test to minimum pressure equal to field test pressure.
      b. Except as otherwise specified herein, no additional shop hydrostatic test will be required on fittings and specials fabricated from successfully tested straight pipe and where new welds are tested as specified.
      c. Hydrostatically test fittings and specials with crotch plates, regardless of whether straight pipe sections used were previously tested.

B. Shop Nondestructive Testing.
   1. Welds: 100 percent visually examined by Certified Weld Inspector to criteria in ASME BPVC SEC VIII, Division 1.
   2. Butt Joint Welds: Spot radiographically examine pipe in accordance with ASME BPVC SEC VIII, Div. 1, Par. UW 52 or 100 percent ultrasonically examined.
   3. Fillet Welds: 100 percent examine using magnetic particle inspection method in accordance with ASME BPVC SEC VIII, Division 1.
   4. Groove Welds: 100 percent ultrasonically examine or radiographically spot examined in accordance with ASME BPVC SEC VIII, Division 1.
   5. Air test collars and wrappers in accordance with AWWA C206.

PART 3 - EXECUTION

3.1 GENERAL

A. Install products in accordance with manufacturer's instructions.

B. Install piping complete with jointing materials and accessories, anchors, and other appurtenances.

C. Prepare trench as specified in Specification Section 31 23 33.

D. Unless otherwise permitted by Contracting Officer's Representative, maximum length of open trench: 300 linear feet or the amount of pipe that can be installed in a single day, whichever is less.

3.2 JOINTING

A. Flanged Joints:
   1. Only install flange joints where shown in Drawings
   2. Thoroughly clean flange faces of all foreign material before the joint is assembled.
   3. Center gasket and draw up connecting flanges watertight without unnecessarily stressing the flanges.
   4. Tighten all bolts in a progressive diametrically opposite sequence and torqued with a suitable, approved, and calibrated torque wrench.
      a. Torque values: as recommended by the pipe manufacturer.
         1) For ASME B16.5 Class 900 flanges: Do not torque bolts more than two times.
            a) Mark each bolt with a single line after bolt has been torqued.
            b) If bolt has been torqued two times, replace all hardware on flange prior to subsequent tightening
   5. Apply all clamping torque to nuts only. Apply wax tape to fully encapsulate all nuts and bolts per manufacturer's recommendations.

B. Mechanical Couplings:
   1. Only install mechanical couplings where shown in Drawings
   2. When installing couplings, take care that the connecting pipe ends, couplings and gaskets are clean and free of all dirt and foreign matter and the surfaces in contact with the gasket are smooth with no surface deformations.
   3. Install in conformity with the recommendation and instruction of the coupling manufacturer and as specified for flanges.
   4. Utilize type and size wrenches as recommended by the coupling manufacturer in bolting couplings.
   5. Tighten coupling bolts to secure a uniform annular space between the follower rings and the body of the pipe with bolts tightened approximately the same amount.
      a. Tighten diametrically opposite bolts progressively and evenly.
      b. For final tightening, use a suitable, approved and calibrated torque wrench set for the torque recommended by the coupling manufacturer.
      c. Apply clamping torque to the nut only.
   6. Apply wax tape to fully encapsulate all nuts and bolts per manufacturer's recommendations.

C. Field Welded Joints.
   1. Perform welds in accordance with AWS D1.1, AWWA C206, approved welding procedures, and referenced welding codes.
      1) In case of conflict, AWS D1.1 shall govern.
   2. Welding Procedures.
      a. Upon completion of each field-welded joint, mark each joint with the welding operator's regular identification number and the last two digits of the year the Work was completed.
         1) Steel stamping directly on piping will not be acceptable unless "low stress" die stamps, such as interrupted dot or round-nose types, are used.
      b. Test field welded joints in accordance with Paragraph entitled "Field Quality Control".

c. Allow Nondestructive testing agency access to field welded joints to perform independent Nondestructive Testing.
d. Repair defective welds at no additional cost to the Government.
e. Following successful tests of the joint, coat the exterior joint spaces as indicated.
1) Holiday test dielectric coating, including heat shrink sleeves, as approved by the Contracting Officer's Representative.
2) After a successful Holiday test, backfilling may be completed.

D. Threaded and Coupled (T/C):
1. Provide T/C end conditions that meet ASME B1.2 requirements.
2. Furnish pipe with factory-made T/C ends.
3. Field cut additional threads full and clean with sharp dies.
4. Leave not more than three (3) pipe threads exposed at each branch connection.
5. Ream ends of pipe after threading and before assembly to remove burrs.
6. Use Teflon thread tape on male thread in mating joints.

3.3 REPAIR OF SHOP APPLIED COATINGS

A. Coatings: In accordance with coating system provided for adjacent main line pipe.

B. All Other Coatings: In accordance with coating manufacturer's recommendations.

3.4 COATING OF FIELD JOINTS

A. In accordance with coating system provided for adjacent main line pipe.

3.5 TEST STATION/PIPELINE MARKERS

A. Pipeline markers shall be installed at locations specified above and test station markers shall be installed where shown on the drawings except where these facilities will not be under asphalt or concrete pavement.

3.6 WARNING TAPE

A. Install warning tape with all new pipelines.
1. See Specification Section 10 14 00.

3.7 FIELD QUALITY CONTROL

A. Field Welding.
1. Tests and inspections indicated in this section to be performed by Certified Welding Inspector.
2. Visual Test Inspect all welds (100 percent inspection) and mark to indicate acceptance or rejection.
3. Inspect 100 percent of butt joint welds with full circumference magnetic particle or ultrasonic testing.
4. Weld Acceptance:
a. Perform visual inspection in accordance with AWS D1.1
b. Perform either magnetic particle testing in accordance with AWS D1.1 or Ultrasonic testing in accordance with ASME BPVC, Section VIII, Division 1
c. Acceptance: in accordance with standards specified above.
5. Remove defective welds in manner that permits proper and complete repair by welding at no additional cost to the Government.
6. Caulking or peening of welds is not permitted.
7. Pay for Certified Welding Inspector retesting of unsatisfactory welds at no additional cost to Government.

3.8 HYDROSTATIC TESTING

A. Hydrostatic testing of steel pipe: See Specification Section 33 11 33.

3.9 PIPELINE DISINFECTION

A. Disinfection of steel pipe: See Specification Section 33 11 33.

END OF SECTION

This page intentionally left blank.

SECTION 33 05 28
PIPE - SPOOLABLE COMPOSITE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Spoolable composite pipe for North Rim Supai Tunnel (Volume 3).

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 33 11 13 - Water Main Construction.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI):
      a. B16.5, Pipe Flanges and Flanged Fittings.
      b. 61, Drinking Water System Components – Health Effects
   2. Code of Federal Regulations (CFR):
      a. 49 CFR 192.121, Design of Plastic Pipe.
      b. 49 CFR 192.123, Design Limitations for Plastic Pipe.
   3. American Society for Testing and Materials (ASTM):
      a. A312, Stainless Steel Pipe Specification Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
   4. American Petroleum Institute (API):
      a. 15S, Spoolable Reinforced Plastic Line Pipe.
      b. 17J, Unbonded Flexible Pipe.

1.3 DEFINITIONS

A. OHTC: Overall Heat Transfer Coefficient.

B. OBC: Operating Bend Radius.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 05 00.
   2. Certifications:
      a. Manufacture installer certification.
      b. ANSI 61 certification status for all components coming into contact with potable water.
   3. Hydrostatic Factory Test:
      a. 1.3 times the maximum pipe pressure rating for an 8-hour period for each reel of spooled pipe.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Spoolable Composite Pipe:
      a. FlexSteel.
      b. No substitutions allowed.

2.2 PIPING

A. General:

1. Composite pipe:
    a. Provide 2,250 psi pressure rated pipe with fittings and appurtenances to locations shown on Drawings.
    b. Furnish materials in accordance with API 17J and API 15S in full compliance to the following material specifications:
        1) Operating Bend Radius: 2.2 FT
        2) Weight: 3.4 LB/FT
        3) Maximum Operating Temperature: 150°F
        4) OHTC: 6.9 BTU/hr·ft²·°F.
        5) Maximum Installation Tension: 10,500 LBS
        6) Liner: NSF 61 compliant

B. Fittings:
    1. ASTM A312 TP 316L.
    2. At North Rim Plateau: ANSI B16.5 Class 150 lapped flanges to connect to other pipeline components.
    3. At Supai Tunnel Rest Area: ANSI B16.5 Class 300 lapped flanges to connect to other pipeline components.

C. Factory Testing:
    1. Provide hydrostatic testing at 1.3 times the pipe pressure rating for an 8-hour period for each reel of spooled pipe.

PART 3 - EXECUTION

3.1 IDENTIFICATION

A. Identify each length of pipe clearly at intervals of 5 FT or less.
    1. Include manufacturer's name and trademark.
    2. Nominal size of pipe, appurtenant information regarding polymer cell classification and critical identifications regarding performance specifications, and "NSF" approvals when applicable.

3.2 INSTALLATION

A. See Specification Section 33 11 13 – Water Main Construction.

B. Storage:
    1. Pipe is to remain on reels until installation.
    2. The storage temperature must remain within the acceptable manufacturer’s recommendations.
    3. Cover and seal the cut ends of the pipe to protect the steel layer from environmental conditions.

C. General:
    1. Prior to installation:
        a. Install permanent and temporary guides to constrain pipe to alignment as shown on the Drawings
        b. Install rock anchors per specification section 31 68 13 – Rock Foundation Anchors
    2. Monitor the power winch unit gauges to determine the pull pressure applied when pulling the pipe.
        a. Pulling force is not to exceed maximum installation tension per manufacturer.
    3. Install the fitting within 1-2 days following installation.
        a. If it is unknown that a fitting will be installed within 1-2 days of installation provide temporary support of the pipe as directed by the Manufacturer’s representative.

D. Fitting Procedures:

1. Prior to installing the fitting, the end of the spoolable pipe is laid out and visually inspected to be clean of debris and inspected for any damage or defects that could compromise a proper seal.
2. Position each reel or coil so that the beginning of the pipe faces the end of the previous pipe, creating an “S” bend.
3. Place the pipe in a straight approach to insert it into the swage machine while maintaining the roundness of the pipe end to properly accept the fitting.
4. The external sleeve of the fitting is swaged down using the swage machine to develop the clamping force and fluid seal.
5. After a fresh cut is made at the end of the pipe, the pipe is deburred, cleaned and continuity wires installed.
6. Welds at end fittings shall be passivated or performed in such a manner as to not create heat tint oxide.

E. Pipe Annulus Pressure Test:
   1. After installation, perform air pressure test of the annular space between the HDPE liner and jacket per manufacturer installation testing requirements.

F. Documentation:
   1. Fitting installation inspection logs (one per fitting) that certifies that installation was performed in accordance with the procedure is completed and emailed to manufacturer within 48 hours following installation to maintain certification.

G. Technicians to supply daily field tickets. Hydrostatic Pipe Conditioning/Depressurization:
   1. See Specification Section 33 11 13 – Water Main Construction

H. Cleaning, Flushing, and Disinfection:
   1. See Specification Section 33 11 13 – Water Main Construction

END OF SECTION

This page intentionally left blank.

SECTION 33 05 31.11
POLYVINYL CHLORIDE GRAVITY SEWER PIPE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. PVC gravity sewer pipe and fittings.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 31 23 33 - Trenching and Backfilling.
   2. Section 33 31 11 - Gravity Sewer Pipeline and Manhole Construction.

1.2 QUALITY ASSURANCE

A. See Specification Section 33 31 11.

B. Referenced Standards:
   1. ASTM International (ASTM):
      a. PVC (polyvinyl chloride) materials:
         1) D1784, Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds.
         2) D2564, Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems
         3) D2729, Standard Specification for Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings
         4) D3034, Standard Specification for Type PSM Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings.
         5) D3212, Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals.
         6) F477, Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe.
         7) F679, Standard Specification for Poly(Vinyl Chloride) (PVC) Large-Diameter Plastic Gravity Sewer Pipe and Fittings.
         8) F1336, Standard Specification for Poly(Vinyl Chloride) (PVC) Gasketed Sewer Fittings
         9) D2321, Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications.

1.3 DEFINITIONS

A. DR: Dimension Ratio, Outside Diameter/Minimum Wall Thickness, both in inches

B. Diametral Deflection: reduction in diameter caused by earth and surcharge loads acting on the installed pipe.

C. PS: Pipe Stiffness, PSI

D. SDR: Standard Dimension Ratio.

1.4 SUBMITTALS

A. Action Submittals:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Copies of manufacturer's written directions regarding material handling, delivery, storage and installation.
      c. Schedule sheet showing compliance of all system components.

1) Attach technical product data on gaskets, pipe, fittings, and other components.

B. Where an option for joint type is specified, submit jointing method(s) selected and where each will be used.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. JM Eagle
   2. Or equal.

2.2 PVC DRAINAGE AND SEWER PIPING

A. Materials:
   1. PVC pipe and fittings: rigid, unplasticized polyvinyl chloride (PVC) made of PVC plastic having a cell classification of 12454-B or 12454-C per ASTM D1784.

B. Pipe:
   1. ASTM D3034
      a. DR: See Drawings
   2. Ensure impact strengths and pipe stiffnesses are in full compliance to these Specifications.

C. Joints
   1. Elastomeric gasket joint meeting requirements of ASTM D3212.
      a. Gaskets (seals) per ASTM F477.

D. Fittings:
   1. Elastomeric Joints: Per ASTM D3034 and ASTM F649

PART 3 - EXECUTION

3.1 IDENTIFICATION

A. Identify each length of pipe clearly at intervals of 5 feet or less per governing ASTM Standard for each type of pipe used.

3.2 INSTALLATION

A. See Section 31 23 33.

B. See Section 33 31 11.

C. Install pipe and fittings in accordance with ASTM D2321, the above referenced Specification Sections and as recommended by the manufacturer.

D. Infiltration and Exfiltration:
   1. Hydrostatic exfiltration and infiltration for sanitary sewers (groundwater level is below the top of pipe):
      a. Leakage rate: 200 GAL per inch diameter per mile of pipe per day at average head on test section of 3 FT.
      b. Average head is defined from groundwater elevation to average pipe crown.

c. Acceptable test head leakage rate for heads greater than 3 FT. Acceptable leakage rate (gallons per inch diameter per mile per day) equals 115 by (actual test head to the ½ power).

2. Observe full instructions of the Contracting Officer's Representative for carrying of testing procedures.
3. Perform tests only during presence of the Contracting Officer's Representative or his authorized representative.
4. Should any test on any section of pipeline disclose either infiltration rates greater than allowed or disclose air loss rate greater than that permitted, locate and repair the defective joints or pipes at no cost to Government and retest until requirements stated are met.

E. Diametral Deflection:
1. After backfilling, check each section of pipe for diametral deflection by pulling a mandrel through the pipe.
2. Pipe with deflection exceeding 3 % of the inside diameter: remove and replace to provide a deflection of less than 3 %.
3. Retest any repaired pipe.

END OF SECTION

This page intentionally left blank.

SECTION 33 05 33.23
HIGH DENSITY POLYETHYLENE (HDPE) PIPE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. High Density Polyethylene (HDPE) pipe, fittings, and appurtenances.

B. Related Sections include but are not necessarily limited to:
   1. Section 31 23 33 – Utilities Trenching and Backfilling.
   2. Section 33 11 13 - Water Main Construction.

1.2 QUALITY ASSURANCE

A. Qualifications:
   1. Manufacturer of HDPE Pipe:
      a. Obtain all HDPE pipe and fittings from one manufacturer, unless otherwise acceptable to Government and Contracting Officer's Representative.
      b. Manufacturer of pipe required by this Section shall be regularly engaged in the business of manufacturing HDPE piping of the size and type(s) required for the Work. Upon Contracting Officer's Representative's request, submit documentation of manufacturer's prior furnishing of PE pipe of the type(s) required for not less than five other projects, similar in size and complexity to the HDPE piping Work of this Project, in North America.
      c. Equipment:
         1) Qualify each extrusion line and molding machine to produce pressure rated products by taking representative production samples and performing sustained pressure tests in accordance with ASTM D1598.
      d. QA and QC Program:
         1) HDPE pipe manufacturer shall have documented quality assurance and quality control programs as part of manufacturer's routine business.
         2) HDPE piping manufacturer shall maintain permanent Quality Control (QC) and Quality Assurance (QA) records.
   2. Installer:
      a. Use one entity for all HDPE pipe Work, unless otherwise acceptable to Government and Contracting Officer's Representative.
      b. Installer shall be experienced with HDPE piping installation work similar in scope and complexity to the Work of this Section. When required by Contracting Officer's Representative, submit documentation of successful completion of not less than five similar projects to the HDPE piping Work completed within the past five years.
      c. Where HDPE pipe joints are made by fusing, fusion machine operators shall have been trained by fusion machine manufacturer within 12 months of starting to perform joint fusion Work. Submit documentation of such training, acceptable to Contracting Officer's Representative, when requested by Contracting Officer's Representative.
      d. Installer's training and qualifications shall be acceptable to HDPE pipe Supplier retained for the Project.

B. Referenced Standards:
   1. American National Standards Institute (ANSI):
      a. B1.1, Unified Inch Screw Threads (UN and UNR Thread Form).
      b. B16.1, Cast-Iron Pipe Flanges and Flanged Fittings, Class 25, 125, 250, and 800.
      c. B16.21, Nonmetallic Flat Gaskets for Pipe Flanges.
   2. ASTM International (ASTM):

a. A182, Standard Specification for Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service.
b. A193, Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications.
c. A194, Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both.
d. A536, Standard Specification for Ductile Iron Castings.
e. B695, Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel.
f. D638, Standard Test Method for Tensile Properties of Plastics.
g. D1248, Specification for Polyethylene Plastics Molding and Extrusion Materials.
h. D1505, Test Method for Density of Plastics by the Density-Gradient Technique.
i. D1598, Test Method for Time-to-Failure of Plastic Pipe Under Constant Internal Pressure.
j. D1599, Test Method for Short-Time Hydraulic Failure Pressure of Plastics Pipe, Tubing and Fittings.
k. D2239, Standard Specification for Polyethylene (PE) Plastic Pipe (SIDR-PR) Based on Controlled Inside Diameter.
l. D2290, Test Method for Apparent Tensile Strength of Ring or Tubular Plastics and Reinforced Plastics by Split Disk Method.
m. D2683, Standard Specification for Socket-Type Polyethylene Fittings for Outside Diameter-Controlled Polyethylene Pipe and Tubing.
n. D2737, Standard Specification for Polyethylene (PE) Plastic Tubing.
o. D2837, Standard Test Method for Obtaining Hydrostatic Design Basis for Thermoplastic Pipe Materials or Pressure Design Basis for Thermoplastic Pipe Products.
p. D3035, Standard Specification for Polyethylene (PE) Plastic Pipe (DR-PR) Based on Controlled Outside Diameter.
q. D3261, Butt Heat Fusion Polyethylene (PE) Plastic Fittings for Polyethylene (PE) Plastic Pipe and Tubing.
r. D3350, Standard Specification for Polyethylene (PE) Plastics Pipe and Fittings Materials.
s. F714, Standard Specification for Polyethylene (PE) Plastic Pipe (SDR-PR) Based on Outside Diameter.
t. F1055, Standard Specification for Electrofusion Type Polyethylene Fittings for Outside Diameter Controlled Polyethylene Pipe and Tubing.
u. F2620, Standard Practice for Heat Fusion Joining of Polyethylene Pipe and Fittings.

3. American Water Works Association (AWWA).
   a. C207, Steel Pipe Flanges for Waterworks Service, Sizes 4 In. through 144 In.
   b. C901, Polyethylene (PE) Pressure Pipe and Tubing, ¾ In. through 3 In. for Water Service.
   c. C906, Polyethylene (PE) Pressure Pipe and Fittings, 4 In. through 65 In. for Waterworks.
4. Plastic Pipe Institute (PPI):
   a. TR-3, Policies and Procedures for Developing Recommended Hydrostatic Design Stresses for Thermoplastic Pipe Material.
   b. PPI TR-4, Recommended Hydrostatic Strengths and Design Stresses for Thermo-plastic Pipe and Fittings Compounds.
5. Military Specification (Mil Spec):
   a. QQ-P-416F, Plating, Cadmium Electro Deposited.
6. National Sanitation Foundation International (NSF):
   a. 14, Plastic Piping Components and Related Materials.
   b. 61, Drinking Water System Components – Health Effects.

1.3 DEFINITIONS

A. DIPS: Ductile Iron Pipe Size

B. CTS: Copper Tube Size.

C. DR: Dimension Ratio, Outside Diameter/Minimum Wall Thickness, in inches.

D. Diametral Deflection: reduction in diameter caused by earth and surcharge loads acting on the installed pipe.

E. ESCR: Environmental Stress Crack Resistance.

F. HDB: Hydrostatic Design Basis per ASTM D2837.

G. IPS: Iron Pipe Size.

H. SDR: Standard Dimension Ratio.

I. SIDR: Standard Inside Diameter Ratio.

1.4 SUBMITTALS

A. Action Submittals: Submit the following:
   1. Shop Drawings:
      a. Dimensioned, scaled drawings showing Contractor's proposed deviations from piping alignment shown on the Drawings.
      b. Where necessary, include both plans and profiles and section views in Shop Drawings.
   2. Product Data:
      a. Manufacturer's published catalog information and specifications for each size and type of HDPE piping required, including pipe, fittings, and jointing materials.
      b. Clearly indicate in product data submittal the reference standards with which pipe and appurtenances comply.
      c. Where couplings are required, submit coupling manufacturer's catalog literature and specifications for the specific couplings proposed.

B. Informational Submittals; Submit the following:
   1. Supplier Instructions:
      a. Submit manufacturer's written instructions for handling, storing, and installing piping furnished.
   2. Source Quality Control:
      a. Submit documentation from manufacturer that piping materials furnished were inspected at the production facility and successfully passed manufacturer's quality control procedures.
      b. Manufacturer's affidavit certifying piping materials furnished comply with AWWA standards C906 and C901.
   3. Fusion machines proposed for joining pipe/fittings.
   4. Field Quality Control:
      a. Documentation of temperature and pressure profiles from data logger for each butt fusion joint.
      b. Results of field quality control testing and inspections required in this Section.
      c. Documentation of temperature and pressure profiles from data logger for each butt fusion joint.
   5. Qualifications:
      a. Piping materials manufacturer, when requested by Contracting Officer's Representative.
      b. Installer, when requested by Contracting Officer's Representative, including documentation of training in use of joint fusion equipment.
      c. Documentation showing fusion machine operators have been trained within 12 months prior to installation.

1.5 DELIVERY, HANDLING, AND STORAGE

A. In addition to requirements elsewhere in the Contract Documents, also comply with the following:
   1. During handling and installation avoid conditions where pipe bends excessively and avoid imparting kinks in the pipe.

PART 2 - PRODUCTS

2.1 MANUFACTURER'S

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. HDPE Pipe and Fittings:
      a. JM Eagle.
      b. Or equal.
   2. Restrained Transition Fitting:
      a. Romac Industries, Alpha Restrained Joint Extended Range Coupling.
      b. Or equal.
   3. Butt Fusion Equipment:
      a. McElroy Manufacturing, Inc.
      b. Or equal.
   4. Adaptor Couplings
      a. Central Plastics Co.
      b. Or equal.

2.2 MATERIALS

A. Pipe:
   1. AWWA C906 for pipe4 IN and larger, C901 for pipe less than 4 IN.
      a. PE 4710 polyethylene material.
      b. ASTM D3350 cell classification 445574C/E.
      c. Minimum Hydrostatic Design Basis (HDB) of 1600 psi at 73 degF when tested in accordance with PPI TR-3.
      d. NSF/ANSI 61/14 approved.

B. Fittings:
   1. Same material and thickness as adjoining pipe.
      a. Fittings may be one DR lower (thicker) than the adjoining pipe.
   2. Molded fittings:
      a. Butt fused: ASTM D3261.
      b. Socket: ASTM D2683.
      c. Electrofusion: ASTM F1055.
   3. Fabricated fittings: Fabricate from same material as pipe.
   4. NSF 61 approved.

C. Back-up flanges:
   1. Stainless steel: ASTM A182, dimensions per AWWA C207, Class 150.
   2. Ductile iron: ASTM A536 Grade 70-50-5, dimensions per ANSI B16.1, Class 150.

D. Nuts and Bolts:

1. Buried:
    a. Tee-Head bolts for mechanical joints: Per AWWA/ANSI C111/A21.11.
    b. Other bolts and nuts: ASTM A307 ASTM A193/A194 Grade B8 Class 1.
    c. Wax Tape Coatings per AWWA C217.
2. Exposed: Mechanical galvanized ASTM B695, Class 40 ASTM A193/A194 Grade B8, Class 1.
3. Heads and dimensions per ASME B1.1.
4. Threaded per ASME B1.1.
5. Project ends 1/4 to 1/2 inches beyond nuts.

2.3 MANUFACTURED UNITS

A. Adaptor Fittings:
    1. Mechanical joint: Adaptor fitting butt fused to pipe.
        a. Capable of developing same pressure rating as specified for pipe.
        b. Restrained.

B. Restrained Adapter Fitting:
    1. Restrained coupling for pipe material transition.

C. Butt Fusion Equipment:
    1. Designed to create a joint at least as strong as the adjacent pipe.
    2. Equipment to contain stops to prevent excessive pressure on pipe ends during fusion process.
    3. Equipment to plane pipe ends prior to fusion process.
    4. Provide a data logger to document temperatures and pressures for each butt fusion joint.

2.4 DESIGN CRITERIA

A. DR:
    1. See Drawings

B. Pipe Outside Diameter (OD):
    1. Minimum allowable pipe outside diameter: Per AWWA C906, Table 4 (DIPS sizing).
    2. Minimum allowable pipe outside diameter: Per AWWA C901, Table 4 (IPS sizing).

C. Stub flanges: Design to develop full pressure specified for pipe with blind flange connected to back up flange.

D. Test Pressure:
    1. Force Main – 150 psig hydraulic
    2. Water and raw water – as shown on drawings

2.5 FABRICATION

A. Pipe:
    1. Clearly mark pipe and fittings per AWWA C906 for 4 IN and larger or AWWA C901 for less than 4 IN.
    2. Coiled HDPE Pipe
    3. Service Indication:
        a. Provide integral color or longitudinal color banding at 180 degrees apart.
            1) Potable Water: Black pipe with blue stripes.
            2) Sanitary Sewer or Force Main: Black pipe with green stripes.
            3) Raw Water: Black pipe

B. Stub flanges:
    1. Design for use with backing flange.
    2. Fabricate per manufacturers recommendations to develop full pressure specified for pipe with blind flange attached.

2.6 SOURCE QUALITY CONTROL

A. Materials:
   1. Incoming polyethylene materials:
      a. Inspect for density per ASTM D1505, melt rate per ASTM D1248, and carbon containment.
      b. Provide certification from material supplier for a minimum of 30% of all incoming polyethylene materials.
      c. HDPE pipe manufacturer shall verify certifications prior to processing into finished pipe or fittings.
   2. Outgoing materials:
      a. Manufacturer to inspect for diameter, wall thickness, length, straightness, out-of-roundness, concentricity, toe-in, inside and outside surface finish, markings, and end cut.
      b. Perform tests to determine density, melt flow rate, carbon content, and carbon dispersion.
      c. Test representative samples of the pipe provided to determine hoop tensile strength and ductility by either quick burst per ASTM D1599 or ring tensile per ASTM D2290.

B. Equipment:
   1. Qualify each extrusion line and molding machine to produce pressure rated products by taking representative production samples and performing sustained pressure tests in accordance with ASTM D1598.

C. Quality Control Program:
   1. Pipe and fitting manufacturer shall maintain permanent Quality Control (QC) and Quality Assurance (QA) records.
   2. Certified copies of the quality control data taken during product manufacture shall be supplied to the Government upon request.

D. Factory Inspection:
   1. HDPE pipe manufacturer's production and quality assurance facilities shall be open and available for inspection by Government and its authorized representatives.

PART 3 - EXECUTION

3.1 GENERAL

1. Locations: Provide HDPE piping at the locations shown on the Drawings or as otherwise shown and indicated in the Contract Documents.
2. Install HDPE piping in accordance with:
   a. Section 31 23 33 - Trenching and Backfilling.
   b. Section 33 11 13 – Water Main Construction.
   c. Other provisions of the Contract Documents.
   d. Laws and Regulations.

3.2 INSTALLATION

A. General:
   1. Locations: Provide HDPE piping at the locations shown on the Drawings or as otherwise shown and indicated in the Contract Documents.
   2. Install HDPE piping in accordance with:
      a. Manufacturer's recommendations.
      b. Section 31 23 33 - Trenching and Backfilling.
      c. Other provisions of the Contract Documents.
      d. Laws and Regulations.
   3. Field repairs of pipe and fittings are not permitted, with the exception that portions of straight pipe which are damaged may be cut off provided this alteration is in accordance with the manufacturer's established procedures.

B. Joining method - Fusion Joints:
   1. Joint Fusing – General:
      a. Comply with ASTM F2620, the Contract Documents, pipe manufacturer's written instructions, and fusion machine manufacturer's written recommendations. Where any of the foregoing are inconsistent, comply with the more-stringent requirement. When unclear, request and obtain from Contracting Officer's Representative written interpretation or clarification.
      b. Fusion joiner shall be appropriately qualified and experienced in making the type of fusion joint required (butt fusion, socket fusion or sidewall fusion). Fusion joiner shall perform fusion jointing only for the types of joints for which such person possesses appropriate qualifications and experience.
      c. Cost of certifying its workers and performing testing required for fused joints is Contractor's responsibility.
      d. Provide appropriate protection for fusing joints based on weather conditions.
      e. Fuse HDPE pipe joints on the surface prior to installation into the trench.
      f. Plane ends of joints in fusion machine prior to heating.
      g. Push pipe ends together in fusion machine and visually inspect compatibility and alignment of the two ends.
      h. Comply with pipe manufacturer's written procedures and recommended heating time for pipe ends based upon pipe diameter and ambient temperature.
      i. Visually inspect joint immediately upon removal of heating element to verify blistering of pipe has not occurred. Blistering of pipe is unacceptable.
      j. Do not remove pipe from fusion machine until pipe has cooled in accordance with the pipe manufacturer's written instructions.
   2. HDPE pipe 1 inch diameter and smaller shall be socket fused.
   3. HDPE pipe joints 1.5 inch diameter and larger shall be butt fused.
   4. Electro-Fusion Fittings:
      a. Comply with pipe fitting manufacturer's written instructions.
      b. Ensure ends of connecting pipe are round when fusing.
      c. Couplings: Provide means to place pipe ends in compression when fusion coupler is heated/cooled until connection process is complete.

C. Joining method – Flanged Joints:
   1. Slide back up flange on HDPE pipe.
   2. Install stub flange on end of pipe with butt fusion.
   3. Make flange faces flat and perpendicular to pipe centerline.
   4. Allow one flange free movement in any direction while bolts are being tightened.
   5. Do not assemble adjoining flexible joints until flanged joints in piping system have been tightened.
   6. Gradually tighten flange bolts uniformly to permit even gasket compression.
   7. Check bolt torque after 24 hours to ensure that stress relief has not occurred.

D. Joining method – Mechanical Joint:
   1. Use only for joining HDPE pipe to DIP or PVC (AWWA C900).
   2. Install MJ adapter on end of HDPE pipe with butt fusion.
   3. Clean components before installation.
      a. Clean bolts and nuts by wire brushing.
      b. Lubricate bolts with vegetable-based oil only.
   4. Insert nuts and bolts (or studs), finger tighten, and progressively tighten diametrically opposite bolts uniformly around the flange to the proper tension.
   5. Execute care when tightening joints to prevent undue strain upon adjoining pipe.
   6. Test joint for leakage.
      a. If joint leaks under pressure testing, loosen or remove nuts and bolts, reset, or replace the gasket, reinstall or retighten bolts and nuts, and retest the joints.
      b. Completed joints shall be watertight.
   7. Check bolt torque after 24 hours to ensure that stress relief has not occurred.

E. Install buried pipe per Section 33 11 13 and per details shown on Drawings.
   1. Allow pipe to stabilize to trench bottom or casing temperature before final tie-in or backfilling.
   2. For field sweeps, the minimum cold bending allowable radius:

DR	Minimum Radius
7-9	20 times pipe OD
>9 - 13.5	25 times pipe OD
>13.5 - 21	27 times pipe OD
>21	30 times pipe OD
With flange or fitting inside sweep	100 times pipe OD

3.3 FIELD QUALITY CONTROL

A. Use only fusion machine operators who have been trained by machine manufacturer no more than 12 months prior to installation of first fused joint.

B. Visually inspect all pipe for gouges.
   1. Gouges in excess of 10% of the pipe wall thickness are not acceptable.
   2. In area where excessive gouges are present, cut out affected pipe section and butt fused to remaining pipe to make a continuous section.

C. Check integrity of the heating plate in the fusion equipment a minimum of twice per each 8-hour work shift for temperature uniformity.

D. Butt-fused specimen joint inspection and testing per ASTM F2620, Appendix X4.
   1. Visually inspect all joints during and after joining to ensure that the joint meets the requirements of the butt fusion welding procedure.
   2. On every day that butt fusions are to be made, prepare a trial butt-fused joint specimen.
   3. Allow specimen to cool completely and cut into test straps.
   4. Test straps:
      a. Length: 12 inches (min) or 30 times the wall thickness.
      b. Location of fusion: Center of strap.
      c. Width: 1 inch (min) or 1.5 times the wall thickness.
   5. Visually examine the straps for voids or discontinuities.
   6. Deform the straps by bending, torque, or impact.
   7. If failure of the joint occurs outside of the joint area, the test is acceptable.

E. Diametral Deflection:
   1. After backfilling, check each section of pipe for deflection by pulling a mandrel through the pipe.
   2. Pipe with deflection exceeding 3% of the inside diameter: remove backfill and replace to provide a deflection of less than 3%.
   3. Retest any repaired pipe.

F. Hydrostatically test installed pipe and fittings in accordance with Section 33 11 13.
   1. Provide butt fused stub flanges, back up flanges and blind flanges on both ends of each pipe string for hydrostatic testing prior to and following installation.

END OF SECTION

SECTION 33 05 61
CONCRETE MANHOLES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Precast concrete round manhole structures and appurtenant items.
      a. Sanitary sewer manholes and appurtenances.
      b. Drain manholes and appurtenances.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 21 00 - Reinforcement.
   2. Section 03 31 30 - Concrete Materials and Proportioning.
   3. Section 31 23 33 – Utilties - Trenching and Backfilling.
   4. Section 09 96 00 - High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. A48/A48M, Standard Specification for Gray Iron Castings.
      b. C150/C150M, Standard Specification for Portland Cement.
      c. C478, Standard Specification for Precast Reinforced Concrete Manhole Sections.
      d. C923, Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes, and Laterals.
      e. D1227, Standard Specification for Emulsified Asphalt Used as a Protective Coating for Roofing.
      f. D4586, Standard Specification for Asphalt Roof Cement, Asbestos-Free.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
   2. Fabrication and/or layout drawings:
      a. Include detailed diagrams of manholes showing typical components and dimensions, reinforcements and other details.
      b. Itemize, on separate schedule, sectional breakdown of each manhole structure with all components and refer to drawing identification number or notation.
      c. Indicate knockout elevations for all piping entering each manhole.
   3. Buoyancy uplift and structural calculations.
   4. Drawings shall be signed and sealed by a Professional Engineer registered in the State of Arizona.

B. Unless approved prior to submittal, submit all products from this Specification Section in one complete submittal package. Include all products and accessories together.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient

characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:

1. Manhole rings, covers and frames:
   a. Deeter Foundry.
   b. Or Equal.
2. Black mastic joint compound:
   a. RAM-NEK.
   b. Or Equal.
3. Premolded joint compound:
   a. RAM-NEK.
   b. Or Equal.
4. Emulsified fibrated asphalt compound:
   a. Sonneborn Hydrocide 700B.
   b. Or Equal.

2.2 SANITARY SEWER, STORM AND DRAIN MANHOLE STRUCTURE COMPONENTS

A. Manhole Components:

1. Reinforcement: ASTM C478.
2. Minimum wall thickness: 5 inches.
3. Minimum base thickness: 12 inches.
4. Provide the following components for each manhole structure:
   a. Base (precast) with integral bottom section or (cast-in-place).
   b. Precast bottom section(s).
   c. Precast barrel section(s).
   d. Precast eccentric transition section.
   e. Precast adjuster ring(s).
   f. Precast concrete transition section.
   g. Precast flat top.
5. Unless dimensioned or specifically noted on Drawings, provide manhole section with minimum 48 inches inside dimensions.

B. Nonpressure Type Frames and Cover:

1. Cast iron frame and covers: ASTM A48/A48M, Class 35 (minimum).
2. Use only cast iron of best quality, free from imperfections and blow holes.
3. Furnish frame and cover of heavy-duty construction a minimum total weight of 450 pounds.
4. Machine all horizontal surfaces.
5. Furnish unit with solid nonventilated lid with concealed pickholes.
   a. Letter covers "SEWER" for all collection system manholes and "WATER" for water systems.
6. Ensure minimum clear opening of 24 inches diameter.

C. Pressure Type Frame and Cover:

1. Provide covers meeting the requirements of the Nonpressure Type Frames and Cover paragraph above and as modified below.
2. Furnish frame and bolted cover of heavy-duty construction.
   a. Equip unit with six stainless steel countersunk 3/8 inches diameter by 1-1/2 inches long bolts with stainless steel washers.
3. Provide solid lid and minimum 1/8 inches thick x 1/2 inches wide continuous strip neoprene gasket.
4. Furnish unit with a minimum of six anchorage holes and six, 6 inches long x 3/4 inches diameter stainless steel anchor bolts.

D. Special Coatings and Joint Treatment:
   1. Joints of precast sections:
      a. Black mastic compound: ASTM D4586.
   2. Vertical wall surfaces:
      a. Emulsified fibrated asphalt compound meeting ASTM D1227 Type II for all exterior and interior vertical wall surfaces.

E. Sanitary Sewer Manhole Concrete:
   1. Provide all sanitary manholes constructed with Portland ASTM C150/C150M, Type I or II cement with a tricalcium aluminate content not to exceed 8%.
   2. Mix aggregate shall be a minimum of 50% crushed limestone.
   3. Provide 3000 psi nonshrink grout.

PART 3 - EXECUTION

3.1 MANHOLE CONSTRUCTION

A. General:
   1. Construct cast-in-place concrete base slabs.
   2. Make inverts with a semi-circular bottom conforming to the inside contour of the adjacent sewer sections.
   3. On all straight runs, lay pipe through manhole and cut out top half of pipe.
      a. See detail on Drawings.
      b. If pipes deflect at manhole, shape as specified in Paragraphs 2 and 4 inches this General Paragraph.
   4. Shape inverts accurately and steel trowel finish.
      a. For changes in direction of the sewer and entering branches into the manhole, make a circular curve in the manhole invert using as large a radius as manhole inside diameter will permit.
      b. Pour base slab integral with bottom barrel section.

B. Build each manhole to dimensions shown on plans and at such elevation that pipe sections built into wall of manhole will be true extensions of line of pipe.

C. For all horizontal mating surfaces between concrete and concrete or concrete and metal, above established high groundwater elevation shown trowel apply to clean surface black mastic joint compound to a minimum wet thickness of 1/4 inches immediately prior to mating the surfaces.

D. For horizontal joints that fall below established high groundwater elevation shown, install a resilient O-ring type gasket or pre-molded joint compound.

E. Seal all pipe penetrations in manhole.
   1. Form pipe openings smooth and well shaped.
   2. After installation, seal cracks with, non shrink grout.
   3. After grout cures, wire brush smooth and apply two coats emulsified fibrated asphalt compound to minimum wet thickness of 1/8 inches to ensure complete seal.

F. Set and adjust frame and cover final 6 inches (minimum) to 18 inches (maximum) to match finished pavement or finished grade elevation using precast adjuster rings.

END OF SECTION

This page intentionally left blank.

SECTION 33 11 13
WATER MAIN CONSTRUCTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. The following requirements for “buried” piping for water transmission, distribution, and yard piping:
      a. Coordination and interface with existing facilities and utilities.
      b. Connections to existing water mains.
      c. Testing, flushing and disinfection.

B. See Division 40 specifications for “interior” process piping requirements.

C. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 50 00 – Temporary Facilities and Controls
   2. Section 32 91 13 - Finish Grading.
   3. Section 33 05 24 – Steel Utility Pipe
   4. Section 33 05 28 – Pipe – Spoolable Composite
   5. Section 33 05 33.23 – Polyethylene Pressure Pipe and Tubing (AWWA C901 and AWWA C906)
   6. Section 33 12 19 - Fire Hydrant.
   7. Section 40 05 00 - Pipe and Pipe Fittings - Basic Requirements.
   8. Section 40 05 51 - Common Requirements for Process and Utility Valves.
   9. Section 40 05 61 - Gate Valves.
   10. Section 40 05 63 – Ball Valves

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Water Work Association (AWWA):
      a. B300, Standard for Hypochlorites.
      b. B301, Standard for Liquid Chlorine.
      c. C651, Standard for Disinfecting Water Mains.

1.3 SUBMITTALS

A. Submit results of the leakage tests, identifying the specific length of pipe tested, the test pressure, the duration of test and the amount of leakage.

B. Submit satisfactory bacteriological test reports on disinfection requirements.

C. Submit qualifications for lab performing disinfection analysis.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Pipe: Refer to the following Specification Sections:
   1. 33 05 24 – Steel Utility Pipe
   2. 33 05 28 – Pipe – Spoolable Composite
   3. 33 05 33.23 – High Density polyethylene (HDPE) Pipe

B. In-Line Valves:
   1. Refer to the following Specification Sections:
      a. 40 05 61 – Gate Valves
      b. 40 05 63 – Ball Valves.

2. Provide adjustable valve boxes.
    a. Include price of valve boxes in price of valve installed complete.

C. Fire Hydrants: Refer to Specification Section 33 12 19.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install water main to the line and grade on the Drawings.
    1. Water mains to be staked at a minimum 100 feet interval with depth of cuts monitored.

C. Field verify depth of utilities that will be crossed.
    1. Adjust water main elevation as required during construction.
    2. No separate payment will be made for field verification or adjustment of main depths as required.

D. Contractor will restore all existing structures or services damaged by Contractor's operations at no cost to Government.

3.2 INTERRUPTION OF SERVICE

A. Interruption of service to water users shall not exceed 4 hours.
    1. Notify property owners of interruption a minimum of 24 hours in advance.

3.3 UNDERGROUND SERVICES

A. Notify utility representative prior to construction to obtain available information on location of existing utilities.
    1. Contractor shall be responsible for locating all utilities.

B. Existing water services are not to be connected to the new water mains.
    1. Damage to existing water service to be repaired, using pipe and union the same size and material equal to existing service or other material approved by Contracting Officer’s Representative.

3.4 FLEXIBLE PAVEMENT AND CONCRETE SURFACE REMOVAL AND REPLACEMENT

A. All Portland cement concrete and asphalt noted for removal and replacement shall be cut prior to removal.
    1. Cut by sawing, vertical cut to be 1 inch minimum.
    2. The remaining depth of section may be broken out in a manner subject to Contracting Officer’s Represtative’s approval.
    3. Width of section removed to be either a width not greater than the outside diameter of the water main plus 4 feet-0 inches or broken out to the nearest joint.

B. Replace Portland cement concrete and asphalt equal to or better than original paving plus 2 inches.

C. Debris resulting from the above operations shall be removed and hauled as directed by the Contracting Officer’s Representative.

D. Include driveway removal and replacement in lump sum price Contract Line Items that include water main construction.

3.5 GRAVEL SURFACED DRIVES AND ROADWAYS

A. Restore all damaged gravel surfaced drives and roadways to a condition equal to or better than original.
    1. Include gravel surfaced drives and roadway restoration in lump sum price of Contract Line Items that include water main construction.

2. Replacement gravel gradation.

3.6 TRAILS

A. Restore all damaged trails to a condition equal to or better than original as directed by Contracting Officer's Representative.

B. Include trail restoration in lump sum price of Contract Line Items that include water main construction.

3.7 PROTECTION OF EXISTING UTILITIES

A. Contractor to verify the location of all underground utilities.
   1. Omission from, or the inclusion of utility locations on the plans is not to be considered as the nonexistence of or a definite location of existing underground utilities.

B. A representative of the underground utilities shall be notified 24 hours in advance of crossings.

3.8 CONNECTIONS TO EXISTING WATER MAINS

A. Make connections to existing water mains as shown on Drawings, by attaching to existing or changed fitting.
   1. Cost for making connections shall include cost of all fittings including flexible couplings and shall be included in the lump sum price of Contract Line Items that include water main construction.

B. Where the connection is made to an existing water main which can be adequately isolated from the distribution system, it shall be termed a "dry connection."

C. Contractor is responsible for controlling and disposing of water in the trench at no additional cost to the Government.

3.9 SEWER CROSSINGS

A. Water mains crossing house sewers, storm sewers or sanitary sewers shall be laid to provide a vertical separation of at least 24 inches between the bottom of the water main and the top of the gravity sewer and a horizontal separation of 6 feet between the outside surface of the water main to outside surface of the gravity sewer, whenever possible.
   1. A water main may be laid closer than 6 feet if the crown of the sewer is at least 24 inches below the water main invert.
   2. In the event 24 inches of vertical separation cannot be provided at a gravity sewer crossing, the sewer shall be removed for a distance of 10 feet on each side of the water main and replaced with one 20 feet length of epoxy coated ductile iron pipe with mechanically restrained joints or internal joint restraint of the same size, or approved equal.
      a. If sewer joints are located 6 feet or less beyond the outside surface of the watermain, then a restrained joint or approved equal shall be used.
      b. Concrete collars shall be provided at each end of the ductile iron pipe to connect to the existing sewer pipe as shown on the Drawings.
   3. In no case shall the watermain be less than 1 foot above or 2 feet below a gravity sewer.

B. Where a pressure sewer line crosses above or less than 6 feet below a water line, the pressure sewer line shall be encased in at least 6 inches of concrete or constructed using mechanical joint ductile iron pipe for 10 feet on either side of the water main.
   1. In no case shall a pressure sewer line be located less than 6 feet horizontally or less than 2 feet below a water main.

C. The minimum horizontal separation between water lines and manholes shall be 6 feet, measured from center of the manhole.

D. Payment for crossings shall be included in the bid unit price of the water main.

3.10 TREES

A. Do not remove trees without written instructions from the Contracting Officer's Representative unless tree removal is shown on drawings.
   1. No separate payment will be made for tree removal and the cost shall be included in lump sum price of the Contract Line Items that include water main construction.

3.11 FENCES, SIGNS, MAILBOXES, ETC.

A. Restore all damaged fences, signs, mailboxes, etc., to their original conditions.
   1. No separate payment will be made for these items.

3.12 FIELD QUALITY CONTROL

A. Hydrostatic Testing:
   1. All valves, hydrants, pipe and fittings shall be hydrostatically tested.
   2. Furnish all necessary apparatus to run hydrostatic test, including necessary taps into the pipe.
   3. Prior to pressure testing, expel air from the pipe.
   4. Install corporation cocks at all high points in water main to allow air to be expelled.
   5. HDPE systems:
      a. HDPE pressure pipe systems: Test HDPE pressure pipe in accordance with the latest version of ASTM F2164. Pressurize the test section to the system test pressure and maintain this pressure for four (4) hours by adding make-up water. After this initial expansion phase, reduce the test pressure by 10 psi and stop adding make-up water. If the test pressure remains steady (within 5% of the target value) for one (1) hour, no leakage is indicated.
   6. Steel systems:
      a. General:
         1) All joints, including welds, are to be left exposed for examination during the test.
         2) Provide additional temporary supports for piping systems designed for vapor or gas to support the weight of the test water.
         3) Provide temporary restraints for expansion joints for additional pressure load under test.
         4) Isolate equipment in piping system with rated pressure lower than pipe test pressure.
         5) Do not coat or insulate exposed piping until successful performance of pressure test.
      b. High pressure pipe systems:
         1) Applies to:
            a) All piping systems normally exceeding 1,000 PSI working pressure.
         2) Submit pressure testing plan for review and approval by Contracting Officer's Representative minimum 14 days prior to commencing testing.
         3) In-person witness testing of pressure test piping not allowed.
      c. Monitor test pressure remotely via live-feed cameras at all critical viewpoint locations.
   7. Ductile Iron Pipe systems:
      a. For ductile iron pipe on this project, this material is used within precast concrete vault structures and is considered to be exposed piping. As such, all ductile iron pipe in exposed situations shall have zero allowable leakage.
   8. Spoolable Composite Pipe systems:
      a. Account for pipe to expansion and reinforcement shifts.
         1) Close the valves and pressurize the pipe up to 25% of hydrostatic test pressure (HTP): See Drawings for hydrostatic test pressures at specified testing locations.
         2) Pressurization of the pipe shall not exceed a rate increase of 145 psi/minute.
      b. Following the 8 hr pre-pressurization, add air if required to bring pressure back up to 25% of the HTP and monitor the pressure drop at least every 15 minutes for a minimum of 1 hour.
      c. Repressurize to near pipe pressure classification and shut-in the pipe again.

d. Monitor the pressure drop from one more hour and compare findings of the previous hour.
    1) If the pressure drop is less than that of the previous hour, the pipe is expanding rather than leaking.
e. If the rate of pressure drop does not taper off, repressure to near pipe pressure classification and test for two additional cycles.
    1) Each cycle should last for a minimum of 1 hour.
f. If the results show diminishing pressure drops for each cycle, hydrostatic testing can begin after a minimum of 8 hours to allow the inner lining to expand and the reinforcement to shift.
g. The pipe shall be pressurized to 25% of maximum of 8 hours prior to hydrostatic testing to account for pipe to expansion and reinforcement shifts.
h. After hydrostatic testing is complete, slowly open the pump manifold bleed valve to depressurize the pipeline in 25% test pressure decrements until the pressure is completely released.
    1) Depressurization of the pipe shall not exceed a rate of 290 psi/minute to prevent the possibility of a vacuum.

B. Cleaning, Flushing, and Disinfection :
1. Maintain interior of all pipes, fittings and other accessories free from dirt and foreign material at all times.
    a. If, in the opinion of the Contracting Officer's Representative, the pipe contains dirt that will not be removed by flushing, the pipe interior shall be cleaned and swabbed with bactericidal solution.
    b. At close of day's work or whenever workmen are absent from jobsite, plug, cap or otherwise provide watertight seal from open ends of pipe to prevent ingress of foreign material.
    c. If water is in trench, seal shall remain in place until trench is pumped dry.
2. After favorable performance of pressure test and prior to final acceptance, thoroughly flush the entire potable and non-potable water piping systems and perform disinfection on the potable system only, as prescribed.
    a. Perform all work including preventative measures during construction in full compliance to AWWA C651.
3. Flush each segment of the system to provide a flushing velocity of not less than 2.5 feet per second.
4. .Drain flushing water to location approved by the Contractor Officer's Representative.
5. Perform disinfection using one of the following forms:
    a. Application of chlorine gas-water mixture by means of solution-feed chlorinating device.
        1) Liquid chlorine shall comply with AWWA B301.
    b. Application of calcium hypochlorite, or sodium hypochlorite.
        1) Chlorine compounds shall comply with AWWA B300.
6. Disinfect pipe with chlorinated water as per AWWA C651.
    a. Method of application of chlorine shall be by continuous feed method or slug method.
    b. During disinfection procedure, ensure that initial and residual chlorine concentrations meet AWWA C651 requirements by testing by an approved method as directed by the Contracting Officer's Representative.
    c. Cost of testing shall be included in the Bid Unit Price for water mains and no separate payment will be made for this item.
7. Tag the system during the disinfection procedure.
8. Following disinfection for required contact period, neutralize chlorine residual in water by treating with reducing agent.
    a. Refer to AWWA C651.
    b. Flush all treated water from pipeline at its extremities until replacement water throughout pipe, upon test is proved comparable in quality to water in existing system.

c. Take two samples to test for bacteriological quality as directed by Contracting Officer's Representative.
d. Repeat disinfection procedure until two satisfactory results are obtained.
e. Quality of water delivered by the new water main to remain satisfactory for a minimum period of two days.

9. Secure satisfactory bacteriological reports on samples from the system.
   a. Ensure all sampling and testing procedures are in full compliance to AWWA C651, and applicable requirements of the State of Arizona.
      1) No separate payment will be made for this item.
10. The Government will provide the water required to fill the main initially and will pay for the water required to flush the main once in accordance with Specification Section 01 50 00 – Temporary Facilities and Controls.
    a. Filling and flushing shall be performed during periods of low usage, as approved by Contracting Officer's Representative.

END OF SECTION

SECTION 33 12 19
FIRE HYDRANT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Dry-barrel fire hydrant.

B. Related Sections include but are not necessarily limited to:
   1. Section 09 96 00 - High Performance Industrial Coatings.
   2. Section 40 05 00 - Pipe and Pipe Fittings - Basic Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Water Works Association (AWWA):
      a. C502, Standard for Dry-Barrel Fire Hydrants.
      b. M17, Installation, Operation and Maintenance of Fire Hydrants.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data:
      a. Acknowledgement that products submitted meet the requirements of the standards referenced.
      b. Manufacturer's installation instructions.
      c. Acknowledge and verify dimensions and provide list of integral parts and materials.
   2. Prior to submission of shop drawings, submit one copy of complete submittal information direct to GCNP Emergency Services Department, Attn: Fire Chief, and request and secure written approval of hydrant selection.
      a. Incorporate copies of written approval letter with submittals.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 FIRE HYDRANT

A. Design and Fabrication:
   1. Conform to AWWA C502.
   2. Provide with either compression or gate design.
   3. Provide with a 5 IN valve opening, nozzle section consisting of two, 2-1/2 IN hose nozzles and one, 4-1/2 IN steamer.
   4. Provide with water passages to permit full flow of water to minimize friction loss.
   5. Furnish with multiple weep holes for positive draining to allow water to escape readily from standpipe when hydrant valve is closed.
   6. Designed to throttle flow when partially opened.
   7. Designed to allow removal of valve and valve stem without digging up hydrant.
   8. Suitable for 5 feet of bury.
   9. Furnish with mechanical (gland type) joint inlet connections.
   10. Design to break off at ground line when struck by a vehicle.

11. Furnish with O-ring packing only.
12. Furnish hose and steamer nozzles with threads conforming to standard threads used by local Fire Department.
13. Furnish with direction of opening as required by local Fire Department with direction of opening cast on dome.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install hydrants at locations indicated in accordance with AWWA M17 and the following:
   1. Remove foreign material from barrel of hydrant before placement.
   2. Install plumb and at same elevation as connecting pipe and main.
   3. Place each hydrant on a slab of concrete not less than 6 inches thick and 18 inches SQ.
   4. Block backside of hydrant, opposite pipe connection, with concrete firmly wedged between hydrant and vertical face of undisturbed trench.
   5. Place granular bedding material around base of hydrant to the dimensions shown in the Drawings.
   6. Firmly tamp carefully compacted backfill around hydrant to surface of ground and to a distance of 5 feet in front of hydrant.

3.2 COATINGS AND FINISHES

A. Provide hydrant with below grade and above grade coatings as per Section 09 96 00.
   1. Paint above grade with color conforming to the requirements of the local Fire Department.

END OF SECTION

SECTION 33 16 33
HYDROPNEUMATIC TANKS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Hydropneumatic tanks:
      a. North Rim WTP (TK-68113 and TK-68114)
      b. Roaring Springs PH (TK-53111).
   2. Automatic level control system.
   3. Miscellaneous instruments and valves.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 - Equipment - Basic Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Water Works Association (AWWA)
      a. C652-11 Disinfection of Water-Storage Facilities.
   2. American Society of Mechanical Engineers (ASME):
      a. Section VIII, Division 1, Boiler and Pressure Vessel Code.
   3. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   4. National Sanitation Foundation International (NSF):
      a. 612, Drinking Water System Components - Health Effects.
   5. National Fire Protection Association (NFPA):
      a. 70, National Electric Code (NEC).

B. Miscellaneous:
   1. Pressure vessels shall be in accordance with the latest revision of the ASME Code for Unfired Pressure Vessels, Section VIII, Division 1.
   2. The NEC shall be used for all wiring.
   3. All local Plumbing Codes shall be met.
   4. The system and anchorage of the tank shall conform to the building code.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. The control system manufacturer shall provide the following submittals as a minimum:
      a. Show all taps, location of control equipment, supports and other appurtenant devices.
         1) Drawings shall include dimensions and cross sectional views of all equipment showing detail of construction.
      b. Equipment weights and anchor bolt designs.
   3. Instrumentation description sheets verifying controls as specified herein including sight glass, air pressure gage.
      a. Source quality control test reports.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.4 QUALIFICATIONS

A. The vendor shall be capable of demonstrating technical competence in the design of hydropneumatic tanks with a minimum of ten (10) years of experience.

B. The Contractor shall arrange for personnel representing the tank manufacturer to check the installations and instruct the Contracting Officer's personnel in the proper operation.

C. Tank Supplier: Provide a minimum of five references for a comparable installation.

D. Tank Welders: ASME certified.

1.5 SYSTEM GUARANTEE

A. All work shall be guaranteed for a minimum of two years after Substantial Completion. Guarantee that the performance of the equipment will meet the specified requirements, and that a safe operating system will result.

B. This guarantee will be in the form of a letter supplemented by a field test report showing compliance with these Specifications.

C. The guarantee letter and report shall be submitted to the Contracting Officer and shall be the sole responsibility of the tank supplier.

D. Just before one year after Substantial Completion, assist the Government in performing an inspection and evaluation of the tank and system performance.

E. Be present for this inspection and shall furnish an experienced foreman, rigging, and assistance for the evaluation.

F. If the system does not meet the specified requirements, perform replacement and repairs.

The contractor must coordinate its repair work with a time convenient for the Government.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements.
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Charlatte America, (Basis of Design).
   2. Or Equal.

C. Submit request for product substitution in accordance with Specification Section 01 33 23.

2.2 GENERAL REQUIREMENTS

A. The control system shall include a vertical bladder-type hydropneumatic tank.

B. North Rim WTP:
   1. The hydropneumatic tank must be designed to maintain system pressure in the Upper Zone.

C. Roaring Springs:
   1. The hydropneumatic tank must be designed to match the dynamics of the emergency wash station.

D. There shall be no less than 10% reserve water in the tank at the maximum outflow conditions.

2.3 DESIGN AND PERFORMANCE REQUIREMENTS

A. Provide hydropneumatic control system. Major components shall include:
   1. Hydropneumatic Tanks.
   2. Miscellaneous instruments and valves.

B. Hydropneumatic tanks shall be designed to limit pump cycling during expected flow rates in the system.
   1. The tank shall be designed to meet the pressure requirements at the flow rate specified.
   2. Tank performance will be verified by operational field tests at pumping rates defined below for piping system.

C. The manufacturer shall conduct an analysis of the piping system, utilizing the specified data. Results of the analysis shall:
   1. Clearly indicate that the design and size of the tank provided will limit pump cycling
   2. Show that the system will meet the performance guarantees outlined in Article 1.5 SYSTEM GUARANTEE.

D. Design for anchor bolts by supplier. Materials for complete anchor bolt assembly including studs, nuts and washers to be provided by Contractor.

2.4 COMPONENTS

A. Hydropneumatic Tanks:
   1. The tank shall be a vertical, bladder-type vessel suitable for use with potable water.
   2. Tanks shall be constructed of carbon steel for a maximum allowable working pressure of 150 PSIG in accordance with the ASME Section VIII.
      a. Tanks shall be provided with a flanged line connection, two support saddles, lifting lugs, drain, safety relief valve.
   3. Size of the tanks is shown in Drawings.
      a. Each tank shall have a minimum volume of:
         1) North Rim WTP: 198 GAL.
         2) Roaring Springs: 53 GAL.
      b. Each tank shall have a maximum outside diameter of:
         1) North Rim WTP: 33 IN.
         2) Roaring Springs: 24 IN.
      c. Each tank shall have a maximum outside height of:
         1) North Rim WTP: 72 IN.
         2) Roaring Springs PH: 43 IN.
   4. Tank finish system: Refer to Specification Section 09 96 00 for finishing of internal and external surfaces.
      a. All painting and coating shall be completed at the factory. Field painting and coating will not be accepted.
         1) The tank interior shall be painted with an NSF 61 epoxy coating with a uniform layer thickness of no less than 6 mils.
         2) The tank exterior shall be painted with an anti-corrosion polyurethane and shall have a uniform layer with a minimum thickness of 10 mils.
   5. Structural Design and Supports shall include the tank, supports, and anchor bolts shall be designed based by the manufacturer based upon local building codes in addition to the following criteria:
      a. Design for a hydrostatic pressure of 150 PSIA and a factory test pressure of 225 psi with no reactive load permitted through the inlet/outlet piping.
      b. Support tank by THREE support legs. Material of construction shall comply with ASTM A 36 or ASTM A 285, Grade C. Weld the support legs to the tank. Bolted type leg extensions shall be provided to raise the tank height sufficient to clear the attached piping as shown on project plans.

B. Level Indication System:
   1. The purpose of the level indication system is to monitor the water level in the pressure vessel within the designed range depending on the number of pumps in operation and the associated dynamic heads.
   2. Provide an Automatic Level Indication System for water level in the tank that includes the following principal components:
      a. Flag-style Magnetic Level Gauge for visual monitoring of tank level. Level gauge shall be furnished with manual isolation valves at the top, at the bottom and for draining the gauge.
         1) Continuous vessel monitoring shall be provided via differential pressure transmitter mounted directly to a 2" side mounted flange on the vessel outlet. Transmitter shall be configured to transmit a 4-20 mA signal corresponding to 0-100% vessel level. Diaphragms and capillary tubing shall be provided for connection to the air side charging connection manifold at the top of the vessel. A 2" flanged stainless steel ball valve shall be supplied for isolation of the transmitter, Apollo standard port or equal. Ball valve provided with bolt kits for mounting.
         2) A 1/2" side mounted theaded connection shall be provided on the vessel outlet for tank drain. Also provided for this function shall be a 1/2" threaded stainless steel ball valve for isolation of the tank drain, Apollo standard port or equal.

2.5 SOURCE QUALITY CONTROL:

A. Hydrostatic test the hydropneumatic tank in accordance with ASME Code for Unfired Pressure Vessels.

B. Submissions shall include Form U-1A "Manufacturers' Data Report for Unfired Pressure Vessels" prepared by the tank manufacturer to certify that the tank was built in accordance with ASME Code Rules for the Construction of Unfired Pressure Vessels and inspected by a certified inspector.
   1. Include copies of this form in the Operation and Maintenance Manual.

2.6 SERVICE CONDITIONS

A. Vessel hydraulic performance conditions and design data shall be as shown below. In the event that the manufacturer's hydraulic analysis of the system yields varying design requirements, the Contracting Officer's Represtative should be contacted.
   1. North Rim WTP (TK-68113 and TK-68114):
      a. Tank Configuration: Vertical.
      b. Minimum Capacity: 185 GAL.
      c. Minimum Operating Pressure: 22 PSIG
      d. Maximum Operating Pressure: 45 PSIG.
      e. Minimum Field Test Pressure: 60 PSIG.
      f. Design Pressure: 100 PSIG.
      g. Factory Hydrostatic Test Pressure 225 PSIG.
      h. Bladder Material: Food Quality Butyl Rubber.
      i. Precharge Pressure Setting: 35 PSIG.
      j. Outlet Flange Size: 2 IN.
      k. Outlet Flange Pressure Rating: ANSI Class 150 per ANSI B16.5.
   2. Roaring Springs (TK-53111):
      a. Tank Configuration: Vertical.
      b. Minimum Capacity: 53 GAL.
      c. Minimum Operating Pressure: 50 PSIG.
      d. Maximum Operating Pressure: 80 PSIG
      e. Minimum Field Test Pressure: 100 PSIG.
      f. Design Pressure: 150 PSIG.
      g. Factory Hydrostatic Test Pressure 225 PSIG.
      h. Bladder Material: Food Quality Butyl Rubber.
      i. Precharge Pressure Setting: 37.1 PSIG.

j. Outlet Flange Size: 2 IN.
k. Outlet Flange Pressure Rating: ANSI Class 150 per ANSI B16.5.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Cast anchor bolts in concrete foundation as embedded items.
   1. Use rigid template for each set of bolts.

B. Furnish and install the tank to include all necessary piping, valves, and other related appurtenances.

C. Tank shall be installed in accordance with manufacturers recommendations.

D. All supports, piping, valves, and related appurtenances shall be provided and installed by the Contractor at no additional cost to the Government

3.2 FIELD QUALITY CONTROL

A. Testing shall be performed by the Contractor in the presence of the Contracting Officer's Representative and a representative of the manufacturer.

B. Employ and pay for services of equipment manufacturer's field service representative(s) to:
   1. Inspect equipment covered by these Specifications.
   2. Supervise any adjustments and installation checks.
   3. Conduct startup of equipment and perform operational checks.
   4. Provide Contracting Officer, through the Contractor, with a written statement that manufacturer's equipment has been installed properly, started up, and is ready for operation by Government's personnel.
   5. Instruct Government's personnel for a minimum of 44 hours at jobsite on operation and maintenance of the piece of equipment.

END OF SECTION

This page intentionally left blank.

SECTION 33 31 11
GRAVITY SEWER PIPELINE AND MANHOLE CONSTRUCTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Installation and testing of sewer pipes, manholes, structures and appurtenances.
   2. Connections to existing sewers.
   3. Coordination and interface with existing facilities and utilities.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 31 23 00 - Earthwork
   2. Section 31 23 33 – Utilities – Trenching and Backfilling
   3. Section 32 12 00 - Flexible Paving.
   4. Section 32 13 13 – Concrete Paving.
   5. Section 32 91 13 - Finish Grading.
   6. Section 33 05 61 - Concrete Manholes.
   7. Section 33 05 31.11 – Polyvinyl Chloride Gravity Sewer Pipe.
   8. Section 33 05 33.23 – High Density Polyethylene (HDPE) Pipe.

C. Alternates:
   1. The following materials may be used at Contractor's option unless otherwise specified or shown on Drawings:

1.2 REFERENCES

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. C1244, Standard Test Method for Concrete Sewer Manholes by the Negative Air Pressure (Vacuum) Test Prior to Backfill.
      b. F1417, Standard Practice for Installation Acceptance of Plastic Non-pressure Sewer Lines Using Low-Pressure Air.
   2. Occupational Safety and Health Administration (OSHA).

1.3 SUBMITTALS

A. Action Submittals: Submit the following:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Catalog data sheets for all materials.
      c. Details of all piping system components confirming that the pipe, fittings, and appurtenances conform to the specified requirements.
      d. Manufacturers' written recommendations for material handling, delivery, storage, installation, and minor repair of materials damaged in shipping.
      e. Recommended details for buoyancy restraint and for manhole/pipe connections.
      f. Fabrication and/or layout drawings as specified in individual pipe material specifications.
   2. See individual pipe material specifications for additional required submittals.

B. Informational Submittals: Submit the following:
   1. Submit Test Report of leakage tests results including the following:
      a. Test procedure.
      b. Pipeline segment tested.
      c. Length of pipe tested.
      d. Test pressure.
      e. Test duration

f. Amount of leakage.
g. Corrective action, if any.

C. Contract Closeout Information:
1. Operation and Maintenance Data.

1.4 DELIVERY, STORAGE, AND HANDLING OF PRODUCTS

A. In addition to the requirements specified in this section, see related paragraphs in individual pipe specifications.

B. Deliver, handle and store products in accordance with manufacturer's instructions.

C. Protect pipeline sections stored at the site from damage.

D. Store all products above the ground upon platforms, pallets, skids, or other supports supplied by the Contractor.
1. Store in a way to permit ready access for identification and inspection by the Contracting Officer.

E. Keep products free from dirt and other foreign matter.

F. Provide suitable quantities of all lifting equipment to handle the pipe.
1. Do not utilize any equipment that is not rated to handle the intended loading or conditions of use to which it will be subjected, or which will damage or gouge the pipe.
2. Do not drag or drop pipe.

G. Place pipe lain directly on the ground prior to placement on an area free of loose stones or sharp objects.

H. Repair or replace any new pipe and fittings damaged before or during installation at Contractor’s expense, before proceeding further.
1. Utilize repairs methods as recommended by the manufacturer.
2. Replace damaged materials as directed by Government’s project representative.

I. Protect PVC pipe from UV degradation if stored outside for more than 60 days.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Pipe:
1. See individual pipe material specifications listed in Paragraph 1.1 B above.

B. Manholes: See Specification Section 33 05 61.

C. Fill and Backfill at manholes: See Specification Section 31 23 33.

D. Embedment and Bedding Materials for pipelines: See Section Specification 31 23 33.

E. Size mandrels to meet deflection requirements specified herein or in the individual pipe material specifications.

F. Pipe Joint Testing Equipment:
1. Utilize joint testing equipment capable of providing sufficient sealing pressure for air bladder to prevent leakage through bladder seals.
2. Include pressure gage to aid in verification of adequate applied pressure and joints ability to withstand the applied pressure without leaking.
3. Utilize joint testing equipment that encapsulates the full 360 degrees circumference of the joint and at least 6 inches each side of the joint.
4. Pipes furnished with an integral "testable" joint, consisting of two gaskets with fittings to pressure test between the two gaskets, will be tested using the test equipment and fittings recommended by the pipe manufacturer.

PART 3 - EXECUTION

3.1 GENERAL REQUIREMENTS

A. Furnish all construction materials and equipment required for installation and backfill.

B. Install the pipeline complete, including bends, stop logs, and other associated work and appurtenances, as shown on the Drawings or as herein specified.

C. Make all necessary connections to sewer lines shown on the Drawings and in accordance with the Specifications.

D. Construct all sewer piping, manholes, and structures to be free of visible ground water inflow.
   1. Install sanitary sewer manholes and structures that are watertight.
      a. Repair or replace any manhole or structure showing infiltration of ground water through joined surfaces, pipe to manhole/structure connections or manhole/structure wall.
   2. Install pipe with water tight pipe joints.
      a. Repair or replace any joint showing infiltration of ground water through pipe joints or pipe to manhole connections.

E. Alignment: Lay gravity sewer lines in straight alignment and uniform grade between manholes.
   1. Install at grade as shown on drawings.

F. Brace and protect pipe sections to prevent deformation during installation and backfill.

G. Deflection: Pipe deflection after final backfill shall not exceed the specified limits detailed in the individual pipe specifications.
   1. Remove and replace any pipe observed to be deflecting in excess of the specified limits.

H. Perform testing in accordance with requirements of this Section.

I. Provide Contracting Officer with free access to work for inspection.
   1. Such inspection shall not relieve the Contractor of his responsibility for performing Work in accordance with the Contract Documents.

3.2 PIPELINE INSTALLATION

A. Lateral shoring of the trench walls or other similar construction methods may be required.
   1. Design and implement all such methods.
   2. When required, install shoring in accordance with all applicable local, State and OSHA regulations.
   3. Remove shoring prior to backfilling.

B. Grade bottoms of trenches such that when bedding is placed between the trench bottom and the pipe, each section of pipe is installed to the specified depth or elevation with uniform support.

C. Determine and fix alignment and grade or elevation of each pipeline from offset stakes or calibrated laser instruments.

D. Install pipelines on the line and grade shown on the drawings.
   1. Calculate required elevation of each pipe joint and survey installed elevation at each joint prior to stabbing the next joint to verify grade.
   2. Relay pipe to proper grade if any joint elevation deviates from the calculation by more than 0.01 feet.

E. Remove material at the bottom of the trench if determined to be unsuitable by the Contracting Officer.
   1. Backfill trench with approved subgrade material or bedding material to the specified depth or elevation as described in Specification Section 31 23 33.

F. Install only clean pipe and fittings.
   1. Provide physical barriers to protect open ends of sections of pipe in place from the entrance of trench water, mud, dirt, or other foreign substances with when pipe installation is not in progress.

G. Begin pipe laying at the lowest elevation with bell ends facing the direction of laying, except when reverse laying is permitted by Contracting Officer.

H. Where the drawings require concrete encasement or flowable fill embedment and backfill, anchor pipe as required to prevent floatation.
   1. Alternatively, Contractor may place concrete or flow fill materials in staged lifts allowing each lift to reach initial set prior to placing the subsequent lift to limit buoyancy effects and prevent floatation of the pipeline.

I. Pipeline may be backfilled as it is installed, provided all inspection and testing requirements are met.

J. Pipe, fittings, and special pieces will be subject to inspection by Contracting Officer, prior to installation.
   1. Report all damages not detected by Contracting Officer but discovered by Contractor during installation to Contracting Officer for corrective action or replacement.

K. Repair of pipe damaged during installation shall conform to the manufacturer's repair procedures; with the concurrence of Contracting Officer.

3.3 RESTORATION

A. Restore all existing structures or services damaged by Contractor's operations at no cost to Government.
   1. Repair or replace culverts that are damaged, removed or interfere with the work as part of restoration at no additional cost to Government.

B. Restore all area disturbed by installation of the pipelined in accordance with the Specifications, the Drawings,
   1. Provide slope protection, re-vegetation, and road restoration as necessary.

C. Driveway Removal and Replacement:
   1. All Portland cement concrete and asphalt noted for removal and replacement shall be cut prior to removal.
      a. Cut by sawing, vertical cut to be 1 inch minimum.
      b. The remaining depth of section may be broken out in a manner subject to Contracting Officer's approval.
      c. Width of section removed to be either a width not greater than the outside diameter of the sewer plus 4 feet-0 inches or broken out to the nearest joint.
   2. Replace Portland cement concrete and asphalt equal to or better than original paving.
   3. Debris resulting from the above operations shall be removed and disposed offsite.
   4. Include driveway removal and replacement in cost of the bid unit price of the sewer pipe.

D. Gravel Surfaced Drives and Roadways:
   1. Restore all damaged gravel surfaced drives and roadways to a condition equal to or better than original.
      a. Payment to be at bid unit price for this item.
   2. Replace using ADOT Class 2 ABC

E. Trees:
   1. Do not remove trees without written instructions from the Contracting Officer unless tree removal is shown on drawings.
   2. No separate payment will be made for tree removal and the cost shall be included in the bid unit price sewer pipe.

F. Fences, Signs, Mailboxes, etc.:
   1. Restore all damaged fences, signs, mailboxes, etc., to their original conditions.
      a. No separate payment will be made for these items.

3.4 PROTECTION OF EXISTING UTILITIES

A. Verify the location of all underground utilities.

1. Omission from, or the inclusion of utility locations on the plans is not to be considered as the nonexistence of or a definite location of existing underground utilities.

B. Notify utility representative prior to construction to obtain available information on location of existing utilities.
   1. Contractor shall be responsible for locating all utilities.

C. Notify representative of the underground utilities 24 hours in advance of crossings.

D. Existing water services and sewer services:
   1. Repair damage to existing water service using copper pipe and union the same size as existing service.

3.5 FIELD QUALITY CONTROL

A. General Testing Requirements:
   1. Furnish necessary personnel, materials, and equipment, including bulkheads, restraints, anchors, temporary connections, pumps, water, pressure gauges, and other means and facilities required to perform tests.
   2. Obtain Contracting Officer's approval of methods and the equipment used for the tests prior to testing.
   3. Provide reasonable facilities and access for Contracting Officer to inspect, test and obtain such information as required with respect to the materials used and the progress and condition of the Work and the results obtained.
      a. Work that is not performed in accordance with the procedure or does not comply with the requirements of the Specifications will be rejected.
   4. Coordinate testing schedules with Contracting Officer.
      a. Perform all specified tests under observation of Contracting Officer.
      b. Provide a minimum of 24 hours advance notice prior to commencing any testing.
   5. Perform testing as work progresses and as required to facilitate connections with existing sewers.
   6. Obtain water for testing and cleaning at no additional cost to the Government.
   7. Test only those portions of pipes that have been installed as part of this Contract.
      a. Test new pipe sections prior to making final connections to existing piping.
      b. Furnish and install plugs, bulkheads, and restraints required to isolate new pipe sections.
   8. Unsuccessful Tests:
      a. Where tests are not successful, correct defects or remove defective piping and appurtenances and install piping and appurtenances that comply with the specified requirements.
      b. Repeat testing until tests are successful at no additional cost to Government.
   9. Deflection:
      a. Check each section of pipe after backfilling for deflection by pulling a mandrel through the pipe.
      b. Conduct test after the final backfill has been in place at least 30 days.
      c. Maximum long term deflection: less than 5% of the initial diameter.

B. Lamping:
   1. Install each section of sewer line between manholes to be straight and uniformly graded.
   2. Each section will be lamped by Contracting Officer.
   3. Furnish suitable assistants, materials and air monitoring to assist Contracting Officer.

C. Low Pressure Air Testing:
   1. Test all gravity sewer pipes up to 30-inch diameter with a low pressure air test.
      a. Comply with ASTM F1417.
      b. Time elapsed for a 1 psi drop in air pressure: Per ASTM F1417.
      c. Maximum air loss: Per ASTM F1417.
   2. Submit schedule to Contracting Officer for approval prior to starting the tests.
   3. Conduct air test after the final backfill has been in place for a minimum of 30 days.

4. Correct pipes failing air test and conduct second test after final backfill has been in place an additional 30 days.
5. If the length of sewer to be tested is fully or partially submerged in groundwater, increase test pressure as necessary to overcome the actual static pressure exerted by the groundwater.
    a. If a test pressure greater than 8 psi results, utilize water infiltration testing in lieu of air testing.
6. Locate leaks by testing short sections of pipe.
    a. Repair leaks and retest affected reach of sewer.

D. Deflection Testing: Test all flexible sewer pipes for deflection.
1. Perform the mandrel test with the Contracting Officer in observance for all pipe sizes.
2. Deflection limits: as indicated in the individual pipe sections or as specified in article titled "General Testing Requirements" above, whichever is more stringent.
3. Conduct test after the final backfill has been in place at least 30 days.
4. Correct pipes failing deflection test and conduct second test after final backfill has been in place an additional 30 days.

E. Manhole Testing:
1. Vacuum test all manholes per ASTM C1244 for leakage after installation, before epoxy lining is placed, and prior to being backfilled.
2. Visually inspect all manholes for leaks and defects prior to vacuum testing.
3. Repair all leaks, defects or cracks discovered by visual inspection prior to vacuum testing.
4. Seal all pipes entering manholes at a point outside the manhole walls so as to include testing of the pipe/manhole joints.
5. Make all necessary repairs and retest the manhole.
    a. Inspect exterior of the manhole during this period for visible evidence of leakage.
    b. All repairs will be subject to acceptance by the Contracting Officer.

END OF SECTION

SECTION 33 52 10
SERVICE PIPING - FUEL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:
   1. Fuel service (diesel) pipe and piping components.
   2. Fuel service containment sumps.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 09 96 00 - High Performance Industrial Coatings.
   2. Division 26 - Electrical.
   3. Section 31 23 33 - Trenching and Backfilling.
   4. Section 33 56 13 - Aboveground Fuel Storage Tanks.
   5. Section 33 56 16 - Fuel Day Tank Systems.
   6. Section 40 41 13 – Heat Tracing Cable.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Association of State Highway and Transportation Officials (AASHTO):
      a. M294, Standard Specification for Corrugated Polyethylene Pipe, 12- to 60-in. Diameter.
   2. American Petroleum Institute (API):
      a. RP 1110, Recommended Practice for the Pressure Testing of Steel Pipelines for the Transportation of Gas, Petroleum Gas, Hazardous Liquids, Highly Volatile Liquids, or Carbon Dioxide.
      b. Spec 5L, Specification for Line Pipe.
      c. Spec 6FB, Specification for Fire Test for End Connections.
      d. Std 607, Fire Test for Quarter-turn Valves and Valves Equipped with Nonmetallic Seats.
   3. American Society of Mechanical Engineers (ASME):
      a. B1.1, Unified Inch Screw Threads (UN and UNR Thread Form).
      b. B16.5, Pipe Flanges and Flanged Fittings.
      c. B16.9, Factory-Made Wrought Steel Butt-Welding Fittings.
      d. B16.10, Face-to-Face and End-to-End Dimensions of Valves.
      e. B16.11, Forged Steel Fittings, Socket Welding and Threaded.
      f. B16.21, Nonmetallic Flat Gaskets for Pipe Flanges.
      g. B16.34, Valves – Flanged, Threaded, and Welding End.
      h. B16.39, Malleable Iron Threaded Pipe Unions: Classes 150, 250, and 300.
      i. B18.2.1, Square and Hex Bolts and Screws (Inch Series).
      j. B18.2.2, Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series).
      k. B31.3, Process Piping.
      l. B36.10, Welded and Seamless Wrought Steel Pipe.
      m. B36.19, Stainless Steel Pipe.
      n. Boiler and Pressure Vessel Code (BPVC), Section V – Nondestructive Examination.
   4. American Society of Nondestructive Testing (ASNT):
      a. SNT-TC-1A – Personnel Qualification and Certification in Nondestructive Testing.
   5. ASTM International (ASTM):
      a. A53, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
      b. A105, Standard Specification for Carbon Steel Forgings for Piping Applications.

c. A182, Standard Specification for Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service.
d. A193, Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications.
e. A194, Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both.
f. A216, Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service.
g. A234, Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service.
h. A269, Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service.
i. A312, Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes.
j. A351, Standard Specification for Castings, Austenitic, for Pressure-Containing Parts.
k. A380, Standard Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems.
l. A403, Standard Specification for Wrought Austenitic Stainless Steel Piping Fittings.
m. A733, Standard Specification for Welded and Seamless Carbon Steel and Austenitic Stainless Steel Pipe Nipples.
n. A743, Standard Specification for Castings, Iron-Chromium, Iron-Chromium-Nickel, Corrosion Resistant, for General Application.
o. D229, Rigid Sheet and Plate Materials Used for Electrical Insulation.
p. D2310, Standard Classification for Machine-Made Fiberglass (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe.
q. D2996, Standard Specification for Filament-Wound Fiberglass (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe.
r. F436, Hardened Steel Washers.

6. Code of Federal Regulations (CFR):
   a. 40 CFR 112, Oil Pollution Prevention.
7. Manufacturers Standardization Society of the Valve and Fittings Industry (MSS):
   a. SP-6, Standard Finishes for Contact Faces of Pipe Flanges and Connecting-End Flanges of Valves and Fittings.
   b. SP-25, Standard Marking Systems for Valves, Fittings, Flanges, and Unions.
   c. SP-58, Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation.
   d. SP-72, Ball Valves with Flanged or Buttwelding Ends for General Service.
8. National Fire Protection Association (NFPA):
   a. 20, Standard for the Installation of Stationary Pumps for Fire Protection.
   b. 30, Flammable and Combustible Liquids Code.
   c. 30A, Motor Fuel Dispensing Facilities and Repair Garages.
   d. 31, Standard for the Installation of Oil-Burning Equipment.
   e. 37, Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines.
   f. 110, Standard for Emergency and Standby Power Systems.
9. Occupational Safety and Health Administration (OSHA):
   a. 29 CFR 1910.106, Flammable Liquids.
10. Society for Protective Coatings (SSPC):
    a. PA 1, Shop, Field, and Maintenance Painting of Steel.
11. Society of Automotive Engineers International (SAE):
    a. AMS3275, Sheet, Acrylonitrile Butadiene (NBR) Rubber and Non-Asbestos Fiber Fuel and Oil Resistant.
    b. J514, Hydraulic Tube Fittings.
12. Underwriters Laboratories, Inc. (UL):
    a. 842, Standard for Valves for Flammable Fluids.

b. 971, Nonmetallic Underground Piping for Flammable Liquids.

B. Regulatory Requirements:
1. Pipe installers must be licensed / certified by the state when the state requires licensed installers.

C. Inspections:
1. Visually inspect pipe and fittings at time of delivery and just prior to installation.
a. Do not use defective materials.
b. Immediately remove rejected material from the project site.

1.3 DEFINITIONS

A. Contracting Officer: Includes the Contracting Officer (CO) and the Contracting Officer Representative (COR).

1.4 SYSTEM DESCRIPTION

A. Design Requirements:
1. Provide the entire fuel distribution system as a complete and fully operational system.
2. Size, select, construct, and install equipment and system components to operate together as a complete system.
3. Pipe and piping components specified herein shall be compatible with the fuel to be handled.

1.5 SUBMITTALS

A. Shop Drawings:
1. Fabrication and/or layout drawings:
a. Below ground piping drawings with information including:
1) Dimensions of piping lengths.
2) Invert or centerline elevations of piping crossings.
3) Acknowledgement of bury depth requirements.
4) Details of fittings, tapping locations, thrust blocks, restrained joint segments, harnessed joint segments, and related appurtenances.
5) Line slopes, vents, and drains.
b. Above ground piping drawings with information including:
1) Dimensions of piping from wall surfaces or column lines.
2) Centerline dimensions of piping.
3) Centerline elevation and size of intersecting conduit/conduit racks, or other potential interferences requiring coordination.
4) Location and type of pipe supports and anchors.
5) Details of fittings, tapping locations, equipment connections, flexible expansion joints, connections to equipment, and related appurtenances.
6) Line slopes, air release vents and low point drains.

B. Product Data:
1. Pipe and piping components:
a. Acknowledgement that products submitted meet requirements of standards referenced.
b. Copies of manufacturer's written directions regarding material handling, delivery, storage, and installation.
c. Copies of manufacturer’s installation, operation, and maintenance (IOM) manuals.
d. Separate schedule sheet for each piping system scheduled in this Section showing compliance of all system components.
1) Attach technical product data on gaskets, pipe, fittings, and other components.

C. Test Reports:
1. Welding NDE test results.
2. Pressure test reports:
a. Copies of pressure test results on all piping systems.

b. Notification of time and date of piping pressure tests.

D. Certificates:
1. Contractor qualifications:
a. State licenses or certificates, if applicable.
b. For fiberglass reinforced plastic (FRP) pipe installation, certification by the FRP manufacturer as a qualified installer of their products is required.
2. Welding qualifications:
a. Welding Procedure Specification (WPS) for each pipe material and process.
b. Procedure Qualification Record (PQR) for each WPS.
c. Welder Performance Qualification (WPQ) for each welder and each specification.

E. Refer to Section 01 81 33 – Cyber Security Requirements for required cyber security related submittals.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Handle, store, and protect pipe and piping components to prevent damage before and during installation in accordance with the manufacturer's recommendations. Replace damaged or defective items.
1. Provide factory-applied plastic end-caps on each length of pipe and tube.
a. Maintain end-caps through shipping, storage, and handling to prevent pipe-end damage and prevent entrance of dirt, debris, and moisture.
2. Protect stored pipes and tubes.
a. Elevate above grade and enclose with durable, waterproof wrapping.
3. Protect flanges, fittings, and specialties from moisture and dirt by inside storage and enclosure, or by packaging with durable, waterproof wrapping.
4. Use of bare cables, chains, hooks, metal bars, or narrow skids in contact with coated, plastic, FRP, or flexible pipe is not permitted.
5. Avoid contact of any ferrous metal with stainless steel.
a. Use tools dedicated for stainless steel.
b. Handle using nylon slings and alloy chains, cable, or straps.
c. Store on racks constructed of non-ferrous metal or line with rubber.

1.7 PROJECT/SITE CONDITIONS

A. Fuel required for the testing, flushing, and cleaning efforts, as specified in this section, will be provided and delivered by the Contractor. Do not flush, clean, or test any system with fuel or liquid not intended for final system operation. Fuel used in the system will become the property of the Contracting Officer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, piping that may be incorporated in the work include, but are not limited to, the following:
1. General Valves:
a. Morrison Brothers.
b. Or Equal.
2. Heat Tracing Equipment:

a. nVent Raychem.
b. Or Equal.

C. Submit requests for substitution in accordance with Specification Section 01 33 23.

2.2 CONDITIONS OF SERVICE

A. Properties of Fuel Stored:
1. Fuel / Fluid: Diesel.
2. Design Pressure: 60 PSIG.
3. Internal temperature: Ambient DEGF.

2.3 ABOVE GRADE FUEL SERVICE PIPING

A. System Description:
1. Fuel Piping (diesel):
a. Single-wall carbon or stainless steel piping.
2. Vapor Piping:
a. Single-wall carbon or stainless steel piping.

B. General Requirements:
1. Standards compliance:
a. NFPA 30.
b. NFPA 37.
c. NFPA 110.
d. ASME B31.3.
2. Pipe and piping components specified herein shall be designed to handle a working pressure of 275 psiG at 100 degrees F.
3. Pipe:
a. Pipe shall meet the material, fabrication, and operating requirements of ASME B31.3, except as modified herein.
4. Fittings:
a. Fittings smaller than 2-1/2 inches shall be Class 3000, socket weld type, unless indicated otherwise, conforming to ASME B16.11.
b. Fittings 2-1/2 inches and larger shall be buttweld type conforming to ASME B16.9 and be of the same wall thickness as the adjoining pipe.
c. Elbows shall be long radius unless otherwise indicated.
5. Flanges:
a. Flanges shall conform to ASME B16.5, Class 150.
b. Gaskets (non-isolating):
1) Provide flange gaskets that are 1/16 inches thick and that conform to ASME B16.21, raised-face type, unless otherwise indicated.
2) Gaskets shall be constructed of nitrile butadiene (Buna-N) or acrylonitrile butadiene rubber (NBR) material conforming to SAE AMS3275.
3) Provide gaskets that are factory cut from one piece of material.
4) Gaskets shall have a fire safe designation such as testing per API 6FB.
5) Maximum operating temperature: 700 degrees F.
6) Spiral-wound gaskets consisting of stainless-steel spiral wound metal, flexible graphite filler, and inner and centering rings of material similar to piping are also acceptable.
c. Gaskets (electrically isolating):
1) Flange gaskets shall conform to ASTM D229 and shall provide an electrical insulating material of 1,000 ohms minimum resistance.
2) Provide full face type gaskets that are chemically compatible with the fuel to be handled.
3) Provide flanges that have a full surface 0.03 inches thick, spiral-wound mylar insulating sleeves between the bolts and the holes in the flanges. Bolts may have reduced shanks of a diameter not less than the diameter at the root of the threads.

4) Provide high strength 1/8 inches thick phenolic insulating washers next to the flanges with flat circular stainless steel washers over the insulating washers and under bolt heads and nuts.
5) Provide bolts long enough to compensate for the insulating gaskets and stainless steel washers.

d. Bolts, nuts, and washers:
1) Bolts and nuts for pipe flanges shall conform to ASME B18.2.1 and ASME B18.2.2, except as otherwise specified.
2) Bolts shall be hexagonal, heavy series type. Bolts shall be threaded in accordance with ASME B1.1, Class 2A fit, Coarse Thread Series, for sizes 1 inch and smaller, and Eight-Pitch Thread Series for sizes larger than 1 inch.
3) Nuts shall be the hexagonal, heavy series type. Nuts shall be threaded in accordance with ASME B1.1, Class 2B fit, Coarse Thread Series, for sizes 1 inch and smaller, and Eight-Pitch Thread Series for sizes larger than 1 inch.
4) Bolts shall be of sufficient length to project at least two full threads beyond the nuts with the bolts tightened to the required torque.

6. Piping Components:
a. Piping components shall meet the material, fabrication, and operating requirements of ASME B31.3, except as modified herein.
b. Pressure design class for piping components shall be Class 150 as defined in ASME B16.5.

7. Chemical compatibility:
a. All pipe and piping components shall be chemically compatible with products to be transported including EPA approved additives.
b. Internals of pipe and piping components that could be exposed to fuel during system operation shall not be constructed of zinc-coated (galvanized) metal, brass, bronze, or other copper-bearing alloys unless noted otherwise.

8. Pipe Heat Trace (Outdoors)
a. Provide heat tracing control panel and heat trace circuits for new outdoor diesel supply, return and tank fill piping designed to maintain the product at a set point of 60 DEGF.
b. Heat Trace control panel shall be in NEMA 4X enclosure with thermostat-controlled enclosure heater.
c. Heat tracing shall be:
1) Raychem Corp.
2) Or Equal

d. Electrical power supply requirement for the heat trace control panel shall be 208 VAC, 3 phase, 60 Hz with 3 Pole Thermal Magnetic main circuit breaker.
e. Electrical power supply requirement for individual heat trace circuits shall be 208 VAC single phase with individual 20A, 2 Pole thermal magnetic breakers for maximum circuit length of 360 feet.
f. Heat tracing system shall include:
1) Self-regulating or power-limiting parallel circuit construction consisting of an inner core of conductive material between parallel copper bus wires, with inverse temperature-conductivity characteristics with metal over-braid with maximum continuous exposure temperature of 185 DEGF.
2) 30 mA Ground Fault Equipment Protection
3) Mod bus RTU/RS485 or TCP/Ethernet Communications
4) Lockout Capable Breakers
5) Thermostats adjustable between 35 and 200 DEGF minimum with maximum differential range of 9 DEGF, furnished complete with NEMA 4X enclosures in all areas, stainless steel temperature bulb and capillary.
6) All necessary or required components and accessories, such as power connection boxes, splice boxes, end seals, straps, tape, and fitting brackets.
7) Polyolefin or Fluoropolymer (Teflon) insulation.

8) All ancillary junction points and system junction boxes including, Splice/Tee Boxes, End Seals and end of circuit monitoring.

9. Pipe Insulation (Outdoors)
   a. Provide preformed fiberglass pipe insulation on all exterior piping that is heat traced.
   b. Minimum insulation thickness shall be 2 IN.
   c. Insulation characteristics shall include:
      1) Density (approximate): 4 LBS/CUFT.
      2) Temperature rating (minimum): 600 DEGF.
      3) Average thermal conductivity not to exceed 0.23 (BTU-IN)/(HR-FT 2-DegF) at a mean temperature of 75 DEGF.
      4) Fire hazard rating in accordance with UL 723, ASTM E84, and/or NFPA 255 with a flame spread rating not exceeding 25 and smoke developed rating not exceeding 50.
      5) Moisture absorption in accordance with ASTM C553 and not greater than 5 PCT moisture by volume when exposed to moisture laden air at 120 DEGF and 96 PCT relative humidity.
      6) Fungi and bacteria resistance in accordance with ASTM C665 and shall not breed or promote growth.
   d. Provide piping jacket of preformed PVC with a minimum thickness of 0.028 IN and proven resistance to ultraviolet degradation when temperatures do not exceed the limits of PVC.
   e. Provide insulation protection shield or calcium silicate insulation insert at pipe support locations to avoid crushing insulation.

C. Carbon Steel Piping:
   1. Pipe:
      a. Carbon steel pipe shall conform to ASTM A53, Grade B, electric resistance welded or seamless (Type E or S).
      b. Pipe smaller than 2-1/2 inches shall be Schedule 80.
      c. Pipe 2-1/2 inches and larger shall be Schedule 40.
   2. Fittings:
      a. Fittings smaller than 2-1/2 inches shall conform to ASTM A105.
      b. Fittings 2-1/2 inches and larger shall conform to ASTM A234, Grade WPB.
   3. Flanges:
      a. Flanges shall conform to ASTM A105.
      b. Bolts shall conform to ASTM A193, Class 2, Grade B7.
      c. Nuts shall conform to ASTM A194, Grade 2H.
      d. Washers shall conform to ASTM F436, Type 1 (carbon steel), flat circular for carbon steel bolts.

D. Stainless Steel Piping:
   1. Pipe:
      a. Stainless steel pipe shall conform to ASTM A312, Grade TP304L, seamless only, pickled and passivated as per ASTM A380.
      b. Pipe smaller than 3 inches shall be Schedule 40S.
      c. Pipe 3 inches and larger shall be Schedule 10S.
   2. Fittings:
      a. Fittings smaller than 2-1/2 inches shall conform to ASTM A182, Grade F304L.
      b. Fittings 2-1/2 inches and larger shall conform to ASTM A403, Grade WP304L, seamless only.
   3. Flanges:
      a. Flanges shall conform to ASTM A182, Grade F304 or F304L, forged type.
      b. Bolts shall conform to ASTM A193, Class 2, Grade B8.
      c. Nuts shall conform to ASTM A194, Grade 8.
      d. Washers shall be flat circular and Type 304 stainless steel.

E. Piping Components:

1. Welded Nipples:
    a. Nipples shall conform to ASTM A733 and be constructed of the same material as the connecting pipe.
2. Steel Couplings:
    a. Couplings shall conform to API Spec 5L, seamless, extra heavy, wrought steel with recessed ends.
3. Threaded Unions:
    a. Unions shall conform to ASME B16.39, Class 150.
4. Flexible Pipe Connector:
    a. Connector shall be the flexible, close pitch, metal hose type that is constructed with exterior annular corrugations and provided with a single layer of braided wire sheath covering.
    b. Connectors shall be constructed entirely of stainless steel and be rated for the system working pressure and temperature.
    c. Connectors shall have flanged end connections.

F. Pipe Hangers and Supports.
1. General requirements:
    a. Hangers and supports shall be the adjustable type conforming to MSS SP-58, except as modified herein.
    b. Provide hot-dipped galvanized finish on rods, nuts, bolts, washers, hangers, and supports.
    c. Provide miscellaneous metal that conforms to ASTM A36, standard mill finished structural steel shapes, hot-dipped galvanized.
    d. Provide PTFE pads between the piping and pipe supports to eliminate contact of dissimilar metals and prevent abrasion of the piping.
2. Pipe protection shields:
    a. Shields shall conform to MSS SP-58, Type 40, except material shall be Type 316 stainless steel. Provide shields at each slide type pipe hanger and support.
3. Low Friction Supports:
    a. Supports must have self-lubricating anti-friction bearing elements composed of 100% virgin tetrafluoroethylene polymer and reinforcing aggregates, prebonded to appropriate backing steel members. The coefficient of static friction between bearing elements shall be 0.06 from initial installation for both vertical and horizontal loads and deformation shall not exceed 0.002 inches under allowable static loads.
    b. Bonds between material and steel shall be heat cured, high temperature epoxy.
    c. Design pipe hangers and support elements for the loads applied.
    d. Provide anti-friction material with a minimum of 0.09 inches thickness.
    e. Provide hot-dipped galvanized steel supports.
    f. Provide supports that are factory designed and manufactured.

2.4 GENERAL VALVES

A. General Requirements:
1. Valves shall have flanged or threaded end connections and conform to ASME B16.34, Class 150.
2. Body, bonnet, and cover materials shall correspond to the piping system materials:
    a. Carbon steel: ASTM A216.
    b. Stainless steel: ASTM A351, Grade CF8M; ASTM A743, Type 304 or 316; or ASTM A216, Grade WCB internally plated with nickel.
3. Provide stainless steel stem and trim.
4. Seats, body seals, and stem seals shall be Viton, Buna-N, or PTFE (except where graphite required by valve manufacturer for fire resistance rating).

B. Ball Valve:
1. Ball valve shall be:
    a. Apollo 88A-100 Series.

b. Flow-Tek RF15.
c. Morrison Brothers Figure 691B.
d. Or equal.

2. Ball valve shall be non-lubricated, double-seated, with lockable handles.
3. Ball valve shall be fire safe in accordance with API Spec. 607or UL 842 listed for flammable liquids.

C. Check Valve:
1. Check valves shall be:
a. Crane Fig. 147.
b. Morrison Brothers Figure 246A.
c. Or equal.
2. Check valve shall be full opening, tilting disc, non-slam, swing type. Disc shall be guided and controlled to contact the entire seating surface.

D. Foot Valve:
1. Foot valves shall be:
a. Morrison Brothers Figure 335A.
b. Or equal.
2. Foot valve shall be double-poppet type with metal-to-metal seating.
3. Foot valve shall include inlet filter screen with stainless steel materials.

E. Relief Valve:
1. Expansion relief valves shall be:
a. Taylor 8200 Series (atmospheric pressure at discharge).
b. Taylor 7700 Series (backpressure at discharge).
c. Or equal.
2. Provide relief valves at each location where segments of pipe can be isolated by valving or blinding. Utilize balanced-type relief valves as required to prevent excessive cascading pressures. Product shall return back to storage tank.

F. Anti-Siphon Valve:
1. Anti-siphon valves shall be:
a. Morrison Brothers Figure 910.
b. Approved equals.
2. Anti-siphon valve shall be normally closed, spring loaded, angle pattern type valve.
3. Provide anti-siphon valve at the high point of the fuel lines from a storage tank.
4. Valve opening pressure shall be selected to correspond to vertical drop from tank.
5. Valve shall include integral thermal expansion relief.
6. Use anti-siphon solenoid valves (below) on remote pump suction lines.

G. Fire Safety Valve:
1. Fire safety valves shall be:
a. Morrison Brothers Figure 446.
b. Or equal.
2. Provide a fire safety valve close to where the fuel supply line enters the building.
3. Valve shall have a fusible link. Valve shall activate and stop flow in the event of a fire.

H. Anti-Siphon Solenoid Valve:
1. Anti-siphon solenoid valves shall be:
a. Morrison Brothers Figure 710.
b. Or equal.
2. Provide anti-siphon solenoid valve at the high point of the fuel supply lines from a storage tank.
3. Valve shall be normally closed and shall open upon receipt of an electronic signal from corresponding pumps (tank-mounted pump, dispenser pump, fuel polisher pump, day tank supply pump, generator fuel pump, etc.) when switched on.
4. Valve shall operate on 120 VAC.

5. Enclosure shall be watertight and rated for hazardous locations – NEMA 3, 4X, 7 and 9; groups C and D.
6. Valve shall include integral thermal expansion relief.
7. Valve shall include manual override option where available.

2.5 ACCESSORIES

A. Pressure Gauge:
1. Pressure gauges shall be:
   a. Ashcroft 1009.
   b. Or equal.
2. Pressure gauge shall be the single style type that conforms to ASME B40.100.
3. Pressure gauge shall have a minimum 4 inches dial size.
4. Pressure gauge materials shall be stainless steel case and tube.
5. Pressure gauge shall be liquid-filled with glycerin.
6. Provide pressure gauge assembly with isolation ball valve, pressure snubber, and scale range as indicated.

B. Line Strainer
1. Strainer shall be:
   a. Morrison Brothers Figure 285B.
   b. Or equal.
2. Provide a Y-type line strainer in each product discharge line.
3. Basket shall have bottom access and be removable for cleaning and/or replacement.
4. Strainer materials shall be cast iron brass body and 100 mesh, stainless steel screen.

C. Cam and Groove Couplers:
1. Coupler shall be:
   a. Morrison Brothers Figure 800A.
   b. OPW 633AST.
   c. Or equal.
2. Coupler shall be aluminum or bronze.
3. Provide coupler with a lockable cap.
   a. Cap shall be:
      1) Morrison Brothers Figure 800DC.
      2) OPW 634BK.
      3) Or equal.
   b. Cap shall mate with the adapter and have a latching mechanism which provides a water tight seal.
   c. Cap shall provide some type of locking provision and be easily attachable and removable.
   d. Cap shall be attached to the fill adapter by a 12 inch section of brass cable or fuel resistant rope.
   e. Cap gasket shall be Buna-N or Viton material.

2.6 FINISHES

A. General Requirements:
1. Ship, store, and handle coating materials as well as apply and cure coatings in accordance with SSPC PA 1.

B. Above Grade Fuel Service Piping:
1. Coat the exterior of above ground carbon steel piping, flanges, fittings, nuts, bolts, washers, and piping components in accordance with Section 09 96 00, using a three-part coating system. Coating system shall include a zinc-rich epoxy primer, epoxy polyamide intermediate coat, and polyurethane top coat. Coating system shall be compatible with fuel to be handled and appropriate for outdoor exposures including UV resistance.
2. Perform surface preparation and coating application in accordance with product manufacturer's instructions.

3. Do not coat stainless steel piping, flanges, fittings, or other components.
4. Do not coat piping components provided with a complete factory coating.
5. Prior to any coating, clean surfaces to remove dust, dirt, rust, oil, and grease.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Procure necessary permits, provide required notifications, and coordinate inspections with Arizona State Fire Marshal's Office, Arizona Department of Environmental Quality, and other agencies as required.

B. Installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing shall be in accordance with ASME B31.3, NFPA 30 , NFPA 37, NFPA 110, and local codes except as modified herein.

C. General Locations and Arrangements:
   1. Drawings (plans, schematics, and diagrams) indicate the general location and arrangement of the piping systems.
   2. Final location and arrangement of piping layout shall take into consideration pipe sizing and friction loss, expansion, pump sizing, and other design considerations.
   3. So far as practical, install piping as indicated.

D. Safety rules as specified in NFPA 30 and OSHA shall be strictly observed.

E. Never direct bury threaded connections, unions, flanges, valves, air vents, or drains. Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible.

F. Piping:
   1. General:
      a. Thoroughly clean pipe of all scale and foreign matter before the piping is assembled.
      b. Cut pipe accurately to measurements established at the jobsite, and work into place without springing or forcing. Cut pipe square and have burrs removed by reaming.
      c. Install pipe to permit free expansion and contraction without causing damage to the building structure, pipe, joints, or hangers.
      d. Cutting or other weakening of the building structure to facilitate piping installation will not be permitted without written approval.
      e. Use reducing fittings for changes in pipe sizes.
      f. Install equipment and piping into space allotted and allow adequate acceptable clearances for installation, replacement, entry, servicing, and maintenance.
      g. Provide electric isolation fittings between dissimilar metals.
      h. Install piping straight and true to bear evenly on supports.
      i. Piping shall be free of sags and traps, shall not be embedded in concrete pavement, and shall drain as indicated.
      j. Make changes in direction with fittings. Mitering or notching pipe or other similar construction to form elbows or tees will not be permitted.
      k. When work is not in progress, securely close open ends of pipe and fittings with an expandable pipe plug so that water, earth, or other substances cannot enter the pipe or fittings.
   2. Welded Connections:
      a. Unless otherwise indicated on the drawings, pipe joints shall be welded.
      b. Welding shall be performed in accordance with the requirements of ASME B31.3.
      c. Construct branch connections with welding tees or forged welding branch outlets.
      d. Do not weld stainless steel pipe to carbon steel pipe.
      e. Make all edges true. Do not leave a shoulder on the inside of the pipe.
      f. Maintain a 1/16 inches clearance between the end of the pipe and the shoulder in the socket during the installation of socket-welded fittings and valves.

g. Welds shall present a smooth, regular workmanship appearance free from defects including undercutting, porosity, and cracking.
h. Perform non-destructive examination (NDE) per the requirements of ASME B31.3 for Normal Fluid Service.
    1) Use inspectors qualified to at least Level II in accordance with ASNT SNT-TC-1A.
    2) Conform to ASME BPVC, Section V.
    3) Examine 100% of all welds using visual examination.
    4) Examine at least 10% of all welds using liquid penetrant examination.
        a) If defects are detected, examine all welds using this procedure.
        b) Remove all penetrant test materials by flushing, washing, or wiping clean with appropriate solvents.
    5) Examine at least 5% of all butt welds using radiographic examination.

3. Existing Piping Systems:
    a. No interruptions or isolation of an existing fuel handling service or system shall be performed unless the actions are approved by the Contracting Officer Representative.
    b. Undertake connections in fashion which will disturb system as little as possible.
    c. Provide suitable equipment and facilities to drain and dispose of liquid removed without damage to NPS property.
    d. Once tie-in to an existing system is initiated, continue work continuously until tie-in is made and tested.
    e. Perform initial cutting of existing fuel pipe with a multiwheel pipe cutter, using a nonflammable lubricant. After cut is made, seal interior of piping with a gas barrier plug. Purge interior of piping with carbon dioxide or nitrogen prior to performing any welding process.
4. Threaded Connections:
    a. Provide threaded end connections only on piping 2 inches in nominal size or smaller and only where necessary for attachment to valves or equipment.
    b. Provide adequate and accessible unions for disassembly and maintenance or replacement of valves and equipment.
    c. Provide threaded connections with PTFE tape or equivalent thread-joint compound applied to the male threads only.
    d. Not more than three threads shall show after the joint is tightened.
5. Bolted Connections:
    a. For each bolted connection of stainless steel components (e.g., pipes, piping components, valves, and equipment) use stainless steel bolts or studs, nuts, and washers.
    b. For each bolted connection of carbon steel components, use carbon steel bolts or studs, nuts, and washers.
    c. Extend bolts, or studs, no less than two full threads beyond their corresponding nut when tightened to the required torque.
    d. Prior to installing nuts, apply a compatible anti-seize compound to the male threads.
6. Flanges:
    a. Except where threaded end connections and/or unions are indicated, provide flanged joints in each line immediately preceding the connection to a piece of equipment or material requiring maintenance such as pumps, general valves, control valves, strainers, and other similar items and as indicated.
    b. Assemble flanged joints square and tight with matched flanges, gaskets, and bolts.
    c. Flange bolt holes shall evenly straddle centerlines.
    d. Use flanged connections between FRP pipe and metal pipe with the metal pipe anchored within 5 feet of the connection.
    e. For flanges, provide washers under each bolt head and nut.
    f. Torque wrenches shall be used to tighten all flange bolts to the torque recommended by the gasket manufacturer. Tightening pattern shall be as recommended by the gasket manufacturer.
    g. Use anti-seize compound on threads for stainless steel bolts.

7. Manual Air Vents:
    a. Provide 1 inch (or match main line size if smaller) air vents at all high points and where indicated to ensure adequate venting of the piping system.
8. Drains:
    a. Provide 1 inch (or match main line size if smaller) drains at all low points and where indicated to ensure complete drainage of the piping. Drains shall be accessible, and shall consist of nipples, ball valves, and caps or plugged tees unless otherwise indicated.
9. Flexible Pipe Connectors:
    a. Attach connectors to components in strict accordance with the latest printed instructions of the manufacturer to ensure a vapor tight joint.
    b. Hangers, when required to suspend the connectors, shall be of the type recommended by the flexible pipe connector manufacturer and shall be provided at the intervals recommended.
10. Grounding:
    a. Provide adequate grounding of all above grade piping and equipment per requirements of Division 26.
    b. Provide approved jumpers where fittings cause a break in the electrical continuity of the system.

G. Pipe Hangers and Supports:
1. Install hangers with a maximum spacing as defined in Table 1 below, except where indicated otherwise. In addition to meeting the requirements of Table 1, provide additional hangers and supports where concentrated piping loads exist (e.g., valves).

Table 1. Maximum Hanger Spacing

Nominal Pipe Size (IN)	0-1	1.5	2	3	4	6	8	10	12
Maximum Hanger Spacing (FT)	7	9	10	12	14	17	19	22	23

3.2 COORDINATION

A. Coordinate the work of the system manufacturer's service personnel during construction, testing, calibration, and acceptance of the system.

B. Coordinate installation of dispenser sumps, containment sumps, and piping with all other trades to ensure piping is installed and dispenser sumps are bolted to the fuel island frames prior to placing of concrete.

3.3 FUEL SUPPLY

A. No fuel will be delivered to the system until the Contractor has satisfactorily completed the initial system cleaning and adjusting.

B. Upon completion and acceptance of construction, examination, and initial testing, the Contractor will deliver fuel to the storage tanks to enable the Contractor to make final adjustments to equipment and controls and complete performance testing.

C. Any fuel found to be contaminated and unsuitable for reuse during subsequent testing and commissioning activities will be the responsibility of the Contractor to remove and dispose, including pumping out the entire fuel supply from the storage tanks and piping system.

3.4 PRESSURE TESTS

A. General Testing Requirements:
1. All tests shall be made in the presence of the Contracting Officer or Contracting Officer Representative. Any preliminary tests that the Contractor may make without such tests being observed by the Contracting Officer or Contracting Officer Representativewill not be

accepted unless prior approval is obtained. The Contractor shall provide notification at least 48 hours before any work is to be inspected or tested.

2. Furnish labor, materials, equipment, electricity, repairs, and retesting necessary for any of the tests required herein.
3. Perform piping test in accordance with the applicable requirements of ASME B31.3 except as modified herein.
4. To facilitate the tests, various sections of the piping system may be isolated and tested separately.
5. Where piping sections terminate at flanged valve points, close the line by means of blind flanges in lieu of relying on the valve.
6. Provide tapped flanges to allow a direct connection between the piping and the air compressor and/or pressurizing pump. Use tapped flanges for gauge connections. Taps in the permanent line will not be permitted. Gauges will be subject to testing and approval.
7. Provide provisions to prevent displacement of the piping during testing.
8. Keep personnel clear of the piping during pneumatic testing. Only authorized personnel shall be permitted in the area during pneumatic and hydrostatic testing.
9. Isolate equipment such as pumps, tanks, and meters from the piping system during the testing.
10. Do not exceed the pressure rating of any component in the piping system during the testing.
11. Following satisfactory completion of each test, relieve the test pressure and seal the pipe immediately.
12. Piping to be installed underground shall not be covered by backfill until the piping has passed the final pneumatic tests described herein.

B. Preliminary Pneumatic Test:
   1. Apply a 25 psiG pneumatic test to product piping. Maintain the pressure while soapsuds or equivalent materials are applied to the exterior of the piping. While applying the soapsuds, visually inspect the entire run of piping, including the bottom surfaces, for leaks (bubble formations).
   2. If leaks are discovered, repair the leaks accordingly and retest.

C. Hydrostatic Test:
   1. Hydrostatically test piping with the fuel to be handled to the lesser of 1-1/2 times operating pressure or 275 psiG in accordance with API RP 1110. Maintain the pressure within the piping for 4 hours with no leakage or reduction in gauge pressure.
   2. If leaks are discovered, repair the leaks accordingly and retest.

D. Exterior Containment Piping Tests:
   1. Apply a minimum pneumatic pressure of 5 psiG to the exterior containment piping (i.e., double wall secondary piping). Maintain the pressure for at least 1 hour while soapsuds or equivalent materials are applied to the exterior of the piping. While applying the soapsuds, visually inspect the entire run of piping, including the bottom surfaces, for leaks (bubble formations).
   2. Repair leaks discovered in accordance with manufacturer's instructions and retest.
   3. Perform testing in compliance with the manufacturer's published installation instructions.

E. Containment Sump Tests:
   1. Test the integrity of the containment sumps using water.
   2. Fill sumps to a level near the top of the sump and at least 4 inches about the highest penetration.
   3. Liquid level shall not lower by more than 1/8 inches over a 1 hour period.
   4. Repair leaks and retest if required.
   5. After the testing, remove all water and clean and dry the sump.

3.5 FLUSHING AND ADJUSTING

A. Flushing:
   1. All new and modified fuel piping shall be flushed with fuel.

2. Fuel shall be pumped at the system's maximum design flow rate from a storage tank, through the fuel supply piping, through dispensers or other equipment, and then into an empty fuel truck or tank as supplied by the Contractor.
3. During flushing procedure, periodically bleed air through high point vents and flush low point drains.
4. Flushing shall continue until the fuel being delivered is free of construction debris. Samples of fuel shall be free of visible contamination or free water.
   a. A minimum of 100 GAL shall be flushed through the system.
   b. The flushing procedure shall be continued until the fuel being delivered is acceptable to the Contracting Officer or Contracting Officer Representative.
5. Replace filters and clean strainers after the flushing operation as needed.

B. Initial System Adjustments:
1. Following the flushing and cleaning operations, each system component shall be initially adjusted, if necessary, to meet the system's final operational requirements.
2. Flow rates and pressures shall be adjusted as required to meet the indicated requirements.
3. The sequence of control for each component shall be adjusted to meet the indicated system requirements.
4. Following the initial system adjustments, the equipment performance tests shall be performed in order to determine any necessary final system adjustments.

3.6 PERFORMANCE TESTS

A. System Performance Tests:
1. After all components of the system have been properly adjusted, test the system to demonstrate that the system meets the performance requirements for which it was designed.
2. If any portion of the system or any piece of equipment fails to pass the tests, the Contractor shall make the necessary repairs or adjustments and the test shall be repeated until satisfactory performance is achieved.
3. Tests shall be performed under all operating scenarios.
4. The system shall be filled with fuel and shall be operable and leak-free prior to acceptance. The Contractor shall be responsible for any leaks in the new or modified portions of the system. Anything wet with fuel is considered to be leaking.

3.7 INITIAL START-UP

A. Contractor shall verify system is ready for service including reinstallation of any manway covers or blind flanges, removal of any blanks, and restoration of any other items modified for the cleaning and testing process.

B. Provide Contracting Officer or Contracting Officer Representative with a written statement that manufacturer's equipment has been installed properly (including conformance with applicable codes and standards), has been started up, and is ready for operation by NPS personnel.

3.8 TRAINING

A. Contractor shall conduct a training course for the operating staff as designated by the Contracting Officer or Contracting Officer Representative.
1. The training period shall be demonstrated to fuel system operators and shall start after the system is functionally completed but prior to final system acceptance.
2. The field instructions shall cover all of the items contained in the operation and maintenance manuals, as well as demonstrations of routine maintenance operations.
3. See Section 01 79 00 for Operator training requirements.

3.9 CLOSEOUT ACTIVITIES

A. Refer to Section 01 81 33 – Cyber-Security Requirements for cyber security related closeout requirements.

END OF SECTION

This page intentionally left blank.

SECTION 33 56 13
ABOVEGROUND FUEL STORAGE TANKS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:
   1. Aboveground fuel storage tank(s) including all components and structures required for a complete operating and approved installation.

B. Related Specification Sections include but are not necessarily limited to:
   1. Division 26 - Electrical.
   2. Section 33 52 10 - Service Piping: Fuel Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Petroleum Institute (API):
      a. 2000, Venting Atmospheric and Low-Pressure Storage Tanks.
      b. MPMS 2.2A, Measurement and Calibration of Upright Cylindrical Tanks by the Manual Strapping Method.
      c. MPMS 2.2E, Petroleum and Liquid Petroleum Products – Calibration of Horizontal Cylindrical Tanks – Part 1: Manual Methods.
   2. American Society of Civil Engineers (ASCE):
      a. 7, Minimum Design Loads and Associated Criteria for Buildings and Other Structures.
   3. American Society of Mechanical Engineers (ASME):
      a. B16.5, Pipe Flanges and Flanged Fittings.
   4. American Society for Testing and Materials (ASTM):
      a. A123, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
   5. Code of Federal Regulations (CFR):
      a. 40 CFR 112, Oil Pollution Prevention.
   6. International Building Code (IBC).
   7. International Fire Code (IFC).
   8. National Fire Protection Association (NFPA):
      a. 20, Standard for the Installation of Stationary Pumps for Fire Protection.
      b. 30, Flammable and Combustible Liquids Code.
      c. 30A, Code for Motor Fuel Dispensing Facilities and Repair Garages.
      d. 31, Standard for the Installation of Oil-Burning Equipment.
      e. 37, Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines.
      f. 70, National Electrical Code.
      g. 110, Standard for Emergency and Standby Power Systems.
      h. 704, Standard System for the Identification of the Hazards of Materials for Emergency Response.
   9. Occupational Safety and Health Administration (OSHA):
      a. 29 CFR 1910.106, Flammable Liquids.
   10. Petroleum Equipment Institute (PEI):
      a. RP200, Recommended Practices for Installation of Aboveground Storage Systems for Motor-Vehicle Fueling.
   11. Steel Tank Institute (STI):
      a. R912, Shop Fabricated Stationary Aboveground Storage Tanks for Flammable, Combustible Liquids. – Installation Instructions.
   12. Underwriters Laboratories, Inc. (UL):
      a. 50, Enclosures for Electrical Equipment, Non-Environmental Considerations.

b. 80, Steel Tanks for Oil-Burner Fuels and Other Combustible Liquids.
c. 142, Steel Aboveground Tanks for Flammable and Combustible Liquids.
d. 443, Steel Auxiliary Tanks for Oil-Burner Fuel.
e. 508A, Standard for Industrial Control Panels.
f. 2085, Protected Aboveground Tanks for Flammable and Combustible Liquids.

B. Regulatory Requirements:
1. Tank installers must be licensed / certified by the state when the state requires licensed installers.

C. Inspections:
1. Visually inspect tanks and associated components at time of delivery and just prior to installation.
a. Do not use defective materials.
b. Immediately remove rejected material from the project site.

D. Tank Manufacturer’s Qualifications:
1. Welders to be ASME qualified.
2. Manufacturer to have its own fabricating plant and must show evidence of a minimum of five installations and five years of experience in the design and manufacturer of aboveground chemical storage tanks of capacity and type specified herein.

1.3 SYSTEM DESCRIPTION

A. Design Requirements:
1. Provide the entire fuel storage and distribution system as a complete and fully operational system.
2. Size, select, construct, and install equipment and system components to operate together as a complete system.
3. Tanks and associated components specified herein shall be compatible with the fuel to be handled.

1.4 SUBMITTALS

A. Shop Drawings:
1. Complete detailed drawings of equipment provided including dimensions and connection locations.
2. Details on nozzles, penetrations, and other openings.
3. Details on platforms, stairs, ladders, handrails, etc., as required.
4. Support loads (including seismic as needed) and anchor bolt design information.
5. Tank manufacturer's calibration charts.

B. Product Data:
1. Acknowledgement that products submitted meet requirements of standards referenced.
2. Copies of manufacturer's standard catalog data (prior to the purchase or installation of the particular component) highlighted to show brand name, model number, size, options, performance charts and curves, etc., in sufficient detail to demonstrate compliance with contract requirements on all parts and equipment.
3. Copies of manufacturer's written directions regarding material handling, delivery, storage, and installation.
4. Copies of manufacturer’s installation, operation, and maintenance (IOM) manuals.

C. Factory Test Reports:
1. Radiographic inspection.
2. Pressure and vacuum tests (both primary tank and secondary containment tank).

D. Field Test Reports:
1. Tank tightness tests.

E. Certificates:
1. Contractor qualifications:

a. State licenses or certificates, if applicable.
2. Mill test reports.
3. Certificates of Compliance to required codes and UL listings.
4. Warranty validation card.
5. Tank strapping table.

1.5 WARRANTY

A. Tank will not fail for a period of 30 years from the date of original purchase due to external corrosion.

B. Tank will not fail for a period of 30 years from the date of original purchase due to internal corrosion provided the tank is used solely with specified fuel.

C. Will not fail for a period of 30 years from the date of original purchase due to structural failure (defined as breaking or collapse) provided the installation is performed and validated by a qualified installation contractor and the tank is used as stated above.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Handle, store, and protect tanks and associated components to prevent damage before and during installation in accordance with the manufacturer's recommendations. Replace damaged or defective items.
1. Do not handle or move tank unless it is empty. Do not drop or drag the tank.
2. Repair abrasions, scars, and blemishes. If repair of satisfactory quality cannot be achieved, replace damaged material immediately
3. Protect flanges, fittings, and specialties from moisture and dirt by inside storage and enclosure, or by packaging with durable, waterproof wrapping.
4. Use of bare cables, chains, hooks, metal bars, or narrow skids in contact with coated tanks or components is not permitted.
5. Avoid contact of any ferrous metal with stainless steel.
a. Use tools dedicated for stainless steel.
b. Handle using nylon slings and alloy chains, cable, or straps.
c. Store on racks constructed of non-ferrous metal or line with rubber.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following Manufacturers are acceptable:
1. Aboveground storage tanks:
a. Highland Tank & Manufacturing Company, Inc.
b. Or equal.
2. Tank components:
a. Franklin Fueling Systems.
b. Or equal.
3. Leak detection, alarm, and electronic level sensing:
a. Veeder-Root Company TLS Series.
b. Or equal.

4. Fuel polishing systems:
    a. Simplex.
    b. Or equal.

C. Submit requests for substitution in accordance with Division 01 Requirements.

2.2 CONDITIONS OF SERVICE

A. Properties of Fuel Stored:
1. Fuel/Fluid: Diesel.
2. Specific gravity: 0.84 at 60 DEGF (Design tank for 1.0 minimum).
3. Reid Vapor Pressure: < 0.1 PSI at 100 DEGF.
4. Flash Point: 126 DEGF.
5. pH: N/A.
6. Internal pressure: Atmospheric (tank vented through pressure/vacuum relief valve with a pressure setting of 8 oz/SQIN and a vacuum setting of 2 oz/SQIN).
7. Internal temperature: Ambient.

B. Structural Design Criteria:
1. Structural design loads for snow, ice, seismic, and wind shall be in accordance with the current edition of the IBC and ASCE 7. Geographic location of the installation is North Rim Grand Canyon Arizona.

C. Environment of Installation:
1. Storage tank will be installed in an above ground high elevation environment.
2. Tank shall be anchored to a concrete pad.

2.3 ABOVEGROUND STORAGE TANKS

A. Tank Configuration:
1. Type: Fireguard® thermally protected, double-wall, steel, aboveground tank.
2. Nominal capacity: 20,000 GAL.
3. Compartments: Single.
4. Configuration: Horizontal, Cylindrical.
5. Tank shall be manufacturer's standard size:
    a. Nominal primary tank diameter: 10 feet.
    b. Nominal primary tank length: 34 feet, 1 inch.
6. Materials:
    a. Primary tank: Carbon steel.
    b. Secondary containment: Carbon steel.
    c. Tank saddles: Carbon steel.
    d. Nozzles: Match primary tank material (using typical pipe fitting alloys).
    e. Interior ladders, drop tubes, and other components: All interior items exposed to the contents of the primary tank shall match material (and coating as applicable) of primary tank unless noted otherwise. This includes vents and manway covers.
    f. Exterior ladders, platforms, and other components: Exterior items shall be fiberglass (such as for grating) and color matched to the tank exterior color.
    g. Coatings: External coating shall be manufacturer's standard system provided it is compatible with the fuel to be handled. Provide internal epoxy lining compatible with the fuel. Color shall be dark green. Contractor shall provide color samples with final color selection by Contracting Officer.
    h. Galvanized materials (zinc coated) shall not be allowed in direct contact with any fuel.

B. Tank Design Criteria:
1. The entire tank assembly shall conform to UL 2085. Tank assembly shall bear the UL 2085 label as a protected tank.
    a. Tank design shall be tested for ballistics, impact, hose stream, and pool fire UL 2085 performance standards.
    b. Tank shall provide at least 2 hours fire protection.

2. The storage tank system must be provided with all safety and overfill prevention devices required by NFPA 30 NFPA 37, NFPA 110, and local codes.
3. The tank assembly and supports shall meet IBC and state building code requirements.
4. Primary tank shall conform to UL 142.
5. Primary tank shall be supported within the containment reservoir with steel tank saddles, or other similar supports, fabricated and installed by the tank manufacturer.
6. The secondary containment reservoir shall be the factory-fabricated, steel type that fully encloses the primary storage tank. The containment reservoir shall conform to UL 142.
7. The interstitial space between the primary tank and the containment reservoir shall be both pressure testable and verifiable. Primary tank and containment reservoir shall be designed to withstand required test pressures per applicable codes.
8. Insulating material in the interstitial space shall be a factory-installed, lightweight, monolithic material and allow fluid migration to the leak detection monitoring point.
9. Tank shall be designed in a manner to provide for monitoring leak detection in the interstitial space.
10. Tank shall be designed to support accessory equipment, as shown on Drawings.

C. Tank Appurtenances:
   1. General:
      a. Provide size, location, and orientation of appurtenances as shown on Drawings.
      b. All openings shall be located in the top of the tank.
         1) Drains located at the bottom of a primary storage tank will not be allowed.
      c. All fill and return lines shall include drop tubes to within 6 inches of tank bottom.
   2. Nameplates:
      a. Nameplate shall list at a minimum manufacturer's name and address, model number, serial number, capacity, code stamps, and UL rating.
      b. Construct plates and fasteners of corrosion resistant materials (such as stainless steel or aluminum) with stamped lettering.
   3. Flanged Nozzles:
      a. Flanges shall conform to ASME B16.5, Class 150.
      b. Flanges shall be raised face and weld neck, slip-on, or socket weld style. Threaded flanges are not to be used.
      c. Gaskets, bolts, nuts, and washers shall conform to requirements of Section 33 52 10.
   4. Threaded Coupling Nozzles:
      a. Threaded MNPT or FNPT as indicated.
   5. Manways:
      a. Tank manways shall have an internal diameter of not less than 24 inches.
      b. Provide tank manways with a matching flanged watertight manway cover including a UL listed gasket.
      c. Piping will not penetrate through access manways (exceptions: primary tank emergency vent assembly, other items specifically shown on Drawings).
   6. Internal Ladder:
      a. Provide internal ladder at access manway to bottom of tank.
      b. Design ladder in accordance with OSHA standards.
   7. Support Saddles:
      a. Shall be sized such that the bottom of the tank is no more than 12 inches above grade.
   8. Lifting Lugs:
      a. Provide lifting lugs for installation.
      b. Lugs shall be capable of supporting the empty tank weight with a 2:1 safety factor.
   9. Striker Plates:
      a. Provide striker plates under each tank opening.
   10. Exterior Access:
      a. Provide platform, stair, ladder, catwalk, and handrail system allowing access to all tank top nozzles and manways as shown on Drawings.
      b. Design all access components in accordance with OSHA standards.
      c. Provide any necessary concrete to support above items.

d. Tank shall have appropriate warning signs as required by NFPA 704 and local regulations.

2.4 TANK COMPONENTS

A. Aboveground Tank Emergency Vent:
1. Provide emergency venting for both primary and secondary containment tanks of a double wall tank.
2. Emergency vent shall be:
   a. Morrison Brothers Figure 244.
   b. Protectoseal Series 7800.
   c. Or equal.
3. Vent shall be the normally-closed, UL listed type that vents outward and upward.
4. Vent capacity and design shall comply with API 2000, NFPA 30 , NFPA 37, NFPA 110, UL 2085, and UL 142 for the size of tank and fuel specified.
5. Vent shall be designed to relieve only at a pressure setting above that of the normal vent.
6. Provide vent with the flow rating (e.g., cubic feet per minute) permanently labeled on the vent's exterior.

B. Updraft Vent:
1. Provide updraft vent designed to provide normal venting for storage tank.
2. Updraft vent shall be:
   a. Morrison Brothers Figure 354.
   b. Or equal.
3. Vent must be used in conjunction with emergency vent.
4. Vent and vent pipe shall comply with API 2000, NFPA 30 , NFPA 37, NFPA 110, UL 142, and UL 2085 for the fuel specified and associated fill / discharge rates.
5. Vent body shall be constructed of either cast steel or aluminum.
6. Body shall be self-draining to protect from condensate and freezing.
7. Vent intake shall be covered with a 40-mesh stainless steel wire screen.
8. Vent pipe sizing shall be not less than 2 inches nominal diameter.

C. Tank Overfill Prevention Valve:
1. Provide tank with an overfill prevention valve installed at the fill port. Valve shall stop flow of product when the tank reaches a preset warning level.
2. Tank overfill prevention valve shall be:
   a. Morrison Brothers Figure 9095AA or 9095DS.
   b. OPW 61fSTOP.
   c. Or equal.
3. Overfill prevention valve shall comply with NFPA 30 , NFPA 37, and NFPA 110.
4. Valve shall be the float-activated, shutoff type that is an integral part of the drop tube used for filling.
5. Valve shall completely stop the flow of fuel into the tank when the liquid level rises above 95% of tank capacity. Fuel shut off shall be cushioned to prevent fluid hammer.
6. Valve shall be constructed of the same material as the fill tube.
7. Use adapters per manufacturer.
8. Provide valve fill port with cap. Cap shall mate with adapter and have a latching mechanism which provides a watertight seal.

2.5 TANK GAUGES

A. Stick Gauge:
1. Provide each tank with two wooden stick gauges.
2. Gauge length shall allow measurement of entire level of fuel in tank.
3. Gauges shall be compatible with the fuel to be measured (no swelling or damage from fuel contact).
4. Provide gauge with non-sparking caps on each end.

5. Mark gauges in feet and inches. The smallest unit of measure on the gauge shall be 1/16 inches.

B. Tank Strapping Table:
   1. Furnish two tank manufacturer's certified strapping tables (calibration charts) for each tank.
   2. Tables shall indicate the liquid contents in gallons for each 1/16 inches of tank depth.
   3. For each tank, provide an electronic media file of each strapping table.

C. Clock Gauge:
   1. Provide tank with clock gauge. Gauge shall be float-activated.
   2. Gauge shall be:
      a. Morrison Brothers Figure 818.
      b. Or equal.
   3. Gauge shall be accessible and viewable when the tank is filled.
      a. Gauge shall be viewable from 20 feet minimum.
      b. Orientate gauge facing towards tank fill point location.
   4. Gauge float and cable shall be stainless steel.
   5. Gauge shall have vapor tight seals to prevent condensation from fogging the viewing glass.

2.6 ADAPTERS, COUPLERS, AND ACCESSORIES

A. Remote Spill Container:
   1. Provide a remote spill container for use during remote filling operations.
   2. Remote spill container shall be:
      a. Morrison Brothers 515 series.
      b. OPW 211-RMOT.
      c. Or equal.
   3. 12-gauge steel construction.
   4. Minimum capacity of 15 GAL.
   5. Container used for diesel or oil loadout shall have a single 4 inches port.
   6. Container used for gasoline or ethanol loadout shall have double 4 inches ports.
   7. Container shall have a pedestal mount, lockable lid, and a 1-inch drain with lockable ball valve.

B. Tank Fill Line:
   1. Fill Adapter:
      a. Provide cam and groove fill adapter in API standard 3 inches size.
      b. Adapter shall be in accordance with cam and groove couplers in Section 33 52 10.
      c. Adapter shall be a top seal cam and groove adapter with a tight-fit connection to prevent vapor emissions during filling.
   2. Locking Ball Valve:
      a. Provide a locking ball valve for each product fill line.
      b. Ball valve shall be in accordance with Section 33 52 10.
   3. Check Valve:
      a. Provide a swing check valve for each product fill line.
      b. Check valves shall be in accordance with Section 33 52 10.

C. Water Draw-Off Pump:
   1. Provide hand-operated water draw-off pump capable of self-priming with sufficient lift to draw contents from the bottom of the fuel storage tank.
   2. Water draw-off pump shall be:
      a. Fill-Rite FR110.
      b. Or equal.
   3. Provide anti-siphon valve near the fuel storage tank connection nozzle.

2.7 MONITORING SYSTEMS

A. Monitoring panel, level sensors, and leak detection sensors shall be from one manufacturer and compatible with each other.

B. Coordinate connection requirements of level and leak detection sensors with tank nozzle configuration.

C. Sensors shall be compatible with the fuel to be handled.

D. Tank Interstitial Space Leak Detection Sensors:
   1. The interstitial space of each tank shall be continuously and automatically monitored to detect breaches in the integrity of the primary tank.
   2. Sensors shall be float or optical, non-discriminating type.
   3. Sensors shall be UL listed and intrinsically safe for use in a Class 1, Division 1, Group D environment as defined by NFPA 70.
   4. Sensors shall be easily removed from the tank.

E. Tank Level Sensors:
   1. Each tank shall be provided with a level gauging sensor.
   2. Sensors shall be temperature sensing, magnetostrictive type with stainless steel guide shaft and length to suit tank height.
   3. Sensors shall be accurate to within +/-1/16 inches and be capable of measuring a liquid level over the tank's full usable capacity.
   4. Sensors shall be capable of measuring water accumulation in inches from 3/4 to 5 inches off the bottom of the storage tank.
   5. Sensors shall be UL listed and intrinsically safe for use in a Class 1, Division 1, Group D environment as defined by NFPA 70.

F. Electronic Monitoring System Panel:
   1. The following tank level sensor setpoints shall be programmed at the monitoring panel:
      a. High-high: 95%.
      b. High: 90%.
      c. Low: 15%.
      d. Low-Low: At minimum pump submergence level.

2.8 FUEL POLISHING SYSTEMS

A. General Description:
   1. Fuel polishing system shall be a self-contained, factory-assembled, automated fuel filtration system.
   2. Fuel polishing system shall be capable of eliminating water, sediment, particulates, and microbial contamination as follows:
      a. Water: 99.9% removal.

B. Power Requirements:
   1. Fuel polishing system shall have a single power connection to feed pump(s), control panel, switches, sensors, etc., within the package.
   2. Power supply shall be 115, 230 VAC, 1 phase, 60 Hz.

C. Enclosure:
   1. All system components shall be contained within a powder coated, weatherproof, outdoor UL 50 listed enclosure with appropriate ventilation.
   2. Enclosure shall be provided with a mounting stand that positions primary components for easy access.
   3. Front access door shall be lockable.
   4. Provide enclosure with internal heater for freeze protection based on a climatic winter design temperature of 0 DEGF.

D. Piping / Tubing:
   1. Piping, tubing, and related accessories shall be primarily stainless steel construction.
   2. Provide containment trough beneath piping and filtration equipment within the housing.

E. Pump:
   1. Provide positive displacement, direct coupled, rotary pump.

2. Pump shall provide a minimum flow rate of 30 GPH.
3. Motor shall be UL listed, TEFC type, with solid-state type thermal overload protection.

F. Initial Filtration/Water Separation:
1. Provide assembly including centrifugal water and particulate separation with integral 30 micron maximum hydrophobic particulate filter.
2. Assembly shall also include integral water collection and drain valve.

G. Final Filtration:
1. Provide 3 micron maximum particulate and/or water adsorbing spin-on filter.

H. Instrumentation:
1. Provide vacuum pressure sensor or switch downstream of the initial filtration / water separation assembly to monitor condition of the assembly.
2. Provide pressure sensor or switch upstream of the final filtration assembly to monitor condition of the assembly.
3. Provide flow sensor or switch which monitors for inadequate flow.
4. Provide water sensor integral to the collection section of the initial filtration / water separation assembly which monitors for high water level.
5. Provide leak detection sensor in the containment trough.

I. Control Panel:
1. Provide control panel within the enclosure capable of automatic control of the filter polishing system as well as monitoring of system functions.
2. Control panel shall be UL 508A listed.
3. Controls, indications, and alarms shall be internal to the enclosure unless noted otherwise.
4. Panel shall provide the following controls:
   a. Pump hand-off-auto control switch.
   b. Programmable timer for controlling automatic cycling.
   c. Alarm reset.
   d. Emergency stop mushroom-top pushbutton (external to enclosure).
5. Panel shall provide the following indications:
   a. Pump operating hour counter.
   b. Power available status light.
   c. Pump running status light.
6. Panel shall provide the following alarms with system shutdown:
   a. High vacuum (indicating a problem with the initial filtration/water separation assembly).
   b. High pressure (indicating a problem with the final filtration assembly).
   c. Inadequate flow.
   d. High water level in the water separation collection.
   e. Leak in enclosure.
   f. Provide beacon light external to the enclosure to provide general indication of system alarm condition.
7. Panel shall accept signal from master emergency stop system controlling the fuel system as a whole. Functionality shall be the same as the panel's emergency stop.

PART 3 - EXECUTION

3.1 FACTORY TESTING

A. Comply with performance testing requirements of UL 142 and UL 2085.

B. Any test failure shall require corrective action and retest.

3.2 SHIPPING & DELIVERY

A. Provide coverings for all tank openings suitable for protecting against exposure to harmful weather conditions for a period of several months.

3.3 INSTALLATION

A. Procure necessary permits, provide required notifications, and coordinate inspections with Arizona State Fire Marshal's Office, Arizona Department of Environmental Quality, and other agencies as required.

B. Handle storage tank with care to prevent damage during placement and install in accordance with the manufacturer's written instructions, STI R912, NFPA 30 , NFPA 37, NFPA 110, and local codes.

C. Inspect the exterior surface of the tank for obvious visual damage prior to the placement of the tank.
   1. Correct surface damage to the tank according to manufacturer's requirements before proceeding with the equipment installation.

D. Erect all stairs, platforms, ladders, and handrails as applicable.

E. Secure tank to concrete footing as specified by manufacturer.

F. Field Painting:
   1. Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory, shall be painted and have identification markings applied as specified in Contract Documents.

G. Install tank with a 1% slope with tank-mounted pumps or issue lines on the high side and drain / water draw-off nozzle on the low side.

H. Install interstitial tank sensors and detectors at the tank's low end.
   1. Sensor installation shall be in accordance with the tank manufacturer's recommendations and shall not compromise the tank's secondary containment in any manner.
   2. Sensors shall be easily removed from a tank.
   3. Connection of metal conduit to steel tanks shall be with dielectric fittings.

I. Attach grounding cabling to the tank in accordance with Division 26 requirements.

3.4 FUEL SUPPLY

A. Refer to Section 33 52 10.

3.5 PRESSURE TESTS

A. General Testing Requirements:
   1. All tests shall be made in the presence of the Contracting Officer or Contracting Officer Representative. Any preliminary tests that the Contractor may make without such tests being observed by the Contracting Officer or Contracting Officer Representative will not be accepted unless prior approval is obtained. The Contractor shall provide notification at least 48 hours before any work is to be inspected or tested.
   2. Furnish labor, materials, equipment, electricity, repairs, and retesting necessary for any of the tests required herein.

B. Tank Tightness Tests:
   1. Perform tightness tests on the tank in presence of Contracting Officer or Contracting Officer Representative, Fire Marshal representative, NPS Staff, and/or tank manufacturer's representative prior to making piping connections or installing tank accessories.
   2. Perform testing in accordance with STI R912 except as modified herein or by manufacturer's written instructions.
   3. Provide test equipment, tools, blind flanges, and instruments necessary to accomplish testing.
   4. Repair leaks discovered during the tightness tests in accordance with tank manufacturer's instructions.
   5. Following any repair, re-test the tank until the tank successfully passes the testing requirements of this paragraph.

6. Tank tightness testing may be waived at Contracting Officer or Contracting Officer Representative discretion for tanks that continue to maintain a factory-applied vacuum at least until the tank is set in its final location.

3.6 CLEANING AND ADJUSTING

A. Prior to system performance testing, the following cleaning and adjustments shall be performed.
   1. Initial System Cleaning:
      a. Visually inspect and clean free of debris the interior of the tank before filling.
      b. In the event of entry into a storage tank, the Contractor shall ensure a safe atmosphere exists.
      c. Contractor shall remove all preservatives and foreign matter from valves, line strainers, pumps, and other equipment coming in contact with fuel.
      d. No fuel will be delivered to the system until the Contractor has satisfactorily completed this initial system cleaning.
   2. Initial System Adjustments:
      a. Each system component shall be initially adjusted, if necessary, to meet the system's final operational requirements.
      b. The sequence of control for each component shall be adjusted to meet the indicated system requirements.
      c. Following the initial system adjustments, the equipment tests shall be performed in order to determine any necessary final system adjustments.

3.7 PERFORMANCE TESTS

A. System Performance Tests:
   1. After all components of the system have been properly adjusted, test the system to demonstrate that the system meets the performance requirements for which it was designed.
   2. Test specific pieces of equipment in accordance with manufacturer’s written instructions and test procedures for ensuring proper operation.
   3. If any portion of the system or any piece of equipment fails to pass the tests, the Contractor shall make the necessary repairs or adjustments and the test shall be repeated until satisfactory performance is achieved.
   4. Tests shall be performed under all operating scenarios.
   5. The tests shall demonstrate the following:
      a. The sensors and alarms are operational and perform as designed.
      b. Overfill valve shuts off at high liquid level.
      c. Vent piping is clear of debris, and the pressure/vacuum relief vent is operating properly.
      d. Vapor recovery systems perform as designed.
      e. Each safety device works properly and is tested per NFPA requirements as necessary.
      f. Fuel polishing system performs per manufacturer’s testing instructions.

3.8 INITIAL START-UP

A. Refer to Section 33 52 10.

3.9 TRAINING

A. Refer to Section 33 52 10 and 01 79 00 for Operator Training Requirements.

END OF SECTION

This page intentionally left blank.

SECTION 33 56 16
FUEL DAY TANK SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:
   1. Packaged fuel day tank system including tank, pump(s), controls, and other components required for a complete operating and approved installation.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 81 33 – Cyber Security Requirements.
   2. Division 26 - Electrical.
   3. Section 33 52 10 - Service Piping - Fuel Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Petroleum Institute (API):
      a. 2000, Venting Atmospheric and Low-Pressure Storage Tanks.
   2. American Society of Mechanical Engineers (ASME):
      a. B16.5, Pipe Flanges and Flanged Fittings.
   3. Code of Federal Regulations (CFR):
      a. 40 CFR 112, Oil Pollution Prevention.
   4. International Building Code (IBC).
   5. International Fire Code (IFC).
   6. National Fire Protection Association (NFPA):
      a. 20, Standard for the Installation of Stationary Pumps for Fire Protection.
      b. 30, Flammable and Combustible Liquids Code.
      c. 31, Standard for the Installation of Oil-Burning Equipment.
      d. 37, Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines.
      e. 70, National Electrical Code.
      f. 110, Standard for Emergency and Standby Power Systems.
      g. 704, Standard System for the Identification of the Hazards of Materials for Emergency Response.
   7. Occupational Safety and Health Administration (OSHA):
      a. 29 CFR 1910.106, Flammable Liquids.
   8. Underwriters Laboratories, Inc. (UL):
      a. 80, Steel Tanks for Oil-Burner Fuels and Other Combustible Liquids.
      b. 142, Steel Aboveground Tanks for Flammable and Combustible Liquids.
      c. 443, Steel Auxiliary Tanks for Oil-Burner Fuel.
      d. 508A, Standard for Industrial Control Panels.
      e. 2085, Protected Aboveground Tanks for Flammable and Combustible Liquids.

B. Regulatory Requirements:
   1. Day tank installers must be licensed / certified by the state when the state requires licensed installers.

C. Inspections:
   1. Visually inspect day tanks and associated components at time of delivery and just prior to installation.
      a. Do not use defective materials.
      b. Immediately remove rejected material from the project site.

1.3 SYSTEM DESCRIPTION

A. Design Requirements:
   1. Provide the entire fuel distribution and storage system as a complete and fully operational system.
   2. Size, select, construct, and install equipment and system components to operate together as a complete system.
   3. Day tanks and associated components specified herein shall be compatible with the fuel to be handled.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Complete detailed drawings of equipment provided including dimensions and connection locations.
   2. Details on nozzles, penetrations, and other openings.
   3. Support loads (including seismic as needed) and anchor bolt design information.
   4. Details on electrical requirements and controls operation.

B. Product Data:
   1. Acknowledgement that products submitted meet requirements of standards referenced.
   2. Copies of manufacturer's standard catalog data (prior to the purchase or installation of the particular component) highlighted to show brand name, model number, size, options, performance charts and curves, etc., in sufficient detail to demonstrate compliance with contract requirements on all parts and equipment.
   3. Copies of manufacturer's written directions regarding material handling, delivery, storage, and installation.
   4. Copies of manufacturer’s installation, operation, and maintenance (IOM) manuals.

C. Factory Test Reports:
   1. Pressure and/or vacuum tests (both primary tank and secondary containment tank).
   2. Operational test of control system.
   3. Operational test of pump(s).

D. Field Test Reports:
   1. Leak tests.

E. Certificates:
   1. Contractor qualifications:
      a. State licenses or certificates, if applicable.
   2. Certificates of Compliance to required codes and UL listings.
   3. Warranty validation card.

1.5 WARRANTY

A. Provide a minimum two-year field service warranty which covers all parts and all labor for guarantee of parts and workmanship.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Handle, store, and protect day tanks and associated components to prevent damage before and during installation in accordance with the manufacturer’s recommendations. Replace damaged or defective items.
   1. Repair abrasions, scars, and blemishes. If repair of satisfactory quality cannot be achieved, replace damaged material immediately.
   2. Protect flanges, fittings, and specialties from moisture and dirt by inside storage and enclosure, or by packaging with durable, waterproof wrapping.
   3. Use of bare cables, chains, hooks, metal bars, or narrow skids in contact with coated tanks or components is not permitted.
   4. Avoid contact of any ferrous metal with stainless steel.
      a. Use tools dedicated for stainless steel.

b. Handle using nylon slings and alloy chains, cable, or straps.
c. Store on racks constructed of non-ferrous metal or line with rubber.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following Manufacturers are acceptable:
   1. Day tank systems:
      a. Pryco, Inc.
      b. Approved Equal.

C. Submit requests for substitution in accordance with Specification Section 01 33 23.

2.2 CONDITIONS OF SERVICE

A. Properties of Fuel Stored:
   1. Fuel: Diesel.
   2. Specific gravity: 0.84 at 60 DEGF (Design tank for 1.0 minimum).
   3. Viscosity: 2.0 to 4.1 cSt.

B. Power Requirements:
   1. Packaged day tank system shall have a single power connection to feed pump(s), control panel, switches, sensors, etc., within the package.
   2. Power supply shall be 115 or 230 VAC, 1 phase, 60 Hz.

2.3 DAY TANKS

A. Tank Configuration:
   1. Type: Thermally protected, double-wall, steel, aboveground tank.
   2. Nominal capacity: 400 GAL hours of engine operation at full-rated load.
   3. Tank shall be manufacturer's standard size.
   4. Materials:
      a. Primary tank: Carbon steel.
      b. Secondary containment: Carbon steel.
      c. Tank saddles/supports: Carbon steel.
      d. Nozzles: Match primary tank material (using typical pipe fitting alloys).
      e. Drop tubes, and other components: All interior items exposed to the contents of the primary tank shall match material (and coating as applicable) of primary tank. This includes vents.
      f. Coatings: External coating shall be manufacturer's standard system provided it is compatible with the fuel to be handled. Provide internal epoxy lining compatible with the fuel if available.

B. Tank Design Criteria:
   1. Primary tank and secondary containment tank shall be all welded construction.
   2. Packaged system shall be included with a welded steel base suitable for bolt attachment to a concrete pad.
   3. The storage tank system must be provided with all safety devices required by NFPA 30 , NFPA 37, NFPA 110, and local codes.
   4. Primary tank shall conform to UL 142.

5. The secondary containment reservoir shall be the factory-fabricated, steel type that fully encloses the primary storage tank. The containment reservoir shall conform to UL 142.
6. The interstitial space between the primary tank and the containment reservoir shall be both pressure testable and verifiable. Primary tank and containment reservoir shall be designed to withstand required test pressures per applicable codes.
7. Tank shall be designed in a manner to provide for monitoring leak detection in the interstitial space.
8. Tank shall be designed to support accessory equipment.

C. Tank Appurtenances:
1. General:
   a. All openings shall be located in the top of the tank.
      1) Drains located at the bottom of a primary storage tank will not be allowed.
   b. All fill and return lines shall include drop tubes to within 6 inches maximum of tank bottom.
2. Flanged Nozzles:
   a. Flanges shall conform to ASME B16.5, Class 150.
   b. Flanges shall be raised face and weld neck, slip-on, or socket weld style. Threaded flanges are not to be used.
   c. Gaskets, bolts, nuts, and washers shall conform to requirements of Section 33 52 10.
3. Threaded Coupling Nozzles:
   a. Threaded couplings shall be standard MNPT or FNPT as required.
4. Lifting Lugs:
   a. Provide lifting lugs for installation.
   b. Lugs shall be capable of supporting the empty tank assembly weight with a 2:1 safety factor.
5. Striker Plates:
   a. Provide striker plates under each tank opening.

D. Galvanized materials (zinc coated) shall not be allowed in direct contact with any fuel.

E. Tank shall have appropriate warning signs as required by NFPA 704 and local regulations.

2.4 TANK COMPONENTS

A. Day Tank Emergency Vents:
1. Provide emergency venting for both primary and secondary containment tanks of a double wall tank.
2. Emergency vent shall be:
   a. Morrison Brothers Figure 244.
   b. Protectoseal Series 7800.
   c. Approved equals.
3. Vent shall be the normally-closed, UL listed type that vents outward and upward.
4. Vent capacity and design shall comply with API 2000, NFPA 30 , NFPA 37 , NFPA 110 , UL 142 for the size of tank and fuel specified.
5. Vent shall be designed to relieve only at a pressure setting above that of the normal vent.
6. Provide vent with the flow rating (e.g., cubic feet per minute) permanently labeled on the vent's exterior.

B. Updraft Vent:
1. Provide updraft vent designed to provide normal venting for storage tank.
2. Updraft vent shall be:
   a. Morrison Brothers Figure 354.
   b. Approved equals.
3. Vent must be used in conjunction with emergency vent.
4. Vent and vent pipe shall comply with API 2000, NFPA 30 , NFPA 37 , NFPA 110 , UL 142 for the fuel specified and associated fill / discharge rates.
5. Vent body shall be constructed of either cast steel or aluminum.

6. Body shall be self-draining to protect from condensate and freezing.
7. Vent intake shall be covered with a 40 mesh stainless steel wire screen.
8. Vent pipe sizing shall be not less than 2 inches nominal diameter.

C. Manual Level Gauge:
1. Provide buoyant force type gauge with visual indicator arranged for tank-top mounting.
2. Calibrate each reading dial or scale for its specific tank to read from empty to full, with intermediate points of 1/4, 1/2, and 3/4.

D. Piping:
1. Factory installed piping shall conform to the day tank manufacturer's standards.
2. Piping between the main storage tank and the day tank, and between the day tank and the serviced item (emergency generator, etc.), shall comply with Section 33 52 10.

2.5 FUEL TRANSFER AND RETURN PUMPS

A. Fuel transfer and return pumps shall be horizontal, positive displacement type. Direct connect pump to motor through a flexible coupling.

B. Fuel supply transfer pumps shall have a minimum rated flow of 4 GPM minimum and suction lift capability of 15 inHg minimum.

C. Fuel supply transfer pumps shall be in a duplex arrangement. Controls shall alternate lead / lag pumps.

D. Fuel return transfer pumps shall have a minimum rated flow of 150 percent of total supply pump capacity and of identical construction.

E. Equip each pump with a bypass relief valve if not provided with an internal relief valve.

F. Pump motors shall be TEFC construction, NEMA Type B, continuous duty at 40 degrees C, and 1.15 service factor.

2.6 CONTROLS

A. Refer to Section 01 81 33 - Cyber Security Requirements for cyber security related product requirements.

B. Control Panel:
1. Provide NEMA Type enclosed local control panel for the day tank.
2. Control panel shall be pre-wired, tested, and shipped as an integrated system.
3. Control panel shall include the following accessories:
   a. Manual-Off-Auto switch for each fuel transfer pump.
   b. Motor starters and controls.
   c. Power available LED.
   d. Critical low fuel alarm contacts for shut down of engine.
   e. Low-low level fuel alarm LED.
   f. Low-low level fuel alarm contacts for remote annunciator.
   g. Critical high level fuel alarm LED.
   h. Leak detection alarm LED.
   i. Leak detection alarm contacts for connection to fuel system monitoring panel
   j. Alarm horn.
   k. Alarm silence button.
   l. Alarm reset button.
4. Control panel shall provide Ethernet connectivity to transmit alarms and status indications to the facility distributed control system (DCS).
5. Control panel shall accept signal from master emergency stop system controlling the fuel system as a whole. When activated, all day tank pumps shall be shut down.

C. Level Controls:
1. Provide tank-top mounted, single-pole, single-throw type.

2. Arrange high level float switches to close on rise of liquid level, and low level float switches to close on fall of liquid level.
3. Alternatively to float switches, provide a UL Listed, integrated design, digital level controller for differential level control for activation of pumps, duplex pump alternator, tank level indication, system alarms, and manual operating controls. Level controller shall be self-contained as a unit within a NEMA 4 enclosure integral with the day tank assembly.
4. Operation shall be as follows:
    a. Critical low level setting shall activate at 5 percent of normal liquid level and activate contacts for shutting off the engine.
    b. Low-low level setting shall activate alarm at 25 percent of normal level and start the lag fuel transfer pump in conjunction with the lead pump.
    c. Low level setting shall open day tank isolation solenoid valves (if applicable), activate contacts for opening anti-siphon solenoid valves elsewhere in the fuel system, and start the fuel transfer pump at 50 percent of normal liquid level.
    d. High level setting shall close day tank isolation solenoid valves (if applicable), deactivate contacts to close anti-siphon solenoid valves elsewhere in the fuel system, and stop the fuel transfer pump at 90 percent of normal liquid level.
    e. High-high level setting shall start the fuel return pump at 95 percent of normal liquid level. Fuel return pump shall continue operation until the return armed setting (80 percent level) clears.

D. Temperature Switch:
1. Provide tank-top mounted, single-pole, single-throw type.
2. Alternatively, provide a UL Listed, integrated design, digital temperature transmitter with a NEMA 4 enclosure mounted on top of tank.
3. Operation shall be as follows:
    a. High temperature setpoint of 120DEGF shall activate a recirculation mode using the fuel transfer pumps and fuel return pump to supply fresh, cool fuel to the day tank and return heated fuel to the main storage tank.
    b. Solenoid valves both part of the day tank system and elsewhere in the fuel system shall be controlled in the same manner as level control.
    c. Recirculation mode shall deactivate at a setpoint of 100DEGF.

E. Leak Detector Switch:
1. Leak detector switch shall actuate when fuel is detected in the containment basin, and shall stop the fuel transfer pump and associated solenoid valves (same as level control).

F. Sensors shall be compatible with the fuel to be handled.

PART 3 - EXECUTION

3.1 FACTORY TESTING

A. Comply with performance testing requirements of UL 142.

B. Any test failure shall require corrective action and retest.

3.2 SHIPPING & DELIVERY

A. Provide coverings for all day tank openings suitable for protecting against exposure to harmful weather conditions for a period of several months.

3.3 INSTALLATION

A. Procure necessary permits, provide required notifications, and coordinate inspections with Arizona State Fire Marshal’s Office, Arizona Department of Environmental Quality, and other agencies as required.

B. Handle day tank with care to prevent damage during placement and install in accordance with the manufacturer's written instructions, NFPA 30 , NFPA 37, NFPA 110, and local codes.

C. Coordinate installation of the day tank system with other parts of the installation including main storage tank, engine-generator sets, etc., as applicable.

D. Inspect the exterior surface of the day tank for obvious visual damage prior to the placement of the tank.
   1. Correct surface damage to the tank according to manufacturer's requirements before proceeding with the equipment installation.

E. Field Painting:
   1. Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory, shall be painted and have identification markings applied as specified in Contract Documents.

F. Attach grounding cabling to the day tank system in accordance with Division 26 requirements and manufacturer instructions.

3.4 FUEL SUPPLY

A. Refer to Section 33 52 10.

3.5 PRESSURE TESTS

A. General Testing Requirements:
   1. All tests shall be made in the presence of the Contracting Officer or Contracting Officer Representative. Any preliminary tests that the Contractor may make without such tests being observed by the Contracting Officer or Contracting Officer Representative will not be accepted unless prior approval is obtained. The Contractor shall provide notification at least 48 hours before any work is to be inspected or tested.
   2. Furnish labor, materials, equipment, electricity, repairs, and retesting necessary for any of the tests required herein.

B. Tank Tightness Tests:
   1. Perform tightness tests on the tank in presence of Contracting Officer, NPS Staff, Fire Marshal representative, and tank manufacturer's representative prior to making piping connections or installing tank accessories.
   2. Perform testing in accordance with STI R912 except as modified herein or by manufacturer's written instructions.
   3. Provide test equipment, tools, blind flanges, and instruments necessary to accomplish testing.
   4. Repair leaks discovered during the tightness tests in accordance with tank manufacturer's instructions.
   5. Following any repair, re-test the tank until the tank successfully passes the testing requirements of this paragraph.
   6. Tank tightness testing may be waived at Contracting Officer Representative discretion for tanks that continue to maintain a factory-applied vacuum at least until the tank is set in its final location.

3.6 CLEANING AND ADJUSTING

A. Prior to system performance testing, the following cleaning and adjustments shall be performed.
   1. Initial System Cleaning:
      a. Visually inspect and clean free of debris the interior of the tank before filling.
      b. Contractor shall remove all preservatives and foreign matter from valves, line strainers, pumps, and other equipment in contact with fuel.
      c. No fuel will be delivered to the system until the Contractor has satisfactorily completed this initial system cleaning.
   2. Initial System Adjustments:
      a. Each system component shall be initially adjusted, if necessary, to meet the system's final operational requirements in accordance with the manufacturer's instructions.

b. Following the initial system adjustments, the equipment tests shall be performed in order to determine any necessary final system adjustments in accordance with the manufacturer's instructions.

3.7 PERFORMANCE TESTS:

A. System Performance Tests:
   1. After all components of the system have been properly adjusted, test the system to demonstrate that the system meets the performance requirements in accordance with manufacturer's instructions.
   2. If any portion of the system or any piece of equipment fails to pass the tests, the Contractor shall make the necessary repairs or adjustments and the test shall be repeated until satisfactory performance is achieved.
   3. The tests shall demonstrate the following:
      a. All day tank pumps and control functions are operational and perform as designed.
      b. Any solenoid valves (e.g., anti-siphon valves) elsewhere in the system that are activated by the day tank controls function correctly.
      c. Vent piping is clear of debris.
      d. Each safety device works properly and is tested per NFPA requirements as necessary.

3.8 INITIAL START-UP

A. Refer to Section 33 52 10.

3.9 TRAINING

A. Refer to Section 33 52 10 and Section 01 79 00 for Operator training requirements

END OF SECTION

SECTION 33 71 16.23
TUBULAR STEEL POLES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Technical requirements for the weathering tubular steel pole structures, complete with as-specified mounting brackets, tubular steel davit arms, structure erection hardware, pole top caps, bearing plates, base plates, anchor bolt assemblies, and grounding appurtenances as required for each structure.

B. In the event of conflict between this specification and the referenced documents, the requirements of this specification shall take precedence. In the case of conflict between several referenced documents, the more stringent requirement shall be followed. If a conflict exists between this specification or the referenced documents and the supplied drawings, the supplied drawings shall be followed. If clarification is necessary, contact the Contracting officer or Contracting Officer representative.

C. Related Sections include but are not necessarily limited to:
   1. 01 65 50 Material Transport - Helicopters
   2. 26 05 46 – Overhead Power Lines
   3. 33 71 23.17 Micropile Foundations

D. Contractor will provide all coordination between selected steel pole manufacturer and selected micropile foundation contractor for integration between steel poles and micropiles.

1.2 QUALITY ASSURANCE

A. Supplier Qualifications:
   1. The Supplier should have a registered ISO 9001 quality management system. The registration certificate and the quality manual shall be provided by the Supplier upon request.
   2. In lieu of a registered quality management system, the Supplier shall have a functional quality program in place. The quality manual will be provided upon request. The quality program may be evaluated by the Contracting officer or Contracting officer representative, as part of the Supplier approval process and at any time during the contract.
   3. The Supplier shall submit a quality plan as soon as possible after approval of the contract, but no later than 1 week prior to production start. After the quality plan is submitted, any revisions to the plan shall be communicated to the Contracting officer or Contracting officer representative immediately.

B. Referenced Standards:
   1. American Concrete Institute (ACI):
      a. 318, Building Code Requirements for Reinforced Concrete.
   2. American Institute of Steel Construction (AISC):
      a. 360, Specification for Structural Steel Buildings.
   3. American Society of Civil Engineers (ASCE):
      a. 48, Design of Steel Transmission Pole Structures.
      b. MOP 74, Guidelines for Electrical Transmission Line Structural Loading.
      c. MOP 91, Design of Guyed Electrical Transmission Structures.
   4. American Welding Society (AWS):
      a. D1.1, Structural Welding Code - Steel.
   5. ASTM International (ASTM):
      a. A6/A6M, Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling.

b. A36/A36M, Standard Specification for Carbon Structural Steel.
c. A123/A123M, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
d. A143/A143M, Standard Practice for Safeguarding Against Embrittlement of Hot-Dip Galvanized Structural Steel Products and Procedure for Detecting Embrittlement.
e. A153/A153M, Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
f. A354, Standard Specification for Quenched and Tempered Alloy Steel Bolts, Studs, and Other Externally Threaded Fasteners.
g. A370, Standard Test Methods and Definitions for Mechanical Testing of Steel Products.
h. A384/A384M, Standard Practice for Safeguarding Against Warpage and Distortion During Hot-Dip Galvanizing of Steel Assemblies.
i. A394, Standard Specification for Steel Transmission Tower Bolts, Zinc-Coated and Bare.
j. A500/500M, Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes.
k. A563, Standard Specification for Carbon and Alloy Steel Nuts.
l. A572/A572M, Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel.
m. A588/A588M, Standard Specification for High-Strength Low-Alloy Structural Steel, up to 50 ksi (345 MPa) Minimum Yield Point, with Atmospheric Corrosion Resistance.
n. A615/A615M, Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement.
o. A633/A633M, Standard Specification for Normalized High-Strength Low-Alloy Structural Steel Plates.
p. A673/A673M. Standard Specification for Sampling Procedure for Impact Testing of Structural Steel.
q. A871/A871M, Standard Specification for High-Strength Low-Alloy Structural Steel Plate with Atmospheric Corrosion Resistance.
r. A992/A992M, Standard Specification for Structural Steel Shapes.
s. B695, Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel.
t. F2329 - Standard Specification for Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts, and Special Threaded Fasteners.
u. F3125/F3125M, Standard Specification for High Strength Structural Bolts, Steel and Alloy Steel, Heat Treated, 120 ksi (830 MPa) and 150 ksi (1040 MPa) Minimum Tensile Strength, Inch and Metric Dimensions.

6. Institute of Electrical and Electronics Engineers (IEEE):
   a. C2, National Electrical Safety Code (NESC).
7. International Organization for Standardization (ISO):
   a. 9001, Quality Management Systems — Requirements.
8. Steel Structures Painting Council (SSPC):
   a. SP1, Solvent Cleaning
   b. SP6, Commercial Blast Cleaning.
   c. SP7, Brush-Off Blast Cleaning.
   d. SP10, Near-White Blast Cleaning.

1.3 DEFINITIONS

A. Cambering: Fabricating of a slight convex curve in a pole or cross-arm.

B. d/t: Ratio of the diameter of a tubular pole to the steel plate thickness.

C. Contractor: The line construction company, and their authorized representatives, typically contracted for by the Contracting officer, responsible for erecting the steel pole structure, with all appurtenances.

D. Ground line: A designated location on the pole where the surface of the ground will be after installation of a direct embedded pole.

E. Overload capacity factors (OCF): Multipliers that are applied to each of the vertical, transverse, and longitudinal structure loads to obtain an ultimate load.

F. Contracting officer representative: reviews technical information (test results), provides technical direction (requires additional tests), approves submittals following A/E review.

G. P-delta moment: Secondary moment created by the vertical loads acting on the structure when the structure deflects from its unloaded position.

H. Point of fixity: Location on the pole at the ground line or below the ground line where the maximum moment occurs.

I. Project: The line within which the steel pole will be used. Typically, the "Project" will be referred to as a proposed or existing high voltage transmission line, but may also only include a single pole replacement or addition.

J. Raking: The practice of erecting a straight pole out of plumb, or at an inclined angle. "Raking" of a pole is usually done to oppose the direction of the resultant vector sum load under that stated load case.

K. Supplier: The pole manufacturer, who is responsible for the design of the pole and all included appurtenances, fabrication and finishing, adherence to both project specifications and industry practices, and delivery of the pole. The "Supplier" may also include their Contracting Officer, or other authorized representative, to respond to questions or other technical or fabrication issues as they may arise.

L. Ultimate load: Maximum design load, which includes the appropriate overload capacity factors.

M. w/t: Ratio of the width of the pole (flat-to-flat) to the plate thickness.

1.4 SUBMITTALS

A. Design Calculations:
   1. Documentation to be supplied by the Contractor for the Contracting officer representative's approval prior to fabrication includes final design calculations for pole shaft, base plate, anchor bolts, and arms, including their connections.
      a. For the loading cases with overload factors, the total shear, axial forces, moments stresses or stress ratios, section moduli, cross-sectional areas, deflections w/t's for polygonal d/t's for round cross sections at all splices, at arm attachment points (top and bottom), and at least every 10 feet along the pole.
      b. For the critical loading case, shear and axial forces, moments, stresses, section moduli, cross-sectional areas at the arm connections, bolt stresses in the arm connection, and deflection at the end of the arm.
      c. Anticipated deflections at the top of the pole and at the ends of the arms shall be indicated for each pole for the deflection case specified on the design drawings.
      d. For all specified loading cases, reactions and ground line moments shall be supplied.
      e. Calculated shipping weight of each structure excluding anchor bolts. Separate weights shall be given for arms and poles.
      f. Ultimate groundline reactions (including overload factors) in poles and guy wires.
      g. Description of pole shaft, including thickness, length, diameter, cross-sectional geometry, and method of fastening each shaft component.
      h. Data showing the design of the arm, arm connections, arm attachment plates and brackets.

B. Shop Drawings:

1. The standard drawing size for shop drawings shall be 22 inches by 34 inches (ANSI D), but the drawings should be prepared in order that they can be reprinted/copied and reviewed as 11 IN by 17 inches plots. Drawings shall be laid out in legible, organized manner, with uniformity in appearance, style, contents, and dimensional proportions. Material lists shall be included on the shop drawings.
2. Provide two sets of hard copy drawings and two sets in electronic (pdf) format prior to fabrication. Shop drawings submittals shall be sent directly by the Contractor to the contracting officer representative, with the contracting officer representative receiving one hard copy set and one electronic set.
3. The contracting officer representative will return to the Contractor, within two weeks after receipt, one set of drawings marked "Revise & Resubmit," "Furnish as Submitted," "Furnish as Noted," "Not Reviewed," or "Rejected," as the case may be. For any resubmittal, all revisions will be clearly shown on the resubmitted sets of drawings. Design revisions, in accordance with corrections shown thereon, may be considered as ready for fabrication. Corrections or revisions shown on "Resubmit" set shall be made. Drawings shall be returned two weeks after notice of requiring resubmittal.
4. After approvals are received and shop drawings finalized, the Contractor shall furnish two hard copy sets and two electronic (pdf) sets of structural steel shop drawings with the contracting officer representative receiving one hard copy set and one electronic set. The drawings will be used for construction purposes.

C. Product Data:
   1. Acknowledgement that products submitted meet requirements of standards referenced.
      a. Type of material of major components.
   2. Certifications (upon request).
      a. Certified Material Test Reports (CMTR) shall be submitted to the contracting officer representative at no cost.
      b. Certified welding reports for each structure.
      c. Incoming, in-process and final quality records.
   3. Test reports:
      a. Anchor bolts, structural plate, and weld material, shall meet the following requirements for Charpy V-notch tests:
         1) The impact properties in the longitudinal direction of the material, for base plates, shafts, attachment lugs and flanges shall be a minimum of 20 Joules (15 feet-LB) for the average of the three specimens and no single specimen shall be less than 16 Joules (12 feet-LB). The test temperature shall be -20 degrees C and testing shall in accordance with ASTM A673.
         2) For plates 1/2 inches or less in thickness and for quenched and tempered or normalized plates of any thickness, heat lot testing may be used to determine whether the material meets the impact property requirements.
         3) Controlled rolled or as-rolled plates and shapes over 1/2 inches in thickness shall be tested on a slab (plate) testing basis.
         4) Anchor bolts shall have minimum Charpy V-notch impact properties in the longitudinal direction of 20 Joules (15 feet-LB) for the average of the three specimens and no single specimen shall be less than 16 Joules (12 feet-LB). The test temperature shall be -20 degrees C and testing shall be in accordance with ASTM A673.
      b. Test reports on coating thickness.
      c. Report of structure testing, when required, including photographs, diagrams, load trees, test equipment calibration records, etc.

PART 2 - PRODUCTS

2.1 MATERIALS

A. All materials shall comply with the applicable requirements of IEEE/NESC and ASTM specifications. Any modifications to the specifications must be approved by the Contracting officer or its representative prior to bidding.

B. All steel shall conform to ASTM A6/A6M.

C. Poles, arms and conductor brackets shall conform to ASTM A572, ASTM A588, and ASTM A871 with a minimum yield strength of 50 ksi.

D. Base plate shall conform to ASTM A572, ASTM A588, ASTM A633, or ASTM A871 with a minimum yield strength of 50 ksi.

E. Other structural steel shall conform to ASTM A500 Grade B (HSS), ASTM A36 (miscellaneous shapes and bars), or ASTM A992 (wide flange).

F. For galvanized structures, all steel used shall have a silicon content less than 0.06%.

G. All structural bolts shall conform, as applicable, to ASTM F3125 Grade A325, ASTM A354, or ASTM A394. Nuts shall conform to ASTM A563, Grade DH. American Nut Company (ANCO) type self-locking nuts shall be provided for each structure bolt.

H. Anchor bolts for self-supporting structures shall be as required by structure design and anchor bolt material shall conform to ASTM, A615 Grade 60, or Grade 75. Each anchor bolt shall be furnished with four nuts (one below the base plate for leveling, one above for securing the pole with base plate to the anchor bolt, one below the steel pile cap and one above the steel pile cap) and four flat washers or per micropile design.

2.2 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

2.3 DESIGN CRITERIA

A. The design, fabrication, allowable stresses, processes, tolerances, and inspection shall conform to the American Society of Civil Engineers (ASCE) Standard 48, latest edition.

B. Pole structures shall consist of constant tapered regular polygonal tube sections with a minimum of 12 sides. The pole diameter measured shall be measured across flats.

C. Structure beams shall be either multi-sided tube sections or standard hollow structural sections (HSS) as listed in AISC 360.

D. Circular or polygonal cross sections for welded steel tubular members are preferred.

E. Pole tube sections shall have nearly uniform taper, and shall be required to be within 0.15 to 0.5 inches per linear foot, unless otherwise approved by the contracting officer representative.

F. Minimum plate thickness for all pole components shall be 3/16 inches.

G. Pole designs shall be prepared from the provided configuration Drawings and design loads. The structure shall be capable of withstanding all specified loading cases including secondary stresses from foundation movements (when foundation movements are specified), and from vertical loads acting on lateral pole deflection (P-delta effect). Design of poles for these secondary stresses shall not consider the possible restraining effect of conductors or shield wires.

The structure shall withstand the loads without failure, permanent distortion, or exceeding any specified deflection limitations. Longitudinal loads specified on the supplied Drawings shall not be reduced due to the flexibility of the structure. Arm designs and base plate designs shall be supported by actual test results.

H. All loads shown in the load and design Drawings include the appropriate overload factors as per NESC Table 253-1 2023 Edition. All load cases are to be considered as shown in the Drawings.

I. Wind pressures shown in the loading criteria shall be multiplied by the appropriate shape factor applied to the poles. Pressures, in Pa, shall be computed as follows:

p = W x Cd
p = pressure on projected area of the pole normal to wind
W = wind pressure
Cd = shape (or drag) factor

Shape factors for computing the wind on poles are:



J. Designs, in addition to the loading conditions to be considered in the Drawings, shall accommodate load allowances for handling, transportation, and erection without failure, permanent deformation, or damage to the pole when handled according to the Suppliers instructions. Furthermore, the structural design shall accommodate an atmospheric temperature range of -40 degrees F to 120 degrees F.

K. Attachment details for overhead ground wires (OHGW) and conductors are shown on the load and design Drawings. Standard pole line hardware will be used for all attachments and will be the responsibility of the Contractor.

L. Deflection Limitations:
   1. Terminal Deadend Structures:
      a. Horizontal deflection of vertical members:
         1) For 40 degrees F, 2 psf wind, initial tension conditions load case: The greater of 1.5 PCT of above ground pole length, or 40% of the pole base diameter.
      b. Combined vertical and horizontal deflection of horizontal members.
         1) For 40 degrees F, 2 psf wind, initial tension condition load case: 0.5% of unsupported arm length.
         2) For all other listed load cases: 6 inches maximum combined (vector sum) deflection.
   2. Suspension, Angle, and In-Line Deadend Structures:
      a. Horizontal deflection of vertical members:
         1) For 40 degrees F, 2 psf wind, initial tension conditions load case: The greater of 1.5 PCT of above ground pole length, or 40% of the pole base diameter.
      b. Combined vertical and horizontal deflection of horizontal members:
         1) For 40 degrees F, 2 psf wind, initial tension condition load case: 0.5% of unsupported arm length.
         2) For all other listed load cases: 6 inches maximum combined (vector sum) deflection.

M. The maximum design unit stress shall be the minimum yield strength as stated in applicable ASTM and IEEE/NESC specifications for the particular application and types of loads, including overload factors.

N. Foundation rotation and deflection shall be included at the pole base for worst case line loading(s) and considered when determining the appropriate pole shaft sizes, base plates, anchor bolts, connections, and other structural members that would be impacted by displacement at the pole base.
   1. Coordinate all foundation work including but not limited to attachment, allowable rotation, and additional requirements with micropile foundation contractor.

O. Cambering of poles shall not be permitted.

P. Joints:
   1. Poles shall be designed with a minimum number of joints. Field welding shall not be allowed as part of the design of a new pole. The shaft joints to be made in the field shall be bolted flange joints.
   2. The weight of a pole section or a single section shall not exceed 10,000 lbs or the maximum lifting capabilities or the contractor selected lifting helicopter whichever is less. Weight shall include arms, base plates, and other attachments.
   3. All pole sections shall have diamond guides for helicopter installation.
   4. Flange plate connections shall be used for all structures and shall be designed in accordance with requirements provided in ASCE/SEI Standard 48. Flexural stress in the flange plate shall not exceed the specified minimum yield stress of the plate material, and the flange plate connections shall be designed to resist the maximum forces and moments at the connection. As a minimum, the flange plate connection shall also be designed to resist 50% of the moment capacity of the lowest strength adjoining tube. The Supplier shall verify upper to lower flange plate fit and connectivity prior to shipment.

Q. Supplier shall provide cap plate at pole top. The cap plate shall be no more than 2 inches larger than the pole top diameter. The plate shall not be vented but fitted flush and seal welded to the top of the pole.

R. Arms:
   1. Arms shall be designed so the end of the arm is at the specified height under the deflection loading case shown on the Drawings.
   2. Arms shall not deflect vertically more than 12 inches at the end of the arm under any loading condition.
   3. Arms shall be upswept or straight, tapered, steel tubular members, of any cross-sectional type, which meet the dimensions shown on the supplied Drawings.
   4. Arm end plate connection details for hardware attachment shall be typical of those shown on the provided configuration Drawings.
   5. The arms shall be hermetically sealed if a painted finish or weathering steel is specified.
   6. Arms and arm attachments shall be designed to avoid trapping or holding moisture.
   7. For davit arms the horizontal offset listed on the load and design Drawings shall be measured from pole face.
   8. If longer than 4 feet, davit arms shall be upswept unless otherwise noted. The arms can be either linear or curved profiles and manufactured as multi-sided or round tubes.
   9. See Drawings for details of arm attachment and vangs. Supplier will provide all necessary arm bracket vangs and hardware to attach the arm.
   10. Davit arm thru vangs and base connections shall be a minimum of 1 inch thick.
   11. Arms shall have hand holds/climbing loops provided as detailed on the Drawings.
   12. Hand holds/climbing loop shall have a 500 pound vertical capacity.

S. Anchor Bolts:
   1. Supplier will furnish structures with base plates, templates, and threaded anchor bolts with four heavy hex nuts and four flat washers. Anchor bolts shall be evenly spaced.

2. Minimum anchor bolt projection to be based on micropile design.
3. Anchor bolts shall be galvanized.
4. The anchor bolt threads shall be protected during shipping.
5. Only one length of anchor bolt shall be used on each pole.
6. Anchor bolts shall be plainly marked to indicate the structure type, structure number, and orientation.
7. The removable template at the top of the anchor bolt assembly shall be marked with a v-notch to show the angle bisector (transverse axis as shown on Drawings) for all structures. Two, 0.5 inches diameter holes shall be drilled in the template at locations 90 degrees to bisector.
8. Template shall be a minimum thickness of 0.5 inches.

T. Lifting Lugs and Attachments:
1. Lifting lugs shall be provided on large pole sections as required as determined by the Supplier and approved by the Contractor. The lugs shall be designed and positioned such that the pole section will hang nearly vertical when suspended from the lugs. Lifting lugs on arms are optional. The Supplier shall supply complete instructions for handling and erection of poles and arms. All pole sections shall have helicopter lifting vangs.
2. In the design of connections for vangs, brackets, or stiffeners attached to the pole shaft, care shall be taken to distribute the loads sufficiently to protect the wall of the pole from local buckling.

U. Climbing Devices:
1. Clips for working/climbing ladders are to be included on one flat the entire length of each pole beginning 10 feet from the designated pole ground line or base plate, and in all four quadrants 5 feet below each wire attachment point.
2. Ladders shall be “McGregor” type or equivalent.
3. Each ladder clip and connecting assembly shall be designed to support a minimum 5000 pound vertical load and a minimum 200 pound horizontal working load per person attached. Ladder clip spacing shall have a maximum allowance of 2 foot freefall.
4. The clips shall be welded to the pole surface. Connectors shall be drop forged, pressed or formed steel, or made of equivalent materials.
5. Ladder clips shall be located to avoid interference between ladders and other attachments.
6. Ladder clips shall provide maximum safety, minimum projection beyond the shaft surface, minimum opportunity for corrosion, and shall not permit air to enter the shaft.
7. Provide removable step bolt lugs with spacing as indicated beginning 8 feet above ground line and extending to the structure top. Each step lug and step bolt (where required) shall be capable of withstanding a minimum of 750 pounds working load, unless otherwise indicated. If specified in the configuration drawings, step bolts may be required to be used for temporary fall arrest anchor points for climbing the poles. Details of this requirement will be provided in the configuration drawings when noted. Step bolts mounting nuts shall be spaced at 1.5 feet and oriented to provide maximum ease of climbing.

V. Grounding Devices:
1. See 26 05 26 Grounding and Bonding.
2. A NEMA-drilled two-hole pad shall be welded to the pole at locations shown on the drawings for grounding purposes.
3. Multi-piece structures shall have two-hole NEMA grounding pads located on both sides of each structure joint, 11 inches from the assembled joint surface (22 inches maximum between NEMA pads). Maintain continuous grounding across joints.
4. Grounding pad face shall not be painted or covered with other coatings.
5. The grounding pad threads shall be protected from coatings.
6. Personal Ground shall be provided 6 inches below each conductor attachment. Personal ground shall be a 0.75 inches diameter, stainless-steel rod. Ground shall be 1 foot long and project from the pole surface 4 inches.

W. Moisture Prevention:

1. Galvanized poles shall have a drain hole at the bottom. When a painted or weathering steel finish is specified, poles shall be hermetically sealed.
2. Structures shall be designed to eliminate water and refuse traps.
3. Factory drilled pole holes shall be plugged to prevent moisture intrusion during shipping.
4. For field drilled poles and factory drilled poles, Supplier shall provide silicon sealant to seal all through-bolt holes.
5. Structures, when assembled, shall be effectively sealed to prevent moisture intrusion.
6. Connections shall be designed to reduce the effect of pack-out by preventing moisture from entering the joint or by designing the connection to allow moisture to easily drain off.
7. Plastic plugs shall be installed in all nuts welded to the structure and all tapped holes.

PART 3 - EXECUTION

3.1 FABRICATION

A. Steel pole structures shall be fabricated as shown on the load and design drawings. Supplier shall assume responsibility as noted. Supplier shall verify all dimensions to verify proper fit up of structure connections prior to fabrication. Any deviation in dimension or materials from that shown on the drawings shall be submitted to the contracting officer representative by the Contractor for written approval prior to fabrication. Such submittals shall be in the form of shop fabrication drawings, per Section 1.4 of this Specification.

B. Prior to being worked in any manner, structural materials shall be cleaned of all rust and foreign particles.

C. Material shall be straight within the tolerance allowed by ASTM A6. If straightening is necessary, it may be done by mechanical means or by application of a limited amount of localized heat. Straightening shall be done in a manner that will not injure the material.

D. All forming or bending during fabrication shall be done by methods that will prevent embrittlement or loss of strength in the material being worked.

E. All formed sections shall have a minimum bend radius in accordance with the Bend Test Requirements as outlined in the ASTM standards.

F. Poles shall have a nearly uniform taper throughout their entire length. The maximum difference in tapers between two pole sections measured by the diameters shall be 0.2 inches/FT for poles with variable taper.

G. Pole height shall be the height of the pole from the top of the baseplate, or designated ground line, to the top of the structure.

H. Shearing and cutting shall be performed carefully and all portions of the work shall be finished neatly. Copes and re-entrant cuts shall be filleted before cutting.

I. Punching and drilling shall be done accurately. Center to center distance between holes of a piece shall not vary more than 1/16 inches from the specified dimension. Holes other than those shown on the drawings shall be approved by the contracting officer representative. Burrs caused by drilling or punching shall be removed.

J. Holes for connection bolts shall be 1/16 inches larger than the nominal diameter of the bolts.

K. Holes in the flange plates for bolted splices shall be 1/8 inches larger than the bolt diameter.

L. Holes in the base plates for anchor bolts shall be 3/8 inches larger than the nominal diameter of the anchor bolts.

M. Holes of any diameter may be drilled in plate of any thickness. Care shall be taken to maintain accuracy when drilling stacks of plates.

N. Holes may be made by use of a machine guided oxygen torch. Flame cut edges shall be reasonably smooth and suitable for the stresses transmitted to them.

O. All holes, blocks, and clips shall be made with sharp tools and shall be clean-cut without torn or ragged edges.

P. Holes in vangs shall be chamfered 1/8 inches.

Q. Punched Holes:
   1. Holes shall be true to size, smooth and cylindrical without excessive tear-out or depressions.
   2. Burrs that remain after punching shall be removed by grinding, reaming, etc.
   3. Holes shall not be punched in material thicker than 0.75 inches.

R. All connections shall be shop welded, except as noted below:
   1. Connections required or provided for disassembly and shipment of structure. Field welding for reassembly will not be permitted.
   2. Bolted connections shown on the specification drawings or approved on the Supplier's shop drawings.

S. Welding:
   1. All welding shall be in accordance with the requirements of the AWS D1.1.
   2. Welding shall be performed by operators who have been qualified by tests as prescribed by AWS D1.1 to perform the type of welding required.
   3. One hundred percent (100%) weld penetration shall be required but not limited to the following areas:
      a. Circumference welds joining structural members.
      b. Longitudinal welds in female portion of slip joint.
      c. Welds at butt joints of back-up strips.
      d. Base plate to shaft welds.
      e. Longitudinal welds adjacent and within 3 inches of circumference welds, flange welds, base welds and ends of tube.
   4. Full penetration or equivalent 90% partial penetration welds with fillet overlay shall be used for arm-to-arm base, vang-to-plate shaft, and arm box joints.
   5. All other penetration welds shall have 60% minimum penetration.
   6. All weld back-up strips shall be continuous the full length of the welds. Care shall be exercised in the design of welded connections to avoid areas of high stress concentration which could be subject to fatigue or brittle fractures.
   7. Welds shall be free of any cracking, either surface or subsurface.
   8. No undercutting of steel is allowed.
   9. Welds shall be free of surface blowholes, flow holes, and spherical inclusions.
   10. Field welding shall not be permitted except with Contracting officer's approval and the Supplier's direction in repairing a pole.

T. The overall length of the assembled structure should not be less than 6 inches of the specified length and not more than one foot.

U. Finish:
   1. Weathering Steel:
      a. All steel components shall conform to ASTM A588 or A871.
      b. After fabrication the poles shall be cleaned of oil, dust, grease, dirt, and any other foreign material (SSPC-SP1).
      c. After surface cleaning, the steel shall be blast cleaned to remove scale, loose rust etc., in accordance with SSPC-SP6 surface preparation specification; to ensure uniform and rapid formation of the protective oxide layer.

*Water-borne coating

3.2 INSPECTION AND TESTING

A. The Contracting officer and its designated agents shall have free entry at all times while work is being carried on, to all parts of the Supplier's plant to observe any part of the production of the poles covered by this Specification.

B. Steel members which are bent or warped or otherwise improperly fabricated shall be properly repaired or replaced.

C. The cost of tests made by the Supplier or Manufacturer, including cost of the certified test reports shall be considered and included in the price.

D. The Supplier or Manufacturer shall make tests in accordance with ASTM A370 and ASTM A673 to verify that the material used in the structures meets the impact properties.

E. Mill test reports showing chemical and physical properties of all material furnished under this specification shall be traceable to the structure.

F. All plates over 1-1/2 inches thick shall be ultrasonically tested to ensure against defects which could lead to laminar tearing.

G. All visual and non-destructive inspection will be carried out by qualified testing personnel.

H. Weld Quality & Inspection:
   1. The Supplier or Manufacturer shall make certified welding reports for each structure. The reports covering welding shall include all welds of a structure. Each weld shall be clearly identified and the report shall consist of the method of testing, whether the weld is acceptable, the identification of the structure, the date, and the name and signature of the inspector.
   2. Quality and acceptability of the complete length of all full penetration welds shall be determined by visual and ultrasonic inspection.
   3. Quality and acceptability of all welds other than full penetration welds shall be determined by visual inspection, supplemented by magnetic particle, ultrasonic or dye penetrant inspection.
   4. Records of welding procedure and welding operator test results shall be kept for 6 years by the Supplier and shall be made available for review by the Contracting officer.
   5. Weld quality shall conform to AWS D1.1, regardless of the method of welding.
   6. Visual Inspection: Work shall be examined to determine that it meets the requirements of this specification. The size and contour of welds shall be measured with suitable gauges. All welds shall be visually inspected for cracks and other defects. Magnifiers may be required for clear visibility.
   7. Ultrasonic Inspection: Ultrasonic inspection shall be performed on all full penetration welds in accordance with the requirements of Section 6, Part F of the AWS D1.1.

3.3 FULL-SCALE LOAD TEST

A. Full-scale load tests have to be performed on the structures which are subject to such tests as noted on the Drawings.

B. The load testing of any specific pole shall be on a full-scale basis at the Supplier’s facility. Costs for such testing shall be the responsibility of the Contractor, shall be separated from the Supplier's bid, and shall be negotiated in advance of any test preparation.

C. Material procurement for test poles shall be identical to material procurement procedures for regular production run poles.

D. The number, location, direction, holding time, sequence, and increments of the test loads along with the number, location, and direction of deflection readings for an individual pole test shall be approved by the contracting officer representative prior to pole testing.

E. The method of attaching the test loads to the pole, applying the test loads, measuring and recording the test loads, and measuring and recording the deflections shall be approved by the contracting officer representative prior to pole testing.

F. A full report listing results shall be submitted after completion of all testing. Copies of mill test reports shall be included in the load test report. The report shall also include a complete description of the load tests with diagrams and photographs. If required, the Supplier shall provide the contracting officer representative with the following testing data:

1. Location of testing.
2. Method of full scale testing: upright or horizontal.
3. The pole tester shall issue the contracting officer representative three copies of the Pole Test Report. This report shall include descriptions, tools, and drawings describing the above test.

G. The Contracting officer or its representative reserves the right to be present during testing and shall be notified 2 weeks prior to the start of structure fabrication.

H. Use of any factory tested poles to meet order requirements shall be determined by the contracting officer representative.

3.4 MARKING

A. Each piece shall be clearly marked or stamped with its respective erection identification mark.

B. At 5 feet from the base plate (for self-supporting poles) or designated ground line (for direct embed poles), the Supplier shall clearly mark the structure type, structure number (if supplied), overall height, ultimate ground line moment, date of fabrication, and name of pole Supplier.

C. Each pole section shall be marked with an alignment arrow properly located on the exterior surface of the pole section for purposes of ensuring each pole section is correctly aligned.

D. Each pole shall be permanently marked on the bottom of baseplate or bearing plate with the following identifying information: structure type, overall height, structure number (if supplied), and date manufactured.

E. V-notch indicators will be marked on the baseplate to represent the transverse axis as shown on the Drawings.

3.5 STRUCTURE PRE-ASSEMBLY

A. All multi-component sub-assemblies should be pre-assembled and match marked by the Supplier and confirmed by the Contractor, before being shipped.

3.6 SHIPPING

A. The Contracting officer and/or Contracting officer's representative shall be notified 10 days prior to shipment. All parts required for any one structure shall be in one shipment, if possible. Nuts, bolts, and small hardware shall be boxed or bundled and identified. Each shipment shall be accompanied by a list of parts, identifiable by structure type and number. Salt-treated wood blocking or urethane foams shall not be used for shipping or storing weathering steel poles. All weathering steel poles must be protected from corrosive contaminants such as road salt.

B. The anchor bolt threads shall be protected during shipping.

C. If bolt clusters are shipped assembled they shall be rigid enough to withstand the normal jolts of shipping and handling with no displacement of bolts from the proper positions within the cluster.

D. Unless otherwise authorized by Contracting officer, the anchor bolt cage shall be shipped at least 30 days prior to pole shipment.

E. All fabricated steel sections shall be suitably braced for handling and shipping.

F. Steelwork shall be properly protected during shipping so that parts will not be bent, broken, deformed, stained or otherwise damaged.

DAMAGE TO ANY PART OF THE WORK DUE TO LACK OF PROPER HANDLING OR LOADING BY THE SUPPLIER SHALL BE REPAIRED OR REPLACED AT THE SUPPLIER'S EXPENSE.

END OF SECTION

SECTION 33 71 23.17
MICROPILE FOUNDATIONS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. This document shall serve as the Performance Specification for the design and construction of micropile foundations as detailed in the construction documents developed by contractor, as approved by the Contracting Officer, and as specified herein.
   2. The Contractor is responsible for furnishing all materials, products, accessories, tools, equipment, services, transportation, labor and supervision, manufacturing techniques, engineering and Quality Control required for design and installation of micropiles, Pile Caps, and miscellaneous connection materials required for this project.
   3. The Contractor is responsible for the strength and stability of the foundation during construction and shall provide temporary shoring, bracing and other elements required to maintain stability until the foundation construction is complete. It is the Contractor's responsibility to be familiar with the work required in the construction documents and the requirements for executing it properly.

B. Related Requirements: Include, but are not necessarily limited to:
   1. Division 0
   2. Division 1
   3. 26 05 46 OVERHEAD POWER LINES
   4. 33 71 16.23 TUBULAR STEEL POLES

1.2 QUALITY ASSURANCE

A. Supplier Qualifications:
   1. Contractor to provide to the Contracting Officer evidence of micropile Contractor’s competence in the installation of micropiles and Supplier’s competence in the supply and fabrication of micropile components for projects of similar size and scope and may include any or all the following:
   2. Resumes for key Contractor personnel, including project manager, Contractor Engineer of Record (EOR), site supervisor, and drill operators. Project manager, Contractor EOR, and site supervisor shall each have at least 5 years of direct experience with micropile work. EOR shall be registered professional licensed in the State of Arizona.
      a. Evidence of steel fabricator’s competence shall be provided to the Contractor EOR’s satisfaction.
      b. Evidence of casing manufacturer’s competence shall be provided to the Contractor EOR’s satisfaction. Details for the threaded casing joint shall be provided at the request of the Contractor EOR.
      c. Where welding is required for fabrication of foundation parts or assemblies, the fabricator shall provide AWS shop certifications, AWS welder certifications, and qualified weld procedures used to complete the work.
   3. A list of at least 3 projects completed within the previous 5 years wherein the installing Contractor installed micropiles similar to those shown in the construction documents. Such list to include names and phone numbers of those project representatives who can verify the installing Contractor’s participation in those projects.

B. Referenced Standards:
   1. In case of a conflict between the construction drawings, specifications, or referenced standards, the Engineer of Record responsible for the foundation designs shall decide which governs.

1.3 DEFINITIONS

A. Casing: steel tube typically introduced during the drilling process in overburden soil to temporarily stabilize the drill hole. This casing is withdrawn as the micropile is grouted, and in most types of micropiles, some casing is permanently left in place for connections, and flexural capacity of the micropile.

B. Centralizer: device used to position the reinforcing steel in the drill hole and/or casing so that a minimum grout cover is provided (typically composed of PVC or steel - not wood).

C. Contractor: unless otherwise described, Contractor shall include the entity/company contracted to design and install foundations and all the entity/company's subcontractors.

D. Coupler: device by which partial lengths of micropile Reinforcement bar are spliced.

E. Design Load (DL): factored controlling load expected to be applied to the micropile during its service life.

F. Engineer of Record (EOR): Contractor's Engineer who is licensed to stamp and sign engineering documents specifically limited to the micropile supported foundations for this Project.

G. Micropile: a small-diameter, bored, cast-in-place, composite pile, in which the applied load is resisted by steel reinforcement, cement grout, and frictional grout-to-ground bond.

H. Contracting Officer Representative: on-site personnel who represents the Contracting Officer.

I. Pile Cap: structural steel member that connects the transmission structure to the micropiles.

J. Reinforcement: high-strength central steel component of the micropile.

1.4 SUBMITTALS

A. Submittals will be provided to the Contracting Officer for record following review and acceptance by the EOR.

B. Micropile Engineer of Record Design Submittals:
   1. Submit the following documents prepared, sealed, and signed by licensed, registered Engineer of Record retained by Contractor, Subcontractor, or Supplier:
      a. Micropile design and construction drawings. Including but not limited to items shown in the Drawings.
      b. Micropile design and construction specifications including but not limited to the items listed in this specification and shown in the Drawings.
      c. Certificate of Compliance.
   2. Testing Plans, Procedures, and Testing Limitations:
      a. Testing procedure, apparatus, and limitations of apparatus and procedure, for:
         1) Source quality control activities indicated in this Section.
         2) Field quality control activities indicated in this Section.
   3. Other Required Engineer of Record Design Submittals:
         1) Calculations bearing EOR seal and signature.
         2) Shop Drawings bearing EOR approval stamp.
         3) Product data bearing EOR approval stamp.
         4) Testing plans (indicating test procedures, testing limitations, and other procedures) bearing EOR approval stamp.
   4. Source Quality Control Submittals:
      a. Contractor to submit Source Quality Control activities.
      b. Results of tests, inspections, and other quality control activities performed at the place of production or fabrication.
   5. Field Quality Control Submittals:
      a. Contractor to submit detailed Contractor Quality Control activities.
      b. Results of tests, inspections, and other quality control activities performed at the Site.
   6. Supplier Site Visit Reports:

a. Report of each visit to the Site by Supplier, summarizing purpose of visit, activities while onsite, problems encountered, advice given to Contractor or Subcontractor, and actions taken.

C. Shop Drawings:
   1. Shop drawings necessary for foundation construction shall be submitted prior to the start of construction for review and acceptance by the EOR. Shop drawings shall include:
      a. Fabrication drawings for steel Pile Cap members and miscellaneous connection elements including but not limited to couplers, nuts, and washers.
   2. Shop drawings shall be clearly identified by creation date and revision status, with changes from a previous submission clouded for identification.

D. Information Submittal:
   1. Materials: Materials for all structural elements shall comply with the requirements of the construction documents. Any request for substitution shall be made by the Contractor as soon as practical, but no less than 60 days prior to incorporation of respective materials into the work. For specific material requirements, see the micropile Construction and Steel Cap Construction section of this Specification.
   2. Mill Specifications: Contractor shall submit certified Mill Test Reports (MTRs) for all structural steel elements and prime casing sections. In certified MTRs, the ultimate strength, elongation, and chemical composition shall be included along with the appropriate ASTM designation which governed the preparation of the MTR. Alternatively, the Contractor may submit Independent Test Reports (ITRs) for mill secondary casing material for review and acceptance by the EOR.
   3. Grout Mix Design: Contractor shall submit a grout mix design at least 60 days prior to incorporation of respective materials into the work. The grout mix design shall include a cement mill certification from the manufacturing plant and break test data supporting the compressive strength specified in the construction documents. Previous test results for the proposed grout mix, completed within one year of the start of work, may be submitted for initial verification of the required compressive strengths for installation of initial production piles.
   4. Installation Records: Contractor shall prepare and submit installation Records for each micropile foundation installed and shall include all information required in the micropile Construction and Steel Cap Construction section of this Specification. A sample Installation Record is required at least 60 days prior to the start of foundation construction.

E. As-Built Survey
   1. Individual piles suspected to fall outside of specified tolerances shall be surveyed, and location details shall be submitted to the EOR and Contracting Officer for review and approval.

F. Micropile Testing
   1. Contractor shall submit detailed plans for the method proposed for testing the micropiles for review and acceptance 60 days prior to beginning testing. This shall include all necessary drawings and details to clearly describe the test method and equipment proposed, and shall include calibration reports, performed by an independent lab within one year of the pile test date, for each hydraulic test ram to be used.

G. Closeout Submittals: Submit the following:
   1. Maintenance Contracts:
   2. Operation and Maintenance Data:
      a. Submit in accordance with Section 01 78 23 - Operation and Maintenance Data.

1.5 SHIPPING, STORAGE, AND HANDLING

A. All materials and hardware for the construction of the foundation shall be free of structural defects and protected from damage. Store materials on wood pallets or supports to prevent

distortion or damage and to keep from contacting the ground. Cementitious materials shall be stored with measures to prevent caking. Fasteners that become dry or rusty before use shall be cleaned and lubricated. Damage to materials shall be cause for rejection.

B. Contractor shall store materials to permit easy access for inspection and identification.

PART 2 - PRODUCTS

A. All components to be weathering steel or compatible with weathering steel

B. Micropile:
   1. Contractor shall furnish materials that are new and without defects and comply with the testing submittal requirements of the EOR. All materials and items needed for the design and construction of micropiles to be determined and provided by Contractor and approved by EOR. Items may include the following:
      a. Centralizers and Spacers: Centralizers and Spacers shall be fabricated from material non-detrimental to the reinforcing steel. Centralizers and Spacers shall be securely attached to the Reinforcement and sized to allow grout to freely flow up the drill hole and casing and between adjacent reinforcing bars.
      b. Grout: EOR to approve grout mixture and testing to meet required compressive strength.
      c. Permanent Casing: Casing sections for micropiles shall have the minimum nominal diameter, wall thickness, and minimum yield strength as specified by EOR.
      d. Reinforcement Bar: Steel micropile Reinforcement bar shall have the minimum dimensions and mechanical properties required by EOR. Corrosion protection shall be determined by the EOR.
      e. Micropile Hex Nuts: Hex nuts for micropiles shall have the minimum dimensions and mechanical properties required to develop the ultimate tensile strength of the micropile. Reinforcement bar and shall be hot-dipped galvanized in accordance with ASTM A153. The supplier is responsible for specifying the material and heat treatment specifications necessary to meet this requirement.
      f. Couplers: Stop-type and thru-thread couplers shall have the minimum dimensions and mechanical properties specified by the EOR and as required to develop the ultimate tensile strength of the micropile reinforcement bar. The supplier is responsible for specifying the material and heat treatment specifications necessary to meet this requirement. Stop-type couplers shall be fabricated to maintain alignment of the bar segments when connected. Corrosion protection shall be determined by the EOR.
      g. Washers: Flat hardened washers shall be fabricated in accordance with ASTM F436 where specified by the EOR.
         1) Flat or beveled micropile washers shall be fabricated to the dimensions, flatness tolerance, and minimum yield strength specified by the EOR. The supplier is responsible for specifying the material and heat treatment specifications necessary to meet the required yield strength.
      h. Bearing Plates: Embedded bearing plates shall be fabricated in accordance with ASTM A36 (36 ksi) or ASTM A572 (50 ksi) as specified by the EOR. Corrosion protection shall be as specified by the EOR.

C. Steel Pile Cap:
   1. Contractor shall furnish materials that are new and without defects and comply with the testing submittal requirements of the EOR. All materials and items needed for the design and construction of steel pile caps to be determined and provided by Contractor and approved by EOR. Items may include the following:
      a. Pile Cap Plate: Steel Pile Caps shall be fabricated in accordance with ASTM A588 (50 ksi) or ASTM A572 (50 ksi) as specified by the EOR. Any corrosion protection shall be as specified by the EOR.
      b. Weld Filler Material: The fabricator shall select weld electrodes in accordance with AWS D1.1 for the materials being welded.

c. Structure-to-Pile Cap Hex Nuts: Hex nuts for structure-to-Pile Cap connections shall meet the requirements and approval of the EOR.
d. Structure-to-Pile Cap Anchor Rods: Threaded anchor rods for structure-to-Pile Cap connections shall be meet the requirements and approval of the EOR.
e. Washers: Flat hardened washers shall be fabricated in accordance with ASTM F436.
f. Pile Cap Bolted Connections: Bolts used for connecting parts of a Pile Cap assembly shall be meet the requirements and approval of the EOR.

PART 3 - EXECUTION

3.1 MICROPILE CONSTRUCTION

A. Micropile construction shall conform to the following requirements and the requirements set forth by the EOR:
1. Construction Site Survey
a. Initial locations and elevations are shown in the Drawings. Contractor shall conduct any necessary excavation to meet clearance or projection requirements specified on the approved construction drawings. Contractor shall reestablish survey locations and elevations during construction and identify any variances from intended layout for review and acceptance by the EOR and Contracting Officer Representative.
2. Locates
a. Contractor shall locate all utilities and structures above and underground in the area of the Work. If the Contractor identifies a utility within 10 feet of the foundation then the EOR shall be immediately notified in order to conduct an evaluation of the possible impact of the utility on the foundation design. Foundation construction shall stop until the EOR confirms the design or provides modifications accordingly. Contractor is responsible for protection of all identified existing utilities and structures. Any modification to the foundation location to be approved by the Contracting Officer Representative.
b. Contractor shall review project documents and complete soil borings to determine subsurface conditions which may impact the installation of the micropiles and shall notify EOR of any condition that would affect proper installation of micropiles immediately after the condition is revealed. Contractor shall stop installation work until the matter can be resolved upon mutual satisfaction of Contractor, EOR, and Contracting Officer Representative.
3. Waste Removal and Site Drainage Control
a. The Contracting Officer shall permit a temporary storage location on site for drill flush and construction related waste, including excess grout. Contractor shall provide transportation to the temporary storage location, and permanent off-site disposal for all waste. Contractor shall provide positive control and discharge of all surface water that will affect construction of the micropile installation, maintain all pipes or conduits used to control surface water during construction, and repair damage caused by surface water.
4. Foundation Installation
a. Contractor shall provide all means and methods for identification and characterization of geotechnical conditions at each site and is responsible for the installation of the proper foundation requirements set forth in the approved construction drawings.
5. Inspection
a. Contractor will provide one on-site field engineer working under responsible charge of the EOR, and an adequate number of experienced inspectors to sample materials, observe critical elements of foundation construction, observe tests, and compile reports based upon on-site observations.
6. Drilling

a. Drilling equipment and methods shall be suitable for drilling through the encountered conditions as determined by the EOR without causing damage to any overlying or adjacent structures or services.

7. Splicing
   a. Splices shall be constructed to develop the required design strength of the pile section as determined by the EOR. Lengths of permanent casing and Reinforcement steel to be spliced shall be secured to maintain proper alignment and in such a manner to avoid eccentricity or angle between the axes of the two lengths to be spliced.
8. Reinforcement Steel Placement
   a. Reinforcement may be placed prior to grouting or placed into the grout-filled drill hole before temporary casing (if used) is withdrawn. The surface of the Reinforcement shall be free of deleterious substances such as soil, mud, grease, or oil that might contaminate the micropile and degrade the steel to grout bond. The position of the reinforcement shall be maintained during the withdrawal of the drill casings.
   b. Centralizers and Spacers shall be provided at spacings determined by the EOR. Centralizers and Spacers shall permit the free flow of grout without misalignment of the Reinforcement steel and permanent casing. The central Reinforcement steel with centralizers shall be lowered into the stabilized drill hole to the desired depth without difficulty. Partially inserted Reinforcement steel shall not be driven or forced into the hole. Contractor shall re-drill, advance casing if necessary, and reinsert Reinforcement steel when necessary to facilitate insertion.
9. Micropile Construction Tolerances
   a. The Contractor shall confirm that micropiles are constructed within tolerances developed by the EOR and approved by the Contracting Officer Representative.
10. Grouting Procedures
    a. Contractor shall be responsible for all means and methods for grouting procedures. Grouting procedures to be determined by the EOR and approved by the Contracting Officer Representative.
11. Grout Sampling and Testing
    a. The Contractor shall sample grout during construction at a frequency of no less than one set of (5) 2-inch diameter grout cylinders or 2-inch grout cubes from each grout mixing plant each day of operation or per every 10 piles, whichever occurs more frequently. Samples shall be cured and tested for compressive strength in accordance with AASHTO T106/ASTM C109/ASTM C39. Of the 5 specimens collected at each sampling, 2 specimens shall be tested at 28 days and 1 specimen shall be held in reserve for use as the EOR requires. The remainder of the specimens may be used to conduct tests earlier or may be held to provide comparison to Contracting Officer Representative's tests. The Contracting Officer may collect and test grout specimens at any interval. Upon conformance of achieving minimum specified compressive strength, reserve samples may be discarded.
    b. Correlation between w/cm ratio and relative density (specific gravity) allows for monitoring of the mixed grout and shall be monitored utilizing a mud balance test. The minimum acceptable relative density for the specified w/cm ratio is 1.85.
12. Installation Records
    a. Contractor shall prepare for the EOR and Contracting Officer a micropile Installation Record for each installed micropile. These Installation Records will include drilling duration and observations, information on encountered geotechnical units, approximate final depth and orientation of piles, description of unusual conditions or installation behavior, grout pressure attained, grout quantities pumped, material sizes, quantities, and product tracking numbers.
    b. The Contractor will submit to the Contracting Officer a letter of conformance stamped by the EOR for each installed foundation within 60 days of installation. The letter of conformance shall include micropile Installation Records, Proof Test reports, grout batch tickets and grout test reports, and inspection checklists.
    c. Verification Test reports shall be submitted prior to foundation construction.
13. Electrical Grounding

a. Micropile shall be bonded to ground. Contractor shall furnish and install grounding rods to meet ground resistivity requirements of no more than 25 ohms. Contractor shall provide ground test results for record.

14. Micropile Load Tests
    a. The Contractor shall provide and conduct a Verification Test and Proof Test program approved by the EOR prior to construction of production micropiles at locations designated by the Contractor and agreed upon by the Contracting Officer and EOR.
    b. Pass and failure criteria for the Verification Test and Proof Test shall be determined by the Contractor and agreed upon by the Contracting Officer and EOR. For failed micropiles the Contractor shall modify the design, construction procedure, or both on subsequent micropile installations at that location. Any modification that requires changes to the micropile design shall be reviewed and accepted by the EOR and Contracting Officer Representative prior to resuming construction.
    c. Designation of test piles shall be at the contractor's discretion, with concurrence from the EOR. The EOR reserves the right to require additional testing given unusual conditions or installation behavior. At a minimum, Verification and Proof testing shall be performed at the following rates:
        1) Tubular steel monopole structure foundations shall have one pile tested for foundations up to and including 8 piles. Two piles shall be tested for foundations with more than 12 piles.

3.2 STEEL PILE CAP CONSTRUCTION

A. Steel Pile Cap construction shall conform to the following requirements and the requirements set forth by the EOR.

B. Contractor to coordinate all construction means and methods for required steel pile cap with fabricators and manufactures. Contractor to coordinate final bolt hole pattern with steel pole manufacture prior to fabrication of caps and steel poles at all locations described in the construction package. EOR to approve final steel pile cap designs prior to release for manufacturing.

C. Installation
    1. Before erection proceeds, and with the steel erector present, verify elevations of micropiles for compliance with requirements set by the EOR. Do not proceed with erection until unsatisfactory conditions have been corrected and reviewed by the EOR and Contracting Officer Representative.
    2. After installation, clean any field welds, bolted connections, and abraded areas and apply repair paint.
    3. Structure base plate connection nuts shall be installed to the requirements of the structure manufacturer.

D. Pile Cap Installation Tolerances
    1. The Contractor shall confirm that Pile Caps are installed to maintain the micropile construction tolerances as set by the EOR and approved by the Contracting Officer Representative.

END OF SECTION

This page intentionally left blank.

SECTION 33 81 19

FOLD OVER ANTENNA TOWER ASSEMBLY

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:
1. For the purposes of this Specification, “Antenna Tower” includes the tower itself, its supporting foundation, cableways that provide a path for communications cables to travel between the tower and an adjacent communications equipment enclosure, and all appurtenances that attach to the tower to support radio communications.
2. Requirements for equipment, materials, and installation of an Antenna Tower.
3. The Contractor shall design, furnish, and install a latticed self-supporting antenna tower, including its foundations, antenna mounting brackets, climbing ladder with safety cable and climbing device, obstruction light(s) as required by rule, RF transmission line bridge extending to a communications enclosure, and appurtenances as required for a complete installation, as specified in the Contract Drawings.
4. The Contract Drawings contain additional information on the site-specific requirements and design parameters that must be addressed in the final design, fabrication, and construction of the towers.
5. All antenna towers must be freestanding with all tower loads transferred into the foundation. No guy wires or other auxiliary support mechanisms are allowed.
6. Tower loading requirements for each site are as shown on the Contract Drawings. Note that tower, foundation, and all supporting elements must be designed for a 25% increase in Effective Projected Area (EPA).
7. Tower anti-climbing safety system must be designed, fabricated, and supplied by the tower manufacturer.
8. All proposed equipment/components must work together to form a fully functioning system without adjustments, derates, or field modification.
9. The Contractor must ensure that antenna tower supplier is licensed to provide services at the site location.
10. Note that this project does not include specification, procurement, construction, or installation of communications equipment. This work will be done by others under a future contract.

B. Related Specification Sections include but are not necessarily limited to:
1. Division 1 – General Requirements
2. Section 01 77 00 – Closeout Procedures
3. Section 09 96 00 – High Performance Industrial Coatings

1.2 CODES STANDARDS AND REFERENCES

A. References:
1. American National Standards Institute (ANSI) and/or Telecommunications Industries Association (TIA)
   a. ANSI/TIA-222-I (eye) Structural Standard for Antenna Supporting Structures, Antennas, and Small Wind Turbine Support Structures
   b. TIA-322 Loading, Analysis, and Design Criteria Related to the Installation, Alteration and Maintenance of Communication Structures
2. International Building Code 2024 Edition
3. Motorola R56 Standards and Guidelines for Communications Sites
4. American Institute for Steel Construction (AISC)
   a. Specification for Structural Steel Buildings – Allowable Stress Design and Plastic Design

5. American Welding Society (AWS)
    a. AWS D1.1 Structural Welding Code
6. American Society for Testing and Materials (ASTM)
    a. A6 Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling
    b. A36 Standard Specification for Carbon Structural Steel
    c. A53 Welded and Seamless Steel Pipe
    d. A123 / A123M – 09 Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
    e. A153/A153-16a Standard Specification for Zing Coating (Hot-Dip) on Iron and Steel Hardware
    f. A307 Carbon Steel Bolts and Studs, 60,000 PSI Tensile Strength
    g. ASTM A325 High-Strength Bolts and Structural Steel Joints
    h. A370 Standard Test Methods and Definitions for Mechanical Testing of Steel Products
    i. A384 Standard Practice for Safeguarding Against Warpage and Distortion During Hot-Dip Galvanizing of Steel Assemblies
    j. A394 Standard Specification for Steel Transmission Tower Bolts, Zinc-Coated and Bare
    k. A449 Standard Specification for Hex Cap Screws, Bolts and Studs, Steel, Heat Treated, 120/105/90 ksi Minimum Tensile Strength, General Use
    l. A475 Zinc Coated Steel Wire Strands
    m. A563 Standard Specification for Carbon and Alloy Steel Nuts
    n. A572 Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel
    o. A586 Zinc Coated Parallel and Helical Steel Wire Structural Strand
    p. A780 Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
    q. ASTM B695: Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel
    r. D4956 Standard Specification for Retroreflective Sheeting for Traffic Control
    s. E376 Standard Practice for Measuring Coating Thickness by Magnetic-Field or Eddy Current (Electromagnetic) Testing Methods
    t. F2329 Standard Specification for Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts, and Special Threaded Fasteners
    u. F3125 Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength
7. Federal Aviation Administration
    a. AC-70/7460-1H Obstruction Marking and Lighting
    b. AC-150/5345-43 Specification for Obstruction Lighting Equipment
    c. AC-1560/5345-1 Approved Airport Lighting Equipment
8. Occupational Safety and Health Administration
    a. OSHA Standards, 29 CFR, Part 1910
9. American Society of Safety Professionals/Engineers (ASSP)
    a. A10.48 Standard Criteria for Safety Practices with the Construction, Demolition, Modification and Maintenance of Communications Structures
    b. Z359 Fall Protection and Arrest Standards Package
10. Safety Equipment Manufacturing Committee (SEMC)
    a. Industry-wide standards for identification tags & signs

1.3 PROJECT CONDITIONS

A. The locations of tower and foundation are shown conceptually on the drawings to establish the intent and character of the installation. As is the foundation plan and subject to modification through the final engineering and design by the contractor after their review of the geotechnical

report, tower loading, and environmental conditions for each site. Exact locations of items of work shall be field coordinated prior to installation.

B. Verify areas of work, conditions, products to match existing, and conflicts between contract documentation and site conditions minimizing the disturbed area to every extent possible.

C. Verify locations of foundation, and ice bridge supports and alignment point to communications enclosure, before construction.

D. No axial rotation of tower from design plans is permitted without approval of the Contracting Officer.

E. Purchaser must verify that the installation is in conformance with local, state and federal requirements for obstruction marking and lighting.

1.4 PROJECT COORDINATION

A. As specified in Contract Documents.

B. Coordinate the work of this Section with that of other sections of Divisions 00, 01, 02, 03, and 31 as required.

1.5 QUALITY ASSURANCE

A. Quality assurance must comply with the requirements of the Contract's Quality Management Plans.

1.6 SUBMITTALS

A. Shop Drawings: See other Specification Sections for requirements for the mechanics and administration process.

B. Product technical data including:
   1. Acknowledgement that products submitted meet requirements of the referenced standards.
   2. Manufacturer’s components, assembly and construction specifications, certifications, drawings, calculations and instructions sealed and signed by an engineer registered in the State of Arizona
   3. All components must come with appropriate manufacturer-applied labeling attached to each component.

1.7 WARRANTY

A. Unless otherwise noted in individual specification sections or elsewhere in this specification section, materials and workmanship shall meet or exceed industry standards and be fully guaranteed for a minimum of twenty (20) years. If the manufacturer's standard warranty is longer than twenty (20) years, then the Contractor shall provide the longer warranty.

B. Contractor shall submit a copy of all manufacturer warranty information.

PART 2 - PRODUCTS

2.1 GENERAL

A. All materials must meet or exceed all applicable referenced standards, federal, state, local, and NPS requirements, and conform to codes and ordinances of authorities having jurisdiction.

2.2 ANTENNA TOWER

A. Manufacturers:
   1. Rohn (55G Fold-Over Tower Assembly).
   2. Or Equal.

B. Height:
   1. 40-foot tower height not including accessories.

C. Materials:
   1. Hot Dip Galvanized Steel (ANSI/TIA-222-G)

D. Antenna Tower System Vendor:
   1. Must provide drawings, specifications and calculations for all towers, foundations, ancillary items and appurtenances for the tower components, assembly and construction sealed and signed by an engineer registered in the State within which the antenna tower site is located. Contractor must construct and install antenna towers per the approved plans and specifications.
   2. In order to specifically define responsibilities, the vendor shall maintain in-house control over the design and fabricating functions. Subcontracting of these responsibilities will be cause for rejection of a vendor.
   3. Each structural member shall be identified by a part number and all parts with the same part number must be interchangeable. This will result in tower sections capable of being installed in any 120-degree rotation. Match marking requirements of tower sections by the manufacturer, for proper assembly, shall not be acceptable.
   4. Tower leg members shall utilize a 50 KSI minimum yield strength. Tubular leg members with flange splices shall maintain an open interior diameter through the flange plate at least as large as the inside diameter of the tube and shall be welded externally and internally.
   5. Flange leg connections shall utilize a minimum of four bolts per leg.
   6. All fabricated tower members shall be hot dip galvanized after fabrication per ASTM Standard A123. Hardware shall be galvanized per ASTM Standard A153 and B695. Other types of coating are not acceptable.
   7. Tower assembly drawings illustrating all component part numbers and their respective locations shall be provided. As a minimum, assembly drawing shall be accompanied by a letter sealed by a registered professional engineer licensed in the state of Arizona, certifying that the tower meets all design requirements per ANSI/TIA-222-I (eye).
   8. The tower manufacturer shall be an AISC Certified Fabricator and shall demonstrate quality steel manufacturing standards for production.
   9. Only AWS Certified Welders shall be employed for tower fabrication with a quality control manual maintained to establish minimum acceptable fabrication standards, procedures and requirements for documentation.
   10. Warning labels are applied at the factory at the locations indicated on the section assembly drawings. Contact vendor for replacement of damaged or missing warning labels.

E. Antenna Tower System Requirements
   1. The Antenna Tower must be self-supporting with a normal overall structural height as per the drawing set.
   2. The Antenna Tower must be designed for a maximum wind speed per ANSI/TIA-222-I (eye) as indicated in the drawing set.
   3. The Antenna Tower must meet ANSI/TIA-222-I (eye) specifications for simultaneous ice loading and wind conditions as indicated in the drawing set.
   4. The Antenna Tower must be designed to accommodate the antenna loads and all appurtenance loads as specified in the drawing set, as well as the appurtenance loads as specified by the manufacturer.
   5. Antenna Tower structure class is III (Public Safety).
   6. Exposure and topographic category shall be determined per the drawing set.
   7. The following appurtenances must be incorporated into the Antenna Tower as required by ANSI/TIA-222-I (eye).
      a. Safety Device
      b. Transmission Line Support Ladders/Brackets
      c. Antenna Mounts
      d. Lightning and Grounding Materials
      e. Conduit
   8. Foundations must be designed to support the reactions shown for the conditions existing at the site.

9. The antennas to be sized for load shall include:
    a. 3ft High Performance Dual-Polar Parabolic (Cambium Networks RDH4509C or Equal).
    b. 450 MHZ Colinear Antenna – 6dB Gain (Telewave.IO ANT450F6 or Equal)
    c. 450 MHZ Omni Antenna – 0 dB Gain (Telwave.IO ANT450F2 or Equal)
10. Antenna Coating:
    a. See Specification Section 09 96 00 – High Performance Industrial Coatings.

F. Antenna Tower Accessories include but are not limited to:
1. Anti-Climb Panel
2. Lightning-Rod
3. Side Arm Brackets for (4) antennas
4. Face Mount

2.3 ANTENNA TOWER SYSTEM GROUNDING

A. Provide products and system as specified in Motorola R56. Site must be considered Type B2 (Extra Duty), per R56.

B. Installation must be grounded in accordance with local and national codes. Resistance to ground must not exceed 10 Ohms. Grounding material must be ordered separately.

C. Tie Grounding System to Tank Lightning protection system.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Tower installation must be in accordance with the approved final drawings and specifications.

B. Contractor must be responsible for the performance of all inspection and testing activities to demonstrate compliance with the drawings, specifications, referenced standards, applicable codes, and industry practices. Installation, operation and dismantling must be performed by experienced and trained personnel only.

C. Ensure electrical installation is secured and properly grounded.

D. Bolted connections are torqued per manufacturer requirements. All field connections are bolted. A nut locking device or self-locking nut must be provided for all bolts.

E. Install warning sign provided with tower (p/n acws) at base of the tower in a visible location.

F. Install locking device in holes provided at the base of the rotating sections to prevent unauthorized use of the tower (locking device to be provided by others).

END OF SECTION

This page intentionally left blank.

DIVISION 40

PROCESS INTERCONNECTIONS

This page intentionally left blank.

SECTION 40 05 00
PIPE AND PIPE FITTINGS - BASIC REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Process piping systems for “interior” piping at:
      a. Roaring Springs Pumphouse (Volume 5).
      b. North Rim WTP (Volume 6).

B. See Division 33 for all “buried” piping specifications including water transmission, distribution piping, yard piping, and sewer.

C. Related Specification Sections include but are not necessarily limited to:
   1. Section 09 96 00 – High Performance Industrial Coatings.
   2. Section 10 14 00 – Identification Devices.
   3. Section 31 23 33 – Trenching, Backfilling, and Compacting for Utilities.
   4. Section 40 05 07 – Pipe Support Systems.
   5. Section 40 05 19 – Pipe - Ductile Iron.
   6. Section 40 05 23 – Pipe - Stainless Steel.
   7. Section 40 05 24 – Pipe - Steel.
   8. Section 40 05 31 – Pipe - Plastic.
   9. Section 40 05 51 – Common Requirements for Process and Utility Valves.
   10. Section 40 42 00 – Pipe, Duct and Equipment Insulation.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Association of State Highway and Transportation Officials (AASHTO):
      a. M36, Corrugated Steel Pipe, Metallic-Coated, for Sewers and Drains (Equivalent ASTM A760).
      b. M190, Standard Specification for Bituminous Coated Corrugated Metal Culvert Pipe and Pipe Arches.
      c. M252, Standard Specification for Corrugated Polyethylene Drainage Tubing.
      d. M278, Standard Specification for Class PS 46 Polyvinyl Chloride (PVC) Pipe.
      e. M294, Interim Specification for Corrugated Polyethylene Pipe 12 to 24 Inch Diameter.
   2. American Iron and Steel Institute (AISI).
   3. American Society of Mechanical Engineers (ASME):
      a. B16.3, Malleable Iron Threaded Fittings.
      b. B16.5, Pipe Flanges and Flanged Fittings.
      c. B16.9, Factory-Made Wrought Steel Butt-Welding Fittings.
      d. B16.22, Wrought Copper and Bronze Solder - Joint Pressure Fittings.
      e. B16.26, Cast Copper Alloy Fittings for Flared Copper Tubes.
      f. B36.19, Stainless Steel Pipe.
      g. B40.100, Pressure Gauges and Gauge Attachments.
   4. ASTM International (ASTM):
      a. A53, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
      b. A74, Standard Specification for Cast Iron Soil Pipe and Fittings.
      c. A106, Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service.
      d. A126, Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings.

e. A182, Standard Specification for Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service.
f. A197, Standard Specification for Cupola Malleable Iron.
g. A234, Standard Specification for Pipe Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service.
h. A269, Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service.
i. A312, Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes.
j. A518, Standard Specification for Corrosion-Resistant High-Silicon Iron Castings.
k. A536, Standard Specification for Ductile Iron Castings.
l. A587, Standard Specification for Electric-Resistance-Welded Low-Carbon Steel Pipe for the Chemical Industry.
m. A760, Standard Specification for Corrugated Steel Pipe, Metallic-Coated for Sewers and Drains.
n. A774, Standard Specification for As-Welded Wrought Austenitic Stainless Steel Fittings for General Corrosive Service at Low and Moderate Temperatures.
o. A778, Standard Specification for Welded, Unannealed Austenitic Stainless Steel Tubular Products.
p. B88, Standard Specification for Seamless Copper Water Tube.
q. C14, Standard Specification for Concrete Sewer, Storm Drain, and Culvert Pipe.
r. C76, Standard Specification for Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe.
s. C425, Standard Specification for Compression Joints for Vitrified Clay Pipe and Fittings.
t. C443, Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets.
u. C564, Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings.
v. C700, Standard Specification for Vitrified Clay Pipe, Extra Strength, Standard Strength and Perforated.
w. D1785, Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120.
x. D2466, Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40.
y. D2467, Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80.
z. D2513, Standard Specification for Thermoplastic Gas Pressure Pipe, Tubing, and Fittings.
aa. D2683, Standard Specification for Socket-Type Polyethylene Fittings for Outside Diameter-Controlled Polyethylene Pipe and Tubing.
bb. D3261, Standard Specification for Butt Heat Fusion Polyethylene (PE) Plastic Fittings for Polyethylene (PE) Plastic Pipe and Tubing.
cc. D4101, Standard Specification for Polypropylene Plastic Injection and Extrusion Materials.
dd. F439, Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80.
ee. F441, Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe, Schedules 40 and 80.

5. American Water Works Association (AWWA):
   a. B300, Standard for Hypochlorites.
   b. C200, Standard for Steel Water Pipe - 6 IN and Larger.
   c. C207, Standard for Steel Pipe Flanges for Waterworks Service - Sizes 4 IN through 144 IN.
   d. C208, Standard for Dimensions for Fabricated Steel Water Pipe Fittings.
   e. C606, Standard for Grooved and Shouldered Joints.

f. C651, Standard for Disinfecting Water Mains.
g. C800, Standard for Underground Service Line Valves and Fittings.

6. American Water Works Association/American National Standards Institute (AWWA/ANSI):
   a. C110/A21.10, Standard for Ductile-Iron and Gray-Iron Fittings.
   b. C111/A21.11, Standard for Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings.
   c. C115/A21.15, Standard for Flanged Ductile-Iron Pipe with Ductile-Iron or Gray-Iron Threaded Flanges.
   d. C151/A21.51, Standard for Ductile-Iron Pipe, Centrifugally Cast, for Water.
   e. C153/A21.53, Standard for Ductile-Iron Compact Fittings for Water Service.
7. Chlorine Institute, Inc. (CI):
   a. Pamphlet 6, Piping Systems for Dry Chlorine.
8. Cast Iron Soil Pipe Institute (CISPI):
   a. 301, Standard Specification for Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications.
9. International Plumbing Code (IPC).
10. National Fire Protection Association (NFPA):
    a. 54, National Fuel Gas Code.
    b. 69, Standard on Explosion Prevention Systems.
11. Underwriters Laboratories, Inc. (UL).

B. Coordinate flange dimensions and drillings between piping, valves, and equipment.

1.3 DEFINITIONS

A. Hazardous Gas Systems: Chlorine gas, sulfur dioxide gas, carbon dioxide gas, lab gases.

B. HPIC: High performance industrial coating.

C. PVDF: Polyvinylidene fluoride.

1.4 SUBMITTALS

A. Shop Drawings:
1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
2. Product technical data including:
   a. Acknowledgement that products submitted meet requirements of standards referenced.
   b. Copies of manufacturer's written directions regarding material handling, delivery, storage and installation.
   c. Separate schedule sheet for each piping system showing compliance of all system components.
      1) Attach technical product data on gaskets, pipe, fittings, and other components.
3. Fabrication and/or Layout Drawings:
   a. Interior piping drawings (minimum scale 1/8 IN equals 1 FT) with information including:
      1) Dimensions of piping from column lines or wall surfaces.
      2) Centerline dimensions of piping.
      3) Centerline elevation and size of intersecting ductwork, conduit/conduit racks, or other potential interferences requiring coordination.
      4) Location and type of pipe supports and anchors.
      5) Locations of valves and valve actuator type.
      6) Details of fittings, tapping locations, equipment connections, flexible expansion joints, connections to equipment, and related appurtenances.
      7) Acknowledgement of valve, equipment and instrument tag numbers.
      8) Provisions for expansion and contraction.
      9) Line slopes and air release vents.
      10) Rough-in data for plumbing fixtures.

b. Schedule of interconnections to existing piping and method of connection.

B. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

C. Informational Submittals:
1. Qualifications of lab performing disinfection analysis on water systems.
2. Test reports:
a. Copies of pressure test results on all piping systems.
b. Reports defining results of dielectric testing and corrective action taken.
c. Disinfection test report.
d. Notification of time and date of piping pressure tests.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Protect pipe coating during handling using methods recommended by manufacturer.
1. Use of bare cables, chains, hooks, metal bars or narrow skids in contact with coated pipe is not permitted.

B. Prevent damage to pipe during transit.
1. Repair abrasions, scars, and blemishes.
2. If repair of satisfactory quality cannot be achieved, replace damaged material immediately.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable if not specified in the individual pipe specifications in Division 40:
1. Insulating unions:
a. "Dielectric" by Epco.
b. Or equal.
2. Dirt strainers (Y type):
a. Mueller (#351).
b. Or equal.
3. Chemical strainers (Y type):
a. Chemtrol.
b. Or equal.
4. Dry disconnect couplings:
a. Kamlock.
b. Or equal.
5. Dielectric flange kit:
a. PSI.
b. Or equal.
6. Pipe saddles (for gage installation):
a. Dresser Style 91 (steel and ductile iron systems).
b. Dresser Style 194 (nonmetallic systems).
c. Or equal.
7. Flexible Expansion joints:

a. Mercer.
b. Or equal.

8. Flexible Braided Connector:
   a. Flexicraft
   b. Or equal
9. Dismantling Joint:
   a. Romac DJ400.
   b. Or equal.
10. Wide Range Couplings and Alpha Restrained Joint Couplings:
    a. Romac Alpha.
    b. Or equal.

2.2 PIPE SCHEDULES

A. Piping system materials, fittings and appurtenances are subject to requirements of specific pipe schedules located in the Drawings (Volumes 5 and 6).

2.3 COMPONENTS AND ACCESSORIES

A. Flanges: See individual piping specifications.

B. Gaskets: See individual piping specifications.

C. Nuts, Bolts, Washers:
   1. Nuts: ASTM A194, Grade 2H.
   2. Bolts: ASTM A193, B7.
   3. Washers: ASTM F436.
   4. Coating: Paint.
   5. Torque: Per manufacturer's recommendations.
      a. Do not torque bolts to specification more than 2 times.
      b. Provide new bolts if more than 2 torque events occur on the same bolts.
   6. Project bolts ¼" to ½" beyond nuts.

D. Insulating Components:
   1. Dielectric flange kits:
      a. Flat faced.
      b. 1/8 IN thick dielectric gasket, phenolic, non-asbestos.
      c. Suitable for 175 PSI, 210 DEG F.
      d. 1/32 IN wall thickness bolt sleeves.
      e. 1/8 IN thick phenolic insulating washers.
   2. Dielectric unions:
      a. Screwed end connections.
      b. Rated at 175 PSI, 210 DEG F.
      c. Provide dielectric gaskets suitable for continuous operation at union rated temperature and pressure.

E. Dirt Strainers:
   1. Y-type.
   2. Composition bronze.
   3. Rated for test pressure and temperature of pipe in which they are installed.
   4. 20 mesh Monel screen.
   5. Threaded bronze plug in the blowoff outlet.
   6. Threaded NPT end connections.

F. Strainers for Chemical Applications:
   1. Y-type.
   2. Strainers of same material, test pressure, and temperature rating as system in which strainer is placed.

G. Reducers:

1. Furnish appropriate size reducers and reducing fittings to mate pipe to equipment connections.
2. Connection size requirements may change from those shown on Drawings depending on equipment furnished. Additional costs due to connection size changes will be responsibility of Contractor.

H. Protective Coating and Lining:
1. Include pipe, fittings, and appurtenances where coatings, linings, coating, tests and other items are specified.
2. Field coat pipe in accordance with Specification Section 09 96 00.

I. Pressure Gages:
1. See Specification Section 01 61 03.

J. Dry Disconnect Couplings:
1. Adapters:
   a. Male adapters: Size shown on Drawings.
   b. Adapters:
      1) Female NPT end connection for sludge and flush applications.
      2) Male NPT end connection for chemical applications.
   c. Construct adapters for sludge applications from cast iron or steel.
   d. Construct adapters for chemical and PVC system applications 3 IN and below from polypropylene.
      1) Above 3 IN size, provide stainless steel units.
2. Couplers:
   a. Built-in valve and spring loaded poppet which close automatically when disconnected.
   b. Designed to remain with only one arm locked in closed position.
   c. Construct couplers for sludge applications fabricated from material utilized for adapters.
   d. Construct couplers for chemical and PVC system applications 3 IN and less from polypropylene with stainless steel arms and pins.
      1) Above 3 IN, provide stainless steel units.
   e. Gasket: Compatible with conveyed liquid.
3. Dust caps: For all adapters.

K. Sacrificial Anode Cathodic Protection:
1. 3 LB magnesium sacrificial anodes, prepackaged in a cloth bag containing 75% hydrated gypsum, 20% bentonite and 5% anhydrous sodium sulphate.
2. TW 600 V or an HMWPE insulated copper lead attached to the anode.

L. Flexible Expansion Joints:
1. Materials:
   a. Bellows and Maximum Temperature Rating:
      1) Chemically compatible with fluid in connecting piping
      2) CIP service: Viton, 250 DEG F
      3) Hot Process Air Service: High Temperature EPDM, 300 DEG F
      4) All Other Service: UV-resistant EPDM, 250 DEG F
   b. Flanges:
      1) Outdoor Service: Ductile iron or stainless steel
      2) Indoor Service: Continuation of bellows material
   c. Connecting hardware: 316 stainless steel.
   d. Reinforcing rings: Stainless steel.
2. Flanges to meet ASME B16.5 Class 150
3. Restraint, where required:
   a. Provide restraint limit bolts (control rods) and nuts to restrain joint at test pressure of piping.
   b. Control rod material: 316 stainless steel.
4. Pressure rating at 70 DEGF:

a. Equal to or greater than test pressure of connecting piping.
b. 70 PSIG minimum.

5. Single Arch Design
   a. Minimum axial compression: 0.75 IN
   b. Minimum axial extension: 0.50 IN
   c. Minimum lateral deflection: 0.50 IN
   d. Minimum angular deflection: 3.8 DEGREES
6. Double Arch Design
   a. Minimum axial compression: 1.5 IN
   b. Minimum axial extension: 1.0 IN
   c. Minimum lateral deflection: 1.0 IN
   d. Minimum angular deflection: 8.8 DEGREES

M. Flexible Braided Connectors:
1. Materials:
   a. Inner hose: 316/316L stainless steel
   b. Braid: 304/304L or 316/316L stainless steel
   c. Flanges: Carbon Steel, ASME B16.5 Class 150 flat face
   d. Connecting hardware: carbon steel, or 316 stainless steel with dielectric isolating sleeves
2. Pressure rating at 70 DEGF: 200 PSIG minimum.
3. Minimum lateral deflection: 3/8”
4. For all Finished Water service: must meet materials requirements of NSF-61

N. Orifice Plates and Assemblies:
1. Provide a series of orifice plates or a multistage orifice plate assembly to reduce pressure where indicated on the Drawings (Volume 5).
2. Flanges: ANSI 150.
   a. Provide threaded ends for pressure gauge connection.
   b. Maximum face-to-face length: 20-IN
3. Materials:
   a. Orifice plate or assembly:
      1) Carbon steel.
      2) 1/4 IN thick plates.
4. Pressure requirements:
   a. Upstream pressure: 150 PSIG.
   b. Downstream pressure: 0 PSIG.
   c. Provide pressure gauges upstream and downstream of the orifice plates or assembly.
5. Does not apply to the orifice plate in the Process Waste Tank in Volume 6. See Drawings for requirements.

PART 3 - EXECUTION

3.1 INTERIOR AND EXPOSED EXTERIOR PIPING INSTALLATION

A. Install piping in vertical and horizontal alignment as shown on Drawings.

B. Alignment of piping smaller than 4 IN may not be shown; however, install according to Drawing intent and with clearance and allowance for:
1. Expansion and contraction.
2. Operation and access to equipment, doors, windows, hoists, moving equipment.
3. Headroom and walking space for working areas and aisles.
4. System drainage and air removal.

C. Enter and exit through structure walls, floor and ceilings using penetrations and seals specified in Specification Section 01 73 20 and as shown on the Drawings.

D. Install vertical piping runs plumb and horizontal piping runs parallel with structure walls.

E. Pipe Support:
   1. Use methods of piping support as shown on Drawings and as required in Specification Section 40 05 07.
   2. Select piping supports are shown on the Drawings.
      a. Contractor is responsible for design of support systems not shown on Drawings per Specification Section 40 05 07.
   3. Where pipes run parallel and at same elevation or grade, they may be grouped and supported from common trapeze-type hanger, provided hanger rods are increased in size as specified for total supported weight.
      a. The pipe in the group requiring the least maximum distance between supports shall set the distance between trapeze hangers.
   4. Size pipe supports with consideration to specific gravity of liquid being piped.

F. Locate and size sleeves and castings required for piping system.
   1. Arrange for chases, recesses, inserts or anchors at proper elevation and location.

G. Use reducing fittings throughout piping systems.
   1. Bushings will not be allowed unless specifically approved.

H. Equipment Drainage and Miscellaneous Piping:
   1. Provide drip pans and piping at equipment where condensation may occur.
   2. Hard pipe stuffing box leakage to nearest floor drain.
   3. Avoid piping over electrical components such as motor control centers, panelboards, etc.
      a. If piping must be routed over electrical equipment, utilize 16 GA, 316 stainless steel drip pan under piping and over full length of electrical equipment.
      b. Hard pipe drainage to nearest floor drain.
   4. Provide drainage for process piping at locations shown on Drawings.
   5. For applications defined above and for other miscellaneous piping, provide 304 stainless steel piping and fittings.
      a. Size to handle application with 3/4 IN being minimum size provided.

I. Unions:
   1. Install in position that will permit valve or equipment to be removed without dismantling adjacent piping.
   2. Mechanical type couplings may serve as unions.
   3. Additional flange unions are not required at flanged connections.

J. Install expansion devices as necessary to allow expansion/contraction movement.

K. Provide full face gaskets on all systems.

L. Anchorage and Blocking:
   1. Block, anchor, or harness exposed piping subjected to forces in which joints are installed to prevent separation of joints and transmission of stress into equipment or structural components not designed to resist those stresses.

M. Equipment Pipe Connections:
   1. Equipment - General:
      a. Exercise care in bolting flanged joints so that there is no restraint on the opposite end of pipe or fitting which would prevent uniform gasket pressure at connection or would cause unnecessary stresses to be transmitted to equipment flanges.
      b. Where push-on joints are used in conjunction with flanged joints, final positioning of push-on joints shall not be made until flange joints have been tightened without strain.
      c. Tighten flange bolts at uniform rate which will result in uniform gasket compression over entire area of joint.
         1) Provide tightening torque in accordance with manufacturer's recommendations.
      d. Support and match flange faces to uniform contact over their entire face area prior to installation of any bolt between the piping flange and equipment connecting flange.

e. Permit piping connected to equipment to freely move in directions parallel to longitudinal centerline when and while bolts in connection flange are tightened.
f. Align, level, and wedge equipment into place during fitting and alignment of connecting piping.
g. Grout equipment into place prior to final bolting of piping, but not before initial fitting and alignment.
h. To provide maximum flexibility and ease of alignment, assemble connecting piping with gaskets in place and minimum of four bolts per joint installed and tightened.
    1) Test alignment by loosening flange bolts to see if there is any change in relationship of piping flange with equipment connecting flange.
    2) Realign as necessary, install flange bolts, and make equipment connection.
i. Provide utility connections to equipment shown on Drawings, scheduled or specified.

2. Plumbing and HVAC equipment:
    a. Make piping connections to plumbing and HVAC equipment, including but not limited to installation of fittings, strainers, pressure reducing valves, flow control valves and relief valves provided with or as integral part of equipment.
    b. Furnish and install sinks, fittings, strainers, pressure reducing valves, flow control valves, pressure relief valves, and shock absorbers that are provided by manufacturer as an integral part of equipment, even when not specified.
    c. For each water supply piping connection to equipment, furnish and install union and gate or angle valve.
        1) Provide wheel handle stop valve at each laboratory sink water supply.
        2) Minimum size: 1/2 IN.
    d. Furnish and install "P" trap for each waste piping connection to equipment if waste is connected directly to building sewer system.
        1) Size trap as required by IPC.
        2) Provide trap guard devices in all floor drains and floor sinks.
    e. Stub piping for equipment, sinks, lavatories, supply and drain fittings, key stops, "P" traps, miscellaneous traps, and miscellaneous brass through wall or floor and cap and protect until such time when later installation is performed.

N. Provide insulating components where dissimilar metals are joined together.

O. Instrument Connections:
    1. See drawing details.

3.2 CONNECTIONS WITH EXISTING PIPING

A. Where connection between new work and existing work is made, use suitable and proper fittings to suit conditions encountered.

B. Undertake connections in fashion which will disturb system as little as possible.

C. Provide suitable equipment and facilities to dewater, drain, and dispose of liquid removed without damage to adjacent property.

D. Where connections to existing systems necessitate employment of past installation methods not currently part of trade practice, utilize necessary special piping components.

E. Where connection involves potable water systems, provide disinfection methods as prescribed in this Specification Section.

F. Once tie-in to each existing system is initiated, continue work continuously until tie-in is made and tested.

3.3 ACCESS PROVISIONS

A. Provide access doors or panels in walls, floors, and ceilings to permit access to valves, piping and piping appurtenances requiring service.

B. Size access panels to allow inspection and removal of items served, minimum 10 x 14 IN size.

C. Fabricate door and frame of minimum 14 GA, stretcher leveled stock, cadmium plated or galvanized after fabrication and fitted with screw driver lock of cam type.

D. Provide with key locks, keyed alike, in public use areas.

E. Furnish panels with prime coat of HPIC. See Specification Section 09 96 00.

F. Style and type as required for material in which door installed.

G. Where door is installed in fire-rated construction, provide door bearing UL label required for condition.

3.4 CATHODIC PROTECTION

A. Isolate, dielectrically, all piping from all other metals including reinforcing bars in concrete slabs, other pipe lines, and miscellaneous metal.

B. Make all connections from wire or cable by Thermit Cadwelding accomplished by operators experienced in this process.

C. Install all cables with a loop and overhead knot around each pipe and slack equal to at least 50% of the straight line length.

D. After cadwelding, coat all exposed metallic surfaces with hot applied tape.

3.5 PRESSURE GAGES

A. Provide at locations shown on the Drawings and as specified.

B. See Specification Section 01 61 03.

3.6 FIELD QUALITY CONTROL

A. Pipe Testing - General:
   1. Test piping systems as follows:
      a. Test exposed, non-insulated piping systems upon completion of system.
      b. Test exposed, insulated piping systems upon completion of system but prior to application of insulation.
      c. Test concealed interior piping systems prior to concealment and, if system is insulated, prior to application of insulation.
      d. Test buried piping (insulated and non-insulated) prior to backfilling and, if insulated, prior to application of insulation.
   2. Isolate equipment which may be damaged by the specified pressure test conditions.
   3. Perform pressure test using calibrated pressure gages and calibrated volumetric measuring equipment to determine leakage rates.
      a. Select each gage so that the specified test pressure falls within the upper half of the gage's range.
      b. Notify the Contracting Officer’s Representative 24 HRS prior to each test.
   4. Completely assemble and test new piping systems prior to connection to existing pipe systems.
   5. Acknowledge satisfactory performance of tests and inspections in writing to Contracting Officer prior to final acceptance.
   6. Bear the cost of all testing and inspecting, locating and remedying of leaks and any necessary retesting and re-examination.

B. Pressure Testing:
   1. Testing medium: Unless otherwise specified in the Drawings, utilize the following test media.
      a. Process systems: Water.
      b. Laboratory gases and natural gas systems: Cylinder nitrogen.
      c. Other liquid systems: Water.
   2. Allowable leakage rates:

a. Hazardous gas systems, all exposed piping systems, all pressure piping systems and all buried, insulated piping systems which are hydrostatically pressure tested shall have zero leakage goal at the specified test pressure throughout the duration of the test.
b. Non-hazardous gas and air systems which are tested with air shall have a maximum pressure drop of 5% of the specified test pressure throughout the duration of the test.
c. For low pressure (less than 25 PSIG) air testing, the acceptable time for loss of 1 PSIG of air pressure shall be:

PIPE SIZE (IN DIA)	TIME, MINUTES/100 FT
4	0.3
6	0.7
8	1.2
10	1.5
12	1.8
15	2.1
18	2.4
21	3.0
24	3.6
27	4.2
30	4.8
33	5.4
36	6.0
42	7.3
48	7.6

3. Hydrostatic pressure testing methodology:
   a. General:
      1) All joints, including welds, are to be left exposed for examination during the test.
      2) Provide additional temporary supports for piping systems designed for vapor or gas to support the weight of the test water.
      3) Provide temporary restraints for expansion joints for additional pressure load under test.
      4) Isolate equipment in piping system with rated pressure lower than pipe test pressure.
      5) Do not coat or insulate exposed piping until successful performance of pressure test.
   b. Soil, waste, drain and vent systems:
      1) Test at completion of installation of each stack or section of piping by filling system with water and checking joints and fittings for leaks.
      2) Eliminate leaks before proceeding with work or concealing piping.
      3) Minimum test heights shall be 10 FT above highest stack inlet.
   c. High pressure pipe systems:
      1) Applies to:
         a) Volume 5 – RSPH – RW5
         b) Volume 5 – RSPH – W5
         c) All other piping systems normally exceeding 1,000 PSI working pressure.
      2) Submit pressure testing plan for review and approval by Contracting Officer's Representative minimum 14 days prior to commencing testing.

3) In-person witness testing of pressure test piping not allowed.
4) Monitor test pressure remotely via live-feed cameras at all critical viewpoint locations.
5) Tighten high pressure flange bolts per torque and sequence in accordance with ASME B16.5.
a) Torque all flanges to specified value using torque wrench.
b) Upon reaching final torque, mark all flange bolts with paint/marker.
c) Bolting hardware of Class 900 and larger flanges cannot be torqued to design setting more than twice. Remove and replace bolted hardware as necessary.
d) Contractor to supply 4 sets of additional hardware and gaskets for each high-pressure flange type/class and size.

4. Liquefied propane systems - testing methodology:
a. Maintain specified test pressure until each joint has been thoroughly examined for leaks by means of soap suds and glycerine.
b. Wipe joints clean after test.
5. Air testing methodology:
a. General:
1) Ensure air is ambient temperature.
b. Low pressure air testing:
1) Place plugs in line and inflate to manufacturer's designated seal pressure.
2) Check plugs for proper sealing.
3) Introduce low pressure air into sealed line segment until air pressure reaches 4 PSIG greater than ground water or allowable limits of ASTM F1417.
a) Use test gage conforming to ASME B40.100 with 0 to 15 PSI scale and accuracy of 1% of full range.
4) Allow 2 minutes for air pressure to stabilize.
5) After stabilization period (3.5 PSIG minimum pressure in pipe) discontinue air supply to line segment.
6) Record pressure at beginning and end of test.

C. Dielectric Testing Methods and Criteria:
1. Provide electrical check between metallic non-ferrous pipe or appurtenances and ferrous elements of construction to assure discontinuity has been maintained.
2. Wherever electrical contact is demonstrated by such test, locate the point or points of continuity and correct the condition.

3.7 CLEANING, DISINFECTION AND PURGING

A. Cleaning:
1. Clean interior of piping systems thoroughly before installing.
2. Maintain pipe in clean condition during installation.
3. Before jointing piping, thoroughly clean and wipe joint contact surfaces and then properly dress and make joint.
a. Pig high pressure air piping before connecting to valves or instruments.
4. At completion of work and prior to Final Acceptance, thoroughly clean work installed under these Specifications.
a. Clean equipment, fixtures, pipe, valves, and fittings of grease, metal cuttings, and sludge which may have accumulated by operation of system, from testing, or from other causes.
b. Repair any stoppage or discoloration or other damage to parts of building, its finish, or furnishings, due to failure to properly clean piping system, without cost to Government.
5. After erection of piping and tubing, but prior to installation of service outlet valves, blow liquefied petroleum gas and digester gas systems clear of free moisture and foreign matter by means of air, nitrogen or carbon dioxide.
a. Oxygen shall never be used.
6. Clean chlorine piping in accordance with CI Pamphlet 6.

B. Disinfection of Potable Water Systems:
   1. After favorable performance of pressure test and prior to Final Acceptance, thoroughly flush entire potable water piping system including supply, source and any appurtenant devices and perform disinfection as prescribed.
   2. Perform work, including preventative measures during construction, in full compliance with AWWA C651.
   3. Perform disinfection using sodium hypochlorite complying with AWWA B300.
   4. Flush each segment of system to provide flushing velocity of not less than 2.5 FT per second.
   5. Drain flushing water to sanitary sewer.
      a. Do not drain flushing water to receiving stream.
   6. Use continuous feed method of application.
      a. Tag system during disinfection procedure to prevent use.
   7. After required contact period, flush system to remove traces of heavily chlorinated water.
   8. After final flushing and before placing water in service, obtain an independent laboratory approved by the Contracting Officer to collect samples and test for bacteriological quality.
      a. Repeat entire disinfection procedures until satisfactory results are obtained.
   9. Secure and deliver to Contracting Officer, satisfactory bacteriological reports on samples taken from system.
      a. Ensure sampling and testing procedures are in full compliance to AWWA C651, local water purveyor and applicable requirements of State of Arizona.

C. Purging Liquefied Petroleum Gas:
   1. Existing piping:
      a. Turn off gas supply.
      b. Vent line pressure outdoors.
      c. If section exceeds the following, then remaining gas shall be displaced with an inert gas.
         1) 50 FT for 2-1/2 IN pipe.
         2) 30 FT for 3 IN pipe.
         3) 15 FT for 4 IN pipe.
         4) 10 FT for 6 IN pipe.
         5) Any length for 8 IN or larger pipe.
   2. New piping:
      a. Including but not limited to:
         1) All fuel gas piping.
         2) Digesters.
         3) Digester gas equipment.
         4) Fuel gas trains.
      b. Purge air filled system with fuel gas:
         1) Providing piping length is less than:
            a) 30 FT for 3 IN pipe.
            b) 15 FT for 4 IN pipe.
            c) 10 FT for 6 IN pipe.
            d) Any length for 8 IN and larger pipe.
         2) Providing a moderately rapid and continuous flow of fuel gas is introduced.
            a) Introduce fuel gas at one end.
            b) Vent air at opposite end.
         3) Provided fuel gas flow is continuous without interruption until vented gas is free of air.
         4) The point of discharge shall not be left unattended during purging.
      c. If the piping is 3 IN or larger and exceeds lengths stated above.
         1) Purge air with inert gas in accordance with NFPA 54 and NFPA 69.
         2) Purge inert gas with fuel gas.
   3. Discharge of purged gases:

a. Open end of piping shall not discharge into confined spaces or areas where there are sources of ignition.

3.8 LOCATION OF BURIED OBSTACLES

A. Furnish exact location and description of buried utilities encountered and thrust block placement.

B. Reference items to definitive reference point locations such as entrances to buildings, existing structure lines, fire hydrants, and related fixed structures.

C. Include such information as location, elevation, coverage, supports and additional pertinent information.

D. Incorporate information on "As-Recorded" Drawings.

3.9 PIPE INSULATION

A. Insulate pipe and pipe fittings in accordance with Specification Section 40 42 00.

END OF SECTION

SECTION 40 05 07
PIPE SUPPORT SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Pipe support and anchor systems.
   2. Design of Pipe Support Systems as specified.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 35 73 – Delegated Design Procedures.
   2. Section 01 81 10 – Wind and Seismic Design Criteria
   3. Section 03 15 19 – Anchorage to Concrete.
   4. Section 05 50 00 – Metal Fabrications.
   5. Section 09 96 00 – High Performance Industrial Coatings.
   6. Section 40 42 00 – Pipe, Duct and Equipment Insulation.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B31.3, Process Piping.
   2. ASTM International (ASTM):
      a. A36, Standard Specification for Carbon Structural Steel.
      b. A123/123M, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
      c. A153/153M, Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
      d. A276, Standard Specification for Stainless Steel Bars and Shapes.
      e. A575, Standard Specification for Steel Bars, Carbon, Merchant Quality, M-Grades.
      f. A576, Standard Specification for Steel Bars, Carbon, Hot-Wrought, Special Quality.
      g. A917, Standard Specification for Steel Sheet, Coated by the Electrolytic Process for Applications Requiring Designation of the Coating Mass on Each Surface (General Requirements).
      h. A918, Standard Specification for Steel Sheet, Zinc-Nickel Alloy Coated by the Electrolytic Process for Applications Requiring Designation of the Coating Mass on Each Surface.
      i. B633, Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.
   3. American Welding Society (AWS):
      a. D1.1, Structural Welding Code - Steel.
      b. D1.6, Structural Welding Code - Stainless Steel.
   4. Manufacturers Standardization Society of the Valve and Fittings Industry Inc. (MSS):
      a. SP-58, Pipe Hangers and Supports - Materials, Design, Manufacture, Selection, Application, and Installation.

B. Responsibility:
   1. Several custom pipe support systems are shown in the Drawings for the following instances:
      a. Roaring Springs Pumphouse (Volume 5) – Pipe supports for high pressure piping, piping with constrained clearances, and standard piping.
      b. North Rim WTP (Volume 6) – Pipe Rack in Process Area for multiple process pipes crossing room.
   2. General piping supports are indicated on the Drawings for all other pipe 4 inches diameter and larger.
      a. Use Pipe Support details in Volume 0.

b. Incorporate those details with the requirements of this Specification Section to provide the piping support system.

3. Pipe support details are not indicated on the Drawings for piping 3 inches and smaller, but shall be provided by Contractor.
   a. Use Pipe Support details in Volume 0.
   b. Incorporate those details with the requirements of this Specification Section to provide the piping support system.
4. Provide all labor, materials, equipment, and incidentals as shown, specified, and required to design, furnish and install the system of hangers, supports, guidance, anchorage and appurtenances.

C. Each type of pipe hanger or support shall be the product of one manufacturer.

D. Pipe Support Designer Qualifications:
   1. Licensed Professional Engineer registered in the State of Arizona.
   2. Minimum of five years experience designing pipe supports for projects of similar size and complexity.

1.3 SUBMITTALS

A. See Section 01 33 23 Submittal Requirements for the mechanics and administration of the submittal process.

B. Action Submittals:
   1. Shop Drawings:
      a. Scaled drawings showing location, installation, material, loads and forces, and deflection of all hangers and supports.
      b. Analyze each pipe system for all loads and forces on hangers and supports and their reaction forces to the structure to which they are fastened.
      c. Where Contract Documents indicate Contractor is to design pipe support systems, submit detail design calculations and scaled drawings signed by Pipe support designer.
   2. Product data:
      a. Acknowledgement that products meet requirements of referenced standards.
      b. Manufacturer's installation instructions.
      c. Itemized list of wall sleeves, anchors, support devices and all other items related to pipe support system.

C. Informational Submittals:
   1. Pipe support designer qualifications.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Product requirements are to conform with the following salient characteristics:

2.2 MANUFACTURED UNITS

A. General:
   1. Galvanized components:
      a. Hot-dip galvanized components:
         1) General: ASTM 123/123M.
         2) Fasteners and hardware: ASTM 153/153M.
   2. Dissimilar metals protection:
      a. Galvanized-to-galvanized and galvanized-to-aluminum: No protection required.
      b. All other galvanized-to-dissimilar metal connections: Neoprene or nylon pads, shims, grommets, etc.
   3. Fiberglass Reinforced Plastic (FRP) components:
      a. Corrosive:
         1) Material: Polyester.

2) Channels: similar to Aickinstrut F2000 or F2100.
3) Pipe Clamps: similar to Aickinstrut FPS or FRPC.
4) Clevis Hangers: similar to Aickinstrut FMCH.
5) Beam Clamp: similar to Aickinstrut FMBC.
6) Column Base: similar to Aickinstrut F2852, F5853 or 5854

b. Highly Corrosive:
1) Material: Vinyl ester.
2) Channels: similar to Aickinstrut F2000 or F2100.
3) Pipe Clamps: similar to Aickinstrut FPS or FRPC.
4) Clevis Hangers: similar to Aickinstrut FMCH.
5) Beam Clamp: similar to Aickinstrut FMBC.
6) Column Base: similar to Aickinstrut F2852, F5853 or 5854

B. Hanger Rods:
1. Material:
a. ASTM A36.
b. ASTM A575, Grade M1020.
c. ASTM A576, Grade 1020.
d. Minimum allowable tensile stress of 12,000 psi at 650 degrees F per MSS SP-58.
e. Corrosion resistant: 304 or 316 stainless steel per ASTM A276.
2. Continuously threaded.
3. For galvanized rods: hot-dip galvanized after threads are cut.
4. Load limit:

NOMINAL ROD DIAMETER	MAXIMUM SAFE LOAD, (LBS)
3/8 inches diameter (min)	610
1/2 inches diameter	1,130
5/8 inches diameter	1,810
3/4 inches diameter	2,710
7/8 inches diameter	3,770
1 inches diameter	4,960

C. Hangers:
1. Materials
a. Corrosive areas: 304 stainless steel.
b. Highly corrosive areas: 316 stainless steel.
c. For use directly on copper pipe: Copper plated or PVC coated.
d. For use other than directly on copper pipe: Hot-dip galvanized.
2. Hanger type schedule:

APPLICATION	PIPE SIZE	HANGER TYPE
All except noted	4 inches and less	MSS Type 1 or 12 Similar to Empire Figure 41.
All except noted	Over 4 inches	MSS Type 1 Similar to Empire Figure 11.
Steam, condensate, and hot water	All	MSS Type 43 Similar to Empire Figure 272 w/ spring hanger.

Service in chemical storage areas and as indicated on drawings for corrosion resistance	All	FRP trapeze hangers or clevis hangers as specified below.

D. Concrete Inserts for Hanger Rods:
   1. Continuous slots: Similar to Unistrut #P1000.
   2. Individual inserts: Similar to ANVIL Figure 281.
   3. See Specification Section 03 15 19, Mechanical Anchors.

E. Beam Clamps for Hanger Rods:
   1. MSS SP-58, Type 21.
   2. Material:
      a. Corrosive areas: 304 stainless steel.
      b. Highly corrosive areas: 316 stainless steel.
      c. All other areas: Hot-dip galvanized steel.
   3. Standard duty.
   4. Similar to Empire Industries Figure 155.

F. Trapeze Hangers for Suspended Piping:
   1. General:
      a. Material:
         1) Corrosive areas: 304 stainless steel.
         2) Highly corrosive areas: 316 stainless steel.
         3) All other areas: Hot-dip galvanized steel.
      b. Angles, channels, or other structural shapes.
      c. Curved roller surfaces at support point corresponding with type of hanger required.
      d. See Specification Section 05 50 00.
   2. In chemical storage and feed areas and as indicated on the Drawings:
      a. Materials: FRP.
      b. Aikinstrut fiberglass channel or equal.

G. Vertical Pipe Supports:
   1. At base of riser.
      a. Material:
         1) Corrosive areas: 304 stainless steel.
         2) Highly corrosive areas: 316 stainless steel.
         3) All other areas: Hot-dip galvanized steel.
   2. Lateral movement:
      a. Clamps or brackets:
         1) MSS SP-58, Type 8 or 42.
         2) Similar to Empire Industries Figure 50.

H. Expanding Pipe Supports:
   1. Spring hanger type.
   2. MSS SP-58, Type 51.

I. Pipe Support Saddle:
   1. General:
      a. Minimum contact area:
         1) 120 Degrees minimum.
         2) 180 Degrees for PVC/PE greater than 12 inches
   2. For pipe located 3 feet or less from floor elevation, except as otherwise indicated on Drawings.
      a. MSS SP-58 Type 38.
      b. Material:
         1) Corrosive areas: 304 stainless steel.
         2) Highly corrosive areas: 316 stainless steel.

3) All other areas: Hot-dip galvanized steel.

c. Similar to Empire Industries Figure 426.

J. Pipe Support Risers/Stanchions:
1. Schedule 40 pipe.
2. Material: Match pipe support saddle.
3. Size: As recommended by saddle manufacturer.

K. Pipe Support Base Plate:
1. 4 inches larger than support.
2. Collar 3/16 inches thickness, circular in shape, and sleeve type connection to pipe.
3. Collar fitted over outside of support pipe and extended 2 inches from floor plate.
4. Collar welded to floor plate.
5. Edges ground smooth.
6. Material: Match pipe support saddle.
7. Provide low carbon stainless steel for welded fabrications.

L. Pipe Covering Protection Saddle:
1. For insulated pipe at point of support.
2. MSS SP-58 Type 40.
3. Material:
   a. Corrosive areas: 304 stainless steel.
   b. Highly corrosive areas: 316 stainless steel.
   c. All other areas: Hot-dip galvanized steel.
4. Stainless steel: Similar to Empire Industries Figure 167.

M. Wall Brackets:
1. For pipe located near walls and 8 feet or more above floor elevation or as otherwise indicated on the Drawings.
2. MSS SP-58, Type 33.
3. Material:
   a. Corrosive areas: 304 stainless steel.
   b. Highly corrosive areas: 316 stainless steel.
   c. All other areas: Hot-dip galvanized steel.
4. Similar to Empire Industries Figure 802.

N. Pipe Anchors:
1. For locations shown on the Drawings.
2. 1/4 inches steel plate construction.
3. Hot-dip galvanized after fabrication.
4. Designed to prevent movement of pipe at point of attachment.

O. Seismic Restraints:
1. Galvanized steel aircraft cables or steel angles or channels.
2. Prestretch steel aircraft cables to establish a certified minimum modulus of elasticity.
   a. Design cable restraints to resist seismic tension loads.
   b. Design steel restraints to resist both tension and compression loads with a minimum safety factor of 2.
3. Design restraint and connection steel assemblies that swivel to the final installation angle.
4. Do not mix cable and steel angle restraints to brace the same system.
5. Clamp steel angles to the threaded hanger rods at the seismic sway restraint locations utilizing a minimum of two ductile iron clamps.
6. Provide transverse bracing at 40 feet-0 inches maximum spacing unless noted otherwise.
7. Provide longitudinal bracing at 80 feet-0 inches maximum spacing unless noted otherwise.
8. Provide fuel gas piping with transverse bracing 20 feet-0 inches on-center maximum and longitudinal bracing 4 0 feet-0 inches on-center maximum.
9. At vertical pipe risers, wherever possible, support the weight of the riser at a point or points above the center of gravity of the riser.

a. Provide lateral guides at the top and bottom of the riser, and at intermediate points not to exceed 30 feet-0 inches on-center.

P. Pipe Guides:
1. For locations on both sides on each expansion joint or loop.
2. To ensure proper alignment of expanding or contracting pipe.
3. Material:
a. Corrosive areas: 304 stainless steel.
b. Highly corrosive areas: 316 stainless steel.
c. All other areas: Hot-dip galvanized steel.
4. Similar to Empire Industries Figure 255.

2.3 DESIGN REQUIREMENTS

A. Supports capable of supporting the pipe for all service and testing conditions.
1. Provide 5 to 1 safety factor.

B. Allow free expansion and contraction of the piping to prevent excessive stress resulting from service and testing conditions or from weight transferred from the piping or attached equipment.

C. Design supports and hangers to allow for proper pitch of pipes.

D. For chemical and waste piping, design, materials of construction and installation of pipe hangers, supports, guides, restraints, and anchors:
1. ASME B31.3.
2. MSS SP-58.
3. Except where modified by this Specification.

E. For steam and hot and cold water process piping, design, materials of construction and installation of pipe hangers, supports, guides, restraints, and anchors:
1. ASME B31.1.
2. MSS SP-58.
3. See Section 22 05 29 for plumbing applications.

F. Check all physical clearances between piping, support system and structure.
1. Provide for vertical adjustment after erection.

G. Support vertical pipe runs in pipe chases at base of riser.
1. Support pipes for lateral movement with clamps or brackets.

H. Place hangers are to be installed on outside of pipe insulation.
1. Use a pipe covering protection saddle for insulated pipe at support point.
2. Insulated piping 1-1/2 inches and less:
a. Provide a 9 inches length of high density perlite or high density calcium silicate at saddle.
3. Insulated piping over 1-1/2 inches: Provide a 12 inches length of high density perlite or high density calcium silicate at saddle.
4. See Specification Section 40 42 00.

I. Provide 20 GA pipe saddle for fiberglass and plastic pipe support points to ensure minimum contact width of 4 inches.
1. Material: match support.

J. Pipe Support Spacing:
1. General:
a. Factor loads by specific weight of liquid conveyed if specific weight is greater than water.
b. Locate pipe supports at maximum spacing scheduled.
c. Provide at least one support for each length of pipe at each change of direction and at each valve.
2. Steel, stainless steel, cast-iron, ductile-ironpipe support schedule:

PIPE SIZES - IN	MAXIMUM SPAN - FT
1-1/2 and less	5
2 thru 4	10
5 thru 8	15
10 and greater	20

3. Copper pipe support schedule:

PIPE SIZES - IN	MAXIMUM SPAN - FT
2-1/2 and less	5
3 thru 6	10
8 and greater	15

4. PVC pipe support schedule:

PIPE SIZES - IN	MAXIMUM SPAN - FT
1-1/4 and less	3
1-1/2 thru 3	4
4 and greater	5

* Maximum fluid temperature of 120 degrees F.

5. Support each length and every fitting:
   a. Bell and spigot piping:
      1) At least one hanger.
      2) Applied at bell.
   b. Mechanical coupling joints:
      1) Place hanger within 2 feet of each side of fittings to keep pipes in alignment.
6. Space supports for soil and waste pipe and other piping systems not included above every 5 feet.
7. Provide continuous support for tubing via containment piping.
8. For PVC and FRP piping:
   a. Provide Unistrut Unicushion (or equal) wrap of pipe at each support.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Provide piping systems exhibiting pulsation, vibration, swaying, or impact with suitable constraints to correct the condition.
   1. Included in this requirement are movements from:
      a. Trap discharge.
      b. Water hammer.
      c. Similar internal forces.

B. Seismic Restraints:
   1. Provide seismic restraints on all piping, with the exception of the following:
      a. Piping less than 1 inch diameter when used for gas and compressed air.
      b. Piping less than 2-1/2 inches diameter for all other cases.

c. Piping suspended by individual hangers where the distance from the top of the pipe to the bottom of the support for the hanger is 12 inches or less.
d. Provide fuel gas piping with transverse bracing 20 feet-0 inches on-center maximum and longitudinal bracing 40 feet-0 inches on-center maximum.

C. Weld Supports:
1. AWS D1.1.
2. Weld anchors to pipe in accordance with ASME B31.3.
3. AWS D1.6 for stainless steel supports.

D. Locate piping and pipe supports as to not interfere with open accesses, walkways, platforms, and with maintenance or disassembly of equipment.

E. Inspect hangers for:
1. Design offset.
2. Adequacy of clearance for piping and supports in the hot and cold positions.
3. Guides to permit movement without binding.
4. Adequacy of anchors.

F. Inspect hangers after erection of piping systems and prior to pipe testing and flushing.

G. Anchorage to Concrete - reference Specification Section 03 15 19.

H. Install individual or continuous slot concrete inserts for use with hangers for piping and equipment.
1. Install concrete inserts as concrete forms are installed.

I. Welding:
1. Welding rods: ASTM and AWS standards.
2. Integral attachments:
a. Include welded-on ears, shoes, plates and angle clips.
b. Ensure material for integral attachments is of good weldable quality.
3. Preheating, welding and post-heat treating: ASME B31.3, Chapter V.

J. Field Painting:
1. Comply with Specification Section 09 96 00.

END OF SECTION

SECTION 40 05 19
PIPE - DUCTILE IRON PROCESS PIPE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Ductile iron piping, fittings, and appurtenances.

B. Related Sections include but are not necessarily limited to:
   1. Section 40 05 00 – Pipe and Pipe Fittings – Basic Requirements.
   2. Section 40 05 07 – Pipe Support Systems.

1.2 REFERENCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B1.1, Unified Inch Screw Threads (UN and UNR Thread Form).
      b. B16.1, Gray Iron Pipe Flanges and Flanged Fittings: Classes 25, 125, and 250.
   2. ASTM International (ASTM):
      a. A307, Standard Specification for Carbon Steel Bolts and Studs, 60,000 psi Tensile Strength.
      b. A536, Standard Specification for Ductile Iron Castings
      c. B695, Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel.
   3. American Water Works Association (AWWA):
      a. C606, Standard for Grooved and Shouldered Joints.
   4. American Water Works Association/American National Standards Institute (AWWA/ANSI):
      a. C104/A21.4, Standard for Cement–Mortar Lining for Ductile-Iron Pipe and Fittings.
      b. C105/A21.5, Standard for Polyethylene Encasement for Ductile-Iron Pipe Systems.
      c. C110/A21.10, Standard for Ductile-Iron and Gray-Iron Fittings.
      d. C111/A21.11, Standard for Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings.
      e. C115/A21.15, Standard for Flanged Ductile-Iron Pipe with Ductile-Iron or Gray-Iron Threaded Flanges.
      f. C150/A21.50, Standard for Thickness Design of Ductile-Iron Pipe.
      g. C151/A21.51, Standard for Ductile-Iron Pipe, Centrifugally Cast, for Water.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Comply with Section 40 05 00.
   2. Certification of factory hydrostatic testing.
   3. If mechanical coupling system is used, submit piping, fittings, and appurtenant items which will be used to meet system requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Flanged Adaptors (flanged coupling adapters):
      a. Romac Style FCA501 (cast iron).
      b. Or equal.
   2. Restrained Flanged Adapters (restrained flanged coupling adapters):
      a. Romac Style Alpha FC (ductile iron).
      b. Or equal.
   3. Compression Sleeve Coupling:
      a. Romac Industries, Style EC501 (ductile iron).
      b. Or equal.
   4. Insulating Couplings:
      a. Romac, Style IC400.
      b. Or equal.
   5. Reducing Couplings:
      a. Romac, Style RC501 (ductile).
      b. Or equal.
   6. Transition Coupling:
      a. George Fischer Waga N.V., Multi-Joint® 3000 Plus, Type 3007.
      b. Or equal.
   7. Ceramic Epoxy Lining:
      a. PROTECTO 401 by Induron.
      b. Or equal.
   8. Polyethylene Encasement Tape:
      a. Chase, Chasekote 750.
      b. Or equal.
   9. Restrained joints:
      a. U.S. Pipe (TR-Flex) - 4 inches to 36 inches.
      b. Or equal.
   10. Expansion joints:
      a. Dresser (Style 63).
      b. Or equal.
   11. Dismantling joints:
      a. Dresser (Style 131).
      b. Or equal.

2.2 MATERIALS

A. Ductile Iron Pipe:
   1. AWWA/ANSI C115/A21.15.
   2. AWWA/ANSI C150/A21.50.
   3. AWWA/ANSI C151/A21.51.

B. Fittings and Flanges:
   1. AWWA/ANSI C110/A21.10.
   2. AWWA/ANSI C115/A21.15.
   3. Flanges drilled and faced per ASME B16.1 for both 125 and 250 psi applications.

C. Nuts and Bolts:
   1. Buried:
      a. Tee-Head bolts for mechanical joints: Per AWWA/ANSI C111/A21.11.
      b. Other bolts and nuts: ASTM A307.
      c. Wax Tape Coatings per AWWA C217.
   2. Exposed: Mechanical galvanized ASTM B695, Class 40.

3. Heads and dimensions per ASME B1.1.
4. Threaded per ASME B1.1.
5. Project ends 1/4 to 1/2 inches beyond nuts.

D. Gaskets:
1. Flanged, push-on and mechanical joints: Rubber, AWWA/ANSI C111/A21.11.
2. Grooved coupling joints: Rubber, AWWA C606.

E. If mechanical coupling system is used, utilize pipe thickness and grade in accordance with AWWA C606.
1. Manufactured from ductile iron conforming to ASTM A536.
2. Gaskets: Flushseal® type or equal, grade to suit the intended service.

F. Polyethylene Encasement: Per AWWA/ANSI C105/A21.5.

2.3 MANUFACTURED UNITS

A. Couplings:
1. Flanged adaptors:
a. Unit consisting of steel or carbon steel body sleeve, flange, followers, Grade 30 rubber gaskets.
b. Provide units specified in the MANUFACTURERS Article.
c. Supply flanges meeting standards of adjoining flanges.
d. Rate entire assembly for test pressure for each respective application.
2. Compression sleeve coupling:
a. Unit consisting of steel sleeve, followers, Grade 30 rubber gaskets.
b. Provide units specified in the MANUFACTURERS Article.
c. Supply flanges meeting standards of adjoining flanges.
d. The working pressure rating of the entire assembly shall be greater than or equal to the test pressure for each respective piping application.
e. Provide field coating for buried couplings per AWWA C203.

2.4 FABRICATION

A. Any exposed pipe scheduled to be painted shall be furnished without bituminous material or other outside coatings.

B. Furnish cast parts with lacquer finish compatible with finish coat.

2.5 LININGS AND COATINGS

A. Cement mortar lining: per AWWA/ANSI C104/A21.4
1. Standard thickness.

B. Ceramic epoxy:
1. Minimum dry film thickness: 40 mil.

C. Glass Lining:
1. Minimum two-coat process.
a. Base coat heated to solidly fuse glass to pipe surface.
b. Subsequent coat(s) heated to form integral bond with preceding coat.
2. Final finish parameters:
a. Thickness: 8-12 mils.
b. Hardness: Above 5 on MOHS scale.
c. Density: 2.5-3.0 grams per cubic centimeter.
d. Metal to lining bonding: Capable of withstanding strain of 0.0001 inches/IN without damage to lining.
3. Complete compatibility between fittings and piping.
4. Provide in accordance with ASTM B1000.

2.6 SOURCE QUALITY CONTROL

A. Factory Test:
   1. Subject pipe to hydrostatic test of not less than 500 psi with the pipe under the full test pressure for at least 10 seconds.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Joining Method - Push-On Mechanical (Gland-Type) Joints:
   1. Install in accordance with AWWA/ANSI C111/A21.11.
   2. Assemble mechanical joints carefully according to manufacturer's recommendations.
   3. If effective sealing is not obtained, disassemble, thoroughly clean, and reassemble the joint.
   4. Do not overstress bolts.
   5. Where piping utilizes mechanical joints with tie rods, align joint holes to permit installation of harness bolts.

B. Joining Method - Push-On Joints:
   1. Install in accordance with AWWA/ANSI C151/A21.51.
   2. Assemble push-on joints in accordance with manufacturer's directions.
   3. Bevel and lubricate spigot end of pipe to facilitate assembly without damage to gasket.
      a. Use lubricant that is non-toxic, does not support the growth of bacteria, has no deteriorating effects on the gasket material, and imparts no taste or odor to water in pipe.
   4. Assure the gasket groove is thoroughly clean.
   5. For cold weather installation, warm gasket prior to placement in bell.
   6. Taper of bevel shall be approximately 30 degrees with centerline of pipe and approximately 1/4 inches back.

C. Joining Method - Flanged Joints:
   1. Install in accordance with AWWA/ANSI C115/A21.15.
   2. Extend pipe completely through screwed-on flanged and machine flange face and pipe in single operation.
   3. Make flange faces flat and perpendicular to pipe centerline.
   4. When bolting flange joints, exercise extreme care to ensure that there is no restraint on opposite end of pipe or fitting which would prevent uniform gasket compression or would cause unnecessary stress, bending or torsional strains to be applied to cast flanges or flanged fittings.
   5. Allow one flange free movement in any direction while bolts are being tightened.
   6. Do not assemble adjoining flexible joints until flanged joints in piping system have been tightened.
   7. Gradually tighten flange bolts uniformly to permit even gasket compression.

D. Flange Adaptors 12 inches and Less:
   1. Locate and drill holes for anchor studs after pipe is in place and bolted tight.
   2. Drill holes not more than 1/8 inches larger than diameter of stud projection.

E. Wax Tape Coating System:
   1. After installation, coat all buried nuts and bolts with a wax tape coating system in accordance with:
      a. AWWA C217.
      b. Manufacturer’s printed instructions.
   2. Cut strips of wax tape and apply them around all bolts and nuts so that there are no voids or spaces under the tape.
   3. Apply a sufficient amount of tape to completely encapsulate all bare metal.

F. Cutting:
   1. Do not damage interior lining material during cutting.

2. Use abrasive wheel cutters or saws.
3. Make square cuts.
4. Bevel and free cut ends of sharp edges after cutting.

G. Support exposed pipe in accordance with Section 40 05 00.

H. Install buried piping in accordance with Section 40 05 00.

I. Install restrained joint systems where specified in Section 40 05 00.

3.2 FIELD QUALITY CONTROL

A. Test piping systems in accordance with the volume-specific pipe schedules.

B. Wax Tape Coating Testing:
1. Inspect each wax tape-wrapped component.
   a. Verify primer was applied to substrate surfaces.
   b. Ensure the wax tape has fully encapsulated all portions of the substrate.
   c. Ensure that the wax tape is in intimate contact with the substrate.
   d. Verify that the wax tape has been applied to the specified thickness.
2. Do not backfill until this inspection is complete and the wax tape application is approved by the Contracting Officer's Representative.

END OF SECTION

This page intentionally left blank.

SECTION 40 05 23
PIPE - STAINLESS STEEL PROCESS PIPE AND TUBING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Stainless steel tubing, piping, fittings, and appurtenances.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 40 05 00 - Pipe and Pipe Fittings – Basic Requirements.

1.2 REFERENCES

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B16.1, Gray Iron Pipe Flanges and Flanged Fittings (Classes 25, 125 and 250).
      b. B31.1, Power Piping.
      c. BPVC.IX, Welding, Brazing, and Fusing Qualifications
   2. ASTM International (ASTM):
      a. A182, Standard Specification for Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service.
      b. A269, Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service.
      c. A312, Standard Specification for Seamless, Welded, and Heavy Cold Worked Austenitic Stainless Steel Pipes.
      d. A320, Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for Low-Temperature Service.
      e. A380/A380M, Standard Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems.
      f. A530, Standard Specification for General Requirements for Specialized Carbon and Alloy Steel Pipe.
      g. A774, Standard Specification for As-Welded Wrought Austenitic Stainless Steel Fittings for General Corrosive Service at Low and Moderate Temperatures.
      h. A778, Standard Specification for Welded, Unannealed Austenitic Stainless Steel Tubular Products.
      i. A967, Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts.
   3. American Water Works Association (AWWA):
      a. C228, Standard for Stainless-Steel Pipe Flange Joints for Water Service - Sizes 2 In. Through 72 In.
   4. American Welding Society:
      a. AWS D10.4M/D10.4, Guide for Welding Austenitic Stainless Steel Piping and Tubing.
   5. Nickel Development Institute (NiDI):
      a. Publication 11 007, Guidelines for the Welded Fabrication of Nickel-Containing Stainless Steels for Corrosion Resistant Services.

1.3 SUBMITTALS

A. Action Submittals: Submit the following:
   1. Shop Drawings:
      a. Product technical data including:
         1) Acknowledgement that products submitted meet requirements of standards referenced.
      b. Free iron contamination test method to be used (to be selected from ASTM A380).

B. Informational Submittals: Submit the following:
    a. Fabrication details and welding procedure specifications for all work to be done under this Specification Section.

C. See Specification Section 40 05 00 for additional piping submittal requirements.

1.4 QUALITY ASSURANCE

A. QA/QC Program
    1. Fabricator shall have QA/QC program and procedures in place to avoid cross contamination between stainless steel and carbon steel products, materials, and tools.
        a. The QA/QC program shall be made available to Government for review and comment.
    2. Weld personnel shall be qualified for all positions that will be welded, qualified by testing in accordance with applicable welding code ASME BPVC.IX or AWS D1.6 requirements.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Tubing:
    1. ASTM A269.
    2. Filler material: Extra low carbon (ELC) with 0.03% maximum carbon.

B. Pipe:
    1. ASTM A778.
    2. ASTM A312.
    3. Diameter tolerance and wall thickness tolerance are to conform to ASTM A530.

C. Pipe Fittings:
    1. ASTM A774.

D. Flanges:
    1. Conventional:
        a. Flat faced.
        b. Welding neck or slip on type.
        c. ASTM A182, Type 316L AWWA C228 Class 2A
    2. Van Stone:
        a. Stainless steel stub end to match pipe material.
        b. Follower flange: stainless steel.
            1) AWWA C228 Class SB

E. Nuts, Bolts and Washers:
    1. ASTM A320, Type 316.
    2. Provide two (2) nuts for 1 inch diameter bolt applications and larger.

F. Gaskets and O-rings:
    1. High temperature EPDM
        a. Temperature rating of 300 degrees F.
        b. Within first 100 feet of positive displacement blower discharge
    2. Rubber or neoprene.
        a. Temperature rating of 250 degrees F.
        b. All other locations

2.2 FABRICATION

A. Shop Welding:
    1. Weld stainless steel in accordance with AWS D10.4-86R and NiDI 11 007.
    2. Provide welds sound and free from embedded scale or slag, and tensile strength at weld not less than pipe.
    3. Perform butt welds only with an inert gas shielded process.

4. Provide adequate inert gas protection to the top and under or backside of the weld to protect from atmospheric contamination.
5. Provide filler metal appropriate for the base material being welded.
6. Use only inert gas shielded welding processes for spool fabrication.
7. Provide butt welds with 100% penetration to the interior or back side of the weld joint.
8. Weld reinforcement on both sides of the weld is to be smooth, uniform and no more than 1/16 inches in height.

B. Joints:
1. Shop welded circumferential buttweld joints.
2. ASME B16.1, Class 150.

C. Expansion Joints:
1. Fabricate for 15 psi internal pressure and 250 degrees F operation unless otherwise noted in 40 05 00.
2. Ensure aerial travel in expansion joints of 3.1 inches minimum for 15,000 cycles or 5.2 inches for 1000 cycles.
3. Furnish each assembly with minimum four control tie rods.
4. Fabricate with 125 pound flanged end connections.
5. For components/assemblies to be used only in dry air service environments:
    a. No post-fabrication passivation treatments are required for external corrosion control.
6. For components/assemblies to be used in outdoor atmospheric service environments:
    a. Following the conclusion of fabrication operations, just prior to shipping, pickle/passivate all fabricated components per the requirements of ASTM A380 to remove dirt, weld heat tint, weld slag, free iron contamination, and other surface contaminants suffered during fabrication.
        1) 100% of all completed welds and material shall be pickled/passivated.
        2) Additional methods listed in ASTM A967 or NiDI 11 007 may be used for cleaning/passivation with written approval from the Contracting Officer.
    b. For assemblies containing crevices, joints, flanges, gaskets, pockets, etc. that cannot be completely rinsed to remove passivating acids/chemicals following the passivating process: disassemble prior to passivation, then reassemble after rinsing/drying.
    c. Protect passivated assemblies from additional carbon steel surface contamination or other contaminants during all subsequent packing, shipping, handling, and installation operations as described in ASTM A380.
7. For components/assemblies to be used in corrosive service environments:
    a. Following the conclusion of fabrication operations, just prior to shipping, pickle/passivate all fabricated components per the requirements of ASTM A380 to remove dirt, weld heat tint, weld slag, free iron contamination, and other surface contaminants suffered during fabrication.
        1) 100% of all completed welds and material shall be pickled/passivated.
        2) Additional methods listed in ASTM A967 or NiDI 11 007 may be used for cleaning/passivation with written approval from the Contracting Officer.
    b. For assemblies containing crevices, joints, pockets, etc. that cannot be completely rinsed to remove passivating acids/chemicals following the passivating process disassemble prior to passivation, then reassemble after rinsing/drying.
    c. Demonstrate that all free iron contamination has been removed from all stainless steel surfaces per one of the free iron test methods listed in ASTM A380.
        1) Submit the free iron contamination test method to be used for approval in advance by the Contracting Officer.
        2) Test results must be approved in writing by the Contracting Officer.
    d. Protect passivated assemblies from additional carbon steel surface contamination or other contaminants during all subsequent packing, shipping, handling, and installation operations as described in ASTM A380.
8. Control and dispose solvents and acid cleaning solutions used by the fabricator according to all applicable federal, state, and local pollution control regulations.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Prior to installation, inspect and verify condition of piping and appurtenances.
   1. Installation constitutes installer's acceptance of condition for satisfactory installation.

3.2 PREPARATION

A. Correct defects or conditions which may interfere with or prevent a satisfactory installation.

B. Ensure ends of pipe to be fitted with flanges have all protrusions ground flush.

3.3 INSTALLATION

A. Ensure all pipe cutting, threading and jointing conforms to requirements of ASME B31.1.
   1. Lubricate all pipe threads with Teflon tape.

B. Field Welding:
   1. Weld stainless steel in accordance with AWS D10.4-86R and NiDI 11 007.
   2. For welding on existing lines, clean and purge lines prior to welding.
   3. Provide welds sound and free from embedded scale or slag, and tensile strength at weld not less than pipe.
   4. Perform butt welds only with an inert gas shielded process.
   5. Provide adequate inert gas protection to the top and under or backside of the weld to protect from atmospheric contamination.
   6. Apply filler metal for all manually-performed welds appropriate for the base material being welded.
   7. Use only inert gas shielded welding processes for spool fabrication.
   8. Provide butt welds with 100% penetration to the interior or back side of the weld joint.
   9. Weld reinforcement on both sides of the weld is to be smooth, uniform and no more than 1/16 inches in height.
   10. Following completion of field welding, remove all dirt, weld heat tint, weld slag, free iron contamination, and other surface contaminants suffered during field welding.
      a. For external corrosion control, follow the pickling/passivation/cleanliness requirements of Article 2.2, Paragraph E of this specification section.
      b. Additional methods listed in ASTM A967, Section 8 may be used for passivation with Contracting Officer approval.

C. Joining Method - Flanges:
   1. Leave 1/8 inches to 3/8 inches flange bolts projection beyond face of nut after tightening.
      a. Coordinate dimensions and drillings of flanges with flanges for valves, equipment, and other systems.
      b. Tighten bolts evenly around pipe until following range of torques is achieved:

BOLT SIZE, IN	RANGES OF TORQUE, FT/LBS
5/8	40 - 60
3/4	60 - 90
1	70 - 100
1-1/4	90 - 120

D. Expansion Joints:
   1. Install in accordance with manufacturer's instructions.
   2. Apply anti-seize compound to all exposed steel threads.

3.4 FIELD QUALITY CONTROL

A. Test piping systems in accordance with the volume-specific pipe schedules .

3.5 CLEANING

A. Clean in accordance with Specification Section 40 05 00.

END OF SECTION

This page intentionally left blank.

SECTION 40 05 24
PIPE - STEEL

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Steel pipe, fittings, and appurtenances.

B. Related Sections include but are not necessarily limited to:
   1. Section 09 96 00 - High Performance Industrial Coatings.
   2. Section 40 05 00 - Pipe and Pipe Fittings – Basic Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B1.1, Unified Inch Screw Threads (UN and UNR Thread Form).
      b. B1.2, Gages and Gaging for Unified Inch Screw Threads.
      c. B16.3, Malleable Iron Threaded Fittings.
      d. B16.5, Pipe Flanges and Flanged Fittings.
      a. B16.9, Factory-Made Wrought Steel Butt-Welding Fittings.
      b. B16.11, Forged Steel Fittings, Socket Welding and Threaded.
      c. B16.47, Large Diameter Steel Flanges NPS 26 Through NPS 60 Metric/Inch Standard
      d. B31.1, Power Piping.
      e. B31.3, Process Piping.
      f. B31.9, Building Services Piping.
      g. Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.
   2. ASTM International (ASTM):
      a. A36, Standard Specification for Carbon Structural Steel.
      b. A53, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
      c. A181, Standard Specification for Carbon Steel Forgings, for General-Purpose Piping.
      d. A193, Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service.
      e. A194/A194M, Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both.
      f. A234, Standard Specification for Pipe Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service.
      g. A283, Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates.
      h. A307, Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 psi Tensile Strength
      i. A572, Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel.
      j. A1011, Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength.
      k. B6, Standard Specification for Zinc.
   3. American Water Works Association (AWWA):
      a. C200, Standard for Steel Water Pipe - 6 inches and Larger.
      b. C203, Standard for Coal-Tar Protective Coatings and Linings for Steel water Pipeline - Enamel and Tape - Hot Applied.

c. C205, Standard for Cement-Mortar Lining and Coating for Steel Water Pipe - 4 inches and Larger - Shop Applied.
d. C206, Standard for Field Welding of Steel Water Pipe.
e. C207, Standard for Steel Pipe Flanges for Waterworks Service - Sizes 4 inches through 144 inches.
f. C208, Standard for Dimensions for Fabricated Steel Water Pipe Fittings.
g. C209, Standard for Cold-Applied Tape Coatings for the Exterior of Special Sections, Connections, and Fittings for Steel Water Pipelines.
h. C210, Standard for Liquid-Epoxy Coating Systems for the Interior and Exterior of Steel Water Pipelines.
i. C606, Standard for Grooved and Shouldered Joints.
j. M11, Steel Pipe - A Guide for Design and Installation.

4. Society of Automotive Engineers (SAE):
   a. AMS-QQ-P-416, Cadmium Plating Electro deposited.

B. Qualifications:
1. Application of coating materials including preparation of surfaces, priming, and lining and coating of pipe, fittings, and specials, in shop, repairs of any damage to lining or coating occurring during shipment or any other time, and field lining and coating of ends where linings or coatings have been held back for welded field joints, shall be done by established and recognized pipe company acceptable to Contracting Officer's Representative.
2. Use only certified welders meeting procedures and performance outlined in ASME Section IX, AWWA C200 Section 3.3.3 and other codes and requirements per local building and utility requirements.

1.3 SUBMITTALS

A. Shop Drawings:
1. See Specification Section 40 05 00.
2. Factory test reports.
3. If mechanical grooved type coupling system is used, submit piping, fittings, and appurtenant items which will be utilized.
4. Coating manufacturer's qualifications.
5. Welders certificates.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. General:
1. Product requirements are to conform to the salient characteristics listed in this Specification Section.

B. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Flanged adaptors:
   a. Victaulic Vic Flange Adapter (Style 741 (steel))
   b. Or Equal
2. Braided Flexible Connectors
   a. Class 150 ASME B16.5 flanged ends

b. Flanges: carbon steel
c. Inner hose: 321/316/304 stainless steel
d. Braiding: 316/304 stainless steel
e. Pressure rating at 70 DEGF: 200 psi min
f. Lateral flexion:
    1) 2.5 IN: 0.125 IN MINIMUM
    2) 5 IN: 0.375 IN MINIMUM
g. Manufacturers:
    1) Flexicraft
    2) Or Eqaul

3. Mechanical couplings and fittings:
    a. Roll grooved (use only on suction pipe within Roaring Springs Pumphouse – Lines DD1 and RW4 in pipe schedule in Drawings):
        1) Victaulic Style 77.
        2) Or Equal
    b. Shouldered:
        1) Victaulic Style 809N.
        2) Or Equal
4. Factory-applied plastic or epoxy coatings:
    a. See Specification Section 09 96 00 – High Performance Industrial Coatings

2.2 MATERIALS

A. All materials used in steel piping systems shall meet or exceed pressure test requirements specified in the volume-specific pipe schedule.

B. Steel Pipe (Fabricated Type):
1. AWWA C200:
    a. ASTM A36, Grade C steel plate.
    b. ASTM A283, Grade D steel plate.
    c. ASTM A572, steel plate.
    d. ASTM A1011, steel sheet.

C. Steel Pipe (Mill Type): ASTM A53, Type E or S.

A. Fittings (For Fabricated Pipe): AWWA C208.

B. Fittings (For Mill Type Pipe):
1. ASTM A234.
2. ASME B16.3, ASME B16.5, ASME B16.9, ASME B16.11.

C. Flanges (Fabricated Pipe):
1. Type
    a. ASME B16.5
2. Flange material: ASTM A283, Grade C or D, ASTM A181, Grade 1.
3. Flange finish: Flat faced.
4. Pressure class: As indicated in Drawings or greater than or equal to pipe test pressure

D. Flanges (Mill Type Pipe):
1. ASME B16.5.
2. Flat faced.
3. Weld-neck flanges.
4. Pressure class: As indicated in Drawings or greater than or equal to pipe test pressure

E. Nuts and Bolts:
1. Buried:
    a. Bolts and nuts: ASTM A307 ASTM A193/A194 Grade B8 Class 1.
    b. Wax Tape Coatings per AWWA C217

2. Exposed: Mechanical galvanized ASTM B695, Class 40 ASTM A193/A194 Grade B8 Class 1.
3. Heads and dimensions per ASME B1.1.
4. Threaded per ASME B1.1.
5. Project ends 1/4 to 1/2 inches beyond nuts.

F. Gaskets:
   a. Grooved coupling joints: Rubber, AWWA C606.
   b. Flanged joints:
      1) Inner Ring: 304 Stainless Steel.
      2) Spiral Winding: 304 Stainless Steel
      3) Filler: PTFE.
      4) Outer Ring: Carbon Steel

G. Insulating Components:
   1. Dielectric Flange Kits:
      a. Type E - Flat faced.
      b. 1/8 IN thick dielectric gasket, phenolic, non-asbestos.
      c. Match or exceed pressure rating of flange
      d. Maximum temperature: 210 DEG F
      e. 1/32 IN wall thickness bolt sleeves
      f. 1/8 IN thick phenolic insulating washers.
   2. Dielectric unions:
      a. Screwed end connections
      b. Match or exceed pressure rating of piping system
      c. Maximum temperature: 210 DEG F
      d. Provide dielectric gaskets suitable for continuous operation at union rated temperature and pressure.

2.3 MANUFACTURED UNITS

A. Couplings:
   1. Flanged adaptors:
      a. Steel or carbon steel body sleeve, flange, followers and Grade 30 rubber gaskets.
      b. Provide units specified in Article 2.1.
      c. Flanges meeting standards of adjoining flanges.
      d. Entire assembly to be rated for test pressure specified on Piping Schedule for each respective application.
   2. Compression sleeve coupling:
      a. Steel sleeve, followers Grade 30 and rubber gaskets.
      b. Provide units specified in Article 2.1.
      c. Flanges meeting standards of adjoining flanges.
      d. Entire assembly to be rated for test pressure specified on Piping Schedule for each respective application.
      e. Provide field coating for buried couplings per AWWA C203.
   3. Mechanical coupling joint:
      a. Use of mechanical grooved (AWWA C606) type couplings and fittings in lieu of flanged joints is acceptable where specifically specified in Section 40 05 00.
      b. Utilize units defined in Article 2.1.

2.4 ACCESSORIES

A. Heating Water Application:
   1. For steel heating lines, provide braided, flanged stainless steel connectors for connection to equipment.
   2. Provide pump connectors with stainless steel construction, rubber filled bellows and flanged end connections.

B. Natural Gas Equipment Isolator: 316L stainless steel, T-321 stainless steel braid with connections compatible with joints in piping system.

2.5 FABRICATION

A. Provide piping (mill or fabricated) for use in this Project with minimum wall thicknesses as indicated on Drawings
   1. Design pipe in accordance with operating pressures shown in Piping Schedules for a design stress limited to 50% of yield.

B. Furnish cast parts with lacquer finish compatible with finish coating.

C. Furnish without outside coating of bituminous material any exposed pipe scheduled to be painted.

D. Fabricated Fittings:
   1. AWWA C208.
   2. Assure ratio of radius of bend to diameter of pipe equal to or greater than 1.0.

E. Protective Coatings:
   1. Factory Applied Coating: See Specification Section 09 96 00 – High Performance Industrial Coatings
   2. Field paint pipe in accordance with Section 09 96 00.

2.6 SOURCE QUALITY CONTROL

A. Testing:
   1. Shop hydrostatic test fabricated steel pipe and fittings.
   2. Field hydrostatic test all pipe as specified in Section 40 05 00.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. See Drawings for locations where the following joining methods are permitted. Use of joining method other than what is shown in Drawings shall only be permitted if approved by Contracting Officer’s Representative.

C. Joining Methods - Flanges:
   1. Facing method:
      a. Insert slip-on flange on pipe.
      b. Assure maximum tolerances for flange faces from normal with respect to axis of pipe is 0.005 inches per foot of flange diameter.
      c. Test flanges after welding to pipe for true to face condition and reface, if necessary, to bring to specified tolerance.
   2. Joining method:
      a. Leave 1/8 to 3/8 inches of flange bolts projecting beyond face of nut after tightening.
      b. Coordinate dimensions and drillings of flanges with flanges for valves, pumps, equipment, tank, and other interconnecting piping systems.
      c. When bolting flange joints, exercise extreme care to assure that there is no restraint on opposite end of pipe or fitting which would prevent uniform gasket compression or cause unnecessary stress, bending or torsional strains being applied to cast flanges or flanged fittings.
         1) Allow one flange free movement in any direction while bolts are being tightened.
      d. Do not assemble adjoining flexible coupled, mechanical coupled or welded joints until flanged joints in piping system have been tightened.
      e. Gradually tighten flange bolts uniformly to permit even gasket compression.
         1) For ASME B16.5 Class 900 flanges: Do not torque bolts more than two times.

a) Mark each bolt with a single line after bolt has been torqued.
b) If bolt has been torqued two times, replace all hardware on flange prior to subsequent tightening

f. Do not overstress bolts to compensate for poor installation.

D. Joining Method - Welded Joints:
1. Perform welding in accordance with AWWA C206 and this Section.
2. For flange attachment perform in accordance with AWWA C207.
3. Have each welding operator affix an assigned symbol to all his welds.
   a. Mark each longitudinal joint at the extent of each operator's welding.
   b. Mark each circumferential joint, nozzle, or other weld into places 180 degrees apart.
4. Welding for all process piping shall conform to ASME B31.3.
   a. Welding of utility piping 125 psi and less shall be welded per ASME B31.9.
   b. Utility piping above 125 psi shall conform to ASME B31.1.
5. Provide caps, tees, elbows, reducers, etc., manufactured for welded applications.
6. Weldolets may be used for 5 inches and larger pipe provided all slag is removed from inside the pipe.
7. Weld-in nozzles may be used for branch connections to mains and where approved by Contracting Officer’s Representative.
8. Use all long radius welding elbows for expansion loops and bends.
9. Use long radius reducing welding elbows 90 degrees bends and size changes are required.

E. Joining Method - Couplings:
1. Mechanical coupling:
   a. Arrange piping so that pipe ends are in full contact.
   b. Groove and shoulder ends of piping in accordance with manufacturer's recommendations.
   c. Provide coupling and grooving technique assuring a connection which passes pressure testing requirements.

F. Joining Method - Threaded and Coupled (T/C):
1. Provide T/C end conditions that meet ASME B1.2 requirements.
2. Furnish pipe with factory-made T/C ends.
3. Field cut additional threads full and clean with sharp dies.
4. Leave not more than three pipe threads exposed at each branch connection.
5. Ream ends of pipe after threading and before assembly to remove burrs.
6. Use Teflon thread tape on male thread in mating joints.

G. Support exposed piping in accordance with Section 40 05 00.

H. Install buried piping per Section 40 05 00.

I. Wax Tape Coating System:
1. After installation, coat all buried nuts and bolts with a wax tape coating system in accordance with:
   a. AWWA C217 NACE International RP0375.
   b. Manufacturer’s printed instructions.
2. Cut strips of wax tape and apply them around all bolts and nuts so that there are no voids or spaces under the tape.
3. Apply a sufficient amount of tape to completely encapsulate all bare metal.

3.2 FIELD QUALITY CONTROL

A. Field Welding.
1. Visually inspect all welds (100 percent inspection) and mark to indicate acceptance or rejection.
2. Inspect 100 percent of butt joint welds with full circumference magnetic particle or ultrasonic testing.
3. Weld Acceptance:

a. Perform visual inspection in accordance with AWS D1.1
b. Perform magnetic particle testing in accordance with AWS D1.1
c. Perform Ultrasonic testing in accordance with ASME BPVC, Section VIII, Division 1
d. Accpetance: in accordance with standards specified above.

4. Remove defective welds in manner that permits proper and complete repair by welding at no additional cost to the Government
5. Caulking or peening of welds is not permitted
6. Retest unsatisfactory welds at no additional cost to Government

B. Hydrostatic testing of steel pipe: See Specification Section 40 05 00

C. Pipeline Disinfection
1. Disinfection of steel pipe: See Ppecification Section 40 05 00

END OF SECTION

This page intentionally left blank.

SECTION 40 05 25
PIPE - PREFABRICATED, INSULATED AND JACKETED

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Underground, prefabricated, insulated and jacketed piping with flexible joints.

B. Related Sections include but are not necessarily limited to the following:
   1. Section 31 23 33 - Trenching and Backfilling.
   2. Section 40 05 00 – Pipe and Pipe Fittings – Basic Requirements
   3. Section 40 05 24 - Pipe - Steel.

1.2 QUALITY ASSURANCE

A. See Section 40 05 00.

B. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B31.1, Power Piping.
   2. ASTM International (ASTM):
      a. A53, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
      b. C272, Standard Test Method for Water Absorption of Core Materials for Sandwich Constructions.
      c. C518, Standard Test Method for Steady-State Thermal Transmission Properties
      d. D638, Standard Test Method for Tensile Properties of Plastics.
      e. D790, Standard Test Method for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials.
      f. D1238, Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer.
      g. D1505, Standard Test Method for Density of Plastics by the Density-Gradient Technique.
      h. D1621, Standard Test Method for Compressive Properties of Rigid Cellular Plastics.
      i. D1622, Standard Test Method for Apparent Density of Rigid Cellular Plastics
      j. D1785, Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120.
      k. D2842, Standard Test Method for Water Absorption of Rigid Cellular Plastics
      l. D6226, Standard Test Method for Open Cell Content of Rigid Cellular Plastics.

1.3 SYSTEM DESCRIPTION

A. Provide underground piping for heating water system piping.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See Section 40 05 00.
   2. Product technical data including:
      a. Component details.
      b. Location of miscellaneous fittings including anchors and seals.
      c. Field closures.
   3. Provide copy of stress analysis for proposed system in accordance with ASME B31.1.
   4. Factory test report.

B. Contract Closeout Information:

1. Operation and Maintenance Data:
    a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements.
    1. Manufactures listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
    2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
    1. GF Urecon (Basis of Design).
    2. Or Equal.

C. Submit request for product substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Insulation:
    1. Field Applied Half-Shells:
        a. Rigid polyisocyanurate or polyurethane foam
            1) "K" factor: 0.14 to 0.154 BTU IN/SQFT HR DEGF, ASTM C518.
            2) Density: 2.0 pound/CUFT, ASTM D1622.
            3) Compressive strength: 18 to 27 lbs/sqin, ASTM D1621.
            4) Thickness: 2 IN for 4 IN SCH 80 Steel pipe.
            5) Closed cell content: >/= 90% minimum, ASTM D6226.
            6) Water absorption: </=4.0% by volume, ASTM C2842.

B. Jacket:
    1. Jacket may be the following:
        a. Extruded Black High Density Polyethylene Copolymer:
            1) Machine coated, factory applied.
            2) Minimum 2% carbon black, well dispersed.
            3) Density > 59.1 to 59.6 pounds/CUFT as per ASTM D1505.
            4) Melt index of compound < 0.00088 to 0.00033 lb as per ASTM D1238.
            5) Flexural modulus of compound < 160,000 to 110,000 psi as per ASTM D790.
            6) Tensile strength at yield of compound < 4,000 to 3,500 psi as per ASTM D638.
            7) Ultraviolet inhibitors added.
            8) Minimum thickness 125 mils.

        b. Cut and rolled galvanized steel :
            1) Thickness = 18 to 22 gauge.
            2) Service temperature range = cryogenic to 200 DEGF.
            3) Minimum ambient installation temperature = -29 DEGF.
            4) Minimum thickness 0.034 inches.
            5) Ultraviolet inhibitors added.
            6) For exposed galvanized steel jacket located in Volume 5 – RSPH provide the following coating:

a) Factory coated with primer and top coated with liquid epoxy in the field per Specification Section 09 96 00 – High Performance Industrial Coatings
b) Color:
   (1) Sherwin Williams SW4003 Pallet Tan
   (2) Or Equal

PART 3 - EXECUTION

3.1 INSTALLATION

A. Observe manufacturer's recommendation for handling, cutting, jointing, installing, and testing.

B. Trenching, Backfilling, and Compaction:
   1. See Section 31 23 33.
   2. Comply with manufacturer's recommendations where they are more stringent.

3.2 FIELD QUALITY CONTROL

A. Tests:
   1. See Section 40 05 00.

B. Manufacturer's Field Service:
   1. Factory-trained field service person present for minimum of three working days including two site visits.
   2. Factory-field service person visits to provide instruction for installation and testing and to verify installation is being performed in accordance with manufacturer's recommendations.

END OF SECTION

This page intentionally left blank.

SECTION 40 05 31
PIPE - PLASTIC

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Plastic pipe:
      a. PVC.
      b. CPVC.
   2. Plastic tubing:
      a. Polyethylene.
      b. Reinforced PVC.
      c. Teflon (PTFE).

B. Related Specification Sections include, but are not necessarily limited to:
   1. Section 40 05 00 - Pipe and Pipe Fittings – Basic Requirements.
   2. Section 40 05 07 – Pipe Support Systems.

1.2 QUALITY ASSURANCE

A. See Specification Section 40 05 00.

B. Reference Standards:
   1. ASTM International (ASTM):
      a. D1784, Standard Classification System and Basis for Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds.
      b. D1785, Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80 and 120.
      c. D2464, Standard Specification for Threaded Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80.
      d. D2466, Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40.
      e. D2467, Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80.
      f. D2564, Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems.
      g. D2665, Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings.
      h. F437, Standard Specification for Threaded Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80.
      i. F439, Standard Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80.
      j. F441, Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe, Schedules 40 and 80.
      k. F493, Standard Specification for Solvent Cements for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe and Fittings.
      l. F593, Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.
      m. F656, Standard Specification for Primers for Use in Solvent Cement Joints of Poly(Vinyl Chloride) (PVC) Plastic Pipe and Fittings.
      n. Installation:
         1) D2321, Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications.
   2. NSF International (NSF).

a. 14, Plastic Piping Components and Related Materials.
b. 61, Drinking Water System Components – Health Effects.

1.3 SUBMITTALS

A. See Specification Section 40 05 00.

PART 2 - PRODUCTS

2.1 GENERAL

A. Identification
1. Identify each length of pipe clearly at intervals of 5 feet or less.
a. Include manufacturer's name and trademark.
b. Nominal size of pipe, appurtenant information regarding polymer cell classification and critical identifications regarding performance specifications and NSF approvals when applicable.

2.2 PVC PRESSURE PIPING

A. Materials:
1. Manufacture pipe, fittings and appurtenances from virgin polyvinyl chloride (PVC) compound which meets the requirements of Type 1, Grade 1 (cell classification 12454-B) per ASTM D1784.
2. Manufacture pipe, fittings and valves from materials that have been tested and approved for conveying potable water by the NSF.
3. Solvent Welded Joints:
a. Use primer and solvent cement recommended by the PVC pipe manufacturer.
b. Primer: ASTM F656.
c. Solvent cement: ASTM D2564 (except sodium hypochlorite and caustic service).
d. Sodium hypochlorite and caustic service:
1) Certified by the manufacturer for sodium hypochlorite and caustic.
2) Solvent Cement: ASTM F493.
3) Manufacturer:
a) Weld-On 724 CPVC.
b) Or equal.

B. PVC Pressure Piping (Exposed)
1. Pipe:
a. Furnish pipe meeting requirements of ASTM D1785, Schedule 80.
b. Pipe 2 inches and less to be solvent welded.
c. Pipe larger than 2 inches may be either flanged or solvent welded unless shown otherwise on Drawings.
2. Fittings: Provide ASTM D2467 PVC socket type fittings, Schedule 80.
3. Flanges/Unions:
a. Furnish flanges and unions at locations shown on Drawings.
b. Provide either flanges or unions at valves, penetrations through structures and equipment connections.
c. For pipe larger than 2 inches, provide 150 pound socket type flat-faced PVC flanges.
d. For pipe 2 inches and less, provide socket type PVC union with elastomeric O-rings.
4. Flanged connections.
a. Use flat, full faced gaskets.
b. Furnish heavy hex head bolts, each with one heavy hex nut, ASTM F593 Type 316 stainless steel.
c. Use spacers supplied by pipe manufacturer when mating raised-faced flanges to other flanges.

5. Flexible Couplings:
    a. Provide flexible couplings at locations shown on the Drawings for pipe flexibility and expansion/contraction purposes.
    b. Unless otherwise specified, provide rubber expansion joints as manufactured by General Rubber Corporation or equal.

2.3 CPVC PRESSURE PIPING

A. Materials:
1. Manufacture pipe, fittings and appurtenances from chlorinated polyvinyl chloride (CPVC) compound per ASTM F441.
    a. CPVC 4120-05 or CPVC 4120-06.
    b. Cell Classification 23447 per ASTM D1784.
2. Pipe: ASTM F441.
    a. Exposed: Schedule 80, IPS.
3. Fittings:
    a. Match or exceed schedule of connecting pipe.
    b. Schedule 80:
        1) Socket: ASTM F439.
4. Solvent Welded Joints:
    a. Use primer and solvent cement recommended by the CPVC pipe manufacturer.
    b. Primer: ASTM F656.
    c. Solvent cement: ASTM D2564 (except for sodium hypochlorite and caustic service).
    d. Sodium hypochlorite and caustic service:
        1) Certified by the manufacturer for sodium hypochlorite and caustic.
        2) Manufacturers:
            a) Weld-On 724.
            b) Or equal.
5. Flanges/Unions:
    a. Furnish flanges and unions at locations shown on Drawings.
    b. Provide either flanges or unions at valves, penetrations through structures, and equipment connections.
    c. For pipe larger than 2 inches, provide 150 pound socket type flat-faced CPVC flanges.
    d. For pipe 2 inches and less, provide socket type CPVC union with Viton O-rings.
6. Use flat, full faced Viton gaskets at flanged connections.
    a. Furnish heavy hex head bolts, each with one heavy hex nut, ASTM F593 Type 316 stainless steel.
7. Use spacers supplied by pipe manufacturer when mating raised-faced flanges to other flanges.
8. Flexible Couplings:
    a. Provide flexible couplings at locations shown on the Drawings for pipe flexibility and expansion/contraction purposes.
    b. Unless otherwise specified, provide rubber expansion joints as manufactured by General Rubber Corporation or equal.

2.4 CLEAR PVC

A. Where indicated on the Drawings, install clear PVC pipe and fittings meeting the following requirements:
1. Schedule 80 pipe and fittings.
2. Manufacture pipe, fittings and appurtenances from polyvinyl chloride (PVC) compound which meets the requirements of Type 1, Grade 1, (12454-B) Polyvinyl Chloride as outlined in ASTM D1784.
3. Pipe per ASTM D1785.
4. Fittings socket type per ASTM D2467.
5. “Near Water Clear” designation for transparency.

2.5 PLASTIC TUBING

A. General: Provide tubing with fittings and appurtenances as shown on Drawings.

B. Polyethylene Tubing:
   1. HDPE tubing.
   2. Provide tubing with working pressure of 100 psi at 70 degrees F.
   3. Design tubing with a minimum safety factor of 4 to 1 ratio of burst pressure to working pressure at all operating temperatures.
   4. Fittings:
      a. Compression type.
      b. Same material and working pressure as tubing.
      c. Use stainless steel barbed type adapters with stainless steel hose clamps for transition to rigid pipe.

C. Clear PVC Braided Tubing:
   1. Furnish clear PVC braided tubing with braid within the walls.
      a. Reinforcing braid.
         1) Pressure applications: polyester, nylon or approved alternate material.
         2) Suction (vacuum) applications: polyester wire.
   2. Working temperature:
      a. Have tubing manufactured of components with working temperatures from 0 to 150 degrees F.
   3. Provide tubing with working pressure of 100 psi at 70 degrees F.
   4. Design tubing with a minimum safety factor of 4 to 1 ratio of burst pressure to working pressure at maximum temperature.
   5. Hosecraft USA PT1 (PT5 for suction) or equal.
   6. Fittings:
      a. Material: PVC, polypropylene or HDPE.
         1) Utilize one material for all PVC tubing.
      b. Use compression fittings.
      c. Provide fittings capable of withstanding temperatures from a -70 to 250 degrees F.
      d. Ensure fittings have the same pressure and temperature rating as the tubing.

D. PTFE Tubing:
   1. Provide tubing with working pressure of 100 psi at 70 degrees F.
   2. Design tubing with a minimum safety factor of 4 to 1 ratio of burst pressure to working pressure at all operating temperatures.
   3. Fittings:
      a. Same material and working pressure as tubing.
      b. Use barbed type adapters with stainless steel hose clamps.

2.6 ELASTOMERS

A. Seals, O-rings and gaskets:
   1. Coagulant (Alum, PACl, ACH): EPDM or Viton.
   2. Citric Acid: EPDM or Viton.
   3. Hydrochloric Acid: PTFE or Viton.
   4. Sodium bisulfite: EPDM or Viton.
   5. Sodium hydroxide (Caustic): EPDM or PTFE.
   6. Sodium hypochlorite: Viton or PTFE.
   7. Water and sewage: natural or synthetic rubber.

PART 3 - EXECUTION

3.1 INSPECTION

A. Inspect pipe materials for defects in material and workmanship.

B. Verify compatibility of pipe and fittings.

3.2 PVC/CPVC/CLEAR PVC PRESSURE PIPING (EXPOSED)

A. Installation:
   1. Field threading PVC will not be permitted.
      a. Provide threaded connections or attachments by the use of factory molded socket by male threaded adapters.
      b. Socket by female threaded adapters are not acceptable.
   2. Employ installation and pipe support practices and solvent welding all in compliance to the manufacturer's printed recommendation.
   3. Perform installation procedures, handling, thrust restraint, connections, and other appurtenant operations in full compliance to the manufacturer's printed recommendations and in full observance to plan details when more stringent.
   4. Pipe Support:
      a. See Drawings and Section 40 05 07 – Pipe Support Systems.
      b. Continuously support PVC/CPVC piping at liquid operating temperatures in excess of 100 DEGF.
      c. For vertical piping, band the pipe at intervals to rigidly support load of twice vertical load.
      d. Support riser clamps on spring hangers.
      e. Do not clamp PVC/CPVC tightly or restrict movement for expansion and contraction.

3.3 TUBING

A. Install per drawings and manufacturer’s recommendations.

END OF SECTION

This page intentionally left blank.

SECTION 40 05 51
COMMON REQUIREMENTS FOR PROCESS AND UTILITY VALVES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Common requirements for all valves and associated appurtenances on the project.
   2. Valve Schedules for:
      a. Roaring Springs Pumphouse – Volume 5.
      b. North Rim WTP – Volume 6.

B. Reference Drawings for Valve Schedules for all other volumes not listed above.

C. Related Sections include, but are not necessarily limited to:
   1. Section 01 61 03 – Equipment - Basic Requirements.
   2. Section 09 96 00 – High Performance Industrial Coatings.
   3. Section 40 05 00 – Pipe and Pipe Fittings – Basic Requirements.
   4. Section 40 05 52 – Miscellaneous Valves.
   5. Section 40 05 57 – Actuators for Process Valves and Gates.
   6. Section 40 05 61 – Gate Valves.
   7. Section 40 05 62 – Plug Valves.
   8. Section 40 05 63 – Ball Valves.
   9. Section 40 05 64 – Butterfly Valves.
   10. Section 40 05 65 – Globe Valves.

1.2 REFERENCES

A. Definitions:
   1. The following are definitions of abbreviations used in this Specification Section or one of the individual valve sections:
      a. CWP: Cold water working pressure.
      b. SWP: Steam working pressure.
      c. WOG: Water, oil, gas working pressure.
      d. WWP: Water working pressure.

B. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B1.20.1, Pipe Threads, General Purpose.
      b. B16.1, Gray Iron Pipe Flanges and Flanged Fittings: Classes 25, 125, and 250.
      c. B16.18, Cast Copper Alloy Solder Joint Pressure Fittings.
      d. B16.34, Valves-Flanged, Threaded and Welding End.
   2. ASTM International (ASTM):
      a. A126, Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings.
      b. D256, Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics.
      c. D638, Standard Test Method for Tensile Properties of Plastics.
      d. D648, Standard Test Method for Deflection Temperature of Plastics Under Flexural Load in the Edgewise Position.
      e. D695, Standard Test Method for Compressive Properties of Rigid Plastics.
      f. D2240, Standard Test Method for Rubber Property-Durometer Hardness.
   3. American Water Works Association (AWWA):
      a. C207, Standard for Steel Pipe Flanges for Waterworks Service - Sizes 4 inches through 144 inches.
      b. C500, Standard for Metal-Seated Gate Valves for Water Supply Service.

c. C504, Standard for Rubber-Seated Butterfly Valves.
d. C507, Standard for Ball Valves, 6 inches through 48 inches (150 mm through 1200 mm).
e. C509, Standard for Resilient-Seated Gate Valves for Water Supply Service.
f. C550, Standard for Protective Coatings for Valves and Hydrants.
g. C606, Standard for Grooved and Shouldered Joints.

4. American Water Works Association/American National Standards Institute (AWWA/ANSI):
   a. C111/A21.11, Standard for Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings.
5. National Electrical Manufacturers Association (NEMA):
   a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   b. MG 1, Motors and Generators.
6. National Fire Protection Association (NFPA):
   a. 70, National Electrical Code (NEC).

1.3 SUBMITTALS

A. Action Submittals:
   1. Shop Drawings:
      a. Product technical data including:
         1) Acknowledgement that products submitted meet requirements of standards referenced.
         2) Manufacturer's installation instructions.
         3) Valve pressure and temperature rating.
         4) Valve material of construction.
         5) Special linings.
         6) Valve dimensions and weight.
         7) Valve flow coefficient.
      b. See Section 40 05 57 - Actuators for Process Valves and Gates for actuator submittal requirements.
         1) Actuators shall be submitted with their respective value as one submittal.

B. Informational Submittals:
   1. Test reports if required in the individual valve specification.

C. Closeout Submittals:
   1. Operation and Maintenance Data:
      a. Submit in accordance with Section 01 78 23 - Operation and Maintenance Data.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of the individual valve Specification Sections, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, refer to individual valve Specification Sections for acceptable manufacturers.

2.2 MATERIALS

A. Refer to individual valve Specification Sections.

2.3 FABRICATION

A. Refer to individual valve Specification Sections for specifications of each type of valve used on Project.

B. All finished water valves shall be NSF 61 certified or for high pressure valves (Volume 5), all materials of components in contact with the finished water shall be common materials used in NSF 61 certified lower pressure valves.

C. End Connections:
   1. Provide the type of end connections for valves as shown on the Drawings.
   2. Comply with the following standards:
      a. Threaded: ASME B1.20.1.
      b. Flanged: ASME B16.1, Class 125 unless otherwise noted or AWWA C207.
      c. Bell and spigot or mechanical (gland) type: AWWA/ANSI C111/A21.11.
      d. Soldered: ASME B16.18.
      e. Grooved: Rigid joints per Table 5 of AWWA C606.

D. Nuts, Bolts, and Washers:
   1. Wetted or internal to be bronze or stainless steel.
      a. Exposed to be zinc or cadmium plated.
   2. Buried:
      a. Tee-Head bolts for mechanical joints: Per AWWA/ANSI C111/A21.11.
      b. Other bolts and nuts: ASTM A307 ASTM A193/A194 Grade B8M, Class 1.
   3. Exposed: Mechanical galvanized ASTM B695, Class 40 ASTM A193/A194 Grade B8M, Class 1.
   4. Heads and dimensions per ASME B1.1.
   5. Threaded per ASME B1.1.
   6. Project ends 1/4 to 1/2 inches beyond nuts.

E. On Insulated Piping: Provide valves with extended stems to permit proper insulation application without interference from handle.

F. Epoxy Interior Coating: Provide epoxy interior coating for all ferrous surfaces in accordance with AWWA C550.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Painting Requirements: Comply with Section 09 96 00 for High Performance Industrial Coatings.

C. Setting Buried Valves:
   1. Locate valves installed in pipe trenches where buried pipe indicated on Drawings.
   2. Set valves and valve boxes plumb.
   3. Place valve boxes directly over valves with top of box set at finished grade.
   4. Install valve in closed position.
   5. Place valve on firm footing in trench to prevent settling and excessive strain on connection to pipe.
   6. After installation, backfill up to top of box for a minimum distance of 4 feet on each side of box.

D. Support exposed valves and piping adjacent to valves independently to eliminate pipe loads being transferred to valve and valve loads being transferred to the piping.

E. For grooved coupling valves (Volume 5), install rigid type couplings or provide separate support to prevent rotation of valve from installed position.

F. Wax Tape Coating System:
   1. After installation, coat all buried nuts and bolts with a wax tape coating system in accordance with:
      a. AWWA C217 NACE International RP0375.
      b. Manufacturer's printed instructions.
   2. Cut strips of wax tape and apply them around all bolts and nuts so that there are no voids or spaces under the tape.
   3. Apply a sufficient amount of tape to completely encapsulate all bare metal.

G. For threaded valves, provide union on one side within 2 feet of valve to allow valve removal.

3.2 FIELD QUALITY CONTROL

A. Wax Tape Coating Testing:
   1. Inspect each wax tape-wrapped component.
      a. Verify primer was applied to substrate surfaces.
      b. Ensure the wax tape has fully encapsulated all portions of the substrate.
      c. Ensure that the wax tape is in full contact with the substrate.
      d. Verify that the wax tape has been applied to the specified thickness.
   2. Do not backfill until the inspection is complete and the wax tape application is approved by the Contracting Officer's Representative.

3.3 ADJUSTMENT

A. See section 40 05 57 for adjustment requirements related to actuators.

3.4 VALVE SCHEDULE

A. The following Valve Schedules apply to:
   1. Roaring Springs Pumphouse – Volume 5.
   2. North Rim WTP – Volume 6.

B. The Valve Schedules include tagging and general criteria for:
   1. All valves 2" and larger.
   2. All valves that are powered, regardless of size.

C. Valves shown on the Drawings, but not included in the Valve Schedules, are still the responsibility of the Contractor.

D. Reference Drawings for Valve Schedules for all other volumes not listed above.

Roaring Springs Pumphouse – Volume 5:

Tag No.	Description	Type[1]	Actuator[2] (Voltage/ Phase)[3]	Size (IN)	Operation[4]	Maximum Operating Pressure (psi)
V-52331	Globe Valve on PD1 (Upstream of 2" Flow Meter)	Angle	MH	3	-	20
FV-52331	Raw Water to NR Butterfly Valve	BFV	E (480/3)	4	O/C	150
V-52131	RW AutoFilter 1 Inlet BUTTERFLY valve	BFV	M	4	-	150
V-52132	RW AutoFilter 2 Inlet BUTTERFLY valve	BFV	M	4	-	150
V-52133	RW AutoFilter 1 Discharge BUTTERFLY valve	BFV	M	4	-	150
V-52134	RW AutoFilter 2 Discharge BUTTERFLY valve	BFV	M	4	-	150
V-52231	Raw Water Pump 1 Suction BUTTERFLY valve	BFV-GRV	M	5	-	150
V-52232	Raw Water Pump 2 Suction BUTTERFLY valve	BFV-GRV	M	5	-	150
ARV-52331	Raw Water Air Release Valve (Autofilter Inlet)	Air Relief	-	1	-	150
ARV-52332	Raw Water Air Release Valve (Autofilter Outlet)	Air Relief	-	1	-	150
CKV-52331	Raw Water Pump 1 CHECK valve	Check	-	4	-	1,700
CKV-52332	Raw Water Pump 2 CHECK valve	Check	-	4	-	1,700
V-52233	Raw Water Pump 1 Discharge BALL valve	Ball	MH	4	-	1,700
V-52234	Raw Water Pump 2 Discharge BALL valve	Ball	MH	4	-	1,700
CKV-52333	Raw Water Pump common CHECK valve	Check	-	4	-	1,700

Tag No.	Description	Type[1]	Actuator[2] (Voltage/ Phase)[3]	Size (IN)	Operation[4]	Maximum Operating Pressure (psi)
V-52031	Raw Water Pump common Discharge BALL valve	Ball	MH	4	-	1,700
V-52032	RW outlet isolation BALL valve (outside in vault/buried)	Ball	MH	4	-	1,700
V-52033	Raw Water Pressure Dissipation BALL valve	Ball	MH	3	-	1,500
EDS-52312	RW Pressure Dissipation Valve	Energy Dissipating Sleeve	MH	3	-	1,500
V-53131	FW Outdoor Inlet Isolation Valve (outside in vault/buried)	Ball	MH	4	-	1,500
EDS-53117	FW Mokveld Inlet control modulating valve	Axial Control	E (120)	2	O/C	1,500
V-53231	FW Recirc Pump suction BALL valve	Ball	M	2	-	5
FV-53331	Chlorine Transfer actuated ball valve	Ball	E (480/3)	1	O/C	80
V-53331	Water softener feed DIAPHRAGM valve	Diaphragm	MH	1	-	80
V-53232	Recirculation Rotameter Control DIAPHRAGM valve	Diaphragm	MH	1.5	-	80
LCV-53332	Day Tank float valve	Float	M	1	-	80
V-53333	Day Tank discharge isolation BALL valve	Ball	M	1/2	-	2.5
V-53133	FW Tank Outlet BUTTERFLY valve (outside, insulated)	BFV	M	4	-	5
V-53134	FW Tank Drain Isolation BUTTERFLY valve	BFV	MH	4	-	5

1. BFV = Butterfly Valve, GRV = Grooved connection
2. M = Manual, MH = Manual w/geared handwheel, E = Electric
3. 480 = 480V, 3P, 60 HZ; 120 = 120V, 1P, 60 HZ; electric actuators only
4. O/C = Open/Close, MOD = Modulating, O = Open

North Rim WTP – Volume 6:

Tag No.	Description	Type[1]	Actuator[2] (Voltage/ Phase)[3]	Size (IN)	Operation[4]	Maximum Operating Pressure (psi)
V-62031	Raw Water Tank 1 Influent	Gate	M	4	-	20
V-62032	Raw Water Tank 2 Influent	Gate	M	4	-	20
V-62033	Raw Water Tank 1 Effluent	Gate	M	8	-	20
V-62034	Raw Water Tank 2 Effluent	Gate	M	8	-	20
V-62035	Raw Water Tank 1 Drain	Gate	M	4	-	20
V-62036	Raw Water Tank 2 Drain	Gate	M	4	-	20
V-62037	Raw Water Recycle Line Flange	BFV	M	4	-	20
FCV-62032	Raw Water Pump Discharge Flow Control Valve	BFV - High Performance	E (480)	4	MOD	20
FCV-62033	Raw Water Flow Control Valve	BFV - High Performance	E (480)	3	MOD	20
FV-62031	Raw Water Pump Suction Valve	Ball	E (480)	4	O/C	20
CKV-62031	Raw Water Pump Discharge Check Valve	Check	M	4	-	20
CKV-62032	Raw Water Pump Bypass Check Valve	Check	M	4	-	20
FV-63031	Reaction Tank Motorized Waste Valve 1	Plug	E (480)	2	O/C	10
FV-63032	Reaction Tank Motorized Waste Valve 2	Plug	E (480)	2	O/C	10
V-63031	Reaction Tank 1 Influent	BFV	M	6	-	20
V-63032	Reaction Tank 2 Influent	BFV	M	6	-	20
V-63033	Reaction Tank 1 Tank Mounted Instrument	Ball	M	6	-	10
V-63034	Reaction Tank 2 Tank Mounted Instrument	Ball	M	6	-	10
SV-64031	UF CIP Feed Valve	Solenoid	Solenoid	1	O/C	10

Tag No.	Description	Type[1]	Actuator[2] (Voltage/ Phase)[3]	Size (IN)	Operation[4]	Maximum Operating Pressure (psi)
V-64031	UF Influent Skid 1 Valve 1	BFV	M	6	-	10
V-64032	UF Influent Skid 1 Valve 2	BFV	M	6	-	10
V-64033	UF Influent Skid 2 Valve 1	BFV	M	6	-	10
V-64034	UF Influent Skid 2 Valve 2	BFV	M	6	-	10
V-64037	UF Drain 1	Ball	M	2	-	10
V-64039	UF Drain 2	Ball	M	2	-	10
V-66031	Chlorine Contact Tank Influent	BFV	M	4	-	75
V-66033	Chlorine Contact Tank Drain to Lift Station	BFV	M	4	-	15
FV-66031	Chlorine Contact Basin to Finished Water Tank Pumps Check	BFV	E (480)	4	-	15
CKV-68132	Finished Water Tank Pump 1 Discharge Check	Check	M	4	-	70
CKV-68133	Finished Water Tank Pump 2 Discharge Check	Check	M	4	-	70
V-68131	Finished Water Tank Pumps to Finished Water Tanks	BFV	M	4	-	70
V-68132	Finished Water Tank Pump 1 Suction	Ball	M	4	-	15
V-68133	Finished Water Tank Pump 2 Suction	Ball	M	4	-	15
V-68134	Finished Water Tank Pump 1 Discharge	BFV	M	4	-	70
V-68135	Finished Water Tank Pump 2 Discharge	BFV	M	4	-	70
V-68136	Finished Water Tank Pumps Flange for Temp Recycle	BFV	M	4	-	70
V-68031	NRPH Effluent Shut Off	Gate	M	4	-	70
V-68032	Finished Water From WTP Pumps	Gate	M	4	-	70
FV-68033	To Finished Water Tank Recirculation	BFV	E (480)	4	-	70

Tag No.	Description	Type[1]	Actuator[2] (Voltage/ Phase)[3]	Size (IN)	Operation[4]	Maximum Operating Pressure (psi)
V-68431	Finished Water Tank A Influent	Gate	M	4	-	70
V-68432	Finished Water Tank B Influent	Gate	M	4	-	70
V-68433	Finished Water Tank A Effluent	Gate	M	8	-	25
V-68434	Finished Water Tank B Effluent	Gate	M	8	-	25
V-68435	Finished Water Tank A Drain	Gate	M	6	-	25
V-68436	Finished Water Tank B Drain	Gate	M	8	-	25
V-68437	Finished Water Tank B Recirc	Gate	M	8	-	25
V-68438	Finished Water Tank B Combined Distribution	Gate	M	12	-	25
V-68532	Finished Water Tank Yard Recirc	Gate	M	8	-	25
V-68533	FW Tank A Influent Line Stop	Line Stop	M	4	-	75
V-68534	FW Tank A Effluent Line Stop	Line Stop	M	8	-	25
V-68535	FW Tank B Influent Line Stop	Line Stop	M	4	-	75
V-68536	FW Tank B Effluent Line Stop	Line Stop	M	8	-	25
V-68537	FW Tank B Recirc Line Stop	Line Stop	M	8	-	25
FV-68031	Backfeed to RSPH Open/Close Flow Valve	BFV	E (480)	4	O/C	25
FV-68032	Backfeed to Supai Open/Close Flow Valve	Ball	E (480)	2	O/C	25
CKV-68231	High Demand Pump 1 Discharge Check	Check	M	8	-	90
CKV-68232	High Demand Pump 2 Discharge Check	Check	M	8	-	90
CKV-68233	High Demand Pump 3 Discharge Check	Check	M	8	-	90
V-68034	To Lower Zone Distribution System	BFV	M	8	-	25

Tag No.	Description	Type[1]	Actuator[2] (Voltage/ Phase)[3]	Size (IN)	Operation[4]	Maximum Operating Pressure (psi)
V-68035	To RSPH Backfeed	Globe	M	4	-	25
V-68036	To Supai Backfeed	Globe	M	2	-	25
V-68039	To Lower Distribution System	BFV	M	8	-	25
V-68231	High Demand Pump 1 Suction	BFV	M	8	-	25
V-68232	High Demand Pump 1 Discharge	BFV	M	8	-	90
V-68233	High Demand Pump 2 Suction	BFV	M	8	-	25
V-68234	High Demand Pump 2 Discharge	BFV	M	8	-	90
V-68235	High Demand Pump 3 Suction	BFV	M	8	-	25
V-68236	High Demand Pump 3 Discharge	BFV	M	8	-	90
V-68237	High Demand Suction Header	BFV	M	12	-	25
CKV-68331	Upper Zone Booster Pump 1 Discharge Check	Silent Globe Check	M	4	-	90
CKV-68332	Upper Zone Booster Pump 2 Discharge Check	Silent Globe Check	M	4	-	90
CKV-68333	Upper Zone Booster Pump 3 Discharge Check	Silent Globe Check	M	4	-	90
V-68037	Hydropneumatic Tank 1 Valve	BFV	M	2	-	90
V-68038	Hydropneumatic Tank 2 Valve	BFV	M	2	-	90
V-68331	Upper Zone Booster Pump 1 Suction	Ball	M	4	-	25
V-68332	Upper Zone Booster Pump 1 Discharge	BFV	M	4	-	90
V-68333	Upper Zone Booster Pump 2 Suction	Ball	M	4	-	25
V-68334	Upper Zone Booster Pump 2 Discharge	BFV	M	4	-	90
V-68335	Upper Zone Booster Pump 3 Suction	Ball	M	4	-	25

Tag No.	Description	Type[1]	Actuator[2] (Voltage/ Phase)[3]	Size (IN)	Operation[4]	Maximum Operating Pressure (psi)
V-68336	Upper Zone Booster Pump 3 Discharge	BFV	M	4	-	90
V-68337	Upper Zone Booster Pumps Combined Line	BFV	M	8	-	90
V-68338	To Upper Distribution System	BFV	M	8	-	90

1. BFV = Butterfly Valve
2. M = Manual, E = Electric
3. 480 = 480V, 3P, 60 HZ; 120 = 120V, 1P, 60 HZ; electric actuators only
4. O/C = Open/Close, MOD = Modulating

END OF SECTION

This page intentionally left blank.

SECTION 40 05 52
MISCELLANEOUS VALVES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Air release and vacuum relief valves.
   2. Energy dissipating valves.
   3. Automatic control valves:
      a. Pressure-sustaining valves.
      b. Pressure-reducing valves.
      c. Flow control valves.
   4. Pressure-reducing valves (2 inches and smaller).
   5. Solenoid valves.
   6. Pinch valves.
   7. Telescoping valves.
   8. Spray nozzles.

B. Related Specification Sections include, but are not necessarily limited to:
   1. Section 01 61 03 - Equipment - Basic Requirements.
   2. Section 40 05 51 - Common Requirements for Process and Utility Valves.
   3. Section 40 05 57 - Actuators for Process Valves and Gates.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B16.1, Gray Iron Pipe Flanges and Flanged Fittings: Classes 25, 125, and 250.
      b. B16.5, Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24, Metric/Inch Standard.
   2. ASTM International (ASTM)
      a. A487, Standard Specification for Steel Castings Suitable for Pressure Service.
   3. American Water Works Association (AWWA):
      a. C512, Standard for Air-Release, Air-Vacuum, and Combination Air Valves for Waterworks Service.
      b. C550, Standard for Protective Interior Coatings for Valves and Hydrants.
   4. Canadian Standards Association (CSA).
   5. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 05 51.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
   2. Use of Manufacturers not listed for "Basis of Design" products may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.
      a. Submit request for product substitution in accordance with Specification Section 01 33 23 for use of manufacturers not listed for "Basis of Design" products.

2.2 AIR RELEASE AND VACUUM RELIEF VALVES

A. Air Valves - General:
   1. Comply with AWWA C512.

B. Type 1 – Cast Iron Air Valves With Stainless Steel Floats:
   1. Water:
      a. Air Release Valves:
         1) Manufacturers:
            a) DeZurik APCO 200A (1" – 2") or 50A (0.5" – 1").
            b) Or equal.
         2) Materials:
            a) Body and cover: Cast iron or semi-steel.
            b) Float: Stainless steel.
            c) Linkage and trim: Bronze or stainless steel.
         3) Design requirements:
            a) Size: As shown on Drawings
            b) Working pressure:
               (1) Raw water service at RSPH: 150 psig
               (2) All other service: 10 psig.
            c) Release at 10 psig differential at operating pressure in the associated piping system.
   2. Combination Air Release and Vacuum Valves:
      a. Manufacturers:
         1) VAG GA Industries, Figure 945
         2) Or equal.
      b. Materials:
         1) Body and cover: Cast iron.
         2) Float, linkage and hardware: Stainless steel.
         3) Seat: Buna-N.
      c. Design requirements:
         1) Size: 2 inches NPT (connection to process piping or other item to which valve is attached)
         2) Working pressure: 2 - 150 psi.
            a) Provide low pressure seat for valves located inside the North Rim WTP (Volume 6).
         3) Release 444 scfm at 2 psig differential at 0 psig in the associated piping.
         4) Air vacuum capacity: 716 scfm at -5 psi differential from atmospheric.
         5) Provide isolation valve, type as shown on the Drawings.

2.3 ENERGY DISSIPATION VALVES

A. Manufacturers:
   1. Mokveld:
      a. For raw water from Roaring Springs Cave: 3” RZD-RM100X Model Type (EDS-52312), (Basis of Design).
      b. For finished water from North Rim: 2” RZD-RM100X Model Type, (EDS-53117), (Basis of Design).
   2. Or Equal.

B. General:
   1. For raw water service from Roaring Springs Cave:
      a. Provide energy dissipation device at downstream side of primary blowoff to reduce flow and head to drain the service line seasonally.
   2. For finished water service from North Rim:
      a. Provide pressure reducing device at upstream side of finished water tank.

C. Type:
   1. For raw water service from Roaring Springs Cave:
      a. Minimum length: 15 IN.
      b. Maximum length: 15 IN.
      c. All flanges and fittings pressure rating: ASME Class 900.
      d. Provide assembly with lifting lugs for ease of removal from the piping system.
      e. Manual geared actuator.
   2. For finished water service from North Rim:
      a. Minimum length: 14.5 IN.
      b. Maximum length: 14.5 IN.
      c. All flanges and fittings pressure rating: ASME Class 900.
      d. Provide assembly with lifting lugs for ease of removal from the piping system.
      e. Electric actuator.

D. Materials:
   1. For raw water service from Roaring Springs Cave:
      a. Body: ASTM A487 CA6NM CI.B stainless steel.
      b. Flanges: Class 900 ASME B16.5 RF integral flanges.
      c. Trim: 13%, 4% Ni stainless steel.
      d. Piston and disc: UNS S42400.
      e. Cage: UNS S42400.
      f. Stem: UNS G41300.
      g. Spring: Inconel.
   2. For finished water service from North Rim:
      a. Body: ASTM A487 CA6NM CI.B stainless steel.
      b. Flanges: Class 900 ASME B16.5 RF integral flanges.
      c. Trim: 13% Cr, 4% Ni stainless steel.
      d. Piston and disc: UNS S42400.
      e. Cage: UNS S42400.
      f. Stem: UNS G41300.
      g. Spring: Inconel.

E. Design Requirements:
   1. For raw water service from Roaring Springs Cave:
      a. Maximum Pressure: 2,250 psi.
      b. Design Inlet Pressure: 1,500 psi.
      c. Design Outlet Pressure: 5 psi
      d. Design Maximum Flow Rate: 330 gallons per minute.
      e. Actuator: Manual handwheel with traveling scale.
   2. For finished water service from North Rim:
      a. Actuator: AUMA/SIPOS SEVEN, see Specification Section 40 05 57

b. Size: 2 IN
c. Max Cv: 2.5
    1) Cv at 70% open = 1.05
d. Pressure requirements:
    1) Inlet pressure: 1530 psi.
    2) Outlet pressure: 5 to 60 psi
e. Flow Rate:
    1) Maximum flow rate: 50 GPM
    2) Minimum flow rate: 8 GPM

F. Testing: Factory test valve to confirm performance requirements are achieved prior to shipment to site.

2.4 AUTOMATIC CONTROL VALVES FOR WATER SERVICE

A. Basic Valve:
1. Type:
    a. Diaphragm-actuated hydraulically operated.
        1) Manufacturers:
            a) Cla-val.
            b) Or Equal
        2) Materials:
            a) Body: Ductile iron.
            b) Seat insert: Stainless steel.
            c) Disc: Buna-N.
            d) Diaphragm: Nylon fabric bonded with synthetic rubber.
        3) Design requirements: Do not use diaphragm as seating surface.
2. Design requirements:
    a. Volume 6 – Combined Backfeed Flow Control Valve (FV-68031):
        1) Size: 4 IN
        2) Operating Pressure:
            a) Maximum: 50 psi
            b) Minimum: 20 psi
        3) Flow Range:
            a) Normal maximum: 60 GPM
            b) Normal minimum: 10 GPM
    b. Volume 6 – Supai Tunnel Backfeed Flow Control Valve (FV-68032):
        1) Size: 2 IN
        2) Operating Pressure:
            a) Maximum: 50 psi
            b) Minimum: 20 psi
        3) Flow Range:
            a) Normal maximum: 40 GPM
            b) Normal minimum: 1 GPM

B. Control:
1. Type:
    a. Flow control:
        1) Manufacturers:
            a) Cla-val, Model 40-01
            b) Or Equal
        2) Design requirements: Modulate basic valve to limit maximum flow rate through valve
            a) Settings:
                (1) Innercanyon back feed pipeline:
                    (a) Maximum flow rate: 60 GPM
                (2) Supai pipeline:

(a) Maximum flow rate: 40 GPM

b. Pressure relief, pressure regulating, or pressure-sustaining control:
   1) Manufacturers:
      a) Cla-val, Model 50-01.
      b) Or equal.
   2) Design requirements: Modulate basic valve to maintain constant upstream pressure by bypassing or relieving excess pressure.
c. Pressure-reducing control:
   1) Manufacturers:
      a) Cla-val, Model 90-01.
      b) Or equal.
   2) Design requirements: Modulate basic valve to maintain a uniform downstream pressure as set on control pilots.

2. Design requirements:
   a. Assembles all control features and hardware on basic valve at factory.
   b. Use corrosion-resistant metal for all exposed portions of the control.
   c. Include with valve control:
      1) Stop valves.
      2) Strainer.
      3) Valves for opening and closing speed control.
      4) Pilot valves.
      5) Solenoid valves.
      6) Pressure switches as necessary to provide control function.
   d. Electric components:
      1) 110 V, 1 PH.
      2) In weatherproof enclosure unless shown otherwise on the Drawings.

2.5 PRESSURE-REDUCING VALVES (2 INCHES AND SMALLER)

A. High Pressure - Pressure-Reducing Valve at Supai Tunnel Rest Area located in the PRV vault adjacent to composting restrooms:
   1. Manufacturers:
      a. TESCOM Corporation (Basis of Design).
      b. Or equal.
   2. Materials:
      a. Body: 316 Stainless steel
      b. Strainer body: Stainless steel.
      c. Strainer screen: Stainless steel.
   3. Design requirements:
      a. Self-contained diaphragm operated.
         1) Spring loaded.
         2) Field adjustable.
      b. Strainer: Y-type on supply.
      c. Size as shown in Drawings
      d. In and Out Port Type: NPT
      e. Orifice Size: 0.375-IN
      f. Inlet and Outlet Port Size: 1-IN
      g. Flow rate: 0-20 gpm
      h. Upstream Pressure: See Drawings
      i. Downstream Pressure: See Drawings

B. Standard Pressure - Pressure-Reducing Valve at Supai Tunnel Rest Area located adjacent to water filling stations:
   1. Manufacturers:
      a. Watts, Series 25AUB-Z3.
      B. Or equal.
   2. Materials:

a. Body: Bronze.
b. Strainer body: Bronze.
c. Strainer screen: Stainless steel.

3. Design requirements:
   a. Self-contained diaphragm operated.
      1) Spring loaded.
      2) Field adjustable.
   b. Strainer: Y-type on supply.
   c. Size as shown in Drawings.
   d. In and Out Port Type: NPT
   e. Inlet and Outlet Port Size: 1-IN
   f. Flow rate: 0-10 gpm
   g. Upstream Pressure: 100 psi
   h. Downstream Pressure: 40 psi

2.6 SOLENOID VALVES (2 INCHES AND SMALLER)

A. General Service (Air - Water):
   1. Manufacturer:
      a. ASCO.
      b. Or equal.
   2. Materials:
      a. Body: Brass.
      b. Seat: Buna-N.
      c. Insulation: Class F.
   3. Design requirements:
      a. 110 VAC.
      b. Two-way, normally closed.
      c. Enclosure: Compatible with area classifications indicated on Drawings.
      d. Working pressure, air and water: 125 PSIG.
   4. Accessories: Provide strainer on supply.

2.7 ACCESSORIES

A. Furnish any accessories required to provide a completely operable valve.

2.8 FABRICATION

A. Completely shop assemble unit including any interconnecting piping, speed control valves, control isolation valves and electrical components.

B. Provide internal epoxy coating suitable for potable water for all iron body valves in accordance with AWWA C550.

2.9 SOURCE QUALITY CONTROL

A. Shop hydrostatically test to piping system test pressure (minimum).

2.10 MAINTENANCE MATERIALS

A. Provide one set of any special tools or wrenches required for operation or maintenance for each type valve.

B. Provide equivalent one year supply of lubricants and gear oil.

2.11 SPARE VALVES

A. Furnish spare valves in accordance with the following:

Description	Number of Spares	Size
Combination Air Release and Vacuum Valve	4	2”

Description	Number of Spares	Size
High Pressure – Pressure Reducing Valve at Supai Tunnel	1	1”
Energy Dissipation Valve – Finished water from North Rim	1	2”

PART 3 - EXECUTION

3.1 INSTALLATION

A. General: See Specification Section 01 61 03 and Specification Section 40 05 51.

B. Air Release, Vacuum Relief, and Pressure Relief Valves:
   1. Pipe exhaust to a suitable disposal point.
   2. Where exhausted to a trapped floor drain, terminate exhaust line 6 inches minimum above floor.

C. Float-Operated Valves: Install baffle around float to minimize turbulence adjacent to float.

D. Energy Dissapation Valves: Valve representative must be present in the field to adjust valve to meet hydraulic conditions.

3.2 FIELD QUALITY CONTROL

A. Clean, inspect, and operate valve to ensure all parts are operable and valve seats properly.

B. Check and adjust valves and accessories in accordance with manufacturer's instructions and place into operation.

END OF SECTION

This page intentionally left blank.

SECTION 40 05 57
ACTUATORS FOR PROCESS VALVES AND GATES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Actuator requirements for valves including:
      a. Manual actuators.
      b. Electric motor actuators.
   2. Special requirements for Energy Dissipating Valve (EDS-54117) actuator at Roaring Springs Pumphouse (Volume 5).

B. Related Sections include, but are not necessarily limited to:
   1. Section 01 61 03 – Equipment - Basic Requirements.
   2. Section 01 81 33 – Cyber Security Requirements.
   3. Section 40 05 51 – Common Requirements for Process and Utility Valves.
   4. Section 40 05 52 – Miscellaneous Valves.
   5. Section 40 05 61 – Gate Valves.
   6. Section 40 05 62 – Plug Valves.
   7. Section 40 05 63 – Ball Valves.
   8. Section 40 05 64 – Butterfly Valves.
   9. Section 40 05 65 – Globe Valves.

1.2 REFERENCES

A. Definitions:
   1. The following are definitions of abbreviations used in this Specification Section:
      a. CPU: Central Processing Unit.
      b. NRTL: Nationally Recognized Testing Laboratory.
      c. OEM: Original Equipment Manufacturer.
      d. OIT: Operator Interface Terminal.

B. Referenced Standards:
   1. American National Standards Institute/Underwriters Laboratory (ANSI/UL):
      a. 429, UL Standard for Safety Electrically Operated Valves.
      b. 50, UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations.
      c. 50E, UL Standard for Safety Enclosures for Electrical Equipment, Environmental Considerations.
   2. ASTM International (ASTM):
      a. A266, Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings.
      b. A269, Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service.
      c. A276, Standard Specification for Stainless Steel Bars and Shapes.
      d. A564, Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes.
      e. A693, Standard Specification for Precipitation-Hardening Stainless and Heat-Resisting Steel Plate, Sheet, and Strip.
   3. American Water Works Association (AWWA):
      a. C500, Standard for Metal-Seated Gate Valves for Water Supply Valves.
      b. C541, Standard for Hydraulic and Pneumatic Cylinder and Vane-Type Actuators for Valves and Slide Gates.
      c. C542, Standard for Electric Motor Actuators for Valves and Slide Gates.

d. C550, Standard for Protective Coatings for Valves and Hydrants.
4. National Electrical Manufacturers Association (NEMA):
   a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   b. MG 1, Motors and Generators.
5. National Fire Protection Association (NFPA):
   a. 70, National Electrical Code (NEC).

1.3 QUALITY ASSURANCE

A. Qualifications:
   1. Manufacturer: shall have at least 100 installations with each type of proposed actuator.

1.4 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Provide actuators that are fully compatible with valve for which it is intended.
   2. Contractor to coordinate valve manufacturer and actuator manufacturer for proper sizing of actuator.
      a. See below for safety factors.
   3. Mounting hardware by actuator manufacturer.
   4. Install actuators at valve manufacturers' facility to allow for testing prior to shipping to site.

1.5 SUBMITTALS

A. Action Submittals:
   1. Submit in conjunction with associated valve submittal. Separate submittals will not be acceptable.
   2. Shop Drawings:
      a. Layout drawings with dimensions.
         1) Include those dimensions necessary for interfacing with valve.
      b. Include wiring and control diagrams for electric actuators.
   3. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Torque or thrust output.
      c. Travel speed.
      d. Input power (voltage, phase, frequency).
      e. Input current draw.
      f. Control methodology.
      g. Control accessories.
      h. Short Circuit Current Rating (SCCR) nameplate marking in accordance with NFPA 70.
         1) Include any required calculations in accordance with Section 01 61 03.
   4. Factory finish colors available at no additional cost to the Government for selection by the Contracting Officer's Representative.
   5. Test reports.

B. Informational Submittals:
   1. Manufacturer's installation instructions.
   2. Acknowledgement that the Manufacturer meets the number of installations criteria.
      a. Submit for each different manufacturer.
      b. Submit detailed installation list if requested by Contracting Officer's Representative.
   3. Recommended storage procedures.
   4. Certificate of Proper Installation that includes verification from actuator manufacturer that actuators have been installed properly, that all limit switches and position potentiometers have been properly adjusted, and that the actuator responds correctly to the position command.
   5. Test Reports.
   6. Additional information For EDS-54117:
      a. Power Consumption requirements for UPS sizing
      b. Modbus TCP/IP tables for programming requirements.

C. Closeout Information:
   1. Operation and Maintenance Data:
      a. See Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
      b. Lubrication Guide.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with specifications.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Manual Valve Actuators:
      a. By valve manufacturer.
   2. Electric Motor Actuators:
      a. Energy Dissipation Valve (EDS-53117)
         1) AUMA Actuators, Inc., SIPOS SEVEN Model Type (Basis of Design)
         2) Or equal.
      b. All Other Valves:
         1) Rotork IQ3 Series (Basis of Design).
         2) Or equal.

C. Submit request for product substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Refer to individual Articles for materials.

2.3 MANUAL VALVE ACTUATORS

A. General:
   1. Refer to Section 01 81 33 - Cyber Security Requirements for cyber security related product requirements.
   2. Counter-clockwise opening as viewed from the top.
   3. Direction of opening and the word OPEN to be cast in handwheel or valve bonnet.
      a. Paint in a contrasting color such as white.
   4. Size actuator to produce required torque with a maximum pull of 80 pounds at the maximum pressure rating of the valve provided, and withstand without damage a pull of 200 pounds on handwheel or chainwheel or 300 foot-pounds torque on the operating nut.
   5. Unless otherwise specified, actuators for valves to be buried, submerged or installed in vaults or manholes shall be sealed to withstand at least 20 FT of submergence.
   6. Valve Lockout Devices:
      a. Device manufactured from same material as valve actuator, preventing access to valve actuator, to accept lock shackle.
   7. Extension stem:
      a. Install where shown or specified.
      b. Solid steel with actuator key and nut.
      c. Diameter not less than stem of valve actuator shaft.
      d. Pin all stem connections.

e. Center in valve box or grating opening band with guide bushing.

B. Buried Valve Actuators:
1. Provide 2 inch standard actuator nuts in accordance with AWWA C500, Section 3.16.
2. Provide at least 2 tee handle keys for actuator nuts, with 5 feet extension between key and handle.
3. Extension stem:
a. Provide for buried valves greater than 4 feet below finish grade.
b. Extend to within 6 inches of finish grade.
4. Cast Iron Valve Box:
a. Provide screw or slide type adjustable cast iron valve box,
b. 5 inch minimum diameter, 3/16 inch minimum thickness, and identifying cast iron cover rated for traffic load.
c. Valve box cover embossed with fluid service such as "WATER", "SEWER", etc.
5. Box base to enclose buried valve gear box or bonnet.
6. Provide concrete pad encasement of valve box as shown on Drawings for all buried valves unless shown otherwise.

C. Exposed Valve Manual Actuators:
1. Provide for all exposed valves not having electric actuators.
2. Provide handwheels for gate, globe, and energy dissipation valves.
a. Size handwheels for valves in accordance with AWWA C500.
3. Provide lever actuators for plug valves, butterfly valves, and ball valves 3 inches diameter and smaller.
a. Provide lever actuators for butterfly valves with a minimum of five intermediate lock positions between full open and full close.
b. Provide at least two levers for each type and size of valve furnished.
4. Provide gear actuators for plug valves, butterfly valves, and ball valves 4 inches diameter and larger.
5. Gear actuators to be totally enclosed, permanently lubricated and with sealed bearings.
6. Provide chain actuators for valves 6 feet or higher from finish floor to valve centerline.
a. Cadmium-plated chain looped to within 3 feet of finish floor.
b. Equip chain wheels with chain guides to permit rapid operation with reasonable side pull without "gagging" the wheel.
c. For smaller valves with lever or handle actuators, provide offset tee handles with attached chain for operation from the operating floor.
d. Provide chain storage device for each chain.
1) Chain-Away Bucket by Chain Away Corporation.
2) Or equal.
7. Provide cast iron floor stands where shown on Drawings.
a. Stands to be furnished by valve manufacturer with actuator.
b. Stands or actuator to include thrust bearings for valve operation and weight of accessories.

D. Submerged Actuators:
1. Install the valve actuator on top of an extension bonnet 3 feet above any adjacent personnel access.
2. Provide flanged, watertight bonnet to valve connection.
3. Provide a top brace support for the bonnet.
a. Install the brace 6 inches below the top of the wall as shown.
4. Materials:
a. Extension bonnet: Cast iron ASTM A126 or steel.
b. Brace and anchor bolts: Type 304 stainless steel.

2.4 ELECTRIC MOTOR ACTUATORS

A. Comply with AWWA C542 as modified herein.

B. Provide actuators with an NRTL label.

C. General:
   1. Provide electric motor actuators for valves indicated: on the Drawings, in valve/gate schedules in the Specifications, or elsewhere in the Contract Documents.
   2. Size actuators for 125% of required torque to operate valve under design pressure/head unless specified otherwise.
   3. Unless otherwise specified, provide each electric motor actuator with integral control devices for operation, including pushbuttons.
      a. When actuator's integral control station would be 5 feet or more above the nearest operating floor, or out of reach of facility personnel, then provide remotely located control station, with pushbuttons, in accordance with this Section.
   4. Locate and orient integral control station for ease of reading.
   5. Locations of remotely located control stations:
      a. If location of remotely located operator station is not shown on the Drawings, coordinate location with Contracting Officer's Representative.
      b. Furnish and install cabling between actuator and remote station if not provided by the actuator manufacturer.
   6. Provide equipment or control panels with Short Circuit Current Rating (SCCR) labeling as required by NFPA 70 and other applicable codes.
      a. See Section 01 61 03 for information on how to determine the available fault current, such that the SCCR rating meets or exceeds the available fault current.

D. Electric Motor Actuators (Standard Duty):
   1. Furnish electric actuator integral with valve consisting of:
      a. Motor.
      b. Gearing.
      c. Handwheel.
      d. Limit and torque switches.
      e. Lubricants.
      f. Heating elements.
      g. Wiring.
      h. Terminals for motor power and controls.
      i. Drive nut.
   2. Quarter and Multi-turn:
      a. For butterfly valves, ball valves, plug valves, and similar, provide quarter turn actuators.
      b. For gate valves, globe valves, and similar, provide multi-turn actuators.
   3. Housing/enclosure:
      a. Provide cast iron gear housing and cast iron load bearing enclosure for motor, gearing and control assembly.
      b. Non load bearing enclosure and housing: Aluminum or cast iron.
      c. Rated for area classification shown on Drawings.
      d. Provide O-ring seals for covers and entries.
      e. Terminal and limit switch compartment covers are to be fastened to gear housing by stainless steel fasteners with capture device to prevent loss.
   4. Motors:
      a. See Valve Schedules in Section 40 05 51 for motor input power.
      b. Provide motors that are totally enclosed, high torque design made expressly for valve/gate actuator service and capable of operating the valve under full differential pressure for complete open-close and reverse cycle of travel at least twice in immediate succession without overheating.
      c. Provide motors rated for continuous duty for modulating actuators.
      d. Provide motors in accordance with NEMA MG 1 standards where applicable, with Class F insulation with a Class B temperature rise, and to operate successfully at any voltage within 10% above or below rated voltage.

e. Provide positive method to ensure motor bearings are permanently lubricated.
f. Provide three thermal switches imbedded in windings:
    1) 120 degrees apart.
    2) Provide motor shutdown at high temperature.
g. Motor housing:
    1) Aluminum or cast iron.
    2) Totally enclosed nonventilated with cooling fins.
h. Provide motor capable of operating in any position.
i. Provide motor sealed from gearcase to allow any mounting position.

5. Gearing:
    a. Provide power gearing consisting of heat treated steel helical gears, carburized and hardened alloy steel worm, and alloy bronze worm gear, all grease or oil bath lubricated, designed for 100% overload, and effectively sealed against entrance of foreign matter.
    b. Provide gearing mechanism constructed to permit field changes of reduction gear ratio.
    c. Design actuators so that motor comes up to speed before stem load is encountered in either opening or closing operation.
    d. Limit switch gearings and feedback device reduction gearing:
        1) Steel or bronze.
    e. Support rotating shafts with anti-friction bearings.
    f. Provide separate drive nut/thrust bearing assembly:
        1) Mounted to base of actuator.
        2) High tensile bronze.
        3) Quarter turn actuator: Provide 90 degrees mounting intervals.
        4) Provide grease fitting on drive assembly.
6. Handwheel:
    a. Permanently attached for manual operation.
    b. Positive declutch mechanism to engage and disengage handwheel.
    c. Design such that handwheel does not rotate during motor operation.
    d. Design such that inoperable motor does not prevent manual operation.
    e. Direction of opening and the word OPEN to be cast in handwheel.
        1) Paint in a contrasting color such as white.
7. Limit torque and thrust loads in both closing and opening directions by torque limit switches or equivalent integral hardware and software programming.
    a. Provide torque switches with micrometer adjustment and reference setting indicator.
        1) Provide adjustment variation of approximately 40% in torque setting.
    b. Provide switches having rating of not less than 6 A at 120 Vac and 2.2 A at 115 Vdc.
    c. Provide limit and torque switches with totally sealed contacts.
8. Furnish electric actuator with two (2) geared limit switch assemblies (or equivalent integral hardware and software programming) with each switch assembly having four (4) separate limit switches:
    a. Assure each limit switch assembly is geared to driving mechanism and is independently adjustable to trip at any point at, and between, the fully open and fully closed valve position.
    b. Provide minimum of two (2) normally open contacts and two (2) normally closed contacts at each end of valve travel.
    c. Provide switches with inductive contact rating of not less than 1.5 A at 120 / 240 Vac, 1.5 A at 24 Vdc.
    d. Limit switches to be fully field adjustable when power is applied to actuator.
9. Size space heating elements to prevent condensation in both motor and geared limit switch compartment(s).
    a. Furnish heating elements rated at 120 Vac with heaters continuously energized.
10. Open-close actuator controls:

a. Provide control assembly with necessary holding relays, reversing starter, control transformers of sufficient capacity to provide control power, space heating element power and valve position transmitter.
b. Sealed, non-intrusive design.
c. Provide integral operator interface enclosure rated for area classification shown on Drawings with:
    1) Open pushbutton.
    2) Close pushbutton.
    3) Stop pushbutton.
    4) Local-Off-Remote switch.
    5) Full open light.
    6) Full close light.
    7) Open and close relays as required.
    8) Provide control enclosure to accept:
        a) Remote open control.
        b) Remote close control.
        c) Remote stop control, when shown in Drawings.
    9) Provide contacts in control enclosure:
        a) Remote status.
        b) Full open status.
        c) Full close status.
        d) Fault status.
    10) Wire all components to an internal terminal strip and include mounted wiring diagram inside enclosure.

11. Additional requirements for modulating valve actuators:
    a. Proportional position servo-amplifier installed integral with the actuator control compartment.
    b. Positioning of valve: Proportional to an input signal.
    c. Servo-amplifier adjustments: Zero, span, gain, and dead-band.
    d. Provide position control as shown on the Drawings that interfaces with the position control/position feedback instrumentation wiring to and from PLC.
12. Input/output protocol. See Paragraph 2.5.
13. Special Requirements for Energy Dissipation Valve (EDS-53117).
    a. Provide factory-mounted electric motor actuator capable of varying the valve rotational speed during the stroke via programmable integral frequency converter.
    b. Actuator shall operate on 120V, 60-hertz, single phase AC power.
    c. Default factory set conditions for actuator settings:
        1) The “fully open” position will be set in the factory at 70% of the valve stroke and field adjusted to achieve the desired flow rate.
        2) During “Normal” operation, actuator shall move the valve from fully-open to fully-closed position or reverse in a minimum of 5.2 minutes.
        3) Actuator supplier shall develop a program within the actuator for valve rotational speed adjustments to create a controlled increase in flow rate per hydraulic transient analysis or other design input. Supplier shall provide and obtain approval of the program prior to bid.
            a) Opening program:
                (1) Open over 270 seconds from 0% to 13% of total stroke.
                (2) Open over 44 seconds from 13% to 70% of total stroke.
            b) Closing Program:
                (1) Close at 44 seconds from 70% to 13% of total stroke.
                (2) Open at 270 seconds from 13% to 0% of total stroke.
    d. Supplier representative shall modify factory programming in the field to accommodate field hydraulic conditions
    e. Provide indicator to show position of valve.

f. Motor frequency converter shall be an integral part of the valve actuator and shall be the foundation upon which the actuator functions. No external frequency converters, drives, third parts hardware or components will be acceptable.
g. Actuator shall have the ability to continuously monitor valve position and provide feedback via selectable 4-20mA analog signal output.
h. Actuator shall have programmable outputs for full open and full closed position as well as other programmable outputs as required and defined within the factory supplied technical data. Supplier representative shall be factory trained to adjust these settings and shall ensure proper setpoints have been obtained once the valves are installed.
i. Actuator shall continuously monitor torque and provide selectable outputs for individual or combined over-torque condition. Provide dry contacts for torque opened and closed.
j. Provide local controls for push button electrical operation of the actuator. Local operation shall follow the programmed speed-time curve.
k. Actuator shall have MODBUS TCP/IP interface for monitoring actuator in SCADA.
l. Supplier representative shall be qualified in the setup, commissioning, programming, testing and training necessary for a successful installation and provide permanent record of all field programmable outputs, valve torque curves (if used) and speed-time curve setpoints in the operations manuals.

14. Special Requirements for CT Tank Submerged Valve – Volume 6 (FV-66031).
    a. Coordinate floor stand and electric actuator for extended stem operation of submerged valve inside tank.

2.5 COMMUNICATIONS PROTOCOL FOR POWERED ACTUATORS

A. Hard Wired.
   1. Dry contacts for discrete input/output.
   2. 4-20 mA for position control/feedback for modulating valves/gates.

B. For EDS-53117, Ethernet port for Modbus TCP/IP communication protocol.

2.6 SOURCE QUALITY CONTROL

A. General:
   1. Prior to shipping equipment from the production or fabrication facility, perform manufacturer's standard factory tests and inspections and submit documentation of successful completion.

B. Standard Duty Electric Actuators:
   1. Proof of Design Test: In conformance with AWWA C542 Section 5.2.
   2. Performance Test: In conformance with AWWA C542 Section 5.3.

2.7 SPARE ACTUATORS AND PARTS

A. Provide one set of any special tools or wrenches required for operation or maintenance for each type of valve.

B. Furnish spare actuators in accordance with the following:

Tag Number	Number of Spares	Size
EDS-53117	1	2"

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Location and Orientation:

1. Provide actuators and operators as shown on the Drawings.
2. When not expressly shown on the Drawings, orient valve actuators so valve operator is in a position reasonably convenient for facility personnel.
3. When orientation of actuator, operator location, or both are unclear, submit a request to the Contracting Officer's Representative for interpretation, and obtain response before furnishing the associated Submittal and in no event, later than purchasing the associated valve and actuator.
4. Contractor is responsible for costs and delays incurred to remedy valve actuators furnished or installed in inconvenient or difficult-to-access positions and locations unless the Contract Documents require otherwise.
5. Contractor is not responsible for providing permanent platforms, stairs, or other extraordinary, permanent access measures unless so required by the Contract Documents.

C. Touch Up Painting Requirements:
1. Repair scuffs and scratches on factory-applied finishes using materials furnished by the actuator manufacturer, applied in accordance with the written instructions of both the actuator manufacturer and the supplier of the touch-up material(s).

3.2 FIELD QUALITY CONTROL

A. Demonstrate that all powered valves operate properly after installation.

B. See Section 01 79 00 for Operator training requirements

3.3 ADJUSTMENT

A. Adjust valves, actuators, and appurtenant equipment to comply with Sections 01 75 00 and 01 75 10, as appropriate.
1. Operate valve, open and close at system pressures.

B. For all 120 Vac and 480 Vac electric actuators, employ and pay for services of valve actuator manufacturer's field service representative to:
1. Inspect valve actuators covered by this Specification Section.
2. Program/configure actuator integral software.
3. Supervise adjustments and installation checks:
   a. Open and close valves electrically under local manual and demonstrate that all limit switches are properly adjusted and that switch contacts are functioning properly by verifying the inputs are received at the remote input/output (RIO) panels or local control panel as appropriate.
   b. Position powered modulating valves under local manual control and demonstrate that the valve position feedback potentiometer is properly adjusted and that the feedback signal is received at the RIO panels or local control panel as appropriate.
   c. Simulate a valve position command signal at the RIO panel or local control panel as appropriate and demonstrate that the valve is controlled to the desired position without excessive hunting.
   d. For Energy Dissipation Valve (EDS-53117):
      1) Actuator representative must be present in the field to adjust actuator to meet hydraulic conditions.
   e. Upon completion, configure all spare actuators with identical settings to in-service actuators.
   f. Submit to Government a Certificate of Proper Installation (COPI) from the valve actuator manufacturer that includes:
      1) Verification that the actuators have been installed properly.
      2) Verification that all limit switches and position potentiometers have been properly adjusted.
      3) Verification that the valve actuator responds correctly to the valve position command.

END OF SECTION

This page intentionally left blank.

SECTION 40 05 61
GATE VALVES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Gate valves.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 40 05 51 - Common Requirements for Process and Utility Valves.
   2. Section 40 05 57 – Actuators for Process Valves and Gates.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. A126, Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings.
   2. American Water Works Association (AWWA):
      a. C500, Standard for Metal-Seated Gate Valves for Water Supply Service.
      b. C504, Standard for Rubber-Seated Butterfly Valves.
      c. C550, Standard for Protective Epoxy Interior Coatings for Valves and Hydrants.
   3. Manufacturers Standardization Society of the Valve and Fittings Industry Inc. (MSS):
      a. SP-9, Spot Facing for Bronze, Iron and Steel Flanges.
      b. SP-70, Cast Iron Gate Valves, Flanged and Threaded Ends.
      c. SP-80, Bronze Gate, Globe, Angle and Check Valves.
   4. NSF International (NSF):
      a. 61, Drinking Water System Components - Health Effects.

1.3 DEFINITIONS

A. OS&Y: Outside Screw and Yoke.

B. NRS: Non-rising Stem.

C. RS: Rising Stem.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 05 51.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

2.2 VALVES: WATER (HOT, COLD, HEATING, COOLING, SERVICE, PROCESS, POTABLE, NON-POTABLE, AND WASTEWATER)

A. Resilient Wedge Gate Valves, 2 to 48 inches (Water, Wastewater Application):
   1. Comply with AWWA C509.
   2. Materials:
      a. Stem and stem nut: Bronze.
         1) Wetted bronze parts in low zinc bronze.
         2) Aluminum bronze components: Heat treated per AWWA C504.
      b. Body, gate: Cast iron.
      c. Resilient wedge: Ethylene Propylene Diene Monomer (EPDM).
   3. Design requirements:
      a. Minimum 200 PSIG working pressure.
      b. Buried: NRS, O-ring stem seal, 2 inches square operating nut.
      c. Exposed: OS&Y, stuffing box stem seal, handwheel.
      d. Counter clockwise open rotation.
      e. Fusion bonded epoxy coating interior and exterior except stainless steel and bearing surfaces.
         1) Comply with AWWA C550.
         2) Wetted bronze parts in low zinc bronze.
         3) Aluminum bronze components: Heat treated per AWWA C504.
   4. Manufacturers:
      a. American Flow Control.
      b. Or equal.

2.3 ACCESSORIES

A. Refer to Drawings and Valve Schedules for type of actuators.
   1. Furnish actuator integral with valve.

B. Refer to Specification Section 40 05 57 for actuator requirements.

2.4 FABRICATION

A. General:
   1. Provide valves with clear waterways the full diameter of the valve.

B. Spot valves in accordance with MSS SP-9.

PART 3 - EXECUTION

3.1 INSTALLATION

A. See Specification Section 40 05 51.

B. Where larger buried valves utilize smaller bypass valves, provide a second valve box installed over the bypass valve operating nut.

C. Do not install gate valves inverted or with the stems sloped more than 45 degrees from the upright unless the valve was ordered and manufactured specifically for this orientation.

END OF SECTION

SECTION 40 05 62
PLUG VALVES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Plug valves at the North Rim WTP (Volume 6).

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 40 05 51 – Common Requirements for Process and Utility Valves.
   2. Section 40 05 57 - Actuators for Process Valves and Gates.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B16.1, Gray Iron Pipe Flanges and Flanged Fittings: Classes 25, 125 and 250.
   2. ASTM International (ASTM):
      a. A126, Standard Specification for Gray Iron Castings for Valves, Flanges and Pipe Fittings.
      b. A536, Standard Specification for Ductile Iron Castings.
      c. D2240, Standard Test Method for Rubber Property-Durometer Hardness.
   3. American Water Works Association (AWWA):
      a. C517 Resilient-Seated Cast-Iron Eccentric Plug Valves

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 05 51.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
   2. See Specification Section 40 05 51.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the manufacturers listed under the specific valve types are acceptable.

2.2 NON-LUBRICATED ECCENTRIC PLUG VALVES (SLUDGE)

A. Manufacturers:
   1. DeZurik.
   2. Or equal.

B. Materials:
   1. Body: Cast-iron ASTM A126, Class B.
   2. Plug: One or two-piece construction ductile iron, ASTM A536 65-45-12 or cast iron, ASTM A126 Class B.
   3. Plug facing: Grease and/or petroleum-resistant resilient Neoprene or Buna-N compound, 70 Type A durometer hardness per ASTM D2240.
   4. Shaft bearing bushings: Permanently lubricated TFE or Delrin sleeve type stainless steel or bronze.
   5. Valve seats: Welded-in overlay of 90% nickel, (minimum 1/8 inches thick).
   6. Stem seal: per AWWA C517, Section 4.4.7.

2.3 ACCESSORIES

A. Refer to Drawings and Valve Schedules for type of actuator.
   1. Furnish actuator integral with valve.

B. Refer to Specification Section 40 05 57 for actuator requirements.

2.4 DESIGN REQUIREMENTS

A. Non-Lubricated Eccentric Plug Valves:
   1. Port area:
      a. Valves 1/2 IN to 3 IN: equal to or exceed 80% of full pipe area.
   2. Valve body: Fitted with bolted bonnet.
   3. End connections: See Specification Section 40 05 51.
   4. Stem seal: Adjustable and replaceable without disassembling valve or bonnet.
   5. Designed for seating drip tight in any flow direction.
   6. Rating:
      a. 1/2 through 12 inches, 175 psi working pressure.
   7. Actuator:
      a. Actuator gearing in enclosure suitable for running in oil with seals on shaft to prevent entry of dirt or water.
      b. Positive identification on actuator indicating valve position.
      c. Adjustable stop to set closing torque.

2.5 FABRICATION

A. See Specification Section 40 05 51.

PART 3 - EXECUTION

3.1 INSTALLATION

A. See Specification Section 40 05 51.

B. Install valves with valve stem horizontal, plug seat on inlet side and with plug rotating up into the open position for valves in horizontal lines.

C. Install valve with actuator above pipe or plug centerline.

END OF SECTION

SECTION 40 05 63
BALL VALVES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Ball valves.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 40 05 51 - Common Requirements for Process and Utility Valves.
   2. Section 40 05 57 - Actuators for Process Valves and Gates.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. A48, Standard Specification for Gray Iron Castings.
      b. A126, Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings.
      c. A216, Standard Specification for Stee; Castings, Carbon, Suitable for Fusion Welding, for High Temperature Service.
      d. A276, Standard Specification for Stainless Steel Bars and Shapes.
      e. A351, Standard Specification for Castings, Austenitic, for Pressure-Containing Parts.
      f. B637, Standard Specification for Precipitation-Hardening and Cold Worked Nickel Alloy Bars, Forgings, and Forging Stock for Moderate or High Temperature Service.
   2. American Water Works Association (AWWA):
      a. C507, Standard for Ball Valves, 6 inches through 48 inches.
   3. Manufacturers Standardization Society of the Valve and Fittings Industry Inc. (MSS):
      a. SP-110, Ball Valves; Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends.

1.3 DEFINITIONS

A. PVDF: Polyvinylidene fluoride.

B. PTFE: Polytetrafluoroethylene.

C. RPTFE: Reinforced PolyTetraFluoroEthylene.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 05 51.
   2. Test results for AWWA valves.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient

characteristics are not characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Volume 5 only:
    a. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
    b. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.
2. Volume 6:
    a. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the manufacturers listed under the specific valve types are acceptable.

C. Volume 5 Only:
1. Submit request for product substitution in accordance with Specification Section 01 33 23.

2.2 METALLIC BALL VALVES 1/4 TO 4 INCHES DIAMETER

A. Volume 3:
1. Comply with MSS SP-110.
2. Manufacturers:
    a. Apollo.
    b. Or Equal.
3. Materials (All Stainless Steel):
    a. Body: Three-part stainless steel, ASTM A351 CF8M.
    b. Ball: Stainless steel ASTM A276.
    c. Seats: RPTFE.
4. Design Requirements:
    a. Rated for a minimum of:
        1) 1500 psi CWP.
        2) 150 psi of saturated steam.
        3) 29 inches vacuum.
    b. Two-position lockable handle.
    c. Stem with blowout-proof design.
    d. Balancing stop for all applications.

B. Bodies with mounting pad for applications requiring actuators

C. Volume 5 only.
1. Comply with MSS SP-110.
2. Manufacturers:
    a. Cameron, WKM 370D6 Model Type (Basis of Design)
    b. Or Equal
3. Materials (All Stainless Steel):
    a. Body: Carbon steel, ASTM A216 Gr. WCC.
    b. Ball: Stainless steel ASTM A216 Gr. WCC/A405 CS ENP.
    c. Seats: RPTFE.
    d. Seat Rings: A216 Gr. WCC/A105 CS ENP
    e. Seat Spring: B637 N07500
4. Design Requirements:
    a. Rated for a minimum of:
        1) 1,700 PSI CWP for 4 IN Class 900 valves.
        2) 1,500 PSI CWP for 3 IN Class 900 valves.
    b. Ball valves in Volume 5 shall be class 900 unless noted otherwise on the drawings.

c. Two-position lockable handle.
d. Stem with blowout-proof design.
e. Balancing stop for all applications.
f. Bodies with mounting pad for applications requiring actuators.

D. Volume 6.
1. Comply with MSS SP-110.
2. Manufacturers:
a. Apollo.
b. Or Equal.
3. Materials (All Stainless Steel):
a. Body: Three-part stainless steel, ASTM A351 CF8M.
b. Ball: Stainless steel ASTM A276.
c. Seats: RPTFE.
4. Design Requirements:
a. Rated for a minimum of:
1) 500 psi CWP.
2) 150 psi of saturated steam.
3) 29 inches vacuum.
b. Two-position lockable handle.
c. Stem with blowout-proof design.
d. Balancing stop for all applications.
e. Bodies with mounting pad for applications requiring actuators.

2.3 PLASTIC BALL VALVES: 1/2 INCHES TO 4 INCHES DIAMETER

A. Manufactuers:
1. Chemtrol/NIBCO.
2. Or Equal.

B. Materials:
1. Body, stem, ball, handle, end connectors:
a. PVC ASTM D1784-12454B or CPVC ASTM D1784-23447-B.
1) Material to match connecting piping.
2. Ball Seat: Teflon.
3. O-rings: Viton

C. Design Requirements:
1. Rated at 150 psi at 75 degrees F.
2. Double or "true union" design.
3. Blocks both directions, upstream and downstream.
4. Union nut capable of compensating for seat wear.
5. Body with mounting pad for actuators where required.
6. Capable of being disconnected at downstream end under full line pressure.
7. Sodium hypochlorite service:
a. Provide "vented" ball valves for all service with greater than 5% concentrated sodium hypochlorite.

2.4 ACCESSORIES

A. Refer to Drawings and valve schedule for type of actuators.
1. Furnish actuator integral with valve.

B. Refer to Specification Section 40 05 57 for actuator requirements.

2.5 SOURCE QUALITY CONTROL

A. Furnish record of test.

B. Product Testing non-AWWA C507 ball valves: MSS SP-110.

2.6 SPARE ACTUATORS AND PARTS

A. Provide one set of any special tools or wrenches required for operation or maintenance for each type valve.

B. Furnish spare actuators in accordance with the following:

Tag Number	Number of Spares	Size
V-52338	1	4”
V-52331	1	3”

PART 3 - EXECUTION

3.1 INSTALLATION

A. See Specification Section 40 05 51.

END OF SECTION

SECTION 40 05 64
BUTTERFLY VALVES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. General use butterfly valves.
   2. High performance butterfly valves – water service – Volume 6.

B. Related Sections include but are not necessarily limited to:
   1. Section 40 05 00 – Pipe and Pipe Fittings – Basic Requirements.
   2. Section 40 05 51 – Common Requirements for Process and Utility Valves.
   3. Section 40 05 57 – Actuators for Process Valves and Gates.

1.2 REFERENCES

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B16.5, Pipe Flanges and Flanged Fittings - NPS 1/2 Through NPS 24.
   2. ASTM International (ASTM):
      a. A48, Standard Specification for Gray Iron Castings.
      b. A126, Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings.
      c. A276, Standard Specification for Stainless Steel Bars and Shapes.
      d. A395, Standard Specification for Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures.
      e. A436, Standard Specification for Austenitic Gray Iron Castings.
      f. A536, Standard Specification for Ductile Iron Castings.
   3. American Water Works Association (AWWA):
      a. C504, Standard for Rubber-Seated Butterfly Valves.
      b. C519, Standard for High-Performance Butterfly Valves – 3 In. through 60 In.
   4. NSF International (NSF).
      a. NSF/ANSI 61 - Drinking Water System Components – Health Effects.

1.3 SUBMITTALS

A. Action Submittals:
   1. See Specification Section 40 05 51.

B. Informational Submittals:
   1. For valves 8 inches and larger, furnish "Affidavit of Compliance" with Government in accordance with AWWA C504 and C519.
   2. For all modulating and control valves:
      a. Cv curves for applicable valve sizes.
      b. Associated actuator information (See Section 40 05 57).

C. Closeout Submittals:
   1. Operation and Maintenance Data:
      a. Submit in accordance with Section 01 78 23 - Operation and Maintenance Data.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient

characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. General Use BFVs (Flanged and MJ):
      a. DeZurik.
      b. Or equal.
   2. General Use BFVs (Grooved- Volume 5 only):
      a. Victaulic.
      b. Or equal.
   3. High Performance BFVs:
      a. DeZurik.
      b. Or equal.

2.2 GENERAL USE BUTTERFLY VALVES

A. For use in all locations, except where high performance butterfly valves are required as indicated in the Valve Schedules located in specification Section 40 05 51.

B. Comply with AWWA C504, except as noted in this Specification Section.

C. Materials:
   1. Valve bodies:
      a. ASTM A126, Class B or ASTM A536 Grade 65-45-12 ductile iron.
   2. Valve shafts:
      a. Shaft and pin material: Type 304 stainless steel.
      b. Bushings/Packing/O-rings: EPDM, Nylatron, RTFE or TFE.
      c. Bearings: Nylatron, Reinforced TFE, or equal.
   3. Valve discs:
      a. Cast iron with welded 304 stainless edge or Type 304 Stainless Steel disk.
   4. Valve seats:
      a. Water: EPDM or Hycar.
         1) EPDM only for valves potentially exposed to chlorinated water.
      b. Compressed air: Teflon, PTFE.
      c. Process air: Peroxide cured EPDM, Viton, RTFE, rated for 300 degrees F minimum or higher if required by service.
      d. Located in valve body.
   5. Shaft bearing: Bronze, TFE-coated stainless steel or reinforced TFE.
   6. Shaft seal in addition to any sealing provided by seat: Suitable synthetic rubber rings or PTFE V-ring suitable for operating conditions.

D. Design Requirements:
   1. Shaft design
      a. Valves 16-inch and less: One-piece.
      b. Valves greater than 16-inch: one-piece or two-piece (stub shaft).
   2. Seat type: Resilient.
   3. Body type (exposed and submerged):
      a. Flanged.
         1) Fully tapped anchor lugs drilled per ASME B16.5.
      b. Grooved ends (Volume 5 only).
         1) Victaulic rolled groove, Type 77, or equal.
      c. Valves 24 inch and greater: short body flanged.
   4. Working Pressure Rating (exposed and submerged): See Valve Schedules in Section 40 05

2.3 HIGH PERFORMANCE BUTTERFLY VALVES

A. In locations where reliability is critical, for automated valves that modulate for flow control or actuate periodically in intervals less than 2 hours, high performance butterfly valves with an offset disc design shall be used.
   1. See Valve Schedules located in Section 40 05 51 for identification of high performance butterfly valves.

B. Design Requirements:
   1. Comply with AWWA C519 except as noted herein.
   2. Shaft:
      a. One-piece shaft.
      b. Separate shaft seal.
      c. Minimum shaft diameter to conform to AWWA C519.
   3. Seat located in valve body.
   4. Valve design tested to 1,000,000 cycles.
   5. Bi-directional drip tight seat.
   6. Double-offset disc design to provide 360-degree sealing.
   7. Packing: Multiple v-ring PTFE or braided carbon graphite including anti-extrusion ring.
   8. Throttling Performance:
      a. Valve flow coefficient (Cv) curves must meet the following:

VALVE SIZE	DEGREE OF VALVE OPENING		
	20	50	90
3-IN	15 +/- 5	80 +/- 10	225 +/- 20
4-IN	25 +/- 5	150 +/- 15	410 +/- 25

C. Materials of construction:
   1. Body: Carbon steel – ASTM 216 Grade WCB.
   2. Disc: Type 316 or CF8M stainless steel.
   3. Shaft and pins: 17-4PH stainless steel or Type 316 stainless steel.
   4. Seals:
      a. Water: PTFE.
      b. Process air and high temperature: Graphite rings.
         1) Designed for continuous operation at 300 DEG F.
   5. Backing ring: Stainless steel.
   6. Bushings/Bearings: TFE/Glass liner with a 316 Stainless steel shell.
   7. Seat:
      a. Two part with encapsulated RTFE or PTFE.
         1) Designed for continuous operation at 300 DEG F (process air service).
         2) Seat Retainer: Titanium or Nickel-Chromium Alloy (for oxidizing service).
         3) Seat Retainer: Match valve body material (for process air service).
   8. End connection: Lug-wafer.
      a. Class 250 valves: Flanged at dead ends.

2.4 ACCESSORIES

A. Refer to Drawings and/or Valve Schedules for type of actuators.
   1. Furnish actuator integral with valve.
   2. Provided by valve manufacturer and installed at valve manufacturer's facility.

B. Refer to Section 40 05 57 for actuator requirements.

PART 3 - EXECUTION

3.1 INSTALLATION

A. See Section 40 05 51.

END OF SECTION

SECTION 40 05 65
GLOBE VALVES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Globe valves.
   2. Specialty, small-diameter diaphragm valves

B. Related Sections include but are not necessarily limited to:
   1. Section 40 05 00 – Pipe and Pipe Fittings – Basic Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Manufacturers Standardization Society of the Valve and Fittings Industry Inc. (MSS):
      a. SP-80, Bronze Gate, Globe, Angle and Check Valves.
      b. SP-85, Cast Iron Globe and Angle Valves, Flanged and Threaded Ends.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 05 51.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

2.2 GLOBE VALVE: WATER 125 TO 200 PSI, 2 INCHES AND LARGER

A. Class 250 iron globe valve.

B. Comply with MSS SP-85.

C. Manufacturers:
   1. Nibco F768B.
   2. Or Equal.

D. Materials:
   1. Body, bonnet: Cast iron.
   2. Stem: Brass ASTM B16 Alloy
   3. Disc and seat: Bronze.

E. Design Requirements:
   1. Bolted bonnet, OS&Y, bronze mounted, renewable seat and disc.
   2. Actuator: Handwheel.

2.3 GLOBE VALVE: WATER 125 TO 200 PSI, 1 1/2 INCH AND SMALLER

A. Comply with MSS SP-80

B. Manufacturers:
   1. Milwaukee Valve UP502.
   2. Or Equal.

C. Materials:
   1. Body, bonnet: Bronze
   2. Stem: Brass
   3. Disc: Bronze
   4. Disc Ring: Brass

D. Design Requirements:
   1. Threaded bonnet.
   2. Actuator: Handwheel.
   3. Lead-free.

2.4 DIAPHRAGM VALVE: WATER UP TO 100 PSI, 1 1/2 INCH AND SMALLER

A. Thermoplastic construction
   1. Multi-turn handwheel
   2. Clear, visual diaphragm position indicator
   3. Connecting ends: true union FNPT (ANSI/ASME B1.20.1)

B. Manufacturers:
   1. Spears
   2. Hayward
   3. Or Equal.

C. Materials:
   1. Body:
      a. PVC (ASTM D1784)
      b. CPVC (ASTM D1784)
   2. Diaphragm: EPDM, Viton/FKM, or PTFE
   3. Body hardware: stainless steel
   4. True union O-rings: EPDM, or Viton/FKM

D. Design Requirements:
   1. Pressure rated to at least 100 PSI at 70 DEG F
   2. Where used to control flow: see associated flow meter design details, Section 40 71 00

PART 3 - EXECUTION

3.1 INSTALLATION

A. See Section 40 05 51.

B. Install globe valves with stem in horizontal position wherever possible.

END OF SECTION

SECTION 40 05 66
CHECK VALVES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Check valves 2.5-inch diameter and smaller.
   2. Swing check valves, three-inch to 24-inch diameter.
   3. Double door check valves.
   4. Silent check valves.
   5. Ball check valves, 0.5-inch to two-inch diameter.
   6. Duckbill check valves.
   7. Silent Globe Check Valves

B. Related Requirements: Include but are not necessarily limited to:
   1. Section 40 05 51 - Common Requirements for Process and Utility Valves.

1.2 REFERENCES

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B16.1, Gray Iron Pipe Flanges and Flanged Fittings: Classes 25, 125, and 250.
   2. American Water Works Association (AWWA):
      a. C508, Standard for Swing-Check Valves for Waterworks Service, 2 in. through 24 in. NPS.
   3. Manufacturers Standardization Society of the Valve and Fittings Industry Inc. (MSS):
      a. SP-71, Gray Iron Swing Check Valves, Flanged and Threaded Ends.
      b. SP-80, Bronze Gate, Globe, Angle and Check Valves.
   4. NSF International (NSF), formerly National Sanitation Foundation:
      a. NSF/ANSI 61 – Drinking Water System Components– Health Effects.

1.3 QUALITY ASSURANCE

A. Qualifications:
   1. Valve Manufacturers:
      a. Manufacturer shall be a business regularly engaged in manufacturing and furnishing check valves of the type required and similar equipment.
      b. Manufacturer shall be able to document having furnished not less than 50check valves, of the type required, of size equal to or larger than those required for the Work, during the past five years.
      c. When requested by Contracting Officer's Representative, submit qualifications statement indicating manufacturer's record of manufacturing and furnishing check valves of the types and sizes required and furnish documentation of compliance with qualifications requirements of this Article.

1.4 SUBMITTALS

A. Action Submittals: Submit the following:
   1. In accordance with Section 40 05 51.

B. Informational Submittals: Submit the following:
   1. In accordance with Section 40 05 51.
   2. Manufacturer's Instructions:
      a. Manufacturer's written instructions for delivery, handling, storage, installation, and startup.

C. Closeout Submittals: Submit the following:
   1. Operation and Maintenance Data:
      a. Submit in accordance with Section 01 78 23 - Operations and Maintenance Manuals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the manufacturers listed under the specific valve types are acceptable.

2.2 MATERIALS AND EQUIPMENT – GENERAL

A. For service in potable water applications or water that will be treated to become potable, materials and equipment shall be in conformance with NSF/ANSI 61.

2.3 CHECK VALVES: 2.5 INCHES AND SMALLER

A. Volume 6
   1. Class 125 Bronze Swing Check Valves (Compressed Air, Water):
      a. Comply with MSS SP-80.
      b. Manufacturers:
         1) Nibco T413-V.
         2) Or Equal.
   2. Materials
      a. Body, bonnet, disc: Bronze.
   3. Design requirements
      a. 125 psi steam to 406 degrees F, 200 psi WOG.
      b. Horizontal swing, renewable disc.

2.4 SWING CHECK VALVES: 3 INCHES TO 24 INCHES

A. Volume 6

B. Swing Check Valves (Water):
   1. Comply with AWWA C508.

C. Manufacturers:
      a. GA Industries.
      b. Or Equal.

D. Materials:
      a. Body and cover: Cast iron.
      b. Seat ring, hinge: Bronze.
      c. Disc:
         1) 3 to 4 inches: Bronze.
         2) 6 to 24 inches: Cast iron with bronze face
      d. Hinge shaft: Stainless steel.
      e. Bearings, connecting hardware: Bronze.

E. Design requirements:
      a. 175 psi working pressure (3 to 12 inches).
      b. 150 psi working pressure (14 to 24 inches).

c. Furnish with outside weight and lever or lever and spring.

2.5 SILENT CHECK VALVES

A. Flanged Center Guided Silent Check Valves
1. Volume 5 – Raw Water High Pressure
2. Manufacturers:
   a. DFT Valves, Excalibur Model Type (Basis of Design).
   b. Or Equal.
3. Type:
   a. Minimum length: 18 IN.
   b. ASME B16.1, Class 900.
   c. Provide assembly with lifting lugs for ease of removal from the piping system.
4. Materials:
   a. Body: Carbon Steel, ASTM A216 Grade WCB.
   b. Plug and seat: Stainless Steel, ASTM A351 CF8M .
   c. Spring: Stainless steel, ASTM A313 316.
5. Design Requirements:
   a. Rated for a minimum of 1,700 psi.
   b. Cv of 242.
   c. Body type: Flanged center guided

B. Globe Body Silent Check Valves
1. Volume 6 – Finished Water
2. Manufacturers:
   a. Flowmatic 402 BTR.
   b. Or Equal.
3. Type:
   a. ASME B16.42, Class 150.
   b. Provide assembly with lifting lugs for ease of removal from the piping system.
4. Materials:
   a. Body: Ductile Iron.
   b. Plug and seat: Bronze .
   c. Spring: Stainless steel.
5. Design Requirements:
   a. Rated for a minimum of 200 psi.
   b. Body type: Flanged center guided, globe style.
   c. NSF 61 certified.

2.6 BALL CHECK VALVES: 1/2 INCHES TO 4 INCHES

A. Manufacturers:
1. R&G Sloane.
2. Or equal.

B. Performance Criteria:
1. Pressure Rating: 150 psi at 73 degrees F.

C. Materials:
1. Body: PVC
2. Ball: Glass filled or polypropylene.
3. Seals: Viton or EPDM.

D. Design Requirements:
1. Connectors: Double union.

2.7 SOURCE QUALITY CONTROL

A. Factory Tests and Inspections:

1. Perform manufacturer's standard factory tests and inspections on materials and equipment furnished. Correct defects prior to shipment to the Site.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Installation – General:
   1. Provide check valves at locations shown and indicated in the Contract Documents.
   2. Install in accordance with the Contract Documents and manufacturer's written instructions. In event of conflict between the Contract Documents and manufacturer's written instructions, obtain written interpretation or clarification from Contracting Officer's Representative.
   3. Comply with:
      a. Section 01 61 03 - Equipment - Basic Requirements.
      b. Section 40 05 00 – Pipe and Pipe Fittings – Basic Requirements
      c. Section 40 05 51 - Common Requirements for Process and Utility Valves.
   4. Before installing, ensure each check valve is clean and free of dirt and debris.

3.2 FIELD QUALITY CONTROL

A. Field Tests and Inspections:
   1. Promptly after installing, before installing connecting pipe, verify proper and free operation of check valve.
   2. Hydrostatically test check valves together with associated piping.
   3. To extent practical, prior to Substantial Completion, verify proper operation of each installed check valve.

END OF SECTION

SECTION 40 41 13
HEAT TRACING CABLE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Heat tracing cable as required for heat tracing of pipes as indicated on the Drawings.

B. Related Sections include but are not necessarily limited to:
   1. Section 26 05 00 - Electrical - Basic Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data:
      a. Power requirements for each circuit based upon actual length of heat trace and maintained temperature.
      b. Circuit breaker rating based upon inrush current at minimum expected start-up temperature.
      c. Length of heat tape for each pipe size and run.
      d. Coordinate and verify length and Watts/FT of heat tape required based upon pipe size and insulation thickness.
         1) Include the calculations to support the heat tape output.
      e. See Section 26 05 00 for additional requirements.
   2. Fabrication and/or layout drawings:
      a. Wiring diagrams showing physical locations of thermostats and heat trace power supply.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

C. Informational Submittals:
   1. Test reports: Megger test results.

1.4 DELIVERY, STORAGE, AND HANDLING

A. Shall be stored such that they are not exposed to sunlight or other UV rays.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following Manufacturers are acceptable:
   1. Raychem Corp.
   2. Or Approved Equal.

2.2 HEAT TRACING

A. Design Parameters:
   1. Pipe diameter, length and material: See Drawings and relevant piping Specifications.
   2. Flange, valve, pipe support size: See Drawings and relevant piping Specifications.
   3. Pipe insulation type and thickness: See Drawings and relevant piping Specifications.
   4. Temperatures requirements:
      a. Low ambient temperature for the specific location: -20 DEGF.
      b. Start-up temperature (alarm thermostat set point):
         1) Water/wastewater lines: 40 DEGF.
         2) Caustic feed lines: 80 DEGF.
      c. Maintain temperature (power thermostat set point):
         1) Water/wastewater lines: 40 DEGF.
         2) Caustic feed lines: 80 DEGF.
      d. High temperature exposure with power off: 185 DEGF.
   5. Wind factor for the specific location: 10 MPH.
   6. Electrical requirements:
      a. Voltage: 120 V or 208V single-phase. Coordinate system voltage with available supply voltage.
      b. Circuit breaker: Field coordinate if other than 20A GFEPCI type.
   7. Safety factor: 10%.

B. Self-regulating or power-limiting parallel circuit construction consisting of an inner core of conductive material between parallel copper bus wires, with inverse temperature - conductivity characteristics with metal overbraid.

C. Thermostats adjustable between 35 and 200 degrees F minimum with maximum differential range of 9 degrees F, furnished complete with NEMA 4 enclosures in all areas, stainless steel temperature bulb and capillary.

D. All necessary or required components and accessories, such as power connection boxes, end seals, straps, tape and fitting brackets.

E. In noncorrosive and nonhazardous locations, insulation shall be Polyolefin.

F. In corrosive, hazardous and hydrocarbon locations insulation shall be Fluoropolymer (Teflon).

2.3 VOLUME 5 HEAT TRACE CONTROLLER

A. Heat trace controller panel to control and monitor electric heat tracing circuits for freeze protection for insulated piping.
   1. Capacities and Characteristics:
      a. Maximum Heat Output at 50 deg F (10 deg C): 8 W/ft. (26 W/m).
      b. Voltage: 120V, Single Phase.
      c. Power consumption: < 24 W
   2. Controls:
      a. Single-Circuit Local Digital Controller.
      b. Control self-regulating heating cable via an energy saving, programmable single-circuit controller to provide adjustable maintained temperatures in the range of 32 to 104 deg F (0 to 40 deg C).
      c. Controller must have three temperature sensor inputs.
      d. Operating Temperature Range: 32 to 104 deg F (0 to 40 deg C).

e. Display screen.
f. Communications:
    1) Alarm relay to Plant PLC Cabinet
    2) Run status to Plant PLC Cabinet

2.4 VOLUME 6 ADDITIONAL REQUIREMENTS

A. Heat tracing in Volume 6 will protect fire suppressing water lines, which necessitate the following additional requirements:
    1. Outer Jacket: Modified polyolefin with ultraviolet inhibitor. Outer jacket to be printed with cable model number, agency listings, batch number, and meter marks (for ease of installation within maximum circuit length).
    2. Capacities and Characteristics:
        a. Maximum Heat Output at 50 deg F (10 deg C): 8 W/ft. (26 W/m).
        b. Voltage: 120V, Single Phase.
    3. Controls:
        a. Single-Circuit Local Digital Controller.
        b. Control self-regulating heating cable via an energy saving, programmable single-circuit controller to provide adjustable maintained temperatures in the range of 32 to 104 deg F (0 to 40 deg C).
        c. Provide one controller for each heat-tracing circuit.
        d. Controller shall have a red faceplate for easy identification as a fire sprinkler heat-tracing controller.
        e. Controller must have two temperature sensor inputs.
        f. Controller shall be able to use two temperature sensors per circuit with ability to assign any of them for high temperature cut-out function.
        g. External thermostatic control devices for high limit or high temperature cutout are not permitted.
        h. Controller shall include user-defined filters for temperature alarms to avoid nuisance alarms.
        i. Controller shall have ground fault current sensing and relaying equipment that complies with UL1053 requirements.
        j. External ground fault devices are not permitted.
        k. Controller shall have user-defined settings for ground fault alarm and trip levels.
        l. Controller shall store at least 100 past events/alarms to aid maintenance.
        m. Controller Capabilities:
            1) Supply Voltage: 100 to 277 V ac.
            2) Enclosure: Type 12 (indoor use).
            3) Operating Temperature Range: 32 to 104 deg F (0 to 40 deg C).
            4) Display: 5” touchscreen color display.
            5) Control: DP mechanical relay type.
            6) Control Algorithms: Ambient On/Off. Line Sensing. Proportional Ambient Sensing Control for energy savings.
            7) Supervisory Relay to provide supervisory signal/fault to Plant PLC Cabinet for any of the following conditions:
                a) Ground fault current
                b) Low system temperature
                c) High system temperature
                d) Temperature sensor failure
                e) Internal error
                f) Loss of continuity
                g) Loss of incoming supply voltage
            8) Temperature Sensor Inputs:
                a) Quantity: Two.
                b) Type: Thermistor 2 K-Ohm / 77°F (25°C), 2-wire
            9) Alarm Outputs:

a) AC Relay: Single pole double throw relay, volt-free; maximum switching capacity (resistive load only) 1 A/24 VDC, 1A/24 VAC
b) Outputs: Normally Open or Normally Closed.

10) Stored Parameters:
a) Time stamp.
b) Warning.
c) Event description.
d) Device ID.
e) Language.
f) Country.
g) Control mode.
h) Cable type.
i) Supply voltage.
j) Sensor 1 and 2.
k) Setpoint.
l) Deadband.
m) Minimum expected ambient temperature.
n) Power adjustment.
o) GFCI current.
p) Load current.
q) Board temperature.

11) Alarm Conditions:
a) Low and High temperature.
b) Ground fault alarm and trip.
c) RTD failure.

12) Communications:
a) Alarm relay to Plant PLC Cabinet

13) Temperature Sensors:
a) Contractor shall be able to program the controller to keep the heating cable powered in case of temperature sensor failure.

14) Approval: The complete heat trace system (heating cable, connection kits, and controller) shall be UL Listed (VGNJ) and marked for intended use of freeze protection of standpipes, mains and branch fire sprinkler piping.

PART 3 - EXECUTION

3.1 PREPARATION

A. Install materials after piping has been tested and approved.

3.2 INSTALLATION

A. Insulate and heat trace wet pipe systems as indicated on Drawings.

B. Install materials in accordance with manufacturer's instructions.
1. Each circuit shall not exceed the manufacturer's recommended maximum length.

C. For Metallic Piping:
1. Heat tracing shall be installed completely wired.
2. Cut heat trace to lengths as required and secure to pipe with glass or polyester fiber tape.

D. For Nonmetallic Piping:
1. Allow for extra heat trace output because nonmetallic pipe has a lower heat transfer.
a. Heat tracing shall be installed completely wired.
2. Cut heat trace to lengths as required and secure to pipe with aluminum tape throughout the length of the trace.

E. Protection and Control Requirements:
1. Protection by a GFEPCI circuit breaker.

a. Breaker amperage rating shall be coordinated with Contractor when different than the Contract Drawings.
2. Provide an ambient sensing thermostat for power and line sensing thermostat for alarm.
3. The alarm thermostat shall be placed on the opposite end of the circuit from the power thermostat or power connection to allow for annunciation of partial failure of a circuit or the loss of power from a tripped GFEPCI circuit breaker.
4. Provide a monitoring module that monitors the voltage (circuit breaker status) to each circuit.
5. The alarm from the alarm thermostat and monitor module shall be annunciated on the indicated control system.

3.3 TESTING

A. Megger the cables at the manufacturers recommended voltage level three times.
1. Before installation.
2. After attachment to pipe but before insulation is installed.
3. After pipe insulation is installed but before energization.

END OF SECTION

This page intentionally left blank.

SECTION 40 42 00
PIPE, DUCT AND EQUIPMENT INSULATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Insulation:
      a. Piping insulation.
      b. Duct insulation.
      c. Equipment insulation.
   2. Adhesives, mastics, sealants, and finishes.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 22 05 03 - Pipe and Pipe Fittings – Plumbing Systems
   2. Section 40 05 07 - Pipe Support Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. C177, Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of Guarded-Hot-Plate Apparatus.
      b. C411, Standard Test Method for Hot-Surface Performance of High-Temperature Thermal Insulation.
      c. C423, Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method.
      d. C553, Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications.
      e. C612, Standard Specification for Mineral Fiber Block and Board Thermal Insulation.
      f. C1071, Standard Specification for Fibrous Glass Duct Lining Insulation (Thermal and Sound Absorbing Material).
      g. D1056, Standard Specification for Flexible Cellular Materials-Sponge or Expanded Rubber.
      h. E96, Standard Test Methods for Water Vapor Transmission of Materials.
      i. F25, Standard Test Method for Sizing and Counting Airborne Particulate Contamination in Cleanrooms and Other Dust-Controlled Areas.
      j. C518, Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus.
      k. E84, Standard Test Method for Surface Burning Characteristics of Building Materials.
      l. E119 Standard Method of Fire Tests of Building Construction, 2 Hour Wall Panel Test, 2 Hour External Total Engulfment Test, hose stream evaluation.
      m. E136, Combustion Characteristics of Building Materials in a Vertical Tube Furnace.
      n. E162, Surface Flammability of Materials.
      o. E814, Through-Penetration, 2-Hour Firestop Test.
      p. E2336: Standard Test Methods Fire Resistive Grease Duct Enclosure Systems.
   2. ISO 6944-1985, Method of Determining Fire Resistance of Ventilation Ducts.
   3. National Fire Protection Association (NFPA):
      a. 255, Standard Method of Test of Surface Burning Characteristics of Building Materials.
   4. Underwriters Laboratories, Inc. (UL):
      a. 723, Standard for Test for Surface Burning Characteristics of Building Materials.
   5. National Commercial and Industrial Insulation Standards (2013 seventh edition).
      a. Published by Midwest Insulation Contractors Association (MICA).
      b. Endorsed by National Insulation Association (NIA).
      c. MICA plate numbers listed in this specification reference this document.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Submit manufacturer cut sheets and specifications for insulation materials, adhesives, cement, together with manufacturer's recommended methods of application and coverage for coatings and adhesives.
   2. Submit itemized schedule by building of proposed insulation systems showing density, thermal conductivity, thickness, adhesive, jackets and vapor barriers.
   3. Certifications: Products will meet the requirements of the Contract Documents.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Elastomeric insulation:
      a. Armacell
      b. Or equal.
   2. Fiberglass insulation:
      a. Johns Manville.
      b. Or equal.
   3. PVC jacket:
      a. Johns Manville.
      b. Or equal.
   4. Equipment insulation:
      a. Johns Manville.
      b. Or equal.
   5. Ductwork insulation:
      a. Johns Manville.
      b. Or equal.
   6. High density perlite:
      a. Johns Manville.
   7. High density calcium silicate:
      a. Johns Manville.
      b. Or equal.

2.2 PIPING INSULATION - ELASTOMERIC

A. General:
   1. Insulation fire and smoke hazard ratings for composite (insulation, jacket or facing, and adhesive used to adhere the facing or jacket to the insulation), listed and labeled as tested by procedure ASTM E84, NFPA 255 or UL 723, not exceeding:
      a. Flame spread: 25.
      b. Smoke developed: 50.
   2. Accessories (adhesives, mastics, cements, and tapes: Same component ratings as listed above.

3. Indicate on product labels or their shipping cartons: NRTL certification that flame and smoke ratings do not exceed above requirements.
4. Permanent treatment of jackets or facings to impart flame and smoke safety is required.
   a. Water-soluble treatments are prohibited.
5. Insulated shields at pipe support points.

B. Pipe, Fitting, and Valve Insulation:
1. Flexible elastomeric closed cell pipe insulation.
   a. Average thermal conductivity not to exceed 0.27 (BTU-IN)/(HR-FT^2-DEGF) at mean temperature of 75 DEGF, temperature range -40 to 220 DEGF; permeability not to exceed 0.05 by ASTM E96; water absorption 3% by ASTM C1763 or ASTM C209 and manufactured without CFCs, HFCs, HCFCs, PBDEs, or formaldehyde.
2. Provide minimum insulation thickness conforming to schedules or as shown on the Drawings.

2.3 PIPING INSULATION - FIBERGLASS

A. Pipe and Fitting Insulation:
1. Preformed fiberglass pipe insulation:
   a. Density: 4 LBS/CUFT.
   b. Temperature rated: 650 DEGF.
   c. Average thermal conductivity not to exceed 0.23 (BTU-IN)/(HR-FT^2-DEGF) at mean temperature of 75 DEGF.
   d. Fire hazard rating:
      1) UL 723, ASTM E84, NFPA 255.
      2) Flame spread not exceeding 25 and smoke developed not exceeding 50.
2. Moisture adsorption:
   a. ASTM C553.
   b. Not greater than 5% moisture by volume when exposed to moisture laden air at 120 DEGF and 96% RH.
3. Fungi and bacteria resistance:
   a. ASTM C665.
   b. Does not breed or promote growth.
   c. Flame attenuated glass fibers bonded with thermosetting resin.
4. Piping jackets (general applications):
   a. Aluminum: 16 MIL embossed aluminum. Provide for insulated piping installed outdoors.
   b. PVC: Preformed 0.028 IN thick PVC jackets fabricated from B.F. Goodrich PVC sheeting V-66 with proven resistance to ultraviolet degradation when temperatures do not exceed the limits of PVC. Provide on all interior insulated exposed piping.
   c. Piping jacket not required on concealed piping.
5. Provide minimum insulation thickness conforming to schedules or as shown on the Drawings.

2.4 PIPE INSULATION INSERTS AT HANGERS

A. High Density Perlite:
1. Pre-formed.
2. Fire hazard rating:
   a. UL 723, ASTM E84, NFPA 255.
   b. Flame spread: Zero.
   c. Smoke developed: Zero.
3. Average density: 13 LBS/CUFT.
4. Compressive strength: 80 PSI to produce 5% compression.
5. Maximum surface temperature: 1,200 DEGF.

B. High Density Calcium Silicate:
1. Pre-formed.

2. Fire hazard rating:
    a. UL 723, ASTM E84, NFPA 255.
    b. Flame spread: Zero.
    c. Smoke developed: Zero.
3. Average density: 14 LBS/CUFT.
4. Compressive strength: 100 PSI to produce 5% compression.
5. Maximum surface temperature: 1,200 DEGF.

2.5 EQUIPMENT INSULATION

A. Insulation for Equipment:
    1. Fire hazard classification:
        a. UL 723, ASTM E84, NFPA 255.
        b. Flame spread not exceeding 25 and smoke developed not exceeding 50.
    2. Provide minimum insulation thickness conforming to Schedules, equipment specifications, or as shown on Drawings.

2.6 DUCTWORK INSULATION: FIBERGLASS

A. Flexible Insulation:
    1. Material: Commercial-grade fiberglass thermal insulation, formaldehyde free.
    2. Scheduled thickness and installed R-value. Installed R-value when compressed to a maximum of 25% following recommended duct wrap stretch outs.
    3. Factory-applied foil scrim vapor barrier facing.
    4. Average thermal conductivity not to exceed 0.27 (BTU-IN)/(HR-FT^2-DEGF) at a mean temperature of 75 DEGF (installed).
    5. Fungi and bacteria resistance:
        a. ASTM C1338.
        b. Does not breed or promote growth.
    6. Fire hazard classification:
        a. UL 723, ASTM E84, NFPA 255.
        b. Flame spread not exceeding 25 and smoke developed not exceeding 50.
    7. Basis of design: Johns Manville Microlite fiberglass duct wrap insulation.

B. Semi-Rigid Insulation for Indoor Installation:
    1. Scheduled thickness and R-value.
    2. Factory applied vapor barrier facing-white scrim foil.
    3. Average thermal conductivity not to exceed 0.23 (BTU-IN)/(HR-FT^2-DEGF) at a mean temperature of 75 DEGF.
    4. Fungi and bacteria resistance:
        a. ASTM C1338.
        b. Does not breed or promote growth.
    5. Moisture adsorption:
        a. ASTM C553.
        b. Not greater than 0.5% moisture by volume when exposed to moisture laden air at 120 DEGF and 96% RH.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. General:
    1. Consider ductwork, piping and equipment as exposed, except as otherwise indicated.
    2. Consider ductwork, piping and equipment in walls, partitions, floors, pipe chases, pipe shafts and duct shafts as concealed.
        a. Consider ductwork, piping and equipment above ceilings as concealed.
    3. Provide release for insulation application after installation and testing is complete.

a. Apply insulation on clean, dry surfaces after inspection.

4. Provide insulation continuous through wall, roof and ceiling openings, pipe hangers, supports and sleeves.
5. Provide insulation with vapor barrier for piping, ductwork and equipment where surfaces may be cooler than surrounding air temperatures.
   a. Provide vapor barrier (0.17 perm-IN; ASTM C553) continuous and unbroken.
   b. Hangers, supports, anchors, and related items that are secured directly to cold surfaces must be adequately insulated and vapor-sealed to prevent condensation.
6. Apply specified adhesives, mastics and coatings at the manufacturer's recommended coverage per unit volume.

C. Piping Insulation - Elastomeric:
   1. Do not insulate until satisfactory completion of required pressure testing.
   2. Apply insulation to clean, dry surfaces.
   3. Slip insulation on pipe prior to connection.
      a. Whenever the slip-on technique is not possible provide insulation neatly slit and snapped over the pipe.
   4. Fabricate and install fitting cover over insulation according to manufacturer's recommendations.
   5. Seal joints, slits, miter-cuts and other exposed edges of insulation with adhesive, recommended by the insulation manufacturer, to ensure complete vapor barrier.

D. Piping Insulation - Fiberglass:
   1. Apply over clean dry pipe.
      a. Butt all joints together firmly.
   2. Seal joints, slits, miter-cuts and other exposed edges of insulation as recommended by the insulation manufacturer.
   3. Insulate fittings, valves, and flanges with insulation thickness equal to adjacent pipe.
   4. PVC pipe jacket:
      a. Apply jacketing with a minimum of 1 IN overlap.
         1) Weld longitudinal and circumferential seams with adhesives as recommended by manufacturer.
      b. Provide slip-joints every 30 FT and between fittings if distance exceeds 8 FT.
         1) Construct slip-joints by overlapping jacket sections 6 to 10 IN.
      c. Provide pre-molded PVC covers of same material and manufacturer as jacket for fittings, valves, flanges, and related items in insulated piping systems.
   5. Aluminum pipe jacket:
      a. Field-applied aluminum jacket with vapor-sealed longitudinal and butt joints.
      b. Provide smooth and straight joint with a minimum 2 IN overlap.
      c. Secure joints with corrosion-resistant screws spaced 0.25 to 0.50 IN back from edge.
      d. Center spacing of screws 5 IN maximum or as required to provide smooth tight-fitted joints.
      e. Place joints on least exposed side of piping to obtain neat appearance.

E. Equipment: Install per manufacturer's instructions.

F. Ductwork Insulation - Fiberglass:
   1. Flexible insulation:
      a. Butt edges tightly.
         1) Secure insulation with adhesive applied in 6 IN strips on 12 IN centers and/or pins, applied on not more than 18 IN centers so that the insulation conforms to the duct surfaces uniformly and firmly.
      b. Seal joints with facing overlap or 4 IN wide strips of like facing material adhered and stapled in place.
      c. Properly seal any penetration in vapor barrier facing with adhesive.
      d. Cut insulation slightly longer than the perimeter of the duct to ensure full thickness at corners.

2. Semi-rigid insulation and duct interior lining board:
   a. Impaling over pins.
      1) Apply insulation with edges tightly butted.
      2) Apply insulation with mechanically welded fasteners to the duct and secured with speed clips.
      3) Clip pins off close to clip.
      4) Space pins as required to hold insulation firmly against duct surface but not less than one pin per 1.5 SQFT.
      5) Seal joints and speed clips with 3 IN wide strip of facing adhered with adhesive.
   b. If the welded pin method is impossible, secure insulation to the duct with adhesive.
      1) Cover the entire surface of duct with adhesive.
      2) Use corner metal angle to protect edge of insulation.
      3) Protect edge of insulation.
      4) Seal joints as above.

3.2 REPAIR

A. Whenever any factory applied insulation or job-applied insulation is removed or damaged, replace with the same quality of material and workmanship.

3.3 SCHEDULES

A. Refrigeration Lines (suction and liquid):
   1. Type: Elastomeric.

B. Pipe, Fittings and Valves:
   1. Type: Fiberglass, unless noted otherwise.

APPLICATION	PIPE SIZE	THICKNESS	JACKET
Cold Water (Potable) and Non-Potable Water located in unheated or concealed spaces	1-1/2 IN and less	1 IN	PVC for piping exposed inside building and less than 8 ft. above finished floor. Jacket not required for concealed piping, or pipe higher than 8ft above finished floor.
	Over 1-1/2 IN	1-1/2 IN	
Cold Water (Potable) and Non-Potable Water located in heated spaces	All sizes	Insulation not required	-
Hot Water (domestic), Hot Water Recirculation, and Tepid water (to emergency showers)	1-1/2 IN and less	1 IN	PVC for piping exposed inside building and less than 8 ft. above finished floor. Jacket not required for concealed piping, or pipe higher than 8ft above finished floor.
	Over 1-1/2 IN	1-1/2 IN	
Refrigeration Lines (35 - 60 DEGF)	1 IN and less Over 1 IN	3/4 IN 1 IN	Aluminum outside of building; PVC for exposed piping inside building and less than 8 ft of finished floor. Jacket not required for concealed piping, or pipe

APPLICATION	PIPE SIZE	THICKNESS	JACKET
			higher than 8ft above finished floor.

C. Equipment:

EQUIPMENT	INSULATION SYSTEM
Below drain pans serving cooling coils	1 IN flexible elastomeric closed cell sheet.
Hydropneumatics tanks	Uninsulated
Domestic hot water expansion tanks	Uninsulated
Hot water recirculation pumps	Uninsulated

D. Ductwork:
   1. Type: Fiberglass.

DUCT SERVICE	INSULATION AND THICKNESS	MINIMUM R-VALUE $(HR\text{-}FT^2\text{-}DEGF)/BTU$
Supply, return and outside air ducts, located in concealed or unconditioned spaces.	3 IN, flexible, with vapor barrier	12.0
Exhaust and outside air ducts, located inside conditioned buildings, between damper and wall louver, including plenums.	3 IN, flexible or semi-rigid, with vapor barrier	12.0
Supply, return, outside air and exhaust ducts located inside buildings and exposed within the conditioned space that it serves.	Uninsulated, unless noted otherwise	N/A

END OF SECTION

This page intentionally left blank.

SECTION 40 61 13
PROCESS CONTROL SYSTEM GENERAL REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Basic requirements for complete instrumentation system for process control.
2. Requirement for Contractor to subcontract the Process Control System portion of the Work to a "Hardware Systems Integrator."
3. Programming is by Others – NOT IN CONTRACT.

B. Related Specification Sections include but are not necessarily limited to:
1. Section 01 81 33- Cyber Security Requirements.
2. Section 10 14 00 – Identification Devices.
3. Section 40 61 21 – Process Control System Testing.
4. Section 40 61 43 – Surge Protection Devices for Instrumentation and Control Equipment.
5. Section 40 61 93 – Process Control System Input-Output List.
6. Section 40 61 96 – Process Control Descriptions.
7. Section 40 62 05 – Computer System Hardware and Ancillaries.
8. Section 40 63 43 – Programmable Logic Controllers.
9. Section 40 66 05 – Network Equipment.
10. Section 40 66 40 – Telemetry Systems.
11. Section 40 67 00 - Control System Equipment Panels and Racks.
12. Section 40 67 63 – Uninterruptible Power Supply.
13. Section 40 71 00 – Flow Instrumentation.
14. Section 40 72 00 – Level Instrumentation.
15. Section 40 73 00 – Pressure Instrumentation.
16. Section 40 74 00 – Temperature Instrumentation.
17. Section 40 75 00 – Process Liquid Analytical Measurement
18. Section 40 76 00 – Process Gas Analytical Measurement
19. Section 40 78 00 – Panel Mounted Instruments.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
1. The International Society of Automation (ISA):
   a. 7.0.01, Quality Standard for Instrument Air.
   b. S5.1, Instrumentation Symbols and Identification.
   c. S5.3, Graphic Symbols for Distributed Control/Shared Display Instrumentation, Logic and Computer Systems.
   d. S20, Standard Specification Forms for Process Measurement and Control Instruments, Primary Elements and Control Valves.
   e. 62443-3-3 Security for Industrial Automation and Control Systems, Part 3-3: System Security Requirements and Security Levels
2. National Electrical Manufacturers Association (NEMA):
   a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
3. National Fire Protection Association (NFPA):
   a. 70, National Electrical Code (NEC).
   b. 79, Electrical Standard for Industrial Machinery
4. Underwriters Laboratories, Inc. (UL):
   a. 913, Standard for Safety, Intrinsically Safe Apparatus and Associated Apparatus for Use in Class I, II, and III, Division 1, Hazardous (Classified) Locations.
5. National Institute of Standards and Technology (NIST)
   a. SP 800-82, Guide to Industrial Control Systems (ICS) Security

B. Qualifications:
   1. Hardware System Integrator:
      a. Experience:
         1) Have satisfactorily provided control system integration for a minimum of five projects of similar magnitude and function.
      b. Certification:
         1) Control Systems Integrators Association (CSIA) Certification.
      c. Qualification:
         1) Regularly engaged in the design and the installation of process control and instrumentation systems and their associated subsystems as they are applied to the municipal water and wastewater industry.
         2) Capable of responding to on-site problems within 24 hours of notice. Provide an on-site response within 24 hours of notification starting at two months before scheduled start up to two months after startup completion.
   2. Panel Fabricator:
      a. Experience:
         1) Have satisfactorily fabricated control panels for a minimum of five projects of similar magnitude and function.
      b. Certification:
         1) UL508A & UL698A Certification.

C. Miscellaneous:
   1. Comply with electrical classifications and NEMA enclosure types shown on Drawings.

1.3 DEFINITIONS

A. Calibrate: To standardize a device so that it provides a specified response to known inputs.

B. Hazardous Areas: Class I, II or III areas as defined in NFPA 70.

C. Highly Corrosive and Corrosive Areas: Rooms or areas identified on the Drawings where there is a varying degree of spillage or splashing of corrosive materials such as water, wastewater or chemical solutions; or chronic exposure to corrosive, caustic or acidic agents, chemicals, chemical fumes or chemical mixtures.

D. Intrinsically Safe Circuit: A circuit in which any spark or thermal effect is incapable of causing ignition of a mixture of flammable or combustible material in air under test conditions as prescribed in UL 913.

E. Hardware System Integrator: A Contractor/Subcontractor who combines instrumentation, control devices, hardware, and networking products from multiple vendors to provide a fully functioning control system.

1.4 SYSTEM DESCRIPTION

A. Hardware Control System Requirements:
   1. This Specification Section provides the general requirements for the control system.
   2. The control system consists of all primary elements, transmitters, switches, controllers, computers, communication devices, recorders, indicators, panels, signal converters, signal boosters, amplifiers, special power supplies, special or shielded cable, special grounding or isolation, auxiliaries, wiring, and other devices required to provide complete control of the plant as specified in the Contract Documents.

B. Utilization of Hardware System Integrator:
   1. Retain a Hardware System Integrator subcontractor to provide a fully functioning control system.
      a. The Hardware System Integrator shall be responsible for the provision of an integrated control system fully functioning in accordance with the requirements of the Contract Documents.
      b. As a minimum, the responsibilities of the Hardware System Integrator shall include:

1) Supervision of installation and final connections.
2) Calibrations.
3) Preparation of Drawings and Operation and Maintenance Manuals.
4) System start-up.
5) Training.
6) Demonstration of substantial completion and all other aspects of the control system.

2. Provide all required coordination of instrumentation with other work to ensure that necessary wiring, conduits, contacts, relays, converters, and incidentals are provided in order to transmit, receive, and control necessary signals to other control elements, to control panels, and to receiving stations.

1.5 SUBMITTALS

A. Shop Drawings:
1. Submittals shall be original printed material or clear unblemished photocopies of original printed material.
   a. Facsimile information is not acceptable.
2. Limit the scope of each submittal to one Specification Section.
   a. Each submittal must be submitted under the Specification Section containing requirements of submittal contents.
   b. Do not provide any submittals for Specification Section 40 61 13.
3. Product technical data including:
   a. Acknowledgement that products submitted meet requirements of standards referenced.
   b. Equipment catalog cut sheets.
   c. Instrument data sheets:
      1) ISA S20 or approved equal.
      2) Separate data sheet for each instrument type.
   d. Materials of construction.
   e. Minimum and maximum flow ranges.
   f. Pressure loss curves.
   g. Physical limits of components including temperature and pressure limits.
   h. Size and weight.
   i. Electrical power requirements and wiring diagrams.
   j. NEMA rating of housings.
   k. Submittals shall be marked with arrows to show exact features to be provided.
4. Comprehensive asset inventory of all networked components:
   a. Provide in Excel spreadsheet format.
   b. Coordinate with the Contracting Officer to determine the preferred method of delivery to assure security of information contained in asset inventory.
   c. Include:
      1) Device ID.
      2) Manufacturer.
      3) Model Number.
      4) Serial Number.
      5) Device Use description.
      6) Firmware.
5. Network Diagrams:
   a. Provide in VISIO, AUTOCAD, and PDF formats.
   b. Coordinate with the Contracting Officer to determine the preferred method of delivery to assure security of information contained in Network Diagrams.
   c. Logical Network Diagram(s):
      1) Depict information flow through network(s), and include:
         a) Major network devices, subnets, and VLANs.
         b) Include all wireless communication devices.
         c) Include the following information for each networked device:

(1) Device ID.
(2) Device description.
(3) Manufacturer/model number.
(4) MAC address.
(5) IP address.
(6) Ports and Protocols

d. Physical Network Diagram(s):
1) Show all network components, ports, protocols, connections and cables.
a) Include all wireless communication devices.

6. Process connected instrument installation details containing the following minimum information:
a. Bill of materials providing as a minimum the following information:
1) Tube material and size.
2) Connection size.
3) Fitting size, material, and rating.
4) Valve type and material.
5) Instrument description.
6) Pipe stand size and material.
b. Tube slope requirements.
c. Required elevations and dimensions.
7. Comprehensive set of wiring diagrams as specified in Section 40 67 00.
8. Panel fabrication drawings as specified in Section 40 67 00.
9. PLC equipment drawings.
10. Nameplate layout drawings.
11. Drawings, systems, and other elements are represented schematically in accordance with ISA S5.1 and ISA S5.3.
a. The nomenclature, tag numbers, equipment numbers, panel numbers, and related series identification contained in the Contract Documents shall be employed exclusively throughout submittals.
12. All panel and wiring drawings shall be provided in both hardcopy and softcopy.
a. Furnish electronic files on Government's designated electronic media.
b. Drawings in AUTOCAD format.
13. Provide a parameter setting summary sheet for each field configurable device.
14. Certifications:
a. Documentation verifying that calibration equipment is certified with NIST traceability.
b. Approvals from independent testing laboratories or approval agencies, such as UL, FM or CSA.
1) Certification documentation is required for all equipment for which the specifications require independent agency approval.
15. Testing reports: Source quality control reports.

B. Qualifications Submittal:
1. Documentation verifying contractor/subcontractor adherence to specified certifications and qualifications.

C. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
2. All Shop Drawings shall be modified with as-built information/corrections.
3. Instrumentation and Control Equipment Operation and Maintenance Manual Content:
a. Provide a printed copy of the following sheets following the Equipment Record sheets or ISA Data Sheets.
1) Loop Check-out Sheet.
2) Instrument Certification Sheet.
3) Final Control Element Certification Sheet.

b. Provide the following detailed information:
    1) Use equipment tag numbers from the Contract Documents to identify equipment and system components.
    2) As-constructed fabrication or layout drawings and wiring diagrams.
    3) As-constructed network asset inventory, and physical and logical network diagrams.
        a) Coordinate with the Contracting Officer to determine the preferred method of delivery to assure security of this information.
c. Password Summary Report:
    1) Provide a Password Summary Report documenting the administrative password for each device and describing the procedure to change the password for each device. Do not provide the Password Summary Report in electronic format. Provide two hardcopies of the Password Summary Report, each copy in its own sealed envelope.
    2) Provide listing of all temporary usernames setup for each device.
d. Additional information as required in the associated equipment or system Specification Section.

4. Warranties: Provide copies of warranties and list of factory authorized service agents.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Do not remove shipping blocks, plugs, caps, and desiccant dryers installed to protect the instrumentation during shipment until the instruments are installed, and permanent connections are made.

PART 2 - PRODUCTS

2.1 NEMA TYPE REQUIREMENTS

A. Provide enclosures/housing for control system components in accordance with the area designations provided on the Drawings.
1. Areas designated as wet: NEMA Type 4.
2. Areas designated as wet and/or corrosive, chemical rooms: NEMA Type 4X.
3. Areas designated as Class I hazardous, Groups A, B, C, or D as defined in NFPA 70:
    a. NEMA Type 7 unless all electrical components within enclosure utilize intrinsically safe circuitry.
        1) Utilize intrinsically safe circuits to the maximum extent practical and as depicted in the Contract Documents.
4. Either architecturally or non-architecturally finished areas designated as dry, noncorrosive, and nonhazardous: NEMA Type 12.
5. Areas designated to be subject to temporary submersion: NEMA 6P.

2.2 PERFORMANCE AND DESIGN REQUIREMENTS

A. Unless stated otherwise, system operating criteria are as follows:
1. Performance: All instruments and control devices shall perform in accordance with manufacturer's specifications.

2.3 ACCESSORIES

A. Provide identification devices for instrumentation system components in accordance with Specification Section 10 14 00.

B. Provide corrosion resistant spacers to maintain 1/4 IN separation between equipment and mounting surface in wet areas, on below grade walls and on walls of liquid containment or processing areas such as Clarifiers, Digesters, Reservoirs, etc.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Wherever feasible, use bottom entry for all conduit entry to instruments and junction boxes.

B. Install electrical components per the requirements of the Electrical design.

3.2 FIELD QUALITY CONTROL

A. See Section 01 75 00 and Section 40 61 21.

END OF SECTION

Loop Check-out Sheet

Project Name:		Contracting Officer's Project No. (if applicable):	Page of
Project Owner:		Regulatory Agency Project No. (if applicable):	
HDR Project No.:		Date:	

LEAK AND TERMINATION/CONTINUITY CHECKS

DESCRIPTION	FIELD					CONTROL CAB	
	LEAK CHECK(1)			TERM/CONT CHECK(2)		TERM/CONT CHECK(2)	
	Device Tag No.	Process Conn.	Signal Tube	Device Tag No.	Termination Ident.	Device Tag No.	Termination Ident.

1. Leak check for pneumatic signal tubing to be per ISA-PR7.1.
2. Termination/continuity check includes check at terminated equipment for: (a) correct polarity, (b) appropriate signal generation, transmission and reception, and (c) correct shield & ground terminations.

OPERATOR INTERFACE CHECK-OUT

MONITORING POINTS OBSERVED

PARAMETER TYPE	TAG NO.	TAG NO.	TAG NO.	TAG NO.	TAG NO.	TAG NO.
PROCESS VAR						
EQUIP STATUS						
ALARM POINT						

OPERATOR CONTROL FUNCTIONS CHECKED

FUNCTION TYPE	TAG NO.	LOCATION	TAG NO.	LOCATION	TAG NO.	LOCATION

FINAL CONFIGURED SETTINGS

TAG NO.	SWITCH & ALARM SP	CONTROLLERS			
		Gain	Reset, rpm	Deriv. (rate), min	PV Set Point

Describe all interlocks checked, equipment started/stopped, valves/operators stroked. Describe modes of operation checked, and location of operator interface (local/remote).

_____

_____

_____

I certify that the control loop referenced on this page has been completely checked and functions in accordance with applicable drawings and specifications.

Certified by:_____ Date:_____
(Work Performed By)

Instrument Certification Sheet

Project Name:	Contracting Officer's Project No. (if applicable):
Project Owner:	Regulatory Agency Project No. (if applicable):
HDR Project No.	Date:
Control Loop No.:	
Instrument Tag No.	Transmitter/gauge span:
Manufacturer:	Switch set-point:
Model No.	Switch dead band:
Serial No.	Switch range:

TRANSMITTERS AND INDICATORS

	INCREASING INPUT			DECREASING INPUT		
% OF SPAN	**INPUT**	**OUTPUT**	**ERROR (% of span)**	**INPUT**	**OUTPUT**	**ERROR (% of span)**
0%						
25%						
50%						
75%						
100%						
Other (if applicable)						
Other (if applicable)						

SWITCHES

	INCREASING INPUT			DECREASING INPUT		
ACTUATION POINT	**INPUT**	**OUTPUT**	**ERROR (% of range)**	**INPUT**	**OUTPUT**	**ERROR (% of range)**
High (Increasing input)						
Low (Decreasing input)						

Maximum allowable error (per Contract Documents): _____

Remarks: _____

_____

_____

CALIBRATION EQUIPMENT UTILIZED

DEVICE TYPE	MFR/MODEL NO.	ACCURACY	NIST TRACEABILITY?

Certified by: _____ Date Certified: _____

Final Control Element Certification Sheet

Project Name:	Contracting Officer's Project No. (if applicable):
Project Owner:	Regulatory Agency Project No. (if applicable):
HDR Project No.	Date:
Control Loop No.:	

Tag No.	
Description:	
Manufacturer:	
Model No.	
Serial No.	

Actuator:	Pneumatic: _____ Electric: _____
Positioner:	Direct: _____ Reverse: _____
Positioner:	Input: _____ Output: _____
I/P Converter:	Input: _____ Output: _____
Valve to	_____ on air failure
Valve to	_____ on power failure

I/P CONVERTER

	INCREASING INPUT			DECREASING INPUT		
% OF SPAN	**INPUT**	**OUTPUT**	**ERROR (% of span)**	**INPUT**	**OUTPUT**	**ERROR (% of span)**
0%						
25%						
50%						
75%						
100%						

Specified I/P converter accuracy: _____ % of span.

FINAL CONTROL ELEMENT

	INCREASING INPUT			DECREASING INPUT		
% OF SPAN	**INPUT**	**TRAVEL**	**ERROR (% of full travel)**	**INPUT**	**TRAVEL**	**ERROR (% of full travel)**
0%						
25%						
50%						
75%						
100%						

Remarks: _____

CALIBRATION EQUIPMENT UTILIZED

DEVICE TYPE	MFR/MODEL NO.	ACCURACY	NIST TRACEABILITY?

Certified by:_____ Date Certified:_____

SECTION 40 61 21
PROCESS CONTROL SYSTEM TESTING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Requirements for testing of the Process Control System (PCS) hardware prior to shipment to the site for installation.
      a. Inspection.
         1) System hardware operational testing.
      b. Test records.
   2. Requirements for field testing of the PCS hardware.
      a. Process Control System checkout.
      b. Instrument calibration.
      c. Loop testing/checkout.
      d. Integrated Site Acceptance Test (SAT).

B. Related Sections include but are not necessarily limited to:
   1. Section 40 66 40 - Telemetry Systems.
   2. Section 40 61 13 - Process Control System General Requirements.
   3. Section 40 61 96 - Process Control Descriptions.
   4. Section 40 63 43 - Programmable Logic Controllers.
   5. Section 40 67 00 - Control System Equipment Panels and Racks.

1.2 SUBMITTALS

A. See Specification Section 40 61 13 - Process Control System General Requirements.

B. Submit factory acceptance test (FAT) procedures not less than 60 calendar days prior to the scheduled start of the FAT. Include the following:
   1. Modular test plans and procedures to allow individual test segments to be repeated if necessary.
   2. Description of individual test segments and steps comprising each segment, particularly the methods and processes to be followed.
   3. Procedures for monitoring, correcting, and retesting variances.
   4. Procedures for controlling and documenting all changes made to the hardware after the start of testing.
   5. Descriptions of test equipment.
   6. Expected results for each segment, including pass/fail criteria.
   7. Documentation record forms to be used to certify results of factory testing.
   8. Block diagrams of the hardware test configuration, such as PLC external communication links, and any test or simulation hardware.

C. Post factory acceptance test:
   1. Final documentation certifying that all Factory tests have passed with satisfactory results.

D. Field Testing Shop Drawings:
   1. Site Acceptance Test (SAT) Plan.
      a. Divide into separate sections based on process systems.
      b. Each section shall include a sequential step-by-step checkout plan.
   2. Typical Loop Checkout Sheets, if other than the form included within contract documents.
   3. Typical Instrument Certification Sheets, if other than the form included within contract documents.
   4. Typical Final Control Element Certification Sheets, if other than the form included within contract documents.

E. Contract Closeout Information:
   1. Site Acceptance Test (SAT) Report.
   2. Completed Loop Checkout Sheets.
   3. Completed Instrument Certification Sheets.
   4. Completed Final Control Element Certification Sheets.
   5. All system startup and checkout records, test records, recommendations and information furnished by Manufacturer's representatives during testing and startup, and amended documentation specified herein.

PART 2 - PRODUCTS

2.1 FACTORY ACCEPTANCE TESTING (FAT) GENERAL:

A. Utilize software-based process control simulation and hardware-based I/O simulation panel when conducting the full system test. Software based process simulation shall provide a real time, virtual environment that interfaces with the PLC system applications software and programming to mimic the behavior of the real process equipment and systems that are monitored and controlled by the PLC system, including the capability to force individual or groups of PLC system I/O.

B. Demonstration of communication between processors and to remote I/O racks shall be included in the Test Procedure, where applicable. Test shall include demonstration that communication performance (speed, error rates, failure and recovery) conforms to specified requirements.

C. Any tested segment or function that does not pass in accordance with agreed upon pass/fail criteria shall be retested until the segment or function passes the test.
   1. Pass/fail criteria shall be in accordance with criteria set forth in an approved shop drawing submittal.
   2. The Contractor shall bear all additional travel/accommodation expenses incurred by the Government due to extension of test dates for retesting of failed segments or functions.

D. Notify the Contracting Officer in writing a minimum of 15 calendar days prior to the Factory Tests.
   1. Contracting Officer has the option to witness all required tests.

E. Testing Documentation
   1. Make following documentation available at test site during the tests:
      a. Contract Documents.
      b. Factory Demonstration Testing procedures.
      c. List of equipment to be testing including make, model, and serial number.
      d. Shop Drawing submittal data for equipment being tested.
   2. Contracting Officer reserves the right to review all documentation at any time during testing.

2.2 FAT INSPECTION

A. Inspect all panels, operator interface terminals and cabinets prior to conducting the FAT to verify quality of workmanship and conformance to the Contract Documents and approved Shop Drawings. Completed inspection forms shall be available for review at time of FAT. Inspection shall include, but not be limited to, the following:
   1. Nameplates, warning labels, and tags including correct spelling, color and size of labeling and characters and methods of attachment.
   2. Enclosure surfaces flatness, finish, and color.
   3. Proper operation of doors, catches, and locks.
   4. Wire and cable type, sizes, tagging, color coding and tag placement in readily viewable position/orientation.
   5. Wiring runs across hinges properly installed.
   6. Proper wiring routing, layout, practices, and grounding.
   7. All electrical circuits checked for continuity.

8. Terminal block contact ratings and numbers.
9. Fuses and circuit breakers are correct size and type.
10. Terminal block, fuse, breaker, and other required installed spares.
11. General arrangement and space allocation, including spare usable space in panel and on panel door.
12. PLC equipment (chassis, power supply, power conditioners, processor, adapters, I/O and specialty modules, installed spares, spare usable space) is appropriate.
13. Proper placement, isolation, and labeling of intrinsic safety devices and associated wiring.
14. Equipment and devices are properly installed and secured to subpanels and doors using appropriate methods and in appropriate orientation.
15. Fans, blowers, heaters, air-conditioning units, and other environmental controls are operational, unobstructed, and installed with sufficient clearances recommended by component manufacturers.

2.3 FAT - SYSTEM HARDWARE OPERATIONAL TESTING

A. Assemble and interconnect the relevant portions of the PLC system as a network that mimics the network that is designed to be installed at the Plant site.
   1. As a minimum, the test shall include the following major components:
      a. All control panels that interface with the control network and are provided by the Hardware System Integrator, including local control panels, remote control panels, hardwired/relay logic and PLC based panels.
      b. Remote I/O Panels.
      c. Operator Interface Terminals (OITs).
      d. Ethernet Networks.
      e. Uninterruptible Power Supplies.
      f. Network switches, Routers, Firewalls.

B. Packaged systems not present at FAT shall be simulated; communications interfaces shall be simulated to/from vendor confirmed interface registers.

C. All system hardware components and equipment shall be tested to verify proper operation as standalone units. Tests shall include, but not be limited to, the following:
   1. AC/DC power checks.
   2. Power fail/restart tests.
   3. Verification of UPS power to required components.
   4. Diagnostics checks.
   5. Verification that all equipment is functioning properly.
   6. All electrical circuits energized simultaneously and continuously for 48 HRS without failures.
   7. All alarm circuits connected to simulated alarm contacts to verify operation.
   8. All interlock and shutdown circuits checked for operability and proper function by means of simulated contact.
   9. All hardwired input/output devices and components, and all network connected I/O shall be tested to verify operability and basic calibration.
   10. Simulate operation of electronic control and receiving instruments and circuits.
   11. Any other test required to place the PCS equipment in an operating state.
   12. Verify all redundant components fail-over hands-free without interruption of control process.
   13. Verify all servers and networking equipment are in an operating state with no unresolved errors in error logs.
   14. Verify that all components are integrated and communicating for full simulation test including any simulation software if applicable.
   15. Compliance with Contract Documents.

2.4 FAT - SYSTEM SOFTWARE DEMONSTRATION

A. Demonstrate the proper operation based on simulation of all the analog inputs/outputs and all of the discrete inputs and outputs, as indicated on the I/O list. In addition to testing allocated I/O, all spare I/O points shall also be simulated and tested to the degree possible. Demonstration shall show that the monitoring and control software and configuration and programming associated with the input/output points (hardwired and network connected) performs as intended.
   1. PLC I/O testing shall include, as a minimum, the following:
      a. Simulate a field digital input for each PLC input point at the terminal strip of the control panel and verify PLC response.
      b. Simulate a field analog input for each PLC analog input point at the terminal strip of the control panel and verify PLC response.
      c. Force an output for each PLC digital output point and verify the signal presence at the terminal strip in the control panel.
      d. Simulate an analog output for each PLC analog output point and verify signal presence at the terminal strip in the control panel.

2.5 FAT TEST RECORDS:

A. Maintain records of all factory test results. As a minimum, include the following items:
   1. Name of test procedure.
   2. Date of the test.
   3. Test results for each test segment including a passed/failed indication.
   4. Signatures of individual(s) performing the test and witnesses thereof.

PART 3 - EXECUTION

3.1 GENERAL

A. Maintain accurate daily log of all testing activities, calibration functions, and final set point adjustments.

B. All testing may be witnessed by the Contracting Officer.

3.2 INSTALLATION AND CHECKOUT

A. Provide the following (after successful completion of the Factory Acceptance Test):
   1. Check and approve the installation of all Process Control System (PCS) components (including but not limited to control panels, field instruments, networking equipment and racks, patch panels and all cable and wiring interconnects between various equipment, panels and devices) prior to placing the various processes and equipment into operation. As a minimum, include:
      a. Verification and documentation that all field instruments function properly and are properly calibrated.
      b. Testing of all final control actions.

B. Provide services of authorized manufacturer's representatives to check and verify the correctness of the installation and operation of equipment.

3.3 PRE-DEMONSTRATION PERIOD TESTING

A. Instrumentation Calibration:
   1. Verify and document that all instruments and control devices are calibrated to provide the performance required by the Contract Documents.
   2. Factory furnished calibration certifications are acceptable for the following:
      a. Flow meters.
      b. Pressure sensors utilized with annular sleeve.
      c. Temperature sensors.
   3. On-site calibration verification is required for all other instruments, including "smart" transmitters that have been factory calibrated.

a. Provide calibration checks at 0 %, 25 %, 50 %, 75 % and 100 % of span for pressure transmitters and gages.
    1) Check for both increasing and decreasing input signals to detect hysteresis.
b. In addition to factory calibration certification, temperature sensors and gages shall be checked at a single point for conformance to required accuracy.
c. Level transducers/transmitters shall be checked at two points in addition to zero.
d. Analytical sensors shall be calibrated in accordance with manufacturer's recommendations.
e. Check operation of all switches to verify actuation occurs in accordance with manufacturer's specified accuracy.
f. Replace any instrument which cannot be properly adjusted.

4. Mark range, date, set point and calibrator's initials on each instrument by means of blue or black ink on a waterproof tag affixed to the instrument.
5. Calibration equipment shall be certified by an independent agency with traceability to NIST.
    a. Certification shall be up to date.
    b. Use of equipment with expired certifications shall not be permitted.
6. Calibration equipment shall be at least three times more accurate as the device being calibrated.

B. Loop Testing:
1. Utilize the Loop Check-Out Sheet located at the end of Section 40 61 13 (or Contracting Officer approved equivalent) to document on-site calibration checks.
    a. Utilize separate loop checkout form for each loop.
2. Minimum loop checkout requirements:
    a. Use actual signals where available.
    b. Check control signal generation, transmission, reception and response for all inputs and outputs for all control loops under simulated operating conditions by imposing a signal on the loop at the instrument connections.
    c. Closely observe controllers, indicators, transmitters, HMI displays, OIT displays, recorders, alarm and trip units, remote set points, ratio systems, and other components.
        1) Verify that readings at all loop components are in agreement.
        2) Make corrections as required.
        3) Following any corrections, retest the loop.
    d. Stroke all control valves, cylinders, drives and connecting linkages from the local control station and from the control room operator interface.
    e. Check all interlocks for proper functionality.
    f. Record all set point and calibration changes on all affected Contract Documents and turn over to Contracting Officer.

C. Integrated Site Acceptance Test (SAT):
1. Following the testing of each individual loop, perform an integrated Site Acceptance Test (SAT) to verify that all equipment is operating as described in the Contract Documents and equipment manufacturers recommendations. Checks to include:
    a. The intended monitoring and control functions of the PCS including all vendor-furnished equipment packaged systems, communications, operator interfaces, statuses, alarms, trends and recording of historical tags.
    b. Integrated testing on a system-by-system basis as described in the submitted Site Acceptance Test (SAT) plan sequential step-by-step checkout plan.
        1) Documentation with initials of the individuals who perform each test.
2. Provide minimum fifteen (15) day notice to the Contracting Officer prior to commencing the SAT.
3. For Roaring Spring Pumphouse: Construct LCP-53117 on the North Rim and test it operates properly with Finished Water Dissipation Valve EDS-53117 before equipment is brought down to RSPH.
4. Promptly address and re-test any malfunctions, issues and incompletions arising during the SAT at no additional expense to Government.

D. At the successful conclusion of all testing specified in this article, submit the “Instrumentation Supplier's Instrumentation Installation Certificate” required in Section 01 75 00 - Checkout and Start-up Procedures.

3.4 DEMONSTRATION PERIOD

A. Demonstrate the functional integrity of the Process Control System for the entire facility in accordance with the demonstration period requirements in Section 01 75 00 - Checkout and Start-up Procedures as evidence of Substantial Completion.

END OF SECTION

SECTION 40 61 43
SURGE PROTECTION DEVICES FOR INSTRUMENTATION AND CONTROL EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Category IC1 SPD - Dedicated 120 VAC circuit, series connection, control panel mounted.
   2. Category IC2 SPD - Individual equipment plug-in device (point of use protection).
   3. Category IC3 SPD - Discrete 120 VAC control signal, control panel mounted.
   4. Category IC4 SPD - Analog instrumentation signal, field mounted.
   5. Category IC5 SPD - Analog instrumentation signal, control panel mounted.
   6. Category IC6 SPD - Combination 120 VAC circuit and analog signal, field mounted.
   7. Category IC7 SPD - Discrete low voltage control signal, control panel mounted.
   8. Category IC8 SPD - Data line, control panel mounted.

B. Related Sections include but are not necessarily limited to:
   1. Section 40 61 13 - Process Control Systems General Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. C62.41.2, Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits.
   2. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   3. Underwriters Laboratories, Inc. (UL):
      a. 497B, Standard for Safety Protectors for Data Communications and Fire-Alarm Circuits.
      b. 1283, Standard for Electromagnetic Interference Filters.
      c. 1363, Standard for Relocatable Power Taps.
      d. 1449, Standard for Surge Protective Devices.

B. Qualifications:
   1. Provide devices for a manufacturer who has been regularly engaged in the development, design, testing, listing and manufacturing of SPDs of the types and ratings required for a period of 10 years or more and whose products have been in satisfactory use in similar service.
   2. Upon request, suppliers or manufacturers shall provide a list of not less than three customer references showing satisfactory operation.

1.3 DEFINITIONS

A. Clamping Voltage: The voltage measured at the end of the 6 IN output leads of the SPD and from the zero-voltage reference to the peak of the surge when the applied surge is induced at the 90-degree phase angle of the applied system frequency voltage.

B. Let-Through Voltage: The voltage measured at the end of the 6 IN output leads of the SPD and from the system peak voltage to the peak of the surge when the applied surge is induced at the 90-degree phase angle of the applied system frequency voltage.

C. Maximum Continuous Operating Voltage (MCOV): The maximum steady state voltage at which the SPD can operate and meet its specification within its rated temperature.

D. Maximum Surge Current:

1. The maximum 8 x 20 microsecond surge current pulse the SPD is capable of surviving on a single-impulse basis without suffering either performance degradation or more than 10% deviation of clamping voltage at a specified surge current.
2. Listed by mode, since number and type of components in any SPD may vary by mode.

E. Protection Modes: This parameter identifies the modes for which the SPD has directly connected protection elements, i.e., line-to-neutral (L-N), line-to-line (L-L), line-to-ground (L-G), neutral-to-ground (N-G).

F. Surge Current per Phase:
1. The per phase rating is the total surge current capacity connected to a given phase conductor.
2. For example, a wye system surge current per phase would equal L-N plus L-G; a delta system surge current per phase would equal L-L plus L-G.
a. The N-G mode is not included in the per phase calculation.

G. System Peak Voltage: The electrical equipment supply voltage sine wave peak (i.e., for a 120 V system the L-N peak voltage is 170 V).

1.4 SUBMITTALS

A. Shop Drawings:
1. For named products, submit only a catalog cut sheet.
a. For all other products, submit the data required below.
2. See Specification Section 40 61 13 for general process control systems requirements.
3. Product technical data for non-specified models:
a. Manufacturer's experience.
b. Standard catalog cut sheet.
c. Electrical and mechanical drawing showing unit dimensions, weights, mounting provisions, connection details and layout diagram of the unit.
d. Create a Product Data Sheet for each different model number of SPD provided.
1) Data in the Product Data Sheet heading:
a) SPD Type per PART 2 of the Specification.
b) Manufacturer's Name.
c) Product model number.
2) Data in the Product Data Sheet body:
a) Column one: Specified value/feature of every paragraph of PART 2 of the Specification.
b) Column two: Manufacturer's certified value confirming the product meets the specified value/feature.
3) Data in the Product Data Sheet closing:
a) Signature of the manufacturer's official (printed and signed).
b) Title of the official.
c) Date of signature.

B. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.5 WARRANTY

A. The manufacturer shall provide a minimum of a five-year Limited Warranty from date of shipment against failure when installed in compliance with applicable national/local electrical codes and the manufacturer's installation, operation and maintenance instructions.

B. See Specification Section 01 67 00 – Product Requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 CATEGORY IC1 SPD

A. Approved Products:
   1. Eaton AGPH 12015.
   2. Or Equal.

B. Standards: UL 1449.

C. Design:
   1. General:
      a. Mounted internally to control panels for point-of-use loads.
      b. MOV based or multi-stage hybrid solid-state high-performance suppression system.
      c. Designed for series connection.
      d. Enclosure: Metallic or plastic, flange or DIN rail mounting.
      e. Field connection: Provide unit with external terminal screws for each phase, neutral and ground that will accept #14 through #12 conductors.
      f. Device monitoring: Long-life, solid state, externally visible indicators that monitor the on-line status of the unit's suppression filter system or power loss in any of the phases.
   2. Operating voltage: 120 VAC.
   3. Operating current: 15 A minimum.
   4. Operating frequency: 45 to 65 Hz.
   5. Modes of protection: All modes, L-N, L-G and N-G.
   6. Maximum continuous operating voltage: Less than 130% of system peak voltage.
   7. Maximum surge current: 20,000 A per phase, 10,000 A per mode minimum.
   8. Minimum repetitive surge current capacity: 1000 impulses with no degradation of more than 10% deviation of the clamping voltage.
   9. Fusing: Optional integral unit level and/or component level short circuit and/or thermal overload protection.
      a. External protection as recommended by manufacturer.
   10. Maximum clamping voltages, dynamic test with voltages measured from the zero-voltage reference and 90 DEG phase angle:

System Voltage	Test Mode	IEEE C62.41.2		UL 1449 (3rd Edition VPRs)
		B Comb. Wave	A Ring Wave	
L-N = 120 V	L-N	400 V	300 V	400 V
	L-G	500 V	400 V	400 V
	N-G	500 V	400 V	500 V

2.3 CATEGORY IC2 SPD

A. Approved Product:
   1. MTL Zone Guardian ZG-43006.
   2. Or Equal.

B. Standards: UL 1283, UL 1363 and UL 1449.

C. Design:
   1. General:
      a. Multi-stage hybrid solid-state high-performance suppression system.
      b. Point of use protection, designed for plug-in devices.
      c. Enclosure: Metallic or impact resistant plastic.
      d. Field connection:
         1) The minimum line cord shall be #14 three conductors, 6 FT in length, with a single piece strain relief NEMA 5-15P plug.
         2) The minimum plug-in strip shall have six single NEMA 5-15R receptacles.
      e. ON/OFF toggle switch.
      f. Device monitoring:
         1) Long-life, solid state, externally visible indicators that monitor the on-line status of the unit's suppression filter system or power loss in any of the phases.
         2) A fuse or circuit breaker status only monitor system is not acceptable.
   2. Operating voltage: 120 VAC.
   3. Modes of protection: All modes, L-N, L-G and N-G.
   4. Maximum continuous operating voltage: Less than 130% of system peak voltage.
   5. Operating frequency: 45 to 65 Hz.
   6. Maximum surge current: 20,000 A per phase, 10,000 A per mode minimum.
   7. Minimum repetitive surge current capacity: 1000 impulses with no degradation of more than 10% deviation of the clamping voltage.
   8. Overcurrent protection: The SPD may contain internal fuses or circuit breaker, but an IEEE B3 combination wave shall not cause the fuse or circuit breaker to open and render the SPD inoperable.
   9. Maximum clamping voltages, dynamic test with voltages measured from the zero-voltage reference and 90 DEG phase angle:

System Voltage	Test Mode	IEEE C62.41.2		UL 1449
		B Comb. Wave	A Ring Wave	
L-N = 120 V	L-N	400 V	300 V	330 V
	L-G	500 V	400 V	400 V
	N-G	500 V	400 V	400 V

   10. EMI-RFI noise rejection: Attenuation greater than 30 dB for frequencies between 100 kHz and 100 MHz

2.4 CATEGORY IC3 SPD

A. Approved Products:
   1. ASCO 205120NF010AN3D0.
   2. Or Equal.

B. Standards: UL 497B or UL 1449.

C. Design:
   1. General:
      a. Mounted internally to control panels for point-of-use loads.
      b. Multi-stage hybrid solid-state high-performance suppression system.
      c. Designed for series connection.

d. Enclosure: Metallic or plastic, flange or DIN rail mounting.
e. Field connection: Provide unit with external terminal screws for each phase, neutral and ground that will accept #14 through #12 conductors.
f. Device monitoring: Long-life, solid state, externally visible indicators that monitors the on-line status of the unit's suppression filter system or power loss in any of the phases.

2. Operating voltage: 120 VAC.
3. Operating current: 3 A minimum.
4. Operating frequency: 45 to 65 Hz.
5. Modes of protection: L-N, when ground conductor is present L-G and N-G.
6. Maximum continuous operating voltage: Less than 130% of system peak voltage.
7. Maximum surge current: 6000 A per phase, 3000 A per mode minimum.
8. Minimum repetitive surge current capacity:
   a. The SPD shall meet one of the following:
      1) 1000 occurrences of a 200 A, 10 x 1000 microsecond waveform.
      2) 400 occurrences of a 500 A, 10 x 1000 microsecond waveform.
      3) 100 occurrences of a 400 A, 10 x 700 microsecond waveform.
      4) 100 occurrences of a 2000 A, 8 x 20 microsecond waveform.
9. Maximum clamping voltages, measured from the zero-voltage reference:
   a. The SPD shall meet one of the following:
      1) 400A, 10x700 microsecond waveform: 200% of system voltage.
      2) IEEE B3 combination wave: 250% of system voltage.
      3) IEEE B3 ring wave: 200% of system peak voltage.
      4) IEEE A3 ring wave: 200% of system peak voltage.
      5) Mode N-G clamping voltage may be 175% higher than the L-G levels.

2.5 CATEGORY IC4 SPD

A. Approved Products:
   1. ASCO 157D036S10KXW5R0, 157D036S10KXW5S0, 157D036S10KXW6M0, or 157D036S10KXW6P0.
   2. Or Equal.

B. Standards: None.

C. Design:
   1. General:
      a. For protection of field mounted equipment connected to 4-20mA analog signal loops.
      b. Mounted directly to an unused conduit entry on a process transmitter housing.
      c. Multi-stage hybrid solid-state high-performance suppression system.
      d. Designed for series connection.
      e. Enclosure: 1/2 IN to 3/4 IN stainless steel conduit pipe nipple.
   2. Operating voltage: 24 VDC or as indicated on the Drawings.
   3. Modes of protection: All modes, L-L and L-G.
   4. Maximum continuous operating voltage: Less than 130% of system peak voltage.
   5. Maximum surge current: 10,000 A.
   6. Minimum repetitive surge current capacity:
      a. The SPD shall meet one (1) of the following:
         1) 1000 occurrences of a 200 A, 10 x 1000 microsecond waveform.
         2) 400 occurrences of a 500 A, 10 x 1000 microsecond waveform.
         3) 100 occurrences of a 400 A, 10 x 700 microsecond waveform.
         4) 100 occurrences of a 2000 A, 8 x 20 microsecond waveform.
         5) 10 occurrences of a 10,000 A, 8 x2 0 microsecond waveform.
   7. Maximum clamping voltages, L-L:
      a. The SPD shall meet one of the following:
         1) 400 A, 10 x 700 microsecond waveform: 400% of system voltage.
         2) 10,000 A, 8 x 20 microsecond waveform: 400% of system voltage.
         3) IEEE B3 combination wave: 250% of system voltage.
   8. Maximum clamping voltages, L-G:

a. The SPD shall meet one of the following:
1) 400 A, 10 x 700 microsecond waveform: 200% of system voltage.
2) 10,000 A, 8 x 20 microsecond waveform: 200% of system voltage.
3) IEEE B3 combination wave: 300% of system voltage.

2.6 CATEGORY IC5 SPD

A. Approved Products:
1. ASCO 105D036S10KAN3D0 or 175D036S10KAPCN0 with PCB1B base.
2. Or Equal.

B. Standards: UL 497B.

C. Design:
1. General:
a. Mounted internally to control panels for protection of equipment connected to analog signal loops.
b. Multi-stage hybrid solid-state high-performance suppression system.
c. Designed for series connection.
d. Enclosure: Metallic or plastic, flange or DIN rail mounting.
e. Field connection: The unit shall have external terminal screws for line and ground conductors.
2. Operating voltage: 24 VDC or as indicated on the Drawings.
3. Modes of protection: All modes, L-L and L-G.
4. Maximum continuous operating voltage: Less than 130% of system peak voltage.
5. Maximum surge current: 10,000 A.
6. Minimum repetitive surge current capacity:
a. The SPD shall meet one of the following:
1) 1000 occurrences of a 200 A, 10 x 1000 microsecond waveform.
2) 400 occurrences of a 500 A, 10 x 1000 microsecond waveform.
3) 100 occurrences of a 400 A, 10 x 700 microsecond waveform.
4) 100 occurrences of a 2000 A, 8 x 20 microsecond waveform.
5) 10 occurrences of a 10,000 A, 8 x 20 microsecond waveform.
7. Maximum clamping voltages, L-L:
a. The SPD shall meet one of the following:
1) 400 A, 10 x 700 microsecond waveform: 400% of system voltage.
2) 10,000 A, 8 x 20 microsecond waveform: 400% of system voltage.
3) IEEE B3 combination wave: 225% of system voltage.
8. Maximum clamping voltages, L-G:
a. The SPD shall meet one of the following:
1) 400 A, 10 x 700 microsecond waveform: 200% of system voltage.
2) 10,000 A, 8 x 20 microsecond waveform: 200% of system voltage.
3) IEEE B3 combination wave: 300% of system voltage.

2.7 CATEGORY IC6 SPD

A. Approved Products:
1. ASCO 265120NF015BD4J0.
2. Or Equal.

B. Product:
1. Field mounted for protection of field mounted equipment connected to 120 V power and 4 - 20 mA analog signal loops.
2. Category IC1 and Category IC5 SPDs mounted in a common enclosure.
3. Enclosure: Metallic or nonmetallic NEMA 4X.

2.8 CATEGORY IC7 SPD

A. Approved Products:
1. ASCO 105 Series.

2. Or Equal.

B. Standards: UL 497B.

C. Design:
   1. General:
      a. Mounted internally to control panels for protection of equipment connected to a discrete signal.
      b. Multi-stage hybrid solid-state high-performance suppression system.
      c. Designed for series connection.
      d. Enclosure: Metallic or plastic, flange or DIN rail mounting.
      e. Field connection: Provide unit with external terminal screws for line and ground conductors.
   2. Operating voltage: 24 VDC or 24 VAC or 120 VAC or as indicated on the Drawings.
   3. Modes of protection: All modes:
      a. AC applications: L-N, L-G, N-G
      b. DC applications: POS-NEG, POS-GND, NEG-GND.
   4. Maximum continuous operating voltage: Less than 130% of system peak voltage.
   5. Maximum surge current: 10,000 A.
   6. Minimum repetitive surge current capacity:
      a. The SPD shall meet one of the following:
         1) 1000 occurrences of a 200 A, 10 x 1000 microsecond waveform.
         2) 400 occurrences of a 500 A, 10 x 1000 microsecond waveform.
         3) 100 occurrences of a 400 A, 10 x 700 microsecond waveform.
         4) 100 occurrences of a 2000 A, 8 x 20 microsecond waveform.
         5) 10 occurrences of a 10,000 A, 8 x 20 microsecond waveform.
   7. Maximum clamping voltages, L-L (POS-NEG):
      a. The SPD shall meet one of the following:
         1) 400 A, 10 x 700 microsecond waveform: 400% of system voltage.
         2) 10,000 A, 8 x 20 microsecond waveform: 400% of system voltage.
         3) IEEE B3 combination wave: 250% of system voltage.
   8. Maximum clamping voltages, L-G:
      a. The SPD shall meet one of the following:
         1) 400 A, 10 x 700 microsecond waveform: 200% of system voltage.
         2) 10,000 A, 8 x 20 microsecond waveform: 200% of system voltage.
         3) IEEE B3 combination wave: 300% of system voltage.

2.9 SOURCE QUALITY CONTROL

A. Performance tests to be performed or independently verified by a certified testing laboratory.

B. The SPD are to be tested as a complete SPD system including Integral unit level and/or component level fusing.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Category IC1 SPD:
   1. Provide on the following applications:
      a. Incoming 120 V power to all control panels.
      b. Line side of 120 V power terminals to equipment (e.g., PLCs, transmitters).
   2. Connected in series with the panel's or equipment’s branch circuit.
   3. Provide fuse protection as recommended by manufacturer.
   4. Flange mount or DIN rail mount in control panel.
   5. Connect all SPDs in the panel to the same grounding point.

C. Category IC2 SPD:
   1. Provide on the following applications:
      a. All desktop or rack mounted equipment that is cord and plug connected (e.g., computers, printers) except where power is sourced from a UPS.
   2. Locate near equipment to be protected.

D. Category IC3 SPD:
   1. Provide on the following applications:
      a. 120 V discrete PLC signals into a control panel from float switches, position switches, etc., where the device is mounted outdoors or in a remote building or structure from the control panel and where the control conductors are routed above grade or underground.
      b. 120 V discrete PLC signals into a control panel from float switches, position switches, etc., where both the device and control panel are mounted outdoors, and the control conductors are routed above grade or underground.
   2. Connected in series with the equipment.
   3. Provide fuse protection as recommended by manufacturer.
   4. Flange mount or DIN rail mount in control panel.
   5. Connect all SPDs in the panel to the same grounding point.

E. Category IC4 SPD:
   1. Provide on the following applications:
      a. Loop powered transmitter (flow, level, etc.) where the transmitter is mounted outdoors or in a remote building or structure from the control panel and the signal conductors are routed above grade or underground.
      b. Loop powered transmitter (flow, level, etc.) where both the transmitter and control panel are mounted outdoors, and the signal conductors are routed above grade or underground.
   2. Connect in series with the equipment.
   3. Attach to spare conduit entry of transmitter or inline of conduit at the transmitter.
   4. Bond transmitter to a grounded structure or provide a ground rod.
   5. Ground shield at control panel end.
   6. Verify SPDs series resistance and capacitance does not interfere with the transmitters signal.

F. Category IC5 SPD:
   1. Provide on the following applications:
      a. Incoming 4-20 mA signals into a control panel from transmitters (flow, level, etc.) where the transmitter is mounted outdoors or in a remote building or structure from the control panel and the signal conductors are routed above grade or underground.
      b. Incoming 4-20 mA signals into a control panel from transmitters (flow, level, etc.) where both the transmitter and control panel are mounted outdoors, and the signal conductors are routed above grade or underground.
   2. Connect in series with the equipment.
   3. Flange mount or DIN rail mount in control panel.
   4. Connect all SPDs in the control panel to the same grounding point.
   5. Verify SPDs series resistance and capacitance does not interfere with the transmitters signal.

G. Category IC6 SPD:
   1. Provide on the following applications:
      a. Outdoor field mounted transmitter (flow, level, etc.) that requires 120 V power and provides a 4-20 mA signal to a control panel where the conductors are routed above grace or underground.
   2. Connect in series with the equipment.
   3. Mounted adjacent to equipment.
   4. Bond transmitter to a grounded structure or provide a ground rod.
   5. Ground shield at control panel end.
   6. Verify SPDs series resistance and capacitance does not interfere with the transmitters signal.

H. Category IC7 SPD:

1. Provide on the following applications:
    a. Low voltage (e.g., 24 VAC, 24 VDC) discrete PLC signals into a control panel from float switches, position switches, etc., where the device is mounted outdoors or in a remote building or structure from the control panel and where the control conductors are routed above grade or underground.
    b. Low voltage (e.g., 24 VAC, 24 VDC) discrete PLC signals into a control panel from float switches, position switches, etc., where both the device and control panel are mounted outdoors, and the control conductors are routed above grade or underground.
2. Connect in series with the equipment.
3. Flange mount or DIN rail mount in control panel.
4. Connect all SPDs in the control panel to the same grounding point.

END OF SECTION

This page intentionally left blank.

SECTION 40 61 93
PROCESS CONTROL SYSTEM INPUT-OUTPUT LIST

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Process Control System (PCS) PLC Input/Output (I/O) List description.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 40 61 13 - Process Control System General Requirements.
   2. Section 40 63 43 - Programmable Logic Controllers.
   3. Section 40 61 96 - Process Control Descriptions.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. The International Society of Automation (ISA):
      a. 5.1, Instrumentation Symbols and Identification.

1.3 SUBMITTALS

A. Action Submittals: Submit the following:
   1. Shop Drawings:
      a. Any proposed deviations from the I/O List format, content and attributes stipulated in this Section shall be submitted for approval. I/O List development shall not proceed until the deviation has been approved.
      b. Pre-fabrication I/O Lists for approval.

B. Informational Submittals: Submit the following:
   1. I/O Lists:
      a. Post-Commissioning Final I/O Lists.
      b. Submit I/O Lists in PDF and native format Microsoft Excel indexed by Area, Panel, and Point Description.

PART 2 - PRODUCTS

2.1 I/O LIST

A. The I/O List in the Appendices 40 61 93A – 40 61 93B of this Section herein contains I/O point information derived from the Contract drawings and specifications.

B. The I/O List shall be used as the starting point in the development of the final PLC I/O database.
   1. The I/O list does not include internal software points generated by the control system and is used solely within the control system.

C. The I/O List is organized in columns as follows:
   1. I/O TAG describes the point name that will be used throughout the control system to identify the point.
   2. P&ID references the Process and Instrumentation Diagram on which the point is depicted in relation to the process and equipment.
   3. I/O TYPE denotes the signal type such as analog input or output, discrete input or output, pulse, etc.
   4. INTERFACE denoted the method of communication (hardwired, via datalink, etc.).
   5. POINT DESCRIPTION is a concise English language description of the point's function in in relation to the process in terms that a user can easily understand.
   6. PANEL indicates the control panel in which the I/O module is located.

7. FIELD DEVICE TAG is the tag assigned to the associated field instrument (if applicable).
8. SIGNAL TYPE is the type of electrical signaling used.
9. CALIBRATED RANGE is the minimum and maximum (range) limit in engineering units of an analog input or output signal.
10. ENGINEERING UNITS lists the units associated with the point value.
11. DIGITAL ZERO STATE is the state descriptor associated with the open or zero (0) state of a discrete signal. Ex. ALARM, OFF, HIGH, FAILED.
12. DIGITAL ONE STATE is the state descriptor associated with the closed or one (1) state of a discrete signal. Ex. NORMAL, ENERGIZED, FLOW.
13. INTERPOSING RELAY indicates if a relay needs to be interposed between the control system and the field signal.
14. NOTES is for any pertinent notations that help in identification or understanding of the signal source, features or new/existing status.
15. REV indicates the number of revisions of that I/O point since the bid.

D. I/O List:
   1. Refer to Appendix 40 61 93A.

PART 3 - EXECUTION

3.1 I/O DATABASE DEVELOPMENT

A. The Systems Integrator shall develop the complete I/O List containing all information needed to facilitate panel building, testing and programming, and the fully document the I/O layout and interconnections.

B. The Systems Integrator shall obtain the Government's existing tag naming conventions, abbreviations, facility codes, standard state descriptors, and other relevant information prior to creating the I/O List.

C. Prior to the start of PLC panel fabrication, the Systems Integrator shall submit an I/O List for review and approval

D. The I/O List shall include for each I/O point, at minimum, all information as listed in Section 2.1.C.

E. Maintain a copy of the complete I/O List with modifications during construction in native file format. I/O List shall be accessible to Government and Contracting Officer upon request.

F. Following successful project Commissioning, submit an "As Installed" final I/O List, with all fields representing the updated information, including all field updated information.

3.2 I/O HARDWARE CONFIGURATION

A. Partition the signals among I/O modules as described in Section 40 63 43 to provide enhanced process control and monitoring resiliency.

3.3 I/O POINT DATA FIELDS

A. Information in the I/O List data fields may be subject to review and modification by the Government or Contracting Officer during the Submittal review phase.
   1. Incorporate changes as directed by the reviewer through the system and associated documentation, at no additional cost to the Government, subject to the following limitations:
      a. Requested modifications shall be limited to 20% of the total number of I/O points.
         1) This 20% shall not include changes to the I/O List prior to the Submittal review.
         2) Corrections for errors by the Systems Integrator shall not count toward the 20% modification limit.
      b. Each unique change shall count as one modification.
         1) For example, modifying the description, range, and engineering unit for one analog input counts as three separate modifications.

c. Analog input alarm limit adjustments shall not count as modifications.

END OF SECTION

APPENDIX 40 61 93A

I/O LIST

REV	P&ID DRAWING	FIELD DEVICE	POINT TAG	PLC/RIO	DESCRIPTION	OFF STATE / RANGE	ON STATE / ENG UNITS	I/O TYPE	RACK	SLOT	POINT / CHANNEL
				RSPH MASTER PLC	CPU			H2-DM1E	0	CPU	
				RSPH MASTER PLC	REMOTE I/O MASTER			H2-ERM100	0	0	
				RSPH MASTER PLC	COMMUNICATIONS MODULE			H2-ECOM	0	1	
	052-Y605	ZS-51101	ZS-51101	RSPH MASTER PLC	PLC CPU MASTER MODE	STANDBY	MASTER	DI	0	2	0
				RSPH MASTER PLC	SPARE			DI	0	2	1
				RSPH MASTER PLC	SPARE			DI	0	2	2
				RSPH MASTER PLC	SPARE			DI	0	2	3
	052-Y605	G-51101	XSM-51101	RSPH MASTER PLC	MASTER PLC MASTER MODE	<NULL>	MASTER	DO	0	2	4
	052-Y605	R-51101	XSS-51101	RSPH MASTER PLC	MASTER PLC STANDBY MODE	<NULL>	STANDBY	DO	0	2	5
				RSPH MASTER PLC	SPARE			DO	0	2	6
				RSPH MASTER PLC	SPARE			DO	0	2	7
				RSPH STANDBY PLC	CPU			H2-DM1E		CPU	
				RSPH STANDBY PLC	REMOTE I/O MASTER			H2-ERM100		0	
				RSPH STANDBY PLC	COMMUNICATIONS MODULE			H2-ECOM		1	
	052-Y605	ZS-51102	ZS-51102	RSPH STANDBY PLC	PLC CPU STANDBY MODE	MASTER	STANDBY	DI	0	2	0
				RSPH STANDBY PLC	SPARE			DI	0	2	1
				RSPH STANDBY PLC	SPARE			DI	0	2	2
				RSPH STANDBY PLC	SPARE			DI	0	2	3
	052-Y605	G-51102	XSM-51102	RSPH STANDBY PLC	STANDBY PLC MASTER MODE	<NULL>	MASTER	DO	0	2	4
	052-Y605	R-51102	XSS-51102	RSPH STANDBY PLC	STANDBY PLC STANDBY MODE	<NULL>	STANDBY	DO	0	2	5
				RSPH STANDBY PLC	SPARE			DO	0	2	6
				RSPH STANDBY PLC	SPARE			DO	0	2	7
				RSPH RIO 1	ETHERNET BASED CONTROLLER			H2-EBC100	1	CPU	
	052-Y601	PMP-52211	TI-52211A	RSPH RIO 1	RAW WATER PUMP 1 BEARING TEMPERATURE	-58 - 500	DEG F	RTD	1	0	0
	052-Y601	PMP-52211	TI-52211B	RSPH RIO 1	RAW WATER PUMP 1 MOTOR WINDING 1 TEMP 1	-58 - 500	DEG F	RTD	1	0	1
	052-Y601	PMP-52211	TI-52211C	RSPH RIO 1	RAW WATER PUMP 1 MOTOR WINDING 1 TEMP 2	-58 - 500	DEG F	RTD	1	0	2
				RSPH RIO 1	SPARE			RTD	1	0	3
	052-Y601	PMP-52211	TI-52211D	RSPH RIO 1	RAW WATER PUMP 1 MOTOR WINDING 2 TEMP 1	-58 - 500	DEG F	RTD	1	1	0
	052-Y601	PMP-52211	TI-52211E	RSPH RIO 1	RAW WATER PUMP 1 MOTOR WINDING 2 TEMP 2	-58 - 500	DEG F	RTD	1	1	1
	052-Y601	PMP-52211	TI-52211F	RSPH RIO 1	RAW WATER PUMP 1 MOTOR WINDING 3 TEMP 1	-58 - 500	DEG F	RTD	1	1	2
				RSPH RIO 1	SPARE			RTD	1	1	3
	052-Y601	PMP-52211	TI-52211G	RSPH RIO 1	RAW WATER PUMP 1 MOTOR WINDING 3 TEMP 2	-58 - 500	DEG F	RTD	1	2	0
	052-Y601	PMP-52211	TI-52211H	RSPH RIO 1	RAW WATER PUMP 1 MOTOR DRIVE END BEARING TEMPERATURE	-58 - 500	DEG F	RTD	1	2	1
	052-Y601	PMP-52211	TI-52211J	RSPH RIO 1	RAW WATER PUMP 1 MOTOR OPPOSITE END BEARING TEMPERATURE	-58 - 500	DEG F	RTD	1	2	2
				RSPH RIO 1	SPARE			RTD	1	2	3
	052-Y601	PMP-52212	TI-52212A	RSPH RIO 1	RAW WATER PUMP 2 BEARING TEMPERATURE	-58 - 500	DEG F	RTD	1	3	0
	052-Y601	PMP-52212	TI-52212B	RSPH RIO 1	RAW WATER PUMP 2 MOTOR WINDING 1 TEMP 1	-58 - 500	DEG F	RTD	1	3	1
	052-Y601	PMP-52212	TI-52212C	RSPH RIO 1	RAW WATER PUMP 2 MOTOR WINDING 1 TEMP 2	-58 - 500	DEG F	RTD	1	3	2
				RSPH RIO 1	SPARE			RTD	1	3	3
	052-Y601	PMP-52212	TI-52212D	RSPH RIO 1	RAW WATER PUMP 2 MOTOR WINDING 2 TEMP 1	-58 - 500	DEG F	RTD	1	4	0
	052-Y601	PMP-52212	TI-52212E	RSPH RIO 1	RAW WATER PUMP 2 MOTOR WINDING 2 TEMP 2	-58 - 500	DEG F	RTD	1	4	1
	052-Y601	PMP-52212	TI-52212F	RSPH RIO 1	RAW WATER PUMP 2 MOTOR WINDING 3 TEMP 1	-58 - 500	DEG F	RTD	1	4	2
				RSPH RIO 1	SPARE			RTD	1	4	3
	052-Y601	PMP-52212	TI-52212G	RSPH RIO 1	RAW WATER PUMP 2 MOTOR WINDING 3 TEMP 2	-58 - 500	DEG F	RTD	1	5	0
	052-Y601	PMP-52212	TI-52212H	RSPH RIO 1	RAW WATER PUMP 2 MOTOR DRIVE END BEARING TEMPERATURE	-58 - 500	DEG F	RTD	1	5	1
	052-Y601	PMP-52212	TI-52212J	RSPH RIO 1	RAW WATER PUMP 2 MOTOR OPPOSITE END BEARING TEMPERATURE	-58 - 500	DEG F	RTD	1	5	2
				RSPH RIO 1	SPARE			RTD	1	5	3
				RSPH RIO 1	SPARE			RTD	1	6	0
				RSPH RIO 1	SPARE			RTD	1	6	1
				RSPH RIO 1	SPARE			RTD	1	6	2

				RSPH RIO 1	SPARE			RTD	1	6	3
	052-Y601	PMP-52211	SI-52211	RSPH RIO 1	RAW WATER PUMP 1 SPEED INDICATION	0-60	HZ	AI	1	7	0
	052-Y601	PMP-52211	VI-52211A	RSPH RIO 1	RAW WATER PUMP 1 PUMP VIBRATION X	0-1	IN/S	AI	1	7	1
	052-Y601	PMP-52211	VI-52211B	RSPH RIO 1	RAW WATER PUMP 1 PUMP VIBRATION Y	0-1	IN/S	AI	1	7	2
	052-Y601	PMP-52211	VI-52211C	RSPH RIO 1	RAW WATER PUMP 1 PUMP VIBRATION Z	0-1	IN/S	AI	1	7	3
	052-Y601	PMP-52211	VI-52211D	RSPH RIO 1	RAW WATER PUMP 1 MOTOR DRIVE END VIBRATION X	0-1	IN/S	AI	1	7	4
	052-Y601	PMP-52211	VI-52211E	RSPH RIO 1	RAW WATER PUMP 1 MOTOR DRIVE END VIBRATION Y	0-1	IN/S	AI	1	7	5
	052-Y601	PMP-52211	VI-52211F	RSPH RIO 1	RAW WATER PUMP 1 MOTOR DRIVE END VIBRATION Z	0-1	IN/S	AI	1	7	6
				RSPH RIO 1	SPARE			AI	1	7	7
				RSPH RIO 2	ETHERNET BASED CONTROLLER			H2-EBC100	2	CPU	
	052-Y601	PMP-52212	SI-52212	RSPH RIO 2	RAW WATER PUMP 2 SPEED INDICATION	0-60	HZ	AI	2	0	0
	052-Y601	PMP-52212	VI-52212A	RSPH RIO 2	RAW WATER PUMP 2 PUMP VIBRATION X	0-1	IN/S	AI	2	0	1
	052-Y601	PMP-52212	VI-52212B	RSPH RIO 2	RAW WATER PUMP 2 PUMP VIBRATION Y	0-1	IN/S	AI	2	0	2
	052-Y601	PMP-52212	VI-52212C	RSPH RIO 2	RAW WATER PUMP 2 PUMP VIBRATION Z	0-1	IN/S	AI	2	0	3
	052-Y601	PMP-52212	VI-52212D	RSPH RIO 2	RAW WATER PUMP 2 MOTOR DRIVE END VIBRATION X	0-1	IN/S	AI	2	0	4
	052-Y601	PMP-52212	VI-52212E	RSPH RIO 2	RAW WATER PUMP 2 MOTOR DRIVE END VIBRATION Y	0-1	IN/S	AI	2	0	5
	052-Y601	PMP-52212	VI-52212F	RSPH RIO 2	RAW WATER PUMP 2 MOTOR DRIVE END VIBRATION Z	0-1	IN/S	AI	2	0	6
				RSPH RIO 2	SPARE			AI	2	0	7
	052-Y601	PMP-52211	VI-52211G	RSPH RIO 2	RAW WATER PUMP 1 MOTOR OPPOSITE END VIBRATION X	0-1	IN/S	AI	2	1	0
	052-Y601	PMP-52211	VI-52211H	RSPH RIO 2	RAW WATER PUMP 1 MOTOR OPPOSITE END VIBRATION Y	0-1	IN/S	AI	2	1	1
	052-Y601	PMP-52211	VI-52211J	RSPH RIO 2	RAW WATER PUMP 1 MOTOR OPPOSITE END VIBRATION Z	0-1	IN/S	AI	2	1	2
	052-Y601	PMP-52212	VI-52212G	RSPH RIO 2	RAW WATER PUMP 2 MOTOR OPPOSITE END VIBRATION X	0-1	IN/S	AI	2	1	3
	052-Y601	PMP-52212	VI-52212H	RSPH RIO 2	RAW WATER PUMP 2 MOTOR OPPOSITE END VIBRATION Y	0-1	IN/S	AI	2	1	4
	052-Y601	PMP-52212	VI-52212J	RSPH RIO 2	RAW WATER PUMP 2 MOTOR OPPOSITE END VIBRATION Z	0-1	IN/S	AI	2	1	5
				RSPH RIO 2	SPARE			AI	2	1	6
				RSPH RIO 2	SPARE			AI	2	1	7
	052-Y601	PIT-52261	PI-52261	RSPH RIO 2	AUTO FILTERS INLET PRESSURE	0-200	PSI	AI	2	2	0
	052-Y601	PIT-52262	PI-52262	RSPH RIO 2	RAW WATER PUMPS SUCTION PRESSURE	0-200	PSI	AI	2	2	1
	052-Y601	PIT-52265	PI-52265	RSPH RIO 2	RAW WATER PUMPS OUTLET PRESSURE	0-2000	PSI	AI	2	2	2
	052-Y601	PIT-52266	PI-52266	RSPH RIO 2	RAW WATER PUMPS COMMON DISCHARGE PRESSURE	0-2000	PSI	AI	2	2	3
	052-Y601	FIT-52371	FI-52371	RSPH RIO 2	RAW WATER PUMPS SUCTION FLOW	0-200	GPM	AI	2	2	4
	052-Y601	FIT-52372	FI-52372	RSPH RIO 2	RAW WATER BYPASS FLOW	0-250	GPM	AI	2	2	5
	052-Y601	AIT-52381	AI-52381	RSPH RIO 2	RAW WATER TURBIDITY	0-10000	NTU	AI	2	2	6
				RSPH RIO 2	SPARE			AI	2	2	7
	052-Y602	EDS-53117	ZI-53117	RSPH RIO 2	FINISHED WATER ENERGY DISSIPATION VALVE POSITION	0-100	%	AI	2	3	0
	052-Y602	LT-53251	LI-53251	RSPH RIO 2	FINISHED WATER TANK LEVEL	0-15	FT	AI	2	3	1
	052-Y602	PIT-53162	PI-53162	RSPH RIO 2	FINISHED WATER PRE-PRESSURE REDUCER PRESSURE	0-2000	PSI	AI	2	3	2
	052-Y602	PIT-53164	PI-53164	RSPH RIO 2	FINISHED WATER POST-PRESSURE REDUCER PRESSURE	0-100	PSI	AI	2	3	3
	052-Y602	FIT-53171	FI-53171	RSPH RIO 2	FINISHED WATER FLOW	0-150	GPM	AI	2	3	4
	052-Y602	FIT-53172	FI-53172	RSPH RIO 2	DISTRIBUTION FLOW	0-100	GPM	AI	2	3	5
	052-Y603	FIT-53371	FI-53371	RSPH RIO 2	MAKEUP WATER FLOW	0-15	GPM	AI	2	3	6
				RSPH RIO 2	SPARE			AI	2	3	7
	052-Y604	PIT-53161	PI-53161	RSPH RIO 2	FINISHED WATER BOOSTER PUMPS SYSTEM PRESSURE	0-100	PSI	AI	2	4	0
	052-Y604	FIT-53173	FI-53173	RSPH RIO 2	FINISHED WATER BOOSTER PUMPS SYSTEM FLOW	0-40	GPM	AI	2	4	1
	052-Y602	AIT-53281	AI-53281	RSPH RIO 2	FINISHED WATER TANK RECIRCULATION CHLORINE RESIDUAL	0-20	PPM	AI	2	4	2
	052-Y602	AIT-53282	AI-53282	RSPH RIO 2	FINISHED WATER TANK pH	5-10	pH	AI	2	4	3
	052-Y602	TIT-53281	TI-53281	RSPH RIO 2	FINISHED WATER TANK TEMPERATURE	0-100	DEG F	AI	2	4	4
	052-Y602	AIT-53282	TI-53282	RSPH RIO 2	FINISHED WATER TANK RECIRCULATION TEMPERATURE	32 - 212	DEG F	AI	2	4	5
				RSPH RIO 2	SPARE			AI	2	4	6
				RSPH RIO 2	SPARE			AI	2	4	7
	052-Y603	PMP-53421	SI-53421	RSPH RIO 2	SODIUM HYPOCHLORITE BATCH METERING PUMP SPEED INDICATION	0-7.67	GPH	AI	2	5	0
	052-Y603	PMP-53321	SI-53321	RSPH RIO 2	DILUTE SODIUM HYPOCHLORITE METERING PUMP 1 SPEED INDICATION	0-7.67	GPH	AI	2	5	1
	052-Y603	PMP-53322	SI-53322	RSPH RIO 2	DILUTE SODIUM HYPOCHLORITE METERING PUMP 2 SPEED INDICATION	0-7.67	GPH	AI	2	5	2
	052-Y603	LIT-53351	LI-53351	RSPH RIO 2	DILUTE SODIUM HYPOCHLORITE DAY TANK LEVEL	0-5	FT	AI	2	5	3
	052-Y603	SCL-53411	WI-53491	RSPH RIO 2	SODIUM HYPOCHLORITE DRUM OR CARBOY WEIGHT	0 - 850	LBS	AI	2	5	4
	052-Y605	TIT-51681	TI-51681	RSPH RIO 2	PUMP ROOM TEMPERATURE	-10-160	DEG F	AI	2	5	5
				RSPH RIO 2	SPARE			AI	2	5	6
				RSPH RIO 2	SPARE			AI	2	5	7
				RSPH RIO 2	SPARE			AI	2	6	0
				RSPH RIO 2	SPARE			AI	2	6	1
				RSPH RIO 2	SPARE			AI	2	6	2
				RSPH RIO 2	SPARE			AI	2	6	3

				RSPH RIO 2	SPARE			AI	2	6	4
				RSPH RIO 2	SPARE			AI	2	6	5
				RSPH RIO 2	SPARE			AI	2	6	6
				RSPH RIO 2	SPARE			AI	2	6	7
	052-Y601	PMP-52211	SC-52211	RSPH RIO 2	RAW WATER PUMP 1 SPEED COMMAND	0-60	HZ	AO	2	7	0
	052-Y601	PMP-52212	SC-52212	RSPH RIO 2	RAW WATER PUMP 2 SPEED COMMAND	0-60	HZ	AO	2	7	1
	052-Y603	PMP-53421	SC-53421	RSPH RIO 2	SODIUM HYPOCHLORITE BATCH METERING PUMP SPEED COMMAND	0-7.67	GPH	AO	2	7	2
	052-Y603	PMP-53321	SC-53321	RSPH RIO 2	DILUTE SODIUM HYPOCHLORITE METERING PUMP 1 SPEED COMMAND	0-7.67	GPH	AO	2	7	3
	052-Y603	PMP-53322	SC-53322	RSPH RIO 2	DILUTE SODIUM HYPOCHLORITE METERING PUMP 2 SPEED COMMAND	0-7.67	GPH	AO	2	7	4
				RSPH RIO 2	SPARE			AO	2	7	5
				RSPH RIO 2	SPARE			AO	2	7	6
				RSPH RIO 2	SPARE			AO	2	7	7
				RSPH RIO 3	ETHERNET BASED CONTROLLER			H2-EBC100	3	CPU	
				RSPH RIO 3	SPARE			AO	3	0	0
				RSPH RIO 3	SPARE			AO	3	0	1
				RSPH RIO 3	SPARE			AO	3	0	2
				RSPH RIO 3	SPARE			AO	3	0	3
				RSPH RIO 3	SPARE			AO	3	0	4
				RSPH RIO 3	SPARE			AO	3	0	5
				RSPH RIO 3	SPARE			AO	3	0	6
				RSPH RIO 3	SPARE			AO	3	0	7
	052-Y601	PMP-52211	HSR-52211	RSPH RIO 3	RAW WATER PUMP 1 REMOTE	LOCAL	REMOTE	DI	3	1	0
	052-Y601	PMP-52211	NI-52211	RSPH RIO 3	RAW WATER PUMP 1 RUNNING	OFF	RUNNING	DI	3	1	1
	052-Y601	PMP-52211	UA-52211	RSPH RIO 3	RAW WATER PUMP 1 FAULT	<NULL>	ALARM	DI	3	1	2
	052-Y601	PMP-52211	PSL-52211	RSPH RIO 3	RAW WATER PUMP 1 SEAL FAIL	<NULL>	ALARM	DI	3	1	3
	052-Y601	PMP-52211	ES-52211	RSPH RIO 3	RAW WATER PUMP 1 E-STOP	ALARM	<NULL>	DI	3	1	4
	052-Y601	PSHH-52263	PSHH-52263	RSPH RIO 3	RAW WATER PUMP 1 PRESSURE SWITCH HIGH HIGH	<NULL>	ALARM	DI	3	1	5
	052-Y601	FV-52331	HSR-52331	RSPH RIO 3	RAW WATER BYPASS VALVE REMOTE	LOCAL	REMOTE	DI	3	1	6
	052-Y601	FV-52331	ZSO-52331	RSPH RIO 3	RAW WATER BYPASS VALVE OPENED	<NULL>	OPENED	DI	3	1	7
	052-Y601	FV-52331	ZSC-52331	RSPH RIO 3	RAW WATER BYPASS VALVE CLOSED	<NULL>	CLOSED	DI	3	1	8
	052-Y601	FV-52331	UA-52331	RSPH RIO 3	RAW WATER BYPASS VALVE FAULT	<NULL>	ALARM	DI	3	1	9
	052-Y601	FIT-52371	FQI-52371	RSPH RIO 3	RAW WATER PUMPS SUCTION FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	1	10
	052-Y601	FIT-52372	FQI-52372	RSPH RIO 3	RAW WATER BYPASS FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	1	11
	052-Y601	FSL-52371	FSL-52371	RSPH RIO 3	TURBIDIMETER DRAIN FLOW SWITCH LOW	<NULL>	ALARM	DI	3	1	12
	052-Y602	PMP-53221	HSR-53221	RSPH RIO 3	FINISHED WATER RECIRCULATION PUMP REMOTE	LOCAL	REMOTE	DI	3	1	13
	052-Y602	PMP-53221	NI-53221	RSPH RIO 3	FINISHED WATER RECIRCULATION PUMP RUNNING	OFF	RUNNING	DI	3	1	14
	052-Y602	PMP-53221	UA-53221	RSPH RIO 3	FINISHED WATER RECIRCULATION PUMP FAULT	<NULL>	ALARM	DI	3	1	15
	052-Y602	FIT-53171	FQI-53171	RSPH RIO 3	FINISHED WATER FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	1	16
	052-Y602	FIT-53172	FQI-53172	RSPH RIO 3	DISTRIBUTION FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	1	17
	052-Y602	HC-53283	NI-53283	RSPH RIO 3	HEAT TRACE CONTROLLER RUNNING STATUS	OFF	ON	DI	3	1	18
	052-Y602	HC-53283	UA-53283	RSPH RIO 3	HEAT TRACE CONTROLLER FAULT	<NULL>	ALARM	DI	3	1	19
	052-Y602	HTR-53241	NI-53284	RSPH RIO 3	IMMERSION HEATER CONTROLLER RUNNING STATUS	OFF	ON	DI	3	1	20
	052-Y602	HTR-53241	UA-53284	RSPH RIO 3	IMMERSION HEATER CONTROLLER FAULT	<NULL>	ALARM	DI	3	1	21
				RSPH RIO 3	SPARE			DI	3	1	22
				RSPH RIO 3	SPARE			DI	3	1	23
				RSPH RIO 3	SPARE			DI	3	1	24
				RSPH RIO 3	SPARE			DI	3	1	25
				RSPH RIO 3	SPARE			DI	3	1	26
				RSPH RIO 3	SPARE			DI	3	1	27
				RSPH RIO 3	SPARE			DI	3	1	28
				RSPH RIO 3	SPARE			DI	3	1	29
				RSPH RIO 3	SPARE			DI	3	1	30
				RSPH RIO 3	SPARE			DI	3	1	31
	052-Y601	PMP-52212	HSR-52212	RSPH RIO 3	RAW WATER PUMP 2 REMOTE	LOCAL	REMOTE	DI	3	2	0
	052-Y601	PMP-52212	NI-52212	RSPH RIO 3	RAW WATER PUMP 2 RUNNING	OFF	RUNNING	DI	3	2	1
	052-Y601	PMP-52212	UA-52212	RSPH RIO 3	RAW WATER PUMP 2 FAULT	<NULL>	ALARM	DI	3	2	2
	052-Y601	PMP-52212	PSL-52212	RSPH RIO 3	RAW WATER PUMP 2 SEAL FAIL	<NULL>	ALARM	DI	3	2	3
	052-Y601	PMP-52212	ES-52212	RSPH RIO 3	RAW WATER PUMP 2 E-STOP	ALARM	<NULL>	DI	3	2	4
	052-Y601	PSHH-52264	PSHH-52264	RSPH RIO 3	RAW WATER PUMP 2 PRESSURE SWITCH HIGH HIGH	<NULL>	ALARM	DI	3	2	5
	052-Y601	AFV-52111	NI-52111	RSPH RIO 3	AUTO FILTER 1 BACKWASH ACTIVE	OFF	RUNNING	DI	3	2	6
	052-Y601	AFV-52112	NI-52112	RSPH RIO 3	AUTO FILTER 2 BACKWASH ACTIVE	OFF	RUNNING	DI	3	2	7
	052-Y601	LCP-52101	UA-52101	RSPH RIO 3	AUTO FILTERS FAULT	<NULL>	ALARM	DI	3	2	8
	052-Y602	EDS-53117	HSR -53117	RSPH RIO 3	ENERGY DISSIPATION VALVE REMOTE	LOCAL	REMOTE	DI	3	2	9
	052-Y602	EDS-53117	ZSO-53117	RSPH RIO 3	ENERGY DISSIPATION VALVE OPENED	<NULL>	OPENED	DI	3	2	10
	052-Y602	EDS-53117	ZSC-53117	RSPH RIO 3	ENERGY DISSIPATION VALVE CLOSED	<NULL>	CLOSED	DI	3	2	11
	052-Y602	EDS-53117	UA-53117	RSPH RIO 3	ENERGY DISSIPATION VALVE FAULT	<NULL>	ALARM	DI	3	2	12

	052-Y602	EDS-53117	TSH-53117	RSPH RIO 3	ENERGY DISSIPATION VALVE MOTOR TEMPERATURE HIGH	<NULL>	ALARM	DI	3	2	13
	052-Y602	EDS-53117	YSO-53117	RSPH RIO 3	ENERGY DISSIPATION VALVE TORQUE OPENED LIMIT	<NULL>	TORQUE OPENED	DI	3	2	14
	052-Y602	EDS-53117	YSC-53117	RSPH RIO 3	ENERGY DISSIPATION VALVE TORQUE CLOSED LIMIT	<NULL>	TORQUE CLOSED	DI	3	2	15
	052-Y602	EDS-53117	EDS-PS1	RSPH RIO 3	ENERGY DISSIPATION VALVE CONTROL PANEL POWER SUPPLY 1	ALARM	OK	DI	3	2	16
	052-Y602	EDS-53117	EDS-PS2	RSPH RIO 3	ENERGY DISSIPATION VALVE CONTROL PANEL POWER SUPPLY 2	ALARM	OK	DI	3	2	17
	052-Y602	EDS-53117	EDS-UPS	RSPH RIO 3	ENERGY DISSIPATION VALVE CONTROL PANEL UPS	ALARM	OK	DI	3	2	18
	052-Y605	LSHH-51651	LSHH-51651	RSPH RIO 3	PUMP ROOM FLOOR FLOOD LEVEL SWITCH HIGH HIGH	<NULL>	ALARM	DI	3	2	19
				RSPH RIO 3	SPARE			DI	3	2	20
				RSPH RIO 3	SPARE			DI	3	2	21
				RSPH RIO 3	SPARE			DI	3	2	22
				RSPH RIO 3	SPARE			DI	3	2	23
				RSPH RIO 3	SPARE			DI	3	2	24
				RSPH RIO 3	SPARE			DI	3	2	25
				RSPH RIO 3	SPARE			DI	3	2	26
				RSPH RIO 3	SPARE			DI	3	2	27
				RSPH RIO 3	SPARE			DI	3	2	28
				RSPH RIO 3	SPARE			DI	3	2	29
				RSPH RIO 3	SPARE			DI	3	2	30
				RSPH RIO 3	SPARE			DI	3	2	31
	052-Y603	PMP-53321	HSR-53321	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING PUMP 1 REMOTE	LOCAL	REMOTE	DI	3	3	0
	052-Y603	PMP-53321	NI-53321	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING PUMP 1 RUNNING	OFF	RUNNING	DI	3	3	1
	052-Y603	PMP-53321	UA-53321	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING PUMP 1 FAULT	<NULL>	ALARM	DI	3	3	2
	052-Y603	PMP-53321	MSH-53321	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING PUMP 1 HOSE LEAK	<NULL>	ALARM	DI	3	3	3
	052-Y603	PSHH-53361	PSHH-53361	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING PUMP 1 PRESSURE SWITCH HIGH HIGH	<NULL>	ALARM	DI	3	3	4
	052-Y603	LSH-53451	LSH-53451	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE CONTAINMENT SUMP LEVEL SWITCH HIGH	<NULL>	ALARM	DI	3	3	5
	052-Y603	FV-53331	ZSO-53331	RSPH RIO 3	MAKEUP WATER FLOW VALVE OPENED	<NULL>	OPENED	DI	3	3	6
	052-Y603	FV-53331	ZSC-53331	RSPH RIO 3	MAKEUP WATER FLOW VALVE CLOSED	<NULL>	CLOSED	DI	3	3	7
	052-Y603	FIT-53371	FQI-53371	RSPH RIO 3	MAKEUP WATER FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	3	8
	052-Y604	PMP-53111	HSR-53111	RSPH RIO 3	FINISHED WATER BOOSTER PUMP 1 REMOTE	LOCAL	REMOTE	DI	3	3	9
	052-Y604	PMP-53111	NI-53111	RSPH RIO 3	FINISHED WATER BOOSTER PUMP 1 RUNNING	OFF	RUNNING	DI	3	3	10
	052-Y604	PMP-53111	UA-53111	RSPH RIO 3	FINISHED WATER BOOSTER PUMP 1 FAULT	<NULL>	ALARM	DI	3	3	11
	052-Y604	PSH-53111	PSH-53111	RSPH RIO 3	FINISHED WATER BOOSTER PUMP 1 PRESSURE SWITCH HIGH	<NULL>	ALARM	DI	3	3	12
	052-Y604	FIT-53173	FQI-53173	RSPH RIO 3	FINISHED WATER BOOSTER PUMPS SYSTEM FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	3	13
	052-Y605	ZS-51491	ZS-51491	RSPH RIO 3	OPERATIONS ROOM EXTERIOR DOOR INTRUSION SWITCH	ALARM	<NULL>	DI	3	3	14
	052-Y605	ZS-51492	ZS-51492	RSPH RIO 3	ELECTRICAL AREA EXTERIOR DOOR INTRUSION SWITCH	ALARM	<NULL>	DI	3	3	15
	052-Y605	ZS-51493	ZS-51493	RSPH RIO 3	PUMP ROOM EXTERIOR DOOR INTRUSION SWITCH	ALARM	<NULL>	DI	3	3	16
	052-Y605	SD-51098	SD-51098	RSPH RIO 3	ELECTRICAL ROOM SMOKE DETECTOR	ALARM	<NULL>	DI	3	3	17
	052-Y605	SD-51099	SD-51099	RSPH RIO 3	PROCESS ROOM SMOKE DETECTOR	ALARM	<NULL>	DI	3	3	18
				RSPH RIO 3	SPARE			DI	3	3	19
				RSPH RIO 3	SPARE			DI	3	3	20
				RSPH RIO 3	SPARE			DI	3	3	21
				RSPH RIO 3	SPARE			DI	3	3	22
				RSPH RIO 3	SPARE			DI	3	3	23
				RSPH RIO 3	SPARE			DI	3	3	24
				RSPH RIO 3	SPARE			DI	3	3	25
				RSPH RIO 3	SPARE			DI	3	3	26
				RSPH RIO 3	SPARE			DI	3	3	27
				RSPH RIO 3	SPARE			DI	3	3	28
				RSPH RIO 3	SPARE			DI	3	3	29
				RSPH RIO 3	SPARE			DI	3	3	30
				RSPH RIO 3	SPARE			DI	3	3	31
	052-Y603	PMP-53421	HSR-53421	RSPH RIO 3	SODIUM HYPOCHLORITE BATCH METERING PUMP REMOTE	LOCAL	REMOTE	DI	3	4	0
	052-Y603	PMP-53421	NI-53421	RSPH RIO 3	SODIUM HYPOCHLORITE BATCH METERING PUMP RUNNING	OFF	RUNNING	DI	3	4	1
	052-Y603	PMP-53421	UA-53421	RSPH RIO 3	SODIUM HYPOCHLORITE BATCH METERING PUMP FAULT	<NULL>	ALARM	DI	3	4	2
	052-Y603	PMP-53421	MSH-53421	RSPH RIO 3	SODIUM HYPOCHLORITE BATCH METERING PUMP HOSE LEAK	<NULL>	ALARM	DI	3	4	3
	052-Y603	PMP-53421	MSH-53421	RSPH RIO 3	SODIUM HYPOCHLORITE BATCH METERING PUMP SEAL LEAK	<NULL>	ALARM	DI	3	4	4
	052-Y603	PMP-53322	HSR-53322	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING PUMP 2 REMOTE	LOCAL	REMOTE	DI	3	4	5
	052-Y603	PMP-53322	NI-53322	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING PUMP 2 RUNNING	OFF	RUNNING	DI	3	4	6
	052-Y603	PMP-53322	UA-53322	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING PUMP 2 FAULT	<NULL>	ALARM	DI	3	4	7
	052-Y603	PSHH-53362	PSHH-53362	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING PUMP 2 PRESSURE SWITCH HIGH HIGH	<NULL>	ALARM	DI	3	4	8
	052-Y604	PMP-53112	HSR-53112	RSPH RIO 3	FINISHED WATER BOOSTER PUMP 2 REMOTE	LOCAL	REMOTE	DI	3	4	9
	052-Y604	PMP-53112	NI-53112	RSPH RIO 3	FINISHED WATER BOOSTER PUMP 2 RUNNING	OFF	RUNNING	DI	3	4	10
	052-Y604	PMP-53112	UA-53112	RSPH RIO 3	FINISHED WATER BOOSTER PUMP 2 FAULT	<NULL>	ALARM	DI	3	4	11
	052-Y604	PSH-53112	PSH-53112	RSPH RIO 3	FINISHED WATER BOOSTER PUMP 2 PRESSURE SWITCH HIGH	<NULL>	ALARM	DI	3	4	12
	052-Y604	FSH-51671	FSH-51671	RSPH RIO 3	EMERGENCY SHOWER EYE WASH STATION IN USE	<NULL>	ALARM	DI	3	4	13
	009-Y621	RS-PLC-PS1	RS-PS1	RSPH RIO 3	RSPH CONTROL PANEL POWER SUPPLY 1	ALARM	OK	DI	3	4	14
	009-Y621	RS-PLC-PS2	RS-PS2	RSPH RIO 3	RSPH CONTROL PANEL POWER SUPPLY 2	ALARM	OK	DI	3	4	15
	009-Y621	RS-PLC-UPS	RS-UPS	RSPH RIO 3	RSPH CONTROL PANEL UPS	ALARM	OK	DI	3	4	16
	009-Y621	RS-PLC-POR	RS-POR	RSPH RIO 3	RSPH POWER OK	ALARM	OK	DI	3	4	17

				RSPH RIO 3	SPARE			DI	3	4	18
				RSPH RIO 3	SPARE			DI	3	4	19
				RSPH RIO 3	SPARE			DI	3	4	20
				RSPH RIO 3	SPARE			DI	3	4	21
				RSPH RIO 3	SPARE			DI	3	4	22
				RSPH RIO 3	SPARE			DI	3	4	23
				RSPH RIO 3	SPARE			DI	3	4	24
				RSPH RIO 3	SPARE			DI	3	4	25
				RSPH RIO 3	SPARE			DI	3	4	26
				RSPH RIO 3	SPARE			DI	3	4	27
				RSPH RIO 3	SPARE			DI	3	4	28
				RSPH RIO 3	SPARE			DI	3	4	29
				RSPH RIO 3	SPARE			DI	3	4	30
				RSPH RIO 3	SPARE			DI	3	4	31
	052-Y601	PMP-52211	XC-52211	RSPH RIO 3	RAW WATER PUMP 1 RUN COMMAND	STOP	START	DO	3	5	0
	052-Y601	PMP-52211	XR-52211	RSPH RIO 3	RAW WATER PUMP 1 VFD RESET COMMAND	<NULL>	RESET	DO	3	5	1
	052-Y601	FV-52331	XVO-52331	RSPH RIO 3	RAW WATER BYPASS VALVE OPEN COMMAND	<NULL>	OPEN	DO	3	5	2
	052-Y601	FV-52331	XVC-52331	RSPH RIO 3	RAW WATER BYPASS VALVE CLOSE COMMAND	<NULL>	CLOSE	DO	3	5	3
	052-Y603	PMP-53321	XC-53321	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING PUMP 1 RUN COMMAND	STOP	START	DO	3	5	4
	052-Y603	FV-53331	XVO-53331	RSPH RIO 3	MAKE WATER FLOW VALVE OPEN COMMAND	<NULL>	OPEN	DO	3	5	5
	052-Y603	FV-53331	XVC-53331	RSPH RIO 3	MAKE WATER FLOW VALVE CLOSED COMMAND	<NULL>	CLOSE	DO	3	5	6
	052-Y604	PMP-53111	XC-53111	RSPH RIO 3	FINISHED WATER BOOSTER PUMP 1 RUN COMMAND	STOP	START	DO	3	5	7
				RSPH RIO 3	SPARE			DO	3	5	8
				RSPH RIO 3	SPARE			DO	3	5	9
				RSPH RIO 3	SPARE			DO	3	5	10
				RSPH RIO 3	SPARE			DO	3	5	11
				RSPH RIO 3	SPARE			DO	3	5	12
				RSPH RIO 3	SPARE			DO	3	5	13
				RSPH RIO 3	SPARE			DO	3	5	14
				RSPH RIO 3	SPARE			DO	3	5	15
	052-Y601	PMP-52212	XC-52212	RSPH RIO 3	RAW WATER PUMP 2 RUN COMMAND	STOP	START	DO	3	6	0
	052-Y601	PMP-52212	XR-52212	RSPH RIO 3	RAW WATER PUMP 2 VFD RESET COMMAND	<NULL>	RESET	DO	3	6	1
	052-Y602	EDS-53117	XVO-53117	RSPH RIO 3	ENERGY DISSIPATION VALVE OPEN COMMAND	<NULL>	OPEN	DO	3	6	2
	052-Y602	PMP-53221	XC-53221	RSPH RIO 3	FINISHED WATER RECIRCULATION PUMP RUN COMMAND	STOP	START	DO	3	6	3
	052-Y603	PMP-53421	XC-53421	RSPH RIO 3	SODIUM HYPOCHLORITE BATCH METERING PUMP RUN COMMAND	STOP	START	DO	3	6	4
	052-Y603	PMP-53322	XC-53322	RSPH RIO 3	DILUTE SODIUM HYPOCHLORITE METERING 2 RUN COMMAND	STOP	START	DO	3	6	5
	052-Y604	PMP-53112	XC-53112	RSPH RIO 3	FINISHED WATER BOOSTER PUMP 2 RUN COMMAND	STOP	START	DO	3	6	6
				RSPH RIO 3	SPARE			DO	3	6	7
				RSPH RIO 3	SPARE			DO	3	6	8
				RSPH RIO 3	SPARE			DO	3	6	9
				RSPH RIO 3	SPARE			DO	3	6	10
				RSPH RIO 3	SPARE			DO	3	6	11
				RSPH RIO 3	SPARE			DO	3	6	12
				RSPH RIO 3	SPARE			DO	3	6	13
				RSPH RIO 3	SPARE			DO	3	6	14
				RSPH RIO 3	SPARE			DO	3	6	15
				RSPH RIO 3	SPARE			DO	3	7	2
				RSPH RIO 3	SPARE			DO	3	7	3
				RSPH RIO 3	SPARE			DO	3	7	4
				RSPH RIO 3	SPARE			DO	3	7	5
				RSPH RIO 3	SPARE			DO	3	7	6
				RSPH RIO 3	SPARE			DO	3	7	7
				RSPH RIO 3	SPARE			DO	3	7	8
				RSPH RIO 3	SPARE			DO	3	7	9
				RSPH RIO 3	SPARE			DO	3	7	10
				RSPH RIO 3	SPARE			DO	3	7	11
				RSPH RIO 3	SPARE			DO	3	7	12
				RSPH RIO 3	SPARE			DO	3	7	13
				RSPH RIO 3	SPARE			DO	3	7	14
				RSPH RIO 3	SPARE			DO	3	7	15
	EXISTING			RSPH RIO 4	ETHERNET BASED CONTROLLER			H4-EBC	4	CPU	
	EXISTING		TT_1001	RSPH RIO 4	ROARING SPRINGS CAVE WATER TEMPERATURE	0 - 409.5	DEG F	RTD	4	0	0
	EXISTING			RSPH RIO 4	SPARE			RTD	4	0	1
	EXISTING			RSPH RIO 4	SPARE			RTD	4	0	2
	EXISTING			RSPH RIO 4	SPARE			RTD	4	0	3
	EXISTING			RSPH RIO 4	SPARE			RTD	4	0	4

	EXISTING			RSPH RIO 4	SPARE			RTD	4	0	5
	EXISTING			RSPH RIO 4	SPARE			RTD	4	0	6
	EXISTING			RSPH RIO 4	SPARE			RTD	4	0	7
	EXISTING		XCL_1001	RSPH RIO 4	ROARING SPRINGS CAVE LIGHTS ON/OFF COMMAND	OFF	ON	DO	4	1	0
	EXISTING			RSPH RIO 4	SPARE			DO	4	1	1
	EXISTING			RSPH RIO 4	SPARE			DO	4	1	2
	EXISTING			RSPH RIO 4	SPARE			DO	4	1	3
	EXISTING			RSPH RIO 4	SPARE			DO	4	1	4
	EXISTING			RSPH RIO 4	SPARE			DO	4	1	5
	EXISTING			RSPH RIO 4	SPARE			DO	4	1	6
	EXISTING			RSPH RIO 4	SPARE			DO	4	1	7
	EXISTING		LI_1051	RSPH RIO 4	ROARING SPRINGS CAVE WATER LEVEL	0 - 166	IN	AI	4	2	0
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	1
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	2
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	3
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	4
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	5
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	6
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	7
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	8
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	9
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	10
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	11
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	12
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	13
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	14
	EXISTING			RSPH RIO 4	SPARE			AI	4	2	15
	EXISTING			RSPH RIO 4	FILL MODULE			D4-FILL	4	3	
	EXISTING			RSPH RIO 4	FILL MODULE			D4-FILL	4	4	
	EXISTING			RSPH RIO 4	FILL MODULE			D4-FILL	4	5	
	EXISTING			RSPH RIO 4	FILL MODULE			D4-FILL	4	6	
	EXISTING			RSPH RIO 4	FILL MODULE			D4-FILL	4	7	

REV	P&ID DRAWING	FIELD DEVICE	POINT TAG	PLC / RIO	DESCRIPTION	OFF STATE / RANGE	ON STATE / ENG UNITS	I/O TYPE	RACK	SLOT	POINT / CHANNEL
				NR WTP MASTER PLC	PLC CPU			H2-DM1E	0	CPU	
				NR WTP MASTER PLC	REMOTE I/O MASTER			H2-ERM100	0	0	
				NR WTP MASTER PLC	COMMUNICATIONS MODULE			H2-ECOM	0	1	
	062-Y615	ZS-61101	ZS-61101	NR WTP MASTER PLC	PLC CPU MASTER MODE SWITCH	STANDBY	MASTER	DI	0	2	0
				NR WTP MASTER PLC	SPARE			DI	0	2	1
				NR WTP MASTER PLC	SPARE			DI	0	2	2
				NR WTP MASTER PLC	SPARE			DI	0	2	3
	062-Y615	G-61101	XSM-61101	NR WTP MASTER PLC	MASTER PLC MASTER MODE	<NULL>	MASTER	DO	0	2	4
	062-Y615	R-61101	XSS-61101	NR WTP MASTER PLC	MASTER PLC STANDBY MODE	<NULL>	STANDBY	DO	0	2	5
				NR WTP MASTER PLC	SPARE			DO	0	2	6
				NR WTP MASTER PLC	SPARE			DO	0	2	7
				NR WTP STANDBY PLC	PLC CPU			H2-DM1E	0	CPU	
				NR WTP STANDBY PLC	REMOTE I/O MASTER			H2-ERM100	0	0	
				NR WTP STANDBY PLC	COMMUNICATIONS MODULE			H2-ECOM	0	1	
	062-Y615	ZS-61102	ZS-61102	NR WTP STANDBY PLC	PLC CPU STANDBY MODE SWITCH	MASTER	STANDBY	DI	0	2	0
				NR WTP STANDBY PLC	SPARE			DI	0	2	1
				NR WTP STANDBY PLC	SPARE			DI	0	2	2
				NR WTP STANDBY PLC	SPARE			DI	0	2	3
	062-Y615	G-61102	XSM-61102	NR WTP STANDBY PLC	STANDBY PLC MASTER MODE	<NULL>	MASTER	DO	0	2	4
	062-Y615	R-61102	XSS-61102	NR WTP STANDBY PLC	STANDBY PLC STANDBY MODE	<NULL>	STANDBY	DO	0	2	5
				NR WTP STANDBY PLC	SPARE			DO	0	2	6
				NR WTP STANDBY PLC	SPARE			DO	0	2	7
				NR WTP RIO 1	ETHERNET BASED CONTROLLER			H2-EBC100	1	CPU	
	062-Y602	PMP-62011	SI-62011	NR WTP RIO 1	RAW WATER PUMP SPEED INDICATION	0-60	HZ	AI	1	0	0
	062-Y602	FCV-62032	ZI-62032	NR WTP RIO 1	RAW WATER PUMP DISCHARGE FLOW CONTROL VALVE POSITION INDICATION	0-100	%	AI	1	0	1
	062-Y603	MX-63012	SI-63012	NR WTP RIO 1	RAPID MIXER 1 SPEED INDICATION	0-60	HZ	AI	1	0	2
	062-Y601	LT-62051	LI-62051	NR WTP RIO 1	RAW WATER TANK 1 LEVEL	0-15	FT	AI	1	0	3
	062-Y603	LIT-63051	LI-63051	NR WTP RIO 1	REACTION TANK 1 LEVEL	0-15	FT	AI	1	0	4
	062-Y602	PIT-62061	PI-62061	NR WTP RIO 1	RAW WATER SUCTION PRESSURE	0-40	PSI	AI	1	0	5
	062-Y602	PIT-62062	PI-62062	NR WTP RIO 1	RAW WATER PUMP DISCHARGE PRESSURE	0-400	PSI	AI	1	0	6
				NR WTP RIO 1	SPARE			AI	1	0	7
	062-Y602	FCV-62033	ZI-62033	NR WTP RIO 1	RAW WATER FLOW CONTROL VALVE POSITION INDICATION	0-100	%	AI	1	1	0
	062-Y603	MX-63013	SI-63013	NR WTP RIO 1	RAPID MIXER 2 SPEED INDICATION	0-60	HZ	AI	1	1	1
	062-Y601	LT-62052	LI-62052	NR WTP RIO 1	RAW WATER TANK 2 LEVEL	0-15	FT	AI	1	1	2
	062-Y603	LIT-63052	LI-63052	NR WTP RIO 1	REACTION TANK 2 LEVEL	0-15	FT	AI	1	1	3
	062-Y602	AIT-62081	AI-62081	NR WTP RIO 1	RAW WATER PH	5-10	pH	AI	1	1	4
	062-Y602	AIT-62081	TI-62081	NR WTP RIO 1	RAW WATER TEMPERATURE	32-212	DEG F	AI	1	1	5
	062-Y602	AIT-62082	AI-62082	NR WTP RIO 1	RAW WATER TURBIDITY	0-700	NTU	AI	1	1	6
				NR WTP RIO 1	SPARE			AI	1	1	7
	062-Y605	LT-66051	LI-66051	NR WTP RIO 1	CHLORINE CONTACT TANK LEVEL	0-13	FT	AI	1	2	0
	062-Y602	FIT-62071	FI-62071	NR WTP RIO 1	RAW WATER FLOW	0-150	GPM	AI	1	2	1
	062-Y605	FIT-64071	FI-64071	NR WTP RIO 1	COMBINED FILTRATE FLOW	0-150	GPM	AI	1	2	2
	062-Y604	AIT-64081	AI-64081	NR WTP RIO 1	UF INFLUENT PH	5-10	pH	AI	1	2	3
	062-Y604	AIT-64081	TI-64081	NR WTP RIO 1	UF INFLUENT TEMPERATURE	32-212	DEG F	AI	1	2	4
	062-Y604	AIT-64082	AI-64082	NR WTP RIO 1	UF INFLUENT CHLORINE RESIDUAL	0-5	PPM	AI	1	2	5
	062-Y604	AIT-64083	AI-64083	NR WTP RIO 1	COMBINED FILTRATE TURBIDITY	0-700	NTU	AI	1	2	6
				NR WTP RIO 1	SPARE			AI	1	2	7
	062-Y605	AIT-68181	AI-68181	NR WTP RIO 1	CHLORINE CONTACT TANK INFLUENT PH	5-10	pH	AI	1	3	0
	062-Y605	AIT-68181	TI-68181	NR WTP RIO 1	CHLORINE CONTACT TANK INFLUENT TEMPERATURE	32-212	DEG F	AI	1	3	1
	062-Y605	AIT-68182	AI-68182	NR WTP RIO 1	CHLORINE CONTACT TANK INFLUENT CHLORINE RESIDUAL	0-5	PPM	AI	1	3	2
	062-Y605	AIT-66082	AI-66082	NR WTP RIO 1	FINISHED WATER PH	5-10	pH	AI	1	3	3
	062-Y605	AIT-66082	TI-66082	NR WTP RIO 1	FINISHED WATER TEMPERATURE	32-212	DEG F	AI	1	3	4
	062-Y605	AIT-66081	AI-66081	NR WTP RIO 1	FINISHED WATER CHLORINE RESIDUAL	0-5	PPM	AI	1	3	5
	062-Y605	AIT-66083	AI-66083	NR WTP RIO 1	FINISHED WATER TURBIDITY	0-700	NTU	AI	1	3	6
				NR WTP RIO 1	SPARE			AI	1	3	7

	062-Y606	PMP-68111	SI-68111	NR WTP RIO 1	FINISHED WATER TANK PUMP 1 SPEED INDICATION	0-60	HZ	AI	1	4	0
	062-Y609	PMP-68311	SI-68311	NR WTP RIO 1	UPPER ZONE BOOSTER PUMP 1 SPEED INDICATION	0-60	HZ	AI	1	4	1
	062-Y609	PMP-68312	SI-68312	NR WTP RIO 1	UPPER ZONE BOOSTER PUMP 2 SPEED INDICATION	0-60	HZ	AI	1	4	2
	062-Y608	PMP-68211	SI-68211	NR WTP RIO 1	HIGH DEMAND PUMP 1 SPEED INDICATION	0-60	HZ	AI	1	4	3
	062-Y607	LT-68451	LI-68451	NR WTP RIO 1	FINISHED WATER TANK A LEVEL	0-40	FT	AI	1	4	4
	062-Y606	PIT-68161	PI-68161	NR WTP RIO 1	FINISHED WATER TANK PUMPS PRESSURE	0-40	PSI	AI	1	4	5
	062-Y608	PIT-68162	PI-68162	NR WTP RIO 1	UPPER ZONE BOOSTER PUMPS SUCTION PRESSURE	0-50	PSI	AI	1	4	6
				NR WTP RIO 1	SPARE			AI	1	4	7
	062-Y606	PMP-68112	SI-68112	NR WTP RIO 1	FINISHED WATER TANK PUMP 2 SPEED INDICATION	0-60	HZ	AI	1	5	0
	062-Y609	PMP-68313	SI-68313	NR WTP RIO 1	UPPER ZONE BOOSTER PUMP 3 SPEED INDICATION	0-60	HZ	AI	1	5	1
	062-Y608	PMP-68212	SI-68212	NR WTP RIO 1	HIGH DEMAND PUMP 2 SPEED INDICATION	0-60	HZ	AI	1	5	2
	062-Y608	PMP-68213	SI-68213	NR WTP RIO 1	HIGH DEMAND PUMP 3 SPEED INDICATION	0-60	HZ	AI	1	5	3
	062-Y607	LT-68452	LI-68452	NR WTP RIO 1	FINISHED WATER TANK B LEVEL	0-40	FT	AI	1	5	4
	062-Y609	PIT-68163	PI-68163	NR WTP RIO 1	UPPER ZONE DISTRIBUTION PRESSURE	0-300	PSI	AI	1	5	5
				NR WTP RIO 1	SPARE			AI	1	5	6
				NR WTP RIO 1	SPARE			AI	1	5	7
	062-Y609	FIT-68171	FI-68171	NR WTP RIO 1	UPPER ZONE DISTRIBUTION FLOW	0-1200	GPM	AI	1	6	0
	062-Y608	FIT-68172	FI-68172	NR WTP RIO 1	LOWER ZONE DISTRIBUTION FLOW	0-1100	GPM	AI	1	6	1
	062-Y606	FIT-68173	FI-68173	NR WTP RIO 1	FINISHED WATER TANKS INFLUENT FLOW	0-150	GPM	AI	1	6	2
	062-Y608	FIT-68174	FI-68174	NR WTP RIO 1	COMBINED BACKFEED FLOW	0-150	GPM	AI	1	6	3
	062-Y608	FIT-68175	FI-68175	NR WTP RIO 1	SUPAI TUNNEL BACKFEED FLOW	0-100	GPM	AI	1	6	4
				NR WTP RIO 1	SPARE			AI	1	6	5
				NR WTP RIO 1	SPARE			AI	1	6	6
				NR WTP RIO 1	SPARE			AI	1	6	7
	062-Y614	LIT-69151	LI-69151	NR WTP RIO 1	PROCESS WASTE TANK LEVEL	0-8	FT	AI	1	7	0
	062-Y606	AIT-66084	AI-66084	NR WTP RIO 1	DISTRIBUTION WATER INSTRUMENTATION PANEL PH	5-10	pH	AI	1	7	1
	062-Y606	AIT-66084	TI-66084	NR WTP RIO 1	DISTRIBUTION WATER INSTRUMENTATION PANEL TEMPERATURE	32-212	DEG F	AI	1	7	2
	062-Y606	AIT-66085	AI-66085	NR WTP RIO 1	DISTRIBUTION WATER INSTRUMENTATION PANEL CHLORINE RESIDUAL	0-5	PPM	AI	1	7	3
	062-Y606	AIT-66086	AI-66086	NR WTP RIO 1	DISTRIBUTION WATER INSTRUMENTATION PANEL TURBIDITY	0-700	NTU	AI	1	7	4
				NR WTP RIO 1	SPARE			AI	1	7	5
				NR WTP RIO 1	SPARE			AI	1	7	6
				NR WTP RIO 1	SPARE			AI	1	7	7
				NR WTP RIO 2	ETHERNET BASED CONTROLLER - REMOTE I/O DROP (9 SLOT RACK)			H2-EBC100	2	CPU	
	062-Y615	TIT-61081	TI-61081	NR WTP RIO 2	PROCESS ROOM TEMPERATURE	-10-160	DEG F	AI	2	0	0
	062-Y615	TIT-61082	TI-61082	NR WTP RIO 2	MECHANICAL ROOM TEMPERATURE	-10-160	DEG F	AI	2	0	1
	062-Y615	TIT-67081	TI-67081	NR WTP RIO 2	ACIDS ROOM TEMPERATURE	-10-160	DEG F	AI	2	0	2
	062-Y615	TIT-67082	TI-67082	NR WTP RIO 2	BASES ROOM TEMPERATURE	-10-160	DEG F	AI	2	0	3
				NR WTP RIO 2	SPARE			AI	2	0	4
				NR WTP RIO 2	SPARE			AI	2	0	5
				NR WTP RIO 2	SPARE			AI	2	0	6
				NR WTP RIO 2	SPARE			AI	2	0	7
				NR WTP RIO 2	SPARE			AI	2	1	0
				NR WTP RIO 2	SPARE			AI	2	1	1
				NR WTP RIO 2	SPARE			AI	2	1	2
				NR WTP RIO 2	SPARE			AI	2	1	3
				NR WTP RIO 2	SPARE			AI	2	1	4
				NR WTP RIO 2	SPARE			AI	2	1	5
				NR WTP RIO 2	SPARE			AI	2	1	6
				NR WTP RIO 2	SPARE			AI	2	1	7
				NR WTP RIO 2	SPARE			AI	2	2	0
				NR WTP RIO 2	SPARE			AI	2	2	1
				NR WTP RIO 2	SPARE			AI	2	2	2
				NR WTP RIO 2	SPARE			AI	2	2	3
				NR WTP RIO 2	SPARE			AI	2	2	4
				NR WTP RIO 2	SPARE			AI	2	2	5
				NR WTP RIO 2	SPARE			AI	2	2	6
				NR WTP RIO 2	SPARE			AI	2	2	7
	062-Y602	PMP-62011	SC-62011	NR WTP RIO 2	RAW WATER PUMP SPEED COMMAND	0-60	HZ	AO	2	3	0
	062-Y603	MX-63012	SC-63012	NR WTP RIO 2	RAPID MIXER 1 SPEED COMMAND	0-60	HZ	AO	2	3	1

	062-Y606	PMP-68111	SC-68111	NR WTP RIO 2	FINISHED WATER TANK PUMP 1 SPEED COMMAND	0-60	HZ	AO	2	3	2
	062-Y609	PMP-68311	SC-68311	NR WTP RIO 2	UPPER ZONE BOOSTER PUMP 1 SPEED COMMAND	0-60	HZ	AO	2	3	3
	062-Y609	PMP-68312	SC-68312	NR WTP RIO 2	UPPER ZONE BOOSTER PUMP 2 SPEED COMMAND	0-60	HZ	AO	2	3	4
	062-Y608	PMP-68211	SC-68211	NR WTP RIO 2	HIGH DEMAND PUMP 1 SPEED COMMAND	0-60	HZ	AO	2	3	5
				NR WTP RIO 2	SPARE			AO	2	3	6
				NR WTP RIO 2	SPARE			AO	2	3	7
	062-Y602	FCV-62033	ZC-62033	NR WTP RIO 2	RAW WATER FLOW CONTROL VALVE POSITION COMMAND	0-100	%	AO	2	4	0
	062-Y602	FCV-62032	ZC-62032	NR WTP RIO 2	RAW WATER PUMP DISCHARGE FLOW CONTROL VALVE POSITION COMMAND	0-100	%	AO	2	4	1
	062-Y603	MX-63013	SC-63013	NR WTP RIO 2	RAPID MIXER 2 SPEED COMMAND	0-60	HZ	AO	2	4	2
	062-Y606	PMP-68112	SC-68112	NR WTP RIO 2	FINISHED WATER TANK PUMP 2 SPEED COMMAND	0-60	HZ	AO	2	4	3
	062-Y609	PMP-68313	SC-68313	NR WTP RIO 2	UPPER ZONE BOOSTER PUMP 3 SPEED COMMAND	0-60	HZ	AO	2	4	4
	062-Y608	PMP-68212	SC-68212	NR WTP RIO 2	HIGH DEMAND PUMP 2 SPEED COMMAND	0-60	HZ	AO	2	4	5
	062-Y608	PMP-68213	SC-68213	NR WTP RIO 2	HIGH DEMAND PUMP 3 SPEED COMMAND	0-60	HZ	AO	2	4	6
				NR WTP RIO 2	SPARE			AO	2	4	7
				NR WTP RIO 2	SPARE			AO	2	5	0
				NR WTP RIO 2	SPARE			AO	2	5	1
				NR WTP RIO 2	SPARE			AO	2	5	2
				NR WTP RIO 2	SPARE			AO	2	5	3
				NR WTP RIO 2	SPARE			AO	2	5	4
				NR WTP RIO 2	SPARE			AO	2	5	5
				NR WTP RIO 2	SPARE			AO	2	5	6
				NR WTP RIO 2	SPARE			AO	2	5	7
	062-Y601	LSHH-62051	LSHH-62051	NR WTP RIO 2	RAW WATER TANK 1 LEVEL SWITCH HIGH	<NULL>	ALARM	DI	2	6	0
	062-Y603	LSH-63051	LSH-63051	NR WTP RIO 2	REACTION TANK 1 LEVEL SWITCH HIGH	<NULL>	ALARM	DI	2	6	1
	062-Y602	FV-62031	HSR-62031	NR WTP RIO 2	RAW WATER PUMP SUCTION VALVE REMOTE	LOCAL	REMOTE	DI	2	6	2
	062-Y602	FV-62031	ZSO-62031	NR WTP RIO 2	RAW WATER PUMP SUCTION VALVE OPEN	<NULL>	OPENED	DI	2	6	3
	062-Y602	FV-62031	ZSC-63031	NR WTP RIO 2	RAW WATER PUMP SUCTION VALVE CLOSED	<NULL>	CLOSED	DI	2	6	4
	062-Y602	FV-62031	UA-62031	NR WTP RIO 2	RAW WATER PUMP SUCTION VALVE FAULT	<NULL>	ALARM	DI	2	6	5
	062-Y602	PMP-62011	HSR-62011	NR WTP RIO 2	RAW WATER PUMP REMOTE	LOCAL	REMOTE	DI	2	6	6
	062-Y602	PMP-62011	NI-62011	NR WTP RIO 2	RAW WATER PUMP RUNNING	OFF	RUNNING	DI	2	6	7
	062-Y602	PMP-62011	UA-62011	NR WTP RIO 2	RAW WATER PUMP FAULT	<NULL>	ALARM	DI	2	6	8
	062-Y602	PMP-62011	TSH-62011	NR WTP RIO 2	RAW WATER PUMP MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	2	6	9
	062-Y603	MX-63012	HSR-63012	NR WTP RIO 2	RAPID MIXER 1 REMOTE	LOCAL	REMOTE	DI	2	6	10
	062-Y603	MX-63012	NI-63012	NR WTP RIO 2	RAPID MIXER 1 RUNNING	OFF	RUNNING	DI	2	6	11
	062-Y603	MX-63012	UA-63012	NR WTP RIO 2	RAPID MIXER 1 FAULT	<NULL>	ALARM	DI	2	6	12
	062-Y603	MX-63012	TSH-63012	NR WTP RIO 2	RAPID MIXER 1 MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	2	6	13
	062-Y603	FV-63031	HSR-63031	NR WTP RIO 2	REACTION TANK 1 WASTE VALVE REMOTE	LOCAL	REMOTE	DI	2	6	14
	062-Y603	FV-63031	ZSO-63031	NR WTP RIO 2	REACTION TANK 1 WASTE VALVE OPENED	<NULL>	OPENED	DI	2	6	15
	062-Y603	FV-63031	ZSC-63031	NR WTP RIO 2	REACTION TANK 1 WASTE VALVE CLOSED	<NULL>	CLOSED	DI	2	6	16
	062-Y603	FV-63031	UA-63031	NR WTP RIO 2	REACTION TANK 1 WASTE VALVE FAULT	<NULL>	ALARM	DI	2	6	17
	062-Y602	PMP-62012	HSR-62012	NR WTP RIO 2	RAW WATER SAMPLE PUMP REMOTE	LOCAL	REMOTE	DI	2	6	18
	062-Y602	PMP-62012	NI-62012	NR WTP RIO 2	RAW WATER SAMPLE PUMP RUNNING	OFF	RUNNING	DI	2	6	19
	062-Y602	PMP-62012	UA-62012	NR WTP RIO 2	RAW WATER SAMPLE PUMP FAULT	<NULL>	ALARM	DI	2	6	20
	062-Y604	PMP-64011	HSR-64011	NR WTP RIO 2	UF INFLUENT SAMPLE PUMP REMOTE	LOCAL	REMOTE	DI	2	6	21
	062-Y604	PMP-64011	NI-64011	NR WTP RIO 2	UF INFLUENT SAMPLE PUMP RUNNING	OFF	RUNNING	DI	2	6	22
	062-Y604	PMP-64011	UA-64011	NR WTP RIO 2	UF INFLUENT SAMPLE PUMP FAULT	<NULL>	ALARM	DI	2	6	23
	062-Y606	PMP-66011	HSR-66011	NR WTP RIO 2	DISTRIBUTION WATER INSTRUMENTATION PANEL SAMPLE PUMP REMOTE	LOCAL	REMOTE	DI	2	6	24
	062-Y606	PMP-66011	NI-66011	NR WTP RIO 2	DISTRIBUTION WATER INSTRUMENTATION PANEL SAMPLE PUMP RUNNING	OFF	RUNNING	DI	2	6	25
	062-Y606	PMP-66011	UA-66011	NR WTP RIO 2	DISTRIBUTION WATER INSTRUMENTATION PANEL SAMPLE PUMP FAULT	<NULL>	ALARM	DI	2	6	26
				NR WTP RIO 2	SPARE			DI	2	6	27
				NR WTP RIO 2	SPARE			DI	2	6	28
				NR WTP RIO 2	SPARE			DI	2	6	29
				NR WTP RIO 2	SPARE			DI	2	6	30
				NR WTP RIO 2	SPARE			DI	2	6	31
	062-Y601	LSHH-62052	LSHH-62052	NR WTP RIO 2	RAW WATER TANK 2 LEVEL SWITCH HIGH	<NULL>	ALARM	DI	2	7	0
	062-Y603	LSH-63052	LSH-63052	NR WTP RIO 2	REACTION TANK 2 LEVEL SWITCH HIGH	<NULL>	ALARM	DI	2	7	1
	062-Y605	LSHH-66051	LSHH-66051	NR WTP RIO 2	CHLORINE CONTACT TANK LEVEL SWITCH HIGH HIGH	<NULL>	ALARM	DI	2	7	2
	062-Y602	FCV-62033	HSR-62033	NR WTP RIO 2	RAW WATER FLOW CONTROL VALVE REMOTE	LOCAL	REMOTE	DI	2	7	3
	062-Y602	FCV-62033	ZSO-62033	NR WTP RIO 2	RAW WATER FLOW CONTROL VALVE OPENED	<NULL>	OPENED	DI	2	7	4
	062-Y602	FCV-62033	ZSC-62033	NR WTP RIO 2	RAW WATER FLOW CONTROL VALVE CLOSED	<NULL>	CLOSED	DI	2	7	5
	062-Y602	FCV-62033	UA-62033	NR WTP RIO 2	RAW WATER FLOW CONTROL VALVE FAULT	<NULL>	ALARM	DI	2	7	6
	062-Y602	FCV-62032	HSR-62032	NR WTP RIO 2	RAW WATER PUMP DISCHARGE FLOW CONTROL VALVE REMOTE	LOCAL	REMOTE	DI	3	7	7
	062-Y602	FCV-62032	ZSO-62032	NR WTP RIO 2	RAW WATER PUMP DISCHARGE FLOW CONTROL VALVE OPENED	<NULL>	OPENED	DI	3	7	8
	062-Y602	FCV-62032	ZSC-62032	NR WTP RIO 2	RAW WATER PUMP DISCHARGE FLOW CONTROL VALVE CLOSED	<NULL>	CLOSED	DI	3	7	9

	062-Y602	FCV-62032	UA-62032	NR WTP RIO 2	RAW WATER PUMP DISCHARGE FLOW CONTROL VALVE FAULT	<NULL>	ALARM	DI	3	7	10
	062-Y603	MX-63013	HSR-63013	NR WTP RIO 2	RAPID MIXER 2 REMOTE	LOCAL	REMOTE	DI	2	7	11
	062-Y603	MX-63013	NI-63013	NR WTP RIO 2	RAPID MIXER 2 RUNNING	OFF	RUNNING	DI	2	7	12
	062-Y603	MX-63013	UA-63013	NR WTP RIO 2	RAPID MIXER 2 FAULT	<NULL>	ALARM	DI	2	7	13
	062-Y603	MX-63013	TSH-63013	NR WTP RIO 2	RAPID MIXER 2 MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	2	7	14
	062-Y603	FV-63032	HSR-63032	NR WTP RIO 2	REACTION TANK 2 WASTE VALVE REMOTE	LOCAL	REMOTE	DI	2	7	15
	062-Y603	FV-63032	ZSO-63032	NR WTP RIO 2	REACTION TANK 2 WASTE VALVE OPENED	<NULL>	OPENED	DI	2	7	16
	062-Y603	FV-63032	ZSC-63032	NR WTP RIO 2	REACTION TANK 2 WASTE VALVE CLOSED	<NULL>	CLOSED	DI	2	7	17
	062-Y603	FV-63032	UA-63032	NR WTP RIO 2	REACTION TANK 2 WASTE VALVE FAULT	<NULL>	ALARM	DI	2	7	18
	062-Y605	FV-66031	HSR-66031	NR WTP RIO 2	CHLORINE CONTACT TANK OUTLET VALVE REMOTE	LOCAL	REMOTE	DI	2	7	19
	062-Y605	FV-66031	ZSO-66031	NR WTP RIO 2	CHLORINE CONTACT TANK OUTLET VALVE OPENED	<NULL>	OPENED	DI	2	7	20
	062-Y605	FV-66031	ZSC-66031	NR WTP RIO 2	CHLORINE CONTACT TANK OUTLET VALVE CLOSED	<NULL>	CLOSED	DI	2	7	21
	062-Y605	FV-66031	UA-66031	NR WTP RIO 2	CHLORINE CONTACT TANK OUTLET VALVE FAULT	<NULL>	ALARM	DI	2	7	22
	062-Y605	PMP-68117	HSR-68117	NR WTP RIO 2	CT TANK INFLUENT SAMPLE PUMP REMOTE	LOCAL	REMOTE	DI	2	7	23
	062-Y605	PMP-68117	NI-68117	NR WTP RIO 2	CT TANK INFLUENT SAMPLE PUMP RUNNING	OFF	RUNNING	DI	2	7	24
	062-Y605	PMP-68117	UA-68117	NR WTP RIO 2	CT TANK INFLUENT SAMPLE PUMP FAULT	<NULL>	ALARM	DI	2	7	25
	062-Y602	FIT-62071	FQI-62071	NR WTP RIO 2	RAW WATER FLOW TOTALIZE	<NULL>	TOTALIZE	DI	2	7	26
	062-Y605	FIT-64071	FQI-64071	NR WTP RIO 2	COMBINED FILTRATE FLOW TOTALIZE	<NULL>	TOTALIZE	DI	2	7	27
				NR WTP RIO 2	SPARE			DI	2	7	28
				NR WTP RIO 2	SPARE			DI	2	7	29
				NR WTP RIO 2	SPARE			DI	2	7	30
				NR WTP RIO 2	SPARE			DI	2	7	31
				NR WTP RIO 3	ETHERNET BASED CONTROLLER - REMOTE I/O DROP (9 SLOT RACK)			H2-EBC100	3	CPU	
	062-Y606	PMP-68111	HSR-68111	NR WTP RIO 3	FINISHED WATER TANK PUMP 1 REMOTE	LOCAL	REMOTE	DI	3	0	0
	062-Y606	PMP-68111	NI-68111	NR WTP RIO 3	FINISHED WATER TANK PUMP 1 RUNNING	OFF	RUNNING	DI	3	0	1
	062-Y606	PMP-68111	UA-68111	NR WTP RIO 3	FINISHED WATER TANK PUMP 1 FAULT	<NULL>	ALARM	DI	3	0	2
	062-Y606	PMP-68111	TSH-68111	NR WTP RIO 3	FINISHED WATER TANK PUMP 1 MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	3	0	3
	062-Y609	PMP-68311	HSR-68311	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 1 REMOTE	LOCAL	REMOTE	DI	3	0	4
	062-Y609	PMP-68311	NI-68311	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 1 RUNNING	OFF	RUNNING	DI	3	0	5
	062-Y609	PMP-68311	UA-68311	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 1 FAULT	<NULL>	ALARM	DI	3	0	6
	062-Y609	PMP-68311	TSH-68311	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 1 MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	3	0	7
	062-Y609	PMP-68312	HSR-68312	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 2 REMOTE	LOCAL	REMOTE	DI	3	0	8
	062-Y609	PMP-68312	NI-68312	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 2 RUNNING	OFF	RUNNING	DI	3	0	9
	062-Y609	PMP-68312	UA-68312	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 2 FAULT	<NULL>	ALARM	DI	3	0	10
	062-Y609	PMP-68312	TSH-68312	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 2 MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	3	0	11
	062-Y608	PMP-68211	HSR-68211	NR WTP RIO 3	HIGH DEMAND PUMP 1 REMOTE	LOCAL	REMOTE	DI	3	0	12
	062-Y608	PMP-68211	NI-68211	NR WTP RIO 3	HIGH DEMAND PUMP 1 RUNNING	OFF	RUNNING	DI	3	0	13
	062-Y608	PMP-68211	UA-68211	NR WTP RIO 3	HIGH DEMAND PUMP 1 FAULT	<NULL>	ALARM	DI	3	0	14
	062-Y608	PMP-68211	TSH-68211	NR WTP RIO 3	HIGH DEMAND PUMP 1 MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	3	0	15
	062-Y608	FV-68031	HSR-68031	NR WTP RIO 2	COMBINED BACKFEED FLOW VALVE REMOTE	LOCAL	REMOTE	DI	3	0	16
	062-Y608	FV-68031	ZSO-68031	NR WTP RIO 2	COMBINED BACKFEED FLOW VALVE OPENED	<NULL>	OPENED	DI	3	0	17
	062-Y608	FV-68031	ZSC-68031	NR WTP RIO 2	COMBINED BACKFEED FLOW VALVE CLOSED	<NULL>	CLOSED	DI	3	0	18
	062-Y608	FV-68031	UA-68031	NR WTP RIO 2	COMBINED BACKFEED FLOW VALVE FAULT	<NULL>	ALARM	DI	3	0	19
	062-Y608	FV-68032	HSR-68032	NR WTP RIO 2	BACKFEED TO SUPAI TUNNEL FLOW VALVE REMOTE	LOCAL	REMOTE	DI	3	0	20
	062-Y608	FV-68032	ZSO-68032	NR WTP RIO 2	BACKFEED TO SUPAI TUNNEL FLOW VALVE OPENED	<NULL>	OPENED	DI	3	0	21
	062-Y608	FV-68032	ZSC-68032	NR WTP RIO 2	BACKFEED TO SUPAI TUNNEL FLOW VALVE CLOSED	<NULL>	CLOSED	DI	3	0	22
	062-Y608	FV-68032	UA-68032	NR WTP RIO 2	BACKFEED TO SUPAI TUNNEL FLOW VALVE FAULT	<NULL>	ALARM	DI	3	0	23
	062-Y609	FIT-68171	FQI-68171	NR WTP RIO 3	UPPER ZONE DISTRIBUTION FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	0	24
	062-Y608	FIT-68172	FQI-68172	NR WTP RIO 3	LOWER ZONE DISTRIBUTION FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	0	25
	062-Y606	FIT-68173	FQI-68173	NR WTP RIO 3	FINISHED WATER PUMPS FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	0	26
	062-Y608	FIT-68174	FQI-68174	NR WTP RIO 3	RSPH BACKFEED FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	0	27
	062-Y608	FIT-68175	FQI-68175	NR WTP RIO 3	SUPAI TUNNEL BACKFEED FLOW TOTALIZE	<NULL>	TOTALIZE	DI	3	0	28
				NR WTP RIO 3	SPARE			DI	3	0	29
				NR WTP RIO 3	SPARE			DI	3	0	30
				NR WTP RIO 3	SPARE			DI	3	0	31
	062-Y606	PMP-68112	HSR-68112	NR WTP RIO 3	FINISHED WATER TANK PUMP 2 REMOTE	LOCAL	REMOTE	DI	3	1	0
	062-Y606	PMP-68112	NI-68112	NR WTP RIO 3	FINISHED WATER TANK PUMP 2 RUNNING	OFF	RUNNING	DI	3	1	1
	062-Y606	PMP-68112	UA-68112	NR WTP RIO 3	FINISHED WATER TANK PUMP 2 FAULT	<NULL>	ALARM	DI	3	1	2
	062-Y606	PMP-68112	TSH-68112	NR WTP RIO 3	FINISHED WATER TANK PUMP 2 MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	3	1	3
	062-Y609	PMP-68313	HSR-68313	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 3 REMOTE	LOCAL	REMOTE	DI	3	1	4
	062-Y609	PMP-68313	NI-68313	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 3 RUNNING	OFF	RUNNING	DI	3	1	5
	062-Y609	PMP-68313	UA-68313	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 3 FAULT	<NULL>	ALARM	DI	3	1	6
	062-Y609	PMP-68313	TSH-68313	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 3 MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	3	1	7
	062-Y608	PMP-68212	HSR-68212	NR WTP RIO 3	HIGH DEMAND PUMP 2 REMOTE	LOCAL	REMOTE	DI	3	1	8
	062-Y608	PMP-68212	NI-68212	NR WTP RIO 3	HIGH DEMAND PUMP 2 RUNNING	OFF	RUNNING	DI	3	1	9

	062-Y608	PMP-68212	UA-68212	NR WTP RIO 3	HIGH DEMAND PUMP 2 FAULT	<NULL>	ALARM	DI	3	1	10
	062-Y608	PMP-68212	TSH-68212	NR WTP RIO 3	HIGH DEMAND PUMP 2 MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	3	1	11
	062-Y608	PMP-68213	HSR-68213	NR WTP RIO 3	HIGH DEMAND PUMP 3 REMOTE	LOCAL	REMOTE	DI	3	1	12
	062-Y608	PMP-68213	NI-68213	NR WTP RIO 3	HIGH DEMAND PUMP 3 RUNNING	OFF	RUNNING	DI	3	1	13
	062-Y608	PMP-68213	UA-68213	NR WTP RIO 3	HIGH DEMAND PUMP 3 FAULT	<NULL>	ALARM	DI	3	1	14
	062-Y608	PMP-68213	TSH-68213	NR WTP RIO 3	HIGH DEMAND PUMP 3 MOTOR TEMPERATURE SWITCH HIGH	<NULL>	ALARM	DI	3	1	15
	062-Y608	FV-68033	HSR-68033	NR WTP RIO 3	FINISHED WATER TANK RECIRCULATION VALVE REMOTE	LOCAL	REMOTE	DI	3	1	16
	062-Y608	FV-68033	ZSO-68033	NR WTP RIO 3	FINISHED WATER TANK RECIRCULATION VALVE OPENED	<NULL>	OPENED	DI	3	1	17
	062-Y608	FV-68033	ZSC-68033	NR WTP RIO 3	FINISHED WATER TANK RECIRCULATION VALVE CLOSED	<NULL>	CLOSED	DI	3	1	18
	062-Y608	FV-68033	UA-68033	NR WTP RIO 3	FINISHED WATER TANK RECIRCULATION VALVE FAULT	<NULL>	ALARM	DI	3	1	19
	062-Y614	LSHH-69151	LSHH-69151	NR WTP RIO 3	PROCESS WASTE TANK LEVEL SWITCH HIGH HIGH	<NULL>	ALARM	DI	3	1	20
	062-Y615	FSH-61641	FSH-61641	NR WTP RIO 3	LABRATORY ESEW IN USE	<NULL>	ALARM	DI	3	1	21
	062-Y615	FSH-61642	FSH-61642	NR WTP RIO 3	ACIDS DRUM ROOM ESEW IN USE	<NULL>	ALARM	DI	3	1	22
	062-Y615	FSH-61643	FSH-61643	NR WTP RIO 3	BASES DRUM ROOM ESEW IN USE	<NULL>	ALARM	DI	3	1	23
				NR WTP RIO 3	SPARE			DI	3	1	24
				NR WTP RIO 3	SPARE			DI	3	1	25
				NR WTP RIO 3	SPARE			DI	3	1	26
				NR WTP RIO 3	SPARE			DI	3	1	27
				NR WTP RIO 3	SPARE			DI	3	1	28
				NR WTP RIO 3	SPARE			DI	3	1	29
				NR WTP RIO 3	SPARE			DI	3	1	30
				NR WTP RIO 3	SPARE			DI	3	1	31
	062-Y615	ZS-61491A	ZS-61491A	NR WTP RIO 3	VESTIBULE EXTERIOR DOOR RIGHT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	0
	062-Y615	ZS-61491B	ZS-61491B	NR WTP RIO 3	VESTIBULE EXTERIOR DOOR LEFT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	1
	062-Y615	ZS-61492A	ZS-61492A	NR WTP RIO 3	MECHANICAL ROOM EXTERIOR DOOR RIGHT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	2
	062-Y615	ZS-61492B	ZS-61492B	NR WTP RIO 3	MECHANICAL ROOM EXTERIOR DOOR LEFT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	3
	062-Y615	ZS-61493	ZS-61493	NR WTP RIO 3	BREAK/KITCHEN ROOM EXTERIOR DOOR INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	4
	062-Y615	ZS-61494	ZS-61494	NR WTP RIO 3	PROCESS AREA SOUTH EXTERIOR DOOR INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	5
	062-Y615	ZS-61495A	ZS-61495A	NR WTP RIO 3	ACIDS ROOM EXTERIOR DOOR RIGHT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	6
	062-Y615	ZS-61495B	ZS-61495B	NR WTP RIO 3	ACIDS ROOM EXTERIOR DOOR LEFT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	7
	062-Y615	ZS-61496A	ZS-61496A	NR WTP RIO 3	PROCESS MATERIALS SHOP EXTERIOR DOOR RIGHT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	8
	062-Y615	ZS-61496B	ZS-61496B	NR WTP RIO 3	PROCESS MATERIALS SHOP EXTERIOR DOOR LEFT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	9
	062-Y615	ZS-61497A	ZS-61497A	NR WTP RIO 3	ELECTRICAL ROOM EXTERIOR DOOR RIGHT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	10
	062-Y615	ZS-61497B	ZS-61497B	NR WTP RIO 3	ELECTRICAL ROOM EXTERIOR DOOR LEFT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	11
	062-Y615	ZS-61498	ZS-61498	NR WTP RIO 3	PROCESS AREA NORTH EXTERIOR DOOR INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	12
	062-Y615	ZS-61499A	ZS-61499A	NR WTP RIO 3	BASES ROOM EXTERIOR DOOR RIGHT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	13
	062-Y615	ZS-61499B	ZS-61499B	NR WTP RIO 3	BASES ROOM EXTERIOR DOOR LEFT INTRUSION SWITCH	ALARM	<NULL>	DI	3	2	14
	062-Y615	SD-61109	SSH-61109	NR WTP RIO 3	ELECTRICAL ROOM SMOKE DETECTOR	<NULL>	ALARM	DI	3	2	15
	062-Y615	HTP-61741	UA-61741	NR WTP RIO 3	HEAT TRACE PANEL FAULT	<NULL>	ALARM	DI	3	2	16
	009-Y620	NR WTP CP PS 1	NR-PLC-PS1	NR WTP RIO 3	NR WTP CONTROL PANEL POWER SUPPLY 1	NOT OK	OK	DI	3	2	17
	009-Y620	NR WTP CP PS 2	NR-PLC-PS2	NR WTP RIO 3	NR WTP CONTROL PANEL POWER SUPPLY 2	NOT OK	OK	DI	3	2	18
	009-Y620	NR WTP CP UPS	NR-PLC-UPS	NR WTP RIO 3	NR WTP CONTROL PANEL UPS	NOT OK	OK	DI	3	2	19
				NR WTP RIO 3	SPARE			DI	3	2	20
				NR WTP RIO 3	SPARE			DI	3	2	21
				NR WTP RIO 3	SPARE			DI	3	2	22
				NR WTP RIO 3	SPARE			DI	3	2	23
				NR WTP RIO 3	SPARE			DI	3	2	24
				NR WTP RIO 3	SPARE			DI	3	2	25
				NR WTP RIO 3	SPARE			DI	3	2	26
				NR WTP RIO 3	SPARE			DI	3	2	27
				NR WTP RIO 3	SPARE			DI	3	2	28
				NR WTP RIO 3	SPARE			DI	3	2	29
				NR WTP RIO 3	SPARE			DI	3	2	30
				NR WTP RIO 3	SPARE			DI	3	2	31
				NR WTP RIO 3	SPARE			DI	3	3	0
				NR WTP RIO 3	SPARE			DI	3	3	1
				NR WTP RIO 3	SPARE			DI	3	3	2
				NR WTP RIO 3	SPARE			DI	3	3	3
				NR WTP RIO 3	SPARE			DI	3	3	4
				NR WTP RIO 3	SPARE			DI	3	3	5
				NR WTP RIO 3	SPARE			DI	3	3	6
				NR WTP RIO 3	SPARE			DI	3	3	7
				NR WTP RIO 3	SPARE			DI	3	3	8
				NR WTP RIO 3	SPARE			DI	3	3	9
				NR WTP RIO 3	SPARE			DI	3	3	10
				NR WTP RIO 3	SPARE			DI	3	3	11

				NR WTP RIO 3	SPARE			DI	3	3	12
				NR WTP RIO 3	SPARE			DI	3	3	13
				NR WTP RIO 3	SPARE			DI	3	3	14
				NR WTP RIO 3	SPARE			DI	3	3	15
				NR WTP RIO 3	SPARE			DI	3	3	16
				NR WTP RIO 3	SPARE			DI	3	3	17
				NR WTP RIO 3	SPARE			DI	3	3	18
				NR WTP RIO 3	SPARE			DI	3	3	19
				NR WTP RIO 3	SPARE			DI	3	3	20
				NR WTP RIO 3	SPARE			DI	3	3	21
				NR WTP RIO 3	SPARE			DI	3	3	22
				NR WTP RIO 3	SPARE			DI	3	3	23
				NR WTP RIO 3	SPARE			DI	3	3	24
				NR WTP RIO 3	SPARE			DI	3	3	25
				NR WTP RIO 3	SPARE			DI	3	3	26
				NR WTP RIO 3	SPARE			DI	3	3	27
				NR WTP RIO 3	SPARE			DI	3	3	28
				NR WTP RIO 3	SPARE			DI	3	3	29
				NR WTP RIO 3	SPARE			DI	3	3	30
				NR WTP RIO 3	SPARE			DI	3	3	31
	062-Y602	PMP-62011	XC-62011	NR WTP RIO 3	RAW WATER PUMP START/STOP COMMAND	STOP	START	DO	3	4	0
	062-Y602	FV-62031	XVO-62031	NR WTP RIO 3	RAW WATER PUMP SUCTION VALVE OPEN COMMAND	CLOSE	OPEN	DO	3	4	1
	062-Y602	FV-62031	XVC-62031	NR WTP RIO 3	RAW WATER PUMP SUCTION VALVE CLOSE COMMAND	CLOSE	OPEN	DO	3	4	2
	062-Y603	MX-63012	XC-63012	NR WTP RIO 3	RAPID MIXER 1 START/STOP COMMAND	STOP	START	DO	3	4	3
	062-Y603	FV-63031	XVO-63031	NR WTP RIO 3	REACTION TANK 1 WASTE VALVE OPEN COMMAND	<NULL>	OPEN	DO	3	4	4
	062-Y603	FV-63031	XVC-63031	NR WTP RIO 3	REACTION TANK 1 WASTE VALVE CLOSE COMMAND	<NULL>	CLOSE	DO	3	4	5
	062-Y603	FV-63031	XVS-63031	NR WTP RIO 3	REACTION TANK 1 WASTE VALVE STOP COMMAND	<NULL>	STOP	DO	3	4	6
	062-Y606	PMP-68111	XC-68111	NR WTP RIO 3	FINISHED WATER TANK PUMP 1 START/STOP COMMAND	STOP	START	DO	3	4	7
	062-Y609	PMP-68311	XC-68311	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 1 START/STOP COMMAND	STOP	START	DO	3	4	8
	062-Y609	PMP-68312	XC-68312	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 2 START/STOP COMMAND	STOP	START	DO	3	4	9
	062-Y608	PMP-68211	XC-68211	NR WTP RIO 3	HIGH DEMAND PUMP 1 START/STOP COMMAND	STOP	START	DO	3	4	10
	062-Y602	PMP-62012	XC-62012	NR WTP RIO 3	RAW WATER SAMPLE PUMP START/STOP COMMAND	STOP	START	DO	3	4	11
	062-Y604	PMP-64011	XC-64011	NR WTP RIO 3	UF INFLUENT SAMPLE PUMP START/STOP COMMAND	STOP	START	DO	3	4	12
	062-Y604	SV-64031	XVO-64031	NR WTP RIO 3	UF CIP TANK SOLENOID VALVE OPEN COMMAND	<NULL>	OPEN	DO	3	4	13
				NR WTP RIO 3	SPARE			DO	3	4	14
				NR WTP RIO 3	SPARE			DO	3	4	15
	062-Y603	MX-63013	XC-63013	NR WTP RIO 3	RAPID MIXER 2 START/STOP COMMAND	STOP	START	DO	3	5	0
	062-Y603	FV-63032	XVO-63032	NR WTP RIO 3	REACTION TANK 2 WASTE VALVE OPEN COMMAND	<NULL>	OPEN	DO	3	5	1
	062-Y603	FV-63032	XVC-63032	NR WTP RIO 3	REACTION TANK 2 WASTE VALVE CLOSE COMMAND	<NULL>	CLOSE	DO	3	5	2
	062-Y603	FV-63032	XVS-63032	NR WTP RIO 3	REACTION TANK 2 WASTE VALVE STOP COMMAND	<NULL>	STOP	DO	3	5	3
	062-Y605	FV-66031	XVO-66031	NR WTP RIO 3	CHLORINE CONTACT TANK OUTLET VALVE OPEN COMMAND	<NULL>	OPEN	DO	3	5	4
	062-Y605	FV-66031	XVC-66031	NR WTP RIO 3	CHLORINE CONTACT TANK OUTLET VALVE CLOSE COMMAND	<NULL>	CLOSE	DO	3	5	5
	062-Y606	PMP-68112	XC-68112	NR WTP RIO 3	FINISHED WATER TANK PUMP 2 START/STOP COMMAND	STOP	START	DO	3	5	6
	062-Y609	PMP-68313	XC-68313	NR WTP RIO 3	UPPER ZONE BOOSTER PUMP 3 START/STOP COMMAND	STOP	START	DO	3	5	7
	062-Y608	PMP-68212	XC-68212	NR WTP RIO 3	HIGH DEMAND PUMP 2 START/STOP COMMAND	STOP	START	DO	3	5	8
	062-Y608	PMP-68213	XC-68213	NR WTP RIO 3	HIGH DEMAND PUMP 3 START/STOP COMMAND	STOP	START	DO	3	5	9
	062-Y606	PMP-66011	XC-66011	NR WTP RIO 3	DISTRIBUTION WATER INSTRUMENTATION PANEL SAMPLE PUMP START/STOP COMMAND	STOP	START	DO	3	5	10
	062-Y605	PMP-68117	XC-68117	NR WTP RIO 3	CT TANK INFLUENT SAMPLE PUMP START/STOP COMMAND	STOP	START	DO	3	5	11
				NR WTP RIO 3	SPARE			DO	3	5	12
				NR WTP RIO 3	SPARE			DO	3	5	13
				NR WTP RIO 3	SPARE			DO	3	5	14
				NR WTP RIO 3	SPARE			DO	3	5	15
	062-Y608	FV-68031	XVO-68031	NR WTP RIO 3	BACKFEED TO RSPH FLOW VALVE OPEN COMMAND	<NULL>	OPEN	DO	3	6	0
	062-Y608	FV-68031	XVC-68031	NR WTP RIO 3	BACKFEED TO RSPH FLOW VALVE CLOSE COMMAND	<NULL>	CLOSE	DO	3	6	1
	062-Y608	FV-68031	XVS-68031	NR WTP RIO 3	BACKFEED TO RSPH FLOW VALVE STOP COMMAND	<NULL>	STOP	DO	3	6	2
	062-Y608	FV-68032	XVO-68032	NR WTP RIO 3	BACKFEED TO SUPAI TUNNEL FLOW VALVE OPEN COMMAND	<NULL>	OPEN	DO	3	6	3
	062-Y608	FV-68032	XVC-68032	NR WTP RIO 3	BACKFEED TO SUPAI TUNNEL FLOW VALVE CLOSE COMMAND	<NULL>	CLOSE	DO	3	6	4
	062-Y608	FV-68032	XVS-68032	NR WTP RIO 3	BACKFEED TO SUPAI TUNNEL FLOW VALVE STOP COMMAND	<NULL>	STOP	DO	3	6	5
	062-Y608	FV-68033	XVO-68033	NR WTP RIO 3	FINISHED WATER TANK RECIRCULATION OPEN COMMAND	<NULL>	OPEN	DO	3	6	6
	062-Y608	FV-68033	XVC-68033	NR WTP RIO 3	FINISHED WATER TANK RECIRCULATION CLOSE COMMAND	<NULL>	CLOSE	DO	3	6	7
	062-Y608	FV-68033	XVS-68033	NR WTP RIO 3	FINISHED WATER TANK RECIRCULATION STOP COMMAND	<NULL>	STOP	DO	3	6	8
				NR WTP RIO 3	SPARE			DO	3	6	9
				NR WTP RIO 3	SPARE			DO	3	6	10
				NR WTP RIO 3	SPARE			DO	3	6	11
				NR WTP RIO 3	SPARE			DO	3	6	12

				NR WTP RIO 3	SPARE			DO	3	6	13
				NR WTP RIO 3	SPARE			DO	3	6	14
				NR WTP RIO 3	SPARE			DO	3	6	15
				NR WTP RIO 3	SPARE			DO	3	7	0
				NR WTP RIO 3	SPARE			DO	3	7	1
				NR WTP RIO 3	SPARE			DO	3	7	2
				NR WTP RIO 3	SPARE			DO	3	7	3
				NR WTP RIO 3	SPARE			DO	3	7	4
				NR WTP RIO 3	SPARE			DO	3	7	5
				NR WTP RIO 3	SPARE			DO	3	7	6
				NR WTP RIO 3	SPARE			DO	3	7	7
				NR WTP RIO 3	SPARE			DO	3	7	8
				NR WTP RIO 3	SPARE			DO	3	7	9
				NR WTP RIO 3	SPARE			DO	3	7	10
				NR WTP RIO 3	SPARE			DO	3	7	11
				NR WTP RIO 3	SPARE			DO	3	7	12
				NR WTP RIO 3	SPARE			DO	3	7	13
				NR WTP RIO 3	SPARE			DO	3	7	14
				NR WTP RIO 3	SPARE			DO	3	7	15
				NR WTP RIO 4	ETHERNET BASED CONTROLLER			H2-EBC100	4	CPU	
	062-Y610	WIT-67191	WI-67191	NR WTP RIO 4	COAGULANT DRUM 1 WEIGHT INDICATION	0-850	LB	AI	4	0	0
	062-Y611	WIT-67291	WI-67291	NR WTP RIO 4	SODIUM HYPOCHLORITE DRUM 1 WEIGHT INDICATION	0-850	LB	AI	4	0	1
	062-Y611	WIT-67292	WI-67292	NR WTP RIO 4	SODIUM HYPOCHLORITE DRUM 2 WEIGHT INDICATION	0-850	LB	AI	4	0	2
	062-Y610	WIT-67391	WI-67391	NR WTP RIO 4	CAUSTIC SODA DRUM 1 WEIGHT INDICATION	0-850	LB	AI	4	0	3
	062-Y610	WIT-67392	WI-67392	NR WTP RIO 4	CAUSTIC SODA DRUM 2 WEIGHT INDICATION	0-850	LB	AI	4	0	4
				NR WTP RIO 4	SPARE			AI	4	0	5
				NR WTP RIO 4	SPARE			AI	4	0	6
				NR WTP RIO 4	SPARE			AI	4	0	7
	062-Y612	WIT-67691	WI-67691	NR WTP RIO 4	SODIUM BISULFITE DRUM WEIGHT INDICATION	0-850	LB	AI	4	1	0
	062-Y612	WIT-67692	WI-67692	NR WTP RIO 4	CITRIC ACID DRUM WEIGHT INDICATION	0-850	LB	AI	4	1	1
	062-Y613	WIT-67491	WI-67491	NR WTP RIO 4	HYDROCHLORIC ACID DRUM WEIGHT INDICATION	0-850	LB	AI	4	1	2
	062-Y610	WIT-67192	WI-67192	NR WTP RIO 4	COAGULANT DRUM 2 WEIGHT INDICATION	0-850	LB	AI	4	1	3
	062-Y611	WIT-67293	WI-67293	NR WTP RIO 4	SODIUM HYPOCHLORITE TOTE WEIGHT INDICATION	0-3500	LB	AI	4	1	4
				NR WTP RIO 4	SPARE			AI	4	1	5
				NR WTP RIO 4	SPARE			AI	4	1	6
				NR WTP RIO 4	SPARE			AI	4	1	7
	062-Y610	PMP-67111	SI-67111	NR WTP RIO 4	COAGULANT METERING PUMP SPEED INDICATION	0-60	HZ	AI	4	2	0
	062-Y611	PMP-67211	SI-67211	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 1 SPEED INDICATION	0-60	HZ	AI	4	2	1
	062-Y611	PMP-67212	SI-67212	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 2 SPEED INDICATION	0-60	HZ	AI	4	2	2
	062-Y611	PMP-67213	SI-67213	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 3 SPEED INDICATION	0-60	HZ	AI	4	2	3
	062-Y611	PMP-67214	SI-67214	NR WTP RIO 4	SODIUM HYPOCHLORITE SWING PUMP SPEED INDICATION	0-60	HZ	AI	4	2	4
	062-Y610	PMP-67311	SI-67311	NR WTP RIO 4	CAUSTIC SODA METERING PUMP SPEED INDICATION	0-60	HZ	AI	4	2	5
	062-Y612	AIT-61881	AI-61881	NR WTP RIO 4	SULFUR DIOXIDE ANALYZER	0-25	PPM	AI	4	2	6
				NR WTP RIO 4	SPARE			AI	4	2	7
				NR WTP RIO 4	SPARE			AI	4	3	0
				NR WTP RIO 4	SPARE			AI	4	3	1
				NR WTP RIO 4	SPARE			AI	4	3	2
				NR WTP RIO 4	SPARE			AI	4	3	3
				NR WTP RIO 4	SPARE			AI	4	3	4
				NR WTP RIO 4	SPARE			AI	4	3	5
				NR WTP RIO 4	SPARE			AI	4	3	6
				NR WTP RIO 4	SPARE			AI	4	3	7
	062-Y610	PMP-67111	SC-67111	NR WTP RIO 4	COAGULANT METERING PUMP SPEED COMMAND	0-60	HZ	AO	4	4	0
	062-Y611	PMP-67211	SC-67211	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 1 SPEED COMMAND	0-60	HZ	AO	4	4	1
	062-Y611	PMP-67212	SC-67212	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 2 SPEED COMMAND	0-60	HZ	AO	4	4	2
	062-Y611	PMP-67213	SC-67213	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 3 SPEED COMMAND	0-60	HZ	AO	4	4	3
	062-Y611	PMP-67214	SC-67214	NR WTP RIO 4	SODIUM HYPOCHLORITE SWING PUMP SPEED COMMAND	0-60	HZ	AO	4	4	4
	062-Y610	PMP-67311	SC-67311	NR WTP RIO 4	CAUSTIC SODA METERING PUMP SPEED COMMAND	0-60	HZ	AO	4	4	5
				NR WTP RIO 4	SPARE			AO	4	4	6
				NR WTP RIO 4	SPARE			AO	4	4	7

	062-Y610	PMP-67111	HSR-67111	NR WTP RIO 4	COAGULANT METERING PUMP REMOTE	LOCAL	REMOTE	DI	4	5	0
	062-Y610	PMP-67111	NI-67111	NR WTP RIO 4	COAGULANT METERING PUMP RUNNING	OFF	RUNNING	DI	4	5	1
	062-Y610	PMP-67111	UA-67111	NR WTP RIO 4	COAGULANT METERING PUMP FAULT	<NULL>	ALARM	DI	4	5	2
	062-Y610	LSH-67151	LSH-67151	NR WTP RIO 4	COAGULANT SUMP LEVEL SWITCH HIGH	<NULL>	ALARM	DI	4	5	3
	062-Y611	PMP-67211	HSR-67211	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 1 REMOTE	LOCAL	REMOTE	DI	4	5	4
	062-Y611	PMP-67211	NI-67211	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 1 RUNNING	OFF	RUNNING	DI	4	5	5
	062-Y611	PMP-67211	UA-67211	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 1 FAULT	<NULL>	ALARM	DI	4	5	6
	062-Y611	PMP-67212	HSR-67212	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 2 REMOTE	LOCAL	REMOTE	DI	4	5	7
	062-Y611	PMP-67212	NI-67212	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 2 RUNNING	OFF	RUNNING	DI	4	5	8
	062-Y611	PMP-67212	UA-67212	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 2 FAULT	<NULL>	ALARM	DI	4	5	9
	062-Y611	PMP-67213	HSR-67213	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 3 REMOTE	LOCAL	REMOTE	DI	4	5	10
	062-Y611	PMP-67213	NI-67213	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 3 RUNNING	OFF	RUNNING	DI	4	5	11
	062-Y611	PMP-67213	UA-67213	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 3 FAULT	<NULL>	ALARM	DI	4	5	12
	062-Y611	PMP-67214	HSR-67214	NR WTP RIO 4	SODIUM HYPOCHLORITE SWING PUMP REMOTE	LOCAL	REMOTE	DI	4	5	13
	062-Y611	PMP-67214	NI-67214	NR WTP RIO 4	SODIUM HYPOCHLORITE SWING PUMP RUNNING	OFF	RUNNING	DI	4	5	14
	062-Y611	PMP-67214	UA-67214	NR WTP RIO 4	SODIUM HYPOCHLORITE SWING PUMP FAULT	<NULL>	ALARM	DI	4	5	15
	062-Y611	LSH-67251	LSH-67251	NR WTP RIO 4	SODIUM HYPOCHLORITE SUMP LEVEL SWITCH HIGH	<NULL>	ALARM	DI	4	5	16
	062-Y610	PMP-67311	HSR-67311	NR WTP RIO 4	CAUSTIC SODA METERING PUMP REMOTE	LOCAL	REMOTE	DI	4	5	17
	062-Y610	PMP-67311	NI-67311	NR WTP RIO 4	CAUSTIC SODA METERING PUMP RUNNING	OFF	RUNNING	DI	4	5	18
	062-Y610	PMP-67311	UA-67311	NR WTP RIO 4	CAUSTIC SODA METERING PUMP FAULT	<NULL>	ALARM	DI	4	5	19
	062-Y610	LSH-67351	LSH-67351	NR WTP RIO 4	CAUSTIC SODA LEVEL SWITCH HIGH	<NULL>	ALARM	DI	4	5	20
	062-Y613	LSH-67451	LSH-67451	NR WTP RIO 4	HYDROCHLORIC ACID SUMP LEVEL SWITCH HIGH	<NULL>	ALARM	DI	4	5	21
	062-Y613	LSH-67551	LSH-67551	NR WTP RIO 4	CORROSION INHIBITOR SUMP LEVEL SWITCH HIGH	<NULL>	ALARM	DI	4	5	22
	062-Y612	LSH-67651	LSH-67651	NR WTP RIO 4	SODIUM BISULFITE SUMP LEVEL SWITCH HIGH	<NULL>	ALARM	DI	4	5	23
	062-Y612	LSH-67652	LSH-67652	NR WTP RIO 4	CITRIC ACID SUMP LEVEL SWITCH HIGH	<NULL>	ALARM	DI	4	5	24
	009-Y621	NR WTP CHEMICAL RIO CP PS 1	NR-RIO-PS1	NR WTP RIO 4	NR WTP CHEMICAL RIO CONTROL PANEL POWER SUPPLY 1	NOT OK	OK	DI	4	5	25
	009-Y621	NR WTP CHEMICAL RIO CP PS 2	NR-RIO-PS2	NR WTP RIO 4	NR WTP CHEMICAL RIO CONTROL PANEL POWER SUPPLY 2	NOT OK	OK	DI	4	5	26
	009-Y621	NR WTP CHEMICAL RIO CP UPS	NR-RIO-UPS	NR WTP RIO 4	NR WTP CHEMICAL RIO CONTROL PANEL UPS	NOT OK	OK	DI	4	5	27
				NR WTP RIO 4	SPARE			DI	4	5	28
				NR WTP RIO 4	SPARE			DI	4	5	29
				NR WTP RIO 4	SPARE			DI	4	5	30
				NR WTP RIO 4	SPARE			DI	4	5	31
				NR WTP RIO 4	SPARE			DI	4	6	0
				NR WTP RIO 4	SPARE			DI	4	6	1
				NR WTP RIO 4	SPARE			DI	4	6	2
				NR WTP RIO 4	SPARE			DI	4	6	3
				NR WTP RIO 4	SPARE			DI	4	6	4
				NR WTP RIO 4	SPARE			DI	4	6	5
				NR WTP RIO 4	SPARE			DI	4	6	6
				NR WTP RIO 4	SPARE			DI	4	6	7
				NR WTP RIO 4	SPARE			DI	4	6	8
				NR WTP RIO 4	SPARE			DI	4	6	9
				NR WTP RIO 4	SPARE			DI	4	6	10
				NR WTP RIO 4	SPARE			DI	4	6	11
				NR WTP RIO 4	SPARE			DI	4	6	12
				NR WTP RIO 4	SPARE			DI	4	6	13
				NR WTP RIO 4	SPARE			DI	4	6	14
				NR WTP RIO 4	SPARE			DI	4	6	15
				NR WTP RIO 4	SPARE			DI	4	6	16
				NR WTP RIO 4	SPARE			DI	4	6	17
				NR WTP RIO 4	SPARE			DI	4	6	18
				NR WTP RIO 4	SPARE			DI	4	6	19
				NR WTP RIO 4	SPARE			DI	4	6	20
				NR WTP RIO 4	SPARE			DI	4	6	21
				NR WTP RIO 4	SPARE			DI	4	6	22
				NR WTP RIO 4	SPARE			DI	4	6	23
				NR WTP RIO 4	SPARE			DI	4	6	24
				NR WTP RIO 4	SPARE			DI	4	6	25
				NR WTP RIO 4	SPARE			DI	4	6	26
				NR WTP RIO 4	SPARE			DI	4	6	27
				NR WTP RIO 4	SPARE			DI	4	6	28
				NR WTP RIO 4	SPARE			DI	4	6	29
				NR WTP RIO 4	SPARE			DI	4	6	30
				NR WTP RIO 4	SPARE			DI	4	6	31
	062-Y610	PMP-67111	XC-67111	NR WTP RIO 4	COAGULANT METERING PUMP START/STOP COMMAND	STOP	START	DO	4	7	0
	062-Y611	PMP-67211	XC-67211	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 1 START/STOP COMMAND	STOP	START	DO	4	7	1

	062-Y611	PMP-67212	XC-67212	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 2 START/STOP COMMAND	STOP	START	DO	4	7	2
	062-Y611	PMP-67213	XC-67213	NR WTP RIO 4	SODIUM HYPOCHLORITE METERING PUMP 3 START/STOP COMMAND	STOP	START	DO	4	7	3
	062-Y611	PMP-67214	XC-67214	NR WTP RIO 4	SODIUM HYPOCHLORITE SWING PUMP START/STOP COMMAND	STOP	START	DO	4	7	4
	062-Y610	PMP-67311	XC-67311	NR WTP RIO 4	CAUSTIC SODA METERING PUMP START/STOP COMMAND	STOP	START	DO	4	7	5
	062-Y612	AIT-61881	XSA-61881	NR WTP RIO 4	SULFUR DIOXIDE ANALYZER HORN/STROBE ON COMMAND	<NULL>	ALARM/ON	DO	4	7	6
				NR WTP RIO 4	SPARE			DO	4	7	7
				NR WTP RIO 4	SPARE			DO	4	7	8
				NR WTP RIO 4	SPARE			DO	4	7	9
				NR WTP RIO 4	SPARE			DO	4	7	10
				NR WTP RIO 4	SPARE			DO	4	7	11
				NR WTP RIO 4	SPARE			DO	4	7	12
				NR WTP RIO 4	SPARE			DO	4	7	13
				NR WTP RIO 4	SPARE			DO	4	7	14
				NR WTP RIO 4	SPARE			DO	4	7	15

This page intentionally left blank.

SECTION 40 61 96
PROCESS CONTROL DESCRIPTIONS

PART 1 - GENERAL

1.1 SUMMARY

A. These operational descriptions are for information only to provide additional information on the system.
B. Programming of non-vendor packages are not in contract.
C. Sections Include:
    1. Process Control Description content and structure.
        a. Process Control Description for each system includes:
            1) Description of System.
            2) Reference Documents.
            3) Major Equipment.
            4) Major Field Instruments.
            5) Detailed Process Control Descriptions.
            6) Response to Instrument or Equipment Failures.
            7) Displays at HMI.
            8) Report Descriptions.
            9) Alarms.

1.2 DEFINITIONS/ABBREVIATIONS

A. Abbreviations are defined as listed below:

A/M	Auto/Manual
ATS	Automatic Transfer Switch
FW	Finished Water
H/O/A	Hand-Off-Auto
H/O/R	Hand-Off-Remote
HMI	Human-Machine Interface
L/O/R	Local-Off-Remote
L/R/O	Local/Remote/Off
L/R	Local/Remote
LCP	Local Control Panel
LCS	Local Control Station
MCC	Motor Control Center
O/C	Open/Close
OI	Operator Interface (valve actuator, VFD)
OIT	Operator Interface Terminal
P&ID	Process and Instrumentation Diagram
PCS	Process Control System
PID	Proportional-Integral-Derivative

PFD	Process Flow Diagram
PLC	Programmable Logic Controller
RIO	Remote DCS or PLC Input/Output Equipment
RTU	Remote Terminal Unit
RVSS	Reduced Voltage Solid-state Starter
RW	Raw Water
S/S	Start/Stop
UF	Ultrafiltration
UPS	Uninterruptible Power Supply
VFD	Variable Frequency Drive
WTP	Water Treatment Plant

PART 2 - EXECUTION

2.1 PROCESS CONTROL DESCRIPTIONS

A. The following Appendices to this Section contain the process control descriptions that define the PCS procedural, and operational requirements of the processes included in the Work:
   1. 40 61 96 - Appendix A - Volume 5 Roaring Springs Pumphouse Process Control Descriptions.
   2. 40 61 96 - Appendix B - Volume 6 North Rim WTP Process Control Descriptions.

END OF SECTION

APPENDIX A - VOLUME 5 RS PUMPHOUSE PROCESS CONTROL DESCRIPTIONS

1.1 DESCRIPTION OF PLANT SYSTEM

A. Description of Plant System
   1. The Roaring Springs Pumphouse directs Raw Water from the Roaring Springs Cave to the North Rim Water Treatment Plant. The Pumphouse receives Finished Water from the North Rim WTP into the Finished Water Tank for distribution into the canyon. A chlorine boosting system ensures chlorine residual levels of the distributed Finished Water are met.

1.2 RAW WATER PUMPING

A. Description of System
   1. Raw Water is received from the Roaring Springs Cave and is filtered by the vendor-supplied Auto Filters. The Raw Water pumps operate to maintain level in the Raw Water Tanks at the North Rim Water Treatment Plant. Raw Water Pumping will stop if the turbidity of the Raw Water is high.

B. Reference Documents
   1. RAW WATER PUMPING P&ID (52-Y601)

C. Major Equipment
   1. RW Bypass Valve (FV-52331)
   2. Auto Filter Vendor Control Panel (LCP-52101)
   3. Auto Filter 1 (AFV-52111)
   4. Auto Filter 2 (AFV-52112)
   5. Raw Water Pump 1 (PMP-52211)
   6. Raw Water Pump 2 (PMP-52212)
   7. Manual Control Valve (EDS-52312)

D. Major Field Instrumentation
   1. RW Turbidity (AE/AIT-52381)
   2. Auto Filters Inlet Pressure (PIT-52261)
   3. RW Pumps Suction Pressure (PIT-52262)
   4. RW Pumps Suction Flow (FIT-52371)
   5. RW Pump 1 Pressure Switch High-High (PSHH-52263)
   6. RW Pump 2 Pressure Switch High-High (PSHH-52264)
   7. RW Pumps Discharge Pressure (PIT-52265)
   8. RW Pumps Common Discharge Pressure (PIT-52266)
   9. RW Bypass Flow (FIT-52372)
   10. Turbidimeter Drain Flow Switch (FSL-52371)

E. Detailed Process Control Descriptions
   1. Raw Water Bypass Valve
      a. Raw Water Bypass Valve opens when RW Turbidity is above an operator-adjustable High Turbidity setpoint. The Bypass Valve will close when RW Turbidity is below the same operator-adjustable High Turbidity setpoint.
   2. Auto Filters
      a. Auto Filter System is vendor-controlled and operates on a Duty/Standby basis. Backwash Active signals for each filter and a common fault alarm are received from the Auto Filter Vendor Control Panel to the main PLC for alarming and monitoring.
   3. Raw Water Pumps
      a. The Raw Water Pumps operate in Duty/Standby where the Standby pump will be called into operation if the Duty pump fails.
      b. The Raw Water Pumps will start when the Raw Water Tanks at the North Rim Water Treatment Plant are below an operator-adjustable Low Level Pumps Start Setpoint. The

pumps will stop when the Raw Water Tanks are above an operator High Level Pumps Stop Setpoint.

c. When called to run, the Duty pump speed will increase by 1Hz every half a second until the Raw Water Pumps Suction Flow is above an operator-adjustable Low Flow Shutdown setpoint, then it will increase by 0.5Hz every half a second until the Raw Water Pumps Suction Flow is greater than an operator-adjustable Raw Water Pumps Normal Operations Flow Setpoint. The Duty pump will then modulate speed to maintain an operator-adjustable Raw Water Pumps Flow Setpoint.

d. The Raw Water Pumps shutdown if the RW Pumps Inlet Pressure Low Alarm is active, if any of the related pump temperature or vibration alarms are active (listed in section A.2.I), if the Raw Water Turbidity is greater than the operator-entered High Turbidity setpoint, or if the Raw Water Tanks High Level Alarms are seen from the North Rim WTP PLC.

e. When called to stop, the Duty Pump will decrease speed by 0.5Hz every half a second until the Raw Water Pumps Suction Flow is below the operator-adjustable Low Flow Shutdown setpoint, then the speed will be set to 0.

F. Response to Instrument or Equipment Failures

1. At all times, the PLC will take the last known level of the NR WTP Raw Water Tanks and calculate how much time it will take to fill the tanks. Upon loss of communication with the North Rim WTP PLC and if Tank Fill Comms Loss mode is enabled by the operator and if a Raw Water Pump is currently running in Auto Mode, the Duty pump will run for that calculated amount of time minus half an hour or will stop when communications have been lost for 1 hour.

G. Displays at HMI/OIT

1. RW Turbidity
2. RW Bypass Valve Remote/Position
3. Auto Filter 1 Backwash Active
4. Auto Filter 2 Backwash Active
5. Auto Filters Inlet Pressure
6. RW Pumps Inlet Pressure
7. RW Pumps Suction Flow and Totalized Flow
8. RW Pump 1 Remote/Running/Speed/Vibrations/Temperatures
9. RW Pump 2 Remote/Running/Speed/Vibrations/Temperatures
10. RW Pumps Discharge Pressure
11. RW Pumps Common Discharge Pressure
12. RW Bypass Flow

H. Report Descriptions

1. RW Turbidity
2. RW Bypass Valve Position
3. Auto Filters Inlet Pressure
4. RW Pumps Inlet Pressure
5. RW Pumps Suction Flow
6. RW Pump 1 Speed
7. RW Pump 1 Pump Vibration X
8. RW Pump 1 Pump Vibration Y
9. RW Pump 1 Pump Vibration Z
10. RW Pump 1 Motor Vibration X
11. RW Pump 1 Motor Vibration Y
12. RW Pump 1 Motor Vibration Z
13. RW Pump 1 Bearing Temperature
14. RW Pump 1 Winding 1 Temperature 1
15. RW Pump 1 Winding 1 Temperature 2
16. RW Pump 1 Winding 2 Temperature 1
17. RW Pump 1 Winding 2 Temperature 2

18. RW Pump 1 Winding 3 Temperature 1
19. RW Pump 1 Winding 3 Temperature 2
20. RW Pump 1 Drive End Bearing Temperature
21. RW Pump 1 Opposite End Bearing Temperature
22. RW Pump 2 Speed
23. RW Pump 2 Pump Vibration X
24. RW Pump 2 Pump Vibration Y
25. RW Pump 2 Pump Vibration Z
26. RW Pump 2 Motor Vibration X
27. RW Pump 2 Motor Vibration Y
28. RW Pump 2 Motor Vibration Z
29. RW Pump 2 Bearing Temperature
30. RW Pump 2 Winding 1 Temperature 1
31. RW Pump 2 Winding 1 Temperature 2
32. RW Pump 2 Winding 2 Temperature 1
33. RW Pump 2 Winding 2 Temperature 2
34. RW Pump 2 Winding 3 Temperature 1
35. RW Pump 2 Winding 3 Temperature 2
36. RW Pump 2 Drive End Bearing Temperature
37. RW Pump 2 Opposite End Bearing Temperature
38. RW Pumps Discharge Pressure
39. RW Pumps Common Discharge Pressure
40. RW Bypass Flow

I. Alarms
    1. Discrete Alarms
        a. RW Bypass Valve Fault
        b. RW Bypass Valve Fail to Open and Fail to Close
        c. Auto Filter Vendor Control Panel Fault
        d. RW Pump 1 Fault
        e. RW Pump 1 Seal Leak
        f. RW Pump 1 E-Stop
        g. RW Pump 1 Pressure Switch High-High
        h. RW Pump 1 Fail to Start
        i. RW Pump 2 Fault
        j. RW Pump 2 Seal Leak
        k. RW Pump 2 E-Stop
        l. RW Pump 2 Pressure Switch High-High
        m. RW Pump 2 Fail to Start
        n. Turbidimeter Drain Flow Switch Low
    2. Analog Alarms
        a. RW Turbidity High and High-High
        b. Auto Filter Inlet Pressure High and Low
        c. RW Pumps Inlet Pressure High and Low
        d. RW Pump 1 Pump Vibration X High and High-High
        e. RW Pump 1 Pump Vibration Y High and High-High
        f. RW Pump 1 Pump Vibration Z High and High-High
        g. RW Pump 1 Motor Vibration X High and High-High
        h. RW Pump 1 Motor Vibration Y High and High-High
        i. RW Pump 1 Motor Vibration Z High and High-High
        j. RW Pump 1 Bearing Temperature High and High-High
        k. RW Pump 1 Motor Winding 1 Temperature 1 High and High-High
        l. RW Pump 1 Motor Winding 1 Temperature 2 High and High-High
        m. RW Pump 1 Motor Winding 2 Temperature 1 High and High-High
        n. RW Pump 1 Motor Winding 2 Temperature 2 High and High-High
        o. RW Pump 1 Motor Winding 3 Temperature 1 High and High-High

p. RW Pump 1 Motor Winding 3 Temperature 2 High and High-High
q. RW Pump 1 Motor Drive End Bearing Temperature High and High-High
r. RW Pump 1 Motor Opposite End Temperature High and High-High
s. RW Pump 2 Pump Vibration X High and High-High
t. RW Pump 2 Pump Vibration Y High and High-High
u. RW Pump 2 Pump Vibration Z High and High-High
v. RW Pump 2 Motor Vibration X High and High-High
w. RW Pump 2 Motor Vibration Y High and High-High
x. RW Pump 2 Motor Vibration Z High and High-High
y. RW Pump 2 Bearing Temperature High and High-High
z. RW Pump 2 Motor Winding 1 Temperature 1 High and High-High
aa. RW Pump 2 Motor Winding 1 Temperature 2 High and High-High
bb. RW Pump 2 Motor Winding 2 Temperature 1 High and High-High
cc. RW Pump 2 Motor Winding 2 Temperature 2 High and High-High
dd. RW Pump 2 Motor Winding 3 Temperature 1 High and High-High
ee. RW Pump 2 Motor Winding 3 Temperature 2 High and High-High
ff. RW Pump 2 Motor Drive End Bearing Temperature High and High-High
gg. RW Pump 2 Motor Opposite End Temperature High and High-High
hh. RW Pumps Discharge Pressure High and Low
ii. RW Common Discharge Pressure High and Low

1.3 FINISHED WATER CONVEYANCE

A. Description of System
1. Finished Water is received from the North Rim WTP and fills the Finished Water Tank at the Roaring Springs Pump House for distribution. The Finished Water Recirculation Pump mixes the water for chlorine boosting. The Tank Heater maintains temperature in the tank.

B. Reference Documents
1. FINISHED WATER CONVEYANCE P&ID (52-Y602)

C. Major Equipment
1. Finished Water Energy Dissipation Valve (EDS-53117)
2. FW Tank (TK-53211)
3. FW Tank Heater (HTR-53241)
4. FW Recirculation Pump (PMP-53221)

D. Major Field Instrumentation
1. FW Pre-Pressure Reducer Pressure (PIT-53162)
2. FW Post-Pressure Reducer Pressure (PIT-53164)
3. Finished Water Flow (FE/FIT-53171)
4. FW Tank Recirculation Chlorine Residual (AE/AIT-53281)
5. FW Tank Recirculation pH/Temperature (AE/AIT-53282)
6. FW Tank Level (LT-53251)
7. FW Tank Heater Controller (HC-53241)
8. FW Tank Temperature (TE-53281)
9. Distribution Flow (FIT-53172)

E. Detailed Process Control Descriptions
1. FW Energy Dissipation Valve
a. The FW Energy Dissipation Valve is controlled by a Level Controller. When the Finished Water Tank Level is below a Finished Water Tank Low Level setpoint, the FW Energy Dissipation Valve will open. When the Finished Water Tank Level is above a Finished Water Tank High Level setpoint, the FW Energy Dissipation Valve will close.
b. The actuator of the Energy Dissipation Valve is programmed to open and close very slowly, over the course of several minutes, to not induce a hydraulic transient wave within the borehole pipeline. To accommodate this, the Finished Water Tank High

Level setpoint is set deliberately low so that when the Energy Dissipation Valve finishes closing, the final resulting water level is still below the overflow level.

c. Due to the specific hydraulic conditions of the borehole pipeline system, the Energy Dissipation Valve cannot be opened past approximately 75- 80% of its full stroke when the borehole pipeline is full. This setting will be pre-programmed into the actuator and will be adjusted in the field after full installation. Depending on programming specifics, the valve actuator may indicate to the PLC that the valve is 100% open when it is only at approximately 75-80% of its stroke. Automated operation of the valve will never let it open past this pre-programmed limit, but any manual override of the actuator's handwheel will allow additional opening range, if called for in an emergency.

2. FW Recirculation Pump
   a. The FW Recirculation Pump will start when the FW Tank level is 1.5 Ft above the tank outlet and will stop if the FW Tank Level is below 1 Ft above the tank outlet.
3. FW Tank Heater
   a. The Tank Heater has a vendor-supplied controller to maintain an operator-adjustable temperature setpoint within the FW Tank. The Tank Heater can be enabled or disabled from the main PLC by the operator.

F. Response to Instrument or Equipment Failures
   1. The Finished Water Energy Dissipation Valve has a UPS that will last for 60-90 minutes and in the event of a power fail, the relay logic inside LCP-53117 will initiate the following sequence of events:
      a. Unless the Finished Water Tank is already at the High Level setpoint, the Finished Water Energy Dissipation Valve will be directed to open.
      b. The Finished Water Tank will fill until it reaches the High Level setpoint, at which point, the Energy Dissipation Valve will be told to close.
      c. After the Energy Dissipation Valve is closed, it will not be allowed to open again until power is restored. Level monitoring within the Finished Water Tank, network communication equipment and other appurtenances wired to LCP-53117 will continue to operate and communicate with the North Rim WTP until the UPS power dies.

G. Displays at HMI/OIT
   1. Finished Water Energy Dissipation Valve Remote/Position
   2. FW Pre-Pressure Reducer Pressure
   3. FW Post-Pressure Reducer Pressure
   4. FW Flow and Totalized Flow
   5. FW Tank Recirculation Chlorine Residual
   6. FW Tank Recirculation pH
   7. FW Tank Recirculation Temperature
   8. FW Recirculation Pump Remote/Running
   9. Heat Trace Controller Running
   10. FW Tank Level
   11. FW Tank Heater Running
   12. FW Tank Temperature
   13. Distribution Flow and Totalized Flow

H. Report Descriptions
   1. Finished Water Energy Dissipation Valve Position
   2. FW Pre-Pressure Reducer Pressure
   3. FW Post-Pressure Reducer Pressure
   4. FW Flow
   5. FW Tank Recirculation Chlorine Residual
   6. FW Tank Recirculation pH
   7. FW Tank Recirculation Temperature
   8. FW Tank Level

9. FW Tank Temperature
10. Distribution Flow

I. Alarms
   1. Discrete Alarms
      a. FW Energy Dissipation Valve Fault
      b. FW Energy Dissipation Valve Motor Temp High
      c. FW Energy Dissipation Valve Fail to Open and Fail to Close
      d. Heat Trace Controller Fault
      e. FW Tank Heater Fault
      f. FW Recirculation Pump Fault
      g. FW Recirculation Pump Fail to Start
   2. Analog Alarms
      a. FW Pre-Pressure Reducer Pressure High and Low
      b. FW Post-Pressure Reducer Pressure High and Low
      c. FW Recirculation Chlorine Residual High and Low
      d. FW Recirculation pH High and Low
      e. FW Recirculation Temperature High
      f. FW Tank Temperature High

1.4 CHLORINE DOSING

A. Description of System
   1. The Chlorine Dosing maintains chlorine levels in the Finished Water Tank. The Transfer Pump maintains a tank of diluted chlorite from a concentrated stock and finished water.

B. Reference Documents
   1. CHLORINE DOSING P&ID (52-Y603)

C. Major Equipment
   1. Makeup Water Flow Valve (FV-53331)
   2. Sodium Hypochlorite Batch Metering Pump (PMP-53421)
   3. Stock Hypochlorite Drum or Carboy Weight Scale (SCL-53411)
   4. Dilute Sodium Hypochlorite Day Tank (TK-53311)
   5. Dilute Sodium Hypochlorite Metering Pump 1 (PMP-53321)
   6. Dilute Sodium Hypochlorite Metering Pump 2 (PMP-53322)

D. Major Field Instrumentation
   1. Makeup Water Flow (FIT-53371)
   2. Stock Hypochlorite Drum or Carboy Weight (WIT-53491)
   3. Dilute Sodium Hypochlorite Day Tank Level (LE/LIT-53351)
   4. Dilute Sodium Hypochlorite Containment Sump Level Switch High (LSH-53451)
   5. Dilute Sodium Hypochlorite Metering Pump 1 Pressure Switch High-High (PSHH-53361)
   6. Dilute Sodium Hypochlorite Metering Pump 2 Pressure Switch High-High (PSHH-53362)

E. Detailed Process Control Descriptions
   1. Sodium Hypochlorite Batch Metering Pump
      a. The Batch Metering Pump initiates operation when the Sodium Hypochlorite Day Tank Level is below an operator-adjustable Low Level setpoint.
      b. When the Sodium Hypochlorite Day Tank is being filled, the measured flow rate of this filling will be used to calculate the speed of the Sodium Hypochlorite Batch Metering Pump required to deliver a finished dilute hypochlorite solution to the Sodium Hypochlorite Day Tank at the operator-adjustable Concentration setpoint.
   2. Makeup Water Flow Valve
      a. Makeup Water Flow Valve opens when the Sodium Hypochlorite Batch Metering Pump is running.
      b. An alarm will generate if the valve is open for over 5 minutes and the Sodium Hypochlorite Batch Metering Pump is not running.
   3. Dilute Sodium Hypochlorite Metering Pumps

a. The Dilute Sodium Hypochlorite Metering Pumps operate in Duty/Standby where the Standby pump will be called into operation if the Duty pump fails.
b. When the FW Tank Recirculation Chlorine Residual value drops below the operator-adjustable Low Chlorine Residual setpoint, the required amount of Dilute Sodium Hypochlorite to be dosed over a set amount of time is calculated.
c. The Dilute Sodium Hypochlorite Metering Pumps will then does into the Recirculation Pump discharge line for the calculated time at the calculated speed. After this timer expires, the Dilute Sodium Hypochlorite Metering Pumps will not be allowed to dose again for several hours while normal recirculation within the FW Tank is allowed to distribute the chlorine addition to the FW Tank uniformly.
d. At the end of this waiting period, if the FW Tank chlorine residual is still below the Low chlorine level, the cycles will be allowed to repeat.

F. Response to Instrument or Equipment Failures
1. If a Chlorine Containment Leak is detected, the Sodium Hypochlorite Batch Metering Pump will not run.

G. Displays at HMI/OIT
1. FW Chlorine Dilute Flow Valve Remote/Position
2. Makeup Water Flow and Totalized Flow
3. Stock Hypochlorite Drum or Carboy Weight
4. Sodium Hypochlorite Batch Metering Pump Remote/Running/Speed
5. Dilute Sodium Hypochlorite Day Tank Level
6. Dilute Sodium Hypochlorite Metering Pump 1 Remote/Running/Speed
7. Dilute Sodium Hypochlorite Metering Pump 2 Remote/Running/Speed

H. Report Descriptions
1. Finished Water Chlorine Dilute Flow
2. Stock Hypochlorite Drum or Carboy Weight
3. Sodium Hypochlorite Batch Metering Pump Speed
4. Dilute Sodium Hypochlorite Day Tank Level
5. Dilute Sodium Hypochlorite Metering Pump 1 Speed
6. Dilute Sodium Hypochlorite Metering Pump 2 Speed

I. Alarms
1. Discrete Alarms
a. FW Chlorine Dilute Flow Valve Fault
b. FW Chlorine Dilute Flow Valve Fail to Open and Fail to Close
c. Sodium Hypochlorite Batch Metering Pump Fault
d. Sodium Hypochlorite Batch Metering Pump Seal Leak
e. Sodium Hypochlorite Batch Metering Pump Fail to Start
f. Dilute Sodium Hypochlorite Containment Sump Level Switch High
g. Dilute Sodium Hypochlorite Metering Pump 1 Fault
h. Dilute Sodium Hypochlorite Metering Pump 1 Seal Leak
i. Dilute Sodium Hypochlorite Metering Pump 1 Pressure Switch High-High
j. Dilute Sodium Hypochlorite Metering Pump 1 Fail to Start
k. Dilute Sodium Hypochlorite Metering Pump 2 Fault
l. Dilute Sodium Hypochlorite Metering Pump 2 Seal Leak
m. Dilute Sodium Hypochlorite Metering Pump 2 Pressure Switch High-High
n. Dilute Sodium Hypochlorite Metering Pump 2 Fail to Start
2. Analog Alarms
a. Stock Hypochlorite Drum or Carboy Weight Low and Low-Low
b. Dilute Sodium Hypochlorite Day Tank Level High and Low

1.5 FINISHED WATER BOOSTER PUMPS

A. Description of System

1. The Booster Pumps maintain pressure in the event of the emergency eyewash station being operated.

B. Reference Documents
   1. FINISHED WATER BOOSTER PUMPS P&ID (52-Y604)
C. Major Equipment
   1. FW Booster Pump 1 (PMP-53111)
   2. FW Booster Pump 2 (PMP-53112)
   3. FW Booster Pump Local Control Panel (LCP-53111)
   4. ESEW Water Heater Expansion Tank (TK-51641)
   5. ESEW Water Heater (WH-51641)
   6. Emergency Eye Wash Station (ESEW-51641)
   7. FW Bladder Tank (TK-53111)
D. Major Field Instrumentation
   1. FW Booster System Flow (FIT-53173)
   2. Emergency Eye Wash Station Flow Switch High (FSH-51671)
   3. Booster Pumps System Pressure (PIT-53161)
   4. Booster Pump 1 Pressure Switch High (PSH-53161)
   5. Booster Pump 2 Pressure Switch High (PSH-53162)
E. Detailed Process Control Descriptions
   1. FW Booster Pumps
      a. The FW Booster Pumps operate in Duty/Standby where the Standby pump will be called into operation if the Duty pump fails. The pumps will rotate roles when the duty pump stops.
      a. The Booster Pumps start to run to maintain pressure in the system. When the low pressure setpoint from the pressure transmitter (PIT-53161) is set, the duty pump shall start. When the high pressure setpoint is reached, the duty pump shall stop.
      b. If the duty pump is running after a time setpoint and Pressure (PIT-53161) is not increasing and/or Flow (FIT-53173) is low or no flow, then an alarm will activate and the pump(s) will stop until an operator acknowledges that valves are open and pumps are not running dry.
F. Response to Instrument or Equipment Failures
   1. N/A
G. Displays at HMI/OIT
   1. FW Booster Pump 1 Remote/Running
   2. FW Booster Pump 2 Remote/Running
   3. FW Booster Pumps System Flow and Totalized Flow
   4. Booster Pumps System Pressure
H. Report Descriptions
   1. FW Booster Pumps Flow
   2. Booster Pumps System Pressure
I. Alarms
   1. Discrete Alarms
      a. FW Booster Pump 1 Fault
      b. FW Booster Pump 1 Pressure High Switch
      c. FW Booster Pump 1 Fail to Start
      d. FW Booster Pump 2 Fault
      e. FW Booster Pump 2 Pressure High Switch
      f. FW Booster Pump 2 Fail to Start
      g. ESEW In Use
   2. Analog Alarms
      a. Booster Pump System Pressure High and Low

1.6 MISCELLANEOUS SYSTEMS

A. Description of System
   1. N/A

B. Reference Documents
   1. MISCELLANEOUS SYSTEMS P&ID (52-Y605)

C. Major Equipment
   1. N/A

D. Major Field Instrumentation
   1. Electrical Room Smoke Detector (SD-51098)
   2. Process Room Smoke Detector (SD-51099)
   3. Pump Room Temperature Transmitter (TIT-51681)
   4. Operations Room Exterior Door Intrusion Switch (ZS-51491)
   5. Electrical Area Exterior Door Intrusion Switch (ZS-51492)
   6. Pump Room Exterior Door Intrusion Switch (ZS-51493)
   7. Pump Room Floor Flood Level Switch High-High (LSHH-51651)
   8. Master PLC Mode Switch (ZS-51101)
   9. Standby PLC Mode Switch (ZS-51102)

E. Detailed Process Control Descriptions
   1. N/A

F. Response to Instrument or Equipment Failures
   1. N/A

G. Displays at HMI/OIT
   1. Pump Room Temperature

H. Report Descriptions
   1. Pump Room Temperature

I. Alarms
   1. Discrete Alarms
      a. Electrical Room Smoke Detected
      b. Process Room Smoke Detected
      c. Operations Room Exterior Door Intrusion
      d. Electrical Area Exterior Door Intrusion
      e. Pump Room Exterior Door Intrusion
      f. Pump Room Floor Flood
      g. Master/Standby Switches Misconfigured
   2. Analog Alarms
      a. Pump Room Temperature Transmitter High and Low

END OF APPENDIX A

This page intentionally left blank.

APPENDIX B - VOLUME 6 NORTH RIM WTP PROCESS CONTROL DESCRIPTIONS

1.1 DESCRIPTION OF PLANT SYSTEM

A. Raw Water is withdrawn from the Raw Water (RW) Tanks and flow is maintained into the Water Treatment Plant (WTP) building. Coagulant and Sodium Hypochlorite can be added upstream of the Reaction Tanks. Flow is split into two Reaction Tanks, each equipped with a mixer. Flocculated water from the Reaction Tanks flows to two ultrafiltration (UF) membrane skids.

B. The packaged UF skids are designed in a duty/standby arrangement, where one skid will produce water at the full treatment capacity until a backwash is initiated, at which point the standby unit will be brought online. Backwash waste is diverted from the UF skids directly to the Process Waste Tank to be discharged to sewer. The UF skids also use compressed air to assist in the backwashing process. The UF supplier is responsible for the programming of the UF system and will provide signals to communicate with the main WTP controls system.

C. Sodium Hypochlorite and Caustic Soda are added to the UF filtrate water before the Chlorine Contact Tank for contact time for proper disinfection. After disinfection, the Finished Water Tank Pumps transfer into the Finished Water Storage Tanks. Finished Water Chlorine Residual, pH, and Temperature are monitored for Contact Time compliance. Chlorine Residual can be boosted with Sodium Hypochlorite as needed.

D. Distribution is split into the following flow paths:
   1. The Lower Zone distribution is gravity-fed and the flow is metered.
   2. Inner-Canyon Backfeed to Supai Tunnel flow is off the Lower Zone distribution and the flow is metered.
   3. Inner-Canyon Backfeed to Roaring Springs Pump House is off the Lower Zone distribution and the flow is found by subtracting the Supai Tunnel Backfeed flow from the Combined Backfeed flow.
   4. Three Upper Zone Booster Pumps maintain pressure in the Upper Zone distribution.
   5. Three High Demand Pumps boost flow to the Upper Zone distribution system during high demand events.

E. Residual streams from the treatment process, including membrane backwash, neutralized CIP waste, and a dilute solids stream from the Reaction Tanks are collected in the Process Waste Tank before flowing by gravity to the North Rim Wastewater Treatment Plant.

1.2 OVERALL PLANT CONTROLS

A. Description of System
   1. The North Rim WTP will be called into production based on the water level in the Finished Water Tanks or based on Time of Day and will run at a constant flow rate. Startup prerequisites must be met before the plant can run. When moving on standby, the Finished Water Tank Pumps will stop first, followed by the UF System and the Raw Water Flow Control Valve. Other conditions that initiate a plant shutdown include Low Finished Water Chlorine Residual, High UF Filtrate Turbidity, Low Raw Water Tank(s) Level, High and Low Contact Time Tank Level, and operator shutdown initiation. The plant will shut down for the winter.

B. Reference Documents
   1. NR WTP P&IDs

C. Major Equipment
   1. N/A

D. Major Field Instrumentation
   1. N/A

E. Detailed Process Control Descriptions
   1. Plant Production/Standby Operations

a. If the Plant Production/Standby controls are set to be in Manual Mode, then the plant will move into production or standby through manual initiation by the Operator.
b. If Plant Production/Standby controls are set to be in Automatic Mode, then the plant will operate based on Level Control or Time of Day Controls, which act independently of one another.
    1) If Plant Level Controls is Enabled, the plant will move into production when the Finished Water Tanks Level is below an operator-adjustable Plant Production Level and will move into standby when the Finished Water Tanks Level is above an operator-adjustable Plant Standby Level.
    2) If Plant Time of Day Controls is Enabled, the plant will move into production at the operator-adjustable Production Time of Day Setpoint and will move into standby at the Standby Time of Day Setpoint.

2. The following prerequisites must be met before the plant goes into full production:
    a. Raw Water Pump and Flow Valves in Remote
    b. Raw Water Flow Control Valve in Remote
    c. Reaction Tank Mixers in Remote
    d. One UF Skid Online and one UF Skid in Ready
    e. At least one Finished Water Tank Pump in Remote
    f. At least one Coagulant Metering Pump in Remote
    g. At least one Sodium Hypochlorite Metering Pump in Remote
    h. At least one Caustic Soda Metering Pump in Remote
    i. At least one Upper Zone Booster Pump in Remote
    j. At least one High Demand Pump in Remote
    k. Coagulant Dose setpoint greater than 0
    l. Sodium Hypochlorite Dose setpoint greater than 0
    m. Caustic Soda Dose setpoint greater than 0
    n. Raw Water Flow setpoint greater than the Minimum Allowable Flow (not operator-adjustable)
3. The plant production sequence will first move the UF System into Production Mode, then open the Raw Water Flow Control Valve or start the Raw Water Pump.
4. The plant standby sequence will first stop the Finished Water Tank Pumps, then the UF membranes, and finally the Raw Water Flow Control Valve and Raw Water Pump.
5. The North Rim WTP will be called to shut down in either Manual or Automatic Production/Standby Mode if any of the following conditions are true:
    a. Any Water on Floor Alarms Active
    b. Low Finished Water Chlorine Residual Alarm Active
    c. High Finished Water Chlorine Residual Alarm Active
    d. Low Finished Water pH Alarm Active
    e. High Finished Water pH Alarm Active
    f. High UF Filtrate Turbidity Alarm Active
    g. Low Raw Water Tanks Level Alarm Active
    h. High Contact Time Tank Level Alarm Active
    i. Low Contact Time Tank Level Alarm Active
    j. High Finished Water Tanks Level Alarm Active
    k. Critical UF system Fault Alarm Active
    l. UF System No Communications Alarm Active
    m. Chemical Systems Plant Shutdowns as defined in 1.10 through 1.16
6. If Winter Mode is enabled, the plant will be set to Manual Production/Standby Mode and Manual Standby will be forced, and all equipment in the plant will be prevented from running.

F. Response to Instrument or Equipment Failures
    1. N/A
G. Displays at HMI
    1. Plant Production/Standby Sequences
    2. Plant Shutdown Conditions
    3. Plant Winter Mode
H. Report Descriptions
    1. N/A
I. Alarms
    1. N/A

1.3 RAW WATER STORAGE TANKS

A. Description of System
    1. The RW tanks are designed to operate in a parallel arrangement and designed to provide Raw Water storage for the NR WTP treatment process. Tank level is measured via level transmitter and will be displayed on the HMI. High level switches in each tank will send an alarm to the HMI and shut off the RSPH pumps in case of an elevated level in the tanks.
B. Reference Documents
    1. RAW WATER TANKS P&ID (062-Y601)
C. Major Equipment
    1. RW Tank 1 (TK-62011)
    2. RW Tank 2 (TK-62012)
D. Major Field Instrumentation
    1. RW Tank 1 Level (LT-62051)
    2. RW Tank 2 Level (LT-62052)
    3. RW Tank 1 Switch (LSH-62051)
    4. RW Tank 2 Switch (LSH-62052)
E. Detailed Process Control Descriptions
    1. Send Raw Water Tanks Levels to Roaring Springs Pump House.
F. Response to Instrument or Equipment Failures
    1. N/A
G. Displays at HMI
    1. RW 1 Tank Level in Feet
    2. RW 2 Tank Level in Feet
H. Report Descriptions
    1. RW Tank 1 Level
    2. RW Tank 2 Level
I. Alarms
    1. Digital Alarms
        a. RW Tank 1 Level Switch High
        b. RW Tank 2 Level Switch High
    2. Analog Alarms
        a. RW Tank 1 Level High and Low
        b. RW Tank 2 Level High and Low

1.4 RAW WATER METERING, PUMPING, AND SAMPLING

A. Description of System
    1. Raw Water will flow via gravity from the RW Tanks to the WTP building. Flow will be maintained at the operator-entered Flow setpoint as measured by the Raw Water Flowmeter by modulating the Raw Water Flow Control Valve. When the water level in the RW Tanks is below the operator-entered Low Level setpoint, the RW Pump will turn on to transfer water. The Raw Water Pump Discharge Flow Control Valve will modulate to create backpressure on the RW Pump and control flow into the treatment process. A RW Sample Pump transfers Raw Water to the RW Instrument Panel for analysis of pH, Temperature, and Turbidity.

B. Reference Documents
   1. RAW WATER METERING AND PUMPING P&ID (062-Y602)

C. Major Equipment
   1. RW Pump (PMP-62011)
   2. RW Pump Suction Flow Valve (FV-62031)
   3. RW Pump Discharge Flow Control Valve (FCV-62032)
   4. RW Flow Control Valve (FCV-62033)
   5. RW Sample Pump (PMP-62012)

D. Major Field Instrumentation
   1. RW Pressure (PIT-62061)
   2. RW Pump Discharge Pressure (PIT-62062)
   3. RW Flow (FIT-62071)
   4. RW pH/Temperature (AE/AIT-62081)
   5. RW Turbidity (AE/AIT-62082)

E. Detailed Process Control Descriptions
   1. Raw Water Pump
      a. The Raw Water Pump will start when the RW Tanks Level is below the operator-adjustable Low Level Pump Start Setpoint for 30 seconds.
      b. The pump speed modulates to maintain the operator-adjustable Raw Water Flow Setpoint.
      c. The pump stops when the RW Tanks Level is above to operator-adjustable Low Level Pump Start Setpoint for 30 seconds.
   2. Raw Water Pump Suction Flow Valve
      a. The RW Pump Suction Flow Valve opens when the RW Pump is running and closes when the RW Pump stops.
   3. Raw Water Pump Discharge Flow Control Valve
      a. The RW Pump Discharge Flow Control Valve opens when the RW Pump is running and closes when the RW Pump stops.
      b. The valve modulates position to maintain an operator-adjustable RW Pump Discharge Pressure Setpoint.
   4. Raw Water Flow Control Valve
      a. The Raw Water Flow Control Valve closes when the RW Pump is running and opens when the RW Pump stops.
      b. The Raw Water Flow Control Valve modulates position to maintain the operator-adjustable Raw Water Flow Setpoint.
   5. Raw Water Sample Pump
      a. The Raw Water Sample Pump runs when the Raw Water Flow is above the Minimum Allowable Flow (not operator-adjustable).

F. Response to Instrument or Equipment Failures
   1. N/A

G. Displays at HMI
   1. RW Pump Remote/Running/Speed
   2. RW Pump Suction Valve Remote/Open/Close
   3. RW Pump Discharge Valve Remote/Position
   4. RW Flow Control Valve Remote/Position
   5. RW Sample Pump Remote/Running
   6. RW Pressure
   7. RW Pump Discharge Pressure
   8. RW Flow and Totalized Flow
   9. RW pH
   10. RW Temperature
   11. RW Turbidity

H. Report Descriptions

1. RW Pressure
2. RW Pump Speed
3. RW Pump Discharge Flow Control Valve Position
4. RW Pump Discharge Pressure
5. RW Flow Control Valve Position
6. RW Flow
7. RW pH
8. RW Temperature
9. RW Turbidity

I. Alarms
   1. Digital Alarms
      a. RW Sample Pump Fault
      b. RW Pump Suction Flow Valve Fault
      c. RW Pump Suction Flow Valve Fail to Open and Fail to Close
      d. RW Pump Fault
      e. RW Pump Motor Temperature Switch High
      f. RW Pump Fail to Start
      g. RW Discharge Flow Control Valve Fault
      h. RW Discharge Flow Control Valve Fail to Open and Fail to Close
      i. RW Flow Control Valve Fault
      j. RW Discharge Flow Control Valve Fail to Open and Fail to Close
   2. Analog Alarms
      a. RW Pressure High and Low
      b. RW Pump Discharge Pressure High and Low
      c. RW Flow Low
      d. RW pH High and Low
      e. RW Temperature High
      f. RW Turbidity High

1.5 REACTION TANKS

A. Description of System
   1. Raw Water is split and enters Reaction Tank 1 and 2, and variable speed mixers in each Tank disperse and mix coagulant and hypochlorite if the plant is set to be in Pretreatment Mode. Raw Water is mixed for approximately 10 minutes and then flows out of the tanks. It is recombined to flow to the UF membrane skids. Motorized valves are provided on the conical bottom drain of both Reaction Tank 1 and 2 to periodically drain settled solids from the tanks to the Process Waste Tank.

B. Reference Documents
   1. REACTION TANKS P&ID (062-Y603)

C. Major Equipment
   1. Static Inline Mixer (MX-63011)
   2. Reaction Tank 1 (TK-63011)
   3. Reaction Tank 2 (TK-63012)
   4. Reaction Tank 1 Mixer (MX-63012)
   5. Reaction Tank 2 Mixer (MX-63013)
   6. Reaction Tank Valve 1 (FV-63031)
   7. Reaction Tank Valve 2 (FV-63032)

D. Major Field Instrumentation
   1. Reaction Tank 1 Level (LIT-63051)
   2. Reaction Tank 2 Level (LIT-63052)
   3. Reaction Tank 1 Level Switch High (LSH-63051)
   4. Reaction Tank 2 Level Switch High (LSH-63052)

E. Detailed Process Control Descriptions

1. Reaction Tank Mixers
    a. The Reaction Tank Mixers will run when the plant is in Pretreatment Mode, and the associated Reaction Tank Level is above an operator-adjustable Level Mixer Start Setpoint and stop if the associated Reaction Tank Level is below this setpoint.
    b. The mixers will run at an operator-adjustable Speed Setpoint.
2. Reaction Tank Sludge Valves
    a. The Reaction Tank Sludge Valves open for an operator-adjustable Waste Time Setpoint then wait for an operator-adjustable Time Between Wastings Setpoint.
    b. The Reaction Tank Sludge Valve(s) will not open when the Process Waste Tank Level High alarm is active.

F. Response to Instrument or Equipment Failures
1. N/A

G. Displays at HMI
1. Reaction Tank 1 Level
2. Reaction Tank 2 Level
3. Reaction Tank 1 Mixer Remote/Running/Speed
4. Reaction Tank 2 Mixer Remote/Running/Speed
5. Reaction Tank 1 Valve Remote/Open/Close
6. Reaction Tank 2 Valve Remote/Open/Close

H. Report Descriptions
1. Reaction Tank 1 Level
2. Reaction Tank 2 Level
3. Reaction Tank 1 Mixer Speed
4. Reaction Tank 2 Mixer Speed

I. Alarms
1. Digital Alarms
    a. Reaction Tank 1 Level Switch High
    b. Reaction Tank 2 Level Switch High
    c. Reaction Tank 1 Mixer Fault
    d. Reaction Tank 1 Mixer Motor Temperature Switch High
    e. Reaction Tank 1 Mixer Fail to Start
    f. Reaction Tank 2 Mixer Fault
    g. Reaction Tank 2 Mixer Motor Temperature Switch High
    h. Reaction Tank 2 Mixer Fail to Start
    i. Reaction Tank 1 Valve Fault
    j. Reaction Tank 1 Valve Fail to Open and Fail to Close
    k. Reaction Tank 2 Valve Fault
    l. Reaction Tank 2 Valve Fail to Open and Fail to Close
2. Analog Alarms
    a. Reaction Tank 1 Level High
    b. Reaction Tank 2 Level High

1.6 MEMBRANES AND SAMPLING

A. Description of System
1. The UF Influent Sample Pump on the membrane feed line sends flow to the UF Influent Sample Panel to measure pH, Temperature, and Chlorine Residual while being distributed to the UF membrane skids from the Reaction Tanks. The skids each have a capacity to produce 100 GPM of filtrate and are designed in a duty/standby arrangement where one skid produces the full treatment capacity until a backwash is initiated, at which time the standby skid is brought online.
2. Each skid is made up of a rack of membrane modules and has one inlet connection to allow UF influent into the skid. Each skid has one outlet connection, and the outlet of each skid is combined.

3. There is a Combined Effluent Sample Panel downstream of the membrane skids where the combined effluent turbidity is measured prior to disinfection.
4. Each skid has a backwash waste connection where backwash waste is piped to the Process Waste Tank for conveyance to the existing sewer system.

B. Reference Documents
1. ULTRAFILTRATION SYSTEM P&ID (062-Y604)

C. Major Equipment
1. UF Influent Sample Pump (PMP-64011)
2. Master UF Control Panel (LCP-64001)
3. Ultrafiltration Membrane Skid 1 (Vendor Provided)
4. Ultrafiltration Membrane Skid 2 (Vendor Provided)
5. UF CIP Tank (Vendor Provided)
6. CIP Drain Line Valve (SV-64031)

D. Major Field Instrumentation
1. UF Feed pH/Temperature (AE/AIT-64081)
2. UF Feed Chlorine Residual (AE/AIT-64082)
3. UF Feed Turbidity (AE/AIT-64083)

E. Detailed Process Control Descriptions
1. UF Membrane System
   a. The UF membrane skids operate with the following modes:
      1) Online/Production: produces filtrate at 100 gpm.
      2) Backwash: backwashing to clear solids inside the module.
      3) CIP: chemical cleaning to maintain membrane integrity.
      4) Standby: waiting for duty unit to initiate backwash before going online. Will be brought online automatically by the control system.
      5) Offline: UF skid is unavailable.
   b. The UF System communicates the following with the main WTP controls system:
      1) UF System to Plant PLC:
         a) Heartbeat
         b) Critical Fault Alarm
         c) UF Membrane Skid 1 Mode
         d) UF Membrane Skid 2 Mode
         e) UF System Filtrate Flow
         f) CIP Drain Valve Open/Close Command
      2) Plant PLC to UF System:
         a) UF System Production/Standby Command
         b) Chlorine Contact Tank Level
         c) UF Feed pH
         d) UF Feed Temperature
         e) UF Feed Chlorine Residual
         f) Online Caustic Soda Drum Weight
         g) Sodium Hypochlorite Drum 1 Weight
         h) Sodium Hypochlorite Drum 2 Weight
         i) Sodium Hypochlorite Tote Weight
         j) Sodium Bisulfite Drum Weight
         k) Citric Acid Drum Weight
         l) Hydrochloric Acid Drum Weight
2. UF Influent Sample Pump
   a. The UF Influent Sample Pump runs when the Raw Water Flow is above the Minimum Allowable Flow (not operator-adjustable).

F. Response to Instrument or Equipment Failures
1. The plant will shut down if the UF System has a Critical Fault Alarm.

G. Displays at HMI
1. UF Influent Sample Pump Remote/Running
2. CIP Drain Line Valve Open/Close

3. UF Feed pH
4. UF Feed Temperature
5. UF Feed Chlorine Residual
6. Combined Filtrate Turbidity

H. Report Descriptions
1. UF Feed pH
2. UF Feed Temperature
3. UF Feed Chlorine Residual
4. Combined Filtrate Turbidity

I. Alarms
1. Digital Alarms
a. UF Influent Sample Pump Fault
b. UF Influent Fail to Start
2. Analog Alarms
a. UF Feed pH High and Low
b. UF Feed Temperature High
c. UF Feed Chlorine Residual High and Low
d. Combined Filtrate Turbidity High

1.7 DISINFECTION AND CHLORINE CONTACT (CT) TANK

A. Description of System
1. Downstream of the membrane skids, there are two chemical injection points for Sodium Hypochlorite and Caustic Soda. The doses are flow paced on the filtrate flow rate. A static mixer is used to provide mixing of the chemicals in the process stream. Downstream of the injection points, the CT Tank Influent Instrumentation Panel analyzers are provided to measure Chlorine Residual, pH, and Temperature. Flow from the membranes is conveyed to the CT Tank that is used for disinfection contact time. The CT Tank is constructed in a two-pass configuration with a center baffle wall open at one end to improve hydraulic conditions for disinfection. The CT Tank is designed to provide the required contact time at the design flow of 100 GPM. The CT Tank Outlet Valve allows for isolation of the CT Tank when the FW Tank Pumps downstream operate in recirculation mode. Downstream of the CT Tank and FW Tank Pumps, the Finished Water Instrumentation Panel analyzers are provided to measure Chlorine Residual, pH, and Temperature.

A. Reference Documents
1. DISINFECTION AND CHLORINE CONTACT TANK P&ID (062-Y605)

B. Major Equipment
1. Chlorine Contact Tank (TK-66011)
2. Filtrate Water Static Mixer (MX-66011)
3. CT Tank Influent Sample Pump (PMP-68117)
4. CT Tank Outlet Valve (FV-66031)

C. Major Field Instrumentation
1. Combined Filtrate Flow (FIT-64071)
2. CT Tank Influent pH/Temperature (AE/AIT-68181)
3. CT Tank Influent Chlorine Residual (AE/AIT-68182)
4. Finished Water Chlorine Residual (AE/AIT-66081)
5. Finished Water pH/Temperature (AE/AIT-66082)
6. Finished Water Turbidity (AE/AIT-66083)
7. CT Tank Level (LT-66051)
8. CT Tank Level Switch High-High (LSHH-66051)

D. Detailed Process Control Descriptions
1. CT Tank Influent Sample Pump
a. The CT Tank Influent Sample Pump runs when the Combined Filtrate Flow is above the Minimum Allowable Flow (not operator-adjustable).
2. CT Tank Outlet Valve

a. The CT Tank Outlet Valve opens when the WTP moves into Production Mode and closes when the WTP goes into Standby or Winter Mode

E. Response to Instrument or Equipment Failures
   1. N/A
F. Displays at HMI
   1. Combined Filtrate Flow and Totalized Flow
   2. CT Tank Influent Sample Pump Remote/Running
   3. CT Tank Influent pH
   4. CT Tank Influent Temperature
   5. CT Tank Influent Chlorine Residual
   6. CT Tank Level
   7. CT Tank Outlet Valve Remote/Open/Close
   8. Finished Water Chlorine Residual
   9. Finished Water pH
   10. Finished Water Temperature
   11. Finished Water Turbidity
G. Report Descriptions
   1. Combined Filtrate Flow
   2. CT Tank Influent pH
   3. CT Tank Influent Temperature
   4. CT Tank Influent Chlorine Residual
   5. CT Tank Level
   6. Finished Water Chlorine Residual
   7. Finished Water pH
   8. Finished Water Temperature
   9. Finished Water Turbidity
A. Alarms
   1. Digital Alarms
      a. CT Tank Influent Sample Pump Fault
      b. CT Tank Influent Sample Pump Fail to Start
      c. CT Tank Level Switch High-High
   2. Analog Alarms
      a. CT Tank Influent pH High and Low
      b. CT Tank Influent Temperature High
      c. CT Tank Influent Chlorine Residual High and Low
      d. CT Tank Level High and Low
      e. Finished Water Chlorine Residual High and Low
      f. Finished Water pH High and Low
      g. Finished Water Temperature High
      h. Finished Water Turbidity High

1.8 FINISHED WATER TANK PUMPS

A. Description of System
   1. Water is conveyed from the CT Tank to the Finished Water (FW) Pumps suction header where two Finished Water Pumps will be installed in a duty/standby configuration. Each variable speed pump is rated at 100 GPM and is designed to transfer FW to the FW Tanks at a rate that maintains a constant level in the CT Tank. The FW Tank Pumps will also be used to recirculate FW through the FW Tanks when the WTP is in standby mode or during Winter Mode when the system demands are met by the tank volume. A secondary chlorine dosing point is provided on the recirculation line to increase the Chlorine Residual if the FW drops below a target setpoint. The Distribution Water Instrumentation Panel Sample Pump transfers FW from the recirculation line to the Distribution Water Instrumentation Panel to measure Chlorine Residual, Turbidity, pH, and Temperature.

B. Reference Documents
   1. FINISHED WATER TANK PUMPS P&ID (062-Y606)

C. Major Equipment
   1. FW Tank Pump 1 (PMP-68111)
   2. FW Tank Pump 2 (PMP-68112)
   3. Distribution Water Instrumentation Panel Sample Pump (PMP-66011)

D. Major Field Instrumentation
   1. Distribution Water pH/Temperature (AE/AIT-66084)
   2. Distribution Water Chlorine Residual (AE/AIT-66085)
   3. Distribution Water Turbidity (AE/AIT-66086)
   4. FW Tank Pumps Discharge Pressure (PIT-68161)
   5. FW Tanks Influent Flow (FIT-68173)

E. Detailed Process Control Descriptions
   1. FW Tank Pumps
      a. The Finished Water Tank Pumps operate in Duty/Standby where the Standby pump will be called into operation if the Duty pump fails.
      a. When the plant is in Production Mode, the duty pump will modulate speed to maintain an operator-adjustable Contact Time Tank Level setpoint.
      b. When the plant is in Standby or Winter Mode, the duty pump will run continuously and maintain a constant operator-adjustable Finished Water Tank Pumps Recirculation Speed Setpoint.
      c. The FW Tank Pumps will stop while the plant is between modes.
   2. Distribution Water Sample Pump
      a. Distribution Water Sample Pump runs continuously.

F. Response to Instrument or Equipment Failures
   1. N/A

G. Displays at HMI
   1. FW Tank Pump 1 Remote/Running/Speed
   2. FW Tank Pump 2 Remote/Running/Speed
   3. Distribution Water Sample Pump Remote/Running
   4. Distribution Water pH
   5. Distribution Water Temperature
   6. Distribution Water Chlorine Residual
   7. Distribution Water Turbidity
   8. FW Tank Pumps Discharge Pressure
   9. FW Tank Influent Flow & Totalized Flow

H. Report Descriptions
   1. Distribution Water pH
   2. Distribution Water Temperature
   3. Distribution Water Chlorine Residual
   4. Distribution Water Turbidity
   5. FW Tank Pumps Discharge Pressure
   6. FW Tanks Influent Flow
   7. FW Tank Pump 1 Speed
   8. FW Tank Pump 2 Speed

I. Alarms
   1. Digital Alarms
      a. FW Tank Pump 1 Fault
      b. FW Tank Pump 1 High Motor Temperature Switch
      c. FW Tank Pump 1 Fail to Start
      d. FW Tank Pump 2 Fault
      e. FW Tank Pump 2 High Motor Temperature Switch
      f. FW Tank Pump 2 Fail to Start
      g. Distribution Water Sample Pump Fault
      h. Distribution Water Sample Pump Fail to Start

2. Analog Alarms
    a. Distribution Water pH High and Low
    b. Distribution Water Temperature Low
3. Distribution Water Chlorine Residual High and Low
4. Distribution Water Turbidity High
5. Finished Water Pumps Pressure High and Low

1.9 FINISHED WATER STORAGE AND DISTRIBUTION

A. Description of System
1. Flow is conveyed from the FW Tanks through a 12-inch line to the WTP building. The FW Tanks operate in series and maintain the same level under normal operations.
2. Inside WTP, the FW is diverted as follows:
    a. FW Tank Recirculation: Flow is diverted back to the FW Tank Pumps to be recirculated through the FW Tanks when the plant is in Standby Mode or Winter Mode.
    b. Distribution System Lower Zone: Flow is conveyed by gravity to the Lower Zone.
    c. Backfeed to Supai Tunnel: Flow is conveyed off the Combined Backfeed line which is split off the Lower Zone.
    d. Backfeed to Innercanyon: Flow is conveyed off the Combined Backfeed line which is split off the Lower Zone. Flow is determined by subtracting the Backfeed to Supai Tunnel flow from the Combined Backfeed flow.
    e. Distribution System Upper Zone: Three Upper Zone Booster Pumps supply flow to the Upper Zone. A pressure transmitter on the discharge of the pumps measures the system pressure in the Upper Zone. Hydropneumatic tanks are provided on the discharge of the pumps to limit pump cycling.
    f. Distribution System High Demand Pumps: Three High Demand Pumps are connected to the Upper Zone to meet high demands. The discharge of these pumps is connected to the discharge of the Booster Pumps. The pressure and flow are measured with the same instruments as the Booster Pumps.

B. Reference Documents
1. FINISHED WATER TANKS P&ID (062-Y607)
2. FINISHED WATER METERING AND PUMPING 1 P&ID (062-Y608)
3. FINISHED WATER METERING AND PUMPING 2 P&ID (062-Y609)

C. Major Equipment
1. FW Tank A (TK-68411)
2. FW Tank B (TK-68412)
3. High Demand Pump 1 (PMP-68211)
4. High Demand Pump 2 (PMP-68212)
5. High Demand Pump 3 (PMP-68213)
6. Upper Zone Booster Pump 1 (PMP-68311)
7. Upper Zone Booster Pump 2 (PMP-68312)
8. Upper Zone Booster Pump 3 (PMP-68313)
9. Backfeed to Innercanyon Flow Valve (FV-68031)
10. Backfeed to Supai Flow Valve (FV-68032)
11. FW Recirculation Valve (FV-68033)
12. Hydropneumatic Tank 1 (TK-68113)
13. Hydropneumatic Tank 2 (TK-68114)

D. Major Field Instrumentation
1. FW Tank A Level (LT-68451)
2. FW Tank B Level (LT-68452)
3. Upper Zone Booster Pumps Suction Pressure (PIT-68162)
4. Upper Zone Distribution Pressure (PIT-68163)
5. Upper Zone Flow (FIT-68171)
6. Lower Zone Flow (FIT-68172)
7. Combined Backfeed Flow (FIT-68174)

8. Backfeed to Supai Flow (FIT-68175)

E. Detailed Process Control Descriptions
   1. Upper Zone Booster Pumps
      a. The Upper Zone Booster Pumps operate in Lead/Lag1/Lag2.
      b. The pumps will modulate speed to maintain an operator-entered Upper Zone Pressure setpoint.
      c. When the pressure drops below the operator-entered Upper Zone Booster Pumps Low Pressure setpoint, an additional pump will be called to run in order of Lead, Lag1, Lag2.
      d. When the pressure rises above an operator-entered Upper Zone Booster Pumps High Pressure setpoint, a pump will stop in order of Lag2, Lag1, Lead.
      e. If a High Demand Pump is running, the Upper Zone Booster Pumps will not run.
   2. High Demand Pumps
      a. The High Demand Pumps will operate in Lead/Lag/Standby.
      b. When the pressure drops below the operator-entered High Demand Pumps Low Pressure setpoint and all three Upper Zone Booster Pumps are running, a High Demand Pump will be called to run in order of Lead then Lag.
      c. When the pressure rises above the operator-entered High Demand Pumps High Pressure setpoint and at least one High Demand Pump is running, the pumps will stop in order of Lag then Lead. If both High Demand Pumps turn off, the Upper Zone Booster Pumps will turn on.
   3. Finished Water Recirculation
      a. The FW Recirculation Valve opens when the WTP moves into Standby or Winter Mode and closes when the WTP moves into Production Mode.
   4. Backfeed to Inner-Canyon
      a. The Backfeed to Inner-Canyon Flow Valve and Backfeed to Supai Tunnel Flow Valve open and close based on operator initiation.
      b. The Backfeed to Supai Tunnel Flow Valve will close automatically when a Supai Tunnel High Flow Alarm is active.
      c. The Backfeed to Inner-Canyon Flow Valve will close automatically when a Inner Canyon High Flow Alarm is active and Supai Tunnel High Flow Alarm is not active.

F. Response to Instrument or Equipment Failures
   1. N/A

G. Displays at HMI
   1. High Demand Pump 1 Remote/Running/Speed
   2. High Demand Pump 2 Remote/Running/Speed
   3. High Demand Pump 3 Remote/Running/Speed
   4. Upper Zone Distribution Pump 1 Remote/Running/Speed
   5. Upper Zone Distribution Pump 2 Remote/Running/Speed
   6. Upper Zone Distribution Pump 3 Remote/Running/Speed
   7. Backfeed to Inner-Canyon Flow Valve Remote/Open/Close
   8. Backfeed to Supai Tunnel Flow Valve Remote/Open/Close
   9. FW Recirculation Flow Valve Remote/Open/Close
   10. FW Tank A Level
   11. FW Tank B Level
   12. Upper Zone Distribution Pumps Suction Pressure
   13. Combined Backfeed Flow and Totalized Flow
   14. Backfeed to Supai Tunnel Flow and Totalized Flow
   15. Lower Zone Distribution Flow and Totalized Flow
   16. Upper Zone Distribution Pressure
   17. Upper Zone Distribution Flow and Totalized Flow

H. Report Descriptions
   1. FW Tank A Level
   2. FW Tank B Level
   3. High Demand Pump 1 Speed

4. High Demand Pump 2 Speed
5. High Demand Pump 3 Speed
6. Upper Zone Distribution Pump 1 Speed
7. Upper Zone Distribution Pump 2 Speed
8. Upper Zone Distribution Pump 3 Speed
9. Upper Zone Distribution Pumps Suction Pressure
10. Upper Zone Distribution Pressure
11. Combined Backfeed to Inner-Canyon Flow
12. Backfeed to Supai Tunnel Flow
13. Lower Zone Distribution Flow
14. Upper Zone Distribution Flow

I. Alarms
   1. Digital Alarms
      a. High Demand Pump 1 Fault
      b. High Demand Pump 1 Motor High Temperature Switch
      c. High Demand Pump 1 Fail to Start
      d. High Demand Pump 2 Fault
      e. High Demand Pump 2 Motor High Temperature Switch
      f. High Demand Pump 2 Fail to Start
      g. High Demand Pump 3 Fault
      h. High Demand Pump 3 Motor High Temperature Switch
      i. High Demand Pump 3 Fail to Start
      j. Combined Backfeed to Inner-Canyon Flow Valve Fault
      k. Combined Backfeed to Inner-Canyon Flow Valve Fail to Open and Fail to Close
      l. Backfeed to Supai Tunnel Flow Valve Fault
      m. Backfeed to Supai Tunnel Flow Valve Fail to Open and Fail to Close
      n. FW Recirculation Flow Valve Fault
      o. FW Recirculation Flow Valve Fail to Open and Fail to Close
      p. Upper Zone Distribution Pump 1 Fault
      q. Upper Zone Distribution Pump 1 Motor High Temperature Switch
      r. Upper Zone Distribution Pump 1 Fail to Start
      s. Upper Zone Distribution Pump 2 Fault
      t. Upper Zone Distribution Pump 2 Motor High Temperature Switch
      u. Upper Zone Distribution Pump 2 Fail to Start
      v. Upper Zone Distribution Pump 3 Fault
      w. Upper Zone Distribution Pump 3 Motor High Temperature Switch
      x. Upper Zone Distribution Pump 3 Fail to Start
   2. Analog Alarms
      a. FW Tank A Level High and Low
      b. FW Tank B Level High and Low
      c. Upper Zone Distribution Pumps Suction Pressure High and Low
      d. Combined Backfeed to Inner-Canyon Flow High
      e. Backfeed to Supai Tunnel Flow High
      f. Lower Zone Distribution Flow High and Low
      g. Upper Zone Distribution Pressure High and Low
      h. Upper Zone Distribution Flow High and Low

1.10 COAGULANT FEED SYSTEM

A. Description of System
   1. The WTP process uses coagulant when the plant is in Pretreatment Mode. Coagulant will be delivered in 55-gallon drums by truck and will be stored inside the Acids Drum Room in the WTP building. One metering pump (with a shelf spare) will deliver coagulant to the RW line upstream of the Reaction Tanks. The control system will adjust the speed of the pump to deliver an adjustable dose of coagulant. This dose is flow paced from the meter on the

RW line. The coagulant drums will sit on top of containment sumps. The room also contains an emergency shower and eyewash.
2. Although scales are provided for two drums, only one drum will be online at a time.

B. Reference Documents
1. DRUM CHEMICAL SYSTEM P&ID (062-Y610)

C. Major Equipment
1. Coagulant Metering Pump (PMP-67111)
2. Coagulant Drum 1 Weight Scale (SCL-67111)
3. Coagulant Drum 2 Weight Scale (SCL-67112)

D. Major Field Instrumentation
1. Coagulant Containment Sump Level Switch High (LSH-67151)
2. Coagulant Drum 1 Weight (WIT-67191)
3. Coagulant Drum 2 Weight (WIT-67192)

E. Detailed Process Control Descriptions
1. Coagulant Metering Pump
   a. The operator will select which drum is in use.
   b. The Coagulant Metering Pump will run when the plant is in production, the plant is in Pretreatment Mode, and Coagulant Dosing is enabled by the operator. It will stop when the plant is in standby or Coagulant Dosing is disabled by the operator.
   c. The pump speed will be determined by the operator-entered Dose setpoint and paced off the Raw Water Flow.
   d. If the Coagulant Metering Pump becomes unavailable or if there is no coagulant to dose while in Pretreatment mode, the Coagulant Metering Pump system will issue a warning for 10 minutes before issuing a plant shutdown command.

F. Response to Instrument or Equipment Failures
1. N/A

G. Displays at HMI
1. Coagulant Pump Remote/Running/Speed
2. Coagulant Drum 1 Weight
3. Coagulant Drum 2 Weight

H. Report Descriptions
1. Coagulant Pump Speed
2. Coagulant Drum 1 Weight
3. Coagulant Drum 2 Weight

I. Alarms
1. Digital Alarms
   a. Coagulant Sump Level Switch High
   b. Coagulant Metering Pump Fault
   c. Coagulant Metering Pump Fail to Start
2. Analog Alarms
   a. Coagulant Drum 1 Weight Low
   b. Coagulant Drum 2 Weight Low

1.11 CAUSTIC SODA FEED SYSTEM

A. Description of System
1. Caustic Soda will be delivered in 55-gallon drums and will be stored inside the Bases Drum Room in the WTP building. One contractor-provided metering pump (with a shelf spare) will deliver caustic soda to the FW line upstream of the CT Tank. The control system will adjust the speed of the pump to deliver an adjustable dose of caustic soda. This dose is adjusted based on the pH measured by the analyzer on the FW line. The UF membrane skids also require Caustic Soda as part of the CIP process. Vendor-provided metering pumps will deliver Caustic Soda to the membrane skids. The membrane control system will adjust the

pump speed to deliver an adjustable dose of Caustic Soda. The caustic soda drums will sit on top of containment sumps. The room also contains an emergency shower and eyewash.
2. Although scales are provided for two drums, only one drum will be online at a time.
3. The Caustic Soda system will issue a plant shutdown command if the Caustic Soda Metering Pump becomes unavailable or if the online drum has a low alarm.

B. Reference Documents
1. DRUM CHEMICAL SYSTEM P&ID (062-Y610)

C. Major Equipment
1. Caustic Soda Metering Pump (PMP-67311)
2. CIP Pump Panel (Vendor Provided)
3. Caustic Soda Drum 1 Weight Scale (SCL-67311)
4. Caustic Soda Drum 2 Weight Scale (SCL-67312)

D. Major Field Instrumentation
1. Caustic Soda Sump Level Switch High (LSH-67351)
2. Caustic Soda Drum 1 Weight (WIT-67391)
3. Caustic Soda Drum 2 Weight (WIT-67392)

E. Detailed Process Control Descriptions
1. Caustic Soda Metering Pump
   a. The Caustic Soda Metering Pump will run when the plant is in production and Caustic Dosing is enabled by the operator. The pump will stop when the plant is in standby or Caustic Dosing is disabled by the operator.
   b. When in Flow Pacing Mode, the pump speed will be determined by the operator-entered Dose setpoint and paced off the Combined Filtrate Flow.
   c. When in PID Mode, the pump speed will modulate speed to maintain an operator-entered CT Tank Influent pH setpoint.

F. Response to Instrument or Equipment Failures
1. N/A

G. Displays at HMI
1. Caustic Soda Metering Pump Remote/Running/Speed
2. Caustic Soda Drum 1 Weight
3. Caustic Soda Drum 2 Weight

H. Report Descriptions
1. Caustic Soda Metering Pump Speed
2. Caustic Soda Drum 1 Weight
3. Caustic Soda Drum 2 Weight

I. Alarms
1. Digital Alarms
   a. Caustic Soda Sump Level Switch High
   b. Caustic Soda Metering Pump Fault
   c. Caustic Soda Metering Pump Fail to Start
2. Analog Alarms
   a. Caustic Soda Drum 1 Weight Low
   b. Caustic Soda Drum 2 Weight Low

1.12 SODIUM HYPOCHLORITE FEED SYSTEM

A. Description of System
1. The Plant will use Sodium Hypochlorite as the primary disinfectant. Sodium Hypochlorite will be delivered in 55-Gallon Drums and 300-Gallon Totes and will be stored inside the Bases Drum Room in the WTP building.
2. Four injection points are included in the WTP design:
   a. Filtrate Line Upstream of CT Tank: This is the primary point of disinfection for WTP. One contractor- provided metering pump (with a shelf-spare and swing pump) will

deliver Sodium Hypochlorite to the Filtrate line. The control system will adjust the speed of the pump to deliver an adjustable dose of Sodium Hypochlorite. Chlorine dose is set by the operator at the HMI, and the control system will pace the dose from the FW flow meter. This dose will normally be biased (trimmed) by the chlorine level in the FW downstream of the CT Tank.

b. RW Line Upstream of Reaction Tanks: If oxidation of iron and manganese is needed in RW, Sodium Hypochlorite will be used at this injection point. One contractor- provided metering pump is provided to deliver Sodium Hypochlorite to the RW line. The control system will adjust the speed of the pump to deliver an adjustable dose of sodium hypochlorite. Chlorine dose is set by the operator at the HMI, and the control system will pace the dose from the RW flow meter.

c. FW Upstream of FW Tank Pumps: Chlorine residual is measured in the recirculation line of the FW Tanks. If the residual is measured to drop below a target residual (adjustable in the HMI), then Sodium Hypochlorite will be used at this injection point to boost the residual in the Tanks. One contractor provided metering pump is provided to deliver Sodium Hypochlorite to the FW line. The control system will adjust the speed of the pump to deliver an adjustable dose of Sodium Hypochlorite. Chlorine dose is set by the operator at the HMI, and the control system will bias (trim) the dose until the target residual is met.

d. UF Membrane Skids for CIP: The UF membrane skids also require Sodium Hypochlorite as part of the CIP process. Vendor-provided metering pumps will deliver Sodium Hypochlorite to the membrane skids. The membrane control system will adjust the pump speed to deliver an adjustable dose of Sodium Hypochlorite.

B. Reference Documents
   1. DRUM CHEMICAL SYSTEM P&ID (062-Y611)

C. Major Equipment
   1. Sodium Hypochlorite Drum 1 Weight Scale (SCL-67211)
   2. Sodium Hypochlorite Drum 2 Weight Scale (SCL-67212)
   3. Sodium Hypochlorite Tote Weight Scale (SCL-67213)
   4. Sodium Hypochlorite Metering Pump 1 (PMP-67211)
   5. Sodium Hypochlorite Metering Pump 2 (PMP-67212)
   6. Sodium Hypochlorite Metering Pump 3 (PMP-67213)
   7. Sodium Hypochlorite Metering Swing Pump (PMP-67214)
   8. CIP Pump Panel (Vendor Provided)

D. Major Field Equipment
   1. Sodium Hypochlorite Containment Sump Level Switch High (LSH-67251)
   2. Sodium Hypochlorite Drum 1 Weight (WIT-67291)
   3. Sodium Hypochlorite Drum 2 Weight (WIT-67292)
   4. Sodium Hypochlorite Tote Weight (WIT-67293)

E. Detailed Process Control Descriptions
   1. Sodium Hypochlorite Metering Pumps
      a. The Sodium Hypochlorite Metering Pumps operate with a single Swing Pump that must be selected into service by the operator to be used in a Duty/Standby formation where the Standby pump will be called into operation if the Duty pump fails.
      b. The Sodium Hypochlorite Duty Pump for the filtrate water will run when the plant is in production. The pump will stop if the plant is in standby. The pump speed will modulate to an operator-entered Dose setpoint paced to the Combined Filtrate Water Flowmeter and trimmed to the Contact Time Tank Chlorine Residual.
      c. The Sodium Hypochlorite Duty Pump for the raw water will run when the plant is in production and enabled by the operator. The pump will stop if the plant is in standby or disabled by the operator. The pump speed will modulate to an operator-entered Dose setpoint paced to the Raw Water Flowmeter.
      d. The Sodium Hypochlorite Duty Pump for the finished water will run when the Finished Water Chlorine Residual is below the operator-entered Low Chlorine Residual setpoint.

The pump speed will modulate to an operator-entered Dose setpoint paced to the Finished Water Tanks Influent Flow that is trimmed to meet the operator-adjustable Finished Water Chlorine Residual Setpoint.

e. The Sodium Hypochlorite system will issue a plant shutdown command if the metering pumps for the Filtrate Water line become unavailable, if the metering pumps for the finished water become unavailable, or all the drums and totes have low alarms.

F. Response to Instrument or Equipment Failures
   1. N/A

G. Displays at HMI/OIT
   1. Sodium Hypochlorite Pump 1 Remote/Running/Speed
   2. Sodium Hypochlorite Pump 2 Remote/Running/Speed
   3. Sodium Hypochlorite Pump 3 Remote/Running/Speed
   4. Sodium Hypochlorite Swing Pump Remote/Running/Speed
   5. Sodium Hypochlorite Drum 1 Weight
   6. Sodium Hypochlorite Drum 2 Weight
   7. Sodium Hypochlorite Tote Weight

H. Report Descriptions
   1. Sodium Hypochlorite Metering Pump 1 Speed
   2. Sodium Hypochlorite Metering Pump 2 Speed
   3. Sodium Hypochlorite Metering Pump 3 Speed
   1. Sodium Hypochlorite Swing Pump Speed
   2. Sodium Hypochlorite Drum 1 Weight
   3. Sodium Hypochlorite Drum 2 Weight
   4. Sodium Hypochlorite Tote Weight

I. Alarms
   1. Digital Alarms
      a. Sodium Hypochlorite Sump Level Switch High
      b. Sodium Hypochlorite Metering Pump 1 Fault
      c. Sodium Hypochlorite Metering Pump 1 Fail to Start
      d. Sodium Hypochlorite Metering Pump 2 Fault
      e. Sodium Hypochlorite Metering Pump 2 Fail to Start
      f. Sodium Hypochlorite Metering Pump 3 Fault
      g. Sodium Hypochlorite Metering Pump 3 Fail to Start
      h. Sodium Hypochlorite Swing Pump Fault
      i. Sodium Hypochlorite Swing Pump Fail to Start
   2. Analog Alarms
      a. Sodium Hypochlorite Drum 1 Weight Low
      b. Sodium Hypochlorite Drum 2 Weight Low
      c. Sodium Hypochlorite Tote Weight Low

1.13 SODIUM BISULFITE AND FEED SYSTEM

A. Description of System
   1. The UF membrane skids require Sodium Bisulfite as part of the CIP process. Sodium Bisulfite will be delivered in 55-gallon drums and will be stored inside the Acids Drum Room in the WTP building. Metering pumps are vendor-provided and will deliver Sodium Bisulfite to the membrane skids. The membrane control system will adjust the pump speed to deliver an adjustable dose of sodium bisulfite. The sodium bisulfite drums will sit on top of containment sumps. The room also contains an emergency shower and eyewash.

B. Reference Documents
   1. DRUM CHEMICAL SYSTEM P&ID (062-Y612)

C. Major Equipment
   1. Sodium Bisulfite Weight Scale (SCL-67611)
   2. CIP Pump Panel (Vendor Provided)

D. Major Field Instrumentation
   1. Sodium Bisulfite Containment Sump Level Switch High (LSH-67651)
   2. Sodium Bisulfite Weight (WIT-67691)
   3. Sulfur Dioxide Gas (AE/AIT-61881)

E. Detailed Process Control Descriptions
   1. The Sodium Bisfulfite system is controlled by the UF system.
   2. The horns and strobes located in the acid drum room and at the chemical room entrance will activate when the Sulfur Dioxide Gas High Alarm is active.
   3. The Sodium Bisulfite system will issue a plant shutdown command if the drum has a low alarm.

F. Response to Instrument or Equipment Failures
   1. N/A

G. Displays at HMI/OIT
   1. Sodium Bisulfite Drum Weight

H. Report Descriptions
   1. Sodium Bisulfite Drum Weight

I. Alarms
   1. Digital Alarms
      a. Sodium Bisulfite Containment Sump Level Switch
   2. Analog Alarms
      a. Sodium Bisulfite Drum Weight Low
      b. Chemical Room Sulfur Dioxide Analyzer High

1.14 CITRIC ACID STORAGE AND FEED SYSTEM

A. Description of System
   1. The UF membrane skids require Citric Acid as part of the CIP process. Citric acid will be delivered in 55-gallon drums and will be stored inside the Acids Chemical Drum Room in the WTP building. Metering pumps are vendor-provided and will deliver Citric Acid to the membrane skids. The membrane control system will adjust the pump speed to deliver an adjustable dose of Citric Acid. The Citric Acid drums will sit on top of a containment sump. The room also contains an emergency shower and eyewash.

B. Reference Documents
   1. DRUM CHEMICAL SYSTEM P&ID (062-Y612)

C. Major Equipment
   1. Citric Acid Drum Weight Scale (SCL-67612)
   2. CIP Pump Panel (Vendor Provided)

D. Major Field Instrumentation
   1. Citric Acid Containment Sump Level Switch High (LSH-67652)
   2. Citric Acid Drum Weight (WIT-67692)

E. Detailed Process Control Descriptions
   1. This system is controlled by the UF system.

F. Response to Instrument or Equipment Failures
   1. N/A

G. Displays at HMI/OIT
   1. Citric Acid Drum Weight

H. Report Descriptions
   1. Citric Acid Drum Weight

I. Alarms
   1. Digital Alarms

a. Citric Acid Containment Sump Level Switch High
2. Analog Alarms
a. Citric Acid Drum Weight Low

1.15 HYDROCHLORIC ACID FEED SYSTEM

A. Description of System
1. The UF membrane skids require Hydrochloric Acid as part of the CIP process. Hydrochloric acid will be delivered in 55-gallon drums and will be stored inside the Acids Drum Room in the WTP building. Metering pumps are vendor-provided and will deliver Hydrochloric Acid to the membrane skids. The membrane control system will adjust the pump speed to deliver an adjustable dose of Hydrochloric Acid. The Hydrochloric Acid drums will sit on top of containment sumps. The room also contains an emergency shower and eyewash.

B. Reference Documents
1. DRUM CHEMICAL SYSTEM P&ID (062-Y613)

C. Major Equipment
1. Hydrochloric Acid Drum Weight Scale (SCL-67411)
2. CIP Pump Panel (Vendor Provided)

A. Major Field Instrumentation
1. Hydrochloric Acid Containment Sump Level Switch High (LSH-67451)
2. Hydrochloric Acid Drum Weight (WIT-67491)

B. Detailed Process Control Descriptions
1. This system is controlled by the UF system.

C. Response to Instrument or Equipment Failures
1. N/A

D. Displays at HMI/OIT
1. Hydrochloric Acid Drum Weight

E. Report Descriptions
1. Hydrochloric Acid Drum Weight

F. Alarms
1. Digital Alarms
a. Hydrochloric Acid Containment Sump Level Switch
2. Analog Alarms
a. Hydrochloric Acid Drum Weight Low

1.16 CORROSION INHIBITOR FEED SYSTEM (FUTURE)

A. Description of System
1. The design of the treatment process allows for the future use of a corrosion inhibitor should the Park require the chemical in the future. Corrosion inhibitor would be delivered in 55-gallon drums and stored inside the Acids Chemical Drum Room in the WTP building. The corrosion inhibitor drums will sit on top of a containment sump.

B. Reference Documents
1. DRUM CHEMICAL SYSTEM P&ID (062-Y613)

C. Major Equipment
1. N/A

D. Major Field Instrumentation
1. Corrosion Inhibitor Containment Sump Level Switch High (LSH-67551)

E. Detailed Process Control Descriptions
1. N/A

F. Response to Instrument or Equipment Failures

1. N/A

G. Displays at HMI/OIT
   1. N/A

H. Report Descriptions
   1. N/A

I. Alarms
   1. Digital Alarms
      a. Corrosion Inhibitor Containment Sump Level Switch High
   2. Analog Alarms
      a. N/A

1.17 PROCESS WASTE TANK

A. Description of System
   1. Process waste is collected in the Process Waste Tank located outside of the WTP Building. Process waste includes reaction tank sludge blowdown, membrane backwash waste, neutralized CIP waste, and the CT Tank drain. The process waste flows by gravity from the Process Waste Tank into a Sanitary Manhole where it combines with WTP building domestic waste flows. The combined waste in the Sanitary Manhole flow by gravity to the existing sewer system. The waste level inside the Process Waste Tank is monitored.

B. Reference Documents
   1. PROCESS WASTE TANK P&ID (062-Y614)

C. Major Equipment
   1. Process Waste Tank (TK-69111)

D. Major Field Instrumentation
   1. Process Waste Tank Level Switch High-High (LSHH-69151)
   2. Process Waste Tank Level (LT-69151)

E. Detailed Process Control Descriptions
   1. N/A

F. Response to Instrument or Equipment Failures
   1. N/A

G. Displays at HMI/OIT
   1. Process Waste Tank Level

H. Report Descriptions
   1. Process Waste Tank Level

I. Alarms
   1. Digital Alarms
      a. Process Waste Tank Level Switch High-High
   2. Analog Alarms
      a. Process Waste Tank Level High

1.18 MISCELLANEOUS SYSTEMS

A. Description of System
   1. N/A

B. Reference Documents
   1. MISCELLANEOUS SYSTEMS P&ID (62-Y615)

C. Major Equipment
   1. N/A

D. Major Field Instrumentation

1. Process Room Temperature (TIT-61081)
2. Acids Drum Room Temperature (TIT-67081)
3. Base Drum Room Temperature (TIT-67082)
4. Mechanical Room Temperature (TIT-61082)
5. Electrical Room Smoke Detector (SD-61109)
6. Heat Trace Panel (HTP-61741)
7. Laboratory Emergency Shower (ESEW-61641)
8. Acids Drum Room Emergency Shower (ESEW-61642)
9. Base Drum Room Emergency Shower (ESEW-61643)
10. Vestibule Exterior Door Right Intrusion Switch (ZS-61491A)
11. Vestibule Exterior Door Left Intrusion Switch (ZS-61491B)
12. Mechanical Room Exterior Door Right Intrusion Switch (ZS-61492A)
13. Mechanical Room Exterior Door Left Intrusion Switch (ZS-61492B)
1. Break/Kitchen Room Exterior Intrusion Switch (ZS-61493)
2. Process Area South Exterior Intrusion Switch (ZS-61494)
3. Acids Room Exterior Door Right Intrusion Switch (ZS-61495A)
4. Acids Room Exterior Door Left Intrusion Switch (ZS-61495B)
5. Process Materials Shop Exterior Door Right Intrusion Switch (ZS-61496A)
6. Process Materials Shop Exterior Door Left Intrusion Switch (ZS-61496B)
7. Electrical Room Exterior Door Right Intrusion Switch (ZS-61497A)
8. Electrical Room Exterior Door Left Intrusion Switch (ZS-61497B)
9. Process Area North Exterior Door Intrusion Switch (ZS-61498)
10. Bases Room Exterior Door Right Intrusion Switch (ZS-61499A)
11. Bases Room Exterior Door Left Intrusion Switch (ZS-61499B)
12. Master PLC Mode Switch (ZS-61101)
13. Standby PLC Mode Switch (ZS-61102)

E. Detailed Process Control Descriptions
1. N/A

F. Response to Instrument or Equipment Failures
1. N/A

G. Displays at HMI/OIT
1. Pump Room Temperature
2. Acids Drum Room Temperature
3. Bases Drum Room Temperature
4. Mechanical Room Temperature

H. Report Descriptions
1. Process Room Temperature
2. Acids Drum Room Temperature
3. Bases Drum Room Temperature
4. Mechanical Room Temperature

I. Alarms
1. Discrete Alarms
    a. Electrical Room Smoke Detector
    b. Heat Trace Panel Fault
    c. Laboratory Emergency Shower Flow Switch High
    d. Acids Drum Room Emergency Shower Flow Switch High
    e. Bases Drum Room Emergency Shower Flow Switch High
    f. Vestibule Exterior Door Right Intrusion Switch
    g. Vestibule Exterior Door Left Intrusion Switch
    h. Mechanical Room Exterior Door Right Intrusion Switch
    i. Mechanical Room Exterior Door Left Intrusion Switch
    j. Break/Kitchen Room Exterior Intrusion Switch
    k. Process Area South Exterior Intrusion Switch

l. Acids Room Exterior Door Right Intrusion Switch
m. Acids Room Exterior Door Left Intrusion Switch
n. Process Materials Shop Exterior Door Right Intrusion Switch
o. Process Materials Shop Exterior Door Left Intrusion Switch
p. Electrical Room Exterior Door Right Intrusion Switch
q. Electrical Room Exterior Door Left Intrusion Switch
r. Process Area North Exterior Door Intrusion Switch
s. Bases Room Exterior Door Right Intrusion Switch
t. Bases Room Exterior Door Left Intrusion Switch
u. Master/Standby Mode Switches Misconfigured

2. Analog Alarms
   a. Process Room Temperature High and Low
   b. Acids Drum Room Temperature High and Low
   c. Bases Drum Room Temperature High and Low
   d. Mechanical Room Temperature High and Low

END OF APPENDIX B

SECTION 40 62 05
COMPUTER SYSTEM HARDWARE AND ANCILLARIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Computer and HMI hardware requirements, which include, but are not necessarily limited to:
      a. LCD Flat Panel Type Monitors.
      b. HMI/Operator Computer Workstations.
      c. Network Servers.
      d. External Data Storage.
      e. Laptops.
      f. Server Racks.
      g. NTP Server.
      h. TVs.
      i. Cameras.
      j. Software.
      k. Accessories and Maintenance Materials.

B. Related Specification Sections include, but are not necessarily limited to:
   1. Section 01 81 33 – Cyber Security Requirements
   2. Section 40 61 13 - Process Control Systems General Requirements.
   3. Section 40 61 21 - Process Control System Testing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. National Institute of Standards and Technologies (NIST):
      a. 800-82, Guide to Industrial Control Systems (ICS) Security.
      b. 800-53, Security and Privacy Controls for Federal Information Systems and Organizations.
   2. International Society of Automation (ISA):
      a. ISA-62443, Cybersecurity Certificate Programs.

1.3 DEFINITIONS

A. HMI: Human Machine Interface.

B. LCD: Liquid Crystal Display.

C. OPC: “OLE for Process Control”, a software standard utilizing a client/server model that makes interoperability possible between automation/control applications and field systems/devices.

D. PC: Personal Computer.

E. RAID: Redundant Array of Independent Disks, a method of storing the same data in different places on multiple hard disks.

F. RAM: Random Access Memory.

G. SCSI: Short for Small Computer System Interface, a parallel interface standard used for attaching peripheral devices to computers.

H. SDRAM: Synchronous Dynamic RAM.

I. SNMP: Simple Network Management Protocol, a set of protocols for managing complex networks.

J. TFT: Thin-Film Transistor, a technology for building LCD screens.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See Section 40 61 13 - Process Control System General Requirements.
   2. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of the referenced standards.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

C. Cyber Security:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 81 33 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 LCD FLAT PANEL TYPE MONITORS

A. Manufacturers:
   1. Dell.
   2. Or Equal.

B. Provide LCD Flat Panel Type Monitors as shown in the Drawings.

C. Design Requirements:
   1. Type of display: Color TFT active-matrix LCD.
   2. Native Resolution:
      a. 27 IN: 2560 X 1440.
   3. Aspect ratio: 16:9 (widescreen).
   4. Image brightness: Minimum 300 CD/M^2.
   5. Display image contrast ratio: 800:1 or higher.
   6. Maximum sync rate (vertical scan rate x horizontal scan rate): At least 75 Hz x 80 KHz.
   7. Viewing angle: 170 DEG both vertical and horizontal.
   8. Adjustable tilt and swivel.
   9. Anti-glare flat screen.
   10. DisplayPort and HDMI video connections. Provide cabling compatible with connected hardware.
   11. Power input: 120 VAC.
   12. Monitor shall be placed on desk. RSPH HMI shall be panel mounted.

D. Configuration:

1. Do not connect USB between monitor and workstation/server.

2.3 HMI/OPERATOR COMPUTER WORKSTATIONS MINI PC

A. Manufacturers:
   1. Dell.
   2. Or Equal.

B. Provide HMI/Operator Computer Workstations as shown on the Drawings.

C. Design:
   1. Processor: Intel® i9, at least 8 Core.
   2. Processing speed: At least 3 GHz.
   3. Cache: At least 6 MB.
   4. RAM: At least 16 GB, sized to run all required applications plus 50%. Expandable to 64 GB.
   5. Hard Disk Drive:
      a. Capacity: At least 1000 GB, minimum 50% usable capacity after all software installed.
      b. Interface: SATA.
      c. Format: SSD.
   6. Chip: Quad-core.
   7. Provide each workstation with the latest edition of the following software:
      a. Operating system: Microsoft Windows™ 11 Pro, 64-bit for workstations or latest Windows™ version supported by Control System software.
      a. Antivirus: McAfee.
   8. DVD-RW drive.
      a. Read speed:
         1) 40x CD.
         2) 16x DVD.
   9. Graphics: Supports number of displays as indicated on Drawings.
   10. Audio: HD with internal speaker.
   11. Expansion Slots: Two.
   12. Communication ports as listed below:
      a. All communication ports as required by functional requirements of Contract Documents.
      b. Minimum four USB Ports 3.0.
      c. Ethernet 10/100/1000 MB/s.
   13. Power input: 120 VAC.
   14. Case style: Tower.
   15. Case color: Black.
   16. Keyboards:
      a. Incorporate Standard QWERTY design with numeric keypad and assigned function keys.
      b. Sculptured keys.
      c. Tactile feedback.
   17. Mouse: High performance mouse with laser sensor and tilt wheel (wireless).
   18. Accessories: Install port locks in all unused ports.

D. Configuration:
   1. Configure workstation to be part of the SCADA Domain.
   2. Install software in least privileged account required to perform functions per ISA-62443 & NIST 800-53.
   3. Remove all unused software and services.
   4. Remove default Administrator accounts.
   5. Configure inactivity session lock for non-operations accounts (maintenance, administrator, etc.).
   6. Disable operating system automatic updates.

2.4 NETWORK SERVERS

A. Manufacturers:
   1. Dell PowerEdge Series.
   2. Or Equal.

B. Provide Network servers as shown on the Drawings.

C. Network Servers – Dedicated Server:
   1. Design and fabrication:
      a. Rack mounted 1U or 2U form factor.
      b. Processor: Minimum 16 logical cores
      c. Processor Speed: Minimum 3.4 GHz
   2. SDRAM:
      a. Minimum 128 GB RDIMM.
      b. Expandable with a minimum of four expansion slots.
      c. Memory type: DDR4
      d. Utilize ECC (Error Correcting Code) memory.
   3. Hot-swappable redundant power supplies and cooling fans.
   4. Hard drives:
      a. RAID 10 configuration.
      b. Type: SSD
      c. Hot-swappable drives, minimum four disks
      d. Total memory capacity: Minimum 3TB for all servers (4 x 3TB SSD HDD).
   5. Power Input: 120 Vac, dual power supplies
   6. Provide each server with the latest edition of the following software:
      a. ESXi V8.03, factory installed by the server manufacturer
      b. Secondary Operating System of Operating system: Microsoft Windows™ Server 2019 or latest Windows™ version supported by Control System software. Secondary OS shall be supplied to support all cores of the CPU(s).
      c. Latest commercially available OS version is compatible with the HMI/SCADA software.
   7. Configuration:
      a. Configure servers to be part of the SCADA Domain.
      b. Install software in least privileged account required to perform functions per ISA-62443 and NIST 800-52.
      c. Remove all unused software and services.
      d. Configure any service accounts required for software function to utilize power user account.
      e. Apply application whitelisting.
      f. Configure audit to report to syslog server per NIST 800-52.
      g. Configure inactivity session lock for privileged accounts. Do not allow concurrent privilege account login.
      h. Disable operating system automatic updates.

2.5 EXTERNAL DATA STORAGE

A. Manufacturers:
   1. Buffalo.
   2. Or Equal.

B. Provide External Data Storage devices as shown on the Drawings.

C. Design Requirements:
   1. Network Attached Storage (NAS)
   2. Rack mounted 1U form factor.
   3. Processor Speed: Minimum 2.2 GHz.
   4. Utilize user friendly, menu driven means for operator to select:
      a. Either manual or automatic data capture and storage.

b. Operator selectable data to include as a minimum:
   1) Measured process variables.
   2) Alarm conditions.
   3) Status conditions monitored by the control system.
5. Hard drives:
   a. RAID 5 configuration.
   b. Total memory: Minimum 100TB.
6. Connectivity: Integrated Dual-Gigabit network adapters.
7. Provide backup/restore utility to prompt user prior to overwriting any existing files.
8. Data retrieval based on either time and date or tag.

D. Configuration:
1. Configure per Contracting Officer's required backup schedule including full backups and incremental data.
2. Password protect read/write access.

2.6 LAPTOPS

A. Manufacturers:
1. Dell
2. HP
3. Or Equal.

B. Design and fabrication:

C. Processor: Intel® Core™ Ultra 7 265U

D. OS: Windows 11 Pro

E. Screen: 16 Inch FHD

F. Memory: 32 GB

G. Hard Drive Size/ Type: 1 TB / SSD

H. Ports
1. USB: (2)Type A and (2) Type C
2. HDMI
3. RJ-45 Ethernet Port

I. Full keyboard with numeric keypad

J. Battery: 55 Wh

K. Chassis: Aluminum

2.7 SERVER RACKS:

A. Manufacturers:
1. Eaton
2. Or Equal.

B. Provide Server Racks as shown on the Drawings.

C. Design and fabrication:
1. Width: 19 IN.
2. Depth: Minimum 30 IN. Provide depth of rack that leaves at least 6 IN clear space behind equipment with rear cabling terminations.
3. Height: As shown on drawings.
4. Style: Open rack.
5. Material: Aluminum or steel with powder coat finish.
6. Leveling feet and rollers.
7. Sliding shelves.
8. Anti-tip stabilizer feet.

9. Rack mounted power strips and cable management.
    a. Two vertical power strips shall be provided at rear of panel. Power strip capacity shall be sized with minimum 25% spare plug space for future expansion.
    b. Cable management is required at bottom and rear of panel for routing of all cables which exit the enclosure.
    c. Cable management is required above and below all switches and patch panels.
10. Rackmount LCD Console KVIM Switches:
    a. 8 port switch
    b. Full-sized keyboard, touchpad mouse, and LCD display
    c. 1U rack.
    d. Power Supply: 120 Vac
11. SNMP capable door monitoring switch.
12. Temperature switch wired to local PLC and configured as alarm in HMI.

2.8 NETWORK TIMP PROTOCOL (NTP) SERVER

A. Manufacturers:
    1. Time Machines, Inc TM1000A
    2. Or Equal.

B. Design Requirements:
    1. Source of Time Served: US GPS Satellites
    2. GPS Lock: 2D
    3. NTP requests per second: >350
    4. NMEA Output: Yes
    5. Protocol: NTP
    6. Network Speed: 10/100 MBS
    7. Web Based Setup: Yes
    8. Accessories
        a. 5 VDC Power Supply
        b. Double Rack mount 1U plate for time servers
        c. Rack Mountable PoE NTP Clock display with 0.5-inch digits

2.9 TVS

A. Manufacturers:
    1. Samsung.
    2. Or Equal.

B. Provide TVs as shown on the Drawings.

C. Design Requirements:
    1. Display Technology: OLED / QLED / QNED / LED
    2. Screen Size: 65 inches
    3. Native Resolution: 4K Ultra HD (3840 x 2160)
    4. Aspect ratio: 16:9 (widescreen)
    5. Refresh Rate: 120Hz / 60Hz
    6. Ports: HDMI 2.1
    7. Power input: 120 VAC
    8. Stand: Wall-mount compatible (VESA 400 x 400)

2.10 CAMERAS

A. Manufacturers:
    1. Axis Communications
    2. Or Equal.

B. Design Requirements:
    1. Pan/Tilt/Zoom Adjustment Range:
        a. Pan/Tilt Angle: 360-degree rotation

b. Zoom Focus: 30x auto-focus zoom lens
2. Image Performance:
a. Minimum Resolution: 1920 x 1080
b. Minimum Frame Rate: 30 fps
c. Dynamic Range:
1) Minimum Wide Dynamic Range: 120 dB
3. Image Control:
a. Day/Night Control: Automatic and Manual
4. Electronic Shutter Speed: 1/1 to 1/10,000 second
5. Connection Type: Ethernet TCP/IP
6. Camera Housing Unit:
a. NEMA 4X
b. IP 66 rated
7. Environmental Operation Range:
a. Operating Temperature: -4°F to 120°F
b. Operating Humidity: 20-80 %
8. Power Source:
a. PoE
b. 11W max

C. Schedule:

LOCATION	DESCRIPTION	TYPE
RSPH	ELECTRICAL AREA	PAN-TILT-ZOOM
RSPH	PROCESS AREA	PAN-TILT-ZOOM
RSPH	RSPH ENTRANCE	PAN-TILT-ZOOM
NR WTP	PROCESS AREA	PAN-TILT-ZOOM
NR WTP	VESTIBLUE MAIN ENTRANCE	PAN-TILT-ZOOM

2.11 SOFTWARE

A. The Contracting Officer will contract separately for HMI software and SCADA programming services. The basis of design for the HMI of the SCADA system is AVEVA Edge (formally Indusoft Web Studio).

2.12 ACCESSORIES AND MAINTENANCE MATERIALS

A. Provide all accessories required to furnish a complete computer-based network for the control system to accomplish the requirements of the Drawings and Specifications.

B. Furnish Contracting Officer with the following extra materials:
1. One spare toner cartridge per laser printer provided.
2. One spare port lock key for each lock type provided.

PART 3 - EXECUTION

3.1 DEMONSTRATION

A. Demonstrate system in accordance with Section 01 75 00 - Checkout and Start-Up Procedures.

3.2 INSTALLATION AND CHECKOUT

A. Provide installation and checkout in accordance with Section 40 61 13 - Process Control System General Requirements and Section 40 61 21 - Process Control System Testing.

END OF SECTION

SECTION 40 63 43
PROGRAMMABLE LOGIC CONTROLLERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Programmable logic controller (PLC) control system(s), including hardware, documentation, and training.
   2. Programming is by Others – NOT IN CONTRACT.

B. Related Sections include but are not necessarily limited to:
   1. Section 01 75 00 - Checkout and Start-Up Procedures.
   2. Section 01 81 33 - Cyber Security Requirements.
   3. Section 26 05 19 - Wire and Cable - 600 Volt and Below.
   4. Section 40 61 13 - Process Control Systems General Requirements.
   5. Section 40 67 00 - Control System Equipment Panels and Racks.
   6. Section 40 61 96 - Process Control Descriptions.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. The Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. C37.90.2, Withstand Capability of Relay Systems to Radiated Electromagnetic Interference from Transceivers.
      b. C62.41.2, Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits.
   2. International Electrotechnical Commission (IEC)
      a. IEC 61131-3 Programmable logic controllers
   3. National Electrical Manufacturers Association (NEMA):
      a. ICS 1, Industrial Control and Systems General Requirements.

B. Qualifications:
   1. Installation supervisor shall have had experience in overseeing installation and startup of at least three similar installations within the last five years.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 61 13.
   2. Product technical data including:
      a. Results of factory testing or simulation procedures.
      b. Bill of materials.
      c. Drawings containing the following information to be submitted as part of Specification Section 40 67 00 (control panels) submittals:
         1) Arrangement drawings for PLC system components.
         2) Panel and enclosure plans, sections and details.
         3) Access opening locations and required clearances for each panel and enclosure.
         4) Enclosure internal wiring and terminal blocks.
      d. DIP Switch and Jumper Settings, if applicable.
      e. I/O Rack and I/O Module Layout.
      f. Wiring and Interconnection diagrams.
         1) Power Supplies.
         2) I/O Points.
         3) Communications.

g. Catalog cut sheets containing information on PLC components to be submitted as part of these Specification Section submittals.

3. Certifications:
   a. Qualifications of installation supervisor.
4. Software Licenses:
   a. Documentation of all software package licenses being assigned to Contracting Officer.
5. Manufacturer support proposal.

B. Operation and Maintenance Data:
   1. See Specification Section 01 78 23 for requirements for:
      a. The mechanics and administration of the submittal process.
      b. The content of Operation and Maintenance Manuals.
   2. Maintenance procedures.
      a. Include the location and phone numbers of service centers (including 24 HR "hot lines").
      b. Provide specific information including operation and maintenance requirements, troubleshooting guide, parts ordering, field service personnel requests, and service contracts.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. AutomationDirect.com H2-DM1E CPU and associated D2 power supplies, bases, modules, network cards, and ZIP-Link wiring.
   2. No substitutions allowed.

2.2 PERFORMANCE AND DESIGN REQUIREMENTS

A. See Specification Section 40 61 13.

B. The PLC system shall meet the control requirements of Section 40 61 96 - Process Control Descriptions, Drawings, and Specifications.

C. The PLC system be capable of operating in ambient conditions of 0 to 55 DEGC temperatures and 30 to 95% relative humidity without the need for purging or air conditioning.

D. Environmental Controls:
   1. Furnish circulation fans in solid state control system enclosures.
   2. Over-temperature switches shall be utilized to provide special cooling if required to maintain operating temperatures within the manufacturer's specified range.
   3. Air conditioning applications shall include means of preventing moisture condensation.

E. Where the PLC is utilized to control multiple trains of equipment and where the equipment in each train operates as a unit relatively independent of other equipment trains (e.g., facility with multiple boiler units or filter trains), the PLC components (I/O modules, power supplies, etc.) shall be designed so that the failure of any one component does not affect equipment on all trains. The following shall also apply.
   1. I/O modules shall be segregated on a train basis unless required for safety reasons.
   2. Where several equipment units operate in parallel, but are not considered assigned to a particular equipment train (e.g., multiple raw water pumps or chemical feed pumps all discharging into a common header), the PLC I/O modules associated with each equipment unit shall be assigned so that the failure of any one I/O module will not affect the other parallel unit/s.

2.3 HARDWARE

A. Processor shall include diagnostic indicators for power, mode, low battery, communications ports, and memory and I/O errors.

B. Redundant processors, if specified, shall have seamless fail-over capability

C. I/O modules shall be capable of being replaced while the PLC is on-line and operating.

D. All I/O modules shall have terminal block failure or removal monitoring.

E. Analog output modules shall have a resolution of at least 12 bits.

F. Electrical isolation shall be provided between logical and field device.

G. I/O Module field wiring shall remain undisturbed when removing or replacing an I/O module.

H. Fail-safe state for each I/O channel shall be capable of being defined as on, off or fail-in-place.

I. Power Supply Units:
   a. Electrical service to PLC system shall be 105 to 125 VAC, 60 Hz, ±10 PCT, 1 PH power.

J. All PLC control system components shall be capable of meeting or exceeding electromagnetic interference tests per IEEE C37.90.2.

K. Incorporate the following minimum safety measures:
   1. Watchdog function to monitor:
      a. Internal processor clock failure.
      b. Processor memory failure.
      c. Loss of communication between processor and I/O modules.
      d. Processor ceases to execute logic program.
   2. Safety function wiring: Emergency safety switches shall not be wired into the controller.
   3. Safe wiring:
      a. Unless otherwise specified, activation of alarms and stopping of equipment shall be based on de-energization rather than the energization of control circuits.
   4. Design PLC system with high noise immunity to prevent occurrence of false logic signals resulting from switching transients, relay and circuit breaker noise or conducted and radiated radio frequency interference.
   5. Operator intervention:
      a. Logic system failure shall not preclude proper operator intervention.
      b. Safety shutdown of equipment or a system shall require manual operator intervention before the equipment or system operation may be reestablished.

2.4 SOFTWARE

A. All software will be furnished and licensed to Contracting Officer.
   1. At Substantial Completion, transfer all licenses and media to Contracting Officer.
   2. At Substantial Completion, all software must be in the latest edition and licensed to the Contracting Officer.
   3. Software shall be lifetime licensed and be OEM supported for at least 10 years from date of license transfer.

B. Programming
   1. By Others – NOT IN CONTRACT.

C. Manufacturer support
   1. Submit a written proposal for a manufacturer support agreement for all products specified herein for a minimum of 12 months starting at final completion of the project. The cost of this manufacturer support agreement shall not be included in the Contract Price. The support agreement shall be executed in the name of, and for the benefit of, the Contracting Officer. At a minimum, this agreement shall provide the Contracting Officer with:
      a. 8 AM to 5 PM, 5 days per week manufacturer telephone support.

b. Access to the manufacturer's technical support website.
c. Software updates and patches.

2.5 COMPONENTS

A. PLC System Central Processor Unit (CPU):
1. The PLC shall have the capability of communicating with other commonly specified PLC and control system protocols such as Modbus/RTU, Modbus/TCP, Profibus, Ethernet/TCP and OPC.
2. CPU shall include capability to modify logic via an Ethernet port without taking processor offline.
3. Memory:
a. Battery-backed RAM.
b. Non-volatile program storage via flash EEPROM:
1) Automatically download to RAM in the event RAM is corrupted.
4. Memory battery backup shall be capable of 180 days memory retention with fresh battery.
a. Provide visual indication of battery status and alarm low battery voltage.
b. Memory battery backup shall be capable of 28 days memory retention after the "Battery Low" indicating LED is on.
5. Plug-in card designed to allow quick field replacement of faulty devices.
a. Provide unit designed for field replacement and expansion of memory without requiring rewiring or use of special tools.
6. 20% minimum spare useable memory capacity after all required programming is in place and operating.
7. Capable of executing all control functions required by the Specifications and Drawings.
8. Built-in three-mode (proportional-integral-derivative) control capabilities.
a. As directly selectable algorithms requiring no user knowledge of programming languages.
9. Lighted status indicators for "RUN".
10. The CPU shall be Host H2-DM1E supplied through AutomationDirect.com. No substitutions or equals to match GRCA Utilities standards.

B. PLC Base Units (Racks):
1. Provide Base units as shown on the drawings
2. Part numbers are as shown supplied through AutomationDirect.com. No substitutions or equals to match GRCA Utilities standards.
3. Base units
a. 4 slot 120 VAC Powered: D2-04B-1
9 slot 120 VAC Powered: D2-09B-1

C. Input/Output (I/O) Modules:
1. Provide plug-in modular-type I/O racks with cables to connect to all other required PLC system components.
2. Provide modules as shown on Drawings
3. Part numbers are as shown supplied through AutomationDirect.com. No substitutions or equals to match GRCA Utilities standards.
4. Provide I/O system with:
a. I/O solid state boards with status lights indicating I/O status.
b. Electric isolation between logic and field device.
c. Capability of withstanding low energy common mode transient to 1000 V without failure.
d. Incorporate noise suppression design.
e. Capable of meeting or exceeding electrical noise tests, NEMA ICS 1-109.60-109.66.
f. Capable of being removed and inserted into the I/O rack under power, without affecting any other I/O modules in the rack.
g. Install 20% spare I/O points for each type.
5. Input/output connection requirements:

a. Make connections to I/O subsystem by terminating all field wiring on terminal blocks within the I/O enclosure.
b. Prewire I/O modules to terminal blocks.
c. Provide terminal blocks with continuous marking strip.
d. Size terminals to accommodate all active data base points and spares.
e. Provide terminals for individual termination of each signal shield.
f. Field wiring shall not be disturbed when removing or replacing an I/O module.

6. Discrete I/O modules:
   a. Interface to ON/OFF devices.
   b. I/O status indicator on module front.
   c. Voltage rating to match circuit voltage.
   d. Output module current rating:
      1) Match maximum circuit current draw.
      2) Minimum 1.0 continuous A/point for 120 VAC applications.
   e. Isolated modules for applications where one module interfaces with devices utilizing different sources of power.
7. Discrete outputs shall be fused:
   a. Provide one fuse per common or per isolated output.
   b. Provide blown fuse indication.
   c. External fusing shall be provided if output module does not possess internal fusing.
   d. Fuses provided external to output model shall:
      1) Be in accordance with module manufacturer's specifications.
      2) Be installed at terminal block.
8. Discrete Input (DI) Modules:
   a. 24 VDC Powered, 32 Point – D2-32ND3 with Zip-link wiring module and terminal block.
   b. 120 VAC Powered, 16 Point – D2-16NA
9. Discrete Output (DO) Modules:
   a. 24 VDC Powered, 16 Point – D2-16TD2-2
      1) All DO modules shall be wired to an interposing DPDT pilot relay.
10. Discrete Input/Relay Output Modules:
    a. 24 VDC Powered, 4 Point DC In / 4 Point Relay Out – D2-08CDR
11. Analog I/O modules:
    a. Input modules to accept signals indicated on Drawings or Specifications.
    b. Minimum 12-bit resolution.
    c. I/O chassis supplied power for powering connected field devices.
    d. Differential inputs and outputs.
    e. User configurable for desired fault-response state.
    f. Provide output signals as indicated on Drawings and Specifications.
    g. Analog Input (AI) Modules
       1) 4-20mA Current, 8 Channels– F2-08AD-1
       2) RTD, 4 Channel – F2-04RTD
    h. Analog Output (AO) Modules – F2-08DA-1
12. Communications Modules:
    a. Provide communications modules as shown on Drawings.
    b. Provide communications modules for each protocol required for a complete communications system as shown on drawings.
    c. Multi-port Ethernet modules may not be utilized as a network switch or to bridge networks.
    d. Communications Modules:
       1) Ethernet RIO Module – H2-ERM100
       2) Ethernet Base Controller – H2-EBC100
       3) Ethernet Communications Module – H2-ECOM100

D. Power Supply Units:
   1. Provide regulated power units:

a. Designed to operate with PLC system and shall provide power to:
   1) All components of PLC system.
   2) All two-wire field instruments.
   3) Other devices as indicated on Drawings or Specifications.
b. Capable of supplying PLC system when all the specified spare capacity is utilized.
c. Each power supply shall be sized such that it will carry no more than 75% of capacity under normal loads.

2. Electrical service to PLC system is 105 to 125 V, 60 Hz, +1%, 1 PH power.
3. Separate AC circuit breakers shall be provided for each power supply.
4. If the PLC system is field expandable beyond the specified spare capacity, and if such expansion requires power supply modification, note such requirements in the submittals and allow room for power supply modification in the PLC system enclosure.
5. Capable of meeting or exceeding electrical noise tests, NEMA ICS 1-109.60-109.66.
6. Power distribution:
   a. Immune to transients and surges resultant from noisy environment.
   b. Shall provide constant voltage level DC distribution to all devices.
7. Provide uninterruptible power supply (UPS) to sustain full power to UPS powered loads listed below for a minimum of 10 minutes following loss of primary power and to ensure that the transient power surges and dips do not affect the operation of the PLC system.
   a. UPS powered loads:
      1) All rack mounted PLC components.
      2) Local operator consoles.
      3) All power supplies furnished with the PLC and associated loads.
   b. Input:
      1) 120 VAC +10%.
      2) 60 Hz.
      3) Line fuse protection.
   c. Output:
      1) 120 VAC 5%.
      2) 60 Hz.
      3) Short circuit protected.
      4) Instantaneous transfer time.
   d. IEEE C62.41 Class A voltage surges of 6000 V attenuated to less than 50 V on the output.
   e. Battery: Maintenance free lead acid.

E. PLC System Enclosure:
1. In accordance with Specification Section 40 67 00 - Control Panels.
2. Component placement:
   a. Mount all controller components within the enclosure to allow maximum convection cooling.
   b. Either install power supplies above all other equipment with manufacturers recommended clearance between the power supply and the enclosure top, or adjacent to other components, but with sufficient spacing for circulation of cooling air.
   c. Locate incoming devices (isolation or constant voltage transformers, local power disconnects, surge suppressors, etc.) to keep power wire running within an enclosure as short as possible.
   d. If items such as magnetic starters, contactors, relays, and other electromagnetic devices must be located within the same enclosure as the PLC system components, place a barrier with at least 6 IN of separation between the magnetic area and the control area.
   e. Place circulating fans close to major heat generating devices.
   f. Segregate input/output modules into groups of identical type.
3. Wiring and grounding to be in accordance with Specification Section 40 67 00.
4. Termination requirements:
   a. In accordance with Specification Section 40 67 00.

b. Make connections to I/O subsystem by terminating all field wiring on terminal blocks within the enclosure.
c. Prewire I/O modules to terminal blocks.
d. Size terminals to accommodate all active database points and spares.
e. Provide terminals for individual termination of each signal shield.
f. Field wiring shall not be disturbed when removing or replacing an I/O module.

2.6 ACCESSORIES

A. Provide all accessories required to install and test a complete PLC control system to accomplish the requirements of the Drawings and Specifications.

B. Provide all programming cables required to configure the PLC logic.

2.7 SOURCE QUALITY CONTROL

A. Perform Factory Acceptance Testing in accordance with Specification Section 40 61 21 - Process Control System Testing.

2.8 MAINTENANCE MATERIALS

A. Furnish Contracting Officer with the following extra spares:
1. One spare CPU for every four field CPUs or fraction thereof installed.
2. One spare communication module by type installed in the Plant.
3. One spare I/O card of each card type for every 10 cards or fraction thereof installed.
4. One spare power supply for every 10 power supplies or fraction thereof installed.
5. A set of spare fuses for all installed types.

B. Include a complete bill of materials itemizing each line item with make and model (part) number.

C. Include a complete set of all special tools required to install, maintain and test the PLCs.

PART 3 - EXECUTION

3.1 FUNCTIONALITY

A. Complete System.
1. Provide interface support with all foreign device interfaces (FDI) communicating with the PLC system.

3.2 INSTALLATION

A. Install PLC control system in accordance with manufacturer's written instructions.

3.3 FIELD QUALITY CONTROL

A. Test system in accordance with Specification Section 40 61 21 - Process Control System Testing.

B. Demonstrate system in accordance with Specification Section 01 75 00 - Checkout and Start-Up Procedures.

3.4 TRAINING

A. Employee of the manufacturer or certified representative to provide (2) sessions of 8 hours of operation and maintenance training at the Project site after the system has successfully undergone all field testing and acceptance procedures.

B. As a minimum, training shall cover:
1. Hardware overview.
2. Documentation.
3. Maintenance.
4. Trouble shooting.

5. Operation, e.g., changing set points, passwords, etc.

3.5 DOCUMENTATION

A. Update O&M manuals to reflect as-built conditions.

END OF SECTION

SECTION 40 66 05
NETWORK EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Computer network requirements, which include, but are not necessarily limited to:
      a. Ethernet Switches.
      b. Routers.

B. Related Sections include but are not necessarily limited to:
   1. Section 40 61 13 - Process Control Systems General Requirements.
   2. Section 40 61 21 - Process Control System Testing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. 802.3, Information Technology - Local and Metropolitan Area Networks - Part 3: Carrier Sense Multiple Access with Collision Detection (CSMA/CD) Access Method and Physical Layer Specifications.
         1) 802.3u: IEEE Standards for Local and Metropolitan Area Networks: Supplement to Carrier Sense Multiple Access with Collision Detection (CSMA/CD) Access Method and Physical Layer Specifications Media Access Control (MAC) Parameters, Physical Layer, Medium Attachment Units, and Repeater for 100 Mb/s Operation, Type 100BASE-T.
         2) 802.3x: IEEE Standards for Local and Metropolitan Area Networks: Specification for 802.3 Full Duplex Operation.
   2. National Institute of Standards and Technologies (NIST):
      a. 800-82, Guide to Industrial Control Systems (ICS) Security.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 61 13 - Process Control System General Requirements.

B. Product Data:
   1. Acknowledgement that products submitted meet requirements of the referenced standards.
   2. Network requirements verification plan.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 ETHERNET SWITCHES

A. Accessories: “Stack Module” as required to create logical switches and reduce network latency.

B. Provide Ethernet switches as shown on the Drawings.

C. Layer 2, 3 – Rack Mounted Ethernet Switches (Managed):
   1. Manufacturers:
      a. Ubiquity EdgeSwitch
      b. Or Equal.
   2. Design and fabrication:
      a. Support Ethernet 100/1000 Mbps.
      b. Backbone ports for connection with SFP.
      c. 16 100/1000 Mbps twisted pair ports (RJ45) as required for communication with devices as depicted in the Contract Documents. 16 port minimum.
         1) Unless otherwise noted, provide at least two spare 100/1000 Mbps port (twisted pair) at each Ethernet switch.
      d. Support SNMP v3 and Web based management.
      e. Rapid Spanning Tree Protocol.
      f. IGMP (Internet Group Management Protocol) supports IP multicast filtering to enable switches to automatically route messages only to appropriate ports.
      g. Check all received data for validity.
         1) Discard invalid and defective frames or fragments.
      h. Monitor connected TP/TX line segments for short-circuit or interrupt using regular link test pulses in accordance with IEEE 802.3.
      i. Monitor attached fiber optic lines for open circuit conditions in accordance with IEEE 802.3.
      j. As applicable, meet requirements of IEEE 802.3.
      k. One rack unit (RU) form factor.
      l. Power switch with 120 VAC or 24 VDC power input per Drawings.
      m. Provide dual redundant power supplies.
      n. Provide LED status lights to indicate:
         1) Power: Supply voltage present.
         2) Fault.
         3) Port status.
      o. Environmental rating:
         1) Operating temperature: 32 DEGF to 122 DEGF.
         2) Humidity: 95% relative humidity, non-condensing.
   3. Function in self-healing ring structure.
      a. If one section in the ring fails, the ring structure changes to a line structure within 0.5 seconds.
   4. 19 IN Rack mount.

D. Layer 2, 3 – Panel Mounted Ethernet Switches (Managed):
   1. Manufacturers:
      a. Cisco IE3500.
      b. Or Equal.
   2. Design and fabrication:
      a. Support Ethernet 10/100 Mbps.
      b. Two backbone ports for connection to single mode fiber via type LC connectors (ST not allowed).
      c. 16 10/100 Mbps twisted pair ports (RJ45) as required for communication with devices as depicted in the Contract Documents. 16 port minimum.
         1) Unless otherwise noted, provide at least two spare 10/100 MBit/s port (twisted pair) at each Ethernet switch.

d. Support SNMP v3 and Web based management.
e. Rapid Spanning Tree Protocol.
f. IGMP (Internet Group Management Protocol) supports IP multicast filtering to enable switches to automatically route messages only to appropriate ports.
g. Check all received data for validity.
    1) Discard invalid and defective frames or fragments.
h. Monitor connected TP/TX line segments for short-circuit or interrupt using regular link test pulses in accordance with IEEE 802.3.
i. Monitor attached fiber optic lines for open circuit conditions in accordance with IEEE 802.3.
j. As applicable, meet requirements of IEEE 802.3.
k. One rack unit (RU) form factor.
l. Power switch with 24 VDC power input.
m. Provide dual redundant power supplies.
n. Provide LED status lights to indicate:
    1) Power: Supply voltage present.
    2) Fault.
    3) Port status.
o. Environmental rating:
    1) Operating temperature: 32 DEGF to 122 DEGF.
    2) Humidity: 95% relative humidity, non-condensing.

3. Function in self-healing ring structure.
    a. If one section in the ring fails, the ring structure changes to a line structure within 0.5 seconds.
4. DIN Rail mounted.

E. Layer 2 – Panel Mounted Ethernet Switches (Unmanaged):
1. Manufacturers:
    a. Stride (Automation Direct)
    b. Phoenix Contact
    c. Or Equal.
2. Design and fabrication:
    a. Support Ethernet 10/100/1000 Mbps.
    b. 4 X Gigabit Ethernet backbone uplink module for connection to single mode fiber via type LC connectors (ST not allowed).
    c. Backbone (trunk) fiber ports shall be via SFP modules.
    d. 24 10/100/1000 Mbps twisted pair ports as required for communication with other devices as depicted in the Contract Documents.
        1) Unless otherwise noted, provide at least two spare 10/100/1000 MBit/s port (twisted pair) at each Ethernet switch.
    e. Support IEEE 802.3bz to go beyond 1 Gbps with Cat5e or Cat6 cables.
    f. Rapid Spanning Tree Protocol.
    g. IGMP (Internet Group Management Protocol) supports IP multicast filtering to enable switches to automatically route messages only to appropriate ports.
    h. Check all received data for validity.
        1) Discard invalid and defective frames or fragments.
    i. Monitor connected TP/TX line segments for short-circuit or interrupt using regular link test pulses in accordance with IEEE 802.3.
    j. Monitor attached fiber optic lines for open circuit conditions in accordance with IEEE 802.3.
    k. As applicable, meet requirements of IEEE 802.3.
    l. One rack unit (RU) form factor.
    m. At least 480 Gbps stacking bandwidth.
    n. Power switch with 24 VDC power input.
    o. Provide dual redundant power supplies.
    p. Provide LED status lights to indicate:

1) Power: Supply voltage present.
2) Fault.
3) Port status.

q. Environmental rating:
1) Operating temperature: 32 DEGF to 122 DEGF.
2) Humidity: 95% relative humidity, non-condensing.

3. Function in self-healing ring structure.
a. If one section in the ring fails, the ring structure changes to a line structure within 0.5 seconds.
4. 19 IN Rack Mount
5. Minimum 16 port

F. Small Form-Factor Pluggable (SFP) Modules:
1. Provide SFPs as required to support the communication capability shown on the Drawings. Use SFPs instead of media converters. Media converter use is prohibited unless approved in writing.
2. Hardware Features:
a. At least two SFP ports per switch supplied with 1 Gbps fiber transceivers for connection to single mode fiber via type LC connectors. Quantity and type as required for communication with devices as depicted in the Contract Documents.
b. Diagnostic data for the associated media type is available through the command line interface of the switch.
c. 10/100/1000 MBit/s twisted pair ports (RJ45) as required for communication with devices as depicted in the Contract Documents.
1) Unless otherwise noted, provide at least two spare 10/100/1000 MBit/s port (twisted pair) at each Ethernet switch.
d. Check all received data for validity.
1) Discard invalid and defective frames or fragments.
e. Monitor connected TP/TX line segments for short-circuit or interrupt using regular link test pulses in accordance with IEEE 802.3.
f. Monitor attached fiber optic lines for open circuit conditions in accordance with IEEE 802.3.
g. As applicable, meet requirements of IEEE 802.3.
h. Power switch with 24 VDC power input.
i. Provide LED status lights to indicate:
1) Power: Supply voltage present.
2) Fault.
3) Port status.
j. Environmental rating:
1) Operating temperature: 32 DEGF to 122 DEGF.
2) Humidity: 95% relative humidity, non-condensing.
k. Accessories: Provide port locks for all unused ports.

G. Configuration:
1. All managed switch unused ports shall be administratively disabled. Sticky MAC or similar port configuration shall be utilized to bind ports to MAC address of connected device per NIST 800-82.
2. Layer 3 switches shall be configured using static routes per NIST 800-82 best practices. Routes between VLANS shall be restricted except where required to achieve contract required functionality.
3. Change default usernames and passwords.

Volume 5 – NR RSPH:

TAG	SERVICE	MANAGED OR UNMANAGED, MOUNTING TYPE

RSPH MSW	RSPH PLC CP	MANAGED, PANEL
RS CAVE MSW	RS CAVE CP (EXISTING)	MANAGED, PANEL
RSPH SW	RSPH PLC CP	UNMANAGED, PANEL
LCP MSW	LCP-53117	MANAGED, PANEL

Volume 6 – NR WTP:

TAG	SERVICE	MANAGED OR UNMANAGED , MOUNTING TYPE
NR WTP SW1	NR WTP PLC CP	UNMANAGED, PANEL
NR WTP SW2	NR WTP CHEMICAL RIO CP	UNMANAGED, PANEL
NR WTP SW3	NPS TELECOM CABINET RW TANK (EXISTING)	UNMANAGED, RACK
NR WTP SW4	NR WTP MCC	UNMANAGED, PANEL
NR WTP MSW1	NR WTP NW RACK	MANAGED, RACK
NR WTP MSW2	NR WTP PLC CP	MANAGED, PANEL

2.3 ROUTERS

A. Manufacturers:
   1. Ubiquity Edge Routers
   2. Or Equal.

B. Provide routers as shown in the Drawings.

C. Router Design Requirements:
   1. Support for multiple, diverse access links: T1/E1, T3/E3, xDSL, Gigabit and 10-Gigabit Ethernet
   2. Modular interfaces with online removal and insertion (OIR) for module upgrades without network disruption.
   3. Provide security protection features. Including packet filtering, stateful firewall, AAA, IPSec VPN, and Auto-discovery VPN.
   4. GUI based device management capability for configuring routing, firewall, VPN, WAN and LAN.
   5. Built-in firewall. Systems Integrator shall configure and enable firewall in the connection to the outside world to provide controlled access policies for the sharing of data between networks.

D. Configuration:
   1. Configure to restrict all routes not required by contract.
   2. Configure dynamic routing.
   3. Change default usernames and passwords.

Volume 5 – NR RSPH:

TAG	SERVICE
RSPH RTR1	RSPH PLC CP
RSPH RTR2	RSPH PLC CP

Volume 6 – NR WTP:

TAG	SERVICE
NR WTP RTR1	NR WTP NW RACK
NR WTP RTR2	NR WTP NW RACK

2.4 SOURCE QUALITY CONTROL

A. Provide Factory testing as required in section 40 61 21 - Process Control System Testing.

2.5 ACCESSORIES AND MAINTENANCE MATERIALS

A. Provide all accessories required to furnish a complete computer-based network for the control system to accomplish the requirements of the Drawings and Specifications.

B. Furnish Contracting Officer with the following extra materials:
   1. One spare Ethernet switch of each type utilized.

PART 3 - EXECUTION

3.1 DEMONSTRATION

A. Demonstrate system in accordance with Section 01 75 00 - Checkout and Start-Up Procedures.

3.2 FIELD QUALITY CONTROL

A. Verify network is configured per requirements herein.

B. Submit verification plan.

3.3 INSTALLATION AND CHECKOUT

A. Provide installation and checkout in accordance with Section 40 61 13 - Process Control System General Requirements and Section 40 61 21 - Process Control System Testing.

END OF SECTION

SECTION 40 66 40
TELEMETRY SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Items required include but are not necessarily limited to the following:
      a. Radios.
      b. Antennas
      c. Transmission Cable and Miscellaneous.

B. Related Sections include but are not necessarily limited to:
   1. Section 10 14 00 - Identification Devices.
   2. Section 33 81 19 – Fold Over Antenna Tower Assembly
   3. Section 40 61 13 - Process Control Systems General Provisions.
   4. Section 40 61 96 - Control Loop Descriptions.
   5. Section 40 67 00 - Control Systems Equipment Panels and Racks.
   6. Section 40 61 43 - Surge Protection Devices for Instrumentation and Control Equipment.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Federal Communications Commission (FCC):
      a. 47 CFR 15 - Radio Frequency Devices, Section 15.247 - Operation within the bands 902 - 928 MHz, 2400 - 2483.5 MHz, and 5725 - 5850 MHz
   2. International Electrotechnical Commission (IEC):
      a. 61131-3 - Programmable Controllers, Part 3: Programming Languages.
   3. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
      a. 802.3, Information Processing Systems - Local Area Networks - Part 3: Carrier Sense Multiple Access with Collision Detection (CSMA/CD) Access Method and Physical Layer Specifications.
   4. Telecommunications Industry Association/Electronic Industries Alliance/American National Standards Institute (TIA/EIA/ANSI):
      a. 222-H, Structural Standards for Steel Antenna Towers and Antenna Supporting Structures.
   5. Underwriters Laboratories, Inc. (UL):
      a. 508A, Standard for Safety Industrial Control Panels.

1.3 DEFINITIONS

A. ARQ (Automatic Resend Query): Method of error correction where the receiver initiates an order to retransmit data blocks that are determined to be corrupted in transmission.

B. BER (Bit Error Rate):
   1. Ratio of the number of bits received in error to the total number of bits transmitted.
   2. Used as a measure of quality for data links.

C. CRC (Cyclic Redundancy Check):
   1. A method utilized for detecting errors in data transmission or storage.
   2. A number is calculated on the originating end based on the message contents and then appended to the message before transmission.
   3. The CRC is re-calculated on the receiving end and compared to the previously calculated value.
   4. A match of the two calculated values indicates a high probability that the message was received correctly.

D. Omnidirectional Antenna: An antenna that radiates maximum power uniformly 360 DEG in the horizontal plane.

E. Spread Spectrum:
   1. Communications techniques whereby a radio frequency signal is spread (modulated) to generate an expanded bandwidth signal.
      a. Frequency Hopping Spread Spectrum: Communications technique where the frequency of the radio signal repeatedly “hops” from one Frequency to another in accordance with a random but predictable sequence.
      b. Direct Sequence Spread Spectrum: Communications technique whereby the stream of information (data signal) is combined with a higher data-rate bit sequence, or chipping code, that divides the user data according to a spreading ratio.

F. Yagi Antenna:
   1. A unidirectional radio antenna, consisting of a driven element, a reflector, and one or more directors.
   2. The antenna is primarily used for frequencies above 10 MHz and are used only in point-to-point applications.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 61 13.
   2. Product technical data:
      a. Panel and enclosure plans, sections and details in accordance with Specification Section 40 67 00.
      b. Enclosure internal wiring and terminal blocks in accordance with Specification Section 40 67 00.
      c. Manufacturer's installation instructions.
   3. Submit all catalogs cut sheets under Specification Section 40 66 40 submittals.
   4. Submit all panel fabrication drawings and associated wiring diagrams under Specification Section 40 67 00 submittals.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

C. Informational Submittals:
   1. Clear listing of all passwords required to edit, modify, or view programming and configuration.
   2. List of all recommended spares for maintenance purposes with each item separately priced.
      a. List shall include all special tools and test equipment necessary for the maintenance of the complete system.

D. Field Quality Control.
   1. Items listed under Part 3, 3.2 of this specification section.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 450 MHZ RADIOS

A. Manufacturers:
   1. GE-MDS SD4.
   2. No substitutions allowed.

B. Design and Fabrication:
   1. Communication:
      a. Type:
         1) Ethernet communication in accordance with IEEE 802.3.
      b. Ports:
         1) Ethernet: 100 Base-Tx Ethernet.
      c. Data throughput: 42 Kbps.
   2. FCC licensed.
   3. Frequency: 450-470 MHz
   4. Encryption: 128-bit.
   5. Diagnostics:
      a. Provide capability for remote diagnostics monitoring.
         1) Provide licensed copies to the Contracting Officer of any software required for remote diagnostics.
   6. Configuration:
      a. Provide remote configuration capability.
         1) Provide licensed copies to the Contracting Officer of any software required for remote configuration.
   7. Receiver sensitivity:
      a. -95 dBm at 512 Kbps with 0.000001 (10^{-6}) BER.
      b. -98 dBm at 230 Kbps with 0.000001 (10^{-6}) BER.
      c. -105 dBm at 115 Kbps with 0.000001 (10^{-6}) BER.
   8. Error detection/correction: 16-bit CRC with ARQ (Automatic Re-send Query).
   9. Temperature: -40 to +158 DEGF.
   10. Humidity: 90% at 104 DEGF noncondensing.
   11. Input power: 10.5-24 VDC.
   12. Case: Aluminum.
   13. Agency approvals: FCC 47 CFR 15.247.

2.3 450 MHZ ANTENNAS

A. Manufacturers:
   1. Telewave.IO ANT450F2
   2. Or Equal

B. Design and Fabrication:
   1. Frequency Range (Continuous) 406-480 MHz
   2. Gain 0 dB
   3. Power Rating (RMS Continuous) 500 watts / 57 dBm
   4. Impedance: 50 Ohms
   5. Connector: Recessed Type N - Female
   6. Pattern: Omnidirectional
   7. VSWR: 1.5:1 or less
   8. Vertical beamwidth: 80° vertical/ 360° horizontal
   9. Radome Dimensions Inches (H" x D") 42 x 2.75
   10. Total Weight 9 lbs.
   11. Temperature Range (Operating or Storage) -58° to +140° -50° to +60°
   12. Wind rating / with 0.5” ice (MPH) 150/ 225
   13. Maximum exposed area (sq ft) 1.0 ft.²

14. Lateral Thrust experienced by the antenna 100 (lb.) 38
15. Bending Moment transfer to top clamp 100 MPH (lb.-ft) 30 ft.lb

C. Options
1. Mounting Hardware: Clamp Kit for Radome Antennas
2. Surge Suppressor: Type N bulkhead
3. Provide antenna surge suppressor and connect to coaxial cable in cabinet housing radio, per manufacturer recommendations.
4. Materials:
   a. Antenna:
      1) Aluminum or aluminum with Teflon impregnated powder coat finish.
   b. Mounting hardware:
      1) 316 stainless steel or aluminum.
5. Provide all supports, lightning suppressors, and other apparatuses required to make a complete and operable radio telemetry system.

2.4 FIXED WIRELESS BACKHAUL RADIOS

A. Manufacturers:
1. Cambium Network PTP-670
2. No substitutions allowed.

B. Design and Fabrication:
1. RF Bands Wide-band operation: 4.9 to 6.05 GHz
2. Channel sizes: 5, 10, 15, 20, 30, 40, and 45 MHz channels
3. Spectral Efficiency: 10 bps/Hz maximum
4. Channel Selection: By Dynamic Spectrum Optimization or manual intervention Automatic selection on start-up and continual self-optimization to avoid interference
5. Maximum Transmit Power: Up to 27 dBm
6. System Gain: Up to 164 dB with Integrated antenna
7. Modulation / Error Correction: Fast Preemptive Adaptive Modulation featuring 13 modulation / FEC coding levels ranging from BPSK to 256 QAM dual payload MIMO
8. Duplex Scheme: Time Division Duplex (TDD) Adaptive or fixed transmit/receive duty cycles Split frequency operation allows separate to transmit and receive frequencies are allowed by regulation. Optional TDD synchronization using PTP-SYNC Module
9. Connectorized Antenna:
   a. Operate with a selection of separately purchased single and dual polarity antennas through 2 x N-type female connectors
10. Range: Up to 155 miles (250 km)
11. Security: FIPS-197 compliant 128/256-bit AES Encryption (optional) HTTPS and SNMPv3 identity-based user accounts Configurable password rules
12. Event logging and management: logging via syslog Disaster recovery and vulnerability management

2.5 FIXED WIRELESS BACKHAUL ANTENNAS

A. Manufacturers:
1. Cambium Network Model RDH4509C
2. Or Equal

B. Design and Fabrication:
1. Type High-Performance Parabolic Reflector
2. Antenna Diameter, nominal .6 m | 2 ft
3. Polarization Dual
4. Standard RF Connector Type 2 x N-Female
5. Size: 2 Foot
6. Frequency Range: 5.25–5.85 GHz

C. Electrical Specifications

1. Operating Frequency Band 5.25 – 5.85 GHz
2. Half Power Beamwidth, Horizontal 6.1º
3. Half Power Beamwidth, Vertical 6.1º
4. Cross-Polar Discrimination 28 dB
5. Front to Back Ratio (F/B) 44 dB Gain,
6. Low Frequency 28.3 dBi Gain,
7. Mid Frequency 28.8 dBi Gain,
8. High Frequency 29.3 dBi
9. Return Loss 14 (Minimum)
10. VSWR 1.5 (Maximum

D. Mechanical Specifications
1. Fine Azimuth Adjustment +/- 10º
2. Fine Elevation Adjustment +/- 30º
3. Mounting Pipe Diameter min/max 2 in (5.08 cm) /1.54.5 in (11.4 cm)
4. Net Weight 27 lbs. (12.2 kg)
5. Wind Velocity, Operational 90 mph (145 km/h)
6. Wind Velocity, Survival Rating 125 mph (200 km/h)
7. Operating Temperature Range -40°C to 60°C
8. Storage Temperature Range -50°C to 70°C

2.6 TRANSMISSION CABLE AND MISCELLANEOUS

A. Cable Manufacturers:
1. Andrew Corp. LDF Series.
2. Or Equal.

B. Provide low-loss foam-dielectric type cable connecting radio antenna port(s) with the antenna.
1. Cable: 1/2 IN or larger diameter as required to prevent signal losses in cable from degrading performance.
2. Weatherproof, suitable for direct environmental exposure.
a. Use "O" ring seals on connections.

C. Provide "super flexible" transmission cable (length as required) at the radio antenna port.
1. Provide standard Type N connectors for connection to a continuous piece of cable extending to the antenna.

D. Provide grounding connections to the antenna mast and cable in accordance with manufacturer’s recommendations.

2.7 COAXIAL RF SURGE PROTECTOR

A. Manufactures:
1. Polyphaser (GTH-NFF-AL)
2. Andrew
3. L-Com
4. Or Equal

B. Design and Fabrication:
1. Design: Gas Discharge Tube type
2. Frequency Range: DC to 6GHZ
3. Max Power 500W
4. DC Operating Voltage: 60 V Max
5. Impedance: 50 Ohm
6. Connectors (In/Out): N Female
7. Directional: Bidirectional
8. Throughput Energy: 2500uJ
9. Mounting: Wall or Bulkhead
10. Enviromental:IP67

2.8 ACCESSORIES

A. Provide all accessories required to furnish a complete telemetry system to accomplish the requirements of the Drawings and Specifications.

B. Security Controls:
   1. Provide door switches per Drawings monitored by discrete input utilizing normally closed contacts.

2.9 MAINTENANCE MATERIALS

A. Furnish Contracting Officer with the following extra materials:
   1. One spare power supply of each type for every 10 power supplies or fraction thereof installed.
   2. One spare radio of each type installed.
   3. One spare surge arrestor.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install telemetry system as shown on Drawings in accordance with manufacturer's written instruction.

3.2 FIELD QUALITY CONTROL

A. Conduct startup of equipment and perform operational checks.

B. Maintain and submit an accurate daily or weekly log of all commissioning and startup functions.
   1. All commissioning/startup functions may be witnessed by the Contracting Officer.
   2. All reports shall be signed and dated by the Contractor.

C. Provide Contracting Officer with a written statement that equipment has been installed properly, started up, and is ready for operation by Contracting Officer’s personnel.

3.3 DEMONSTRATION

A. Demonstrate system in accordance with Specification Section 01 75 00 and 40 61 21.

3.4 TRAINING

A. On-site Training:
   1. Provide certified manufacturer’s representative to provide 4 HRS of operation and maintenance training at the Project site after the system has successfully undergone all field testing and acceptance procedures.
      a. As a minimum, training shall cover:
         1) Hardware overview.
         2) Maintenance.
         3) Troubleshooting.
         4) Operation, e.g., changing set points, passwords, etc.

END OF SECTION

SECTION 40 67 00
CONTROL SYSTEM EQUIPMENT PANELS AND RACKS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Requirements for control panels and enclosures utilized as follows:
      a. Unless noted otherwise, all control panels and enclosures housing control components that are specified in Section 40 66 40, Section 40 71 00, Section 40 72 00, Section 40 73 00, Section 40 63 43, Section 40 78 00, or Section 40 61 43.
      b. This Section is only applicable to panels furnished with equipment specified in other specification divisions when so stated in the applicable equipment specification section.

B. Related Sections include but are not necessarily limited to:
   1. Section 10 14 00 - Identification Devices.
   2. Division 11 - Equipment.
   3. Division 26 - Electrical.
   4. Section 40 66 40 - Telemetry Systems.
   5. Section 40 61 13 - Process Controls Systems General Requirements.
   6. Section 40 61 21 - Process Control System Testing.
   7. Section 40 61 43 - Surge Protection Devices for Instrumentation and Control Equipment.
   8. Section 40 63 43 - Programmable Logic Controllers.
   9. Section 40 71 00 - Flow Measurement.
   10. Section 40 72 00 - Level Measurement.
   11. Section 40 73 00 - Pressure, Strain, and Force Measurement.
   12. Section 40 74 00 - Temperature Measurement.
   13. Section 40 75 00 - Process Liquid Analytical Measurement.
   14. Section 40 78 00 - Panel Mounted Instruments.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI).
   2. ASTM International (ASTM):
      a. B75, Standard Specification for Seamless Copper Tube.
   3. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. ICS 4, Industrial Control and Systems: Terminal Blocks.
   4. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC):
         1) Article 409, Industrial Control Panels.
         2) Article 504, Intrinsically Safe Systems.
      b. 79, Electrical Standard for Industrial Machinery
   5. Underwriters Laboratories, Inc. (UL):
      a. 508A, Standard for Safety Industrial Control Panels.
      b. 698A, Standard for Industrial Control Panels Relating to Hazardous (Classified) Locations.
      c. 913, Standard for Safety Intrinsically Safe Apparatus and Associated Apparatus for Use in Class I, II, and III, Division 1, Hazardous (Classified) Locations.

B. Qualifications:
   1. See Section 40 61 13 - Process Control Systems General Requirements.

C. Miscellaneous:
   1. Approved supplier of Industrial Control Panels under provisions of UL 508A or UL 698A.

a. Entire assembly shall be affixed with a UL 508A or UL 698A label "Listed Enclosed Industrial Control Panel" prior to shipment to the jobsite.
b. Control panel(s) without an affixed UL 508A or UL 698A label shall be rejected and sent back to the Contractor's factory.

1.3 DEFINITIONS

A. Panel: Control panels or enclosures listed in the schedule included in this Specification Section.

B. Foreign Voltages: Voltages that may be present in circuits when the panel's main power is disconnected.

C. Intrinsically Safe:
1. A device, instrument or component that will not produce sparks or thermal effects under normal or abnormal conditions that will ignite a specified gas mixture.
2. Designed such that electrical and thermal energy limits inherently are at levels incapable of causing ignition.

D. Intrinsically Safe Circuit: A circuit in which any spark or thermal effect is incapable of causing ignition of a mixture of flammable or combustible material in air under test conditions as prescribed in UL 913.

E. Cable: Multi-conductor, insulated, with outer sheath containing either building wire or instrumentation wire.

F. Instrumentation Cable:
1. Multiple conductors, insulated, twisted or untwisted, with outer sheath.
2. Instrumentation cable is typically either TSP (twisted-shielded pair) or TST (twisted-shielded triad) and is used for the transmission of low current or low voltage signals.

G. Ground Fault Circuit Interrupter (GFCI): A type of device (e.g., circuit breaker or receptacle) which detects an abnormal current flow to ground and opens the circuit preventing a hazardous situation.

H. Programmable Logic Controller (PLC): A specialized industrial computer using programmed, custom instructions to provide automated monitoring and control functions by interfacing software control strategies to input/output devices.

I. Input/Output (I/O): Hardware for the moving of control signals into and/or out of a PLC or RTU.

J. Supervisory Control and Data Acquisition (SCADA): Used in process control applications, where programmable logic controllers (PLCs) perform control functions but are monitored and supervised by computer workstations.

K. Highway Addressable Remote Transducer (HART): A bi-directional communication protocol that provides data access between intelligent field instruments and host systems.

L. Digital Signal Cable: Used for the transmission of digital communication signals between computers, PLCs, RTUs, etc.

M. Uninterruptible Power Supply (UPS): A backup power unit that provides continuous power when the normal power supply is interrupted.

N. Loop Calibrator: Portable testing and measurement tool capable of accurately generating and measuring 4-20ma DC analog signals.

1.4 SUBMITTALS

A. Shop Drawings:
1. See Section 40 61 13.
2. Prepared with computer aided design (CAD) software.
3. Printed on 11 by 17 IN sheets.
4. Drawings shall include a title block containing the following:
a. Plant or facility name where panel(s) are to be installed.

b. Drawing title.
c. Drawing number.
d. Revision list with revision number and date
e. Drawing date.
f. Drawing scale.
g. Manufacturer name, address, and telephone number.

5. Cover sheet for each drawing set shall indicate the following:
   a. Plant or facility name.
   b. Project name.
   c. Submittal description.
   d. Revision number.
   e. Issue date.
6. Table of contents sheet(s) shall indicate the following for each drawing in the set:
   a. Drawing number.
   b. Drawing title.
   c. Sheet number.
7. Legend and abbreviation sheet shall indicate the following:
   a. Description of symbols and abbreviations used.
   b. Panel construction notes including enclosure NEMA rating, finish type and color, wire type, wire color strategy, conductor sizes, and wire labeling strategy.
   c. Confirmation that the panel(s) are to be affixed with a UL 508A or UL 698A label prior to shipment from the factory.
8. Bill of Materials for each panel shall include the following component information:
   a. Instrument tag number.
   b. Quantity.
   c. Functional name or description.
   d. Manufacturer.
   e. Complete model number.
   f. Size or rating.
9. Panel exterior layout drawings to scale and shall indicate the following:
   a. Panel materials of construction, dimensions, and total assembled weight.
   b. Panel access openings.
   c. Conduit access locations.
   d. Front panel device layout.
   e. Nameplate schedule:
      1) Nameplate location.
      2) Legend which indicates text, letter height and color, and background color.
      3) Short Circuit Current Rating (SCCR) marking per NFPA 70 or statement of exception. Include any required calculations.
   f. Alarm annunciator window engraving schedule.
   g. Layouts of graphic panels or mosaic displays.
10. Panel interior layout drawings shall be drawn to scale and shall indicate the following:
    a. Sub-panel or mounting pan dimensions.
    b. Interior device layouts.
    c. PLC/RTU general arrangement layouts.
    d. Wire-way locations, purpose, fill calculations, and dimensions.
    e. Terminal strip designations.
    f. Location of external wiring and/or piping connections.
    g. Location of lighting fixtures, switches and receptacles.
11. Wiring diagrams shall consist of the following:
    a. Panel power distribution diagrams.
    b. Control and instrumentation wiring diagrams.
    c. PLC/RTU I/O information:
       1) Model number of I/O module.
       2) Description of I/O module type and function.

3) Rack and slot number.
4) Terminal number on module.
5) Point or channel number.
6) Programmed point addresses.
7) Signal function and type.

d. Internal network connections diagram
e. Wiring diagrams shall identify each wire as it is to be labeled.

B. Manufacturer catalog cut sheets for enclosure, finish, panel devices, control auxiliaries, and accessories.

C. Electrical load calculations for each panel:
1. Total connected load.
2. Peak electrical demand for each panel.

D. Climate control calculations for each panel.
1. Verify that sufficient dissipation and/or generation of heat is provided to maintain interior panel temperatures within the rated operating temperatures of panel components.

E. Unwitnessed Factory Testing confirmation of completion.

F. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.
2. See Section 40 61 13.

G. Informational Submittals:
1. Record Drawings:
a. Updated panel drawings delivered with the panel(s) from the Contractor's factory.
b. Drawings shall be enclosed in transparent plastic and firmly secured within each panel.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Enclosures:
a. Hoffman
b. Or Equal.
2. Panel heaters:
a. Hoffman
b. Or Equal.
3. Heat exchangers and air conditioners:
a. Hoffman
b. Or Equal.
4. Cooling fans and exhaust packages:
a. Hoffman
b. Or Equal.
5. Internal corrosion inhibitors:

a. Hoffman ; Model A-HCI.
b. Or Equal.

2.2 ACCESSORIES

A. Panel Nameplates and Identification:
1. See Section 10 14 00.

2.3 FABRICATION

A. General:
1. Coordinate panel dimensions with available space and surrounding equipment as identified in the Contract Documents.
2. Fabricate panels with instrument arrangements as identified in the Contract Documents.
3. Provide panel(s) with the required enclosure rating per NEMA 250 to meet classifications identified in the Contract Documents.
4. Devices installed in panel openings shall have a NEMA enclosure rating at least equal to the panel enclosure rating.
   a. Devices that cannot be obtained with an adequate NEMA rating shall be installed behind a transparent viewing window.
   b. The window shall maintain the required NEMA rating of the enclosure.
5. Panel(s) shall be completely assembled at the Contractor’s factory.
   a. No fabrication other than correction of minor defects or minor transit damage shall be performed on panels at the jobsite.
6. Painting:
   a. Panels fabricated from steel shall have their internal and external surfaces prepared, cleaned, primed, and painted.
      1) Mechanically abrade all surfaces to remove rust, scale, and surface imperfections.
      2) Provide final surface treatment with 120 grit abrasives or finer, followed by spot putty to fill all voids.
      3) Utilize solvent or chemical methods to clean panel surfaces.
      4) Apply surface conversion of zinc phosphate prior to painting to improve paint adhesion and to increase corrosion resistance.
      5) Electrostatically apply polyester urethane powder coating to all inside and outside surfaces.
      6) Bake powder coating at high temperatures to bond coating to enclosure surface.
         a) Panel interior shall be white with semi-gloss finish.
         b) Panel exterior shall be ANSI #61 gray with flat finish.
      7) Application of alkyd liquid enamel coating shall be allowed in lieu of polyester urethane powder for wall mounted NEMA 1 or NEMA 12 rated panels.
   b. Panels fabricated from stainless steel, aluminum, or fiberglass shall not be painted.
7. Finish opening edges of panel cutouts to smooth and true surface conditions.
   a. Panels fabricated from steel shall have the opening edges finished with the panel exterior paint.
8. Panels shall meet all requirements of UL 508A or UL 698A.
   a. If more than one disconnect switch is required to disconnect all power within a panel or enclosure, provide a cautionary marking with the word "CAUTION" and the following or equivalent, "Risk of Electric Shock-More than one disconnect switch required to de-energize the equipment before servicing."
9. Provide control panel in accordance with NFPA 70, Article 409.
   a. In the event of any conflict between NFPA 70, Article 409 and UL 508A or UL 698A, the more stringent requirement shall apply.
10. Provide equipment or control panels with Short Circuit Current Rating (SCCR) labeling as required by NFPA 70 and other applicable codes.
    a. Determine the SCCR rating by one of the following methods:
       1) Method 1: SCCR rating meets or exceeds the available fault current of the source equipment when indicated on the Drawings.

2) Method 2: SCCR rating meets or exceeds the source equipment's Amp Interrupting Current (AIC) rating as indicated on the Drawings.
3) Method 3: SCCR rating meets or exceeds the calculated available short circuit current at the control panel.

b. The source equipment is the switchboard, panelboard, motor control center or similar equipment where the control panel circuit originates.
c. For Method 3, provide calculations justifying the SCCR rating. Utilize source equipment available fault current or AIC rating as indicated on the Drawings.

B. Free-Standing Panels:
1. Welded construction.
2. Completely enclosed, self-supporting, and gasketed, dust tight.
3. Rolled lip around all sides of enclosure door opening.
4. Seams and corners welded and ground smooth to touch and smooth in visual appearance.
5. Full height, fully gasketed flush pan doors.
6. Full length piano hinges rated for 1.5 times door plus instrument weight.
7. Doors with keyed alike locking handles and three-point catch.
8. Appropriate conduit, wiring, and instrument openings shall be provided.
9. Lifting eyebolts to allow simple, safe rigging and lifting of panel during installation.

C. Wall Mounted Panels:
1. Seams continuously welded and ground smooth.
2. Rolled lip around all sides of enclosure door opening.
3. Gasketed dust tight.
4. Three-point latching mechanism operated by oil tight key-locking handle.
5. Key doors alike.
6. Continuous heavy GA hinge pin on doors.
a. Hinges rated for 1.5 times door plus instrument weight.
7. Front full opening door.
8. Brackets for wall mounting.

D. Internal Panel Wiring:
1. Panel wire duct shall be installed between each row of components, and adjacent to each terminal strip.
a. Route wiring within the panel in wire-duct neatly tied and bundled with tie wraps.
b. Follow wire-duct manufacturer's recommended fill limits.
c. Wire-ducts shall have removable snap-on covers and perforated walls for easy wire entrance.
d. Wire-duct shall be constructed of nonmetallic materials with rating more than the maximum voltage carried therein.
2. Wiring shall be installed such that if wires are removed from one device, source of power will not be disrupted to other devices.
3. Splicing and tapping of wires permitted only at terminal blocks.
4. Wire bunches to doors shall be secured at each end so that bending or twisting will be around longitudinal axis of wire.
a. Protect bend area with sleeve.
5. Arrange wiring neatly, cut to proper length, with surplus wire removed.
a. Arrange wiring with sufficient clearance.
b. Provide abrasion protection for wire bundles that pass-through openings or across edges of sheet metal.
6. AC circuits shall be routed separate from analog signal cables and digital signal cables.
a. Separate by at least 6 IN, except at unavoidable crossover points and at device terminations.
7. Separation of intrinsically safe circuit conductors and non-intrinsically safe circuit conductors:
a. Secure conductors so that any intrinsically safe circuit conductor that might come loose from a terminal is unlikely to encounter another terminal.

b. Separate non-intrinsically safe circuit conductors from intrinsically safe circuit conductors by one of the following methods:
    1) Separation of non-intrinsically safe circuit conductors from intrinsically safe circuit conductors by at least 2 IN (50 MM).
    2) Separation of non-intrinsically safe circuit conductors from intrinsically safe circuit conductors by use of a grounded metal partition 0.0359 IN (0.91 MM) or thicker.
    3) Separation of non-intrinsically safe circuit conductors from intrinsically safe circuit conductors by use of an approved insulating partition that extends to within 0.0625 IN (1.5 MM) of the enclosure walls.
    4) Where either (1) all the intrinsically safe circuit conductors or (2) all the non-intrinsically safe circuit conductors are in grounded metal-sheathed or metal-clad cables where the sheathing or cladding capable of carrying fault current to ground.

8. Separate different intrinsically safe circuit conductors from each other by one of the following means:
    a. The conductors of each circuit are within a grounded metal shield.
    b. The conductors of each circuit have insulation with a minimum thickness of 0.01 IN (0.25 MM).
9. Provide minimum clearance of 0.125 IN (3 MM) between uninsulated parts of intrinsically safe field wiring conductors connected to terminals and grounded metal or other conducting parts.
10. Wiring to pilot devices or rotary switches shall be individually bundled and installed with a "flexible loop" of sufficient length to permit the component to be removed from panel for maintenance without removing terminations.
11. Conductors for AC and DC circuits shall be type MTW stranded copper listed for operation with 600 V at 90 DEGC.
    a. Conductor size shall be as required for load and 16 AWG minimum.
    b. Internal panel wiring color code:
        1) AC circuits:
            a) Power wiring: Black.
            b) Control interconnections: Yellow.
            c) Neutral: White.
            d) Ground: Green.
        2) Low voltage DC circuits:
            a) Power wiring: Blue.
            b) Control interconnections: Blue and white striped.
        3) Foreign voltage circuits: Orange.
        4) Annunciator circuits: Red.
        5) Intrinsically safe circuits: Light Blue.
12. Analog signal cables shall be of 600 V insulation, stranded copper, twisted-shielded pairs.
    a. Conductor size: 18 AWG minimum.
    b. Terminate shield drain conductors to ground only at one end of the cable.
13. High precision 250-ohm resistors with 0.25 % accuracy shall be used where 4 - 20 mA DC analog signals are converted to 1 - 5 VDC signals.
    a. Resistors located at terminal strips.
    b. Resistors terminated using individual terminal blocks and with no other conductors.
    c. Resistor leads shall be uninsulated and of sufficient length to allow test or calibration equipment (e.g., HART communicator, loop calibrator) to be properly attached to the circuit with clamped test leads.
14. Analog signals for devices in separate enclosures shall not be wired in series.
    a. Loop isolators shall be used where analog signals are transmitted between control enclosures.
15. Wire and cable identification:
    a. Wire and cables numbered and tagged at each termination.
    b. Wire tags:
        1) Slip-on, PVC wire sleeves with legible, machine-printed markings.

2) Adhesive, snap-on, or adhesive type labels are not acceptable.

c. Markings as identified in the Shop Drawings.

E. Grounding Requirements:

1. Equipment grounding conductors shall be separated from incoming power conductors at the point of entry.
2. Minimize grounding conductor length within the enclosure by locating the ground reference point as close as practical to the incoming power point of entry.
3. Bond electrical racks, chassis and machine elements to a central ground bus.
   a. Nonconductive materials, such as paint, shall be removed from the area where the equipment contacts the enclosure.
4. Bond the enclosure to the ground bus.
5. It is imperative that good electrical connections are made at the point between the ground bus and enclosure.
6. Panel-mounted devices shall be bonded to the panel enclosure or the panel grounding system by means of locknuts or pressure mounting methods.
7. Sub-panels and doors shall be bonded to ground.
8. Associated apparatus (connected to intrinsically safe circuits) and associated cable shields:
   a. Ground in accordance with the associated control drawing (drawing provided for the intrinsically safe circuit and which contains manufacturer's entity parameters).

F. Termination Requirements:

1. Wiring to circuits external to the panel connected to interposing terminal blocks.
2. Terminal blocks rigidly mounted on DIN rail mounting channels.
3. Terminal strips are located to provide adequate space for entrance and termination of the field conductors.
4. One side of each strip of terminal blocks reserved exclusively for the termination of field conductors.
5. Terminal block markings:
   a. Marking shall be the same as associated wire marking.
   b. Legible, machine-printed markings.
   c. Markings as identified in the shop drawings.
6. Terminal block mechanical characteristics, and electrical characteristics shall be in accordance with NEMA ICS 4.
7. Terminal blocks with continuous marking strips.
   a. Each terminal block shall be identified with machine printed labels.
8. Terminals shall facilitate wire sizes as follows:
   a. 120 VAC applications: Conductor size 12 AWG minimum.
   b. Other: Conductor size 14 AWG minimum.
9. Analog signal cable shield drain conductors shall be individually terminated.
10. Install minimum of 20 % spare terminals.
11. Bladed, knife switch, isolating type terminal blocks where control voltages enter or leave the panel.
12. Fused terminal blocks shall be used in the following circuits:
    a. Control voltage is used to energize a solenoid valve.
    b. DC power is connected to 2-wire, loop-powered instruments.
13. Fused terminal blocks shall be provided with blown fuse indicators.
14. When control circuits require more than one field conductor connected to a single wiring point, enough terminal points shall be connected internally to allow termination of only one field conductor per terminal block.
15. DIN rail mounting channels shall be installed along full length of the terminal strip areas to facilitate future expansion.
16. Connections to devices with screw type terminals shall be made using spade-tongue, insulated, compression terminators.
17. Intrinsically safe circuit termination:

a. Provide at least 0.25 IN (6 MM) clearance between two terminals for connection of field wiring of different intrinsically safe circuits, unless this clearance is permitted to be reduced by the control drawing this is provided for the intrinsically safe circuit and which contains manufacturer's entity parameters.
b. Identify intrinsically safe circuits at terminal and junction locations in a manner that is intended to prevent unintentional interference with the circuits during testing and servicing as required by NEC, Article 504.

G. Component Mounting and Placement:
1. Components shall be installed per manufacturer instructions.
2. Mount and wire so removal or replacement may be accomplished without interruption of service to adjacent devices.
3. Locate all devices mounted inside enclosures so terminals and adjustment devices are readily accessible without use of special tools and with terminal markings clearly visible.
4. Control relays and other control auxiliaries shall be mounted on DIN rail mounting channels where practical.
5. Front panel devices shall be mounted within a range of 40 to 70 IN above the finished floor, unless otherwise shown in the Contract Documents.
6. PLC/RTU and I/O rack installation:
a. Located such that the LED indicators and switches are readily visible with the panel door open.
b. Located such that repair and/or replacement of component can be accomplished without the need to remove wire terminations or other installed components.
7. Locate power supplies with sufficient spacing for circulation of air.
8. Where components such as magnetic starters, contactors, relays, and other electromagnetic devices are installed within the same enclosure as the PLC/RTU system components, provide a barrier of at least 6 IN of separation between the "power area containing the electromagnetic devices" and the "control area".
9. Components mounted in the panel interior shall be fastened to an interior sub-panel using machine screws.
a. Fastening devices shall not project through the outer surface of the panel enclosure.
10. Excess mounting space of at least 20 % for component types listed below to facilitate future expansion:
a. Fuse holders.
b. Circuit breakers.
c. Control relays.
d. Time delay relays.
e. Intrinsically safe barriers and relays.
11. Components installed on sub-panels shall be provided with a minimum spacing between component and wire duct of 1 IN.
a. Minimum of 2 IN separation between terminal strips and wire ducts.

H. Power Distribution:
1. Main incoming power circuits shall be protected with a thermal magnetic circuit breaker.
a. Limit load to maximum of 80 % of circuit breaker rating.
2. Component types listed below shall be individually DIN mounted branch circuit breakers so that they may be individually de-energized for maintenance:
a. PLC/RTU power supply modules.
b. I/O cards – 16 points.
c. Network and communications equipment.
d. HMIs or Operator Interfaces.
e. Single-loop controllers.
f. Recorders.
g. Alarm annunciators.
3. Each control panel with PLC/RTU components shall be furnished with power protection in the form of a double conversion UPS.

4. Equip each panel with necessary power supplies with ratings required for installed equipment and with minimum 25 % spare capacity.
5. Constant voltage transformers, balancing potentiometers, and rectifiers as necessary for specific instrument requirements.

I. Internal Panel Lighting and Service Receptacles:
1. Panels less than or equal to 4 FT wide:
   a. One electrical GFCI duplex receptacle.
   b. One LED light fixture with manual switch(es).
2. Panels or panel faces greater than 4 FT wide:
   a. One duplex electrical GFCI receptacle per 6 FT of length.
   b. Continuous LED lighting strip with manual switches.

J. Security Controls:
1. Provide all control panels and enclosures with door switch monitored by discrete input utilizing normally closed contacts.

K. Environmental Controls:
1. Indoor panels located in a designated electrical room or control room:
   a. Thermostat controlled cooling fans with exhaust louvers if required to maintain temperature inside panel(s) below the maximum operating temperature rating of the internal components.
   b. Internal corrosion inhibitors.
2. Indoor panels not located within a designated electrical room or control room:
   a. Thermostat controlled heaters to maintain temperature approximately 10 DEGF above ambient for condensation prevention inside the panels.
   b. Automatically controlled, closed-loop heat exchangers or closed-loop air conditioners where required to maintain temperature inside each enclosure below the maximum operating temperature rating of the components inside the panel(s).
   c. Internal corrosion inhibitors.
3. Outdoor panels:
   a. Outdoor temperature ranges from 0 DEGF to 140 DEGF.
   b. Thermostat controlled heaters to maintain temperature approximately 10 DEGF above ambient dew point for condensation prevention inside the panels.
   c. Outdoor temperature ranges from 0 DEGF to 140 DEGF.
   d. Thermostat controlled closed-loop heat exchangers or closed-loop air conditioners if required to maintain temperature inside each enclosure below the maximum operating temperature rating of the components inside the panel.
   e. Internal corrosion inhibitors.
4. Environmental control components:
   a. Panel heaters:
      1) Thermostat controlled.
      2) Fan driven.
      3) Components are mounted in an anodized aluminum housing.
      4) Designed for sub-panel mounting.
      5) Powered from 120 VAC and protected with a dedicated circuit breaker.
   b. Cooling fans and exhaust packages:
      1) Cooling fan with louver or grill and replaceable filter.
      2) Designed to be mounted within a panel cutout to provide positive airflow through the panel.
      3) Cooling fan and exhaust louvers shall be designed and listed to maintain a NEMA 12 enclosure rating.
      4) Fitted with replaceable, high-density foam or synthetic fiber.
      5) Cooling fan controlled with a separately mounted thermostat with bi-metal sensor and adjustable dial for temperature setting.
      6) Powered from 120 VAC and protected with a dedicated circuit breaker.
   c. Heat exchangers and air conditioners:

1) Dual loops are designed to isolate panel interior air from exterior air.
2) Thermostat controlled.
3) Operate from 120 VAC and protected with a dedicated circuit breaker.

d. Internal corrosion inhibitors:
1) Contains chemicals which vaporize and condense on surfaces in the enclosure.
2) Inhibitor shall be applied in accordance with manufacturer instructions for the enclosure volume.
3) Inhibitor shall be applied in the panel(s) prior to shipment from the Contractor's factory.

2.4 UNWITNESSED FACTORY TESTING

A. Inspect and test entire panel assembly to verify readiness for shipment or readiness for the Contracting Officer to attend factory testing if specified in Section 40 61 21 - Process Control System Testing.

B. Location: Panel fabricator factory.

C. Tests shall be fully documented and signed by the panel fabricator's factory supervisor.

D. The panel shop shall fully test the control panel for correct wiring.
1. Each I/O point shall be checked by measuring or connecting circuits at the field terminal blocks.

E. Burn-in test: Panel(s) shall be fully energized for a minimum period of 48 HRS.

F. A PLC Central Processing Unit (CPU) shall be obtained and connected to the panel(s) if necessary for testing purposes.

G. Testing equipment (such as digital multi-meters, analog loop calibrators, and laptop computers with PLC programming software) shall be used as required for testing.

H. The following functions shall be tested as a minimum:
1. Verify functions of the panel(s) required by the Contract Documents.
2. Correctness of wiring from all panel field terminals to all I/O points and to all panel components.
3. Simulate and test each discrete signal at the field terminal strips.
4. Simulate and test each analog signal using loop calibrators.
5. Correct operation of communications between PLC system Central Processing Units (CPUs) and Remote I/O bases.
6. Correct operation of single-loop controllers (including digital communication to microprocessor-based devices).
7. Correct operation of all digital communication devices.
8. Verify online and offline diagnostic tests and procedures.

I. Deficiencies shall be corrected prior to requesting the Contracting Officer to attend factory testing if specified, or prior to shipment from the Contractor's factory.

2.5 FACTORY ACCEPTANCE TESTING

A. Perform Factory Acceptance Testing witnessed by the Contracting Officer if specified in Section 40 61 21 - Process Control System Testing.

2.6 MAINTENANCE MATERIALS

A. Extra Materials:
1. Quantity of 25 % replacement lamps for each type installed (minimum of 12 of each type).
2. Minimum 12 replacement filters for each type installed.
3. 1 QT of exterior finish touch-up paint.
4. One complete set of replacement corrosion inhibitors in sealed packages for each panel.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install freestanding panels on 4 IN high concrete housekeeping pads.

B. Anchor panels in a manner to prevent the enclosure from racking, which may cause the access doors to become misaligned.

C. Obtain approved panel layouts prior to installation of conduits.

D. Install products in accordance with manufacturer's instructions.

END OF SECTION

SECTION 40 67 63
UNINTERRUPTIBLE POWER SUPPLY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Uninterruptible Power Supplies (UPS).

B. Related Sections include but are not necessarily limited to:
   1. Section 40 61 13 - Process Control Systems General Requirements.
   2. Section 40 67 00 – Control System Equipment Panels and Racks

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Federal Communications Commission (FCC):
      a. Code of Federal Regulations, 47 CFR 15, Radio Frequency Devices.
   2. Underwriters Laboratories, Inc. (UL):
      a. 1778, Standard for Uninterruptible Power Supply Equipment.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Section 40 61 13 - Process Control System General Requirements.
   2. Provide load calculations and manufacturer's runtime chart to substantiate UPS sizing.

B. Contract Closeout Information:
   1. Operation and Maintenance Manuals:
      a. See Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 UNINTERRUPTIBLE POWER SUPPLIES (UPS)

A. Manufacturers:
   1. APC Smart-UPS.
   2. Or Equal.

B. Design and fabrication:
   1. Double conversion, true online type.
   2. Tower or Rack type format as required for installation.
      a. Tower for control panels.

b. Rack type for network racks.
3. Waveform: Pure sine wave.
4. Power factor correction.
5. Minimum 10 minutes power ride through of 100% of connected load without incoming power.
   a. Provide extended battery or batteries as necessary to achieve the specified battery run time.
6. For UPS and related components located within a server rack, install UPS and battery components at bottom of cabinet.
7. Frequency range: 45-65 Hz.
8. Input protection:
   a. Fuse or circuit breaker.
9. Output voltage regulation:
   a. ±1% online.
   b. ±2% on battery mode.
10. Battery: Replaceable, sealed, lead-acid; maintenance free.
11. Three stage battery charging for prolonged battery life.
12. Battery over discharge protection.
13. Input power cord.
14. Output receptacles.
15. Efficiency:
   a. Normal mode, minimum: 89%.
   b. Efficiency mode, minimum: 95%.
   c. Battery mode, minimum: 83%.
16. Operating temperature: 32 to 104 DEGF.
17. Relative humidity: 5-95% non-condensing.
18. Integral bypass to automatically bypass UPS on selected fault conditions.
19. Front panel indication of UPS status and alarm conditions.
   a. UPS Fault.
   b. UPS on battery.
   c. UPS is online and operating normally.
   d. Battery low.
   e. UPS in bypass.
20. Utilize alarm contacts to enable remote annunciation of the following conditions.
   a. UPS Fault.
   b. UPS on battery.
   c. Low battery.
   d. UPS in bypass.
21. Accessories:
   a. Communications Modules
      1) Provide an Ethernet communications module for network connectivity.
      2) Communication Protocol to be Modbus TCP/IP
22. Agency Approvals:
   a. Safety: UL 1778.
   b. Emissions: FCC Part 15 (Class A).

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

3.2 FIELD QUALITY CONTROL

A. Test UPS to verify the UPS switches power upon incoming AC power loss correctly while keeping all UPS powered loads online.

B. Perform testing of the UPS to determine if each UPS is sized correctly to maintain the specified run time.

END OF SECTION

This page intentionally left blank.

SECTION 40 71 00
FLOW INSTRUMENTATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Flow Transmitters:
      a. Magnetic Flow Meters (Inline).
   2. Flow Indicators:
      a. Rotameters
   3. Flow Switch:
      a. Thermal Dispersion

B. Related Specification Sections include but are not necessarily limited to:
   1. See Section 01 79 00 – Demonstration and Training, for operator training requirements. Section 40 61 13 – Process Control Systems General Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. A126, Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings.
      b. A240, Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications.
      c. B16.5, Pipe Flanges and Flanged Fittings.
      d. B626, Standard Specification for Welded Nickel and Nickel-Cobalt Alloy Tube.
      e. PTC 19.5, Application of Fluid Meters, Part 2.
   2. ASTM International (ASTM):
      a. A126, Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings.
   3. American Water Works Association (AWWA).
   4. National Sanitation Foundation (NSF).
   5. US Department of Interior Bureau of Reclamation (USDIBR):
      a. Water Measurement Manual.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 61 13.
   2. Performance criteria – thermal mass gas flow:
      a. Pressure drop charts for all flow conditioners at operating conditions.

B. Operation and Maintenance Manuals:
   1. See Specification Section 01 78 23 for requirements for:
      a. The mechanics and administration of the submittal process.
      b. The content of Operation and Maintenance Manuals.

1.4 SYSTEM DESCRIPTION

A. The instruments specified in this Specification Section are the primary element components for the control loops specified in Specification Section 40 61 96.

B. These instruments are integrated with other control system components specified under Specification Section 40 61 13 series to produce the functional control defined in the Contract Documents.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 FLOW TRANSMITTERS

A. Magnetic Flow Meters (Inline):
   1. Acceptable manufacturers:
      a. ABB (WaterMaster).
      b. Or Equal.
   2. Accessories:
   3. Design and fabrication:
      a. Utilize characterized field principle of electromagnetic induction to produce signal directly proportional to flow rate.
      b. High input impedance pre-amplifiers.
         1) Minimum impedance: 10^{10} ohms.
      c. Provide flanged end connections per ASME B16.5 up to 24 IN; meters 30 IN and larger shall be AWWA, rated for piping system operating and test conditions. Meter body shall be rated to same pressure as the flanges.
      d. Grounding requirements:
         1) Nonmetallic or lined pipe:
            a) Inlet and outlet grounding rings of same material as electrode or as recommended by manufacturer to meet process requirements.
         2) Conductive piping:
            a) Conductive path between the meter and the piping flanges.
      e. Provide cable between magnetic flow meter and transmitter.
         1) Cable shall be potted and fitted by manufacturer at the factory.
      f. Pulsed DC magnetic field excitation.
      g. Automatic zero.
      h. Adjustable low flow cutoff.
      i. Minimum signal lock (empty tube zero) to prevent false measurement when tube is empty.
      j. Accuracy: ±0.4% of rate.
      k. 4-20 mA DC isolated output into maximum 800 ohms.
      l. Discrete 24VDC totalized flow pulse output.
      m. Power supply: 117 V ±10%, 60 Hz.
      n. Indication of flow rate and totalized flow at transmitter.
      o. Meter operable as specified in liquids with 5.0 micro mho/cm or more conductivity.
      p. Transmitter electronics shall use microprocessor-based architecture and be configured using parameters.
      q. All meters for drinking water service shall be NSF 61 certified.
   4. Schedule:

TAG NUMBER	SERVICE	FLOW RANGE (GPM)	METER SIZE (IN)	NEMA (IP) RATING	INTEGRAL OR REMOTE TRANSMITTER
FE/FIT-52371	RSPH Raw Water Pumps Suction	10 - 200	4”	4X	Integral
FE/FIT-52372	RSPH Raw Water Bypass	10 - 250	2”	4X	Remote
FE/FIT-53171	RSPH Finished Water	10 - 150	2”	4X	Integral
FE/FIT-53172	RSPH Distribution	10 - 100	2”	4X	Remote
FE/FIT-53173	RSPH Booster Pump System	10 - 40	2”	4X	Integral
FE/FIT-53371	RSPH Makeup Water	0 - 15	1”	4X	Integral
FE/FIT-62071	WTP Raw Water	10-150	4”	4X	Integral
FE/FIT-64071	WTP Combined Filtrate	10-150	4”	4X	Remote
FE/FIT-68171	WTP Upper Zone Distribution	50-1200	8”	4X	Integral
FE/FIT-68172	WTP Lower Zone Distribution	50-1100	8”	4X	Integral
FE/FIT-68173	WTP FW Tanks Influent	10-150	4”	4X	Integral
FE/FIT-68174	WTP Combined Back-Feed	10-150	4”	4X	Integral
FE/FIT-68175	WTP Back-Feed to Supai Tunnel	10-100	2”	4X	Integral

2.3 FLOW SWITCHES

A. Thermal Dispersion Type Flow Switches:
   1. Acceptable manufacturer:
      a. Fluid Components, Inc.
         1) For Low Flow: FLT93L
      b. Or Equal.
   2. Materials:
      a. All wetted surfaces: 316 stainless steel.
      b. Enclosure: Cast iron or aluminum.
   3. Design and fabrication:
      a. Solid state electronics.
      b. Inaccuracy, liquids: ±0.5% of reading or ±0.4 FPS.
      c. Inaccuracy, gas: ±0.5% of reading or ±2 SFPS.
      d. Repeatability: ±1% of full signal.
      e. Response time: Adjustable down to 1 second.
      f. Utilize two platinum RTD's in thermowells in flow stream for differential temperature measurement.
      g. SPDT switch points.
         1) Contacts rated:
         2) 1 amp inductive at 125 VDC.
         3) 5 amp inductive at 120 VAC.
      h. Process connections: 3/4 inches MNPT.
      i. Power supply: 120 VAC, 60 Hz.
      j. Process temperature: -40 to 350 degrees F.
   4. Schedule:

TAG NUMBER	SERVICE	FLOW RANGE (GPM)	PROCESS			PIPE SIZE (IN)	NEMA (IP) RATING
			MEDIUM	TEMP	PRESSURE		

FSH-51671	RSPH Emergency Shower/Eye Wash Station	0.5 – 30	Water	72 degF	100 psi	2”	4X
FSL-52371	RSPH Turbidimeter Drain Line	0.25 – 0.79	Water	72 degF	0.1 psi	½”	4X
FSH-61641	WTP Laboratory Emergency Shower/Eye Wash Station	0.5 – 30	Water	72 degF	100 psi	2”	4X
FSH-61642	WTP Acids Drum Room Emergency Shower/Eye Wash Station	0.5 – 30	Water	72 degF	100 psi	2”	4X
FSH-61643	WTP Bases Drum Room Emergency Shower/Eye Wash Station	0.5 – 30	Water	72 degF	100 psi	2”	4X

2.4 FLOW INDICATORS

A. Rotameters - General Performance:
   1. Acceptable manufacturers:
      a. Blue-White
      b. Dwyer
      c. Omega
      d. Or Equal.
   2. Materials:
      a. Tube/Body: Acrylic plastic. or Polysulfone
      b. Float: Glass or 316 stainless steel.
      c. End fittings: 316 stainless steel or PVC.
      d. RSPH: Elastomers used should be Viton, if possible.

   3. Design and fabrication:
      a. Integral needle valve unless otherwise noted.
      b. Diaphragm valve on finished water recirculation line provided by contractor with the following:
         1) Cv at 100% open of 28±1
         2) Cv at 50% open of 23±1
      c. Direct reading scale.
      d. Clean-out plugs.
      e. Accuracy: ±10%.
   4. Schedule:

TAG NUMBER	SERVICE	FLUID	SIZE (IN)	FLOW RANGE (GPM)	DESIGN PRESSURE (PSI)
FI-52371	RSPH Raw Water Instrumentation Panel	Water	1/2”	0.1 – 1**	1 – 4
FI-53172	RSPH Finished Water Instrumentation Panel	Water	1”	0.1 – 1**	1 – 5
FI-53271	RSPH Finished Water Recirculation Line*	Water	1 ½”	3.0 - 30	4 - 10

TAG NUMBER	SERVICE	FLUID	SIZE (IN)	FLOW RANGE (GPM)	DESIGN PRESSURE (PSI)
FI-62071	WTP RW Instrumentation Panel	Water	¼"	Per mfr standard	5-40
FI-64071	WTP Combined Filtrate Instrumentation Panel	Water	¼"	Per mfr standard	5-40
FI-64072	WTP UF Feed Instrumentation Panel	Water	¼"	Per mfr standard	5-40
FI-66071	WTP FW Instrumentation Panel	Water	¼"	Per mfr standard	5-40
FI-66072	WTP Distribution Water Instrumentation Panel	Water	¼"	Per mfr standard	5-40
FI-66073	WTP Distribution Water Instrumentation Panel	Water	¼"	Per mfr standard	5-40
FI-66074	WTP CT Tank Influent Instrumentation Panel	Water	¼"	Per mfr standard	5-40

* No needle valve. See associated diaphragm valve
** Unless otherwise noted by instrument manufacturer, coordinate flow range with all instrument panel component specifications.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

3.2 TRAINING

A. Provide on-site training in accordance with Specification Section 01 75 00.

END OF SECTION

This page intentionally left blank.

SECTION 40 72 00
LEVEL INSTRUMENTATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Level Transmitters:
      a. Differential Pressure (Level) Sensor and Transmitter.
      b. Submersible Level Sensor and Transmitter.
      c. Radar Level Sensor and Transmitter.
   2. Level Switches:
      a. Float-Tilt Type Level Switch.
      b. Displacement Type Level Switch.

B. Related Specification Sections include but are not necessarily limited to:
   1. See Section 01 79 00 – Demonstration and Training, for operator training requirements.
   2. Section 40 61 13 - Process Control Systems General Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B16.5, Pipe Flanges and Flanged Fittings.
   2. ASTM International (ASTM):
      a. A106, Standard Specification for Seamless Carbon Steel Pipe for High Temperature Service.
      b. A743, Standard Specification for Castings, Iron-Chromium, Iron-Chromium-Nickel, Corrosion Resistant, for General Application

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 61 13.

B. Operation and Maintenance Manuals:
   1. See Specification Section 01 78 23 for requirements for:
      a. The mechanics and administration of the submittal process.
      b. The content of Operation and Maintenance Manuals.

1.4 SYSTEM DESCRIPTION

A. The instruments specified in this Specification Section are the primary element components for the control loops specified in Specification Section 40 61 96.

B. These instruments are integrated with other control system components specified under Specification Section 40 61 13 series to produce the functional control defined in the Contract Documents.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, the Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 LEVEL TRANSMITTERS

A. Differential Pressure (Level) Transmitter:
   1. Manufacturers:
      a. Emerson Rosemount 2051L.
      b. Or Equal.
   2. Specifications:
      a. General:
         1) Measurement Principle: Differential-pressure (hydrostatic) based on the height of the liquid head.
            a) Open (to atmosphere) or Closed Tanks.
            b) Uses a process isolating diaphragm.
         2) Minimum/Maximum Span: ±0.15 to 300 PSIG.
         3) Accuracy: ±0.075% of span.
      b. Process Connection:
         1) Isolation Diaphragm: 3 IN.
         2) Flange Rating: Class 600.
      c. Display and Configuration:
         1) Integral Display for live measurement and configuration.
         2) Adjustable zero and span.
         3) Damping: Adjustable from 0 to 32 seconds.
         4) Output variables: Level; optional: Volume, Mass.
         5) Output Units: psi, feet, inches, meters, or millimeters (mm).
      d. Electrical:
         1) Signal Power: Loop-powered, 2-wire, 24 VDC.
         2) Current Output: Analog 4-20 Ma into a 400-ohm loop.
         3) Configuration: With remote hand-held configurator.
         4) Cable entry: 1/2 IN NPT connection.
      e. Materials of Construction:
         1) Process flanges and adapters: 316SS.
         2) Wetted O-rings: Viton or PTFE glass.
         3) Vent/Drain valves: 316SS.
         4) Isolation Diaphragm: 316SS.
            a) Fill Fluid: Glycerin and water.
         5) Housing: Polyurethane-covered aluminum or cast aluminum.
      f. Environment:
         1) Ambient Temperature: -40 to 158 DEGF (-40 to 70 DEGC).
         2) Humidity: up to 99%.
         3) Process Temperature: -4 to 185 DEGF (-20 to 85 DEGC); depends on fill fluid.
         4) Process Pressure: 0 to 35 PSIG.
         5) Protection: Refer to Area Classification Drawings.
   3. Schedule (or Instrument List):

TAG NUMBER	SERVICE	RANGE (FT)	PROCESS CONNECTION
LIT-63051	WTP Reaction Tank 1	0-15	FLANGE

TAG NUMBER	SERVICE	RANGE (FT)	PROCESS CONNECTION
LIT-63052	WTP Reaction Tank 2	0-15	FLANGE

B. Submersible Level Sensor and Transmitter:
   1. Manufacturers:
      a. Mercoid (SBLT2 & PBLT2)
      b. Or Equal.
   2. Specifications:
      a. General:
         1) Measurement Principle: Differential-pressure (hydrostatic) based on the height of the liquid head with the use an integrated strain gauge or Ceramic Capacitive Sensor (CCS).
            a) Open (to atmosphere) or Closed Tanks.
            b) Fully submersible.
            c) Vented cable.
         2) Minimum/Maximum Span: 1 to 100 PSIG (2.31 to 230.67 FT).
         3) Accuracy: ±0.25% of span.
      b. Process Connection:
         1) None: Mount vertically at the bottom of tank.
         2) Option: 1/4 IN NPT connection on tank.
      c. Display and Configuration:
         1) Output variables: Pressure (converted to level).
         2) Output Units: psi.
      d. Electrical:
         1) Signal Power: Loop-powered, 2-wire, 24 VDC.
         2) Current Output: Analog 4-20 Ma into a 600-ohm loop.
         3) Cable entry: 1/4 IN NPT connection to attached jacketed cable.
      e. Materials of Construction:
         1) Probe: 316SS, Viton, Titanium.
         2) Cable: Hytrel, FEP, HDPE
      f. Environment:
         1) Ambient Temperature: -40 to 212 DEGF (-40 to 1000 DEGC).
         2) Humidity: Up to 100%, non-condensing.
         3) Process Temperature: -4 to 104 DEGF (-20 to 60 DEGC).
         4) Process Pressure: 0 to 35 PSIG.
         5) Protection: Refer to Area Classification Drawings.
   3. Spare:
      a. Provide 1 Spare Submersible Level Sensor and Transmitter with 0-40 Feet range.
   4. Schedule (or Instrument List):

TAG NUMBER	SERVICE	RANGE (FT)
LT-53251	RSPH Finished Water Tank	0-15
LT-62051	WTP RW Tank 1	0-15
LT-62052	WTP RW Tank 2	0-15
LT-66051	WTP Chlorine Contact Tank	0-13
LT-68451	WTP FW Tank A	0-40

TAG NUMBER	SERVICE	RANGE (FT)
LT-68452	WTP FW Tank B	0-40
LT-69151	WTP Process Waste Tank	0-8

C. Radar Level Sensor and Transmitter:
   1. Manufacturers:
      a. Endress+Hauser (Micropilot FMR51 & FMR52)
      b. Or Equal.
   2. Specifications:
      a. General.
         1) Measurement Principle: Radar pulses guided down the antenna, reflected off the process media and received again by the radar system; Radar system uses 'time-of-flight' method.
            a) Level and Interface Measurement.
            b) Ability to protect against vapor and turbulence.
            c) Connection on top of tank.
            d) Applications: Liquids- Sodium Hypochlorite and Magnesium Hydroxide.
            e) No compensation needed for process changes such as pressure, temperature, density, or conductivity.
         2) Measurement Range (see Schedule):
            a) Horn Antenna version: up to 131 feet.
            b) Flush Mounted with Horn Antenna version: Up to 131 feet.
         3) Accuracy: 0.1% of measurement.
         4) Repeatability: 0.04 inches.
         5) Temperature Stability: 0.01% of measurement.
      b. Design and fabrication:
         1) Measured error:
            a) Horn Antenna: 0.2 inches
            b) Flush Mounted with Horn Antenna version: ±0.2 inches
         2) Resolution: 0.1 inches.
         3) Repeatability: 0.1 inches.
         4) Response time: 1 second.
      c. Process Connection:
         1) Threaded (NPT): See Schedule.
         2) Flange (ASME): see Schedule
      d. Display and Configuration:
         1) Integral Display for live measurement and configuration.
         2) Adjustable zero and span.
         3) Output variable: Level.
         4) Output Units: Feet, inches, meters, or millimeters (mm).
      e. Electrical:
         1) Signal Power: Loop-powered, 2-wire, 24 VDC.
         2) Current Output: Analog 4-20 Ma into a 500-ohm loop.
         3) Communication: HART.
         4) Configuration: With remote hand-held configurator.
         5) Cable entry: 1/2 IN NPT connection.
         6) Complies with FCC Part 15.
      f. Mechanical:

1) Horn Antenna:
a) Material: 316L/SST.
2) Enclosure: Aluminum, polyester-coated.

g. O-rings: Viton.
h. Flanges: 316 stainless steel.
i. Environment:
1) Ambient Temperature: -40 to 176 degrees F (-40 to 80 degrees C).
2) Humidity: Up to 99%.
3) Process Temperature: -4 to 176 degrees F (-20 to 80 degrees C).
4) Process Pressure: 0 to 160 bar.
5) Protection: Refer to Area Classification Drawings.

3. Schedule (or Instrument List):

TAG NUMBER	SERVICE	RANGE (FT)	ANTENNA SIZE
LE/LIT-53351	RSPH Dilute Sodium Hypochlorite Tank	0-5	2”

2.3 LEVEL SWITCHES

A. Float-Tilt Type Level Switch:
1. Manufacturers:
a. Anchor Scientific Inc. (Roto-Float)
b. Or Equal.
2. Materials:
a. Float material: Polypropylene or Teflon coated type 316 stainless steel.
b. Cable jacket: PVC, neoprene.
c. Cable clamp: Polypropylene or 316 stainless steel.
3. Design and fabrication:
a. Mercury-free switch.
b. Provide switch complete with flexible electrical cables.
c. SPDT contact rated at 1 amp at 120 VAC.
d. Direct acting float switch:
1) Switch actuates on rising level.
2) Switch deactivates when liquid falls 1 IN below actuation level.
e. Terminate cables in junction box.
f. Process temperature: max. 120 DEGF.
g. Install floats per drawing details.
4. Schedule (or Instrument List):

TAG NUMBER	SERVICE	CONTACT NO / NC / NO-NC	MOUNTING ELEVATION	MOUNTING (Suspended or Pipe)
LSHH-62051	WTP Raw Water Tank 1	NO	8387.50’	Suspended
LSHH-62052	WTP Raw Water Tank 2	NO	8387.50’	Suspended
LSH-63051	WTP Reaction Tank 1	NO	8369.50’	Suspended
LSH-63052	WTP Reaction Tank 2	NO	8369.50’	Suspended
LSHH-66051	WTP Chlorine Contact Tank	NO	8372.00’	Suspended

TAG NUMBER	SERVICE	CONTACT NO / NC / NO-NC	MOUNTING ELEVATION	MOUNTING (Suspended or Pipe)
LSHH-69151	WTP Process Waste Tank	NO	8349.71’	Suspended

B. Displacement Type Level Switches:
   1. Acceptable manufacturer:
      a. Innovative Components PTFE LS-15-600.
      b. Or Equal.
   2. Materials:
      a. Displacer: PTFE (polytetrafluoroethylene).
      b. Support rod or cable: PTFE or Polypropylene or 316 stainless steel.
      c. All materials shall be compatible with associated chemicals.
   3. Design and fabrication:
      a. Displacer supported by rod or cable suspended on spring.
      b. Switch type: Dry contact.
      c. Mercury-free switch.
      d. Cable: 10 FT maximum.
      e. Connection 1/8” NPT thread
      f. Float 1.1” Diameter
      g. Switch contacts:
         1) NO.
         2) Rated 5 amps inductive at 120 VAC.
   4. Schedule (or Instrument List):

TAG NUMBER	SERVICE
LSH-53451	RSPH Sodium Hypochlorite Tank Containment
LSHH-51651	RSPH Pump Room Floor
LSH-67151	WTP Coagulant Feed System Containment
LSH-67351	WTP Caustic Feed System Containment
LSH-67251	WTP Sodium Hypochlorite System Containment
LSH-67651	WTP Sodium Bisulfite System Containment
LSH-67652	WTP Citric Acid System Containment
LSH-67451	WTP Hydrochloric Acid System Containment
LSH-67551	WTP Corrosion Inhibitor System Containment

2.4 ACCESSORIES

A. Furnish all mounting brackets, hardware and appurtenances required for mounting primary elements and transmitters.
   1. Materials, unless otherwise specified, shall be as follows:
      a. Bolts, nuts, washers, expansion anchors: 316 stainless steel.

b. Mounting brackets:
   1) Standard: 316 stainless steel.
   2) Highly corrosive areas: 316 stainless steel.
c. Mounting plates, angles:
   1) Standard: Carbon steel.
   2) Corrosive areas: 316 stainless steel.
d. Instrument pipe stands:
   1) Standard: Hot-dip galvanized 2 IN schedule 40, ASTM A106, Grade B carbon steel.
   2) Corrosive areas: 316 stainless steel.

B. Provide handheld communicator compatible for all intelligent transmitters furnished.
   1. Hand held communicator shall provide capability to check calibration, change transmitter range, and provide diagnostics.
   2. If these features are provided with the intelligent transmitter that is accessible, the hand held communicator is not required.

C. Cable lengths between sensors and transmitters shall be continuous (without splices) and as required to accommodate locations as shown on Drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install instrument mounting pipe stands level and plumb.

C. Instrument Valves:
   1. Orient stems for proper operation.
   2. Install arrays orderly and neat in appearance with true horizontal and vertical lines.
   3. Provide a minimum of 2 IN clearance between valve handle turning radii where there are multiple valve handles appearing in a straight line.
   4. Valves shall have bonnets and any soft seals removed during welding or soldering into the line.
      a. When cool, reassemble the valves.
   5. Support each valve individually.
      a. The tubing system does not qualify as support for the valve.

D. Locate instrument piping and tubing to be free of vibration and interference with other piping, conduit, or equipment.

E. Keep foreign matter out of the system.

F. Remove all oil on piping and tubing with solvent before piping and tubing installation.

G. Plug all open ends and connections to keep out contaminants.

H. Threaded Connection Seals:
   1. Use Tite-Seal or acceptable alternate.
   2. Use of lead base pipe dope or Teflon tape is not acceptable.
   3. Do not apply Tite-Seal to tubing threads of compression fittings.

I. Instrument Mounting:
   1. Mount all instruments where they will be accessible from fixed ladders, platforms, or grade.
   2. Mount all local indicating instruments with face forward toward the normal operating area, within reading distance, and in the line of sight.
   3. Mount instruments level, plumb, and support rigidly.
   4. Mount to provide:
      a. Protect from heat, shock, and vibrations.

b. Provide accessibility for maintenance.
c. Free from interference with piping, conduit and equipment.

3.2 TRAINING

A. Provide on-site training in accordance with Specification Section 01 75 00.

END OF SECTION

SECTION 40 73 00
PRESSURE INSTRUMENTATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Pressure Transmitters.
      a. Pressure Transmitters (Gauge and Absolute).
      b. Differential Pressure Transmitters
   2. Pressure Switches.
      a. Electro-Mechanical.
   3. Pressure Indicators.
      a. Pressure Gauges – Mechanical.
   4. Isolation Devices.
      a. Diaphragm Seals.

B. Related Specification Sections include but are not necessarily limited to:
   1. See Section 01 79 00 – Demonstration and Training, for operator training requirements.
   2. Section 40 61 13 - Process Control Systems General Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B16.5, Pipe Flanges and Flanged Fittings
   2. ASTM International (ASTM):
      a. A106/A106M, Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 61 13.

B. Operation and Maintenance Manuals:
   1. See Specification Section 01 78 23 for requirements for:
      a. The mechanics and administration of the submittal process.
      b. The content of Operation and Maintenance Manuals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 PRESSURE TRANSMITTERS

A. Pressure Transmitters (Gauge and Absolute):

1. Manufacturers:
    a. Emerson Rosemount (3051).
    b. Or Equal.
2. Materials:
    a. Process flanges and adapters: 316 stainless steel.
    b. Vent/drain valve: 316 stainless steel.
    c. Isolating diaphragm: 316 stainless steel.
    d. Housing: Aluminum.
    e. Fill fluid:
        1) Utilize halocarbon fill for process applications involving strong oxidizing agents.
            a) Agents include but are not limited to: Cl2, KMN04, FeCl, Na0H, and Na0Cl.
        2) Utilize manufacturer's standard fill for other applications.
            a) Ensure fill is suitable for application temperatures.
3. Design and fabrication:
    a. Smart transmitters utilizing microprocessor-based electronics.
    b. Signal Output: 4-20 mA DC proportional to pressure.
    c. Nonvolatile EEPROM memory.
    d. Power supply: 24 VDC Loop Power.
    e. Adjustable zero and span.
    f. Temperature limits:
        1) -20 to 180 DEGF (without digital display).
        2) -4 to 175 DEGF (with digital display).
    g. Overpressure limits: Withstand 150% of stated maximum service pressure without damage.
    h. Humidity limits: 0 to 100% relative humidity.
    i. Damping: Adjustable between 0 and 60 seconds.
    j. Accuracy (includes effects of linearity, repeatability and hysteresis): ±0.10% of calibrated span for 15:1 rangeability.
    k. Stability: ±0.2% of upper range limit for 12 months.
    l. Temperature effect:
        1) Total effect including span and zero errors: + 0.2% of upper range limit per 100 DEGF for minimum 15:1 rangeability.
    m. Minimum 1/2 IN pressure connection.
    n. Equip with test jacks or accessible terminals for testing output.
    o. Equip with isolation valve and test connections with isolation valves and/or plugs.
4. Schedule:

TAG NUMBER	SERVICE	RANGE (PSIG)	PROCESS CONNECTION (IN-LINE OR FLANGE)
PIT-52261	RSPH Raw Water Auto Filters Inlet	0 - 200	In-line
PIT-52262	RSPH Raw Water Pumps Suction	0 - 200	In-line
PIT-52265	RSPH Raw Water Pumps Outlet	0 – 2,000	In-line
PIT-52266	RSPH Raw Water Pumps Common Discharge	0 – 2,000	In-line
PIT-53161	RSPH Booster System	0 - 100	In-line
PIT-53162	RSPH Finished Water Pre-Pressure Reducer	0 – 2,000	In-line

PIT-53164	RSPH Finished Water Post-Pressure Reducer	0 – 100	In-line
PIT-62061	WTP Raw Water Suction	0-40	In-line
PIT-62062	WTP Raw Water Pump Discharge	0-400	In-line
PIT-68161	WTP FW Tank Pumps	0-40	In-line
PIT-68162	WTP Upper Zone Distribution Pumps Suction	0-50	In-line
PIT-68163	WTP Upper Zone Distribution	0-300	In-line

2.3 PRESSURE SWITCHES

A. Electro-Mechanical:
   1. Manufacturers:
      a. Allen-Bradley (Model 836).
      b. Or Equal.
   2. Materials:
      a. Wetted switch elements: 316 stainless steel.
      b. Diaphragm seal housing: 316 stainless steel.
      c. Pressure snubber:
         1) Filter disc: 316 stainless steel.
         2) Housing: 316 stainless steel.
   3. Accessories:
      a. Provide ball valve to isolate pressure switch from source.
      b. Utilize pressure snubber with porous metal discs to provide pulsation dampening on pressure switch as shown on schedule.
      c. On applications where a pressure switch and a pressure gauge are used at the same location, it is permissible to utilize one pulsation dampener and diaphragm seal to isolate both elements from the process fluid.
   4. Design and fabrication:
      a. Utilize “Snap Action” type contact switches.
      b. Trip scale, Operation Range and Differential Adjustment Screws are field adjustable.
      c. No external power needed.
      d. One SPDT contact rated:
         1) 0.5 amps inductive at 125 VDC.
         2) 5 amps inductive at 120 VAC.
      e. Switch set points:
         1) Above 1,000 PSI:
            a) Between 30 and 35% of switch rated working range.
            b) Operating pressure range not to exceed 35% of switch rated working pressure.
         2) Below 1,000 PSI:
            a) Set points between 30 and 70% of switch rated working range.
            b) Operating pressure not to exceed 75% of switch rated working range.
      f. Accuracy: ±1% of full scale.
      g. Process connection: Minimum of 1/4 IN.
      h. Conduit connection: Minimum of 1/2 IN.
   5. Schedule:

TAG NUMBER	SERVICE	SETTING (PSIG)	ISOLATION REQ'D
PSHH-52263	RSPH Raw Water Pump 1 Outlet	1798	No

PSHH-52264	RSPH Raw Water Pump 2 Outlet	1798	No
PSH-53111	RSPH Booster Pump 1	85	No
PSH-53112	RSPH Booster Pump 2	85	No
PSHH-53361	RSPH Sodium Hypochlorite Metering Pump 1	50	Yes
PSHH-53362	RSPH Sodium Hypochlorite Metering Pump 2	50	Yes

2.4 PRESSURE INDICATORS

A. Pressure Gauge - Mechanical:
   1. Manufacturers:
      a. Ametek / USGauge (Solfrunt Model 1980).
      b. Or Equal.
   2. Materials:
      a. Bourdon tube, socket, connecting tube: 316 stainless steel.
      b. Case: Phenolic.
      c. Pressure snubber:
         1) Filter disc: 316 stainless steel.
         2) Housing: 316 stainless steel.
   3. Accessories:
      a. Provide valve at point of connection to equipment and at panel if panel mounted.
      b. Utilize pressure snubber with porous metal discs to provide pulsation dampening on gauge applications as shown on schedule.
      c. Provide 1/2 IN stainless steel anti-siphon pigtail inlet connection for hot water and steam applications.
   4. Design and fabrication:
      a. All components suitable for service at:
         1) 250 DEGF.
         2) The maximum process temperature to which the gauge is to be exposed.
      b. Provide viewer protection from element rupture.
      c. Calibrate gauges at jobsite for pressure and temperature in accordance with manufacturer's instructions.
      d. Unless otherwise required by codes, provide stem mounted or flush mounted, as required, with dial diameter as follows:

PIPE SIZE	DIAL SIZE	GAUGE CONNECTION
1-1/2 IN or less	2-1/2 IN	¼ IN
Larger than 1-1/2 IN	4-1/2 IN	½ IN

      e. Equip with white faces, black numerals and black pointers.
      f. Gauge tapping position to be clear of equipment functions and movements and protected from maintenance and operation of equipment.
         1) Gauge to be readable from an accessible standing position.
      g. Gauge accuracy: 1% of full range.
      h. Select gauge range so that:
         1) The normal operating value is in the middle third of the dial.
         2) Maximum operating pressure does not exceed 75% of the full-scale range.
   5. Schedule:

TAG NUMBER	SERVICE	PRESSURE RANGE (PSI)	ISOLATION REQ'D
PI-52361	RSPH RW Instrumentation Panel	0 - 100	No
PI-52362	RSPH RW Instrumentation Panel	0 - 100	No
PI-52363	RSPH RW Instrumentation Panel	0 - 10	No
PI-52364	RSPH Upstream Orifice Plate	0 - 200	No
PI-52365	RSPH Downstream Orifice Plate	0 - 50	No
PI-52366	RSPH Process Drain	0 – 1600 psi	No
PI-52367	RSPH Process Drain	0 – 100 psi	No
PI-53161	RSPH Finished Water Pre-Pressure Reducer	0 – 1600 psi	No
PI-53163	RSPH Finished Water Post-Pressure Reducer	0 - 25	No
PI-53165	RSPH Bladder Tank Line	0 - 100	No
PI-53361	RSPH RW Instrumentation Panel	0 - 100	No
PI-53166	RSPH Bladder Tank	0 - 100	No
PI-53361	RSPH Sodium Hypochlorite Metering Pump 1	0 - 100	Yes
PI-53362	RSPH Sodium Hypochlorite Metering Pump 2	0 - 100	Yes
PI-62061	WTP RW Sample Pump	0-50	No
PI-64064	WTP UF Influent Sample Pump	0-50	No
PI-68163	WTP FW Tank Pump 1	0-50	No
PI-68164	WTP FW Tank Pump 2	0-50	No
PI-66064	WTP Distribution Water Instrumentation Panel Sample Pump	0-50	No
PI-68161	WTP Hydro-Pneumatic Tank 1	0-150	No
PI-68162	WTP Hydro-Pneumatic Tank 2	0-150	No
PI-68165	WTP CT Tank Influent Instrumentation Panel Sample Pump	0-50	No
PI-67161	WTP Coagulant Metering Pump	0-30	Yes
PI-67261	WTP Sodium Hypochlorite Metering Pump 1	0-30	Yes

TAG NUMBER	SERVICE	PRESSURE RANGE (PSI)	ISOLATION REQ'D
PI-67262	WTP Sodium Hypochlorite Metering Pump 2	0-30	Yes
PI-67263	WTP Sodium Hypochlorite Metering Pump 3	0-30	Yes
PI-67264	WTP Sodium Hypochlorite Swing Pump	0-30	Yes
PI-67361	WTP Caustic Soda Metering Pump	0-30	Yes
PI-68261	WTP High Demand Pump 1	0-300	No
PI-68262	WTP High Demand Pump 2	0-300	No
PI-68263	WTP High Demand Pump 3	0-300	No
PI-68361	WTP Upper Zone Booster Pump 1	0-300	No
PI-68362	WTP Upper Zone Booster Pump 2	0-300	No
PI-68363	WTP Upper Zone Booster Pump 3	0-300	No
N/A	Volume 3 – Supai Tunnel PRV Vault Upstream of PRV	0-750	Yes
N/A	Volume 3 – Supai Tunnel PRV Vault Downstream of PRV	0-150	Yes

2.5 ISOLATION DEVICES

A. Diaphragm Seal:
   1. Manufacturers:
      a. Ametek.
      b. Or Equal.
   2. Materials:
      a. Lower housing: 316 stainless steel.
      b. Diaphragm material: 316 stainless steel.
   3. Design and fabrication:
      a. Isolates instrument from process fluids which are corrosive or contain solids.
      b. Upper housing with bleed screw.
      c. Lower housing with flushing connection.
      d. Fill fluid:
         1) Utilize halocarbon fill for process applications involving strong oxidizing agents.
            a) Agents include but are not limited to: Cl2, KMNO4, FeCl, NaOH, and NaOCl.
         2) Utilize manufacturer's standard fill for other applications.
            a) Ensure fill is suitable for application temperatures.
      e. Process connections:
         1) Instrument: 0.25 IN female NPT.
         2) Process: 0.5 IN female NPT.
         3) PVC pipe applications: Use a socket weld connection.
   4. Installed where specified or shown on Drawings.

B. Sludge or Scum Service Instrument Isolators:
   1. Materials:
      a. Diaphragm: 316 stainless steel.

b. Lower Housing: 316 stainless steel.

2. Design and fabrication:
   a. Isolates instruments from process fluids which are corrosive or contain solids.
   b. Upper housing with bleed screw.
   c. Lower housing with flushing connection.
   d. Fill fluid:
      1) Utilize halocarbon fill for process applications involving strong oxidizing agents.
         a) Agents include but are not limited to: Cl_2, $KMNO_4$, FeCl, NaOH, and NaOCl.
      2) Utilize manufacturer's standard fill for other applications.
         a) Ensure fill is suitable for application temperatures.
   e. Process connections:
      1) 0.5 IN female NPT.
      2) Process: In-lined flanged, match line size
3. Provide as shown on Drawings.

2.6 ACCESSORIES

A. Furnish all mounting brackets, hardware and appurtenances required for mounting primary elements and transmitters.
   1. Materials, unless otherwise specified, shall be as follows:
      a. Bolts, nuts, washers, expansion anchors: 316 stainless steel.
      b. Mounting brackets:
         1) Standard: 316 stainless steel.
         2) Highly corrosive areas: 316 stainless steel.
      c. Mounting plates, angles:
         1) Standard: Carbon steel.
         2) Corrosive areas: 316 stainless steel.
      d. Instrument pipe stands:
         1) Standard: Hot-dip galvanized 2 IN schedule 40, ASTM A106, Grade B carbon steel.
         2) Corrosive areas: 316 stainless steel.

B. Tubing Support Angles and Brackets:
   1. Any of the following materials are acceptable:
      a. Aluminum support with dielectric material between support and tubing.
      b. Type 316 stainless steel.
      c. Fiberglass.

C. Tubing Tray or Channel:
   1. Aluminum.
   2. Provide dielectric material between tray or channel and tubing.

D. Provide handheld communicator compatible with all intelligent transmitters furnished.
   1. Handheld communicator shall provide capability to check calibration, change transmitter range, and provide diagnostics.
   2. If these features are provided with the intelligent transmitter, the handheld communicator is not required.

E. Cable lengths between sensors and transmitters shall be continuous (without splices) and as required to accommodate locations as shown on Drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install instrument mounting pipe stands level and plumb.

C. Instrument Valves:

1. Orient stems for proper operation.
2. Install arrays orderly and neat in appearance with true horizontal and vertical lines.
3. Provide a minimum of 2 IN clearance between valve handle turning radii where there are multiple valve handles appearing in a straight line.
4. Valves shall have bonnets and any soft seals removed during welding or soldering into the line.
    a. When cool, reassemble the valves.
5. Support each valve individually.
    a. The tubing system does not qualify as support for the valve.

D. Locate instrument piping and tubing to be free of vibration and interference with other piping, conduit, or equipment.

E. Keep foreign matter out of the system.

F. Remove all oil on piping and tubing with solvent before piping and tubing installation.

G. Plug all open ends and connections to keep out contaminants.

H. Threaded Connection Seals:
1. Use Tite-Seal or acceptable alternate.
2. Use of lead base pipe dope or Teflon tape is not acceptable.
3. Do not apply Tite-Seal to tubing threads of compression fittings.

I. Instrument Mounting:
1. Mount all instruments where they will be accessible from fixed ladders, platforms, or grade.
2. Mount all local indicating instruments with face forward toward the normal operating area, within reading distance, and in the line of sight.
3. Mount instruments level, plumb, and support rigidly.
4. Mount to provide:
    a. Protection from heat, shock, and vibrations.
    b. Accessibility for maintenance.
    c. Freedom from interference with piping, conduit and equipment.

3.2 TRAINING

A. Provide on-site training in accordance with Specification Section 01 75 00.

END OF SECTION

SECTION 40 74 00
TEMPERATURE INSTRUMENTATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Temperature Transmitters.
   2. Room Temperature Transmitters.

B. The instruments specified in this Specification Section are the primary element components for the control loops which are specified in Specification Section 40 61 96 and shown on the Control System Drawings.
   1. These instruments are integrated with other control system components specified under Specification Section 40 61 13 to produce the functional control system defined in the Contract Documents

C. Related Specification Sections include but are not necessarily limited to:
   1. Section 40 61 13 - Process Control System General Requirements.
   2. Section 40 61 96 - Process Control Descriptions.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Iron and Steel Institute (AISI):
      a. Type 316 Stainless Steel.
   2. American Society of Mechanical Engineers (ASME):
      a. PTC 19.3, TW – 2016 Thermowells.
      b. B40.200, Thermometers, Direct Reading and Remote Reading.
   3. ASTM International (ASTM):
      a. A182/A182M, Standard Specification or Forged or Rolled Alloy and Stainless-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service.
      b. E230/E230M-17 Standard Specification for Temperature-Electromotive Force (emf) Tables for Standardized Thermocouples
      c. F316, Austenitic Stainless Steel.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 61 13 - Process Control Systems General Requirements.

B. Operation and Maintenance Manuals:
   1. See Specification Section 01 78 23 for requirements for:
      a. The mechanics and administration of the submittal process.
      b. The content of Operation and Maintenance Manuals.

PART 2 - PRODUCTS

2.1 TEMPERATURE TRANSMITTER

A. Temperature Transmitters:
   1. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

2. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.
3. Subject to compliance with the Contract Documents, the following manufacturers listed are acceptable:
    a. Emerson Rosemount (3144).
    b. Or Equal.
4. Materials:
    a. Housing: 316SS.
5. Design and fabrication:
    a. Smart transmitters utilizing microprocessor-based electronics.
    b. Input: RTD, thermocouple or millivolt, as specified in schedule.
    c. Signal Output: 4-20 mA DC signal linearly proportional to temperature.
    d. Optional Communication: HART.
    e. Power supply: 24 VDC Loop Power.
    f. Adjustable zero and span.
    g. Temperature limits:
        1) -20 to 180 degrees F (without digital display)
        2) -20 to 158 degrees F (with digital display)
    h. Humidity limits: 0 to 99% relative humidity.
    i. Damping: Adjustable between 0 and 60 seconds.
    j. D/A accuracy: ±0.02%.
    k. Transmitter accuracy (including linearity, repeatability, and hysteresis) shall be in accordance with the following:
        1) RTD (100-ohm platinum): ±0.18 degrees F (Range: -328 to 1022 degrees F).
        2) Thermocouple (Type E): ±0.36 degrees F (Range: -58 to 1832 degrees F).
        3) Thermocouple (Type J): ±0.45 degrees F (Range: -292 to 1400 degrees F).
        4) Thermocouple (Type K): ±0.45 degrees F (Range: -292 to 2501 degrees F).
        5) Thermocouple (Type N): ±0.72 degrees F (Range: -328 to 2372 degrees F).
        6) Thermocouple (Type R): ±1.08 degrees F (Range: 32 to 3214 degrees F).
        7) Thermocouple (Type S): ±0.90 degrees F (Range: 32 to 3214 degrees F).
        8) Thermocouple (Type T): ±0.45 degrees F (Range: -328 to 752 degrees F).
        9) Millivolt input: ±0.015 mV (Range: -10 to 100 mV).
    l. Stability:
        1) Any of the following drift limits are acceptable:
            a) 0.05% of input reading plus 0.043% of span per 12 months.
            b) 0.05% of maximum span per 12 months.
    m. Ambient temperature effects (including digital, D/A conversion, and cold junction effects):
        1) Any of the following effects per 50 degrees F change are acceptable:
            a) One-half reference inaccuracy +0.18 degrees F.
            b) Effects in accordance with the following inputs:
                (1) RTD (100 platinum): ±0.0027 degrees F.
                (2) Thermocouple (Type E): ±0.0072 degrees F +0.000043% of reading.
                (3) Thermocouple (Type J):
                    (a) Readings >32 degrees F: ±0.0072 degrees F +0.00029% of reading.
                    (b) Readings <32 degrees F: ±0.0072 degrees F +0.02% of absolute value of reading.
                (4) Thermocouple (Type K):
                    (a) Readings >32 degrees F: ±0.0090 degrees F +0.00054% of reading.
                    (b) Readings <32 degrees F: ±0.0090 degrees F +0.0002% of absolute value of reading.
                (5) Thermocouple (Type N): ±0.0090 degrees F +0.00036%.
                (6) Thermocouple (Type R):
                    (a) Readings >392 degrees F: ±0.027 degrees F.

(b) Readings <392 degrees F: ±0.038DegF - 0.0032% of reading.
(7) Thermocouple (Type S):
(a) Readings >392 degrees F: ±0.027 degrees F.
(b) Readings <392 degrees F: ±0.038DegF - 0.0032% of reading
(8) Thermocouple (Type T):
(a) Readings >32 degrees F: ±0.009 degrees F.
(b) Readings <32 degrees F: ±0.009 degrees F + 0.0036% of absolute value of reading.
(9) Millivolt input: ±0.00025 Mv.

6. Schedule:

TAG NUMBER	SERVICE	INPUT TYPE	OUTPUT SIGNAL	TEMPERATURE RANGE (degrees F)
TIT-53281	RSPH Finished Water Tank	RTD	4-20 mA	0-100

2.2 ROOM TEMPERATURE AND HUMIDITY TRANSMITTERS

A. Room Temperature and Humidity Transmitters:
1. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
2. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.
3. Subject to compliance with the Contract Documents, the following manufacturers listed are acceptable:
a. Comet (T3110).
b. Or Equal.
4. Design and Fabrication:
a. Measurement: Room Temp and/or Humidity.
b. Range:
1) Temperature: -10 to 160 degrees F (-23 to 71 degrees C).
2) Humidity: 0 to 100% RH.
c. Accuracy:
1) Temperature: ±0.4 degrees F (0.2 degrees C).
2) Humidity: ±2.5% RH.
d. Output: 4-20 mA.
e. Supply Voltage: 24 VDC (9 – 30 VDC).
f. Environment Protection: IP65.
5. Schedule:

TAG NUMBER	SERVICE	MEASUREMENT (Temp, RH, Both)	WALL or OUTDOOR MOUNT
TIT-51681	RSPH Pump Room	Temp	Wall
TIT-61081	WTP Process Room	Temp	Wall
TIT-67081	WTP Acids Room	Temp	Wall
TIT-67082	WTP Base Drum Room	Temp	Wall
TIT-61082	WTP Mechanical Room	Temp	Wall

2.3 ACCESSORIES

A. Furnish all mounting brackets, hardware and appurtenances required for mounting primary elements and transmitters.
   1. Materials, unless otherwise specified, shall be as follows:
      a. Bolts, nuts, washers, expansion anchors: 316 stainless steel.
      b. Mounting brackets:
         1) Standard: Carbon steel.
         2) Corrosive areas: 316 stainless steel.
      c. Mounting plates, angles:
         1) Standard: Carbon steel.
         2) Corrosive areas: 316 stainless steel.
      d. Instrument pipe stands:
         1) Standard: carbon steel.
         2) Corrosive areas: 316 stainless steel.

B. Provide handheld communicator compatible with all intelligent transmitters furnished.
   1. Handheld communicators shall provide capability to check calibration, change transmitter range, and provide diagnostics.
   2. If these features are provided with the intelligent transmitter, the handheld communicator is not required.

C. Cable lengths between sensors and transmitters shall be continuous (without splices) and as required to accommodate locations as shown on Drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install instrument mounting pipe stands level and plumb.

C. Keep foreign matter out of the system.

D. Threaded Connection Seals:
   1. Use Tite-Seal or acceptable alternate.
   2. Use of lead base pipe dope or Teflon tape is not acceptable.
   3. Do not apply Tite-Seal to tubing threads of compression fittings.

E. Capillary Tubing:
   1. Route capillary tubing in tubing tray.
   2. Install capillary tubing with a 2 inches minimum bend radius which does not kink or pinch the capillaries.
   3. Do not cut or disconnect at any point.
   4. Coil excess capillary tubing and secure at the instrument.

F. Temperature Elements:
   1. Assemble in the following sequence:
      a. Remove temperature sensor sheaths and terminal blocks from the head and nipple assembly.
      b. Connect nipple and head to thermowell installed in the pipe.
      c. Insert sheath and terminal block until it seats in the thermowell.
      d. Connect to the head.

G. Instrument Mounting:
   1. Mount all instruments where they will be accessible from fixed ladders, platforms, or grade.
   2. Mount all local indicating instruments with face forward toward the normal operating area, within reading distance, and in the line of sight.
   3. Mount instruments level, plumb, and support rigidly.

4. Mount to provide:
    a. Protection from heat, shock, and vibrations.
    b. Accessibility for maintenance.
    c. Freedom from interference with piping, conduit and equipment.

3.2 TRAINING

A. Provide on-site training in accordance with Specification Section 01 79 00.

END OF SECTION

This page intentionally left blank.

SECTION 40 75 00
PROCESS LIQUID ANALYTICAL MEASUREMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. pH Sensor and Analyzer.
   2. Chlorine Residual (Total) Analyzer – with Reagent.
   3. Chlorine Residual (Total) Analyzer – Reagentless.
   4. Turbidity Analyzer.

B. Related Specification Sections include but are not necessarily limited to:
   1. See Section 01 79 00 – Demonstration and Training, for operator training requirements.
   2. Section 40 61 13 - Process Control Systems General Requirements.

1.2 QUALITY ASSURANCE

1. National Electrical Manufacturers Association (NEMA):
   a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).

1.3 SYSTEM DESCRIPTION

A. These instruments are integrated with other control system components specified under Specification Section 40 61 13 series to produce the functional control defined in the Contract Documents.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 61 13.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 ANALYTICAL ELEMENTS

A. pH Sensor and Analyzer:
   1. Manufacturers:
      a. Hach DPS1 sensor and sc200 controller (transmitter)
      b. No Substitutions allowed.
   2. Materials:

a. Sensor: Polyether sulfone (PES), KYNAR, (polyvinylidene fluoride (PVDF)), liquid crystal polymer, epoxy (less than 140F).
b. O-rings: VITON.

3. Design and fabrication:
   a. Sensor:
      1) Digital technology enabling calibration of sensor with a laboratory analyzer/transmitter prior to field installation of sensor with the field-mounted analyzer/transmitter.
      2) Integral pre-amplifier to convert high impedance pH into stable, noise-free signal.
      3) Automatic temperature compensation.
      4) Shielded cable.
      5) Sensitivity: 0.01 pH
      6) Range: See schedule below.
   b. Hach SC200 Transmitter:
      1) Local digital display.
         a) LCD or LED.
         b) Minimum 3-1/2 digit.
      2) Current output: 2 isolated 4-20 mA DC.
         a) One for pH, one for temperature.
      3) Capable of HART communication.
      4) Power supply: 120VAC.
      5) Ambient temperature: 0: 0 to 122 DEGF.
      6) Ambient humidity: 0: 0-95%.
      7) Accuracy: +: +0.01 pH
      8) Repeatability: +: +0.1 pH
      9) Temperature compensated: 32 to 212 DEGF.
      10) Temperature coefficient:
         a) Input: +: +0.0017 pH/DEGF.
         b) Output: +: +0.0033 pH/DEGF.

4. Schedule

TAG NUMBER	SERVICE	MEASUREMENT RANGE (pH)	NEMA RATING
AE/AIT-53282	RSPH Finished Water Tank	5-10	4X
AE/AIT-62081	WTP RW Instrumentation Panel	5-10	4X
AE/AIT-64081	WTP UF Feed Instrumentation Panel	5-10	4X
AE/AIT-66082	WTP FW Instrumentation Panel	5-10	4X
AE/AIT-68181	WTP CT Tank Influent Instrumentation Panel	5-10	4X
AE/AIT-66084	WTP Distribution Water Instrumentation Panel	5-10	4X

**Temperature sensors are integral with pH sensors.

B. Chlorine Residual Analyzers – with Reagent:
   1. Manufacturers:
      a. Hach CL17 with sc4500 Controller.
      b. No Substitutions Allowed.
   2. Materials:
      a. Enclosure: ABS plastic with clear polycarbonate windows.

3. Design and fabrication:
    a. Utilize colorimeter with a seal-free, solid state mixing system, including self-cleaning stir bar.
    b. Amperometry method is also acceptable with pH compensation.
    c. Accuracy: ±5% or ±0.035 MG/L as Cl_2, whichever is greater.
    d. Operate from 120 VAC power.
    e. Provided with amplifiers, signal converting circuitry, range and zero adjustment, all with coupling and voltage regulation circuitry as required to enable chlorine residual transmitter to meet criteria specified.
    f. Provided with one-month supply of reagents.
4. SC4500 Controller
    1) The controller shall be a microprocessor-based instrument.
    2) Connections between the sensors and the controller shall be "plug and play."
    3) The Interface unit shall allow operators to control sensors and interface functions with menu-driven software.
    4) The interface unit shall have a built-in data logger with the capacity to store data for 15-minute intervals for up to 6 months with two sensors per controller.
    5) The interface unit shall include two analog 4-20 mA outputs and 3 SPDT form 'C' alarm contacts.
    6) The interface unit shall include two independent PID control functions.
    7) The interface unit shall be housed in a NEMA-4X/IP66 metal enclosure with corrosion-resistant finish.
    8) The controller shall be mounted horizontal or vertical on surface, panel, or pipe.
    9) Power supply: 120VAC.
5. Schedule:

TAG NUMBER	SERVICE	MEASUREMENT RANGE (mg/L)	NEMA RATING
AE/AIT-64082	WTP UF Feed Instrumentation Panel	0-5.0	4X
AE/AIT-66081	WTP FW Instrumentation Panel	0-5.0	4X
AE/AIT-68182	WTP CT Tank Influent Instrumentation Panel	0-5.0	4X
AE/AIT-66085	WTP Distribution Water Instrumentation Panel	0-5.0	4X

C. Chlorine Residual Analyzers – Reagentless:
1. Manufacturers
    a. Hach CL10Fsc with sc4500 Controller.
    b. No Substitutions Allowed.
2. Materials:
    a. Corrosion-resistant materials (stainless steel, PVC, silicon rubber and polycarbonate).
3. Design:
    a. Measurement Range: 0-20 ppm.
    b. Accuracy:
        1) Free Chlorine:
            a) ±3% of the reference test at constant pH less than 7.2 (±0.2 pH unit)
            b) ±10% of the reference test at stable pH less than 8.5 (±0.5 pH unit from the pH at calibration)
    c. Operating Temperature: 0 to 45-deg C.
4. Power Requirements:
    a. Sensor: 12 VDC, 30 mA maximum.

b. Panel: 12 VDC ±10%, at 100 mA maximum.

5. SC4500 Controller
    1) The controller shall be a microprocessor-based instrument.
    2) Connections between the sensors and the controller shall be “plug and play.”
    3) The Interface unit shall allow operators to control sensors and interface functions with menu-driven software.
    4) The interface unit shall have a built-in data logger with the capacity to store data for 15-minute intervals for up to 6 months with two sensors per controller.
    5) The interface unit shall include two analog 4-20 mA outputs and 3 SPDT form ‘C’ alarm contacts.
    6) The interface unit shall include two independent PID control functions.
    7) The interface unit shall be housed in a NEMA-4X/IP66 metal enclosure with corrosion-resistant finish.
    8) The controller shall be mounted horizontal or vertical on surface, panel, or pipe.
    9) Power supply: 120VAC.
6. Schedule:

TAG NUMBER	SERVICE	MEASUREMENT RANGE (mg/L)	NEMA RATING
AE/AIT-53281	RSPH Finished Water Tank	0-20	4X

D. Turbidity Analyzer:
1. Manufacturers:
    a. Roaring Spring Pumphouse:
        1) Hach Surface Scatter 7sc with sc4500 Controller
    b. North Rim WTP:
        1) TU5300sc with sc200 Controller.
    c. No Substitutions Allowed.
2. Surface Scatter 7sc Design:
    a. Design and Fabrication:
        1) Type: Continuous monitoring system of particulate turbidity. The system uses the Nephelometric method of measurement.
        2) Performance:
            a) Accuracy: ± 5 percent of reading or ± 0.1 NTU (whichever is greater) from 0 - 2000 NTU. ±10 percent of reading from 2000 to 9999 NTU.
            b) Repeatability: Better than ± 1.0 percent of reading or ± 0.04 NTU, whichever is greater for each range.
            c) Response Time: Initial response in 45 seconds.
            d) Resolution: <1000 NTU: 0.01 NTU. 1000 – 9999.9 NTU: 0.1 NTU.
            e) Sample Flow Rate: 15 to 30 gph
            f) Sample Temperature Range: 32 - 122 °F
            g) Output: 4- 20 mA DC programmable over any portion of the 0.01 – 9999.9 NTU range.
    b. Construction Features:
        1) Turbidimeter Sensor:
            a) Optical and hydraulic components housed in turbidimeter body..
            b) Body is constructed of corrosion-resistant structural plastic.
            c) Fittings: Match tubing size to equipment port size.
            d) Mounting: panel.
    c. Hach SC4500 Controller:
        1) The controller shall be a microprocessor-based instrument.
        2) Connections between the sensors and the controller shall be “plug and play.”
        3) The Interface unit shall allow operators to control sensors and interface functions with menu-driven software.

4) The interface unit shall have a built-in data logger with the capacity to store data for 15-minute intervals for up to 6 months with two sensors per controller.
5) The interface unit shall include two analog 4-20 mA outputs and 3 SPDT form 'C' alarm contacts.
6) The interface unit shall include two independent PID control functions.
7) The interface unit shall be housed in a NEMA-4X/IP66 metal enclosure with corrosion-resistant finish.
8) The controller shall be mounted horizontal or vertical on surface, panel, or pipe.
9) Power supply: 120VAC.

3. TU5300sc Design and Fabrication:
   a. Type: Continuous monitoring system of low levels of particulate turbidity. The system uses the Nephelometric method of measurement. The system consists of turbidimeter and a control unit.
   b. Performance:
      1) Accuracy: ± 2 percent or 0.01 NTU (the larger value) from 0 - 40 NTU, ±10 percent of reading from–40 - 700 NTU
      2) Repeatability: 0.002 NTU or 1 percent
      3) Response Time: < 30 seconds at 100 mL/min
      4) Resolution: 0.0001 NTU(0.0001 to 0.9999/1.000 to 9.999/10.00 to 99.99/100.0 to 700 NTU)
      5) Sample Flow Rate: 100 to 1000 mL/min; optimal flow rate 200 to 500mL/min
      6) Sample Temperature Range: 2 to 60 °C (36 to 140 °F)
      7) Output: 4- 20 mA DC programmable over any portion of the 0-100 NTU range.
   c. Construction Features:
      1) Turbidimeter Sensor:
         a) Optical and hydraulic components housed in turbidimeter body. Optical components mounted on a head assembly for easy removal for servicing without disturbing sample flow.
         b) Body is constructed of corrosion-resistant structural plastic.
         c) Tubing Requirements: ¼-inch OD.
         d) Mounting: wall.
   d. Hach SC200 Transmitter:
      1) Local digital display.
         a) LCD or LED.
         b) Minimum 3-1/2 digit.
      2) Current output: 2: 2 isolated 4-20 mA DC.
      3) Power supply: 120VAC.
      4) Ambient temperature: 0: 0 to 122 DEGF.
      5) Ambient humidity: 0: 0-95%.
   e. Temperature compensated: 32: 32 to 212 DEGF.
   f. General Requirements
      1) Automatic Cleaning Module.
      2) Accessories: Include sufficient quantities for a one-year operation assuming routine calibration. Include calibration cylinder, formazin calibration kit, and bubble trap extraction tool.
4. Schedule:

TAG NUMBER	SERVICE	MEASUREMENT RANGE (NTU)	NEMA RATING	MODEL
AE/AIT-52381	RSPH Raw Water	0.01-9999.9	4X	Surface Scatter 7sc
AE/AIT-62082	WTP RW Instrumentation Panel	0 - 700	4X	TU5300sc
AE/AIT-64083	WTP Combined Filtrate Instrumentation Panel	0 - 700	4X	TU5300sc
AE/AIT-66083	WTP FW Instrumentation Panel	0 - 700	4X	TU5300sc
AE/AIT-66086	WTP Distribution Water Instrumentation Panel	0 - 700	4X	TU5300sc

2.3 ACCESSORIES

A. Furnish all mounting brackets, hardware and appurtenances required for mounting primary elements and transmitters.
   1. Materials, unless otherwise specified, shall be as follows:
      a. Bolts, nuts, washers, expansion anchors: 316 stainless steel.
      b. Mounting brackets:
         1) Standard: 316 stainless steel.
         2) Highly corrosive areas: 316 stainless steel.
      c. Mounting plates, angles:
         1) Standard: Carbon steel.
         2) Corrosive areas: 316 stainless steel.
      d. Instrument pipe stands:
         1) Standard: Hot dip galvanized 2 IN schedule 40, ASTM A106, Grade B carbon steel.
         2) Corrosive areas: 316 stainless steel.
      e. Sensor Mounting: Per schedule and Contract Documents. Provide the following mounting brackets:
         1) Hach Pole Mount Assembly for 1" NPT Sensors # 9253000
         2) Hach Float Mount Assembly for 1" NPT Sensors # 9253100

B. Tubing Support Angles and Brackets:
   1. Any of the following materials are acceptable:
      a. Aluminum support with dielectric material between support and tubing.
      b. Type 316 stainless steel.
      c. Fiberglass.

C. Tubing Tray or Channel:
   1. Aluminum.
   2. Provide dielectric material between tray or channel and tubing.

D. Provide handheld communicator compatible with all intelligent transmitters furnished.
   1. Handheld communicators shall provide capability to check calibration, change transmitter range, and provide diagnostics.
   2. If these features are provided with the intelligent transmitter, the handheld communicator is not required (provided the transmitter is easily accessible by an operator without the need for a ladder/step stool).

E. Cable lengths between sensors and transmitters shall be continuous (without splices) and as required to accommodate locations as shown on Drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install instrument mounting pipe stands level and plumb.

C. Instrument Mounting:
   1. Mount all instruments where they will be accessible from fixed ladders, platforms, or grade.
   2. Mount all local indicating instruments with face forward toward the normal operating area, within reading distance, and in the line of sight.
   3. Mount instruments level, plumb, and support rigidly.
   4. Mount to provide:
      a. Protection from heat, shock, and vibrations.
      b. Accessibility for maintenance.
      c. Freedom from interference with piping, conduit and equipment.

3.2 TRAINING

A. Provide on-site training in accordance with Specification Section 01 75 00.

END OF SECTION

This page intentionally left blank.

SECTION 40 76 00
PROCESS GAS ANALYTICAL MEASUREMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Analytical components.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 40 61 13 - Process Control System General Requirements.
   2. Section 40 61 96 - Process Control Descriptions.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. A106, Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service.

1.3 SYSTEM DESCRIPTION

A. These instruments are integrated with other control system components specified under Specification Section 40 61 13 to produce the functional control defined in the Contract Documents.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 40 61 13.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 ANALYTICAL ELEMENTS

A. Combustible and Toxic Gas Detectors:
   1. Acceptable manufacturers:
      a. MSA Instruments.
      b. Or Equal.
   2. Control unit – Gas Monitoring Control Panel:
      a. Front mounted indication.

1) Minimum three-digit display of gas concentration associated with each sensor.
2) Alarm status indicators for each gas sensing channel:
a) Trouble.
b) High gas level detected.
c) High-high gas level detected.

b. Alarm relay outputs:
1) Separate contacts for each alarm or trouble condition associated with each gas sensing channel.
2) Separate "system trouble" contact to indicate trouble in the event any of the following conditions are true:
a) System power loss.
b) Signal loss from any sensor.
c) Signal out of appropriate range.
d) Control module malfunction or removal.
3) Each output contact shall be Form C, DPDT, rated for 3 amps resistive at 120 VAC.

c. Output signals:
1) 4-20 mA signal representing gas concentration for each gas sensor (up to 6 sensors).
2) Relay signals for Go-No-Go panels and Horn/Strobe panels.

d. Temperature range: 32 to 140 degrees F.

e. Relative humidity range: 0-95% non-condensing.

3. Sensor and transmitter design and fabrication:
a. Sensor mounting type shall be as indicated on schedule: Either diffusion mounted, duct mounted, or sample draw mounted.

b. For sensors required by schedule to be sample draw type:
1) Provide a compressed air aspirator or motorized pump to draw a sample past the sensor.
2) Utilize a flow switch to provide annunciation of low sample flow rate to the sensor.

c. Duct mounted gas sensor shall be able to monitor gas flow rates up to 85 fps.

d. LEL gas sensor shall be catalytic bead type with demonstrated resistance to poisoning by silicones and hydrogen sulfide gases.

e. Toxic gas sensor shall be the electrochemical type and shall not require the periodic addition of reagents.

f. Interconnect wiring from sensor to transmitter (if not integral) or control unit shall be 3-wire shielded cable.

g. Sensing element shall have minimum useful life of one year.

h. Transmitter output: 4-20 mA signal proportional to measured gas level.
1) Capable of driving 600-ohm load at 24 VDC supply voltage.

i. Accuracy:
1) LEL gas detection:
a) +3% LEL to 50% full scale.
b) +5% LEL, 50 to 100% full scale.
2) Toxic gas detection:
a) +10% full scale or 2 PPM, whichever is greater.

j. Environmental:
1) Ambient operating temperature: -40 to 158 degrees F.
2) Relative humidity: 0-95% non-condensing.

k. Housing: In accordance with the area classification shown on Drawings.

l. Provide nonintrusive means of calibration.

m. Local displays:
1) 3-1/2digit LCD or LED display of measured gas level.
2) Fault LED.

n. Standalone sensors and transmitters (without central control unit):

1) Provide relay contacts rated at 1/2 amps at 120 VAC for each of the following conditions:
    a) High gas level (warning level).
    b) High-high gas level (alarm level).
    c) Sensor fault condition.

o. Relay contacts shall be normally energized (normally closed); contacts shall open in the event of a warning, alarm or trouble condition.

p. Minimum detector response time when exposed to 100% LEL gas concentration:
    1) 10 seconds to 50% LEL.
    2) 30 seconds to 90% LEL.

q. Store calibration data in nonvolatile memory or back up with battery.

r. Safety approvals:
    1) CSA Certification.
    2) FM approved.

4. Automatic calibration:
    a. Provide automatic calibration capabilities with the following features:
        1) Capability to automatically calibrate gas sensor once a day, once a month, or anytime in between.
        2) Status and fault indication.
        3) Relay contacts to provide alarm signal in the event of low calibration gas.
        4) Environmental:
            a) Ambient operating temperature: -4 to 122 degrees F.
            b) Relative humidity: 15-95% non-condensing.
        5) During a fault condition, the sensor uses previous calibration parameters.
        6) During calibration cycle, the sensor alarms are disabled to prevent false alarms or activation of equipment connected to the relays.
5. Provide one calibration kit for each type of gas monitored.
    a. Calibration kits shall be furnished completely with all tubing, regulators, fittings, communication devices, and accessories required to calibrate sensors.
    b. Calibration kits shall utilize nonintrusive means of calibrating sensors/transmitters.
6. Provide two full cylinders for each type of calibration check gas.
    a. Cylinder size: 17 liters.
7. Provide the same quantity of zero air cylinders as the total required number of calibration check gas cylinders (of all types).
8. Schedule:

TAG NO	SERVICE	GAS	MOUNT TYPE	RANGE	SETPOINTS	
					WARN (HI)	ALARM (HI HI)
AE/AIT-61881	WTP Chemical Storage Room	Sulfur Dioxide	Wall	0-25ppm	HIGH	HIGH-HIGH

2.3 ACCESSORIES

A. Furnish all mounting brackets, hardware and appurtenances required for mounting primary elements and transmitters.
    1. Materials, unless otherwise specified, shall be as follows:
        a. Bolts, nuts, washers, expansion anchors: 316 stainless steel.
        b. Mounting brackets:
            1) Standard: 316 stainless steel.
            2) Highly corrosive areas: Aluminum.
        c. Mounting plates, angles:
            1) Standard: Carbon steel.
            2) Corrosive areas: 316 stainless steel.
        d. Instrument pipe stands:

1) Standard: Hot dip galvanized 2 inches schedule 40, ASTM A106, Grade B carbon steel.
2) Corrosive areas: 316 stainless steel.

B. Cable lengths between sensors and transmitters shall be continuous (without splices) and as required to accommodate locations as shown on Drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Instrument Mounting:
1. Mount all instruments where they will be accessible from fixed ladders, platforms, or grade.
2. Mount all local indicating instruments with face forward toward the normal operating area, within reading distance, and in the line of sight.
3. Mount instruments level, plumb, and support rigidly.
4. Mount to provide:
   a. Protection from heat, shock, and vibrations.
   b. Accessibility for maintenance.
   c. Freedom from interference with piping, conduit and equipment.

3.2 TRAINING

A. Provide on-site training in accordance with Specification Section 01 75 00.

END OF SECTION

SECTION 40 78 00
PANEL MOUNTED INSTRUMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Pilot devices:
      a. Selector switches.
      b. Pushbuttons.
      c. Indicating lights.
   2. Relays/timers:
      a. Control relay.
      b. Time delay relays.
   3. Termination equipment:
      a. Terminal blocks.
      b. Fuse holders.
   4. Power supplies:
      a. DC power supplies.
      b. Isolation transformers.
   5. Voltage surge protection devices.
   6. Running time indicator.
   7. Intrinsic Safety Isolators.
   8. Alarm Beacons.
   9. Horns.
   10. Intrusion Switches.
   11. Level Indicating Controller.

B. Related Sections include but are not necessarily limited to:
   1. Section 26 43 13 - Surge Protection Devices for Low-Voltage Electrical Power Circuits.
   2. Section 40 61 13 - Process Control Systems General Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. The International Society of Automation (ISA):
      a. S18.1, Annunciator Sequences and Specifications.
      b. ANSI/ISA-12.02.02-2014, Recommendations for the Preparation, Content, and Organization of Intrinsic Safety Control Drawings.
   2. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
      b. ICS 2, Industrial Control and Systems: Controllers, Contactors, and Overload Relays Rated 600 Volts.

B. Miscellaneous:
   1. Assure units comply with electrical area classifications and NEMA enclosure type shown on Drawings.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Control Drawings for intrinsically safe systems:
      a. Print on 8-1/2 x 11 IN sheets.
      b. In accordance with recommendations of ANSI/ISA-12.02.02-2014.
      c. One control drawing per sheet.

d. Identify model numbers of both the associated apparatus and the intrinsically safe apparatus.
e. Include wiring diagram showing interconnections of the intrinsically safe apparatus and the associated apparatus.
f. Provide entity parameters for both the associated apparatus and the intrinsically safe apparatus.
g. Identify line of demarcation between classified (hazardous) and unclassified (nonhazardous) locations and identify equipment that is installed in each location.
h. Identify hazardous areas by class, groups, and divisions.
i. Show maximum nonhazardous location voltage that may be used with the associated apparatus.
j. Include any specific conditions that are necessary to maintain the intrinsic safety protection.

B. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers listed in the subsections are acceptable.

2.2 PILOT DEVICES

A. Selector Switches:
1. Manufacturers:
a. Eaton.
b. Or Equal.
2. Design and fabrication:
a. Heavy-duty type.
b. Oiltight.
c. Rotary cam units conform to NEMA ICS 2-216.22.
d. Mounting hole: 30.5 MM (Unless otherwise Mounted)
e. Supply switches have number of positions required with contact blocks to fulfill functions shown and specified. At a minimum provide DPDT contact blocks with two normally open (NO) and two normally closed (NC) contacts.
f. UL listed.
g. Maintained contact type.
h. Lever type operators.
i. Black colored operators.
j. Designed with cam and contact block with approximate area of 2 IN SQ.
k. Legend plate marked per Contract Documents.
l. Contact block requirements:
1) Dry and indoor locations: Standard contact blocks rated for 10 A continuous current.
2) Wet or outside locations: Hermetically sealed contact blocks.

3. DIN Rail Mounting Bracket
    a. As Required on Plans
    b. Mounting hole: 22 MM
    c. Manufacturers:
        1) Eaton.
        2) Or Equal.

B. Pushbuttons:
1. Manufacturers:
    a. Eaton.
    b. Or Equal.
2. Materials:
    a. Backing diaphragm: Buna-N.
3. Design and fabrication:
    a. Heavy-duty type.
    b. Oiltight.
    c. Conforming to NEMA ICS 2-216.22.
    d. Mounting hole: 30.5 MM.
    e. Diaphragm backed.
    f. UL listed.
    g. Emergency stop pushbuttons to have mushroom head operator and two sets of maintained normally open (NO) and normally closed (NC) contacts. Emergency stop pushbutton shall be NEMA 4X rated. E-stops shall be push to engage and twist and pull to release.
    h. Non-illuminated type:
        1) Momentary contact with necessary contact blocks.
        2) Molded, solid color melamine buttons.
        3) Standard flush operators with half shroud.
        4) Green colored buttons for START or ON and red color for STOP or OFF.
        5) Appropriate contact blocks to fulfill functions shown or specified. At a minimum provide DPDT contact blocks with two normally open (NO) and two normally closed (NC) contacts.
    i. Contact block requirements:
        1) Dry and indoor locations: Standard contact blocks rated for 10 A continuous current.
        2) Wet or outside locations: Hermetically sealed contact blocks.
        3) Legend plate marked per Contract Documents.
    j. Illuminating type:
        1) Momentary contact with necessary contact blocks.
        2) Serves as both pushbutton control and indicating light.
        3) Green colored lenses for start or on and red for STOP or OFF.
        4) Resistor-type full voltage light unit with lens and panel gasket.
        5) Legend plate marked per Contract Documents.
        6) Appropriate contact blocks to fulfill functions shown or specified.

C. Indicating Lights:
1. Manufacturers:
    a. Eaton.
    b. Or Equal.
2. Design and fabrication:
    a. Heavy duty.
    b. Oiltight.
    c. Type allowing replacement of bulb without removal from control panel.
    d. LED.
    e. UL listed.

f. 24 V lamp.
g. Legend plate marked per Contract Documents.
h. Nominal 2 IN SQ face.
i. Mounting hole: 30.5 MM.
j. Push-to-test indicating lights.
k. Plastic lens.
l. Color code lights are as follows:
    1) Green: ON or running; valve open.
    2) Amber: Standby; auto mode; ready.
    3) Red: OFF or stopped; valve closed.
    4) White: Power ON.

2.3 RELAYS/TIMERS

A. Control Relays:
1. Manufacturers:
    a. Magnacraft/ Schneider 782 series with 70-782EL11-1 socket
    b. Or Equal.
2. Design and fabrication:
    a. Plug-in general purpose ice cube relay.
    b. Blade connector type.
    c. Switching capacity: 10 A.
    d. Contact material: Silver cadmium oxide.
    e. Provide relays with a minimum of 4 DPDT contacts.
    f. Coil voltage: 24 VDC.
    g. Relay sockets are DIN rail mounted.
    h. Internal neon or LED indicator is lit when coil is energized.
    i. Clear polycarbonate dust cover with clip fastener.
    j. Check button.
    k. Temperature rise:
        1) Coil: 85 DEGF max.
        2) Contact: 65 DEGF max.
    l. Insulation resistance: 100 Meg min.
    m. Frequency response: 1800 operations/hour.
    n. Operating temperature: -20 to +150 DEGF.
    o. Life expectancy:
        1) Electrical: 500,000 operations or more.
        2) Mechanical: 50,000,000 operations or more.
    p. UL listed.
    q. Socket
        1) DIN rail mounted
        2) 300 VAC, 10 A rating
        3) Shall separate coil terminals on one side and form C contacts.
        4) Terminations shall be finger safe and form c contacts in an elevated step configuration.

B. Time Delay Relays:
1. Manufacturers:
    a. Rockwell Automation (Allen Bradley) Bulletin 700-HNC.
    b. Or Equal.
2. Design and fabrication:
    a. Melt design test and performance requirements of NEMA ICS 2-218.
    b. Heavy-duty.
    c. Solid-state construction.
    d. DPDT contacts.
    e. External adjusting dial.
    f. Auxiliary relays as required to perform functions specified or shown on Drawings.

g. Operates on 117 VAC (±10%) power source.
h. Contact rating: A150 per NEMA ICS 2-125.
i. Furnish with "on" and "timing out" indicators.

2.4 TERMINATION EQUIPMENT

A. Terminal Blocks:
1. Manufacturers:
a. Phoenix Contact UT-4.
b. Or Equal.
2. Design and fabrication:
a. Modular type feed-through with screw compression clamp.
b. Screws: Stainless steel.
c. Current bar: Nickel-plated Copper Alloy.
d. Thermoplastic insulation rated at -40 to +90 DEGC.
e. Wire insertion area: Funnel-shaped to guide all conductor strands into terminal.
f. Install end sections and end stop at each end of terminal strip.
g. Install machine-printed terminal markers on both sides of block.
h. Spacing: 6 MM.
i. Wire size: 22-12 AWG.
j. Rated voltage: 600 V.
k. Din rail mounting.
l. UL listed.
3. Standard-type block:
a. Rated current: 30 A.
b. Color: Gray body.
4. Bladed-type block:
a. Terminal block with knife blade disconnects which connect or isolate the two (2) sides of the block.
b. Rated current: 10 A.
c. Color:
1) Panel control voltage leaves enclosure - normal: Gray body, orange switch.
2) Foreign voltage entering enclosure: Orange body, orange switch.
5. Grounded-type block:
a. Electrically grounded to mounting rail.
b. Use to terminal ground wires and analog cable shields.
c. Color: Green and yellow body.

B. Fuse Holders:
1. Manufacturers:
a. Phoenix Contact.
b. Or Equal.
2. Design and fabrication:
a. Modular-Type with screw compression clamp.
b. Screws: Stainless steel.
c. Current bar: Nickel-plated copper alloy.
d. Thermoplastic insulation rated as -40 to +105 DEGC.
e. Wire insertion area: Funnel-shaped to guide all conductor strands into terminal.
f. Blocks can be ganged for multi-pole operation.
g. Install end sections and end stop at each end of terminal strip.
h. Install machine-printed terminal markers on both sides of block.
i. Spacing: 9.1 MM.
j. Wire size: 30-12 AWG.
k. Rated voltage: 300 V.
l. Rated current: 12 A.
m. Fuse size: 1/4 x 1-1/4.
n. Blown fuse indication.

o. DIN rail mounting.
p. UL listed.

2.5 POWER SUPPLIES

A. DC Power Supplies:
1. Manufacturers:
a. Sola Hevi-Duty.
b. Or Equal.
2. Design and fabrication:
a. Converts 120 VAC input to DC power at required voltage.
b. DIN rail mount with enclosure (i.e., not open frame).
c. Switching type.
d. Provide redundant 24VDC modules with diode redundancy module for automatic switchover to standby module on failure of primary module.
e. Hardwire module fault dry contact to associated PLC input for alarm at Plant SCADA.
f. AC input: 120 VAC ±15%, nominal 60 Hz.
g. Efficiency: Minimum 86%.
h. Rated mean time between failure (MTBF): 500,000 HRS.
i. Voltage regulation:
1) Static: Less than 1.0% V_{out}.
2) Dynamic: ±2% V_{out} overall.
j. Output ripple/noise: Less than 100 mV peak to peak (20 MHz).
k. Overload, short circuit and open circuit protection.
l. Temperature rating: 0 to 60 DEGC full rated, derated linearly to 50% at 70 DEGC.
m. Humidity rating: Up to 90%, non-condensing.
n. LED status indication for DC power.
o. UL listed.
3. Redundancy Module: Required
4. Sizing: 125% of maximum load per power supply.

B. Isolation Transformers:
1. Manufacturers:
a. Topaz Noise Suppressor Noise Isolator.
b. Or Equal.
2. Design and fabrication:
a. Protects sensitive electronic equipment from electrical noise.
b. Common-mode noise attenuation: 146 dB at 0.0005 pF coupling capacitance.
c. Normal-mode attenuation: 60 dB
d. Input voltage range: ±10% of rated.
e. Regulation: 3.5% or less from full-load to no-load.
f. Dielectric strength: 2,500 VAC minimum.
g. Harmonic distortion: 1% maximum.
h. Electromagnetic interference: 0-1 gauss maximum at 18 IN.
i. UL listed.

2.6 VOLTAGE SURGE PROTECTION DEVICES

A. See Specification Section 26 43 13.

2.7 RUNNING TIME INDICATORS

A. Acceptable Manufacturer:
1. Eagle Signal Controls.
2. Or Equal.

B. Design and Fabrication:
1. Six-digit wheels include a 1/10 digit.
2. Non-reset type.

3. Time ranges in hours.
4. Automatic recycling at zero.
5. Accuracy: 1%.
6. Sealed against dirt and moisture.
7. Tamper-proof.

2.8 INTRINSIC SAFETY ISOLATORS

A. Manufacturers:
1. Phoenix Contact.
2. Or Equal.

B. Design and fabrication:
1. Utilize in all circuits is required to be intrinsically safe.
2. All instruments installed in hazardous locations per Section 40 67 00 shall be provided with intrinsic safety isolators.
3. DIN rail mountable.
4. Provide galvanic isolation; use of passive Zener diode type barriers are not permitted.
5. Limit voltage and current so that no spark or thermal effect can cause a potentially explosive atmosphere to ignite.
6. Provide redundant 24Vdc power supplies in accordance with Paragraph 2.5 A of this Specification. Power supplies shall be installed in the same enclosure as the intrinsic safety isolators. Power supplies shall be sized based on the active load of the associated loop powered field instruments.
7. Furnish and install in accordance with requirements of NEC Article 504, Intrinsically Safe Systems.
8. Certified to be in accordance with latest edition of UL 913 for application.
9. Operating temperature: 0 to 122 DEGF.

2.9 ALARM BEACONS

A. Manufacturers:
1. Edwards Signaling.
2. Or Equal.

B. Design and fabrication:
1. LED beacons with snap-on high impact polycarbonate base.
2. NEMA 4X or NEMA 7 enclosure.
3. 24 VDC powered.
4. Beacons shall be suitable for indoor and outdoor installations and shall be panel or conduit mountable.
5. Beacon housing shall be capable of withstanding vibration.
6. Dual-mode design which shall be field adjustable between steady-on OR flashing mode by changing a dipswitch inside the beacon housing.
7. Shatter resistant polycarbonate lens. Lens color shall be shown on the Drawings.
8. For outdoor locations, provide manufacturer supplied gasket kit Edwards Signaling cat. no. GSK-KIT for weatherproofing.
9. For wall mount installations, provide manufacturers supply wall mount bracket Edwards Signaling cat. no. WBR.
10. For panel mount installations, provide manufacturers supply mounting accessories.
11. UL listed.

2.10 HORNS

A. Manufacturers:
1. Edwards Signaling.
2. Or Equal.

B. Design and fabrication:
1. PLC compatible model.

2. NEMA 7 enclosure.
3. Volume adjustable.
4. Coded or sustained tones.
5. Shall be completely assembled. NO site assembly required.
6. Corrosion resistant finish.
7. Designed for semi-flush panel mount.
8. Manufacturers supplied mounting accessories.
9. Minimum 100 dBA sound at 10 FT.
10. 120 VAC powered.
11. Beacons shall be suitable for indoor and outdoor installations and shall be panel or conduit mountable.
12. Beacon housing shall be capable of withstanding vibration.
13. Dual-mode design which shall be field adjustable between steady-on OR flashing mode by changing a dipswitch inside the beacon housing.
14. Shatter resistant polycarbonate lens. Lens color shall be as shown on the Drawings.
15. For outdoor locations, provide manufacturer supplied gasket kit Edwards Signaling cat. no. GSK-KIT for weatherproofing.
16. For wall mount installations, provide manufacturers supply wall mount bracket Edwards Signaling cat. no. WBR.
17. For panel mount installations, provide manufacturers supply mounting accessories.
18. UL listed.

2.11 INTRUSION SWITCHES

A. Manufacturers:
1. GE Interlogix
2. Or Equal.

B. Design and fabrications:
1. Enclosure: NEMA 4X, IP 67 or NEMA 7.
2. Hermetically sealed reed switch, encapsulated in polyurethane.
3. Response time: 1msec.
4. Must be compatible with door material.
5. Polarity-sensitive with reference to magnet direction.
6. Form C.
7. 24VDC.
8. UL Listed.
9. Temperature range: -40 °F to 150 °F
10. Provide as shown on drawings.

2.12 LEVEL INDICATING CONTROLLER (LIC)

A. Manufacturers:
1. Precision Digital PD6001
2. Or Equal.

B. Design and fabrications:
1. Provide LIC for LCP-53117 enclosure.
2. Enclosure: NEMA 4X.
3. UL listed.
4. 24VDC power.
5. Level Input 4-20mA loop powered.
6. Level Output 4-20mA.
7. Minimum of 2 Setpoint Replays with NO and NC (SPDT) contacts.
8. LCD panel.
9. Mounted on control front panel enclosure door.
10. Display in Feet and Inches.

11. Modbus TCP/IP communication protocol. Provide M-Link gateway and Modbus RTU to TCP/IP gateway to change to this communication protocol.

C. Provide 1 Spare LIC for LCP-53117. Preprogram Spare LIC with same parameters once SAT is complete.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Preprogram spare LIC with same parameters once SAT is complete.

END OF SECTION

This page intentionally left blank.

DIVISION 41

MATERIAL PROCESSING AND HANDLING EQUIPMENT

This page intentionally left blank.

SECTION 41 22 23
HOISTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Hoists.
2. Trolleys.
3. Monorails and stops.
4. Bridge and trolley electrification systems.
5. Appurtenances.

B. Related Requirements: Include, but are not necessarily limited to:
1. Section 01 61 03 - Common Work Results for Equipment.
2. Section 05 12 00 - Structural Steel.
3. Section 10 14 00 - Identification Devices.

1.2 REFERENCES

A. Reference Standards: Standards referenced in this section include, but are not necessarily limited to, the following:
1. American Gear Manufacturers Association (AGMA):
   a. 6013-A - Standard for Industrial Enclosed Gear Drives.
2. American Institute of Steel Construction (AISC):
   a. 303 - Code of Standard Practice for Steel Buildings and Bridges.
   b. 360 - Specification for Structural Steel Buildings.
3. American National Standards Institute (ANSI):
   a. MH 27.1 - Specifications for Patented Track Underhung Cranes and Monorail Systems.
4. American Society of Mechanical Engineers (ASME):
   a. B30.11 - Monorails and Underhung Cranes -Safety Standard for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings.
   b. B30.16 - Overhead Underhung and Stationary Hoists.
   c. HST-1 - Performance Standard for Electric Chain Hoists.
   d. HST-2 - Performance Standard For Hand Chain Manually Operated Chain Hoists.
   e. HST-4 - Performance Standard for Overhead Electric Wire Rope Hoists.
5. American Welding Society (AWS):
   a. D1.1/D1.1M - Structural Welding Code – Steel.
   b. D14.1/D14.1M - Specification for Welding of Industrial and Mill Cranes and other Material Handling Equipment.
6. ASTM International (ASTM):
   a. A36/A36M - Specification for Carbon Structural Steel.
   b. A563/A563M - Standard Specification for Carbon and Alloy Steel Nuts (Inch and Metric).
   c. A572/A572M - Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel.
   d. F436/F436M - Standard Specification for Hardened Steel Washers Inch and Metric Dimensions.
   e. F3125/F3125M - Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength.
7. National Fire Protection Association:
   a. NFPA 820 - Standard for Fire Protection in Wastewater Treatment and Collection Facilities.

8. Research Council on Structural Connections (RCSC), also known as the Bolt Council:
    a. Specification for Structural Joints Using ASTM A325 or A490 Bolts.

1.3 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
1. Coordinate hoisting systems required under this Section with other construction and equipment in the same space as hoisting systems. Hoisting systems shall have free travel, while hoist hook is retracted to maximum extent, across entire run of trolley travel (when trolleys are required), without impacting or conflicting with other items, unless expressly shown or indicated as occupying the same space as hoisting system components (including hook) in the Contract Documents. Unless otherwise shown or indicated, hoisting systems, including hooks when chain or wire rope is retracted to maximum extent possible, that will impact or conflict with other equipment or construction are unacceptable.
2. When Contractor is required to provide equipment or other item, for which hoisting system will be used for Government's or facility manager's (if other than Government) operation and maintenance of the facility, coordinate lifting requirements of such items so that they can be properly hoisted using hoisting system at final, installed location.
3. Review installation procedures under other Specifications and coordinate installation of items that will be installed with, or before, hoisting system Work.
4. Notify other contractors, if any, whether on the Project or other projects, in advance of installing hoisting system Work to provide other contractors with sufficient time to install items included in their contracts that will be installed with, or before, hoisting system Work.

B. Sequencing:
1. Do not commence installing hoisting systems, including monorails (if any), until supporting structural elements are in place and capable of bearing required loads.
2. Do not begin installing operable hoisting equipment (including hoists and trolleys) until building or structure is enclosed, weathertight, and environmental controls are in place and operational.

1.4 QUALITY ASSURANCE

A. Regulatory Requirements:
1. Selected Laws and Regulations with which hoisting system Work shall comply, including building codes and electrical codes, are indicated in Section 01 42 00 - References.

B. Qualifications:
1. Manufacturers of Hoisting Equipment and Monorails:
    a. Manufacturers shall possess not less than five years of experience producing substantially similar equipment to that required and shall be able to furnish documentation of not less than five installations in satisfactory operation for not less than five years each in the United States.
    b. Upon Contracting Officer's request, submit documentation of manufacturer's compliance with requirements of this provision.
    c. Retain one hoisting equipment manufacturer to furnish all hoisting systems for the Work. Such Supplier shall properly coordinate hoisting equipment and monorails, to furnish complete, operational, coordinated hoisting systems.
2. Installer:
    a. Engage to perform the Work under this Section an installer experienced in installing hoisting equipment similar to that required for the Project and is acceptable to hoisting equipment manufacturer.
    b. Submit proposed installer's company name, address, and qualifications, and the following information for not less than three successful, completed projects:
        1) Entity name, contact person (currently employed by such entity and familiar with the completed project), and telephone numbers for contact personnel of each owner and architect or Contracting Officer responsible for project.
        2) Installer's approximate compensation for the associated hoisting system work.

3) Month and year each hoisting system was substantially completed.
4) Brief description of each hoisting system, including hoisting capacity, and other pertinent information.

3. Welders:
   a. Welding personnel shall be certified in accordance with AWS D1.1/D1.1M and D14.1/D14.1M.
   b. Upon Contracting Officer's request, submit documentation of welders' compliance with requirements of this provision.

1.5 SUBMITTALS

A. Action Submittals:
   1. Shop Drawings:
      a. Submit schedule (in tabular or narrative format) of hoisting systems, indicating specific space and distinctive characteristics (such as type, hoisting capacity,, and other information) for each.
      b. Monorail manufacturer's erection and layout drawings for each monorail system required, indicating dimensions, details, and materials. Show and indicate supports and components of monorail system. Indicate weight of each monorail system component. Indicate specific reference standards and construction codes with which monorail system components comply.
      c. Certified clearance diagram showing arrangement of each hoisting system and clearances, including plan and sections.
   2. Product Data:
      a. Manufacturer's literature, illustrations, specifications identifying materials of construction, rated capacities, dimensions of individual components, and finishes.
      b. Manufacturer's standard drawings for each model and size of hoisting equipment required, indicating dimensions, details, and materials. Indicate specific reference standards and construction codes with which hoisting equipment complies.
      c. Manufacturer, model, weight, and horsepower of each hoisting equipment component.
   3. Testing Plans:
      a. Plan for load testing at the Site, in accordance with the "Field Quality Control" Article of this Section.
      b. Obtain Contracting Officer's approval of testing plan not later than five days prior to load test.
      c. Testing plan shall indicate proposed dates and time, together with anticipated duration, for load testing each hoisting system provided. Indicate specific types of operating tests and load tests required, together with brief narrative of how such testing will be performed. Also indicate types of test loads and weight of each, together with required certification thereof in accordance with this Section's "Field Quality Control" Article. Include in testing plan proposed forms for recording results of required field tests and inspections.

B. Informational Submittals:
   1. Manufacturer's Instructions:
      a. Special shipping, storage, protection, and handling instructions.
      b. Routine maintenance requirements prior to startup.
   2. Field Quality Control Submittals:
      a. Load test report, including copy of certifications of test weights used.
   3. Supplier Site Visit Reports:
      a. Report of each visit to the Site by Supplier, summarizing purpose of visit, activities while onsite, problems encountered, advice given to Contractor or Subcontractor, and actions taken.
      b. Manufacturer's field service reports, in accordance with Section 01 61 03.
   4. Qualifications Statements:
      a. Manufacturer, when requested by Contracting Officer.
      b. Installer.

c. Welders, when requested by Contracting Officer.

C. Closeout Submittals: Submit the following:
1. Operation and Maintenance Data:
a. Submit operation and maintenance manuals in accordance with Section 01 78 23.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Storage and Handling Requirements:
1. Store monorails and appurtenances indoors in environmentally controlled space.
2. Handle monorails in manner that does not result in bending, warping, or other deformation of monorails and appurtenances.

PART 2 - PRODUCTS

2.1 HOISTING EQUIPMENT

A. Manufacturers:
1. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
a. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.
2. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
a. Columbus McKinnon.
b. Or equal.

B. Description:
1. Provide all labor, materials, equipment, construction equipment and machinery, tools, services, accessories, and incidentals necessary or required to furnish and install hoisting systems of the types required, complete and operational.
2. Hoisting systems shall comply with headroom, hook elevation, and side clearance requirements shown and indicated in the Contract Documents, and shall provide required available lift height and capacity.
3. Service Conditions:
a. Electrical equipment and controls shall comply with applicable NEMA requirements for equipment environment specified in the Contract Documents.
b. Electric motors shall comply with Section 01 61 03.

C. Performance Criteria:
1. Hoisting Systems with Electrically operated Hoists:

Designation	Hoisting System A	Hoisting System B
General Location:	NOT USED	NOT USED
Hoist Type:		
Ambient Conditions (deg F, Indoor/Outdoor):		
Quantity Required:		
Capacity (tons):		
Required Lift (feet):		
Max Required Hook Elevation:		

Lift Speed (fpm):		
Combined Hoist and Trolley Connected Motor Demand (HP)		
Trolley Type:		
Trolley Speed (fpm):		
Volts:		
Phase:		
Frequency (Hertz):		
Control:		
Electrification:		
Notes: 1. "Not Applicable" (or similar indication), when entered for speed or horsepower, means manually-geared is required.		

2. Hoisting Systems with Manually Operated Trolley and Chain Hoists:

Designation	**Hoisting System C**	**Hoisting System** D	**Hoisting System E**	**Hoisting System F**	**Hoisting System G**
General Location:	Above Reaction Tanks (TK-63011, TK-63102) – North Rim WTP	Above High Demand Pumps (PMP-68211, PMP-68212, PMP-68213) – North Rim WTP	NOT USED	Roaring Springs Pump House – High Demand Pumps (PMP-52212, PMP52211)	NOT USED
Hoist Type:	Chain	Chain		Chain, Low headroom	
Ambient Conditions (deg. F, Indoor/Outdoor):	55-100 deg F Indoor	55-100 deg F Indoor		Indoor	
Quantity Required:	One	One		Two	
Capacity (tons):	0.50	1.0		2.0	
Required Lift (feet):	20	10		5.5	
Max Required Hook Elevation:	8379.00	8370.96		4865.18	
Maximum Chain Effort to Lift Full Load (lbs.):	60	79		60	
Lifting Ratio, (Feet of Chain Overhauled to Feet Lifted):	19:1	31:1		71:1	

Trolley Type:	Manual Geared	Manual Geared		Manual Geared	

D. Materials, Fabrication, and Assembly:
   1. Materials, Fabrication, and Assembly – General:
      a. Design Stresses: Provide load suspension components so that stresses at rated load do not exceed 20 percent of material's average ultimate strength.
      b. Shop welding shall be in accordance with AWS D14.1/D14.1M.
      c. Selection and fabrication of equipment furnished shall include special consideration of available headroom and vertical clearances.
      d. Provide hoisting equipment with configuration and weight that does not result in load exceeding design capacity of associated monorail and supporting elements.
      e. Trolley shall be appropriate for impact allowance required in Article 2.2.C of this Section, relative to trolley impact loads.
   2. Manual Chain Hoists:
      a. Comply with ASME B30.16 and ASME HST-2. Welding shall comply with AWS D14.1/D14.1M.
      b. Hooks: Provide latch-type hooks, free to rotate through 360 degrees under all loading conditions. Hooks shall be heat-treated, drop forged steel.
      c. Hoisting Chain: Hardened alloy steel.
      d. Lift Wheel: Machined and heat-treated alloy steel with hardened steel chain guides.
      e. Gearing: Machined and heat-treated. Hoist gear box shall be in accordance with AGMA 6013-A. Provide means for adequate lubrication of gearing.
      f. Load Brake: Hoist shall include Weston-type load brake, providing instant brake and release action for positive control of load.
      g. Provide overload protection device preventing lifting of loads beyond rated capacity.
      h. Housing and Covers: Cast aluminum.
      i. Provide supports, fasteners, brackets, chain container, and necessary accessories.
   3. Manual Geared Trolleys:
      a. Wheels: Fabricated of spark-resistant bronze or stainless steel. Wheels shall have uniform surface hardness and be capable of carrying maximum applied load.
      b. Trolley shall be geared to provide hand chain driven locomotion.
      c. Gearing: Machined and heat-treated. Hoist gear box shall be in accordance with AGMA 6013-A. Provide means for adequate lubrication of gearing.

E. Operation and Controls
   1. Hoist Speed Control:
      a. Hoist motor shall be single-speed, two-speed, or variable speed, as indicated in Paragraph 2.1.C ("Performance Criteria") of this Section, with magnetic control.
      b. Each magnetic control shall have contactors sized for specified service class. Reversing contactors shall be mechanically and electrically interlocked to prevent line-to-line faults and shall be provided with automatic reset thermal overloads for hoist motor.
   2. Trolley Speed Control:
      a. Trolley motor shall be single-speed, two-speed, or variable speed, as indicated in Paragraph 2.1.C ("Performance Criteria") of this Section, with magnetic control.
      b. Each magnetic control shall have contactors sized for specified service class. Reversing contactors shall be mechanically and electrically interlocked to prevent line-to-line faults and shall be provided with automatic reset thermal overloads for trolley motor.
   3. Pendant Control: Control of hoist and trolley shall be by pendant pushbutton control station.
      1) Pendant mounting shall include steel cable to provide strain relief for pendant's electric control cable. Provide isolating transformer to reduce voltage to 120 volts (or less) in control circuits.

2) Mount control pendant and cable on 12-gauge stainless steel C-track festoon system allowing hoist and trolley to move independent of control station.

F. Finishes:
1. Shop-Applied Finish Materials:
a. Primer:
1) Hoisting system manufacturer's standard for items in the installation and service environment indicated.
b. Finish Coat:
1) Hoisting system manufacturer's standard for items in the installation and service environment indicated.
2. Shop Finishing Methods:
a. Surface preparation and shop painting is required for ferrous metals, equipment, and accessories. Do not paint stainless steel and machined surfaces. Application of paint and coating materials shall be in accordance with hoisting system manufacturer's standard procedures for applying the subject paint or coatings for the service environment required.
b. Gears, bearing surfaces, and other machined surfaces shall receive heavy application of rust-inhibiting coating that shall be maintained during storage and until equipment is substantially complete.

G. Identification and Nameplates:
1. Provide on each subassembly data plate (nameplate) in accordance with Section 01 61 03.

2.2 MONORAILS

A. Manufacturers:
1. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
a. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.
2. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
a. Railmaster by American Crane & Hoist.
b. Or equal.

B. Description:
1. Provide all labor, materials, equipment, construction equipment and machinery, tools, services, accessories, and incidentals necessary or required to furnish and install monorail systems of the types required, complete and operational.
2. Monorail shall comply with headroom, elevation, and side clearance requirements shown and indicated in the Contract Documents, and shall provide required available lift height and capacity.
3. Monorails shall comply with ASME B30.11.
4. Location and arrangement of required monorails are shown or indicated on the Drawings.
5. Monorails provided shall be appropriate for associated trolleys and hoists. Ensure complete compatibility of hoisting system components and complete operability of hoisting systems.

C. Performance and Design Criteria:
1. Connect monorail to supporting structural system at locations shown on the Drawings.
2. Monorails shall be capable of supporting required capacity of associated hoists plus 25 percent for impact load plus weight of hoists, trolley, and accessories without exceeding allowable working stress of track material, with maximum deflection of 1/450 of span.

3. Include in design calculations an impact allowance for trolley. Impact allowance shall be not less than 0.5 percent of rated load for each foot per minute of hoisting speed with minimum allowance of 15 percent and maximum allowance of 50 percent.
4. Track splices shall be designed by track Supplier and shall be provided at support points.

D. Materials and Fabrication:
1. Materials: Monorail track, hangers, supports, plates and connection elements shall be ASTM A36 or ASTM A572, Grade 50 steel.
2. Patented Track Systems:
   a. Provide patented track system, comprised of specially designed track with wide upper flange and ”narrow” lower flange, connected by vertical steel segment, specifically intended for use as monorail for trolleys equipped with hoists. Lower load carrying (tension) flange portion of the track shall have Brinell hardness of not less than 195.
   b. Provide 4-inch patented track system.
   c. Patented track monorail systems, including bracing and attachment to supporting structure, shall comply with ANSI MH 27.1, ANSI/AISC 303, AISC 360, Section 05 12 00 - Structural Steel, and other, pertinent requirements of the Contract Documents.
3. Brace track to prevent sideways movement under full load conditions.
4. Welding shall comply with AWS D1.1/D1.1M.
5. Bolts shall comply with ASTM F3125, with ASTM A563 nuts, and ASTM F436 hardened washers. Connections shall comply with RCSC Specification for Structural Joints Using ASTM A325 or ASTM A490 Bolts.
6. Provide end stops at all track ends.

2.3 ACCESSORIES

A. Warning Signage:
1. Unless otherwise required in this Section, signage required in this provision shall comply with Section 10 14 00 - Identification Devices.
2. Install on hoist’s lower load block or controls in readable position durable label or labels displaying:
   a. The word WARNING or other legend designed to bring the label to the attention of an operator.
   b. Clear and obvious indication of rated capacity of hoist.
   c. Cautionary language against:
      1) Lifting more than rated load.
      2) Operating hoist when hook is not centered under hoist.
      3) Operating hoist with twisted, kinked or damaged rope or chain.
      4) Operating damaged or malfunctioning hoist.
      5) Operating hoist with rope that is not properly seated in its groove (if applicable).
      6) Lifting people or lifting loads over people.
      7) Removing or obscuring warning labels.
3. Provide appropriate warning signage at intervals not greater than 10 feet along monorails with electrical conductors.
4. Provide appropriate warning signage at stairs and ladders located six feet or less from monorail(s).
5. Warning signage legend and colors:
   a. “DANGER” (red).
   b. “HIGH VOLTAGE” (black).
   c. “480 VAC” (black).
   d. “KEEP OFF” (red).

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verification of Conditions:

1. Prior to commencing hoisting system Work, verify supporting structural elements and systems are in place, properly installed, and appropriate for bearing loads imposed by hoisting systems under rated loads. Verify the following:
    a. Design strength of cast-in-place concrete supporting elements has been attained.
    b. Structural steel or other support framing is properly erected, plumb and level, in accordance with the Contract Documents, adequately bolted, braced and welded, and ready for installation of hoisting system.
2. Clearances:
    a. Prior to commencing hoisting system Work, measure vertical and horizontal clearances along entire length of monorail. Verify that there is at least three inches clear between topmost part of hoisting system and lowest overhead obstruction, including structural supports of roof decks and overhead slabs, roof drain piping and other plumbing systems, HVAC ducting and equipment, process piping, electrical systems, including light fixtures, electrical conduits and junction boxes, and others.
    b. To the extent practicable, verify clearances between existing equipment and systems located below each hoisting system.
    c. In accordance with the Contract Documents, give Contracting Officer written notice of conditions that would interfere with proper installation and operation of hoisting system Work. Submit request for interpretation or clarification when Contractor believes hoisting system will conflict with other elements of the Work or existing construction to remain. Such notices and requests for interpretation or clarification shall be clear and specific, indicating each individual, potential obstruction and its associated dimensions and elevations. Include dimensioned sketches and photographic documentation as necessary.
    d. Do not proceed with the associated hoisting system Work until conflicts are resolved.

B. Installing the Work constitutes Contractor's approval of underlying work, substrates, and field conditions prevailing at the time of the Work.

3.2 PREPARATION

A. Supporting Work:
1. Provide structural supports and elements, for hoisting system Work, in accordance with Section 05 12 00 - Structural Steel. Field welding to building or structure shall be in accordance with AWS D1.1/D1.1M.

3.3 INSTALLATION

A. Installation Requirements:
1. Install hoisting systems in accordance with the Contract Documents, manufacturer's written instructions, ASME B30.11, ASME B30.16, ASME HST-1, ASME HST-2, and ASME HST-4 (as applicable to the equipment provided), and Laws and Regulations. Where such requirements conflict, obtain interpretation or clarification from Contracting Officer prior to commencing the associated Work.
2. Monorails:
    a. Hoisting system installer shall install monorails.
    b. Support monorails as shown and indicated on the Drawings.
    c. Provide supports necessary for easy removal of monorails for repair or replacement.
    d. Align monorails true and level.
    e. Provide flush joints at bottom of monorails.
    f. Installation tolerance for monorails shall be in accordance with ASME B30.11.
3. Comply with Section 01 61 03.
4. Install signage for hoisting systems, including warning signage, in accordance with the Contract Documents, including Section 10 14 00.

3.4 FIELD QUALITY CONTROL

A. Field Tests and Inspections:

1. After installing equipment and associated controls, when required, perform at the Site operating tests for hoisting system and appurtenances. Should testing indicate malfunction, make repairs and adjustments as necessary. Repeat testing and adjusting until, in Contracting Officer's opinion, installation is complete and equipment is functioning properly and accurately, and is substantially complete. Equipment will not be substantially complete until required field quality control activities are successfully completed and acceptable to Contracting Officer.
2. Load Test:
    a. Perform load tests under supervision of manufacturer's factory-trained service technician, in presence of Contracting Officer or Government's Site Representative.
    b. Weights used in load testing shall be certified by a state or local bureau of weights and measures. Submit weight certification as part of both load testing plan and load test report.
    c. Load testing shall comply with ASME B30.11 or ASME B30.16, as applicable, and the following:
        1) Trolley travel full length of monorail with rated load, while verifying that all functions operate properly.
        2) Hoist brake drift test with rated load: Lift weight, measure distance to floor, allow five minutes to elapse, and re-measure. Record the results measured. Criteria for Acceptance: No difference in measurements.
        3) Upper/lower limit switch test with no load.
        4) Emergency stop test with no load.
        5) Deflection Test: With hoist positioned at center of longest monorail span, and at tip of longest cantilever (if any), while hoisting rated load, measure distance from hook to floor. Remove load and measure distance from hook to floor. Criteria for Acceptance: Difference in measurements shall comply with manufacturer's specifications and deflection criteria indicated in Article 2.2 of this Section
    d. Load Test Report: Submit results of load testing in report indicating tests performed, data collected, results of each test, and corrective actions taken (if any). Test report shall be signed by manufacturer's service technician present during testing and installer's site superintendent or foreman as applicable.

B. Manufacturer's Services: Provide services of qualified, factory-trained service technician to perform the following for each hoisting system furnished:
1. Inspect materials and equipment upon delivery to the Site, determine whether items are as required by the Contract Documents and Submittals approved by Contracting Officer, and are in appropriate quantity and condition. Instruct Contractor's personnel in handling and unloading items from delivery vehicles, and advise Contractor during unloading and initial handling of items. Instruct Contractor's personnel regarding proper storage of delivered items.
2. Instruct Contractor in installing equipment and provide appropriate advice during equipment installation.
3. Inspect and adjust equipment after installation, including electrification and initial, post-installation checkout.
4. Upon completion of installation, together with Contractor, perform checkout and startup of equipment. Verify proper operation, including complete functionality of required controls. Assist Contractor with performing other required field tests and inspections, including load tests.
5. Instruct facility personnel in operating and maintaining the equipment, in accordance with Section 01 79 00 – Demonstration and Training.
6. Technician shall revisit the Site as often as necessary until installation is acceptable.
7. Manufacturer's Installation Certification: Submit certification that manufacturer's technician has checked completed installation and equipment, as specified in the Contract Documents, has been provided in accordance with manufacturer's recommendations, and operation of equipment is satisfactory. Certification shall be signed by manufacturer's technician present at the Site and Contractor.

8. All costs, including expenses for travel, lodging, meals and incidentals, and cost of travel time, for visits to the Site shall be included in the Contract Price.

3.5 PROTECTION

A. Do not use permanent hoisting systems provided as part of the Work unless otherwise expressly allowed in Section 01 11 00 – Summary of Work.

END OF SECTION

This page intentionally left blank.

DIVISION 43

PROCESS GAS AND LIQUID HANDLING, PURIFICATION, AND STORAGE EQUIPMENT

This page intentionally left blank.

SECTION 43 21 00
PUMPING EQUIPMENT - BASIC REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Basic requirements for all pumping equipment.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 - Equipment - Basic Requirements.
   2. Section 09 96 00 - High Performance Industrial Coatings.
   3. Section 43 23 22 – Pumping Equipment – Horizontal Split Case Centrifugal Pumps.
   4. Section 43 23 31.29 – Pumping Equipment – Vertical Inline Centrifugal Pumps.
   5. Section 43 24 17 – Pumping Equipment - Sample Pumps.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ANSI/Hydraulic Institute (ANSI/HI):
      a. 9.6.3, Rotodynamic (Centrifugal and Vertical) Pumps – Guideline for Allowable Operating Region.
      b. 9.6.4, Rotodynamic Pumps for Vibration Measurements and Allowable Values.
      c. 9.6.6, Rotodynamic Pumps for Pump Piping.
      d. 11.6, Rotodynamic Submersible Pump for Hydraulic Performance, Hydrostatic Pressure, Mechanical, and Electrical Acceptance Tests.
      e. 14.6, Rotodynamic Pumps for Hydraulic Performance Acceptance Tests.

B. Coordinate all mechanical seal systems specified to ensure pump and seal compatibility.

C. Pump/motor and VFD coordination: See Specification Section 01 61 03.

1.3 DEFINITIONS

A. The abbreviations used in this section are defined as follows:
   1. AOR: Allowable Operating Range.
   2. BEP: Best Efficiency Point.
   3. IPS: Iron Pipe Size.
   4. NPSH3: Net Positive Suction Head for 3% head loss.
   5. POR: Preferred Operating Range.
   6. TDH: Total Dynamic Head.
   7. TEFC: Totally Enclosed Fan Cooled.
   8. VFD: Variable Frequency Drive.

B. Pump Service Category: Pump or pumps having identical names (not tag numbers) used for specific pumping service.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 01 61 03.
   2. Product technical data including:
      a. Performance data and curves with flow (GPM), head (FT), horsepower, hydraulic efficiency, rotating speed (RPM), AOR, BEP, POR, NPSH3 requirements, minimum bowl submergence requirements for vertical mixed flow, axial and turbine pumps.
      b. Pump accessory data.
      c. Bearing supports, shafting details and lubrication provisions.
         1) Bearing life calculations.

2) Critical speed calculations.
d. Solids passage information.
3. Certifications:
a. Certified pump performance curves as described in the SOURCE QUALITY CONTROL Article.
b. Verification of Primary and Secondary conditions in POR and AOR.
4. Test reports:
a. Factory hydrostatic test as specified in individual specification sections.

B. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

C. Informational Submittals:
1. Certifications:
a. Provide a written statement that manufacturer's equipment has been installed properly, started up and is ready for operation by Contracting Officer's personnel.

D. Anchorage calculations per Section 01 81 10

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Pumps:
a. See individual pump Specification Sections.
2. Mechanical seals:
a. See individual pump Specification Sections.
b. If not specified in pump Specification Section:
1) Garlock.
2) Or Equal.

2.2 CENTRIFUGAL PUMP DESIGN

A. Provide units with increasing head characteristics from the end run out portion of the curve to:
1. Shut-off condition.
2. For Mixed flow, Axial Flow and Propeller pumps to the right-hand edge of the AOR.
3. All pumps except those for sewage service shall comply with NSF-61.

2.3 ACCESSORIES

A. See Specification Section 01 61 03.

B. Each Unit:
1. Lifting eye bolts or lugs.
2. Plugged gage cock connection at suction and discharge nozzles.
3. Tapped and plugged openings for casing and bearing housing vents and drains.
4. Fittings for properly adding flushing lubricant.
5. Pressure relief fittings for grease lubrication.

C. Mechanical Seals:
   1. Provide as specified in the narrow-scope pump sections.
   2. Provide stationary balanced O-ring type.
   3. Provide water lubrication - cooling.
   4. Materials:
      a. Provide as specified in narrow-scope pump sections
      b. If not included in narrow-scope pump sections use the following materials:
         1) Metal parts except springs: 316 stainless steel.
         2) Springs: Hastelloy C.
         3) Seal faces: Unfilled carbon graphite versus silica-free Grade 99.5 ceramic.
         4) Elastomers: Viton.

2.4 FABRICATION

A. Pump Support:
   1. Design base to support weight of drive, shafting and pump.
   2. Comply with HI vibration limitations.
   3. Mount horizontal pump, motor and coupling on single piece drip lip type baseplate.
   4. Mount vertical pumps on single piece pedestal baseplate.
   5. Fabricate to withstand all operating loads transmitted from the pump and drive.
   6. On vertically configured end suction centrifugal pumps when supplied with a fabricated steel mounting frame and suction elbow, the suction elbow shall be a long radius reducing elbow with greater than 50% area reduction to comply with Table 9.6.6.3.2 of ANSI/HI 9.6.6 standard for straight pipe lengths.

2.5 SOURCE QUALITY CONTROL

A. Verification primary design condition in POR.

B. Verification secondary design condition in AOR.

C. Factory hydrostatic test all pumps at 150% of shut-off head for a minimum of five minutes.

D. If specifically required in the individual pump specification sections, provide factory tests:
   1. All units:
      a. Conduct tests in accordance with HI.
         1) Shut-off head and design condition: Positive unilateral performance tolerance meeting Grade 1U per ANSI/HI 14.6 for Rotodynamic Pumps.
         2) Shut-off head and design conditions: Positive unilateral performance tolerances meeting Grade 1U per ANSI/HI 11.6 for Rotodynamic Submersible Pumps.
   2. All pumps:
      a. Head (FT) versus flow (GPM) pump curves:
         1) Efficiencies along curve.
         2) Brake horsepower along each curve.
   3. Results certified by a registered professional engineer.

E. Statically and dynamically balance each pump per ANSI/HI standards.
   1. If specifically required in the individual pump specification sections or in Specification Section 01 61 03, field vibration test pumps:
      a. To meet requirements of ANSI/HI 9.6.4 for Rotodynamic Pumps at any point on the pumps and motor.

F. To meet requirements of ANSI/HI 11.6 for Submersible Pumps.

PART 3 - EXECUTION

3.1 INSTALLATION

A. See Specification Section 01 61 03.

B. Floor or Pad-Mounted Units (Non-Submersible):
   1. Align vertically and horizontally level, wedge and plumb units to match piping interfaces.
   2. Assure no unnecessary stresses are transmitted to equipment flanges.
   3. Tighten flange bolts at uniform rate and manufacturer's recommended torque for uniform gasket compression.
      a. For ASME B16.5 Class 900 flanged connections: Do not torque bolts more than two times.
         1) Mark each bolt with a single line after bolt has been torqued.
         2) If bolt has been torqued two times, replace all hardware on flange prior to subsequent tightening.
   4. Support and match flange faces to uniform contact over entire face area prior to bolting pipe flange and equipment.
   5. Permit piping connecting to equipment to freely move in directions parallel to longitudinal centerline when and while bolts in connection flange are tightened.
   6. Grout equipment into place prior to final bolting of piping but not before initial fitting and alignment.
   7. Assemble connecting piping with gaskets in place and minimum of four bolts per joint installed and tightened.
      a. Test alignment by loosening flange bolts to see if there is any change in relationship of piping flange with equipment connecting flange.
      b. Realign as necessary, install flange bolts and make equipment connection.
   8. Field paint units as defined in Specification Section 09 96 00.
   9. Provide pressure gage, visible from grade or operating floor, on discharge of all pumps and on suction and discharge of all non-submersible units.

C. Submersible Units:
   1. Assemble connecting piping with gaskets in place and minimum of four bolts per joint installed and tightened.
      a. Test alignment by loosening flange bolts to see if there is any change in relationship of piping flange with equipment connecting flange.
      b. Realign as necessary, install flange bolts and make equipment connection.
   2. Field paint units as defined in Specification Section 09 96 00.
   3. Provide discharge pressure gage visible from grade or operating floor.

3.2 EQUIPMENT VIBRATION MONITORING AND TESTING PROGRAM:

A. Provide vibration testing on all rotating and reciprocating equipment having driver 50 HP and greater.

B. Additional requirements for vibration monitoring and testing equipment at 25 DegC:
   1. Requirements for analyzer:
      a. Frequency range: 10 Hz to 20kHz
      b. Frequency accuracy: 0.02 percent
      c. Non-integrated spectral amplitude accuracy: 5 percent, 3Hz to 65 kHz
      d. Single integrated spectral amplitude accuracy: 5 percent, 10 Hz to 20kHz
      e. Supports measurements of acceleration, velocity, displacement, envelop demodulation for bearing defect detection.
      f. Capable of two-place computer balancing
   2. Requirements for vibration sensor at 25 DegC:
      a. Sensitivity: ±5 percent = 100 mV/g
      b. Acceleration range: ±50 g
      c. Amplitude nonlinearity: ±1 percent
      d. Frequency response: ±10 Hz to 7 kHz (±3 dB)
      e. Permanently attach vibration test and monitoring mounting pads to mechanical equipment at location recommended by the equipment manufacturer or as recommended by the testing firm

3. Acceptability of equipment conditions, except pumps, based on ISO 1940-1 Balance Quality Grade G2.5 criteria

C. Acceptability of pumping equipment to be based on current ANSI/HI criteria:
   1. ANSI/HI 9.6.4-2022 for all pumps

D. Repair or replace equipment shown to be out of range of acceptable tolerance until the equipment meets or exceeds acceptability standards

E. Utilize a Contracting Officer approved testing agency to perform vibration monitoring and testing on the following equipment:
   1. RSPH Pumphouse:
      a. PMP-52211 and PMP-52212

F. Permanently attach vibration test and monitoring mounting pads to the equipment at locations recommended by the equipment manufacturer or as recommended by the vibration testing agency.

G. Utilize mounting pads suitable for permanent installation and for incorporation into a predictive maintenance program.

H. For variable speed equipment provide vibration testing at 1 Hz increments of VFD speed throughout entire operating range.

I. Diagnosis to include, but is not limited to the following:
   1. Unbalance
   2. Misalignment
   3. Bent shaft
   4. Journal bearing related problems
   5. Rolling contract bearing problems
   6. Mechanical looseness
   7. Resonance
   8. Foundation flexibility
   9. Electrically induced problems
   10. Pump problems
   11. Fan problems
   12. Coupling problems
   13. Drive belt problems
   14. Gear problems
   15. Electric motor induced vibration from VFD or VFD carrier frequency
   16. Natural frequency of the installed equipment

J. Provide machinery condition diagnosis based on an acceptable machinery vibration severity guide or machinery fault guide analysis provided by the testing agency, ISO 1940 Balance Quality Grade 6.3 as a minimum.

K. Tolerances for pumping equipment shall be per HI published standards.

L. Immediately notify Contracting Officer if equipment is shown to be out of range of the specified tolerance until the equipment meets the specified normal operating range required in the machinery fault guide analysis.

M. Document testing with written report.
   1. Report to include initial testing results, acceptance criteria, corrective action taken to meet acceptance, verification of corrective action and acceptance report and baseline.
   2. Natural frequency of installed equipment utilizing and impact hammer.

N. Report to include graphical plots of vibration signature for each test point at a scale which illustrates all vibration levels greater than 0.025 ips RMS.

O. Electrical Equipment and Connections Testing Program:

1. Perform testing on Electrical equipment and connections in accordance with the Electrical specifications requirements.
2. Testing of motors:
    a. After installation and prior to energizing the motor, perform inspections and tests per NETA ATS 7.15 for all motors.
    b. Verify motor has been lubricated.
    c. Bump motor to check for correct rotation.
3. Immediately notify Contracting Officer if equipment shown to be out of range of the acceptable tolerance until the equipment meets or exceed acceptability standards.

P. Other Testing:
1. Perform tests and inspections not specifically listed but required to verify equipment is safe to energize and operate.
2. Subbase that supports the equipment base and that is made in the form of a cast iron or steel structure that has supporting beams, legs, and cross members that are cast, welded, or bolted shall be tested for a natural frequency of vibration after equipment is mounted.
    a. The ratio of the natural frequency of the structure to the frequency of the disturbing force shall not be between 0.5 and 1.5.
    b. Immediately notify Contracting Officer if equipment shown to be out of range of the acceptable tolerance.

Q. Infrared Thermography Testing Program:
1. Perform infrared thermography testing for equipment specified in other Divisions during the Equipment Demonstration Period.
    a. Perform on rotating and reciprocating equipment having drivers 50 HP or greater.
2. Additional requirements for infrared thermography monitoring and testing equipment:
    a. Temperature range: -10 to 350 DegC
    b. Accuracy: ±2 percent or 2 DegC, whichever is greater
    c. Repeatability: ±1 percent or 1 DegC, whichever is greater
    d. Temperature indication resolution: 0.1 DegC
    e. Minimum focus distance: 0.3 meters
    f. Output in color palettes: JPEG, BMP, or other digital format compatible with Windows
3. Perform inspection per ASTM E1934.
    a. Operate VFD driven equipment at 100 percent speed during thermographic inspection.
4. Acceptability of electrical connections and components based on temperature comparison between components and ambient air temperatures not greater than 10 DegC per ASTM E1934.
5. Acceptability of motors and equipment bearings based on temperature rise not greater than 5 DegC above the equipment and/or bearing manufacturer’s published criteria.
6. Immediately notify Contracting Officer if equipment shown to be out of range of the acceptable tolerance.

3.3 FIELD QUALITY CONTROL

A. Provide services of equipment manufacturer's field service representative(s) to:
1. Inspect equipment covered by this Specification Section.
2. Supervise pre-start adjustments and installation checks.
3. Conduct initial start-up of equipment and perform operational checks.
4. Provide performance testing and vibration testing that meet Hydraulic Institute standards, with written statement of correct installation supplied by manufacturer.

END OF SECTION

SECTION 43 23 22
PUMPING EQUIPMENT - HORIZONTAL MULTI-STAGE CENTRIFUGAL PUMPS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Horizontal multi-stage centrifugal pumps

B. Section solely applies to: Volume 5 – Roaring Springs Pumphouse (RSPH)

C. Related Sections include but are not necessarily limited to:
   1. Section 01 67 00 – Product Requirements.
   2. Section 43 21 00 - Pumping Equipment - Basic Requirements.

1.2 REFERENCES

A. Referenced Standards:
   1. American Bearing Manufacturers Association (ABMA).
   2. American Iron and Steel Institute (AISI):
      a. Steel Products Manual.
   3. American National Standards Institute (ANSI).
   4. ASTM International (ASTM):
      a. A48, Standard Specification for Gray Iron Castings.
      b. B505, Standard Specification for Copper Alloy Continuous Castings.
      c. B584, Standard Specification for Copper Alloy Sand Castings for General Applications.
   5. American Society of Mechanical Engineers (ASME):
      a. B16.1, Pipe Flanges and Flanged Fittings
   6. Hydraulic Institute (HI):
      a. Standards for Centrifugal, Rotary and Reciprocating Pumps.

1.3 SUBMITTALS

A. Action Submittals: Submit the following:
   1. Shop Drawings:
      a. See Specification Section 43 21 00.
      b. Product technical data including:

      Complete fabrication and assembly drawings, together with detailed specifications and data covering materials, drive unit, parts, devices, and accessories forming a part of the equipment furnished, shall be submitted in accordance with the Submittals Procedures section. The data and specifications for each unit shall include, but not limited to:

      c. Pump technical information including certified pump curve and operating data.
      d. Type and model.
      e. Tag number.
      f. Pump location.
      g. Rotative speed.
      h. Size of suction flange
      i. Size of discharge flange.
      j. Maximum design thrust in all three axes on each suction and discharge flange.
      k. Net weight of pump only.
      l. Net weight of drive/motor only
      m. Net weight with motor, baseplate, and coupling.
      n. Diagrams of lifting point locations and recommended strap arrangements for:
         1) Entire pump assembly
         2) Pump assembly without drive/motor
         3) Drive/motor only

o. Complete performance curves showing capacity versus head, NPSH required, pump efficiency, wire to water efficiency, and pump input power.
p. Data of coupling.
q. Data on shop painting.
r. Base and anchor bolt details.
s. All available wiring and design information on pump accessories
    1) RTDs
    2) Accelerometers/vibration sensors
    3) Pressure sensors

B. Closeout Submittals: Submit the following:
1. Operation and Maintenance Data:
a. Submit in accordance with Section 01 78 23 - Operation and Maintenance Data.

1.4 QUALITY ASSURANCE

A. Balance:
1. All rotating parts to be accurately machined.
2. All rotating parts to be dynamically balanced.
3. Avoid resonance at normal operating speed.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Horizontal multi-stage pumps:
a. Flowserve.
b. No substitutions.

B. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Couplings
a. Falk Steelflex T31
b. Or equal
2. Mechanical seals:
a. Garlock
b. Or equal
3. Motors
a. Baldor
b. Or equal

2.2 MATERIALS

A. Materials:
1. Roaring Springs Pumphouse Raw Water High Pressure Pumps:
a. Casing: Chrome Steel
b. Casing Rings: None
c. Impeller: Stainless Steel
d. Impeller Wear Ring: None

e. Shaft: Chrome Steel
f. Shaft Sleeve: Stainless Steel
g. Bearings:
    1) Suction End: Sleeve, SiC/CrO
    2) Discharge End: Roller, Oil
    3) Intermediate: Sleeve, SiC/CrO
h. Drive Shaft Coupling: Flexible-Falk, Woods, or Equal.
i. Coupling Guard: Carbon Steel
j. Seal Housing: Chrome Steel
k. Baseplate: Fabricated Steel.

2.3 EQUIPMENT

A. Performance and Configuration Requirements:
    1. Roaring Springs Pumphouse Raw Water High Pressure Pumps:
        a. Tags: PMP-52211 and PMP-52212
        b. Primary design condition: 150 GPM at 3,604 feet TDH at a motor frequency of 55 Hz and with minimum pump efficiency of 50%.
        c. Secondary condition: 150 GPM at 4400 feet TDH at a motor frequency of 60 Hz with minimum pump efficiency of 50%.
        d. Motor efficiency at design conditions: 90%, minimum
        e. Motor Power Factor at design conditions: 0.9, minimum
        f. Shutoff condition: 0 gpm at 4550 feet TDH, minimum.
        g. Pump configuration: Horizontal.
        h. Maximum pump speed: 3600 RPM.
        i. Nameplate driver horsepower: 450 HP, maximum.
        j. Motor type: Inverted-Duty rated for use with variable speed drive
        k. Drive configuration: Flexible coupled
        l. Minimum solids passage: 50 microns.
        m. Net positive suction head required (NPSHR): 10 feet. maximum.
        n. Suction flange:
            1) ANSI/ASME B16.5 150# Flat Face/Raised Face or ANSI/ASME B16.5 300# Raised Face
            2) Axial (viewed from motor end)
        o. Discharge flange:
            1) ANSI/ASME B16.5 900# Raised Face Flange
            2) Radial Top, (viewed from motor end)

B. Assure increasing head characteristic from secondary design condition to shutoff condition.

C. Provide pumps with net positive suction head requirements that are less than the available net positive suction head at design condition specified.

D. Pumping units to be capable of withstanding all thrust conditions.

E. Rotating parts of the pumps to operate without excessive end thrust, vibration or noise.

F. Maximum vibration measured at the top of the motor not to exceed 90% maximum peak to peak amplitude as required by HI standards.

G. Maximum vibration measured at the top of the pump not to exceed 90% maximum peak to peak amplitude.

H. Dimensions:
    1. Maximum baseplate length: 12-FT
    2. Maximum motor weight: 3,800 LBS
    3. Maximum weight of individual pump assembly minus motor (dry): 3,800 LBS

2.4 ACCESSORIES

A. See Section 43 21 00.

B. Each Unit:
   1. Lifting eye bolts or lugs
      a. Provide means of connection for lifting and replacing motor independently of pump assembly
      b. Provide lifting connections on base to allow for uniform lifting of entire pump assembly, minus pump motor
      c. Provide lifting connections for entirely complete pump assembly (including motor) to allow for helicopter transport
   2. Plugged gage cock connection at suction and discharge nozzles
   3. Tapped and plugged openings for casing and bearing housing vents and drains
   4. Fittings for properly adding flushing lubricant
   5. Pressure relief fittings for grease lubricaiton

C. Mechanical seal pressure switch

D. Temperature Monitoring RTDs

E. Accelerometers

2.5 COMPONENTS AND FABRICATION

A. Casings
   1. Furnish casing to withstand 200% of maximum working pressure.
   2. Provide inter-stage casings with interchangeable parts

B. Pump Shaft
   1. Heat treated and machined over its entire length.
   2. Stagger impeller keyways

C. Bearings
   1. Provide anti-friction type bearings rated for ABMA L-10 life of 50,000 hours based on 24 hour continuous operation
   2. Provide product lubricated radial, food oil lubricated anti-friction thrust bearing (axial suction)
   3. Option: Provide food oil lubricated radial and thrust anti-friction bearings (radial suction)
   4. Insulated Bearing: Required

D. Impellers:
   1. Balance per HI standards.
   2. One piece casting and machined for smooth finish.
   3. Keyed to shaft and held in center of volute.

E. Mechanical seal pressure switch
   1. Factory-set pressure switch to monitor mechanical seal pressure.
   2. Close contact when pressure drops below factory-determined setpoint.
   3. Clearly indicate setpoint on pressure instrument labeling or pump nameplate.
   4. Wiring and design shall conform to Specification Section 40 73 00, Paragraph 2.3.A.4

F. Provide one-piece cast bearing housing with catch basin reservoir.

G. Furnish baseplate with drip lip to mount motor and pump.

H. Temperature Monitoring:
   1. 100 Ohm Platinum – Dual Element RTD
   2. Temperature range: -50 to 260°C (-58 to 500°F).
   3. Case: Stainless steel.
   4. Babbitt tip: Factory applied babbitt tip.
   5. Leads: Stranded copper with PTFE insulation; stainless steel over-braid
   6. Must be provided on both motor and pump
   7. Locations:
      a. Motor Windings, 2 per phase

b. Motor Drive End Bearing
c. Motor Outboard Bearing
d. Pump Discharge Bearing

I. Accelerometers:
1. 2 wire 4-20mA acceleration output
2. Connector: M12
3. Case Material: Stainless Steel
4. Sensing Element/Construction: PZT/Shear
5. Mounting Torque: 5.9ft. lbs
6. Mounting Bolt Provided
7. Shielded Cable Assembly
8. Mounting Threads: - ¼-28" UNF Male
9. Must be provided on both motor and pump
10. Locations:
a. Motor, every bearing – X/Y/Z combined axis
b. Pump, discharge bearing – X/Y/Z combined axis

J. Mechanical Seals
1. Type: Single Cartridge, Stationary Balanced O-Ring
2. Flushing/lubrication: Plan 11/13 (radial suction) product lubricated.
3. Materials
a. Metal parts except springs: 316 stainless steel.
b. Springs: Hastelloy C.
c. Seal faces: Unfilled carbon graphite versus silica-free Grade 99.5 ceramic.
d. Elastomers: Viton.

K. Couplings:
1. Heavy duty, grid-type.
2. Sized to handle drive shaft torque.
3. Able to compensate for minor angular, parallel, and axial misalignment between motor and pump.
4. Coupling to have 2 hubs, a spacer and 2 flexible discs.
5. Able to remove and replace flexible discs without disturbing drive unit.
6. Coupling to have a 1.25 service factor.
7. Allowable stresses in coupling to be 1.5 times the value used to size the pump shaft.
8. Manufacturer to provide computational analysis of the torsion of the pump/driver system.
9. The torsional critical speed shall not occur below the maximum operating speed nor within 20% above maximum operating speed.
10. Encase coupling assembly in rigidly mounted coupling guard meeting the minimum industrial safety standards define by OSHA.
11. Coupling guard to be securely attached to pump base.

L. Motors:
1. Reduced voltage starting, solid shaft, squirrel cage, induction type.
2. Inverter Duty.
3. 460 V, 60 Hz, 3 PH.
4. Motor terminal box mounted left side (viewed from shaft end)
5. ODP type
a. Maximum driver weight: 3000 lbs
6. Size motor to drive pump continuously over the complete head - capacity range without the load exceeding the nameplate rating.
7. Design motor for 85 DegC ambient.
8. Ground Rings: Required
9. Insulation class: F
10. Temperature Monitoring:
a. 100 Ohm Platinum – Dual Element RTD
b. Temperature range: -50 to 260°C (-58 to 500°F).

c. Case: Stainless steel.
d. Babbitt tip: Factory applied babbitt tip.
e. Leads: Stranded copper with PTFE insulation; stainless steel over-braid
f. Locations:
1) Motor Windings, 2 per phase
2) Motor Case

11. Terminal box

M. Pump Support:
1. Design base to support weight of drive, shafting and pump.
2. Comply with HI vibration limitations.
3. Mount horizontal pump, motor and coupling on single piece drip lip type baseplate.
4. Fabricate to withstand all operating loads transmitted from the pump and drive.
5. Provide horizontal motor positioning screws and baseplate leveling screws.

N. Data Plates:
1. Provide stainless steel data plate securely attached to pump.
2. Include manufacturer's name, pump size and type, serial number, speed, impeller diameter, capacity and head rating, and other pertinent data.

O. Pump rotation to conform with piping arrangement shown on Drawings.

P. Seismic:
1. Comply with Section 01 81 10 – Wind and Seismic Design Criteria.

2.6 SOURCE QUALITY CONTROL

A. Factory test each pump 100 hp or greater:
1. One pump of each service category.
2. Measure capacity, power, and efficiency.
3. Provide certified performance curves for 60 Hz and 55 Hz operation.
4. Submit test procedures and test results.
5. Package shipped equipment to facilitate field disassembly and reassembly
a. Include mechanical provisions on equipment to hoist-lift the following items independently or in combination:
1) Motor
2) Pump
3) Coupling
4) Pump, coupling, and baseplate
b. Provide any specialty tools required
c. Include full instructions on disassembly and reassembly of individual components, as well as lifting requirements

2.7 MAINTENANCE MATERIALS

A. Extra Materials:
1. Furnish Government the following extra parts for each pump service category:
a. Impellers: Two
b. Mechanical seal: One unit
c. Any additional spare parts recommended by manufacturer not noted above

PART 3 - EXECUTION

3.1 FIELD QUALITY CONTROL

A. See Section 43 21 00.

B. Upon delivery to laydown area, disassemble units as necessary for helicopter delivery to final location

C. Contract field services of manufacturer representative for all field tasks where direct input may be necessary

D. Reassembly of each pump at final location must be supervised and approved by manufacturer representative in person

E. Provide copy of manufacturer's standard commissioning plan to COR for review a minimum of 2 weeks prior to pump startup.

F. Start up and commissioning of each pump to be witnessed and supervised by manufacturer’s representative in person

END OF SECTION

This page intentionally left blank.

SECTION 43 23 31.29
PUMPING EQUIPMENT - VERTICAL INLINE CENTRIFUGAL PUMPS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes the following pumps:
   1. Roaring Spring Pumphouse (Volume 5):
      a. Finished Water Booster Pumps:
         1) PMP-53111 and appurtenances.
         2) PMP-53112 and appurtenances.
   2. North Rim WTP (Volume 6):
      a. Raw Water Pump:
         1) PMP-62011 and appurtenances.
      b. Finished Water Tank Pumps:
         1) PMP-68111 and appurtenances.
         2) PMP-68112 and appurtenances.
      c. High Demand Pumps:
         1) PMP-68211 and appurtenances.
         2) PMP-68212 and appurtenances.
         3) PMP-68213 and appurtenances.
      d. Upper Zone Booster Pumps:
         1) PMP-68311 and appurtenances.
         2) PMP-68312 and appurtenances.
         3) PMP-68313 and appurtenances.

B. Related Sections include, but are not necessarily limited to:
   1. Section 01 61 03 – Equipment – Basic Requirements.
   2. Section 01 79 00 – Demonstration and Training.
   3. Section 26 29 23 – Variable Frequency Drives – Low Voltage
   4. Section 40 05 00 - Pipe and Pipe Fittings - Basic Requirements.
   5. Section 43 21 00 - Pumping Equipment - Basic Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B16.1, Gray Iron Pipe Flanges and Flanged Fittings: Classes 25, 125, 250, and 300.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product technical data including:
      a. Complete fabrication and assembly drawings, together with detailed specifications and data covering materials, drive unit, parts, devices, and accessories forming a part of the equipment furnished. The data and specifications for each unit shall include:
         1) Complete performance curves showing capacity versus head, NPSH required, pump efficiency, wire to water efficiency, and pump input power.
            a) Include exact values at primary and secondary operating conditions.
         2) Type and model.
         3) Tag number.
         4) Pump location.
         5) Rotative speed.
         6) Suction nozzle size.

7) Discharge nozzle size.
8) Net weight of pump only.
9) Net weight with motor, baseplate, and coupling.
10) Coupling data.
11) Shop painting data.
12) Base and anchor bolt details.

B. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Data submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements.
1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Grundfos
2. Or Equal.

C. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 PERFORMANCE AND DESIGN REQUIREMENTS

A. Site Conditions:
1. Roaring Springs Pumphouse Elevation: 4859.55 FT.
2. North Rim WTP Elevation: 8359.00 FT.

B. Performance Parameters:
1. Roaring Springs Pumphouse:
a. Finished Water Booster Pump (PMP-53111 and PMP-53112):
1) Primary design condition: 20 GPM at 155 FT TDH.
2) Secondary design condition: 6-10 GPM at 192 FT TDH.
3) Shutoff head (minimum): 210 FT TDH.
4) NPSHr at primary design condition (maximum): 9.7 FT.
5) Speed (maximum): 3,600 RPM.
6) Drive: Constant Speed.
7) Motor horsepower (maximum): 3 HP.
8) Suction Flange: Stainless Steel, ANSI Class 150.
9) Discharge flange: Stainless Steel, ANSI Class 150.
2. North Rim WTP:
a. Raw Water Pump (PMP-62011):
1) Primary design condition: 108 GPM at 9.5 FT TDH.
2) Secondary design condition: 50 GPM at 12.2 FT TDH.
a) Shutoff head (minimum): 14 FT TDH.
3) NPSHr at primary design condition (maximum): 5.4 FT.
4) Speed (maximum): 1,800 RPM.
5) Drive: Variable frequency.
6) Motor horsepower (maximum): 0.5 HP.
7) Suction Flange: Cast Iron, ANSI Class 150.

8) Discharge flange: Cast Iron, ANSI Class 150.

b. Finished Water Tank Pumps (PMP-68111 and PMP-68112):
   1) Primary design condition: 100 GPM at 37 FT TDH.
   2) Secondary design condition: 50 GPM at 59 FT TDH.
      a) Shutoff head (minimum): 65 FT.
   3) NPSHr at primary design condition (maximum): 11.0 FT.
   4) Maximum speed: 3,600 RPM.
   5) Drive: Variable frequency.
   6) Motor horsepower (maximum): 2 HP.
   7) Suction Flange: Cast Iron, ANSI Class 150.
   8) Discharge flange: Cast Iron, ANSI Class 150.

c. High Demand Pumps (PMP-68211, PMP-68212, and PMP-68213):
   1) Primary design condition: 1,000 GPM at 104 FT TDH.
   2) Secondary design condition: 650 GPM at 115 FT TDH.
      a) Shutoff head (minimum): 190 FT.
   3) NPSHr at primary design condition (maximum): 24.6 FT.
   4) Maximum speed: 3,600 RPM.
   5) Drive: Variable frequency.
   6) Motor horsepower (maximum): 40 HP.
   7) Suction Flange: Cast Iron, ANSI Class 150.
   8) Discharge flange: Cast Iron, ANSI Class 150.

d. Upper Zone Booster Pumps (Pumps PMP-68311, PMP-68312, and PMP-68313):
   1) Primary design condition: 45 GPM at 104 FT TDH.
   2) Secondary design condition: 30 GPM at 142 FT TDH.
   3) Shutoff head (minimum): 145 FT.
   4) NPSHr at primary design condition (maximum): 7.7 FT.
      a) Maximum speed: 3,600 RPM.
      b) Drive: Variable frequency.
   5) Motor horsepower (maximum): 3 HP.
      a) Suction Flange: Cast Iron, ANSI Class 300.
   6) Discharge flange: Cast Iron, ANSI Class 300.

3. Provide pumps with increasing head characteristics from secondary design conditions to shutoff condition.
4. Provide motor with temperature switch for all pumps.

2.3 COMPONENTS

A. Suction/Discharge Base:
   1. Cast iron, ASTM 80-55-06.

B. Impellers:
   1. Type: Enclosed.
   2. Material: 304 SS.
   3. Wear Rings: 304 SS.
   4. Secure to shaft by means of a split cone and nut design.

C. Impeller Wear Rings:
   1. 304 SS.

D. Shaft:
   1. Stainless Steel AISI 431.

E. Mechanical Seals:
   1. Type: Cartridge.
   2. Seal replacement shall be possible without removal of pump components other than coupling guard, motor couplings, motor seal, and seal cover.

F. Intermediate Bearing Journals:
   1. Silicon Carbide.

G. Chamber Bushings:
   1. Graphite filled PTFE.

H. O-rings:
   1. EPDM or FKM.

I. Data Plates:
   1. Provide stainless steel data plate securely attached to pump.
   2. Include manufacturer's name, pump size and type, serial number, speed, impeller diameter, capacity and head rating at the primary design condition, and other pertinent data.

J. Motors:
   1. Roaring Springs Pumphouse:
      a. Reduced voltage starting, solid shaft, squirrel cage, induction type.
      b. Inverter Duty.
      c. 208 V, 60 Hz, 1 PH.
      d. TEFC type.
      e. Motor temperature switch (thermostatic type)
      f. Size motor to drive pump continuously over the complete head - capacity range without the load exceeding the nameplate rating.
      g. Design motor for 85 Deg C ambient.
   2. North Rim WTP:
      a. Reduced voltage starting, solid shaft, squirrel cage, induction type.
      b. Inverter Duty.
      c. 460 V, 60 Hz, 3 PH.
      d. TEFC type.
      e. Motor temperature switch (thermostatic type)
      f. Size motor to drive pump continuously over the complete head - capacity range without the load exceeding the nameplate rating.
      g. Design motor for 85 Deg C ambient.

2.4 MAINTENANCE MATERIALS

A. Extra Materials:
   1. Furnish the Government the following extra parts for each pump service category:
      a. Roaring Springs Pumphouse:
         1) Finished Water Booster Pumps:
            a) Wearing rings: Two sets.
            b) Mechanical Seal: Two cartridges.
      b. North Rim WTP:
         1) Raw Water Pump:
            a) Wearing rings: One set.
            b) Mechanical Seal: One cartridge.
         2) Finished Water Tank Pumps:
            a) Wearing rings: Two sets.
            b) Mechanical Seal: Two cartridges.
         3) High Demand Pumps:
            a) Wearing rings: One set.
            b) Mechanical Seal: One cartridge.
         4) Upper Zone Booster Pumps:
            a) Wearing rings: Two sets.
            b) Mechanical Seal: Two cartridges.

PART 3 - EXECUTION

3.1 INSTALLATION

A. See Section 43 21 00.

B. Install products in accordance with manufacturer's instructions.

3.2 FIELD QUALITY CONTROL

A. See Section 43 21 00.

B. See Section 01 79 00 for operator training requirements.

END OF SECTION

This page intentionally left blank.

SECTION 43 24 17
PUMPING EQUIPMENT - SAMPLE PUMPS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:
   1. Centrifugal type sample pumps located at:
      a. Roaring Springs Pumphouse finished water recirculation pump – Volume 5 (PMP-53221).
      b. North Rim WTP sample pumps – Volume 6 (PMP-62012, PMP-64011, PMP-66011, PMP-68117).

B. Related Sections include but are not necessarily limited to:
   1. Section 43 21 00 – Pumping Equipment: Basic Requirements.

1.2 QUALITY ASSURANCE

A. See Section 43 21 00.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. See Section 43 21 00.
   3. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Dimensional drawings.
      d. Materials of construction.
      e. Pump curves.

B. Operation and Maintenance Manuals:
   1. See Section 01 78 23 for requirements for:
      a. The mechanics and administration of the submittal process.
      b. The content of Operation and Maintenance Manuals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Roaring Springs Pumphouse:
      a. Finish Thompson Inc. Model DB6.
      b. Or Equal.

2. North Rim WTP:
    a. Dayton Model 2ZWR5A.
    b. Or Equal.

2.2 EQUIPMENT

A. Furnish and install at locations shown on Drawings and as described herein.
    1. Roaring Springs Pumphouse:
        a. Pumps:
            1) Finished Water; PMP-53221.
        b. Operating Conditions:
            1) 20 GPM @ 20 FT TDH.
            2) Shutoff condition: 26 FT TDH (minimum).
        c. Pump type: Close coupled mag-drive centrifugal.
        d. Motor: 115 volt, single phase, 60 hertz, open fan-cooled (TEFC motor).
            1) Continuous operation with thermal overload protection.
        e. Local Control Stations.
            1) Furnish one local control station for pump.
            2) Stainless steel, NEMA 4 rated enclosure.
            3) Input Power: 120 VAC, 1-phase.
            4) See Drawings for controls.
    2. North Rim WTP:
        a. Sample Pumps:
            1) Raw Water; PMP-62012.
            2) UF Feed; PMP-64011.
            3) CT Tank Influent; PMP-68117.
            4) Distribution Water; PMP-66011.
        b. Operating Conditions:
            1) 3 GPM @ 30 FT TDH.
            2) Shutoff condition: 44 FT TDH (minimum).
        c. Pump type: Close coupled centrifugal.
        d. Motor: 115 volt, single phase, 60 hertz, open fan-cooled.
            1) Continuous operation with thermal overload protection.
        e. Local Control Stations.
            1) Furnish one local control station for each sample pump.
            2) Stainless steel, NEMA 4 rated enclosure.
            3) Input Power: 120 VAC, 1-phase.
            4) See Drawings for controls.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Coordinate pumps and motors to ensure an operable system is provided to meet the specified condition.

3.2 EQUIPMENT INSTALLATION

A. Testing:
    1. Upon completion of the installation, the equipment shall be field tested to certify the system meets performance requirements.
    2. All testing costs shall be covered by the Contractor.

B. If equipment fails test:
    1. Modify the unit until passes the test, at Contractor’s expense.
    2. Replace the unit with a larger model that will meet requirements.

3. The Contractor shall pay for additional testing required to verify that performance criteria are met.

END OF SECTION

This page intentionally left blank.

SECTION 43 41 15
FIELD ERECTED WELDED STEEL TANKS

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, design services, and construction services for design, supply, and erection of new Raw Water Tanks at the North Rim WTP site (Volume 6).
   1. The work consists of two, new welded carbon steel tanks and tank appurtenances as indicated in this Specification, the tank Drawings, and in accordance with provisions of the Contract Documents.

B. Furnish labor, materials, tools, equipment, design services, and construction services for the rehabilitation of the existing Finished Water Tanks at the North Rim WTP site (Volume 6).
   1. The work consists of assessing the condition of and conducting rehabilitation of two, welded steel potable water tanks and tank appurtenances as indicated in this Specification, the tank Drawings, and in accordance with provisions of the Contract Documents.

C. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 35 73 – Delegated Design Procedure.
   2. Section 01 74 15 – Disinfection of Facilities.
   3. Section 05 12 00 – Structural Steel.
   4. Section 05 50 00 – Metal Fabrications.
   5. Section 09 96 00 – High Performance Industrial Coatings.
   6. Section 13 47 13.17 – Cathodic Protection of Steel Water Tanks – Internal Galvanic.
   7. Section 40 41 13 – Heat Tracing Cable

1.2 QUALITY ASSURANCE

A. Referenced Standard:
   1. American Petroleum Institute (API):
      a. STD 650, Welded Tanks for Oil Storage.
      b. STD 653, Tank Inspection, Repair, Alteration and Reconstruction.
   2. The American Society for Nondestructive Testing, Inc. (ANST):
      a. SNT-TC-1A, Personnel Qualification and Certification in Nondestructive Testing, Inc.
   3. American Society of Civil Engineers (ASCE):
      a. 7, Minimum Design Loads and Associated Criteria for Buildings and Other Structures.
   4. American Society of Safety Engineers (ASSE/SAFE):
      a. A1264.1, Safety Requirements for Workplace Walking/Working Surfaces and Their Access; Workplace, Floor, Wall and Roof Openings; Stairs and Guardrails Systems.
   5. ASME International (ASME):
      a. B16.5, Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard.
      b. B16.11, Forged Fittings, Socket-Welding and Threaded.
      c. B16.25, Buttwelding Ends.
   6. Arizona Administrative Code:
      a. R18-4-119 – Standards for Additives, Materials, and Equipment.
   7. ASTM International (ASTM):
      a. A6/A6M, Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling.
      b. A36/A36M, Standard Specification for Carbon Structural Steel.
      c. A53/A53M, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
      d. A992/A992M, Standard Specification for Structural Steel Shapes.

e. C1107, Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink).

8. American Water Works Association (AWWA):
   a. C652, Standard for Disinfection of Water Storage Facilities.
   b. D100, Standard for Welded Steel Tanks for Water Storage.
   c. D102, Standard for Painting Steel Water Storage Tanks.
9. American Welding Society (AWS):
   a. D1.1, Structural Welding Code - Steel.
   b. QC1, Specification for AWS Certification of Welding Inspectors.
10. Environmental Protection Agency (EPA):
    a. The Emergency Planning and Community Right-to-Know Act (EPCRA).
    b. The Resource Conservation and Recovery Act (RCRA).
11. NACE International (NACE):
    a. SP0178, Design, Fabrication, and Surface Finish Practices for Tank and Vessels to be Lined for Immersion Service.
12. National Institute of Steel Detailing (NISD).
13. NSF International Standard (NSF) / American National Standards Institute (ANSI):
    a. 61, Drinking Water System Components - Health Effects.
14. Occupational Safety and Health Administration (OSHA):
    a. 29 CFR 1910, Occupational and Safety and Health Standards, referred to herein as OSHA Standards.
15. Steel Structures Painting Council (SSPC):
    a. PA 1, Shop, Field, and Maintenance Coating of Metals.
    b. SP 6, Commercial Blast Cleaning.
    c. SP 10, Near-white Blast Cleaning.
    d. Vis 1, Surface Preparation Standards for Painting Steel Surfaces.

B. Qualifications:
1. Contractor shall have a minimum of five years experience in constructing similar tanks and shall be able to show evidence of at least five installations in satisfactory operation for at least five years.
2. Tank supplier shall have a minimum of 10 years experience in fabrication and erection of field erected, welded steel potable water storage tanks.
   a. Tank supplier's experience shall provide evidence of at least ten installations that have been in satisfactory operations for at least 10 years.
3. Qualify welding procedures and operators in accordance with AWS.
4. Provide services of a professional engineer, registered in the State of Arizona and experienced in the design of field erected welded steel potable water storage tanks and tank appurtenances who will be in responsible engineering charge of the Work associated with the welded steel potable water storage tanks specified herein.
   a. All calculations and detailed tank drawings shall be sealed and signed by a professional engineer, registered in the State of Arizona.

C. Quality Control:
1. Except as otherwise shown on the Drawings or specified, all design, materials, joints, workmanship and all other aspects of the tank design and fabrication shall conform to AWWA D100.
   a. Any required field repairs or alterations after initial erection shall conform to the most stringent requirements of AWWA D100, API 650, and API 653.
2. Testing agency:
   a. Testing of materials and field welds and water samples for compliance with technical requirements of the Specifications shall be the duty of a testing laboratory provided by the Contractor as described in Section 01 40 00.
3. Testing services:
   a. Test proposed materials in the laboratory and/or field for compliance with the Specifications, as described in Section 01 40 00.

b. Perform field radiograph tests to assure that the proper quality of field welding has been obtained.
c. Test water samples for bacteria for compliance with the Specifications.
d. Report all test results to the Contracting Officer's Representative.

4. Authority and duties of testing agency:
   a. Technicians representing the testing laboratory shall inspect and test the materials in the field and report their findings to the Contracting Officer's Representative.
      1) When the materials furnished or the Work performed fails to fulfill the Specification requirements, the technician shall inform the Contracting Officer to such failure.
   b. The technician shall not act as foreman or perform other duties for Contractor.
      1) Work will be checked as it progresses, but failure to detect any defective work or materials shall not in any way prevent later rejection when such defect is discovered, nor shall it obligate the Contracting Officer for final acceptance.
      2) Technicians are not authorized to revoke, alter, relax, enlarge, or release any requirements of the Specifications, nor to approve or accept any portion of the Work.
5. Responsibilities and duties of Contractor:
   a. The use of testing services shall in no way relieve the Contractor of their responsibility to furnish materials and construction in full compliance with the Contract Documents.
   b. Provide labor as necessary to obtain and handle samples at the Project site or at other locations where the material is sourced.
   c. Provide advance notification to the testing agency of operations to allow for completion of quality tests and for the assignment of personnel.

1.3 SUBMITTALS

A. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Finished Water Tank Condition Assessment Report
   1. Compile and submit all condition assessment reports on corrosion and coatings of the Finished Water Tanks to the Contracting Officer's Representative.

C. Quality Control:
   1. The supplier shall submit a copy of their quality assurance/quality control (QA/QC) procedure for review.

D. Shop Drawings:
   1. Product technical data including:
      a. Manufacturer's installation instructions.
      b. Acknowledgement that tanks as designed and fabricated meet the requirements of the standards referenced.
      c. Descriptions, drawings and manufacturer's cut-sheets of pre-manufactured pipe supports, ladders, safety cage, safety railing, ladder safety device, external water level gage, cane fiber filler, sealant, and other manufactured items.
   2. Fabrication and/or layout drawings:
      a. Prepare Shop Drawings under NISD Quality Procedures Program certification.
      b. Complete detailed Shop Drawings for all of the Work showing clearly all pieces, sizes, dimensions, details, connections, materials, and shop coatings.
         1) All Shop Drawings must be checked and signed "approved" before submittal.
         2) Show interior and exterior surface areas of the tank including details of all required accessories.
         3) Show all cuts, copes, and holes.
         4) Indicate all shop and field bolts.
         5) Indicate all shop and field welds using AWS symbols.
         6) Submit anchor bolt sizing and location drawing (for installation by contractor), if required by tank supplier's design.

7) Grounding lug locations relative to other tank features.

c. Fabrication shall not be started until Shop Drawings are approved.

E. Certifications:
1. Provide the name and experience record of the tank supplier or sub-contractor.
2. Copies of current welding certificates for each welder assigned to perform welding indicating compliance with testing specified by AWS.
3. Welder qualification data and prequalified welding procedures.
4. Special Inspections reports.
5. Submit certificate of compliance with NSF/ANSI 61 standard and with Arizona Administrative Code R18-4-119 for all materials or products which will be in contact with potable water.

F. Tank Supplier's Design Calculations:
1. Submit tank supplier's design calculations signed and sealed by a Registered Professional Engineer licensed in the State of Arizona, who may be an employee of the tank supplier. The state of registration, registration number and name on the seal shall be clearly legible.
a. Tank supplier's design calculations will be for information only and no direct action will be taken by the Contracting Officer's Representative.

G. Test Reports:
1. Furnish certified copies of test results from radiographs and inspections of test segments.

H. Record Drawings:
1. Immediately upon completion of the Work, submit Record Drawings showing actual in-place installation of the Work specified in this Section. Record Drawings shall show locations of all radiographs and test segments.

1.4 DELIVERY, STORAGE, AND HANDLING

A. Protect materials, shop fabricated components and accessories from damage during shipping. Cover all flanged nozzles with wooden blanks and provide plastic caps for all threaded or socket weld openings.

B. Store all materials and shop fabricated components in a suitable manner on site to avoid damage and prevent corrosion.

1.5 EQUIPMENT PERFORMANCE GUARANTEES

A. Tank supplier shall guarantee that material, equipment, and components in its scope of supply are free of defect and that workmanship is suitable for intended usage and environment.
1. Any failures or underperformance due to material defects or poor workmanship shall be corrected free of charge.
2. The guarantee shall be in effect for a period of 12 months after acceptance of the installation.
3. Acceptance shall be given in writing by Contracting Officer after successful start-up and commissioning.
4. All replacement parts or repairs shall be warranted for 12 months from date of replacement or repair.

PART 2 - PRODUCTS

2.1 NEW RAW WATER TANKS

A. Tank Requirements
1. Construct two (2) welded carbon steel potable water storage tanks conforming with the following requirements for each tank:
a. Size:
1) Diameter: 32 feet.
2) Nominal Capacity: 150,000 gallons.

3) Height from floor to crest of overflow: As indicated on Drawings
4) Maximum overall height to the top of shell plates: As indicated on Drawings

b. Roof Type:
1) Column supported conical roof with structural steel rafters and girders.
a) Rafters and girders shall be wide-flange sections conforming to ASTM A992
(1) Channel sections are not permitted
b) Interior columns shall conform to ASTM A53, unless otherwise approved by the Contracting Officer.
(1) Center column shall be configured to have a halo style support for the rafters.
(a) Dollar plate configuration is not allowed.
c) No structural members shall project below the water line at the interior of the tank.

c. Bottom Type:
1) Essentially a flat bottom tank.
a) Provide a nominal slope as indicated on the Drawings which slopes downward from the center of the tank to the perimeter shell of the tank.

d. Minimum Steel Thickness:
1) The minimum thickness of any part of the tank structure, except floor and roof plates, shall be the larger of the following:
a) 1/16 IN greater than the minimum values specified in Section 3.10 of AWWA D100.
b) 1/16 IN greater than the values determined from design criteria specified in Paragraph 1.2B above.
2) Floor plates shall be a minimum of 5/16 IN thick.
3) Roof plates shall be a minimum of 1/4 IN thick.

e. Additional Requirements:
1) It is the responsibility of the tank supplier to design the tanks, platforms and catwalks.
2) All grating, handrails, and ladders shall be designed to meet OSHA and Building Code requirements.
3) All tanks shall have external grounding lugs attached at the lowest point. Four grounding lugs shall be provided, located 90-degrees apart around the tank circumference.

B. Seismic Design Load Criteria
1. Meet requirements of Section 01 81 10 – Wind and Seismic Design Criteria.

C. Wind Design Load Criteria
1. Meet requirements of Section 01 81 10 – Wind and Seismic Design Criteria.
2. Wind loading design shall be for empty tank conditions.

D. Water Flow Rates (Each Tank)
1. Maximum Inlet Flow: 175 gpm.
2. Maximum Discharge Flow: 200 gpm.

E. Site Temperatures
a. Minimum: -20 DEGF.
b. Maximum: 120 DEGF.

F. Snow Load
1. Roof snow load: 256 PSF.

G. Site Elevation: 8361 FT above MSL.

H. Tank and Appurtenances Design Loads
1. Platforms: 40 PSF.
2. Ladders, stairs, and handrail: per OSHA standards.
3. Tank roofs: 25 PSF minimum.

I. Tank Accessories
   1. Shell Manholes (Manways):
      a. Install in bottom course of tank shell at locations identified on the Drawings.
      b. Manholes shall be 36 IN in diameter.
      c. The shell manhole requirements shall be in accordance with API 650.
         1) Provide tombstone shape reinforcing plate according to API 650, Figure 5-6.
         2) Provide manhole component sizes and dimensions according to API 650, Figure 5-7A.
      d. Provide heavy extruded EPDM rubber gasket between manhole cover and bolting flange.
   2. Piping Connections:
      a. Provide inlet, outlet, and drain piping connections as shown in Drawings.
      b. Pipe and connections shall conform to the applicable requirements of Division 40 – Process Interconnections.
   3. Overflow:
      a. Provide a steel overflow pipe, of the size shown on the Drawings, with welded or flanged connections.
         1) Attach pipe to tank shell with suitable brackets spaced at proper intervals.
         2) See Drawings for required air gap at discharge of overflow pipe.
         3) Pipe shall conform to the requirements of Division 40 – Process Interconnections.
         4) Provide #24 mesh screen on overflow at air gap.
   4. Ladders:
      a. Provide an outside vertical steel ladder with a hinged vertical security door.
      b. Ladders shall be located as shown on the Drawings.
      c. Ladders shall have square rungs with an anti-slip surface:
         1) Provide painted steel ladder rungs with an anti-slip surface permanently bonded or imprinted to the rung.
         2) Paint applied to anti-slip surface shall be limited to a maximum of 10 mils DFT, unless specified otherwise by the manufacturer.
         3) Anti-slip material shall be UV rated for continuous sun exposure.
         4) Anti-slip surface shall provide a commercial surface. Do not provide industrial or other surfacing that can cause injury to unprotected hands.
      d. The security door shall be fabricated from a minimum of 3/16 IN smooth metal plate, be a minimum of 8 FT in height and have provisions for a pad lock:
         1) Do not include any exterior bracing, protrusions or other appurtenances that could allow scaling or climbing of the security door.
         2) Security door shall be painted to match final tank exterior color.
      e. Provide one inside vertical steel ladder from the tank bottom to the roof hatch.
   5. Ladder Safety Devices:
      a. Provide a fall prevention system in the following locations:
         1) Vertical exterior ladders.
      b. Fall prevention systems shall consist of a trolley device that slides on a stainless steel cable system mounted to the ladder:
         1) The cable system shall be compatible with the existing Government's safety climb trolley devices.
         2) Provide 3M DBI-SALA Lad-Saf Top Bracket, Model No. 6116280.
         3) Provide 3M DBI-SALA Lad-Saf X3 Detachable Cable Sleeve, Model No. 6160054.
         4) Provide 3M DBI-SALA Lad-Saf Cable Vertical Safety System, Model No. 6119060, Stainless Steel.
         5) Provide 3M DBI-SALA Lad-Saf Cable Guide, Model No. 6100400.
         6) Provide 3M DBI-SALA Lad-Saf Cable Vertical Safety System Bracketry, Model No. 6116633.
   6. Vents:
      a. Furnish tanks with a vent located at the center of the roof.

1) Velocity of air flow through the vent shall allow reasonable natural air movement for the head space to be ventilated to protect the tank's interior coating system.
2) Vent shall be designed and constructed to exclude birds and insects using #24 mesh screen and to allow for anchorage of painter's rigging.
3) Bottom of tank vent opening shall be 36 IN minimum above tank roof.
4) Overflow pipe shall not be considered a tank vent.

7. Roof Hatches:
   a. At least one roof hatch not less than 36 IN by 30 IN shall be located on the roof of each tank as shown on the Drawings.
   b. The hatches shall be equipped with suitable hinges and a hasp to permit locking. The hinge shall be located on the 30 IN side of the hatch.
   c. The opening shall have a curb at least 4 IN high. The cover shall be arranged to provide an insect tight closure and shall overlap the curb at least 2 IN.
   d. The roof hatch shall be single leaf. The roof hatch shall be pre-assembled from the manufacturer.
   e. Minimum performance characteristics:
      1) Cover shall be reinforced to support a minimum live load of 40 PSF with a maximum deflection of 1/150th of the span or a maximum design pressure of 70 PSF with a factor of safety of 2.
      2) Operation of the cover shall be smooth and easy with controlled operation throughout the entire arc of opening and closing.
      3) Operation of the cover shall not be affected by temperature.
      4) Entire hatch shall be weathertight with fully welded corner joints on cover and curb.
   f. Cover shall be minimum of 11 gauge aluminum with a 3 IN beaded flange with formed reinforcing members. Cover shall have a heavy extruded EPDM rubber gasket that is bonded to the cover interior to assure a continuous seal when compressed to the top surface of the curb.
   g. Lifting mechanisms: Manufacturer shall provide compression spring operators enclosed in telescopic tubes to provide, smooth, easy, and controlled cover operation throughout the entire arc of opening and closing. The upper tube shall be the outer tube to prevent accumulation of moisture, grit, and debris inside the lower tube assembly.
   h. Hardware:
      1) Heavy pintle hinges shall be provided.
      2) Cover shall be equipped with a spring latch with interior and exterior turn handles.
      3) Roof hatch shall be equipped with interior and exterior padlock hasps.
      4) The latch strike shall be a stamped component bolted to the curb assembly.
      5) Cover shall automatically lock in the open position with a rigid hold open arm.
      6) All hardware shall be fabricated from Type 316 stainless steel.
   i. All aluminum material shall be dielectrically isolated from the tank.
8. Roof Openings:
   a. Where indicated on the Drawings, openings and couplings shall be provided in the roof. The top of the opening or coupling shall be set level.
9. Cathodic Protection System:
   a. See Drawings for layout of system.
   b. Meet requirements of Section 13 47 13.17 – Cathodic Protection of Steel Water Tanks Internal Galvanic.
10. Nameplate:
    a. Provide an engraved or embossed nameplate that lists the following:
       1) Tank name and SCADA identification number.
       2) Year tank was built.
       3) Tank supplier name.
       4) Coating subcontractor name.
       5) Interior coating type and number of coats.
       6) Exterior coating type and number of coats.

7) Nominal capacity in gallons.
8) Nominal diameter and nominal height, in feet and inches.
9) Name of Contractor.

b. The nameplate material shall be suitable for outdoor continuous sun exposure.
c. The dimensions shall be a minimum of 4 IN tall and 12 IN wide.
d. The nameplate shall be mounted near the tank using stainless or galvanized strut support material. The nameplate shall be continuously supported.

11. Tank Level Indicators:
    a. Provide tank level indicators on each tank as shown on the Drawings.
    b. Manufacturers:
        1) Versa-Steel.
        2) Or approved equal.
    c. Style:
        1) Indicator shall be half travel where the bottom of the indicator will be attached halfway up the tank wall. The indicator will read the full depth of the tank.
        2) Indicator shall be direct read with the bottom showing 0 FT and the top showing the maximum tank depth.
    d. Materials:
        1) Any materials required for installation not provided by the manufacturer shall be furnished by the Contractor.
12. Grounding Lugs:
        1) All tanks shall have external grounding lugs attached at the lowest point. Four grounding lugs shall be provided, located 90-degrees apart around the tank circumference.
13. Lightning Protection Air Terminal Mounts:
    a. Welded brackets shall be provided for mounting of lightning protection air terminals by the lightning protection supplier. Coordinate location and quantity of brackets with lightning protection designer/supplier.
14. Post-Construction Survey:
    a. Contractor shall pay for and employ the services of a registered land surveyor to provide post-construction survey that includes the northing, easting, and elevation of the following locations:
        1) Tank floor elevation at the tank shell.
        2) Tank floor elevation at the tank center.
        3) Tank roof elevation at the tank shell.
        4) Tank roof elevation at the tank center.
        5) Overflow weir crest elevation.
    b. Surveyor shall be licensed in the State of Arizona.

J. Non-Shrink Grout
1. Use non-shrink grout ASTM D 1107/C 1107M, approximately 3/4-inch thick. Grout shall be installed below the bottom plates, wherever they rest on concrete, as shown on the Drawings.

K. Polysulfide Sealant
1. Sealant shall be installed as shown on the Drawings or as approved by the Contracting Officer's Representative.
    a. Sealant shall be premium-grade, high-performance, moisture-cured, 1-component, polyurethane-based, non-sag elastomeric sealant meeting Federal specification TT-S-00230C, Type II, Class A, and ASTM C-920, Type S, Grade NS, Class 35, use T, NT, O, M, G, I.
    b. All sealant must be NSF 61 certified.

L. Tank Coatings
1. See Specification Section 09 96 00 – High Performance Industrial Coatings.

M. Foundation

1. The tank foundation shall consist of an oiled (liquid asphalt) sand cushion sitting on a reinforced concrete ringwall footing. The tank supplier shall supply the oil sand cushion along with protection of the sand cushion from erosion per AWWA D100.

2.2 EXISTING FINISHED WATER TANK

A. Tank Characteristics
   1. Tank A
      a. Size
         1) Diameter: 93 FT.
         2) Nominal Capacity: 2 Million Gallons.
         3) Height from floor to crest of overflow: As indicated on Drawings.
         4) Maximum overall height to the top of shell plates: As indicated on Drawings.
      b. Year Constructed: 1963.
   2. Tank B
      a. Size
         1) Diameter: 93 FT.
         2) Nominal Capacity: 2 Million Gallons.
         3) Height from floor to crest of overflow: As indicated on Drawings.
         4) Maximum overall height to the top of shell plates: As indicated on Drawings.
      b. Year Constructed: 1977.

B. Tank Accessories
   1. Shell Manholes (Manways):
      a. Replace 24” side access manway seal with heavy extruded EPDM rubber gasket between manhole cover and bolting flange.
   2. Vents:
      a. Replace existing tank vents with new vents in the same location.
         1) Velocity of air flow through the vent shall allow reasonable natural air movement for the head space to be ventilated to protect the tank’s interior coating system.
         2) Vent shall be designed and constructed to exclude birds and insects using #24 mesh screen and to allow for anchorage of painter’s rigging.
         3) Bottom of tank vent opening shall be 36 IN minimum above tank roof.
         4) Overflow pipe shall not be considered a tank vent.
   3. Cathodic Protection System:
      a. Remove existing cathodic protection system and replace with new system.
      b. See Drawings for layout of new system.
      c. Meet requirements of Section 13 47 13.17 – Cathodic Protection of Steel Water Tanks Internal Galvanic.
   4. Tank Level Indicators
      a. Remove and replace tank level indicators on each tank.
      b. Manufacturers:
         1) Versa-Steel.
         2) Or approved equal.
      c. Style:
         1) Indicator shall be half travel where the bottom of the indicator will be attached halfway up the tank wall. The indicator will read the full depth of the tank.
         2) Indicator shall be direct read with the bottom showing 0 FT and the top showing the maximum tank depth.
      d. Materials:
         1) Any materials required for installation not provided by the manufacturer shall be furnished by the Contractor.

C. Polysulfide Sealant
   1. Sealant shall be installed as shown on the Drawings or as approved by the Contracting Officer’s Representative.

a. Sealant shall be premium-grade, high-performance, moisture-cured, 1-component, polyurethane-based, non-sag elastomeric sealant meeting Federal specification TT-S-00230C, Type II, Class A, and ASTM C-920, Type S, Grade NS, Class 35, use T, NT, O, M, G, I.
b. All sealant must be NSF 61 certified.

D. Tank Coatings
1. Recoat interior and exterior of both tanks and interior piping.
2. See Specification Section 09 96 00 – High Performance Industrial Coatings.

PART 3 - EXECUTION

3.1 NEW RAW WATER TANKS

A. Installation
1. Except as otherwise shown on Drawings or specified, erect tank in accordance with the requirements of AWWA D100 and this specification.
2. All tank fabrications, connections, and appurtenances shall be welded connections unless shown otherwise on the Drawings.
a. All welded connections shall be continuous seal welds.
b. The use of fasteners is acceptable only for temporary erection and assembly purposes.
c. All fasteners used for temporary erection and assembly purposes shall be removed or completely seal welded on all sides of the bolts, nuts, and washers. If fasteners are completely removed, the mating edges of connections at bolt holes shall be seal welded.
d. Perform all welding in accordance with the applicable design code/standard and AWS procedures.
3. Install all ancillary equipment, platforms, and devices in accordance with the tank design codes/standards and manufacturer's instructions for buy out items.
4. Asphalt-impregnated Sand Cushion:
a. Sand shall be thoroughly mixed with liquid asphalt prior to placement on a reinforced ringwall footing. Asphalt content of the mix shall be 6 to 7% by weight. Sand shall be placed and compacted by rolling with pneumatic tired rollers so finished surface is true to grade and the edge is approximately 1/4 IN above concrete mat foundation.
5. Install cane fiber filler between bottom plates and concrete pipe encasements.
6. Leveling and Sealing:
a. Provide a 4 IN by 4 IN steel shim of varying thicknesses at 4 FT apart on ringwall to level tank and to provide a 3/4 IN minimum space between tank bottom and top of ringwall foundation for grouting. Other methods of leveling may be used if approved by Contracting Officer. The entire outer edge of the bottom shall be sealed to ringwall with a 1/2 IN width of a two-component polysulfide sealant per specifications.
7. Pipe Connections:
a. Connections between tank and piping shall be made by the Contractor. Method of connection shall be approved by the Contracting Officer's Representative and, when completed, shall provide a completely watertight joint.
b. After connection is made to tank bottom, lip of drain pipe shall be ground flush with bottom of tank.
c. Piping installation shall conform to the applicable requirements of Division 40 – Process Interconnections.
8. Roof Plate Interior Lap Joints:
a. Seal laps on the underside of the roof plate using continuous seal welds.

B. Field Quality Control
1. Qualifications of Welders and Welding Procedures:
a. Welding shall be performed only by welders qualified in accordance with AWWA D100.
b. Welding shall be performed in accordance with AWWA D100.
2. Inspection:

a. Inspections shall be per the requirements of the project Specifications and per AWWA D100.
b. Weld inspections shall include both spot radiographs and test segments.
    1) The first welding inspection shall be made in the presence of the Contracting Officer's Representative, and at such other times as the Contracting Officer's Representative may require after initial inspection.
    2) Contracting Officer's Representative shall designate locations where spot radiographs are to be made and where test segments are to be removed.
c. Provide and operate the necessary equipment for taking test plugs and spot radiographs required by the above specification. Remove, prepare, and test as many plugs and spot radiographs as Contracting Officer's Representative may require, in accordance with Section 11 of AWWA D100. Cost of taking spot radiographs and plugs, preparing them, and testing them shall be paid for by the Contractor.

3. Testing and Repair:
    a. Bottom Plate:
        1) Perform weld inspections as required by Paragraph 3.2.B.
        2) After tank has been completely erected, bottom welds shall be tested for leakage by covering weld seams with soap suds and applying a partial vacuum of at least 2 PSI by means of a vacuum box with a gage and an observation window.
        3) All bottom welds shall be tested and all tests shall be witnessed by the Contracting Officer's Representative, unless otherwise permitted and approved in writing by the Contracting Officer.
    b. Shell Plate:
        1) Perform weld inspections as required by Paragraph 3.2.B.
        2) After tank has been completely erected, but before surface preparation for painting, it shall be filled with water furnished at the tank site, as directed by the Contracting Officer's Representative, at a sufficient volume to fill the tank to its maximum level and inspected for leaks.
            a) Dispose of test water as directed by the Contracting Officer's Representative. Obtaining proper permitting for the discharge and disposal of the test water is the responsibility of the Contractor.
    c. Piping:
        1) Pipe shall be tested as required by applicable requirements of Division 40 – Process Interconnections.
    d. Repair and retesting:
        1) Any leaks or defective welds shown by testing shall be repaired in accordance with requirements of AWWA D100 and to complete satisfaction of Contracting Officer.
        2) After repairs have been made, water level shall be raised and repaired area shall be retested.
        3) If leakage from tank is of a magnitude that entire tank must be retested, responsibility belongs to Contractor for any additional water costs required. Contracting Officer to determine the water rate schedule to be used in computing water cost.
        4) During the filling of the tank, the tank shall be inspected for leaks. Leaks occurring during or after filling of the tank, shall be repaired by drilling, chipping or gas gouging out all defective welds and rewelding.
        5) No repair work shall be done on any joint unless the water is at least 2 FT below the area being repaired.
        6) If the number and/or severity of the leaks are, in the opinion of the Contracting Officer, too extensive to correct while the tank is filled with water, the Contracting Officer may direct that the tank be emptied and the entire disinfection and re-filling procedure be repeated until all leaks are sealed and the tank is properly disinfected.
        7) All defects and leaks shall be repaired by the Contractor in accordance with the requirements of AWWA D 100 and to the complete satisfaction of the Contracting

Officer. The interior and exterior paint coatings shall be completely repaired, including repeat of surface preparation of the steel before re-coating.

C. Coating
   1. General:
      a. Except as noted herein, preparation of all surfaces and application of complete coating systems shall be performed in the field.
      b. All exposed steel surfaces, inside and outside of tank shall be coated per Specification Section 09 96 00 – High Performance Industrial Coatings.
      c. Contact surfaces of shop assemblies, which must be cleaned before forming or welding, and surfaces that will be otherwise inaccessible after installation shall be prepared in the shop. Immediately after shop sand blasting and cleaning, contact surfaces of shop assemblies shall be coated with appropriate prime coat.
      d. Top of rafters and girders shall be prepared, primed, and coated prior to erection.
      e. All coatings and coating products used to be utilized for surfaces in contact with the tank contents shall be NSF/ANSI 61 approved for use in potable water service.

D. Cleaning
   1. Painted surfaces shall be protected from discoloration, scratching or other damage. Any damaged areas requiring touch-up painting prior to project completion and acceptance shall be the responsibility of the Contractor.

E. First Anniversary Inspection
   1. Inside and outside surfaces of tank shall be inspected by Contracting Officer and Contractor approximately 11 months after the painting Work has been completed. Inspection, remedial work, if required, and report shall be as required by AWWA D102, Section 9.

F. Monitoring and Reporting
   1. Contractor shall submit monthly EPRCA tabulations of abrasives, solvents, gases, and bulk material by weight and dates on site.
   2. Contractor shall submit monthly RCRA solid waste and/or hazardous waste reports. Contractor required to have waste stream for solid waste and/or hazardous waste determined and sample results of waste streams.

3.2 EXISTING FINISHED WATER TANKS

A. Sediment Removal
   1. Remove up to 6 IN of sediment from the floor of both tanks.

B. Corrosion Condition Assessment
   1. Provide a corrosion assessment on the full interior of both tanks.
   2. Assessment shall include non-destructive magnetic flux leakage testing using a floor scanner on the entire floor of both tanks.
      a. Identify areas of greater than 50% section loss for additional investigation.
      b. If any floor areas are not accessible, perform spot checks using an Ultrasonic Thickness Gauge.
   3. Any repair work deemed necessary by the Corrosion Condition Assessment is not part of this scope of work.

C. Coating
   1. Existing coatings on all exposed steel surfaces, inside and outside of the tanks shall be removed and the surfaces prepared per Specification Section 09 96 00 – High Performance Industrial Coatings.
   2. Existing coatings on all exposed piping inside the tanks shall be removed and the surfaces prepared per Specification Section 09 96 00 – High Performance Industrial Coatings.
   3. All exposed steel surfaces, inside and outside of the tanks shall be coated per Specification Section 09 96 00 – High Performance Industrial Coatings.
   4. All piping inside the tanks shall be coated per Specification Section 09 96 00 – High Performance Industrial Coatings.

a. All coatings and coating products used to be utilized for surfaces in contact with the tank contents shall be NSF/ANSI 61 approved for use in potable water service.

D. Installation
   1. Pipe Connections:
      a. A new 8 IN steel pipe with heat trace shall be connected to Finished Water Tank A as shown in the Drawings.
      b. New piping connections shall provide completely watertight joints.
      c. Piping installation shall conform to the applicable requirements of Division 40 – Process Interconnections.
      d. See Section 40 41 13 for heat trace requirements

E. Disinfection
   1. Disinfection shall conform to all applicable requirements of AWWA C652, and as specified in this Section.
      a. After tank has been painted and interior surfaces have thoroughly dried, remove all visible dirt and contaminating materials. Interior of tank shall be disinfected by spraying all surfaces, including underside of roof and roof support members, with a 200 ppm chlorine solution. Solution shall remain in contact with surfaces for a minimum of 30 minutes. Technique shall be such that a sterile tank will result. After spray disinfection, tank shall be flushed to drain for at least ten minutes, then tank shall be filled and allowed to overflow for another ten minutes. Samples for bacteriological tests will then be taken by Contractor. Disinfection shall continue until bacteria-free samples are obtained.
      b. Water for initial disinfection and for filling and for any additional disinfection during the initial disinfection procedure if required to obtain satisfactory bacteriological samples, will be furnished by the Project, to the site boundary. Responsibility belongs to Contractor for pumps, hoses and other temporary equipment required to fill the tank, all chlorine required, and for obtaining proper disinfection as determined by bacteriological tests made by the testing laboratory.
      c. If the tank must be emptied, re-sprayed, flushed and refilled to obtain satisfactory bacteriological samples, or because of extensive leakage, the Project will furnish additional water for the Work at the expense of Contractor. Additional chlorine required costs shall belong to Contractor.
      d. After the chlorination procedure is completed and accepted by the Contracting Officer, the tank shall be emptied.
      e. Supply all necessary pumps, hoses and other required equipment each time the tank is to be emptied.

F. Cleaning
   1. Painted surfaces shall be protected from discoloration, scratching or other damage. Any damaged areas requiring touch-up painting prior to project completion and acceptance shall be the responsibility of the Contractor.

G. Monitoring and Reporting
   1. Contractor shall submit monthly EPRCA tabulations of abrasives, solvents, gases, and bulk material by weight and dates on site.
   2. Contractor shall submit monthly RCRA solid waste and/or hazardous waste reports. Contractor required to have waste stream for solid waste and/or hazardous waste determined and sample results of waste streams.

END OF SECTION

This page intentionally left blank.

SECTION 43 41 16

CARBON STEEL TANKS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Reaction Tanks and associated accessories at North Rim WTP (TK-63011, TK-63012) – Volume 6.

B. Related Specification Sections include, but are not necessarily limited to:
   1. Section 01 61 03 – Equipment – Basic Requirements.
   2. Section 01 81 10 – Wind and Seismic Design Criteria.
   3. Section 09 96 00 – High Performance Industrial Coatings.
   4. Section 46 41 11 – Vertical Mixers.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Petroleum Institute (API):
      a. STD 2000, Venting Atmospheric and Low Pressure Storage Tanks.
   2. American Society of Civil Engineers (ASCE):
      a. 7, Minimum Design Loads for Buildings and Other Structures.
   3. ASTM International (ASTM):
      a. A36, Standard Specification for Carbon Structural Steel.
      b. A635, Standard Specification for Steel, Sheet and Strip, Heavy-Thickness Coils, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, and High-Strength Low-Alloy with Improved Formability, General Requirements for.
      c. A1011, Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength.
   4. National Fire Protection Association (NFPA):
      a. 30, Flammable and Combustible Liquids Code.
   5. Petroleum Equipment Institute (PEI):
      a. RP200, Recommended Practices for Installation of Aboveground Storage Systems.
   6. Society for Protective Coatings/NACE International (SSPC/NACE):
      a. SP 6/NACE No. 3, Commercial Blast Cleaning.
   7. Underwriters Laboratories, Inc. (UL):
      a. 142, Standard for Safety Steel Aboveground Tanks for Flammable and Combustible Liquids.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product technical data including:
      a. Acknowledgement that product submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Seismic, wind and anchorage calcs per Specification Section 01 81 10.
   3. Fabrication and/or Layout Drawings:
      a. Complete detailed Drawings of equipment furnished.
      b. Complete construction details.
         1) Anchor bolt locations for installation by Contractor.

2) All nozzles and fittings.

4. Certifications:
   a. Certificates of compliance with standards specified for all major components incorporated into work.
   b. Certification the equipment was installed in accordance with manufacturer's written Specifications.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.4 QUALITY ASSURANCE

A. Manufacturer shall be regularly engaged in the business of providing carbon steel tanks meeting API-650, and shall have successfully provided steel tanks for a minimumof ten (10) installations in the last three (3) years.

1.5 WARRANTY

A. Tanks shall be warranted for a minimum of three (3) years against any defects in material or workmanship. A written warranty shall be provided.

B. Manufacturer shall replace tank inwhole, or any individual component, found to be defective during the warranty period.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Eaton.
   2. Or Equal.

2.2 MATERIALS

A. Storage Tank:
   1. Carbon Steel (ASTM A36, ASTM A635 or ASTM A1011).

B. Tank Fittings, Nozzles and Vents:
   1. Conform to the requirements of NFPA 30, UL 142, API STD 2000, and PEI RP200.

2.3 TANK DESIGN REQUIREMENTS

A. Construction:
   1. Vertical, above-ground, vented, cylindrical shaped, welded steel tank with 30 degree cone bottom and dometop, with stand, meeting requirements of API-650.
   2. Meet requirements of Section 01 81 10 – Wind and Seismic Design Criteria
   3. Tank installation:
      a. Tanks installed on concrete pad in Process Floor of North Rim WTP.
   4. Capacity:
      a. 2,500 GAL (nominal).
      b. Shell diameter: 7.5 FT.

c. Straight sidewall height: 6.5 FT.

5. Welds:
   a. Use low molybdenum content welds that are compatible with the concentrations and temperatures of the solution being stored.
   b. Welded on both the interior and exterior sides of the tank.
   c. Welds completed by certified welders.
6. Minimum Steel Thickness:
   a. Top and bottom= 3/8-inch
   b. Sidewalls= 5/16-inch
7. Nozzles:
   a. Flanged, 150#.
8. Nuts and Bolts.
   a. All nuts and bolts shall be 316 SST, Hastelloy C or titanium.

B. Operating Pressure Conditions:
   1. Normal operation- Atmospheric.

C. Design Loads:
   1. See Specification Section 01 81 10 and Drawings.
   2. Contractor shall coordinate mixer design loads with tank manufacturer.

D. Tank manufacturer to provide stainless steel anchor bolts and other materials as required for installation of tank on foundation.

E. Nozzle Schedule: See Drawings.

2.4 TANK ACCESSORIES

1. Lift lugs as required to lift tank and set it in place vertically.
2. Tie-downs as required to meet Design Loads.
3. Ladder and safety cage (TK-63011 only).
   a. Provide ladder fabricated of carbon steel matching the material for the tank.
   b. Ladder and safety cage shall be delivered separately and bolted into place.
      1) Ladder and cage shall be removable.
   c. See Drawings for additional support requirements.
4. Catwalk, platform, and mixer supports:
   a. Carbon steel framing.
   b. Grating: Aluminum.
      1) For loading, see structural drawings.
   c. Coordinate with mixer manufacturer.
5. Provide external overflow pipe with lateral supports.
   a. See Drawings for extents of overflow pipe.
6. Provide exterior signs:
   a. Tank manufacturing details.

2.5 SHOP OR FACTORY FINISHING

A. Reaction Tank Interior and Exterior Finishing and Coatings:
   1. Preparation:
      a. Thoroughly clean interior and exterior of tanks surfaces of rust, slag, foreign debris, oil, prepare surface profile per coating product data sheet.
   2. Coatings:
      a. Interior surfaces, exterior side walls, exterior cone, tank top surface, support frame, ladder, catwalk, and platform:
         1) Tnemec Epoxoline Series FC22.
         2) Or equal.
      b. Exterior color shall match raw water piping.

PART 3 - EXECUTION

3.1 DELIVERY, STORAGE AND HANDLING

A. Unload, handle, and store tanks per the manufacturers' recommendations.

B. Prevent foreign debris from entering the tank nozzles and fittings throughout shipment and installation.

3.2 INSTALLATION

A. Erect and install tanks in accordance with the manufacturer's written instructions.

B. The tank pad or mounting surface shall be level to within 1/8-inch over a six-foot span, or as recommended by the tank manufacturer.

3.3 FIELD QUALITY CONTROL

A. Reaction Tank:
   1. Testing:
      a. Perform low-pressure air test of tank and fittings, including temporary blind flanges at end of tank overflow, fill and vent lines, and up to the closed valve on the outlet line.
      b. Maintain pressure of 3 PSIG and apply soap solution to all exterior fittings. Any visible air bubbles are to be considered a leak.
      c. Repair any leaks found and retest until there are no leaks.
   2. Exterior Coating:
      a. Properly restore any coating damaged during shipping and installation by applying the same coating that manufacturer used in shop applications.
      b. Provide coating for field welds.

B. Employ and pay for services of equipment manufacturer's field service representative.
   1. Inspect equipment covered by this Specification Section and the equipment installation.
   2. Provide Contracting Officer's Representative with a written statement that manufacturer's equipment has been installed properly and in accordance with the requirements of UL 142.

END OF SECTION

SECTION 43 41 45
FIBERGLASS REINFORCED PLASTIC (FRP) TANKS

PART 1 - GENERAL

1.1 DESCRIPTION

A. Description of Work:
   1. Roaring Springs Pumphouse:
      a. Finished Water Tank: One (1) TK-53211.
      b. Finished Water Tank Appurtenances: Immersion Heater (HTR-53241).

B. Related Specification Sections include, but are not limited to:
   1. See Section 01 79 00 – Demonstration and Training, for operator training requirements.
   2. Section 01 81 10 – Wind and Seismic Design Criteria
   3. Section 01 61 03 – Equipment – Basic Requirements
   4. Section 46 33 11 – Chemical Feed – Liquid Systems
   5. Division 26 – Electrical
   6. Division 40 – Process Interconnections

1.2 QUALITY ASSURANCE / QUALITY CONTROL

A. Manufacturer's Qualifications: Manufacturer of the FRP tanks shall have experience in providing similar type equipment and shall show evidence with his submittal of at least five (5) installations where equipment of the same material and design and similar application as specified herein have been in satisfactory operation for at least five (5) years.
   1. Manufacturer shall have FRPI certification.

B. Reference Standards:
   1. The material employed in items fabricated of fiberglass reinforced plastic shall be capable of withstanding maximum calculated stresses that may occur during fabrication, installation, and continuous operation, with allowance for an adequate safety factor. The Manufacturer shall provide a certification that materials employed in items of fabrication of fiberglass reinforced plastic shall meet the materials testing requirements defined in the latest revisions of the following ASTM standards as applicable for each FRP tank to be furnished under this Section.
      a. ASTM C 581, Determining Chemical Resistance of Thermosetting Resins Used in Glass Fiber Reinforced Structures Intended for Liquid Service
      b. ASTM C 582, Standard Specification for Contract-Molded Reinforced Thermaretting Plastic Laminates for Corrosion-Resistant Equipment
      c. ASTM D 638, Test for Tensile Properties of Plastics
      d. ASTM D 695, Test for Compressive Properties of Rigid Plastics
      e. ASTM D 746, Test Method for Brittleness Temperature of Plastics and Elastomers by Impact
      f. ASTM D 790, Test for Flexural Properties of Plastic
      g. ASTM D 883, Standard Nomenclature Relating to Plastics
      h. ASTM D 1505, Test Method for Density of Plastics by the Density Gradient Technique
      i. ASTM D 1693, Test Method for Environmental Stress Cracking of Ethylene Plastics
      j. ASTM D 2563, Classifying Visual Defects in Glass Reinforced Laminate Parts
      k. ASTM D 2583, Test for Indentation Hardness of Plastics by Means of a Barcol Impresser
      l. ASTM D 2996, Filament Wound, Reinforced Thermosetting Resin Pipe
      m. ASTM D 3299, Filament Wound Glass Reinforced Thermoset Resin Chemical Resistant Tanks
   2. NSF/ANSI Standards:
      a. NSF-61/ANSI-61, Drinking Water System Components

3. American Water Works Association (AWWA)
    a. D120 – Thermosetting Fiberglass-Reinforced Plastic Tanks.

1.3 SUBMITTALS

A. Shop Drawings:
1. Shop drawings for the tank shall include as a minimum the following:
    a. Resin and materials of construction including thickness of each layer.
    b. Service conditions for chemical environment and temperature.
    c. A statement from the manufacturer that fabrication shall be in accordance with these Specifications.
    d. A statement from the manufacturer that the materials and resin used for each tank are designed to meet all requirements for the intended service as specified herein.
    e. Tabulation in checklist form to indicate compliance with applicable ASTM standards for each tank and service condition.
    f. Certifications and design calculations required by the specifications.
    g. Details of tank anchoring and tie-down systems for each tank for the service and load conditions specified herein.
    h. Layouts and assembly schedules for tank identifying the location and elevation from the bottom of the tank for all inlet, outlet, and other integrally molded connections, baffles and appurtenances specified herein and shown on the approved Shop Drawings.
    i. Details of access ladders, and top access platforms and handrails for all tanks requiring top access with design calculations.
    j. A certificate from resin manufacturer listing nomenclature, composition, and characteristics of the resin.
    k. Paint color.
    l. Tank insulation materials.
    m. Total shipping weight, per tank.

B. Calculations
1. Seismic, wind and anchorage calcs per 01 81 10 stamped by Arizona PE.

C. Factory Test Reports:
1. Each tank shall be given a hydrostatic test by the manufacturer before shipment.
2. Each tank shall be inspected for defects in accordance with the requirements of ASTM C582.
3. Each tank shall be completely shop-fabricated and shall be shipped to the Point of Delivery as a single unit.
    a. Ladders shall be installed in the field by Contractor.

D. Installation Instructions
1. Submit with the delivery of the FRP tank detailed Installation Instructions.

E. Certificates Warranties and Guarantees
1. Certification of Design: The Certification of Design shall be signed by a Licensed Professional Engineer currently registered in the State of Arizona. The Manufacturer shall provide with the submittals, a Certification of Design with calculations from the tank manufacturer verifying that:
    a. The FRP storage tanks to be furnished under this Section meet the applicable ASTM and International Building Code (IBC) standards, and requirements specified herein for the conditions of service for each tank.
    b. The design of the tanks and supports to be furnished under this Section meet the loading conditions, lateral restraint, and seismic force requirements specified herein.

1.4 WARRANTY

A. Provide with a manufacturer’s warranty certificate for defects in workmanship and materials for of period of five (5) years from the date of delivery to the project site.

1. This warranty shall specifically state that if any leaks occur in the tank, the manufacturer shall repair the leaks at their own cost.
2. If three (3) leaks occur in the tank during the warranty period, the tank manufacturer shall remove the non-conforming tank, and install a new tank at no cost to the Government.

B. See Specification Section 01 67 00 – Product Requirements.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the salient characteristics.

B. Subject to compliance with the Contract Documents the following manufacturers are acceptable:
   1. Design Tanks.
   2. Or Equal.

2.2 PROCESS SYSTEM DESIGN AND PERFORMANCE

A. Site Temperatures:
   1. Minimum: 20 DegF.
   2. Maximum: 90 DegF.

B. Snow Load:
   1. Tank Roof Snow Load: 147 LB / SQ. FT
      a. In accordance with IBC 2024.

2.3 DESIGN CRITERIA

A. Finished Water Storage Tank:
   1. Provide FRP storage tanks as follows. The values are nominal.
   2. Tag: TK-53211
   3. Location: Outside at Roaring Springs Pumphouse.
   4. Tank description: Finished Water Storage Tank.
   5. Min. storage capacity: 7,100 gallons.
   6. Nominal diameter (without insulation): 11 FT.
   7. Height: 11 FT.
   8. Liquid: Water.
   9. Specific Gravity: 1.0
   10. See Sheet 052 D310.

2.4 EQUIPMENT DESIGN AND FABRICATION

A. General
   1. Configuration:
      a. Finished water storage tank:
         1) Flat bottomed, vertical cylindrical, vented tank with flat top.
   2. Include only that volume in the vessel below the overflow pipe invert elevation.
      a. Nominal capacity or the inclusion of volumes above the overflow pipe invert as additional capacity shall not be acceptable.
   3. Design finished water storage tank for a concentrated top load of 250 pounds distributed over a 4 x 4 square inch area at the top of the tank.
   4. Design storage tanks for seismic forces in accordance with Specification 01 81 10 - Wind and Seismic Design Criteria.

5. Insulation:
    a. Tanks to include 2” thick foam board insulation with FRP overwrap.
    b. Tanks should be rated for a 30 year service life in high-heat, strong UV, arid conditions.
6. Paint/Color:
    a. Exterior of tank/insulation (whichever layer is on the exterior): painted a neutral, natural color that blends with the surrounding environment.
    b. Submit available color palette to Contracting Officer for selection.

B. Equipment Design Requirements
1. Meet requirements of Section 01 81 10 – Wind and Seismic Design Criteria.
2. Tank must be NSF-61 certified.
3. Details of Construction:
    a. Filament wound main structure shall meet the requirements of ASTM D3299.
    b. All tanks shall meet or exceed Visual Acceptance Level II in accordance with ASTM D2563.
    c. The laminate comprising the structural tank (bottom, cylindrical shell, top head) shall consist of a corrosion resistant barrier comprised of an inner surface, interior layer, and structural layer.
        1) Inner Surface: The inner surface exposed to the chemical environment shall be a resin rich layer 15 to 30 mils thick, reinforced with 2 ply of synthetic fiber surface mat. Resin shall be compatible with stored chemical.
        2) Interior Layer: The inner surface layer exposed to the corrosive environment shall be followed with a layer composed of resin, reinforced only with noncontinuous glass fiber strands applied to a minimum thickness of 100 mils. The combined thickness of the inner surface and interior layer shall be 115 to 130 mils and in no case less than 110 mils.
        3) Glass content of the inner liner and the interior combined shall be 27 percent ±5 percent by weight.
        4) The degree of cure, after post cure shall be such as to exhibit a Barcol hardness on the inner surface of at least 90 percent of the resin manufacturer’s minimum specified hardness for the cured laminate with a synthetic surface mat.
        5) Structural Layer: Subsequent reinforcement shall be continuous strand roving that, in combination with the inner surface and interior layers, is needed to satisfy the design requirements.
            a) Glass content of this filament wound structural layer shall be 50 to 80 percent by weight.
            b) Only those constructions evaluated for design properties shall be used.
        6) Overall wall thickness: a minimum of 0.25 inches.
    d. Post Cure: post-cure complete tank with dry heat in accordance with the resin manufacturer’s recommendations.
        1) Complete any internal repairs or rework prior to post curing.
        2) If any repairs are made following post cure, an additional posture cycle is required.
    e. Head: The head knuckle radius: 1.5 IN, minimum.
    f. Bottom: The bottom knuckle radius: 2 IN, minimum.
    g. Extend reinforcement of the knuckle radius area up the vertical tank shell a minimum of 12 IN.
4. Flanged Nozzles:
    a. Hand lay-up construction with the pipe stub molded integrally with the pipe flange and gusseted, constructed in conformance with requirements of the referenced standard.
    b. Connections to FRP tank nozzles: provide flexible connections, with spring constants rated for FRP.
    c. Flanged Nozzles:
        1) FRP gusseted, flat face, conforming to ANSI B16.5 ISO-pound drilling.
        2) Contact molded to the tank wall with inside and outside layups as required.
        3) Located and oriented as shown on the approved shop drawing submittal.

4) Protrude sufficiently to allow clear distance between the back of the tank flange and the outer surface of the tank (4-in minimum) to allow access to flange bolts.
5) Threaded (NPT) coupling tank penetrations are not permitted.
6) Manufacturer responsible for the flanged nozzle coordination to ensure that the flange positions and orientations shall be per specific tank locations, and that all tank connections shall match the installation requirements according to specific site locations and positions.

d. Suction nozzles per drawings and of the
1) Full drain type with centerline of the nozzle positioned at the elevation of the tank bottom.
a) If a flat bottom tank is provided, notch tank concrete pad as required by the Manufacturer to accommodate the full drain flanged nozzle.
b) If the internal bottom surface is sloped, provide minimum of three inches across the diameter, toward the drain/pump outlet.

e. Top of overflow pipes: located as shown in attached nozzle schedules and on drawings. Overflow connections: flanged and positioned as shown on the drawings.

f. Nozzle Schedule
1) Finished Water Storage Tank – Roaring Springs Pumphouse (TK-53211)
a) Nozzle schedule shown on Sheet 052 D503.

5. Manways:
a. Finished Water Storage Tank:
1) Sidewall Manways:
a) 24 inch tank opening.
b) Water-tight.
2) Top Manways:
a) Flat top.
b) 24 inch tank opening.
c) Post and bar type hinge.
d) Drip lip.
e) Raised collar laminated into the tank top
f) Water-tight.
g) Lock-able.

6. Lifting Lugs:
a. Capable of withstanding weight of empty tank with safety factor of 5 to 1.
b. Provide a minimum of four (4) lifting lugs on all storage tanks.
c. Type 3I6L stainless steel
d. Attached to tank wall with hand-layed up laminate equal to or greater than the tank wall thickness.

7. Tie-down Ring/Mounting Lugs:
a. Meet requirements of Section 01 81 10 – Wind and Seismic Design Criteria
b. Design housing for full bottom support and provide with a minimum of four (4) Type 316 stainless steel hold down lugs.
c. Furnish all necessary Type 316 stainless steel adhesive type anchor bolts, nuts and washers as required to meet the design requirements specified.
d. Design tank housing and base with ample supports so that the whole assembly is self supporting in any operation condition and in accordance with all external loading specified herein.
e. Provide several layers (as required by the manufacturer) of 30-pound roofing felt under the tank equal in diameter to the outside diameter of the tank plus 2 inches to resist uneven loadings on the tank and anchor system.

8. Pipe Supports:
a. FRP
b. Laminated or resin bonded to the tank's exterior wall.

9. Tank Pad Requirements
a. Finished Water Tank must be placed on concrete pad per structural drawings.

10. Tank Tie Down Requirements
    a. Tank vendor to supply required tank tie down equipment as required to meet seismic and wind requirements.

C. Fabrication and Material Requirements:
1. Fiberglass reinforced polyester or vinylester complying with ASTM D 3299.
2. Resin:
    a. Commercial grade, premium, corrosion resistant vinylester that has been evaluated in a laminate by test in accordance with ASTM C-581 for the intended service and recommended for this service by the resin manufacturer.
    b. Strictly adhere to the resin manufacturer's recommendations for surface veil materials, resin cure systems, and post cure requirements.
    c. Contain no pigments, dyes, colorants, or filler, except as follows:
        1) A thixotropic agent that does not interfere with visual inspection of laminate quality may only be added for viscosity control in resins that are not to be used in the inner corrosion barrier, interior layers, interior secondary layers, and interior top coats.
        2) Resin pastes used to fill crevices may contain thixotropic agents provided that all such areas are subsequently covered with a full corrosion-resistant barrier laminate.
        3) Provide a pigmented exterior gel coat provided on the tank to reduce ultraviolet decomposition of the sodium hypochlorite.
        4) Add ultraviolet absorbers to the exterior surface for improved weather resistance.
        5) Utilize the same premium grade resin throughout wall section of the tank.
    d. Acceptable commercially available resins:
        1) Vinylester, such as Derakane 411, Derakane 470; epoxy that does not contain Bisphenol "A."
        2) Chlorinated polyester, such as Hetron 197; epoxy that does not contain Bisphenol "A."
    e. Surface Veils:
        1) Provide surface veils on all interior tank wall surfaces.
        2) Where "C" glass surface veil is not suitable, replace with compatible synthetic materials such as Nexus 10-12 as manufactured by Burlington Industries, or equal.
3. Reinforcement:
    a. Chopped Strand Mat: Chopped commercial grade E-type glass strands bonded together using a binder.
        1) Treated with a sizing that is chemically compatible with the resin system used.
    b. Continuous Roving: Commercial grade of E-type glass fiber with a sizing that is chemically compatible with the resin system used.
    c. Woven Roving: In accordance with ASTM specification D2150.
    d. Surface Mat: The reinforcement used for the inner surface: either a commercial grade chemical resistant glass surface mat or an organic fiber surface mat as recommended by the resin manufacturer for the intended service.

D. Linings:
1. Line tanks with a chemically resistant lining that is compatible with chemical being stored or has the potential of being introduced to the tank.

2.5 EQUIPMENT AND COMPONENTS

A. Immersion Tank Heater (HTR-53241)
1. Not provided by Finished Water Tank Manufacturer.
2. See Drawing: 052 Y602.
3. Manufacturer:
    a. Watlow WATROD
    b. Or Equal.
4. Style: flanged immersion heater.
5. Connection: 304 SS 4-IN ANSI 150# FF flange.

6. Sheath: Passivated 316 SS.
7. Capacity: 3kW.
8. Electrical requirements: 480V, 3 ph, 60 Hz.
9. Control Panel:
    a. Control panel enclosure: NEMA 4.
    b. Classification: Outdoor rated.
    c. Ambient Temperature: 20 degF to 90 degF
    d. Controls:
        1) Heater Circuit Contactor.
        2) Limit Over Temperature Disconnect.
        3) Operator Enable Switch.
        4) Input/Output Signal Capability to PLC:
            a) See 052 Y602.

B. Finished Water Storage Tank.
1. Access Ladder:
    a. Equip with non-caged ladder.
    b. Provide hinged and lockable security gate to prevent access.
    c. Removable and bolted to integrally mounted supports.
        1) Mounting bolts: stainless steel.
    d. Total length: Provide clearance from finished grade to allow for thermal expansion to top of each tank.
    e. Maximum center-to-center distance on mounting supports is 5 FT.
    f. Design in accordance with OSHA Standards.
    g. Material: FRP.
2. Fall Prevention System:
    a. Provide safety hooks at tank top to which safety lines may be attached for personnel accessing tank top.
3. Vent:
    a. Material: SCH 10 Carbon Steel.
    b. Equip with #24 mesh stainless steel screen.

C. Identification:
1. Label: Permanently attach either Type 316 stainless steel or fiberglass label to each tank at an elevation 5 feet above the tank support pad with the following minimum information:
    a. Type of liner reinforcement:
    b. Corrosion-resistant surface veil.
    c. Resin.
    d. Tank capacity.
    e. Date of manufacture.

PART 3 - EXECUTION

3.1 SHOP INSPECTION

A. The tank may be inspected prior to gelcoat by the Contracting Officer's Representative, when applicable, and are not to be coated or shipped without written approval by the Contracting Officer.
1. Shop drawings, nozzle cutouts, material documents and material usage records are to be made available to the inspector upon such visit.

B. All nozzle cutouts are to be marked for location of origin, and shipped with the respective tank.
1. At the manufacturer's discretion, a representative sample may be sent to the Contracting Officer Representative for inspection following removal from the mandrel, to assure that the liner and structural thicknesses will be judged as adequate prior to further manufacturing efforts.

C. Employ services of independent testing laboratory qualified to perform quality control testing.
   1. Samples:
      a. Number: 1 per 1,000 SF of surface area, minimum of 5 per item or tank.
      b. Locations: As selected by testing lab, cut-outs from nozzles acceptable.
   2. Tests:
      a. Glass content.
      b. Tensile strength.
      c. Flexural strength.
      d. Modules of elasticity.
      e. Hardness.
   3. Tolerances:
      a. Products that do not meet or exceed the specifications will be rejected.
      b. Out-of-round or parallel: ±1 percent of ID.
      c. Flanges:
         1) Thickness: +1/32 IN.
         2) Flat and true: +1/32 IN.
      d. Surface characteristics: As specified.

3.2 DELIVERY AND INSTALLATION

A. FRP tanks will be installed by the Contractor as shown on the drawings and as recommended by the Manufacturer's representative.

B. Cleaning:
   1. With mild soap solution.
   2. Rinse with 180 DegF water.

3.3 MANUFACTURERS FIELD SERVICES

A. Provide the following services from the Manufacturer in addition to any other services specified herein, and required by these Specifications.
   1. A factory trained field representative, shall be provided for a minimum of one (1) trip and a minimum of one (1) eight (8) hour day to provide installation assistance and field testing services. The field representative may be an employee of the Manufacturer. The installation services and field testing services shall be coordinated with the Contractor.
   2. See Section 01 79 00 for additional training requirements.

B. All costs, including travel, lodging, meals, and incidentals for manufacturer services shall be included in the Manufacturer's proposal.

END OF SECTION

DIVISION 46

WATER AND WASTEWATER EQUIPMENT

This page intentionally left blank.

SECTION 46 13 15
ALUMINUM COVERS - FORMED PANEL TYPE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Engineered aluminum flat-formed panel covers and appurtenances for Secondary Treatment Building over process trains, aerobic digester, and areas designated on the Drawings. The primary function of the system is to support loads exceeding that of the standard aluminum grating and help control odor.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 15 19 - Anchorage to Concrete
   2. Section 05 50 00 - Metal Fabrications.
   3. Section 09 96 00 - High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. The Aluminum Association, Inc. (AA):
      a. ADM, Aluminum Design Manual, current edition.
      b. Specifications for Aluminum Structures.
   2. American Society of Civil Engineers (ASCE):
      a. 7, Minimum Design Loads for Buildings and Other Structures.
   3. American Welding Society (AWS):
      a. D1.2, Structural Welding Code - Aluminum.

B. Qualifications:
   1. Manufacturer's Qualifications: Manufacturer shall have experience in manufacturing aluminum covers of similar size and type to the system specified. For a manufacturer to be determined acceptable for providing aluminum covers on this project, they must show evidence of a minimum of five installations and five years' experience in the design and manufacture of systems of similar size and type as specified herein. The manufacturer must design and fabricate the system at its own facilities.
   2. Engineer for aluminum covers: Professional Engineer licensed in the State of Arizona.
      a. Professional Engineer to have minimum five years' experience in design of aluminum covers with scope similar to this Project.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Fabrication and layout drawings showing dimensions, sizes, thicknesses, gauges, materials, finishes, typical details, hatches, attachments and appurtenances.
   2. Erection procedures and directions.
   3. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
   4. Certifications:
      a. Complete set of design drawings and calculations showing the governing stresses in all members and connections signed by a Professional Engineer registered in the State of Arizona. Calculations are submitted for record purposes only, but will be checked for compliance with contract documents.

B. Project Information:
   1. Manufacturer's qualifications to show compliance with Paragraph 1.2.B.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Hallsten.
   2. CST Industries.
   3. Or Equal.

2.2 DESCRIPTION

A. Covers shall be an engineered, removable, custom fit, flat formed panel aluminum cover system specifically designed for the requirements specified herein and the areas shown on the Drawings. The covers shall be made of formed sheet aluminum nested panels.

B. The covers shall be clear-spanning and designed to be self-supporting from the periphery concrete walls and slabs without the use of sub-framing. Anchor to concrete per Section 03 15 19 - Anchorage to Concrete. Provisions shall be made to allow for thermal expansion.

C. The cover system shall consist of individual panels capable of being removed without needing to remove more than the two adjacent panels. The required lifting force per panel shall not exceed the dead weight of that panel.

D. The panels shall utilize vertical bends that act as stiffeners to achieve structural rigidity. The bottom of the vertical stiffener shall be further bent to form a joint that allows panels to interlock to each other in order to form a complete cover system. The panel interlock arrangement shall be designed so that no threaded fasteners are required. The ends of the panels shall be sealed with flashing that attaches to the basin wall.

E. The cover design shall prevent water ponding which may result in excessive infiltration or over-stressing the flat cover. The cover surface paneling shall be designed as a substantially air and water tight system under the specified design loading conditions.

2.3 DESIGN AND MATERIALS

A. Materials:
   1. Panels, other extrusions and miscellaneous shapes: Alloy 6061-T6. Standard mill finish. Minimum thickness of 0.09 inches.
   2. Fasteners and anchor bolts: 316 stainless steel.
   3. Sealant: Urethane or Silicone and resistant to ozone and ultraviolet light and conform to Federal Specification TT-S-00230C.
   4. Gaskets: Neoprene or Santoprene and shall be resistant to ozone and shielded from exposure to ultraviolet light.

B. Criteria:
   1. Design covers in accordance with the Aluminum Design Manual, ADM-1 as a minimum.
      a. All loads are unfactored unless otherwise noted.
   2. Dead Load: Defined as the weight of the structure and all permanently attached items that are supported by the structure.
   3. Live Load:
      a. Uniform 250 psf on entire cover.
      b. Concentrated Load: 1000 pounds on a 12 inches x 12 inches area.
      c. All components to be adequate for the uniform load or the concentrated load, whichever requires the stronger component.
   4. Maximum Weight of any individual removable or operable section: 150 pounds.
   5. Vacuum/Pressure Load: Not Applicable.
   6. Load Combinations: As required per ASCE 7.
   7. Clear Spans: As shown on the Drawings. No intermediate supports unless otherwise noted.

8. Deflection: For the above loads and load combinations, the deflection of the structural members shall not exceed L/180, and the cladding shall not exceed L/60 as defined by the Aluminum Association.
9. Material Design Temperature Range: -40 degrees F to 160 degrees F.

C. Hinged Equipment Access Hatches:
1. Provide hinged access panels where required for equipment access and at locations shown on the Drawings.
2. Provide stainless steel hinges and accessories including quick release connection device and latches to hold in place when open. Install neoprene gasket to prevent air leakage.

D. Sampling Hatches:
1. Provide sampling hatches at locations indicated on the Drawings.
2. Hatches shall be factory fabricated into the covers. Field cutting for hatch shall not be allowed.
3. Provide lifting handle and 316 stainless steel full length hinge and full gasketing.

E. Instrumentation Connections:
1. Provide instrumentation connections at locations and per details shown on the Drawings.
2. Connections shall be factory-fabricated into the covers. Field cutting or fabrication shall not be allowed.
3. Seal all penetrations.

F. Covers shall be designed specifically for installation in wastewater treatment plant applications and capable of withstanding corrosion, pitting and degradation from the action of wastewater and associated gases, including hydrogen sulfide in concentrations of up to 200 PPM. The covers shall provide a complete seal for use in an odor control application. Odor control takeoff flanged connections shall be provided at the locations shown on the Drawings. All splice joints shall be interlocked and gasketed.

G. Dissimilar materials shall be separated from each other through the use of approved gaskets or coatings.

PART 3 - EXECUTION

3.1 ERECTION

A. Erect all components in accordance with manufacturer's recommendations.

B. Do not force components during erection.

C. All work shall be completed by installers skilled and experienced in the erection of similar aluminum products. Covers shall be erected plumb and level and in proper alignment.

3.2 FIELD QUALITY CONTROL

A. Field re-fabrication of structural components or covers will not be accepted.

END OF SECTION

This page intentionally left blank.

SECTION 46 33 11
CHEMICAL FEED - LIQUID SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Chemical metering pumps for the following:
      a. Roaring Springs Pumphouse - Volume 5:
         1) Sodium hypochlorite.
         2) Dilute sodium hypochlorite.
      b. North Rim WTP - Volume 6:
         1) Sodium hypochlorite.
         2) Sodium hydroxide.
         3) Hydrochloric acid.
         4) Sodium bisulfite.
         5) Citric acid.
         6) Alum or Aluminum Chlorohydrate (Coagulant).
   2. Chemical feed system accessories.
   3. See Specification Sections 46 53 49 – Membrane Bio-Reactor System and 46 61 33 – UF Membrane Equipment for clean in place chemical systems furnished with membrane treatment systems.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 – Equipment - Basic Requirements.
   2. Section 01 75 00 – Checkout and Start-Up Procedures for Water Treatment Facilities.
   3. Section 01 79 00 – Demonstration and Training.
   4. Section 40 05 00 – Pipe and Pipe Fittings - Basic Requirements.
   5. Section 40 05 51 – Valves - Basic Requirements.
   6. Section 40 61 96 – Process Control Descriptions.
   7. Section 43 21 00 – Pumping Equipment - Basic Requirements.
   8. Section 43 41 16 – Carbon Steel Tanks.
   9. Section 43 41 45 – Fiberglass Reinforced Plastic – Chemical Tanks.
   10. Section 46 41 11 – Vertical Mixers.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).

B. Pumps must be manufactured in compliance with most current ISO 9001 standards.

C. Pumps shall meet CE, NSF 61 standards.

D. Secure and coordinate entire system including but not necessarily limited to metering pumps, electric equipment, controls, hardware, valving, and piping.
   1. If chemical metering panels and/or components installed at injection point are not supplied by the pump manufacturer, provide chemical metering panel layout drawings and equipment cut sheets to pump manufacture for review and comment.

1.3 SYSTEM DESCRIPTION

A. System shall include all materials and components required for a fully operational system.

1.4 SUBMITTALS

A. Shop Drawings:

1. See Specification Section 01 61 03.
2. Drawings and product data:
    a. See Specification Section 43 21 00.
    b. Pumps:
        1) Chemical resistance data for materials used.
        2) Complete performance information:
            a) Characteristic performance curve showing flow rate as a function of pump operation in relation to discharge pressure or head.
            b) Pressure rating.
            c) NPSH required.
            d) Power demand or full load amps.
            e) Plunger diameter (as applicable).
    c. Control modes.
    d. Valves:
        1) See Specification Section 40 05 51.
    e. Piping:
        1) See Specification Section 40 05 00.

B. Contract Closeout Information:
1. Operation and Maintenance Data:
    a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.5 PROJECT CONDITIONS

A. Operating conditions and configuration.
1. Site Elevations:
    a. Roaring Springs Pumphouse - Volume 5: 4,860 FT.
    b. North Rim WTP - Volume 6: 8,359 FT.
2. System Configuration:
    a. Discharge piping configuration from chemical feed panels to dosing point as shown in Drawings.
    b. Chemical storage and suction conditions:
        1) Roaring Springs Pumphouse Tank and Carboy Systems – General Description
            a) Chemical metering pump panel mounted next to bulk tank with suction lines plumbed to the bulk tank
            b) Chemical transfer pump mounted above carboy(s) with flexible suction line extending into carboy
        2) North Rim WTP Drum and Tote Systems – General Description
            a) Chemical metering panels mounted above drums/totes with suction lines extending into the drum/tote.

B. Pumped Liquid:

LIQUID	PLANTS USED	% STRENGTH	SPECIFIC GRAVITY	TEMP DEGF
Coagulant (COAG) (Aluminum Sulfate assumed)	1	40 – 50 (alum)	1.3	50-85
Sodium hypochlorite (HYPO)	1, 2	6 – 15	1.15	50-85
Dilute sodium hypochlorite (HYPO-DL)	2	0.5 – 1	1.00	50-85
Sodium hydroxide (CA)	1	20 – 25	1.28	50-85
Hydrochloric acid (HCL)	1	30 – 33	1.10	50-85

LIQUID	PLANTS USED	% STRENGTH	SPECIFIC GRAVITY	TEMP DEGF
Sodium bisulfite (SBS)	1	35 – 40	1.33	50-85
Citric Acid (CIT)	1	50	1.20	50-85

1. North Rim Water Treatment Plant
2. Roaring Springs Pumphouse

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
2. Use of Manufacturers not listed for "Basis of Design" products may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.
    a. Submit request for product substitution in accordance with Specification Section 01 33 23 for use of manufacturers not listed for "Basis of Design" products.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Peristaltic metering pumps and accessories:
    a. Watson-Marlow - Qdos/Qdos H-FLO (Basis of Design).
    b. Or equal.
2. Diaphragm metering pumps and accessories:
    a. ProMinent -GammaX (Basis of Design).
    b. Or equal.

2.2 METERING PUMPS

A. General design requirements:
1. Variable speed drive.
2. Performance requirements:
    a. Accuracy over metering range: ±0.5 percent.
    b. Repeatability over full operating range: ±0.5 percent.
    c. Pump output at full speed, or stroke length and speed, within 3% of rated value based on submitted characteristic performance curves.
    d. Contractor and manufacturer shall be responsible for servicing, adjusting, and replacing equipment until installed operational performance requirements are met.
        1) In the event that performance requirements cannot be met, Contracting Officer reserves the right to require that all equipment be removed from the site and replaced.
3. Components must be NSF Standard 61 Drinking Water System approved.

B. Peristaltic pump design requirements:
1. Rotation speed: Automatic.
2. Provide leak detection/tube failure alarm internal to the pump head.

3. Self-priming up to 30 feet of suction lift (water).
4. Units shall have the following characteristics:
    a. Tag numbers: PMP-53421, PMP-53321, PMP-53322.
        1) Pumps must be identical model type from the same manufacturer.
        2) Chemical: Sodium hypochlorite or dilute sodium hypochlorite.
        3) Heads/unit: 1.
        4) Service: Sodium hypochlorite transfer and dosing.
            a) Motor minimum FLA: 2.92
            b) Variable speed DC

Volume 5 – Roaring Springs Pumphouse:

Equipment Tag	Chemical	Pump Type	Tube Type	Dosing Capacity (gph)	Operating Condition (psi)
PMP-53421	HYPO	Peristaltic	Per pump manufacturer recommendation	0.001–7.67	35
PMP-53321	HYPO-DL	Peristaltic	Per pump manufacturer recommendation	0.001–7.67	35
PMP-53322	HYPO-DL	Peristaltic	Per pump manufacturer recommendation	0.001–7.67	35

C. Diaphragm pump design requirements:
    1. Self-priming up to 10 FT of suction lift (water).
        a. Provide foot valves in chemical drums.
    2. Stroke speed: Automatic.
    3. Stroke adjustment: Manual.
    4. Stroke rate: Minimum 10 strokes per minute or no less than 15% of maximum.
    5. Stroke length: 15% minimum.
    6. Units shall have the following characteristics:
        a. Tag number(s): PMP-67111, PMP-67211, PMP-67212, PMP-67213, PMP-67214, and PMP-67312.
        b. Chemical: Coagulant, sodium hypochlorite, and sodium hydroxide (caustic).
        c. Heads/unit: 1.
        d. Capacity/head (GPH): See tables below.
        e. Discharge pressure (PSI): See tables below.
        f. Type diaphragm:
            1) Mechanical disc.
        g. Maximum stroke rate (SPM): 120.
        h. Drive:
            1) Solenoid.
            2) Motor minimum HP: 0.5.
            3) Variable speed.
        i. Electrical requirements: 120V 60hz.

D. Each metering pump shall meet the listed pump materials and capacities:

Volume 6 - North Rim WTP:

Equipment Tag	Chemical	Pump Type	Diaphragm Type	Dosing Capacity (gph)	Operating Condition (psi)
PMP-67111	COAG	Diaphragm	vPTFE	0.004-0.1	5-10
PMP-67211	HYPO	Diaphragm	vPTFE	0.004-0.14	5-10
PMP-67212	HYPO	Diaphragm	vPTFE	0.004-0.14	5-10
PMP-67213	HYPO	Diaphragm	vPTFE	0.004-0.14	5-10
PMP-67214	HYPO	Diaphragm	vPTFE	0.004-0.14	5-10
PMP-67312	CA	Diaphragm	vPTFE	0.01-0.1	5-10

2.3 FABRICATION AND MANUFACTURE

A. Metering Pumps
   1. General:
      a. Furnish pumps capable of operation with pump head rotation in either direction.
      b. Furnish pumps capable to run dry for an indefinite time without damage.
      c. Design so the process fluid is only in contact with the inside of the pump tubing or hose.
      d. Design with one compression roller or shoe fully engaged at all times with the tubing providing complete compression to prevent backflow and siphoning.
      e. Provide tubes comprised of material compatible with the chemical being pumped and select thickness for the longest use life.
      f. No special tools required for regular maintenance.
      g. Permanent nameplate identifying pump by chemical service and tag number.
      h. Peristaltic metering pumps
         1) Field-replaceable cartridge style tubing chamber
   2. Motor:
      a. Reversible, brushless DC gear motor rated for continuous duty.
      b. Include overload protection.
      c. Permanently lubricated speed reducers furnished.
      d. Enclosed in a NEMA 4X housing.
      e. 120 V, 1 phase, 60 Hz.
         1) Power cord: 3-conductor with SJTW jacket.
   3. Controls:
      a. Provide each pump with a multi-color VGA graphic LCD display for menu driven configuration settings, pump output value, service alerts, tube failure detection (TFD)

system and flow verification system (FVS) alarms status, remote input signal values, and tubing life indicator.

b. Provide a touchpad control for stop/start, configuration menu access and navigation, operating mode selection, auto priming, display options selection, tube life data, and reverse direction.

c. Plant Control System interface:
   1) Status and Alarm dry relay contacts:
      a) Running status.
      b) Fault status.
      c) Tube failure/Leak alarm.
   2) Remote Start/Stop dry contact.
   3) Inputs:
      a) Analog, 4-20ma speed command signal.
      b) Digital (on/off) run command.
   4) Outputs:
      a) Analog, 4-20ma speed command signal.
      b) Digital (on/off) run status.
      c) Digital (on/off) general alarm status.
      d) Digital (on/off) in-remote status.
   5) Custom Adders:
      a) H-O-A switch in “hand” status output.
      b) Pump Running lights.
      c) Pushbutton reset switch.
      d) Alarm reset push buttons.
      e) Two general alarm outputs.

d. Cybersecurity Requirements:
   1) Refer to Section 01 81 33 Cyber Security Requirements for full requirements.

2.4 SYSTEM ACCESSORIES

1. General:
   a. Provide system accessories relevant to each system. Refer to process drawings and P&IDs.
   b. Furnish all materials of construction compatible with the associated chemical.
2. Drum Scales:
   a. Corrosion resistant materials.
      1) Vary corrosion resistant materials selection to match chemical type for full chemical compatibility
   b. One scale per operational drum as indicated on P&IDs.
   c. Controls: local readout and 4-20 mA output with alarm contacts.
   d. Power: 120 VAC, 60 Hz.
   e. Scale Range: 0-850 pounds.
   f. Scale Size: 24”x24”.
   g. Coordinate with manufacturer for installation and anchoring requirements.
3. Tote Scales:
   a. Corrosion resistant materials.
      1) Vary corrosion resistant materials selection to match chemical type for full chemical compatibility
   b. One scale per operational tote as indicated on P&IDs.
   c. Controls: local readout and 4-20 mA output with alarm contacts.
   d. Power: 120 VAC, 60 Hz.
   e. Scale Size: 48”x48”.
   f. Scale Range: 0-3500 pounds.
   g. Coordinate with manufacturer for installation and anchoring requirements.
4. Spill Containment Pads (Volume 5):
   a. Refer to process drawings for location and size of spill containment pads.

b. Must be capable of containing the capacity of the largest single container within the containment area.

5. Pressure Relief Valves:
   a. Provide one (1) for each metering pump.
   b. Sized for the flow rate of its associated pump.
   c. Factory-adjusted to pressure recommended for pump.
   d. Pressure setting indicated on nameplate.
   e. Field adjustable without disconnecting piping or fittings.
   f. Flow through, diaphragm type design.
   g. Spring loaded ball provides positive seating of diaphragm.
   h. Materials: Compatible with material pumped.
   i. Manufacturer: Plast-o-matic or equal.
6. Pressure Regulating Valves:
   a. Provide (1) backpressure valve for each feed point as indicated on the Drawings.
   b. Check valve and anti-siphoning capabilities.
   c. Field adjustable back pressure valve without disconnecting piping or fittings.
   d. Flow through, diaphragm type design.
   e. Sized for the flow rate of its associated pump.
   f. Factory-adjusted to pressure recommended for pump.
   g. Pressure setting indicated on nameplate.
   h. Spring loaded ball provides positive seating of diaphragm.
   i. Materials: Compatible with material pumped.
   j. Manufacturer: Plast-o-matic or equal.
7. Float Valves (Volume 5):
   a. Materials:
      1) Body: PVC or Polypropylene
      2) Float: PVC or Polypropylene
      3) Use corrosion-resistant metal for all exposed portions.
   b. Design Requirements:
      1) Maximum working pressure: 60 PSI.
      2) Float operated valve will automatically fill tank as float is lowered.
8. Chemical metering panels
   a. Material: FRP.
   b. Stainless steel fasteners.
   c. Floor or wall mounted.
      1) See Drawings for mount style and maximum panel size.
9. Calibration columns:
   a. Provide calibration columns as indicated on drawings (size and amount).
   b. Graduated in standard divisions.
   c. Units constructed of PVC or material compatible with liquid being pumped.
   d. Manufacturer: Chemcon, Inc. or equal.
10. Isolation Ball Valves, Isolation Diaphragm Valves, Check Valves, and True Unions
    a. From same fittings manufacturer. Use of union fittings from different manufacturers not allowed.
    b. PVC or CPVC to match chemical service
    c. True unions on both ends
       1) Union elastomers fully compatible with chemical service specific to each application
    d. For sodium hypochlorite service, all ball valves must be vented type.
11. Chemical Injection Quills (Volume 6):
    a. Provide injection quills as indicated in Drawings.
    b. Stainless steel restraint hook and chain.
    c. Standard tip configuration with spring loaded ball check valve.
    d. Body rated at 150 psig.
    e. Supply shut-off valve matching the injection quill material.

1) Design valve to allow for the removal of the injection quill while the process line remains in operation.

f. Manufacturer: Primary Fluid Systems, Inc. or equal.

g. Fabrication:
   1) Quick release injection nozzles with a quick disconnect fitting on both sides of housing of which poppet valves seal instantly when disconnected to prevent accidental draining from chemical feed line and injector body.
   2) 0.5-inch NPT(F) by 0.5-inch NPT(M), angled discharge point.
   3) Project into pipe one-third to one-half the diameter.

h. Spare Injection Quills:
   1) Provide one (1) spare injection quill and seal for each dosage point.

PART 3 - EXECUTION

3.1 INSTALLATION

A. See Specification Section 01 61 03.

B. System shall be installed per manufacturer's standard procedures.

C. Level, plumb and align unit.

3.2 FIELD QUALITY CONTROL

A. See Specification Section 01 75 00.

B. Each manufacturer's representative shall provide:
   1. Eight hours of equipment installation, start up and commissioning services.

C. See Section 01 79 00 for operator training requirements.

END OF SECTION

SECTION 46 41 11
VERTICAL MIXERS

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes:
   1. Vertical Mixers:
      a. Reaction Tank mixers at North Rim WTP (MX-63011, MX-63012) – Volume 6.

B. Related Sections include, but are not necessarily limited to:
   1. Section 01 79 00 – Demonstration and Training.
   2. Section 01 61 03 – Equipment: Basic Requirements.
   3. Section 26 29 23 – Variable Frequency Drives.
   4. Section 43 41 16 – Carbon Steel Tanks.
   5. Section 46 33 11 – Chemical Feed – Liquid Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Bearing Manufacturers Association (ABMA):
      a. 9, Load Ratings and Fatigue Life for Ball Bearings.
      b. 11, Load Ratings and Fatigue Life for Roller Bearings.
   2. American Gear Manufacturers Association (AGMA):
      a. 2000-A88, Gear Classification and Inspection Handbook - Tolerances and Measuring Methods for Unassembled Spur and Helical Gears.
   3. American Iron and Steel Institute (AISI).
   4. NSF International (NSF)
      a. Standard 61 – Drinking Water System Components – Health Effects.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product technical data:
      a. Provide calculations supporting all shaft and impeller design criteria (stresses and critical speed).
         1) Calculations must be stamped by a Professional Engineer.
      b. Submit all loadings, torques and forces imposed by operating equipment.
         1) Calculations must be stamped by a Professional Engineer.
      c. Pumping capacity and water horsepower at delivered power.
      d. Dimension drawings of mixer showing plan and section views.
      e. Impeller size.
      f. Mixing direction.
      g. Annotated materials of construction.

B. Operation and Maintenance Manuals:
   1. See Specification Section 01 78 23 for requirements for:
      a. The mechanics and administration of the submittal process.
      b. The content of Operation and Maintenance Manuals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following Manufacturers are acceptable:
   1. SPX Flow, Lightnin Mixers.
   2. Or equal.

2.2 MATERIALS

A. Mixer Housings: Fabricated steel or cast iron.

B. Impeller Shaft: Stainless steel, AISI Type 316.

C. Impeller: Stainless steel, AISI Type 316.

D. Hardware and Wetted Parts: Stainless steel, AISI Type 316.

2.3 SITE CONDITIONS

A. North Rim WTP:
   1. Elevation: 8359.00 FT
   2. Water temperature range: 5 – 20 ˚C.

2.4 EQUIPMENT

A. North Rim WTP
   1. Vertical Mixers (MX-63011, MX-63012):
      a. Performance Requirements:
         1) Design Treatment Flow Rate: 50 gpm
         2) Velocity gradient (G) at Design Treatment Flow Rate: 250 Seconds^-1 (at minimum water temperature of 5 degC).
         3) Materials: 316SS for wetted parts.
         4) Vessel: North Rim WTP Reaction Tanks (TK-63011, TK-63012)
         5) Number of impellers per mixer: Minimum of 1 and maximum of 2.
         6) Impeller diameter: Determined by Manufacturer.
         7) Impeller type: Determined by Manufacturer.
         8) Maximum impeller spacing from floor (lowest impeller): Determined by Manufacturer, but no lower than bottom of tank straight wall (i.e. not in the cone).
         9) Motor size: Maximum 5 HP.
         10) Motor speed: Maximum 2000 RPM.
         11) Drive type: Variable Frequency.
            a) VFDs shall be provided by the electrical subcontractor and installed in the MCC in the Electrical Room.
            b) Mixer motor shall be inverter duty rated for VFD operation.
         12) Shaft diameter: Determined by Manufacturer.
         13) Shaft length: Determined by Manufacturer.
         14) Baffles: Provide number, size and location to contactor. Contractor shall coordinate with tank supplier to include baffles in tank design.
         15) Loads:
            a) Mixer Weight: 350 lbs (estimated).
            b) Vertical Downward Load: 1,100 LBS (estimated).

c) Contractor shall coordinate the actual loads with tank supplier to include in tank design.

16) Mixer shall be mounted to mixer frame designed and provided by tank manufacturer.

2.5 FABRICATION

A. Vertical Mixers (MX-63011 and MX-63012):

1. Speed Reducer:
   a. Designed and rated per AGMA standards with AGMA nameplate.
   b. Type:
      1) Right angle gear drive.
      2) Hollow quill-type output shaft, independent bearing support with flexible coupling connection of output shaft to agitator shaft above the reducer.
      3) Separate agitator shaft bearings located above and below the main drive bearings.
      4) Combination helical, spiral bevel, and pinion drive gears.
      5) Totally enclosed.
      6) Worm gears not acceptable.
   c. Service factor: 1.15 of nameplate on motor.
   d. Gearing: AGMA quality number 10 or better AGMA 2000-A88.
   e. Bearing design:
      1) Antifriction type, ball or roller, oil or grease lubricated.
      2) All shaft bearings located outside of tankage.
      3) All bearings minimum ABMA L-10 life of 100,000 HRS at full motor nameplate HP at design speed.
   f. Lifting lugs: Included.
   g. Lubrication:
      1) Oil or grease lubricated (food grade).
      2) Positive lubrication of all gears and bearings.
      3) Maintain recommended thermal rating of reducer.
      4) Dipstick oil level indicators.
      5) Adequate separation between oil and grease lubricated parts.
      6) Oil breather located above possible oil foam level.
      7) Provide food grade lubricant.
   h. Lubricating provisions:
      1) Oil drain.
      2) Grease fittings protected by removable neoprene cover and easily accessible.
      3) Greased bearings with seal to retain grease.
   i. Reducer agitator and output shaft and impeller shaft.
      1) Agitator shaft oversized for direct, rigid coupling to impeller shaft.
      2) Maximum indicated runout at lower end of impeller shaft not to exceed 1/8 IN for every 10 FT of overhang, as measured when turning by hand.
      3) Maximum stress: 9,000 psi at maximum load.
      4) Independent bearing support of agitator shaft and hollow quill output shaft provides isolation of speed reducer from shock loads, impeller bending loads, and the weight of impeller shaft.
      5) Torsionally resilient, flexible coupling connects hollow quill output shaft and agitator shaft above the reducer.
      6) Full load noise level not to exceed 85 dBa at 3 FT from any part of the drive assembly.
2. Impeller:
   a. Dynamically and hydraulically stable.
   b. Blades bolted to hub, for removal from tank.
   c. Impeller hook keyed to impeller shaft.
   d. Maximum stress on impeller component not to exceed 12,000 psi under maximum operation load.
   e. Use of stabilizing rings or fins will not influence stress limitations specified.

f. Operating speed does not exceed 75 percent of first critical speed.

3. Shaft Coupling:
   a. Rigid flange on agitator shaft located for ease of access, or single piece shaft connected by means of a split removable coupling accessible from outside of tank.
   b. Designed to transmit 200 percent of full torque and 150 percent of axial load.
   c. Located outside tankage.
   d. Faces provided with rabbeted male and female piloted connection.
   e. No match marks required for alignment.
   f. Allow agitator shaft to be split into sections for maintenance removal using two flanged couplings. Sections are not to exceed 6ft.
4. Motor:
   a. See Section 01 61 03.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install units in accordance with Manufacturer's recommendations.
   1. Confirm units are balanced, level and adequately supported at deck.
   2. Confirm drives do not oscillate or move during operation.
   3. Anchor bolts provided by Contractor and set per requirements of Manufacturer.

3.2 FIELD QUALITY CONTROL

A. Employ and pay for services of authorized Manufacturer's field service representative to:
   1. Inspect equipment to be installed under these Specifications.
   2. Supervise any adjustments and perform modifications as necessary.
   3. Conduct start-up of equipment.

B. See Section 01 79 00 for Operator training requirements.

END OF SECTION

SECTION 46 41 17
STATIC MIXERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. In-line static mixers including injectors and nozzles for chemical injection located at:
      a. North Rim WTP static mixers (MX-63011, MX-66011) – Volume 6.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 40 05 00 – Pipe and Pipe Fittings- Basic Requirements.
   2. Section 46 33 11 – Chemical Feed- Liquid Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI)
      a. B16.1, Cast Iron Pipe Flanges and Flanged Fittings

B. Related Specification Sections include, but are not necessarily limited to:
   1. Section 40 05 00 – Pipe and Pipe Fittings- Basic Requirements.
   2. Section 46 33 11 – Chemical Feed- Liquid Systems.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Requirements in Specification Section 01 61 03.
   2. Submit dimensional drawings and installation details.

B. Operation and Maintenance Manuals:
   1. See Specification Section 01 78 23 for requirements for:
      a. The mechanics and administration of the submittal process.
      b. The content of Operation and Maintenance Manuals.

C. Calculations:
   1. Flow characteristics, flow velocity, pressure drop, and mixing efficiency.

D. Testing/Certification
   1. Factory test results and certification.
   2. Statement that mixer components are compatible with chemicals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Komax.

2. Or Equal.

2.2 PERFORMANCE AND DESIGN REQUIREMENTS

A. General
1. The mixers shall be the static in-line type with no moving parts. The static mixers shall be flanged metal alloy and stainless steel.

B. Performance
1. North Rim WTP
   a. Raw Water Piping (MX-63011)
      1) Chemical Application
         a) Sodium Hypochlorite:
            (1) Percentage: 12.5%.
            (2) Specific Gravity: 1.15.
            (3) Feed Rate Min/Max: 0.004 gph/0.14 gph.
         b) Alum or Aluminum Chlorohydrate (ACH):
            (1) Percentage: 50%.
            (2) Specific Gravity: 1.3.
            (3) Feed Rate Min/Max: 0.004 gph/0.10 gph.
   b. Finished Water Piping (MX-66011)
      1) Chemical Application
         a) Sodium Hydroxide (Caustic):
            (1) Percentage: 33%.
            (2) Specific Gravity: 1.28.
            (3) Feed Rate Min/Max: 0.01 gph/0.10 gph.
         b) Sodium Hypochlorite:
            (1) Percentage: 12.5%.
            (2) Specific Gravity: 1.15.
            (3) Feed Rate Min/Max: 0.004 gph/0.14 gph.
   c. Material of Construction:
      1) Body: Ductile Iron with Epoxy Coating or Cast Iron with Epoxy Coating.
      2) Mixing Elements: Stainless Steel.
   d. Pipe Diameter:
      1) See Drawings.
   e. Maximum Length: 28-inches.
   f. Design Flows:
      1) Maximum: 100 gpm.
      2) Minimum: 20 gpm.
   g. Thoroughness of Mixing at Minimum Flow Rate: 70%.
   h. Maximum Pressure Loss: 2.0 psi at 100 gpm.
   i. Design Pressure:
      1) MX-63011: 5 psi.
      2) MX-66011: 10 psi.
   j. Mixing by stationary, rigid elements arranged to split and recombine component streams to provide a homogeneous stream at the mixer outlet.
   k. Continuously seal elements to the pipe walls.
   l. Fabricate each mixer in one piece.
   m. Flanged ends per ANSI B16.1, Class 125, flat face.
   n. Epoxy coating: NSF 61 compliant, 6-8 mils thick.
   o. Locate injectors and nozzles such that the edges are at least 3-inches from any seams in the shell or housing.

Part 3 - EXECUTION

3.1 FACTORY TESTING

A. After completion of fabrication, pressure test mixer to a pressure of 100 psi.
   1. Test duration: Minimum one hour.
   2. Exhibit no leaks when subjected to the above hydrostatic test pressure.

3.2 INSTALLATION

A. Install mixer as shown on the Drawings and in accordance with the manufacturer's recommendations.

B. Provide adequate pipe supports in accordance with Sections 40 05 07.

END OF SECTION

This page intentionally left blank.

SECTION 46 61 33
ULTRAFILTRATION MEMBRANE EQUIPMENT

PART 1 - GENERAL

1.1 SECTION INCLUDES

A. Ultrafiltration (UF) Membrane Treatment System at the North Rim WTP (Volume 6).
   1. System includes feed pumps, strainers, UF module rack skids, compressed air system and clean-in-place (CIP) circulation, neutralization, chemical feed systems, and tank.
   2. Provide system with an automated control system.

1.2 RELATED SECTIONS

A. Section 01 61 03 – Equipment – Basic Requirements.

B. Section 01 74 15 – Disinfection of Facilities.

C. Section 01 75 00 – Checkout and Start-up Procedures for Water Treatment Facilities.

D. See Section 01 79 00 – Demonstration and Training.

E. Section 01 81 10 – Wind and Seismic Design Criteria.

F. Section 01 81 33 – Cybersecurity Requirements.

G. Section 26 24 19 – Motor Control Equipment.

H. Section 26 29 23 – Variable Frequency Drive – Low Voltage.

I. Section 40 05 00 – Pipe and Pipe Fittings – Basic Requirements.

J. Section 40 61 96 – Process Control Descriptions.

1.3 DEFINITIONS

A. CIP: Clean in Place.

B. GFD: Gallons per square foot per day.

C. I/O: Input/Output.

D. LRV: Log Removal Value.

E. NPSH: Net Positive Suction Head.

F. PLC: Programmable Logic Controller.

1.4 SUBMITTALS

A. All submittal requirements must be submitted as a single package for review, excluding Operation and Maintenance Data.

B. Shop drawings consistent with Section 01 33 23 and Section 01 61 03.

C. Operation and Maintenance Data consistent with Section 01 78 23.

D. Submit the following information for all equipment, valves, instrumentation, and other items in this specification section:
   1. Equipment name and identification number (e.g., equipment number).
   2. Item detailed description and specifications.
   3. Item weight(s) both empty and filled with the service fluid.
   4. Electrical data, including control and wiring diagrams, as applicable.
   5. Complete electrical schematic diagrams, as applicable.

a. Show all field connections with identification of terminations between control panels, junction terminal boxes, and equipment items.

6. Complete dimensioned fabrication, foundation/anchor bolt placement, assembly,and installation drawings.
7. Catalog cut sheets and brochures.
8. Bill of materials.
9. Recommended list of spare parts (in addition to those listed at the end of this Section).
10. Elevation of proposed control panels and solenoid banks showing panel-mounted devices, details of enclosure type, load list, and current draw of panel, and list of all terminals required to receive inputs or to transmit outputs from the control panel.

E. Submit detailed and dimensioned drawings of the UF Skid Assemblies including:
1. Plan drawing showing all the UF Skid Assemblies.
2. Plan drawing showing the foundation plan for the UF Skid Assemblies.
3. Plan drawing showing the CIP Tank attachment requirements.

F. Submit detailed process and instrumentation diagrams (P&IDs) specific to the Project showing the complete UF treatment system, including all the items specified in this section as well as the required interconnecting piping. Include, at a minimum, the following components:
1. Symbol Legend and Abbreviation Index.
2. Feed Pumps and Feed Strainers.
3. Membrane Filter Rack Array and Filtered Water Assemblies.
4. Membrane Filter Rack Details.
5. Reverse Filtration Systems.
6. Chemical CIP and Neutralization System.
7. Compressed Air System.
8. Piping System Configuration and Requirements, including connection points to Contractor-supplied piping.

G. Submit detailed Control Panel drawings including the following:
1. Enclosure fabrication notes.
2. Dimensioned enclosure door layout and side profile, including the locations of all lights, latches, and other appurtenances.
3. Backpanel layout showing all input and output modules, control and instrumentation devices, power transformers/supplies, electrical receptacles, and all other installed items.
4. Power Distribution Schematics.
5. Wiring Schematics.
6. Communications Schematics, including Ethernet module connections and DeviceNet module connections.
7. Uninterrupted Power Supply (UPS).
8. Nameplate Details.
9. Terminal Strip Detail.

H. Submit detailed CIP System I/O & Valve Manifold drawings including the following:
1. I/O & Valve Manifold layout showing all input and output modules, solenoid valves, pressure-regulators w/ gage, control and instrumentation devices,electrical receptacles, cabling connections, and all other installed items.
2. Power Distribution Schematics.
3. Wiring Schematics.
4. Ethernet/IP Module Connection Details.
5. Nameplate Details.

I. Submit detailed Off-Skid Equipment drawings including the following:
1. Enclosure Fabrication Notes.
2. I/O & Valve Manifold layout showing all input and output modules, solenoid valves, pressure-regulators w/ gage, control and instrumentation devices, electricalreceptacles, cabling connections, and all other installed items.
3. Power Distribution Schematics.

4. Wiring Schematics.
5. Ethernet/IP Module Connection Details.
6. Nameplate Details.

J. Submit detailed Electrical Interconnection Drawings including the following:
   1. Power One Line Diagrams.
   2. Network Diagrams showing communications between the Control Panel, Plant Controls, and other items via Ethernet, Devicenet or other cabling systems.

K. Submit performance data curves for all pumps including head, capacity, horsepower demand, NPSH required, and pump efficiency over the entire operating range of the pump.

L. Submit assembly and installation drawings for all pumps including shaft size, seal, coupling, bearings, anchor bolt plan, part nomenclature, material list, outline dimensions, and shipping weights.

M. Submit the following information for the Compressed Air System:
   1. Performance data.
   2. Noise data including noise suppression data on noise enclosures.
   3. Integral Refrigerated Dryer data.
   4. Filter data.
   5. Assembly and installation drawings including outline dimensions, installation guidelines and shipping weights.

N. Submit the following information for the CIP Tank to be provided under this specification system:
   1. Dimensions.
   2. Nozzle schedule including size, location (per Drawings), and rating.
   3. Tank capacity (gallons).
   4. Maximum design specific gravity.
   5. Materials of construction.
   6. Installation, mounting and securing requirements.

O. Submit the following information for the CIP Tank Heater to be provided under this specification system:
   1. Manufacturer and model.
   2. Dimensions and mounting requirements.
   3. Power requirements.
   4. Wiring and controls.
   5. Manufacturer’s specific part numbers.
   6. Submit the following information for the instruments to be provided under this specification system:
      a. Manufacturer and model.
      b. Dimensions.
      c. Power requirements.
      d. Operating range.
      e. Materials of construction.
      f. Manufacturer’s specific part numbers.

P. Submit the following additional information shall be submitted for the UPS’s to be provided under this specification system:
   1. Dimensions.
   2. Power requirements.
   3. Materials of construction.
   4. Manufacturer’s specific part numbers.

Q. Submit the following information for the Instrumentation and Control System:
   1. Network diagram:
      a. A drawing showing all network attachments between equipment and control panels.

2. One Line Diagram:
    a. A one-line power diagram detailing the system loads.
3. Input/Output List:
    a. A list of each of the supplied control panel's I/O with the following details:
        1) Control Panel the point is located in (main, CIP, Rack, etc.)
        2) PLC rack number (if applicable).
        3) PLC rack slot number the card is in (if applicable).
        4) PLC address.
        5) Type of I/O (Digital Input - DI, Digital Output - DO, Analog Input - AI, Analog Output - AO, etc.)
        6) I/O module terminal number.
        7) I/O point tag.
        8) Signal type or description.
        9) Signal power requirements.
        10) Schematic diagram line number.
4. Wire List
    a. This list details the wires required to be run from each control panel to other control panels and to field devices.
    b. The list includes the following details:
        1) Wire number.
        2) Panel the wire is from.
        3) Schematic line number of wire starting point.
        4) Panel terminal number of wire starting point.
        5) Panel or device the wire terminates in.
        6) Schematic line number of wire termination point.
        7) Device terminal number of wire termination.
        8) Recommended wire type (size and insulation type and color).
        9) Cable number (if applicable).
        10) Recommended cable make and model (if applicable).
        11) Cable description (if applicable).
5. Instrument List
    a. The instrument list details all the instruments supplied with the UF system.
    b. The list includes the following details:
        1) Instrument tag number.
        2) P&ID location.
        3) Function.
        4) Manufacturer and model number.
        5) Span.
        6) Process connection.
        7) Field install or factory install.
        8) Power required.
        9) Subsystem instrument is in (rack, CIP, etc.).
6. Instrument Catalog Cut Sheets
    a. Provide an annotated catalog cut sheet for each type of instrument.
        1) Indicate on the cover sheet for each cut sheet the instruments that the cut sheet described.
        2) Provide annotations that indicate the specifications and pertinent information for each instrument.
7. System Functional Description (i.e Control Narrative)
    a. The System Functional Description is a detailed description of the system process and how it is controlled.
    b. Provide a description of each system cycle (filter forward, backwash/reverse flow, chemical cleanings, etc.), description of setpoints and alarms and a description of the process loops.
8. HMI Screen Shots

a. Provide HMI screen shots in a word document that shows examples of each type of HMI screen used in the project.

1.5 CYBER SECURITY REQUIREMENTS

A. Refer to Section 01 81 33 Cyber Security Requirements for full requirements.

B. The supplier agrees to assist in cybersecurity basic hygiene practices as dictated by best practices included in this section, government regulation at time of bid, and as provided by the Contracting Officer's cyber security team. At a minimum, the supplier will coordinate with Contracting Officer and assign IP addresses and subnet masking, utilized of managed switches with unused ports disabled, provide all equipment with latest firmware at time of field installation and coordinate with Contracting Officer to remove default usernames and passwords, and to remove Contractor passwords at time of system acceptance.

C. The Contracting Officer will have full and unrestricted access to upgrade, modify, or change the software, networking, firmware, or other areas of micro-processor-based systems except for parameters that may result in voided warranty if changed outside of prescribed range without approval from supplier.

D. Unless specifically included within the plans and specifications, remote access appliances for interaction with programmable devices will not be accepted during construction or operation. Where remote access is required, supplier shall submit RFI for allowable methods and describe the specific remote access needs (i.e. programming, view only data, emergency support) and coordinate with Contracting Officer/engineer on acceptable solution which achieves the Contracting Officer's cybersecurity risk tolerance.

E. Only Government-provided programming laptops will be allowed to connect/access on site networks.

F. Supplier shall coordinate programming efforts to allow for data to be passed from PLC to PLC for operations and coordination between supplier and balance of plant or other PLCs.

1.6 QUALITY ASSURANCE

A. Supply all equipment from a single manufacturer.

B. See Seismic Design criteria contained in Drawings and in Section 01 81 10.

C. Manufacturers Qualifications and Experience
   1. Technical Requirements
      a. Factory where equipment is manufactured and assembled shall be ISO 9001 certified.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Veolia Environnement S.A.

B. No substitutions allowed.

2.2 SYSTEM DESCRIPTION

A. The UF Membrane System is composed of parallel UF skids. The skids operate independently with dedicated feed pumps and strainers, but in conjunction to rotate service when a skid goes down for cleaning, thereby preventing multiple skids from backwashing at the same time, etc.

B. Membranes:
   1. PVDF, hollow-fiber, flow from outside to inside.
   2. Membranes are on the California Department of Public Health (CDPH) list of Accepted Alternative Membrane Filtration Technologies.

C. The UF skid independently alternate between producing filtrate and short cleaning phases to remove the particles that collect on the membrane surface during filtrate production.
   1. The most frequent cleaning process is a backwash or reverse filtration cycle, consisting of combinations of water and air cleaning cycles. The cycle is fully automated.
   2. Chemical cleaning cycles using various chemical and various durations for maintenance cleaning and/or heaving cleaning cycles.
      a. Cleaning cycles shall use the following chemicals:
         1) Sodium Hypochlorite: 10-12.5%.
         2) Sodium Hydroxide: 25-38%.
         3) Citric Acid: 35-50%.
         4) Sodium Bisulfite: 30-40%.
         5) Hydrochloric Acid: 25-40%.
   3. Chemical cleaning cycles including generation of cleaning solutions, chemical circulation, and chemical neutralization are fully automated.

D. Provide system capable of performing integrity testing using air hold tests consistent with EPA LG-2 MFGM requirements.
   1. At an adjustable interval the UF membrane system automatically performs an air pressure hold test on a UF skid.
      a. Air displaces the water in the membrane modules until they are empty, at which time the pressure attains its set point.
      b. The air is then shut off and the rate of decay of the pressure is monitored.
      c. If the decay rate is faster than acceptable, the system creates an alarm.
   2. When the membrane train approaches the LRV/decay rate limit, the UF module is isolated for a brief period of time while an air leak test is performed on all membrane modules in that UF skid.
      a. For the air leak test, the feed side of the module is drained and air is introduced to the inside/product side of the membrane elements.
      b. Visual and audible tests can be performed while air is introduced in the feed.
      c. When air on the filtrate side is observed or heard, the damaged module is repaired through fiber plugs or epoxy and the UF skid is put back online.
   3. Program system to calculate and display the log removal credit based on the Integrity testing.

E. Provide turbidity monitoring of each skid's filtrate to continuously assesses the membrane integrity by using effluent turbidity as an indicator of system performance.
   1. Connect on-line turbidimeters to the membrane filtration effluent lines to monitor the performance of the membrane system.
   2. Record data from the on-line turbidimeters on the Membrane System PLC.
      a. Upon high turbidity from a skid, change status of that skid from Production to Standby and do not return to service until authorized by operator override.

2.3 SCOPE OF SUPPLY

A. North Rim
   1. Furnish all UF membrane system components as specified herein from one supplier.
      a. Ultrafiltration (UF) Membrane System:
         1) Two (2) factory-assembled 100 gpm UF skids.
            a) System consists of one duty and one standby UF skid.
            b) Maximum instantaneous flux: 20 gfd at 20 degrees Celsius.
         2) Meet all specific requirements of ADEQ related to the use of alternative membrane filtration technologies.
      b. Raw Water Strainers:
         1) Provide raw water strainer on each skid.
         2) Type 304 or 316 stainless steel wire mesh screens or wedge wire screens.
         3) Self-cleaning.
         4) Locate on the discharge side of the Feed Pumps.

5) Furnish strainer system capable of maintaining forward water production while cleaning.

c. Provide a backwash system that is integral to the membrane skid.
   1) Maximum backwash cycle duration: 180 seconds to complete.

d. CIP/ Maintenance Wash Equipment and Neutralization Equipment:
   1) Provide supplier's standard CIP / Maintenance Wash system capable of cleaning the UF membranes in-situ to assist with reducing the system's overall TMP.
   2) Automate CIP using chemical feed equipment as required for the system.
   3) Provide system capable of heating the cleaning solutions to 104 degrees F, consisting of:
      a) CIP Circulation Pump and CIP Hot Water Tank (either FRP or polyethylene).
      b) Inline CIP solution heater.
      c) Chemical metering pumps to transfer bulk chemicals as required for cleans.
      d) Dosing system controls, valves, pH probe, chlorine analyzer, and other instrumentation and controls.
   4) Maximum duration skid offline for maintenance wash cycle: 60 minutes.
   5) Maximum duration skid offline for a full CIP clean: 300 minutes with 60 minutes to neutralize the first CIP solution and discharge it to sewer.

e. A Compressed Air System, including air compressors, a receiver, and related valves, instruments and appurtenances.
   1) Include one duty and one standby compressor.

f. On-line Membrane Integrity Test System: Provide with anautomated membrane integrity testing that meets EPA LT2/MFGM requirements.

g. Minimum instrumentation components:
   1) Feed manifold pressure transmitter (one on each skid/rack).
   2) Feed turbidity transmitter (one panel mounted skid on combined feedwater).
   3) Skid flow rate transmitter (one per skid/rack).
   4) Filtrate turbidity transmitter (one per skid/rack).
   5) Other instrumentation as required for a fully functional and automatedsystem.

2. Equipment and Services provided by Contractor and/or not in UF supplier's scope.
   a. All other works and equipment necessary to complete the project, including but not limited to:
      1) Items as identified in the UF supplier's P&IDs.
      2) Construction of equipment pads, supports and grouting of equipment.
      3) Unloading, unpacking, storage, installation, assembly, and field erection of the system.
      4) Interconnecting piping between the UF skids, between the UF skids and the ancillary equipment being provided by the UF supplier, (including compressed air, and chemical cleaning systems), and between the UF skids or ancillary equipment being supplied by others.
         a) See UF supplier P&IDs for further information.
      5) Pipe supports for all piping connected to the UF skids.
      6) Supply and storage of all chemicals required for the membrane cleaning system, maintenance, and/or operation.
      7) Supply and installation of VFDs, motor control centers and disconnects.
      8) Supply and installation of control wiring, power cabling including cabletrays, conduits, fittings and supports as necessary.
      9) Instrumentation not specifically listed in the UF supplier's scope as specified herein.
      10) Lubricants.
      11) Supervision of installation.
      12) Supply and installation of anchor bolts.

2.4 PROJECT CHARACTERISTICS AND PERFORMANCE REQUIREMENTS

A. North Rim

1. Provide UF membrane system capable of meeting and incorporating the following raw and finished water characteristics:
   a. Feed Water Quality
      1) The raw water supply source is Roaring Springs Cave.
      2) A coagulation and sweep flocculation Reaction Tank provides pretreatment.
      3) During times of low turbidity and TOC levels in the raw water, the coagulant feed may be shut off. However, water would still pass through the Reaction Tank.
      4) Assume turbidity is not reduced through the pretreatment process since the system does not include actual sedimentation.
      5) The UF system will be operated at reduced flux/production rates during high raw water turbidity events.
      6) North Rim WTP UF Feed Water Quality

Parameter	Units	Range	Average	Design Max Value
Temperature	(deg C)		14.4	5 – 25
pH	(s.u.)	7.45 – 8.27	7.91	6.7 – 9.0
Turbidity	(NTU)	0.2 – 14.5	2.49	See Paragraph 2.4.A.1.b.1
Alkalinity	(mg/L $CaCO_3$)	73 - 180	152	230
TDS	(mg/L)	101 - 177	148	900
TOC	(mg/L)	1.00 – 4.61	1.52	7.0
Hardness	(mg/L $CaCO_3$)	74 - 170	147	240
Iron	(mg/L)	0.05 – 0.80	0.16	1.25
Manganese	(mg/L)	0.02 – 0.02	0.02	0.15
Chloride	(mg/L)	0.83 – 5.00	1.32	20
Sulfate	(mg/L)	1.49 – 5.00	2.13	30

   b. Filtrate Water Quality
      1) Permeate Turbidity: Less than or equal to 0.1 NTU 95% of the time, and less than 0.3 NTU 100% of the time.
      2) Average Permeate Particle Count: 5 particles larger than 2 microns per mL of filtrate water (as measured over a 24-hr period).
      3) Maximum Permeate Particle Count: 10 particles larger than 2 microns per mL of filtrate water (as measured over a 24-hr period).
      4) Minimum Cryptosporidium Removal: 3.0-log.
      5) Minimum Giardia Removal: 3.0-log.
2. Provide a UF membrane system meeting the following water productionand hydraulic design conditions:
   a. Design system to provide a variable maximum flux and production rate based on raw water turbidity.
      1) At a feed water turbidity of 10 NTU or less, design skid to have a nominal capacity of 100 GPM of filtrate through one independent duty skid (100 GPM firm capacity per UF skid).
      2) Each skid minimum permeate production rate: 20 GPM.
      3) Scaled water production will be provided between turbidity ranges of 10-200 NTU.
      4) Provide skids capable of operating independently of one another, including only one skid being in production at a time

5) Design such that one skid is capable of staying in production while the other is in backwash or cleaning cycles.

b. Water from the Reaction Tank will be fed to the UF system by gravity.
   1) The finished floor elevation of the UF area is at 8359.00 FT MSL, and the hydraulic grade line will be maintained above the feed pump impeller on the UF skids.

c. Design UF system to be capable of providing sufficient driving head to push filtrate through the membranes, downstream treatment processes and into the Chlorine Contact Tank.
   1) High water level of the Chlorine Contact Tank: approximately 8370.64 FT MSL (20 ft of static head on the filtrate side of the UF system).
   2) Assume an additional 25 FT of dynamic losses through piping, valves, and downstream treatment processes for a total required residual pressure on the discharge side of the UF system of approximately 45 FT.

3. Treatment System Operating Conditions
   a. Site Elevation: 8350 feet above Mean Sea Level.
   b. Maximum Instantaneous Flux Rate at 20 deg C: 35 gallons per square foot per day (GFD).
   c. UF Train Recovery: Greater than or equal to 85% as calculated overall 24 hour periods.
   d. Overall UF System Recovery: Greater than or equal to 95% as calculated overall 24 hour periods.
   e. CIP Frequency: 1 every 30 days (or less frequent);
   f. Maximum allowable trans membrane pressure (TMP) between CIP's: 90% of maximum allowable TMP as listed on the California Dept of Public Health list of Accepted Alternative Membrane Filtration Technologies for the membrane element.
   g. Maintenance Wash Frequency: 1 every 4 days (or less frequent).
   h. Maximum allowable TMP between maintenance washes: 85% of maximum allowable TMP as listed on the California Dept of Public Health list of Accepted Alternative Membrane Filtration Technologies for the membrane element.
4. Related Treatment Processes
   a. The North Rim WTP will use direct filtration on the raw water with an in-line flocculation/reaction tank providing ten minutes of sweep coagulation time, and with no sedimentation, prior to the UF system.
      1) Coagulant feed may be turned off when raw water turbidity is less than 10 NTU and TOC concentrations are less than 1.5 mg/L.
      2) Assume the majority of solids will be carried onto the membranes.
   b. Chemical Feed Systems: The following chemicals may or may not be fed to the process stream prior to the flocculation basins and/or subsequent membrane system:
      1) Sodium hypochlorite prior to the membrane system to maintain a total chlorine residual of up to 1.0 mg/L to help reduce biological fouling.
      2) 9 to 40 mg/L dose of Alum or Aluminum Chlorohydrate (ACH) fed prior to the Reaction Tank to provide coagulation.

2.5 GENERAL INTEGRATION AND COORDINATION REQUIREMENTS

A. General
   1. Provide, in coordination with the UF equipment supplier, a complete, fully functional UF treatment system and all appurtenances as shown and specified.
   2. Ensure that all UF treatment system equipment is rigidly and accurately anchored into position and all necessary foundation bolts, plates, nuts and washers shall be furnished and installed.
      a. Anchor bolt materials: Type 316 stainless steel.

B. UF control system
   1. Provide a single PLC/OIT in a shared control panel with remote I/O to skids using Ethernet.
      a. Mount directly to one of the skids.
      b. Enclosure: NEMA 4X.

1) Include air conditioning skid for operation in an area at 50 deg C.

c. Provide a flat panel operator interface panel or micro PC (with industrial monitor) on the primary control panel.
   1) OIT: Indusoft V8.1 SP4 (or latest) in a CE or IWS runtime.
   2) PLCs: Allen Bradley ControlLogix or CompactLogix, Modicon M580/M340 or Host H2-DM1E PLC families are allowed.

d. The UF control system will be brought into the NPS Indusoft HMI SCADA system by others and is not included in the contract.

e. Design UF PLC programs to be locked (Password to be provided to Government under secure means) so as to allow full access to code and data files for the Government's SCADA controls integrator work.

f. Design UF control system to be able to be controlled from either the local Master Control Panel or from HMIs in the Control Room.

g. Design UF control system to work together with SCADA systems (by others) to the extent practical.
   1) Coordinate UF system controls with the Government's SCADA controls integrator.
   2) Transfer UF control system files to the Government's integrator following factory testing.

2. Electrical Gear
   a. Submit information on Contractor supplied VFDs and motor control center to UF supplier for verification of system compatibility.
3. Data Exchange / Plant and UF System
   a. UF System to Plant
      1) UF system Ready (permissives cleared).
      2) UF system faults and alarms.
      3) Number of UF skids available for operation.
      4) Number of UF skids in Production.
4. UF System Permissives
   a. Verify water level in Reaction Tank is above Minimum Operating Level.
      1) Measure water level prior to placing the UF system in Forward Production from Standby.
   b. Verify water level in Chlorine Contact Tank is below High Level prior to placing the UF system in Forward Production from Standby.
   c. Upon UF system alarm that takes the system from Forward Production to Standby, Plant will go from On-line to Standby or otherwise takes the system out of service.

PART 3 - EXECUTION

3.1 INSPECTION

A. Check all components and equipment upon delivery to the site.
   1. Remove any components or equipment that are found to be damaged or defective from the site.

3.2 INSTALLATION

A. General Requirements
   1. Install UF membrane system, associated equipment, instrumentation, and appurtenances in strict accordance with manufacturer's written instructions and recommendations, and with the Contract Documents referenced herein.
   2. Provide the grades of oil and grease for all pumps in accordance with the recommendations of the manufacturer.
   3. Set anchor bolts in accordance with the manufacturer's approved shop drawings.

3.3 FACTORY TESTING AND INSPECTION

A. At the option of the Contracting Officer, the Contracting Officer and/or Contracting Officer's Representative may attend factory testing.

1. Provide Contracting Officer a minimum of two weeks notification prior to the start of testing.
2. All costs associated with the Contracting Officer and/or Contracting Officer's Representative attending factory testing will be borne by the Government.

B. Submit certified test reports to the Contracting Officer to document the performance and results of the factory testing.

3.4 FUNCTIONAL TEST

A. Once written certification of proper installation has been issued, Coordinate with UF supplier to perform functional testing of the UF system.
   1. The UF supplier is responsible for functional testing with assistance from the Contractor.

B. The UF supplier shall operate all valves, controls, and other devices to ensure they are functional and ready for performance testing.

C. Complete all functional testing to the satisfaction of the Contracting Officer prior to commencing the Initial Performance Test.

D. The purpose of the Functional Test is to demonstrate the effectiveness of thefollowing system components and features:
   1. Automatic start/stop and flow control of MF/UF trains using SCADA.
   2. Manual flow control using SCADA.
   3. Automatic cleaning of raw water strainers.
   4. Automatic backwashing under various programmable functions (time, volume, TMP).
   5. Automatic shutoff and alarm for various failure modes for each UF train and for entire UF System.
   6. Start and stop of air system.
   7. Automatic membrane integrity test system.
   8. Bubble point testing of the membrane modules, if not factory performed. If factory performed, submit documentation noting the testing and results.
   9. Determination of clean water permeability of each train and temperature correction of the clean water permeability for each train.
   10. Monitoring and recovery of operating data.
   11. All control functions.
   12. Operation of chemical cleaning system.
   13. Operation of all monitoring instruments.

E. Submit a testing protocol from UF supplier for the Functional Test a minimum of 7 days prior to the start of the test.
   1. Include a space next to each item on the protocol for the initials of the UF manufacturer's onsite representative, indicating each item has been satisfactorily completed.
   2. Submit a final, initialed copy of the protocol a minimum of 7 days prior to the Initial Performance Test.

3.5 INITIAL PERFORMANCE TEST

A. Following completion of the Functional Test and calibration of all instruments, conduct the Initial Performance Test (IPT).
   1. Provide UF supplier on-site time as needed, and at their discretion, to successfully complete the IPT,
   2. IPT is the responsibility of the UF supplier.

B. Substantial completion of the UF system shall occur upon successful completion of the IPT.

C. Provide the IPT Report from UF supplier within 3 weeks of completion of the test period.

D. To perform the test, operate the system continuously over a 14-day test period, and collect and summarize data to demonstrate that the system meets the performance test requirements for the parameters listed below.

1. Operation by UF supplier on site.

E. IPT testing: performed using chlorinated water from Roaring Springs source.
   1. In all cases, compliance with the requirements of the specification will be determined for each calendar day.
   2. To successfully pass the test, the UF system must comply with requirements for each of the 14 days of the IPT.
   3. Test Criteria:
      a. Production Capacity: System meets production capacity requirements when operated at or below the average flux rate.
      b. Recovery: System meets requirements for membrane system recovery.
      c. Pressure Limitations: System operates within the TMP limit that is specified, and system provides adequate pressure to discharge to the Chlorine Contact Tank.
      d. Average Flux Rate: Flux rate does not exceed the specified limits.
      e. Membrane Integrity Test System: Operate integrity test at a frequency of once every 24 hours to determine integrity of the membranes and proper functionality of the test system.
      f. Membrane Permeate Water Quality: Membrane permeate water meets requirements as specified.
         1) Particle count performance will be checked 2-3 times with water samples being sent to a commercial laboratory for confirmation of performance requirements.
      g. Maintenance Clean: Perform the maintenance clean operation at least once on each UF train during the 14-day test period.
      h. Clean-in-Place: Perform the clean-in-place operation at least once on each UF train during the 14-day test period.
      i. Clean Water Permeability: Following each cleaning operation, determine the clean water permeability, temperature-corrected to 20°C.
         1) The temperature-corrected clean water permeability of each MF/UF train must comply with the required specified value.
      j. Cleaning System Neutralization: Following maintenance and recovery chemical cleaning of each UF train, dechlorinate and/or neutralize the cleaning solutions as required.
         1) pH of the discharge: Between 6.5 and 9.0 at all times.
         2) Total chlorine concentration of the discharge: Less than 1.5 mg/L at all times.
      k. Energy Usage: Monitor average power consumption and maximum power demand for the UF system for the duration of the test period.

F. Criteria for successful completion of the IPT: 14 continuous days of operation without a major failure in the system equipment supplied by the UF supplier and demonstration that the UF system meets all performance requirements established herein.
   1. A major failure in the system is one that interrupts system operation for more than 4 hours.

G. If due to the equipment supplied by the UF supplier, the UF system fails to successfully complete the IPT, the UF supplier shall have the option of repeating the test over a second 14-day period.

H. If due to the equipment supplied by the UF supplier, the system fails to successfully complete the IPT during the second test period, submit written plan prepared by UF supplier for modifying the system to meet all test requirements.
   1. Submit the written plan within 7 days after the second unsuccessful test period ends.

I. If in the event the UF system has not been able to meet the performance requirements specified herein after the second test period, then the UF supplier can choose as its sole obligation to pay either of the following options:
   1. Liquidated Damages in the amount of 25% of the Contract Value.
   2. Expenses to continue to remedy the system with modifications up to the Contract Value.

J. During the IPT, Contracting Officer will have the option of collecting samples for independent analyses to confirm measurements and analyses conducted by the UF supplier.

K. The Contracting Officer has the option of witnessing all testing performed by the UF supplier.
   1. Notify Contracting Officer a minimum of two weeks in advance of testing.

3.6 DEMONSTRATION TESTING

A. Following successful completion of the IPT, Contractor may commence Demonstration testing for entire plant.
   1. See Section 01 75 00.

3.7 MONITORED TEST PERIOD

A. The Monitored Test Period (MTP) will begin and end at Contracting Officer's discretion based on date of Demonstration completion (see Section 01 75 00).

B. Compliance with the requirements for recovery, production capacity, chemical cleaning interval and membrane permeate water quality will be determined during the MTP.
   1. It is anticipated that either the Government will hire operators through or retain third party operators to operate the treatment facility during this time period.
   2. UF supplier shall be responsible for reviewing data and coordinating activities.
   3. UF supplier may provide onsite assistance at Contractor's and Supplier's discretion.

C. MTP duration: 90 days.
   1. Include a minimum of one (1) high flow period (100 gpm) as determined by Contracting Officer.
   2. Contracting Officer also reserves the right to confirm filtrate throughput capacity at the minimum specified temperature within one year of the Demonstration completion (if the minimum temperatures are not available during the 90 day MTP).
      a. Throughput capacity verification will be limited to one day.

D. Conduct MTP at average and maximum allowable flux rates specified.

E. The UF supplier to be responsible for monitoring the operating conditions and performance during the MTP.
   1. Remote monitoring of the system by the UF supplier will not be allowed.
   2. Government will download operational data and transfer to the UF supplier for review.
   3. Government may conduct their own monitoring and record keeping during the MTP.

F. Submit a summary of the data at the end of the MTP and a report of the results prepared by the UF supplier to the Contracting Officer.
   1. Include a narrative description, tables and graphs of flux, temperature corrected permeability, TMP versus time, energy use, recovery, CIP frequency, and other parameters to document the performance of thesystem.
   2. Submit the report within 30 days following the end of the MTP.

G. The following parameters will be evaluated to determine compliance with the UF supplier's stated performance. Performance Liquidated Damages that will be owed to Government by the UF supplier for system failure due to equipment supplied by UF supplier, are as described below:
   1. Membrane Filtrate Water Quality:
      a. If the system fails to comply with the requirements for membrane permeate water quality, provide to Contracting Officer a written plan of modifications to the UF system (such as, but not limited to, repairing damaged fibers, replacing seals, complete replacement of the membrane system) to achieve compliance with the requirements from the UF supplier.
      b. Upon implementation of the plan of modifications, recommence the MTP in its entirety.
   2. Recovery:
      a. Report membrane system recovery and UF train recovery to the nearest tenth of a percentage point.

b. For each one-tenth of a percentage point that the membrane system or train recovery for the duration of the MTP is below the specified recovery requirement, the UF supplier shall pay Government $5,000.
    1) This amount is based on feedwater pumping costs (capital and annualized energy costs), chemical usage and the cost of water.

3. CIP and Maintenance Wash Interval:
    a. If the average interval between chemical cleanings during the MTP is less than specified based on maximum allowable TMP, the UF supplier shall pay Government $2,000 for each additional chemical cleaning per year that will be needed.
        1) This amount accounts for the approximate present worth of the additional chemicals and labor over a 20-year period.
4. Production Capacity: Production capacity shall be reported in units of GPM.
    a. For the North Rim WTP, for each 2 GPM that the production capacity is less than the required production capacity the UF equipment supplier shall pay Government $5,000.
5. All Performance Liquidated Damages will be cumulative.
6. All Performance Liquidated Damages assessed against the UF supplier, in aggregate, will not exceed twenty-five (25) percent of the UF supplier's contract price.

3.8 ULTRAFILTRATION SYSTEM SUPPLIER'S FIELD SERVICES

A. Provide qualified Field Service personnel on site to perform the work identified in 3.4, 3.5 and 3.6.
    1. Provide additional days at no additional cost to Government if a test fails and needs to be repeated.
    2. Provide operator training – See Section 01 79 00 for requirements.

B. Contractor is responsible for coordinating with UF supplier and Government to make sure that required work items are complete and ready prior to service days provided by the UF supplier.

3.9 PERFORMANCE GUARANTEES AND WARRANTIES

A. See the attached warranty and performance guarantees at the end of this section.

B. See Section 3.6.

C. Unless otherwise noted, all the warranties start upon successful completion of the Initial Performance Test.

Warranty Item	**Warranty Provision**
Mechanical Warranty	
Warranty on Equipment	Twenty-four (24) months for all components except where otherwise specified.
Membrane Warranty	
Warranty on Membrane Modules	A ten (10) year warranty on membrane modules with the first thirty-six (36) months offered as full replacement warranty and the remaining eighty-four (84) months as a prorated warranty.
Capacity Guarantee	Design daily net capacity with all trains operational at a temperature ≥ 7°C: 100 GPM at North Rim WTP. Design daily net capacity with one (1) train out of service at a temperature of ≥ 7°C: 100 gpm for North Rim WTP.
Water Quality Guarantee	≤ 0.1 NTU 95% of the time.

Warranty Item	Warranty Provision
	≤ 0.3 NTU 100% of the time.
Membrane Integrity Warranty	Maximum of 0.3% repair events per module per year (as percentage of the total number of fibers per module). Maximum of 0.9% repair events per module over the warranty duration (as percentage of the total number of fibers per module).
Membrane Permeability Warranty	Minimum clean membrane permeability at 20°C: 1.67 GFD/PSI at North Rim WTP.
Performance Warranty	
General Performance Warranty	Applicable for the duration of the Initial Performance Test. Warranty to begin within thirty (30) days of completion of commissioning of the system.
Guaranteed Recovery	Design daily UF train recovery of ≥ 95% Design daily plant recovery of ≥ 95%
Cleaning Interval Guarantee	Maintenance clean interval: one (1) every 4 days Recovery clean interval: ≥ 30 days
Membrane Element Replacement Costs	
Membrane Element Price	$1,175 US dollars per element for a 20 year period from start date of the warranty, with price adjustment per the Consumer Price Index from time of Contractor's bid.

3.10 SPARE PARTS

A. UF Supplier shall provide the following spare parts identified in the attached table below:

END OF SECTION

This page intentionally left blank.

Item	Description	SIZE	CON. TYPE	Tag #	original # supplied	Part Number	Unit Type	Recommended QTY	Criticality Rating
PUMP	SPARE PROCESS PUMP, NO MOTOR						Pump only	1	C
PUMP	SPARE PROCESS PUMP, NO MOTOR						Complete Unit	1	C
INSTRUMENT	GAUGE-PRESS,SS,30PSI	1/4"	M-NPT				Complete Unit	1	C
INSTRUMENT	TRANS-FLOW,CI,PTFE	8"	FLG				Complete Unit	1	C
MEMBRANE REPAIR	BUBBLE/PRESSURE DECAY TEST KIT						Kit	1	R
INSTRUMENT	ANLZR-MULTI,230VAC,ATI,0-200PPM						Complete Unit	1	C
INSTRUMENT	TRANS,DUAL,4-20MA,100-240,CM442,E+H						Complete Unit	1	C
INSTRUMENT	ANLZR-TURBID						Complete Unit	1	C
AIR DRYER	AIR DRYER	3/4"	F-NPT				Complete Unit	1	R
INSTRUMENT	SWITCH-FLOW,BRS,24VDC,KO,3.3-66'/SEC	1/2"	M-NPT				Complete Unit	1	R
INSTRUMENT	TRANS-LEVEL,FMB50	1 1/2"	M-NPT				Complete Unit	1	R
PLC PROCESSOR	Allen Bradley				1			1	C
HMI	Allen Bradley				1			1	C
ETHERNET SWITCH					1			1	C

This page intentionally left blank.

SECTION 46 61 73
AUTOMATIC SELF-CLEANING FILTER

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Auto Filters - automatic self-cleaning screening filters.

B. Related Specification Sections include but are not necessarily limited to:
   1. Division 00 - Procurement and Contracting Requirements.
   2. Division 01 - General Requirements.
   3. Section 01 61 03 - Equipment - Basic Requirements.
   4. Section 03 31 30– Concrete Materials and Proportioning.
   5. Section 40 05 51 – Common Requirements for Process and Utility Valves.
   6. Section 40 05 57 – Actuators for Process Valves and Gates.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ANSI/ASME B16.1, Pipe Flanges and Flanged Fittings
   2. ANSI/ASME B1.20.1, American National Standard Tapered Pipe Thread fittings
   3. ASME Boiler and Pressure Vessel Code
   4. AWWA C207 Steel Pipe Flanges for Waterworks Service
   5. AWWA C228 Stainless Steel Pipe Flanges for Water Service

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. See Specification Section 01 61 03.
   3. Submit complete erection, installation, and operation and maintenance information provided by the manufacturer.
   4. Submit, as a minimum, complete construction details, materials of construction, drawings of the Auto Filter vessels, backwash pumping and piping system, electrically operated valves, control panel layout, and other pertinent information.
      a. Include dimensions of recommended clearances for routine maintenance and parts replacement
   5. Submit copy of report verifying completion of start-up and related field services.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

C. Informational Submittals:
   1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Design capabilities.
   3. Size, make and type of electric motors and drive systems.
   4. Information on piping and accessories supplied by manufacturer.
   5. Operation manual for Auto-Filter units matching selected model
   6. Operation manual for associated control panel
   7. List of spare parts to be included

1.4 CYBER SECURITY REQUIREMENTS

A. Refer to Section 01 81 33 for full requirements.

B. The supplier agrees to assist in effective cyber security practices as dictated by the Contracting Officer's cyber security team. This may include changing IP addresses, subnet masking, use of managed switches, firmware upgrades and other industry standard practices.

C. The Contracting Officer shall have full and unrestricted access to upgrade, modify, or change the software, networking, firmware, or other areas of micro-processor-based systems.

D. The ability of a supplier to remotely access the system will be severely restricted and will be allowed after a risk evaluation and only with Contracting Officer's prior permission on a case-by-case basis only, not a blanket allowance.

1.5 DELIVERY, STORAGE AND HANDLING

A. Submit with the Shop Drawings, any special instructions for unloading at the jobsite, handling, storage and routine maintenance prior to installation and start-up.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are now satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Tekleen Automatic Filters, LLC – Model ABW4-LP
   2. Or equal

2.2 EQUIPMENT

A. Service Conditions:
   1. Source of influent flow: Pressurized gravity line.
   2. Peak flow: 150 GPM through a 10 micron fine screen.

Parameter	Strainers
Location	Roaring Springs Pumphouse (Volume 5)
Quantity	Two
Type of Fluid	Raw Water
Fluid pH	6 to 8
Filter Element Rating	10 micron
Inlet and Outlet Diameters	4-inch diameter
Required Operating Flow	150 gpm per filter, not concurrently
Typical Operating Pressure	145 psig
Maximum Operating/Test Pressure	200 psig
Upstream Pressure	150 psig
Max. Operating Temperature	105 deg. Fahrenheit
Max. Operating Pressure Drop Across Clean Strainer	0.5 psig @ 150 gpm

3. Average flow: 150 GPM each filter.
4. Minimum flow: 50 GPM each filter.
5. Maximum flow: 250 each filter
6. Influent TSS level:
   a. 20 MG/L average.

B. Performance:
1. Maximum effluent particle size equal to filter element micron rating when Auto Filter is operating at a differential pressure drop of 7 PSID, excluding hydraulic pressure differential resulting from clean water flow

C. General Description:
1. Auto Filters shall be straight-flow, automatic, self-cleaning, hydraulically actuated, and equipped for cleaning using internal suction and raking mechanism.
2. Auto Filters shall be suitable for use with raw cave water
3. Auto Filter units preassembled into, tested, and shipped as a skid assembly with common inlet and outlet connections.

D. Dimensions:
1. Maximum skid length: 60-IN
2. Maximum skid width: 40-IN
3. Maximum filter unit length: 60-IN
4. Maximum filter unit cylinder diameter: 12.75-IN
5. Maximum filter unit service area length: 35-IN

2.3 COMPONENTS

A. Body and Cover:
1. 316/316L Stainless Steel, constructed to withstand the greater of the specified maximum working pressure or 150 psig.

B. Piping Connections:
1. For principal connections to process fluid inlet and discharge, provide inlet and outlet flanges in accordance with Class 150 ANSI B16.5, flat face. Size inlet and outlet connections as per Drawings.
2. Provide a minimum 1.5-IN nominal diameter flushing discharge/drain connection in accordance with ANSI B1.20.1 (NPT). Provide threaded bushing or reducer as needed. Furnish 1.5-IN electrically-actuated ball valve for installation in backwash wastewater discharge line; electrically-actuated ball valve shall conform to Section 40 05 63 – Ball Valves.
   a. Connection to ball valve shall be on the same side of the filter unit cylinder body and follow the same angular alignment as the inlet and outlet connections.
   b. Ball valve shall be powered and electronically operated solely by manufacturer-supplied control panel
   c. Ball valve shall be rated for a minimum of 200 psig
   d. Ball valve actuator shall be capable of manual operation via manual lever, hand wheel, or crank mechanism, even if no power is applied.
   e. Ball valve actuator shall have visual indicator or valve's OPEN/CLOSE operating state
   f. Ball valve construction shall be type 304/304L or 316/316L stainless steel
3. Provide integral connections not less than 1/4-IN NPT at inlet and outlet for measuring pressure across each Auto-Filter vessel.
   a. Provide stainless steel isolating valves at these connection points for flushing and servicing pressure indicator(s) and associated piping.
4. Provide vent, not less than 1/4-IN NPT-M, in strainer cover. Vent plug shall be stainless steel.

C. Seals and O-rings shall be Buna-N.

D. Filter Elements:
   1. Coarse/pre-strainer screen shall be Type 316/316L stainless steel
   2. Filtering element shall be Type 316/316L stainless steel with uniformly spaced 10 micron openings.
   3. The filtering element shall be fabricated to allow for maximum open area and strength.

E. Cleaning Mechanism
   1. The filter cleaning mechanism shall provide a controlled spiral path for the suction scanner nozzles across the inside surface of the filtering screen.
   2. The opening of the backwash flush valve shall create suction forces at each of the 4 suction scanner nozzles creating a reversed flow through the filtering screen and into the nozzle orifice.
   3. The reversed flow through the filtering screen shall force the contaminants from the filtering screen, back through the suction scanner nozzles and out the flush valve.
   4. The surface area of each suction scanner nozzle shall not exceed 0.20 square inches to provide minimal flush waste.
   5. The optimum distance of the suction scanner nozzle to the screen shall be fixed by the manufacturer and shall not be adjustable to prevent the reduction of manufacturer's designed cleaning performance.
   6. The flush flow shall not exceed 160 GPM at 105 PSIG.
   7. Each cleaning cycle duration shall be user-configurable from 10 to at least 120 seconds.
   8. The minimum pressure required shall be 40 PISG to assure high cleaning efficiency.
   9. The cleaning mechanism assembly shall be constructed of 316/316L stainless steel and glass filled Nylon.
   10. The suction scanner nozzle assembly shall be constructed of CPVC.

F. Drive Unit:
   1. The suction scanner shall be driven by an internal Bidirectional Hydrodynamic Drive (BHD) that is directly attached to the suction scanner.
   2. The BHD shall utilize the flush waste flow during the flush process to provide forces that rotate the suction scanner.
   3. The suction scanner shall rotate on a double axial spline that allows the suction scanner nozzles to traverse across the screen and return to their start position without interrupting the flush process and without the aid of an external return mechanism such as electric motors, limit switches and/or hydraulically actuated pistons.
   4. Each suction scanner nozzle's rotation shall overlap with its prior rotation by no less than 10%.
   5. The patented drive mechanism shall include a pressure balance line. This pressure balance line shall balance forces on each end of the suction scanner to reduce forces on the drive mechanism, thereby minimizing mechanical wear and excessive flush waste.

G. Skid Assembly
   1. Holds two filter units in vertical stack arrangement.
      a. Filter braces and all structural components designed to meet seismic requirements
      b. Filter unit attachments capable of being field-disassembled and reassembled for servicing and filter unit replacement
      c. Provide dedicated strap and/or lift points for lifting at site by crane or helicopter
   2. All piping filter unit piping connections on one side of skid
   3. Inlet and Outlet connections
      a. Flanged, in conformance with ASME B16.5, Class 150 flat face
      b. Along the short sides of skid
   4. Inlet and Outlet manifolds
      a. Orient in accordance with Drawings
      b. Stainless steel Type 304/304L or 316/316L
      c. Supported with integral pipe supports
      d. Capable of being field-disassembled and reassembled for servicing
         1) Do not affix piping to skid via welds or other permanent connections

e. Include manual bypass butterfly valve between Inlet and Outlet manifolds
    a) Rated for skid assembly design pressure
    b) Geared handwheel operator, shall not protrude beyond extents of skid footprint
f. Include butterfly isolation valves for inlet and outlet connections to each filter unit
    1) In conformance with 40 05 64
    2) Wafer or fully-lugged style
    3) Lever handle operator
    4) Tagged in accordance with process and instrumentation Drawings
g. Flush outlet manifold
    1) Provided by Contractor in accordance with Drawings

H. Control System:
1. The filter control system shall be provided by the filter manufacturer.
2. Provide one (1) filter control system for all Auto Filter vessels.
3. Control system not mounted on skid. Mounted on wall per Drawings.
4. The filter control system shall control all aspects of each filter's operation, including but not limited to flushing by:
    a. Differential pressure (DP) Only (Default)
    b. Time & DP (DP will always override Time if there is a DP flush is required.)
    c. Continuous flush
    d. Manual flush
5. Control interface:
    a. Digital backlit display indicating, but not limited to: status, selection, input/output, pressures, alarms
    b. Interaction with backlit display via arrow keys, alphaneumeric keys, or touch screen display
6. Signals
    a. Discrete outputs:
        1) Flushing valve OPEN or CLOSED – 120VAC
        2) General FAULT or ALARM – 120VAC
7. Power service shall be 120V 60Hz 1-PH input fed by 20 Amp breaker
8. Construction
    a. NEMA 3S enclosure
    b. Waterproof connections for all other electrical components of manufacturer's system
9. Controls – General:
    a. Regardless of operating mode, when flushing cycle is initiated, the following shall occur:
        1) Control system will enable flushing valve to OPEN
        2) Pressure differential between vessel inlet and discharge will initiate suction of accumulated solids from inside face of the fine filter and transport them through suction nozzles to the drain port
        3) Hydraulic drive pressure will initiate rotation of the suction nozzle assembly about the vessel's long axis
        4) Suction nozzles shall move backwards and forwards along length of fine filter element such that entirety of the filter element is uniformly contacted by the suction nozzles during a flushing cycle
        5) Upon closing of the flushing valve, Auto Filter will return to regular forward filtration
    b. Flushing Valve Control
        1) To avoid hydraulic shock to Non-Potable Water system, flushing valve actuator OPEN and CLOSE times shall be independent user-selectable settings within the filter control system
        2) Manufacturer shall state minimum recommended OPEN and CLOSE times in operating manual as pertaining to preferred operation of the Auto Filters
10. Automatic Mode:

a. In automatic mode, operation of an Auto Filter's cleaning systems shall be intermittent, controlled by either adjustable cycle timer or manufacturer-provided differential pressure element that measures across each vessel.
b. When facility operator selects automatic control via timer, the Auto Filter shall self-clean automatically at time interval field-selected by facility operations and maintenance personnel at the filter control system panel.
    1) If a DP flush is required, it will always override the Time setting when operating in this mode.
    2) In this occurrence, timer will reset
c. When facility operator selects automatic control via differential pressure, the Auto Filter will self-clean automatically when pressure drop across the strainer element, as measured by differential pressure switch provided with each strainer unit, exceeds the differential pressure setpoint. Differential pressure setpoint shall be initially factory-set at differential pressure recommended by manufacturer, and shall be field-adjustable by facility operations and maintenance personnel.
d. At no point must the filter control system be allowed to automatically signal both Auto Filters to go into a flushing cycle at the same time.

11. Manual Mode:
    a. When manual mode is selected at the control panel by facility operations and maintenance personnel, filter self-cleaning will be initiated and performed when manually initiated at the filter system control panel.
    b. In manual mode, Auto Filter cleaning shall be initiated and stopped by manual command at the filter system control panel.
    c. Manual mode shall allow continuous backwashing, as required by facility operations and maintenance personnel, during commissioning or troubleshooting.

2.4 OPERATIONAL SEQUENCES

A. Operation Sequence: Manufacturer's standard.

B. Manual Operation:
    1. Control panel shall provide for manual control of all controlled components, including valves.

C. See control loop descriptions for the interfacing with other components.

2.5 MAINTENANCE MATERIALS

A. Extra Materials:
    1. Furnish the Contract Officer the following spare parts:
        a. One (1) fine filter element
        b. Two (2) cover seals
        c. One (1) lower bulkhead seal washer
        d. Two (2) internal assembly O-rings
        e. One (1) pressure instrument isolation valve
        f. One (1) flushing Ball Valve

PART 3 - EXECUTION

3.1 INSTALLATION

A. See Specification Section 01 61 03.

B. Install the filter equipment and related components according to manufacturer's recommendations.

3.2 FIELD QUALITY CONTROL

A. Prior to start-up, inspect all equipment including electrically operated valves and Auto Filters for proper operation, lubrication, quiet operation and satisfactory performance.

B. Verify operation of controls with the controlled components (valves, etc.) prior to start-up.
   1. Include verifications of all control points.

C. Employ and pay for services of equipment manufacturer's field service representative to:
   1. Inspect equipment covered by this Specification Section.
   2. Supervise adjustments and installation checks.
   3. Conduct start-up and operational checks.
   4. Provide Contract Officer with a written statement that manufacturer's equipment has been installed properly, started up, and is ready for operation by Contract Offcier's personnel.
   5. Provide services of field representative for a period of 8 man-hours to perform the work items described above and to instruct Contract Officer's personnel in operation of the equipment.

D. Operational Test:
   1. Fill the Auto Filter with finished water completely
   2. Operate the unit in filter mode and flushing mode
   3. Following these operational modes, allow the unit to operate normally in the auto mode for a period of 48 HRS.
   4. Test unit flushing mode with source water under field conditions.
   5. If any problems occur during this testing period, make the necessary adjustments or modifications and repeat the test.

END OF SECTION

This page intentionally left blank.